Undergraduate Texts in Mathematics

UTM

Sheldon Axler

# Linear Algebra Done Right

*Fourth Edition*

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# Undergraduate Texts in Mathematics

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**Undergraduate Texts in Mathematics** are generally aimed at third- and fourth-year undergraduate mathematics students at North American universities. These texts strive to provide students and teachers with new perspectives and novel approaches. The books include motivation that guides the reader to an appreciation of interrelations among different aspects of the subject. They feature examples that illustrate key concepts as well as exercises that strengthen understanding.

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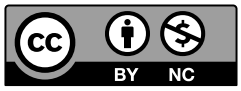
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# About the Author

Sheldon Axler received his undergraduate degree from Princeton University, followed by a PhD in mathematics from the University of California at Berkeley.

As a postdoctoral Moore Instructor at MIT, Axler received a university-wide teaching award. He was then an assistant professor, associate professor, and professor at Michigan State University, where he received the first J. Sutherland Frame Teaching Award and the Distinguished Faculty Award.

Axler received the Lester R. Ford Award for expository writing from the Mathematical Association of America in 1996, for a paper that eventually expanded into this book. In addition to publishing numerous research papers, he is the author of six mathematics textbooks, ranging from freshman to graduate level. Previous editions of this book have been adopted as a textbook at over 375 universities and colleges and have been translated into three languages.

Axler has served as Editor-in-Chief of the *Mathematical Intelligencer* and Associate Editor of the *American Mathematical Monthly*. He has been a member of the Council of the American Mathematical Society and of the Board of Trustees of the Mathematical Sciences Research Institute. He has also served on the editorial board of Springer's series Undergraduate Texts in Mathematics, Graduate Texts in Mathematics, Universitext, and Springer Monographs in Mathematics.

Axler is a Fellow of the American Mathematical Society and has been a recipient of numerous grants from the National Science Foundation.

Axler joined San Francisco State University as chair of the Mathematics Department in 1997. He served as dean of the College of Science & Engineering from 2002 to 2015, when he returned to a regular faculty appointment as a professor in the Mathematics Department.



Carrie Heeter, Bishnu Sarangi

*The author and his cat Moon.*

*Cover equation*: Formula for the $n^{\text{th}}$ Fibonacci number. Exercise 21 in Section 5D uses linear algebra to derive this formula.



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# *Preface for Students*

You are probably about to begin your second exposure to linear algebra. Unlike your first brush with the subject, which probably emphasized Euclidean spaces and matrices, this encounter will focus on abstract vector spaces and linear maps. These terms will be defined later, so don't worry if you do not know what they mean. This book starts from the beginning of the subject, assuming no knowledge of linear algebra. The key point is that you are about to immerse yourself in serious mathematics, with an emphasis on attaining a deep understanding of the definitions, theorems, and proofs.

You cannot read mathematics the way you read a novel. If you zip through a page in less than an hour, you are probably going too fast. When you encounter the phrase "as you should verify", you should indeed do the verification, which will usually require some writing on your part. When steps are left out, you need to supply the missing pieces. You should ponder and internalize each definition. For each theorem, you should seek examples to show why each hypothesis is necessary. Discussions with other students should help.

As a visual aid, definitions are in yellow boxes and theorems are in blue boxes (in color versions of the book). Each theorem has an infomal descriptive name.

Please check the website below for additional information about the book, including a link to videos that are freely available to accompany the book.

Your suggestions, comments, and corrections are most welcome.

Best wishes for success and enjoyment in learning linear algebra!

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# *Preface for Instructors*

You are about to teach a course that will probably give students their second exposure to linear algebra. During their first brush with the subject, your students probably worked with Euclidean spaces and matrices. In contrast, this course will emphasize abstract vector spaces and linear maps.

The title of this book deserves an explanation. Most linear algebra textbooks use determinants to prove that every linear operator on a finite-dimensional complex vector space has an eigenvalue. Determinants are difficult, nonintuitive, and often defined without motivation. To prove the theorem about existence of eigenvalues on complex vector spaces, most books must define determinants, prove that a linear operator is not invertible if and only if its determinant equals 0, and then define the characteristic polynomial. This tortuous (torturous?) path gives students little feeling for why eigenvalues exist.

In contrast, the simple determinant-free proofs presented here (for example, see 5.19) offer more insight. Once determinants have been moved to the end of the book, a new route opens to the main goal of linear algebra—understanding the structure of linear operators.

This book starts at the beginning of the subject, with no prerequisites other than the usual demand for suitable mathematical maturity. A few examples and exercises involve calculus concepts such as continuity, differentiation, and integration. You can easily skip those examples and exercises if your students have not had calculus. If your students have had calculus, then those examples and exercises can enrich their experience by showing connections between different parts of mathematics.

Even if your students have already seen some of the material in the first few chapters, they may be unaccustomed to working exercises of the type presented here, most of which require an understanding of proofs.

Here is a chapter-by-chapter summary of the highlights of the book:

- Chapter 1: Vector spaces are defined in this chapter, and their basic properties are developed.
- Chapter 2: Linear independence, span, basis, and dimension are defined in this chapter, which presents the basic theory of finite-dimensional vector spaces.
- Chapter 3: This chapter introduces linear maps. The key result here is the fundamental theorem of linear maps: if $T$ is a linear map on $V$, then $\dim V = \dim \operatorname{null} T + \dim \operatorname{range} T$. Quotient spaces and duality are topics in this chapter at a higher level of abstraction than most of the book; these topics can be skipped (except that duality is needed for tensor products in Section 9D).



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- Chapter 4: The part of the theory of polynomials that will be needed to understand linear operators is presented in this chapter. This chapter contains no linear algebra. It can be covered quickly, especially if your students are already familiar with these results.
- Chapter 5: The idea of studying a linear operator by restricting it to small subspaces leads to eigenvectors in the early part of this chapter. The highlight of this chapter is a simple proof that on complex vector spaces, eigenvalues always exist. This result is then used to show that each linear operator on a complex vector space has an upper-triangular matrix with respect to some basis. The minimal polynomial plays an important role here and later in the book. For example, this chapter gives a characterization of the diagonalizable operators in terms of the minimal polynomial. Section 5E can be skipped if you want to save some time.
- Chapter 6: Inner product spaces are defined in this chapter, and their basic properties are developed along with tools such as orthonormal bases and the Gram–Schmidt procedure. This chapter also shows how orthogonal projections can be used to solve certain minimization problems. The pseudoinverse is then introduced as a useful tool when the inverse does not exist. The material on the pseudoinverse can be skipped if you want to save some time.
- Chapter 7: The spectral theorem, which characterizes the linear operators for which there exists an orthonormal basis consisting of eigenvectors, is one of the highlights of this book. The work in earlier chapters pays off here with especially simple proofs. This chapter also deals with positive operators, isometries, unitary operators, matrix factorizations, and especially the singular value decomposition, which leads to the polar decomposition and norms of linear maps.
- Chapter 8: This chapter shows that for each operator on a complex vector space, there is a basis of the vector space consisting of generalized eigenvectors of the operator. Then the generalized eigenspace decomposition describes a linear operator on a complex vector space. The multiplicity of an eigenvalue is defined as the dimension of the corresponding generalized eigenspace. These tools are used to prove that every invertible linear operator on a complex vector space has a square root. Then the chapter gives a proof that every linear operator on a complex vector space can be put into Jordan form. The chapter concludes with an investigation of the trace of operators.
- Chapter 9: This chapter begins by looking at bilinear forms and showing that the vector space of bilinear forms is the direct sum of the subspaces of symmetric bilinear forms and alternating bilinear forms. Then quadratic forms are diagonalized. Moving to multilinear forms, the chapter shows that the subspace of alternating $n$-linear forms on an $n$-dimensional vector space has dimension one. This result leads to a clean basis-free definition of the determinant of an operator. For complex vector spaces, the determinant turns out to equal the product of the eigenvalues, with each eigenvalue included in the product as many times as its multiplicity. The chapter concludes with an introduction to tensor products.

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This book usually develops linear algebra simultaneously for real and complex vector spaces by letting $\mathbf{F}$ denote either the real or the complex numbers. If you and your students prefer to think of $\mathbf{F}$ as an arbitrary field, then see the comments at the end of Section 1A. I prefer avoiding arbitrary fields at this level because they introduce extra abstraction without leading to any new linear algebra. Also, students are more comfortable thinking of polynomials as functions instead of the more formal objects needed for polynomials with coefficients in finite fields. Finally, even if the beginning part of the theory were developed with arbitrary fields, inner product spaces would push consideration back to just real and complex vector spaces.

You probably cannot cover everything in this book in one semester. Going through all the material in the first seven or eight chapters during a one-semester course may require a rapid pace. If you must reach Chapter 9, then consider skipping the material on quotient spaces in Section 3E, skipping Section 3F on duality (unless you intend to cover tensor products in Section 9D), covering Chapter 4 on polynomials in a half hour, skipping Section 5E on commuting operators, and skipping the subsection in Section 6C on the pseudoinverse.

A goal more important than teaching any particular theorem is to develop in students the ability to understand and manipulate the objects of linear algebra. Mathematics can be learned only by doing. Fortunately, linear algebra has many good homework exercises. When teaching this course, during each class I usually assign as homework several of the exercises, due the next class. Going over the homework might take up significant time in a typical class.

Some of the exercises are intended to lead curious students into important topics beyond what might usually be included in a basic second course in linear algebra.

### The author's top ten

Listed below are the author's ten favorite results in the book, in order of their appearance in the book. Students who leave your course with a good understanding of these crucial results will have an excellent foundation in linear algebra.

- any two bases of a vector space have the same length (2.34)
- fundamental theorem of linear maps (3.21)
- existence of eigenvalues if $\mathbf{F} = \mathbf{C}$ (5.19)
- upper-triangular form always exists if $\mathbf{F} = \mathbf{C}$ (5.47)
- Cauchy–Schwarz inequality (6.14)
- Gram–Schmidt procedure (6.32)
- spectral theorem (7.29 and 7.31)
- singular value decomposition (7.70)
- generalized eigenspace decomposition theorem when $\mathbf{F} = \mathbf{C}$ (8.22)
- dimension of alternating $n$-linear forms on $V$ is 1 if $\dim V = n$ (9.37)

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### Major improvements and additions for the fourth edition

- Over 250 new exercises and over 70 new examples.
- Increasing use of the minimal polynomial to provide cleaner proofs of multiple results, including necessary and sufficient conditions for an operator to have an upper-triangular matrix with respect to some basis (see Section 5C), necessary and sufficient conditions for diagonalizability (see Section 5D), and the real spectral theorem (see Section 7B).
- New section on commuting operators (see Section 5E).
- New subsection on pseudoinverse (see Section 6C).
- New subsections on QR factorization/Cholesky factorization (see Section 7D).
- Singular value decomposition now done for linear maps from an inner product space to another (possibly different) inner product space, rather than only dealing with linear operators from an inner product space to itself (see Section 7E).
- Polar decomposition now proved from singular value decomposition, rather than in the opposite order; this has led to cleaner proofs of both the singular value decomposition (see Section 7E) and the polar decomposition (see Section 7F).
- New subsection on norms of linear maps on finite-dimensional inner product spaces, using the singular value decomposition to avoid even mentioning supremum in the definition of the norm of a linear map (see Section 7F).
- New subsection on approximation by linear maps with lower-dimensional range (see Section 7F).
- New elementary proof of the important result that if $T$ is an operator on a finite-dimensional complex vector space $V$, then there exists a basis of $V$ consisting of generalized eigenvectors of $T$ (see 8.9).
- New Chapter 9 on multilinear algebra, including bilinear forms, quadratic forms, multilinear forms, and tensor products. Determinants now are defined using a basis-free approach via alternating multilinear forms.
- New formatting to improve the student-friendly appearance of the book. For example, the definition and result boxes now have rounded corners instead of right-angle corners, for a gentler look. The main font size has been reduced from 11 point to 10.5 point.

Please check the website below for additional links and information about the book. Your suggestions, comments, and corrections are most welcome.

Best wishes for teaching a successful linear algebra class!

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# *Acknowledgments*

I owe a huge intellectual debt to all the mathematicians who created linear algebra over the past two centuries. The results in this book belong to the common heritage of mathematics. A special case of a theorem may first have been proved long ago, then sharpened and improved by many mathematicians in different time periods. Bestowing proper credit on all contributors would be a difficult task that I have not undertaken. In no case should the reader assume that any result presented here represents my original contribution.

Many people helped make this a better book. The three previous editions of this book were used as a textbook at over 375 universities and colleges around the world. I received thousands of suggestions and comments from faculty and students who used the book. Many of those suggestions led to improvements in this edition. The manuscript for this fourth edition was class tested at 30 universities. I am extremely grateful for the useful feedback that I received from faculty and students during this class testing.

The long list of people who should be thanked for their suggestions would fill up many pages. Lists are boring to read. Thus to represent all contributors to this edition, I will mention only Noel Hinton, a graduate student at Australian National University, who sent me more suggestions and corrections for this fourth edition than anyone else. To everyone who contributed suggestions, let me say how truly grateful I am to all of you. Many many thanks!

I thank Springer for providing me with help when I needed it and for allowing me the freedom to make the final decisions about the content and appearance of this book. Special thanks to the two terrific mathematics editors at Springer who worked with me on this project—Loretta Bartolini during the first half of my work on the fourth edition, and Elizabeth Loew during the second half of my work on the fourth edition. I am deeply indebted to David Kramer, who did a magnificent job of copyediting and prevented me from making many mistakes.

Extra special thanks to my fantastic partner Carrie Heeter. Her understanding and encouragement enabled me to work intensely on this new edition. Our wonderful cat Moon, whose picture appears on the *About the Author* page, provided sweet breaks throughout the writing process. Moon died suddenly due to a blood clot as this book was being finished. We are grateful for five precious years with him.

Sheldon Axler

Chapter 1

# *Vector Spaces*

Linear algebra is the study of linear maps on finite-dimensional vector spaces. Eventually we will learn what all these terms mean. In this chapter we will define vector spaces and discuss their elementary properties.

In linear algebra, better theorems and more insight emerge if complex numbers are investigated along with real numbers. Thus we will begin by introducing the complex numbers and their basic properties.

We will generalize the examples of a plane and of ordinary space to $\mathbf{R}^n$ and $\mathbf{C}^n$, which we then will generalize to the notion of a vector space. As we will see, a vector space is a set with operations of addition and scalar multiplication that satisfy natural algebraic properties.

Then our next topic will be subspaces, which play a role for vector spaces analogous to the role played by subsets for sets. Finally, we will look at sums of subspaces (analogous to unions of subsets) and direct sums of subspaces (analogous to unions of disjoint sets).



Pierre Louis Dumesnil, Nils Forsberg

*René Descartes explaining his work to Queen Christina of Sweden. Vector spaces are a generalization of the description of a plane using two coordinates, as published by Descartes in 1637.*

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# 1A $\mathbf{R}^n$ and $\mathbf{C}^n$

## Complex Numbers

You should already be familiar with basic properties of the set $\mathbf{R}$ of real numbers. Complex numbers were invented so that we can take square roots of negative numbers. The idea is to assume we have a square root of $-1$, denoted by $i$, that obeys the usual rules of arithmetic. Here are the formal definitions.

**1.1 definition: *complex numbers*, $\mathbf{C}$**

- A *complex number* is an ordered pair $(a, b)$, where $a, b \in \mathbf{R}$, but we will write this as $a + bi$.
- The set of all complex numbers is denoted by $\mathbf{C}$:

$$\mathbf{C} = \{a + bi : a, b \in \mathbf{R}\}.$$

- *Addition* and *multiplication* on $\mathbf{C}$ are defined by

$$(a + bi) + (c + di) = (a + c) + (b + d)i,$$
$$(a + bi)(c + di) = (ac - bd) + (ad + bc)i;$$

here $a, b, c, d \in \mathbf{R}$.

If $a \in \mathbf{R}$, we identify $a + 0i$ with the real number $a$. Thus we think of $\mathbf{R}$ as a subset of $\mathbf{C}$. We usually write $0 + bi$ as just $bi$, and we usually write $0 + 1i$ as just $i$.

*The symbol i was first used to denote $\sqrt{-1}$ by Leonhard Euler in 1777.*

To motivate the definition of complex multiplication given above, pretend that we knew that $i^2 = -1$ and then use the usual rules of arithmetic to derive the formula above for the product of two complex numbers. Then use that formula to verify that we indeed have

$$i^2 = -1.$$

Do not memorize the formula for the product of two complex numbers—you can always rederive it by recalling that $i^2 = -1$ and then using the usual rules of arithmetic (as given by 1.3). The next example illustrates this procedure.

**1.2 example: *complex arithmetic***

The product $(2 + 3i)(4 + 5i)$ can be evaluated by applying the distributive and commutative properties from 1.3:

$$\begin{aligned}(2 + 3i)(4 + 5i) &= 2 \cdot (4 + 5i) + (3i)(4 + 5i)\\ &= 2 \cdot 4 + 2 \cdot 5i + 3i \cdot 4 + (3i)(5i)\\ &= 8 + 10i + 12i - 15\\ &= -7 + 22i.\end{aligned}$$

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Our first result states that complex addition and complex multiplication have the familiar properties that we expect.

1.3 *properties of complex arithmetic*

**commutativity**
$\alpha + \beta = \beta + \alpha$ and $\alpha\beta = \beta\alpha$ for all $\alpha, \beta \in \mathbf{C}$.

**associativity**
$(\alpha + \beta) + \lambda = \alpha + (\beta + \lambda)$ and $(\alpha\beta)\lambda = \alpha(\beta\lambda)$ for all $\alpha, \beta, \lambda \in \mathbf{C}$.

**identities**
$\lambda + 0 = \lambda$ and $\lambda 1 = \lambda$ for all $\lambda \in \mathbf{C}$.

**additive inverse**
For every $\alpha \in \mathbf{C}$, there exists a unique $\beta \in \mathbf{C}$ such that $\alpha + \beta = 0$.

**multiplicative inverse**
For every $\alpha \in \mathbf{C}$ with $\alpha \neq 0$, there exists a unique $\beta \in \mathbf{C}$ such that $\alpha\beta = 1$.

**distributive property**
$\lambda(\alpha + \beta) = \lambda\alpha + \lambda\beta$ for all $\lambda, \alpha, \beta \in \mathbf{C}$.

The properties above are proved using the familiar properties of real numbers and the definitions of complex addition and multiplication. The next example shows how commutativity of complex multiplication is proved. Proofs of the other properties above are left as exercises.

1.4 example: *commutativity of complex multiplication*

To show that $\alpha\beta = \beta\alpha$ for all $\alpha, \beta \in \mathbf{C}$, suppose

$$\alpha = a + bi \quad \text{and} \quad \beta = c + di,$$

where $a, b, c, d \in \mathbf{R}$. Then the definition of multiplication of complex numbers shows that

$$\begin{aligned}\alpha\beta &= (a + bi)(c + di)\\ &= (ac - bd) + (ad + bc)i\end{aligned}$$

and

$$\begin{aligned}\beta\alpha &= (c + di)(a + bi)\\ &= (ca - db) + (cb + da)i.\end{aligned}$$

The equations above and the commutativity of multiplication and addition of real numbers show that $\alpha\beta = \beta\alpha$.

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Next, we define the additive and multiplicative inverses of complex numbers, and then use those inverses to define subtraction and division operations with complex numbers.

**1.5 definition: $-\alpha$, *subtraction*, $1/\alpha$, *division***

Suppose $\alpha, \beta \in \mathbf{C}$.

- Let $-\alpha$ denote the additive inverse of $\alpha$. Thus $-\alpha$ is the unique complex number such that
$$\alpha + (-\alpha) = 0.$$
- *Subtraction* on $\mathbf{C}$ is defined by
$$\beta - \alpha = \beta + (-\alpha).$$
- For $\alpha \neq 0$, let $1/\alpha$ and $\frac{1}{\alpha}$ denote the multiplicative inverse of $\alpha$. Thus $1/\alpha$ is the unique complex number such that
$$\alpha(1/\alpha) = 1.$$
- For $\alpha \neq 0$, *division* by $\alpha$ is defined by
$$\beta/\alpha = \beta(1/\alpha).$$

So that we can conveniently make definitions and prove theorems that apply to both real and complex numbers, we adopt the following notation.

**1.6 notation: $\mathbf{F}$**

Throughout this book, $\mathbf{F}$ stands for either $\mathbf{R}$ or $\mathbf{C}$.

Thus if we prove a theorem involving $\mathbf{F}$, we will know that it holds when $\mathbf{F}$ is replaced with $\mathbf{R}$ and when $\mathbf{F}$ is replaced with $\mathbf{C}$.

*The letter* $\mathbf{F}$ *is used because* $\mathbf{R}$ *and* $\mathbf{C}$ *are examples of what are called* ***fields****.*

Elements of $\mathbf{F}$ are called *scalars*. The word "scalar" (which is just a fancy word for "number") is often used when we want to emphasize that an object is a number, as opposed to a vector (vectors will be defined soon).

For $\alpha \in \mathbf{F}$ and $m$ a positive integer, we define $\alpha^m$ to denote the product of $\alpha$ with itself $m$ times:
$$\alpha^m = \underbrace{\alpha \cdots \alpha}_{m \text{ times}}.$$
This definition implies that
$$(\alpha^m)^n = \alpha^{mn} \quad \text{and} \quad (\alpha\beta)^m = \alpha^m \beta^m$$
for all $\alpha, \beta \in \mathbf{F}$ and all positive integers $m, n$.

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## *Lists*

Before defining $\mathbf{R}^n$ and $\mathbf{C}^n$, we look at two important examples.

1.7 example: $\mathbf{R}^2$ *and* $\mathbf{R}^3$

- The set $\mathbf{R}^2$, which you can think of as a plane, is the set of all ordered pairs of real numbers:
$$\mathbf{R}^2 = \{(x, y) : x, y \in \mathbf{R}\}.$$
- The set $\mathbf{R}^3$, which you can think of as ordinary space, is the set of all ordered triples of real numbers:
$$\mathbf{R}^3 = \{(x, y, z) : x, y, z \in \mathbf{R}\}.$$

To generalize $\mathbf{R}^2$ and $\mathbf{R}^3$ to higher dimensions, we first need to discuss the concept of lists.

1.8 definition: *list, length*

- Suppose $n$ is a nonnegative integer. A *list* of *length* $n$ is an ordered collection of $n$ elements (which might be numbers, other lists, or more abstract objects).
- Two lists are equal if and only if they have the same length and the same elements in the same order.

*Many mathematicians call a list of length $n$ an $n$-**tuple**.*

Lists are often written as elements separated by commas and surrounded by parentheses. Thus a list of length two is an ordered pair that might be written as $(a, b)$. A list of length three is an ordered triple that might be written as $(x, y, z)$. A list of length $n$ might look like this:

$$(z_1, \ldots, z_n).$$

Sometimes we will use the word *list* without specifying its length. Remember, however, that by definition each list has a finite length that is a nonnegative integer. Thus an object that looks like $(x_1, x_2, \ldots)$, which might be said to have infinite length, is not a list.

A list of length 0 looks like this: $(\,)$. We consider such an object to be a list so that some of our theorems will not have trivial exceptions.

Lists differ from sets in two ways: in lists, order matters and repetitions have meaning; in sets, order and repetitions are irrelevant.

1.9 example: *lists versus sets*

- The lists $(3, 5)$ and $(5, 3)$ are not equal, but the sets $\{3, 5\}$ and $\{5, 3\}$ are equal.
- The lists $(4, 4)$ and $(4, 4, 4)$ are not equal (they do not have the same length), although the sets $\{4, 4\}$ and $\{4, 4, 4\}$ both equal the set $\{4\}$.

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## $\mathbf{F}^n$

To define the higher-dimensional analogues of $\mathbf{R}^2$ and $\mathbf{R}^3$, we will simply replace $\mathbf{R}$ with $\mathbf{F}$ (which equals $\mathbf{R}$ or $\mathbf{C}$) and replace the 2 or 3 with an arbitrary positive integer.

**1.10 notation: $n$**

Fix a positive integer $n$ for the rest of this chapter.

**1.11 definition: $\mathbf{F}^n$, *coordinate***

$\mathbf{F}^n$ is the set of all lists of length $n$ of elements of $\mathbf{F}$:

$$\mathbf{F}^n = \{(x_1, ..., x_n) : x_k \in \mathbf{F} \text{ for } k = 1, ..., n\}.$$

For $(x_1, ..., x_n) \in \mathbf{F}^n$ and $k \in \{1, ..., n\}$, we say that $x_k$ is the $k^{\text{th}}$ *coordinate* of $(x_1, ..., x_n)$.

If $\mathbf{F} = \mathbf{R}$ and $n$ equals 2 or 3, then the definition above of $\mathbf{F}^n$ agrees with our previous notions of $\mathbf{R}^2$ and $\mathbf{R}^3$.

**1.12 example: $\mathbf{C}^4$**

$\mathbf{C}^4$ is the set of all lists of four complex numbers:

$$\mathbf{C}^4 = \{(z_1, z_2, z_3, z_4) : z_1, z_2, z_3, z_4 \in \mathbf{C}\}.$$

If $n \geq 4$, we cannot visualize $\mathbf{R}^n$ as a physical object. Similarly, $\mathbf{C}^1$ can be thought of as a plane, but for $n \geq 2$, the human brain cannot provide a full image of $\mathbf{C}^n$. However, even if $n$ is large, we can perform algebraic manipulations in $\mathbf{F}^n$ as easily as in $\mathbf{R}^2$ or $\mathbf{R}^3$. For example, addition in $\mathbf{F}^n$ is defined as follows.

*Read **Flatland: A Romance of Many Dimensions**, by Edwin A. Abbott, for an amusing account of how $\mathbf{R}^3$ would be perceived by creatures living in $\mathbf{R}^2$. This novel, published in 1884, may help you imagine a physical space of four or more dimensions.*

**1.13 definition: *addition in* $\mathbf{F}^n$**

*Addition* in $\mathbf{F}^n$ is defined by adding corresponding coordinates:

$$(x_1, ..., x_n) + (y_1, ..., y_n) = (x_1 + y_1, ..., x_n + y_n).$$

Often the mathematics of $\mathbf{F}^n$ becomes cleaner if we use a single letter to denote a list of $n$ numbers, without explicitly writing the coordinates. For example, the next result is stated with $x$ and $y$ in $\mathbf{F}^n$ even though the proof requires the more cumbersome notation of $(x_1, ..., x_n)$ and $(y_1, ..., y_n)$.

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**1.14** *commutativity of addition in* $\mathbf{F}^n$

If $x, y \in \mathbf{F}^n$, then $x + y = y + x$.

Proof Suppose $x = (x_1, \ldots, x_n) \in \mathbf{F}^n$ and $y = (y_1, \ldots, y_n) \in \mathbf{F}^n$. Then

$$\begin{aligned} x + y &= (x_1, \ldots, x_n) + (y_1, \ldots, y_n) \\ &= (x_1 + y_1, \ldots, x_n + y_n) \\ &= (y_1 + x_1, \ldots, y_n + x_n) \\ &= (y_1, \ldots, y_n) + (x_1, \ldots, x_n) \\ &= y + x, \end{aligned}$$

where the second and fourth equalities above hold because of the definition of addition in $\mathbf{F}^n$ and the third equality holds because of the usual commutativity of addition in $\mathbf{F}$. ∎

*The symbol ∎ means "end of proof".*

If a single letter is used to denote an element of $\mathbf{F}^n$, then the same letter with appropriate subscripts is often used when coordinates must be displayed. For example, if $x \in \mathbf{F}^n$, then letting $x$ equal $(x_1, \ldots, x_n)$ is good notation, as shown in the proof above. Even better, work with just $x$ and avoid explicit coordinates when possible.

**1.15** notation: 0

Let 0 denote the list of length $n$ whose coordinates are all 0:

$$0 = (0, \ldots, 0).$$

Here we are using the symbol 0 in two different ways—on the left side of the equation above, the symbol 0 denotes a list of length $n$, which is an element of $\mathbf{F}^n$, whereas on the right side, each 0 denotes a number. This potentially confusing practice actually causes no problems because the context should always make clear which 0 is intended.

**1.16** example: *context determines which* 0 *is intended*

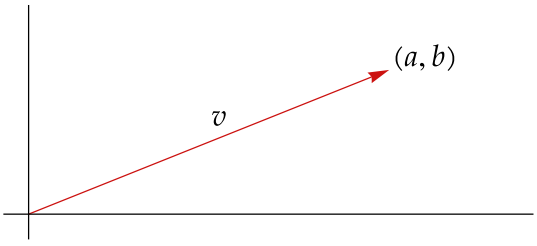
Consider the statement that 0 is an additive identity for $\mathbf{F}^n$:

$$x + 0 = x \quad \text{for all } x \in \mathbf{F}^n.$$

Here the 0 above is the list defined in 1.15, not the number 0, because we have not defined the sum of an element of $\mathbf{F}^n$ (namely, $x$) and the number 0.

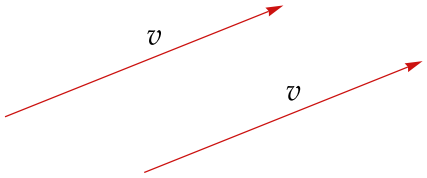
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A picture can aid our intuition. We will draw pictures in $\mathbf{R}^2$ because we can sketch this space on two-dimensional surfaces such as paper and computer screens. A typical element of $\mathbf{R}^2$ is a point $v = (a, b)$. Sometimes we think of $v$ not as a point but as an arrow starting at the origin and ending at $(a, b)$, as shown here. When we think of an element of $\mathbf{R}^2$ as an arrow, we refer to it as a *vector*.




*Elements of $\mathbf{R}^2$ can be thought of as points or as vectors.*

When we think of vectors in $\mathbf{R}^2$ as arrows, we can move an arrow parallel to itself (not changing its length or direction) and still think of it as the same vector. With that viewpoint, you will often gain better understanding by dispensing with the coordinate axes and the explicit coordinates and just thinking of the vector, as shown in the figure here. The two arrows shown here have the same length and same direction, so we think of them as the same vector.




*A vector.*

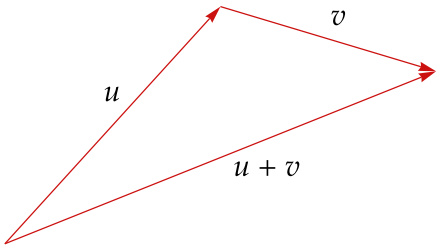
Whenever we use pictures in $\mathbf{R}^2$ or use the somewhat vague language of points and vectors, remember that these are just aids to our understanding, not substitutes for the actual mathematics that we will develop. Although we cannot draw good pictures in high-dimensional spaces, the elements of these spaces are as rigorously defined as elements of $\mathbf{R}^2$.

*Mathematical models of the economy can have thousands of variables, say $x_1, \ldots, x_{5000}$, which means that we must work in $\mathbf{R}^{5000}$. Such a space cannot be dealt with geometrically. However, the algebraic approach works well. Thus our subject is called **linear algebra**.*

For example, $\left(2, -3, 17, \pi, \sqrt{2}\right)$ is an element of $\mathbf{R}^5$, and we may casually refer to it as a point in $\mathbf{R}^5$ or a vector in $\mathbf{R}^5$ without worrying about whether the geometry of $\mathbf{R}^5$ has any physical meaning.

Recall that we defined the sum of two elements of $\mathbf{F}^n$ to be the element of $\mathbf{F}^n$ obtained by adding corresponding coordinates; see 1.13. As we will now see, addition has a simple geometric interpretation in the special case of $\mathbf{R}^2$.

Suppose we have two vectors $u$ and $v$ in $\mathbf{R}^2$ that we want to add. Move the vector $v$ parallel to itself so that its initial point coincides with the end point of the vector $u$, as shown here. The sum $u + v$ then equals the vector whose initial point equals the initial point of $u$ and whose end point equals the end point of the vector $v$, as shown here.




*The sum of two vectors.*

In the next definition, the 0 on the right side of the displayed equation is the list $0 \in \mathbf{F}^n$.

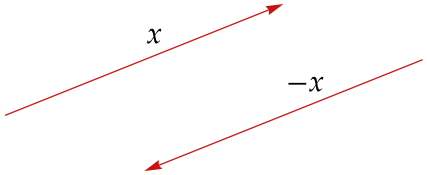
**1.17 definition: *additive inverse in* $\mathbf{F}^n$, $-x$**

For $x \in \mathbf{F}^n$, the *additive inverse* of $x$, denoted by $-x$, is the vector $-x \in \mathbf{F}^n$ such that

$$x + (-x) = 0.$$

Thus if $x = (x_1, ..., x_n)$, then $-x = (-x_1, ..., -x_n)$.

The additive inverse of a vector in $\mathbf{R}^2$ is the vector with the same length but pointing in the opposite direction. The figure here illustrates this way of thinking about the additive inverse in $\mathbf{R}^2$. As you can see, the vector labeled $-x$ has the same length as the vector labeled $x$ but points in the opposite direction.




*A vector and its additive inverse.*

Having dealt with addition in $\mathbf{F}^n$, we now turn to multiplication. We could define a multiplication in $\mathbf{F}^n$ in a similar fashion, starting with two elements of $\mathbf{F}^n$ and getting another element of $\mathbf{F}^n$ by multiplying corresponding coordinates. Experience shows that this definition is not useful for our purposes. Another type of multiplication, called scalar multiplication, will be central to our subject. Specifically, we need to define what it means to multiply an element of $\mathbf{F}^n$ by an element of $\mathbf{F}$.

**1.18 definition: *scalar multiplication in* $\mathbf{F}^n$**

The *product* of a number $\lambda$ and a vector in $\mathbf{F}^n$ is computed by multiplying each coordinate of the vector by $\lambda$:

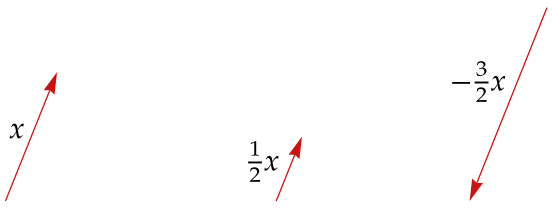
$$\lambda(x_1, ..., x_n) = (\lambda x_1, ..., \lambda x_n);$$

here $\lambda \in \mathbf{F}$ and $(x_1, ..., x_n) \in \mathbf{F}^n$.

Scalar multiplication has a nice geometric interpretation in $\mathbf{R}^2$. If $\lambda > 0$ and $x \in \mathbf{R}^2$, then $\lambda x$ is the vector that points in the same direction as $x$ and whose length is $\lambda$ times the length of $x$. In other words, to get $\lambda x$, we shrink or stretch $x$ by a factor of $\lambda$, depending on whether $\lambda < 1$ or $\lambda > 1$.

If $\lambda < 0$ and $x \in \mathbf{R}^2$, then $\lambda x$ is the vector that points in the direction opposite to that of $x$ and whose length is $|\lambda|$ times the length of $x$, as shown here.

*Scalar multiplication in* $\mathbf{F}^n$ *multiplies together a scalar and a vector, getting a vector. In contrast, the dot product in* $\mathbf{R}^2$ *or* $\mathbf{R}^3$ *multiplies together two vectors and gets a scalar. Generalizations of the dot product will become important in Chapter 6.*




*Scalar multiplication.*

### *Digression on Fields*

A *field* is a set containing at least two distinct elements called 0 and 1, along with operations of addition and multiplication satisfying all properties listed in 1.3. Thus $\mathbf{R}$ and $\mathbf{C}$ are fields, as is the set of rational numbers along with the usual operations of addition and multiplication. Another example of a field is the set $\{0, 1\}$ with the usual operations of addition and multiplication except that $1 + 1$ is defined to equal 0.

In this book we will not deal with fields other than $\mathbf{R}$ and $\mathbf{C}$. However, many of the definitions, theorems, and proofs in linear algebra that work for the fields $\mathbf{R}$ and $\mathbf{C}$ also work without change for arbitrary fields. If you prefer to do so, throughout much of this book (except for Chapters 6 and 7, which deal with inner product spaces) you can think of $\mathbf{F}$ as denoting an arbitrary field instead of $\mathbf{R}$ or $\mathbf{C}$. For results (except in the inner product chapters) that have as a hypothesis that $\mathbf{F}$ is $\mathbf{C}$, you can probably replace that hypothesis with the hypothesis that $\mathbf{F}$ is an algebraically closed field, which means that every nonconstant polynomial with coefficients in $\mathbf{F}$ has a zero. A few results, such as Exercise 13 in Section 1C, require the hypothesis on $\mathbf{F}$ that $1 + 1 \neq 0$.

## *Exercises 1A*

**1** Show that $\alpha + \beta = \beta + \alpha$ for all $\alpha, \beta \in \mathbf{C}$.

**2** Show that $(\alpha + \beta) + \lambda = \alpha + (\beta + \lambda)$ for all $\alpha, \beta, \lambda \in \mathbf{C}$.

**3** Show that $(\alpha\beta)\lambda = \alpha(\beta\lambda)$ for all $\alpha, \beta, \lambda \in \mathbf{C}$.

**4** Show that $\lambda(\alpha + \beta) = \lambda\alpha + \lambda\beta$ for all $\lambda, \alpha, \beta \in \mathbf{C}$.

**5** Show that for every $\alpha \in \mathbf{C}$, there exists a unique $\beta \in \mathbf{C}$ such that $\alpha + \beta = 0$.

**6** Show that for every $\alpha \in \mathbf{C}$ with $\alpha \neq 0$, there exists a unique $\beta \in \mathbf{C}$ such that $\alpha\beta = 1$.

**7** Show that

$$\frac{-1 + \sqrt{3}i}{2}$$

is a cube root of 1 (meaning that its cube equals 1).

**8** Find two distinct square roots of $i$.

**9** Find $x \in \mathbf{R}^4$ such that

$$(4, -3, 1, 7) + 2x = (5, 9, -6, 8).$$

**10** Explain why there does not exist $\lambda \in \mathbf{C}$ such that

$$\lambda(2 - 3i, 5 + 4i, -6 + 7i) = (12 - 5i, 7 + 22i, -32 - 9i).$$

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**11** Show that $(x + y) + z = x + (y + z)$ for all $x, y, z \in \mathbf{F}^n$.

**12** Show that $(ab)x = a(bx)$ for all $x \in \mathbf{F}^n$ and all $a, b \in \mathbf{F}$.

**13** Show that $1x = x$ for all $x \in \mathbf{F}^n$.

**14** Show that $\lambda(x + y) = \lambda x + \lambda y$ for all $\lambda \in \mathbf{F}$ and all $x, y \in \mathbf{F}^n$.

**15** Show that $(a + b)x = ax + bx$ for all $a, b \in \mathbf{F}$ and all $x \in \mathbf{F}^n$.

"Can you do addition?" the White Queen asked. "What's one and one and one and one and one and one and one and one and one and one?"
"I don't know," said Alice. "I lost count."

—*Through the Looking Glass*, Lewis Carroll

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## 1B Definition of Vector Space

The motivation for the definition of a vector space comes from properties of addition and scalar multiplication in $\mathbf{F}^n$: Addition is commutative, associative, and has an identity. Every element has an additive inverse. Scalar multiplication is associative. Scalar multiplication by 1 acts as expected. Addition and scalar multiplication are connected by distributive properties.

We will define a vector space to be a set $V$ with an addition and a scalar multiplication on $V$ that satisfy the properties in the paragraph above.

**1.19 definition:** *addition, scalar multiplication*

- An *addition* on a set $V$ is a function that assigns an element $u + v \in V$ to each pair of elements $u, v \in V$.
- A *scalar multiplication* on a set $V$ is a function that assigns an element $\lambda v \in V$ to each $\lambda \in \mathbf{F}$ and each $v \in V$.

Now we are ready to give the formal definition of a vector space.

**1.20 definition:** *vector space*

A *vector space* is a set $V$ along with an addition on $V$ and a scalar multiplication on $V$ such that the following properties hold.

**commutativity**
$u + v = v + u$ for all $u, v \in V$.

**associativity**
$(u + v) + w = u + (v + w)$ and $(ab)v = a(bv)$ for all $u, v, w \in V$ and for all $a, b \in \mathbf{F}$.

**additive identity**
There exists an element $0 \in V$ such that $v + 0 = v$ for all $v \in V$.

**additive inverse**
For every $v \in V$, there exists $w \in V$ such that $v + w = 0$.

**multiplicative identity**
$1v = v$ for all $v \in V$.

**distributive properties**
$a(u + v) = au + av$ and $(a + b)v = av + bv$ for all $a, b \in \mathbf{F}$ and all $u, v \in V$.

The following geometric language sometimes aids our intuition.

**1.21 definition:** *vector, point*

Elements of a vector space are called *vectors* or *points*.

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The scalar multiplication in a vector space depends on $\mathbf{F}$. Thus when we need to be precise, we will say that $V$ is a *vector space over* $\mathbf{F}$ instead of saying simply that $V$ is a vector space. For example, $\mathbf{R}^n$ is a vector space over $\mathbf{R}$, and $\mathbf{C}^n$ is a vector space over $\mathbf{C}$.

**1.22 definition:** *real vector space, complex vector space*

- A vector space over $\mathbf{R}$ is called a *real vector space*.
- A vector space over $\mathbf{C}$ is called a *complex vector space*.

Usually the choice of $\mathbf{F}$ is either clear from the context or irrelevant. Thus we often assume that $\mathbf{F}$ is lurking in the background without specifically mentioning it.

With the usual operations of addition and scalar multiplication, $\mathbf{F}^n$ is a vector space over $\mathbf{F}$, as you should verify. The example of $\mathbf{F}^n$ motivated our definition of vector space.

*The simplest vector space is* $\{0\}$*, which contains only one point.*

**1.23 example:** $\mathbf{F}^\infty$

$\mathbf{F}^\infty$ is defined to be the set of all sequences of elements of $\mathbf{F}$:

$$\mathbf{F}^\infty = \{(x_1, x_2, \dots) : x_k \in \mathbf{F} \text{ for } k = 1, 2, \dots\}.$$

Addition and scalar multiplication on $\mathbf{F}^\infty$ are defined as expected:

$$(x_1, x_2, \dots) + (y_1, y_2, \dots) = (x_1 + y_1, x_2 + y_2, \dots),$$
$$\lambda(x_1, x_2, \dots) = (\lambda x_1, \lambda x_2, \dots).$$

With these definitions, $\mathbf{F}^\infty$ becomes a vector space over $\mathbf{F}$, as you should verify. The additive identity in this vector space is the sequence of all 0's.

Our next example of a vector space involves a set of functions.

**1.24 notation:** $\mathbf{F}^S$

- If $S$ is a set, then $\mathbf{F}^S$ denotes the set of functions from $S$ to $\mathbf{F}$.
- For $f, g \in \mathbf{F}^S$, the *sum* $f + g \in \mathbf{F}^S$ is the function defined by

$$(f + g)(x) = f(x) + g(x)$$

for all $x \in S$.

- For $\lambda \in \mathbf{F}$ and $f \in \mathbf{F}^S$, the *product* $\lambda f \in \mathbf{F}^S$ is the function defined by

$$(\lambda f)(x) = \lambda f(x)$$

for all $x \in S$.

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As an example of the notation above, if $S$ is the interval $[0, 1]$ and $\mathbf{F} = \mathbf{R}$, then $\mathbf{R}^{[0,1]}$ is the set of real-valued functions on the interval $[0, 1]$.

You should verify all three bullet points in the next example.

**1.25 example: $\mathbf{F}^S$ *is a vector space***

- If $S$ is a nonempty set, then $\mathbf{F}^S$ (with the operations of addition and scalar multiplication as defined above) is a vector space over $\mathbf{F}$.
- The additive identity of $\mathbf{F}^S$ is the function $0 : S \to \mathbf{F}$ defined by

$$0(x) = 0$$

for all $x \in S$.

- For $f \in \mathbf{F}^S$, the additive inverse of $f$ is the function $-f : S \to \mathbf{F}$ defined by

$$(-f)(x) = -f(x)$$

for all $x \in S$.

*The elements of the vector space $\mathbf{R}^{[0,1]}$ are real-valued functions on $[0, 1]$, not lists. In general, a vector space is an abstract entity whose elements might be lists, functions, or weird objects.*

The vector space $\mathbf{F}^n$ is a special case of the vector space $\mathbf{F}^S$ because each $(x_1, \ldots, x_n) \in \mathbf{F}^n$ can be thought of as a function $x$ from the set $\{1, 2, \ldots, n\}$ to $\mathbf{F}$ by writing $x(k)$ instead of $x_k$ for the $k^{\text{th}}$ coordinate of $(x_1, \ldots, x_n)$. In other words, we can think of $\mathbf{F}^n$ as $\mathbf{F}^{\{1,2,\ldots,n\}}$. Similarly, we can think of $\mathbf{F}^\infty$ as $\mathbf{F}^{\{1,2,\ldots\}}$.

Soon we will see further examples of vector spaces, but first we need to develop some of the elementary properties of vector spaces.

The definition of a vector space requires it to have an additive identity. The next result states that this identity is unique.

**1.26 *unique additive identity***

A vector space has a unique additive identity.

Proof Suppose $0$ and $0'$ are both additive identities for some vector space $V$. Then

$$0' = 0' + 0 = 0 + 0' = 0,$$

where the first equality holds because $0$ is an additive identity, the second equality comes from commutativity, and the third equality holds because $0'$ is an additive identity. Thus $0' = 0$, proving that $V$ has only one additive identity. ∎

Each element $v$ in a vector space has an additive inverse, an element $w$ in the vector space such that $v + w = 0$. The next result shows that each element in a vector space has only one additive inverse.

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**1.27 *unique additive inverse***

Every element in a vector space has a unique additive inverse.

Proof Suppose $V$ is a vector space. Let $v \in V$. Suppose $w$ and $w'$ are additive inverses of $v$. Then

$$w = w + 0 = w + (v + w') = (w + v) + w' = 0 + w' = w'.$$

Thus $w = w'$, as desired. ∎

Because additive inverses are unique, the following notation now makes sense.

**1.28 notation: $-v$, $w - v$**

Let $v, w \in V$. Then

- $-v$ denotes the additive inverse of $v$;
- $w - v$ is defined to be $w + (-v)$.

Almost all results in this book involve some vector space. To avoid having to restate frequently that $V$ is a vector space, we now make the necessary declaration once and for all.

**1.29 notation: $V$**

For the rest of this book, $V$ denotes a vector space over $\mathbf{F}$.

In the next result, 0 denotes a scalar (the number $0 \in \mathbf{F}$) on the left side of the equation and a vector (the additive identity of $V$) on the right side of the equation.

**1.30 *the number 0 times a vector***

$0v = 0$ for every $v \in V$.

Proof For $v \in V$, we have

$$0v = (0 + 0)v = 0v + 0v.$$

Adding the additive inverse of $0v$ to both sides of the equation above gives $0 = 0v$, as desired. ∎

*The result in 1.30 involves the additive identity of $V$ and scalar multiplication. The only part of the definition of a vector space that connects vector addition and scalar multiplication is the distributive property. Thus the distributive property must be used in the proof of 1.30.*

In the next result, 0 denotes the additive identity of $V$. Although their proofs are similar, 1.30 and 1.31 are not identical. More precisely, 1.30 states that the product of the scalar 0 and any vector equals the vector 0, whereas 1.31 states that the product of any scalar and the vector 0 equals the vector 0.

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**1.31** *a number times the vector* 0

$a0 = 0$ for every $a \in \mathbf{F}$.

Proof For $a \in \mathbf{F}$, we have

$$a0 = a(0 + 0) = a0 + a0.$$

Adding the additive inverse of $a0$ to both sides of the equation above gives $0 = a0$, as desired. ∎

Now we show that if an element of $V$ is multiplied by the scalar $-1$, then the result is the additive inverse of the element of $V$.

**1.32** *the number* $-1$ *times a vector*

$(-1)v = -v$ for every $v \in V$.

Proof For $v \in V$, we have

$$v + (-1)v = 1v + (-1)v = (1 + (-1))v = 0v = 0.$$

This equation says that $(-1)v$, when added to $v$, gives 0. Thus $(-1)v$ is the additive inverse of $v$, as desired. ∎

## *Exercises 1B*

**1** Prove that $-(-v) = v$ for every $v \in V$.

**2** Suppose $a \in \mathbf{F}$, $v \in V$, and $av = 0$. Prove that $a = 0$ or $v = 0$.

**3** Suppose $v, w \in V$. Explain why there exists a unique $x \in V$ such that $v + 3x = w$.

**4** The empty set is not a vector space. The empty set fails to satisfy only one of the requirements listed in the definition of a vector space (1.20). Which one?

**5** Show that in the definition of a vector space (1.20), the additive inverse condition can be replaced with the condition that

$$0v = 0 \text{ for all } v \in V.$$

Here the 0 on the left side is the number 0, and the 0 on the right side is the additive identity of $V$.

*The phrase a "condition can be replaced" in a definition means that the collection of objects satisfying the definition is unchanged if the original condition is replaced with the new condition.*

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**6** Let $\infty$ and $-\infty$ denote two distinct objects, neither of which is in $\mathbf{R}$. Define an addition and scalar multiplication on $\mathbf{R} \cup \{\infty, -\infty\}$ as you could guess from the notation. Specifically, the sum and product of two real numbers is as usual, and for $t \in \mathbf{R}$ define

$$t\infty = \begin{cases} -\infty & \text{if } t < 0, \\ 0 & \text{if } t = 0, \\ \infty & \text{if } t > 0, \end{cases} \qquad t(-\infty) = \begin{cases} \infty & \text{if } t < 0, \\ 0 & \text{if } t = 0, \\ -\infty & \text{if } t > 0, \end{cases}$$

and

$$\begin{aligned} t + \infty = \infty + t = \infty + \infty = \infty, \\ t + (-\infty) = (-\infty) + t = (-\infty) + (-\infty) = -\infty, \\ \infty + (-\infty) = (-\infty) + \infty = 0. \end{aligned}$$

With these operations of addition and scalar multiplication, is $\mathbf{R} \cup \{\infty, -\infty\}$ a vector space over $\mathbf{R}$? Explain.

**7** Suppose $S$ is a nonempty set. Let $V^S$ denote the set of functions from $S$ to $V$. Define a natural addition and scalar multiplication on $V^S$, and show that $V^S$ is a vector space with these definitions.

**8** Suppose $V$ is a real vector space.

- The *complexification* of $V$, denoted by $V_{\mathbf{C}}$, equals $V \times V$. An element of $V_{\mathbf{C}}$ is an ordered pair $(u, v)$, where $u, v \in V$, but we write this as $u + iv$.
- Addition on $V_{\mathbf{C}}$ is defined by

  $$(u_1 + iv_1) + (u_2 + iv_2) = (u_1 + u_2) + i(v_1 + v_2)$$

  for all $u_1, v_1, u_2, v_2 \in V$.
- Complex scalar multiplication on $V_{\mathbf{C}}$ is defined by

  $$(a + bi)(u + iv) = (au - bv) + i(av + bu)$$

  for all $a, b \in \mathbf{R}$ and all $u, v \in V$.

Prove that with the definitions of addition and scalar multiplication as above, $V_{\mathbf{C}}$ is a complex vector space.

*Think of $V$ as a subset of $V_{\mathbf{C}}$ by identifying $u \in V$ with $u + i0$. The construction of $V_{\mathbf{C}}$ from $V$ can then be thought of as generalizing the construction of $\mathbf{C}^n$ from $\mathbf{R}^n$.*

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## 1C Subspaces

By considering subspaces, we can greatly expand our examples of vector spaces.

**1.33 definition: *subspace***

A subset $U$ of $V$ is called a *subspace* of $V$ if $U$ is also a vector space with the same additive identity, addition, and scalar multiplication as on $V$.

The next result gives the easiest way to check whether a subset of a vector space is a subspace.

*Some people use the terminology **linear subspace**, which means the same as subspace.*

**1.34 *conditions for a subspace***

A subset $U$ of $V$ is a subspace of $V$ if and only if $U$ satisfies the following three conditions.

**additive identity**
$0 \in U$.

**closed under addition**
$u, w \in U$ implies $u + w \in U$.

**closed under scalar multiplication**
$a \in \mathbf{F}$ and $u \in U$ implies $au \in U$.

Proof If $U$ is a subspace of $V$, then $U$ satisfies the three conditions above by the definition of vector space.

*The additive identity condition above could be replaced with the condition that $U$ is nonempty (because then taking $u \in U$ and multiplying it by 0 would imply that $0 \in U$). However, if a subset $U$ of $V$ is indeed a subspace, then usually the quickest way to show that $U$ is nonempty is to show that $0 \in U$.*

Conversely, suppose $U$ satisfies the three conditions above. The first condition ensures that the additive identity of $V$ is in $U$. The second condition ensures that addition makes sense on $U$. The third condition ensures that scalar multiplication makes sense on $U$.

If $u \in U$, then $-u$ [which equals $(-1)u$ by 1.32] is also in $U$ by the third condition above. Hence every element of $U$ has an additive inverse in $U$.

The other parts of the definition of a vector space, such as associativity and commutativity, are automatically satisfied for $U$ because they hold on the larger space $V$. Thus $U$ is a vector space and hence is a subspace of $V$. ∎

The three conditions in the result above usually enable us to determine quickly whether a given subset of $V$ is a subspace of $V$. You should verify all assertions in the next example.

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1.35 example: *subspaces*

(a) If $b \in \mathbf{F}$, then
$$\{(x_1, x_2, x_3, x_4) \in \mathbf{F}^4 : x_3 = 5x_4 + b\}$$
is a subspace of $\mathbf{F}^4$ if and only if $b = 0$.

(b) The set of continuous real-valued functions on the interval $[0, 1]$ is a subspace of $\mathbf{R}^{[0,1]}$.

(c) The set of differentiable real-valued functions on $\mathbf{R}$ is a subspace of $\mathbf{R}^{\mathbf{R}}$.

(d) The set of differentiable real-valued functions $f$ on the interval $(0, 3)$ such that $f'(2) = b$ is a subspace of $\mathbf{R}^{(0,3)}$ if and only if $b = 0$.

(e) The set of all sequences of complex numbers with limit 0 is a subspace of $\mathbf{C}^{\infty}$.

*The set $\{0\}$ is the smallest subspace of $V$, and $V$ itself is the largest subspace of $V$. The empty set is not a subspace of $V$ because a subspace must be a vector space and hence must contain at least one element, namely, an additive identity.*

Verifying some of the items above shows the linear structure underlying parts of calculus. For example, (b) above requires the result that the sum of two continuous functions is continuous. As another example, (d) above requires the result that for a constant $c$, the derivative of $cf$ equals $c$ times the derivative of $f$.

The subspaces of $\mathbf{R}^2$ are precisely $\{0\}$, all lines in $\mathbf{R}^2$ containing the origin, and $\mathbf{R}^2$. The subspaces of $\mathbf{R}^3$ are precisely $\{0\}$, all lines in $\mathbf{R}^3$ containing the origin, all planes in $\mathbf{R}^3$ containing the origin, and $\mathbf{R}^3$. To prove that all these objects are indeed subspaces is straightforward—the hard part is to show that they are the only subspaces of $\mathbf{R}^2$ and $\mathbf{R}^3$. That task will be easier after we introduce some additional tools in the next chapter.

## *Sums of Subspaces*

When dealing with vector spaces, we are usually interested only in subspaces, as opposed to arbitrary subsets. The notion of the sum of subspaces will be useful.

*The union of subspaces is rarely a subspace (see Exercise 12), which is why we usually work with sums rather than unions.*

1.36 definition: *sum of subspaces*

Suppose $V_1, \ldots, V_m$ are subspaces of $V$. The *sum* of $V_1, \ldots, V_m$, denoted by $V_1 + \cdots + V_m$, is the set of all possible sums of elements of $V_1, \ldots, V_m$. More precisely,
$$V_1 + \cdots + V_m = \{v_1 + \cdots + v_m : v_1 \in V_1, \ldots, v_m \in V_m\}.$$

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Let's look at some examples of sums of subspaces.

1.37 example: *a sum of subspaces of* $\mathbf{F}^3$

Suppose $U$ is the set of all elements of $\mathbf{F}^3$ whose second and third coordinates equal 0, and $W$ is the set of all elements of $\mathbf{F}^3$ whose first and third coordinates equal 0:

$$U = \{(x, 0, 0) \in \mathbf{F}^3 : x \in \mathbf{F}\} \quad \text{and} \quad W = \{(0, y, 0) \in \mathbf{F}^3 : y \in \mathbf{F}\}.$$

Then

$$U + W = \{(x, y, 0) \in \mathbf{F}^3 : x, y \in \mathbf{F}\},$$

as you should verify.

1.38 example: *a sum of subspaces of* $\mathbf{F}^4$

Suppose

$$U = \{(x, x, y, y) \in \mathbf{F}^4 : x, y \in \mathbf{F}\} \quad \text{and} \quad W = \{(x, x, x, y) \in \mathbf{F}^4 : x, y \in \mathbf{F}\}.$$

Using words rather than symbols, we could say that $U$ is the set of elements of $\mathbf{F}^4$ whose first two coordinates equal each other and whose third and fourth coordinates equal each other. Similarly, $W$ is the set of elements of $\mathbf{F}^4$ whose first three coordinates equal each other.

To find a description of $U + W$, consider a typical element $(a, a, b, b)$ of $U$ and a typical element $(c, c, c, d)$ of $W$, where $a, b, c, d \in \mathbf{F}$. We have

$$(a, a, b, b) + (c, c, c, d) = (a + c, a + c, b + c, b + d),$$

which shows that every element of $U + W$ has its first two coordinates equal to each other. Thus

1.39 $$U + W \subseteq \{(x, x, y, z) \in \mathbf{F}^4 : x, y, z \in \mathbf{F}\}.$$

To prove the inclusion in the other direction, suppose $x, y, z \in \mathbf{F}$. Then

$$(x, x, y, z) = (x, x, y, y) + (0, 0, 0, z - y),$$

where the first vector on the right is in $U$ and the second vector on the right is in $W$. Thus $(x, x, y, z) \in U + W$, showing that the inclusion 1.39 also holds in the opposite direction. Hence

$$U + W = \{(x, x, y, z) \in \mathbf{F}^4 : x, y, z \in \mathbf{F}\},$$

which shows that $U + W$ is the set of elements of $\mathbf{F}^4$ whose first two coordinates equal each other.

The next result states that the sum of subspaces is a subspace, and is in fact the smallest subspace containing all the summands (which means that every subspace containing all the summands also contains the sum).

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**1.40** *sum of subspaces is the smallest containing subspace*

Suppose $V_1, ..., V_m$ are subspaces of $V$. Then $V_1 + \cdots + V_m$ is the smallest subspace of $V$ containing $V_1, ..., V_m$.

Proof The reader can verify that $V_1 + \cdots + V_m$ contains the additive identity 0 and is closed under addition and scalar multiplication. Thus 1.34 implies that $V_1 + \cdots + V_m$ is a subspace of $V$.

The subspaces $V_1, ..., V_m$ are all contained in $V_1 + \cdots + V_m$ (to see this, consider sums $v_1 + \cdots + v_m$ where all except one of the $v_k$'s are 0). Conversely, every subspace of $V$ containing $V_1, ..., V_m$ contains $V_1 + \cdots + V_m$ (because subspaces must contain all finite sums of their elements). Thus $V_1 + \cdots + V_m$ is the smallest subspace of $V$ containing $V_1, ..., V_m$. ∎

*Sums of subspaces in the theory of vector spaces are analogous to unions of subsets in set theory. Given two subspaces of a vector space, the smallest subspace containing them is their sum. Analogously, given two subsets of a set, the smallest subset containing them is their union.*

## *Direct Sums*

Suppose $V_1, ..., V_m$ are subspaces of $V$. Every element of $V_1 + \cdots + V_m$ can be written in the form

$$v_1 + \cdots + v_m,$$

where each $v_k \in V_k$. Of special interest are cases in which each vector in $V_1 + \cdots + V_m$ can be represented in the form above in only one way. This situation is so important that it gets a special name (direct sum) and a special symbol ($\oplus$).

**1.41** definition: *direct sum,* $\oplus$

Suppose $V_1, ..., V_m$ are subspaces of $V$.

- The sum $V_1 + \cdots + V_m$ is called a *direct sum* if each element of $V_1 + \cdots + V_m$ can be written in only one way as a sum $v_1 + \cdots + v_m$, where each $v_k \in V_k$.
- If $V_1 + \cdots + V_m$ is a direct sum, then $V_1 \oplus \cdots \oplus V_m$ denotes $V_1 + \cdots + V_m$, with the $\oplus$ notation serving as an indication that this is a direct sum.

**1.42** example: *a direct sum of two subspaces*

Suppose $U$ is the subspace of $\mathbf{F}^3$ of those vectors whose last coordinate equals 0, and $W$ is the subspace of $\mathbf{F}^3$ of those vectors whose first two coordinates equal 0:

$$U = \{(x, y, 0) \in \mathbf{F}^3 : x, y \in \mathbf{F}\} \quad \text{and} \quad W = \{(0, 0, z) \in \mathbf{F}^3 : z \in \mathbf{F}\}.$$

Then $\mathbf{F}^3 = U \oplus W$, as you should verify.

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1.43 example: *a direct sum of multiple subspaces*

Suppose $V_k$ is the subspace of $\mathbf{F}^n$ of those vectors whose coordinates are all 0, except possibly in the $k^{\text{th}}$ slot; for example, $V_2 = \{(0, x, 0, ..., 0) \in \mathbf{F}^n : x \in \mathbf{F}\}$. Then

$$\mathbf{F}^n = V_1 \oplus \cdots \oplus V_n,$$

as you should verify.

*To produce ⊕ in TEX, type* `\oplus`.

---

Sometimes nonexamples add to our understanding as much as examples.

1.44 example: *a sum that is not a direct sum*

Suppose

$$V_1 = \{(x, y, 0) \in \mathbf{F}^3 : x, y \in \mathbf{F}\},$$

$$V_2 = \{(0, 0, z) \in \mathbf{F}^3 : z \in \mathbf{F}\},$$

$$V_3 = \{(0, y, y) \in \mathbf{F}^3 : y \in \mathbf{F}\}.$$

Then $\mathbf{F}^3 = V_1 + V_2 + V_3$ because every vector $(x, y, z) \in \mathbf{F}^3$ can be written as

$$(x, y, z) = (x, y, 0) + (0, 0, z) + (0, 0, 0),$$

where the first vector on the right side is in $V_1$, the second vector is in $V_2$, and the third vector is in $V_3$.

However, $\mathbf{F}^3$ does not equal the direct sum of $V_1, V_2, V_3$, because the vector $(0, 0, 0)$ can be written in more than one way as a sum $v_1 + v_2 + v_3$, with each $v_k \in V_k$. Specifically, we have

$$(0, 0, 0) = (0, 1, 0) + (0, 0, 1) + (0, -1, -1)$$

and, of course,

$$(0, 0, 0) = (0, 0, 0) + (0, 0, 0) + (0, 0, 0),$$

where the first vector on the right side of each equation above is in $V_1$, the second vector is in $V_2$, and the third vector is in $V_3$. Thus the sum $V_1 + V_2 + V_3$ is not a direct sum.

---

The definition of direct sum requires every vector in the sum to have a unique representation as an appropriate sum. The next result shows that when deciding whether a sum of subspaces is a direct sum, we only need to consider whether 0 can be uniquely written as an appropriate sum.

*The symbol ⊕, which is a plus sign inside a circle, reminds us that we are dealing with a special type of sum of subspaces—each element in the direct sum can be represented in only one way as a sum of elements from the specified subspaces.*

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**1.45** *condition for a direct sum*

Suppose $V_1, \dots, V_m$ are subspaces of $V$. Then $V_1 + \cdots + V_m$ is a direct sum if and only if the only way to write 0 as a sum $v_1 + \cdots + v_m$, where each $v_k \in V_k$, is by taking each $v_k$ equal to 0.

Proof First suppose $V_1 + \cdots + V_m$ is a direct sum. Then the definition of direct sum implies that the only way to write 0 as a sum $v_1 + \cdots + v_m$, where each $v_k \in V_k$, is by taking each $v_k$ equal to 0.

Now suppose that the only way to write 0 as a sum $v_1 + \cdots + v_m$, where each $v_k \in V_k$, is by taking each $v_k$ equal to 0. To show that $V_1 + \cdots + V_m$ is a direct sum, let $v \in V_1 + \cdots + V_m$. We can write

$$v = v_1 + \cdots + v_m$$

for some $v_1 \in V_1, \dots, v_m \in V_m$. To show that this representation is unique, suppose we also have

$$v = u_1 + \cdots + u_m,$$

where $u_1 \in V_1, \dots, u_m \in V_m$. Subtracting these two equations, we have

$$0 = (v_1 - u_1) + \cdots + (v_m - u_m).$$

Because $v_1 - u_1 \in V_1, \dots, v_m - u_m \in V_m$, the equation above implies that each $v_k - u_k$ equals 0. Thus $v_1 = u_1, \dots, v_m = u_m$, as desired. ∎

The next result gives a simple condition for testing whether a sum of two subspaces is a direct sum.

*The symbol $\iff$ used below means "if and only if"; this symbol could also be read to mean "is equivalent to".*

**1.46** *direct sum of two subspaces*

Suppose $U$ and $W$ are subspaces of $V$. Then

$$U + W \text{ is a direct sum} \iff U \cap W = \{0\}.$$

Proof First suppose that $U + W$ is a direct sum. If $v \in U \cap W$, then $0 = v + (-v)$, where $v \in U$ and $-v \in W$. By the unique representation of 0 as the sum of a vector in $U$ and a vector in $W$, we have $v = 0$. Thus $U \cap W = \{0\}$, completing the proof in one direction.

To prove the other direction, now suppose $U \cap W = \{0\}$. To prove that $U + W$ is a direct sum, suppose $u \in U$, $w \in W$, and

$$0 = u + w.$$

To complete the proof, we only need to show that $u = w = 0$ (by 1.45). The equation above implies that $u = -w \in W$. Thus $u \in U \cap W$. Hence $u = 0$, which by the equation above implies that $w = 0$, completing the proof. ∎

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The result above deals only with the case of two subspaces. When asking about a possible direct sum with more than two subspaces, it is not enough to test that each pair of the subspaces intersect only at 0. To see this, consider Example 1.44. In that nonexample of a direct sum, we have $V_1 \cap V_2 = V_1 \cap V_3 = V_2 \cap V_3 = \{0\}$.

*Sums of subspaces are analogous to unions of subsets. Similarly, direct sums of subspaces are analogous to disjoint unions of subsets. No two subspaces of a vector space can be disjoint, because both contain 0. So disjointness is replaced, at least in the case of two subspaces, with the requirement that the intersection equal* $\{0\}$.

## *Exercises 1C*

**1** For each of the following subsets of $\mathbf{F}^3$, determine whether it is a subspace of $\mathbf{F}^3$.
(a) $\{(x_1, x_2, x_3) \in \mathbf{F}^3 : x_1 + 2x_2 + 3x_3 = 0\}$
(b) $\{(x_1, x_2, x_3) \in \mathbf{F}^3 : x_1 + 2x_2 + 3x_3 = 4\}$
(c) $\{(x_1, x_2, x_3) \in \mathbf{F}^3 : x_1 x_2 x_3 = 0\}$
(d) $\{(x_1, x_2, x_3) \in \mathbf{F}^3 : x_1 = 5x_3\}$

**2** Verify all assertions about subspaces in Example 1.35.

**3** Show that the set of differentiable real-valued functions $f$ on the interval $(-4, 4)$ such that $f'(-1) = 3f(2)$ is a subspace of $\mathbf{R}^{(-4,4)}$.

**4** Suppose $b \in \mathbf{R}$. Show that the set of continuous real-valued functions $f$ on the interval $[0, 1]$ such that $\int_0^1 f = b$ is a subspace of $\mathbf{R}^{[0,1]}$ if and only if $b = 0$.

**5** Is $\mathbf{R}^2$ a subspace of the complex vector space $\mathbf{C}^2$?

**6** (a) Is $\{(a, b, c) \in \mathbf{R}^3 : a^3 = b^3\}$ a subspace of $\mathbf{R}^3$?
(b) Is $\{(a, b, c) \in \mathbf{C}^3 : a^3 = b^3\}$ a subspace of $\mathbf{C}^3$?

**7** Prove or give a counterexample: If $U$ is a nonempty subset of $\mathbf{R}^2$ such that $U$ is closed under addition and under taking additive inverses (meaning $-u \in U$ whenever $u \in U$), then $U$ is a subspace of $\mathbf{R}^2$.

**8** Give an example of a nonempty subset $U$ of $\mathbf{R}^2$ such that $U$ is closed under scalar multiplication, but $U$ is not a subspace of $\mathbf{R}^2$.

**9** A function $f\colon \mathbf{R} \to \mathbf{R}$ is called *periodic* if there exists a positive number $p$ such that $f(x) = f(x + p)$ for all $x \in \mathbf{R}$. Is the set of periodic functions from $\mathbf{R}$ to $\mathbf{R}$ a subspace of $\mathbf{R}^{\mathbf{R}}$? Explain.

**10** Suppose $V_1$ and $V_2$ are subspaces of $V$. Prove that the intersection $V_1 \cap V_2$ is a subspace of $V$.

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**11** Prove that the intersection of every collection of subspaces of $V$ is a subspace of $V$.

**12** Prove that the union of two subspaces of $V$ is a subspace of $V$ if and only if one of the subspaces is contained in the other.

**13** Prove that the union of three subspaces of $V$ is a subspace of $V$ if and only if one of the subspaces contains the other two.

*This exercise is surprisingly harder than Exercise 12, possibly because this exercise is not true if we replace* $\mathbf{F}$ *with a field containing only two elements.*

**14** Suppose

$$U = \{(x, -x, 2x) \in \mathbf{F}^3 : x \in \mathbf{F}\} \quad \text{and} \quad W = \{(x, x, 2x) \in \mathbf{F}^3 : x \in \mathbf{F}\}.$$

Describe $U + W$ using symbols, and also give a description of $U + W$ that uses no symbols.

**15** Suppose $U$ is a subspace of $V$. What is $U + U$?

**16** Is the operation of addition on the subspaces of $V$ commutative? In other words, if $U$ and $W$ are subspaces of $V$, is $U + W = W + U$?

**17** Is the operation of addition on the subspaces of $V$ associative? In other words, if $V_1, V_2, V_3$ are subspaces of $V$, is

$$(V_1 + V_2) + V_3 = V_1 + (V_2 + V_3)?$$

**18** Does the operation of addition on the subspaces of $V$ have an additive identity? Which subspaces have additive inverses?

**19** Prove or give a counterexample: If $V_1, V_2, U$ are subspaces of $V$ such that

$$V_1 + U = V_2 + U,$$

then $V_1 = V_2$.

**20** Suppose

$$U = \{(x, x, y, y) \in \mathbf{F}^4 : x, y \in \mathbf{F}\}.$$

Find a subspace $W$ of $\mathbf{F}^4$ such that $\mathbf{F}^4 = U \oplus W$.

**21** Suppose

$$U = \{(x, y, x + y, x - y, 2x) \in \mathbf{F}^5 : x, y \in \mathbf{F}\}.$$

Find a subspace $W$ of $\mathbf{F}^5$ such that $\mathbf{F}^5 = U \oplus W$.

**22** Suppose

$$U = \{(x, y, x + y, x - y, 2x) \in \mathbf{F}^5 : x, y \in \mathbf{F}\}.$$

Find three subspaces $W_1, W_2, W_3$ of $\mathbf{F}^5$, none of which equals $\{0\}$, such that $\mathbf{F}^5 = U \oplus W_1 \oplus W_2 \oplus W_3$.

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**23** Prove or give a counterexample: If $V_1, V_2, U$ are subspaces of $V$ such that

$$V = V_1 \oplus U \quad \text{and} \quad V = V_2 \oplus U,$$

then $V_1 = V_2$.

*Hint: When trying to discover whether a conjecture in linear algebra is true or false, it is often useful to start by experimenting in* $\mathbf{F}^2$.

**24** A function $f\colon \mathbf{R} \to \mathbf{R}$ is called *even* if

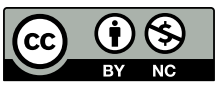
$$f(-x) = f(x)$$

for all $x \in \mathbf{R}$. A function $f\colon \mathbf{R} \to \mathbf{R}$ is called *odd* if

$$f(-x) = -f(x)$$

for all $x \in \mathbf{R}$. Let $V_e$ denote the set of real-valued even functions on $\mathbf{R}$ and let $V_o$ denote the set of real-valued odd functions on $\mathbf{R}$. Show that $\mathbf{R}^{\mathbf{R}} = V_e \oplus V_o$.

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Chapter 2

# Finite-Dimensional Vector Spaces

In the last chapter we learned about vector spaces. Linear algebra focuses not on arbitrary vector spaces, but on finite-dimensional vector spaces, which we introduce in this chapter.

We begin this chapter by considering linear combinations of lists of vectors. This leads us to the crucial concept of linear independence. The linear dependence lemma will become one of our most useful tools.

A list of vectors in a vector space that is small enough to be linearly independent and big enough so the linear combinations of the list fill up the vector space is called a basis of the vector space. We will see that every basis of a vector space has the same length, which will allow us to define the dimension of a vector space.

This chapter ends with a formula for the dimension of the sum of two subspaces.

*standing assumptions for this chapter*

- **F** denotes **R** or **C**.
- $V$ denotes a vector space over **F**.



*The main building of the Institute for Advanced Study, in Princeton, New Jersey. Paul Halmos (1916–2006) wrote the first modern linear algebra book in this building. Halmos's linear algebra book was published in 1942 (second edition published in 1958). The title of Halmos's book was the same as the title of this chapter.*

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# 2A Span and Linear Independence

We have been writing lists of numbers surrounded by parentheses, and we will continue to do so for elements of $\mathbf{F}^n$; for example, $(2, -7, 8) \in \mathbf{F}^3$. However, now we need to consider lists of vectors (which may be elements of $\mathbf{F}^n$ or of other vector spaces). To avoid confusion, we will usually write lists of vectors without surrounding parentheses. For example, $(4, 1, 6), (9, 5, 7)$ is a list of length two of vectors in $\mathbf{R}^3$.

**2.1 notation: *list of vectors***

We will usually write lists of vectors without surrounding parentheses.

## *Linear Combinations and Span*

A sum of scalar multiples of the vectors in a list is called a linear combination of the list. Here is the formal definition.

**2.2 definition: *linear combination***

A *linear combination* of a list $v_1, \ldots, v_m$ of vectors in $V$ is a vector of the form

$$a_1 v_1 + \cdots + a_m v_m,$$

where $a_1, \ldots, a_m \in \mathbf{F}$.

**2.3 example: *linear combinations in* $\mathbf{R}^3$**

- $(17, -4, 2)$ is a linear combination of $(2, 1, -3), (1, -2, 4)$, which is a list of length two of vectors in $\mathbf{R}^3$, because

$$(17, -4, 2) = 6(2, 1, -3) + 5(1, -2, 4).$$

- $(17, -4, 5)$ is not a linear combination of $(2, 1, -3), (1, -2, 4)$, which is a list of length two of vectors in $\mathbf{R}^3$, because there do not exist numbers $a_1, a_2 \in \mathbf{F}$ such that

$$(17, -4, 5) = a_1(2, 1, -3) + a_2(1, -2, 4).$$

  In other words, the system of equations

$$\begin{aligned} 17 &= 2a_1 + a_2 \\ -4 &= a_1 - 2a_2 \\ 5 &= -3a_1 + 4a_2 \end{aligned}$$

  has no solutions (as you should verify).

**2.4 definition: *span***

The set of all linear combinations of a list of vectors $v_1, \ldots, v_m$ in $V$ is called the *span* of $v_1, \ldots, v_m$, denoted by $\operatorname{span}(v_1, \ldots, v_m)$. In other words,

$$\operatorname{span}(v_1, \ldots, v_m) = \{a_1 v_1 + \cdots + a_m v_m : a_1, \ldots, a_m \in \mathbf{F}\}.$$

The span of the empty list $( )$ is defined to be $\{0\}$.

**2.5 example: span**

The previous example shows that in $\mathbf{F}^3$,

- $(17, -4, 2) \in \operatorname{span}((2, 1, -3), (1, -2, 4))$;
- $(17, -4, 5) \notin \operatorname{span}((2, 1, -3), (1, -2, 4))$.

Some mathematicians use the term ***linear span***, which means the same as span.

**2.6 *span is the smallest containing subspace***

The span of a list of vectors in $V$ is the smallest subspace of $V$ containing all vectors in the list.

Proof Suppose $v_1, \ldots, v_m$ is a list of vectors in $V$.

First we show that $\operatorname{span}(v_1, \ldots, v_m)$ is a subspace of $V$. The additive identity is in $\operatorname{span}(v_1, \ldots, v_m)$ because

$$0 = 0v_1 + \cdots + 0v_m.$$

Also, $\operatorname{span}(v_1, \ldots, v_m)$ is closed under addition because

$$(a_1 v_1 + \cdots + a_m v_m) + (c_1 v_1 + \cdots + c_m v_m) = (a_1 + c_1)v_1 + \cdots + (a_m + c_m)v_m.$$

Furthermore, $\operatorname{span}(v_1, \ldots, v_m)$ is closed under scalar multiplication because

$$\lambda(a_1 v_1 + \cdots + a_m v_m) = \lambda a_1 v_1 + \cdots + \lambda a_m v_m.$$

Thus $\operatorname{span}(v_1, \ldots, v_m)$ is a subspace of $V$ (by 1.34).

Each $v_k$ is a linear combination of $v_1, \ldots, v_m$ (to show this, set $a_k = 1$ and let the other $a$'s in 2.2 equal 0). Thus $\operatorname{span}(v_1, \ldots, v_m)$ contains each $v_k$. Conversely, because subspaces are closed under scalar multiplication and addition, every subspace of $V$ that contains each $v_k$ contains $\operatorname{span}(v_1, \ldots, v_m)$. Thus $\operatorname{span}(v_1, \ldots, v_m)$ is the smallest subspace of $V$ containing all the vectors $v_1, \ldots, v_m$. ∎

**2.7 definition: *spans***

If $\operatorname{span}(v_1, \ldots, v_m)$ equals $V$, we say that the list $v_1, \ldots, v_m$ *spans* $V$.

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2.8 example: *a list that spans* $\mathbf{F}^n$

Suppose $n$ is a positive integer. We want to show that

$$(1, 0, ..., 0), (0, 1, 0, ..., 0), ..., (0, ..., 0, 1)$$

spans $\mathbf{F}^n$. Here the $k^{\text{th}}$ vector in the list above has 1 in the $k^{\text{th}}$ slot and 0 in all other slots.

Suppose $(x_1, ..., x_n) \in \mathbf{F}^n$. Then

$$(x_1, ..., x_n) = x_1(1, 0, ..., 0) + x_2(0, 1, 0, ..., 0) + \cdots + x_n(0, ..., 0, 1).$$

Thus $(x_1, ..., x_n) \in \text{span}((1, 0, ..., 0), (0, 1, 0, ..., 0), ..., (0, ..., 0, 1))$, as desired.

Now we can make one of the key definitions in linear algebra.

2.9 definition: *finite-dimensional vector space*

A vector space is called *finite-dimensional* if some list of vectors in it spans the space.

Example 2.8 above shows that $\mathbf{F}^n$ is a finite-dimensional vector space for every positive integer $n$.

*Recall that by definition every list has finite length.*

The definition of a polynomial is no doubt already familiar to you.

2.10 definition: *polynomial,* $\mathcal{P}(\mathbf{F})$

- A function $p : \mathbf{F} \to \mathbf{F}$ is called a *polynomial* with coefficients in $\mathbf{F}$ if there exist $a_0, ..., a_m \in \mathbf{F}$ such that

$$p(z) = a_0 + a_1 z + a_2 z^2 + \cdots + a_m z^m$$

  for all $z \in \mathbf{F}$.
- $\mathcal{P}(\mathbf{F})$ is the set of all polynomials with coefficients in $\mathbf{F}$.

With the usual operations of addition and scalar multiplication, $\mathcal{P}(\mathbf{F})$ is a vector space over $\mathbf{F}$, as you should verify. Hence $\mathcal{P}(\mathbf{F})$ is a subspace of $\mathbf{F}^{\mathbf{F}}$, the vector space of functions from $\mathbf{F}$ to $\mathbf{F}$.

If a polynomial (thought of as a function from $\mathbf{F}$ to $\mathbf{F}$) is represented by two sets of coefficients, then subtracting one representation of the polynomial from the other produces a polynomial that is identically zero as a function on $\mathbf{F}$ and hence has all zero coefficients (if you are unfamiliar with this fact, just believe it for now; we will prove it later—see 4.8). **Conclusion:** the coefficients of a polynomial are uniquely determined by the polynomial. Thus the next definition uniquely defines the degree of a polynomial.

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**2.11 definition: *degree of a polynomial*, $\deg p$**

- A polynomial $p \in \mathcal{P}(\mathbf{F})$ is said to have *degree* $m$ if there exist scalars $a_0, a_1, \ldots, a_m \in \mathbf{F}$ with $a_m \neq 0$ such that for every $z \in \mathbf{F}$, we have
$$p(z) = a_0 + a_1 z + \cdots + a_m z^m.$$
- The polynomial that is identically 0 is said to have degree $-\infty$.
- The degree of a polynomial $p$ is denoted by $\deg p$.

In the next definition, we use the convention that $-\infty < m$, which means that the polynomial 0 is in $\mathcal{P}_m(\mathbf{F})$.

**2.12 notation: $\mathcal{P}_m(\mathbf{F})$**

For $m$ a nonnegative integer, $\mathcal{P}_m(\mathbf{F})$ denotes the set of all polynomials with coefficients in $\mathbf{F}$ and degree at most $m$.

If $m$ is a nonnegative integer, then $\mathcal{P}_m(\mathbf{F}) = \operatorname{span}(1, z, \ldots, z^m)$ [here we slightly abuse notation by letting $z^k$ denote a function]. Thus $\mathcal{P}_m(\mathbf{F})$ is a finite-dimensional vector space for each nonnegative integer $m$.

**2.13 definition: *infinite-dimensional vector space***

A vector space is called *infinite-dimensional* if it is not finite-dimensional.

**2.14 example: $\mathcal{P}(\mathbf{F})$ *is infinite-dimensional.***

Consider any list of elements of $\mathcal{P}(\mathbf{F})$. Let $m$ denote the highest degree of the polynomials in this list. Then every polynomial in the span of this list has degree at most $m$. Thus $z^{m+1}$ is not in the span of our list. Hence no list spans $\mathcal{P}(\mathbf{F})$. Thus $\mathcal{P}(\mathbf{F})$ is infinite-dimensional.

## *Linear Independence*

Suppose $v_1, \ldots, v_m \in V$ and $v \in \operatorname{span}(v_1, \ldots, v_m)$. By the definition of span, there exist $a_1, \ldots, a_m \in \mathbf{F}$ such that
$$v = a_1 v_1 + \cdots + a_m v_m.$$
Consider the question of whether the choice of scalars in the equation above is unique. Suppose $c_1, \ldots, c_m$ is another set of scalars such that
$$v = c_1 v_1 + \cdots + c_m v_m.$$
Subtracting the last two equations, we have
$$0 = (a_1 - c_1)v_1 + \cdots + (a_m - c_m)v_m.$$

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Thus we have written 0 as a linear combination of $(v_1, ..., v_m)$. If the only way to do this is by using 0 for all the scalars in the linear combination, then each $a_k - c_k$ equals 0, which means that each $a_k$ equals $c_k$ (and thus the choice of scalars was indeed unique). This situation is so important that we give it a special name—linear independence—which we now define.

**2.15 definition: *linearly independent***

- A list $v_1, ..., v_m$ of vectors in $V$ is called *linearly independent* if the only choice of $a_1, ..., a_m \in \mathbf{F}$ that makes

$$a_1 v_1 + \cdots + a_m v_m = 0$$

  is $a_1 = \cdots = a_m = 0$.
- The empty list $( )$ is also declared to be linearly independent.

The reasoning above shows that $v_1, ..., v_m$ is linearly independent if and only if each vector in $\operatorname{span}(v_1, ..., v_m)$ has only one representation as a linear combination of $v_1, ..., v_m$.

**2.16 example: *linearly independent lists***

(a) To see that the list $(1, 0, 0, 0), (0, 1, 0, 0), (0, 0, 1, 0)$ is linearly independent in $\mathbf{F}^4$, suppose $a_1, a_2, a_3 \in \mathbf{F}$ and

$$a_1(1, 0, 0, 0) + a_2(0, 1, 0, 0) + a_3(0, 0, 1, 0) = (0, 0, 0, 0).$$

Thus

$$(a_1, a_2, a_3, 0) = (0, 0, 0, 0).$$

Hence $a_1 = a_2 = a_3 = 0$. Thus the list $(1, 0, 0, 0), (0, 1, 0, 0), (0, 0, 1, 0)$ is linearly independent in $\mathbf{F}^4$.

(b) Suppose $m$ is a nonnegative integer. To see that the list $1, z, ..., z^m$ is linearly independent in $\mathcal{P}(\mathbf{F})$, suppose $a_0, a_1, ..., a_m \in \mathbf{F}$ and

$$a_0 + a_1 z + \cdots + a_m z^m = 0,$$

where we think of both sides as elements of $\mathcal{P}(\mathbf{F})$. Then

$$a_0 + a_1 z + \cdots + a_m z^m = 0$$

for all $z \in \mathbf{F}$. As discussed earlier (and as follows from 4.8), this implies that $a_0 = a_1 = \cdots = a_m = 0$. Thus $1, z, ..., z^m$ is a linearly independent list in $\mathcal{P}(\mathbf{F})$.

(c) A list of length one in a vector space is linearly independent if and only if the vector in the list is not 0.

(d) A list of length two in a vector space is linearly independent if and only if neither of the two vectors in the list is a scalar multiple of the other.

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If some vectors are removed from a linearly independent list, the remaining list is also linearly independent, as you should verify.

**2.17 definition: *linearly dependent***

- A list of vectors in $V$ is called *linearly dependent* if it is not linearly independent.
- In other words, a list $v_1, \ldots, v_m$ of vectors in $V$ is linearly dependent if there exist $a_1, \ldots, a_m \in \mathbf{F}$, not all 0, such that $a_1 v_1 + \cdots + a_m v_m = 0$.

**2.18 example: *linearly dependent lists***

- $(2, 3, 1), (1, -1, 2), (7, 3, 8)$ is linearly dependent in $\mathbf{F}^3$ because
$$2(2, 3, 1) + 3(1, -1, 2) + (-1)(7, 3, 8) = (0, 0, 0).$$
- The list $(2, 3, 1), (1, -1, 2), (7, 3, c)$ is linearly dependent in $\mathbf{F}^3$ if and only if $c = 8$, as you should verify.
- If some vector in a list of vectors in $V$ is a linear combination of the other vectors, then the list is linearly dependent. (Proof: After writing one vector in the list as equal to a linear combination of the other vectors, move that vector to the other side of the equation, where it will be multiplied by $-1$.)
- Every list of vectors in $V$ containing the 0 vector is linearly dependent. (This is a special case of the previous bullet point.)

The next lemma is a terrific tool. It states that given a linearly dependent list of vectors, one of the vectors is in the span of the previous ones. Furthermore, we can throw out that vector without changing the span of the original list.

**2.19 *linear dependence lemma***

Suppose $v_1, \ldots, v_m$ is a linearly dependent list in $V$. Then there exists $k \in \{1, 2, \ldots, m\}$ such that
$$v_k \in \operatorname{span}(v_1, \ldots, v_{k-1}).$$
Furthermore, if $k$ satisfies the condition above and the $k^{\text{th}}$ term is removed from $v_1, \ldots, v_m$, then the span of the remaining list equals $\operatorname{span}(v_1, \ldots, v_m)$.

Proof Because the list $v_1, \ldots, v_m$ is linearly dependent, there exist numbers $a_1, \ldots, a_m \in \mathbf{F}$, not all 0, such that
$$a_1 v_1 + \cdots + a_m v_m = 0.$$
Let $k$ be the largest element of $\{1, \ldots, m\}$ such that $a_k \neq 0$. Then
$$v_k = -\frac{a_1}{a_k} v_1 - \cdots - \frac{a_{k-1}}{a_k} v_{k-1},$$
which proves that $v_k \in \operatorname{span}(v_1, \ldots, v_{k-1})$, as desired.

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Now suppose $k$ is any element of $\{1, ..., m\}$ such that $v_k \in \text{span}(v_1, ..., v_{k-1})$. Let $b_1, ..., b_{k-1} \in \mathbf{F}$ be such that

$$v_k = b_1 v_1 + \cdots + b_{k-1} v_{k-1}. \tag{2.20}$$

Suppose $u \in \text{span}(v_1, ..., v_m)$. Then there exist $c_1, ..., c_m \in \mathbf{F}$ such that

$$u = c_1 v_1 + \cdots + c_m v_m.$$

In the equation above, we can replace $v_k$ with the right side of 2.20, which shows that $u$ is in the span of the list obtained by removing the $k^{\text{th}}$ term from $v_1, ..., v_m$. Thus removing the $k^{\text{th}}$ term of the list $v_1, ..., v_m$ does not change the span of the list. ∎

If $k = 1$ in the linear dependence lemma, then $v_k \in \text{span}(v_1, ..., v_{k-1})$ means that $v_1 = 0$, because $\text{span}(\,) = \{0\}$. Note also that parts of the proof of the linear dependence lemma need to be modified if $k = 1$. In general, the proofs in the rest of the book will not call attention to special cases that must be considered involving lists of length 0, the subspace $\{0\}$, or other trivial cases for which the result is true but needs a slightly different proof. Be sure to check these special cases yourself.

2.21 example: *smallest $k$ in linear dependence lemma*

Consider the list

$$(1, 2, 3), (6, 5, 4), (15, 16, 17), (8, 9, 7)$$

in $\mathbf{R}^3$. This list of length four is linearly dependent, as we will soon see. Thus the linear dependence lemma implies that there exists $k \in \{1, 2, 3, 4\}$ such that the $k^{\text{th}}$ vector in this list is a linear combination of the previous vectors in the list. Let's see how to find the smallest value of $k$ that works.

Taking $k = 1$ in the linear dependence lemma works if and only if the first vector in the list equals 0. Because $(1, 2, 3)$ is not the 0 vector, we cannot take $k = 1$ for this list.

Taking $k = 2$ in the linear dependence lemma works if and only if the second vector in the list is a scalar multiple of the first vector. However, there does not exist $c \in \mathbf{R}$ such that $(6, 5, 4) = c(1, 2, 3)$. Thus we cannot take $k = 2$ for this list.

Taking $k = 3$ in the linear dependence lemma works if and only if the third vector in the list is a linear combination of the first two vectors. Thus for the list in this example, we want to know whether there exist $a, b \in \mathbf{R}$ such that

$$(15, 16, 17) = a(1, 2, 3) + b(6, 5, 4).$$

The equation above is equivalent to a system of three linear equations in the two unknowns $a, b$. Using Gaussian elimination or appropriate software, we find that $a = 3, b = 2$ is a solution of the equation above, as you can verify. Thus for the list in this example, taking $k = 3$ is the smallest value of $k$ that works in the linear dependence lemma.

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Now we come to a key result. It says that no linearly independent list in $V$ is longer than a spanning list in $V$.

**2.22** *length of linearly independent list $\leq$ length of spanning list*

In a finite-dimensional vector space, the length of every linearly independent list of vectors is less than or equal to the length of every spanning list of vectors.

Proof Suppose that $u_1, ..., u_m$ is linearly independent in $V$. Suppose also that $w_1, ..., w_n$ spans $V$. We need to prove that $m \leq n$. We do so through the process described below with $m$ steps; note that in each step we add one of the $u$'s and remove one of the $w$'s.

**Step 1**

Let $B$ be the list $w_1, ..., w_n$, which spans $V$. Adjoining $u_1$ at the beginning of this list produces a linearly dependent list (because $u_1$ can be written as a linear combination of $w_1, ..., w_m$). In other words, the list

$$u_1, w_1, ..., w_n$$

is linearly dependent.

Thus by the linear dependence lemma (2.19), one of the vectors in the list above is a linear combination of the previous vectors in the list. We know that $u_1 \neq 0$ because the list $u_1, ..., u_m$ is linearly independent. Thus $u_1$ is not in the span of the previous vectors in the list above (because $u_1$ is not in $\{0\}$, which is the span of the empty list). Hence the linear dependence lemma implies that we can remove one of the $w$'s so that the new list $B$ (of length $n$) consisting of $u_1$ and the remaining $w$'s spans $V$.

**Step $k$, for $k = 2, ..., m$**

The list $B$ (of length $n$) from step $k-1$ spans $V$. In particular, $u_k$ is in the span of the list $B$. Thus the list of length $(n+1)$ obtained by adjoining $u_k$ to $B$, placing it just after $u_1, ..., u_{k-1}$, is linearly dependent. By the linear dependence lemma (2.19), one of the vectors in this list is in the span of the previous ones, and because $u_1, ..., u_k$ is linearly independent, this vector cannot be one of the $u$'s.

Hence there still must be at least one remaining $w$ at this step. We can remove from our new list (after adjoining $u_k$ in the proper place) a $w$ that is a linear combination of the previous vectors in the list, so that the new list $B$ (of length $n$) consisting of $u_1, ..., u_k$ and the remaining $w$'s spans $V$.

After step $m$, we have added all the $u$'s and the process stops. At each step as we add a $u$ to $B$, the linear dependence lemma implies that there is some $w$ to remove. Thus there are at least as many $w$'s as $u$'s. ∎

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The next two examples show how the result above can be used to show, without any computations, that certain lists are not linearly independent and that certain lists do not span a given vector space.

2.23 example: *no list of length 4 is linearly independent in* $\mathbf{R}^3$

The list $(1, 0, 0), (0, 1, 0), (0, 0, 1)$, which has length three, spans $\mathbf{R}^3$. Thus no list of length larger than three is linearly independent in $\mathbf{R}^3$.

For example, we now know that $(1, 2, 3), (4, 5, 8), (9, 6, 7), (-3, 2, 8)$, which is a list of length four, is not linearly independent in $\mathbf{R}^3$.

2.24 example: *no list of length 3 spans* $\mathbf{R}^4$

The list $(1, 0, 0, 0), (0, 1, 0, 0), (0, 0, 1, 0), (0, 0, 0, 1)$, which has length four, is linearly independent in $\mathbf{R}^4$. Thus no list of length less than four spans $\mathbf{R}^4$.

For example, we now know that $(1, 2, 3, -5), (4, 5, 8, 3), (9, 6, 7, -1)$, which is a list of length three, does not span $\mathbf{R}^4$.

Our intuition suggests that every subspace of a finite-dimensional vector space should also be finite-dimensional. We now prove that this intuition is correct.

**2.25** ***finite-dimensional subspaces***

Every subspace of a finite-dimensional vector space is finite-dimensional.

Proof Suppose $V$ is finite-dimensional and $U$ is a subspace of $V$. We need to prove that $U$ is finite-dimensional. We do this through the following multistep construction.

**Step 1**

If $U = \{0\}$, then $U$ is finite-dimensional and we are done. If $U \neq \{0\}$, then choose a nonzero vector $u_1 \in U$.

**Step *k***

If $U = \operatorname{span}(u_1, \ldots, u_{k-1})$, then $U$ is finite-dimensional and we are done. If $U \neq \operatorname{span}(u_1, \ldots, u_{k-1})$, then choose a vector $u_k \in U$ such that

$$u_k \notin \operatorname{span}(u_1, \ldots, u_{k-1}).$$

After each step, as long as the process continues, we have constructed a list of vectors such that no vector in this list is in the span of the previous vectors. Thus after each step we have constructed a linearly independent list, by the linear dependence lemma (2.19). This linearly independent list cannot be longer than any spanning list of $V$ (by 2.22). Thus the process eventually terminates, which means that $U$ is finite-dimensional. ∎

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## *Exercises 2A*

**1** Find a list of four distinct vectors in $\mathbf{F}^3$ whose span equals

$$\{(x, y, z) \in \mathbf{F}^3 : x + y + z = 0\}.$$

**2** Prove or give a counterexample: If $v_1, v_2, v_3, v_4$ spans $V$, then the list

$$v_1 - v_2, v_2 - v_3, v_3 - v_4, v_4$$

also spans $V$.

**3** Suppose $v_1, ..., v_m$ is a list of vectors in $V$. For $k \in \{1, ..., m\}$, let

$$w_k = v_1 + \cdots + v_k.$$

Show that $\text{span}(v_1, ..., v_m) = \text{span}(w_1, ..., w_m)$.

**4** (a) Show that a list of length one in a vector space is linearly independent if and only if the vector in the list is not 0.

(b) Show that a list of length two in a vector space is linearly independent if and only if neither of the two vectors in the list is a scalar multiple of the other.

**5** Find a number $t$ such that

$$(3, 1, 4), (2, -3, 5), (5, 9, t)$$

is not linearly independent in $\mathbf{R}^3$.

**6** Show that the list $(2, 3, 1), (1, -1, 2), (7, 3, c)$ is linearly dependent in $\mathbf{F}^3$ if and only if $c = 8$.

**7** (a) Show that if we think of $\mathbf{C}$ as a vector space over $\mathbf{R}$, then the list $1 + i, 1 - i$ is linearly independent.

(b) Show that if we think of $\mathbf{C}$ as a vector space over $\mathbf{C}$, then the list $1 + i, 1 - i$ is linearly dependent.

**8** Suppose $v_1, v_2, v_3, v_4$ is linearly independent in $V$. Prove that the list

$$v_1 - v_2, v_2 - v_3, v_3 - v_4, v_4$$

is also linearly independent.

**9** Prove or give a counterexample: If $v_1, v_2, ..., v_m$ is a linearly independent list of vectors in $V$, then

$$5v_1 - 4v_2, v_2, v_3, ..., v_m$$

is linearly independent.

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**10** Prove or give a counterexample: If $v_1, v_2, ..., v_m$ is a linearly independent list of vectors in $V$ and $\lambda \in \mathbf{F}$ with $\lambda \neq 0$, then $\lambda v_1, \lambda v_2, ..., \lambda v_m$ is linearly independent.

**11** Prove or give a counterexample: If $v_1, ..., v_m$ and $w_1, ..., w_m$ are linearly independent lists of vectors in $V$, then the list $v_1 + w_1, ..., v_m + w_m$ is linearly independent.

**12** Suppose $v_1, ..., v_m$ is linearly independent in $V$ and $w \in V$. Prove that if $v_1 + w, ..., v_m + w$ is linearly dependent, then $w \in \operatorname{span}(v_1, ..., v_m)$.

**13** Suppose $v_1, ..., v_m$ is linearly independent in $V$ and $w \in V$. Show that

$$v_1, ..., v_m, w \text{ is linearly independent} \iff w \notin \operatorname{span}(v_1, ..., v_m).$$

**14** Suppose $v_1, ..., v_m$ is a list of vectors in $V$. For $k \in \{1, ..., m\}$, let

$$w_k = v_1 + \cdots + v_k.$$

Show that the list $v_1, ..., v_m$ is linearly independent if and only if the list $w_1, ..., w_m$ is linearly independent.

**15** Explain why there does not exist a list of six polynomials that is linearly independent in $\mathcal{P}_4(\mathbf{F})$.

**16** Explain why no list of four polynomials spans $\mathcal{P}_4(\mathbf{F})$.

**17** Prove that $V$ is infinite-dimensional if and only if there is a sequence $v_1, v_2, ...$ of vectors in $V$ such that $v_1, ..., v_m$ is linearly independent for every positive integer $m$.

**18** Prove that $\mathbf{F}^\infty$ is infinite-dimensional.

**19** Prove that the real vector space of all continuous real-valued functions on the interval $[0, 1]$ is infinite-dimensional.

**20** Suppose $p_0, p_1, ..., p_m$ are polynomials in $\mathcal{P}_m(\mathbf{F})$ such that $p_k(2) = 0$ for each $k \in \{0, ..., m\}$. Prove that $p_0, p_1, ..., p_m$ is not linearly independent in $\mathcal{P}_m(\mathbf{F})$.

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## 2B Bases

In the previous section, we discussed linearly independent lists and we also discussed spanning lists. Now we bring these concepts together by considering lists that have both properties.

**2.26 definition: *basis***

A *basis* of $V$ is a list of vectors in $V$ that is linearly independent and spans $V$.

**2.27 example: *bases***

(a) The list $(1, 0, \ldots, 0), (0, 1, 0, \ldots, 0), \ldots, (0, \ldots, 0, 1)$ is a basis of $\mathbf{F}^n$, called the *standard basis* of $\mathbf{F}^n$.

(b) The list $(1, 2), (3, 5)$ is a basis of $\mathbf{F}^2$. Note that this list has length two, which is the same as the length of the standard basis of $\mathbf{F}^2$. In the next section, we will see that this is not a coincidence.

(c) The list $(1, 2, -4), (7, -5, 6)$ is linearly independent in $\mathbf{F}^3$ but is not a basis of $\mathbf{F}^3$ because it does not span $\mathbf{F}^3$.

(d) The list $(1, 2), (3, 5), (4, 13)$ spans $\mathbf{F}^2$ but is not a basis of $\mathbf{F}^2$ because it is not linearly independent.

(e) The list $(1, 1, 0), (0, 0, 1)$ is a basis of $\{(x, x, y) \in \mathbf{F}^3 : x, y \in \mathbf{F}\}$.

(f) The list $(1, -1, 0), (1, 0, -1)$ is a basis of

$$\{(x, y, z) \in \mathbf{F}^3 : x + y + z = 0\}.$$

(g) The list $1, z, \ldots, z^m$ is a basis of $\mathcal{P}_m(\mathbf{F})$, called the *standard basis* of $\mathcal{P}_m(\mathbf{F})$.

In addition to the standard basis, $\mathbf{F}^n$ has many other bases. For example,

$$(7, 5), (-4, 9) \quad \text{and} \quad (1, 2), (3, 5)$$

are both bases of $\mathbf{F}^2$.

The next result helps explain why bases are useful. Recall that "uniquely" means "in only one way".

**2.28 *criterion for basis***

A list $v_1, \ldots, v_n$ of vectors in $V$ is a basis of $V$ if and only if every $v \in V$ can be written uniquely in the form

2.29
$$v = a_1 v_1 + \cdots + a_n v_n,$$

where $a_1, \ldots, a_n \in \mathbf{F}$.

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*This proof is essentially a repetition of the ideas that led us to the definition of linear independence.*

**Proof** First suppose that $v_1, \ldots, v_n$ is a basis of $V$. Let $v \in V$. Because $v_1, \ldots, v_n$ spans $V$, there exist $a_1, \ldots, a_n \in \mathbf{F}$ such that 2.29 holds. To show that the representation in 2.29 is unique, suppose $c_1, \ldots, c_n$ are scalars such that we also have

$$v = c_1 v_1 + \cdots + c_n v_n.$$

Subtracting the last equation from 2.29, we get

$$0 = (a_1 - c_1)v_1 + \cdots + (a_n - c_n)v_n.$$

This implies that each $a_k - c_k$ equals 0 (because $v_1, \ldots, v_n$ is linearly independent). Hence $a_1 = c_1, \ldots, a_n = c_n$. We have the desired uniqueness, completing the proof in one direction.

For the other direction, suppose every $v \in V$ can be written uniquely in the form given by 2.29. This implies that the list $v_1, \ldots, v_n$ spans $V$. To show that $v_1, \ldots, v_n$ is linearly independent, suppose $a_1, \ldots, a_n \in \mathbf{F}$ are such that

$$0 = a_1 v_1 + \cdots + a_n v_n.$$

The uniqueness of the representation 2.29 (taking $v = 0$) now implies that $a_1 = \cdots = a_n = 0$. Thus $v_1, \ldots, v_n$ is linearly independent and hence is a basis of $V$. ∎

A spanning list in a vector space may not be a basis because it is not linearly independent. Our next result says that given any spanning list, some (possibly none) of the vectors in it can be discarded so that the remaining list is linearly independent and still spans the vector space.

As an example in the vector space $\mathbf{F}^2$, if the procedure in the proof below is applied to the list $(1, 2), (3, 6), (4, 7), (5, 9)$, then the second and fourth vectors will be removed. This leaves $(1, 2), (4, 7)$, which is a basis of $\mathbf{F}^2$.

**2.30** *every spanning list contains a basis*

Every spanning list in a vector space can be reduced to a basis of the vector space.

**Proof** Suppose $v_1, \ldots, v_n$ spans $V$. We want to remove some of the vectors from $v_1, \ldots, v_n$ so that the remaining vectors form a basis of $V$. We do this through the multistep process described below.

Start with $B$ equal to the list $v_1, \ldots, v_n$.

**Step 1**

If $v_1 = 0$, then delete $v_1$ from $B$. If $v_1 \neq 0$, then leave $B$ unchanged.

**Step *k***

If $v_k$ is in $\operatorname{span}(v_1, \ldots, v_{k-1})$, then delete $v_k$ from the list $B$. If $v_k$ is not in $\operatorname{span}(v_1, \ldots, v_{k-1})$, then leave $B$ unchanged.

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Stop the process after step $n$, getting a list $B$. This list $B$ spans $V$ because our original list spanned $V$ and we have discarded only vectors that were already in the span of the previous vectors. The process ensures that no vector in $B$ is in the span of the previous ones. Thus $B$ is linearly independent, by the linear dependence lemma (2.19). Hence $B$ is a basis of $V$.

We now come to an important corollary of the previous result.

**2.31** *basis of finite-dimensional vector space*

Every finite-dimensional vector space has a basis.

Proof By definition, a finite-dimensional vector space has a spanning list. The previous result tells us that each spanning list can be reduced to a basis.

Our next result is in some sense a dual of 2.30, which said that every spanning list can be reduced to a basis. Now we show that given any linearly independent list, we can adjoin some additional vectors (this includes the possibility of adjoining no additional vectors) so that the extended list is still linearly independent but also spans the space.

**2.32** *every linearly independent list extends to a basis*

Every linearly independent list of vectors in a finite-dimensional vector space can be extended to a basis of the vector space.

Proof Suppose $u_1, \ldots, u_m$ is linearly independent in a finite-dimensional vector space $V$. Let $w_1, \ldots, w_n$ be a list of vectors in $V$ that spans $V$. Thus the list

$$u_1, \ldots, u_m, w_1, \ldots, w_n$$

spans $V$. Applying the procedure of the proof of 2.30 to reduce this list to a basis of $V$ produces a basis consisting of the vectors $u_1, \ldots, u_m$ and some of the $w$'s (none of the $u$'s get deleted in this procedure because $u_1, \ldots, u_m$ is linearly independent).

As an example in $\mathbf{F}^3$, suppose we start with the linearly independent list $(2, 3, 4), (9, 6, 8)$. If we take $w_1, w_2, w_3$ to be the standard basis of $\mathbf{F}^3$, then applying the procedure in the proof above produces the list

$$(2, 3, 4), (9, 6, 8), (0, 1, 0),$$

which is a basis of $\mathbf{F}^3$.

As an application of the result above, we now show that every subspace of a finite-dimensional vector space can be paired with another subspace to form a direct sum of the whole space.

*Using the same ideas but more advanced tools, the next result can be proved without the hypothesis that $V$ is finite-dimensional.*

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**2.33** *every subspace of $V$ is part of a direct sum equal to $V$*

Suppose $V$ is finite-dimensional and $U$ is a subspace of $V$. Then there is a subspace $W$ of $V$ such that $V = U \oplus W$.

Proof Because $V$ is finite-dimensional, so is $U$ (see 2.25). Thus there is a basis $u_1, \dots, u_m$ of $U$ (by 2.31). Of course $u_1, \dots, u_m$ is a linearly independent list of vectors in $V$. Hence this list can be extended to a basis $u_1, \dots, u_m, w_1, \dots, w_n$ of $V$ (by 2.32). Let $W = \operatorname{span}(w_1, \dots, w_n)$.

To prove that $V = U \oplus W$, by 1.46 we only need to show that

$$V = U + W \quad \text{and} \quad U \cap W = \{0\}.$$

To prove the first equation above, suppose $v \in V$. Then, because the list $u_1, \dots, u_m, w_1, \dots, w_n$ spans $V$, there exist $a_1, \dots, a_m, b_1, \dots, b_n \in \mathbf{F}$ such that

$$v = \underbrace{a_1 u_1 + \cdots + a_m u_m}_{u} + \underbrace{b_1 w_1 + \cdots + b_n w_n}_{w}.$$

We have $v = u + w$, where $u \in U$ and $w \in W$ are defined as above. Thus $v \in U + W$, completing the proof that $V = U + W$.

To show that $U \cap W = \{0\}$, suppose $v \in U \cap W$. Then there exist scalars $a_1, \dots, a_m, b_1, \dots, b_n \in \mathbf{F}$ such that

$$v = a_1 u_1 + \cdots + a_m u_m = b_1 w_1 + \cdots + b_n w_n.$$

Thus

$$a_1 u_1 + \cdots + a_m u_m - b_1 w_1 - \cdots - b_n w_n = 0.$$

Because $u_1, \dots, u_m, w_1, \dots, w_n$ is linearly independent, this implies that

$$a_1 = \cdots = a_m = b_1 = \cdots = b_n = 0.$$

Thus $v = 0$, completing the proof that $U \cap W = \{0\}$. ∎

## *Exercises 2B*

**1** Find all vector spaces that have exactly one basis.

**2** Verify all assertions in Example 2.27.

**3** (a) Let $U$ be the subspace of $\mathbf{R}^5$ defined by

$$U = \{(x_1, x_2, x_3, x_4, x_5) \in \mathbf{R}^5 : x_1 = 3x_2 \text{ and } x_3 = 7x_4\}.$$

Find a basis of $U$.

(b) Extend the basis in (a) to a basis of $\mathbf{R}^5$.

(c) Find a subspace $W$ of $\mathbf{R}^5$ such that $\mathbf{R}^5 = U \oplus W$.

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**4** (a) Let $U$ be the subspace of $\mathbf{C}^5$ defined by

$$U = \{(z_1, z_2, z_3, z_4, z_5) \in \mathbf{C}^5 : 6z_1 = z_2 \text{ and } z_3 + 2z_4 + 3z_5 = 0\}.$$

Find a basis of $U$.

(b) Extend the basis in (a) to a basis of $\mathbf{C}^5$.

(c) Find a subspace $W$ of $\mathbf{C}^5$ such that $\mathbf{C}^5 = U \oplus W$.

**5** Suppose $V$ is finite-dimensional and $U, W$ are subspaces of $V$ such that $V = U + W$. Prove that there exists a basis of $V$ consisting of vectors in $U \cup W$.

**6** Prove or give a counterexample: If $p_0, p_1, p_2, p_3$ is a list in $\mathcal{P}_3(\mathbf{F})$ such that none of the polynomials $p_0, p_1, p_2, p_3$ has degree 2, then $p_0, p_1, p_2, p_3$ is not a basis of $\mathcal{P}_3(\mathbf{F})$.

**7** Suppose $v_1, v_2, v_3, v_4$ is a basis of $V$. Prove that

$$v_1 + v_2, v_2 + v_3, v_3 + v_4, v_4$$

is also a basis of $V$.

**8** Prove or give a counterexample: If $v_1, v_2, v_3, v_4$ is a basis of $V$ and $U$ is a subspace of $V$ such that $v_1, v_2 \in U$ and $v_3 \notin U$ and $v_4 \notin U$, then $v_1, v_2$ is a basis of $U$.

**9** Suppose $v_1, \ldots, v_m$ is a list of vectors in $V$. For $k \in \{1, \ldots, m\}$, let

$$w_k = v_1 + \cdots + v_k.$$

Show that $v_1, \ldots, v_m$ is a basis of $V$ if and only if $w_1, \ldots, w_m$ is a basis of $V$.

**10** Suppose $U$ and $W$ are subspaces of $V$ such that $V = U \oplus W$. Suppose also that $u_1, \ldots, u_m$ is a basis of $U$ and $w_1, \ldots, w_n$ is a basis of $W$. Prove that

$$u_1, \ldots, u_m, w_1, \ldots, w_n$$

is a basis of $V$.

**11** Suppose $V$ is a real vector space. Show that if $v_1, \ldots, v_n$ is a basis of $V$ (as a real vector space), then $v_1, \ldots, v_n$ is also a basis of the complexification $V_{\mathbf{C}}$ (as a complex vector space).

*See Exercise 8 in Section 1B for the definition of the complexification $V_{\mathbf{C}}$.*

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## 2C Dimension

Although we have been discussing finite-dimensional vector spaces, we have not yet defined the dimension of such an object. How should dimension be defined? A reasonable definition should force the dimension of $\mathbf{F}^n$ to equal $n$. Notice that the standard basis

$$(1, 0, \ldots, 0), (0, 1, 0, \ldots, 0), \ldots, (0, \ldots, 0, 1)$$

of $\mathbf{F}^n$ has length $n$. Thus we are tempted to define the dimension as the length of a basis. However, a finite-dimensional vector space in general has many different bases, and our attempted definition makes sense only if all bases in a given vector space have the same length. Fortunately that turns out to be the case, as we now show.

**2.34** *basis length does not depend on basis*

Any two bases of a finite-dimensional vector space have the same length.

Proof Suppose $V$ is finite-dimensional. Let $B_1$ and $B_2$ be two bases of $V$. Then $B_1$ is linearly independent in $V$ and $B_2$ spans $V$, so the length of $B_1$ is at most the length of $B_2$ (by 2.22). Interchanging the roles of $B_1$ and $B_2$, we also see that the length of $B_2$ is at most the length of $B_1$. Thus the length of $B_1$ equals the length of $B_2$, as desired. ∎

Now that we know that any two bases of a finite-dimensional vector space have the same length, we can formally define the dimension of such spaces.

**2.35** definition: *dimension,* $\dim V$

- The *dimension* of a finite-dimensional vector space is the length of any basis of the vector space.
- The dimension of a finite-dimensional vector space $V$ is denoted by $\dim V$.

**2.36** example: *dimensions*

- $\dim \mathbf{F}^n = n$ because the standard basis of $\mathbf{F}^n$ has length $n$.
- $\dim \mathcal{P}_m(\mathbf{F}) = m + 1$ because the standard basis $1, z, \ldots, z^m$ of $\mathcal{P}_m(\mathbf{F})$ has length $m + 1$.
- If $U = \{(x, x, y) \in \mathbf{F}^3 : x, y \in \mathbf{F}\}$, then $\dim U = 2$ because $(1, 1, 0), (0, 0, 1)$ is a basis of $U$.
- If $U = \{(x, y, z) \in \mathbf{F}^3 : x + y + z = 0\}$, then $\dim U = 2$ because the list $(1, -1, 0), (1, 0, -1)$ is a basis of $U$.

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Every subspace of a finite-dimensional vector space is finite-dimensional (by 2.25) and so has a dimension. The next result gives the expected inequality about the dimension of a subspace.

**2.37** *dimension of a subspace*

If $V$ is finite-dimensional and $U$ is a subspace of $V$, then $\dim U \leq \dim V$.

Proof Suppose $V$ is finite-dimensional and $U$ is a subspace of $V$. Think of a basis of $U$ as a linearly independent list in $V$, and think of a basis of $V$ as a spanning list in $V$. Now use 2.22 to conclude that $\dim U \leq \dim V$.

To check that a list of vectors in $V$ is a basis of $V$, we must, according to the definition, show that the list in question satisfies two properties: it must be linearly independent and it must span $V$. The next two results show that if the list in question has the right length, then we only need to check that it satisfies one of the two required properties. First we prove that every linearly independent list of the right length is a basis.

*The real vector space $\mathbf{R}^2$ has dimension two; the complex vector space $\mathbf{C}$ has dimension one. As sets, $\mathbf{R}^2$ can be identified with $\mathbf{C}$ (and addition is the same on both spaces, as is scalar multiplication by real numbers). Thus when we talk about the dimension of a vector space, the role played by the choice of $\mathbf{F}$ cannot be neglected.*

**2.38** *linearly independent list of the right length is a basis*

Suppose $V$ is finite-dimensional. Then every linearly independent list of vectors in $V$ of length $\dim V$ is a basis of $V$.

Proof Suppose $\dim V = n$ and $v_1, \ldots, v_n$ is linearly independent in $V$. The list $v_1, \ldots, v_n$ can be extended to a basis of $V$ (by 2.32). However, every basis of $V$ has length $n$, so in this case the extension is the trivial one, meaning that no elements are adjoined to $v_1, \ldots, v_n$. Thus $v_1, \ldots, v_n$ is a basis of $V$, as desired.

The next result is a useful consequence of the previous result.

**2.39** *subspace of full dimension equals the whole space*

Suppose that $V$ is finite-dimensional and $U$ is a subspace of $V$ such that $\dim U = \dim V$. Then $U = V$.

Proof Let $u_1, \ldots, u_n$ be a basis of $U$. Thus $n = \dim U$, and by hypothesis we also have $n = \dim V$. Thus $u_1, \ldots, u_n$ is a linearly independent list of vectors in $V$ (because it is a basis of $U$) of length $\dim V$. From 2.38, we see that $u_1, \ldots, u_n$ is a basis of $V$. In particular every vector in $V$ is a linear combination of $u_1, \ldots, u_n$. Thus $U = V$.

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2.40 example: *a basis of* $\mathbf{F}^2$

Consider the list $(5, 7), (4, 3)$ of vectors in $\mathbf{F}^2$. This list of length two is linearly independent in $\mathbf{F}^2$ (because neither vector is a scalar multiple of the other). Note that $\mathbf{F}^2$ has dimension two. Thus 2.38 implies that the linearly independent list $(5, 7), (4, 3)$ of length two is a basis of $\mathbf{F}^2$ (we do not need to bother checking that it spans $\mathbf{F}^2$).

2.41 example: *a basis of a subspace of* $\mathcal{P}_3(\mathbf{R})$

Let $U$ be the subspace of $\mathcal{P}_3(\mathbf{R})$ defined by

$$U = \{p \in \mathcal{P}_3(\mathbf{R}) : p'(5) = 0\}.$$

To find a basis of $U$, first note that each of the polynomials 1, $(x-5)^2$, and $(x-5)^3$ is in $U$.

Suppose $a, b, c \in \mathbf{R}$ and

$$a + b(x-5)^2 + c(x-5)^3 = 0$$

for every $x \in \mathbf{R}$. Without explicitly expanding the left side of the equation above, we can see that the left side has a $cx^3$ term. Because the right side has no $x^3$ term, this implies that $c = 0$. Because $c = 0$, we see that the left side has a $bx^2$ term, which implies that $b = 0$. Because $b = c = 0$, we can also conclude that $a = 0$. Thus the equation above implies that $a = b = c = 0$. Hence the list $1, (x-5)^2, (x-5)^3$ is linearly independent in $U$. Thus $3 \le \dim U$. Hence

$$3 \le \dim U \le \dim \mathcal{P}_3(\mathbf{R}) = 4,$$

where we have used 2.37.

The polynomial $x$ is not in $U$ because its derivative is the constant function 1. Thus $U \ne \mathcal{P}_3(\mathbf{R})$. Hence $\dim U \ne 4$ (by 2.39). The inequality above now implies that $\dim U = 3$. Thus the linearly independent list $1, (x-5)^2, (x-5)^3$ in $U$ has length $\dim U$ and hence is a basis of $U$ (by 2.38).

Now we prove that a spanning list of the right length is a basis.

2.42 *spanning list of the right length is a basis*

Suppose $V$ is finite-dimensional. Then every spanning list of vectors in $V$ of length $\dim V$ is a basis of $V$.

Proof Suppose $\dim V = n$ and $v_1, \ldots, v_n$ spans $V$. The list $v_1, \ldots, v_n$ can be reduced to a basis of $V$ (by 2.30). However, every basis of $V$ has length $n$, so in this case the reduction is the trivial one, meaning that no elements are deleted from $v_1, \ldots, v_n$. Thus $v_1, \ldots, v_n$ is a basis of $V$, as desired. ∎

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The next result gives a formula for the dimension of the sum of two subspaces of a finite-dimensional vector space. This formula is analogous to a familiar counting formula: the number of elements in the union of two finite sets equals the number of elements in the first set, plus the number of elements in the second set, minus the number of elements in the intersection of the two sets.

**2.43** *dimension of a sum*

If $V_1$ and $V_2$ are subspaces of a finite-dimensional vector space, then

$$\dim(V_1 + V_2) = \dim V_1 + \dim V_2 - \dim(V_1 \cap V_2).$$

Proof Let $v_1, \ldots, v_m$ be a basis of $V_1 \cap V_2$; thus $\dim(V_1 \cap V_2) = m$. Because $v_1, \ldots, v_m$ is a basis of $V_1 \cap V_2$, it is linearly independent in $V_1$. Hence this list can be extended to a basis $v_1, \ldots, v_m, u_1, \ldots, u_j$ of $V_1$ (by 2.32). Thus $\dim V_1 = m + j$. Also extend $v_1, \ldots, v_m$ to a basis $v_1, \ldots, v_m, w_1, \ldots, w_k$ of $V_2$; thus $\dim V_2 = m + k$.

We will show that

2.44 $$v_1, \ldots, v_m, u_1, \ldots, u_j, w_1, \ldots, w_k$$

is a basis of $V_1 + V_2$. This will complete the proof, because then we will have

$$\begin{aligned}\dim(V_1 + V_2) &= m + j + k \\ &= (m + j) + (m + k) - m \\ &= \dim V_1 + \dim V_2 - \dim(V_1 \cap V_2).\end{aligned}$$

The list 2.44 is contained in $V_1 \cup V_2$ and thus is contained in $V_1 + V_2$. The span of this list contains $V_1$ and contains $V_2$ and hence is equal to $V_1 + V_2$. Thus to show that 2.44 is a basis of $V_1 + V_2$ we only need to show that it is linearly independent.

To prove that 2.44 is linearly independent, suppose

$$a_1 v_1 + \cdots + a_m v_m + b_1 u_1 + \cdots + b_j u_j + c_1 w_1 + \cdots + c_k w_k = 0,$$

where all the $a$'s, $b$'s, and $c$'s are scalars. We need to prove that all the $a$'s, $b$'s, and $c$'s equal 0. The equation above can be rewritten as

2.45 $$c_1 w_1 + \cdots + c_k w_k = -a_1 v_1 - \cdots - a_m v_m - b_1 u_1 - \cdots - b_j u_j,$$

which shows that $c_1 w_1 + \cdots + c_k w_k \in V_1$. All the $w$'s are in $V_2$, so this implies that $c_1 w_1 + \cdots + c_k w_k \in V_1 \cap V_2$. Because $v_1, \ldots, v_m$ is a basis of $V_1 \cap V_2$, we have

$$c_1 w_1 + \cdots + c_k w_k = d_1 v_1 + \cdots + d_m v_m$$

for some scalars $d_1, \ldots, d_m$. But $v_1, \ldots, v_m, w_1, \ldots, w_k$ is linearly independent, so the last equation implies that all the $c$'s (and $d$'s) equal 0. Thus 2.45 becomes the equation

$$a_1 v_1 + \cdots + a_m v_m + b_1 u_1 + \cdots + b_j u_j = 0.$$

Because the list $v_1, \ldots, v_m, u_1, \ldots, u_j$ is linearly independent, this equation implies that all the $a$'s and $b$'s are 0, completing the proof. ∎

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For $S$ a finite set, let $\#S$ denote the number of elements of $S$. The table below compares finite sets with finite-dimensional vector spaces, showing the analogy between $\#S$ (for sets) and $\dim V$ (for vector spaces), as well as the analogy between unions of subsets (in the context of sets) and sums of subspaces (in the context of vector spaces).

| **sets** | **vector spaces** |
|---|---|
| $S$ is a finite set | $V$ is a finite-dimensional vector space |
| $\#S$ | $\dim V$ |
| for subsets $S_1, S_2$ of $S$, the union $S_1 \cup S_2$ is the smallest subset of $S$ containing $S_1$ and $S_2$ | for subspaces $V_1, V_2$ of $V$, the sum $V_1 + V_2$ is the smallest subspace of $V$ containing $V_1$ and $V_2$ |
| $\#(S_1 \cup S_2)$ $= \#S_1 + \#S_2 - \#(S_1 \cap S_2)$ | $\dim(V_1 + V_2)$ $= \dim V_1 + \dim V_2 - \dim(V_1 \cap V_2)$ |
| $\#(S_1 \cup S_2) = \#S_1 + \#S_2$ $\iff S_1 \cap S_2 = \emptyset$ | $\dim(V_1 + V_2) = \dim V_1 + \dim V_2$ $\iff V_1 \cap V_2 = \{0\}$ |
| $S_1 \cup \cdots \cup S_m$ is a disjoint union $\iff$ $\#(S_1 \cup \cdots \cup S_m) = \#S_1 + \cdots + \#S_m$ | $V_1 + \cdots + V_m$ is a direct sum $\iff$ $\dim(V_1 + \cdots + V_m)$ $= \dim V_1 + \cdots + \dim V_m$ |

The last row above focuses on the analogy between disjoint unions (for sets) and direct sums (for vector spaces). The proof of the result in the last box above will be given in 3.94.

You should be able to find results about sets that correspond, via analogy, to the results about vector spaces in Exercises 12 through 18.

## *Exercises 2C*

**1** Show that the subspaces of $\mathbf{R}^2$ are precisely $\{0\}$, all lines in $\mathbf{R}^2$ containing the origin, and $\mathbf{R}^2$.

**2** Show that the subspaces of $\mathbf{R}^3$ are precisely $\{0\}$, all lines in $\mathbf{R}^3$ containing the origin, all planes in $\mathbf{R}^3$ containing the origin, and $\mathbf{R}^3$.

**3** (a) Let $U = \{p \in \mathcal{P}_4(\mathbf{F}) : p(6) = 0\}$. Find a basis of $U$.
(b) Extend the basis in (a) to a basis of $\mathcal{P}_4(\mathbf{F})$.
(c) Find a subspace $W$ of $\mathcal{P}_4(\mathbf{F})$ such that $\mathcal{P}_4(\mathbf{F}) = U \oplus W$.

**4** (a) Let $U = \{p \in \mathcal{P}_4(\mathbf{R}) : p''(6) = 0\}$. Find a basis of $U$.
(b) Extend the basis in (a) to a basis of $\mathcal{P}_4(\mathbf{R})$.
(c) Find a subspace $W$ of $\mathcal{P}_4(\mathbf{R})$ such that $\mathcal{P}_4(\mathbf{R}) = U \oplus W$.

**5** (a) Let $U = \{p \in \mathcal{P}_4(\mathbf{F}) : p(2) = p(5)\}$. Find a basis of $U$.
(b) Extend the basis in (a) to a basis of $\mathcal{P}_4(\mathbf{F})$.
(c) Find a subspace $W$ of $\mathcal{P}_4(\mathbf{F})$ such that $\mathcal{P}_4(\mathbf{F}) = U \oplus W$.

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**6** (a) Let $U = \{p \in \mathcal{P}_4(\mathbf{F}) : p(2) = p(5) = p(6)\}$. Find a basis of $U$.
(b) Extend the basis in (a) to a basis of $\mathcal{P}_4(\mathbf{F})$.
(c) Find a subspace $W$ of $\mathcal{P}_4(\mathbf{F})$ such that $\mathcal{P}_4(\mathbf{F}) = U \oplus W$.

**7** (a) Let $U = \left\{p \in \mathcal{P}_4(\mathbf{R}) : \int_{-1}^{1} p = 0\right\}$. Find a basis of $U$.
(b) Extend the basis in (a) to a basis of $\mathcal{P}_4(\mathbf{R})$.
(c) Find a subspace $W$ of $\mathcal{P}_4(\mathbf{R})$ such that $\mathcal{P}_4(\mathbf{R}) = U \oplus W$.

**8** Suppose $v_1, \ldots, v_m$ is linearly independent in $V$ and $w \in V$. Prove that

$$\dim \operatorname{span}(v_1 + w, \ldots, v_m + w) \geq m - 1.$$

**9** Suppose $m$ is a positive integer and $p_0, p_1, \ldots, p_m \in \mathcal{P}(\mathbf{F})$ are such that each $p_k$ has degree $k$. Prove that $p_0, p_1, \ldots, p_m$ is a basis of $\mathcal{P}_m(\mathbf{F})$.

**10** Suppose $m$ is a positive integer. For $0 \leq k \leq m$, let

$$p_k(x) = x^k(1 - x)^{m-k}.$$

Show that $p_0, \ldots, p_m$ is a basis of $\mathcal{P}_m(\mathbf{F})$.

*The basis in this exercise leads to what are called* ***Bernstein polynomials****. You can do a web search to learn how Bernstein polynomials are used to approximate continuous functions on* $[0, 1]$.

**11** Suppose $U$ and $W$ are both four-dimensional subspaces of $\mathbf{C}^6$. Prove that there exist two vectors in $U \cap W$ such that neither of these vectors is a scalar multiple of the other.

**12** Suppose that $U$ and $W$ are subspaces of $\mathbf{R}^8$ such that $\dim U = 3$, $\dim W = 5$, and $U + W = \mathbf{R}^8$. Prove that $\mathbf{R}^8 = U \oplus W$.

**13** Suppose $U$ and $W$ are both five-dimensional subspaces of $\mathbf{R}^9$. Prove that $U \cap W \neq \{0\}$.

**14** Suppose $V$ is a ten-dimensional vector space and $V_1, V_2, V_3$ are subspaces of $V$ with $\dim V_1 = \dim V_2 = \dim V_3 = 7$. Prove that $V_1 \cap V_2 \cap V_3 \neq \{0\}$.

**15** Suppose $V$ is finite-dimensional and $V_1, V_2, V_3$ are subspaces of $V$ with $\dim V_1 + \dim V_2 + \dim V_3 > 2 \dim V$. Prove that $V_1 \cap V_2 \cap V_3 \neq \{0\}$.

**16** Suppose $V$ is finite-dimensional and $U$ is a subspace of $V$ with $U \neq V$. Let $n = \dim V$ and $m = \dim U$. Prove that there exist $n - m$ subspaces of $V$, each of dimension $n - 1$, whose intersection equals $U$.

**17** Suppose that $V_1, \ldots, V_m$ are finite-dimensional subspaces of $V$. Prove that $V_1 + \cdots + V_m$ is finite-dimensional and

$$\dim(V_1 + \cdots + V_m) \leq \dim V_1 + \cdots + \dim V_m.$$

*The inequality above is an equality if and only if $V_1 + \cdots + V_m$ is a direct sum, as will be shown in 3.94.*

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**18** Suppose $V$ is finite-dimensional, with $\dim V = n \geq 1$. Prove that there exist one-dimensional subspaces $V_1, \ldots, V_n$ of $V$ such that

$$V = V_1 \oplus \cdots \oplus V_n.$$

**19** Explain why you might guess, motivated by analogy with the formula for the number of elements in the union of three finite sets, that if $V_1, V_2, V_3$ are subspaces of a finite-dimensional vector space, then

$$\begin{aligned}\dim(V_1 + V_2 + V_3)&\\ &= \dim V_1 + \dim V_2 + \dim V_3\\ &\quad - \dim(V_1 \cap V_2) - \dim(V_1 \cap V_3) - \dim(V_2 \cap V_3)\\ &\quad + \dim(V_1 \cap V_2 \cap V_3).\end{aligned}$$

Then either prove the formula above or give a counterexample.

**20** Prove that if $V_1, V_2$, and $V_3$ are subspaces of a finite-dimensional vector space, then

$$\begin{aligned}&\dim(V_1 + V_2 + V_3)\\ &= \dim V_1 + \dim V_2 + \dim V_3\\ &\quad - \frac{\dim(V_1 \cap V_2) + \dim(V_1 \cap V_3) + \dim(V_2 \cap V_3)}{3}\\ &\quad - \frac{\dim((V_1+V_2)\cap V_3) + \dim((V_1+V_3)\cap V_2) + \dim((V_2+V_3)\cap V_1)}{3}.\end{aligned}$$

*The formula above may seem strange because the right side does not look like an integer.*

I at once gave up my former occupations, set down natural history and all its progeny as a deformed and abortive creation, and entertained the greatest disdain for a would-be science which could never even step within the threshold of real knowledge. In this mood I betook myself to the mathematics and the branches of study appertaining to that science as being built upon secure foundations, and so worthy of my consideration.

—*Frankenstein*, Mary Wollstonecraft Shelley

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Chapter 3

# Linear Maps

So far our attention has focused on vector spaces. No one gets excited about vector spaces. The interesting part of linear algebra is the subject to which we now turn—linear maps.

We will frequently use the powerful fundamental theorem of linear maps, which states that the dimension of the domain of a linear map equals the dimension of the subspace that gets sent to 0 plus the dimension of the range. This will imply the striking result that a linear map from a finite-dimensional vector space to itself is one-to-one if and only if its range is the whole space.

A major concept that we will introduce in this chapter is the matrix associated with a linear map and with a basis of the domain space and a basis of the target space. This correspondence between linear maps and matrices provides much insight into key aspects of linear algebra.

This chapter concludes by introducing product, quotient, and dual spaces.

In this chapter we will need additional vector spaces, which we call $U$ and $W$, in addition to $V$. Thus our standing assumptions are now as follows.

*standing assumptions for this chapter*

- **F** denotes **R** or **C**.
- $U$, $V$, and $W$ denote vector spaces over **F**.



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*The twelfth-century Dankwarderode Castle in Brunswick (Braunschweig), where Carl Friedrich Gauss (1777–1855) was born and grew up. In 1809 Gauss published a method for solving systems of linear equations. This method, now called Gaussian elimination, was used in a Chinese book written over 1600 years earlier.*

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# 3A Vector Space of Linear Maps

## Definition and Examples of Linear Maps

Now we are ready for one of the key definitions in linear algebra.

**3.1 definition: *linear map***

A *linear map* from $V$ to $W$ is a function $T: V \to W$ with the following properties.

**additivity**
$T(u + v) = Tu + Tv$ for all $u, v \in V$.

**homogeneity**
$T(\lambda v) = \lambda(Tv)$ for all $\lambda \in \mathbf{F}$ and all $v \in V$.

Note that for linear maps we often use the notation $Tv$ as well as the usual function notation $T(v)$.

*Some mathematicians use the phrase **linear transformation**, which means the same as linear map.*

**3.2 notation: $\mathcal{L}(V, W)$, $\mathcal{L}(V)$**

- The set of linear maps from $V$ to $W$ is denoted by $\mathcal{L}(V, W)$.
- The set of linear maps from $V$ to $V$ is denoted by $\mathcal{L}(V)$. In other words, $\mathcal{L}(V) = \mathcal{L}(V, V)$.

Let's look at some examples of linear maps. Make sure you verify that each of the functions defined in the next example is indeed a linear map:

**3.3 example: *linear maps***

**zero**
In addition to its other uses, we let the symbol 0 denote the linear map that takes every element of some vector space to the additive identity of another (or possibly the same) vector space. To be specific, $0 \in \mathcal{L}(V, W)$ is defined by

$$0v = 0.$$

The 0 on the left side of the equation above is a function from $V$ to $W$, whereas the 0 on the right side is the additive identity in $W$. As usual, the context should allow you to distinguish between the many uses of the symbol 0.

**identity operator**
The *identity operator*, denoted by $I$, is the linear map on some vector space that takes each element to itself. To be specific, $I \in \mathcal{L}(V)$ is defined by

$$Iv = v.$$

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**differentiation**
Define $D \in \mathcal{L}(\mathcal{P}(\mathbf{R}))$ by

$$Dp = p'.$$

The assertion that this function is a linear map is another way of stating a basic result about differentiation: $(f+g)' = f' + g'$ and $(\lambda f)' = \lambda f'$ whenever $f, g$ are differentiable and $\lambda$ is a constant.

**integration**
Define $T \in \mathcal{L}(\mathcal{P}(\mathbf{R}), \mathbf{R})$ by

$$Tp = \int_0^1 p.$$

The assertion that this function is linear is another way of stating a basic result about integration: the integral of the sum of two functions equals the sum of the integrals, and the integral of a constant times a function equals the constant times the integral of the function.

**multiplication by $x^2$**
Define a linear map $T \in \mathcal{L}(\mathcal{P}(\mathbf{R}))$ by

$$(Tp)(x) = x^2 p(x)$$

for each $x \in \mathbf{R}$.

**backward shift**
Recall that $\mathbf{F}^\infty$ denotes the vector space of all sequences of elements of $\mathbf{F}$. Define a linear map $T \in \mathcal{L}(\mathbf{F}^\infty)$ by

$$T(x_1, x_2, x_3, \dots) = (x_2, x_3, \dots).$$

**from $\mathbf{R}^3$ to $\mathbf{R}^2$**
Define a linear map $T \in \mathcal{L}(\mathbf{R}^3, \mathbf{R}^2)$ by

$$T(x, y, z) = (2x - y + 3z, 7x + 5y - 6z).$$

**from $\mathbf{F}^n$ to $\mathbf{F}^m$**
To generalize the previous example, let $m$ and $n$ be positive integers, let $A_{j,k} \in \mathbf{F}$ for each $j = 1, \dots, m$ and each $k = 1, \dots, n$, and define a linear map $T \in \mathcal{L}(\mathbf{F}^n, \mathbf{F}^m)$ by

$$T(x_1, \dots, x_n) = (A_{1,1}x_1 + \cdots + A_{1,n}x_n, \dots, A_{m,1}x_1 + \cdots + A_{m,n}x_n).$$

Actually every linear map from $\mathbf{F}^n$ to $\mathbf{F}^m$ is of this form.

**composition**
Fix a polynomial $q \in \mathcal{P}(\mathbf{R})$. Define a linear map $T \in \mathcal{L}(\mathcal{P}(\mathbf{R}))$ by

$$(Tp)(x) = p(q(x)).$$

---

The existence part of the next result means that we can find a linear map that takes on whatever values we wish on the vectors in a basis. The uniqueness part of the next result means that a linear map is completely determined by its values on a basis.

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**3.4** *linear map lemma*

Suppose $v_1, ..., v_n$ is a basis of $V$ and $w_1, ..., w_n \in W$. Then there exists a unique linear map $T\colon V \to W$ such that

$$Tv_k = w_k$$

for each $k = 1, ..., n$.

Proof First we show the existence of a linear map $T$ with the desired property. Define $T\colon V \to W$ by

$$T(c_1v_1 + \cdots + c_nv_n) = c_1w_1 + \cdots + c_nw_n,$$

where $c_1, ..., c_n$ are arbitrary elements of $\mathbf{F}$. The list $v_1, ..., v_n$ is a basis of $V$. Thus the equation above does indeed define a function $T$ from $V$ to $W$ (because each element of $V$ can be uniquely written in the form $c_1v_1 + \cdots + c_nv_n$).

For each $k$, taking $c_k = 1$ and the other $c$'s equal to 0 in the equation above shows that $Tv_k = w_k$.

If $u, v \in V$ with $u = a_1v_1 + \cdots + a_nv_n$ and $v = c_1v_1 + \cdots + c_nv_n$, then

$$\begin{aligned} T(u + v) &= T((a_1 + c_1)v_1 + \cdots + (a_n + c_n)v_n) \\ &= (a_1 + c_1)w_1 + \cdots + (a_n + c_n)w_n \\ &= (a_1w_1 + \cdots + a_nw_n) + (c_1w_1 + \cdots + c_nw_n) \\ &= Tu + Tv. \end{aligned}$$

Similarly, if $\lambda \in \mathbf{F}$ and $v = c_1v_1 + \cdots + c_nv_n$, then

$$\begin{aligned} T(\lambda v) &= T(\lambda c_1v_1 + \cdots + \lambda c_nv_n) \\ &= \lambda c_1w_1 + \cdots + \lambda c_nw_n \\ &= \lambda(c_1w_1 + \cdots + c_nw_n) \\ &= \lambda Tv. \end{aligned}$$

Thus $T$ is a linear map from $V$ to $W$.

To prove uniqueness, now suppose that $T \in \mathcal{L}(V, W)$ and that $Tv_k = w_k$ for each $k = 1, ..., n$. Let $c_1, ..., c_n \in \mathbf{F}$. Then the homogeneity of $T$ implies that $T(c_kv_k) = c_kw_k$ for each $k = 1, ..., n$. The additivity of $T$ now implies that

$$T(c_1v_1 + \cdots + c_nv_n) = c_1w_1 + \cdots + c_nw_n.$$

Thus $T$ is uniquely determined on $\operatorname{span}(v_1, ..., v_n)$ by the equation above. Because $v_1, ..., v_n$ is a basis of $V$, this implies that $T$ is uniquely determined on $V$, as desired. ∎

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## *Algebraic Operations on $\mathcal{L}(V, W)$*

We begin by defining addition and scalar multiplication on $\mathcal{L}(V, W)$.

**3.5 definition: *addition and scalar multiplication on $\mathcal{L}(V, W)$***

Suppose $S, T \in \mathcal{L}(V, W)$ and $\lambda \in \mathbf{F}$. The *sum* $S + T$ and the *product* $\lambda T$ are the linear maps from $V$ to $W$ defined by

$$(S + T)(v) = Sv + Tv \quad \text{and} \quad (\lambda T)(v) = \lambda(Tv)$$

for all $v \in V$.

You should verify that $S + T$ and $\lambda T$ as defined above are indeed linear maps. In other words, if $S, T \in \mathcal{L}(V, W)$ and $\lambda \in \mathbf{F}$, then $S + T \in \mathcal{L}(V, W)$ and $\lambda T \in \mathcal{L}(V, W)$.

Because we took the trouble to define addition and scalar multiplication on $\mathcal{L}(V, W)$, the next result should not be a surprise.

*Linear maps are pervasive throughout mathematics. However, they are not as ubiquitous as imagined by people who seem to think* cos *is a linear map from* $\mathbf{R}$ *to* $\mathbf{R}$ *when they incorrectly write that* $\cos(x+y)$ *equals* $\cos x + \cos y$ *and that* $\cos 2x$ *equals* $2 \cos x$.

**3.6 *$\mathcal{L}(V, W)$ is a vector space***

With the operations of addition and scalar multiplication as defined above, $\mathcal{L}(V, W)$ is a vector space.

The routine proof of the result above is left to the reader. Note that the additive identity of $\mathcal{L}(V, W)$ is the zero linear map defined in Example 3.3.

Usually it makes no sense to multiply together two elements of a vector space, but for some pairs of linear maps a useful product exists, as in the next definition.

**3.7 definition: *product of linear maps***

If $T \in \mathcal{L}(U, V)$ and $S \in \mathcal{L}(V, W)$, then the *product* $ST \in \mathcal{L}(U, W)$ is defined by

$$(ST)(u) = S(Tu)$$

for all $u \in U$.

Thus $ST$ is just the usual composition $S \circ T$ of two functions, but when both functions are linear, we usually write $ST$ instead of $S \circ T$. The product notation $ST$ helps make the distributive properties (see next result) seem natural.

Note that $ST$ is defined only when $T$ maps into the domain of $S$. You should verify that $ST$ is indeed a linear map from $U$ to $W$ whenever $T \in \mathcal{L}(U, V)$ and $S \in \mathcal{L}(V, W)$.

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**3.8** *algebraic properties of products of linear maps*

**associativity**
$(T_1T_2)T_3 = T_1(T_2T_3)$ whenever $T_1$, $T_2$, and $T_3$ are linear maps such that the products make sense (meaning $T_3$ maps into the domain of $T_2$, and $T_2$ maps into the domain of $T_1$).

**identity**
$TI = IT = T$ whenever $T \in \mathcal{L}(V, W)$; here the first $I$ is the identity operator on $V$, and the second $I$ is the identity operator on $W$.

**distributive properties**
$(S_1 + S_2)T = S_1T + S_2T$ and $S(T_1 + T_2) = ST_1 + ST_2$ whenever $T, T_1, T_2 \in \mathcal{L}(U, V)$ and $S, S_1, S_2 \in \mathcal{L}(V, W)$.

The routine proof of the result above is left to the reader.

Multiplication of linear maps is not commutative. In other words, it is not necessarily true that $ST = TS$, even if both sides of the equation make sense.

**3.9** example: *two noncommuting linear maps from $\mathcal{P}(\mathbf{R})$ to $\mathcal{P}(\mathbf{R})$*

Suppose $D \in \mathcal{L}(\mathcal{P}(\mathbf{R}))$ is the differentiation map defined in Example 3.3 and $T \in \mathcal{L}(\mathcal{P}(\mathbf{R}))$ is the multiplication by $x^2$ map defined earlier in this section. Then

$$((TD)p)(x) = x^2p'(x) \quad \text{but} \quad ((DT)p)(x) = x^2p'(x) + 2xp(x).$$

Thus $TD \neq DT$—differentiating and then multiplying by $x^2$ is not the same as multiplying by $x^2$ and then differentiating.

**3.10** *linear maps take 0 to 0*

Suppose $T$ is a linear map from $V$ to $W$. Then $T(0) = 0$.

Proof By additivity, we have

$$T(0) = T(0 + 0) = T(0) + T(0).$$

Add the additive inverse of $T(0)$ to each side of the equation above to conclude that $T(0) = 0$.

Suppose $m, b \in \mathbf{R}$. The function $f\colon \mathbf{R} \to \mathbf{R}$ defined by

$$f(x) = mx + b$$

is a linear map if and only if $b = 0$ (use 3.10). Thus the linear functions of high school algebra are not the same as linear maps in the context of linear algebra.

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## *Exercises 3A*

**1** Suppose $b, c \in \mathbf{R}$. Define $T\colon \mathbf{R}^3 \to \mathbf{R}^2$ by

$$T(x, y, z) = (2x - 4y + 3z + b, 6x + cxyz).$$

Show that $T$ is linear if and only if $b = c = 0$.

**2** Suppose $b, c \in \mathbf{R}$. Define $T\colon \mathcal{P}(\mathbf{R}) \to \mathbf{R}^2$ by

$$Tp = \Big(3p(4) + 5p'(6) + bp(1)p(2), \int_{-1}^{2} x^3 p(x)\, dx + c \sin p(0)\Big).$$

Show that $T$ is linear if and only if $b = c = 0$.

**3** Suppose that $T \in \mathcal{L}(\mathbf{F}^n, \mathbf{F}^m)$. Show that there exist scalars $A_{j,k} \in \mathbf{F}$ for $j = 1, \ldots, m$ and $k = 1, \ldots, n$ such that

$$T(x_1, \ldots, x_n) = (A_{1,1}x_1 + \cdots + A_{1,n}x_n, \ldots, A_{m,1}x_1 + \cdots + A_{m,n}x_n)$$

for every $(x_1, \ldots, x_n) \in \mathbf{F}^n$.

*This exercise shows that the linear map $T$ has the form promised in the second to last item of Example 3.3.*

**4** Suppose $T \in \mathcal{L}(V, W)$ and $v_1, \ldots, v_m$ is a list of vectors in $V$ such that $Tv_1, \ldots, Tv_m$ is a linearly independent list in $W$. Prove that $v_1, \ldots, v_m$ is linearly independent.

**5** Prove that $\mathcal{L}(V, W)$ is a vector space, as was asserted in 3.6.

**6** Prove that multiplication of linear maps has the associative, identity, and distributive properties asserted in 3.8.

**7** Show that every linear map from a one-dimensional vector space to itself is multiplication by some scalar. More precisely, prove that if $\dim V = 1$ and $T \in \mathcal{L}(V)$, then there exists $\lambda \in \mathbf{F}$ such that $Tv = \lambda v$ for all $v \in V$.

**8** Give an example of a function $\varphi\colon \mathbf{R}^2 \to \mathbf{R}$ such that

$$\varphi(av) = a\varphi(v)$$

for all $a \in \mathbf{R}$ and all $v \in \mathbf{R}^2$ but $\varphi$ is not linear.

*This exercise and the next exercise show that neither homogeneity nor additivity alone is enough to imply that a function is a linear map.*

**9** Give an example of a function $\varphi\colon \mathbf{C} \to \mathbf{C}$ such that

$$\varphi(w + z) = \varphi(w) + \varphi(z)$$

for all $w, z \in \mathbf{C}$ but $\varphi$ is not linear. (Here $\mathbf{C}$ is thought of as a complex vector space.)

*There also exists a function $\varphi\colon \mathbf{R} \to \mathbf{R}$ such that $\varphi$ satisfies the additivity condition above but $\varphi$ is not linear. However, showing the existence of such a function involves considerably more advanced tools.*

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**10** Prove or give a counterexample: If $q \in \mathcal{P}(\mathbf{R})$ and $T\colon \mathcal{P}(\mathbf{R}) \to \mathcal{P}(\mathbf{R})$ is defined by $Tp = q \circ p$, then $T$ is a linear map.

*The function $T$ defined here differs from the function $T$ defined in the last bullet point of 3.3 by the order of the functions in the compositions.*

**11** Suppose $V$ is finite-dimensional and $T \in \mathcal{L}(V)$. Prove that $T$ is a scalar multiple of the identity if and only if $ST = TS$ for every $S \in \mathcal{L}(V)$.

**12** Suppose $U$ is a subspace of $V$ with $U \neq V$. Suppose $S \in \mathcal{L}(U, W)$ and $S \neq 0$ (which means that $Su \neq 0$ for some $u \in U$). Define $T\colon V \to W$ by

$$Tv = \begin{cases} Sv & \text{if } v \in U, \\ 0 & \text{if } v \in V \text{ and } v \notin U. \end{cases}$$

Prove that $T$ is not a linear map on $V$.

**13** Suppose $V$ is finite-dimensional. Prove that every linear map on a subspace of $V$ can be extended to a linear map on $V$. In other words, show that if $U$ is a subspace of $V$ and $S \in \mathcal{L}(U, W)$, then there exists $T \in \mathcal{L}(V, W)$ such that $Tu = Su$ for all $u \in U$.

*The result in this exercise is used in the proof of 3.125.*

**14** Suppose $V$ is finite-dimensional with $\dim V > 0$, and suppose $W$ is infinite-dimensional. Prove that $\mathcal{L}(V, W)$ is infinite-dimensional.

**15** Suppose $v_1, \ldots, v_m$ is a linearly dependent list of vectors in $V$. Suppose also that $W \neq \{0\}$. Prove that there exist $w_1, \ldots, w_m \in W$ such that no $T \in \mathcal{L}(V, W)$ satisfies $Tv_k = w_k$ for each $k = 1, \ldots, m$.

**16** Suppose $V$ is finite-dimensional with $\dim V > 1$. Prove that there exist $S, T \in \mathcal{L}(V)$ such that $ST \neq TS$.

**17** Suppose $V$ is finite-dimensional. Show that the only two-sided ideals of $\mathcal{L}(V)$ are $\{0\}$ and $\mathcal{L}(V)$.

*A subspace $\mathcal{E}$ of $\mathcal{L}(V)$ is called a **two-sided ideal** of $\mathcal{L}(V)$ if $TE \in \mathcal{E}$ and $ET \in \mathcal{E}$ for all $E \in \mathcal{E}$ and all $T \in \mathcal{L}(V)$.*

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# 3B Null Spaces and Ranges

## Null Space and Injectivity

In this section we will learn about two subspaces that are intimately connected with each linear map. We begin with the set of vectors that get mapped to 0.

**3.11 definition: *null space*, null $T$**

For $T \in \mathcal{L}(V, W)$, the *null space* of $T$, denoted by $\operatorname{null} T$, is the subset of $V$ consisting of those vectors that $T$ maps to 0:

$$\operatorname{null} T = \{v \in V : Tv = 0\}.$$

**3.12 example: *null space***

- If $T$ is the zero map from $V$ to $W$, meaning that $Tv = 0$ for every $v \in V$, then $\operatorname{null} T = V$.
- Suppose $\varphi \in \mathcal{L}(\mathbf{C}^3, \mathbf{C})$ is defined by $\varphi(z_1, z_2, z_3) = z_1 + 2z_2 + 3z_3$. Then $\operatorname{null} \varphi$ equals $\{(z_1, z_2, z_3) \in \mathbf{C}^3 : z_1 + 2z_2 + 3z_3 = 0\}$, which is a subspace of the domain of $\varphi$. We will soon see that the null space of each linear map is a subspace of its domain.
- Suppose $D \in \mathcal{L}(\mathcal{P}(\mathbf{R}))$ is the differentiation map defined by $Dp = p'$. The only functions whose derivative equals the zero function are the constant functions. Thus the null space of $D$ equals the set of constant functions.

*The word "null" means zero. Thus the term "null space" should remind you of the connection to 0. Some mathematicians use the term **kernel** instead of null space.*

- Suppose that $T \in \mathcal{L}(\mathcal{P}(\mathbf{R}))$ is the multiplication by $x^2$ map defined by $(Tp)(x) = x^2 p(x)$. The only polynomial $p$ such that $x^2 p(x) = 0$ for all $x \in \mathbf{R}$ is the 0 polynomial. Thus $\operatorname{null} T = \{0\}$.
- Suppose $T \in \mathcal{L}(\mathbf{F}^\infty)$ is the backward shift defined by

$$T(x_1, x_2, x_3, \dots) = (x_2, x_3, \dots).$$

Then $T(x_1, x_2, x_3, \dots)$ equals 0 if and only if the numbers $x_2, x_3, \dots$ are all 0. Thus $\operatorname{null} T = \{(a, 0, 0, \dots) : a \in \mathbf{F}\}$.

The next result shows that the null space of each linear map is a subspace of the domain. In particular, 0 is in the null space of every linear map.

**3.13 *the null space is a subspace***

Suppose $T \in \mathcal{L}(V, W)$. Then $\operatorname{null} T$ is a subspace of $V$.

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Proof Because $T$ is a linear map, $T(0) = 0$ (by 3.10). Thus $0 \in \operatorname{null} T$.

Suppose $u, v \in \operatorname{null} T$. Then

$$T(u + v) = Tu + Tv = 0 + 0 = 0.$$

Hence $u + v \in \operatorname{null} T$. Thus $\operatorname{null} T$ is closed under addition.

Suppose $u \in \operatorname{null} T$ and $\lambda \in \mathbf{F}$. Then

$$T(\lambda u) = \lambda Tu = \lambda 0 = 0.$$

Hence $\lambda u \in \operatorname{null} T$. Thus $\operatorname{null} T$ is closed under scalar multiplication.

We have shown that $\operatorname{null} T$ contains 0 and is closed under addition and scalar multiplication. Thus $\operatorname{null} T$ is a subspace of $V$ (by 1.34). ∎

As we will soon see, for a linear map the next definition is closely connected to the null space.

**3.14 definition: *injective***

A function $T : V \to W$ is called *injective* if $Tu = Tv$ implies $u = v$.

*The term **one-to-one** means the same as injective.*

We could rephrase the definition above to say that $T$ is injective if $u \neq v$ implies that $Tu \neq Tv$. Thus $T$ is injective if and only if it maps distinct inputs to distinct outputs.

The next result says that we can check whether a linear map is injective by checking whether 0 is the only vector that gets mapped to 0. As a simple application of this result, we see that of the linear maps whose null spaces we computed in 3.12, only multiplication by $x^2$ is injective (except that the zero map is injective in the special case $V = \{0\}$).

**3.15 *injectivity* ⟺ *null space equals* $\{0\}$**

Let $T \in \mathcal{L}(V, W)$. Then $T$ is injective if and only if $\operatorname{null} T = \{0\}$.

Proof First suppose $T$ is injective. We want to prove that $\operatorname{null} T = \{0\}$. We already know that $\{0\} \subseteq \operatorname{null} T$ (by 3.10). To prove the inclusion in the other direction, suppose $v \in \operatorname{null} T$. Then

$$T(v) = 0 = T(0).$$

Because $T$ is injective, the equation above implies that $v = 0$. Thus we can conclude that $\operatorname{null} T = \{0\}$, as desired.

To prove the implication in the other direction, now suppose $\operatorname{null} T = \{0\}$. We want to prove that $T$ is injective. To do this, suppose $u, v \in V$ and $Tu = Tv$. Then

$$0 = Tu - Tv = T(u - v).$$

Thus $u - v$ is in $\operatorname{null} T$, which equals $\{0\}$. Hence $u - v = 0$, which implies that $u = v$. Hence $T$ is injective, as desired. ∎

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### *Range and Surjectivity*

Now we give a name to the set of outputs of a linear map.

**3.16 definition: *range***

For $T \in \mathcal{L}(V, W)$, the *range* of $T$ is the subset of $W$ consisting of those vectors that are equal to $Tv$ for some $v \in V$:

$$\text{range } T = \{Tv : v \in V\}.$$

**3.17 example: *range***

- If $T$ is the zero map from $V$ to $W$, meaning that $Tv = 0$ for every $v \in V$, then $\text{range } T = \{0\}$.
- Suppose $T \in \mathcal{L}(\mathbf{R}^2, \mathbf{R}^3)$ is defined by $T(x, y) = (2x, 5y, x + y)$. Then

  $$\text{range } T = \{(2x, 5y, x + y) : x, y \in \mathbf{R}\}.$$

  Note that $\text{range } T$ is a subspace of $\mathbf{R}^3$. We will soon see that the range of each element of $\mathcal{L}(V, W)$ is a subspace of $W$.
- Suppose $D \in \mathcal{L}(\mathcal{P}(\mathbf{R}))$ is the differentiation map defined by $Dp = p'$. Because for every polynomial $q \in \mathcal{P}(\mathbf{R})$ there exists a polynomial $p \in \mathcal{P}(\mathbf{R})$ such that $p' = q$, the range of $D$ is $\mathcal{P}(\mathbf{R})$.

The next result shows that the range of each linear map is a subspace of the vector space into which it is being mapped.

**3.18 *the range is a subspace***

If $T \in \mathcal{L}(V, W)$, then $\text{range } T$ is a subspace of $W$.

Proof Suppose $T \in \mathcal{L}(V, W)$. Then $T(0) = 0$ (by 3.10), which implies that $0 \in \text{range } T$.

If $w_1, w_2 \in \text{range } T$, then there exist $v_1, v_2 \in V$ such that $Tv_1 = w_1$ and $Tv_2 = w_2$. Thus

$$T(v_1 + v_2) = Tv_1 + Tv_2 = w_1 + w_2.$$

Hence $w_1 + w_2 \in \text{range } T$. Thus $\text{range } T$ is closed under addition.

If $w \in \text{range } T$ and $\lambda \in \mathbf{F}$, then there exists $v \in V$ such that $Tv = w$. Thus

$$T(\lambda v) = \lambda Tv = \lambda w.$$

Hence $\lambda w \in \text{range } T$. Thus $\text{range } T$ is closed under scalar multiplication.

We have shown that $\text{range } T$ contains 0 and is closed under addition and scalar multiplication. Thus $\text{range } T$ is a subspace of $W$ (by 1.34). ∎

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**3.19 definition: *surjective***

A function $T: V \to W$ is called *surjective* if its range equals $W$.

To illustrate the definition above, note that of the ranges we computed in 3.17, only the differentiation map is surjective (except that the zero map is surjective in the special case $W = \{0\}$).

Whether a linear map is surjective depends on what we are thinking of as the vector space into which it maps.

*Some people use the term **onto**, which means the same as surjective.*

**3.20 example: *surjectivity depends on the target space***

The differentiation map $D \in \mathcal{L}(\mathcal{P}_5(\mathbf{R}))$ defined by $Dp = p'$ is not surjective, because the polynomial $x^5$ is not in the range of $D$. However, the differentiation map $S \in \mathcal{L}(\mathcal{P}_5(\mathbf{R}), \mathcal{P}_4(\mathbf{R}))$ defined by $Sp = p'$ is surjective, because its range equals $\mathcal{P}_4(\mathbf{R})$, which is the vector space into which $S$ maps.

## *Fundamental Theorem of Linear Maps*

The next result is so important that it gets a dramatic name.

**3.21 *fundamental theorem of linear maps***

Suppose $V$ is finite-dimensional and $T \in \mathcal{L}(V, W)$. Then range $T$ is finite-dimensional and

$$\dim V = \dim \operatorname{null} T + \dim \operatorname{range} T.$$

Proof Let $u_1, \ldots, u_m$ be a basis of null $T$; thus $\dim \operatorname{null} T = m$. The linearly independent list $u_1, \ldots, u_m$ can be extended to a basis

$$u_1, \ldots, u_m, v_1, \ldots, v_n$$

of $V$ (by 2.32). Thus $\dim V = m + n$. To complete the proof, we need to show that range $T$ is finite-dimensional and $\dim \operatorname{range} T = n$. We will do this by proving that $Tv_1, \ldots, Tv_n$ is a basis of range $T$.

Let $v \in V$. Because $u_1, \ldots, u_m, v_1, \ldots, v_n$ spans $V$, we can write

$$v = a_1 u_1 + \cdots + a_m u_m + b_1 v_1 + \cdots + b_n v_n,$$

where the $a$'s and $b$'s are in $\mathbf{F}$. Applying $T$ to both sides of this equation, we get

$$Tv = b_1 Tv_1 + \cdots + b_n Tv_n,$$

where the terms of the form $Tu_k$ disappeared because each $u_k$ is in null $T$. The last equation implies that the list $Tv_1, \ldots, Tv_n$ spans range $T$. In particular, range $T$ is finite-dimensional.

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To show $Tv_1, \dots, Tv_n$ is linearly independent, suppose $c_1, \dots, c_n \in \mathbf{F}$ and

$$c_1 Tv_1 + \cdots + c_n Tv_n = 0.$$

Then

$$T(c_1 v_1 + \cdots + c_n v_n) = 0.$$

Hence

$$c_1 v_1 + \cdots + c_n v_n \in \operatorname{null} T.$$

Because $u_1, \dots, u_m$ spans $\operatorname{null} T$, we can write

$$c_1 v_1 + \cdots + c_n v_n = d_1 u_1 + \cdots + d_m u_m,$$

where the $d$'s are in $\mathbf{F}$. This equation implies that all the $c$'s (and $d$'s) are 0 (because $u_1, \dots, u_m, v_1, \dots, v_n$ is linearly independent). Thus $Tv_1, \dots, Tv_n$ is linearly independent and hence is a basis of range $T$, as desired. ∎

Now we can show that no linear map from a finite-dimensional vector space to a "smaller" vector space can be injective, where "smaller" is measured by dimension.

**3.22** *linear map to a lower-dimensional space is not injective*

Suppose $V$ and $W$ are finite-dimensional vector spaces such that $\dim V > \dim W$. Then no linear map from $V$ to $W$ is injective.

Proof Let $T \in \mathcal{L}(V, W)$. Then

$$\begin{aligned} \dim \operatorname{null} T &= \dim V - \dim \operatorname{range} T \\ &\geq \dim V - \dim W \\ &> 0, \end{aligned}$$

where the first line above comes from the fundamental theorem of linear maps (3.21) and the second line follows from 2.37. The inequality above states that $\dim \operatorname{null} T > 0$. This means that $\operatorname{null} T$ contains vectors other than 0. Thus $T$ is not injective (by 3.15). ∎

**3.23** example: *linear map from $\mathbf{F}^4$ to $\mathbf{F}^3$ is not injective*

Define a linear map $T\colon \mathbf{F}^4 \to \mathbf{F}^3$ by

$$T(z_1, z_2, z_3, z_4) = \left(\sqrt{7} z_1 + \pi z_2 + z_4, 97 z_1 + 3 z_2 + 2 z_3, z_2 + 6 z_3 + 7 z_4\right).$$

Because $\dim \mathbf{F}^4 > \dim \mathbf{F}^3$, we can use 3.22 to assert that $T$ is not injective, without doing any calculations.

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The next result shows that no linear map from a finite-dimensional vector space to a "bigger" vector space can be surjective, where "bigger" is measured by dimension.

**3.24** *linear map to a higher-dimensional space is not surjective*

Suppose $V$ and $W$ are finite-dimensional vector spaces such that $\dim V < \dim W$. Then no linear map from $V$ to $W$ is surjective.

Proof Let $T \in \mathcal{L}(V, W)$. Then

$$\begin{aligned} \dim \operatorname{range} T &= \dim V - \dim \operatorname{null} T \\ &\le \dim V \\ &< \dim W, \end{aligned}$$

where the equality above comes from the fundamental theorem of linear maps (3.21). The inequality above states that $\dim \operatorname{range} T < \dim W$. This means that $\operatorname{range} T$ cannot equal $W$. Thus $T$ is not surjective. ∎

As we will soon see, 3.22 and 3.24 have important consequences in the theory of linear equations. The idea is to express questions about systems of linear equations in terms of linear maps. Let's begin by rephrasing in terms of linear maps the question of whether a homogeneous system of linear equations has a nonzero solution.

Fix positive integers $m$ and $n$, and let $A_{j,k} \in \mathbf{F}$ for $j = 1, \dots, m$ and $k = 1, \dots, n$. Consider the homogeneous system of linear equations

***Homogeneous**, in this context, means that the constant term on the right side of each equation below is* 0.

$$\begin{aligned} \sum_{k=1}^{n} A_{1,k} x_k &= 0 \\ &\vdots \\ \sum_{k=1}^{n} A_{m,k} x_k &= 0. \end{aligned}$$

Clearly $x_1 = \cdots = x_n = 0$ is a solution of the system of equations above; the question here is whether any other solutions exist.

Define $T : \mathbf{F}^n \to \mathbf{F}^m$ by

3.25 $$T(x_1, \dots, x_n) = \left( \sum_{k=1}^{n} A_{1,k} x_k, \dots, \sum_{k=1}^{n} A_{m,k} x_k \right).$$

The equation $T(x_1, \dots, x_n) = 0$ (the 0 here is the additive identity in $\mathbf{F}^m$, namely, the list of length $m$ of all 0's) is the same as the homogeneous system of linear equations above.

Thus we want to know if $\operatorname{null} T$ is strictly bigger than $\{0\}$, which is equivalent to $T$ not being injective (by 3.15). The next result gives an important condition for ensuring that $T$ is not injective.

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**3.26** *homogeneous system of linear equations*

A homogeneous system of linear equations with more variables than equations has nonzero solutions.

Proof Use the notation and result from the discussion above. Thus $T$ is a linear map from $\mathbf{F}^n$ to $\mathbf{F}^m$, and we have a homogeneous system of $m$ linear equations with $n$ variables $x_1, \ldots, x_n$. From 3.22 we see that $T$ is not injective if $n > m$.

Example of the result above: a homogeneous system of four linear equations with five variables has nonzero solutions.

Now we consider the question of whether an inhomogeneous system of linear equations has no solutions for some choice of the constant terms. To rephrase this question in terms of a linear map, fix positive integers $m$ and $n$, and let $A_{j,k} \in \mathbf{F}$ for all $j = 1, \ldots, m$ and all $k = 1, \ldots, n$. For $c_1, \ldots, c_m \in \mathbf{F}$, consider the system of linear equations

***Inhomogeneous**, as used in this context, means that the constant term on the right side of at least one equation below does not equal* 0.

$$\begin{aligned} \sum_{k=1}^{n} A_{1,k} x_k &= c_1 \\ &\vdots \\ \sum_{k=1}^{n} A_{m,k} x_k &= c_m. \end{aligned} \tag{3.27}$$

The question here is whether there is some choice of $c_1, \ldots, c_m \in \mathbf{F}$ such that no solution exists to the system above.

Define $T\colon \mathbf{F}^n \to \mathbf{F}^m$ as in 3.25. The equation $T(x_1, \ldots, x_n) = (c_1, \ldots, c_m)$ is the same as the system of equations 3.27. Thus we want to know if range $T \neq \mathbf{F}^m$. Hence we can rephrase our question about not having a solution for some choice of $c_1, \ldots, c_m \in \mathbf{F}$ as follows: What condition ensures that $T$ is not surjective? The next result gives one such condition.

*The results 3.26 and 3.28, which compare the number of variables and the number of equations, can also be proved using Gaussian elimination. The abstract approach taken here seems to provide cleaner proofs.*

**3.28** *inhomogeneous system of linear equations*

An inhomogeneous system of linear equations with more equations than variables has no solution for some choice of the constant terms.

Proof Use the notation and result from the example above. Thus $T$ is a linear map from $\mathbf{F}^n$ to $\mathbf{F}^m$, and we have a system of $m$ equations with $n$ variables $x_1, \ldots, x_n$. From 3.24 we see that $T$ is not surjective if $n < m$.

Example of the result above: an inhomogeneous system of five linear equations with four variables has no solution for some choice of the constant terms.

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## *Exercises 3B*

**1** Give an example of a linear map $T$ with $\dim \operatorname{null} T = 3$ and $\dim \operatorname{range} T = 2$.

**2** Suppose $S, T \in \mathcal{L}(V)$ are such that $\operatorname{range} S \subseteq \operatorname{null} T$. Prove that $(ST)^2 = 0$.

**3** Suppose $v_1, \dots, v_m$ is a list of vectors in $V$. Define $T \in \mathcal{L}(\mathbf{F}^m, V)$ by

$$T(z_1, \dots, z_m) = z_1 v_1 + \cdots + z_m v_m.$$

(a) What property of $T$ corresponds to $v_1, \dots, v_m$ spanning $V$?
(b) What property of $T$ corresponds to the list $v_1, \dots, v_m$ being linearly independent?

**4** Show that $\{T \in \mathcal{L}(\mathbf{R}^5, \mathbf{R}^4) : \dim \operatorname{null} T > 2\}$ is not a subspace of $\mathcal{L}(\mathbf{R}^5, \mathbf{R}^4)$.

**5** Give an example of $T \in \mathcal{L}(\mathbf{R}^4)$ such that $\operatorname{range} T = \operatorname{null} T$.

**6** Prove that there does not exist $T \in \mathcal{L}(\mathbf{R}^5)$ such that $\operatorname{range} T = \operatorname{null} T$.

**7** Suppose $V$ and $W$ are finite-dimensional with $2 \le \dim V \le \dim W$. Show that $\{T \in \mathcal{L}(V, W) : T \text{ is not injective}\}$ is not a subspace of $\mathcal{L}(V, W)$.

**8** Suppose $V$ and $W$ are finite-dimensional with $\dim V \ge \dim W \ge 2$. Show that $\{T \in \mathcal{L}(V, W) : T \text{ is not surjective}\}$ is not a subspace of $\mathcal{L}(V, W)$.

**9** Suppose $T \in \mathcal{L}(V, W)$ is injective and $v_1, \dots, v_n$ is linearly independent in $V$. Prove that $Tv_1, \dots, Tv_n$ is linearly independent in $W$.

**10** Suppose $v_1, \dots, v_n$ spans $V$ and $T \in \mathcal{L}(V, W)$. Show that $Tv_1, \dots, Tv_n$ spans $\operatorname{range} T$.

**11** Suppose that $V$ is finite-dimensional and that $T \in \mathcal{L}(V, W)$. Prove that there exists a subspace $U$ of $V$ such that

$$U \cap \operatorname{null} T = \{0\} \quad \text{and} \quad \operatorname{range} T = \{Tu : u \in U\}.$$

**12** Suppose $T$ is a linear map from $\mathbf{F}^4$ to $\mathbf{F}^2$ such that

$$\operatorname{null} T = \{(x_1, x_2, x_3, x_4) \in \mathbf{F}^4 : x_1 = 5x_2 \text{ and } x_3 = 7x_4\}.$$

Prove that $T$ is surjective.

**13** Suppose $U$ is a three-dimensional subspace of $\mathbf{R}^8$ and that $T$ is a linear map from $\mathbf{R}^8$ to $\mathbf{R}^5$ such that $\operatorname{null} T = U$. Prove that $T$ is surjective.

**14** Prove that there does not exist a linear map from $\mathbf{F}^5$ to $\mathbf{F}^2$ whose null space equals $\{(x_1, x_2, x_3, x_4, x_5) \in \mathbf{F}^5 : x_1 = 3x_2 \text{ and } x_3 = x_4 = x_5\}$.

**15** Suppose there exists a linear map on $V$ whose null space and range are both finite-dimensional. Prove that $V$ is finite-dimensional.

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**16** Suppose $V$ and $W$ are both finite-dimensional. Prove that there exists an injective linear map from $V$ to $W$ if and only if $\dim V \le \dim W$.

**17** Suppose $V$ and $W$ are both finite-dimensional. Prove that there exists a surjective linear map from $V$ onto $W$ if and only if $\dim V \ge \dim W$.

**18** Suppose $V$ and $W$ are finite-dimensional and that $U$ is a subspace of $V$. Prove that there exists $T \in \mathcal{L}(V, W)$ such that $\operatorname{null} T = U$ if and only if $\dim U \ge \dim V - \dim W$.

**19** Suppose $W$ is finite-dimensional and $T \in \mathcal{L}(V, W)$. Prove that $T$ is injective if and only if there exists $S \in \mathcal{L}(W, V)$ such that $ST$ is the identity operator on $V$.

**20** Suppose $W$ is finite-dimensional and $T \in \mathcal{L}(V, W)$. Prove that $T$ is surjective if and only if there exists $S \in \mathcal{L}(W, V)$ such that $TS$ is the identity operator on $W$.

**21** Suppose $V$ is finite-dimensional, $T \in \mathcal{L}(V, W)$, and $U$ is a subspace of $W$. Prove that $\{v \in V : Tv \in U\}$ is a subspace of $V$ and

$$\dim\{v \in V : Tv \in U\} = \dim \operatorname{null} T + \dim(U \cap \operatorname{range} T).$$

**22** Suppose $U$ and $V$ are finite-dimensional vector spaces and $S \in \mathcal{L}(V, W)$ and $T \in \mathcal{L}(U, V)$. Prove that

$$\dim \operatorname{null} ST \le \dim \operatorname{null} S + \dim \operatorname{null} T.$$

**23** Suppose $U$ and $V$ are finite-dimensional vector spaces and $S \in \mathcal{L}(V, W)$ and $T \in \mathcal{L}(U, V)$. Prove that

$$\dim \operatorname{range} ST \le \min\{\dim \operatorname{range} S, \dim \operatorname{range} T\}.$$

**24** (a) Suppose $\dim V = 5$ and $S, T \in \mathcal{L}(V)$ are such that $ST = 0$. Prove that $\dim \operatorname{range} TS \le 2$.
(b) Give an example of $S, T \in \mathcal{L}(\mathbf{F}^5)$ with $ST = 0$ and $\dim \operatorname{range} TS = 2$.

**25** Suppose that $W$ is finite-dimensional and $S, T \in \mathcal{L}(V, W)$. Prove that $\operatorname{null} S \subseteq \operatorname{null} T$ if and only if there exists $E \in \mathcal{L}(W)$ such that $T = ES$.

**26** Suppose that $V$ is finite-dimensional and $S, T \in \mathcal{L}(V, W)$. Prove that $\operatorname{range} S \subseteq \operatorname{range} T$ if and only if there exists $E \in \mathcal{L}(V)$ such that $S = TE$.

**27** Suppose $P \in \mathcal{L}(V)$ and $P^2 = P$. Prove that $V = \operatorname{null} P \oplus \operatorname{range} P$.

**28** Suppose $D \in \mathcal{L}(\mathcal{P}(\mathbf{R}))$ is such that $\deg Dp = (\deg p) - 1$ for every non-constant polynomial $p \in \mathcal{P}(\mathbf{R})$. Prove that $D$ is surjective.

*The notation $D$ is used above to remind you of the differentiation map that sends a polynomial $p$ to $p'$.*

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**29** Suppose $p \in \mathcal{P}(\mathbf{R})$. Prove that there exists a polynomial $q \in \mathcal{P}(\mathbf{R})$ such that $5q'' + 3q' = p$.

*This exercise can be done without linear algebra, but it's more fun to do it using linear algebra.*

**30** Suppose $\varphi \in \mathcal{L}(V, \mathbf{F})$ and $\varphi \neq 0$. Suppose $u \in V$ is not in $\operatorname{null} \varphi$. Prove that

$$V = \operatorname{null} \varphi \oplus \{au : a \in \mathbf{F}\}.$$

**31** Suppose $V$ is finite-dimensional, $X$ is a subspace of $V$, and $Y$ is a finite-dimensional subspace of $W$. Prove that there exists $T \in \mathcal{L}(V, W)$ such that $\operatorname{null} T = X$ and $\operatorname{range} T = Y$ if and only if $\dim X + \dim Y = \dim V$.

**32** Suppose $V$ is finite-dimensional with $\dim V > 1$. Show that if $\varphi : \mathcal{L}(V) \to \mathbf{F}$ is a linear map such that $\varphi(ST) = \varphi(S)\varphi(T)$ for all $S, T \in \mathcal{L}(V)$, then $\varphi = 0$.

*Hint: The description of the two-sided ideals of $\mathcal{L}(V)$ given by Exercise 17 in Section 3A might be useful.*

**33** Suppose that $V$ and $W$ are real vector spaces and $T \in \mathcal{L}(V, W)$. Define $T_{\mathbf{C}} : V_{\mathbf{C}} \to W_{\mathbf{C}}$ by

$$T_{\mathbf{C}}(u + iv) = Tu + iTv$$

for all $u, v \in V$.

(a) Show that $T_{\mathbf{C}}$ is a (complex) linear map from $V_{\mathbf{C}}$ to $W_{\mathbf{C}}$.
(b) Show that $T_{\mathbf{C}}$ is injective if and only if $T$ is injective.
(c) Show that $\operatorname{range} T_{\mathbf{C}} = W_{\mathbf{C}}$ if and only if $\operatorname{range} T = W$.

*See Exercise 8 in Section 1B for the definition of the complexification $V_{\mathbf{C}}$. The linear map $T_{\mathbf{C}}$ is called the **complexification** of the linear map $T$.*

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# 3C Matrices

## Representing a Linear Map by a Matrix

We know that if $v_1, \ldots, v_n$ is a basis of $V$ and $T\colon V \to W$ is linear, then the values of $Tv_1, \ldots, Tv_n$ determine the values of $T$ on arbitrary vectors in $V$—see the linear map lemma (3.4). As we will soon see, matrices provide an efficient method of recording the values of the $Tv_k$'s in terms of a basis of $W$.

**3.29 definition: *matrix*, $A_{j,k}$**

Suppose $m$ and $n$ are nonnegative integers. An $m$-by-$n$ *matrix* $A$ is a rectangular array of elements of $\mathbf{F}$ with $m$ rows and $n$ columns:

$$A = \begin{pmatrix} A_{1,1} & \cdots & A_{1,n} \\ \vdots & & \vdots \\ A_{m,1} & \cdots & A_{m,n} \end{pmatrix}.$$

The notation $A_{j,k}$ denotes the entry in row $j$, column $k$ of $A$.

**3.30 example: $A_{j,k}$ *equals entry in row $j$, column $k$ of $A$***

Suppose $A = \begin{pmatrix} 8 & 4 & 5-3i \\ 1 & 9 & 7 \end{pmatrix}$. Thus $A_{2,3}$ refers to the entry in the second row, third column of $A$, which means that $A_{2,3} = 7$.

*When dealing with matrices, the first index refers to the row number; the second index refers to the column number.*

Now we come to the key definition in this section.

**3.31 definition: *matrix of a linear map*, $\mathcal{M}(T)$**

Suppose $T \in \mathcal{L}(V, W)$ and $v_1, \ldots, v_n$ is a basis of $V$ and $w_1, \ldots, w_m$ is a basis of $W$. The *matrix of* $T$ with respect to these bases is the $m$-by-$n$ matrix $\mathcal{M}(T)$ whose entries $A_{j,k}$ are defined by

$$Tv_k = A_{1,k}w_1 + \cdots + A_{m,k}w_m.$$

If the bases $v_1, \ldots, v_n$ and $w_1, \ldots, w_m$ are not clear from the context, then the notation $\mathcal{M}(T, (v_1, \ldots, v_n), (w_1, \ldots, w_m))$ is used.

The matrix $\mathcal{M}(T)$ of a linear map $T \in \mathcal{L}(V, W)$ depends on the basis $v_1, \ldots, v_n$ of $V$ and the basis $w_1, \ldots, w_m$ of $W$, as well as on $T$. However, the bases should be clear from the context, and thus they are often not included in the notation.

To remember how $\mathcal{M}(T)$ is constructed from $T$, you might write across the top of the matrix the basis vectors $v_1, \ldots, v_n$ for the domain and along the left the basis vectors $w_1, \ldots, w_m$ for the vector space into which $T$ maps, as follows:

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$$\mathcal{M}(T) = \begin{matrix} \\ w_1 \\ \vdots \\ w_m \end{matrix} \begin{matrix} \begin{matrix} v_1 & \cdots & v_k & \cdots & v_n \end{matrix} \\ \begin{pmatrix} & A_{1,k} & & \\ & \vdots & & \\ & A_{m,k} & & \end{pmatrix} \end{matrix}.$$

In the matrix above only the $k^{\text{th}}$ column is shown. Thus the second index of each displayed entry of the matrix above is $k$. The picture above should remind you that $Tv_k$ can be computed from $\mathcal{M}(T)$ by multiplying each entry in the $k^{\text{th}}$ column by the corresponding $w_j$ from the left column, and then adding up the resulting vectors.

*The $k^{th}$ column of $\mathcal{M}(T)$ consists of the scalars needed to write $Tv_k$ as a linear combination of $w_1, \ldots, w_m$:*

$$Tv_k = \sum_{j=1}^{m} A_{j,k} w_j.$$

If $T$ is a linear map from $\mathbf{F}^n$ to $\mathbf{F}^m$, then unless stated otherwise, assume the bases in question are the standard ones (where the $k^{\text{th}}$ basis vector is 1 in the $k^{\text{th}}$ slot and 0 in all other slots). If you think of elements of $\mathbf{F}^m$ as columns of $m$ numbers, then you can think of the $k^{\text{th}}$ column of $\mathcal{M}(T)$ as $T$ applied to the $k^{\text{th}}$ standard basis vector.

*If $T$ is a linear map from an $n$-dimensional vector space to an $m$-dimensional vector space, then $\mathcal{M}(T)$ is an $m$-by-$n$ matrix.*

3.32 example: *the matrix of a linear map from $\mathbf{F}^2$ to $\mathbf{F}^3$*

Suppose $T \in \mathcal{L}(\mathbf{F}^2, \mathbf{F}^3)$ is defined by

$$T(x, y) = (x + 3y, 2x + 5y, 7x + 9y).$$

Because $T(1, 0) = (1, 2, 7)$ and $T(0, 1) = (3, 5, 9)$, the matrix of $T$ with respect to the standard bases is the 3-by-2 matrix below:

$$\mathcal{M}(T) = \begin{pmatrix} 1 & 3 \\ 2 & 5 \\ 7 & 9 \end{pmatrix}.$$

When working with $\mathcal{P}_m(\mathbf{F})$, use the standard basis $1, x, x^2, \ldots, x^m$ unless the context indicates otherwise.

3.33 example: *matrix of the differentiation map from $\mathcal{P}_3(\mathbf{R})$ to $\mathcal{P}_2(\mathbf{R})$*

Suppose $D \in \mathcal{L}(\mathcal{P}_3(\mathbf{R}), \mathcal{P}_2(\mathbf{R}))$ is the differentiation map defined by $Dp = p'$. Because $(x^n)' = nx^{n-1}$, the matrix of $D$ with respect to the standard bases is the 3-by-4 matrix below:

$$\mathcal{M}(D) = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 3 \end{pmatrix}.$$

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### *Addition and Scalar Multiplication of Matrices*

For the rest of this section, assume that $U$, $V$, and $W$ are finite-dimensional and that a basis has been chosen for each of these vector spaces. Thus for each linear map from $V$ to $W$, we can talk about its matrix (with respect to the chosen bases).

Is the matrix of the sum of two linear maps equal to the sum of the matrices of the two maps? Right now this question does not yet make sense because although we have defined the sum of two linear maps, we have not defined the sum of two matrices. Fortunately, the natural definition of the sum of two matrices has the right properties. Specifically, we make the following definition.

**3.34 definition: *matrix addition***

The *sum of two matrices of the same size* is the matrix obtained by adding corresponding entries in the matrices:

$$\begin{pmatrix} A_{1,1} & \cdots & A_{1,n} \\ \vdots & & \vdots \\ A_{m,1} & \cdots & A_{m,n} \end{pmatrix} + \begin{pmatrix} C_{1,1} & \cdots & C_{1,n} \\ \vdots & & \vdots \\ C_{m,1} & \cdots & C_{m,n} \end{pmatrix}$$

$$= \begin{pmatrix} A_{1,1} + C_{1,1} & \cdots & A_{1,n} + C_{1,n} \\ \vdots & & \vdots \\ A_{m,1} + C_{m,1} & \cdots & A_{m,n} + C_{m,n} \end{pmatrix}.$$

In the next result, the assumption is that the same bases are used for all three linear maps $S + T$, $S$, and $T$.

**3.35 *matrix of the sum of linear maps***

Suppose $S, T \in \mathcal{L}(V, W)$. Then $\mathcal{M}(S + T) = \mathcal{M}(S) + \mathcal{M}(T)$.

The verification of the result above follows from the definitions and is left to the reader.

Still assuming that we have some bases in mind, is the matrix of a scalar times a linear map equal to the scalar times the matrix of the linear map? Again, the question does not yet make sense because we have not defined scalar multiplication on matrices. Fortunately, the natural definition again has the right properties.

**3.36 definition: *scalar multiplication of a matrix***

The product of a scalar and a matrix is the matrix obtained by multiplying each entry in the matrix by the scalar:

$$\lambda \begin{pmatrix} A_{1,1} & \cdots & A_{1,n} \\ \vdots & & \vdots \\ A_{m,1} & \cdots & A_{m,n} \end{pmatrix} = \begin{pmatrix} \lambda A_{1,1} & \cdots & \lambda A_{1,n} \\ \vdots & & \vdots \\ \lambda A_{m,1} & \cdots & \lambda A_{m,n} \end{pmatrix}.$$

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3.37 example: *addition and scalar multiplication of matrices*

$$2\begin{pmatrix} 3 & 1 \\ -1 & 5 \end{pmatrix} + \begin{pmatrix} 4 & 2 \\ 1 & 6 \end{pmatrix} = \begin{pmatrix} 6 & 2 \\ -2 & 10 \end{pmatrix} + \begin{pmatrix} 4 & 2 \\ 1 & 6 \end{pmatrix} = \begin{pmatrix} 10 & 4 \\ -1 & 16 \end{pmatrix}$$

In the next result, the assumption is that the same bases are used for both the linear maps $\lambda T$ and $T$.

3.38 *the matrix of a scalar times a linear map*

Suppose $\lambda \in \mathbf{F}$ and $T \in \mathcal{L}(V, W)$. Then $\mathcal{M}(\lambda T) = \lambda \mathcal{M}(T)$.

The verification of the result above is also left to the reader.

Because addition and scalar multiplication have now been defined for matrices, you should not be surprised that a vector space is about to appear. First we introduce a bit of notation so that this new vector space has a name, and then we find the dimension of this new vector space.

3.39 notation: $\mathbf{F}^{m,n}$

For $m$ and $n$ positive integers, the set of all $m$-by-$n$ matrices with entries in $\mathbf{F}$ is denoted by $\mathbf{F}^{m,n}$.

3.40 $\dim \mathbf{F}^{m,n} = mn$

Suppose $m$ and $n$ are positive integers. With addition and scalar multiplication defined as above, $\mathbf{F}^{m,n}$ is a vector space of dimension $mn$.

Proof The verification that $\mathbf{F}^{m,n}$ is a vector space is left to the reader. Note that the additive identity of $\mathbf{F}^{m,n}$ is the $m$-by-$n$ matrix all of whose entries equal 0.

The reader should also verify that the list of distinct $m$-by-$n$ matrices that have 0 in all entries except for a 1 in one entry is a basis of $\mathbf{F}^{m,n}$. There are $mn$ such matrices, so the dimension of $\mathbf{F}^{m,n}$ equals $mn$. ∎

## *Matrix Multiplication*

Suppose, as previously, that $v_1, \ldots, v_n$ is a basis of $V$ and $w_1, \ldots, w_m$ is a basis of $W$. Suppose also that $u_1, \ldots, u_p$ is a basis of $U$.

Consider linear maps $T\colon U \to V$ and $S\colon V \to W$. The composition $ST$ is a linear map from $U$ to $W$. Does $\mathcal{M}(ST)$ equal $\mathcal{M}(S)\mathcal{M}(T)$? This question does not yet make sense because we have not defined the product of two matrices. We will choose a definition of matrix multiplication that forces this question to have a positive answer. Let's see how to do this.

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Suppose $\mathcal{M}(S) = A$ and $\mathcal{M}(T) = B$. For $1 \le k \le p$, we have

$$\begin{aligned}(ST)u_k &= S\Big(\sum_{r=1}^{n} B_{r,k} v_r\Big) \\ &= \sum_{r=1}^{n} B_{r,k} S v_r \\ &= \sum_{r=1}^{n} B_{r,k} \sum_{j=1}^{m} A_{j,r} w_j \\ &= \sum_{j=1}^{m} \Big(\sum_{r=1}^{n} A_{j,r} B_{r,k}\Big) w_j.\end{aligned}$$

Thus $\mathcal{M}(ST)$ is the $m$-by-$p$ matrix whose entry in row $j$, column $k$, equals

$$\sum_{r=1}^{n} A_{j,r} B_{r,k}.$$

Now we see how to define matrix multiplication so that the desired equation $\mathcal{M}(ST) = \mathcal{M}(S)\mathcal{M}(T)$ holds.

**3.41 definition: *matrix multiplication***

Suppose $A$ is an $m$-by-$n$ matrix and $B$ is an $n$-by-$p$ matrix. Then $AB$ is defined to be the $m$-by-$p$ matrix whose entry in row $j$, column $k$, is given by the equation

$$(AB)_{j,k} = \sum_{r=1}^{n} A_{j,r} B_{r,k}.$$

Thus the entry in row $j$, column $k$, of $AB$ is computed by taking row $j$ of $A$ and column $k$ of $B$, multiplying together corresponding entries, and then summing.

Note that we define the product of two matrices only when the number of columns of the first matrix equals the number of rows of the second matrix.

*You may have learned this definition of matrix multiplication in an earlier course, although you may not have seen this motivation for it.*

**3.42 example: *matrix multiplication***

Here we multiply together a 3-by-2 matrix and a 2-by-4 matrix, obtaining a 3-by-4 matrix:

$$\begin{pmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{pmatrix} \begin{pmatrix} 6 & 5 & 4 & 3 \\ 2 & 1 & 0 & -1 \end{pmatrix} = \begin{pmatrix} 10 & 7 & 4 & 1 \\ 26 & 19 & 12 & 5 \\ 42 & 31 & 20 & 9 \end{pmatrix}.$$

Matrix multiplication is not commutative—$AB$ is not necessarily equal to $BA$ even if both products are defined (see Exercise 10). Matrix multiplication is distributive and associative (see Exercises 11 and 12).

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In the next result, we assume that the same basis of $V$ is used in considering $T \in \mathcal{L}(U,V)$ and $S \in \mathcal{L}(V,W)$, the same basis of $W$ is used in considering $S \in \mathcal{L}(V,W)$ and $ST \in \mathcal{L}(U,W)$, and the same basis of $U$ is used in considering $T \in \mathcal{L}(U,V)$ and $ST \in \mathcal{L}(U,W)$.

**3.43** *matrix of product of linear maps*

If $T \in \mathcal{L}(U,V)$ and $S \in \mathcal{L}(V,W)$, then $\mathcal{M}(ST) = \mathcal{M}(S)\mathcal{M}(T)$.

The proof of the result above is the calculation that was done as motivation before the definition of matrix multiplication.

In the next piece of notation, note that as usual the first index refers to a row and the second index refers to a column, with a vertically centered dot used as a placeholder.

**3.44** notation: $A_{j,\cdot}$, $A_{\cdot,k}$

Suppose $A$ is an $m$-by-$n$ matrix.

- If $1 \le j \le m$, then $A_{j,\cdot}$ denotes the 1-by-$n$ matrix consisting of row $j$ of $A$.
- If $1 \le k \le n$, then $A_{\cdot,k}$ denotes the $m$-by-1 matrix consisting of column $k$ of $A$.

**3.45** example: $A_{j,\cdot}$ *equals* $j^{th}$ *row of* $A$ *and* $A_{\cdot,k}$ *equals* $k^{th}$ *column of* $A$

The notation $A_{2,\cdot}$ denotes the second row of $A$ and $A_{\cdot,2}$ denotes the second column of $A$. Thus if $A = \begin{pmatrix} 8 & 4 & 5 \\ 1 & 9 & 7 \end{pmatrix}$, then

$$A_{2,\cdot} = \begin{pmatrix} 1 & 9 & 7 \end{pmatrix} \quad \text{and} \quad A_{\cdot,2} = \begin{pmatrix} 4 \\ 9 \end{pmatrix}.$$

The product of a 1-by-$n$ matrix and an $n$-by-1 matrix is a 1-by-1 matrix. However, we will frequently identify a 1-by-1 matrix with its entry. For example,

$$\begin{pmatrix} 3 & 4 \end{pmatrix}\begin{pmatrix} 6 \\ 2 \end{pmatrix} = \begin{pmatrix} 26 \end{pmatrix}$$

because $3 \cdot 6 + 4 \cdot 2 = 26$. However, we can identify $\begin{pmatrix} 26 \end{pmatrix}$ with 26, writing $\begin{pmatrix} 3 & 4 \end{pmatrix}\begin{pmatrix} 6 \\ 2 \end{pmatrix} = 26$.

The next result uses the convention discussed in the paragraph above to give another way to think of matrix multiplication. For example, the next result and the calculation in the paragraph above explain why the entry in row 2, column 1, of the product in Example 3.42 equals 26.

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**3.46** *entry of matrix product equals row times column*

Suppose $A$ is an $m$-by-$n$ matrix and $B$ is an $n$-by-$p$ matrix. Then

$$(AB)_{j,k} = A_{j,\cdot} B_{\cdot,k}$$

if $1 \le j \le m$ and $1 \le k \le p$. In other words, the entry in row $j$, column $k$, of $AB$ equals (row $j$ of $A$) times (column $k$ of $B$).

Proof Suppose $1 \le j \le m$ and $1 \le k \le p$. The definition of matrix multiplication states that

$$(AB)_{j,k} = A_{j,1}B_{1,k} + \cdots + A_{j,n}B_{n,k}. \tag{3.47}$$

The definition of matrix multiplication also implies that the product of the 1-by-$n$ matrix $A_{j,\cdot}$ and the $n$-by-1 matrix $B_{\cdot,k}$ is the 1-by-1 matrix whose entry is the number on the right side of the equation above. Thus the entry in row $j$, column $k$, of $AB$ equals (row $j$ of $A$) times (column $k$ of $B$). ∎

The next result gives yet another way to think of matrix multiplication. In the result below, $(AB)_{\cdot,k}$ is column $k$ of the $m$-by-$p$ matrix $AB$. Thus $(AB)_{\cdot,k}$ is an $m$-by-1 matrix. Also, $AB_{\cdot,k}$ is an $m$-by-1 matrix because it is the product of an $m$-by-$n$ matrix and an $n$-by-1 matrix. Thus the two sides of the equation in the result below have the same size, making it reasonable that they might be equal.

**3.48** *column of matrix product equals matrix times column*

Suppose $A$ is an $m$-by-$n$ matrix and $B$ is an $n$-by-$p$ matrix. Then

$$(AB)_{\cdot,k} = AB_{\cdot,k}$$

if $1 \le k \le p$. In other words, column $k$ of $AB$ equals $A$ times column $k$ of $B$.

Proof As discussed above, $(AB)_{\cdot,k}$ and $AB_{\cdot,k}$ are both $m$-by-1 matrices. If $1 \le j \le m$, then the entry in row $j$ of $(AB)_{\cdot,k}$ is the left side of 3.47 and the entry in row $j$ of $AB_{\cdot,k}$ is the right side of 3.47. Thus $(AB)_{\cdot,k} = AB_{\cdot,k}$. ∎

Our next result will give another way of thinking about the product of an $m$-by-$n$ matrix and an $n$-by-1 matrix, motivated by the next example.

**3.49** example: *product of a* 3*-by-*2 *matrix and a* 2*-by-*1 *matrix*

Use our definitions and basic arithmetic to verify that

$$\begin{pmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{pmatrix} \begin{pmatrix} 5 \\ 1 \end{pmatrix} = \begin{pmatrix} 7 \\ 19 \\ 31 \end{pmatrix} = 5 \begin{pmatrix} 1 \\ 3 \\ 5 \end{pmatrix} + 1 \begin{pmatrix} 2 \\ 4 \\ 6 \end{pmatrix}.$$

Thus in this example, the product of a 3-by-2 matrix and a 2-by-1 matrix is a linear combination of the columns of the 3-by-2 matrix, with the scalars (5 and 1) that multiply the columns coming from the 2-by-1 matrix.

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The next result generalizes the example above.

**3.50** *linear combination of columns*

Suppose $A$ is an $m$-by-$n$ matrix and $b = \begin{pmatrix} b_1 \\ \vdots \\ b_n \end{pmatrix}$ is an $n$-by-1 matrix. Then

$$Ab = b_1 A_{\cdot,1} + \cdots + b_n A_{\cdot,n}.$$

In other words, $Ab$ is a linear combination of the columns of $A$, with the scalars that multiply the columns coming from $b$.

Proof If $k \in \{1, \ldots, m\}$, then the definition of matrix multiplication implies that the entry in row $k$ of the $m$-by-1 matrix $Ab$ is

$$A_{k,1}b_1 + \cdots + A_{k,n}b_n.$$

The entry in row $k$ of $b_1 A_{\cdot,1} + \cdots + b_n A_{\cdot,n}$ also equals the number displayed above.

Because $Ab$ and $b_1 A_{\cdot,1} + \cdots + b_n A_{\cdot,n}$ have the same entry in row $k$ for each $k \in \{1, \ldots, m\}$, we conclude that $Ab = b_1 A_{\cdot,1} + \cdots + b_n A_{\cdot,n}$. ∎

Our two previous results focus on the columns of a matrix. Analogous results hold for the rows of a matrix. Specifically, see Exercises 8 and 9, which can be proved using appropriate modifications of the proofs of 3.48 and 3.50.

The next result is the main tool used in the next subsection to prove the column–row factorization (3.56) and to prove that the column rank of a matrix equals the row rank (3.57). To be consistent with the notation often used with the column–row factorization, including in the next subsection, the matrices in the next result are called $C$ and $R$ instead of $A$ and $B$.

**3.51** *matrix multiplication as linear combinations of columns*

Suppose $C$ is an $m$-by-$c$ matrix and $R$ is a $c$-by-$n$ matrix.

(a) If $k \in \{1, \ldots, n\}$, then column $k$ of $CR$ is a linear combination of the columns of $C$, with the coefficients of this linear combination coming from column $k$ of $R$.

(b) If $j \in \{1, \ldots, m\}$, then row $j$ of $CR$ is a linear combination of the rows of $R$, with the coefficients of this linear combination coming from row $j$ of $C$.

Proof Suppose $k \in \{1, \ldots, n\}$. Then column $k$ of $CR$ equals $CR_{\cdot,k}$ (by 3.48), which equals the linear combination of the columns of $C$ with coefficients coming from $R_{\cdot,k}$ (by 3.50). Thus (a) holds.

To prove (b), follow the pattern of the proof of (a) but use rows instead of columns and use Exercises 8 and 9 instead of 3.48 and 3.50. ∎

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## *Column–Row Factorization and Rank of a Matrix*

We begin by defining two nonnegative integers associated with each matrix.

**3.52 definition: *column rank, row rank***

Suppose $A$ is an $m$-by-$n$ matrix with entries in $\mathbf{F}$.

- The *column rank* of $A$ is the dimension of the span of the columns of $A$ in $\mathbf{F}^{m,1}$.
- The *row rank* of $A$ is the dimension of the span of the rows of $A$ in $\mathbf{F}^{1,n}$.

If $A$ is an $m$-by-$n$ matrix, then the column rank of $A$ is at most $n$ (because $A$ has $n$ columns) and the column rank of $A$ is also at most $m$ (because $\dim \mathbf{F}^{m,1} = m$). Similarly, the row rank of $A$ is also at most $\min\{m, n\}$.

**3.53 example: *column rank and row rank of a 2-by-4 matrix***

Suppose

$$A = \begin{pmatrix} 4 & 7 & 1 & 8 \\ 3 & 5 & 2 & 9 \end{pmatrix}.$$

The column rank of $A$ is the dimension of

$$\operatorname{span}\left(\begin{pmatrix} 4 \\ 3 \end{pmatrix}, \begin{pmatrix} 7 \\ 5 \end{pmatrix}, \begin{pmatrix} 1 \\ 2 \end{pmatrix}, \begin{pmatrix} 8 \\ 9 \end{pmatrix}\right)$$

in $\mathbf{F}^{2,1}$. Neither of the first two vectors listed above in $\mathbf{F}^{2,1}$ is a scalar multiple of the other. Thus the span of this list of length four has dimension at least two. The span of this list of vectors in $\mathbf{F}^{2,1}$ cannot have dimension larger than two because $\dim \mathbf{F}^{2,1} = 2$. Thus the span of this list has dimension two, which means that the column rank of $A$ is two.

The row rank of $A$ is the dimension of

$$\operatorname{span}\left(\begin{pmatrix} 4 & 7 & 1 & 8 \end{pmatrix}, \begin{pmatrix} 3 & 5 & 2 & 9 \end{pmatrix}\right)$$

in $\mathbf{F}^{1,4}$. Neither of the two vectors listed above in $\mathbf{F}^{1,4}$ is a scalar multiple of the other. Thus the span of this list of length two has dimension two, which means that the row rank of $A$ is two.

---

We now define the transpose of a matrix.

**3.54 definition: *transpose*, $A^t$**

The *transpose* of a matrix $A$, denoted by $A^t$, is the matrix obtained from $A$ by interchanging rows and columns. Specifically, if $A$ is an $m$-by-$n$ matrix, then $A^t$ is the $n$-by-$m$ matrix whose entries are given by the equation

$$(A^t)_{k,j} = A_{j,k}.$$

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3.55 example: *transpose of a matrix*

If $A = \begin{pmatrix} 5 & -7 \\ 3 & 8 \\ -4 & 2 \end{pmatrix}$, then $A^t = \begin{pmatrix} 5 & 3 & -4 \\ -7 & 8 & 2 \end{pmatrix}$.

Note that here $A$ is a 3-by-2 matrix and $A^t$ is a 2-by-3 matrix.

The transpose has nice algebraic properties: $(A+B)^t = A^t + B^t$, $(\lambda A)^t = \lambda A^t$, and $(AC)^t = C^t A^t$ for all $m$-by-$n$ matrices $A, B$, all $\lambda \in \mathbf{F}$, and all $n$-by-$p$ matrices $C$ (see Exercises 14 and 15).

The next result will be the main tool used to prove that the column rank equals the row rank (see 3.57).

3.56 *column–row factorization*

Suppose $A$ is an $m$-by-$n$ matrix with entries in $\mathbf{F}$ and column rank $c \geq 1$. Then there exist an $m$-by-$c$ matrix $C$ and a $c$-by-$n$ matrix $R$, both with entries in $\mathbf{F}$, such that $A = CR$.

Proof Each column of $A$ is an $m$-by-1 matrix. The list $A_{\cdot,1}, \ldots, A_{\cdot,n}$ of columns of $A$ can be reduced to a basis of the span of the columns of $A$ (by 2.30). This basis has length $c$, by the definition of the column rank. The $c$ columns in this basis can be put together to form an $m$-by-$c$ matrix $C$.

If $k \in \{1, \ldots, n\}$, then column $k$ of $A$ is a linear combination of the columns of $C$. Make the coefficients of this linear combination into column $k$ of a $c$-by-$n$ matrix that we call $R$. Then $A = CR$, as follows from 3.51(a). ∎

In Example 3.53, the column rank and row rank turned out to equal each other. The next result states that this happens for all matrices.

3.57 *column rank equals row rank*

Suppose $A \in \mathbf{F}^{m,n}$. Then the column rank of $A$ equals the row rank of $A$.

Proof Let $c$ denote the column rank of $A$. Let $A = CR$ be the column–row factorization of $A$ given by 3.56, where $C$ is an $m$-by-$c$ matrix and $R$ is a $c$-by-$n$ matrix. Then 3.51(b) tells us that every row of $A$ is a linear combination of the rows of $R$. Because $R$ has $c$ rows, this implies that the row rank of $A$ is less than or equal to the column rank $c$ of $A$.

To prove the inequality in the other direction, apply the result in the previous paragraph to $A^t$, getting

$$\begin{aligned} \text{column rank of } A &= \text{row rank of } A^t \\ &\leq \text{column rank of } A^t \\ &= \text{row rank of } A. \end{aligned}$$

Thus the column rank of $A$ equals the row rank of $A$. ∎

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Because the column rank equals the row rank, the last result allows us to dispense with the terms "column rank" and "row rank" and just use the simpler term "rank".

3.58 definition: *rank*

The *rank* of a matrix $A \in \mathbf{F}^{m,n}$ is the column rank of $A$.

See 3.133 and Exercise 8 in Section 7A for alternative proofs that the column rank equals the row rank.

## *Exercises 3C*

**1** Suppose $T \in \mathcal{L}(V, W)$. Show that with respect to each choice of bases of $V$ and $W$, the matrix of $T$ has at least $\dim \operatorname{range} T$ nonzero entries.

**2** Suppose $V$ and $W$ are finite-dimensional and $T \in \mathcal{L}(V, W)$. Prove that $\dim \operatorname{range} T = 1$ if and only if there exist a basis of $V$ and a basis of $W$ such that with respect to these bases, all entries of $\mathcal{M}(T)$ equal 1.

**3** Suppose $v_1, \ldots, v_n$ is a basis of $V$ and $w_1, \ldots, w_m$ is a basis of $W$.

(a) Show that if $S, T \in \mathcal{L}(V, W)$, then $\mathcal{M}(S + T) = \mathcal{M}(S) + \mathcal{M}(T)$.

(b) Show that if $\lambda \in \mathbf{F}$ and $T \in \mathcal{L}(V, W)$, then $\mathcal{M}(\lambda T) = \lambda \mathcal{M}(T)$.

*This exercise asks you to verify 3.35 and 3.38.*

**4** Suppose that $D \in \mathcal{L}(\mathcal{P}_3(\mathbf{R}), \mathcal{P}_2(\mathbf{R}))$ is the differentiation map defined by $Dp = p'$. Find a basis of $\mathcal{P}_3(\mathbf{R})$ and a basis of $\mathcal{P}_2(\mathbf{R})$ such that the matrix of $D$ with respect to these bases is

$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix}.$$

*Compare with Example 3.33. The next exercise generalizes this exercise.*

**5** Suppose $V$ and $W$ are finite-dimensional and $T \in \mathcal{L}(V, W)$. Prove that there exist a basis of $V$ and a basis of $W$ such that with respect to these bases, all entries of $\mathcal{M}(T)$ are 0 except that the entries in row $k$, column $k$, equal 1 if $1 \le k \le \dim \operatorname{range} T$.

**6** Suppose $v_1, \ldots, v_m$ is a basis of $V$ and $W$ is finite-dimensional. Suppose $T \in \mathcal{L}(V, W)$. Prove that there exists a basis $w_1, \ldots, w_n$ of $W$ such that all entries in the first column of $\mathcal{M}(T)$ [with respect to the bases $v_1, \ldots, v_m$ and $w_1, \ldots, w_n$] are 0 except for possibly a 1 in the first row, first column.

*In this exercise, unlike Exercise 5, you are given the basis of $V$ instead of being able to choose a basis of $V$.*

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**7** Suppose $w_1, \dots, w_n$ is a basis of $W$ and $V$ is finite-dimensional. Suppose $T \in \mathcal{L}(V, W)$. Prove that there exists a basis $v_1, \dots, v_m$ of $V$ such that all entries in the first row of $\mathcal{M}(T)$ [with respect to the bases $v_1, \dots, v_m$ and $w_1, \dots, w_n$] are 0 except for possibly a 1 in the first row, first column.

*In this exercise, unlike Exercise 5, you are given the basis of $W$ instead of being able to choose a basis of $W$.*

**8** Suppose $A$ is an $m$-by-$n$ matrix and $B$ is an $n$-by-$p$ matrix. Prove that

$$(AB)_{j,.} = A_{j,.} B$$

for each $1 \le j \le m$. In other words, show that row $j$ of $AB$ equals (row $j$ of $A$) times $B$.

*This exercise gives the row version of 3.48.*

**9** Suppose $a = \begin{pmatrix} a_1 & \cdots & a_n \end{pmatrix}$ is a 1-by-$n$ matrix and $B$ is an $n$-by-$p$ matrix. Prove that

$$aB = a_1 B_{1,.} + \cdots + a_n B_{n,.} .$$

In other words, show that $aB$ is a linear combination of the rows of $B$, with the scalars that multiply the rows coming from $a$.

*This exercise gives the row version of 3.50.*

**10** Give an example of 2-by-2 matrices $A$ and $B$ such that $AB \ne BA$.

**11** Prove that the distributive property holds for matrix addition and matrix multiplication. In other words, suppose $A$, $B$, $C$, $D$, $E$, and $F$ are matrices whose sizes are such that $A(B + C)$ and $(D + E)F$ make sense. Explain why $AB + AC$ and $DF + EF$ both make sense and prove that

$$A(B + C) = AB + AC \quad \text{and} \quad (D + E)F = DF + EF.$$

**12** Prove that matrix multiplication is associative. In other words, suppose $A$, $B$, and $C$ are matrices whose sizes are such that $(AB)C$ makes sense. Explain why $A(BC)$ makes sense and prove that

$$(AB)C = A(BC).$$

*Try to find a clean proof that illustrates the following quote from Emil Artin: "It is my experience that proofs involving matrices can be shortened by 50% if one throws the matrices out."*

**13** Suppose $A$ is an $n$-by-$n$ matrix and $1 \le j, k \le n$. Show that the entry in row $j$, column $k$, of $A^3$ (which is defined to mean $AAA$) is

$$\sum_{p=1}^{n} \sum_{r=1}^{n} A_{j,p} A_{p,r} A_{r,k}.$$

**14** Suppose $m$ and $n$ are positive integers. Prove that the function $A \mapsto A^t$ is a linear map from $\mathbf{F}^{m,n}$ to $\mathbf{F}^{n,m}$.

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**15** Prove that if $A$ is an $m$-by-$n$ matrix and $C$ is an $n$-by-$p$ matrix, then

$$(AC)^t = C^t A^t.$$

*This exercise shows that the transpose of the product of two matrices is the product of the transposes in the opposite order.*

**16** Suppose $A$ is an $m$-by-$n$ matrix with $A \neq 0$. Prove that the rank of $A$ is 1 if and only if there exist $(c_1, ..., c_m) \in \mathbf{F}^m$ and $(d_1, ..., d_n) \in \mathbf{F}^n$ such that $A_{j,k} = c_j d_k$ for every $j = 1, ..., m$ and every $k = 1, ..., n$.

**17** Suppose $T \in \mathcal{L}(V)$, and $u_1, ..., u_n$ and $v_1, ..., v_n$ are bases of $V$. Prove that the following are equivalent.

(a) $T$ is injective.

(b) The columns of $\mathcal{M}(T)$ are linearly independent in $\mathbf{F}^{n,1}$.

(c) The columns of $\mathcal{M}(T)$ span $\mathbf{F}^{n,1}$.

(d) The rows of $\mathcal{M}(T)$ span $\mathbf{F}^{1,n}$.

(e) The rows of $\mathcal{M}(T)$ are linearly independent in $\mathbf{F}^{1,n}$.

Here $\mathcal{M}(T)$ means $\mathcal{M}(T, (u_1, ..., u_n), (v_1, ..., v_n))$.

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# 3D Invertibility and Isomorphisms

## Invertible Linear Maps

We begin this section by defining the notions of invertible and inverse in the context of linear maps.

**3.59 definition: *invertible, inverse***

- A linear map $T \in \mathcal{L}(V, W)$ is called *invertible* if there exists a linear map $S \in \mathcal{L}(W, V)$ such that $ST$ equals the identity operator on $V$ and $TS$ equals the identity operator on $W$.
- A linear map $S \in \mathcal{L}(W, V)$ satisfying $ST = I$ and $TS = I$ is called an *inverse* of $T$ (note that the first $I$ is the identity operator on $V$ and the second $I$ is the identity operator on $W$).

The definition above mentions "an inverse". However, the next result shows that we can change this terminology to "the inverse".

**3.60 *inverse is unique***

An invertible linear map has a unique inverse.

Proof Suppose $T \in \mathcal{L}(V, W)$ is invertible and $S_1$ and $S_2$ are inverses of $T$. Then

$$S_1 = S_1 I = S_1(TS_2) = (S_1 T)S_2 = IS_2 = S_2.$$

Thus $S_1 = S_2$. ∎

Now that we know that the inverse is unique, we can give it a notation.

**3.61 notation: $T^{-1}$**

If $T$ is invertible, then its inverse is denoted by $T^{-1}$. In other words, if $T \in \mathcal{L}(V, W)$ is invertible, then $T^{-1}$ is the unique element of $\mathcal{L}(W, V)$ such that $T^{-1}T = I$ and $TT^{-1} = I$.

**3.62 example: *inverse of a linear map from* $\mathbf{R}^3$ *to* $\mathbf{R}^3$**

Suppose $T \in \mathcal{L}(\mathbf{R}^3)$ is defined by $T(x, y, z) = (-y, x, 4z)$. Thus $T$ is a counterclockwise rotation by 90° in the $xy$-plane and a stretch by a factor of 4 in the direction of the $z$-axis.

Hence the inverse map $T^{-1} \in \mathcal{L}(\mathbf{R}^3)$ is the clockwise rotation by 90° in the $xy$-plane and a stretch by a factor of $\frac{1}{4}$ in the direction of the $z$-axis:

$$T^{-1}(x, y, z) = (y, -x, \tfrac{1}{4}z).$$

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The next result shows that a linear map is invertible if and only if it is one-to-one and onto.

**3.63 *invertibility* $\iff$ *injectivity and surjectivity***

A linear map is invertible if and only if it is injective and surjective.

Proof Suppose $T \in \mathcal{L}(V, W)$. We need to show that $T$ is invertible if and only if it is injective and surjective.

First suppose $T$ is invertible. To show that $T$ is injective, suppose $u, v \in V$ and $Tu = Tv$. Then

$$u = T^{-1}(Tu) = T^{-1}(Tv) = v,$$

so $u = v$. Hence $T$ is injective.

We are still assuming that $T$ is invertible. Now we want to prove that $T$ is surjective. To do this, let $w \in W$. Then $w = T(T^{-1}w)$, which shows that $w$ is in the range of $T$. Thus range $T = W$. Hence $T$ is surjective, completing this direction of the proof.

Now suppose $T$ is injective and surjective. We want to prove that $T$ is invertible. For each $w \in W$, define $S(w)$ to be the unique element of $V$ such that $T(S(w)) = w$ (the existence and uniqueness of such an element follow from the surjectivity and injectivity of $T$). The definition of $S$ implies that $T \circ S$ equals the identity operator on $W$.

To prove that $S \circ T$ equals the identity operator on $V$, let $v \in V$. Then

$$T((S \circ T)v) = (T \circ S)(Tv) = I(Tv) = Tv.$$

This equation implies that $(S \circ T)v = v$ (because $T$ is injective). Thus $S \circ T$ equals the identity operator on $V$.

To complete the proof, we need to show that $S$ is linear. To do this, suppose $w_1, w_2 \in W$. Then

$$T(S(w_1) + S(w_2)) = T(S(w_1)) + T(S(w_2)) = w_1 + w_2.$$

Thus $S(w_1) + S(w_2)$ is the unique element of $V$ that $T$ maps to $w_1 + w_2$. By the definition of $S$, this implies that $S(w_1 + w_2) = S(w_1) + S(w_2)$. Hence $S$ satisfies the additive property required for linearity.

The proof of homogeneity is similar. Specifically, if $w \in W$ and $\lambda \in \mathbf{F}$, then

$$T(\lambda S(w)) = \lambda T(S(w)) = \lambda w.$$

Thus $\lambda S(w)$ is the unique element of $V$ that $T$ maps to $\lambda w$. By the definition of $S$, this implies that $S(\lambda w) = \lambda S(w)$. Hence $S$ is linear, as desired. ∎

For a linear map from a vector space to itself, you might wonder whether injectivity alone, or surjectivity alone, is enough to imply invertibility. On infinite-dimensional vector spaces, neither condition alone implies invertibility, as illustrated by the next example, which uses two familiar linear maps from Example 3.3.

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3.64 example: *neither injectivity nor surjectivity implies invertibility*

- The multiplication by $x^2$ linear map from $\mathcal{P}(\mathbf{R})$ to $\mathcal{P}(\mathbf{R})$ (see 3.3) is injective but it is not invertible because it is not surjective (the polynomial 1 is not in the range).
- The backward shift linear map from $\mathbf{F}^\infty$ to $\mathbf{F}^\infty$ (see 3.3) is surjective but it is not invertible because it is not injective [the vector $(1, 0, 0, 0, \dots)$ is in the null space].

In view of the example above, the next result is remarkable—it states that for a linear map from a finite-dimensional vector space to a vector space of the same dimension, either injectivity or surjectivity alone implies the other condition. Note that the hypothesis below that $\dim V = \dim W$ is automatically satisfied in the important special case where $V$ is finite-dimensional and $W = V$.

3.65 *injectivity is equivalent to surjectivity* (*if* $\dim V = \dim W < \infty$)

Suppose that $V$ and $W$ are finite-dimensional vector spaces, $\dim V = \dim W$, and $T \in \mathcal{L}(V, W)$. Then

$$T \text{ is invertible} \iff T \text{ is injective} \iff T \text{ is surjective.}$$

Proof The fundamental theorem of linear maps (3.21) states that

3.66 $$\dim V = \dim \operatorname{null} T + \dim \operatorname{range} T.$$

If $T$ is injective (which by 3.15 is equivalent to the condition $\dim \operatorname{null} T = 0$), then the equation above implies that

$$\dim \operatorname{range} T = \dim V - \dim \operatorname{null} T = \dim V = \dim W,$$

which implies that $T$ is surjective (by 2.39).

Conversely, if $T$ is surjective, then 3.66 implies that

$$\dim \operatorname{null} T = \dim V - \dim \operatorname{range} T = \dim V - \dim W = 0,$$

which implies that $T$ is injective.

Thus we have shown that $T$ is injective if and only if $T$ is surjective. Thus if $T$ is either injective or surjective, then $T$ is both injective and surjective, which implies that $T$ is invertible. Hence $T$ is invertible if and only if $T$ is injective if and only if $T$ is surjective. ∎

The next example illustrates the power of the previous result. Although it is possible to prove the result in the example below without using linear algebra, the proof using linear algebra is cleaner and easier.

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3.67 example: *there exists a polynomial $p$ such that $((x^2 + 5x + 7)p)'' = q$*

The linear map

$$p \mapsto ((x^2 + 5x + 7)p)''$$

from $\mathcal{P}(\mathbf{R})$ to itself is injective, as you can show. Thus we are tempted to use 3.65 to show that this map is surjective. However, Example 3.64 shows that the magic of 3.65 does not apply to the infinite-dimensional vector space $\mathcal{P}(\mathbf{R})$. We will get around this problem by restricting attention to the finite-dimensional vector space $\mathcal{P}_m(\mathbf{R})$.

Suppose $q \in \mathcal{P}(\mathbf{R})$. There exists a nonnegative integer $m$ such that $q \in \mathcal{P}_m(\mathbf{R})$. Define $T\colon \mathcal{P}_m(\mathbf{R}) \to \mathcal{P}_m(\mathbf{R})$ by

$$Tp = ((x^2 + 5x + 7)p)''.$$

Multiplying a nonzero polynomial by $(x^2 + 5x + 7)$ increases the degree by 2, and then differentiating twice reduces the degree by 2. Thus $T$ is indeed a linear map from $\mathcal{P}_m(\mathbf{R})$ to itself.

Every polynomial whose second derivative equals 0 is of the form $ax + b$, where $a, b \in \mathbf{R}$. Thus $\operatorname{null} T = \{0\}$. Hence $T$ is injective.

Thus $T$ is surjective (by 3.65), which means that there exists a polynomial $p \in \mathcal{P}_m(\mathbf{R})$ such that $((x^2 + 5x + 7)p)'' = q$, as claimed in the title of this example.

Exercise 35 in Section 6A gives a similar but more spectacular example of using 3.65.

The hypothesis in the result below that $\dim V = \dim W$ holds in the important special case in which $V$ is finite-dimensional and $W = V$. Thus in that case, the equation $ST = I$ implies that $ST = TS$, even though we do not have multiplicative commutativity of arbitrary linear maps from $V$ to $V$.

3.68 $ST = I \iff TS = I$ (*on vector spaces of the same dimension*)

Suppose $V$ and $W$ are finite-dimensional vector spaces of the same dimension, $S \in \mathcal{L}(V, W)$, and $T \in \mathcal{L}(W, V)$. Then $ST = I$ if and only if $TS = I$.

Proof First suppose $ST = I$. If $v \in V$ and $Tv = 0$, then

$$v = Iv = (ST)v = S(Tv) = S(0) = 0.$$

Thus $T$ is injective (by 3.15). Because $V$ and $W$ have the same dimension, this implies that $T$ is invertible (by 3.65).

Now multiply both sides of the equation $ST = I$ by $T^{-1}$ on the right, getting

$$S = T^{-1}.$$

Thus $TS = TT^{-1} = I$, as desired.

To prove the implication in the other direction, simply reverse the roles of $S$ and $T$ (and $V$ and $W$) in the direction we have already proved, showing that if $TS = I$, then $ST = I$.

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## *Isomorphic Vector Spaces*

The next definition captures the idea of two vector spaces that are essentially the same, except for the names of their elements.

**3.69 definition: *isomorphism, isomorphic***

- An *isomorphism* is an invertible linear map.
- Two vector spaces are called *isomorphic* if there is an isomorphism from one vector space onto the other one.

Think of an isomorphism $T: V \to W$ as relabeling $v \in V$ as $Tv \in W$. This viewpoint explains why two isomorphic vector spaces have the same vector space properties. The terms "isomorphism" and "invertible linear map" mean the same thing. Use "isomorphism" when you want to emphasize that the two spaces are essentially the same.

It can be difficult to determine whether two mathematical structures (such as groups or topological spaces) are essentially the same, differing only in the names of the elements of underlying sets. However, the next result shows that we need to look at only a single number (the dimension) to determine whether two vector spaces are isomorphic.

**3.70 *dimension shows whether vector spaces are isomorphic***

Two finite-dimensional vector spaces over $\mathbf{F}$ are isomorphic if and only if they have the same dimension.

Proof First suppose $V$ and $W$ are isomorphic finite-dimensional vector spaces. Thus there exists an isomorphism $T$ from $V$ onto $W$. Because $T$ is invertible, we have $\operatorname{null} T = \{0\}$ and $\operatorname{range} T = W$. Thus

$$\dim \operatorname{null} T = 0 \quad \text{and} \quad \dim \operatorname{range} T = \dim W.$$

The formula

$$\dim V = \dim \operatorname{null} T + \dim \operatorname{range} T$$

(the fundamental theorem of linear maps, which is 3.21) thus becomes the equation $\dim V = \dim W$, completing the proof in one direction.

To prove the other direction, suppose $V$ and $W$ are finite-dimensional vector spaces of the same dimension. Let $v_1, \ldots, v_n$ be a basis of $V$ and $w_1, \ldots, w_n$ be a basis of $W$. Let $T \in \mathcal{L}(V, W)$ be defined by

$$T(c_1 v_1 + \cdots + c_n v_n) = c_1 w_1 + \cdots + c_n w_n.$$

Then $T$ is a well-defined linear map because $v_1, \ldots, v_n$ is a basis of $V$. Also, $T$ is surjective because $w_1, \ldots, w_n$ spans $W$. Furthermore, $\operatorname{null} T = \{0\}$ because $w_1, \ldots, w_n$ is linearly independent. Thus $T$ is injective. Because $T$ is injective and surjective, it is an isomorphism (see 3.63). Hence $V$ and $W$ are isomorphic. ∎

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The previous result implies that each finite-dimensional vector space $V$ is isomorphic to $\mathbf{F}^n$, where $n = \dim V$. For example, if $m$ is a nonnegative integer, then $\mathcal{P}_m(\mathbf{F})$ is isomorphic to $\mathbf{F}^{m+1}$.

Recall that the notation $\mathbf{F}^{m,n}$ denotes the vector space of $m$-by-$n$ matrices with entries in $\mathbf{F}$. If $v_1, \ldots, v_n$ is a basis of $V$ and $w_1, \ldots, w_m$ is a basis of $W$, then for each $T \in \mathcal{L}(V, W)$, we have a matrix $\mathcal{M}(T) \in \mathbf{F}^{m,n}$. Thus once bases have been fixed for $V$ and $W$, $\mathcal{M}$ becomes a function from $\mathcal{L}(V, W)$ to $\mathbf{F}^{m,n}$. Notice that 3.35 and 3.38 show that $\mathcal{M}$ is a linear map. This linear map is actually an isomorphism, as we now show.

*Every finite-dimensional vector space is isomorphic to some $\mathbf{F}^n$. Thus why not just study $\mathbf{F}^n$ instead of more general vector spaces? To answer this question, note that an investigation of $\mathbf{F}^n$ would soon lead to other vector spaces. For example, we would encounter the null space and range of linear maps. Although each of these vector spaces is isomorphic to some $\mathbf{F}^m$, thinking of them that way often adds complexity but no new insight.*

**3.71 $\mathcal{L}(V, W)$ and $\mathbf{F}^{m,n}$ are isomorphic**

Suppose $v_1, \ldots, v_n$ is a basis of $V$ and $w_1, \ldots, w_m$ is a basis of $W$. Then $\mathcal{M}$ is an isomorphism between $\mathcal{L}(V, W)$ and $\mathbf{F}^{m,n}$.

Proof We already noted that $\mathcal{M}$ is linear. We need to prove that $\mathcal{M}$ is injective and surjective.

We begin with injectivity. If $T \in \mathcal{L}(V, W)$ and $\mathcal{M}(T) = 0$, then $Tv_k = 0$ for each $k = 1, \ldots, n$. Because $v_1, \ldots, v_n$ is a basis of $V$, this implies $T = 0$. Thus $\mathcal{M}$ is injective (by 3.15).

To prove that $\mathcal{M}$ is surjective, suppose $A \in \mathbf{F}^{m,n}$. By the linear map lemma (3.4), there exists $T \in \mathcal{L}(V, W)$ such that

$$Tv_k = \sum_{j=1}^{m} A_{j,k} w_j$$

for each $k = 1, \ldots, n$. Because $\mathcal{M}(T)$ equals $A$, the range of $\mathcal{M}$ equals $\mathbf{F}^{m,n}$, as desired. ∎

Now we can determine the dimension of the vector space of linear maps from one finite-dimensional vector space to another.

**3.72 $\dim \mathcal{L}(V, W) = (\dim V)(\dim W)$**

Suppose $V$ and $W$ are finite-dimensional. Then $\mathcal{L}(V, W)$ is finite-dimensional and

$$\dim \mathcal{L}(V, W) = (\dim V)(\dim W).$$

Proof The desired result follows from 3.71, 3.70, and 3.40. ∎

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## *Linear Maps Thought of as Matrix Multiplication*

Previously we defined the matrix of a linear map. Now we define the matrix of a vector.

**3.73 definition:** *matrix of a vector,* $\mathcal{M}(v)$

Suppose $v \in V$ and $v_1, ..., v_n$ is a basis of $V$. The *matrix of* $v$ with respect to this basis is the $n$-by-1 matrix

$$\mathcal{M}(v) = \begin{pmatrix} b_1 \\ \vdots \\ b_n \end{pmatrix},$$

where $b_1, ..., b_n$ are the scalars such that

$$v = b_1 v_1 + \cdots + b_n v_n.$$

The matrix $\mathcal{M}(v)$ of a vector $v \in V$ depends on the basis $v_1, ..., v_n$ of $V$, as well as on $v$. However, the basis should be clear from the context and thus it is not included in the notation.

**3.74 example:** *matrix of a vector*

- The matrix of the polynomial $2 - 7x + 5x^3 + x^4$ with respect to the standard basis of $\mathcal{P}_4(\mathbf{R})$ is

$$\begin{pmatrix} 2 \\ -7 \\ 0 \\ 5 \\ 1 \end{pmatrix}.$$

- The matrix of a vector $x \in \mathbf{F}^n$ with respect to the standard basis is obtained by writing the coordinates of $x$ as the entries in an $n$-by-1 matrix. In other words, if $x = (x_1, ..., x_n) \in \mathbf{F}^n$, then

$$\mathcal{M}(x) = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}.$$

Occasionally we want to think of elements of $V$ as relabeled to be $n$-by-1 matrices. Once a basis $v_1, ..., v_n$ is chosen, the function $\mathcal{M}$ that takes $v \in V$ to $\mathcal{M}(v)$ is an isomorphism of $V$ onto $\mathbf{F}^{n,1}$ that implements this relabeling.

Recall that if $A$ is an $m$-by-$n$ matrix, then $A_{\cdot,k}$ denotes the $k^{\text{th}}$ column of $A$, thought of as an $m$-by-1 matrix. In the next result, $\mathcal{M}(Tv_k)$ is computed with respect to the basis $w_1, ..., w_m$ of $W$.

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**3.75** $\mathcal{M}(T)_{\cdot,k} = \mathcal{M}(Tv_k)$.

Suppose $T \in \mathcal{L}(V,W)$ and $v_1,\dots,v_n$ is a basis of $V$ and $w_1,\dots,w_m$ is a basis of $W$. Let $1 \le k \le n$. Then the $k^{\text{th}}$ column of $\mathcal{M}(T)$, which is denoted by $\mathcal{M}(T)_{\cdot,k}$, equals $\mathcal{M}(Tv_k)$.

Proof The desired result follows immediately from the definitions of $\mathcal{M}(T)$ and $\mathcal{M}(Tv_k)$. ∎

The next result shows how the notions of the matrix of a linear map, the matrix of a vector, and matrix multiplication fit together.

**3.76** *linear maps act like matrix multiplication*

Suppose $T \in \mathcal{L}(V,W)$ and $v \in V$. Suppose $v_1,\dots,v_n$ is a basis of $V$ and $w_1,\dots,w_m$ is a basis of $W$. Then

$$\mathcal{M}(Tv) = \mathcal{M}(T)\mathcal{M}(v).$$

Proof Suppose $v = b_1v_1 + \cdots + b_nv_n$, where $b_1,\dots,b_n \in \mathbf{F}$. Thus

3.77 $$Tv = b_1Tv_1 + \cdots + b_nTv_n.$$

Hence

$$\begin{aligned}\mathcal{M}(Tv) &= b_1\mathcal{M}(Tv_1) + \cdots + b_n\mathcal{M}(Tv_n)\\ &= b_1\mathcal{M}(T)_{\cdot,1} + \cdots + b_n\mathcal{M}(T)_{\cdot,n}\\ &= \mathcal{M}(T)\mathcal{M}(v),\end{aligned}$$

where the first equality follows from 3.77 and the linearity of $\mathcal{M}$, the second equality comes from 3.75, and the last equality comes from 3.50. ∎

Each $m$-by-$n$ matrix $A$ induces a linear map from $\mathbf{F}^{n,1}$ to $\mathbf{F}^{m,1}$, namely the matrix multiplication function that takes $x \in \mathbf{F}^{n,1}$ to $Ax \in \mathbf{F}^{m,1}$. The result above can be used to think of every linear map (from a finite-dimensional vector space to another finite-dimensional vector space) as a matrix multiplication map after suitable relabeling via the isomorphisms given by $\mathcal{M}$. Specifically, if $T \in \mathcal{L}(V,W)$ and we identify $v \in V$ with $\mathcal{M}(v) \in \mathbf{F}^{n,1}$, then the result above says that we can identify $Tv$ with $\mathcal{M}(T)\mathcal{M}(v)$.

Because the result above allows us to think (via isomorphisms) of each linear map as multiplication on $\mathbf{F}^{n,1}$ by some matrix $A$, keep in mind that the specific matrix $A$ depends not only on the linear map but also on the choice of bases. One of the themes of many of the most important results in later chapters will be the choice of a basis that makes the matrix $A$ as simple as possible.

In this book, we concentrate on linear maps rather than on matrices. However, sometimes thinking of linear maps as matrices (or thinking of matrices as linear maps) gives important insights that we will find useful.

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Notice that no bases are in sight in the statement of the next result. Although $\mathcal{M}(T)$ in the next result depends on a choice of bases of $V$ and $W$, the next result shows that the column rank of $\mathcal{M}(T)$ is the same for all such choices (because range $T$ does not depend on a choice of basis).

**3.78 *dimension of* range $T$ *equals column rank of* $\mathcal{M}(T)$**

Suppose $V$ and $W$ are finite-dimensional and $T \in \mathcal{L}(V, W)$. Then dim range $T$ equals the column rank of $\mathcal{M}(T)$.

Proof Suppose $v_1, \ldots, v_n$ is a basis of $V$ and $w_1, \ldots, w_m$ is a basis of $W$. The linear map that takes $w \in W$ to $\mathcal{M}(w)$ is an isomorphism from $W$ onto the space $\mathbf{F}^{m,1}$ of $m$-by-1 column vectors. The restriction of this isomorphism to range $T$ [which equals $\operatorname{span}(Tv_1, \ldots, Tv_n)$ by Exercise 10 in Section 3B] is an isomorphism from range $T$ onto $\operatorname{span}(\mathcal{M}(Tv_1), \ldots, \mathcal{M}(Tv_n))$. For each $k \in \{1, \ldots, n\}$, the $m$-by-1 matrix $\mathcal{M}(Tv_k)$ equals column $k$ of $\mathcal{M}(T)$. Thus

$$\dim \text{range } T = \text{ the column rank of } \mathcal{M}(T),$$

as desired. ∎

## *Change of Basis*

In Section 3C we defined the matrix

$$\mathcal{M}(T, (v_1, \ldots, v_n), (w_1, \ldots, w_m))$$

of a linear map $T$ from $V$ to a possibly different vector space $W$, where $v_1, \ldots, v_n$ is a basis of $V$ and $w_1, \ldots, w_m$ is a basis of $W$. For linear maps from a vector space to itself, we usually use the same basis for both the domain vector space and the target vector space. When using a single basis in both capacities, we often write the basis only once. In other words, if $T \in \mathcal{L}(V)$ and $v_1, \ldots, v_n$ is a basis of $V$, then the notation $\mathcal{M}(T, (v_1, \ldots, v_n))$ is defined by the equation

$$\mathcal{M}(T, (v_1, \ldots, v_n)) = \mathcal{M}(T, (v_1, \ldots, v_n), (v_1, \ldots, v_n)).$$

If the basis $v_1, \ldots, v_n$ is clear from the context, then we can write just $\mathcal{M}(T)$.

**3.79 definition: *identity matrix, $I$***

Suppose $n$ is a positive integer. The $n$-by-$n$ matrix

$$\begin{pmatrix} 1 & & 0 \\ & \ddots & \\ 0 & & 1 \end{pmatrix}$$

with 1's on the diagonal (the entries where the row number equals the column number) and 0's elsewhere is called the *identity matrix* and is denoted by $I$.

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In the definition above, the 0 in the lower left corner of the matrix indicates that all entries below the diagonal are 0, and the 0 in the upper right corner indicates that all entries above the diagonal are 0.

With respect to each basis of $V$, the matrix of the identity operator $I \in \mathcal{L}(V)$ is the identity matrix $I$. Note that the symbol $I$ is used to denote both the identity operator and the identity matrix. The context indicates which meaning of $I$ is intended. For example, consider the equation $\mathcal{M}(I) = I$; on the left side $I$ denotes the identity operator, and on the right side $I$ denotes the identity matrix.

If $A$ is a square matrix (with entries in $\mathbf{F}$, as usual) of the same size as $I$, then $AI = IA = A$, as you should verify.

**3.80 definition: *invertible, inverse, $A^{-1}$***

A square matrix $A$ is called *invertible* if there is a square matrix $B$ of the same size such that $AB = BA = I$; we call $B$ the *inverse* of $A$ and denote it by $A^{-1}$.

The same proof as used in 3.60 shows that if $A$ is an invertible square matrix, then there is a unique matrix $B$ such that $AB = BA = I$ (and thus the notation $B = A^{-1}$ is justified).

*Some mathematicians use the terms **nonsingular** and **singular**, which mean the same as invertible and non-invertible.*

If $A$ is an invertible matrix, then $(A^{-1})^{-1} = A$ because

$$A^{-1}A = AA^{-1} = I.$$

Also, if $A$ and $C$ are invertible square matrices of the same size, then $AC$ is invertible and $(AC)^{-1} = C^{-1}A^{-1}$ because

$$\begin{aligned}(AC)(C^{-1}A^{-1}) &= A(CC^{-1})A^{-1}\\ &= AIA^{-1}\\ &= AA^{-1}\\ &= I,\end{aligned}$$

and similarly $(C^{-1}A^{-1})(AC) = I$.

The next result holds because we defined matrix multiplication to make it true—see 3.43 and the material preceding it. Now we are just being more explicit about the bases involved.

**3.81 *matrix of product of linear maps***

Suppose $T \in \mathcal{L}(U, V)$ and $S \in \mathcal{L}(V, W)$. If $u_1, \ldots, u_m$ is a basis of $U$, $v_1, \ldots, v_n$ is a basis of $V$, and $w_1, \ldots, w_p$ is a basis of $W$, then

$$\mathcal{M}(ST, (u_1, \ldots, u_m), (w_1, \ldots, w_p)) = \mathcal{M}(S, (v_1, \ldots, v_n), (w_1, \ldots, w_p))\mathcal{M}(T, (u_1, \ldots, u_m), (v_1, \ldots, v_n)).$$

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The next result deals with the matrix of the identity operator $I$ with respect to two different bases. Note that the $k^{\text{th}}$ column of $\mathcal{M}(I, (u_1, ..., u_n), (v_1, ..., v_n))$ consists of the scalars needed to write $u_k$ as a linear combination of the basis $v_1, ..., v_n$.

In the statement of the next result, $I$ denotes the identity operator from $V$ to $V$. In the proof, $I$ also denotes the $n$-by-$n$ identity matrix.

**3.82** *matrix of identity operator with respect to two bases*

Suppose that $u_1, ..., u_n$ and $v_1, ..., v_n$ are bases of $V$. Then the matrices

$$\mathcal{M}(I, (u_1, ..., u_n), (v_1, ..., v_n)) \quad \text{and} \quad \mathcal{M}(I, (v_1, ..., v_n), (u_1, ..., u_n))$$

are invertible, and each is the inverse of the other.

Proof In 3.81, replace $w_k$ with $u_k$, and replace $S$ and $T$ with $I$, getting

$$I = \mathcal{M}(I, (v_1, ..., v_n), (u_1, ..., u_n))\mathcal{M}(I, (u_1, ..., u_n), (v_1, ..., v_n)).$$

Now interchange the roles of the $u$'s and $v$'s, getting

$$I = \mathcal{M}(I, (u_1, ..., u_n), (v_1, ..., v_n))\mathcal{M}(I, (v_1, ..., v_n), (u_1, ..., u_n)).$$

These two equations above give the desired result. ∎

**3.83** example: *matrix of identity on* $\mathbf{F}^2$ *with respect to two bases*

Consider the bases $(4, 2), (5, 3)$ and $(1, 0), (0, 1)$ of $\mathbf{F}^2$. Because $I(4, 2) = 4(1, 0) + 2(0, 1)$ and $I(5, 3) = 5(1, 0) + 3(0, 1)$, we have

$$\mathcal{M}\big(I, ((4, 2), (5, 3)), ((1, 0), (0, 1))\big) = \begin{pmatrix} 4 & 5 \\ 2 & 3 \end{pmatrix}.$$

The inverse of the matrix above is

$$\begin{pmatrix} \frac{3}{2} & -\frac{5}{2} \\ -1 & 2 \end{pmatrix},$$

as you should verify. Thus 3.82 implies that

$$\mathcal{M}\big(I, ((1, 0), (0, 1)), ((4, 2), (5, 3))\big) = \begin{pmatrix} \frac{3}{2} & -\frac{5}{2} \\ -1 & 2 \end{pmatrix}.$$

Our next result shows how the matrix of $T$ changes when we change bases. In the next result, we have two different bases of $V$, each of which is used as a basis for the domain space and as a basis for the target space. Recall our shorthand notation that allows us to display a basis only once when it is used in both capacities:

$$\mathcal{M}(T, (u_1, ..., u_n)) = \mathcal{M}(T, (u_1, ..., u_n), (u_1, ..., u_n)).$$

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**3.84** *change-of-basis formula*

Suppose $T \in \mathcal{L}(V)$. Suppose $u_1, ..., u_n$ and $v_1, ..., v_n$ are bases of $V$. Let

$$A = \mathcal{M}(T, (u_1, ..., u_n)) \quad \text{and} \quad B = \mathcal{M}(T, (v_1, ..., v_n))$$

and $C = \mathcal{M}(I, (u_1, ..., u_n), (v_1, ..., v_n))$. Then

$$A = C^{-1}BC.$$

Proof In 3.81, replace $w_k$ with $u_k$ and replace $S$ with $I$, getting

3.85 $$A = C^{-1}\mathcal{M}(T, (u_1, ..., u_n), (v_1, ..., v_n)),$$

where we have used 3.82.

Again use 3.81, this time replacing $w_k$ with $v_k$. Also replace $T$ with $I$ and replace $S$ with $T$, getting

$$\mathcal{M}(T, (u_1, ..., u_n), (v_1, ..., v_n)) = BC.$$

Substituting the equation above into 3.85 gives the equation $A = C^{-1}BC$. ∎

The proof of the next result is left as an exercise.

**3.86** *matrix of inverse equals inverse of matrix*

Suppose that $v_1, ..., v_n$ is a basis of $V$ and $T \in \mathcal{L}(V)$ is invertible. Then $\mathcal{M}(T^{-1}) = (\mathcal{M}(T))^{-1}$, where both matrices are with respect to the basis $v_1, ..., v_n$.

## *Exercises 3D*

**1** Suppose $T \in \mathcal{L}(V, W)$ is invertible. Show that $T^{-1}$ is invertible and

$$(T^{-1})^{-1} = T.$$

**2** Suppose $T \in \mathcal{L}(U, V)$ and $S \in \mathcal{L}(V, W)$ are both invertible linear maps. Prove that $ST \in \mathcal{L}(U, W)$ is invertible and that $(ST)^{-1} = T^{-1}S^{-1}$.

**3** Suppose $V$ is finite-dimensional and $T \in \mathcal{L}(V)$. Prove that the following are equivalent.

(a) $T$ is invertible.

(b) $Tv_1, ..., Tv_n$ is a basis of $V$ for every basis $v_1, ..., v_n$ of $V$.

(c) $Tv_1, ..., Tv_n$ is a basis of $V$ for some basis $v_1, ..., v_n$ of $V$.

**4** Suppose $V$ is finite-dimensional and $\dim V > 1$. Prove that the set of noninvertible linear maps from $V$ to itself is not a subspace of $\mathcal{L}(V)$.

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**5** Suppose $V$ is finite-dimensional, $U$ is a subspace of $V$, and $S \in \mathcal{L}(U, V)$. Prove that there exists an invertible linear map $T$ from $V$ to itself such that $Tu = Su$ for every $u \in U$ if and only if $S$ is injective.

**6** Suppose that $W$ is finite-dimensional and $S, T \in \mathcal{L}(V, W)$. Prove that $\operatorname{null} S = \operatorname{null} T$ if and only if there exists an invertible $E \in \mathcal{L}(W)$ such that $S = ET$.

**7** Suppose that $V$ is finite-dimensional and $S, T \in \mathcal{L}(V, W)$. Prove that $\operatorname{range} S = \operatorname{range} T$ if and only if there exists an invertible $E \in \mathcal{L}(V)$ such that $S = TE$.

**8** Suppose $V$ and $W$ are finite-dimensional and $S, T \in \mathcal{L}(V, W)$. Prove that there exist invertible $E_1 \in \mathcal{L}(V)$ and $E_2 \in \mathcal{L}(W)$ such that $S = E_2 T E_1$ if and only if $\dim \operatorname{null} S = \dim \operatorname{null} T$.

**9** Suppose $V$ is finite-dimensional and $T \colon V \to W$ is a surjective linear map of $V$ onto $W$. Prove that there is a subspace $U$ of $V$ such that $T|_U$ is an isomorphism of $U$ onto $W$.

*Here $T|_U$ means the function $T$ restricted to $U$. Thus $T|_U$ is the function whose domain is $U$, with $T|_U$ defined by $T|_U(u) = Tu$ for every $u \in U$.*

**10** Suppose $V$ and $W$ are finite-dimensional and $U$ is a subspace of $V$. Let

$$\mathcal{E} = \{T \in \mathcal{L}(V, W) : U \subseteq \operatorname{null} T\}.$$

(a) Show that $\mathcal{E}$ is a subspace of $\mathcal{L}(V, W)$.
(b) Find a formula for $\dim \mathcal{E}$ in terms of $\dim V$, $\dim W$, and $\dim U$.

*Hint: Define $\Phi \colon \mathcal{L}(V, W) \to \mathcal{L}(U, W)$ by $\Phi(T) = T|_U$. What is $\operatorname{null} \Phi$? What is $\operatorname{range} \Phi$?*

**11** Suppose $V$ is finite-dimensional and $S, T \in \mathcal{L}(V)$. Prove that

$$ST \text{ is invertible} \iff S \text{ and } T \text{ are invertible.}$$

**12** Suppose $V$ is finite-dimensional and $S, T, U \in \mathcal{L}(V)$ and $STU = I$. Show that $T$ is invertible and that $T^{-1} = US$.

**13** Show that the result in Exercise 12 can fail without the hypothesis that $V$ is finite-dimensional.

**14** Prove or give a counterexample: If $V$ is a finite-dimensional vector space and $R, S, T \in \mathcal{L}(V)$ are such that $RST$ is surjective, then $S$ is injective.

**15** Suppose $T \in \mathcal{L}(V)$ and $v_1, \ldots, v_m$ is a list in $V$ such that $Tv_1, \ldots, Tv_m$ spans $V$. Prove that $v_1, \ldots, v_m$ spans $V$.

**16** Prove that every linear map from $\mathbf{F}^{n,1}$ to $\mathbf{F}^{m,1}$ is given by a matrix multiplication. In other words, prove that if $T \in \mathcal{L}(\mathbf{F}^{n,1}, \mathbf{F}^{m,1})$, then there exists an $m$-by-$n$ matrix $A$ such that $Tx = Ax$ for every $x \in \mathbf{F}^{n,1}$.

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**17** Suppose $V$ is finite-dimensional and $S \in \mathcal{L}(V)$. Define $\mathcal{A} \in \mathcal{L}(\mathcal{L}(V))$ by

$$\mathcal{A}(T) = ST$$

for $T \in \mathcal{L}(V)$.

(a) Show that $\dim \operatorname{null} \mathcal{A} = (\dim V)(\dim \operatorname{null} S)$.
(b) Show that $\dim \operatorname{range} \mathcal{A} = (\dim V)(\dim \operatorname{range} S)$.

**18** Show that $V$ and $\mathcal{L}(\mathbf{F}, V)$ are isomorphic vector spaces.

**19** Suppose $V$ is finite-dimensional and $T \in \mathcal{L}(V)$. Prove that $T$ has the same matrix with respect to every basis of $V$ if and only if $T$ is a scalar multiple of the identity operator.

**20** Suppose $q \in \mathcal{P}(\mathbf{R})$. Prove that there exists a polynomial $p \in \mathcal{P}(\mathbf{R})$ such that

$$q(x) = (x^2 + x)p''(x) + 2xp'(x) + p(3)$$

for all $x \in \mathbf{R}$.

**21** Suppose $n$ is a positive integer and $A_{j,k} \in \mathbf{F}$ for all $j, k = 1, \ldots, n$. Prove that the following are equivalent (note that in both parts below, the number of equations equals the number of variables).

(a) The trivial solution $x_1 = \cdots = x_n = 0$ is the only solution to the homogeneous system of equations

$$\begin{aligned} \sum_{k=1}^{n} A_{1,k} x_k &= 0 \\ &\vdots \\ \sum_{k=1}^{n} A_{n,k} x_k &= 0. \end{aligned}$$

(b) For every $c_1, \ldots, c_n \in \mathbf{F}$, there exists a solution to the system of equations

$$\begin{aligned} \sum_{k=1}^{n} A_{1,k} x_k &= c_1 \\ &\vdots \\ \sum_{k=1}^{n} A_{n,k} x_k &= c_n. \end{aligned}$$

**22** Suppose $T \in \mathcal{L}(V)$ and $v_1, \ldots, v_n$ is a basis of $V$. Prove that

$$\mathcal{M}(T, (v_1, \ldots, v_n)) \text{ is invertible} \iff T \text{ is invertible.}$$

**23** Suppose that $u_1, \ldots, u_n$ and $v_1, \ldots, v_n$ are bases of $V$. Let $T \in \mathcal{L}(V)$ be such that $Tv_k = u_k$ for each $k = 1, \ldots, n$. Prove that

$$\mathcal{M}(T, (v_1, \ldots, v_n)) = \mathcal{M}(I, (u_1, \ldots, u_n), (v_1, \ldots, v_n)).$$

**24** Suppose $A$ and $B$ are square matrices of the same size and $AB = I$. Prove that $BA = I$.

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# 3E Products and Quotients of Vector Spaces

## Products of Vector Spaces

As usual when dealing with more than one vector space, all vector spaces in use should be over the same field.

**3.87 definition: *product of vector spaces***

Suppose $V_1, \dots, V_m$ are vector spaces over $\mathbf{F}$.

- The *product* $V_1 \times \cdots \times V_m$ is defined by

$$V_1 \times \cdots \times V_m = \{(v_1, \dots, v_m) : v_1 \in V_1, \dots, v_m \in V_m\}.$$

- Addition on $V_1 \times \cdots \times V_m$ is defined by

$$(u_1, \dots, u_m) + (v_1, \dots, v_m) = (u_1 + v_1, \dots, u_m + v_m).$$

- Scalar multiplication on $V_1 \times \cdots \times V_m$ is defined by

$$\lambda(v_1, \dots, v_m) = (\lambda v_1, \dots, \lambda v_m).$$

**3.88 example: *product of the vector spaces $\mathcal{P}_5(\mathbf{R})$ and $\mathbf{R}^3$***

Elements of $\mathcal{P}_5(\mathbf{R}) \times \mathbf{R}^3$ are lists of length two, with the first item in the list an element of $\mathcal{P}_5(\mathbf{R})$ and the second item in the list an element of $\mathbf{R}^3$.

For example, $(5 - 6x + 4x^2, (3, 8, 7))$ and $(x + 9x^5, (2, 2, 2))$ are elements of $\mathcal{P}_5(\mathbf{R}) \times \mathbf{R}^3$. Their sum is defined by

$$\begin{aligned}(5 - 6x + 4x^2, (3, 8, 7)) + (x + 9x^5, (2, 2, 2)) \\ = (5 - 5x + 4x^2 + 9x^5, (5, 10, 9)).\end{aligned}$$

Also, $2(5 - 6x + 4x^2, (3, 8, 7)) = (10 - 12x + 8x^2, (6, 16, 14))$.

The next result should be interpreted to mean that the product of vector spaces is a vector space with the operations of addition and scalar multiplication as defined by 3.87.

**3.89 *product of vector spaces is a vector space***

Suppose $V_1, \dots, V_m$ are vector spaces over $\mathbf{F}$. Then $V_1 \times \cdots \times V_m$ is a vector space over $\mathbf{F}$.

The proof of the result above is left to the reader. Note that the additive identity of $V_1 \times \cdots \times V_m$ is $(0, \dots, 0)$, where the 0 in the $k^{\text{th}}$ slot is the additive identity of $V_k$. The additive inverse of $(v_1, \dots, v_m) \in V_1 \times \cdots \times V_m$ is $(-v_1, \dots, -v_m)$.

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3.90 example: $\mathbf{R}^2 \times \mathbf{R}^3 \neq \mathbf{R}^5$ *but* $\mathbf{R}^2 \times \mathbf{R}^3$ *is isomorphic to* $\mathbf{R}^5$

Elements of the vector space $\mathbf{R}^2 \times \mathbf{R}^3$ are lists

$$((x_1, x_2), (x_3, x_4, x_5)),$$

where $x_1, x_2, x_3, x_4, x_5 \in \mathbf{R}$. Elements of $\mathbf{R}^5$ are lists

$$(x_1, x_2, x_3, x_4, x_5),$$

where $x_1, x_2, x_3, x_4, x_5 \in \mathbf{R}$.

Although elements of $\mathbf{R}^2 \times \mathbf{R}^3$ and $\mathbf{R}^5$ look similar, they are not the same kind of object. Elements of $\mathbf{R}^2 \times \mathbf{R}^3$ are lists of length two (with the first item itself a list of length two and the second item a list of length three), and elements of $\mathbf{R}^5$ are lists of length five. Thus $\mathbf{R}^2 \times \mathbf{R}^3$ does not equal $\mathbf{R}^5$.

The linear map

$$((x_1, x_2), (x_3, x_4, x_5)) \mapsto (x_1, x_2, x_3, x_4, x_5)$$

is an isomorphism of the vector space $\mathbf{R}^2 \times \mathbf{R}^3$ onto the vector space $\mathbf{R}^5$. Thus these two vector spaces are isomorphic, although they are not equal.

*This isomorphism is so natural that we should think of it as a relabeling. Some people informally say that* $\mathbf{R}^2 \times \mathbf{R}^3$ *equals* $\mathbf{R}^5$*, which is not technically correct but which captures the spirit of identification via relabeling.*

The next example illustrates the idea that we will use in the proof of 3.92.

3.91 example: *a basis of* $\mathcal{P}_2(\mathbf{R}) \times \mathbf{R}^2$

Consider this list of length five of elements of $\mathcal{P}_2(\mathbf{R}) \times \mathbf{R}^2$:

$$(1, (0, 0)), (x, (0, 0)), (x^2, (0, 0)), (0, (1, 0)), (0, (0, 1)).$$

The list above is linearly independent and it spans $\mathcal{P}_2(\mathbf{R}) \times \mathbf{R}^2$. Thus it is a basis of $\mathcal{P}_2(\mathbf{R}) \times \mathbf{R}^2$.

3.92 *dimension of a product is the sum of dimensions*

Suppose $V_1, \ldots, V_m$ are finite-dimensional vector spaces. Then $V_1 \times \cdots \times V_m$ is finite-dimensional and

$$\dim(V_1 \times \cdots \times V_m) = \dim V_1 + \cdots + \dim V_m.$$

Proof Choose a basis of each $V_k$. For each basis vector of each $V_k$, consider the element of $V_1 \times \cdots \times V_m$ that equals the basis vector in the $k^{\text{th}}$ slot and 0 in the other slots. The list of all such vectors is linearly independent and spans $V_1 \times \cdots \times V_m$. Thus it is a basis of $V_1 \times \cdots \times V_m$. The length of this basis is $\dim V_1 + \cdots + \dim V_m$, as desired.

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In the next result, the map $\Gamma$ is surjective by the definition of $V_1 + \cdots + V_m$. Thus the last word in the result below could be changed from "injective" to "invertible".

**3.93** *products and direct sums*

Suppose that $V_1, \ldots, V_m$ are subspaces of $V$. Define a linear map $\Gamma : V_1 \times \cdots \times V_m \to V_1 + \cdots + V_m$ by

$$\Gamma(v_1, \ldots, v_m) = v_1 + \cdots + v_m.$$

Then $V_1 + \cdots + V_m$ is a direct sum if and only if $\Gamma$ is injective.

Proof By 3.15, $\Gamma$ is injective if and only if the only way to write 0 as a sum $v_1 + \cdots + v_m$, where each $v_k$ is in $V_k$, is by taking each $v_k$ equal to 0. Thus 1.45 shows that $\Gamma$ is injective if and only if $V_1 + \cdots + V_m$ is a direct sum, as desired. ∎

**3.94** *a sum is a direct sum if and only if dimensions add up*

Suppose $V$ is finite-dimensional and $V_1, \ldots, V_m$ are subspaces of $V$. Then $V_1 + \cdots + V_m$ is a direct sum if and only if

$$\dim(V_1 + \cdots + V_m) = \dim V_1 + \cdots + \dim V_m.$$

Proof The map $\Gamma$ in 3.93 is surjective. Thus by the fundamental theorem of linear maps (3.21), $\Gamma$ is injective if and only if

$$\dim(V_1 + \cdots + V_m) = \dim(V_1 \times \cdots \times V_m).$$

Combining 3.93 and 3.92 now shows that $V_1 + \cdots + V_m$ is a direct sum if and only if

$$\dim(V_1 + \cdots + V_m) = \dim V_1 + \cdots + \dim V_m,$$

as desired. ∎

In the special case $m = 2$, an alternative proof that $V_1 + V_2$ is a direct sum if and only if $\dim(V_1 + V_2) = \dim V_1 + \dim V_2$ can be obtained by combining 1.46 and 2.43.

## *Quotient Spaces*

We begin our approach to quotient spaces by defining the sum of a vector and a subset.

**3.95** notation: $v + U$

Suppose $v \in V$ and $U \subseteq V$. Then $v + U$ is the subset of $V$ defined by

$$v + U = \{v + u : u \in U\}.$$

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3.96 example: *sum of a vector and a one-dimensional subspace of* $\mathbf{R}^2$

Suppose

$$U = \{(x, 2x) \in \mathbf{R}^2 : x \in \mathbf{R}\}.$$

Hence $U$ is the line in $\mathbf{R}^2$ through the origin with slope 2. Thus

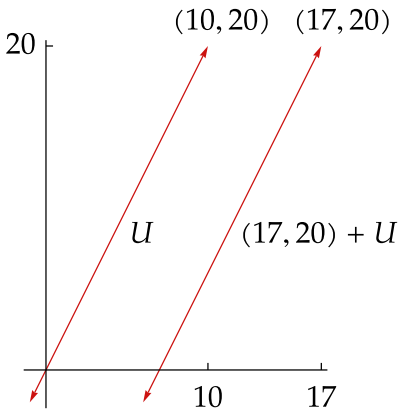
$$(17, 20) + U$$

is the line in $\mathbf{R}^2$ that contains the point $(17, 20)$ and has slope 2.

Because

$$(10, 20) \in U \quad \text{and} \quad (17, 20) \in (17, 20) + U,$$

we see that $(17, 20) + U$ is obtained by moving $U$ to the right by 7 units.




$(17, 20) + U$ *is parallel to the subspace* $U$.

3.97 definition: *translate*

For $v \in V$ and $U$ a subset of $V$, the set $v + U$ is said to be a *translate* of $U$.

3.98 example: *translates*

- If $U$ is the line in $\mathbf{R}^2$ defined by $U = \{(x, 2x) \in \mathbf{R}^2 : x \in \mathbf{R}\}$, then all lines in $\mathbf{R}^2$ with slope 2 are translates of $U$. See Example 3.96 above for a drawing of $U$ and one of its translates.
- More generally, if $U$ is a line in $\mathbf{R}^2$, then the set of all translates of $U$ is the set of all lines in $\mathbf{R}^2$ that are parallel to $U$.
- If $U = \{(x, y, 0) \in \mathbf{R}^3 : x, y \in \mathbf{R}\}$, then the translates of $U$ are the planes in $\mathbf{R}^3$ that are parallel to the $xy$-plane $U$.
- More generally, if $U$ is a plane in $\mathbf{R}^3$, then the set of all translates of $U$ is the set of all planes in $\mathbf{R}^3$ that are parallel to $U$ (see, for example, Exercise 7).

3.99 definition: *quotient space,* $V/U$

Suppose $U$ is a subspace of $V$. Then the *quotient space* $V/U$ is the set of all translates of $U$. Thus

$$V/U = \{v + U : v \in V\}.$$

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3.100 example: *quotient spaces*

- If $U = \{(x, 2x) \in \mathbf{R}^2 : x \in \mathbf{R}\}$, then $\mathbf{R}^2/U$ is the set of all lines in $\mathbf{R}^2$ that have slope 2.
- If $U$ is a line in $\mathbf{R}^3$ containing the origin, then $\mathbf{R}^3/U$ is the set of all lines in $\mathbf{R}^3$ parallel to $U$.
- If $U$ is a plane in $\mathbf{R}^3$ containing the origin, then $\mathbf{R}^3/U$ is the set of all planes in $\mathbf{R}^3$ parallel to $U$.

Our next goal is to make $V/U$ into a vector space. To do this, we will need the next result.

3.101 *two translates of a subspace are equal or disjoint*

Suppose $U$ is a subspace of $V$ and $v, w \in V$. Then

$$v - w \in U \iff v + U = w + U \iff (v + U) \cap (w + U) \neq \emptyset.$$

Proof First suppose $v - w \in U$. If $u \in U$, then

$$v + u = w + ((v - w) + u) \in w + U.$$

Thus $v + U \subseteq w + U$. Similarly, $w + U \subseteq v + U$. Thus $v + U = w + U$, completing the proof that $v - w \in U$ implies $v + U = w + U$.

The equation $v + U = w + U$ implies that $(v + U) \cap (w + U) \neq \emptyset$.

Now suppose $(v + U) \cap (w + U) \neq \emptyset$. Thus there exist $u_1, u_2 \in U$ such that

$$v + u_1 = w + u_2.$$

Thus $v - w = u_2 - u_1$. Hence $v - w \in U$, showing that $(v + U) \cap (w + U) \neq \emptyset$ implies $v - w \in U$, which completes the proof.

Now we can define addition and scalar multiplication on $V/U$.

3.102 definition: *addition and scalar multiplication on $V/U$*

Suppose $U$ is a subspace of $V$. Then *addition* and *scalar multiplication* are defined on $V/U$ by

$$(v + U) + (w + U) = (v + w) + U$$
$$\lambda(v + U) = (\lambda v) + U$$

for all $v, w \in V$ and all $\lambda \in \mathbf{F}$.

As part of the proof of the next result, we will show that the definitions above make sense.

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3.103 *quotient space is a vector space*

Suppose $U$ is a subspace of $V$. Then $V/U$, with the operations of addition and scalar multiplication as defined above, is a vector space.

Proof The potential problem with the definitions above of addition and scalar multiplication on $V/U$ is that the representation of a translate of $U$ is not unique. Specifically, suppose $v_1, v_2, w_1, w_2 \in V$ are such that

$$v_1 + U = v_2 + U \quad \text{and} \quad w_1 + U = w_2 + U.$$

To show that the definition of addition on $V/U$ given above makes sense, we must show that $(v_1 + w_1) + U = (v_2 + w_2) + U$.

By 3.101, we have

$$v_1 - v_2 \in U \quad \text{and} \quad w_1 - w_2 \in U.$$

Because $U$ is a subspace of $V$ and thus is closed under addition, this implies that $(v_1 - v_2) + (w_1 - w_2) \in U$. Thus $(v_1 + w_1) - (v_2 + w_2) \in U$. Using 3.101 again, we see that

$$(v_1 + w_1) + U = (v_2 + w_2) + U,$$

as desired. Thus the definition of addition on $V/U$ makes sense.

Similarly, suppose $\lambda \in \mathbf{F}$. We are still assuming that $v_1 + U = v_2 + U$. Because $U$ is a subspace of $V$ and thus is closed under scalar multiplication, we have $\lambda(v_1 - v_2) \in U$. Thus $\lambda v_1 - \lambda v_2 \in U$. Hence 3.101 implies that

$$(\lambda v_1) + U = (\lambda v_2) + U.$$

Thus the definition of scalar multiplication on $V/U$ makes sense.

Now that addition and scalar multiplication have been defined on $V/U$, the verification that these operations make $V/U$ into a vector space is straightforward and is left to the reader. Note that the additive identity of $V/U$ is $0 + U$ (which equals $U$) and that the additive inverse of $v + U$ is $(-v) + U$. ∎

The next concept will lead to a computation of the dimension of $V/U$.

3.104 definition: *quotient map, $\pi$*

Suppose $U$ is a subspace of $V$. The *quotient map* $\pi\colon V \to V/U$ is the linear map defined by

$$\pi(v) = v + U$$

for each $v \in V$.

The reader should verify that $\pi$ is indeed a linear map. Although $\pi$ depends on $U$ as well as $V$, these spaces are left out of the notation because they should be clear from the context.

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**3.105** *dimension of quotient space*

Suppose $V$ is finite-dimensional and $U$ is a subspace of $V$. Then

$$\dim V/U = \dim V - \dim U.$$

Proof Let $\pi$ denote the quotient map from $V$ to $V/U$. If $v \in V$, then $v+U = 0+U$ if and only if $v \in U$ (by 3.101), which implies that $\operatorname{null} \pi = U$. The definition of $\pi$ implies $\operatorname{range} \pi = V/U$. The fundamental theorem of linear maps (3.21) now implies $\dim V = \dim U + \dim V/U$, which gives the desired result. ∎

Each linear map $T$ on $V$ induces a linear map $\widetilde{T}$ on $V/(\operatorname{null} T)$, which we now define.

**3.106** notation: $\widetilde{T}$

Suppose $T \in \mathcal{L}(V, W)$. Define $\widetilde{T}\colon V/(\operatorname{null} T) \to W$ by

$$\widetilde{T}(v + \operatorname{null} T) = Tv.$$

To show that the definition of $\widetilde{T}$ makes sense, suppose $u, v \in V$ are such that $u + \operatorname{null} T = v + \operatorname{null} T$. By 3.101, we have $u - v \in \operatorname{null} T$. Thus $T(u - v) = 0$. Hence $Tu = Tv$. Thus the definition of $\widetilde{T}$ indeed makes sense. The routine verification that $\widetilde{T}$ is a linear map from $V/(\operatorname{null} T)$ to $W$ is left to the reader.

The next result shows that we can think of $\widetilde{T}$ as a modified version of $T$, with a domain that produces a one-to-one map.

**3.107** *null space and range of* $\widetilde{T}$

Suppose $T \in \mathcal{L}(V, W)$. Then

(a) $\widetilde{T} \circ \pi = T$, where $\pi$ is the quotient map of $V$ onto $V/(\operatorname{null} T)$;

(b) $\widetilde{T}$ is injective;

(c) $\operatorname{range} \widetilde{T} = \operatorname{range} T$;

(d) $V/(\operatorname{null} T)$ and $\operatorname{range} T$ are isomorphic vector spaces.

Proof

(a) If $v \in V$, then $(\widetilde{T} \circ \pi)(v) = \widetilde{T}(\pi(v)) = \widetilde{T}(v + \operatorname{null} T) = Tv$, as desired.

(b) Suppose $v \in V$ and $\widetilde{T}(v + \operatorname{null} T) = 0$. Then $Tv = 0$. Thus $v \in \operatorname{null} T$. Hence 3.101 implies that $v + \operatorname{null} T = 0 + \operatorname{null} T$. This implies that $\operatorname{null} \widetilde{T} = \{0 + \operatorname{null} T\}$. Hence $\widetilde{T}$ is injective, as desired.

(c) The definition of $\widetilde{T}$ shows that $\operatorname{range} \widetilde{T} = \operatorname{range} T$.

(d) Now (b) and (c) imply that if we think of $\widetilde{T}$ as mapping into $\operatorname{range} T$, then $\widetilde{T}$ is an isomorphism from $V/(\operatorname{null} T)$ onto $\operatorname{range} T$. ∎

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## *Exercises 3E*

**1** Suppose $T$ is a function from $V$ to $W$. The *graph* of $T$ is the subset of $V \times W$ defined by

$$\text{graph of } T = \{(v, Tv) \in V \times W : v \in V\}.$$

Prove that $T$ is a linear map if and only if the graph of $T$ is a subspace of $V \times W$.

*Formally, a function $T$ from $V$ to $W$ is a subset $T$ of $V \times W$ such that for each $v \in V$, there exists exactly one element $(v, w) \in T$. In other words, formally a function is what is called above its graph. We do not usually think of functions in this formal manner. However, if we do become formal, then this exercise could be rephrased as follows: Prove that a function $T$ from $V$ to $W$ is a linear map if and only if $T$ is a subspace of $V \times W$.*

**2** Suppose that $V_1, \ldots, V_m$ are vector spaces such that $V_1 \times \cdots \times V_m$ is finite-dimensional. Prove that $V_k$ is finite-dimensional for each $k = 1, \ldots, m$.

**3** Suppose $V_1, \ldots, V_m$ are vector spaces. Prove that $\mathcal{L}(V_1 \times \cdots \times V_m, W)$ and $\mathcal{L}(V_1, W) \times \cdots \times \mathcal{L}(V_m, W)$ are isomorphic vector spaces.

**4** Suppose $W_1, \ldots, W_m$ are vector spaces. Prove that $\mathcal{L}(V, W_1 \times \cdots \times W_m)$ and $\mathcal{L}(V, W_1) \times \cdots \times \mathcal{L}(V, W_m)$ are isomorphic vector spaces.

**5** For $m$ a positive integer, define $V^m$ by

$$V^m = \underbrace{V \times \cdots \times V}_{m \text{ times}}.$$

Prove that $V^m$ and $\mathcal{L}(\mathbf{F}^m, V)$ are isomorphic vector spaces.

**6** Suppose that $v, x$ are vectors in $V$ and that $U, W$ are subspaces of $V$ such that $v + U = x + W$. Prove that $U = W$.

**7** Let $U = \{(x, y, z) \in \mathbf{R}^3 : 2x + 3y + 5z = 0\}$. Suppose $A \subseteq \mathbf{R}^3$. Prove that $A$ is a translate of $U$ if and only if there exists $c \in \mathbf{R}$ such that

$$A = \{(x, y, z) \in \mathbf{R}^3 : 2x + 3y + 5z = c\}.$$

**8** (a) Suppose $T \in \mathcal{L}(V, W)$ and $c \in W$. Prove that $\{x \in V : Tx = c\}$ is either the empty set or is a translate of null $T$.

(b) Explain why the set of solutions to a system of linear equations such as 3.27 is either the empty set or is a translate of some subspace of $\mathbf{F}^n$.

**9** Prove that a nonempty subset $A$ of $V$ is a translate of some subspace of $V$ if and only if $\lambda v + (1 - \lambda)w \in A$ for all $v, w \in A$ and all $\lambda \in \mathbf{F}$.

**10** Suppose $A_1 = v + U_1$ and $A_2 = w + U_2$ for some $v, w \in V$ and some subspaces $U_1, U_2$ of $V$. Prove that the intersection $A_1 \cap A_2$ is either a translate of some subspace of $V$ or is the empty set.

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**11** Suppose $U = \{(x_1, x_2, \dots) \in \mathbf{F}^\infty : x_k \neq 0 \text{ for only finitely many } k\}$.

(a) Show that $U$ is a subspace of $\mathbf{F}^\infty$.

(b) Prove that $\mathbf{F}^\infty / U$ is infinite-dimensional.

**12** Suppose $v_1, \dots, v_m \in V$. Let

$$A = \{\lambda_1 v_1 + \cdots + \lambda_m v_m : \lambda_1, \dots, \lambda_m \in \mathbf{F} \text{ and } \lambda_1 + \cdots + \lambda_m = 1\}.$$

(a) Prove that $A$ is a translate of some subspace of $V$.

(b) Prove that if $B$ is a translate of some subspace of $V$ and $\{v_1, \dots, v_m\} \subseteq B$, then $A \subseteq B$.

(c) Prove that $A$ is a translate of some subspace of $V$ of dimension less than $m$.

**13** Suppose $U$ is a subspace of $V$ such that $V/U$ is finite-dimensional. Prove that $V$ is isomorphic to $U \times (V/U)$.

**14** Suppose $U$ and $W$ are subspaces of $V$ and $V = U \oplus W$. Suppose $w_1, \dots, w_m$ is a basis of $W$. Prove that $w_1 + U, \dots, w_m + U$ is a basis of $V/U$.

**15** Suppose $U$ is a subspace of $V$ and $v_1 + U, \dots, v_m + U$ is a basis of $V/U$ and $u_1, \dots, u_n$ is a basis of $U$. Prove that $v_1, \dots, v_m, u_1, \dots, u_n$ is a basis of $V$.

**16** Suppose $\varphi \in \mathcal{L}(V, \mathbf{F})$ and $\varphi \neq 0$. Prove that $\dim V/(\operatorname{null} \varphi) = 1$.

**17** Suppose $U$ is a subspace of $V$ such that $\dim V/U = 1$. Prove that there exists $\varphi \in \mathcal{L}(V, \mathbf{F})$ such that $\operatorname{null} \varphi = U$.

**18** Suppose that $U$ is a subspace of $V$ such that $V/U$ is finite-dimensional.

(a) Show that if $W$ is a finite-dimensional subspace of $V$ and $V = U + W$, then $\dim W \geq \dim V/U$.

(b) Prove that there exists a finite-dimensional subspace $W$ of $V$ such that $\dim W = \dim V/U$ and $V = U \oplus W$.

**19** Suppose $T \in \mathcal{L}(V, W)$ and $U$ is a subspace of $V$. Let $\pi$ denote the quotient map from $V$ onto $V/U$. Prove that there exists $S \in \mathcal{L}(V/U, W)$ such that $T = S \circ \pi$ if and only if $U \subseteq \operatorname{null} T$.

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# 3F Duality

## Dual Space and Dual Map

Linear maps into the scalar field $\mathbf{F}$ play a special role in linear algebra, and thus they get a special name.

**3.108 definition: *linear functional***

A *linear functional* on $V$ is a linear map from $V$ to $\mathbf{F}$. In other words, a linear functional is an element of $\mathcal{L}(V, \mathbf{F})$.

**3.109 example: *linear functionals***

- Define $\varphi : \mathbf{R}^3 \to \mathbf{R}$ by $\varphi(x, y, z) = 4x - 5y + 2z$. Then $\varphi$ is a linear functional on $\mathbf{R}^3$.
- Fix $(c_1, \ldots, c_n) \in \mathbf{F}^n$. Define $\varphi : \mathbf{F}^n \to \mathbf{F}$ by $\varphi(x_1, \ldots, x_n) = c_1 x_1 + \cdots + c_n x_n$. Then $\varphi$ is a linear functional on $\mathbf{F}^n$.
- Define $\varphi : \mathcal{P}(\mathbf{R}) \to \mathbf{R}$ by
$$\varphi(p) = 3p''(5) + 7p(4).$$
Then $\varphi$ is a linear functional on $\mathcal{P}(\mathbf{R})$.
- Define $\varphi : \mathcal{P}(\mathbf{R}) \to \mathbf{R}$ by
$$\varphi(p) = \int_0^1 p$$
for each $p \in \mathcal{P}(\mathbf{R})$. Then $\varphi$ is a linear functional on $\mathcal{P}(\mathbf{R})$.

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The vector space $\mathcal{L}(V, \mathbf{F})$ also gets a special name and special notation.

**3.110 definition: *dual space*, $V'$**

The *dual space* of $V$, denoted by $V'$, is the vector space of all linear functionals on $V$. In other words, $V' = \mathcal{L}(V, \mathbf{F})$.

**3.111 $\dim V' = \dim V$**

Suppose $V$ is finite-dimensional. Then $V'$ is also finite-dimensional and
$$\dim V' = \dim V.$$

Proof By 3.72 we have
$$\dim V' = \dim \mathcal{L}(V, \mathbf{F}) = (\dim V)(\dim \mathbf{F}) = \dim V,$$
as desired.

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In the following definition, the linear map lemma (3.4) implies that each $\varphi_j$ is well defined.

**3.112 definition: *dual basis***

If $v_1, ..., v_n$ is a basis of $V$, then the *dual basis* of $v_1, ..., v_n$ is the list $\varphi_1, ..., \varphi_n$ of elements of $V'$, where each $\varphi_j$ is the linear functional on $V$ such that

$$\varphi_j(v_k) = \begin{cases} 1 & \text{if } k = j, \\ 0 & \text{if } k \neq j. \end{cases}$$

**3.113 example: *the dual basis of the standard basis of* $\mathbf{F}^n$**

Suppose $n$ is a positive integer. For $1 \leq j \leq n$, define $\varphi_j$ to be the linear functional on $\mathbf{F}^n$ that selects the $j^{\text{th}}$ coordinate of a vector in $\mathbf{F}^n$. Thus

$$\varphi_j(x_1, ..., x_n) = x_j$$

for each $(x_1, ..., x_n) \in \mathbf{F}^n$.

Let $e_1, ..., e_n$ be the standard basis of $\mathbf{F}^n$. Then

$$\varphi_j(e_k) = \begin{cases} 1 & \text{if } k = j, \\ 0 & \text{if } k \neq j. \end{cases}$$

Thus $\varphi_1, ..., \varphi_n$ is the dual basis of the standard basis $e_1, ..., e_n$ of $\mathbf{F}^n$.

The next result shows that the dual basis of a basis of $V$ consists of the linear functionals on $V$ that give the coefficients for expressing a vector in $V$ as a linear combination of the basis vectors.

**3.114 *dual basis gives coefficients for linear combination***

Suppose $v_1, ..., v_n$ is a basis of $V$ and $\varphi_1, ..., \varphi_n$ is the dual basis. Then

$$v = \varphi_1(v)v_1 + \cdots + \varphi_n(v)v_n$$

for each $v \in V$.

Proof Suppose $v \in V$. Then there exist $c_1, ..., c_n \in \mathbf{F}$ such that

3.115 $$v = c_1v_1 + \cdots + c_nv_n.$$

If $j \in \{1, ..., n\}$, then applying $\varphi_j$ to both sides of the equation above gives

$$\varphi_j(v) = c_j.$$

Substituting the values for $c_1, ..., c_n$ given by the equation above into 3.115 shows that $v = \varphi_1(v)v_1 + \cdots + \varphi_n(v)v_n$.

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The next result shows that the dual basis is indeed a basis of the dual space. Thus the terminology "dual basis" is justified.

**3.116** *dual basis is a basis of the dual space*

Suppose $V$ is finite-dimensional. Then the dual basis of a basis of $V$ is a basis of $V'$.

Proof Suppose $v_1, \dots, v_n$ is a basis of $V$. Let $\varphi_1, \dots, \varphi_n$ denote the dual basis.

To show that $\varphi_1, \dots, \varphi_n$ is a linearly independent list of elements of $V'$, suppose $a_1, \dots, a_n \in \mathbf{F}$ are such that

$$a_1\varphi_1 + \cdots + a_n\varphi_n = 0. \tag{3.117}$$

Now

$$(a_1\varphi_1 + \cdots + a_n\varphi_n)(v_k) = a_k$$

for each $k = 1, \dots, n$. Thus 3.117 shows that $a_1 = \cdots = a_n = 0$. Hence $\varphi_1, \dots, \varphi_n$ is linearly independent.

Because $\varphi_1, \dots, \varphi_n$ is a linearly independent list in $V'$ whose length equals $\dim V'$ (by 3.111), we can conclude that $\varphi_1, \dots, \varphi_n$ is a basis of $V'$ (see 2.38). ∎

In the definition below, note that if $T$ is a linear map from $V$ to $W$ then $T'$ is a linear map from $W'$ to $V'$.

**3.118** definition: *dual map, $T'$*

Suppose $T \in \mathcal{L}(V, W)$. The *dual map* of $T$ is the linear map $T' \in \mathcal{L}(W', V')$ defined for each $\varphi \in W'$ by

$$T'(\varphi) = \varphi \circ T.$$

If $T \in \mathcal{L}(V, W)$ and $\varphi \in W'$, then $T'(\varphi)$ is defined above to be the composition of the linear maps $\varphi$ and $T$. Thus $T'(\varphi)$ is indeed a linear map from $V$ to $\mathbf{F}$; in other words, $T'(\varphi) \in V'$.

The following two bullet points show that $T'$ is a linear map from $W'$ to $V'$.

- If $\varphi, \psi \in W'$, then

$$T'(\varphi + \psi) = (\varphi + \psi) \circ T = \varphi \circ T + \psi \circ T = T'(\varphi) + T'(\psi).$$

- If $\lambda \in \mathbf{F}$ and $\varphi \in W'$, then

$$T'(\lambda\varphi) = (\lambda\varphi) \circ T = \lambda(\varphi \circ T) = \lambda T'(\varphi).$$

The prime notation appears with two unrelated meanings in the next example: $D'$ denotes the dual of the linear map $D$, and $p'$ denotes the derivative of a polynomial $p$.

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3.119 example: *dual map of the differentiation linear map*

Define $D: \mathcal{P}(\mathbf{R}) \to \mathcal{P}(\mathbf{R})$ by $Dp = p'$.

- Suppose $\varphi$ is the linear functional on $\mathcal{P}(\mathbf{R})$ defined by $\varphi(p) = p(3)$. Then $D'(\varphi)$ is the linear functional on $\mathcal{P}(\mathbf{R})$ given by

$$(D'(\varphi))(p) = (\varphi \circ D)(p) = \varphi(Dp) = \varphi(p') = p'(3).$$

  Thus $D'(\varphi)$ is the linear functional on $\mathcal{P}(\mathbf{R})$ taking $p$ to $p'(3)$.

- Suppose $\varphi$ is the linear functional on $\mathcal{P}(\mathbf{R})$ defined by $\varphi(p) = \int_0^1 p$. Then $D'(\varphi)$ is the linear functional on $\mathcal{P}(\mathbf{R})$ given by

$$\begin{aligned}(D'(\varphi))(p) &= (\varphi \circ D)(p)\\ &= \varphi(Dp)\\ &= \varphi(p')\\ &= \int_0^1 p'\\ &= p(1) - p(0).\end{aligned}$$

  Thus $D'(\varphi)$ is the linear functional on $\mathcal{P}(\mathbf{R})$ taking $p$ to $p(1) - p(0)$.

In the next result, (a) and (b) imply that the function that takes $T$ to $T'$ is a linear map from $\mathcal{L}(V, W)$ to $\mathcal{L}(W', V')$.

In (c) below, note the reversal of order from $ST$ on the left to $T'S'$ on the right.

3.120 *algebraic properties of dual maps*

Suppose $T \in \mathcal{L}(V, W)$. Then

(a) $(S + T)' = S' + T'$ for all $S \in \mathcal{L}(V, W)$;

(b) $(\lambda T)' = \lambda T'$ for all $\lambda \in \mathbf{F}$;

(c) $(ST)' = T'S'$ for all $S \in \mathcal{L}(W, U)$.

Proof The proofs of (a) and (b) are left to the reader.

To prove (c), suppose $\varphi \in U'$. Then

$$(ST)'(\varphi) = \varphi \circ (ST) = (\varphi \circ S) \circ T = T'(\varphi \circ S) = T'(S'(\varphi)) = (T'S')(\varphi),$$

where the first, third, and fourth equalities above hold because of the definition of the dual map, the second equality holds because composition of functions is associative, and the last equality follows from the definition of composition.

The equation above shows that $(ST)'(\varphi) = (T'S')(\varphi)$ for all $\varphi \in U'$. Thus $(ST)' = T'S'$. ∎

*Some books use the notation $V^*$ and $T^*$ for duality instead of $V'$ and $T'$. However, here we reserve the notation $T^*$ for the adjoint, which will be introduced when we study linear maps on inner product spaces in Chapter 7.*

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## *Null Space and Range of Dual of Linear Map*

Our goal in this subsection is to describe null $T'$ and range $T'$ in terms of range $T$ and null $T$. To do this, we will need the next definition.

**3.121 definition: *annihilator*, $U^0$**

For $U \subseteq V$, the *annihilator* of $U$, denoted by $U^0$, is defined by

$$U^0 = \{\varphi \in V' : \varphi(u) = 0 \text{ for all } u \in U\}.$$

**3.122 example: *element of an annihilator***

Suppose $U$ is the subspace of $\mathcal{P}(\mathbf{R})$ consisting of polynomial multiples of $x^2$. If $\varphi$ is the linear functional on $\mathcal{P}(\mathbf{R})$ defined by $\varphi(p) = p'(0)$, then $\varphi \in U^0$.

For $U \subseteq V$, the annihilator $U^0$ is a subset of the dual space $V'$. Thus $U^0$ depends on the vector space containing $U$, so a notation such as $U^0_V$ would be more precise. However, the containing vector space will always be clear from the context, so we will use the simpler notation $U^0$.

**3.123 example: *the annihilator of a two-dimensional subspace of* $\mathbf{R}^5$**

Let $e_1, e_2, e_3, e_4, e_5$ denote the standard basis of $\mathbf{R}^5$; let $\varphi_1, \varphi_2, \varphi_3, \varphi_4, \varphi_5 \in (\mathbf{R}^5)'$ denote the dual basis of $e_1, e_2, e_3, e_4, e_5$. Suppose

$$U = \operatorname{span}(e_1, e_2) = \{(x_1, x_2, 0, 0, 0) \in \mathbf{R}^5 : x_1, x_2 \in \mathbf{R}\}.$$

We want to show that $U^0 = \operatorname{span}(\varphi_3, \varphi_4, \varphi_5)$.

Recall (see 3.113) that $\varphi_j$ is the linear functional on $\mathbf{R}^5$ that selects the $j^{\text{th}}$ coordinate: $\varphi_j(x_1, x_2, x_3, x_4, x_5) = x_j$.

First suppose $\varphi \in \operatorname{span}(\varphi_3, \varphi_4, \varphi_5)$. Then there exist $c_3, c_4, c_5 \in \mathbf{R}$ such that $\varphi = c_3\varphi_3 + c_4\varphi_4 + c_5\varphi_5$. If $(x_1, x_2, 0, 0, 0) \in U$, then

$$\varphi(x_1, x_2, 0, 0, 0) = (c_3\varphi_3 + c_4\varphi_4 + c_5\varphi_5)(x_1, x_2, 0, 0, 0) = 0.$$

Thus $\varphi \in U^0$. Hence we have shown that $\operatorname{span}(\varphi_3, \varphi_4, \varphi_5) \subseteq U^0$.

To show the inclusion in the other direction, suppose that $\varphi \in U^0$. Because the dual basis is a basis of $(\mathbf{R}^5)'$, there exist $c_1, c_2, c_3, c_4, c_5 \in \mathbf{R}$ such that $\varphi = c_1\varphi_1 + c_2\varphi_2 + c_3\varphi_3 + c_4\varphi_4 + c_5\varphi_5$. Because $e_1 \in U$ and $\varphi \in U^0$, we have

$$0 = \varphi(e_1) = (c_1\varphi_1 + c_2\varphi_2 + c_3\varphi_3 + c_4\varphi_4 + c_5\varphi_5)(e_1) = c_1.$$

Similarly, $e_2 \in U$ and thus $c_2 = 0$. Hence $\varphi = c_3\varphi_3 + c_4\varphi_4 + c_5\varphi_5$. Thus $\varphi \in \operatorname{span}(\varphi_3, \varphi_4, \varphi_5)$, which shows that $U^0 \subseteq \operatorname{span}(\varphi_3, \varphi_4, \varphi_5)$.

Thus $U^0 = \operatorname{span}(\varphi_3, \varphi_4, \varphi_5)$.

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**3.124** *the annihilator is a subspace*

Suppose $U \subseteq V$. Then $U^0$ is a subspace of $V'$.

Proof Note that $0 \in U^0$ (here 0 is the zero linear functional on $V$) because the zero linear functional applied to every vector in $U$ is the zero vector in $V$.

Suppose $\varphi, \psi \in U^0$. Thus $\varphi, \psi \in V'$ and $\varphi(u) = \psi(u) = 0$ for every $u \in U$. If $u \in U$, then

$$(\varphi + \psi)(u) = \varphi(u) + \psi(u) = 0 + 0 = 0.$$

Thus $\varphi + \psi \in U^0$.

Similarly, $U^0$ is closed under scalar multiplication. Thus 1.34 implies that $U^0$ is a subspace of $V'$. ∎

The next result shows that $\dim U^0$ is the difference of $\dim V$ and $\dim U$. For example, this shows that if $U$ is a two-dimensional subspace of $\mathbf{R}^5$, then $U^0$ is a three-dimensional subspace of $(\mathbf{R}^5)'$, as in Example 3.123.

The next result can be proved following the pattern of Example 3.123: choose a basis $u_1, \ldots, u_m$ of $U$, extend to a basis $u_1, \ldots, u_m, \ldots, u_n$ of $V$, let $\varphi_1, \ldots, \varphi_m, \ldots, \varphi_n$ be the dual basis of $V'$, and then show that $\varphi_{m+1}, \ldots, \varphi_n$ is a basis of $U^0$, which implies the desired result. You should construct the proof just outlined, even though a slicker proof is presented here.

**3.125** *dimension of the annihilator*

Suppose $V$ is finite-dimensional and $U$ is a subspace of $V$. Then

$$\dim U^0 = \dim V - \dim U.$$

Proof Let $i \in \mathcal{L}(U, V)$ be the inclusion map defined by $i(u) = u$ for each $u \in U$. Thus $i'$ is a linear map from $V'$ to $U'$. The fundamental theorem of linear maps (3.21) applied to $i'$ shows that

$$\dim \operatorname{range} i' + \dim \operatorname{null} i' = \dim V'.$$

However, $\operatorname{null} i' = U^0$ (as can be seen by thinking about the definitions) and $\dim V' = \dim V$ (by 3.111), so we can rewrite the equation above as

3.126 $$\dim \operatorname{range} i' + \dim U^0 = \dim V.$$

If $\varphi \in U'$, then $\varphi$ can be extended to a linear functional $\psi$ on $V$ (see, for example, Exercise 13 in Section 3A). The definition of $i'$ shows that $i'(\psi) = \varphi$. Thus $\varphi \in \operatorname{range} i'$, which implies that $\operatorname{range} i' = U'$. Hence

$$\dim \operatorname{range} i' = \dim U' = \dim U,$$

and then 3.126 becomes the equation $\dim U + \dim U^0 = \dim V$, as desired. ∎

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The next result can be a useful tool to show that a subspace is as big as possible—see (a)—or to show that a subspace is as small as possible—see (b).

**3.127** *condition for the annihilator to equal* $\{0\}$ *or the whole space*

Suppose $V$ is finite-dimensional and $U$ is a subspace of $V$. Then

(a) $U^0 = \{0\} \iff U = V$;

(b) $U^0 = V' \iff U = \{0\}$.

Proof To prove (a), we have

$$\begin{aligned} U^0 = \{0\} &\iff \dim U^0 = 0 \\ &\iff \dim U = \dim V \\ &\iff U = V, \end{aligned}$$

where the second equivalence follows from 3.125 and the third equivalence follows from 2.39.

Similarly, to prove (b) we have

$$\begin{aligned} U^0 = V' &\iff \dim U^0 = \dim V' \\ &\iff \dim U^0 = \dim V \\ &\iff \dim U = 0 \\ &\iff U = \{0\}, \end{aligned}$$

where one direction of the first equivalence follows from 2.39, the second equivalence follows from 3.111, and the third equivalence follows from 3.125. ∎

The proof of (a) in the next result does not use the hypothesis that $V$ and $W$ are finite-dimensional.

**3.128** *the null space of* $T'$

Suppose $V$ and $W$ are finite-dimensional and $T \in \mathcal{L}(V, W)$. Then

(a) $\operatorname{null} T' = (\operatorname{range} T)^0$;

(b) $\dim \operatorname{null} T' = \dim \operatorname{null} T + \dim W - \dim V$.

Proof

(a) First suppose $\varphi \in \operatorname{null} T'$. Thus $0 = T'(\varphi) = \varphi \circ T$. Hence

$$0 = (\varphi \circ T)(v) = \varphi(Tv) \quad \text{for every } v \in V.$$

Thus $\varphi \in (\operatorname{range} T)^0$. This implies that $\operatorname{null} T' \subseteq (\operatorname{range} T)^0$.

To prove the inclusion in the opposite direction, now suppose $\varphi \in (\operatorname{range} T)^0$. Thus $\varphi(Tv) = 0$ for every vector $v \in V$. Hence $0 = \varphi \circ T = T'(\varphi)$. In other words, $\varphi \in \operatorname{null} T'$, which shows that $(\operatorname{range} T)^0 \subseteq \operatorname{null} T'$, completing the proof of (a).

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(b) We have

$$\begin{aligned}\dim \operatorname{null} T' &= \dim(\operatorname{range} T)^0 \\ &= \dim W - \dim \operatorname{range} T \\ &= \dim W - (\dim V - \dim \operatorname{null} T) \\ &= \dim \operatorname{null} T + \dim W - \dim V,\end{aligned}$$

where the first equality comes from (a), the second equality comes from 3.125, and the third equality comes from the fundamental theorem of linear maps (3.21). ∎

The next result can be useful because sometimes it is easier to verify that $T'$ is injective than to show directly that $T$ is surjective.

**3.129** *$T$ surjective is equivalent to $T'$ injective*

Suppose $V$ and $W$ are finite-dimensional and $T \in \mathcal{L}(V, W)$. Then

$$T \text{ is surjective} \iff T' \text{ is injective.}$$

Proof We have

$$\begin{aligned}T \in \mathcal{L}(V, W) \text{ is surjective} &\iff \operatorname{range} T = W \\ &\iff (\operatorname{range} T)^0 = \{0\} \\ &\iff \operatorname{null} T' = \{0\} \\ &\iff T' \text{ is injective,}\end{aligned}$$

where the second equivalence comes from 3.127(a) and the third equivalence comes from 3.128(a). ∎

**3.130** *the range of $T'$*

Suppose $V$ and $W$ are finite-dimensional and $T \in \mathcal{L}(V, W)$. Then

(a) $\dim \operatorname{range} T' = \dim \operatorname{range} T$;

(b) $\operatorname{range} T' = (\operatorname{null} T)^0$.

Proof

(a) We have

$$\begin{aligned}\dim \operatorname{range} T' &= \dim W' - \dim \operatorname{null} T' \\ &= \dim W - \dim(\operatorname{range} T)^0 \\ &= \dim \operatorname{range} T,\end{aligned}$$

where the first equality comes from 3.21, the second equality comes from 3.111 and 3.128(a), and the third equality comes from 3.125.

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(b) First suppose $\varphi \in \operatorname{range} T'$. Thus there exists $\psi \in W'$ such that $\varphi = T'(\psi)$. If $v \in \operatorname{null} T$, then

$$\varphi(v) = (T'(\psi))v = (\psi \circ T)(v) = \psi(Tv) = \psi(0) = 0.$$

Hence $\varphi \in (\operatorname{null} T)^0$. This implies that $\operatorname{range} T' \subseteq (\operatorname{null} T)^0$.

We will complete the proof by showing that $\operatorname{range} T'$ and $(\operatorname{null} T)^0$ have the same dimension. To do this, note that

$$\begin{aligned} \dim \operatorname{range} T' &= \dim \operatorname{range} T \\ &= \dim V - \dim \operatorname{null} T \\ &= \dim (\operatorname{null} T)^0, \end{aligned}$$

where the first equality comes from (a), the second equality comes from 3.21, and the third equality comes from 3.125. ∎

The next result should be compared to 3.129.

**3.131** *$T$ injective is equivalent to $T'$ surjective*

Suppose $V$ and $W$ are finite-dimensional and $T \in \mathcal{L}(V, W)$. Then

$$T \text{ is injective} \iff T' \text{ is surjective.}$$

Proof We have

$$\begin{aligned} T \text{ is injective} &\iff \operatorname{null} T = \{0\} \\ &\iff (\operatorname{null} T)^0 = V' \\ &\iff \operatorname{range} T' = V', \end{aligned}$$

where the second equivalence follows from 3.127(b) and the third equivalence follows from 3.130(b). ∎

## *Matrix of Dual of Linear Map*

The setting for the next result is the assumption that we have a basis $v_1, \ldots, v_n$ of $V$, along with its dual basis $\varphi_1, \ldots, \varphi_n$ of $V'$. We also have a basis $w_1, \ldots, w_m$ of $W$, along with its dual basis $\psi_1, \ldots, \psi_m$ of $W'$. Thus $\mathcal{M}(T)$ is computed with respect to the bases just mentioned of $V$ and $W$, and $\mathcal{M}(T')$ is computed with respect to the dual bases just mentioned of $W'$ and $V'$. Using these bases gives the following pretty result.

**3.132** *matrix of $T'$ is transpose of matrix of $T$*

Suppose $V$ and $W$ are finite-dimensional and $T \in \mathcal{L}(V, W)$. Then

$$\mathcal{M}(T') = (\mathcal{M}(T))^t.$$

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Proof Let $A = \mathcal{M}(T)$ and $C = \mathcal{M}(T')$. Suppose $1 \le j \le m$ and $1 \le k \le n$.

From the definition of $\mathcal{M}(T')$ we have

$$T'(\psi_j) = \sum_{r=1}^{n} C_{r,j}\varphi_r.$$

The left side of the equation above equals $\psi_j \circ T$. Thus applying both sides of the equation above to $v_k$ gives

$$\begin{aligned}(\psi_j \circ T)(v_k) &= \sum_{r=1}^{n} C_{r,j}\varphi_r(v_k)\\ &= C_{k,j}.\end{aligned}$$

We also have

$$\begin{aligned}(\psi_j \circ T)(v_k) &= \psi_j(Tv_k)\\ &= \psi_j\Big(\sum_{r=1}^{m} A_{r,k}w_r\Big)\\ &= \sum_{r=1}^{m} A_{r,k}\psi_j(w_r)\\ &= A_{j,k}.\end{aligned}$$

Comparing the last line of the last two sets of equations, we have $C_{k,j} = A_{j,k}$. Thus $C = A^t$. In other words, $\mathcal{M}(T') = (\mathcal{M}(T))^t$, as desired. ∎

Now we use duality to give an alternative proof that the column rank of a matrix equals the row rank of the matrix. This result was previously proved using different tools—see 3.57.

**3.133** *column rank equals row rank*

Suppose $A \in \mathbf{F}^{m,n}$. Then the column rank of $A$ equals the row rank of $A$.

Proof Define $T\colon \mathbf{F}^{n,1} \to \mathbf{F}^{m,1}$ by $Tx = Ax$. Thus $\mathcal{M}(T) = A$, where $\mathcal{M}(T)$ is computed with respect to the standard bases of $\mathbf{F}^{n,1}$ and $\mathbf{F}^{m,1}$. Now

$$\begin{aligned}\text{column rank of } A &= \text{column rank of } \mathcal{M}(T)\\ &= \dim \text{range } T\\ &= \dim \text{range } T'\\ &= \text{column rank of } \mathcal{M}(T')\\ &= \text{column rank of } A^t\\ &= \text{row rank of } A,\end{aligned}$$

where the second equality comes from 3.78, the third equality comes from 3.130(a), the fourth equality comes from 3.78, the fifth equality comes from 3.132, and the last equality follows from the definitions of row and column rank. ∎

See Exercise 8 in Section 7A for another alternative proof of the result above.

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## *Exercises 3F*

**1** Explain why each linear functional is surjective or is the zero map.

**2** Give three distinct examples of linear functionals on $\mathbf{R}^{[0,1]}$.

**3** Suppose $V$ is finite-dimensional and $v \in V$ with $v \neq 0$. Prove that there exists $\varphi \in V'$ such that $\varphi(v) = 1$.

**4** Suppose $V$ is finite-dimensional and $U$ is a subspace of $V$ such that $U \neq V$. Prove that there exists $\varphi \in V'$ such that $\varphi(u) = 0$ for every $u \in U$ but $\varphi \neq 0$.

**5** Suppose $T \in \mathcal{L}(V, W)$ and $w_1, \ldots, w_m$ is a basis of range $T$. Hence for each $v \in V$, there exist unique numbers $\varphi_1(v), \ldots, \varphi_m(v)$ such that

$$Tv = \varphi_1(v)w_1 + \cdots + \varphi_m(v)w_m,$$

thus defining functions $\varphi_1, \ldots, \varphi_m$ from $V$ to $\mathbf{F}$. Show that each of the functions $\varphi_1, \ldots, \varphi_m$ is a linear functional on $V$.

**6** Suppose $\varphi, \beta \in V'$. Prove that $\operatorname{null} \varphi \subseteq \operatorname{null} \beta$ if and only if there exists $c \in \mathbf{F}$ such that $\beta = c\varphi$.

**7** Suppose that $V_1, \ldots, V_m$ are vector spaces. Prove that $(V_1 \times \cdots \times V_m)'$ and $V_1' \times \cdots \times V_m'$ are isomorphic vector spaces.

**8** Suppose $v_1, \ldots, v_n$ is a basis of $V$ and $\varphi_1, \ldots, \varphi_n$ is the dual basis of $V'$. Define $\Gamma \colon V \to \mathbf{F}^n$ and $\Lambda \colon \mathbf{F}^n \to V$ by

$$\Gamma(v) = (\varphi_1(v), \ldots, \varphi_n(v)) \quad \text{and} \quad \Lambda(a_1, \ldots, a_n) = a_1v_1 + \cdots + a_nv_n.$$

Explain why $\Gamma$ and $\Lambda$ are inverses of each other.

**9** Suppose $m$ is a positive integer. Show that the dual basis of the basis $1, x, \ldots, x^m$ of $\mathcal{P}_m(\mathbf{R})$ is $\varphi_0, \varphi_1, \ldots, \varphi_m$, where

$$\varphi_k(p) = \frac{p^{(k)}(0)}{k!}.$$

*Here $p^{(k)}$ denotes the $k^{th}$ derivative of $p$, with the understanding that the $0^{th}$ derivative of $p$ is $p$.*

**10** Suppose $m$ is a positive integer.

(a) Show that $1, x - 5, \ldots, (x - 5)^m$ is a basis of $\mathcal{P}_m(\mathbf{R})$.

(b) What is the dual basis of the basis in (a)?

**11** Suppose $v_1, \ldots, v_n$ is a basis of $V$ and $\varphi_1, \ldots, \varphi_n$ is the corresponding dual basis of $V'$. Suppose $\psi \in V'$. Prove that

$$\psi = \psi(v_1)\varphi_1 + \cdots + \psi(v_n)\varphi_n.$$

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**12** Suppose $S, T \in \mathcal{L}(V, W)$.

(a) Prove that $(S + T)' = S' + T'$.

(b) Prove that $(\lambda T)' = \lambda T'$ for all $\lambda \in \mathbf{F}$.

*This exercise asks you to verify (a) and (b) in 3.120.*

**13** Show that the dual map of the identity operator on $V$ is the identity operator on $V'$.

**14** Define $T: \mathbf{R}^3 \to \mathbf{R}^2$ by

$$T(x, y, z) = (4x + 5y + 6z, 7x + 8y + 9z).$$

Suppose $\varphi_1, \varphi_2$ denotes the dual basis of the standard basis of $\mathbf{R}^2$ and $\psi_1, \psi_2, \psi_3$ denotes the dual basis of the standard basis of $\mathbf{R}^3$.

(a) Describe the linear functionals $T'(\varphi_1)$ and $T'(\varphi_2)$.

(b) Write $T'(\varphi_1)$ and $T'(\varphi_2)$ as linear combinations of $\psi_1, \psi_2, \psi_3$.

**15** Define $T: \mathcal{P}(\mathbf{R}) \to \mathcal{P}(\mathbf{R})$ by

$$(Tp)(x) = x^2 p(x) + p''(x)$$

for each $x \in \mathbf{R}$.

(a) Suppose $\varphi \in \mathcal{P}(\mathbf{R})'$ is defined by $\varphi(p) = p'(4)$. Describe the linear functional $T'(\varphi)$ on $\mathcal{P}(\mathbf{R})$.

(b) Suppose $\varphi \in \mathcal{P}(\mathbf{R})'$ is defined by $\varphi(p) = \int_0^1 p$. Evaluate $(T'(\varphi))(x^3)$.

**16** Suppose $W$ is finite-dimensional and $T \in \mathcal{L}(V, W)$. Prove that

$$T' = 0 \iff T = 0.$$

**17** Suppose $V$ and $W$ are finite-dimensional and $T \in \mathcal{L}(V, W)$. Prove that $T$ is invertible if and only if $T' \in \mathcal{L}(W', V')$ is invertible.

**18** Suppose $V$ and $W$ are finite-dimensional. Prove that the map that takes $T \in \mathcal{L}(V, W)$ to $T' \in \mathcal{L}(W', V')$ is an isomorphism of $\mathcal{L}(V, W)$ onto $\mathcal{L}(W', V')$.

**19** Suppose $U \subseteq V$. Explain why

$$U^0 = \{\varphi \in V' : U \subseteq \operatorname{null} \varphi\}.$$

**20** Suppose $V$ is finite-dimensional and $U$ is a subspace of $V$. Show that

$$U = \{v \in V : \varphi(v) = 0 \text{ for every } \varphi \in U^0\}.$$

**21** Suppose $V$ is finite-dimensional and $U$ and $W$ are subspaces of $V$.

(a) Prove that $W^0 \subseteq U^0$ if and only if $U \subseteq W$.

(b) Prove that $W^0 = U^0$ if and only if $U = W$.

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**22** Suppose $V$ is finite-dimensional and $U$ and $W$ are subspaces of $V$.

(a) Show that $(U + W)^0 = U^0 \cap W^0$.
(b) Show that $(U \cap W)^0 = U^0 + W^0$.

**23** Suppose $V$ is finite-dimensional and $\varphi_1, ..., \varphi_m \in V'$. Prove that the following three sets are equal to each other.

(a) $\operatorname{span}(\varphi_1, ..., \varphi_m)$
(b) $((\operatorname{null} \varphi_1) \cap \cdots \cap (\operatorname{null} \varphi_m))^0$
(c) $\{\varphi \in V' : (\operatorname{null} \varphi_1) \cap \cdots \cap (\operatorname{null} \varphi_m) \subseteq \operatorname{null} \varphi\}$

**24** Suppose $V$ is finite-dimensional and $v_1, ..., v_m \in V$. Define a linear map $\Gamma : V' \to \mathbf{F}^m$ by $\Gamma(\varphi) = (\varphi(v_1), ..., \varphi(v_m))$.

(a) Prove that $v_1, ..., v_m$ spans $V$ if and only if $\Gamma$ is injective.
(b) Prove that $v_1, ..., v_m$ is linearly independent if and only if $\Gamma$ is surjective.

**25** Suppose $V$ is finite-dimensional and $\varphi_1, ..., \varphi_m \in V'$. Define a linear map $\Gamma : V \to \mathbf{F}^m$ by $\Gamma(v) = (\varphi_1(v), ..., \varphi_m(v))$.

(a) Prove that $\varphi_1, ..., \varphi_m$ spans $V'$ if and only if $\Gamma$ is injective.
(b) Prove that $\varphi_1, ..., \varphi_m$ is linearly independent if and only if $\Gamma$ is surjective.

**26** Suppose $V$ is finite-dimensional and $\Omega$ is a subspace of $V'$. Prove that

$$\Omega = \{v \in V : \varphi(v) = 0 \text{ for every } \varphi \in \Omega\}^0.$$

**27** Suppose $T \in \mathcal{L}(\mathcal{P}_5(\mathbf{R}))$ and $\operatorname{null} T' = \operatorname{span}(\varphi)$, where $\varphi$ is the linear functional on $\mathcal{P}_5(\mathbf{R})$ defined by $\varphi(p) = p(8)$. Prove that

$$\operatorname{range} T = \{p \in \mathcal{P}_5(\mathbf{R}) : p(8) = 0\}.$$

**28** Suppose $V$ is finite-dimensional and $\varphi_1, ..., \varphi_m$ is a linearly independent list in $V'$. Prove that

$$\dim((\operatorname{null} \varphi_1) \cap \cdots \cap (\operatorname{null} \varphi_m)) = (\dim V) - m.$$

**29** Suppose $V$ and $W$ are finite-dimensional and $T \in \mathcal{L}(V, W)$.

(a) Prove that if $\varphi \in W'$ and $\operatorname{null} T' = \operatorname{span}(\varphi)$, then $\operatorname{range} T = \operatorname{null} \varphi$.
(b) Prove that if $\psi \in V'$ and $\operatorname{range} T' = \operatorname{span}(\psi)$, then $\operatorname{null} T = \operatorname{null} \psi$.

**30** Suppose $V$ is finite-dimensional and $\varphi_1, ..., \varphi_n$ is a basis of $V'$. Show that there exists a basis of $V$ whose dual basis is $\varphi_1, ..., \varphi_n$.

**31** Suppose $U$ is a subspace of $V$. Let $i : U \to V$ be the inclusion map defined by $i(u) = u$. Thus $i' \in \mathcal{L}(V', U')$.

(a) Show that $\operatorname{null} i' = U^0$.
(b) Prove that if $V$ is finite-dimensional, then $\operatorname{range} i' = U'$.
(c) Prove that if $V$ is finite-dimensional, then $\tilde{i}'$ is an isomorphism from $V'/U^0$ onto $U'$.

*The isomorphism in (c) is natural in that it does not depend on a choice of basis in either vector space.*

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**32** The *double dual space* of $V$, denoted by $V''$, is defined to be the dual space of $V'$. In other words, $V'' = (V')'$. Define $\Lambda : V \to V''$ by

$$(\Lambda v)(\varphi) = \varphi(v)$$

for each $v \in V$ and each $\varphi \in V'$.

(a) Show that $\Lambda$ is a linear map from $V$ to $V''$.

(b) Show that if $T \in \mathcal{L}(V)$, then $T'' \circ \Lambda = \Lambda \circ T$, where $T'' = (T')'$.

(c) Show that if $V$ is finite-dimensional, then $\Lambda$ is an isomorphism from $V$ onto $V''$.

*Suppose $V$ is finite-dimensional. Then $V$ and $V'$ are isomorphic, but finding an isomorphism from $V$ onto $V'$ generally requires choosing a basis of $V$. In contrast, the isomorphism $\Lambda$ from $V$ onto $V''$ does not require a choice of basis and thus is considered more natural.*

**33** Suppose $U$ is a subspace of $V$. Let $\pi : V \to V/U$ be the usual quotient map. Thus $\pi' \in \mathcal{L}((V/U)', V')$.

(a) Show that $\pi'$ is injective.

(b) Show that range $\pi' = U^0$.

(c) Conclude that $\pi'$ is an isomorphism from $(V/U)'$ onto $U^0$.

*The isomorphism in (c) is natural in that it does not depend on a choice of basis in either vector space. In fact, there is no assumption here that any of these vector spaces are finite-dimensional.*

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Chapter 4

# *Polynomials*

This chapter contains material on polynomials that we will use to investigate linear maps from a vector space to itself. Many results in this chapter will already be familiar to you from other courses; they are included here for completeness.

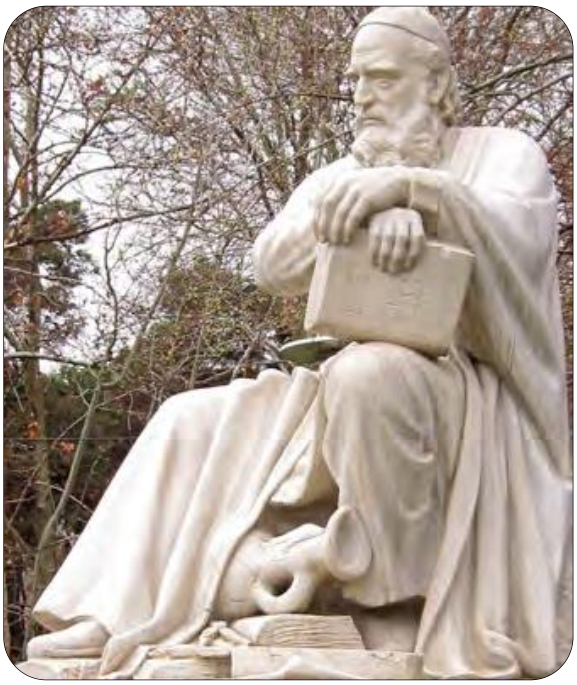
Because this chapter is not about linear algebra, your instructor may go through it rapidly. You may not be asked to scrutinize all the proofs. Make sure, however, that you at least read and understand the statements of all results in this chapter—they will be used in later chapters.

This chapter begins with a brief discussion of algebraic properties of the complex numbers. Then we prove that a nonconstant polynomial cannot have more zeros than its degree. We also give a linear-algebra-based proof of the division algorithm for polynomials, which is worth reading even if you are already familiar with a proof that does not use linear algebra.

As we will see, the fundamental theorem of algebra leads to a factorization of every polynomial into degree-one factors if the scalar field is **C** or to factors of degree at most two if the scalar field is **R**.

*standing assumption for this chapter*

- **F** denotes **R** or **C**.



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*Statue of mathematician and poet Omar Khayyam (1048–1131), whose algebra book written in 1070 contained the first serious study of cubic polynomials.*

© Sheldon Axler 2024
S. Axler, *Linear Algebra Done Right*, Undergraduate Texts in Mathematics,
https://doi.org/10.1007/978-3-031-41026-0_4

Before discussing polynomials with complex or real coefficients, we need to learn a bit more about the complex numbers.

**4.1 definition:** *real part,* $\operatorname{Re} z$*, imaginary part,* $\operatorname{Im} z$

Suppose $z = a + bi$, where $a$ and $b$ are real numbers.

- The *real part* of $z$, denoted by $\operatorname{Re} z$, is defined by $\operatorname{Re} z = a$.
- The *imaginary part* of $z$, denoted by $\operatorname{Im} z$, is defined by $\operatorname{Im} z = b$.

Thus for every complex number $z$, we have

$$z = \operatorname{Re} z + (\operatorname{Im} z)i.$$

**4.2 definition:** *complex conjugate,* $\bar{z}$*, absolute value,* $|z|$

Suppose $z \in \mathbf{C}$.

- The *complex conjugate* of $z \in \mathbf{C}$, denoted by $\bar{z}$, is defined by

$$\bar{z} = \operatorname{Re} z - (\operatorname{Im} z)i.$$

- The *absolute value* of a complex number $z$, denoted by $|z|$, is defined by

$$|z| = \sqrt{(\operatorname{Re} z)^2 + (\operatorname{Im} z)^2}.$$

**4.3 example:** *real and imaginary part, complex conjugate, absolute value*

Suppose $z = 3 + 2i$. Then

- $\operatorname{Re} z = 3$ and $\operatorname{Im} z = 2$;
- $\bar{z} = 3 - 2i$;
- $|z| = \sqrt{3^2 + 2^2} = \sqrt{13}$.

Identifying a complex number $z \in \mathbf{C}$ with the ordered pair $(\operatorname{Re} z, \operatorname{Im} z) \in \mathbf{R}^2$ identifies $\mathbf{C}$ with $\mathbf{R}^2$. Note that $\mathbf{C}$ is a one-dimensional complex vector space, but we can also think of $\mathbf{C}$ (identified with $\mathbf{R}^2$) as a two-dimensional real vector space.

The absolute value of each complex number is a nonnegative number. Specifically, if $z \in \mathbf{C}$, then $|z|$ equals the distance from the origin in $\mathbf{R}^2$ to the point $(\operatorname{Re} z, \operatorname{Im} z) \in \mathbf{R}^2$.

The real and imaginary parts, complex conjugate, and absolute value have the properties listed in the following multipart result.

*You should verify that $z = \bar{z}$ if and only if $z$ is a real number.*

### 4.4 *properties of complex numbers*

Suppose $w, z \in \mathbf{C}$. Then the following equalities and inequalities hold.

**sum of $z$ and $\overline{z}$**
$z + \overline{z} = 2 \operatorname{Re} z$.

**difference of $z$ and $\overline{z}$**
$z - \overline{z} = 2(\operatorname{Im} z)i$.

**product of $z$ and $\overline{z}$**
$z\overline{z} = |z|^2$.

**additivity and multiplicativity of complex conjugate**
$\overline{w + z} = \overline{w} + \overline{z}$ and $\overline{wz} = \overline{w}\,\overline{z}$.

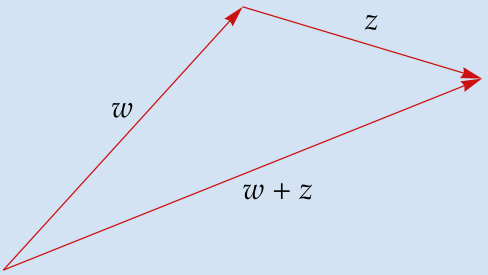
**double complex conjugate**
$\overline{\overline{z}} = z$.

**real and imaginary parts are bounded by $|z|$**
$|\operatorname{Re} z| \le |z|$ and $|\operatorname{Im} z| \le |z|$.

**absolute value of the complex conjugate**
$|\overline{z}| = |z|$.

**multiplicativity of absolute value**
$|wz| = |w|\,|z|$.

**triangle inequality**
$|w + z| \le |w| + |z|$.




*Geometric interpretation of triangle inequality: The length of each side of a triangle is less than or equal to the sum of the lengths of the two other sides.*

Proof Except for the last item above, the routine verifications of the assertions above are left to the reader. To verify the triangle inequality, we have

$$\begin{aligned}
|w + z|^2 &= (w + z)(\overline{w} + \overline{z}) \\
&= w\overline{w} + z\overline{z} + w\overline{z} + z\overline{w} \\
&= |w|^2 + |z|^2 + w\overline{z} + \overline{w\overline{z}} \\
&= |w|^2 + |z|^2 + 2\operatorname{Re}(w\overline{z}) \\
&\le |w|^2 + |z|^2 + 2|w\overline{z}| \\
&= |w|^2 + |z|^2 + 2|w|\,|z| \\
&= (|w| + |z|)^2.
\end{aligned}$$

Taking square roots now gives the desired inequality $|w + z| \le |w| + |z|$. ∎

*See Exercise 2 for the reverse triangle inequality.*

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## *Zeros of Polynomials*

Recall that a function $p: \mathbf{F} \to \mathbf{F}$ is called a polynomial of degree $m$ if there exist $a_0, \ldots, a_m \in \mathbf{F}$ with $a_m \neq 0$ such that

$$p(z) = a_0 + a_1 z + \cdots + a_m z^m$$

for all $z \in \mathbf{F}$. A polynomial could have more than one degree if the representation of $p$ in the form above were not unique. Our first task is to show that this cannot happen.

The solutions to the equation $p(z) = 0$ play a crucial role in the study of a polynomial $p \in \mathcal{P}(\mathbf{F})$. Thus these solutions have a special name.

**4.5 definition: *zero of a polynomial***

A number $\lambda \in \mathbf{F}$ is called a *zero* (or *root*) of a polynomial $p \in \mathcal{P}(\mathbf{F})$ if

$$p(\lambda) = 0.$$

The next result is the key tool that we will use to show that the degree of a polynomial is unique.

**4.6 *each zero of a polynomial corresponds to a degree-one factor***

Suppose $m$ is a positive integer and $p \in \mathcal{P}(\mathbf{F})$ is a polynomial of degree $m$. Suppose $\lambda \in \mathbf{F}$. Then $p(\lambda) = 0$ if and only if there exists a polynomial $q \in \mathcal{P}(\mathbf{F})$ of degree $m - 1$ such that

$$p(z) = (z - \lambda)q(z)$$

for every $z \in \mathbf{F}$.

Proof First suppose $p(\lambda) = 0$. Let $a_0, a_1, \ldots, a_m \in \mathbf{F}$ be such that

$$p(z) = a_0 + a_1 z + \cdots + a_m z^m$$

for all $z \in \mathbf{F}$. Then

$$p(z) = p(z) - p(\lambda) = a_1(z - \lambda) + \cdots + a_m(z^m - \lambda^m) \tag{4.7}$$

for all $z \in \mathbf{F}$. For each $k \in \{1, \ldots, m\}$, the equation

$$z^k - \lambda^k = (z - \lambda) \sum_{j=1}^{k} \lambda^{j-1} z^{k-j}$$

shows that $z^k - \lambda^k$ equals $z - \lambda$ times some polynomial of degree $k - 1$. Thus 4.7 shows that $p$ equals $z - \lambda$ times some polynomial of degree $m - 1$, as desired.

To prove the implication in the other direction, now suppose that there is a polynomial $q \in \mathcal{P}(\mathbf{F})$ such that $p(z) = (z - \lambda)q(z)$ for every $z \in \mathbf{F}$. Then $p(\lambda) = (\lambda - \lambda)q(\lambda) = 0$, as desired. ∎

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Now we can prove that polynomials do not have too many zeros.

**4.8** *degree $m$ implies at most $m$ zeros*

Suppose $m$ is a positive integer and $p \in \mathcal{P}(\mathbf{F})$ is a polynomial of degree $m$. Then $p$ has at most $m$ zeros in $\mathbf{F}$.

Proof We will use induction on $m$. The desired result holds if $m = 1$ because if $a_1 \neq 0$ then the polynomial $a_0 + a_1 z$ has only one zero (which equals $-a_0/a_1$). Thus assume that $m > 1$ and the desired result holds for $m - 1$.

If $p$ has no zeros in $\mathbf{F}$, then the desired result holds and we are done. Thus suppose $p$ has a zero $\lambda \in \mathbf{F}$. By 4.6, there is polynomial $q \in \mathcal{P}(\mathbf{F})$ of degree $m - 1$ such that

$$p(z) = (z - \lambda)q(z)$$

for every $z \in \mathbf{F}$. Our induction hypothesis implies that $q$ has at most $m - 1$ zeros in $\mathbf{F}$. The equation above shows that the zeros of $p$ in $\mathbf{F}$ are exactly the zeros of $q$ in $\mathbf{F}$ along with $\lambda$. Thus $p$ has at most $m$ zeros in $\mathbf{F}$. ∎

The result above implies that the coefficients of a polynomial are uniquely determined (because if a polynomial had two different sets of coefficients, then subtracting the two representations of the polynomial would give a polynomial with some nonzero coefficients but infinitely many zeros). In particular, the degree of a polynomial is uniquely defined.

Recall that the degree of the 0 polynomial is defined to be $-\infty$. When necessary, use the expected arithmetic with $-\infty$. For example, $-\infty < m$ and $-\infty + m = -\infty$ for every integer $m$.

*The 0 polynomial is declared to have degree $-\infty$ so that exceptions are not needed for various reasonable results such as* $\deg(pq) = \deg p + \deg q$.

## *Division Algorithm for Polynomials*

If $p$ and $s$ are nonnegative integers, with $s \neq 0$, then there exist nonnegative integers $q$ and $r$ such that

$$p = sq + r$$

and $r < s$. Think of dividing $p$ by $s$, getting quotient $q$ with remainder $r$. Our next result gives an analogous result for polynomials. Thus the next result is often called the division algorithm for polynomials, although as stated here it is not really an algorithm, just a useful result.

The division algorithm for polynomials could be proved without using any linear algebra. However, as is appropriate for a linear algebra textbook, the proof given here uses linear algebra techniques and makes nice use of a basis of $\mathcal{P}_n(\mathbf{F})$, which is the $(n + 1)$-dimensional vector space of polynomials with coefficients in $\mathbf{F}$ and of degree at most $n$.

*Think of the division algorithm for polynomials as giving a remainder polynomial $r$ when the polynomial $p$ is divided by the polynomial $s$.*

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**4.9** *division algorithm for polynomials*

Suppose that $p, s \in \mathcal{P}(\mathbf{F})$, with $s \neq 0$. Then there exist unique polynomials $q, r \in \mathcal{P}(\mathbf{F})$ such that

$$p = sq + r$$

and $\deg r < \deg s$.

Proof Let $n = \deg p$ and let $m = \deg s$. If $n < m$, then take $q = 0$ and $r = p$ to get the desired equation $p = sq + r$ with $\deg r < \deg s$. Thus we now assume that $n \geq m$.

The list

$$1, z, \ldots, z^{m-1}, s, zs, \ldots, z^{n-m}s \tag{4.10}$$

is linearly independent in $\mathcal{P}_n(\mathbf{F})$ because each polynomial in this list has a different degree. Also, the list 4.10 has length $n + 1$, which equals $\dim \mathcal{P}_n(\mathbf{F})$. Hence 4.10 is a basis of $\mathcal{P}_n(\mathbf{F})$ [by 2.38].

Because $p \in \mathcal{P}_n(\mathbf{F})$ and 4.10 is a basis of $\mathcal{P}_n(\mathbf{F})$, there exist unique constants $a_0, a_1, \ldots, a_{m-1} \in \mathbf{F}$ and $b_0, b_1, \ldots, b_{n-m} \in \mathbf{F}$ such that

$$\begin{aligned} p &= a_0 + a_1 z + \cdots + a_{m-1}z^{m-1} + b_0 s + b_1 zs + \cdots + b_{n-m}z^{n-m}s \\ &= \underbrace{a_0 + a_1 z + \cdots + a_{m-1}z^{m-1}}_{r} + s(\underbrace{b_0 + b_1 z + \cdots + b_{n-m}z^{n-m}}_{q}). \end{aligned} \tag{4.11}$$

With $r$ and $q$ as defined above, we see that $p$ can be written as $p = sq + r$ with $\deg r < \deg s$, as desired.

The uniqueness of $q, r \in \mathcal{P}(\mathbf{F})$ satisfying these conditions follows from the uniqueness of the constants $a_0, a_1, \ldots, a_{m-1} \in \mathbf{F}$ and $b_0, b_1, \ldots, b_{n-m} \in \mathbf{F}$ satisfying 4.11. ∎

## *Factorization of Polynomials over* C

W have been handling polynomials with complex coefficients and polynomials with real coefficients simultaneously, letting $\mathbf{F}$ denote $\mathbf{R}$ or $\mathbf{C}$. Now we will see differences between these two cases. First we treat polynomials with complex coefficients. Then we will use those results to prove corresponding results for polynomials with real coefficients.

*The fundamental theorem of algebra is an existence theorem. Its proof does not lead to a method for finding zeros. The quadratic formula gives the zeros explicitly for polynomials of degree* 2. *Similar but more complicated formulas exist for polynomials of degree* 3 *and* 4. *No such formulas exist for polynomials of degree* 5 *and above.*

Our proof of the fundamental theorem of algebra implicitly uses the result that a continuous real-valued function on a closed disk in $\mathbf{R}^2$ attains a minimum value. A web search can lead you to several

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other proofs of the fundamental theorem of algebra. The proof using Liouville's theorem is particularly nice if you are comfortable with analytic functions. All proofs of the fundamental theorem of algebra need to use some analysis, because the result is not true if $\mathbf{C}$ is replaced, for example, with the set of numbers of the form $c + di$ where $c, d$ are rational numbers.

**4.12** *fundamental theorem of algebra, first version*

Every nonconstant polynomial with complex coefficients has a zero in $\mathbf{C}$.

Proof De Moivre's theorem, which you can prove using induction on $k$ and the addition formulas for cosine and sine, states that if $k$ is a positive integer and $\theta \in \mathbf{R}$, then

$$(\cos\theta + i\sin\theta)^k = \cos k\theta + i\sin k\theta.$$

Suppose $w \in \mathbf{C}$ and $k$ is a positive integer. Using polar coordinates, we know that there exist $r \geq 0$ and $\theta \in \mathbf{R}$ such that

$$r(\cos\theta + i\sin\theta) = w.$$

De Moivre's theorem implies that

$$\left(r^{1/k}\left(\cos\tfrac{\theta}{k} + i\sin\tfrac{\theta}{k}\right)\right)^k = w.$$

Thus every complex number has a $k^{\text{th}}$ root, a fact that we will soon use.

Suppose $p$ is a nonconstant polynomial with complex coefficients and highest-order nonzero term $c_m z^m$. Then $|p(z)| \to \infty$ as $|z| \to \infty$ (because $|p(z)|/|z^m| \to |c_m|$ as $|z| \to \infty$). Thus the continuous function $z \mapsto |p(z)|$ has a global minimum at some point $\zeta \in \mathbf{C}$. To show that $p(\zeta) = 0$, suppose that $p(\zeta) \neq 0$.

Define a new polynomial $q$ by

$$q(z) = \frac{p(z+\zeta)}{p(\zeta)}.$$

The function $z \mapsto |q(z)|$ has a global minimum value of 1 at $z = 0$. Write

$$q(z) = 1 + a_k z^k + \cdots + a_m z^m,$$

where $k$ is the smallest positive integer such that the coefficient of $z^k$ is nonzero; in other words, $a_k \neq 0$.

Let $\beta \in \mathbf{C}$ be such that $\beta^k = -\dfrac{1}{a_k}$. There is a constant $c > 1$ such that if $t \in (0,1)$, then

$$\begin{aligned} |q(t\beta)| &\leq |1 + a_k t^k \beta^k| + t^{k+1}c \\ &= 1 - t^k(1 - tc). \end{aligned}$$

Thus taking $t$ to be $1/(2c)$ in the inequality above, we have $|q(t\beta)| < 1$, which contradicts the assumption that the global minimum of $z \mapsto |q(z)|$ is 1. This contradiction implies that $p(\zeta) = 0$, showing that $p$ has a zero, as desired. ∎

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Computers can use clever numerical methods to find good approximations to the zeros of any polynomial, even when exact zeros cannot be found. For example, no one will ever give an exact formula for a zero of the polynomial $p$ defined by

$$p(x) = x^5 - 5x^4 - 6x^3 + 17x^2 + 4x - 7.$$

However, a computer can find that the zeros of $p$ are approximately the five numbers $-1.87,\ -0.74,\ 0.62,\ 1.47,\ 5.51$.

The first version of the fundamental theorem of algebra leads to the following factorization result for polynomials with complex coefficients. Note that in this factorization, the zeros of $p$ are the numbers $\lambda_1, \ldots, \lambda_m$, which are the only values of $z$ for which the right side of the equation in the next result equals 0.

**4.13** *fundamental theorem of algebra, second version*

If $p \in \mathcal{P}(\mathbf{C})$ is a nonconstant polynomial, then $p$ has a unique factorization (except for the order of the factors) of the form

$$p(z) = c(z - \lambda_1)\cdots(z - \lambda_m),$$

where $c, \lambda_1, \ldots, \lambda_m \in \mathbf{C}$.

Proof Let $p \in \mathcal{P}(\mathbf{C})$ and let $m = \deg p$. We will use induction on $m$. If $m = 1$, then the desired factorization exists and is unique. So assume that $m > 1$ and that the desired factorization exists and is unique for all polynomials of degree $m - 1$.

First we will show that the desired factorization of $p$ exists. By the first version of the fundamental theorem of algebra (4.12), $p$ has a zero $\lambda \in \mathbf{C}$. By 4.6, there is a polynomial $q$ of degree $m - 1$ such that

$$p(z) = (z - \lambda)q(z)$$

for all $z \in \mathbf{C}$. Our induction hypothesis implies that $q$ has the desired factorization, which when plugged into the equation above gives the desired factorization of $p$.

Now we turn to the question of uniqueness. The number $c$ is uniquely determined as the coefficient of $z^m$ in $p$. So we only need to show that except for the order, there is only one way to choose $\lambda_1, \ldots, \lambda_m$. If

$$(z - \lambda_1)\cdots(z - \lambda_m) = (z - \tau_1)\cdots(z - \tau_m)$$

for all $z \in \mathbf{C}$, then because the left side of the equation above equals 0 when $z = \lambda_1$, one of the $\tau$'s on the right side equals $\lambda_1$. Relabeling, we can assume that $\tau_1 = \lambda_1$. Now if $z \neq \lambda_1$, we can divide both sides of the equation above by $z - \lambda_1$, getting

$$(z - \lambda_2)\cdots(z - \lambda_m) = (z - \tau_2)\cdots(z - \tau_m)$$

for all $z \in \mathbf{C}$ except possibly $z = \lambda_1$. Actually the equation above holds for all $z \in \mathbf{C}$, because otherwise by subtracting the right side from the left side we would get a nonzero polynomial that has infinitely many zeros. The equation above and our induction hypothesis imply that except for the order, the $\lambda$'s are the same as the $\tau$'s, completing the proof of uniqueness. ∎

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## *Factorization of Polynomials over* **R**

A polynomial with real coefficients may have no real zeros. For example, the polynomial $1 + x^2$ has no real zeros.

To obtain a factorization theorem over **R**, we will use our factorization theorem over **C**. We begin with the next result.

*The failure of the fundamental theorem of algebra for* **R** *accounts for the differences between linear algebra on real and complex vector spaces, as we will see in later chapters.*

**4.14** *polynomials with real coefficients have nonreal zeros in pairs*

Suppose $p \in \mathcal{P}(\mathbf{C})$ is a polynomial with real coefficients. If $\lambda \in \mathbf{C}$ is a zero of $p$, then so is $\overline{\lambda}$.

Proof Let

$$p(z) = a_0 + a_1 z + \cdots + a_m z^m,$$

where $a_0, \ldots, a_m$ are real numbers. Suppose $\lambda \in \mathbf{C}$ is a zero of $p$. Then

$$a_0 + a_1\lambda + \cdots + a_m\lambda^m = 0.$$

Take the complex conjugate of both sides of this equation, obtaining

$$a_0 + a_1\overline{\lambda} + \cdots + a_m\overline{\lambda}^m = 0,$$

where we have used basic properties of the complex conjugate (see 4.4). The equation above shows that $\overline{\lambda}$ is a zero of $p$. ∎

We want a factorization theorem for polynomials with real coefficients. We begin with the following result.

*Think about the quadratic formula in connection with the result below.*

**4.15** *factorization of a quadratic polynomial*

Suppose $b, c \in \mathbf{R}$. Then there is a polynomial factorization of the form

$$x^2 + bx + c = (x - \lambda_1)(x - \lambda_2)$$

with $\lambda_1, \lambda_2 \in \mathbf{R}$ if and only if $b^2 \geq 4c$.

Proof Notice that

$$x^2 + bx + c = \left(x + \frac{b}{2}\right)^2 + \left(c - \frac{b^2}{4}\right).$$

First suppose $b^2 < 4c$. Then the right side of the equation above is positive for every $x \in \mathbf{R}$. Hence the polynomial $x^2 + bx + c$ has no real zeros and thus cannot be factored in the form $(x - \lambda_1)(x - \lambda_2)$ with $\lambda_1, \lambda_2 \in \mathbf{R}$.

*The equation above is the basis of the technique called completing the square.*

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Conversely, now suppose $b^2 \geq 4c$. Then there is a real number $d$ such that $d^2 = \frac{b^2}{4} - c$. From the displayed equation above, we have

$$\begin{aligned} x^2 + bx + c &= \left(x + \frac{b}{2}\right)^2 - d^2 \\ &= \left(x + \frac{b}{2} + d\right)\left(x + \frac{b}{2} - d\right), \end{aligned}$$

which gives the desired factorization. ∎

The next result gives a factorization of a polynomial over $\mathbf{R}$. The idea of the proof is to use the second version of the fundamental theorem of algebra (4.13), which gives a factorization of $p$ as a polynomial with complex coefficients. Complex but nonreal zeros of $p$ come in pairs; see 4.14. Thus if the factorization of $p$ as an element of $\mathcal{P}(\mathbf{C})$ includes terms of the form $(x - \lambda)$ with $\lambda$ a nonreal complex number, then $(x - \overline{\lambda})$ is also a term in the factorization. Multiplying together these two terms, we get

$$(x^2 - 2(\operatorname{Re} \lambda)x + |\lambda|^2),$$

which is a quadratic term of the required form.

The idea sketched in the paragraph above almost provides a proof of the existence of our desired factorization. However, we need to be careful about one point. Suppose $\lambda$ is a nonreal complex number and $(x - \lambda)$ is a term in the factorization of $p$ as an element of $\mathcal{P}(\mathbf{C})$. We are guaranteed by 4.14 that $(x - \overline{\lambda})$ also appears as a term in the factorization, but 4.14 does not state that these two factors appear the same number of times, as needed to make the idea above work. However, the proof works around this point.

In the next result, either $m$ or $M$ may equal 0. The numbers $\lambda_1, \ldots, \lambda_m$ are precisely the real zeros of $p$, for these are the only real values of $x$ for which the right side of the equation in the next result equals 0.

**4.16** *factorization of a polynomial over* $\mathbf{R}$

Suppose $p \in \mathcal{P}(\mathbf{R})$ is a nonconstant polynomial. Then $p$ has a unique factorization (except for the order of the factors) of the form

$$p(x) = c(x - \lambda_1)\cdots(x - \lambda_m)(x^2 + b_1x + c_1)\cdots(x^2 + b_Mx + c_M),$$

where $c, \lambda_1, \ldots, \lambda_m, b_1, \ldots, b_M, c_1, \ldots, c_M \in \mathbf{R}$, with $b_k^2 < 4c_k$ for each $k$.

Proof First we will prove that the desired factorization exists, and after that we will prove the uniqueness.

Think of $p$ as an element of $\mathcal{P}(\mathbf{C})$. If all (complex) zeros of $p$ are real, then we have the desired factorization by 4.13. Thus suppose $p$ has a zero $\lambda \in \mathbf{C}$ with $\lambda \notin \mathbf{R}$. By 4.14, $\overline{\lambda}$ is a zero of $p$. Thus we can write

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$$\begin{aligned} p(x) &= (x-\lambda)(x-\overline{\lambda})q(x) \\ &= (x^2 - 2(\operatorname{Re}\lambda)x + |\lambda|^2)q(x) \end{aligned}$$

for some polynomial $q \in \mathcal{P}(\mathbf{C})$ of degree two less than the degree of $p$. If we can prove that $q$ has real coefficients, then using induction on the degree of $p$ completes the proof of the existence part of this result.

To prove that $q$ has real coefficients, we solve the equation above for $q$, getting

$$q(x) = \frac{p(x)}{x^2 - 2(\operatorname{Re}\lambda)x + |\lambda|^2}$$

for all $x \in \mathbf{R}$. The equation above implies that $q(x) \in \mathbf{R}$ for all $x \in \mathbf{R}$. Writing

$$q(x) = a_0 + a_1 x + \cdots + a_{n-2}x^{n-2},$$

where $n = \deg p$ and $a_0, \ldots, a_{n-2} \in \mathbf{C}$, we thus have

$$0 = \operatorname{Im} q(x) = (\operatorname{Im} a_0) + (\operatorname{Im} a_1)x + \cdots + (\operatorname{Im} a_{n-2})x^{n-2}$$

for all $x \in \mathbf{R}$. This implies that $\operatorname{Im} a_0, \ldots, \operatorname{Im} a_{n-2}$ all equal 0 (by 4.8). Thus all coefficients of $q$ are real, as desired. Hence the desired factorization exists.

Now we turn to the question of uniqueness of our factorization. A factor of $p$ of the form $x^2 + b_k x + c_k$ with $b_k^2 < 4c_k$ can be uniquely written as $(x-\lambda_k)(x-\overline{\lambda_k})$ with $\lambda_k \in \mathbf{C}$. A moment's thought shows that two different factorizations of $p$ as an element of $\mathcal{P}(\mathbf{R})$ would lead to two different factorizations of $p$ as an element of $\mathcal{P}(\mathbf{C})$, contradicting 4.13. ∎

## *Exercises 4*

**1** Suppose $w, z \in \mathbf{C}$. Verify the following equalities and inequalities.

(a) $z + \overline{z} = 2\operatorname{Re} z$

(b) $z - \overline{z} = 2(\operatorname{Im} z)i$

(c) $z\overline{z} = |z|^2$

(d) $\overline{w+z} = \overline{w} + \overline{z}$ and $\overline{wz} = \overline{w}\,\overline{z}$

(e) $\overline{\overline{z}} = z$

(f) $|\operatorname{Re} z| \le |z|$ and $|\operatorname{Im} z| \le |z|$

(g) $|\overline{z}| = |z|$

(h) $|wz| = |w|\,|z|$

*The results above are the parts of 4.4 that were left to the reader.*

**2** Prove that if $w, z \in \mathbf{C}$, then $\big|\,|w| - |z|\,\big| \le |w - z|$.

*The inequality above is called the* ***reverse triangle inequality****.*

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**3** Suppose $V$ is a complex vector space and $\varphi \in V'$. Define $\sigma\colon V \to \mathbf{R}$ by $\sigma(v) = \operatorname{Re}\varphi(v)$ for each $v \in V$. Show that

$$\varphi(v) = \sigma(v) - i\sigma(iv)$$

for all $v \in V$.

**4** Suppose $m$ is a positive integer. Is the set

$$\{0\} \cup \{p \in \mathcal{P}(\mathbf{F}) : \deg p = m\}$$

a subspace of $\mathcal{P}(\mathbf{F})$?

**5** Is the set

$$\{0\} \cup \{p \in \mathcal{P}(\mathbf{F}) : \deg p \text{ is even}\}$$

a subspace of $\mathcal{P}(\mathbf{F})$?

**6** Suppose that $m$ and $n$ are positive integers with $m \le n$, and suppose $\lambda_1, \ldots, \lambda_m \in \mathbf{F}$. Prove that there exists a polynomial $p \in \mathcal{P}(\mathbf{F})$ with $\deg p = n$ such that $0 = p(\lambda_1) = \cdots = p(\lambda_m)$ and such that $p$ has no other zeros.

**7** Suppose that $m$ is a nonnegative integer, $z_1, \ldots, z_{m+1}$ are distinct elements of $\mathbf{F}$, and $w_1, \ldots, w_{m+1} \in \mathbf{F}$. Prove that there exists a unique polynomial $p \in \mathcal{P}_m(\mathbf{F})$ such that

$$p(z_k) = w_k$$

for each $k = 1, \ldots, m+1$.

*This result can be proved without using linear algebra. However, try to find the clearer, shorter proof that uses some linear algebra.*

**8** Suppose $p \in \mathcal{P}(\mathbf{C})$ has degree $m$. Prove that $p$ has $m$ distinct zeros if and only if $p$ and its derivative $p'$ have no zeros in common.

**9** Prove that every polynomial of odd degree with real coefficients has a real zero.

**10** For $p \in \mathcal{P}(\mathbf{R})$, define $Tp\colon \mathbf{R} \to \mathbf{R}$ by

$$(Tp)(x) = \begin{cases} \dfrac{p(x) - p(3)}{x - 3} & \text{if } x \ne 3, \\ p'(3) & \text{if } x = 3 \end{cases}$$

for each $x \in \mathbf{R}$. Show that $Tp \in \mathcal{P}(\mathbf{R})$ for every polynomial $p \in \mathcal{P}(\mathbf{R})$ and also show that $T\colon \mathcal{P}(\mathbf{R}) \to \mathcal{P}(\mathbf{R})$ is a linear map.

**11** Suppose $p \in \mathcal{P}(\mathbf{C})$. Define $q\colon \mathbf{C} \to \mathbf{C}$ by

$$q(z) = p(z)\,\overline{p(\overline{z})}.$$

Prove that $q$ is a polynomial with real coefficients.

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**12** Suppose $m$ is a nonnegative integer and $p \in \mathcal{P}_m(\mathbf{C})$ is such that there are distinct real numbers $x_0, x_1, \ldots, x_m$ with $p(x_k) \in \mathbf{R}$ for each $k = 0, 1, \ldots, m$. Prove that all coefficients of $p$ are real.

**13** Suppose $p \in \mathcal{P}(\mathbf{F})$ with $p \neq 0$. Let $U = \{pq : q \in \mathcal{P}(\mathbf{F})\}$.

(a) Show that $\dim \mathcal{P}(\mathbf{F})/U = \deg p$.

(b) Find a basis of $\mathcal{P}(\mathbf{F})/U$.

**14** Suppose $p, q \in \mathcal{P}(\mathbf{C})$ are nonconstant polynomials with no zeros in common. Let $m = \deg p$ and $n = \deg q$. Use linear algebra as outlined below in (a)–(c) to prove that there exist $r \in \mathcal{P}_{n-1}(\mathbf{C})$ and $s \in \mathcal{P}_{m-1}(\mathbf{C})$ such that

$$rp + sq = 1.$$

(a) Define $T\colon \mathcal{P}_{n-1}(\mathbf{C}) \times \mathcal{P}_{m-1}(\mathbf{C}) \to \mathcal{P}_{m+n-1}(\mathbf{C})$ by

$$T(r, s) = rp + sq.$$

Show that the linear map $T$ is injective.

(b) Show that the linear map $T$ in (a) is surjective.

(c) Use (b) to conclude that there exist $r \in \mathcal{P}_{n-1}(\mathbf{C})$ and $s \in \mathcal{P}_{m-1}(\mathbf{C})$ such that $rp + sq = 1$.

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Chapter 5

# *Eigenvalues and Eigenvectors*

Linear maps from one vector space to another vector space were the objects of study in Chapter 3. Now we begin our investigation of operators, which are linear maps from a vector space to itself. Their study constitutes the most important part of linear algebra.

To learn about an operator, we might try restricting it to a smaller subspace. Asking for that restriction to be an operator will lead us to the notion of invariant subspaces. Each one-dimensional invariant subspace arises from a vector that the operator maps into a scalar multiple of the vector. This path will lead us to eigenvectors and eigenvalues.

We will then prove one of the most important results in linear algebra: every operator on a finite-dimensional nonzero complex vector space has an eigenvalue. This result will allow us to show that for each operator on a finite-dimensional complex vector space, there is a basis of the vector space with respect to which the matrix of the operator has at least almost half its entries equal to 0.

*standing assumptions for this chapter*

- **F** denotes **R** or **C**.
- $V$ denotes a vector space over **F**.



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*Statue of Leonardo of Pisa (1170–1250, approximate dates), also known as Fibonacci. Exercise 21 in Section 5D shows how linear algebra can be used to find the explicit formula for the Fibonacci sequence shown on the front cover.*

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# 5A Invariant Subspaces

## Eigenvalues

**5.1 definition: *operator***

A linear map from a vector space to itself is called an *operator*.

Suppose $T \in \mathcal{L}(V)$. If $m \geq 2$ and

$$V = V_1 \oplus \cdots \oplus V_m,$$

where each $V_k$ is a nonzero subspace of $V$, then to understand the behavior of $T$ we only need to understand the behavior of each $T|_{V_k}$; here $T|_{V_k}$ denotes the restriction of $T$ to the smaller domain $V_k$. Dealing with $T|_{V_k}$ should be easier than dealing with $T$ because $V_k$ is a smaller vector space than $V$.

*Recall that we defined the notation $\mathcal{L}(V)$ to mean $\mathcal{L}(V, V)$.*

However, if we intend to apply tools useful in the study of operators (such as taking powers), then we have a problem: $T|_{V_k}$ may not map $V_k$ into itself; in other words, $T|_{V_k}$ may not be an operator on $V_k$. Thus we are led to consider only decompositions of $V$ of the form above in which $T$ maps each $V_k$ into itself. Hence we now give a name to subspaces of $V$ that get mapped into themselves by $T$.

**5.2 definition: *invariant subspace***

Suppose $T \in \mathcal{L}(V)$. A subspace $U$ of $V$ is called *invariant* under $T$ if $Tu \in U$ for every $u \in U$.

Thus $U$ is invariant under $T$ if $T|_U$ is an operator on $U$.

**5.3 example: *subspace invariant under differentiation operator***

Suppose that $T \in \mathcal{L}(\mathcal{P}(\mathbf{R}))$ is defined by $Tp = p'$. Then $\mathcal{P}_4(\mathbf{R})$, which is a subspace of $\mathcal{P}(\mathbf{R})$, is invariant under $T$ because if $p \in \mathcal{P}(\mathbf{R})$ has degree at most 4, then $p'$ also has degree at most 4.

**5.4 example: *four invariant subspaces, not necessarily all different***

If $T \in \mathcal{L}(V)$, then the following subspaces of $V$ are all invariant under $T$.

- $\{0\}$ The subspace $\{0\}$ is invariant under $T$ because if $u \in \{0\}$, then $u = 0$ and hence $Tu = 0 \in \{0\}$.
- $V$ The subspace $V$ is invariant under $T$ because if $u \in V$, then $Tu \in V$.
- $\operatorname{null} T$ The subspace $\operatorname{null} T$ is invariant under $T$ because if $u \in \operatorname{null} T$, then $Tu = 0$, and hence $Tu \in \operatorname{null} T$.
- $\operatorname{range} T$ The subspace $\operatorname{range} T$ is invariant under $T$ because if $u \in \operatorname{range} T$, then $Tu \in \operatorname{range} T$.

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Must an operator $T \in \mathcal{L}(V)$ have any invariant subspaces other than $\{0\}$ and $V$? Later we will see that this question has an affirmative answer if $V$ is finite-dimensional and $\dim V > 1$ (for $\mathbf{F} = \mathbf{C}$) or $\dim V > 2$ (for $\mathbf{F} = \mathbf{R}$); see 5.19 and Exercise 29 in Section 5B.

The previous example noted that $\operatorname{null} T$ and $\operatorname{range} T$ are invariant under $T$. However, these subspaces do not necessarily provide easy answers to the question above about the existence of invariant subspaces other than $\{0\}$ and $V$, because $\operatorname{null} T$ may equal $\{0\}$ and $\operatorname{range} T$ may equal $V$ (this happens when $T$ is invertible).

We will return later to a deeper study of invariant subspaces. Now we turn to an investigation of the simplest possible nontrivial invariant subspaces—invariant subspaces of dimension one.

Take any $v \in V$ with $v \neq 0$ and let $U$ equal the set of all scalar multiples of $v$:

$$U = \{\lambda v : \lambda \in \mathbf{F}\} = \operatorname{span}(v).$$

Then $U$ is a one-dimensional subspace of $V$ (and every one-dimensional subspace of $V$ is of this form for an appropriate choice of $v$). If $U$ is invariant under an operator $T \in \mathcal{L}(V)$, then $Tv \in U$, and hence there is a scalar $\lambda \in \mathbf{F}$ such that

$$Tv = \lambda v.$$

Conversely, if $Tv = \lambda v$ for some $\lambda \in \mathbf{F}$, then $\operatorname{span}(v)$ is a one-dimensional subspace of $V$ invariant under $T$.

The equation $Tv = \lambda v$, which we have just seen is intimately connected with one-dimensional invariant subspaces, is important enough that the scalars $\lambda$ and vectors $v$ satisfying it are given special names.

**5.5 definition: *eigenvalue***

Suppose $T \in \mathcal{L}(V)$. A number $\lambda \in \mathbf{F}$ is called an *eigenvalue* of $T$ if there exists $v \in V$ such that $v \neq 0$ and $Tv = \lambda v$.

*The word **eigenvalue** is half-German, half-English. The German prefix **eigen** means "own" in the sense of characterizing an intrinsic property.*

In the definition above, we require that $v \neq 0$ because every scalar $\lambda \in \mathbf{F}$ satisfies $T0 = \lambda 0$.

The comments above show that $V$ has a one-dimensional subspace invariant under $T$ if and only if $T$ has an eigenvalue.

**5.6 example: *eigenvalue***

Define an operator $T \in \mathcal{L}(\mathbf{F}^3)$ by

$$T(x, y, z) = (7x + 3z, 3x + 6y + 9z, -6y)$$

for $(x, y, z) \in \mathbf{F}^3$. Then $T(3, 1, -1) = (18, 6, -6) = 6(3, 1, -1)$. Thus 6 is an eigenvalue of $T$.

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The equivalences in the next result, along with many deep results in linear algebra, are valid only in the context of finite-dimensional vector spaces.

**5.7 *equivalent conditions to be an eigenvalue***

Suppose $V$ is finite-dimensional, $T \in \mathcal{L}(V)$, and $\lambda \in \mathbf{F}$. Then the following are equivalent.

(a) $\lambda$ is an eigenvalue of $T$.

(b) $T - \lambda I$ is not injective.

(c) $T - \lambda I$ is not surjective.

(d) $T - \lambda I$ is not invertible.

*Reminder: $I \in \mathcal{L}(V)$ is the identity operator. Thus $Iv = v$ for all $v \in V$.*

Proof Conditions (a) and (b) are equivalent because the equation $Tv = \lambda v$ is equivalent to the equation $(T - \lambda I)v = 0$. Conditions (b), (c), and (d) are equivalent by 3.65. ∎

**5.8 definition: *eigenvector***

Suppose $T \in \mathcal{L}(V)$ and $\lambda \in \mathbf{F}$ is an eigenvalue of $T$. A vector $v \in V$ is called an *eigenvector* of $T$ corresponding to $\lambda$ if $v \neq 0$ and $Tv = \lambda v$.

In other words, a nonzero vector $v \in V$ is an eigenvector of an operator $T \in \mathcal{L}(V)$ if and only if $Tv$ is a scalar multiple of $v$. Because $Tv = \lambda v$ if and only if $(T - \lambda I)v = 0$, a vector $v \in V$ with $v \neq 0$ is an eigenvector of $T$ corresponding to $\lambda$ if and only if $v \in \operatorname{null}(T - \lambda I)$.

**5.9 example: *eigenvalues and eigenvectors***

Suppose $T \in \mathcal{L}(\mathbf{F}^2)$ is defined by $T(w, z) = (-z, w)$.

(a) First consider the case $\mathbf{F} = \mathbf{R}$. Then $T$ is a counterclockwise rotation by $90°$ about the origin in $\mathbf{R}^2$. An operator has an eigenvalue if and only if there exists a nonzero vector in its domain that gets sent by the operator to a scalar multiple of itself. A $90°$ counterclockwise rotation of a nonzero vector in $\mathbf{R}^2$ cannot equal a scalar multiple of itself. Conclusion: if $\mathbf{F} = \mathbf{R}$, then $T$ has no eigenvalues (and thus has no eigenvectors).

(b) Now consider the case $\mathbf{F} = \mathbf{C}$. To find eigenvalues of $T$, we must find the scalars $\lambda$ such that $T(w, z) = \lambda(w, z)$ has some solution other than $w = z = 0$. The equation $T(w, z) = \lambda(w, z)$ is equivalent to the simultaneous equations

5.10 $$-z = \lambda w, \quad w = \lambda z.$$

Substituting the value for $w$ given by the second equation into the first equation gives

$$-z = \lambda^2 z.$$

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Now $z$ cannot equal 0 [otherwise 5.10 implies that $w = 0$; we are looking for solutions to 5.10 such that $(w, z)$ is not the 0 vector], so the equation above leads to the equation

$$-1 = \lambda^2.$$

The solutions to this equation are $\lambda = i$ and $\lambda = -i$.

You can verify that $i$ and $-i$ are eigenvalues of $T$. Indeed, the eigenvectors corresponding to the eigenvalue $i$ are the vectors of the form $(w, -wi)$, with $w \in \mathbf{C}$ and $w \neq 0$. Furthermore, the eigenvectors corresponding to the eigenvalue $-i$ are the vectors of the form $(w, wi)$, with $w \in \mathbf{C}$ and $w \neq 0$.

In the next proof, we again use the equivalence

$$Tv = \lambda v \iff (T - \lambda I)v = 0.$$

**5.11** *linearly independent eigenvectors*

Suppose $T \in \mathcal{L}(V)$. Then every list of eigenvectors of $T$ corresponding to distinct eigenvalues of $T$ is linearly independent.

Proof Suppose the desired result is false. Then there exists a smallest positive integer $m$ such that there exists a linearly dependent list $v_1, \ldots, v_m$ of eigenvectors of $T$ corresponding to distinct eigenvalues $\lambda_1, \ldots, \lambda_m$ of $T$ (note that $m \geq 2$ because an eigenvector is, by definition, nonzero). Thus there exist $a_1, \ldots, a_m \in \mathbf{F}$, none of which are 0 (because of the minimality of $m$), such that

$$a_1 v_1 + \cdots + a_m v_m = 0.$$

Apply $T - \lambda_m I$ to both sides of the equation above, getting

$$a_1(\lambda_1 - \lambda_m)v_1 + \cdots + a_{m-1}(\lambda_{m-1} - \lambda_m)v_{m-1} = 0.$$

Because the eigenvalues $\lambda_1, \ldots, \lambda_m$ are distinct, none of the coefficients above equal 0. Thus $v_1, \ldots, v_{m-1}$ is a linearly dependent list of $m - 1$ eigenvectors of $T$ corresponding to distinct eigenvalues, contradicting the minimality of $m$. This contradiction completes the proof. ∎

The result above leads to a short proof of the result below, which puts an upper bound on the number of distinct eigenvalues that an operator can have.

**5.12** *operator cannot have more eigenvalues than dimension of vector space*

Suppose $V$ is finite-dimensional. Then each operator on $V$ has at most $\dim V$ distinct eigenvalues.

Proof Let $T \in \mathcal{L}(V)$. Suppose $\lambda_1, \ldots, \lambda_m$ are distinct eigenvalues of $T$. Let $v_1, \ldots, v_m$ be corresponding eigenvectors. Then 5.11 implies that the list $v_1, \ldots, v_m$ is linearly independent. Thus $m \leq \dim V$ (see 2.22), as desired. ∎

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### *Polynomials Applied to Operators*

The main reason that a richer theory exists for operators (which map a vector space into itself) than for more general linear maps is that operators can be raised to powers. In this subsection we define that notion and the concept of applying a polynomial to an operator. This concept will be the key tool that we use in the next section when we prove that every operator on a nonzero finite-dimensional complex vector space has an eigenvalue.

If $T$ is an operator, then $TT$ makes sense (see 3.7) and is also an operator on the same vector space as $T$. We usually write $T^2$ instead of $TT$. More generally, we have the following definition of $T^m$.

**5.13 notation: $T^m$**

Suppose $T \in \mathcal{L}(V)$ and $m$ is a positive integer.

- $T^m \in \mathcal{L}(V)$ is defined by $T^m = \underbrace{T \cdots T}_{m \text{ times}}$.
- $T^0$ is defined to be the identity operator $I$ on $V$.
- If $T$ is invertible with inverse $T^{-1}$, then $T^{-m} \in \mathcal{L}(V)$ is defined by

$$T^{-m} = (T^{-1})^m.$$

You should verify that if $T$ is an operator, then

$$T^m T^n = T^{m+n} \quad \text{and} \quad (T^m)^n = T^{mn},$$

where $m$ and $n$ are arbitrary integers if $T$ is invertible and are nonnegative integers if $T$ is not invertible.

Having defined powers of an operator, we can now define what it means to apply a polynomial to an operator.

**5.14 notation: $p(T)$**

Suppose $T \in \mathcal{L}(V)$ and $p \in \mathcal{P}(\mathbf{F})$ is a polynomial given by

$$p(z) = a_0 + a_1 z + a_2 z^2 + \cdots + a_m z^m$$

for all $z \in \mathbf{F}$. Then $p(T)$ is the operator on $V$ defined by

$$p(T) = a_0 I + a_1 T + a_2 T^2 + \cdots + a_m T^m.$$

This is a new use of the symbol $p$ because we are applying $p$ to operators, not just elements of $\mathbf{F}$. The idea here is that to evaluate $p(T)$, we simply replace $z$ with $T$ in the expression defining $p$. Note that the constant term $a_0$ in $p(z)$ becomes the operator $a_0 I$ (which is a reasonable choice because $a_0 = a_0 z^0$ and thus we should replace $a_0$ with $a_0 T^0$, which equals $a_0 I$).

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5.15 example: *a polynomial applied to the differentiation operator*

Suppose $D \in \mathcal{L}(\mathcal{P}(\mathbf{R}))$ is the differentiation operator defined by $Dq = q'$ and $p$ is the polynomial defined by $p(x) = 7 - 3x + 5x^2$. Then $p(D) = 7I - 3D + 5D^2$. Thus

$$(p(D))q = 7q - 3q' + 5q''$$

for every $q \in \mathcal{P}(\mathbf{R})$.

---

If we fix an operator $T \in \mathcal{L}(V)$, then the function from $\mathcal{P}(\mathbf{F})$ to $\mathcal{L}(V)$ given by $p \mapsto p(T)$ is linear, as you should verify.

5.16 definition: *product of polynomials*

If $p, q \in \mathcal{P}(\mathbf{F})$, then $pq \in \mathcal{P}(\mathbf{F})$ is the polynomial defined by

$$(pq)(z) = p(z)q(z)$$

for all $z \in \mathbf{F}$.

The order does not matter in taking products of polynomials of a single operator, as shown by (b) in the next result.

5.17 *multiplicative properties*

Suppose $p, q \in \mathcal{P}(\mathbf{F})$ and $T \in \mathcal{L}(V)$. Then

(a) $(pq)(T) = p(T)q(T)$;

(b) $p(T)q(T) = q(T)p(T)$.

*Informal proof: When a product of polynomials is expanded using the distributive property, it does not matter whether the symbol is z or T.*

Proof

(a) Suppose $p(z) = \sum_{j=0}^{m} a_j z^j$ and $q(z) = \sum_{k=0}^{n} b_k z^k$ for all $z \in \mathbf{F}$. Then

$$(pq)(z) = \sum_{j=0}^{m} \sum_{k=0}^{n} a_j b_k z^{j+k}.$$

Thus

$$\begin{aligned}(pq)(T) &= \sum_{j=0}^{m} \sum_{k=0}^{n} a_j b_k T^{j+k} \\ &= \Big(\sum_{j=0}^{m} a_j T^j\Big)\Big(\sum_{k=0}^{n} b_k T^k\Big) \\ &= p(T)q(T).\end{aligned}$$

(b) Using (a) twice, we have $p(T)q(T) = (pq)(T) = (qp)(T) = q(T)p(T)$. ∎

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We observed earlier that if $T \in \mathcal{L}(V)$, then the subspaces null $T$ and range $T$ are invariant under $T$ (see 5.4). Now we show that the null space and the range of every polynomial of $T$ are also invariant under $T$.

**5.18** *null space and range of $p(T)$ are invariant under $T$*

Suppose $T \in \mathcal{L}(V)$ and $p \in \mathcal{P}(\mathbf{F})$. Then null $p(T)$ and range $p(T)$ are invariant under $T$.

Proof Suppose $u \in \text{null } p(T)$. Then $p(T)u = 0$. Thus

$$(p(T))(Tu) = T(p(T)u) = T(0) = 0.$$

Hence $Tu \in \text{null } p(T)$. Thus null $p(T)$ is invariant under $T$, as desired.

Suppose $u \in \text{range } p(T)$. Then there exists $v \in V$ such that $u = p(T)v$. Thus

$$Tu = T(p(T)v) = p(T)(Tv).$$

Hence $Tu \in \text{range } p(T)$. Thus range $p(T)$ is invariant under $T$, as desired. ∎

## *Exercises 5A*

**1** Suppose $T \in \mathcal{L}(V)$ and $U$ is a subspace of $V$.

(a) Prove that if $U \subseteq \text{null } T$, then $U$ is invariant under $T$.

(b) Prove that if range $T \subseteq U$, then $U$ is invariant under $T$.

**2** Suppose that $T \in \mathcal{L}(V)$ and $V_1, \ldots, V_m$ are subspaces of $V$ invariant under $T$. Prove that $V_1 + \cdots + V_m$ is invariant under $T$.

**3** Suppose $T \in \mathcal{L}(V)$. Prove that the intersection of every collection of subspaces of $V$ invariant under $T$ is invariant under $T$.

**4** Prove or give a counterexample: If $V$ is finite-dimensional and $U$ is a subspace of $V$ that is invariant under every operator on $V$, then $U = \{0\}$ or $U = V$.

**5** Suppose $T \in \mathcal{L}(\mathbf{R}^2)$ is defined by $T(x, y) = (-3y, x)$. Find the eigenvalues of $T$.

**6** Define $T \in \mathcal{L}(\mathbf{F}^2)$ by $T(w, z) = (z, w)$. Find all eigenvalues and eigenvectors of $T$.

**7** Define $T \in \mathcal{L}(\mathbf{F}^3)$ by $T(z_1, z_2, z_3) = (2z_2, 0, 5z_3)$. Find all eigenvalues and eigenvectors of $T$.

**8** Suppose $P \in \mathcal{L}(V)$ is such that $P^2 = P$. Prove that if $\lambda$ is an eigenvalue of $P$, then $\lambda = 0$ or $\lambda = 1$.

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**9** Define $T: \mathcal{P}(\mathbf{R}) \to \mathcal{P}(\mathbf{R})$ by $Tp = p'$. Find all eigenvalues and eigenvectors of $T$.

**10** Define $T \in \mathcal{L}(\mathcal{P}_4(\mathbf{R}))$ by $(Tp)(x) = xp'(x)$ for all $x \in \mathbf{R}$. Find all eigenvalues and eigenvectors of $T$.

**11** Suppose $V$ is finite-dimensional, $T \in \mathcal{L}(V)$, and $\alpha \in \mathbf{F}$. Prove that there exists $\delta > 0$ such that $T - \lambda I$ is invertible for all $\lambda \in \mathbf{F}$ such that $0 < |\alpha - \lambda| < \delta$.

**12** Suppose $V = U \oplus W$, where $U$ and $W$ are nonzero subspaces of $V$. Define $P \in \mathcal{L}(V)$ by $P(u + w) = u$ for each $u \in U$ and each $w \in W$. Find all eigenvalues and eigenvectors of $P$.

**13** Suppose $T \in \mathcal{L}(V)$. Suppose $S \in \mathcal{L}(V)$ is invertible.

(a) Prove that $T$ and $S^{-1}TS$ have the same eigenvalues.

(b) What is the relationship between the eigenvectors of $T$ and the eigenvectors of $S^{-1}TS$?

**14** Give an example of an operator on $\mathbf{R}^4$ that has no (real) eigenvalues.

**15** Suppose $V$ is finite-dimensional, $T \in \mathcal{L}(V)$, and $\lambda \in \mathbf{F}$. Show that $\lambda$ is an eigenvalue of $T$ if and only if $\lambda$ is an eigenvalue of the dual operator $T' \in \mathcal{L}(V')$.

**16** Suppose $v_1, \dots, v_n$ is a basis of $V$ and $T \in \mathcal{L}(V)$. Prove that if $\lambda$ is an eigenvalue of $T$, then

$$|\lambda| \le n \max\{|\mathcal{M}(T)_{j,k}| : 1 \le j, k \le n\},$$

where $\mathcal{M}(T)_{j,k}$ denotes the entry in row $j$, column $k$ of the matrix of $T$ with respect to the basis $v_1, \dots, v_n$.

*See Exercise 19 in Section 6A for a different bound on $|\lambda|$.*

**17** Suppose $\mathbf{F} = \mathbf{R}$, $T \in \mathcal{L}(V)$, and $\lambda \in \mathbf{R}$. Prove that $\lambda$ is an eigenvalue of $T$ if and only if $\lambda$ is an eigenvalue of the complexification $T_{\mathbf{C}}$.

*See Exercise 33 in Section 3B for the definition of $T_{\mathbf{C}}$.*

**18** Suppose $\mathbf{F} = \mathbf{R}$, $T \in \mathcal{L}(V)$, and $\lambda \in \mathbf{C}$. Prove that $\lambda$ is an eigenvalue of the complexification $T_{\mathbf{C}}$ if and only if $\overline{\lambda}$ is an eigenvalue of $T_{\mathbf{C}}$.

**19** Show that the forward shift operator $T \in \mathcal{L}(\mathbf{F}^\infty)$ defined by

$$T(z_1, z_2, \dots) = (0, z_1, z_2, \dots)$$

has no eigenvalues.

**20** Define the backward shift operator $S \in \mathcal{L}(\mathbf{F}^\infty)$ by

$$S(z_1, z_2, z_3, \dots) = (z_2, z_3, \dots).$$

(a) Show that every element of $\mathbf{F}$ is an eigenvalue of $S$.

(b) Find all eigenvectors of $S$.

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**21** Suppose $T \in \mathcal{L}(V)$ is invertible.

(a) Suppose $\lambda \in \mathbf{F}$ with $\lambda \neq 0$. Prove that $\lambda$ is an eigenvalue of $T$ if and only if $\frac{1}{\lambda}$ is an eigenvalue of $T^{-1}$.

(b) Prove that $T$ and $T^{-1}$ have the same eigenvectors.

**22** Suppose $T \in \mathcal{L}(V)$ and there exist nonzero vectors $u$ and $w$ in $V$ such that

$$Tu = 3w \quad \text{and} \quad Tw = 3u.$$

Prove that 3 or $-3$ is an eigenvalue of $T$.

**23** Suppose $V$ is finite-dimensional and $S, T \in \mathcal{L}(V)$. Prove that $ST$ and $TS$ have the same eigenvalues.

**24** Suppose $A$ is an $n$-by-$n$ matrix with entries in $\mathbf{F}$. Define $T \in \mathcal{L}(\mathbf{F}^n)$ by $Tx = Ax$, where elements of $\mathbf{F}^n$ are thought of as $n$-by-1 column vectors.

(a) Suppose the sum of the entries in each row of $A$ equals 1. Prove that 1 is an eigenvalue of $T$.

(b) Suppose the sum of the entries in each column of $A$ equals 1. Prove that 1 is an eigenvalue of $T$.

**25** Suppose $T \in \mathcal{L}(V)$ and $u, w$ are eigenvectors of $T$ such that $u + w$ is also an eigenvector of $T$. Prove that $u$ and $w$ are eigenvectors of $T$ corresponding to the same eigenvalue.

**26** Suppose $T \in \mathcal{L}(V)$ is such that every nonzero vector in $V$ is an eigenvector of $T$. Prove that $T$ is a scalar multiple of the identity operator.

**27** Suppose that $V$ is finite-dimensional and $k \in \{1, \ldots, \dim V - 1\}$. Suppose $T \in \mathcal{L}(V)$ is such that every subspace of $V$ of dimension $k$ is invariant under $T$. Prove that $T$ is a scalar multiple of the identity operator.

**28** Suppose $V$ is finite-dimensional and $T \in \mathcal{L}(V)$. Prove that $T$ has at most $1 + \dim \operatorname{range} T$ distinct eigenvalues.

**29** Suppose $T \in \mathcal{L}(\mathbf{R}^3)$ and $-4$, 5, and $\sqrt{7}$ are eigenvalues of $T$. Prove that there exists $x \in \mathbf{R}^3$ such that $Tx - 9x = (-4, 5, \sqrt{7})$.

**30** Suppose $T \in \mathcal{L}(V)$ and $(T - 2I)(T - 3I)(T - 4I) = 0$. Suppose $\lambda$ is an eigenvalue of $T$. Prove that $\lambda = 2$ or $\lambda = 3$ or $\lambda = 4$.

**31** Give an example of $T \in \mathcal{L}(\mathbf{R}^2)$ such that $T^4 = -I$.

**32** Suppose $T \in \mathcal{L}(V)$ has no eigenvalues and $T^4 = I$. Prove that $T^2 = -I$.

**33** Suppose $T \in \mathcal{L}(V)$ and $m$ is a positive integer.

(a) Prove that $T$ is injective if and only if $T^m$ is injective.

(b) Prove that $T$ is surjective if and only if $T^m$ is surjective.

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**34** Suppose $V$ is finite-dimensional and $v_1, ..., v_m \in V$. Prove that the list $v_1, ..., v_m$ is linearly independent if and only if there exists $T \in \mathcal{L}(V)$ such that $v_1, ..., v_m$ are eigenvectors of $T$ corresponding to distinct eigenvalues.

**35** Suppose that $\lambda_1, ..., \lambda_n$ is a list of distinct real numbers. Prove that the list $e^{\lambda_1 x}, ..., e^{\lambda_n x}$ is linearly independent in the vector space of real-valued functions on $\mathbf{R}$.

*Hint: Let $V = \operatorname{span}(e^{\lambda_1 x}, ..., e^{\lambda_n x})$, and define an operator $D \in \mathcal{L}(V)$ by $Df = f'$. Find eigenvalues and eigenvectors of $D$.*

**36** Suppose that $\lambda_1, ..., \lambda_n$ is a list of distinct positive numbers. Prove that the list $\cos(\lambda_1 x), ..., \cos(\lambda_n x)$ is linearly independent in the vector space of real-valued functions on $\mathbf{R}$.

**37** Suppose $V$ is finite-dimensional and $T \in \mathcal{L}(V)$. Define $\mathcal{A} \in \mathcal{L}(\mathcal{L}(V))$ by

$$\mathcal{A}(S) = TS$$

for each $S \in \mathcal{L}(V)$. Prove that the set of eigenvalues of $T$ equals the set of eigenvalues of $\mathcal{A}$.

**38** Suppose $V$ is finite-dimensional, $T \in \mathcal{L}(V)$, and $U$ is a subspace of $V$ invariant under $T$. The *quotient operator* $T/U \in \mathcal{L}(V/U)$ is defined by

$$(T/U)(v + U) = Tv + U$$

for each $v \in V$.

(a) Show that the definition of $T/U$ makes sense (which requires using the condition that $U$ is invariant under $T$) and show that $T/U$ is an operator on $V/U$.
(b) Show that each eigenvalue of $T/U$ is an eigenvalue of $T$.

**39** Suppose $V$ is finite-dimensional and $T \in \mathcal{L}(V)$. Prove that $T$ has an eigenvalue if and only if there exists a subspace of $V$ of dimension $\dim V - 1$ that is invariant under $T$.

**40** Suppose $S, T \in \mathcal{L}(V)$ and $S$ is invertible. Suppose $p \in \mathcal{P}(\mathbf{F})$ is a polynomial. Prove that

$$p(STS^{-1}) = Sp(T)S^{-1}.$$

**41** Suppose $T \in \mathcal{L}(V)$ and $U$ is a subspace of $V$ invariant under $T$. Prove that $U$ is invariant under $p(T)$ for every polynomial $p \in \mathcal{P}(\mathbf{F})$.

**42** Define $T \in \mathcal{L}(\mathbf{F}^n)$ by $T(x_1, x_2, x_3, ..., x_n) = (x_1, 2x_2, 3x_3, ..., nx_n)$.

(a) Find all eigenvalues and eigenvectors of $T$.
(b) Find all subspaces of $\mathbf{F}^n$ that are invariant under $T$.

**43** Suppose that $V$ is finite-dimensional, $\dim V > 1$, and $T \in \mathcal{L}(V)$. Prove that $\{p(T) : p \in \mathcal{P}(\mathbf{F})\} \neq \mathcal{L}(V)$.

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# 5B The Minimal Polynomial

## Existence of Eigenvalues on Complex Vector Spaces

Now we come to one of the central results about operators on finite-dimensional complex vector spaces.

**5.19 *existence of eigenvalues***

Every operator on a finite-dimensional nonzero complex vector space has an eigenvalue.

Proof Suppose $V$ is a finite-dimensional complex vector space of dimension $n > 0$ and $T \in \mathcal{L}(V)$. Choose $v \in V$ with $v \neq 0$. Then

$$v, Tv, T^2v, ..., T^nv$$

is not linearly independent, because $V$ has dimension $n$ and this list has length $n + 1$. Hence some linear combination (with not all the coefficients equal to 0) of the vectors above equals 0. Thus there exists a nonconstant polynomial $p$ of smallest degree such that

$$p(T)v = 0.$$

By the first version of the fundamental theorem of algebra (see 4.12), there exists $\lambda \in \mathbf{C}$ such that $p(\lambda) = 0$. Hence there exists a polynomial $q \in \mathcal{P}(\mathbf{C})$ such that

$$p(z) = (z - \lambda)q(z)$$

for every $z \in \mathbf{C}$ (see 4.6). This implies (using 5.17) that

$$0 = p(T)v = (T - \lambda I)(q(T)v).$$

Because $q$ has smaller degree than $p$, we know that $q(T)v \neq 0$. Thus the equation above implies that $\lambda$ is an eigenvalue of $T$ with eigenvector $q(T)v$. ∎

The proof above makes crucial use of the fundamental theorem of algebra. The comment following Exercise 16 helps explain why the fundamental theorem of algebra is so tightly connected to the result above.

The hypothesis in the result above that $\mathbf{F} = \mathbf{C}$ cannot be replaced with the hypothesis that $\mathbf{F} = \mathbf{R}$, as shown by Example 5.9. The next example shows that the finite-dimensional hypothesis in the result above also cannot be deleted.

**5.20 example: *an operator on a complex vector space with no eigenvalues***

Define $T \in \mathcal{L}(\mathcal{P}(\mathbf{C}))$ by $(Tp)(z) = zp(z)$. If $p \in \mathcal{P}(\mathbf{C})$ is a nonzero polynomial, then the degree of $Tp$ is one more than the degree of $p$, and thus $Tp$ cannot equal a scalar multiple of $p$. Hence $T$ has no eigenvalues.

Because $\mathcal{P}(\mathbf{C})$ is infinite-dimensional, this example does not contradict the result above.

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## *Eigenvalues and the Minimal Polynomial*

In this subsection we introduce an important polynomial associated with each operator. We begin with the following definition.

**5.21 definition: *monic polynomial***

A *monic polynomial* is a polynomial whose highest-degree coefficient equals 1.

For example, the polynomial $2 + 9z^2 + z^7$ is a monic polynomial of degree 7.

**5.22 *existence, uniqueness, and degree of minimal polynomial***

Suppose $V$ is finite-dimensional and $T \in \mathcal{L}(V)$. Then there is a unique monic polynomial $p \in \mathcal{P}(\mathbf{F})$ of smallest degree such that $p(T) = 0$. Furthermore, $\deg p \le \dim V$.

Proof If $\dim V = 0$, then $I$ is the zero operator on $V$ and thus we take $p$ to be the constant polynomial 1.

Now use induction on $\dim V$. Thus assume that $\dim V > 0$ and that the desired result is true for all operators on all complex vector spaces of smaller dimension. Let $v \in V$ be such that $v \ne 0$. The list $v, Tv, \dots, T^{\dim V}v$ has length $1 + \dim V$ and thus is linearly dependent. By the linear dependence lemma (2.19), there is a smallest positive integer $m \le \dim V$ such that $T^m v$ is a linear combination of $v, Tv, \dots, T^{m-1}v$. Thus there exist scalars $c_0, c_1, c_2, \dots, c_{m-1} \in \mathbf{F}$ such that

$$c_0 v + c_1 Tv + \cdots + c_{m-1}T^{m-1}v + T^m v = 0. \tag{5.23}$$

Define a monic polynomial $q \in \mathcal{P}_m(\mathbf{F})$ by

$$q(z) = c_0 + c_1 z + \cdots + c_{m-1}z^{m-1} + z^m.$$

Then 5.23 implies that $q(T)v = 0$.

If $k$ is a nonnegative integer, then

$$q(T)(T^k v) = T^k(q(T)v) = T^k(0) = 0.$$

The linear dependence lemma (2.19) shows that $v, Tv, \dots, T^{m-1}v$ is linearly independent. Thus the equation above implies that $\dim \operatorname{null} q(T) \ge m$. Hence

$$\dim \operatorname{range} q(T) = \dim V - \dim \operatorname{null} q(T) \le \dim V - m.$$

Because $\operatorname{range} q(T)$ is invariant under $T$ (by 5.18), we can apply our induction hypothesis to the operator $T|_{\operatorname{range} q(T)}$ on the vector space $\operatorname{range} q(T)$. Thus there is a monic polynomial $s \in \mathcal{P}(\mathbf{F})$ with

$$\deg s \le \dim V - m \quad \text{and} \quad s(T|_{\operatorname{range} q(T)}) = 0.$$

Hence for all $v \in V$ we have

$$(sq)(T)(v) = s(T)(q(T)v) = 0$$

because $q(T)v \in \operatorname{range} q(T)$ and $s(T)|_{\operatorname{range} q(T)} = s(T|_{\operatorname{range} q(T)}) = 0$. Thus $sq$ is a monic polynomial such that $\deg sq \le \dim V$ and $(sq)(T) = 0$.

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The paragraph above shows that there is a monic polynomial of degree at most $\dim V$ that when applied to $T$ gives the 0 operator. Thus there is a monic polynomial of smallest degree with this property, completing the existence part of this result.

Let $p \in \mathcal{P}(\mathbf{F})$ be a monic polynomial of smallest degree such that $p(T) = 0$. To prove the uniqueness part of the result, suppose $r \in \mathcal{P}(\mathbf{F})$ is a monic polynomial of the same degree as $p$ and $r(T) = 0$. Then $(p - r)(T) = 0$ and also $\deg(p - r) < \deg p$. If $p - r$ were not equal to 0, then we could divide $p - r$ by the coefficient of the highest-order term in $p - r$ to get a monic polynomial (of smaller degree than $p$) that when applied to $T$ gives the 0 operator. Thus $p - r = 0$, as desired.

The previous result justifies the following definition.

**5.24 definition: *minimal polynomial***

Suppose $V$ is finite-dimensional and $T \in \mathcal{L}(V)$. Then the *minimal polynomial* of $T$ is the unique monic polynomial $p \in \mathcal{P}(\mathbf{F})$ of smallest degree such that $p(T) = 0$.

To compute the minimal polynomial of an operator $T \in \mathcal{L}(V)$, we need to find the smallest positive integer $m$ such that the equation

$$c_0 I + c_1 T + \cdots + c_{m-1} T^{m-1} = -T^m$$

has a solution $c_0, c_1, \ldots, c_{m-1} \in \mathbf{F}$. If we pick a basis of $V$ and replace $T$ in the equation above with the matrix of $T$, then the equation above can be thought of as a system of $(\dim V)^2$ linear equations in the $m$ unknowns $c_0, c_1, \ldots, c_{m-1} \in \mathbf{F}$. Gaussian elimination or another fast method of solving systems of linear equations can tell us whether a solution exists, testing successive values $m = 1, 2, \ldots$ until a solution exists. By 5.22, a solution exists for some smallest positive integer $m \le \dim V$. The minimal polynomial of $T$ is then $c_0 + c_1 z + \cdots + c_{m-1} z^{m-1} + z^m$.

Even faster (usually), pick $v \in V$ and consider the equation

$$c_0 v + c_1 Tv + \cdots + c_{\dim V - 1} T^{\dim V - 1} v = -T^{\dim V} v. \tag{5.25}$$

Use a basis of $V$ to convert the equation above to a system of $\dim V$ linear equations in $\dim V$ unknowns $c_0, c_1, \ldots, c_{\dim V - 1}$. If this system of equations has a unique solution $c_0, c_1, \ldots, c_{\dim V - 1}$ (as happens most of the time), then the scalars $c_0, c_1, \ldots, c_{\dim V - 1}, 1$ are the coefficients of the minimal polynomial of $T$ (because 5.22 states that the degree of the minimal polynomial is at most $\dim V$).

Consider operators on $\mathbf{R}^4$ (thought of as 4-by-4 matrices with respect to the standard basis), and take $v = (1, 0, 0, 0)$ in the paragraph above. The faster method described above works on over 99.8% of the 4-by-4 matrices with integer entries in the interval $[-10, 10]$ and on over 99.999% of the 4-by-4 matrices with integer entries in $[-100, 100]$.

*These estimates are based on testing millions of random matrices.*

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The next example illustrates the faster procedure discussed above.

5.26 example: *minimal polynomial of an operator on* $\mathbf{F}^5$

Suppose $T \in \mathcal{L}(\mathbf{F}^5)$ and

$$\mathcal{M}(T) = \begin{pmatrix} 0 & 0 & 0 & 0 & -3 \\ 1 & 0 & 0 & 0 & 6 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \end{pmatrix}$$

with respect to the standard basis $e_1, e_2, e_3, e_4, e_5$. Taking $v = e_1$ for 5.25, we have

$$\begin{aligned} Te_1 &= e_2, & T^4e_1 &= T(T^3e_1) = Te_4 = e_5, \\ T^2e_1 &= T(Te_1) = Te_2 = e_3, & T^5e_1 &= T(T^4e_1) = Te_5 = -3e_1 + 6e_2. \\ T^3e_1 &= T(T^2e_1) = Te_3 = e_4, \end{aligned}$$

Thus $3e_1 - 6Te_1 = -T^5e_1$. The list $e_1, Te_1, T^2e_1, T^3e_1, T^4e_1$, which equals the list $e_1, e_2, e_3, e_4, e_5$, is linearly independent, so no other linear combination of this list equals $-T^5e_1$. Hence the minimal polynomial of $T$ is $3 - 6z + z^5$.

Recall that by definition, eigenvalues of operators on $V$ and zeros of polynomials in $\mathcal{P}(\mathbf{F})$ must be elements of $\mathbf{F}$. In particular, if $\mathbf{F} = \mathbf{R}$, then eigenvalues and zeros must be real numbers.

5.27 *eigenvalues are the zeros of the minimal polynomial*

Suppose $V$ is finite-dimensional and $T \in \mathcal{L}(V)$.

(a) The zeros of the minimal polynomial of $T$ are the eigenvalues of $T$.

(b) If $V$ is a complex vector space, then the minimal polynomial of $T$ has the form

$$(z - \lambda_1) \cdots (z - \lambda_m),$$

where $\lambda_1, \ldots, \lambda_m$ is a list of all eigenvalues of $T$, possibly with repetitions.

Proof Let $p$ be the minimal polynomial of $T$.

(a) First suppose $\lambda \in \mathbf{F}$ is a zero of $p$. Then $p$ can be written in the form

$$p(z) = (z - \lambda)q(z),$$

where $q$ is a monic polynomial with coefficients in $\mathbf{F}$ (see 4.6). Because $p(T) = 0$, we have

$$0 = (T - \lambda I)(q(T)v)$$

for all $v \in V$. Because the degree of $q$ is less than the degree of the minimal polynomial $p$, there exists at least one vector $v \in V$ such that $q(T)v \neq 0$. The equation above thus implies that $\lambda$ is an eigenvalue of $T$, as desired.

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To prove that every eigenvalue of $T$ is a zero of $p$, now suppose $\lambda \in \mathbf{F}$ is an eigenvalue of $T$. Thus there exists $v \in V$ with $v \neq 0$ such that $Tv = \lambda v$. Repeated applications of $T$ to both sides of this equation show that $T^k v = \lambda^k v$ for every nonnegative integer $k$. Thus

$$p(T)v = p(\lambda)v.$$

Because $p$ is the minimal polynomial of $T$, we have $p(T)v = 0$. Hence the equation above implies that $p(\lambda) = 0$. Thus $\lambda$ is a zero of $p$, as desired.

(b) To get the desired result, use (a) and the second version of the fundamental theorem of algebra (see 4.13). ∎

A nonzero polynomial has at most as many distinct zeros as its degree (see 4.8). Thus (a) of the previous result, along with the result that the minimal polynomial of an operator on $V$ has degree at most $\dim V$, gives an alternative proof of 5.12, which states that an operator on $V$ has at most $\dim V$ distinct eigenvalues.

Every monic polynomial is the minimal polynomial of some operator, as shown by Exercise 16, which generalizes Example 5.26. Thus 5.27(a) shows that finding exact expressions for the eigenvalues of an operator is equivalent to the problem of finding exact expressions for the zeros of a polynomial (and thus is not possible for some operators).

5.28 example: *An operator whose eigenvalues cannot be found exactly*

Let $T \in \mathcal{L}(\mathbf{C}^5)$ be the operator defined by

$$T(z_1, z_2, z_3, z_4, z_5) = (-3z_5, z_1 + 6z_5, z_2, z_3, z_4).$$

The matrix of $T$ with respect to the standard basis of $\mathbf{C}^5$ is the 5-by-5 matrix in Example 5.26. As we showed in that example, the minimal polynomial of $T$ is the polynomial

$$3 - 6z + z^5.$$

No zero of the polynomial above can be expressed using rational numbers, roots of rational numbers, and the usual rules of arithmetic (a proof of this would take us considerably beyond linear algebra). Because the zeros of the polynomial above are the eigenvalues of $T$ [by 5.27(a)], we cannot find an exact expression for any eigenvalue of $T$ in any familiar form.

Numeric techniques, which we will not discuss here, show that the zeros of the polynomial above, and thus the eigenvalues of $T$, are approximately the following five complex numbers:

$$-1.67, \quad 0.51, \quad 1.40, \quad -0.12 + 1.59i, \quad -0.12 - 1.59i.$$

Note that the two nonreal zeros of this polynomial are complex conjugates of each other, as we expect for a polynomial with real coefficients (see 4.14).

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The next result completely characterizes the polynomials that when applied to an operator give the 0 operator.

**5.29 $q(T) = 0 \iff q$ is a polynomial multiple of the minimal polynomial**

Suppose $V$ is finite-dimensional, $T \in \mathcal{L}(V)$, and $q \in \mathcal{P}(\mathbf{F})$. Then $q(T) = 0$ if and only if $q$ is a polynomial multiple of the minimal polynomial of $T$.

Proof Let $p$ denote the minimal polynomial of $T$.

First suppose $q(T) = 0$. By the division algorithm for polynomials (4.9), there exist polynomials $s, r \in \mathcal{P}(\mathbf{F})$ such that

$$q = ps + r \tag{5.30}$$

and $\deg r < \deg p$. We have

$$0 = q(T) = p(T)s(T) + r(T) = r(T).$$

The equation above implies that $r = 0$ (otherwise, dividing $r$ by its highest-degree coefficient would produce a monic polynomial that when applied to $T$ gives 0; this polynomial would have a smaller degree than the minimal polynomial, which would be a contradiction). Thus 5.30 becomes the equation $q = ps$. Hence $q$ is a polynomial multiple of $p$, as desired.

To prove the other direction, now suppose $q$ is a polynomial multiple of $p$. Thus there exists a polynomial $s \in \mathcal{P}(\mathbf{F})$ such that $q = ps$. We have

$$q(T) = p(T)s(T) = 0\, s(T) = 0,$$

as desired. ∎

The next result is a nice consequence of the result above.

**5.31 *minimal polynomial of a restriction operator***

Suppose $V$ is finite-dimensional, $T \in \mathcal{L}(V)$, and $U$ is a subspace of $V$ that is invariant under $T$. Then the minimal polynomial of $T$ is a polynomial multiple of the minimal polynomial of $T|_U$.

Proof Suppose $p$ is the minimal polynomial of $T$. Thus $p(T)v = 0$ for all $v \in V$. In particular,

$$p(T)u = 0 \text{ for all } u \in U.$$

Thus $p(T|_U) = 0$. Now 5.29, applied to the operator $T|_U$ in place of $T$, implies that $p$ is a polynomial multiple of the minimal polynomial of $T|_U$. ∎

See Exercise 25 for a result about quotient operators that is analogous to the result above.

The next result shows that the constant term of the minimal polynomial of an operator determines whether the operator is invertible.

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**5.32** *$T$ not invertible $\iff$ constant term of minimal polynomial of $T$ is 0*

Suppose $V$ is finite-dimensional and $T \in \mathcal{L}(V)$. Then $T$ is not invertible if and only if the constant term of the minimal polynomial of $T$ is 0.

Proof Suppose $T \in \mathcal{L}(V)$ and $p$ is the minimal polynomial of $T$. Then

$$\begin{aligned} T \text{ is not invertible} &\iff 0 \text{ is an eigenvalue of } T \\ &\iff 0 \text{ is a zero of } p \\ &\iff \text{the constant term of } p \text{ is } 0, \end{aligned}$$

where the first equivalence holds by 5.7, the second equivalence holds by 5.27(a), and the last equivalence holds because the constant term of $p$ equals $p(0)$. ∎

## *Eigenvalues on Odd-Dimensional Real Vector Spaces*

The next result will be the key tool that we use to show that every operator on an odd-dimensional real vector space has an eigenvalue.

**5.33** *even-dimensional null space*

Suppose $\mathbf{F} = \mathbf{R}$ and $V$ is finite-dimensional. Suppose also that $T \in \mathcal{L}(V)$ and $b, c \in \mathbf{R}$ with $b^2 < 4c$. Then $\dim \operatorname{null}(T^2 + bT + cI)$ is an even number.

Proof Recall that $\operatorname{null}(T^2 + bT + cI)$ is invariant under $T$ (by 5.18). By replacing $V$ with $\operatorname{null}(T^2 + bT + cI)$ and replacing $T$ with $T$ restricted to $\operatorname{null}(T^2 + bT + cI)$, we can assume that $T^2 + bT + cI = 0$; we now need to prove that $\dim V$ is even.

Suppose $\lambda \in \mathbf{R}$ and $v \in V$ are such that $Tv = \lambda v$. Then

$$0 = (T^2 + bT + cI)v = (\lambda^2 + b\lambda + c)v = \left(\left(\lambda + \tfrac{b}{2}\right)^2 + c - \tfrac{b^2}{4}\right)v.$$

The term in large parentheses above is a positive number. Thus the equation above implies that $v = 0$. Hence we have shown that $T$ has no eigenvectors.

Let $U$ be a subspace of $V$ that is invariant under $T$ and has the largest dimension among all subspaces of $V$ that are invariant under $T$ and have even dimension. If $U = V$, then we are done; otherwise assume there exists $w \in V$ such that $w \notin U$.

Let $W = \operatorname{span}(w, Tw)$. Then $W$ is invariant under $T$ because $T(Tw) = -bTw - cw$. Furthermore, $\dim W = 2$ because otherwise $w$ would be an eigenvector of $T$. Now

$$\dim(U + W) = \dim U + \dim W - \dim(U \cap W) = \dim U + 2,$$

where $U \cap W = \{0\}$ because otherwise $U \cap W$ would be a one-dimensional subspace of $V$ that is invariant under $T$ (impossible because $T$ has no eigenvectors).

Because $U + W$ is invariant under $T$, the equation above shows that there exists a subspace of $V$ invariant under $T$ of even dimension larger than $\dim U$. Thus the assumption that $U \neq V$ was incorrect. Hence $V$ has even dimension. ∎

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The next result states that on odd-dimensional vector spaces, every operator has an eigenvalue. We already know this result for finite-dimensional complex vectors spaces (without the odd hypothesis). Thus in the proof below, we will assume that $\mathbf{F} = \mathbf{R}$.

**5.34** *operators on odd-dimensional vector spaces have eigenvalues*

Every operator on an odd-dimensional vector space has an eigenvalue.

Proof Suppose $\mathbf{F} = \mathbf{R}$ and $V$ is finite-dimensional. Let $n = \dim V$, and suppose $n$ is an odd number. Let $T \in \mathcal{L}(V)$. We will use induction on $n$ in steps of size two to show that $T$ has an eigenvalue. To get started, note that the desired result holds if $\dim V = 1$ because then every nonzero vector in $V$ is an eigenvector of $T$.

Now suppose that $n \geq 3$ and the desired result holds for all operators on all odd-dimensional vector spaces of dimension less than $n$. Let $p$ denote the minimal polynomial of $T$. If $p$ is a polynomial multiple of $x - \lambda$ for some $\lambda \in \mathbf{R}$, then $\lambda$ is an eigenvalue of $T$ [by 5.27(a)] and we are done. Thus we can assume that there exist $b, c \in \mathbf{R}$ such that $b^2 < 4c$ and $p$ is a polynomial multiple of $x^2 + bx + c$ (see 4.16).

There exists a monic polynomial $q \in \mathcal{P}(\mathbf{R})$ such that $p(x) = q(x)(x^2 + bx + c)$ for all $x \in \mathbf{R}$. Now

$$0 = p(T) = (q(T))(T^2 + bT + cI),$$

which means that $q(T)$ equals 0 on $\operatorname{range}(T^2 + bT + cI)$. Because $\deg q < \deg p$ and $p$ is the minimal polynomial of $T$, this implies that $\operatorname{range}(T^2 + bT + cI) \neq V$.

The fundamental theorem of linear maps (3.21) tells us that

$$\dim V = \dim \operatorname{null}(T^2 + bT + cI) + \dim \operatorname{range}(T^2 + bT + cI).$$

Because $\dim V$ is odd (by hypothesis) and $\dim \operatorname{null}(T^2 + bT + cI)$ is even (by 5.33), the equation above shows that $\dim \operatorname{range}(T^2 + bT + cI)$ is odd.

Hence $\operatorname{range}(T^2 + bT + cI)$ is a subspace of $V$ that is invariant under $T$ (by 5.18) and that has odd dimension less than $\dim V$. Our induction hypothesis now implies that $T$ restricted to $\operatorname{range}(T^2 + bT + cI)$ has an eigenvalue, which means that $T$ has an eigenvalue. ∎

See Exercise 23 in Section 8B and Exercise 10 in Section 9C for alternative proofs of the result above.

## *Exercises 5B*

**1** Suppose $T \in \mathcal{L}(V)$. Prove that 9 is an eigenvalue of $T^2$ if and only if 3 or $-3$ is an eigenvalue of $T$.

**2** Suppose $V$ is a complex vector space and $T \in \mathcal{L}(V)$ has no eigenvalues. Prove that every subspace of $V$ invariant under $T$ is either $\{0\}$ or infinite-dimensional.

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**3** Suppose $n$ is a positive integer and $T \in \mathcal{L}(\mathbf{F}^n)$ is defined by

$$T(x_1, \dots, x_n) = (x_1 + \cdots + x_n, \dots, x_1 + \cdots + x_n).$$

(a) Find all eigenvalues and eigenvectors of $T$.
(b) Find the minimal polynomial of $T$.

*The matrix of $T$ with respect to the standard basis of $\mathbf{F}^n$ consists of all 1's.*

**4** Suppose $\mathbf{F} = \mathbf{C}$, $T \in \mathcal{L}(V)$, $p \in \mathcal{P}(\mathbf{C})$, and $\alpha \in \mathbf{C}$. Prove that $\alpha$ is an eigenvalue of $p(T)$ if and only if $\alpha = p(\lambda)$ for some eigenvalue $\lambda$ of $T$.

**5** Give an example of an operator on $\mathbf{R}^2$ that shows the result in Exercise 4 does not hold if $\mathbf{C}$ is replaced with $\mathbf{R}$.

**6** Suppose $T \in \mathcal{L}(\mathbf{F}^2)$ is defined by $T(w, z) = (-z, w)$. Find the minimal polynomial of $T$.

**7** (a) Give an example of $S, T \in \mathcal{L}(\mathbf{F}^2)$ such that the minimal polynomial of $ST$ does not equal the minimal polynomial of $TS$.
(b) Suppose $V$ is finite-dimensional and $S, T \in \mathcal{L}(V)$. Prove that if at least one of $S, T$ is invertible, then the minimal polynomial of $ST$ equals the minimal polynomial of $TS$.

*Hint: Show that if $S$ is invertible and $p \in \mathcal{P}(\mathbf{F})$, then $p(TS) = S^{-1}p(ST)S$.*

**8** Suppose $T \in \mathcal{L}(\mathbf{R}^2)$ is the operator of counterclockwise rotation by $1^\circ$. Find the minimal polynomial of $T$.

*Because $\dim \mathbf{R}^2 = 2$, the degree of the minimal polynomial of $T$ is at most 2. Thus the minimal polynomial of $T$ is not the tempting polynomial $x^{180} + 1$, even though $T^{180} = -I$.*

**9** Suppose $T \in \mathcal{L}(V)$ is such that with respect to some basis of $V$, all entries of the matrix of $T$ are rational numbers. Explain why all coefficients of the minimal polynomial of $T$ are rational numbers.

**10** Suppose $V$ is finite-dimensional, $T \in \mathcal{L}(V)$, and $v \in V$. Prove that

$$\operatorname{span}(v, Tv, \dots, T^m v) = \operatorname{span}(v, Tv, \dots, T^{\dim V - 1} v)$$

for all integers $m \geq \dim V - 1$.

**11** Suppose $V$ is a two-dimensional vector space, $T \in \mathcal{L}(V)$, and the matrix of $T$ with respect to some basis of $V$ is $\begin{pmatrix} a & c \\ b & d \end{pmatrix}$.

(a) Show that $T^2 - (a + d)T + (ad - bc)I = 0$.
(b) Show that the minimal polynomial of $T$ equals

$$\begin{cases} z - a & \text{if } b = c = 0 \text{ and } a = d, \\ z^2 - (a + d)z + (ad - bc) & \text{otherwise.} \end{cases}$$

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**12** Define $T \in \mathcal{L}(\mathbf{F}^n)$ by $T(x_1, x_2, x_3, ..., x_n) = (x_1, 2x_2, 3x_3, ..., nx_n)$. Find the minimal polynomial of $T$.

**13** Suppose $T \in \mathcal{L}(V)$ and $p \in \mathcal{P}(\mathbf{F})$. Prove that there exists a unique $r \in \mathcal{P}(\mathbf{F})$ such that $p(T) = r(T)$ and $\deg r$ is less than the degree of the minimal polynomial of $T$.

**14** Suppose $V$ is finite-dimensional and $T \in \mathcal{L}(V)$ has minimal polynomial $4 + 5z - 6z^2 - 7z^3 + 2z^4 + z^5$. Find the minimal polynomial of $T^{-1}$.

**15** Suppose $V$ is a finite-dimensional complex vector space with $\dim V > 0$ and $T \in \mathcal{L}(V)$. Define $f\colon \mathbf{C} \to \mathbf{R}$ by

$$f(\lambda) = \dim \operatorname{range}(T - \lambda I).$$

Prove that $f$ is not a continuous function.

**16** Suppose $a_0, ..., a_{n-1} \in \mathbf{F}$. Let $T$ be the operator on $\mathbf{F}^n$ whose matrix (with respect to the standard basis) is

$$\begin{pmatrix} 0 & & & & -a_0 \\ 1 & 0 & & & -a_1 \\ & 1 & \ddots & & -a_2 \\ & & \ddots & & \vdots \\ & & & 0 & -a_{n-2} \\ & & & 1 & -a_{n-1} \end{pmatrix}.$$

Here all entries of the matrix are 0 except for all 1's on the line under the diagonal and the entries in the last column (some of which might also be 0). Show that the minimal polynomial of $T$ is the polynomial

$$a_0 + a_1 z + \cdots + a_{n-1} z^{n-1} + z^n.$$

*The matrix above is called the **companion matrix** of the polynomial above. This exercise shows that every monic polynomial is the minimal polynomial of some operator. Hence a formula or an algorithm that could produce exact eigenvalues for each operator on each $\mathbf{F}^n$ could then produce exact zeros for each polynomial [by 5.27(a)]. Thus there is no such formula or algorithm. However, efficient numeric methods exist for obtaining very good approximations for the eigenvalues of an operator.*

**17** Suppose $V$ is finite-dimensional, $T \in \mathcal{L}(V)$, and $p$ is the minimal polynomial of $T$. Suppose $\lambda \in \mathbf{F}$. Show that the minimal polynomial of $T - \lambda I$ is the polynomial $q$ defined by $q(z) = p(z + \lambda)$.

**18** Suppose $V$ is finite-dimensional, $T \in \mathcal{L}(V)$, and $p$ is the minimal polynomial of $T$. Suppose $\lambda \in \mathbf{F} \backslash \{0\}$. Show that the minimal polynomial of $\lambda T$ is the polynomial $q$ defined by $q(z) = \lambda^{\deg p} p\left(\frac{z}{\lambda}\right)$.

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**19** Suppose $V$ is finite-dimensional and $T \in \mathcal{L}(V)$. Let $\mathcal{E}$ be the subspace of $\mathcal{L}(V)$ defined by

$$\mathcal{E} = \{q(T) : q \in \mathcal{P}(\mathbf{F})\}.$$

Prove that $\dim \mathcal{E}$ equals the degree of the minimal polynomial of $T$.

**20** Suppose $T \in \mathcal{L}(\mathbf{F}^4)$ is such that the eigenvalues of $T$ are $3, 5, 8$. Prove that $(T - 3I)^2(T - 5I)^2(T - 8I)^2 = 0$.

**21** Suppose $V$ is finite-dimensional and $T \in \mathcal{L}(V)$. Prove that the minimal polynomial of $T$ has degree at most $1 + \dim \operatorname{range} T$.

*If $\dim \operatorname{range} T < \dim V - 1$, then this exercise gives a better upper bound than 5.22 for the degree of the minimal polynomial of $T$.*

**22** Suppose $V$ is finite-dimensional and $T \in \mathcal{L}(V)$. Prove that $T$ is invertible if and only if $I \in \operatorname{span}(T, T^2, \ldots, T^{\dim V})$.

**23** Suppose $V$ is finite-dimensional and $T \in \mathcal{L}(V)$. Let $n = \dim V$. Prove that if $v \in V$, then $\operatorname{span}(v, Tv, \ldots, T^{n-1}v)$ is invariant under $T$.

**24** Suppose $V$ is a finite-dimensional complex vector space. Suppose $T \in \mathcal{L}(V)$ is such that 5 and 6 are eigenvalues of $T$ and that $T$ has no other eigenvalues. Prove that $(T - 5I)^{\dim V - 1}(T - 6I)^{\dim V - 1} = 0$.

**25** Suppose $V$ is finite-dimensional, $T \in \mathcal{L}(V)$, and $U$ is a subspace of $V$ that is invariant under $T$.

(a) Prove that the minimal polynomial of $T$ is a polynomial multiple of the minimal polynomial of the quotient operator $T/U$.

(b) Prove that

$$(\text{minimal polynomial of } T|_U) \times (\text{minimal polynomial of } T/U)$$

is a polynomial multiple of the minimal polynomial of $T$.

*The quotient operator $T/U$ was defined in Exercise 38 in Section 5A.*

**26** Suppose $V$ is finite-dimensional, $T \in \mathcal{L}(V)$, and $U$ is a subspace of $V$ that is invariant under $T$. Prove that the set of eigenvalues of $T$ equals the union of the set of eigenvalues of $T|_U$ and the set of eigenvalues of $T/U$.

**27** Suppose $\mathbf{F} = \mathbf{R}$, $V$ is finite-dimensional, and $T \in \mathcal{L}(V)$. Prove that the minimal polynomial of $T_{\mathbf{C}}$ equals the minimal polynomial of $T$.

*The complexification $T_{\mathbf{C}}$ was defined in Exercise 33 of Section 3B.*

**28** Suppose $V$ is finite-dimensional and $T \in \mathcal{L}(V)$. Prove that the minimal polynomial of $T' \in \mathcal{L}(V')$ equals the minimal polynomial of $T$.

*The dual map $T'$ was defined in Section 3F.*

**29** Show that every operator on a finite-dimensional vector space of dimension at least two has an invariant subspace of dimension two.

*Exercise 6 in Section 5C will give an improvement of this result when $\mathbf{F} = \mathbf{C}$.*

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## 5C Upper-Triangular Matrices

In Chapter 3 we defined the matrix of a linear map from a finite-dimensional vector space to another finite-dimensional vector space. That matrix depends on a choice of basis of each of the two vector spaces. Now that we are studying operators, which map a vector space to itself, the emphasis is on using only one basis.

**5.35 definition: *matrix of an operator, $\mathcal{M}(T)$***

Suppose $T \in \mathcal{L}(V)$. The *matrix of* $T$ with respect to a basis $v_1, \ldots, v_n$ of $V$ is the $n$-by-$n$ matrix

$$\mathcal{M}(T) = \begin{pmatrix} A_{1,1} & \cdots & A_{1,n} \\ \vdots & & \vdots \\ A_{n,1} & \cdots & A_{n,n} \end{pmatrix}$$

whose entries $A_{j,k}$ are defined by

$$Tv_k = A_{1,k}v_1 + \cdots + A_{n,k}v_n.$$

The notation $\mathcal{M}(T, (v_1, \ldots, v_n))$ is used if the basis is not clear from the context.

Operators have square matrices (meaning that the number of rows equals the number of columns), rather than the more general rectangular matrices that we considered earlier for linear maps.

If $T$ is an operator on $\mathbf{F}^n$ and no basis is specified, assume that the basis in question is the standard one (where the $k^{\text{th}}$ basis vector is 1 in the $k^{\text{th}}$ slot and 0 in all other slots). You can then think of the $k^{\text{th}}$ column of $\mathcal{M}(T)$ as $T$ applied to the $k^{\text{th}}$ basis vector, where we identify $n$-by-1 column vectors with elements of $\mathbf{F}^n$.

*The $k^{\text{th}}$ column of the matrix $\mathcal{M}(T)$ is formed from the coefficients used to write $Tv_k$ as a linear combination of the basis $v_1, \ldots, v_n$.*

**5.36 example: *matrix of an operator with respect to standard basis***

Define $T \in \mathcal{L}(\mathbf{F}^3)$ by $T(x, y, z) = (2x + y, 5y + 3z, 8z)$. Then the matrix of $T$ with respect to the standard basis of $\mathbf{F}^3$ is

$$\mathcal{M}(T) = \begin{pmatrix} 2 & 1 & 0 \\ 0 & 5 & 3 \\ 0 & 0 & 8 \end{pmatrix},$$

as you should verify.

---

A central goal of linear algebra is to show that given an operator $T$ on a finite-dimensional vector space $V$, there exists a basis of $V$ with respect to which $T$ has a reasonably simple matrix. To make this vague formulation a bit more precise, we might try to choose a basis of $V$ such that $\mathcal{M}(T)$ has many 0's.

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If $V$ is a finite-dimensional complex vector space, then we already know enough to show that there is a basis of $V$ with respect to which the matrix of $T$ has 0's everywhere in the first column, except possibly the first entry. In other words, there is a basis of $V$ with respect to which the matrix of $T$ looks like

$$\begin{pmatrix} \lambda & \\ 0 & * \\ \vdots & \\ 0 & \end{pmatrix};$$

here $*$ denotes the entries in all columns other than the first column. To prove this, let $\lambda$ be an eigenvalue of $T$ (one exists by 5.19) and let $v$ be a corresponding eigenvector. Extend $v$ to a basis of $V$. Then the matrix of $T$ with respect to this basis has the form above. Soon we will see that we can choose a basis of $V$ with respect to which the matrix of $T$ has even more 0's.

**5.37 definition: *diagonal of a matrix***

The *diagonal* of a square matrix consists of the entries on the line from the upper left corner to the bottom right corner.

For example, the diagonal of the matrix

$$\mathcal{M}(T) = \begin{pmatrix} 2 & 1 & 0 \\ 0 & 5 & 3 \\ 0 & 0 & 8 \end{pmatrix}$$

from Example 5.36 consists of the entries $2, 5, 8$, which are shown in red in the matrix above.

**5.38 definition: *upper-triangular matrix***

A square matrix is called *upper triangular* if all entries below the diagonal are 0.

For example, the 3-by-3 matrix above is upper triangular.

Typically we represent an upper-triangular matrix in the form

$$\begin{pmatrix} \lambda_1 & & * \\ & \ddots & \\ 0 & & \lambda_n \end{pmatrix};$$

the 0 in the matrix above indicates that all entries below the diagonal in this $n$-by-$n$ matrix equal 0. Upper-triangular matrices can be considered reasonably simple—if $n$ is large, then at least almost half the entries in an $n$-by-$n$ upper-triangular matrix are 0.

*We often use $*$ to denote matrix entries that we do not know or that are irrelevant to the questions being discussed.*

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The next result provides a useful connection between upper-triangular matrices and invariant subspaces.

**5.39** *conditions for upper-triangular matrix*

Suppose $T \in \mathcal{L}(V)$ and $v_1, ..., v_n$ is a basis of $V$. Then the following are equivalent.

(a) The matrix of $T$ with respect to $v_1, ..., v_n$ is upper triangular.

(b) $\text{span}(v_1, ..., v_k)$ is invariant under $T$ for each $k = 1, ..., n$.

(c) $Tv_k \in \text{span}(v_1, ..., v_k)$ for each $k = 1, ..., n$.

Proof First suppose (a) holds. To prove that (b) holds, suppose $k \in \{1, ..., n\}$. If $j \in \{1, ..., n\}$, then

$$Tv_j \in \text{span}(v_1, ..., v_j)$$

because the matrix of $T$ with respect to $v_1, ..., v_n$ is upper triangular. Because $\text{span}(v_1, ..., v_j) \subseteq \text{span}(v_1, ..., v_k)$ if $j \le k$, we see that

$$Tv_j \in \text{span}(v_1, ..., v_k)$$

for each $j \in \{1, ..., k\}$. Thus $\text{span}(v_1, ..., v_k)$ is invariant under $T$, completing the proof that (a) implies (b).

Now suppose (b) holds, so $\text{span}(v_1, ..., v_k)$ is invariant under $T$ for each $k = 1, ..., n$. In particular, $Tv_k \in \text{span}(v_1, ..., v_k)$ for each $k = 1, ..., n$. Thus (b) implies (c).

Now suppose (c) holds, so $Tv_k \in \text{span}(v_1, ..., v_k)$ for each $k = 1, ..., n$. This means that when writing each $Tv_k$ as a linear combination of the basis vectors $v_1, ..., v_n$, we need to use only the vectors $v_1, ..., v_k$. Hence all entries under the diagonal of $\mathcal{M}(T)$ are 0. Thus $\mathcal{M}(T)$ is an upper-triangular matrix, completing the proof that (c) implies (a).

We have shown that (a) $\implies$ (b) $\implies$ (c) $\implies$ (a), which shows that (a), (b), and (c) are equivalent. ∎

The next result tells us that if $T \in \mathcal{L}(V)$ and with respect to some basis of $V$ we have

$$\mathcal{M}(T) = \begin{pmatrix} \lambda_1 & & * \\ & \ddots & \\ 0 & & \lambda_n \end{pmatrix},$$

then $T$ satisfies a simple equation depending on $\lambda_1, ..., \lambda_n$.

**5.40** *equation satisfied by operator with upper-triangular matrix*

Suppose $T \in \mathcal{L}(V)$ and $V$ has a basis with respect to which $T$ has an upper-triangular matrix with diagonal entries $\lambda_1, ..., \lambda_n$. Then

$$(T - \lambda_1 I) \cdots (T - \lambda_n I) = 0.$$

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Proof Let $v_1, ..., v_n$ denote a basis of $V$ with respect to which $T$ has an upper-triangular matrix with diagonal entries $\lambda_1, ..., \lambda_n$. Then $Tv_1 = \lambda_1 v_1$, which means that $(T - \lambda_1 I)v_1 = 0$, which implies that $(T - \lambda_1 I)\cdots(T - \lambda_m I)v_1 = 0$ for $m = 1, ..., n$ (using the commutativity of each $T - \lambda_j I$ with each $T - \lambda_k I$).

Note that $(T - \lambda_2 I)v_2 \in \operatorname{span}(v_1)$. Thus $(T - \lambda_1 I)(T - \lambda_2 I)v_2 = 0$ (by the previous paragraph), which implies that $(T - \lambda_1 I)\cdots(T - \lambda_m I)v_1 = 0$ for $m = 2, ..., n$ (using the commutativity of each $T - \lambda_j I$ with each $T - \lambda_k I$).

Note that $(T - \lambda_3 I)v_3 \in \operatorname{span}(v_1, v_2)$. Thus by the previous paragraph, $(T-\lambda_1 I)(T-\lambda_2 I)(T-\lambda_3 I)v_3 = 0$, which implies that $(T-\lambda_1 I)\cdots(T-\lambda_m I)v_1 = 0$ for $m = 3, ..., n$ (using the commutativity of each $T - \lambda_j I$ with each $T - \lambda_k I$).

Continuing this pattern, we see that $(T - \lambda_1 I)\cdots(T - \lambda_n I)v_k = 0$ for each $k = 1, ..., n$. Thus $(T - \lambda_1 I)\cdots(T - \lambda_n I)$ is the 0 operator because it is 0 on each vector in a basis of $V$. ∎

Unfortunately no method exists for exactly computing the eigenvalues of an operator from its matrix. However, if we are fortunate enough to find a basis with respect to which the matrix of the operator is upper triangular, then the problem of computing the eigenvalues becomes trivial, as the next result shows.

**5.41** *determination of eigenvalues from upper-triangular matrix*

Suppose $T \in \mathcal{L}(V)$ has an upper-triangular matrix with respect to some basis of $V$. Then the eigenvalues of $T$ are precisely the entries on the diagonal of that upper-triangular matrix.

Proof Suppose $v_1, ..., v_n$ is a basis of $V$ with respect to which $T$ has an upper-triangular matrix

$$\mathcal{M}(T) = \begin{pmatrix} \lambda_1 & & * \\ & \ddots & \\ 0 & & \lambda_n \end{pmatrix}.$$

Because $Tv_1 = \lambda_1 v_1$, we see that $\lambda_1$ is an eigenvalue of $T$.

Suppose $k \in \{2, ..., n\}$. Then $(T - \lambda_k I)v_k \in \operatorname{span}(v_1, ..., v_{k-1})$. Thus $T - \lambda_k I$ maps $\operatorname{span}(v_1, ..., v_k)$ into $\operatorname{span}(v_1, ..., v_{k-1})$. Because

$$\dim \operatorname{span}(v_1, ..., v_k) = k \quad \text{and} \quad \dim \operatorname{span}(v_1, ..., v_{k-1}) = k - 1,$$

this implies that $T - \lambda_k I$ restricted to $\operatorname{span}(v_1, ..., v_k)$ is not injective (by 3.22). Thus there exists $v \in \operatorname{span}(v_1, ..., v_k)$ such that $v \neq 0$ and $(T - \lambda_k I)v = 0$. Thus $\lambda_k$ is an eigenvalue of $T$. Hence we have shown that every entry on the diagonal of $\mathcal{M}(T)$ is an eigenvalue of $T$.

To prove $T$ has no other eigenvalues, let $q$ be the polynomial defined by $q(z) = (z - \lambda_1)\cdots(z - \lambda_n)$. Then $q(T) = 0$ (by 5.40). Hence $q$ is a polynomial multiple of the minimal polynomial of $T$ (by 5.29). Thus every zero of the minimal polynomial of $T$ is a zero of $q$. Because the zeros of the minimal polynomial of $T$ are the eigenvalues of $T$ (by 5.27), this implies that every eigenvalue of $T$ is a zero of $q$. Hence the eigenvalues of $T$ are all contained in the list $\lambda_1, ..., \lambda_n$. ∎

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5.42 example: *eigenvalues via an upper-triangular matrix*

Define $T \in \mathcal{L}(\mathbf{F}^3)$ by $T(x, y, z) = (2x + y, 5y + 3z, 8z)$. The matrix of $T$ with respect to the standard basis is

$$\mathcal{M}(T) = \begin{pmatrix} 2 & 1 & 0 \\ 0 & 5 & 3 \\ 0 & 0 & 8 \end{pmatrix}.$$

Now 5.41 implies that the eigenvalues of $T$ are 2, 5, and 8.

The next example illustrates 5.44: an operator has an upper-triangular matrix with respect to some basis if and only if the minimal polynomial of the operator is the product of polynomials of degree 1.

5.43 example: *whether $T$ has an upper-triangular matrix can depend on* $\mathbf{F}$

Define $T \in \mathcal{L}(\mathbf{F}^4)$ by

$$T(z_1, z_2, z_3, z_4) = (-z_2, z_1, 2z_1 + 3z_3, z_3 + 3z_4).$$

Thus with respect to the standard basis of $\mathbf{F}^4$, the matrix of $T$ is

$$\begin{pmatrix} 0 & -1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 2 & 0 & 3 & 0 \\ 0 & 0 & 1 & 3 \end{pmatrix}.$$

You can ask a computer to verify that the minimal polynomial of $T$ is the polynomial $p$ defined by

$$p(z) = 9 - 6z + 10z^2 - 6z^3 + z^4.$$

First consider the case $\mathbf{F} = \mathbf{R}$. Then the polynomial $p$ factors as

$$p(z) = (z^2 + 1)(z - 3)(z - 3),$$

with no further factorization of $z^2 + 1$ as the product of two polynomials of degree 1 with real coefficients. Thus 5.44 states that there does not exist a basis of $\mathbf{R}^4$ with respect to which $T$ has an upper-triangular matrix.

Now consider the case $\mathbf{F} = \mathbf{C}$. Then the polynomial $p$ factors as

$$p(z) = (z - i)(z + i)(z - 3)(z - 3),$$

where all factors above have the form $z - \lambda_k$. Thus 5.44 states that there is a basis of $\mathbf{C}^4$ with respect to which $T$ has an upper-triangular matrix. Indeed, you can verify that with respect to the basis $(4 - 3i, -3 - 4i, -3 + i, 1)$, $(4 + 3i, -3 + 4i, -3 - i, 1)$, $(0, 0, 0, 1)$, $(0, 0, 1, 0)$ of $\mathbf{C}^4$, the operator $T$ has the upper-triangular matrix

$$\begin{pmatrix} i & 0 & 0 & 0 \\ 0 & -i & 0 & 0 \\ 0 & 0 & 3 & 1 \\ 0 & 0 & 0 & 3 \end{pmatrix}.$$

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**5.44** *necessary and sufficient condition to have an upper-triangular matrix*

Suppose $V$ is finite-dimensional and $T \in \mathcal{L}(V)$. Then $T$ has an upper-triangular matrix with respect to some basis of $V$ if and only if the minimal polynomial of $T$ equals $(z-\lambda_1)\cdots(z-\lambda_m)$ for some $\lambda_1, \ldots, \lambda_m \in \mathbf{F}$.

Proof First suppose $T$ has an upper-triangular matrix with respect to some basis of $V$. Let $\alpha_1, \ldots, \alpha_n$ denote the diagonal entries of that matrix. Define a polynomial $q \in \mathcal{P}(\mathbf{F})$ by

$$q(z) = (z-\alpha_1)\cdots(z-\alpha_n).$$

Then $q(T) = 0$, by 5.40. Hence $q$ is a polynomial multiple of the minimal polynomial of $T$, by 5.29. Thus the minimal polynomial of $T$ equals $(z-\lambda_1)\cdots(z-\lambda_m)$ for some $\lambda_1, \ldots, \lambda_m \in \mathbf{F}$ with $\{\lambda_1, \ldots, \lambda_m\} \subseteq \{\alpha_1, \ldots, \alpha_n\}$.

To prove the implication in the other direction, now suppose the minimal polynomial of $T$ equals $(z-\lambda_1)\cdots(z-\lambda_m)$ for some $\lambda_1, \ldots, \lambda_m \in \mathbf{F}$. We will use induction on $m$. To get started, if $m = 1$ then $z - \lambda_1$ is the minimal polynomial of $T$, which implies that $T = \lambda_1 I$, which implies that the matrix of $T$ (with respect to any basis of $V$) is upper triangular.

Now suppose $m > 1$ and the desired result holds for all smaller positive integers. Let

$$U = \operatorname{range}(T - \lambda_m I).$$

Then $U$ is invariant under $T$ [this is a special case of 5.18 with $p(z) = z - \lambda_m$]. Thus $T|_U$ is an operator on $U$.

If $u \in U$, then $u = (T - \lambda_m I)v$ for some $v \in V$ and

$$(T-\lambda_1 I)\cdots(T-\lambda_{m-1}I)u = (T-\lambda_1 I)\cdots(T-\lambda_m I)v = 0.$$

Hence $(z-\lambda_1)\cdots(z-\lambda_{m-1})$ is a polynomial multiple of the minimal polynomial of $T|_U$, by 5.29. Thus the minimal polynomial of $T|_U$ is the product of at most $m-1$ terms of the form $z - \lambda_k$.

By our induction hypothesis, there is a basis $u_1, \ldots, u_M$ of $U$ with respect to which $T|_U$ has an upper-triangular matrix. Thus for each $k \in \{1, \ldots, M\}$, we have (using 5.39)

$$Tu_k = (T|_U)(u_k) \in \operatorname{span}(u_1, \ldots, u_k). \tag{5.45}$$

Extend $u_1, \ldots, u_M$ to a basis $u_1, \ldots, u_M, v_1, \ldots, v_N$ of $V$. For each $k \in \{1, \ldots, N\}$, we have

$$Tv_k = (T - \lambda_m I)v_k + \lambda_m v_k.$$

The definition of $U$ shows that $(T - \lambda_m I)v_k \in U = \operatorname{span}(u_1, \ldots, u_M)$. Thus the equation above shows that

$$Tv_k \in \operatorname{span}(u_1, \ldots, u_M, v_1, \ldots, v_k). \tag{5.46}$$

From 5.45 and 5.46, we conclude (using 5.39) that $T$ has an upper-triangular matrix with respect to the basis $u_1, \ldots, u_M, v_1, \ldots, v_N$ of $V$, as desired. ∎

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The set of numbers $\{\lambda_1, ..., \lambda_m\}$ from the previous result equals the set of eigenvalues of $T$ (because the set of zeros of the minimal polynomial of $T$ equals the set of eigenvalues of $T$, by 5.27), although the list $\lambda_1, ..., \lambda_m$ in the previous result may contain repetitions.

In Chapter 8 we will improve even the wonderful result below; see 8.37 and 8.46.

**5.47** *if* $\mathbf{F} = \mathbf{C}$*, then every operator on* $V$ *has an upper-triangular matrix*

Suppose $V$ is a finite-dimensional complex vector space and $T \in \mathcal{L}(V)$. Then $T$ has an upper-triangular matrix with respect to some basis of $V$.

Proof The desired result follows immediately from 5.44 and the second version of the fundamental theorem of algebra (see 4.13). ∎

For an extension of the result above to two operators $S$ and $T$ such that

$$ST = TS,$$

see 5.80. Also, for an extension to more than two operators, see Exercise 9(b) in Section 5E.

**Caution:** If an operator $T \in \mathcal{L}(V)$ has a upper-triangular matrix with respect to some basis $v_1, ..., v_n$ of $V$, then the eigenvalues of $T$ are exactly the entries on the diagonal of $\mathcal{M}(T)$, as shown by 5.41, and furthermore $v_1$ is an eigenvector of $T$. However, $v_2, ..., v_n$ need not be eigenvectors of $T$. Indeed, a basis vector $v_k$ is an eigenvector of $T$ if and only if all entries in the $k^{\text{th}}$ column of the matrix of $T$ are 0, except possibly the $k^{\text{th}}$ entry.

You may recall from a previous course that every matrix of numbers can be changed to a matrix in what is called row echelon form. If one begins with a square matrix, the matrix in row echelon form will be an upper-triangular matrix. Do not confuse this upper-triangular matrix with the upper-triangular matrix of an operator with respect to some basis whose existence is proclaimed by 5.47 (if $\mathbf{F} = \mathbf{C}$)—there is no connection between these upper-triangular matrices.

*The row echelon form of the matrix of an operator does not give us a list of the eigenvalues of the operator. In contrast, an upper-triangular matrix with respect to some basis gives us a list of all the eigenvalues of the operator. However, there is no method for computing exactly such an upper-triangular matrix, even though 5.47 guarantees its existence if* $\mathbf{F} = \mathbf{C}$.

## *Exercises 5C*

**1** Prove or give a counterexample: If $T \in \mathcal{L}(V)$ and $T^2$ has an upper-triangular matrix with respect to some basis of $V$, then $T$ has an upper-triangular matrix with respect to some basis of $V$.

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**2** Suppose $A$ and $B$ are upper-triangular matrices of the same size, with $\alpha_1, \ldots, \alpha_n$ on the diagonal of $A$ and $\beta_1, \ldots, \beta_n$ on the diagonal of $B$.

(a) Show that $A + B$ is an upper-triangular matrix with $\alpha_1 + \beta_1, \ldots, \alpha_n + \beta_n$ on the diagonal.

(b) Show that $AB$ is an upper-triangular matrix with $\alpha_1\beta_1, \ldots, \alpha_n\beta_n$ on the diagonal.

*The results in this exercise are used in the proof of 5.81.*

**3** Suppose $T \in \mathcal{L}(V)$ is invertible and $v_1, \ldots, v_n$ is a basis of $V$ with respect to which the matrix of $T$ is upper triangular, with $\lambda_1, \ldots, \lambda_n$ on the diagonal. Show that the matrix of $T^{-1}$ is also upper triangular with respect to the basis $v_1, \ldots, v_n$, with

$$\frac{1}{\lambda_1}, \ldots, \frac{1}{\lambda_n}$$

on the diagonal.

**4** Give an example of an operator whose matrix with respect to some basis contains only 0's on the diagonal, but the operator is invertible.

*This exercise and the exercise below show that 5.41 fails without the hypothesis that an upper-triangular matrix is under consideration.*

**5** Give an example of an operator whose matrix with respect to some basis contains only nonzero numbers on the diagonal, but the operator is not invertible.

**6** Suppose $\mathbf{F} = \mathbf{C}$, $V$ is finite-dimensional, and $T \in \mathcal{L}(V)$. Prove that if $k \in \{1, \ldots, \dim V\}$, then $V$ has a $k$-dimensional subspace invariant under $T$.

**7** Suppose $V$ is finite-dimensional, $T \in \mathcal{L}(V)$, and $v \in V$.

(a) Prove that there exists a unique monic polynomial $p_v$ of smallest degree such that $p_v(T)v = 0$.

(b) Prove that the minimal polynomial of $T$ is a polynomial multiple of $p_v$.

**8** Suppose $V$ is finite-dimensional, $T \in \mathcal{L}(V)$, and there exists a nonzero vector $v \in V$ such that $T^2v + 2Tv = -2v$.

(a) Prove that if $\mathbf{F} = \mathbf{R}$, then there does not exist a basis of $V$ with respect to which $T$ has an upper-triangular matrix.

(b) Prove that if $\mathbf{F} = \mathbf{C}$ and $A$ is an upper-triangular matrix that equals the matrix of $T$ with respect to some basis of $V$, then $-1 + i$ or $-1 - i$ appears on the diagonal of $A$.

**9** Suppose $B$ is a square matrix with complex entries. Prove that there exists an invertible square matrix $A$ with complex entries such that $A^{-1}BA$ is an upper-triangular matrix.

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**10** Suppose $T \in \mathcal{L}(V)$ and $v_1, ..., v_n$ is a basis of $V$. Show that the following are equivalent.

(a) The matrix of $T$ with respect to $v_1, ..., v_n$ is lower triangular.

(b) $\operatorname{span}(v_k, ..., v_n)$ is invariant under $T$ for each $k = 1, ..., n$.

(c) $Tv_k \in \operatorname{span}(v_k, ..., v_n)$ for each $k = 1, ..., n$.

*A square matrix is called* ***lower triangular*** *if all entries above the diagonal are* 0.

**11** Suppose $\mathbf{F} = \mathbf{C}$ and $V$ is finite-dimensional. Prove that if $T \in \mathcal{L}(V)$, then there exists a basis of $V$ with respect to which $T$ has a lower-triangular matrix.

**12** Suppose $V$ is finite-dimensional, $T \in \mathcal{L}(V)$ has an upper-triangular matrix with respect to some basis of $V$, and $U$ is a subspace of $V$ that is invariant under $T$.

(a) Prove that $T|_U$ has an upper-triangular matrix with respect to some basis of $U$.

(b) Prove that the quotient operator $T/U$ has an upper-triangular matrix with respect to some basis of $V/U$.

*The quotient operator $T/U$ was defined in Exercise 38 in Section 5A.*

**13** Suppose $V$ is finite-dimensional and $T \in \mathcal{L}(V)$. Suppose there exists a subspace $U$ of $V$ that is invariant under $T$ such that $T|_U$ has an upper-triangular matrix with respect to some basis of $U$ and also $T/U$ has an upper-triangular matrix with respect to some basis of $V/U$. Prove that $T$ has an upper-triangular matrix with respect to some basis of $V$.

**14** Suppose $V$ is finite-dimensional and $T \in \mathcal{L}(V)$. Prove that $T$ has an upper-triangular matrix with respect to some basis of $V$ if and only if the dual operator $T'$ has an upper-triangular matrix with respect to some basis of the dual space $V'$.

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# 5D Diagonalizable Operators

## Diagonal Matrices

**5.48 definition: *diagonal matrix***

A *diagonal matrix* is a square matrix that is 0 everywhere except possibly on the diagonal.

**5.49 example: *diagonal matrix***

$$\begin{pmatrix} 8 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 5 \end{pmatrix}$$

is a diagonal matrix.

If an operator has a diagonal matrix with respect to some basis, then the entries on the diagonal are precisely the eigenvalues of the operator; this follows from 5.41 (or find an easier direct proof for diagonal matrices).

*Every diagonal matrix is upper triangular. Diagonal matrices typically have many more 0's than most upper-triangular matrices of the same size.*

**5.50 definition: *diagonalizable***

An operator on $V$ is called *diagonalizable* if the operator has a diagonal matrix with respect to some basis of $V$.

**5.51 example: *diagonalization may require a different basis***

Define $T \in \mathcal{L}(\mathbf{R}^2)$ by

$$T(x, y) = (41x + 7y, -20x + 74y).$$

The matrix of $T$ with respect to the standard basis of $\mathbf{R}^2$ is

$$\begin{pmatrix} 41 & 7 \\ -20 & 74 \end{pmatrix},$$

which is not a diagonal matrix. However, $T$ is diagonalizable. Specifically, the matrix of $T$ with respect to the basis $(1, 4), (7, 5)$ is

$$\begin{pmatrix} 69 & 0 \\ 0 & 46 \end{pmatrix}$$

because $T(1, 4) = (69, 276) = 69(1, 4)$ and $T(7, 5) = (322, 230) = 46(7, 5)$.

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For $\lambda \in \mathbf{F}$, we will find it convenient to have a name and a notation for the set of vectors that an operator $T$ maps to $\lambda$ times the vector.

**5.52 definition: *eigenspace*, $E(\lambda, T)$**

Suppose $T \in \mathcal{L}(V)$ and $\lambda \in \mathbf{F}$. The *eigenspace* of $T$ corresponding to $\lambda$ is the subspace $E(\lambda, T)$ of $V$ defined by

$$E(\lambda, T) = \operatorname{null}(T - \lambda I) = \{v \in V : Tv = \lambda v\}.$$

Hence $E(\lambda, T)$ is the set of all eigenvectors of $T$ corresponding to $\lambda$, along with the 0 vector.

For $T \in \mathcal{L}(V)$ and $\lambda \in \mathbf{F}$, the set $E(\lambda, T)$ is a subspace of $V$ because the null space of each linear map on $V$ is a subspace of $V$. The definitions imply that $\lambda$ is an eigenvalue of $T$ if and only if $E(\lambda, T) \neq \{0\}$.

**5.53 example: *eigenspaces of an operator***

Suppose the matrix of an operator $T \in \mathcal{L}(V)$ with respect to a basis $v_1, v_2, v_3$ of $V$ is the matrix in Example 5.49. Then

$$E(8, T) = \operatorname{span}(v_1), \quad E(5, T) = \operatorname{span}(v_2, v_3).$$

If $\lambda$ is an eigenvalue of an operator $T \in \mathcal{L}(V)$, then $T$ restricted to $E(\lambda, T)$ is just the operator of multiplication by $\lambda$.

**5.54 *sum of eigenspaces is a direct sum***

Suppose $T \in \mathcal{L}(V)$ and $\lambda_1, \ldots, \lambda_m$ are distinct eigenvalues of $T$. Then

$$E(\lambda_1, T) + \cdots + E(\lambda_m, T)$$

is a direct sum. Furthermore, if $V$ is finite-dimensional, then

$$\dim E(\lambda_1, T) + \cdots + \dim E(\lambda_m, T) \leq \dim V.$$

Proof To show that $E(\lambda_1, T) + \cdots + E(\lambda_m, T)$ is a direct sum, suppose

$$v_1 + \cdots + v_m = 0,$$

where each $v_k$ is in $E(\lambda_k, T)$. Because eigenvectors corresponding to distinct eigenvalues are linearly independent (by 5.11), this implies that each $v_k$ equals 0. Thus $E(\lambda_1, T) + \cdots + E(\lambda_m, T)$ is a direct sum (by 1.45), as desired.

Now suppose $V$ is finite-dimensional. Then

$$\begin{aligned} \dim E(\lambda_1, T) + \cdots + \dim E(\lambda_m, T) &= \dim(E(\lambda_1, T) \oplus \cdots \oplus E(\lambda_m, T)) \\ &\leq \dim V, \end{aligned}$$

where the first line follows from 3.94 and the second line follows from 2.37. ∎

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### *Conditions for Diagonalizability*

The following characterizations of diagonalizable operators will be useful.

**5.55** *conditions equivalent to diagonalizability*

Suppose $V$ is finite-dimensional and $T \in \mathcal{L}(V)$. Let $\lambda_1, \ldots, \lambda_m$ denote the distinct eigenvalues of $T$. Then the following are equivalent.

(a) $T$ is diagonalizable.

(b) $V$ has a basis consisting of eigenvectors of $T$.

(c) $V = E(\lambda_1, T) \oplus \cdots \oplus E(\lambda_m, T)$.

(d) $\dim V = \dim E(\lambda_1, T) + \cdots + \dim E(\lambda_m, T)$.

Proof An operator $T \in \mathcal{L}(V)$ has a diagonal matrix

$$\begin{pmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \end{pmatrix}$$

with respect to a basis $v_1, \ldots, v_n$ of $V$ if and only if $Tv_k = \lambda_k v_k$ for each $k$. Thus (a) and (b) are equivalent.

Suppose (b) holds; thus $V$ has a basis consisting of eigenvectors of $T$. Hence every vector in $V$ is a linear combination of eigenvectors of $T$, which implies that

$$V = E(\lambda_1, T) + \cdots + E(\lambda_m, T).$$

Now 5.54 shows that (c) holds, proving that (b) implies (c).

That (c) implies (d) follows immediately from 3.94.

Finally, suppose (d) holds; thus

$$\dim V = \dim E(\lambda_1, T) + \cdots + \dim E(\lambda_m, T). \tag{5.56}$$

Choose a basis of each $E(\lambda_k, T)$; put all these bases together to form a list $v_1, \ldots, v_n$ of eigenvectors of $T$, where $n = \dim V$ (by 5.56). To show that this list is linearly independent, suppose

$$a_1 v_1 + \cdots + a_n v_n = 0,$$

where $a_1, \ldots, a_n \in \mathbf{F}$. For each $k = 1, \ldots, m$, let $u_k$ denote the sum of all the terms $a_j v_j$ such that $v_j \in E(\lambda_k, T)$. Thus each $u_k$ is in $E(\lambda_k, T)$, and

$$u_1 + \cdots + u_m = 0.$$

Because eigenvectors corresponding to distinct eigenvalues are linearly independent (see 5.11), this implies that each $u_k$ equals 0. Because each $u_k$ is a sum of terms $a_j v_j$, where the $v_j$'s were chosen to be a basis of $E(\lambda_k, T)$, this implies that all $a_j$'s equal 0. Thus $v_1, \ldots, v_n$ is linearly independent and hence is a basis of $V$ (by 2.38). Thus (d) implies (b), completing the proof. ∎

For additional conditions equivalent to diagonalizability, see 5.62, Exercises 5 and 15 in this section, Exercise 24 in Section 7B, and Exercise 15 in Section 8A.

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As we know, every operator on a finite-dimensional complex vector space has an eigenvalue. However, not every operator on a finite-dimensional complex vector space has enough eigenvectors to be diagonalizable, as shown by the next example.

5.57 example: *an operator that is not diagonalizable*

Define an operator $T \in \mathcal{L}(\mathbf{F}^3)$ by $T(a, b, c) = (b, c, 0)$. The matrix of $T$ with respect to the standard basis of $\mathbf{F}^3$ is

$$\begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix},$$

which is an upper-triangular matrix but is not a diagonal matrix.

As you should verify, 0 is the only eigenvalue of $T$ and furthermore

$$E(0, T) = \{(a, 0, 0) \in \mathbf{F}^3 : a \in \mathbf{F}\}.$$

Hence conditions (b), (c), and (d) of 5.55 fail (of course, because these conditions are equivalent, it is sufficient to check that only one of them fails). Thus condition (a) of 5.55 also fails. Hence $T$ is not diagonalizable, regardless of whether $\mathbf{F} = \mathbf{R}$ or $\mathbf{F} = \mathbf{C}$.

The next result shows that if an operator has as many distinct eigenvalues as the dimension of its domain, then the operator is diagonalizable.

5.58 *enough eigenvalues implies diagonalizability*

Suppose $V$ is finite-dimensional and $T \in \mathcal{L}(V)$ has $\dim V$ distinct eigenvalues. Then $T$ is diagonalizable.

Proof Suppose $T$ has distinct eigenvalues $\lambda_1, \ldots, \lambda_{\dim V}$. For each $k$, let $v_k \in V$ be an eigenvector corresponding to the eigenvalue $\lambda_k$. Because eigenvectors corresponding to distinct eigenvalues are linearly independent (see 5.11), $v_1, \ldots, v_{\dim V}$ is linearly independent.

A linearly independent list of $\dim V$ vectors in $V$ is a basis of $V$ (see 2.38); thus $v_1, \ldots, v_{\dim V}$ is a basis of $V$. With respect to this basis consisting of eigenvectors, $T$ has a diagonal matrix. ∎

In later chapters we will find additional conditions that imply that certain operators are diagonalizable. For example, see the real spectral theorem (7.29) and the complex spectral theorem (7.31).

The result above gives a sufficient condition for an operator to be diagonalizable. However, this condition is not necessary. For example, the operator $T$ on $\mathbf{F}^3$ defined by $T(x, y, z) = (6x, 6y, 7z)$ has only two eigenvalues (6 and 7) and $\dim \mathbf{F}^3 = 3$, but $T$ is diagonalizable (by the standard basis of $\mathbf{F}^3$).

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The next example illustrates the importance of diagonalization, which can be used to compute high powers of an operator, taking advantage of the equation $T^k v = \lambda^k v$ if $v$ is an eigenvector of $T$ with eigenvalue $\lambda$.

*For a spectacular application of these techniques, see Exercise 21, which shows how to use diagonalization to find an exact formula for the $n^{th}$ term of the Fibonacci sequence.*

5.59 example: *using diagonalization to compute $T^{100}$*

Define $T \in \mathcal{L}(\mathbf{F}^3)$ by $T(x, y, z) = (2x + y, 5y + 3z, 8z)$. With respect to the standard basis, the matrix of $T$ is

$$\begin{pmatrix} 2 & 1 & 0 \\ 0 & 5 & 3 \\ 0 & 0 & 8 \end{pmatrix}.$$

The matrix above is an upper-triangular matrix but it is not a diagonal matrix. By 5.41, the eigenvalues of $T$ are 2, 5, and 8. Because $T$ is an operator on a vector space of dimension three and $T$ has three distinct eigenvalues, 5.58 assures us that there exists a basis of $\mathbf{F}^3$ with respect to which $T$ has a diagonal matrix.

To find this basis, we only have to find an eigenvector for each eigenvalue. In other words, we have to find a nonzero solution to the equation

$$T(x, y, z) = \lambda(x, y, z)$$

for $\lambda = 2$, then for $\lambda = 5$, and then for $\lambda = 8$. Solving these simple equations shows that for $\lambda = 2$ we have an eigenvector $(1, 0, 0)$, for $\lambda = 5$ we have an eigenvector $(1, 3, 0)$, and for $\lambda = 8$ we have an eigenvector $(1, 6, 6)$.

Thus $(1, 0, 0), (1, 3, 0), (1, 6, 6)$ is a basis of $\mathbf{F}^3$ consisting of eigenvectors of $T$, and with respect to this basis the matrix of $T$ is the diagonal matrix

$$\begin{pmatrix} 2 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 8 \end{pmatrix}.$$

To compute $T^{100}(0, 0, 1)$, for example, write $(0, 0, 1)$ as a linear combination of our basis of eigenvectors:

$$(0, 0, 1) = \tfrac{1}{6}(1, 0, 0) - \tfrac{1}{3}(1, 3, 0) + \tfrac{1}{6}(1, 6, 6).$$

Now apply $T^{100}$ to both sides of the equation above, getting

$$\begin{aligned} T^{100}(0, 0, 1) &= \tfrac{1}{6}\big(T^{100}(1, 0, 0)\big) - \tfrac{1}{3}\big(T^{100}(1, 3, 0)\big) + \tfrac{1}{6}\big(T^{100}(1, 6, 6)\big) \\ &= \tfrac{1}{6}\big(2^{100}(1, 0, 0) - 2 \cdot 5^{100}(1, 3, 0) + 8^{100}(1, 6, 6)\big) \\ &= \tfrac{1}{6}\big(2^{100} - 2 \cdot 5^{100} + 8^{100}, 6 \cdot 8^{100} - 6 \cdot 5^{100}, 6 \cdot 8^{100}\big). \end{aligned}$$

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We saw earlier that an operator $T$ on a finite-dimensional vector space $V$ has an upper-triangular matrix with respect to some basis of $V$ if and only if the minimal polynomial of $T$ equals $(z-\lambda_1)\cdots(z-\lambda_m)$ for some $\lambda_1,\ldots,\lambda_m \in \mathbf{F}$ (see 5.44). As we previously noted (see 5.47), this condition is always satisfied if $\mathbf{F}=\mathbf{C}$.

Our next result 5.62 states that an operator $T \in \mathcal{L}(V)$ has a diagonal matrix with respect to some basis of $V$ if and only if the minimal polynomial of $T$ equals $(z-\lambda_1)\cdots(z-\lambda_m)$ for some ***distinct*** $\lambda_1,\ldots,\lambda_m \in \mathbf{F}$. Before formally stating this result, we give two examples of using it.

5.60 example: *diagonalizable, but with no known exact eigenvalues*

Define $T \in \mathcal{L}(\mathbf{C}^5)$ by

$$T(z_1,z_2,z_3,z_4,z_5) = (-3z_5, z_1+6z_5, z_2, z_3, z_4).$$

The matrix of $T$ is shown in Example 5.26, where we showed that the minimal polynomial of $T$ is $3-6z+z^5$.

As mentioned in Example 5.28, no exact expression is known for any of the zeros of this polynomial, but numeric techniques show that the zeros of this polynomial are approximately $-1.67,\ 0.51,\ 1.40,\ -0.12+1.59i,\ -0.12-1.59i$.

The software that produces these approximations is accurate to more than three digits. Thus these approximations are good enough to show that the five numbers above are distinct. The minimal polynomial of $T$ equals the fifth degree monic polynomial with these zeros. Now 5.62 shows that $T$ is diagonalizable.

5.61 example: *showing that an operator is not diagonalizable*

Define $T \in \mathcal{L}(\mathbf{F}^3)$ by

$$T(z_1,z_2,z_3) = (6z_1+3z_2+4z_3, 6z_2+2z_3, 7z_3).$$

The matrix of $T$ with respect to the standard basis of $\mathbf{F}^3$ is

$$\begin{pmatrix} 6 & 3 & 4 \\ 0 & 6 & 2 \\ 0 & 0 & 7 \end{pmatrix}.$$

The matrix above is an upper-triangular matrix but is not a diagonal matrix. Might $T$ have a diagonal matrix with respect to some other basis of $\mathbf{F}^3$?

To answer this question, we will find the minimal polynomial of $T$. First note that the eigenvalues of $T$ are the diagonal entries of the matrix above (by 5.41). Thus the zeros of the minimal polynomial of $T$ are $6, 7$ [by 5.27(a)]. The diagonal of the matrix above tells us that $(T-6I)^2(T-7I)=0$ (by 5.40). The minimal polynomial of $T$ has degree at most 3 (by 5.22). Putting all this together, we see that the minimal polynomial of $T$ is either $(z-6)(z-7)$ or $(z-6)^2(z-7)$.

A simple computation shows that $(T-6I)(T-7I) \neq 0$. Thus the minimal polynomial of $T$ is $(z-6)^2(z-7)$.

Now 5.62 shows that $T$ is not diagonalizable.

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**5.62** *necessary and sufficient condition for diagonalizability*

Suppose $V$ is finite-dimensional and $T \in \mathcal{L}(V)$. Then $T$ is diagonalizable if and only if the minimal polynomial of $T$ equals $(z - \lambda_1)\cdots(z - \lambda_m)$ for some list of distinct numbers $\lambda_1, \ldots, \lambda_m \in \mathbf{F}$.

Proof First suppose $T$ is diagonalizable. Thus there is a basis $v_1, \ldots, v_n$ of $V$ consisting of eigenvectors of $T$. Let $\lambda_1, \ldots, \lambda_m$ be the distinct eigenvalues of $T$. Then for each $v_j$, there exists $\lambda_k$ with $(T - \lambda_k I)v_j = 0$. Thus

$$(T - \lambda_1 I)\cdots(T - \lambda_m I)v_j = 0,$$

which implies that the minimal polynomial of $T$ equals $(z - \lambda_1)\cdots(z - \lambda_m)$.

To prove the implication in the other direction, now suppose the minimal polynomial of $T$ equals $(z - \lambda_1)\cdots(z - \lambda_m)$ for some list of distinct numbers $\lambda_1, \ldots, \lambda_m \in \mathbf{F}$. Thus

5.63 $$(T - \lambda_1 I)\cdots(T - \lambda_m I) = 0.$$

We will prove that $T$ is diagonalizable by induction on $m$. To get started, suppose $m = 1$. Then $T - \lambda_1 I = 0$, which means that $T$ is a scalar multiple of the identity operator, which implies that $T$ is diagonalizable.

Now suppose that $m > 1$ and the desired result holds for all smaller values of $m$. The subspace $\operatorname{range}(T - \lambda_m I)$ is invariant under $T$ [this is a special case of 5.18 with $p(z) = z - \lambda_m$]. Thus $T$ restricted to $\operatorname{range}(T - \lambda_m I)$ is an operator on $\operatorname{range}(T - \lambda_m I)$.

If $u \in \operatorname{range}(T - \lambda_m I)$, then $u = (T - \lambda_m I)v$ for some $v \in V$, and 5.63 implies

5.64 $$(T - \lambda_1 I)\cdots(T - \lambda_{m-1} I)u = (T - \lambda_1 I)\cdots(T - \lambda_m I)v = 0.$$

Hence $(z - \lambda_1)\cdots(z - \lambda_{m-1})$ is a polynomial multiple of the minimal polynomial of $T$ restricted to $\operatorname{range}(T - \lambda_m I)$ [by 5.29]. Thus by our induction hypothesis, there is a basis of $\operatorname{range}(T - \lambda_m I)$ consisting of eigenvectors of $T$.

Suppose that $u \in \operatorname{range}(T - \lambda_m I) \cap \operatorname{null}(T - \lambda_m I)$. Then $Tu = \lambda_m u$. Now 5.64 implies that

$$\begin{aligned} 0 &= (T - \lambda_1 I)\cdots(T - \lambda_{m-1} I)u \\ &= (\lambda_m - \lambda_1)\cdots(\lambda_m - \lambda_{m-1})u. \end{aligned}$$

Because $\lambda_1, \ldots, \lambda_m$ are distinct, the equation above implies that $u = 0$. Hence $\operatorname{range}(T - \lambda_m I) \cap \operatorname{null}(T - \lambda_m I) = \{0\}$.

Thus $\operatorname{range}(T - \lambda_m I) + \operatorname{null}(T - \lambda_m I)$ is a direct sum (by 1.46) whose dimension is $\dim V$ (by 3.94 and 3.21). Hence $\operatorname{range}(T - \lambda_m I) \oplus \operatorname{null}(T - \lambda_m I) = V$. Every vector in $\operatorname{null}(T - \lambda_m I)$ is an eigenvector of $T$ with eigenvalue $\lambda_m$. Earlier in this proof we saw that there is a basis of $\operatorname{range}(T - \lambda_m I)$ consisting of eigenvectors of $T$. Adjoining to that basis a basis of $\operatorname{null}(T - \lambda_m I)$ gives a basis of $V$ consisting of eigenvectors of $T$. The matrix of $T$ with respect to this basis is a diagonal matrix, as desired. ∎

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No formula exists for the zeros of polynomials of degree 5 or greater. However, the previous result can be used to determine whether an operator on a complex vector space is diagonalizable without even finding approximations of the zeros of the minimal polynomial—see Exercise 15.

The next result will be a key tool when we prove a result about the simultaneous diagonalization of two operators; see 5.76. Note how the use of the characterization of diagonalizable operators in terms of the minimal polynomial (see 5.62) leads to a short proof of the next result.

**5.65** *restriction of diagonalizable operator to invariant subspace*

Suppose $T \in \mathcal{L}(V)$ is diagonalizable and $U$ is a subspace of $V$ that is invariant under $T$. Then $T|_U$ is a diagonalizable operator on $U$.

Proof Because the operator $T$ is diagonalizable, the minimal polynomial of $T$ equals $(z - \lambda_1) \cdots (z - \lambda_m)$ for some list of distinct numbers $\lambda_1, \ldots, \lambda_m \in \mathbf{F}$ (by 5.62). The minimal polynomial of $T$ is a polynomial multiple of the minimal polynomial of $T|_U$ (by 5.31). Hence the minimal polynomial of $T|_U$ has the form required by 5.62, which shows that $T|_U$ is diagonalizable. ∎

## *Gershgorin Disk Theorem*

**5.66** definition: *Gershgorin disks*

Suppose $T \in \mathcal{L}(V)$ and $v_1, \ldots, v_n$ is a basis of $V$. Let $A$ denote the matrix of $T$ with respect to this basis. A *Gershgorin disk* of $T$ with respect to the basis $v_1, \ldots, v_n$ is a set of the form

$$\left\{ z \in \mathbf{F} : |z - A_{j,j}| \le \sum_{\substack{k=1 \\ k \ne j}}^{n} |A_{j,k}| \right\},$$

where $j \in \{1, \ldots, n\}$.

Because there are $n$ choices for $j$ in the definition above, $T$ has $n$ Gershgorin disks. If $\mathbf{F} = \mathbf{C}$, then for each $j \in \{1, \ldots, n\}$, the corresponding Gershgorin disk is a closed disk in $\mathbf{C}$ centered at $A_{j,j}$, which is the $j^{\text{th}}$ entry on the diagonal of $A$. The radius of this closed disk is the sum of the absolute values of the entries in row $j$ of $A$, excluding the diagonal entry. If $\mathbf{F} = \mathbf{R}$, then the Gershgorin disks are closed intervals in $\mathbf{R}$.

In the special case that the square matrix $A$ above is a diagonal matrix, each Gershgorin disk consists of a single point that is a diagonal entry of $A$ (and each eigenvalue of $T$ is one of those points, as required by the next result). One consequence of our next result is that if the nondiagonal entries of $A$ are small, then each eigenvalue of $T$ is near a diagonal entry of $A$.

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**5.67 *Gershgorin disk theorem***

Suppose $T \in \mathcal{L}(V)$ and $v_1, ..., v_n$ is a basis of $V$. Then each eigenvalue of $T$ is contained in some Gershgorin disk of $T$ with respect to the basis $v_1, ..., v_n$.

Proof Suppose $\lambda \in \mathbf{F}$ is an eigenvalue of $T$. Let $w \in V$ be a corresponding eigenvector. There exist $c_1, ..., c_n \in \mathbf{F}$ such that

5.68 $$w = c_1 v_1 + \cdots + c_n v_n.$$

Let $A$ denote the matrix of $T$ with respect to the basis $v_1, ..., v_n$. Applying $T$ to both sides of the equation above gives

5.69 $$\begin{aligned} \lambda w &= \sum_{k=1}^{n} c_k T v_k \\ &= \sum_{k=1}^{n} c_k \sum_{j=1}^{n} A_{j,k} v_j \end{aligned}$$

5.70 $$= \sum_{j=1}^{n} \Big( \sum_{k=1}^{n} A_{j,k} c_k \Big) v_j.$$

Let $j \in \{1, ..., n\}$ be such that

$$|c_j| = \max\{|c_1|, ..., |c_n|\}.$$

Using 5.68, we see that the coefficient of $v_j$ on the left side of 5.69 equals $\lambda c_j$, which must equal the coefficient of $v_j$ on the right side of 5.70. In other words,

$$\lambda c_j = \sum_{k=1}^{n} A_{j,k} c_k.$$

Subtract $A_{j,j} c_j$ from each side of the equation above and then divide both sides by $c_j$ to get

$$\begin{aligned} |\lambda - A_{j,j}| &= \left| \sum_{\substack{k=1 \\ k \neq j}}^{n} A_{j,k} \frac{c_k}{c_j} \right| \\ &\leq \sum_{\substack{k=1 \\ k \neq j}}^{n} |A_{j,k}|. \end{aligned}$$

Thus $\lambda$ is in the $j^{\text{th}}$ Gershgorin disk with respect to the basis $v_1, ..., v_n$. ∎

Exercise 22 gives a nice application of the Gershgorin disk theorem.

Exercise 23 states that the radius of each Gershgorin disk could be changed to the sum of the absolute values of corresponding column entries (instead of row entries), excluding the diagonal entry, and the theorem above would still hold.

*The Gershgorin disk theorem is named for Semyon Aronovich Gershgorin, who published this result in 1931.*

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## *Exercises 5D*

**1** Suppose $V$ is a finite-dimensional complex vector space and $T \in \mathcal{L}(V)$.

(a) Prove that if $T^4 = I$, then $T$ is diagonalizable.

(b) Prove that if $T^4 = T$, then $T$ is diagonalizable.

(c) Give an example of an operator $T \in \mathcal{L}(\mathbf{C}^2)$ such that $T^4 = T^2$ and $T$ is not diagonalizable.

**2** Suppose $T \in \mathcal{L}(V)$ has a diagonal matrix $A$ with respect to some basis of $V$. Prove that if $\lambda \in \mathbf{F}$, then $\lambda$ appears on the diagonal of $A$ precisely $\dim E(\lambda, T)$ times.

**3** Suppose $V$ is finite-dimensional and $T \in \mathcal{L}(V)$. Prove that if the operator $T$ is diagonalizable, then $V = \operatorname{null} T \oplus \operatorname{range} T$.

**4** Suppose $V$ is finite-dimensional and $T \in \mathcal{L}(V)$. Prove that the following are equivalent.

(a) $V = \operatorname{null} T \oplus \operatorname{range} T$.

(b) $V = \operatorname{null} T + \operatorname{range} T$.

(c) $\operatorname{null} T \cap \operatorname{range} T = \{0\}$.

**5** Suppose $V$ is a finite-dimensional complex vector space and $T \in \mathcal{L}(V)$. Prove that $T$ is diagonalizable if and only if

$$V = \operatorname{null}(T - \lambda I) \oplus \operatorname{range}(T - \lambda I)$$

for every $\lambda \in \mathbf{C}$.

**6** Suppose $T \in \mathcal{L}(\mathbf{F}^5)$ and $\dim E(8, T) = 4$. Prove that $T - 2I$ or $T - 6I$ is invertible.

**7** Suppose $T \in \mathcal{L}(V)$ is invertible. Prove that

$$E(\lambda, T) = E\Big(\tfrac{1}{\lambda}, T^{-1}\Big)$$

for every $\lambda \in \mathbf{F}$ with $\lambda \neq 0$.

**8** Suppose $V$ is finite-dimensional and $T \in \mathcal{L}(V)$. Let $\lambda_1, \ldots, \lambda_m$ denote the distinct nonzero eigenvalues of $T$. Prove that

$$\dim E(\lambda_1, T) + \cdots + \dim E(\lambda_m, T) \leq \dim \operatorname{range} T.$$

**9** Suppose $R, T \in \mathcal{L}(\mathbf{F}^3)$ each have 2, 6, 7 as eigenvalues. Prove that there exists an invertible operator $S \in \mathcal{L}(\mathbf{F}^3)$ such that $R = S^{-1}TS$.

**10** Find $R, T \in \mathcal{L}(\mathbf{F}^4)$ such that $R$ and $T$ each have 2, 6, 7 as eigenvalues, $R$ and $T$ have no other eigenvalues, and there does not exist an invertible operator $S \in \mathcal{L}(\mathbf{F}^4)$ such that $R = S^{-1}TS$.

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**11** Find $T \in \mathcal{L}(\mathbf{C}^3)$ such that 6 and 7 are eigenvalues of $T$ and such that $T$ does not have a diagonal matrix with respect to any basis of $\mathbf{C}^3$.

**12** Suppose $T \in \mathcal{L}(\mathbf{C}^3)$ is such that 6 and 7 are eigenvalues of $T$. Furthermore, suppose $T$ does not have a diagonal matrix with respect to any basis of $\mathbf{C}^3$. Prove that there exists $(z_1, z_2, z_3) \in \mathbf{C}^3$ such that

$$T(z_1, z_2, z_3) = (6 + 8z_1, 7 + 8z_2, 13 + 8z_3).$$

**13** Suppose $A$ is a diagonal matrix with distinct entries on the diagonal and $B$ is a matrix of the same size as $A$. Show that $AB = BA$ if and only if $B$ is a diagonal matrix.

**14** (a) Give an example of a finite-dimensional complex vector space and an operator $T$ on that vector space such that $T^2$ is diagonalizable but $T$ is not diagonalizable.

(b) Suppose $\mathbf{F} = \mathbf{C}$, $k$ is a positive integer, and $T \in \mathcal{L}(V)$ is invertible. Prove that $T$ is diagonalizable if and only if $T^k$ is diagonalizable.

**15** Suppose $V$ is a finite-dimensional complex vector space, $T \in \mathcal{L}(V)$, and $p$ is the minimal polynomial of $T$. Prove that the following are equivalent.

(a) $T$ is diagonalizable.

(b) There does not exist $\lambda \in \mathbf{C}$ such that $p$ is a polynomial multiple of $(z - \lambda)^2$.

(c) $p$ and its derivative $p'$ have no zeros in common.

(d) The greatest common divisor of $p$ and $p'$ is the constant polynomial 1.

*The **greatest common divisor** of $p$ and $p'$ is the monic polynomial $q$ of largest degree such that $p$ and $p'$ are both polynomial multiples of $q$. The Euclidean algorithm for polynomials (look it up) can quickly determine the greatest common divisor of two polynomials, without requiring any information about the zeros of the polynomials. Thus the equivalence of (a) and (d) above shows that we can determine whether $T$ is diagonalizable without knowing anything about the zeros of $p$.*

**16** Suppose that $T \in \mathcal{L}(V)$ is diagonalizable. Let $\lambda_1, \ldots, \lambda_m$ denote the distinct eigenvalues of $T$. Prove that a subspace $U$ of $V$ is invariant under $T$ if and only if there exist subspaces $U_1, \ldots, U_m$ of $V$ such that $U_k \subseteq E(\lambda_k, T)$ for each $k$ and $U = U_1 \oplus \cdots \oplus U_m$.

**17** Suppose $V$ is finite-dimensional. Prove that $\mathcal{L}(V)$ has a basis consisting of diagonalizable operators.

**18** Suppose that $T \in \mathcal{L}(V)$ is diagonalizable and $U$ is a subspace of $V$ that is invariant under $T$. Prove that the quotient operator $T/U$ is a diagonalizable operator on $V/U$.

*The quotient operator $T/U$ was defined in Exercise 38 in Section 5A.*

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**19** Prove or give a counterexample: If $T \in \mathcal{L}(V)$ and there exists a subspace $U$ of $V$ that is invariant under $T$ such that $T|_U$ and $T/U$ are both diagonalizable, then $T$ is diagonalizable.

*See Exercise 13 in Section 5C for an analogous statement about upper-triangular matrices.*

**20** Suppose $V$ is finite-dimensional and $T \in \mathcal{L}(V)$. Prove that $T$ is diagonalizable if and only if the dual operator $T'$ is diagonalizable.

**21** The *Fibonacci sequence* $F_0, F_1, F_2, \ldots$ is defined by

$$F_0 = 0, \ F_1 = 1, \text{ and } F_n = F_{n-2} + F_{n-1} \text{ for } n \geq 2.$$

Define $T \in \mathcal{L}(\mathbf{R}^2)$ by $T(x, y) = (y, x + y)$.

(a) Show that $T^n(0, 1) = (F_n, F_{n+1})$ for each nonnegative integer $n$.
(b) Find the eigenvalues of $T$.
(c) Find a basis of $\mathbf{R}^2$ consisting of eigenvectors of $T$.
(d) Use the solution to (c) to compute $T^n(0, 1)$. Conclude that

$$F_n = \frac{1}{\sqrt{5}}\left[\left(\frac{1+\sqrt{5}}{2}\right)^n - \left(\frac{1-\sqrt{5}}{2}\right)^n\right]$$

for each nonnegative integer $n$.
(e) Use (d) to conclude that if $n$ is a nonnegative integer, then the Fibonacci number $F_n$ is the integer that is closest to

$$\frac{1}{\sqrt{5}}\left(\frac{1+\sqrt{5}}{2}\right)^n.$$

*Each $F_n$ is a nonnegative integer, even though the right side of the formula in (d) does not look like an integer. The number*

$$\frac{1+\sqrt{5}}{2}$$

*is called the* ***golden ratio****.*

**22** Suppose $T \in \mathcal{L}(V)$ and $A$ is an $n$-by-$n$ matrix that is the matrix of $T$ with respect to some basis of $V$. Prove that if

$$|A_{j,j}| > \sum_{\substack{k=1 \\ k \neq j}}^{n} |A_{j,k}|$$

for each $j \in \{1, \ldots, n\}$, then $T$ is invertible.

*This exercise states that if the diagonal entries of the matrix of $T$ are large compared to the nondiagonal entries, then $T$ is invertible.*

**23** Suppose the definition of the Gershgorin disks is changed so that the radius of the $k^{\text{th}}$ disk is the sum of the absolute values of the entries in column (instead of row) $k$ of $A$, excluding the diagonal entry. Show that the Gershgorin disk theorem (5.67) still holds with this changed definition.

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## 5E Commuting Operators

**5.71 definition: *commute***

- Two operators $S$ and $T$ on the same vector space *commute* if $ST = TS$.
- Two square matrices $A$ and $B$ of the same size *commute* if $AB = BA$.

For example, if $T$ is an operator and $p, q \in \mathcal{P}(\mathbf{F})$, then $p(T)$ and $q(T)$ commute [see 5.17(b)].

As another example, if $I$ is the identity operator on $V$, then $I$ commutes with every operator on $V$.

**5.72 example: *partial differentiation operators commute***

Suppose $m$ is a nonnegative integer. Let $\mathcal{P}_m(\mathbf{R}^2)$ denote the real vector space of polynomials (with real coefficients) in two real variables and of degree at most $m$, with the usual operations of addition and scalar multiplication of real-valued functions. Thus the elements of $\mathcal{P}_m(\mathbf{R}^2)$ are functions $p$ on $\mathbf{R}^2$ of the form

$$p = \sum_{j+k \le m} a_{j,k} x^j y^k, \tag{5.73}$$

where the indices $j$ and $k$ take on all nonnegative integer values such that $j + k \le m$, each $a_{j,k}$ is in $\mathbf{R}$, and $x^j y^k$ denotes the function on $\mathbf{R}^2$ defined by $(x, y) \mapsto x^j y^k$.

Define operators $D_x, D_y \in \mathcal{L}(\mathcal{P}_m(\mathbf{R}^2))$ by

$$D_x p = \frac{\partial p}{\partial x} = \sum_{j+k \le m} j a_{j,k} x^{j-1} y^k \quad \text{and} \quad D_y p = \frac{\partial p}{\partial y} = \sum_{j+k \le m} k a_{j,k} x^j y^{k-1},$$

where $p$ is as in 5.73. The operators $D_x$ and $D_y$ are called partial differentiation operators because each of these operators differentiates with respect to one of the variables while pretending that the other variable is a constant.

The operators $D_x$ and $D_y$ commute because if $p$ is as in 5.73, then

$$(D_x D_y)p = \sum_{j+k \le m} jk a_{j,k} x^{j-1} y^{k-1} = (D_y D_x)p.$$

The equation $D_x D_y = D_y D_x$ on $\mathcal{P}_m(\mathbf{R}^2)$ illustrates a more general result that the order of partial differentiation does not matter for nice functions.

---

Commuting matrices are unusual. For example, there are 214,358,881 pairs of 2-by-2 matrices all of whose entries are integers in the interval $[-5, 5]$. Only about 0.3% of these pairs of matrices commute.

*All 214,358,881 (which equals $11^8$) pairs of the 2-by-2 matrices under consideration were checked by a computer to discover that only 674,609 of these pairs of matrices commute.*

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The next result shows that two operators commute if and only if their matrices (with respect to the same basis) commute.

**5.74** *commuting operators correspond to commuting matrices*

Suppose $S, T \in \mathcal{L}(V)$ and $v_1, \ldots, v_n$ is a basis of $V$. Then $S$ and $T$ commute if and only if $\mathcal{M}(S, (v_1, \ldots, v_n))$ and $\mathcal{M}(T, (v_1, \ldots, v_n))$ commute.

Proof We have

$$\begin{aligned} S \text{ and } T \text{ commute} &\iff ST = TS \\ &\iff \mathcal{M}(ST) = \mathcal{M}(TS) \\ &\iff \mathcal{M}(S)\mathcal{M}(T) = \mathcal{M}(T)\mathcal{M}(S) \\ &\iff \mathcal{M}(S) \text{ and } \mathcal{M}(T) \text{ commute,} \end{aligned}$$

as desired. ∎

The next result shows that if two operators commute, then every eigenspace for one operator is invariant under the other operator. This result, which we will use several times, is one of the main reasons why a pair of commuting operators behaves better than a pair of operators that does not commute.

**5.75** *eigenspace is invariant under commuting operator*

Suppose $S, T \in \mathcal{L}(V)$ commute and $\lambda \in \mathbf{F}$. Then $E(\lambda, S)$ is invariant under $T$.

Proof Suppose $v \in E(\lambda, S)$. Then

$$S(Tv) = (ST)v = (TS)v = T(Sv) = T(\lambda v) = \lambda Tv.$$

The equation above shows that $Tv \in E(\lambda, S)$. Thus $E(\lambda, S)$ is invariant under $T$. ∎

Suppose we have two operators, each of which is diagonalizable. If we want to do computations involving both operators (for example, involving their sum), then we want the two operators to be diagonalizable by the same basis, which according to the next result is possible when the two operators commute.

**5.76** *simultaneous diagonalizablity ⟺ commutativity*

Two diagonalizable operators on the same vector space have diagonal matrices with respect to the same basis if and only if the two operators commute.

Proof First suppose $S, T \in \mathcal{L}(V)$ have diagonal matrices with respect to the same basis. The product of two diagonal matrices of the same size is the diagonal matrix obtained by multiplying the corresponding elements of the two diagonals. Thus any two diagonal matrices of the same size commute. Thus $S$ and $T$ commute, by 5.74.

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To prove the implication in the other direction, now suppose that $S, T \in \mathcal{L}(V)$ are diagonalizable operators that commute. Let $\lambda_1, \ldots, \lambda_m$ denote the distinct eigenvalues of $S$. Because $S$ is diagonalizable, 5.55(c) shows that

5.77 $$V = E(\lambda_1, S) \oplus \cdots \oplus E(\lambda_m, S).$$

For each $k = 1, \ldots, m$, the subspace $E(\lambda_k, S)$ is invariant under $T$ (by 5.75). Because $T$ is diagonalizable, 5.65 implies that $T|_{E(\lambda_k, S)}$ is diagonalizable for each $k$. Hence for each $k = 1, \ldots, m$, there is a basis of $E(\lambda_k, S)$ consisting of eigenvectors of $T$. Putting these bases together gives a basis of $V$ (because of 5.77), with each vector in this basis being an eigenvector of both $S$ and $T$. Thus $S$ and $T$ both have diagonal matrices with respect to this basis, as desired. ∎

See Exercise 2 for an extension of the result above to more than two operators.

Suppose $V$ is a finite-dimensional nonzero complex vector space. Then every operator on $V$ has an eigenvector (see 5.19). The next result shows that if two operators on $V$ commute, then there is a vector in $V$ that is an eigenvector for both operators (but the two commuting operators might not have a common eigenvalue). For an extension of the next result to more than two operators, see Exercise 9(a).

**5.78** *common eigenvector for commuting operators*

Every pair of commuting operators on a finite-dimensional nonzero complex vector space has a common eigenvector.

Proof Suppose $V$ is a finite-dimensional nonzero complex vector space and $S, T \in \mathcal{L}(V)$ commute. Let $\lambda$ be an eigenvalue of $S$ (5.19 tells us that $S$ does indeed have an eigenvalue). Thus $E(\lambda, S) \neq \{0\}$. Also, $E(\lambda, S)$ is invariant under $T$ (by 5.75).

Thus $T|_{E(\lambda, S)}$ has an eigenvector (again using 5.19), which is an eigenvector for both $S$ and $T$, completing the proof. ∎

**5.79** example: *common eigenvector for partial differentiation operators*

Let $\mathcal{P}_m(\mathbf{R}^2)$ be as in Example 5.72 and let $D_x, D_y \in \mathcal{L}\big(\mathcal{P}_m(\mathbf{R}^2)\big)$ be the commuting partial differentiation operators in that example. As you can verify, 0 is the only eigenvalue of each of these operators. Also

$$E(0, D_x) = \left\{ \sum_{k=0}^{m} a_k y^k : a_0, \ldots, a_m \in \mathbf{R} \right\},$$

$$E(0, D_y) = \left\{ \sum_{j=0}^{m} c_j x^j : c_0, \ldots, c_m \in \mathbf{R} \right\}.$$

The intersection of these two eigenspaces is the set of common eigenvectors of the two operators. Because $E(0, D_x) \cap E(0, D_y)$ is the set of constant functions, we see that $D_x$ and $D_y$ indeed have a common eigenvector, as promised by 5.78.

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The next result extends 5.47 (the existence of a basis that gives an upper-triangular matrix) to two commuting operators.

**5.80** *commuting operators are simultaneously upper triangularizable*

Suppose $V$ is a finite-dimensional complex vector space and $S, T$ are commuting operators on $V$. Then there is a basis of $V$ with respect to which both $S$ and $T$ have upper-triangular matrices.

Proof Let $n = \dim V$. We will use induction on $n$. The desired result holds if $n = 1$ because all 1-by-1 matrices are upper triangular. Now suppose $n > 1$ and the desired result holds for all complex vector spaces whose dimension is $n - 1$.

Let $v_1$ be any common eigenvector of $S$ and $T$ (using 5.78). Hence $Sv_1 \in \operatorname{span}(v_1)$ and $Tv_1 \in \operatorname{span}(v_1)$. Let $W$ be a subspace of $V$ such that

$$V = \operatorname{span}(v_1) \oplus W;$$

see 2.33 for the existence of $W$. Define a linear map $P : V \to W$ by

$$P(av_1 + w) = w$$

for each $a \in \mathbf{C}$ and each $w \in W$. Define $\hat{S}, \hat{T} \in \mathcal{L}(W)$ by

$$\hat{S}w = P(Sw) \quad \text{and} \quad \hat{T}w = P(Tw)$$

for each $w \in W$. To apply our induction hypothesis to $\hat{S}$ and $\hat{T}$, we must first show that these two operators on $W$ commute. To do this, suppose $w \in W$. Then there exists $a \in \mathbf{C}$ such that

$$(\hat{S}\hat{T})w = \hat{S}(P(Tw)) = \hat{S}(Tw - av_1) = P(S(Tw - av_1)) = P((ST)w),$$

where the last equality holds because $v_1$ is an eigenvector of $S$ and $Pv_1 = 0$. Similarly,

$$(\hat{T}\hat{S})w = P((TS)w).$$

Because the operators $S$ and $T$ commute, the last two displayed equations show that $(\hat{S}\hat{T})w = (\hat{T}\hat{S})w$. Hence $\hat{S}$ and $\hat{T}$ commute.

Thus we can use our induction hypothesis to state that there exists a basis $v_2, \ldots, v_n$ of $W$ such that $\hat{S}$ and $\hat{T}$ both have upper-triangular matrices with respect to this basis. The list $v_1, \ldots, v_n$ is a basis of $V$.

If $k \in \{2, \ldots, n\}$, then there exist $a_k, b_k \in \mathbf{C}$ such that

$$Sv_k = a_k v_1 + \hat{S}v_k \quad \text{and} \quad Tv_k = b_k v_1 + \hat{T}v_k.$$

Because $\hat{S}$ and $\hat{T}$ have upper-triangular matrices with respect to $v_2, \ldots, v_n$, we know that $\hat{S}v_k \in \operatorname{span}(v_2, \ldots, v_k)$ and $\hat{T}v_k \in \operatorname{span}(v_2, \ldots, v_k)$. Hence the equations above imply that

$$Sv_k \in \operatorname{span}(v_1, \ldots, v_k) \quad \text{and} \quad Tv_k \in \operatorname{span}(v_1, \ldots, v_k).$$

Thus $S$ and $T$ have upper-triangular matrices with respect to $v_1, \ldots, v_n$, as desired. ∎

Exercise 9(b) extends the result above to more than two operators.

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In general, it is not possible to determine the eigenvalues of the sum or product of two operators from the eigenvalues of the two operators. However, the next result shows that something nice happens when the two operators commute.

**5.81** *eigenvalues of sum and product of commuting operators*

Suppose $V$ is a finite-dimensional complex vector space and $S, T$ are commuting operators on $V$. Then

- every eigenvalue of $S + T$ is an eigenvalue of $S$ plus an eigenvalue of $T$,
- every eigenvalue of $ST$ is an eigenvalue of $S$ times an eigenvalue of $T$.

Proof There is a basis of $V$ with respect to which both $S$ and $T$ have upper-triangular matrices (by 5.80). With respect to that basis,

$$\mathcal{M}(S + T) = \mathcal{M}(S) + \mathcal{M}(T) \quad \text{and} \quad \mathcal{M}(ST) = \mathcal{M}(S)\mathcal{M}(T),$$

as stated in 3.35 and 3.43.

The definition of matrix addition shows that each entry on the diagonal of $\mathcal{M}(S + T)$ equals the sum of the corresponding entries on the diagonals of $\mathcal{M}(S)$ and $\mathcal{M}(T)$. Similarly, because $\mathcal{M}(S)$ and $\mathcal{M}(T)$ are upper-triangular matrices, the definition of matrix multiplication shows that each entry on the diagonal of $\mathcal{M}(ST)$ equals the product of the corresponding entries on the diagonals of $\mathcal{M}(S)$ and $\mathcal{M}(T)$. Furthermore, $\mathcal{M}(S + T)$ and $\mathcal{M}(ST)$ are upper-triangular matrices (see Exercise 2 in Section 5B).

Every entry on the diagonal of $\mathcal{M}(S)$ is an eigenvalue of $S$, and every entry on the diagonal of $\mathcal{M}(T)$ is an eigenvalue of $T$ (by 5.41). Every eigenvalue of $S + T$ is on the diagonal of $\mathcal{M}(S + T)$, and every eigenvalue of $ST$ is on the diagonal of $\mathcal{M}(ST)$ (these assertions follow from 5.41). Putting all this together, we conclude that every eigenvalue of $S + T$ is an eigenvalue of $S$ plus an eigenvalue of $T$, and every eigenvalue of $ST$ is an eigenvalue of $S$ times an eigenvalue of $T$. ∎

## *Exercises 5E*

**1** Give an example of two commuting operators $S, T$ on $\mathbf{F}^4$ such that there is a subspace of $\mathbf{F}^4$ that is invariant under $S$ but not under $T$ and there is a subspace of $\mathbf{F}^4$ that is invariant under $T$ but not under $S$.

**2** Suppose $\mathcal{E}$ is a subset of $\mathcal{L}(V)$ and every element of $\mathcal{E}$ is diagonalizable. Prove that there exists a basis of $V$ with respect to which every element of $\mathcal{E}$ has a diagonal matrix if and only if every pair of elements of $\mathcal{E}$ commutes.

*This exercise extends 5.76, which considers the case in which $\mathcal{E}$ contains only two elements. For this exercise, $\mathcal{E}$ may contain any number of elements, and $\mathcal{E}$ may even be an infinite set.*

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**3** Suppose $S, T \in \mathcal{L}(V)$ are such that $ST = TS$. Suppose $p \in \mathcal{P}(\mathbf{F})$.

(a) Prove that $\operatorname{null} p(S)$ is invariant under $T$.

(b) Prove that $\operatorname{range} p(S)$ is invariant under $T$.

*See 5.18 for the special case $S = T$.*

**4** Prove or give a counterexample: If $A$ is a diagonal matrix and $B$ is an upper-triangular matrix of the same size as $A$, then $A$ and $B$ commute.

**5** Prove that a pair of operators on a finite-dimensional vector space commute if and only if their dual operators commute.

*See 3.118 for the definition of the dual of an operator.*

**6** Suppose $V$ is a finite-dimensional complex vector space and $S, T \in \mathcal{L}(V)$ commute. Prove that there exist $\alpha, \lambda \in \mathbf{C}$ such that

$$\operatorname{range}(S - \alpha I) + \operatorname{range}(T - \lambda I) \neq V.$$

**7** Suppose $V$ is a complex vector space, $S \in \mathcal{L}(V)$ is diagonalizable, and $T \in \mathcal{L}(V)$ commutes with $S$. Prove that there is a basis of $V$ such that $S$ has a diagonal matrix with respect to this basis and $T$ has an upper-triangular matrix with respect to this basis.

**8** Suppose $m = 3$ in Example 5.72 and $D_x, D_y$ are the commuting partial differentiation operators on $\mathcal{P}_3(\mathbf{R}^2)$ from that example. Find a basis of $\mathcal{P}_3(\mathbf{R}^2)$ with respect to which $D_x$ and $D_y$ each have an upper-triangular matrix.

**9** Suppose $V$ is a finite-dimensional nonzero complex vector space. Suppose that $\mathcal{E} \subseteq \mathcal{L}(V)$ is such that $S$ and $T$ commute for all $S, T \in \mathcal{E}$.

(a) Prove that there is a vector in $V$ that is an eigenvector for every element of $\mathcal{E}$.

(b) Prove that there is a basis of $V$ with respect to which every element of $\mathcal{E}$ has an upper-triangular matrix.

*This exercise extends 5.78 and 5.80, which consider the case in which $\mathcal{E}$ contains only two elements. For this exercise, $\mathcal{E}$ may contain any number of elements, and $\mathcal{E}$ may even be an infinite set.*

**10** Give an example of two commuting operators $S, T$ on a finite-dimensional real vector space such that $S + T$ has a eigenvalue that does not equal an eigenvalue of $S$ plus an eigenvalue of $T$ and $ST$ has a eigenvalue that does not equal an eigenvalue of $S$ times an eigenvalue of $T$.

*This exercise shows that 5.81 does not hold on real vector spaces.*

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Chapter 6

# *Inner Product Spaces*

In making the definition of a vector space, we generalized the linear structure (addition and scalar multiplication) of $\mathbf{R}^2$ and $\mathbf{R}^3$. We ignored geometric features such as the notions of length and angle. These ideas are embedded in the concept of inner products, which we will investigate in this chapter.

Every inner product induces a norm, which you can think of as a length. This norm satisfies key properties such as the Pythagorean theorem, the triangle inequality, the parallelogram equality, and the Cauchy–Schwarz inequality.

The notion of perpendicular vectors in Euclidean geometry gets renamed to orthogonal vectors in the context of an inner product space. We will see that orthonormal bases are tremendously useful in inner product spaces. The Gram–Schmidt procedure constructs such bases. This chapter will conclude by putting together these tools to solve minimization problems.

*standing assumptions for this chapter*

- $\mathbf{F}$ denotes $\mathbf{R}$ or $\mathbf{C}$.
- $V$ and $W$ denote vector spaces over $\mathbf{F}$.



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*The George Peabody Library, now part of Johns Hopkins University, opened while James Sylvester (1814–1897) was the university's first mathematics professor. Sylvester's publications include the first use of the word* ***matrix*** *in mathematics.*

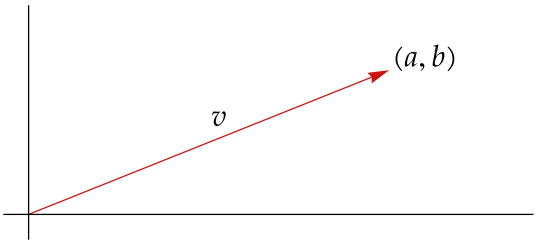
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# 6A Inner Products and Norms

## Inner Products

To motivate the concept of inner product, think of vectors in $\mathbf{R}^2$ and $\mathbf{R}^3$ as arrows with initial point at the origin. The length of a vector $v$ in $\mathbf{R}^2$ or $\mathbf{R}^3$ is called the *norm* of $v$ and is denoted by $\|v\|$. Thus for $v = (a, b) \in \mathbf{R}^2$, we have

$$\|v\| = \sqrt{a^2 + b^2}.$$




*This vector $v$ has norm $\sqrt{a^2 + b^2}$.*

Similarly, if $v = (a, b, c) \in \mathbf{R}^3$, then $\|v\| = \sqrt{a^2 + b^2 + c^2}$.

Even though we cannot draw pictures in higher dimensions, the generalization to $\mathbf{R}^n$ is easy: we define the norm of $x = (x_1, \ldots, x_n) \in \mathbf{R}^n$ by

$$\|x\| = \sqrt{x_1^2 + \cdots + x_n^2}.$$

The norm is not linear on $\mathbf{R}^n$. To inject linearity into the discussion, we introduce the dot product.

**6.1 definition: *dot product***

For $x, y \in \mathbf{R}^n$, the *dot product* of $x$ and $y$, denoted by $x \cdot y$, is defined by

$$x \cdot y = x_1 y_1 + \cdots + x_n y_n,$$

where $x = (x_1, \ldots, x_n)$ and $y = (y_1, \ldots, y_n)$.

The dot product of two vectors in $\mathbf{R}^n$ is a number, not a vector. Notice that $x \cdot x = \|x\|^2$ for all $x \in \mathbf{R}^n$. Furthermore, the dot product on $\mathbf{R}^n$ has the following properties.

*If we think of a vector as a point instead of as an arrow, then $\|x\|$ should be interpreted to mean the distance from the origin to the point $x$.*

- $x \cdot x \geq 0$ for all $x \in \mathbf{R}^n$.
- $x \cdot x = 0$ if and only if $x = 0$.
- For $y \in \mathbf{R}^n$ fixed, the map from $\mathbf{R}^n$ to $\mathbf{R}$ that sends $x \in \mathbf{R}^n$ to $x \cdot y$ is linear.
- $x \cdot y = y \cdot x$ for all $x, y \in \mathbf{R}^n$.

An inner product is a generalization of the dot product. At this point you may be tempted to guess that an inner product is defined by abstracting the properties of the dot product discussed in the last paragraph. For real vector spaces, that guess is correct. However, so that we can make a definition that will be useful for both real and complex vector spaces, we need to examine the complex case before making the definition.

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Recall that if $\lambda = a + bi$, where $a, b \in \mathbf{R}$, then

- the absolute value of $\lambda$, denoted by $|\lambda|$, is defined by $|\lambda| = \sqrt{a^2 + b^2}$;
- the complex conjugate of $\lambda$, denoted by $\overline{\lambda}$, is defined by $\overline{\lambda} = a - bi$;
- $|\lambda|^2 = \lambda\overline{\lambda}$.

See Chapter 4 for the definitions and the basic properties of the absolute value and complex conjugate.

For $z = (z_1, \ldots, z_n) \in \mathbf{C}^n$, we define the norm of $z$ by

$$\|z\| = \sqrt{|z_1|^2 + \cdots + |z_n|^2}.$$

The absolute values are needed because we want $\|z\|$ to be a nonnegative number. Note that

$$\|z\|^2 = z_1\overline{z_1} + \cdots + z_n\overline{z_n}.$$

We want to think of $\|z\|^2$ as the inner product of $z$ with itself, as we did in $\mathbf{R}^n$. The equation above thus suggests that the inner product of the vector $w = (w_1, \ldots, w_n) \in \mathbf{C}^n$ with $z$ should equal

$$w_1\overline{z_1} + \cdots + w_n\overline{z_n}.$$

If the roles of the $w$ and $z$ were interchanged, the expression above would be replaced with its complex conjugate. Thus we should expect that the inner product of $w$ with $z$ equals the complex conjugate of the inner product of $z$ with $w$. With that motivation, we are now ready to define an inner product on $V$, which may be a real or a complex vector space.

One comment about the notation used in the next definition:

- For $\lambda \in \mathbf{C}$, the notation $\lambda \geq 0$ means $\lambda$ is real and nonnegative.

**6.2 definition: *inner product***

An *inner product* on $V$ is a function that takes each ordered pair $(u, v)$ of elements of $V$ to a number $\langle u, v\rangle \in \mathbf{F}$ and has the following properties.

**positivity**
$\langle v, v\rangle \geq 0$ for all $v \in V$.

**definiteness**
$\langle v, v\rangle = 0$ if and only if $v = 0$.

**additivity in first slot**
$\langle u + v, w\rangle = \langle u, w\rangle + \langle v, w\rangle$ for all $u, v, w \in V$.

**homogeneity in first slot**
$\langle \lambda u, v\rangle = \lambda\langle u, v\rangle$ for all $\lambda \in \mathbf{F}$ and all $u, v \in V$.

**conjugate symmetry**
$\langle u, v\rangle = \overline{\langle v, u\rangle}$ for all $u, v \in V$.

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Every real number equals its complex conjugate. Thus if we are dealing with a real vector space, then in the last condition above we can dispense with the complex conjugate and simply state that $\langle u, v\rangle = \langle v, u\rangle$ for all $u, v \in V$.

*Most mathematicians define inner products as above, but many physicists use a definition that requires homogeneity in the second slot instead of the first slot.*

6.3 example: *inner products*

(a) The *Euclidean inner product* on $\mathbf{F}^n$ is defined by

$$\big\langle (w_1, ..., w_n), (z_1, ..., z_n)\big\rangle = w_1\overline{z_1} + \cdots + w_n\overline{z_n}$$

for all $(w_1, ..., w_n), (z_1, ..., z_n) \in \mathbf{F}^n$.

(b) If $c_1, ..., c_n$ are positive numbers, then an inner product can be defined on $\mathbf{F}^n$ by

$$\big\langle (w_1, ..., w_n), (z_1, ..., z_n)\big\rangle = c_1 w_1\overline{z_1} + \cdots + c_n w_n\overline{z_n}$$

for all $(w_1, ..., w_n), (z_1, ..., z_n) \in \mathbf{F}^n$.

(c) An inner product can be defined on the vector space of continuous real-valued functions on the interval $[-1, 1]$ by

$$\langle f, g\rangle = \int_{-1}^{1} fg$$

for all $f, g$ continuous real-valued functions on $[-1, 1]$.

(d) An inner product can be defined on $\mathcal{P}(\mathbf{R})$ by

$$\langle p, q\rangle = p(0)q(0) + \int_{-1}^{1} p'q'$$

for all $p, q \in \mathcal{P}(\mathbf{R})$.

(e) An inner product can be defined on $\mathcal{P}(\mathbf{R})$ by

$$\langle p, q\rangle = \int_{0}^{\infty} p(x)q(x)e^{-x}\,dx$$

for all $p, q \in \mathcal{P}(\mathbf{R})$.

6.4 definition: *inner product space*

An *inner product space* is a vector space $V$ along with an inner product on $V$.

The most important example of an inner product space is $\mathbf{F}^n$ with the Euclidean inner product given by (a) in the example above. When $\mathbf{F}^n$ is referred to as an inner product space, you should assume that the inner product is the Euclidean inner product unless explicitly told otherwise.

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So that we do not have to keep repeating the hypothesis that $V$ and $W$ are inner product spaces, we make the following assumption.

**6.5 notation: $V$, $W$**

For the rest of this chapter, $V$ and $W$ denote inner product spaces over $\mathbf{F}$.

Note the slight abuse of language here. An inner product space is a vector space along with an inner product on that vector space. When we say that a vector space $V$ is an inner product space, we are also thinking that an inner product on $V$ is lurking nearby or is clear from the context (or is the Euclidean inner product if the vector space is $\mathbf{F}^n$).

**6.6 *basic properties of an inner product***

(a) For each fixed $v \in V$, the function that takes $u \in V$ to $\langle u, v\rangle$ is a linear map from $V$ to $\mathbf{F}$.

(b) $\langle 0, v\rangle = 0$ for every $v \in V$.

(c) $\langle v, 0\rangle = 0$ for every $v \in V$.

(d) $\langle u, v + w\rangle = \langle u, v\rangle + \langle u, w\rangle$ for all $u, v, w \in V$.

(e) $\langle u, \lambda v\rangle = \overline{\lambda}\langle u, v\rangle$ for all $\lambda \in \mathbf{F}$ and all $u, v \in V$.

Proof

(a) For $v \in V$, the linearity of $u \mapsto \langle u, v\rangle$ follows from the conditions of additivity and homogeneity in the first slot in the definition of an inner product.

(b) Every linear map takes 0 to 0. Thus (b) follows from (a).

(c) If $v \in V$, then the conjugate symmetry property in the definition of an inner product and (b) show that $\langle v, 0\rangle = \overline{\langle 0, v\rangle} = \overline{0} = 0$.

(d) Suppose $u, v, w \in V$. Then

$$\begin{aligned}\langle u, v + w\rangle &= \overline{\langle v + w, u\rangle}\\ &= \overline{\langle v, u\rangle + \langle w, u\rangle}\\ &= \overline{\langle v, u\rangle} + \overline{\langle w, u\rangle}\\ &= \langle u, v\rangle + \langle u, w\rangle.\end{aligned}$$

(e) Suppose $\lambda \in \mathbf{F}$ and $u, v \in V$. Then

$$\begin{aligned}\langle u, \lambda v\rangle &= \overline{\langle \lambda v, u\rangle}\\ &= \overline{\lambda\langle v, u\rangle}\\ &= \overline{\lambda}\,\overline{\langle v, u\rangle}\\ &= \overline{\lambda}\langle u, v\rangle.\end{aligned}$$

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## *Norms*

Our motivation for defining inner products came initially from the norms of vectors on $\mathbf{R}^2$ and $\mathbf{R}^3$. Now we see that each inner product determines a norm.

**6.7 definition: *norm*, $\|v\|$**

For $v \in V$, the *norm* of $v$, denoted by $\|v\|$, is defined by

$$\|v\| = \sqrt{\langle v, v\rangle}.$$

**6.8 example: *norms***

(a) If $(z_1, \ldots, z_n) \in \mathbf{F}^n$ (with the Euclidean inner product), then

$$\|(z_1, \ldots, z_n)\| = \sqrt{|z_1|^2 + \cdots + |z_n|^2}.$$

(b) For $f$ in the vector space of continuous real-valued functions on $[-1, 1]$ and with inner product given as in 6.3(c), we have

$$\|f\| = \sqrt{\int_{-1}^{1} f^2}.$$

**6.9 *basic properties of the norm***

Suppose $v \in V$.

(a) $\|v\| = 0$ if and only if $v = 0$.

(b) $\|\lambda v\| = |\lambda| \, \|v\|$ for all $\lambda \in \mathbf{F}$.

Proof

(a) The desired result holds because $\langle v, v\rangle = 0$ if and only if $v = 0$.

(b) Suppose $\lambda \in \mathbf{F}$. Then

$$\begin{aligned} \|\lambda v\|^2 &= \langle \lambda v, \lambda v\rangle \\ &= \lambda \langle v, \lambda v\rangle \\ &= \lambda \overline{\lambda} \langle v, v\rangle \\ &= |\lambda|^2 \, \|v\|^2. \end{aligned}$$

Taking square roots now gives the desired equality. ∎

The proof of (b) in the result above illustrates a general principle: working with norms squared is usually easier than working directly with norms.

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Now we come to a crucial definition.

**6.10 definition: *orthogonal***

Two vectors $u, v \in V$ are called *orthogonal* if $\langle u, v \rangle = 0$.

*The word **orthogonal** comes from the Greek word **orthogonios**, which means right-angled.*

In the definition above, the order of the two vectors does not matter, because $\langle u, v \rangle = 0$ if and only if $\langle v, u \rangle = 0$. Instead of saying $u$ and $v$ are orthogonal, sometimes we say $u$ is orthogonal to $v$.

Exercise 15 asks you to prove that if $u, v$ are nonzero vectors in $\mathbf{R}^2$, then

$$\langle u, v \rangle = \|u\| \, \|v\| \cos \theta,$$

where $\theta$ is the angle between $u$ and $v$ (thinking of $u$ and $v$ as arrows with initial point at the origin). Thus two nonzero vectors in $\mathbf{R}^2$ are orthogonal (with respect to the Euclidean inner product) if and only if the cosine of the angle between them is 0, which happens if and only if the vectors are perpendicular in the usual sense of plane geometry. Thus you can think of the word *orthogonal* as a fancy word meaning *perpendicular*.

We begin our study of orthogonality with an easy result.

**6.11 *orthogonality and* 0**

(a) 0 is orthogonal to every vector in $V$.

(b) 0 is the only vector in $V$ that is orthogonal to itself.

Proof

(a) Recall that 6.6(b) states that $\langle 0, v \rangle = 0$ for every $v \in V$.

(b) If $v \in V$ and $\langle v, v \rangle = 0$, then $v = 0$ (by definition of inner product). ∎

For the special case $V = \mathbf{R}^2$, the next theorem was known over 3,500 years ago in Babylonia and then rediscovered and proved over 2,500 years ago in Greece. Of course, the proof below is not the original proof.

**6.12 *Pythagorean theorem***

Suppose $u, v \in V$. If $u$ and $v$ are orthogonal, then

$$\|u + v\|^2 = \|u\|^2 + \|v\|^2.$$

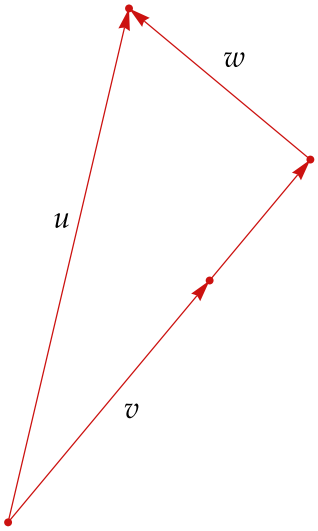
Proof Suppose $\langle u, v \rangle = 0$. Then

$$\begin{aligned} \|u + v\|^2 &= \langle u + v, u + v \rangle \\ &= \langle u, u \rangle + \langle u, v \rangle + \langle v, u \rangle + \langle v, v \rangle \\ &= \|u\|^2 + \|v\|^2. \end{aligned}$$

∎

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Suppose $u, v \in V$, with $v \neq 0$. We would like to write $u$ as a scalar multiple of $v$ plus a vector $w$ orthogonal to $v$, as suggested in the picture here.




*An orthogonal decomposition:*
*$u$ expressed as a scalar multiple of $v$ plus a vector orthogonal to $v$.*

To discover how to write $u$ as a scalar multiple of $v$ plus a vector orthogonal to $v$, let $c \in \mathbf{F}$ denote a scalar. Then

$$u = cv + (u - cv).$$

Thus we need to choose $c$ so that $v$ is orthogonal to $(u - cv)$. Hence we want

$$0 = \langle u - cv, v \rangle = \langle u, v \rangle - c\|v\|^2.$$

The equation above shows that we should choose $c$ to be $\langle u, v \rangle / \|v\|^2$. Making this choice of $c$, we can write

$$u = \frac{\langle u, v \rangle}{\|v\|^2} v + \left( u - \frac{\langle u, v \rangle}{\|v\|^2} v \right).$$

As you should verify, the equation displayed above explicitly writes $u$ as a scalar multiple of $v$ plus a vector orthogonal to $v$. Thus we have proved the following key result.

**6.13** *an orthogonal decomposition*

Suppose $u, v \in V$, with $v \neq 0$. Set $c = \dfrac{\langle u, v \rangle}{\|v\|^2}$ and $w = u - \dfrac{\langle u, v \rangle}{\|v\|^2} v$. Then

$$u = cv + w \quad \text{and} \quad \langle w, v \rangle = 0.$$

The orthogonal decomposition 6.13 will be used in the proof of the Cauchy–Schwarz inequality, which is our next result and is one of the most important inequalities in mathematics.

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**6.14 *Cauchy–Schwarz inequality***

Suppose $u, v \in V$. Then

$$|\langle u, v\rangle| \le \|u\| \, \|v\|.$$

This inequality is an equality if and only if one of $u, v$ is a scalar multiple of the other.

Proof If $v = 0$, then both sides of the desired inequality equal 0. Thus we can assume that $v \neq 0$. Consider the orthogonal decomposition

$$u = \frac{\langle u, v\rangle}{\|v\|^2} v + w$$

given by 6.13, where $w$ is orthogonal to $v$. By the Pythagorean theorem,

$$\begin{aligned} \|u\|^2 &= \left\| \frac{\langle u, v\rangle}{\|v\|^2} v \right\|^2 + \|w\|^2 \\ &= \frac{|\langle u, v\rangle|^2}{\|v\|^2} + \|w\|^2 \\ &\ge \frac{|\langle u, v\rangle|^2}{\|v\|^2}. \end{aligned} \tag{6.15}$$

Multiplying both sides of this inequality by $\|v\|^2$ and then taking square roots gives the desired inequality.

The proof in the paragraph above shows that the Cauchy–Schwarz inequality is an equality if and only if 6.15 is an equality. This happens if and only if $w = 0$. But $w = 0$ if and only if $u$ is a multiple of $v$ (see 6.13). Thus the Cauchy–Schwarz inequality is an equality if and only if $u$ is a scalar multiple of $v$ or $v$ is a scalar multiple of $u$ (or both; the phrasing has been chosen to cover cases in which either $u$ or $v$ equals 0).

*Augustin-Louis Cauchy (1789–1857) proved 6.16(a) in 1821. In 1859, Cauchy's student Viktor Bunyakovsky (1804–1889) proved integral inequalities like the one in 6.16(b). A few decades later, similar discoveries by Hermann Schwarz (1843–1921) attracted more attention and led to the name of this inequality.*

**6.16 example: *Cauchy–Schwarz inequality***

(a) If $x_1, \dots, x_n, y_1, \dots, y_n \in \mathbf{R}$, then

$$(x_1 y_1 + \cdots + x_n y_n)^2 \le (x_1^2 + \cdots + x_n^2)(y_1^2 + \cdots + y_n^2),$$

as follows from applying the Cauchy–Schwarz inequality to the vectors $(x_1, \dots, x_n), (y_1, \dots, y_n) \in \mathbf{R}^n$, using the usual Euclidean inner product.

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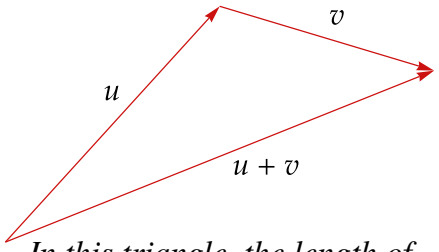
(b) If $f, g$ are continuous real-valued functions on $[-1, 1]$, then

$$\left|\int_{-1}^{1} fg\right|^2 \le \left(\int_{-1}^{1} f^2\right)\left(\int_{-1}^{1} g^2\right),$$

as follows from applying the Cauchy–Schwarz inequality to Example 6.3(c).

The next result, called the triangle inequality, has the geometric interpretation that the length of each side of a triangle is less than the sum of the lengths of the other two sides.

Note that the triangle inequality implies that the shortest polygonal path between two points is a single line segment (a polygonal path consists of line segments).




*In this triangle, the length of $u + v$ is less than the length of $u$ plus the length of $v$.*

**6.17 *triangle inequality***

Suppose $u, v \in V$. Then

$$\|u + v\| \le \|u\| + \|v\|.$$

This inequality is an equality if and only if one of $u, v$ is a nonnegative real multiple of the other.

Proof We have

$$\begin{aligned}
\|u + v\|^2 &= \langle u + v, u + v\rangle \\
&= \langle u, u\rangle + \langle v, v\rangle + \langle u, v\rangle + \langle v, u\rangle \\
&= \langle u, u\rangle + \langle v, v\rangle + \langle u, v\rangle + \overline{\langle u, v\rangle} \\
&= \|u\|^2 + \|v\|^2 + 2\,\mathrm{Re}\langle u, v\rangle \\
&\le \|u\|^2 + \|v\|^2 + 2|\langle u, v\rangle| \qquad 6.18 \\
&\le \|u\|^2 + \|v\|^2 + 2\|u\|\,\|v\| \qquad 6.19 \\
&= (\|u\| + \|v\|)^2,
\end{aligned}$$

where 6.19 follows from the Cauchy–Schwarz inequality (6.14). Taking square roots of both sides of the inequality above gives the desired inequality.

The proof above shows that the triangle inequality is an equality if and only if we have equality in 6.18 and 6.19. Thus we have equality in the triangle inequality if and only if

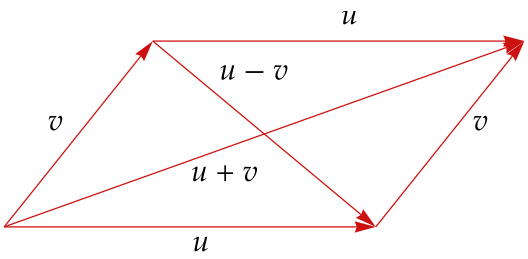
$$\langle u, v\rangle = \|u\|\,\|v\|. \qquad 6.20$$

If one of $u, v$ is a nonnegative real multiple of the other, then 6.20 holds. Conversely, suppose 6.20 holds. Then the condition for equality in the Cauchy–Schwarz inequality (6.14) implies that one of $u, v$ is a scalar multiple of the other. This scalar must be a nonnegative real number, by 6.20, completing the proof. ∎

For the reverse triangle inequality, see Exercise 20.

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The next result is called the parallelogram equality because of its geometric interpretation: in every parallelogram, the sum of the squares of the lengths of the diagonals equals the sum of the squares of the lengths of the four sides. Note that the proof here is more straightforward than the usual proof in Euclidean geometry.




*The diagonals of this parallelogram are $u + v$ and $u - v$.*

**6.21** *parallelogram equality*

Suppose $u, v \in V$. Then

$$\|u + v\|^2 + \|u - v\|^2 = 2(\|u\|^2 + \|v\|^2).$$

Proof We have

$$\begin{aligned}\|u + v\|^2 + \|u - v\|^2 &= \langle u + v, u + v\rangle + \langle u - v, u - v\rangle \\ &= \|u\|^2 + \|v\|^2 + \langle u, v\rangle + \langle v, u\rangle \\ &\quad + \|u\|^2 + \|v\|^2 - \langle u, v\rangle - \langle v, u\rangle \\ &= 2(\|u\|^2 + \|v\|^2),\end{aligned}$$

as desired. ∎

## *Exercises 6A*

**1** Prove or give a counterexample: If $v_1, \ldots, v_m \in V$, then

$$\sum_{j=1}^{m} \sum_{k=1}^{m} \langle v_j, v_k\rangle \geq 0.$$

**2** Suppose $S \in \mathcal{L}(V)$. Define $\langle \cdot, \cdot \rangle_1$ by

$$\langle u, v\rangle_1 = \langle Su, Sv\rangle$$

for all $u, v \in V$. Show that $\langle \cdot, \cdot \rangle_1$ is an inner product on $V$ if and only if $S$ is injective.

**3** (a) Show that the function taking an ordered pair $((x_1, x_2), (y_1, y_2))$ of elements of $\mathbf{R}^2$ to $|x_1 y_1| + |x_2 y_2|$ is not an inner product on $\mathbf{R}^2$.

(b) Show that the function taking an ordered pair $((x_1, x_2, x_3), (y_1, y_2, y_3))$ of elements of $\mathbf{R}^3$ to $x_1 y_1 + x_3 y_3$ is not an inner product on $\mathbf{R}^3$.

**4** Suppose $T \in \mathcal{L}(V)$ is such that $\|Tv\| \leq \|v\|$ for every $v \in V$. Prove that $T - \sqrt{2}\, I$ is injective.

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**5** Suppose $V$ is a real inner product space.

(a) Show that $\langle u+v, u-v\rangle = \|u\|^2 - \|v\|^2$ for every $u, v \in V$.

(b) Show that if $u, v \in V$ have the same norm, then $u+v$ is orthogonal to $u-v$.

(c) Use (b) to show that the diagonals of a rhombus are perpendicular to each other.

**6** Suppose $u, v \in V$. Prove that $\langle u, v\rangle = 0 \iff \|u\| \le \|u + av\|$ for all $a \in \mathbf{F}$.

**7** Suppose $u, v \in V$. Prove that $\|au + bv\| = \|bu + av\|$ for all $a, b \in \mathbf{R}$ if and only if $\|u\| = \|v\|$.

**8** Suppose $a, b, c, x, y \in \mathbf{R}$ and $a^2 + b^2 + c^2 + x^2 + y^2 \le 1$. Prove that $a + b + c + 4x + 9y \le 10$.

**9** Suppose $u, v \in V$ and $\|u\| = \|v\| = 1$ and $\langle u, v\rangle = 1$. Prove that $u = v$.

**10** Suppose $u, v \in V$ and $\|u\| \le 1$ and $\|v\| \le 1$. Prove that

$$\sqrt{1 - \|u\|^2}\sqrt{1 - \|v\|^2} \le 1 - |\langle u, v\rangle|.$$

**11** Find vectors $u, v \in \mathbf{R}^2$ such that $u$ is a scalar multiple of $(1, 3)$, $v$ is orthogonal to $(1, 3)$, and $(1, 2) = u + v$.

**12** Suppose $a, b, c, d$ are positive numbers.

(a) Prove that $(a+b+c+d)\left(\frac{1}{a} + \frac{1}{b} + \frac{1}{c} + \frac{1}{d}\right) \ge 16$.

(b) For which positive numbers $a, b, c, d$ is the inequality above an equality?

**13** Show that the square of an average is less than or equal to the average of the squares. More precisely, show that if $a_1, \ldots, a_n \in \mathbf{R}$, then the square of the average of $a_1, \ldots, a_n$ is less than or equal to the average of $a_1^2, \ldots, a_n^2$.

**14** Suppose $v \in V$ and $v \ne 0$. Prove that $v/\|v\|$ is the unique closest element on the unit sphere of $V$ to $v$. More precisely, prove that if $u \in V$ and $\|u\| = 1$, then

$$\left\| v - \frac{v}{\|v\|} \right\| \le \|v - u\|,$$

with equality only if $u = v/\|v\|$.

**15** Suppose $u, v$ are nonzero vectors in $\mathbf{R}^2$. Prove that

$$\langle u, v\rangle = \|u\|\,\|v\| \cos\theta,$$

where $\theta$ is the angle between $u$ and $v$ (thinking of $u$ and $v$ as arrows with initial point at the origin).

*Hint: Use the law of cosines on the triangle formed by u, v, and u − v.*

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**16** The angle between two vectors (thought of as arrows with initial point at the origin) in $\mathbf{R}^2$ or $\mathbf{R}^3$ can be defined geometrically. However, geometry is not as clear in $\mathbf{R}^n$ for $n > 3$. Thus the angle between two nonzero vectors $x, y \in \mathbf{R}^n$ is defined to be

$$\arccos \frac{\langle x, y \rangle}{\|x\| \, \|y\|},$$

where the motivation for this definition comes from Exercise 15. Explain why the Cauchy–Schwarz inequality is needed to show that this definition makes sense.

**17** Prove that

$$\Big( \sum_{k=1}^{n} a_k b_k \Big)^2 \le \Big( \sum_{k=1}^{n} k a_k^2 \Big) \Big( \sum_{k=1}^{n} \frac{b_k^2}{k} \Big)$$

for all real numbers $a_1, \ldots, a_n$ and $b_1, \ldots, b_n$.

**18** (a) Suppose $f \colon [1, \infty) \to [0, \infty)$ is continuous. Show that

$$\Big( \int_1^\infty f \Big)^2 \le \int_1^\infty x^2 (f(x))^2 \, dx.$$

(b) For which continuous functions $f \colon [1, \infty) \to [0, \infty)$ is the inequality in (a) an equality with both sides finite?

**19** Suppose $v_1, \ldots, v_n$ is a basis of $V$ and $T \in \mathcal{L}(V)$. Prove that if $\lambda$ is an eigenvalue of $T$, then

$$|\lambda|^2 \le \sum_{j=1}^{n} \sum_{k=1}^{n} |\mathcal{M}(T)_{j,k}|^2,$$

where $\mathcal{M}(T)_{j,k}$ denotes the entry in row $j$, column $k$ of the matrix of $T$ with respect to the basis $v_1, \ldots, v_n$.

**20** Prove that if $u, v \in V$, then $\big| \|u\| - \|v\| \big| \le \|u - v\|$.

*The inequality above is called the* ***reverse triangle inequality****. For the reverse triangle inequality when $V = \mathbf{C}$, see Exercise 2 in Chapter 4.*

**21** Suppose $u, v \in V$ are such that

$$\|u\| = 3, \quad \|u + v\| = 4, \quad \|u - v\| = 6.$$

What number does $\|v\|$ equal?

**22** Show that if $u, v \in V$, then

$$\|u + v\| \, \|u - v\| \le \|u\|^2 + \|v\|^2.$$

**23** Suppose $v_1, \ldots, v_m \in V$ are such that $\|v_k\| \le 1$ for each $k = 1, \ldots, m$. Show that there exist $a_1, \ldots, a_m \in \{1, -1\}$ such that

$$\|a_1 v_1 + \cdots + a_m v_m\| \le \sqrt{m}.$$

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**24** Prove or give a counterexample: If $\|\cdot\|$ is the norm associated with an inner product on $\mathbf{R}^2$, then there exists $(x, y) \in \mathbf{R}^2$ such that $\|(x, y)\| \neq \max\{|x|, |y|\}$.

**25** Suppose $p > 0$. Prove that there is an inner product on $\mathbf{R}^2$ such that the associated norm is given by

$$\|(x, y)\| = (|x|^p + |y|^p)^{1/p}$$

for all $(x, y) \in \mathbf{R}^2$ if and only if $p = 2$.

**26** Suppose $V$ is a real inner product space. Prove that

$$\langle u, v\rangle = \frac{\|u + v\|^2 - \|u - v\|^2}{4}$$

for all $u, v \in V$.

**27** Suppose $V$ is a complex inner product space. Prove that

$$\langle u, v\rangle = \frac{\|u + v\|^2 - \|u - v\|^2 + \|u + iv\|^2 i - \|u - iv\|^2 i}{4}$$

for all $u, v \in V$.

**28** A norm on a vector space $U$ is a function

$$\|\cdot\| \colon U \to [0, \infty)$$

such that $\|u\| = 0$ if and only if $u = 0$, $\|\alpha u\| = |\alpha|\|u\|$ for all $\alpha \in \mathbf{F}$ and all $u \in U$, and $\|u + v\| \leq \|u\| + \|v\|$ for all $u, v \in U$. Prove that a norm satisfying the parallelogram equality comes from an inner product (in other words, show that if $\|\cdot\|$ is a norm on $U$ satisfying the parallelogram equality, then there is an inner product $\langle\cdot, \cdot\rangle$ on $U$ such that $\|u\| = \langle u, u\rangle^{1/2}$ for all $u \in U$).

**29** Suppose $V_1, \ldots, V_m$ are inner product spaces. Show that the equation

$$\big\langle (u_1, \ldots, u_m), (v_1, \ldots, v_m)\big\rangle = \langle u_1, v_1\rangle + \cdots + \langle u_m, v_m\rangle$$

defines an inner product on $V_1 \times \cdots \times V_m$.

*In the expression above on the right, for each $k = 1, \ldots, m$, the inner product $\langle u_k, v_k\rangle$ denotes the inner product on $V_k$. Each of the spaces $V_1, \ldots, V_m$ may have a different inner product, even though the same notation is used here.*

**30** Suppose $V$ is a real inner product space. For $u, v, w, x \in V$, define

$$\langle u + iv, w + ix\rangle_{\mathbf{C}} = \langle u, w\rangle + \langle v, x\rangle + (\langle v, w\rangle - \langle u, x\rangle)i.$$

(a) Show that $\langle\cdot, \cdot\rangle_{\mathbf{C}}$ makes $V_{\mathbf{C}}$ into a complex inner product space.
(b) Show that if $u, v \in V$, then

$$\langle u, v\rangle_{\mathbf{C}} = \langle u, v\rangle \quad \text{and} \quad \|u + iv\|_{\mathbf{C}}^2 = \|u\|^2 + \|v\|^2.$$

*See Exercise 8 in Section 1B for the definition of the complexification $V_{\mathbf{C}}$.*

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**31** Suppose $u, v, w \in V$. Prove that

$$\|w - \tfrac{1}{2}(u+v)\|^2 = \frac{\|w-u\|^2 + \|w-v\|^2}{2} - \frac{\|u-v\|^2}{4}.$$

**32** Suppose that $E$ is a subset of $V$ with the property that $u, v \in E$ implies $\frac{1}{2}(u+v) \in E$. Let $w \in V$. Show that there is at most one point in $E$ that is closest to $w$. In other words, show that there is at most one $u \in E$ such that

$$\|w-u\| \le \|w-x\|$$

for all $x \in E$.

**33** Suppose $f, g$ are differentiable functions from $\mathbf{R}$ to $\mathbf{R}^n$.

(a) Show that

$$\langle f(t), g(t)\rangle' = \langle f'(t), g(t)\rangle + \langle f(t), g'(t)\rangle.$$

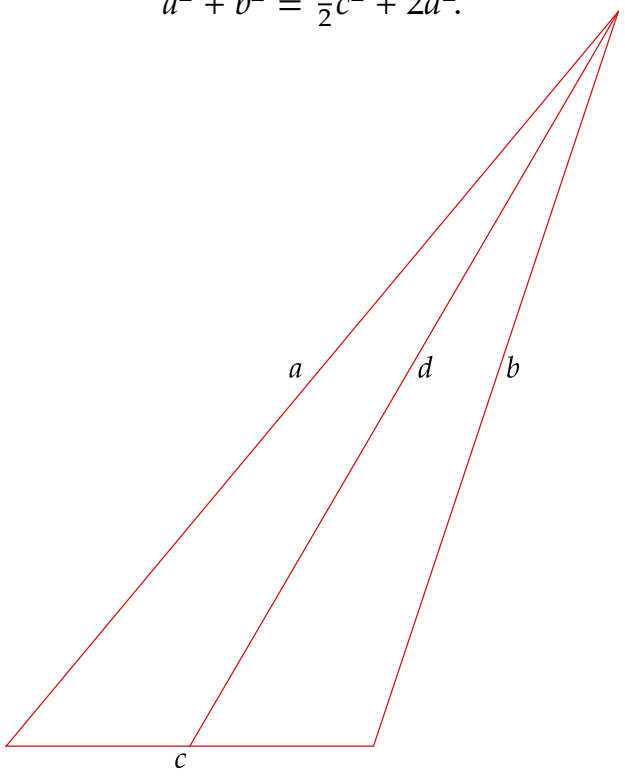
(b) Suppose $c$ is a positive number and $\|f(t)\| = c$ for every $t \in \mathbf{R}$. Show that $\langle f'(t), f(t)\rangle = 0$ for every $t \in \mathbf{R}$.

(c) Interpret the result in (b) geometrically in terms of the tangent vector to a curve lying on a sphere in $\mathbf{R}^n$ centered at the origin.

*A function $f\colon \mathbf{R} \to \mathbf{R}^n$ is called differentiable if there exist differentiable functions $f_1, \ldots, f_n$ from $\mathbf{R}$ to $\mathbf{R}$ such that $f(t) = (f_1(t), \ldots, f_n(t))$ for each $t \in \mathbf{R}$. Furthermore, for each $t \in \mathbf{R}$, the derivative $f'(t) \in \mathbf{R}^n$ is defined by $f'(t) = (f_1'(t), \ldots, f_n'(t))$.*

**34** Use inner products to prove Apollonius's identity: In a triangle with sides of length $a$, $b$, and $c$, let $d$ be the length of the line segment from the midpoint of the side of length $c$ to the opposite vertex. Then

$$a^2 + b^2 = \tfrac{1}{2}c^2 + 2d^2.$$




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**35** Fix a positive integer $n$. The *Laplacian* $\Delta p$ of a twice differentiable real-valued function $p$ on $\mathbf{R}^n$ is the function on $\mathbf{R}^n$ defined by

$$\Delta p = \frac{\partial^2 p}{\partial x_1^2} + \cdots + \frac{\partial^2 p}{\partial x_n^2}.$$

The function $p$ is called *harmonic* if $\Delta p = 0$.

A *polynomial* on $\mathbf{R}^n$ is a linear combination (with coefficients in $\mathbf{R}$) of functions of the form $x_1^{m_1} \cdots x_n^{m_n}$, where $m_1, \ldots, m_n$ are nonnegative integers.

Suppose $q$ is a polynomial on $\mathbf{R}^n$. Prove that there exists a harmonic polynomial $p$ on $\mathbf{R}^n$ such that $p(x) = q(x)$ for every $x \in \mathbf{R}^n$ with $\|x\| = 1$.

*The only fact about harmonic functions that you need for this exercise is that if $p$ is a harmonic function on $\mathbf{R}^n$ and $p(x) = 0$ for all $x \in \mathbf{R}^n$ with $\|x\| = 1$, then $p = 0$.*

*Hint: A reasonable guess is that the desired harmonic polynomial $p$ is of the form $q + (1 - \|x\|^2)r$ for some polynomial $r$. Prove that there is a polynomial $r$ on $\mathbf{R}^n$ such that $q + (1 - \|x\|^2)r$ is harmonic by defining an operator $T$ on a suitable vector space by*

$$Tr = \Delta((1 - \|x\|^2)r)$$

*and then showing that $T$ is injective and hence surjective.*

In realms of numbers, where the secrets lie,
A noble truth emerges from the deep,
Cauchy and Schwarz, their wisdom they apply,
An inequality for all to keep.

Two vectors, by this bond, are intertwined,
As inner products weave a gilded thread,
Their magnitude, by providence, confined,
A bound to which their destiny is wed.

Though shadows fall, and twilight dims the day,
This inequality will stand the test,
To guide us in our quest, to light the way,
And in its truth, our understanding rest.

So sing, ye muses, of this noble feat,
Cauchy–Schwarz, the bound that none can beat.

—written by ChatGPT with input *Shakespearean sonnet on Cauchy–Schwarz inequality*

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# 6B Orthonormal Bases

## Orthonormal Lists and the Gram–Schmidt Procedure

**6.22 definition: *orthonormal***

- A list of vectors is called *orthonormal* if each vector in the list has norm 1 and is orthogonal to all the other vectors in the list.
- In other words, a list $e_1, \ldots, e_m$ of vectors in $V$ is orthonormal if

$$\langle e_j, e_k \rangle = \begin{cases} 1 & \text{if } j = k, \\ 0 & \text{if } j \neq k \end{cases}$$

for all $j, k \in \{1, \ldots, m\}$.

**6.23 example: *orthonormal lists***

(a) The standard basis of $\mathbf{F}^n$ is an orthonormal list.

(b) $\left(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}\right), \left(-\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, 0\right)$ is an orthonormal list in $\mathbf{F}^3$.

(c) $\left(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}\right), \left(-\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, 0\right), \left(\frac{1}{\sqrt{6}}, \frac{1}{\sqrt{6}}, -\frac{2}{\sqrt{6}}\right)$ is an orthonormal list in $\mathbf{F}^3$.

(d) Suppose $n$ is a positive integer. Then, as Exercise 4 asks you to verify,

$$\frac{1}{\sqrt{2\pi}}, \frac{\cos x}{\sqrt{\pi}}, \frac{\cos 2x}{\sqrt{\pi}}, \ldots, \frac{\cos nx}{\sqrt{\pi}}, \frac{\sin x}{\sqrt{\pi}}, \frac{\sin 2x}{\sqrt{\pi}}, \ldots, \frac{\sin nx}{\sqrt{\pi}}$$

is an orthonormal list of vectors in $C[-\pi, \pi]$, the vector space of continuous real-valued functions on $[-\pi, \pi]$ with inner product

$$\langle f, g \rangle = \int_{-\pi}^{\pi} fg.$$

The orthonormal list above is often used for modeling periodic phenomena, such as tides.

(e) Suppose we make $\mathcal{P}_2(\mathbf{R})$ into an inner product space using the inner product given by

$$\langle p, q \rangle = \int_{-1}^{1} pq$$

for all $p, q \in \mathcal{P}_2(\mathbf{R})$. The standard basis $1, x, x^2$ of $\mathcal{P}_2(\mathbf{R})$ is not an orthonormal list because the vectors in that list do not have norm 1. Dividing each vector by its norm gives the list $1/\sqrt{2}, \sqrt{3/2}x, \sqrt{5/2}x^2$, in which each vector has norm 1, and the second vector is orthogonal to the first and third vectors. However, the first and third vectors are not orthogonal. Thus this is not an orthonormal list. Soon we will see how to construct an orthonormal list from the standard basis $1, x, x^2$ (see Example 6.34).

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Orthonormal lists are particularly easy to work with, as illustrated by the next result.

**6.24** *norm of an orthonormal linear combination*

Suppose $e_1, \dots, e_m$ is an orthonormal list of vectors in $V$. Then

$$\|a_1 e_1 + \cdots + a_m e_m\|^2 = |a_1|^2 + \cdots + |a_m|^2$$

for all $a_1, \dots, a_m \in \mathbf{F}$.

Proof Because each $e_k$ has norm 1, this follows from repeated applications of the Pythagorean theorem (6.12). ∎

The result above has the following important corollary.

**6.25** *orthonormal lists are linearly independent*

Every orthonormal list of vectors is linearly independent.

Proof Suppose $e_1, \dots, e_m$ is an orthonormal list of vectors in $V$ and $a_1, \dots, a_m \in \mathbf{F}$ are such that

$$a_1 e_1 + \cdots + a_m e_m = 0.$$

Then $|a_1|^2 + \cdots + |a_m|^2 = 0$ (by 6.24), which means that all the $a_k$'s are 0. Thus $e_1, \dots, e_m$ is linearly independent. ∎

Now we come to an important inequality.

**6.26** *Bessel's inequality*

Suppose $e_1, \dots, e_m$ is an orthonormal list of vectors in $V$. If $v \in V$ then

$$|\langle v, e_1 \rangle|^2 + \cdots + |\langle v, e_m \rangle|^2 \le \|v\|^2.$$

Proof Suppose $v \in V$. Then

$$v = \underbrace{\langle v, e_1 \rangle e_1 + \cdots + \langle v, e_m \rangle e_m}_{u} + \underbrace{v - \langle v, e_1 \rangle e_1 - \cdots - \langle v, e_m \rangle e_m}_{w}.$$

Let $u$ and $w$ be defined as in the equation above. If $k \in \{1, \dots, m\}$, then $\langle w, e_k \rangle = \langle v, e_k \rangle - \langle v, e_k \rangle \langle e_k, e_k \rangle = 0$. This implies that $\langle w, u \rangle = 0$. The Pythagorean theorem now implies that

$$\begin{aligned} \|v\|^2 &= \|u\|^2 + \|w\|^2 \\ &\ge \|u\|^2 \\ &= |\langle v, e_1 \rangle|^2 + \cdots + |\langle v, e_m \rangle|^2, \end{aligned}$$

where the last line comes from 6.24. ∎

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The next definition introduces one of the most useful concepts in the study of inner product spaces.

**6.27 definition: *orthonormal basis***

An *orthonormal basis* of $V$ is an orthonormal list of vectors in $V$ that is also a basis of $V$.

For example, the standard basis is an orthonormal basis of $\mathbf{F}^n$.

**6.28 *orthonormal lists of the right length are orthonormal bases***

Suppose $V$ is finite-dimensional. Then every orthonormal list of vectors in $V$ of length $\dim V$ is an orthonormal basis of $V$.

Proof By 6.25, every orthonormal list of vectors in $V$ is linearly independent. Thus every such list of the right length is a basis—see 2.38. ∎

**6.29 example: *an orthonormal basis of* $\mathbf{F}^4$**

As mentioned above, the standard basis is an orthonormal basis of $\mathbf{F}^4$. We now show that

$$\left(\tfrac{1}{2},\tfrac{1}{2},\tfrac{1}{2},\tfrac{1}{2}\right),\left(\tfrac{1}{2},\tfrac{1}{2},-\tfrac{1}{2},-\tfrac{1}{2}\right),\left(\tfrac{1}{2},-\tfrac{1}{2},-\tfrac{1}{2},\tfrac{1}{2}\right),\left(-\tfrac{1}{2},\tfrac{1}{2},-\tfrac{1}{2},\tfrac{1}{2}\right)$$

is also an orthonormal basis of $\mathbf{F}^4$.

We have

$$\left\|\left(\tfrac{1}{2},\tfrac{1}{2},\tfrac{1}{2},\tfrac{1}{2}\right)\right\| = \sqrt{\tfrac{1}{2^2}+\tfrac{1}{2^2}+\tfrac{1}{2^2}+\tfrac{1}{2^2}} = 1.$$

Similarly, the other three vectors in the list above also have norm 1.

Note that

$$\left\langle\left(\tfrac{1}{2},\tfrac{1}{2},\tfrac{1}{2},\tfrac{1}{2}\right),\left(\tfrac{1}{2},\tfrac{1}{2},-\tfrac{1}{2},-\tfrac{1}{2}\right)\right\rangle = \tfrac{1}{2}\cdot\tfrac{1}{2}+\tfrac{1}{2}\cdot\tfrac{1}{2}+\tfrac{1}{2}\cdot\left(-\tfrac{1}{2}\right)+\tfrac{1}{2}\cdot\left(-\tfrac{1}{2}\right) = 0.$$

Similarly, the inner product of any two distinct vectors in the list above also equals 0.

Thus the list above is orthonormal. Because we have an orthonormal list of length four in the four-dimensional vector space $\mathbf{F}^4$, this list is an orthonormal basis of $\mathbf{F}^4$ (by 6.28).

---

In general, given a basis $e_1,\ldots,e_n$ of $V$ and a vector $v \in V$, we know that there is some choice of scalars $a_1,\ldots,a_n \in \mathbf{F}$ such that

$$v = a_1e_1 + \cdots + a_ne_n.$$

Computing the numbers $a_1,\ldots,a_n$ that satisfy the equation above can be a long computation for an arbitrary basis of $V$. The next result shows, however, that this is easy for an orthonormal basis—just take $a_k = \langle v, e_k\rangle$.

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Notice how the next result makes each inner product space of dimension $n$ behave like $\mathbf{F}^n$, with the role of the coordinates of a vector in $\mathbf{F}^n$ played by $\langle v, e_1\rangle, \ldots, \langle v, e_n\rangle$.

*The formula below for $\|v\|$ is called Parseval's identity. It was published in 1799 in the context of Fourier series.*

**6.30** *writing a vector as a linear combination of an orthonormal basis*

Suppose $e_1, \ldots, e_n$ is an orthonormal basis of $V$ and $u, v \in V$. Then

(a) $v = \langle v, e_1\rangle e_1 + \cdots + \langle v, e_n\rangle e_n$,

(b) $\|v\|^2 = |\langle v, e_1\rangle|^2 + \cdots + |\langle v, e_n\rangle|^2$,

(c) $\langle u, v\rangle = \langle u, e_1\rangle\overline{\langle v, e_1\rangle} + \cdots + \langle u, e_n\rangle\overline{\langle v, e_n\rangle}$.

Proof Because $e_1, \ldots, e_n$ is a basis of $V$, there exist scalars $a_1, \ldots, a_n$ such that

$$v = a_1 e_1 + \cdots + a_n e_n.$$

Because $e_1, \ldots, e_n$ is orthonormal, taking the inner product of both sides of this equation with $e_k$ gives $\langle v, e_k\rangle = a_k$. Thus (a) holds.

Now (b) follows immediately from (a) and 6.24.

Taking the inner product of $u$ with each side of (a) and then using the conjugate symmetry of the inner product gives (c). ∎

**6.31** example: *finding coefficients for a linear combination*

Suppose we want to write the vector $(1, 2, 4, 7) \in \mathbf{F}^4$ as a linear combination of the orthonormal basis

$$\left(\tfrac{1}{2}, \tfrac{1}{2}, \tfrac{1}{2}, \tfrac{1}{2}\right), \left(\tfrac{1}{2}, \tfrac{1}{2}, -\tfrac{1}{2}, -\tfrac{1}{2}\right), \left(\tfrac{1}{2}, -\tfrac{1}{2}, -\tfrac{1}{2}, \tfrac{1}{2}\right), \left(-\tfrac{1}{2}, \tfrac{1}{2}, -\tfrac{1}{2}, \tfrac{1}{2}\right)$$

of $\mathbf{F}^4$ from Example 6.29. Instead of solving a system of four linear equations in four unknowns, as typically would be required if we were working with a nonorthonormal basis, we simply evaluate four inner products and use 6.30(a), getting that $(1, 2, 4, 7)$ equals

$$7\left(\tfrac{1}{2}, \tfrac{1}{2}, \tfrac{1}{2}, \tfrac{1}{2}\right) - 4\left(\tfrac{1}{2}, \tfrac{1}{2}, -\tfrac{1}{2}, -\tfrac{1}{2}\right) + \left(\tfrac{1}{2}, -\tfrac{1}{2}, -\tfrac{1}{2}, \tfrac{1}{2}\right) + 2\left(-\tfrac{1}{2}, \tfrac{1}{2}, -\tfrac{1}{2}, \tfrac{1}{2}\right).$$

Now that we understand the usefulness of orthonormal bases, how do we go about finding them? For example, does $\mathcal{P}_m(\mathbf{R})$ with inner product as in 6.3(c) have an orthonormal basis? The next result will lead to answers to these questions.

The algorithm used in the next proof is called the *Gram–Schmidt procedure*. It gives a method for turning a linearly independent list into an orthonormal list with the same span as the original list.

*Jørgen Gram (1850–1916) and Erhard Schmidt (1876–1959) popularized this algorithm that constructs orthonormal lists.*

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**6.32** *Gram–Schmidt procedure*

Suppose $v_1, \dots, v_m$ is a linearly independent list of vectors in $V$. Let $f_1 = v_1$. For $k = 2, \dots, m$, define $f_k$ inductively by

$$f_k = v_k - \frac{\langle v_k, f_1 \rangle}{\|f_1\|^2} f_1 - \cdots - \frac{\langle v_k, f_{k-1} \rangle}{\|f_{k-1}\|^2} f_{k-1}.$$

For each $k = 1, \dots, m$, let $e_k = \dfrac{f_k}{\|f_k\|}$. Then $e_1, \dots, e_m$ is an orthonormal list of vectors in $V$ such that

$$\operatorname{span}(v_1, \dots, v_k) = \operatorname{span}(e_1, \dots, e_k)$$

for each $k = 1, \dots, m$.

Proof We will show by induction on $k$ that the desired conclusion holds. To get started with $k = 1$, note that because $e_1 = f_1/\|f_1\|$, we have $\|e_1\| = 1$; also, $\operatorname{span}(v_1) = \operatorname{span}(e_1)$ because $e_1$ is a nonzero multiple of $v_1$.

Suppose $1 < k \le m$ and the list $e_1, \dots, e_{k-1}$ generated by 6.32 is an orthonormal list such that

6.33 $$\operatorname{span}(v_1, \dots, v_{k-1}) = \operatorname{span}(e_1, \dots, e_{k-1}).$$

Because $v_1, \dots, v_m$ is linearly independent, we have $v_k \notin \operatorname{span}(v_1, \dots, v_{k-1})$. Thus $v_k \notin \operatorname{span}(e_1, \dots, e_{k-1}) = \operatorname{span}(f_1, \dots, f_{k-1})$, which implies that $f_k \neq 0$. Hence we are not dividing by 0 in the definition of $e_k$ given in 6.32. Dividing a vector by its norm produces a new vector with norm 1; thus $\|e_k\| = 1$.

Let $j \in \{1, \dots, k-1\}$. Then

$$\begin{aligned}
\langle e_k, e_j \rangle &= \frac{1}{\|f_k\| \, \|f_j\|} \langle f_k, f_j \rangle \\
&= \frac{1}{\|f_k\| \, \|f_j\|} \left\langle v_k - \frac{\langle v_k, f_1 \rangle}{\|f_1\|^2} f_1 - \cdots - \frac{\langle v_k, f_{k-1} \rangle}{\|f_{k-1}\|^2} f_{k-1}, f_j \right\rangle \\
&= \frac{1}{\|f_k\| \, \|f_j\|} (\langle v_k, f_j \rangle - \langle v_k, f_j \rangle) \\
&= 0.
\end{aligned}$$

Thus $e_1, \dots, e_k$ is an orthonormal list.

From the definition of $e_k$ given in 6.32, we see that $v_k \in \operatorname{span}(e_1, \dots, e_k)$. Combining this information with 6.33 shows that

$$\operatorname{span}(v_1, \dots, v_k) \subseteq \operatorname{span}(e_1, \dots, e_k).$$

Both lists above are linearly independent (the $v$'s by hypothesis, and the $e$'s by orthonormality and 6.25). Thus both subspaces above have dimension $k$, and hence they are equal, completing the induction step and thus completing the proof. ∎

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6.34 example: *an orthonormal basis of* $\mathcal{P}_2(\mathbf{R})$

Suppose we make $\mathcal{P}_2(\mathbf{R})$ into an inner product space using the inner product given by

$$\langle p, q\rangle = \int_{-1}^{1} pq$$

for all $p, q \in \mathcal{P}_2(\mathbf{R})$. We know that $1, x, x^2$ is a basis of $\mathcal{P}_2(\mathbf{R})$, but it is not an orthonormal basis. We will find an orthonormal basis of $\mathcal{P}_2(\mathbf{R})$ by applying the Gram–Schmidt procedure with $v_1 = 1$, $v_2 = x$, and $v_3 = x^2$.

To get started, take $f_1 = v_1 = 1$. Thus $\|f_1\|^2 = \int_{-1}^{1} 1 = 2$. Hence the formula in 6.32 tells us that

$$f_2 = v_2 - \frac{\langle v_2, f_1\rangle}{\|f_1\|^2} f_1 = x - \frac{\langle x, 1\rangle}{\|f_1\|^2} = x,$$

where the last equality holds because $\langle x, 1\rangle = \int_{-1}^{1} t\,dt = 0$.

The formula above for $f_2$ implies that $\|f_2\|^2 = \int_{-1}^{1} t^2\,dt = \frac{2}{3}$. Now the formula in 6.32 tells us that

$$f_3 = v_3 - \frac{\langle v_3, f_1\rangle}{\|f_1\|^2} f_1 - \frac{\langle v_3, f_2\rangle}{\|f_2\|^2} f_2 = x^2 - \tfrac{1}{2}\langle x^2, 1\rangle - \tfrac{3}{2}\langle x^2, x\rangle x = x^2 - \tfrac{1}{3}.$$

The formula above for $f_3$ implies that

$$\|f_3\|^2 = \int_{-1}^{1} \left(t^2 - \tfrac{1}{3}\right)^2 dt = \int_{-1}^{1} \left(t^4 - \tfrac{2}{3}t^2 + \tfrac{1}{9}\right) dt = \tfrac{8}{45}.$$

Now dividing each of $f_1, f_2, f_3$ by its norm gives us the orthonormal list

$$\sqrt{\tfrac{1}{2}},\ \sqrt{\tfrac{3}{2}}x,\ \sqrt{\tfrac{45}{8}}\left(x^2 - \tfrac{1}{3}\right).$$

The orthonormal list above has length three, which is the dimension of $\mathcal{P}_2(\mathbf{R})$. Hence this orthonormal list is an orthonormal basis of $\mathcal{P}_2(\mathbf{R})$ [by 6.28].

Now we can answer the question about the existence of orthonormal bases.

6.35 *existence of orthonormal basis*

Every finite-dimensional inner product space has an orthonormal basis.

Proof Suppose $V$ is finite-dimensional. Choose a basis of $V$. Apply the Gram–Schmidt procedure (6.32) to it, producing an orthonormal list of length $\dim V$. By 6.28, this orthonormal list is an orthonormal basis of $V$. ∎

Sometimes we need to know not only that an orthonormal basis exists, but also that every orthonormal list can be extended to an orthonormal basis. In the next corollary, the Gram–Schmidt procedure shows that such an extension is always possible.

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**6.36** *every orthonormal list extends to an orthonormal basis*

Suppose $V$ is finite-dimensional. Then every orthonormal list of vectors in $V$ can be extended to an orthonormal basis of $V$.

Proof Suppose $e_1, \dots, e_m$ is an orthonormal list of vectors in $V$. Then $e_1, \dots, e_m$ is linearly independent (by 6.25). Hence this list can be extended to a basis $e_1, \dots, e_m, v_1, \dots, v_n$ of $V$ (see 2.32). Now apply the Gram–Schmidt procedure (6.32) to $e_1, \dots, e_m, v_1, \dots, v_n$, producing an orthonormal list

$$e_1, \dots, e_m, f_1, \dots, f_n;$$

here the formula given by the Gram–Schmidt procedure leaves the first $m$ vectors unchanged because they are already orthonormal. The list above is an orthonormal basis of $V$ by 6.28. ∎

Recall that a matrix is called upper triangular if it looks like this:

$$\begin{pmatrix} * & & * \\ & \ddots & \\ 0 & & * \end{pmatrix},$$

where the 0 in the matrix above indicates that all entries below the diagonal equal 0, and asterisks are used to denote entries on and above the diagonal.

In the last chapter, we gave a necessary and sufficient condition for an operator to have an upper-triangular matrix with respect to some basis (see 5.44). Now that we are dealing with inner product spaces, we would like to know whether there exists an *orthonormal* basis with respect to which we have an upper-triangular matrix. The next result shows that the condition for an operator to have an upper-triangular matrix with respect to some orthonormal basis is the same as the condition to have an upper-triangular matrix with respect to an arbitrary basis.

**6.37** *upper-triangular matrix with respect to some orthonormal basis*

Suppose $V$ is finite-dimensional and $T \in \mathcal{L}(V)$. Then $T$ has an upper-triangular matrix with respect to some orthonormal basis of $V$ if and only if the minimal polynomial of $T$ equals $(z - \lambda_1) \cdots (z - \lambda_m)$ for some $\lambda_1, \dots, \lambda_m \in \mathbf{F}$.

Proof Suppose $T$ has an upper-triangular matrix with respect to some basis $v_1, \dots, v_n$ of $V$. Thus $\operatorname{span}(v_1, \dots, v_k)$ is invariant under $T$ for each $k = 1, \dots, n$ (see 5.39).

Apply the Gram–Schmidt procedure to $v_1, \dots, v_n$, producing an orthonormal basis $e_1, \dots, e_n$ of $V$. Because

$$\operatorname{span}(e_1, \dots, e_k) = \operatorname{span}(v_1, \dots, v_k)$$

for each $k$ (see 6.32), we conclude that $\operatorname{span}(e_1, \dots, e_k)$ is invariant under $T$ for each $k = 1, \dots, n$. Thus, by 5.39, $T$ has an upper-triangular matrix with respect to the orthonormal basis $e_1, \dots, e_n$. Now use 5.44 to complete the proof. ∎

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*Issai Schur (1875–1941) published a proof of the next result in 1909.*

For complex vector spaces, the next result is an important application of the result above. See Exercise 20 for a version of Schur's theorem that applies simultaneously to more than one operator.

**6.38 *Schur's theorem***

Every operator on a finite-dimensional complex inner product space has an upper-triangular matrix with respect to some orthonormal basis.

Proof The desired result follows from the second version of the fundamental theorem of algebra (4.13) and 6.37. ∎

## *Linear Functionals on Inner Product Spaces*

Because linear maps into the scalar field $\mathbf{F}$ play a special role, we defined a special name for them and their vector space in Section 3F. Those definitions are repeated below in case you skipped Section 3F.

**6.39 definition: *linear functional, dual space, $V'$***

- A *linear functional* on $V$ is a linear map from $V$ to $\mathbf{F}$.
- The *dual space* of $V$, denoted by $V'$, is the vector space of all linear functionals on $V$. In other words, $V' = \mathcal{L}(V, \mathbf{F})$.

**6.40 example: *linear functional on $\mathbf{F}^3$***

The function $\varphi : \mathbf{F}^3 \to \mathbf{F}$ defined by

$$\varphi(z_1, z_2, z_3) = 2z_1 - 5z_2 + z_3$$

is a linear functional on $\mathbf{F}^3$. We could write this linear functional in the form

$$\varphi(z) = \langle z, w \rangle$$

for every $z \in \mathbf{F}^3$, where $w = (2, -5, 1)$.

**6.41 example: *linear functional on $\mathcal{P}_5(\mathbf{R})$***

The function $\varphi : \mathcal{P}_5(\mathbf{R}) \to \mathbf{R}$ defined by

$$\varphi(p) = \int_{-1}^{1} p(t)(\cos(\pi t))\, dt$$

is a linear functional on $\mathcal{P}_5(\mathbf{R})$.

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If $v \in V$, then the map that sends $u$ to $\langle u, v\rangle$ is a linear functional on $V$. The next result states that every linear functional on $V$ is of this form. For example, we can take $v = (2, -5, 1)$ in Example 6.40.

*The next result is named in honor of Frigyes Riesz (1880–1956), who proved several theorems early in the twentieth century that look very much like the result below.*

Suppose we make the vector space $\mathcal{P}_5(\mathbf{R})$ into an inner product space by defining $\langle p, q\rangle = \int_{-1}^{1} pq$. Let $\varphi$ be as in Example 6.41. It is not obvious that there exists $q \in \mathcal{P}_5(\mathbf{R})$ such that

$$\int_{-1}^{1} p(t)(\cos(\pi t))\, dt = \langle p, q\rangle$$

for every $p \in \mathcal{P}_5(\mathbf{R})$ [we cannot take $q(t) = \cos(\pi t)$ because that choice of $q$ is not an element of $\mathcal{P}_5(\mathbf{R})$]. The next result tells us the somewhat surprising result that there indeed exists a polynomial $q \in \mathcal{P}_5(\mathbf{R})$ such that the equation above holds for all $p \in \mathcal{P}_5(\mathbf{R})$.

**6.42** *Riesz representation theorem*

Suppose $V$ is finite-dimensional and $\varphi$ is a linear functional on $V$. Then there is a unique vector $v \in V$ such that

$$\varphi(u) = \langle u, v\rangle$$

for every $u \in V$.

Proof First we show that there exists a vector $v \in V$ such that $\varphi(u) = \langle u, v\rangle$ for every $u \in V$. Let $e_1, \ldots, e_n$ be an orthonormal basis of $V$. Then

$$\begin{aligned}\varphi(u) &= \varphi(\langle u, e_1\rangle e_1 + \cdots + \langle u, e_n\rangle e_n)\\ &= \langle u, e_1\rangle \varphi(e_1) + \cdots + \langle u, e_n\rangle \varphi(e_n)\\ &= \left\langle u, \overline{\varphi(e_1)} e_1 + \cdots + \overline{\varphi(e_n)} e_n\right\rangle\end{aligned}$$

for every $u \in V$, where the first equality comes from 6.30(a). Thus setting

6.43 $$v = \overline{\varphi(e_1)} e_1 + \cdots + \overline{\varphi(e_n)} e_n,$$

we have $\varphi(u) = \langle u, v\rangle$ for every $u \in V$, as desired.

Now we prove that only one vector $v \in V$ has the desired behavior. Suppose $v_1, v_2 \in V$ are such that

$$\varphi(u) = \langle u, v_1\rangle = \langle u, v_2\rangle$$

for every $u \in V$. Then

$$0 = \langle u, v_1\rangle - \langle u, v_2\rangle = \langle u, v_1 - v_2\rangle$$

for every $u \in V$. Taking $u = v_1 - v_2$ shows that $v_1 - v_2 = 0$. Thus $v_1 = v_2$, completing the proof of the uniqueness part of the result. ∎

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6.44 example: *computation illustrating Riesz representation theorem*

Suppose we want to find a polynomial $q \in \mathcal{P}_2(\mathbf{R})$ such that

$$\int_{-1}^{1} p(t)(\cos(\pi t))\,dt = \int_{-1}^{1} pq \tag{6.45}$$

for every polynomial $p \in \mathcal{P}_2(\mathbf{R})$. To do this, we make $\mathcal{P}_2(\mathbf{R})$ into an inner product space by defining $\langle p, q\rangle$ to be the right side of the equation above for $p, q \in \mathcal{P}_2(\mathbf{R})$. Note that the left side of the equation above does not equal the inner product in $\mathcal{P}_2(\mathbf{R})$ of $p$ and the function $t \mapsto \cos(\pi t)$ because this last function is not a polynomial.

Define a linear functional $\varphi$ on $\mathcal{P}_2(\mathbf{R})$ by letting

$$\varphi(p) = \int_{-1}^{1} p(t)(\cos(\pi t))\,dt$$

for each $p \in \mathcal{P}_2(\mathbf{R})$. Now use the orthonormal basis from Example 6.34 and apply formula 6.43 from the proof of the Riesz representation theorem to see that if $p \in \mathcal{P}_2(\mathbf{R})$, then $\varphi(p) = \langle p, q\rangle$, where

$$q(x) = \left(\int_{-1}^{1} \sqrt{\tfrac{1}{2}}\cos(\pi t)\,dt\right)\sqrt{\tfrac{1}{2}} + \left(\int_{-1}^{1} \sqrt{\tfrac{3}{2}}\,t\cos(\pi t)\,dt\right)\sqrt{\tfrac{3}{2}}x$$

$$+ \left(\int_{-1}^{1} \sqrt{\tfrac{45}{8}}\left(t^2 - \tfrac{1}{3}\right)\cos(\pi t)\,dt\right)\sqrt{\tfrac{45}{8}}\left(x^2 - \tfrac{1}{3}\right).$$

A bit of calculus applied to the equation above shows that

$$q(x) = \tfrac{15}{2\pi^2}(1 - 3x^2).$$

The same procedure shows that if we want to find $q \in \mathcal{P}_5(\mathbf{R})$ such that 6.45 holds for all $p \in \mathcal{P}_5(\mathbf{R})$, then we should take

$$q(x) = \tfrac{105}{8\pi^4}\left((27 - 2\pi^2) + (24\pi^2 - 270)x^2 + (315 - 30\pi^2)x^4\right).$$

Suppose $V$ is finite-dimensional and $\varphi$ a linear functional on $V$. Then 6.43 gives a formula for the vector $v$ that satisfies

$$\varphi(u) = \langle u, v\rangle$$

for all $u \in V$. Specifically, we have

$$v = \overline{\varphi(e_1)}e_1 + \cdots + \overline{\varphi(e_n)}e_n.$$

The right side of the equation above seems to depend on the orthonormal basis $e_1, \ldots, e_n$ as well as on $\varphi$. However, 6.42 tells us that $v$ is uniquely determined by $\varphi$. Thus the right side of the equation above is the same regardless of which orthonormal basis $e_1, \ldots, e_n$ of $V$ is chosen.

For two additional different proofs of the Riesz representation theorem, see 6.58 and also Exercise 13 in Section 6C.

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## *Exercises 6B*

**1** Suppose $e_1, \dots, e_m$ is a list of vectors in $V$ such that

$$\|a_1 e_1 + \cdots + a_m e_m\|^2 = |a_1|^2 + \cdots + |a_m|^2$$

for all $a_1, \dots, a_m \in \mathbf{F}$. Show that $e_1, \dots, e_m$ is an orthonormal list.

*This exercise provides a converse to 6.24.*

**2** (a) Suppose $\theta \in \mathbf{R}$. Show that both

$$(\cos\theta, \sin\theta), (-\sin\theta, \cos\theta) \quad \text{and} \quad (\cos\theta, \sin\theta), (\sin\theta, -\cos\theta)$$

are orthonormal bases of $\mathbf{R}^2$.

(b) Show that each orthonormal basis of $\mathbf{R}^2$ is of the form given by one of the two possibilities in (a).

**3** Suppose $e_1, \dots, e_m$ is an orthonormal list in $V$ and $v \in V$. Prove that

$$\|v\|^2 = |\langle v, e_1\rangle|^2 + \cdots + |\langle v, e_m\rangle|^2 \iff v \in \operatorname{span}(e_1, \dots, e_m).$$

**4** Suppose $n$ is a positive integer. Prove that

$$\frac{1}{\sqrt{2\pi}}, \frac{\cos x}{\sqrt{\pi}}, \frac{\cos 2x}{\sqrt{\pi}}, \dots, \frac{\cos nx}{\sqrt{\pi}}, \frac{\sin x}{\sqrt{\pi}}, \frac{\sin 2x}{\sqrt{\pi}}, \dots, \frac{\sin nx}{\sqrt{\pi}}$$

is an orthonormal list of vectors in $C[-\pi, \pi]$, the vector space of continuous real-valued functions on $[-\pi, \pi]$ with inner product

$$\langle f, g\rangle = \int_{-\pi}^{\pi} fg.$$

*Hint: The following formulas should help.*

$$(\sin x)(\cos y) = \frac{\sin(x-y) + \sin(x+y)}{2}$$

$$(\sin x)(\sin y) = \frac{\cos(x-y) - \cos(x+y)}{2}$$

$$(\cos x)(\cos y) = \frac{\cos(x-y) + \cos(x+y)}{2}$$

**5** Suppose $f\colon [-\pi, \pi] \to \mathbf{R}$ is continuous. For each nonnegative integer $k$, define

$$a_k = \frac{1}{\sqrt{\pi}} \int_{-\pi}^{\pi} f(x)\cos(kx)\,dx \quad \text{and} \quad b_k = \frac{1}{\sqrt{\pi}} \int_{-\pi}^{\pi} f(x)\sin(kx)\,dx.$$

Prove that

$$\frac{a_0^2}{2} + \sum_{k=1}^{\infty} (a_k^2 + b_k^2) \le \int_{-\pi}^{\pi} f^2.$$

*The inequality above is actually an equality for all continuous functions $f\colon [-\pi, \pi] \to \mathbf{R}$. However, proving that this inequality is an equality involves Fourier series techniques beyond the scope of this book.*

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**6** Suppose $e_1, \dots, e_n$ is an orthonormal basis of $V$.

(a) Prove that if $v_1, \dots, v_n$ are vectors in $V$ such that

$$\|e_k - v_k\| < \frac{1}{\sqrt{n}}$$

for each $k$, then $v_1, \dots, v_n$ is a basis of $V$.

(b) Show that there exist $v_1, \dots, v_n \in V$ such that

$$\|e_k - v_k\| \le \frac{1}{\sqrt{n}}$$

for each $k$, but $v_1, \dots, v_n$ is not linearly independent.

*This exercise states in (a) that an appropriately small perturbation of an orthonormal basis is a basis. Then (b) shows that the number $1/\sqrt{n}$ on the right side of the inequality in (a) cannot be improved upon.*

**7** Suppose $T \in \mathcal{L}(\mathbf{R}^3)$ has an upper-triangular matrix with respect to the basis $(1, 0, 0), (1, 1, 1), (1, 1, 2)$. Find an orthonormal basis of $\mathbf{R}^3$ with respect to which $T$ has an upper-triangular matrix.

**8** Make $\mathcal{P}_2(\mathbf{R})$ into an inner product space by defining $\langle p, q \rangle = \int_0^1 pq$ for all $p, q \in \mathcal{P}_2(\mathbf{R})$.

(a) Apply the Gram–Schmidt procedure to the basis $1, x, x^2$ to produce an orthonormal basis of $\mathcal{P}_2(\mathbf{R})$.

(b) The differentiation operator (the operator that takes $p$ to $p'$) on $\mathcal{P}_2(\mathbf{R})$ has an upper-triangular matrix with respect to the basis $1, x, x^2$, which is not an orthonormal basis. Find the matrix of the differentiation operator on $\mathcal{P}_2(\mathbf{R})$ with respect to the orthonormal basis produced in (a) and verify that this matrix is upper triangular, as expected from the proof of 6.37.

**9** Suppose $e_1, \dots, e_m$ is the result of applying the Gram–Schmidt procedure to a linearly independent list $v_1, \dots, v_m$ in $V$. Prove that $\langle v_k, e_k \rangle > 0$ for each $k = 1, \dots, m$.

**10** Suppose $v_1, \dots, v_m$ is a linearly independent list in $V$. Explain why the orthonormal list produced by the formulas of the Gram–Schmidt procedure (6.32) is the only orthonormal list $e_1, \dots, e_m$ in $V$ such that $\langle v_k, e_k \rangle > 0$ and $\operatorname{span}(v_1, \dots, v_k) = \operatorname{span}(e_1, \dots, e_k)$ for each $k = 1, \dots, m$.

*The result in this exercise is used in the proof of 7.58.*

**11** Find a polynomial $q \in \mathcal{P}_2(\mathbf{R})$ such that $p\left(\frac{1}{2}\right) = \int_0^1 pq$ for every $p \in \mathcal{P}_2(\mathbf{R})$.

**12** Find a polynomial $q \in \mathcal{P}_2(\mathbf{R})$ such that

$$\int_0^1 p(x) \cos(\pi x)\, dx = \int_0^1 pq$$

for every $p \in \mathcal{P}_2(\mathbf{R})$.

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**13** Show that a list $v_1, ..., v_m$ of vectors in $V$ is linearly dependent if and only if the Gram–Schmidt formula in 6.32 produces $f_k = 0$ for some $k \in \{1, ..., m\}$.

*This exercise gives an alternative to Gaussian elimination techniques for determining whether a list of vectors in an inner product space is linearly dependent.*

**14** Suppose $V$ is a real inner product space and $v_1, ..., v_m$ is a linearly independent list of vectors in $V$. Prove that there exist exactly $2^m$ orthonormal lists $e_1, ..., e_m$ of vectors in $V$ such that

$$\operatorname{span}(v_1, ..., v_k) = \operatorname{span}(e_1, ..., e_k)$$

for all $k \in \{1, ..., m\}$.

**15** Suppose $\langle \cdot, \cdot \rangle_1$ and $\langle \cdot, \cdot \rangle_2$ are inner products on $V$ such that $\langle u, v \rangle_1 = 0$ if and only if $\langle u, v \rangle_2 = 0$. Prove that there is a positive number $c$ such that $\langle u, v \rangle_1 = c\langle u, v \rangle_2$ for every $u, v \in V$.

*This exercise shows that if two inner products have the same pairs of orthogonal vectors, then each of the inner products is a scalar multiple of the other inner product.*

**16** Suppose $V$ is finite-dimensional. Suppose $\langle \cdot, \cdot \rangle_1$, $\langle \cdot, \cdot \rangle_2$ are inner products on $V$ with corresponding norms $\| \cdot \|_1$ and $\| \cdot \|_2$. Prove that there exists a positive number $c$ such that $\|v\|_1 \le c\|v\|_2$ for every $v \in V$.

**17** Suppose $\mathbf{F} = \mathbf{C}$ and $V$ is finite-dimensional. Prove that if $T$ is an operator on $V$ such that 1 is the only eigenvalue of $T$ and $\|Tv\| \le \|v\|$ for all $v \in V$, then $T$ is the identity operator.

**18** Suppose $u_1, ..., u_m$ is a linearly independent list in $V$. Show that there exists $v \in V$ such that $\langle u_k, v \rangle = 1$ for all $k \in \{1, ..., m\}$.

**19** Suppose $v_1, ..., v_n$ is a basis of $V$. Prove that there exists a basis $u_1, ..., u_n$ of $V$ such that

$$\langle v_j, u_k \rangle = \begin{cases} 0 & \text{if } j \ne k, \\ 1 & \text{if } j = k. \end{cases}$$

**20** Suppose $\mathbf{F} = \mathbf{C}$, $V$ is finite-dimensional, and $\mathcal{E} \subseteq \mathcal{L}(V)$ is such that

$$ST = TS$$

for all $S, T \in \mathcal{E}$. Prove that there is an orthonormal basis of $V$ with respect to which every element of $\mathcal{E}$ has an upper-triangular matrix.

*This exercise strengthens Exercise 9(b) in Section 5E (in the context of inner product spaces) by asserting that the basis in that exercise can be chosen to be orthonormal.*

**21** Suppose $\mathbf{F} = \mathbf{C}$, $V$ is finite-dimensional, $T \in \mathcal{L}(V)$, and all eigenvalues of $T$ have absolute value less than 1. Let $\epsilon > 0$. Prove that there exists a positive integer $m$ such that $\|T^m v\| \le \epsilon\|v\|$ for every $v \in V$.

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**22** Suppose $C[-1,1]$ is the vector space of continuous real-valued functions on the interval $[-1,1]$ with inner product given by

$$\langle f,g\rangle = \int_{-1}^{1} fg$$

for all $f,g \in C[-1,1]$. Let $\varphi$ be the linear functional on $C[-1,1]$ defined by $\varphi(f) = f(0)$. Show that there does not exist $g \in C[-1,1]$ such that

$$\varphi(f) = \langle f,g\rangle$$

for every $f \in C[-1,1]$.

*This exercise shows that the Riesz representation theorem (6.42) does not hold on infinite-dimensional vector spaces without additional hypotheses on $V$ and $\varphi$.*

**23** For all $u, v \in V$, define $d(u,v) = \|u - v\|$.

(a) Show that $d$ is a metric on $V$.

(b) Show that if $V$ is finite-dimensional, then $d$ is a complete metric on $V$ (meaning that every Cauchy sequence converges).

(c) Show that every finite-dimensional subspace of $V$ is a closed subset of $V$ (with respect to the metric $d$).

*This exercise requires familiarity with metric spaces.*

### *orthogonality at the Supreme Court*

Law professor Richard Friedman presenting a case before the U.S. Supreme Court in 2010:

*Mr. Friedman*: I think that issue is entirely orthogonal to the issue here because the Commonwealth is acknowledging—
*Chief Justice Roberts*: I'm sorry. Entirely what?
*Mr. Friedman*: Orthogonal. Right angle. Unrelated. Irrelevant.
*Chief Justice Roberts*: Oh.
*Justice Scalia*: What was that adjective? I liked that.
*Mr. Friedman*: Orthogonal.
*Chief Justice Roberts*: Orthogonal.
*Mr. Friedman*: Right, right.
*Justice Scalia*: Orthogonal, ooh. (Laughter.)
*Justice Kennedy*: I knew this case presented us a problem. (Laughter.)

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# 6C Orthogonal Complements and Minimization Problems

## Orthogonal Complements

**6.46 definition: *orthogonal complement*, $U^\perp$**

If $U$ is a subset of $V$, then the *orthogonal complement* of $U$, denoted by $U^\perp$, is the set of all vectors in $V$ that are orthogonal to every vector in $U$:

$$U^\perp = \{v \in V : \langle u, v\rangle = 0 \text{ for every } u \in U\}.$$

The orthogonal complement $U^\perp$ depends on $V$ as well as on $U$. However, the inner product space $V$ should always be clear from the context and thus it can be omitted from the notation.

**6.47 example: *orthogonal complements***

- If $V = \mathbf{R}^3$ and $U$ is the subset of $V$ consisting of the single point $(2, 3, 5)$, then $U^\perp$ is the plane $\{(x, y, z) \in \mathbf{R}^3 : 2x + 3y + 5z = 0\}$.
- If $V = \mathbf{R}^3$ and $U$ is the plane $\{(x, y, z) \in \mathbf{R}^3 : 2x + 3y + 5z = 0\}$, then $U^\perp$ is the line $\{(2t, 3t, 5t) : t \in \mathbf{R}\}$.
- More generally, if $U$ is a plane in $\mathbf{R}^3$ containing the origin, then $U^\perp$ is the line containing the origin that is perpendicular to $U$.
- If $U$ is a line in $\mathbf{R}^3$ containing the origin, then $U^\perp$ is the plane containing the origin that is perpendicular to $U$.
- If $V = \mathbf{F}^5$ and $U = \{(a, b, 0, 0, 0) \in \mathbf{F}^5 : a, b \in \mathbf{F}\}$, then
$$U^\perp = \{(0, 0, x, y, z) \in \mathbf{F}^5 : x, y, z \in \mathbf{F}\}.$$
- If $e_1, \ldots, e_m, f_1, \ldots, f_n$ is an orthonormal basis of $V$, then
$$(\operatorname{span}(e_1, \ldots, e_m))^\perp = \operatorname{span}(f_1, \ldots, f_n).$$

We begin with some straightforward consequences of the definition.

**6.48 *properties of orthogonal complement***

(a) If $U$ is a subset of $V$, then $U^\perp$ is a subspace of $V$.

(b) $\{0\}^\perp = V$.

(c) $V^\perp = \{0\}$.

(d) If $U$ is a subset of $V$, then $U \cap U^\perp \subseteq \{0\}$.

(e) If $G$ and $H$ are subsets of $V$ and $G \subseteq H$, then $H^\perp \subseteq G^\perp$.

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Proof

(a) Suppose $U$ is a subset of $V$. Then $\langle u, 0\rangle = 0$ for every $u \in U$; thus $0 \in U^\perp$.

Suppose $v, w \in U^\perp$. If $u \in U$, then

$$\langle u, v + w\rangle = \langle u, v\rangle + \langle u, w\rangle = 0 + 0 = 0.$$

Thus $v + w \in U^\perp$, which shows that $U^\perp$ is closed under addition.

Similarly, suppose $\lambda \in \mathbf{F}$ and $v \in U^\perp$. If $u \in U$, then

$$\langle u, \lambda v\rangle = \overline{\lambda}\langle u, v\rangle = \overline{\lambda} \cdot 0 = 0.$$

Thus $\lambda v \in U^\perp$, which shows that $U^\perp$ is closed under scalar multiplication. Thus $U^\perp$ is a subspace of $V$.

(b) Suppose that $v \in V$. Then $\langle 0, v\rangle = 0$, which implies that $v \in \{0\}^\perp$. Thus $\{0\}^\perp = V$.

(c) Suppose that $v \in V^\perp$. Then $\langle v, v\rangle = 0$, which implies that $v = 0$. Thus $V^\perp = \{0\}$.

(d) Suppose $U$ is a subset of $V$ and $u \in U \cap U^\perp$. Then $\langle u, u\rangle = 0$, which implies that $u = 0$. Thus $U \cap U^\perp \subseteq \{0\}$.

(e) Suppose $G$ and $H$ are subsets of $V$ and $G \subseteq H$. Suppose $v \in H^\perp$. Then $\langle u, v\rangle = 0$ for every $u \in H$, which implies that $\langle u, v\rangle = 0$ for every $u \in G$. Hence $v \in G^\perp$. Thus $H^\perp \subseteq G^\perp$. ∎

Recall that if $U$ and $W$ are subspaces of $V$, then $V$ is the direct sum of $U$ and $W$ (written $V = U \oplus W$) if each element of $V$ can be written in exactly one way as a vector in $U$ plus a vector in $W$ (see 1.41). Furthermore, this happens if and only if $V = U + W$ and $U \cap W = \{0\}$ (see 1.46).

The next result shows that every finite-dimensional subspace of $V$ leads to a natural direct sum decomposition of $V$. See Exercise 16 for an example showing that the result below can fail without the hypothesis that the subspace $U$ is finite-dimensional.

**6.49** *direct sum of a subspace and its orthogonal complement*

Suppose $U$ is a finite-dimensional subspace of $V$. Then

$$V = U \oplus U^\perp.$$

Proof First we will show that

$$V = U + U^\perp.$$

To do this, suppose that $v \in V$. Let $e_1, \ldots, e_m$ be an orthonormal basis of $U$. We want to write $v$ as the sum of a vector in $U$ and a vector orthogonal to $U$.

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We have

6.50 $$v = \underbrace{\langle v, e_1\rangle e_1 + \cdots + \langle v, e_m\rangle e_m}_{u} + \underbrace{v - \langle v, e_1\rangle e_1 - \cdots - \langle v, e_m\rangle e_m}_{w}.$$

Let $u$ and $w$ be defined as in the equation above (as was done in the proof of 6.26). Because each $e_k \in U$, we see that $u \in U$. Because $e_1, \ldots, e_m$ is an orthonormal list, for each $k = 1, \ldots, m$ we have

$$\begin{aligned}\langle w, e_k\rangle &= \langle v, e_k\rangle - \langle v, e_k\rangle \\ &= 0.\end{aligned}$$

Thus $w$ is orthogonal to every vector in $\operatorname{span}(e_1, \ldots, e_m)$, which shows that $w \in U^\perp$. Hence we have written $v = u + w$, where $u \in U$ and $w \in U^\perp$, completing the proof that $V = U + U^\perp$.

From 6.48(d), we know that $U \cap U^\perp = \{0\}$. Now equation $V = U + U^\perp$ implies that $V = U \oplus U^\perp$ (see 1.46). ∎

Now we can see how to compute $\dim U^\perp$ from $\dim U$.

**6.51** *dimension of orthogonal complement*

Suppose $V$ is finite-dimensional and $U$ is a subspace of $V$. Then

$$\dim U^\perp = \dim V - \dim U.$$

Proof The formula for $\dim U^\perp$ follows immediately from 6.49 and 3.94. ∎

The next result is an important consequence of 6.49.

**6.52** *orthogonal complement of the orthogonal complement*

Suppose $U$ is a finite-dimensional subspace of $V$. Then

$$U = (U^\perp)^\perp.$$

Proof First we will show that

6.53 $$U \subseteq (U^\perp)^\perp.$$

To do this, suppose $u \in U$. Then $\langle u, w\rangle = 0$ for every $w \in U^\perp$ (by the definition of $U^\perp$). Because $u$ is orthogonal to every vector in $U^\perp$, we have $u \in (U^\perp)^\perp$, completing the proof of 6.53.

To prove the inclusion in the other direction, suppose $v \in (U^\perp)^\perp$. By 6.49, we can write $v = u + w$, where $u \in U$ and $w \in U^\perp$. We have $v - u = w \in U^\perp$. Because $v \in (U^\perp)^\perp$ and $u \in (U^\perp)^\perp$ (from 6.53), we have $v - u \in (U^\perp)^\perp$. Thus $v - u \in U^\perp \cap (U^\perp)^\perp$, which implies that $v - u = 0$ [by 6.48(d)], which implies that $v = u$, which implies that $v \in U$. Thus $(U^\perp)^\perp \subseteq U$, which along with 6.53 completes the proof. ∎

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Suppose $U$ is a subspace of $V$ and we want to show that $U = V$. In some situations, the easiest way to do this is to show that the only vector orthogonal to $U$ is 0, and then use the result below. For example, the result below is useful for Exercise 4.

*Exercise 16(a) shows that the result below is not true without the hypothesis that $U$ is finite-dimensional.*

**6.54 $U^\perp = \{0\} \iff U = V$ (for $U$ a finite-dimensional subspace of $V$)**

Suppose $U$ is a finite-dimensional subspace of $V$. Then

$$U^\perp = \{0\} \iff U = V.$$

Proof First suppose $U^\perp = \{0\}$. Then by 6.52, $U = (U^\perp)^\perp = \{0\}^\perp = V$, as desired.

Conversely, if $U = V$, then $U^\perp = V^\perp = \{0\}$ by 6.48(c). ∎

We now define an operator $P_U$ for each finite-dimensional subspace $U$ of $V$.

**6.55 definition: *orthogonal projection*, $P_U$**

Suppose $U$ is a finite-dimensional subspace of $V$. The *orthogonal projection* of $V$ onto $U$ is the operator $P_U \in \mathcal{L}(V)$ defined as follows: For each $v \in V$, write $v = u + w$, where $u \in U$ and $w \in U^\perp$. Then let $P_U v = u$.

The direct sum decomposition $V = U \oplus U^\perp$ given by 6.49 shows that each $v \in V$ can be uniquely written in the form $v = u + w$ with $u \in U$ and $w \in U^\perp$. Thus $P_U v$ is well defined. See the figure that accompanies the proof of 6.61 for the picture describing $P_U v$ that you should keep in mind.

**6.56 example: *orthogonal projection onto one-dimensional subspace***

Suppose $u \in V$ with $u \neq 0$ and $U$ is the one-dimensional subspace of $V$ defined by $U = \operatorname{span}(u)$.

If $v \in V$, then

$$v = \frac{\langle v, u\rangle}{\|u\|^2} u + \left(v - \frac{\langle v, u\rangle}{\|u\|^2} u\right),$$

where the first term on the right is in $\operatorname{span}(u)$ (and thus is in $U$) and the second term on the right is orthogonal to $u$ (and thus is in $U^\perp$). Thus $P_U v$ equals the first term on the right. In other words, we have the formula

$$P_U v = \frac{\langle v, u\rangle}{\|u\|^2} u$$

for every $v \in V$.

Taking $v = u$, the formula above becomes $P_U u = u$, as expected. Furthermore, taking $v \in \{u\}^\perp$, the formula above becomes $P_U v = 0$, also as expected.

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**6.57** *properties of orthogonal projection $P_U$*

Suppose $U$ is a finite-dimensional subspace of $V$. Then

(a) $P_U \in \mathcal{L}(V)$;

(b) $P_U u = u$ for every $u \in U$;

(c) $P_U w = 0$ for every $w \in U^\perp$;

(d) range $P_U = U$;

(e) null $P_U = U^\perp$;

(f) $v - P_U v \in U^\perp$ for every $v \in V$;

(g) $P_U^2 = P_U$;

(h) $\|P_U v\| \le \|v\|$ for every $v \in V$;

(i) if $e_1, \ldots, e_m$ is an orthonormal basis of $U$ and $v \in V$, then

$$P_U v = \langle v, e_1 \rangle e_1 + \cdots + \langle v, e_m \rangle e_m.$$

Proof

(a) To show that $P_U$ is a linear map on $V$, suppose $v_1, v_2 \in V$. Write

$$v_1 = u_1 + w_1 \quad \text{and} \quad v_2 = u_2 + w_2$$

with $u_1, u_2 \in U$ and $w_1, w_2 \in U^\perp$. Thus $P_U v_1 = u_1$ and $P_U v_2 = u_2$. Now

$$v_1 + v_2 = (u_1 + u_2) + (w_1 + w_2),$$

where $u_1 + u_2 \in U$ and $w_1 + w_2 \in U^\perp$. Thus

$$P_U(v_1 + v_2) = u_1 + u_2 = P_U v_1 + P_U v_2.$$

Similarly, suppose $\lambda \in \mathbf{F}$ and $v \in V$. Write $v = u + w$, where $u \in U$ and $w \in U^\perp$. Then $\lambda v = \lambda u + \lambda w$ with $\lambda u \in U$ and $\lambda w \in U^\perp$. Thus $P_U(\lambda v) = \lambda u = \lambda P_U v$.

Hence $P_U$ is a linear map from $V$ to $V$.

(b) Suppose $u \in U$. We can write $u = u + 0$, where $u \in U$ and $0 \in U^\perp$. Thus $P_U u = u$.

(c) Suppose $w \in U^\perp$. We can write $w = 0 + w$, where $0 \in U$ and $w \in U^\perp$. Thus $P_U w = 0$.

(d) The definition of $P_U$ implies that range $P_U \subseteq U$. Furthermore, (b) implies that $U \subseteq$ range $P_U$. Thus range $P_U = U$.

(e) The inclusion $U^\perp \subseteq$ null $P_U$ follows from (c). To prove the inclusion in the other direction, note that if $v \in$ null $P_U$ then the decomposition given by 6.49 must be $v = 0 + v$, where $0 \in U$ and $v \in U^\perp$. Thus null $P_U \subseteq U^\perp$.

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(f) If $v \in V$ and $v = u + w$ with $u \in U$ and $w \in U^\perp$, then

$$v - P_U v = v - u = w \in U^\perp.$$

(g) If $v \in V$ and $v = u + w$ with $u \in U$ and $w \in U^\perp$, then

$$(P_U^2)v = P_U(P_U v) = P_U u = u = P_U v.$$

(h) If $v \in V$ and $v = u + w$ with $u \in U$ and $w \in U^\perp$, then

$$\|P_U v\|^2 = \|u\|^2 \le \|u\|^2 + \|w\|^2 = \|v\|^2,$$

where the last equality comes from the Pythagorean theorem.

(i) The formula for $P_U v$ follows from equation 6.50 in the proof of 6.49. ∎

In the previous section we proved the Riesz representation theorem (6.42), whose key part states that every linear functional on a finite-dimensional inner product space is given by taking the inner product with some fixed vector. Seeing a different proof often provides new insight. Thus we now give a new proof of the key part of the Riesz representation theorem using orthogonal complements instead of orthonormal bases as in our previous proof.

The restatement below of the Riesz representation theorem provides an identification of $V$ with $V'$. We will prove only the "onto" part of the result below because the more routine "one-to-one" part of the result can be proved as in 6.42.

Intuition behind this new proof: If $\varphi \in V'$, $v \in V$, and $\varphi(u) = \langle u, v\rangle$ for all $u \in V$, then $v \in (\operatorname{null}\varphi)^\perp$. However, $(\operatorname{null}\varphi)^\perp$ is a one-dimensional subspace of $V$ (except for the trivial case in which $\varphi = 0$), as follows from 6.51 and 3.21. Thus we can obtain $v$ be choosing any nonzero element of $(\operatorname{null}\varphi)^\perp$ and then multiplying by an appropriate scalar, as is done in the proof below.

**6.58** *Riesz representation theorem, revisited*

Suppose $V$ is finite-dimensional. For each $v \in V$, define $\varphi_v \in V'$ by

$$\varphi_v(u) = \langle u, v\rangle$$

for each $u \in V$. Then $v \mapsto \varphi_v$ is a one-to-one function from $V$ onto $V'$.

*Proof* To show that $v \mapsto \varphi_v$ is surjective, suppose $\varphi \in V'$. If $\varphi = 0$, then $\varphi = \varphi_0$. Thus assume $\varphi \ne 0$. Hence $\operatorname{null}\varphi \ne V$, which implies that $(\operatorname{null}\varphi)^\perp \ne \{0\}$ (by 6.49 with $U = \operatorname{null}\varphi$).

***Caution:*** *The function $v \mapsto \varphi_v$ is a linear mapping from $V$ to $V'$ if $\mathbf{F} = \mathbf{R}$. However, this function is not linear if $\mathbf{F} = \mathbf{C}$ because $\varphi_{\lambda v} = \overline{\lambda}\varphi_v$ if $\lambda \in \mathbf{C}$.*

Let $w \in (\operatorname{null}\varphi)^\perp$ be such that $w \ne 0$. Let

$$v = \frac{\overline{\varphi(w)}}{\|w\|^2} w. \tag{6.59}$$

Then $v \in (\operatorname{null}\varphi)^\perp$. Also, $v \ne 0$ (because $w \notin \operatorname{null}\varphi$).

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Taking the norm of both sides of 6.59 gives

6.60
$$\|v\| = \frac{|\varphi(w)|}{\|w\|}.$$

Applying $\varphi$ to both sides of 6.59 and then using 6.60, we have

$$\varphi(v) = \frac{|\varphi(w)|^2}{\|w\|^2} = \|v\|^2.$$

Now suppose $u \in V$. Using the equation above, we have

$$u = \left(u - \frac{\varphi(u)}{\varphi(v)}v\right) + \frac{\varphi(u)}{\|v\|^2}v.$$

The first term in parentheses above is in null $\varphi$ and hence is orthogonal to $v$. Thus taking the inner product of both sides of the equation above with $v$ shows that

$$\langle u, v\rangle = \frac{\varphi(u)}{\|v\|^2}\langle v, v\rangle = \varphi(u).$$

Thus $\varphi = \varphi_v$, showing that $v \mapsto \varphi_v$ is surjective, as desired. ∎

See Exercise 13 for yet another proof of the Riesz representation theorem.

## *Minimization Problems*

The following problem often arises: Given a subspace $U$ of $V$ and a point $v \in V$, find a point $u \in U$ such that $\|v - u\|$ is as small as possible. The next result shows that $u = P_U v$ is the unique solution of this minimization problem.

*The remarkable simplicity of the solution to this minimization problem has led to many important applications of inner product spaces outside of pure mathematics.*

**6.61** *minimizing distance to a subspace*

Suppose $U$ is a finite-dimensional subspace of $V$, $v \in V$, and $u \in U$. Then

$$\|v - P_U v\| \le \|v - u\|.$$

Furthermore, the inequality above is an equality if and only if $u = P_U v$.

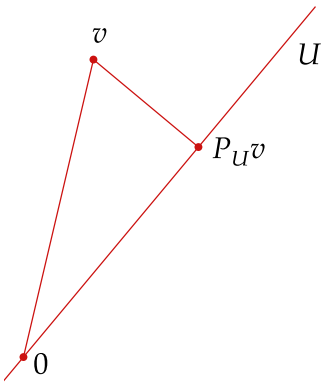
Proof We have

6.62
$$\begin{aligned}\|v - P_U v\|^2 &\le \|v - P_U v\|^2 + \|P_U v - u\|^2\\ &= \|(v - P_U v) + (P_U v - u)\|^2\\ &= \|v - u\|^2,\end{aligned}$$

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where the first line above holds because $0 \le \|P_U v - u\|^2$, the second line above comes from the Pythagorean theorem [which applies because $v - P_U v \in U^\perp$ by 6.57(f), and $P_U v - u \in U$], and the third line above holds by simple algebra. Taking square roots gives the desired inequality.

The inequality proved above is an equality if and only if 6.62 is an equality, which happens if and only if $\|P_U v - u\| = 0$, which happens if and only if $u = P_U v$. ∎




*$P_U v$ is the closest point in $U$ to $v$.*

The last result is often combined with the formula 6.57(i) to compute explicit solutions to minimization problems, as in the following example.

6.63 example: *using linear algebra to approximate the sine function*

Suppose we want to find a polynomial $u$ with real coefficients and of degree at most 5 that approximates the sine function as well as possible on the interval $[-\pi, \pi]$, in the sense that

$$\int_{-\pi}^{\pi} |\sin x - u(x)|^2 \, dx$$

is as small as possible.

Let $C[-\pi, \pi]$ denote the real inner product space of continuous real-valued functions on $[-\pi, \pi]$ with inner product

6.64 $$\langle f, g \rangle = \int_{-\pi}^{\pi} fg.$$

Let $v \in C[-\pi, \pi]$ be the function defined by $v(x) = \sin x$. Let $U$ denote the subspace of $C[-\pi, \pi]$ consisting of the polynomials with real coefficients and of degree at most 5. Our problem can now be reformulated as follows:

Find $u \in U$ such that $\|v - u\|$ is as small as possible.

*A computer that can integrate is useful here.*

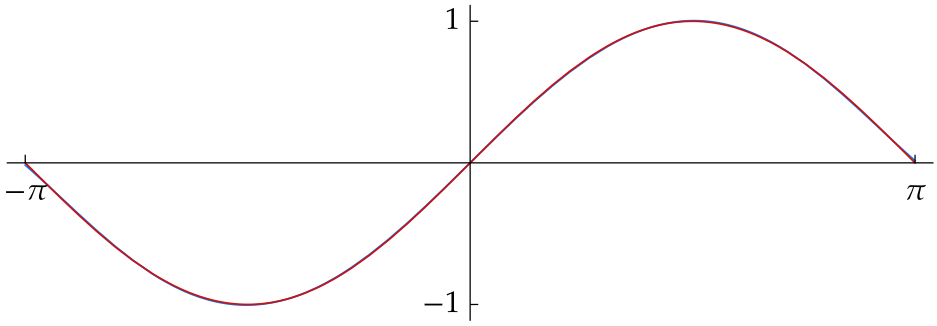
To compute the solution to our approximation problem, first apply the Gram–Schmidt procedure (using the inner product given by 6.64) to the basis $1, x, x^2, x^3, x^4, x^5$ of $U$, producing an orthonormal basis $e_1, e_2, e_3, e_4, e_5, e_6$ of $U$. Then, again using the inner product given by 6.64, compute $P_U v$ using 6.57(i) (with $m = 6$). Doing this computation shows that $P_U v$ is the function $u$ defined by

6.65 $$u(x) = 0.987862x - 0.155271x^3 + 0.00564312x^5,$$

where the $\pi$'s that appear in the exact answer have been replaced with a good decimal approximation. By 6.61, the polynomial $u$ above is the best approximation to the sine function on $[-\pi, \pi]$ using polynomials of degree at most 5 (here "best approximation" means in the sense of minimizing $\int_{-\pi}^{\pi} |\sin x - u(x)|^2 \, dx$).

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To see how good this approximation is, the next figure shows the graphs of both the sine function and our approximation $u$ given by 6.65 over the interval $[-\pi, \pi]$.



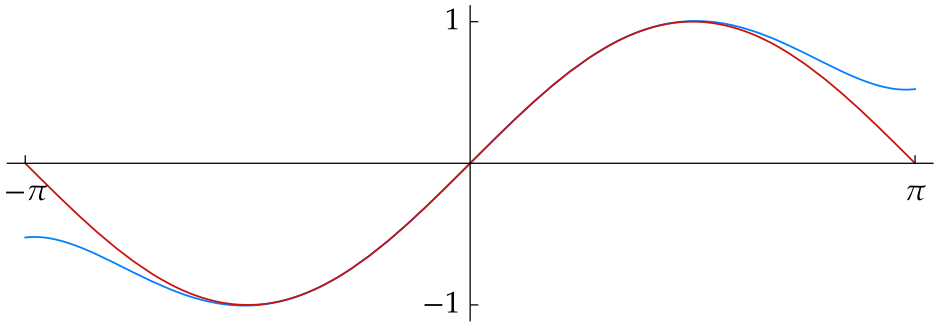

*Graphs on $[-\pi, \pi]$ of the sine function (red) and its best fifth degree polynomial approximation $u$ (blue) from 6.65.*

Our approximation 6.65 is so accurate that the two graphs are almost identical—our eyes may see only one graph! Here the red graph is placed almost exactly over the blue graph. If you are viewing this on an electronic device, enlarge the picture above by 400% near $\pi$ or $-\pi$ to see a small gap between the two graphs.

Another well-known approximation to the sine function by a polynomial of degree 5 is given by the Taylor polynomial $p$ defined by

6.66
$$p(x) = x - \frac{x^3}{3!} + \frac{x^5}{5!}.$$

To see how good this approximation is, the next picture shows the graphs of both the sine function and the Taylor polynomial $p$ over the interval $[-\pi, \pi]$.




*Graphs on $[-\pi, \pi]$ of the sine function (red) and the Taylor polynomial (blue) from 6.66.*

The Taylor polynomial is an excellent approximation to $\sin x$ for $x$ near 0. But the picture above shows that for $|x| > 2$, the Taylor polynomial is not so accurate, especially compared to 6.65. For example, taking $x = 3$, our approximation 6.65 estimates $\sin 3$ with an error of approximately 0.001, but the Taylor series 6.66 estimates $\sin 3$ with an error of approximately 0.4. Thus at $x = 3$, the error in the Taylor series is hundreds of times larger than the error given by 6.65. Linear algebra has helped us discover an approximation to the sine function that improves upon what we learned in calculus!

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## *Pseudoinverse*

Suppose $T \in \mathcal{L}(V, W)$ and $b \in W$. Consider the problem of finding $x \in V$ such that

$$Tx = b.$$

For example, if $V = \mathbf{F}^n$ and $W = \mathbf{F}^m$, then the equation above could represent a system of $m$ linear equations in $n$ unknowns.

If $T$ is invertible, then the unique solution to the equation above is $x = T^{-1}b$. However, if $T$ is not invertible, then for some $b \in W$ there may not exist any solutions of the equation above, and for some $b \in W$ there may exist infinitely many solutions of the equation above.

If $T$ is not invertible, then we can still try to do as well as possible with the equation above. For example, if the equation above has no solutions, then instead of solving the equation $Tx - b = 0$, we can try to find $x \in V$ such that $\|Tx - b\|$ is as small as possible. As another example, if the equation above has infinitely many solutions $x \in V$, then among all those solutions we can try to find one such that $\|x\|$ is as small as possible.

The pseudoinverse will provide the tool to solve the equation above as well as possible, even when $T$ is not invertible. We need the next result to define the pseudoinverse.

In the next two proofs, we will use without further comment the result that if $V$ is finite-dimensional and $T \in \mathcal{L}(V, W)$, then $\operatorname{null} T$, $(\operatorname{null} T)^\perp$, and $\operatorname{range} T$ are all finite-dimensional.

**6.67** *restriction of a linear map to obtain a one-to-one and onto map*

Suppose $V$ is finite-dimensional and $T \in \mathcal{L}(V, W)$. Then $T|_{(\operatorname{null} T)^\perp}$ is a one-to-one map of $(\operatorname{null} T)^\perp$ onto $\operatorname{range} T$.

Proof Suppose that $v \in (\operatorname{null} T)^\perp$ and $T|_{(\operatorname{null} T)^\perp} v = 0$. Hence $Tv = 0$ and thus $v \in (\operatorname{null} T) \cap (\operatorname{null} T)^\perp$, which implies that $v = 0$ [by 6.48(d)]. Hence $\operatorname{null} T|_{(\operatorname{null} T)^\perp} = \{0\}$, which implies that $T|_{(\operatorname{null} T)^\perp}$ is injective, as desired.

Clearly $\operatorname{range} T|_{(\operatorname{null} T)^\perp} \subseteq \operatorname{range} T$. To prove the inclusion in the other direction, suppose $w \in \operatorname{range} T$. Hence there exists $v \in V$ such that $w = Tv$. There exist $u \in \operatorname{null} T$ and $x \in (\operatorname{null} T)^\perp$ such that $v = u + x$ (by 6.49). Now

$$T|_{(\operatorname{null} T)^\perp} x = Tx = Tv - Tu = w - 0 = w,$$

which shows that $w \in \operatorname{range} T|_{(\operatorname{null} T)^\perp}$. Hence $\operatorname{range} T \subseteq \operatorname{range} T|_{(\operatorname{null} T)^\perp}$, completing the proof that $\operatorname{range} T|_{(\operatorname{null} T)^\perp} = \operatorname{range} T$. ∎

Now we can define the pseudoinverse $T^\dagger$ (pronounced "$T$ dagger") of a linear map $T$. In the next definition (and from now on), think of $T|_{(\operatorname{null} T)^\perp}$ as an invertible linear map from $(\operatorname{null} T)^\perp$ onto $\operatorname{range} T$, as is justified by the result above.

*To produce the pseudoinverse notation $T^\dagger$ in TEX, type* `T^\dagger`.

**6.68 definition: *pseudoinverse*, $T^\dagger$**

Suppose that $V$ is finite-dimensional and $T \in \mathcal{L}(V, W)$. The *pseudoinverse* $T^\dagger \in \mathcal{L}(W, V)$ of $T$ is the linear map from $W$ to $V$ defined by

$$T^\dagger w = (T|_{(\operatorname{null} T)^\perp})^{-1} P_{\operatorname{range} T}\, w$$

for each $w \in W$.

Recall that $P_{\operatorname{range} T}\, w = 0$ if $w \in (\operatorname{range} T)^\perp$ and $P_{\operatorname{range} T}\, w = w$ if $w \in \operatorname{range} T$. Thus if $w \in (\operatorname{range} T)^\perp$, then $T^\dagger w = 0$, and if $w \in \operatorname{range} T$, then $T^\dagger w$ is the unique element of $(\operatorname{null} T)^\perp$ such that $T(T^\dagger w) = w$.

The pseudoinverse behaves much like an inverse, as we will see.

**6.69 *algebraic properties of the pseudoinverse***

Suppose $V$ is finite-dimensional and $T \in \mathcal{L}(V, W)$.

(a) If $T$ is invertible, then $T^\dagger = T^{-1}$.

(b) $TT^\dagger = P_{\operatorname{range} T}$ = the orthogonal projection of $W$ onto $\operatorname{range} T$.

(c) $T^\dagger T = P_{(\operatorname{null} T)^\perp}$ = the orthogonal projection of $V$ onto $(\operatorname{null} T)^\perp$.

Proof

(a) Suppose $T$ is invertible. Then $(\operatorname{null} T)^\perp = V$ and $\operatorname{range} T = W$. Thus $T|_{(\operatorname{null} T)^\perp} = T$ and $P_{\operatorname{range} T}$ is the identity operator on $W$. Hence $T^\dagger = T^{-1}$.

(b) Suppose $w \in \operatorname{range} T$. Thus

$$TT^\dagger w = T(T|_{(\operatorname{null} T)^\perp})^{-1} w = w = P_{\operatorname{range} T}\, w.$$

If $w \in (\operatorname{range} T)^\perp$, then $T^\dagger w = 0$. Hence $TT^\dagger w = 0 = P_{\operatorname{range} T}\, w$. Thus $TT^\dagger$ and $P_{\operatorname{range} T}$ agree on $\operatorname{range} T$ and on $(\operatorname{range} T)^\perp$. Hence these two linear maps are equal (by 6.49).

(c) Suppose $v \in (\operatorname{null} T)^\perp$. Because $Tv \in \operatorname{range} T$, the definition of $T^\dagger$ shows that

$$T^\dagger(Tv) = (T|_{(\operatorname{null} T)^\perp})^{-1}(Tv) = v = P_{(\operatorname{null} T)^\perp} v.$$

If $v \in \operatorname{null} T$, then $T^\dagger Tv = 0 = P_{(\operatorname{null} T)^\perp} v$. Thus $T^\dagger T$ and $P_{(\operatorname{null} T)^\perp}$ agree on $(\operatorname{null} T)^\perp$ and on $\operatorname{null} T$. Hence these two linear maps are equal (by 6.49). ∎

*The pseudoinverse is also called the Moore–Penrose inverse.*

Suppose that $T \in \mathcal{L}(V, W)$. If $T$ is surjective, then $TT^\dagger$ is the identity operator on $W$, as follows from (b) in the result above. If $T$ is injective, then $T^\dagger T$ is the identity operator on $V$, as follows from (c) in the result above. For additional algebraic properties of the pseudoinverse, see Exercises 19–23.

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Suppose $T \in \mathcal{L}(V, W)$, $b \in W$, and we want to find $x \in V$ that solves the equation

$$Tx = b.$$

If $T$ is invertible, then $x = T^{-1}b$ is the unique solution. If $T$ is not invertible, then $T^{-1}$ is not defined. However, the pseudoinverse $T^\dagger$ is defined. Taking $x = T^\dagger b$ makes $Tx$ as close to $b$ as possible, as shown by (a) of the next result. Thus the pseudoinverse provides what is called a *best fit* to the equation above.

Among all vectors $x \in V$ that make $Tx$ as close as possible to $b$, the vector $T^\dagger b$ has the smallest norm, as shown by combining (b) in the next result with the condition for equality in (a).

**6.70** *pseudoinverse provides best approximate solution or best solution*

Suppose $V$ is finite-dimensional, $T \in \mathcal{L}(V, W)$, and $b \in W$.

(a) If $x \in V$, then

$$\|T(T^\dagger b) - b\| \le \|Tx - b\|,$$

with equality if and only if $x \in T^\dagger b + \operatorname{null} T$.

(b) If $x \in T^\dagger b + \operatorname{null} T$, then

$$\|T^\dagger b\| \le \|x\|,$$

with equality if and only if $x = T^\dagger b$.

Proof

(a) Suppose $x \in V$. Then

$$Tx - b = (Tx - TT^\dagger b) + (TT^\dagger b - b).$$

The first term in parentheses above is in range $T$. Because the operator $TT^\dagger$ is the orthogonal projection of $W$ onto range $T$ [by 6.69(b)], the second term in parentheses above is in $(\operatorname{range} T)^\perp$ [see 6.57(f)].

Thus the Pythagorean theorem implies the desired inequality that the norm of the second term in parentheses above is less than or equal to $\|Tx - b\|$, with equality if and only if the first term in parentheses above equals 0. Hence we have equality if and only if $x - T^\dagger b \in \operatorname{null} T$, which is equivalent to the statement that $x \in T^\dagger b + \operatorname{null} T$, completing the proof of (a).

(b) Suppose $x \in T^\dagger b + \operatorname{null} T$. Hence $x - T^\dagger b \in \operatorname{null} T$. Now

$$x = (x - T^\dagger b) + T^\dagger b.$$

The definition of $T^\dagger$ implies that $T^\dagger b \in (\operatorname{null} T)^\perp$. Thus the Pythagorean theorem implies that $\|T^\dagger b\| \le \|x\|$, with equality if and only if $x = T^\dagger b$. ∎

A formula for $T^\dagger$ will be given in the next chapter (see 7.78).

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6.71 example: *pseudoinverse of a linear map from* $\mathbf{F}^4$ *to* $\mathbf{F}^3$

Suppose $T \in \mathcal{L}(\mathbf{F}^4, \mathbf{F}^3)$ is defined by

$$T(a, b, c, d) = (a + b + c, 2c + d, 0).$$

This linear map is neither injective nor surjective, but we can compute its pseudoinverse. To do this, first note that $\operatorname{range} T = \{(x, y, 0) : x, y \in \mathbf{F}\}$. Thus

$$P_{\operatorname{range} T}(x, y, z) = (x, y, 0)$$

for each $(x, y, z) \in \mathbf{F}^3$. Also,

$$\operatorname{null} T = \{(a, b, c, d) \in \mathbf{F}^4 : a + b + c = 0 \text{ and } 2c + d = 0\}.$$

The list $(-1, 1, 0, 0), (-1, 0, 1, -2)$ of two vectors in $\operatorname{null} T$ spans $\operatorname{null} T$ because if $(a, b, c, d) \in \operatorname{null} T$ then

$$(a, b, c, d) = b(-1, 1, 0, 0) + c(-1, 0, 1, -2).$$

Because the list $(-1, 1, 0, 0), (-1, 0, 1, -2)$ is linearly independent, this list is a basis of $\operatorname{null} T$.

Now suppose $(x, y, z) \in \mathbf{F}^3$. Then

$$T^\dagger(x, y, z) = (T|_{(\operatorname{null} T)^\perp})^{-1} P_{\operatorname{range} T}(x, y, z) = (T|_{(\operatorname{null} T)^\perp})^{-1}(x, y, 0). \tag{6.72}$$

The right side of the equation above is the vector $(a, b, c, d) \in \mathbf{F}^4$ such that $T(a, b, c, d) = (x, y, 0)$ and $(a, b, c, d) \in (\operatorname{null} T)^\perp$. In other words, $a$, $b$, $c$, $d$ must satisfy the following equations:

$$\begin{aligned} a + b + c &= x \\ 2c + d &= y \\ -a + b &= 0 \\ -a + c - 2d &= 0, \end{aligned}$$

where the first two equations are equivalent to the equation $T(a, b, c, d) = (x, y, 0)$ and the last two equations come from the condition for $(a, b, c, d)$ to be orthogonal to each of the basis vectors $(-1, 1, 0, 0), (-1, 0, 1, -2)$ in this basis of $\operatorname{null} T$. Thinking of $x$ and $y$ as constants and $a$, $b$, $c$, $d$ as unknowns, we can solve the system above of four equations in four unknowns, getting

$$a = \tfrac{1}{11}(5x - 2y),\ b = \tfrac{1}{11}(5x - 2y),\ c = \tfrac{1}{11}(x + 4y),\ d = \tfrac{1}{11}(-2x + 3y).$$

Hence 6.72 tells us that

$$T^\dagger(x, y, z) = \tfrac{1}{11}(5x - 2y, 5x - 2y, x + 4y, -2x + 3y).$$

The formula above for $T^\dagger$ shows that $TT^\dagger(x, y, z) = (x, y, 0)$ for all $(x, y, z) \in \mathbf{F}^3$, which illustrates the equation $TT^\dagger = P_{\operatorname{range} T}$ from 6.69(b).

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## *Exercises 6C*

**1** Suppose $v_1, ..., v_m \in V$. Prove that

$$\{v_1, ..., v_m\}^\perp = (\operatorname{span}(v_1, ..., v_m))^\perp.$$

**2** Suppose $U$ is a subspace of $V$ with basis $u_1, ..., u_m$ and

$$u_1, ..., u_m, v_1, ..., v_n$$

is a basis of $V$. Prove that if the Gram–Schmidt procedure is applied to the basis of $V$ above, producing a list $e_1, ..., e_m, f_1, ..., f_n$, then $e_1, ..., e_m$ is an orthonormal basis of $U$ and $f_1, ..., f_n$ is an orthonormal basis of $U^\perp$.

**3** Suppose $U$ is the subspace of $\mathbf{R}^4$ defined by

$$U = \operatorname{span}((1, 2, 3, -4), (-5, 4, 3, 2)).$$

Find an orthonormal basis of $U$ and an orthonormal basis of $U^\perp$.

**4** Suppose $e_1, ..., e_n$ is a list of vectors in $V$ with $\|e_k\| = 1$ for each $k = 1, ..., n$ and

$$\|v\|^2 = |\langle v, e_1\rangle|^2 + \cdots + |\langle v, e_n\rangle|^2$$

for all $v \in V$. Prove that $e_1, ..., e_n$ is an orthonormal basis of $V$.

*This exercise provides a converse to 6.30(b).*

**5** Suppose that $V$ is finite-dimensional and $U$ is a subspace of $V$. Show that $P_{U^\perp} = I - P_U$, where $I$ is the identity operator on $V$.

**6** Suppose $V$ is finite-dimensional and $T \in \mathcal{L}(V, W)$. Show that

$$T = TP_{(\operatorname{null} T)^\perp} = P_{\operatorname{range} T}T.$$

**7** Suppose that $X$ and $Y$ are finite-dimensional subspaces of $V$. Prove that $P_X P_Y = 0$ if and only if $\langle x, y\rangle = 0$ for all $x \in X$ and all $y \in Y$.

**8** Suppose $U$ is a finite-dimensional subspace of $V$ and $v \in V$. Define a linear functional $\varphi : U \to \mathbf{F}$ by

$$\varphi(u) = \langle u, v\rangle$$

for all $u \in U$. By the Riesz representation theorem (6.42) as applied to the inner product space $U$, there exists a unique vector $w \in U$ such that

$$\varphi(u) = \langle u, w\rangle$$

for all $u \in U$. Show that $w = P_U v$.

**9** Suppose $V$ is finite-dimensional. Suppose $P \in \mathcal{L}(V)$ is such that $P^2 = P$ and every vector in $\operatorname{null} P$ is orthogonal to every vector in $\operatorname{range} P$. Prove that there exists a subspace $U$ of $V$ such that $P = P_U$.

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**10** Suppose $V$ is finite-dimensional and $P \in \mathcal{L}(V)$ is such that $P^2 = P$ and

$$\|Pv\| \le \|v\|$$

for every $v \in V$. Prove that there exists a subspace $U$ of $V$ such that $P = P_U$.

**11** Suppose $T \in \mathcal{L}(V)$ and $U$ is a finite-dimensional subspace of $V$. Prove that

$$U \text{ is invariant under } T \iff P_U T P_U = T P_U.$$

**12** Suppose $V$ is finite-dimensional, $T \in \mathcal{L}(V)$, and $U$ is a subspace of $V$. Prove that

$$U \text{ and } U^\perp \text{ are both invariant under } T \iff P_U T = T P_U.$$

**13** Suppose $\mathbf{F} = \mathbf{R}$ and $V$ is finite-dimensional. For each $v \in V$, let $\varphi_v$ denote the linear functional on $V$ defined by

$$\varphi_v(u) = \langle u, v \rangle$$

for all $u \in V$.

(a) Show that $v \mapsto \varphi_v$ is an injective linear map from $V$ to $V'$.

(b) Use (a) and a dimension-counting argument to show that $v \mapsto \varphi_v$ is an isomorphism from $V$ onto $V'$.

*The purpose of this exercise is to give an alternative proof of the Riesz representation theorem (6.42 and 6.58) when* $\mathbf{F} = \mathbf{R}$. *Thus you should not use the Riesz representation theorem as a tool in your solution.*

**14** Suppose that $e_1, \ldots, e_n$ is an orthonormal basis of $V$. Explain why the dual basis (see 3.112) of $e_1, \ldots, e_n$ is $e_1, \ldots, e_n$ under the identification of $V'$ with $V$ provided by the Riesz representation theorem (6.58).

**15** In $\mathbf{R}^4$, let

$$U = \operatorname{span}((1, 1, 0, 0), (1, 1, 1, 2)).$$

Find $u \in U$ such that $\|u - (1, 2, 3, 4)\|$ is as small as possible.

**16** Suppose $C[-1, 1]$ is the vector space of continuous real-valued functions on the interval $[-1, 1]$ with inner product given by

$$\langle f, g \rangle = \int_{-1}^{1} fg$$

for all $f, g \in C[-1, 1]$. Let $U$ be the subspace of $C[-1, 1]$ defined by

$$U = \{f \in C[-1, 1] : f(0) = 0\}.$$

(a) Show that $U^\perp = \{0\}$.

(b) Show that 6.49 and 6.52 do not hold without the finite-dimensional hypothesis.

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**17** Find $p \in \mathcal{P}_3(\mathbf{R})$ such that $p(0) = 0$, $p'(0) = 0$, and $\int_0^1 |2 + 3x - p(x)|^2 \, dx$ is as small as possible.

**18** Find $p \in \mathcal{P}_5(\mathbf{R})$ that makes $\int_{-\pi}^{\pi} |\sin x - p(x)|^2 \, dx$ as small as possible.

*The polynomial 6.65 is an excellent approximation to the answer to this exercise, but here you are asked to find the exact solution, which involves powers of $\pi$. A computer that can perform symbolic integration should help.*

**19** Suppose $V$ is finite-dimensional and $P \in \mathcal{L}(V)$ is an orthogonal projection of $V$ onto some subspace of $V$. Prove that $P^\dagger = P$.

**20** Suppose $V$ is finite-dimensional and $T \in \mathcal{L}(V, W)$. Show that

$$\operatorname{null} T^\dagger = (\operatorname{range} T)^\perp \quad \text{and} \quad \operatorname{range} T^\dagger = (\operatorname{null} T)^\perp.$$

**21** Suppose $T \in \mathcal{L}(\mathbf{F}^3, \mathbf{F}^2)$ is defined by

$$T(a, b, c) = (a + b + c, 2b + 3c).$$

(a) For $(x, y) \in \mathbf{F}^2$, find a formula for $T^\dagger(x, y)$.

(b) Verify that the equation $TT^\dagger = P_{\operatorname{range} T}$ from 6.69(b) holds with the formula for $T^\dagger$ obtained in (a).

(c) Verify that the equation $T^\dagger T = P_{(\operatorname{null} T)^\perp}$ from 6.69(c) holds with the formula for $T^\dagger$ obtained in (a).

**22** Suppose $V$ is finite-dimensional and $T \in \mathcal{L}(V, W)$. Prove that

$$TT^\dagger T = T \quad \text{and} \quad T^\dagger T T^\dagger = T^\dagger.$$

*Both formulas above clearly hold if $T$ is invertible because in that case we can replace $T^\dagger$ with $T^{-1}$.*

**23** Suppose $V$ and $W$ are finite-dimensional and $T \in \mathcal{L}(V, W)$. Prove that

$$(T^\dagger)^\dagger = T.$$

*The equation above is analogous to the equation $(T^{-1})^{-1} = T$ that holds if $T$ is invertible.*

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Chapter 7

# Operators on Inner Product Spaces

The deepest results related to inner product spaces deal with the subject to which we now turn—linear maps and operators on inner product spaces. As we will see, good theorems can be proved by exploiting properties of the adjoint.

The hugely important spectral theorem will provide a complete description of self-adjoint operators on real inner product spaces and of normal operators on complex inner product spaces. We will then use the spectral theorem to help understand positive operators and unitary operators, which will lead to unitary matrices and matrix factorizations. The spectral theorem will also lead to the popular singular value decomposition, which will lead to the polar decomposition.

The most important results in the rest of this book are valid only in finite dimensions. Thus from now on we assume that $V$ and $W$ are finite-dimensional.

*standing assumptions for this chapter*

- $\mathbf{F}$ denotes $\mathbf{R}$ or $\mathbf{C}$.
- $V$ and $W$ are nonzero finite-dimensional inner product spaces over $\mathbf{F}$.



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*Market square in Lviv, a city that has had several names and has been in several countries because of changing international borders. From 1772 until 1918, the city was in Austria and was called Lemberg. Between World War I and World War II, the city was in Poland and was called Lwów. During this time, mathematicians in Lwów, particularly Stefan Banach (1892–1945) and his colleagues, developed the basic results of modern functional analysis, using tools of analysis to study infinite-dimensional vector spaces. Since the end of World War II, Lviv has been in Ukraine, which was part of the Soviet Union until Ukraine became an independent country in 1991.*

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# 7A Self-Adjoint and Normal Operators

## Adjoints

**7.1 definition: *adjoint*, $T^*$**

Suppose $T \in \mathcal{L}(V, W)$. The *adjoint* of $T$ is the function $T^* : W \to V$ such that

$$\langle Tv, w \rangle = \langle v, T^* w \rangle$$

for every $v \in V$ and every $w \in W$.

*The word **adjoint** has another meaning in linear algebra. In case you encounter the second meaning elsewhere, be warned that the two meanings for adjoint are unrelated to each other.*

To see why the definition above makes sense, suppose $T \in \mathcal{L}(V, W)$. Fix $w \in W$. Consider the linear functional

$$v \mapsto \langle Tv, w \rangle$$

on $V$ that maps $v \in V$ to $\langle Tv, w \rangle$; this linear functional depends on $T$ and $w$. By the Riesz representation theorem (6.42), there exists a unique vector in $V$ such that this linear functional is given by taking the inner product with it. We call this unique vector $T^* w$. In other words, $T^* w$ is the unique vector in $V$ such that

$$\langle Tv, w \rangle = \langle v, T^* w \rangle$$

for every $v \in V$.

In the equation above, the inner product on the left takes place in $W$ and the inner product on the right takes place in $V$. However, we use the same notation $\langle \cdot, \cdot \rangle$ for both inner products.

**7.2 example: *adjoint of a linear map from* $\mathbf{R}^3$ *to* $\mathbf{R}^2$**

Define $T : \mathbf{R}^3 \to \mathbf{R}^2$ by

$$T(x_1, x_2, x_3) = (x_2 + 3x_3, 2x_1).$$

To compute $T^*$, suppose $(x_1, x_2, x_3) \in \mathbf{R}^3$ and $(y_1, y_2) \in \mathbf{R}^2$. Then

$$\begin{aligned} \langle T(x_1, x_2, x_3), (y_1, y_2) \rangle &= \langle (x_2 + 3x_3, 2x_1), (y_1, y_2) \rangle \\ &= x_2 y_1 + 3x_3 y_1 + 2x_1 y_2 \\ &= \langle (x_1, x_2, x_3), (2y_2, y_1, 3y_1) \rangle. \end{aligned}$$

The equation above and the definition of the adjoint imply that

$$T^*(y_1, y_2) = (2y_2, y_1, 3y_1).$$

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7.3 example: *adjoint of a linear map with range of dimension at most* 1

Fix $u \in V$ and $x \in W$. Define $T \in \mathcal{L}(V, W)$ by

$$Tv = \langle v, u \rangle x$$

for each $v \in V$. To compute $T^*$, suppose $v \in V$ and $w \in W$. Then

$$\begin{aligned} \langle Tv, w \rangle &= \big\langle \langle v, u \rangle x, w \big\rangle \\ &= \langle v, u \rangle \langle x, w \rangle \\ &= \big\langle v, \langle w, x \rangle u \big\rangle. \end{aligned}$$

Thus

$$T^* w = \langle w, x \rangle u.$$

In the two examples above, $T^*$ turned out to be not just a function from $V$ to $W$ but a linear map from $V$ to $W$. This behavior is true in general, as shown by the next result.

*The two examples above and the proof below use a common technique for computing $T^*$: start with a formula for $\langle Tv, w \rangle$ then manipulate it to get just $v$ in the first slot; the entry in the second slot will then be $T^* w$.*

**7.4 *adjoint of a linear map is a linear map***

If $T \in \mathcal{L}(V, W)$, then $T^* \in \mathcal{L}(W, V)$.

Proof Suppose $T \in \mathcal{L}(V, W)$. If $v \in V$ and $w_1, w_2 \in W$, then

$$\begin{aligned} \langle Tv, w_1 + w_2 \rangle &= \langle Tv, w_1 \rangle + \langle Tv, w_2 \rangle \\ &= \big\langle v, T^* w_1 \big\rangle + \big\langle v, T^* w_2 \big\rangle \\ &= \big\langle v, T^* w_1 + T^* w_2 \big\rangle. \end{aligned}$$

The equation above shows that

$$T^*(w_1 + w_2) = T^* w_1 + T^* w_2.$$

If $v \in V$, $\lambda \in \mathbf{F}$, and $w \in W$, then

$$\begin{aligned} \langle Tv, \lambda w \rangle &= \overline{\lambda} \langle Tv, w \rangle \\ &= \overline{\lambda} \big\langle v, T^* w \big\rangle \\ &= \big\langle v, \lambda T^* w \big\rangle. \end{aligned}$$

The equation above shows that

$$T^*(\lambda w) = \lambda T^* w.$$

Thus $T^*$ is a linear map, as desired. ∎

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**7.5** *properties of the adjoint*

Suppose $T \in \mathcal{L}(V, W)$. Then

(a) $(S + T)^* = S^* + T^*$ for all $S \in \mathcal{L}(V, W)$;

(b) $(\lambda T)^* = \overline{\lambda} T^*$ for all $\lambda \in \mathbf{F}$;

(c) $(T^*)^* = T$;

(d) $(ST)^* = T^* S^*$ for all $S \in \mathcal{L}(W, U)$ (here $U$ is a finite-dimensional inner product space over $\mathbf{F}$);

(e) $I^* = I$, where $I$ is the identity operator on $V$;

(f) if $T$ is invertible, then $T^*$ is invertible and $(T^*)^{-1} = (T^{-1})^*$.

Proof Suppose $v \in V$ and $w \in W$.

(a) If $S \in \mathcal{L}(V, W)$, then

$$\begin{aligned} \langle (S + T)v, w \rangle &= \langle Sv, w \rangle + \langle Tv, w \rangle \\ &= \langle v, S^* w \rangle + \langle v, T^* w \rangle \\ &= \langle v, S^* w + T^* w \rangle. \end{aligned}$$

Thus $(S + T)^* w = S^* w + T^* w$, as desired.

(b) If $\lambda \in \mathbf{F}$, then

$$\langle (\lambda T)v, w \rangle = \lambda \langle Tv, w \rangle = \lambda \langle v, T^* w \rangle = \langle v, \overline{\lambda} T^* w \rangle.$$

Thus $(\lambda T)^* w = \overline{\lambda} T^* w$, as desired.

(c) We have

$$\langle T^* w, v \rangle = \overline{\langle v, T^* w \rangle} = \overline{\langle Tv, w \rangle} = \langle w, Tv \rangle.$$

Thus $(T^*)^* v = Tv$, as desired.

(d) Suppose $S \in \mathcal{L}(W, U)$ and $u \in U$. Then

$$\langle (ST)v, u \rangle = \langle S(Tv), u \rangle = \langle Tv, S^* u \rangle = \langle v, T^*(S^* u) \rangle.$$

Thus $(ST)^* u = T^*(S^* u)$, as desired.

(e) Suppose $u \in V$. Then

$$\langle Iu, v \rangle = \langle u, v \rangle.$$

Thus $I^* v = v$, as desired.

(f) Suppose $T$ is invertible. Take adjoints of both sides of the equation $T^{-1}T = I$, then use (d) and (e) to show that $T^*(T^{-1})^* = I$. Similarly, the equation $TT^{-1} = I$ implies $(T^{-1})^* T^* = I$. Thus $(T^{-1})^*$ is the inverse of $T^*$, as desired. ∎

If $\mathbf{F} = \mathbf{R}$, then the map $T \mapsto T^*$ is a linear map from $\mathcal{L}(V, W)$ to $\mathcal{L}(W, V)$, as follows from (a) and (b) of the result above. However, if $\mathbf{F} = \mathbf{C}$, then this map is not linear because of the complex conjugate that appears in (b).

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The next result shows the relationship between the null space and the range of a linear map and its adjoint.

**7.6 *null space and range of $T^*$***

Suppose $T \in \mathcal{L}(V, W)$. Then

(a) $\operatorname{null} T^* = (\operatorname{range} T)^\perp$;

(b) $\operatorname{range} T^* = (\operatorname{null} T)^\perp$;

(c) $\operatorname{null} T = (\operatorname{range} T^*)^\perp$;

(d) $\operatorname{range} T = (\operatorname{null} T^*)^\perp$.

Proof We begin by proving (a). Let $w \in W$. Then

$$\begin{aligned} w \in \operatorname{null} T^* &\iff T^* w = 0 \\ &\iff \langle v, T^* w \rangle = 0 \text{ for all } v \in V \\ &\iff \langle Tv, w \rangle = 0 \text{ for all } v \in V \\ &\iff w \in (\operatorname{range} T)^\perp. \end{aligned}$$

Thus $\operatorname{null} T^* = (\operatorname{range} T)^\perp$, proving (a).

If we take the orthogonal complement of both sides of (a), we get (d), where we have used 6.52. Replacing $T$ with $T^*$ in (a) gives (c), where we have used 7.5(c). Finally, replacing $T$ with $T^*$ in (d) gives (b). ∎

As we will soon see, the next definition is intimately connected to the matrix of the adjoint of a linear map.

**7.7 definition: *conjugate transpose, $A^*$***

The *conjugate transpose* of an $m$-by-$n$ matrix $A$ is the $n$-by-$m$ matrix $A^*$ obtained by interchanging the rows and columns and then taking the complex conjugate of each entry. In other words, if $j \in \{1, \ldots, n\}$ and $k \in \{1, \ldots, m\}$, then

$$(A^*)_{j,k} = \overline{A_{k,j}}.$$

**7.8 example: *conjugate transpose of a 2-by-3 matrix***

The conjugate transpose of the 2-by-3 matrix $\begin{pmatrix} 2 & 3+4i & 7 \\ 6 & 5 & 8i \end{pmatrix}$ is the 3-by-2 matrix

$$\begin{pmatrix} 2 & 6 \\ 3-4i & 5 \\ 7 & -8i \end{pmatrix}.$$

*If a matrix $A$ has only real entries, then $A^* = A^t$, where $A^t$ denotes the transpose of $A$ (the matrix obtained by interchanging the rows and the columns).*

The next result shows how to compute the matrix of $T^*$ from the matrix of $T$. **Caution:** With respect to nonorthonormal bases, the matrix of $T^*$ does not necessarily equal the conjugate transpose of the matrix of $T$.

*The adjoint of a linear map does not depend on a choice of basis. Thus we frequently emphasize adjoints of linear maps instead of transposes or conjugate transposes of matrices.*

**7.9 *matrix of $T^*$ equals conjugate transpose of matrix of $T$***

Let $T \in \mathcal{L}(V, W)$. Suppose $e_1, \ldots, e_n$ is an orthonormal basis of $V$ and $f_1, \ldots, f_m$ is an orthonormal basis of $W$. Then $\mathcal{M}(T^*, (f_1, \ldots, f_m), (e_1, \ldots, e_n))$ is the conjugate transpose of $\mathcal{M}(T, (e_1, \ldots, e_n), (f_1, \ldots, f_m))$. In other words,

$$\mathcal{M}(T^*) = (\mathcal{M}(T))^*.$$

Proof In this proof, we will write $\mathcal{M}(T)$ and $\mathcal{M}(T^*)$ instead of the longer expressions $\mathcal{M}(T, (e_1, \ldots, e_n), (f_1, \ldots, f_m))$ and $\mathcal{M}(T^*, (f_1, \ldots, f_m), (e_1, \ldots, e_n))$.

Recall that we obtain the $k^{\text{th}}$ column of $\mathcal{M}(T)$ by writing $Te_k$ as a linear combination of the $f_j$'s; the scalars used in this linear combination then become the $k^{\text{th}}$ column of $\mathcal{M}(T)$. Because $f_1, \ldots, f_m$ is an orthonormal basis of $W$, we know how to write $Te_k$ as a linear combination of the $f_j$'s [see 6.30(a)]:

$$Te_k = \langle Te_k, f_1 \rangle f_1 + \cdots + \langle Te_k, f_m \rangle f_m.$$

Thus

$$\text{the entry in row } j, \text{ column } k, \text{ of } \mathcal{M}(T) \text{ is } \langle Te_k, f_j \rangle.$$

In the statement above, replace $T$ with $T^*$ and interchange $e_1, \ldots, e_n$ and $f_1, \ldots, f_m$. This shows that the entry in row $j$, column $k$, of $\mathcal{M}(T^*)$ is $\langle T^* f_k, e_j \rangle$, which equals $\langle f_k, Te_j \rangle$, which equals $\overline{\langle Te_j, f_k \rangle}$, which equals the complex conjugate of the entry in row $k$, column $j$, of $\mathcal{M}(T)$. Thus $\mathcal{M}(T^*) = (\mathcal{M}(T))^*$. ∎

The Riesz representation theorem as stated in 6.58 provides an identification of $V$ with its dual space $V'$ defined in 3.110. Under this identification, the orthogonal complement $U^\perp$ of a subset $U \subseteq V$ corresponds to the annihilator $U^0$ of $U$. If $U$ is a subspace of $V$, then the formulas for the dimensions of $U^\perp$ and $U^0$ become identical under this identification—see 3.125 and 6.51.

Suppose $T: V \to W$ is a linear map. Under the identification of $V$ with $V'$ and the identification of $W$ with $W'$, the adjoint map $T^*: W \to V$ corresponds to the dual map $T': W' \to V'$ defined in 3.118, as Exercise 32 asks you to verify. Under this identification, the formulas for null $T^*$ and range $T^*$ [7.6(a) and (b)] then become identical to the formulas for null $T'$ and range $T'$ [3.128(a) and 3.130(b)]. Furthermore, the theorem about the matrix of $T^*$ (7.9) is analogous to the theorem about the matrix of $T'$ (3.132).

*Because orthogonal complements and adjoints are easier to deal with than annihilators and dual maps, there is no need to work with annihilators and dual maps in the context of inner product spaces.*

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## *Self-Adjoint Operators*

Now we switch our attention to operators on inner product spaces. Instead of considering linear maps from $V$ to $W$, we will focus on linear maps from $V$ to $V$; recall that such linear maps are called operators.

**7.10 definition: *self-adjoint***

An operator $T \in \mathcal{L}(V)$ is called *self-adjoint* if $T = T^*$.

If $T \in \mathcal{L}(V)$ and $e_1, \ldots, e_n$ is an orthonormal basis of $V$, then $T$ is self-adjoint if and only if $\mathcal{M}(T, (e_1, \ldots, e_n)) = \mathcal{M}(T, (e_1, \ldots, e_n))^*$, as follows from 7.9.

**7.11 example: *determining whether $T$ is self-adjoint from its matrix***

Suppose $c \in \mathbf{F}$ and $T$ is the operator on $\mathbf{F}^2$ whose matrix (with respect to the standard basis) is

$$\mathcal{M}(T) = \begin{pmatrix} 2 & c \\ 3 & 7 \end{pmatrix}.$$

The matrix of $T^*$ (with respect to the standard basis) is

$$\mathcal{M}(T^*) = \begin{pmatrix} 2 & 3 \\ \overline{c} & 7 \end{pmatrix}.$$

Thus $\mathcal{M}(T) = \mathcal{M}(T^*)$ if and only if $c = 3$. Hence the operator $T$ is self-adjoint if and only if $c = 3$.

---

A good analogy to keep in mind is that the adjoint on $\mathcal{L}(V)$ plays a role similar to that of the complex conjugate on $\mathbf{C}$. A complex number $z$ is real if and only if $z = \overline{z}$; thus a self-adjoint operator ($T = T^*$) is analogous to a real number.

We will see that the analogy discussed above is reflected in some important properties of self-adjoint operators, beginning with eigenvalues in the next result.

If $\mathbf{F} = \mathbf{R}$, then by definition every eigenvalue is real, so the next result is interesting only when $\mathbf{F} = \mathbf{C}$.

*An operator $T \in \mathcal{L}(V)$ is self-adjoint if and only if*

$$\langle Tv, w \rangle = \langle v, Tw \rangle$$

*for all $v, w \in V$.*

**7.12 *eigenvalues of self-adjoint operators***

Every eigenvalue of a self-adjoint operator is real.

Proof Suppose $T$ is a self-adjoint operator on $V$. Let $\lambda$ be an eigenvalue of $T$, and let $v$ be a nonzero vector in $V$ such that $Tv = \lambda v$. Then

$$\lambda \|v\|^2 = \langle \lambda v, v \rangle = \langle Tv, v \rangle = \langle v, Tv \rangle = \langle v, \lambda v \rangle = \overline{\lambda} \|v\|^2.$$

Thus $\lambda = \overline{\lambda}$, which means that $\lambda$ is real, as desired. ∎

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The next result is false for real inner product spaces. As an example, consider the operator $T \in \mathcal{L}(\mathbf{R}^2)$ that is a counterclockwise rotation of 90° around the origin; thus $T(x, y) = (-y, x)$. Notice that $Tv$ is orthogonal to $v$ for every $v \in \mathbf{R}^2$, even though $T \neq 0$.

**7.13** *$Tv$ is orthogonal to $v$ for all $v$ $\iff$ $T = 0$ (assuming $\mathbf{F} = \mathbf{C}$)*

Suppose $V$ is a complex inner product space and $T \in \mathcal{L}(V)$. Then

$$\langle Tv, v\rangle = 0 \text{ for every } v \in V \iff T = 0.$$

Proof If $u, w \in V$, then

$$\langle Tu, w\rangle = \frac{\langle T(u+w), u+w\rangle - \langle T(u-w), u-w\rangle}{4} + \frac{\langle T(u+iw), u+iw\rangle - \langle T(u-iw), u-iw\rangle}{4} i,$$

as can be verified by computing the right side. Note that each term on the right side is of the form $\langle Tv, v\rangle$ for appropriate $v \in V$.

Now suppose $\langle Tv, v\rangle = 0$ for every $v \in V$. Then the equation above implies that $\langle Tu, w\rangle = 0$ for all $u, w \in V$, which then implies that $Tu = 0$ for every $u \in U$ (take $w = Tu$). Hence $T = 0$, as desired. ∎

The next result is false for real inner product spaces, as shown by considering any operator on a real inner product space that is not self-adjoint.

*The next result provides another good example of how self-adjoint operators behave like real numbers.*

**7.14** *$\langle Tv, v\rangle$ is real for all $v$ $\iff$ $T$ is self-adjoint (assuming $\mathbf{F} = \mathbf{C}$)*

Suppose $V$ is a complex inner product space and $T \in \mathcal{L}(V)$. Then

$$T \text{ is self-adjoint} \iff \langle Tv, v\rangle \in \mathbf{R} \text{ for every } v \in V.$$

Proof If $v \in V$, then

$$\langle T^*v, v\rangle = \overline{\langle v, T^*v\rangle} = \overline{\langle Tv, v\rangle}. \tag{7.15}$$

Now

$$\begin{aligned} T \text{ is self-adjoint} &\iff T - T^* = 0 \\ &\iff \langle (T - T^*)v, v\rangle = 0 \text{ for every } v \in V \\ &\iff \langle Tv, v\rangle - \overline{\langle Tv, v\rangle} = 0 \text{ for every } v \in V \\ &\iff \langle Tv, v\rangle \in \mathbf{R} \text{ for every } v \in V, \end{aligned}$$

where the second equivalence follows from 7.13 as applied to $T - T^*$ and the third equivalence follows from 7.15. ∎

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On a real inner product space $V$, a nonzero operator $T$ might satisfy $\langle Tv, v\rangle = 0$ for all $v \in V$. However, the next result shows that this cannot happen for a self-adjoint operator.

**7.16 $T$ *self-adjoint and* $\langle Tv, v\rangle = 0$ *for all* $v \iff T = 0$**

Suppose $T$ is a self-adjoint operator on $V$. Then

$$\langle Tv, v\rangle = 0 \text{ for every } v \in V \iff T = 0.$$

Proof We have already proved this (without the hypothesis that $T$ is self-adjoint) when $V$ is a complex inner product space (see 7.13). Thus we can assume that $V$ is a real inner product space. If $u, w \in V$, then

$$\langle Tu, w\rangle = \frac{\langle T(u+w), u+w\rangle - \langle T(u-w), u-w\rangle}{4}, \tag{7.17}$$

as can be proved by computing the right side using the equation

$$\langle Tw, u\rangle = \langle w, Tu\rangle = \langle Tu, w\rangle,$$

where the first equality holds because $T$ is self-adjoint and the second equality holds because we are working in a real inner product space.

Now suppose $\langle Tv, v\rangle = 0$ for every $v \in V$. Because each term on the right side of 7.17 is of the form $\langle Tv, v\rangle$ for appropriate $v$, this implies that $\langle Tu, w\rangle = 0$ for all $u, w \in V$. This implies that $Tu = 0$ for every $u \in V$ (take $w = Tu$). Hence $T = 0$, as desired. ∎

## *Normal Operators*

**7.18 definition: *normal***

- An operator on an inner product space is called *normal* if it commutes with its adjoint.
- In other words, $T \in \mathcal{L}(V)$ is normal if $TT^* = T^*T$.

Every self-adjoint operator is normal, because if $T$ is self-adjoint then $T^* = T$ and hence $T$ commutes with $T^*$.

**7.19 example: *an operator that is normal but not self-adjoint***

Let $T$ be the operator on $\mathbf{F}^2$ whose matrix (with respect to the standard basis) is

$$\begin{pmatrix} 2 & -3 \\ 3 & 2 \end{pmatrix}.$$

Thus $T(w, z) = (2w - 3z, 3w + 2z)$.

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This operator $T$ is not self-adjoint because the entry in row 2, column 1 (which equals 3) does not equal the complex conjugate of the entry in row 1, column 2 (which equals $-3$).

The matrix of $TT^*$ equals

$$\begin{pmatrix} 2 & -3 \\ 3 & 2 \end{pmatrix}\begin{pmatrix} 2 & 3 \\ -3 & 2 \end{pmatrix}, \text{ which equals } \begin{pmatrix} 13 & 0 \\ 0 & 13 \end{pmatrix}.$$

Similarly, the matrix of $T^*T$ equals

$$\begin{pmatrix} 2 & 3 \\ -3 & 2 \end{pmatrix}\begin{pmatrix} 2 & -3 \\ 3 & 2 \end{pmatrix}, \text{ which equals } \begin{pmatrix} 13 & 0 \\ 0 & 13 \end{pmatrix}.$$

Because $TT^*$ and $T^*T$ have the same matrix, we see that $TT^* = T^*T$. Thus $T$ is normal.

---

In the next section we will see why normal operators are worthy of special attention. The next result provides a useful characterization of normal operators.

**7.20** *$T$ is normal if and only if $Tv$ and $T^*v$ have the same norm*

Suppose $T \in \mathcal{L}(V)$. Then

$$T \text{ is normal} \iff \|Tv\| = \|T^*v\| \text{ for every } v \in V.$$

Proof We have

$$\begin{aligned} T \text{ is normal} &\iff T^*T - TT^* = 0 \\ &\iff \langle (T^*T - TT^*)v, v\rangle = 0 \text{ for every } v \in V \\ &\iff \langle T^*Tv, v\rangle = \langle TT^*v, v\rangle \text{ for every } v \in V \\ &\iff \langle Tv, Tv\rangle = \langle T^*v, T^*v\rangle \text{ for every } v \in V \\ &\iff \|Tv\|^2 = \|T^*v\|^2 \text{ for every } v \in V \\ &\iff \|Tv\| = \|T^*v\| \text{ for every } v \in V, \end{aligned}$$

where we used 7.16 to establish the second equivalence (note that the operator $T^*T - TT^*$ is self-adjoint). ∎

The next result presents several consequences of the result above. Compare (e) of the next result to Exercise 3. That exercise states that the eigenvalues of the adjoint of each operator are equal (as a set) to the complex conjugates of the eigenvalues of the operator. The exercise says nothing about eigenvectors, because an operator and its adjoint may have different eigenvectors. However, (e) of the next result implies that a normal operator and its adjoint have the same eigenvectors.

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**7.21** *range, null space, and eigenvectors of a normal operator*

Suppose $T \in \mathcal{L}(V)$ is normal. Then

(a) $\operatorname{null} T = \operatorname{null} T^*$;

(b) $\operatorname{range} T = \operatorname{range} T^*$;

(c) $V = \operatorname{null} T \oplus \operatorname{range} T$;

(d) $T - \lambda I$ is normal for every $\lambda \in \mathbf{F}$;

(e) if $v \in V$ and $\lambda \in \mathbf{F}$, then $Tv = \lambda v$ if and only if $T^*v = \overline{\lambda} v$.

Proof

(a) Suppose $v \in V$. Then

$$v \in \operatorname{null} T \iff \|Tv\| = 0 \iff \|T^*v\| = 0 \iff v \in \operatorname{null} T^*,$$

where the middle equivalence above follows from 7.20. Thus $\operatorname{null} T = \operatorname{null} T^*$.

(b) We have

$$\operatorname{range} T = (\operatorname{null} T^*)^\perp = (\operatorname{null} T)^\perp = \operatorname{range} T^*,$$

where the first equality comes from 7.6(d), the second equality comes from (a) in this result, and the third equality comes from 7.6(b).

(c) We have

$$V = (\operatorname{null} T) \oplus (\operatorname{null} T)^\perp = \operatorname{null} T \oplus \operatorname{range} T^* = \operatorname{null} T \oplus \operatorname{range} T,$$

where the first equality comes from 6.49, the second equality comes from 7.6(b), and the third equality comes from (b) in this result.

(d) Suppose $\lambda \in \mathbf{F}$. Then

$$\begin{aligned}
(T - \lambda I)(T - \lambda I)^* &= (T - \lambda I)(T^* - \overline{\lambda} I) \\
&= TT^* - \overline{\lambda} T - \lambda T^* + |\lambda|^2 I \\
&= T^*T - \overline{\lambda} T - \lambda T^* + |\lambda|^2 I \\
&= (T^* - \overline{\lambda} I)(T - \lambda I) \\
&= (T - \lambda I)^*(T - \lambda I).
\end{aligned}$$

Thus $T - \lambda I$ commutes with its adjoint. Hence $T - \lambda I$ is normal.

(e) Suppose $v \in V$ and $\lambda \in \mathbf{F}$. Then (d) and 7.20 imply that

$$\|(T - \lambda I)v\| = \|(T - \lambda I)^* v\| = \|(T^* - \overline{\lambda} I)v\|.$$

Thus $\|(T - \lambda I)v\| = 0$ if and only if $\|(T^* - \overline{\lambda} I)v\| = 0$. Hence $Tv = \lambda v$ if and only if $T^*v = \overline{\lambda} v$.

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Because every self-adjoint operator is normal, the next result applies in particular to self-adjoint operators.

**7.22 *orthogonal eigenvectors for normal operators***

Suppose $T \in \mathcal{L}(V)$ is normal. Then eigenvectors of $T$ corresponding to distinct eigenvalues are orthogonal.

Proof Suppose $\alpha, \beta$ are distinct eigenvalues of $T$, with corresponding eigenvectors $u, v$. Thus $Tu = \alpha u$ and $Tv = \beta v$. From 7.21(e) we have $T^*v = \overline{\beta} v$. Thus

$$\begin{aligned}(\alpha - \beta)\langle u, v\rangle &= \langle \alpha u, v\rangle - \langle u, \overline{\beta} v\rangle \\ &= \langle Tu, v\rangle - \langle u, T^*v\rangle \\ &= 0.\end{aligned}$$

Because $\alpha \neq \beta$, the equation above implies that $\langle u, v\rangle = 0$. Thus $u$ and $v$ are orthogonal, as desired. ∎

As stated here, the next result makes sense only when $\mathbf{F} = \mathbf{C}$. However, see Exercise 12 for a version that makes sense when $\mathbf{F} = \mathbf{C}$ and when $\mathbf{F} = \mathbf{R}$.

Suppose $\mathbf{F} = \mathbf{C}$ and $T \in \mathcal{L}(V)$. Under the analogy between $\mathcal{L}(V)$ and $\mathbf{C}$, with the adjoint on $\mathcal{L}(V)$ playing a similar role to that of the complex conjugate on $\mathbf{C}$, the operators $A$ and $B$ as defined by 7.24 correspond to the real and imaginary parts of $T$. Thus the informal title of the result below should make sense.

**7.23 *$T$ is normal $\iff$ the real and imaginary parts of $T$ commute***

Suppose $\mathbf{F} = \mathbf{C}$ and $T \in \mathcal{L}(V)$. Then $T$ is normal if and only if there exist commuting self-adjoint operators $A$ and $B$ such that $T = A + iB$.

Proof First suppose $T$ is normal. Let

$$A = \frac{T + T^*}{2} \quad \text{and} \quad B = \frac{T - T^*}{2i}. \tag{7.24}$$

Then $A$ and $B$ are self-adjoint and $T = A + iB$. A quick computation shows that

$$AB - BA = \frac{T^*T - TT^*}{2i}. \tag{7.25}$$

Because $T$ is normal, the right side of the equation above equals 0. Thus the operators $A$ and $B$ commute, as desired.

To prove the implication in the other direction, now suppose there exist commuting self-adjoint operators $A$ and $B$ such that $T = A + iB$. Then $T^* = A - iB$. Adding the last two equations and then dividing by 2 produces the equation for $A$ in 7.24. Subtracting the last two equations and then dividing by $2i$ produces the equation for $B$ in 7.24. Now 7.24 implies 7.25. Because $B$ and $A$ commute, 7.25 implies that $T$ is normal, as desired. ∎

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## *Exercises 7A*

**1** Suppose $n$ is a positive integer. Define $T \in \mathcal{L}(\mathbf{F}^n)$ by

$$T(z_1, \dots, z_n) = (0, z_1, \dots, z_{n-1}).$$

Find a formula for $T^*(z_1, \dots, z_n)$.

**2** Suppose $T \in \mathcal{L}(V, W)$. Prove that

$$T = 0 \iff T^* = 0 \iff T^*T = 0 \iff TT^* = 0.$$

**3** Suppose $T \in \mathcal{L}(V)$ and $\lambda \in \mathbf{F}$. Prove that

$$\lambda \text{ is an eigenvalue of } T \iff \overline{\lambda} \text{ is an eigenvalue of } T^*.$$

**4** Suppose $T \in \mathcal{L}(V)$ and $U$ is a subspace of $V$. Prove that

$$U \text{ is invariant under } T \iff U^\perp \text{ is invariant under } T^*.$$

**5** Suppose $T \in \mathcal{L}(V, W)$. Suppose $e_1, \dots, e_n$ is an orthonormal basis of $V$ and $f_1, \dots, f_m$ is an orthonormal basis of $W$. Prove that

$$\|Te_1\|^2 + \cdots + \|Te_n\|^2 = \|T^*f_1\|^2 + \cdots + \|T^*f_m\|^2.$$

*The numbers $\|Te_1\|^2, \dots, \|Te_n\|^2$ in the equation above depend on the orthonormal basis $e_1, \dots, e_n$, but the right side of the equation does not depend on $e_1, \dots, e_n$. Thus the equation above shows that the sum on the left side does not depend on which orthonormal basis $e_1, \dots, e_n$ is used.*

**6** Suppose $T \in \mathcal{L}(V, W)$. Prove that
(a) $T$ is injective $\iff$ $T^*$ is surjective;
(b) $T$ is surjective $\iff$ $T^*$ is injective.

**7** Prove that if $T \in \mathcal{L}(V, W)$, then
(a) $\dim \operatorname{null} T^* = \dim \operatorname{null} T + \dim W - \dim V$;
(b) $\dim \operatorname{range} T^* = \dim \operatorname{range} T$.

**8** Suppose $A$ is an $m$-by-$n$ matrix with entries in $\mathbf{F}$. Use (b) in Exercise 7 to prove that the row rank of $A$ equals the column rank of $A$.

*This exercise asks for yet another alternative proof of a result that was previously proved in 3.57 and 3.133.*

**9** Prove that the product of two self-adjoint operators on $V$ is self-adjoint if and only if the two operators commute.

**10** Suppose $\mathbf{F} = \mathbf{C}$ and $T \in \mathcal{L}(V)$. Prove that $T$ is self-adjoint if and only if

$$\langle Tv, v \rangle = \langle T^*v, v \rangle$$

for all $v \in V$.

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**11** Define an operator $S: \mathbf{F}^2 \to \mathbf{F}^2$ by $S(w, z) = (-z, w)$.

(a) Find a formula for $S^*$.

(b) Show that $S$ is normal but not self-adjoint.

(c) Find all eigenvalues of $S$.

*If $\mathbf{F} = \mathbf{R}$, then $S$ is the operator on $\mathbf{R}^2$ of counterclockwise rotation by $90°$.*

**12** An operator $B \in \mathcal{L}(V)$ is called *skew* if

$$B^* = -B.$$

Suppose that $T \in \mathcal{L}(V)$. Prove that $T$ is normal if and only if there exist commuting operators $A$ and $B$ such that $A$ is self-adjoint, $B$ is a skew operator, and $T = A + B$.

**13** Suppose $\mathbf{F} = \mathbf{R}$. Define $\mathcal{A} \in \mathcal{L}(\mathcal{L}(V))$ by $\mathcal{A}T = T^*$ for all $T \in \mathcal{L}(V)$.

(a) Find all eigenvalues of $\mathcal{A}$.

(b) Find the minimal polynomial of $\mathcal{A}$.

**14** Define an inner product on $\mathcal{P}_2(\mathbf{R})$ by $\langle p, q \rangle = \int_0^1 pq$. Define an operator $T \in \mathcal{L}(\mathcal{P}_2(\mathbf{R}))$ by

$$T(ax^2 + bx + c) = bx.$$

(a) Show that with this inner product, the operator $T$ is not self-adjoint.

(b) The matrix of $T$ with respect to the basis $1, x, x^2$ is

$$\begin{pmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

This matrix equals its conjugate transpose, even though $T$ is not self-adjoint. Explain why this is not a contradiction.

**15** Suppose $T \in \mathcal{L}(V)$ is invertible. Prove that

(a) $T$ is self-adjoint $\iff$ $T^{-1}$ is self-adjoint;

(b) $T$ is normal $\iff$ $T^{-1}$ is normal.

**16** Suppose $\mathbf{F} = \mathbf{R}$.

(a) Show that the set of self-adjoint operators on $V$ is a subspace of $\mathcal{L}(V)$.

(b) What is the dimension of the subspace of $\mathcal{L}(V)$ in (a) [in terms of $\dim V$]?

**17** Suppose $\mathbf{F} = \mathbf{C}$. Show that the set of self-adjoint operators on $V$ is not a subspace of $\mathcal{L}(V)$.

**18** Suppose $\dim V \geq 2$. Show that the set of normal operators on $V$ is not a subspace of $\mathcal{L}(V)$.

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**19** Suppose $T \in \mathcal{L}(V)$ and $\|T^*v\| \le \|Tv\|$ for every $v \in V$. Prove that $T$ is normal.

*This exercise fails on infinite-dimensional inner product spaces, leading to what are called hyponormal operators, which have a well-developed theory.*

**20** Suppose $P \in \mathcal{L}(V)$ is such that $P^2 = P$. Prove that the following are equivalent.

(a) $P$ is self-adjoint.

(b) $P$ is normal.

(c) There is a subspace $U$ of $V$ such that $P = P_U$.

**21** Suppose $D: \mathcal{P}_8(\mathbf{R}) \to \mathcal{P}_8(\mathbf{R})$ is the differentiation operator defined by $Dp = p'$. Prove that there does not exist an inner product on $\mathcal{P}_8(\mathbf{R})$ that makes $D$ a normal operator.

**22** Give an example of an operator $T \in \mathcal{L}(\mathbf{R}^3)$ such that $T$ is normal but not self-adjoint.

**23** Suppose $T$ is a normal operator on $V$. Suppose also that $v, w \in V$ satisfy the equations

$$\|v\| = \|w\| = 2, \quad Tv = 3v, \quad Tw = 4w.$$

Show that $\|T(v + w)\| = 10$.

**24** Suppose $T \in \mathcal{L}(V)$ and

$$a_0 + a_1 z + a_2 z^2 + \cdots + a_{m-1} z^{m-1} + z^m$$

is the minimal polynomial of $T$. Prove that the minimal polynomial of $T^*$ is

$$\overline{a_0} + \overline{a_1}\, z + \overline{a_2}\, z^2 + \cdots + \overline{a_{m-1}}\, z^{m-1} + z^m.$$

*This exercise shows that the minimal polynomial of $T^*$ equals the minimal polynomial of $T$ if $\mathbf{F} = \mathbf{R}$.*

**25** Suppose $T \in \mathcal{L}(V)$. Prove that $T$ is diagonalizable if and only if $T^*$ is diagonalizable.

**26** Fix $u, x \in V$. Define $T \in \mathcal{L}(V)$ by $Tv = \langle v, u \rangle x$ for every $v \in V$.

(a) Prove that if $V$ is a real vector space, then $T$ is self-adjoint if and only if the list $u, x$ is linearly dependent.

(b) Prove that $T$ is normal if and only if the list $u, x$ is linearly dependent.

**27** Suppose $T \in \mathcal{L}(V)$ is normal. Prove that

$$\operatorname{null} T^k = \operatorname{null} T \quad \text{and} \quad \operatorname{range} T^k = \operatorname{range} T$$

for every positive integer $k$.

**28** Suppose $T \in \mathcal{L}(V)$ is normal. Prove that if $\lambda \in \mathbf{F}$, then the minimal polynomial of $T$ is not a polynomial multiple of $(x - \lambda)^2$.

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**29** Prove or give a counterexample: If $T \in \mathcal{L}(V)$ and there is an orthonormal basis $e_1, \ldots, e_n$ of $V$ such that $\|Te_k\| = \|T^*e_k\|$ for each $k = 1, \ldots, n$, then $T$ is normal.

**30** Suppose that $T \in \mathcal{L}(\mathbf{F}^3)$ is normal and $T(1,1,1) = (2,2,2)$. Suppose $(z_1, z_2, z_3) \in \operatorname{null} T$. Prove that $z_1 + z_2 + z_3 = 0$.

**31** Fix a positive integer $n$. In the inner product space of continuous real-valued functions on $[-\pi, \pi]$ with inner product $\langle f, g \rangle = \int_{-\pi}^{\pi} fg$, let

$$V = \operatorname{span}(1, \cos x, \cos 2x, \ldots, \cos nx, \sin x, \sin 2x, \ldots, \sin nx).$$

(a) Define $D \in \mathcal{L}(V)$ by $Df = f'$. Show that $D^* = -D$. Conclude that $D$ is normal but not self-adjoint.
(b) Define $T \in \mathcal{L}(V)$ by $Tf = f''$. Show that $T$ is self-adjoint.

**32** Suppose $T : V \to W$ is a linear map. Show that under the standard identification of $V$ with $V'$ (see 6.58) and the corresponding identification of $W$ with $W'$, the adjoint map $T^* : W \to V$ corresponds to the dual map $T' : W' \to V'$. More precisely, show that

$$T'(\varphi_w) = \varphi_{T^*w}$$

for all $w \in W$, where $\varphi_w$ and $\varphi_{T^*w}$ are defined as in 6.58.

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## 7B Spectral Theorem

Recall that a diagonal matrix is a square matrix that is 0 everywhere except possibly on the diagonal. Recall that an operator on $V$ is called diagonalizable if the operator has a diagonal matrix with respect to some basis of $V$. Recall also that this happens if and only if there is a basis of $V$ consisting of eigenvectors of the operator (see 5.55).

The nicest operators on $V$ are those for which there is an *orthonormal* basis of $V$ with respect to which the operator has a diagonal matrix. These are precisely the operators $T \in \mathcal{L}(V)$ such that there is an orthonormal basis of $V$ consisting of eigenvectors of $T$. Our goal in this section is to prove the spectral theorem, which characterizes these operators as the self-adjoint operators when $\mathbf{F} = \mathbf{R}$ and as the normal operators when $\mathbf{F} = \mathbf{C}$.

The spectral theorem is probably the most useful tool in the study of operators on inner product spaces. Its extension to certain infinite-dimensional inner product spaces (see, for example, Section 10D of the author's book *Measure, Integration & Real Analysis*) plays a key role in functional analysis.

Because the conclusion of the spectral theorem depends on $\mathbf{F}$, we will break the spectral theorem into two pieces, called the real spectral theorem and the complex spectral theorem.

### Real Spectral Theorem

To prove the real spectral theorem, we will need two preliminary results. These preliminary results hold on both real and complex inner product spaces, but they are not needed for the proof of the complex spectral theorem.

You could guess that the next result is true and even discover its proof by thinking about quadratic polynomials with real coefficients. Specifically, suppose $b, c \in \mathbf{R}$ and $b^2 < 4c$. Let $x$ be a real number. Then

*This completing-the-square technique can be used to derive the quadratic formula.*

$$x^2 + bx + c = \left(x + \frac{b}{2}\right)^2 + \left(c - \frac{b^2}{4}\right) > 0.$$

In particular, $x^2 + bx + c$ is an invertible real number (a convoluted way of saying that it is not 0). Replacing the real number $x$ with a self-adjoint operator (recall the analogy between real numbers and self-adjoint operators) leads to the next result.

**7.26** *invertible quadratic expressions*

Suppose $T \in \mathcal{L}(V)$ is self-adjoint and $b, c \in \mathbf{R}$ are such that $b^2 < 4c$. Then

$$T^2 + bT + cI$$

is an invertible operator.

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Proof Let $v$ be a nonzero vector in $V$. Then

$$\begin{aligned}\langle (T^2 + bT + cI)v, v\rangle &= \langle T^2 v, v\rangle + b\langle Tv, v\rangle + c\langle v, v\rangle \\ &= \langle Tv, Tv\rangle + b\langle Tv, v\rangle + c\|v\|^2 \\ &\geq \|Tv\|^2 - |b|\,\|Tv\|\,\|v\| + c\|v\|^2 \\ &= \left(\|Tv\| - \frac{|b|\,\|v\|}{2}\right)^2 + \left(c - \frac{b^2}{4}\right)\|v\|^2 \\ &> 0,\end{aligned}$$

where the third line above holds by the Cauchy–Schwarz inequality (6.14). The last inequality implies that $(T^2 + bT + cI)v \neq 0$. Thus $T^2 + bT + cI$ is injective, which implies that it is invertible (see 3.65). ∎

The next result will be a key tool in our proof of the real spectral theorem.

**7.27** *minimal polynomial of self-adjoint operator*

Suppose $T \in \mathcal{L}(V)$ is self-adjoint. Then the minimal polynomial of $T$ equals $(z - \lambda_1)\cdots(z - \lambda_m)$ for some $\lambda_1, \ldots, \lambda_m \in \mathbf{R}$.

Proof First suppose $\mathbf{F} = \mathbf{C}$. The zeros of the minimal polynomial of $T$ are the eigenvalues of $T$ [by 5.27(a)]. All eigenvalues of $T$ are real (by 7.12). Thus the second version of the fundamental theorem of algebra (see 6.69) tells us that the minimal polynomial of $T$ has the desired form.

Now suppose $\mathbf{F} = \mathbf{R}$. By the factorization of a polynomial over $\mathbf{R}$ (see 4.16) there exist $\lambda_1, \ldots, \lambda_m \in \mathbf{R}$ and $b_1, \ldots, b_N, c_1, \ldots, c_N \in \mathbf{R}$ with $b_k^2 < 4c_k$ for each $k$ such that the minimal polynomial of $T$ equals

$$(z - \lambda_1)\cdots(z - \lambda_m)(z^2 + b_1 z + c_1)\cdots(z^2 + b_N z + c_N); \tag{7.28}$$

here either $m$ or $N$ might equal 0, meaning that there are no terms of the corresponding form. Now

$$(T - \lambda_1 I)\cdots(T - \lambda_m I)(T^2 + b_1 T + c_1 I)\cdots(T^2 + b_N T + c_N I) = 0.$$

If $N > 0$, then we could multiply both sides of the equation above on the right by the inverse of $T^2 + b_N T + c_N I$ (which is an invertible operator by 7.26) to obtain a polynomial expression of $T$ that equals 0. The corresponding polynomial would have degree two less than the degree of 7.28, violating the minimality of the degree of the polynomial with this property. Thus we must have $N = 0$, which means that the minimal polynomial in 7.28 has the form $(z - \lambda_1)\cdots(z - \lambda_m)$, as desired. ∎

The result above along with 5.27(a) implies that every self-adjoint operator has an eigenvalue. In fact, as we will see in the next result, self-adjoint operators have enough eigenvectors to form a basis.

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The next result, which gives a complete description of the self-adjoint operators on a real inner product space, is one of the major theorems in linear algebra.

**7.29** *real spectral theorem*

Suppose $\mathbf{F} = \mathbf{R}$ and $T \in \mathcal{L}(V)$. Then the following are equivalent.

(a) $T$ is self-adjoint.

(b) $T$ has a diagonal matrix with respect to some orthonormal basis of $V$.

(c) $V$ has an orthonormal basis consisting of eigenvectors of $T$.

Proof First suppose (a) holds, so $T$ is self-adjoint. Our results on minimal polynomials, specifically 6.37 and 7.27, imply that $T$ has an upper-triangular matrix with respect to some orthonormal basis of $V$. With respect to this orthonormal basis, the matrix of $T^*$ is the transpose of the matrix of $T$. However, $T^* = T$. Thus the transpose of the matrix of $T$ equals the matrix of $T$. Because the matrix of $T$ is upper-triangular, this means that all entries of the matrix above and below the diagonal are 0. Hence the matrix of $T$ is a diagonal matrix with respect to the orthonormal basis. Thus (a) implies (b).

Conversely, now suppose (b) holds, so $T$ has a diagonal matrix with respect to some orthonormal basis of $V$. That diagonal matrix equals its transpose. Thus with respect to that basis, the matrix of $T^*$ equals the matrix of $T$. Hence $T^* = T$, proving that (b) implies (a).

The equivalence of (b) and (c) follows from the definitions [or see the proof that (a) and (b) are equivalent in 5.55]. ∎

**7.30** example: *an orthonormal basis of eigenvectors for an operator*

Consider the operator $T$ on $\mathbf{R}^3$ whose matrix (with respect to the standard basis) is

$$\begin{pmatrix} 14 & -13 & 8 \\ -13 & 14 & 8 \\ 8 & 8 & -7 \end{pmatrix}.$$

This matrix with real entries equals its transpose; thus $T$ is self-adjoint. As you can verify,

$$\frac{(1,-1,0)}{\sqrt{2}}, \frac{(1,1,1)}{\sqrt{3}}, \frac{(1,1,-2)}{\sqrt{6}}$$

is an orthonormal basis of $\mathbf{R}^3$ consisting of eigenvectors of $T$. With respect to this basis, the matrix of $T$ is the diagonal matrix

$$\begin{pmatrix} 27 & 0 & 0 \\ 0 & 9 & 0 \\ 0 & 0 & -15 \end{pmatrix}.$$

See Exercise 17 for a version of the real spectral theorem that applies simultaneously to more than one operator.

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## *Complex Spectral Theorem*

The next result gives a complete description of the normal operators on a complex inner product space.

**7.31** *complex spectral theorem*

Suppose $\mathbf{F} = \mathbf{C}$ and $T \in \mathcal{L}(V)$. Then the following are equivalent.

(a) $T$ is normal.

(b) $T$ has a diagonal matrix with respect to some orthonormal basis of $V$.

(c) $V$ has an orthonormal basis consisting of eigenvectors of $T$.

Proof First suppose (a) holds, so $T$ is normal. By Schur's theorem (6.38), there is an orthonormal basis $e_1, \ldots, e_n$ of $V$ with respect to which $T$ has an upper-triangular matrix. Thus we can write

7.32
$$\mathcal{M}(T, (e_1, \ldots, e_n)) = \begin{pmatrix} a_{1,1} & \cdots & a_{1,n} \\ & \ddots & \vdots \\ 0 & & a_{n,n} \end{pmatrix}.$$

We will show that this matrix is actually a diagonal matrix.

We see from the matrix above that

$$\|Te_1\|^2 = |a_{1,1}|^2,$$
$$\|T^*e_1\|^2 = |a_{1,1}|^2 + |a_{1,2}|^2 + \cdots + |a_{1,n}|^2.$$

Because $T$ is normal, $\|Te_1\| = \|T^*e_1\|$ (see 7.20). Thus the two equations above imply that all entries in the first row of the matrix in 7.32, except possibly the first entry $a_{1,1}$, equal 0.

Now 7.32 implies

$$\|Te_2\|^2 = |a_{2,2}|^2$$

(because $a_{1,2} = 0$, as we showed in the paragraph above) and

$$\|T^*e_2\|^2 = |a_{2,2}|^2 + |a_{2,3}|^2 + \cdots + |a_{2,n}|^2.$$

Because $T$ is normal, $\|Te_2\| = \|T^*e_2\|$. Thus the two equations above imply that all entries in the second row of the matrix in 7.32, except possibly the diagonal entry $a_{2,2}$, equal 0.

Continuing in this fashion, we see that all nondiagonal entries in the matrix 7.32 equal 0. Thus (b) holds, completing the proof that (a) implies (b).

Now suppose (b) holds, so $T$ has a diagonal matrix with respect to some orthonormal basis of $V$. The matrix of $T^*$ (with respect to the same basis) is obtained by taking the conjugate transpose of the matrix of $T$; hence $T^*$ also has a diagonal matrix. Any two diagonal matrices commute; thus $T$ commutes with $T^*$, which means that $T$ is normal. In other words, (a) holds, completing the proof that (b) implies (a).

The equivalence of (b) and (c) follows from the definitions (also see 5.55). ∎

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See Exercises 13 and 20 for alternative proofs that (a) implies (b) in the previous result.

Exercises 14 and 15 interpret the real spectral theorem and the complex spectral theorem by expressing the domain space as an orthogonal direct sum of eigenspaces.

See Exercise 16 for a version of the complex spectral theorem that applies simultaneously to more than one operator.

The main conclusion of the complex spectral theorem is that every normal operator on a complex finite-dimensional inner product space is diagonalizable by an orthonormal basis, as illustrated by the next example.

7.33 example: *an orthonormal basis of eigenvectors for an operator*

Consider the operator $T \in \mathcal{L}(\mathbf{C}^2)$ defined by $T(w, z) = (2w - 3z, 3w + 2z)$. The matrix of $T$ (with respect to the standard basis) is

$$\begin{pmatrix} 2 & -3 \\ 3 & 2 \end{pmatrix}.$$

As we saw in Example 7.19, $T$ is a normal operator.

As you can verify,

$$\tfrac{1}{\sqrt{2}}(i, 1), \tfrac{1}{\sqrt{2}}(-i, 1)$$

is an orthonormal basis of $\mathbf{C}^2$ consisting of eigenvectors of $T$, and with respect to this basis the matrix of $T$ is the diagonal matrix

$$\begin{pmatrix} 2 + 3i & 0 \\ 0 & 2 - 3i \end{pmatrix}.$$

## *Exercises 7B*

**1** Prove that a normal operator on a complex inner product space is self-adjoint if and only if all its eigenvalues are real.

*This exercise strengthens the analogy (for normal operators) between self-adjoint operators and real numbers.*

**2** Suppose $\mathbf{F} = \mathbf{C}$. Suppose $T \in \mathcal{L}(V)$ is normal and has only one eigenvalue. Prove that $T$ is a scalar multiple of the identity operator.

**3** Suppose $\mathbf{F} = \mathbf{C}$ and $T \in \mathcal{L}(V)$ is normal. Prove that the set of eigenvalues of $T$ is contained in $\{0, 1\}$ if and only if there is a subspace $U$ of $V$ such that $T = P_U$.

**4** Prove that a normal operator on a complex inner product space is skew (meaning it equals the negative of its adjoint) if and only if all its eigenvalues are purely imaginary (meaning that they have real part equal to 0).

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**5** Prove or give a counterexample: If $T \in \mathcal{L}(\mathbf{C}^3)$ is a diagonalizable operator, then $T$ is normal (with respect to the usual inner product).

**6** Suppose $V$ is a complex inner product space and $T \in \mathcal{L}(V)$ is a normal operator such that $T^9 = T^8$. Prove that $T$ is self-adjoint and $T^2 = T$.

**7** Give an example of an operator $T$ on a complex vector space such that $T^9 = T^8$ but $T^2 \neq T$.

**8** Suppose $\mathbf{F} = \mathbf{C}$ and $T \in \mathcal{L}(V)$. Prove that $T$ is normal if and only if every eigenvector of $T$ is also an eigenvector of $T^*$.

**9** Suppose $\mathbf{F} = \mathbf{C}$ and $T \in \mathcal{L}(V)$. Prove that $T$ is normal if and only if there exists a polynomial $p \in \mathcal{P}(\mathbf{C})$ such that $T^* = p(T)$.

**10** Suppose $V$ is a complex inner product space. Prove that every normal operator on $V$ has a square root.

*An operator $S \in \mathcal{L}(V)$ is called a **square root** of $T \in \mathcal{L}(V)$ if $S^2 = T$. We will discuss more about square roots of operators in Sections 7C and 8C.*

**11** Prove that every self-adjoint operator on $V$ has a cube root.

*An operator $S \in \mathcal{L}(V)$ is called a **cube root** of $T \in \mathcal{L}(V)$ if $S^3 = T$.*

**12** Suppose $V$ is a complex vector space and $T \in \mathcal{L}(V)$ is normal. Prove that if $S$ is an operator on $V$ that commutes with $T$, then $S$ commutes with $T^*$.

*The result in this exercise is called Fuglede's theorem.*

**13** Without using the complex spectral theorem, use the version of Schur's theorem that applies to two commuting operators (take $\mathcal{E} = \{T, T^*\}$ in Exercise 20 in Section 6B) to give a different proof that if $\mathbf{F} = \mathbf{C}$ and $T \in \mathcal{L}(V)$ is normal, then $T$ has a diagonal matrix with respect to some orthonormal basis of $V$.

**14** Suppose $\mathbf{F} = \mathbf{R}$ and $T \in \mathcal{L}(V)$. Prove that $T$ is self-adjoint if and only if all pairs of eigenvectors corresponding to distinct eigenvalues of $T$ are orthogonal and $V = E(\lambda_1, T) \oplus \cdots \oplus E(\lambda_m, T)$, where $\lambda_1, \ldots, \lambda_m$ denote the distinct eigenvalues of $T$.

**15** Suppose $\mathbf{F} = \mathbf{C}$ and $T \in \mathcal{L}(V)$. Prove that $T$ is normal if and only if all pairs of eigenvectors corresponding to distinct eigenvalues of $T$ are orthogonal and $V = E(\lambda_1, T) \oplus \cdots \oplus E(\lambda_m, T)$, where $\lambda_1, \ldots, \lambda_m$ denote the distinct eigenvalues of $T$.

**16** Suppose $\mathbf{F} = \mathbf{C}$ and $\mathcal{E} \subseteq \mathcal{L}(V)$. Prove that there is an orthonormal basis of $V$ with respect to which every element of $\mathcal{E}$ has a diagonal matrix if and only if $S$ and $T$ are commuting normal operators for all $S, T \in \mathcal{E}$.

*This exercise extends the complex spectral theorem to the context of a collection of commuting normal operators.*

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**17** Suppose $\mathbf{F} = \mathbf{R}$ and $\mathcal{E} \subseteq \mathcal{L}(V)$. Prove that there is an orthonormal basis of $V$ with respect to which every element of $\mathcal{E}$ has a diagonal matrix if and only if $S$ and $T$ are commuting self-adjoint operators for all $S, T \in \mathcal{E}$.

*This exercise extends the real spectral theorem to the context of a collection of commuting self-adjoint operators.*

**18** Give an example of a real inner product space $V$, an operator $T \in \mathcal{L}(V)$, and real numbers $b, c$ with $b^2 < 4c$ such that

$$T^2 + bT + cI$$

is not invertible.

*This exercise shows that the hypothesis that $T$ is self-adjoint cannot be deleted in 7.26, even for real vector spaces.*

**19** Suppose $T \in \mathcal{L}(V)$ is self-adjoint and $U$ is a subspace of $V$ that is invariant under $T$.

(a) Prove that $U^\perp$ is invariant under $T$.
(b) Prove that $T|_U \in \mathcal{L}(U)$ is self-adjoint.
(c) Prove that $T|_{U^\perp} \in \mathcal{L}(U^\perp)$ is self-adjoint.

**20** Suppose $T \in \mathcal{L}(V)$ is normal and $U$ is a subspace of $V$ that is invariant under $T$.

(a) Prove that $U^\perp$ is invariant under $T$.
(b) Prove that $U$ is invariant under $T^*$.
(c) Prove that $(T|_U)^* = (T^*)|_U$.
(d) Prove that $T|_U \in \mathcal{L}(U)$ and $T|_{U^\perp} \in \mathcal{L}(U^\perp)$ are normal operators.

*This exercise can be used to give yet another proof of the complex spectral theorem (use induction on* $\dim V$ *and the result that $T$ has an eigenvector).*

**21** Suppose that $T$ is a self-adjoint operator on a finite-dimensional inner product space and that 2 and 3 are the only eigenvalues of $T$. Prove that

$$T^2 - 5T + 6I = 0.$$

**22** Give an example of an operator $T \in \mathcal{L}(\mathbf{C}^3)$ such that 2 and 3 are the only eigenvalues of $T$ and $T^2 - 5T + 6I \neq 0$.

**23** Suppose $T \in \mathcal{L}(V)$ is self-adjoint, $\lambda \in \mathbf{F}$, and $\epsilon > 0$. Suppose there exists $v \in V$ such that $\|v\| = 1$ and

$$\|Tv - \lambda v\| < \epsilon.$$

Prove that $T$ has an eigenvalue $\lambda'$ such that $|\lambda - \lambda'| < \epsilon$.

*This exercise shows that for a self-adjoint operator, a number that is close to satisfying an equation that would make it an eigenvalue is close to an eigenvalue.*

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**24** Suppose $U$ is a finite-dimensional vector space and $T \in \mathcal{L}(U)$.

(a) Suppose $\mathbf{F} = \mathbf{R}$. Prove that $T$ is diagonalizable if and only if there is a basis of $U$ such that the matrix of $T$ with respect to this basis equals its transpose.

(b) Suppose $\mathbf{F} = \mathbf{C}$. Prove that $T$ is diagonalizable if and only if there is a basis of $U$ such that the matrix of $T$ with respect to this basis commutes with its conjugate transpose.

*This exercise adds another equivalence to the list of conditions equivalent to diagonalizability in 5.55.*

**25** Suppose that $T \in \mathcal{L}(V)$ and there is an orthonormal basis $e_1, \ldots, e_n$ of $V$ consisting of eigenvectors of $T$, with corresponding eigenvalues $\lambda_1, \ldots, \lambda_n$. Show that if $k \in \{1, \ldots, n\}$, then the pseudoinverse $T^\dagger$ satisfies the equation

$$T^\dagger e_k = \begin{cases} \frac{1}{\lambda_k} e_k & \text{if } \lambda_k \neq 0, \\ 0 & \text{if } \lambda_k = 0. \end{cases}$$

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## 7C Positive Operators

**7.34 definition: *positive operator***

An operator $T \in \mathcal{L}(V)$ is called *positive* if $T$ is self-adjoint and

$$\langle Tv, v \rangle \geq 0$$

for all $v \in V$.

If $V$ is a complex vector space, then the requirement that $T$ be self-adjoint can be dropped from the definition above (by 7.14).

**7.35 example: *positive operators***

(a) Let $T \in \mathcal{L}(\mathbf{F}^2)$ be the operator whose matrix (using the standard basis) is $\begin{pmatrix} 2 & -1 \\ -1 & 1 \end{pmatrix}$. Then $T$ is self-adjoint and $\langle T(w,z), (w,z) \rangle = 2|w|^2 - 2\operatorname{Re}(w\bar{z}) + |z|^2 = |w-z|^2 + |w|^2 \geq 0$ for all $(w,z) \in \mathbf{F}^2$. Thus $T$ is a positive operator.

(b) If $U$ is a subspace of $V$, then the orthogonal projection $P_U$ is a positive operator, as you should verify.

(c) If $T \in \mathcal{L}(V)$ is self-adjoint and $b, c \in \mathbf{R}$ are such that $b^2 < 4c$, then $T^2 + bT + cI$ is a positive operator, as shown by the proof of 7.26.

**7.36 definition: *square root***

An operator $R$ is called a *square root* of an operator $T$ if $R^2 = T$.

**7.37 example: *square root of an operator***

If $T \in \mathcal{L}(\mathbf{F}^3)$ is defined by $T(z_1, z_2, z_3) = (z_3, 0, 0)$, then the operator $R \in \mathcal{L}(\mathbf{F}^3)$ defined by $R(z_1, z_2, z_3) = (z_2, z_3, 0)$ is a square root of $T$ because $R^2 = T$, as you can verify.

*Because positive operators correspond to nonnegative numbers, better terminology would use the term nonnegative operators. However, operator theorists consistently call these positive operators, so we follow that custom. Some mathematicians use the term* ***positive semidefinite operator****, which means the same as positive operator.*

The characterizations of the positive operators in the next result correspond to characterizations of the nonnegative numbers among $\mathbf{C}$. Specifically, a number $z \in \mathbf{C}$ is nonnegative if and only if it has a nonnegative square root, corresponding to condition (d). Also, $z$ is nonnegative if and only if it has a real square root, corresponding to condition (e). Finally, $z$ is nonnegative if and only if there exists $w \in \mathbf{C}$ such that $z = \bar{w}w$, corresponding to condition (f). See Exercise 20 for another condition that is equivalent to being a positive operator.

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**7.38** *characterization of positive operators*

Let $T \in \mathcal{L}(V)$. Then the following are equivalent.

(a) $T$ is a positive operator.

(b) $T$ is self-adjoint and all eigenvalues of $T$ are nonnegative.

(c) With respect to some orthonormal basis of $V$, the matrix of $T$ is a diagonal matrix with only nonnegative numbers on the diagonal.

(d) $T$ has a positive square root.

(e) $T$ has a self-adjoint square root.

(f) $T = R^*R$ for some $R \in \mathcal{L}(V)$.

Proof We will prove that (a) $\Rightarrow$ (b) $\Rightarrow$ (c) $\Rightarrow$ (d) $\Rightarrow$ (e) $\Rightarrow$ (f) $\Rightarrow$ (a).

First suppose (a) holds, so that $T$ is positive, which implies that $T$ is self-adjoint (by definition of positive operator). To prove the other condition in (b), suppose $\lambda$ is an eigenvalue of $T$. Let $v$ be an eigenvector of $T$ corresponding to $\lambda$. Then

$$0 \le \langle Tv, v\rangle = \langle \lambda v, v\rangle = \lambda\langle v, v\rangle.$$

Thus $\lambda$ is a nonnegative number. Hence (b) holds, showing that (a) implies (b).

Now suppose (b) holds, so that $T$ is self-adjoint and all eigenvalues of $T$ are nonnegative. By the spectral theorem (7.29 and 7.31), there is an orthonormal basis $e_1, \ldots, e_n$ of $V$ consisting of eigenvectors of $T$. Let $\lambda_1, \ldots, \lambda_n$ be the eigenvalues of $T$ corresponding to $e_1, \ldots, e_n$; thus each $\lambda_k$ is a nonnegative number. The matrix of $T$ with respect to $e_1, \ldots, e_n$ is the diagonal matrix with $\lambda_1, \ldots, \lambda_n$ on the diagonal, which shows that (b) implies (c).

Now suppose (c) holds. Suppose $e_1, \ldots, e_n$ is an orthonormal basis of $V$ such that the matrix of $T$ with respect to this basis is a diagonal matrix with nonnegative numbers $\lambda_1, \ldots, \lambda_n$ on the diagonal. The linear map lemma (3.4) implies that there exists $R \in \mathcal{L}(V)$ such that

$$Re_k = \sqrt{\lambda_k}e_k$$

for each $k = 1, \ldots, n$. As you should verify, $R$ is a positive operator. Furthermore, $R^2e_k = \lambda_k e_k = Te_k$ for each $k$, which implies that $R^2 = T$. Thus $R$ is a positive square root of $T$. Hence (d) holds, which shows that (c) implies (d).

Every positive operator is self-adjoint (by definition of positive operator). Thus (d) implies (e).

Now suppose (e) holds, meaning that there exists a self-adjoint operator $R$ on $V$ such that $T = R^2$. Then $T = R^*R$ (because $R^* = R$). Hence (e) implies (f).

Finally, suppose (f) holds. Let $R \in \mathcal{L}(V)$ be such that $T = R^*R$. Then $T^* = (R^*R)^* = R^*(R^*)^* = R^*R = T$. Hence $T$ is self-adjoint. To complete the proof that (a) holds, note that

$$\langle Tv, v\rangle = \langle R^*Rv, v\rangle = \langle Rv, Rv\rangle \ge 0$$

for every $v \in V$. Thus $T$ is positive, showing that (f) implies (a).

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Every nonnegative number has a unique nonnegative square root. The next result shows that positive operators enjoy a similar property.

**7.39** *each positive operator has only one positive square root*

Every positive operator on $V$ has a unique positive square root.

*A positive operator can have infinitely many square roots (although only one of them can be positive). For example, the identity operator on $V$ has infinitely many square roots if* $\dim V > 1$.

Proof Suppose $T \in \mathcal{L}(V)$ is positive. Suppose $v \in V$ is an eigenvector of $T$. Hence there exists a real number $\lambda \geq 0$ such that $Tv = \lambda v$.

Let $R$ be a positive square root of $T$. We will prove that $Rv = \sqrt{\lambda}v$. This will imply that the behavior of $R$ on the eigenvectors of $T$ is uniquely determined. Because there is a basis of $V$ consisting of eigenvectors of $T$ (by the spectral theorem), this will imply that $R$ is uniquely determined.

To prove that $Rv = \sqrt{\lambda}v$, note that the spectral theorem asserts that there is an orthonormal basis $e_1, \ldots, e_n$ of $V$ consisting of eigenvectors of $R$. Because $R$ is a positive operator, all its eigenvalues are nonnegative. Thus there exist nonnegative numbers $\lambda_1, \ldots, \lambda_n$ such that $Re_k = \sqrt{\lambda_k}e_k$ for each $k = 1, \ldots, n$.

Because $e_1, \ldots, e_n$ is a basis of $V$, we can write

$$v = a_1 e_1 + \cdots + a_n e_n$$

for some numbers $a_1, \ldots, a_n \in \mathbf{F}$. Thus

$$Rv = a_1\sqrt{\lambda_1}e_1 + \cdots + a_n\sqrt{\lambda_n}e_n.$$

Hence

$$\lambda v = Tv = R^2 v = a_1\lambda_1 e_1 + \cdots + a_n\lambda_n e_n.$$

The equation above implies that

$$a_1\lambda e_1 + \cdots + a_n\lambda e_n = a_1\lambda_1 e_1 + \cdots + a_n\lambda_n e_n.$$

Thus $a_k(\lambda - \lambda_k) = 0$ for each $k = 1, \ldots, n$. Hence

$$v = \sum_{\{k\,:\,\lambda_k = \lambda\}} a_k e_k.$$

Thus

$$Rv = \sum_{\{k\,:\,\lambda_k = \lambda\}} a_k\sqrt{\lambda}e_k = \sqrt{\lambda}v,$$

as desired. ∎

The notation defined below makes sense thanks to the result above.

**7.40** notation: $\sqrt{T}$

For $T$ a positive operator, $\sqrt{T}$ denotes the unique positive square root of $T$.

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7.41 example: *square root of positive operators*

Define operators $S, T$ on $\mathbf{R}^2$ (with the usual Euclidean inner product) by

$$S(x, y) = (x, 2y) \quad \text{and} \quad T(x, y) = (x + y, x + y).$$

Then with respect to the standard basis of $\mathbf{R}^2$ we have

7.42
$$\mathcal{M}(S) = \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix} \quad \text{and} \quad \mathcal{M}(T) = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}.$$

Each of these matrices equals its transpose; thus $S$ and $T$ are self-adjoint.

If $(x, y) \in \mathbf{R}^2$, then

$$\langle S(x, y), (x, y) \rangle = x^2 + 2y^2 \geq 0$$

and

$$\langle T(x, y), (x, y) \rangle = x^2 + 2xy + y^2 = (x + y)^2 \geq 0.$$

Thus $S$ and $T$ are positive operators.

The standard basis of $\mathbf{R}^2$ is an orthonormal basis consisting of eigenvectors of $S$. Note that

$$\left(\tfrac{1}{\sqrt{2}}, \tfrac{1}{\sqrt{2}}\right), \left(\tfrac{1}{\sqrt{2}}, -\tfrac{1}{\sqrt{2}}\right)$$

is an orthonormal basis of eigenvectors of $T$, with eigenvalue 2 for the first eigenvector and eigenvalue 0 for the second eigenvector. Thus $\sqrt{T}$ has the same eigenvectors, with eigenvalues $\sqrt{2}$ and 0.

You can verify that

$$\mathcal{M}\left(\sqrt{S}\right) = \begin{pmatrix} 1 & 0 \\ 0 & \sqrt{2} \end{pmatrix} \quad \text{and} \quad \mathcal{M}\left(\sqrt{T}\right) = \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{pmatrix}$$

with respect to the standard basis by showing that the squares of the matrices above are the matrices in 7.42 and that each matrix above is the matrix of a positive operator.

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The statement of the next result does not involve a square root, but the clean proof makes nice use of the square root of a positive operator.

**7.43** *$T$ positive and $\langle Tv, v \rangle = 0 \implies Tv = 0$*

Suppose $T$ is a positive operator on $V$ and $v \in V$ is such that $\langle Tv, v \rangle = 0$. Then $Tv = 0$.

Proof We have

$$0 = \langle Tv, v \rangle = \left\langle \sqrt{T}\sqrt{T}v, v \right\rangle = \left\langle \sqrt{T}v, \sqrt{T}v \right\rangle = \left\| \sqrt{T}v \right\|^2.$$

Hence $\sqrt{T}v = 0$. Thus $Tv = \sqrt{T}\left(\sqrt{T}v\right) = 0$, as desired. ∎

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## *Exercises 7C*

**1** Suppose $T \in \mathcal{L}(V)$. Prove that if both $T$ and $-T$ are positive operators, then $T = 0$.

**2** Suppose $T \in \mathcal{L}(\mathbf{F}^4)$ is the operator whose matrix (with respect to the standard basis) is

$$\begin{pmatrix} 2 & -1 & 0 & 0 \\ -1 & 2 & -1 & 0 \\ 0 & -1 & 2 & -1 \\ 0 & 0 & -1 & 2 \end{pmatrix}.$$

Show that $T$ is an invertible positive operator.

**3** Suppose $n$ is a positive integer and $T \in \mathcal{L}(\mathbf{F}^n)$ is the operator whose matrix (with respect to the standard basis) consists of all 1's. Show that $T$ is a positive operator.

**4** Suppose $n$ is an integer with $n > 1$. Show that there exists an $n$-by-$n$ matrix $A$ such that all of the entries of $A$ are positive numbers and $A = A^*$, but the operator on $\mathbf{F}^n$ whose matrix (with respect to the standard basis) equals $A$ is not a positive operator.

**5** Suppose $T \in \mathcal{L}(V)$ is self-adjoint. Prove that $T$ is a positive operator if and only if for every orthonormal basis $e_1, \ldots, e_n$ of $V$, all entries on the diagonal of $\mathcal{M}(T, (e_1, \ldots, e_n))$ are nonnegative numbers.

**6** Prove that the sum of two positive operators on $V$ is a positive operator.

**7** Suppose $S \in \mathcal{L}(V)$ is an invertible positive operator and $T \in \mathcal{L}(V)$ is a positive operator. Prove that $S + T$ is invertible.

**8** Suppose $T \in \mathcal{L}(V)$. Prove that $T$ is a positive operator if and only if the pseudoinverse $T^\dagger$ is a positive operator.

**9** Suppose $T \in \mathcal{L}(V)$ is a positive operator and $S \in \mathcal{L}(W, V)$. Prove that $S^*TS$ is a positive operator on $W$.

**10** Suppose $T$ is a positive operator on $V$. Suppose $v, w \in V$ are such that

$$Tv = w \quad \text{and} \quad Tw = v.$$

Prove that $v = w$.

**11** Suppose $T$ is a positive operator on $V$ and $U$ is a subspace of $V$ invariant under $T$. Prove that $T|_U \in \mathcal{L}(U)$ is a positive operator on $U$.

**12** Suppose $T \in \mathcal{L}(V)$ is a positive operator. Prove that $T^k$ is a positive operator for every positive integer $k$.

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**13** Suppose $T \in \mathcal{L}(V)$ is self-adjoint and $\alpha \in \mathbf{R}$.

(a) Prove that $T - \alpha I$ is a positive operator if and only if $\alpha$ is less than or equal to every eigenvalue of $T$.

(b) Prove that $\alpha I - T$ is a positive operator if and only if $\alpha$ is greater than or equal to every eigenvalue of $T$.

**14** Suppose $T$ is a positive operator on $V$ and $v_1, ..., v_m \in V$. Prove that

$$\sum_{j=1}^{m} \sum_{k=1}^{m} \langle T v_k, v_j \rangle \geq 0.$$

**15** Suppose $T \in \mathcal{L}(V)$ is self-adjoint. Prove that there exist positive operators $A, B \in \mathcal{L}(V)$ such that

$$T = A - B \quad \text{and} \quad \sqrt{T^* T} = A + B \quad \text{and} \quad AB = BA = 0.$$

**16** Suppose $T$ is a positive operator on $V$. Prove that

$$\operatorname{null} \sqrt{T} = \operatorname{null} T \quad \text{and} \quad \operatorname{range} \sqrt{T} = \operatorname{range} T.$$

**17** Suppose that $T \in \mathcal{L}(V)$ is a positive operator. Prove that there exists a polynomial $p$ with real coefficients such that $\sqrt{T} = p(T)$.

**18** Suppose $S$ and $T$ are positive operators on $V$. Prove that $ST$ is a positive operator if and only if $S$ and $T$ commute.

**19** Show that the identity operator on $\mathbf{F}^2$ has infinitely many self-adjoint square roots.

**20** Suppose $T \in \mathcal{L}(V)$ and $e_1, ..., e_n$ is an orthonormal basis of $V$. Prove that $T$ is a positive operator if and only if there exist $v_1, ..., v_n \in V$ such that

$$\langle T e_k, e_j \rangle = \langle v_k, v_j \rangle$$

for all $j, k = 1, ..., n$.

*The numbers $\{\langle T e_k, e_j \rangle\}_{j,k=1,...,n}$ are the entries in the matrix of $T$ with respect to the orthonormal basis $e_1, ..., e_n$.*

**21** Suppose $n$ is a positive integer. The $n$-by-$n$ *Hilbert matrix* is the $n$-by-$n$ matrix whose entry in row $j$, column $k$ is $\frac{1}{j+k-1}$. Suppose $T \in \mathcal{L}(V)$ is an operator whose matrix with respect to some orthonormal basis of $V$ is the $n$-by-$n$ Hilbert matrix. Prove that $T$ is a positive invertible operator.

*Example: The 4-by-4 Hilbert matrix is*

$$\begin{pmatrix} 1 & \frac{1}{2} & \frac{1}{3} & \frac{1}{4} \\ \frac{1}{2} & \frac{1}{3} & \frac{1}{4} & \frac{1}{5} \\ \frac{1}{3} & \frac{1}{4} & \frac{1}{5} & \frac{1}{6} \\ \frac{1}{4} & \frac{1}{5} & \frac{1}{6} & \frac{1}{7} \end{pmatrix}.$$

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**22** Suppose $T \in \mathcal{L}(V)$ is a positive operator and $u \in V$ is such that $\|u\| = 1$ and $\|Tu\| \geq \|Tv\|$ for all $v \in V$ with $\|v\| = 1$. Show that $u$ is an eigenvector of $T$ corresponding to the largest eigenvalue of $T$.

**23** For $T \in \mathcal{L}(V)$ and $u, v \in V$, define $\langle u, v\rangle_T$ by $\langle u, v\rangle_T = \langle Tu, v\rangle$.

(a) Suppose $T \in \mathcal{L}(V)$. Prove that $\langle \cdot, \cdot \rangle_T$ is an inner product on $V$ if and only if $T$ is an invertible positive operator (with respect to the original inner product $\langle \cdot, \cdot \rangle$).

(b) Prove that every inner product on $V$ is of the form $\langle \cdot, \cdot \rangle_T$ for some positive invertible operator $T \in \mathcal{L}(V)$.

**24** Suppose $S$ and $T$ are positive operators on $V$. Prove that

$$\operatorname{null}(S + T) = \operatorname{null} S \cap \operatorname{null} T.$$

**25** Let $T$ be the second derivative operator in Exercise 31(b) in Section 7A. Show that $-T$ is a positive operator.

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# 7D Isometries, Unitary Operators, and Matrix Factorization

## Isometries

Linear maps that preserve norms are sufficiently important to deserve a name.

**7.44 definition: *isometry***

A linear map $S \in \mathcal{L}(V, W)$ is called an *isometry* if

$$\|Sv\| = \|v\|$$

for every $v \in V$. In other words, a linear map is an isometry if it preserves norms.

If $S \in \mathcal{L}(V, W)$ is an isometry and $v \in V$ is such that $Sv = 0$, then

$$\|v\| = \|Sv\| = \|0\| = 0,$$

which implies that $v = 0$. Thus every isometry is injective.

*The Greek word **isos** means equal; the Greek word **metron** means measure. Thus **isometry** literally means equal measure.*

**7.45 example: *orthonormal basis maps to orthonormal list $\Longrightarrow$ isometry***

Suppose $e_1, \ldots, e_n$ is an orthonormal basis of $V$ and $g_1, \ldots, g_n$ is an orthonormal list in $W$. Let $S \in \mathcal{L}(V, W)$ be the linear map such that $Se_k = g_k$ for each $k = 1, \ldots, n$. To show that $S$ is an isometry, suppose $v \in V$. Then

$$v = \langle v, e_1 \rangle e_1 + \cdots + \langle v, e_n \rangle e_n \tag{7.46}$$

and

$$\|v\|^2 = |\langle v, e_1 \rangle|^2 + \cdots + |\langle v, e_n \rangle|^2, \tag{7.47}$$

where we have used 6.30(b). Applying $S$ to both sides of 7.46 gives

$$Sv = \langle v, e_1 \rangle Se_1 + \cdots + \langle v, e_n \rangle Se_n = \langle v, e_1 \rangle g_1 + \cdots + \langle v, e_n \rangle g_n.$$

Thus

$$\|Sv\|^2 = |\langle v, e_1 \rangle|^2 + \cdots + |\langle v, e_n \rangle|^2. \tag{7.48}$$

Comparing 7.47 and 7.48 shows that $\|v\| = \|Sv\|$. Thus $S$ is an isometry.

The next result gives conditions equivalent to being an isometry. The equivalence of (a) and (c) shows that a linear map is an isometry if and only if it preserves inner products. The equivalence of (a) and (d) shows that a linear map is an isometry if and only if it maps some orthonormal basis to an orthonormal list. Thus the isometries given by Example 7.45 include all isometries. Furthermore, a linear map is an isometry if and only if it maps every orthonormal basis to an orthonormal list [because whether or not (a) holds does not depend on the basis $e_1, \ldots, e_n$].

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The equivalence of (a) and (e) in the next result shows that a linear map is an isometry if and only if the columns of its matrix (with respect to any orthonormal bases) form an orthonormal list. Here we are identifying the columns of an $m$-by-$n$ matrix with elements of $\mathbf{F}^m$ and then using the Euclidean inner product on $\mathbf{F}^m$.

**7.49** *characterization of isometries*

Suppose $S \in \mathcal{L}(V, W)$. Suppose $e_1, \ldots, e_n$ is an orthonormal basis of $V$ and $f_1, \ldots, f_m$ is an orthonormal basis of $W$. Then the following are equivalent.

(a) $S$ is an isometry.

(b) $S^*S = I$.

(c) $\langle Su, Sv\rangle = \langle u, v\rangle$ for all $u, v \in V$.

(d) $Se_1, \ldots, Se_n$ is an orthonormal list in $W$.

(e) The columns of $\mathcal{M}(S, (e_1, \ldots, e_n), (f_1, \ldots, f_m))$ form an orthonormal list in $\mathbf{F}^m$ with respect to the Euclidean inner product.

Proof First suppose (a) holds, so $S$ is an isometry. If $v \in V$ then

$$\big\langle (I - S^*S)v, v\big\rangle = \langle v, v\rangle - \big\langle S^*Sv, v\big\rangle = \|v\|^2 - \langle Sv, Sv\rangle = \|v\|^2 - \|Sv\|^2 = 0.$$

Hence the self-adjoint operator $I - S^*S$ equals 0 (by 7.16). Thus $S^*S = I$, proving that (a) implies (b).

Now suppose (b) holds, so $S^*S = I$. If $u, v \in V$ then

$$\langle Su, Sv\rangle = \big\langle S^*Su, v\big\rangle = \langle Iu, v\rangle = \langle u, v\rangle,$$

proving that (b) implies (c).

Now suppose that (c) holds, so $\langle Su, Sv\rangle = \langle u, v\rangle$ for all $u, v \in V$. Thus if $j, k \in \{1, \ldots, n\}$, then

$$\langle Se_j, Se_k\rangle = \langle e_j, e_k\rangle.$$

Hence $Se_1, \ldots, Se_n$ is an orthonormal list in $W$, proving that (c) implies (d).

Now suppose that (d) holds, so $Se_1, \ldots, Se_n$ is an orthonormal list in $W$. Let $A = \mathcal{M}(S, (e_1, \ldots, e_n), (f_1, \ldots, f_m))$. If $k, r \in \{1, \ldots, n\}$, then

$$\text{7.50} \qquad \sum_{j=1}^{m} A_{j,k}\overline{A_{j,r}} = \Big\langle \sum_{j=1}^{m} A_{j,k} f_j, \sum_{j=1}^{m} A_{j,r} f_j \Big\rangle = \langle Se_k, Se_r\rangle = \begin{cases} 1 & \text{if } k = r, \\ 0 & \text{if } k \neq r. \end{cases}$$

The left side of 7.50 is the inner product in $\mathbf{F}^m$ of columns $k$ and $r$ of $A$. Thus the columns of $A$ form an orthonormal list in $\mathbf{F}^m$, proving that (d) implies (e).

Now suppose (e) holds, so the columns of the matrix $A$ defined in the paragraph above form an orthonormal list in $\mathbf{F}^n$. Then 7.50 shows that $Se_1, \ldots, Se_n$ is an orthonormal list in $W$. Thus Example 7.45, with $Se_1, \ldots, Se_n$ playing the role of $g_1, \ldots, g_n$, shows that $S$ is an isometry, proving that (e) implies (a). ∎

See Exercises 1 and 11 for additional conditions that are equivalent to being an isometry.

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## *Unitary Operators*

In this subsection, we confine our attention to linear maps from a vector space to itself. In other words, we will be working with operators.

**7.51 definition: *unitary operator***

An operator $S \in \mathcal{L}(V)$ is called *unitary* if $S$ is an invertible isometry.

*Although the words "unitary" and "isometry" mean the same thing for operators on finite-dimensional inner product spaces, remember that a unitary operator maps a vector space to itself, while an isometry maps a vector space to another (possibly different) vector space.*

As previously noted, every isometry is injective. Every injective operator on a finite-dimensional vector space is invertible (see 3.65). A standing assumption for this chapter is that $V$ is a finite-dimensional inner product space. Thus we could delete the word "invertible" from the definition above without changing the meaning. The unnecessary word "invertible" has been retained in the definition above for consistency with the definition readers may encounter when learning about inner product spaces that are not necessarily finite-dimensional.

**7.52 example: *rotation of* $\mathbf{R}^2$**

Suppose $\theta \in \mathbf{R}$ and $S$ is the operator on $\mathbf{F}^2$ whose matrix with respect to the standard basis of $\mathbf{F}^2$ is

$$\begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}.$$

The two columns of this matrix form an orthonormal list in $\mathbf{F}^2$; hence $S$ is an isometry [by the equivalence of (a) and (e) in 7.49]. Thus $S$ is a unitary operator.

If $\mathbf{F} = \mathbf{R}$, then $S$ is the operator of counterclockwise rotation by $\theta$ radians around the origin of $\mathbf{R}^2$. This observation gives us another way to think about why $S$ is an isometry, because each rotation around the origin of $\mathbf{R}^2$ preserves norms.

The next result (7.53) lists several conditions that are equivalent to being a unitary operator. All the conditions equivalent to being an isometry in 7.49 should be added to this list. The extra conditions in 7.53 arise because of limiting the context to linear maps from a vector space to itself. For example, 7.49 shows that a linear map $S \in \mathcal{L}(V, W)$ is an isometry if and only if $S^*S = I$, while 7.53 shows that an operator $S \in \mathcal{L}(V)$ is a unitary operator if and only if $S^*S = SS^* = I$.

Another difference is that 7.49(d) mentions an orthonormal list, while 7.53(d) mentions an orthonormal basis. Also, 7.49(e) mentions the columns of $\mathcal{M}(T)$, while 7.53(e) mentions the rows of $\mathcal{M}(T)$. Furthermore, $\mathcal{M}(T)$ in 7.49(e) is with respect to an orthonormal basis of $V$ and an orthonormal basis of $W$, while $\mathcal{M}(T)$ in 7.53(e) is with respect to a single basis of $V$ doing double duty.

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**7.53** *characterization of unitary operators*

Suppose $S \in \mathcal{L}(V)$. Suppose $e_1, \ldots, e_n$ is an orthonormal basis of $V$. Then the following are equivalent.

(a) $S$ is a unitary operator.

(b) $S^*S = SS^* = I$.

(c) $S$ is invertible and $S^{-1} = S^*$.

(d) $Se_1, \ldots, Se_n$ is an orthonormal basis of $V$.

(e) The rows of $\mathcal{M}(S, (e_1, \ldots, e_n))$ form an orthonormal basis of $\mathbf{F}^n$ with respect to the Euclidean inner product.

(f) $S^*$ is a unitary operator.

Proof First suppose (a) holds, so $S$ is a unitary operator. Hence

$$S^*S = I$$

by the equivalence of (a) and (b) in 7.49. Multiply both sides of this equation by $S^{-1}$ on the right, getting $S^* = S^{-1}$. Thus $SS^* = SS^{-1} = I$, as desired, proving that (a) implies (b).

The definitions of invertible and inverse show that (b) implies (c).

Now suppose (c) holds, so $S$ is invertible and $S^{-1} = S^*$. Thus $S^*S = I$. Hence $Se_1, \ldots, Se_n$ is an orthonormal list in $V$, by the equivalence of (b) and (d) in 7.49. The length of this list equals $\dim V$. Thus $Se_1, \ldots, Se_n$ is an orthonormal basis of $V$, proving that (c) implies (d).

Now suppose (d) holds, so $Se_1, \ldots, Se_n$ is an orthonormal basis of $V$. The equivalence of (a) and (d) in 7.49 shows that $S$ is a unitary operator. Thus

$$(S^*)^*S^* = SS^* = I,$$

where the last equation holds because we have already shown that (a) implies (b) in this result. The equation above and the equivalence of (a) and (b) in 7.49 show that $S^*$ is an isometry. Thus the columns of $\mathcal{M}(S^*, (e_1, \ldots, e_n))$ form an orthonormal basis of $\mathbf{F}^n$ [by the equivalence of (a) and (e) of 7.49]. The rows of $\mathcal{M}(S, (e_1, \ldots, e_n))$ are the complex conjugates of the columns of $\mathcal{M}(S^*, (e_1, \ldots, e_n))$. Thus the rows of $\mathcal{M}(S, (e_1, \ldots, e_n))$ form an orthonormal basis of $\mathbf{F}^n$, proving that (d) implies (e).

Now suppose (e) holds. Thus the columns of $\mathcal{M}(S^*, (e_1, \ldots, e_n))$ form an orthonormal basis of $\mathbf{F}^n$. The equivalence of (a) and (e) in 7.49 shows that $S^*$ is an isometry, proving that (e) implies (f).

Now suppose (f) holds, so $S^*$ is a unitary operator. The chain of implications we have already proved in this result shows that (a) implies (f). Applying this result to $S^*$ shows that $(S^*)^*$ is a unitary operator, proving that (f) implies (a).

We have shown that (a) $\Rightarrow$ (b) $\Rightarrow$ (c) $\Rightarrow$ (d) $\Rightarrow$ (e) $\Rightarrow$ (f) $\Rightarrow$ (a), completing the proof. ∎

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Recall our analogy between $\mathbf{C}$ and $\mathcal{L}(V)$. Under this analogy, a complex number $z$ corresponds to an operator $S \in \mathcal{L}(V)$, and $\bar{z}$ corresponds to $S^*$. The real numbers ($z = \bar{z}$) correspond to the self-adjoint operators ($S = S^*$), and the nonnegative numbers correspond to the (badly named) positive operators.

Another distinguished subset of $\mathbf{C}$ is the unit circle, which consists of the complex numbers $z$ such that $|z| = 1$. The condition $|z| = 1$ is equivalent to the condition $\bar{z}z = 1$. Under our analogy, this corresponds to the condition $S^*S = I$, which is equivalent to $S$ being a unitary operator. Hence the analogy shows that the unit circle in $\mathbf{C}$ corresponds to the set of unitary operators. In the next two results, this analogy appears in the eigenvalues of unitary operators. Also see Exercise 15 for another example of this analogy.

**7.54** *eigenvalues of unitary operators have absolute value 1*

Suppose $\lambda$ is an eigenvalue of a unitary operator. Then $|\lambda| = 1$.

Proof Suppose $S \in \mathcal{L}(V)$ is a unitary operator and $\lambda$ is an eigenvalue of $S$. Let $v \in V$ be such that $v \neq 0$ and $Sv = \lambda v$. Then

$$|\lambda| \, \|v\| = \|\lambda v\| = \|Sv\| = \|v\|.$$

Thus $|\lambda| = 1$, as desired. ∎

The next result characterizes unitary operators on finite-dimensional complex inner product spaces, using the complex spectral theorem as the main tool.

**7.55** *description of unitary operators on complex inner product spaces*

Suppose $\mathbf{F} = \mathbf{C}$ and $S \in \mathcal{L}(V)$. Then the following are equivalent.

(a) $S$ is a unitary operator.

(b) There is an orthonormal basis of $V$ consisting of eigenvectors of $S$ whose corresponding eigenvalues all have absolute value 1.

Proof Suppose (a) holds, so $S$ is a unitary operator. The equivalence of (a) and (b) in 7.53 shows that $S$ is normal. Thus the complex spectral theorem (7.31) shows that there is an orthonormal basis $e_1, \ldots, e_n$ of $V$ consisting of eigenvectors of $S$. Every eigenvalue of $S$ has absolute value 1 (by 7.54), completing the proof that (a) implies (b).

Now suppose (b) holds. Let $e_1, \ldots, e_n$ be an orthonormal basis of $V$ consisting of eigenvectors of $S$ whose corresponding eigenvalues $\lambda_1, \ldots, \lambda_n$ all have absolute value 1. Then $Se_1, \ldots, Se_n$ is also an orthonormal basis of $V$ because

$$\langle Se_j, Se_k \rangle = \langle \lambda_j e_j, \lambda_k e_k \rangle = \lambda_j \overline{\lambda_k} \langle e_j, e_k \rangle = \begin{cases} 0 & \text{if } j \neq k, \\ 1 & \text{if } j = k \end{cases}$$

for all $j, k = 1, \ldots, n$. Thus the equivalence of (a) and (d) in 7.53 shows that $S$ is unitary, proving that (b) implies (a). ∎

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## *QR Factorization*

In this subsection, we shift our attention from operators to matrices. This switch should give you good practice in identifying an operator with a square matrix (after picking a basis of the vector space on which the operator is defined). You should also become more comfortable with translating concepts and results back and forth between the context of operators and the context of square matrices.

When starting with $n$-by-$n$ matrices instead of operators, unless otherwise specified assume that the associated operators live on $\mathbf{F}^n$ (with the Euclidean inner product) and that their matrices are computed with respect to the standard basis of $\mathbf{F}^n$.

We begin by making the following definition, transferring the notion of a unitary operator to a unitary matrix.

**7.56 definition: *unitary matrix***

An $n$-by-$n$ matrix is called *unitary* if its columns form an orthonormal list in $\mathbf{F}^n$.

In the definition above, we could have replaced "orthonormal list in $\mathbf{F}^n$" with "orthonormal basis of $\mathbf{F}^n$" because every orthonormal list of length $n$ in an $n$-dimensional inner product space is an orthonormal basis. If $S \in \mathcal{L}(V)$ and $e_1, \ldots, e_n$ and $f_1, \ldots, f_n$ are orthonormal bases of $V$, then $S$ is a unitary operator if and only if $\mathcal{M}(S, (e_1, \ldots, e_n), (f_1, \ldots, f_n))$ is a unitary matrix, as shown by the equivalence of (a) and (e) in 7.49. Also note that we could also have replaced "columns" in the definition above with "rows" by using the equivalence between conditions (a) and (e) in 7.53.

The next result, whose proof will be left as an exercise for the reader, gives some equivalent conditions for a square matrix to be unitary. In (c), $Qv$ denotes the matrix product of $Q$ and $v$, identifying elements of $\mathbf{F}^n$ with $n$-by-1 matrices (sometimes called column vectors). The norm in (c) below is the usual Euclidean norm on $\mathbf{F}^n$ that comes from the Euclidean inner product. In (d), $Q^*$ denotes the conjugate transpose of the matrix $Q$, which corresponds to the adjoint of the associated operator.

**7.57 *characterizations of unitary matrices***

Suppose $Q$ is an $n$-by-$n$ matrix. Then the following are equivalent.

(a) $Q$ is a unitary matrix.

(b) The rows of $Q$ form an orthonormal list in $\mathbf{F}^n$.

(c) $\|Qv\| = \|v\|$ for every $v \in \mathbf{F}^n$.

(d) $Q^*Q = QQ^* = I$, the $n$-by-$n$ matrix with 1's on the diagonal and 0's elsewhere.

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The QR factorization stated and proved below is the main tool in the widely used QR algorithm (not discussed here) for finding good approximations to eigenvalues and eigenvectors of square matrices. In the result below, if the matrix $A$ is in $\mathbf{F}^{n,n}$, then the matrices $Q$ and $R$ are also in $\mathbf{F}^{n,n}$.

**7.58** *QR factorization*

Suppose $A$ is a square matrix with linearly independent columns. Then there exist unique matrices $Q$ and $R$ such that $Q$ is unitary, $R$ is upper triangular with only positive numbers on its diagonal, and

$$A = QR.$$

Proof Let $v_1, \dots, v_n$ denote the columns of $A$, thought of as elements of $\mathbf{F}^n$. Apply the Gram–Schmidt procedure (6.32) to the list $v_1, \dots, v_n$, getting an orthonormal basis $e_1, \dots, e_n$ of $\mathbf{F}^n$ such that

$$\text{span}(v_1, \dots, v_k) = \text{span}(e_1, \dots, e_k) \tag{7.59}$$

for each $k = 1, \dots, n$. Let $R$ be the $n$-by-$n$ matrix defined by

$$R_{j,k} = \langle v_k, e_j \rangle,$$

where $R_{j,k}$ denotes the entry in row $j$, column $k$ of $R$. If $j > k$, then $e_j$ is orthogonal to $\text{span}(e_1, \dots, e_k)$ and hence $e_j$ is orthogonal to $v_k$ (by 7.59). In other words, if $j > k$ then $\langle v_k, e_j \rangle = 0$. Thus $R$ is an upper-triangular matrix.

Let $Q$ be the unitary matrix whose columns are $e_1, \dots, e_n$. If $k \in \{1, \dots, n\}$, then the $k^{\text{th}}$ column of $QR$ equals a linear combination of the columns of $Q$, with the coefficients for the linear combination coming from the $k^{\text{th}}$ column of $R$—see 3.51(a). Hence the $k^{\text{th}}$ column of $QR$ equals

$$\langle v_k, e_1 \rangle e_1 + \cdots + \langle v_k, e_k \rangle e_k,$$

which equals $v_k$ [by 6.30(a)], the $k^{\text{th}}$ column of $A$. Thus $A = QR$, as desired.

The equations defining the Gram–Schmidt procedure (see 6.32) show that each $v_k$ equals a positive multiple of $e_k$ plus a linear combination of $e_1, \dots, e_{k-1}$. Thus each $\langle v_k, e_k \rangle$ is a positive number. Hence all entries on the diagonal of $R$ are positive numbers, as desired.

Finally, to show that $Q$ and $R$ are unique, suppose we also have $A = \widehat{Q}\,\widehat{R}$, where $\widehat{Q}$ is unitary and $\widehat{R}$ is upper triangular with only positive numbers on its diagonal. Let $q_1, \dots, q_n$ denote the columns of $\widehat{Q}$. Thinking of matrix multiplication as above, we see that each $v_k$ is a linear combination of $q_1, \dots, q_k$, with the coefficients coming from the $k^{\text{th}}$ column of $\widehat{R}$. This implies that $\text{span}(v_1, \dots, v_k) = \text{span}(q_1, \dots, q_k)$ and $\langle v_k, q_k \rangle > 0$. The uniqueness of the orthonormal lists satisfying these conditions (see Exercise 10 in Section 6B) now shows that $q_k = e_k$ for each $k = 1, \dots, n$. Hence $\widehat{Q} = Q$, which then implies that $\widehat{R} = R$, completing the proof of uniqueness. ∎

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The proof of the QR factorization shows that the columns of the unitary matrix can be computed by applying the Gram–Schmidt procedure to the columns of the matrix to be factored. The next example illustrates the computation of the QR factorization based on the proof that we just completed.

7.60 example: *QR factorization of a* 3*-by-*3 *matrix*

To find the QR factorization of the matrix

$$A = \begin{pmatrix} 1 & 2 & 1 \\ 0 & 1 & -4 \\ 0 & 3 & 2 \end{pmatrix},$$

follow the proof of 7.58. Thus set $v_1, v_2, v_3$ equal to the columns of $A$:

$$v_1 = (1, 0, 0), \quad v_2 = (2, 1, 3), \quad v_3 = (1, -4, 2).$$

Apply the Gram–Schmidt procedure to $v_1, v_2, v_3$, producing the orthonormal list

$$e_1 = (1, 0, 0), \quad e_2 = \Big(0, \tfrac{1}{\sqrt{10}}, \tfrac{3}{\sqrt{10}}\Big), \quad e_3 = \Big(0, -\tfrac{3}{\sqrt{10}}, \tfrac{1}{\sqrt{10}}\Big).$$

Still following the proof of 7.58, let $Q$ be the unitary matrix whose columns are $e_1, e_2, e_3$:

$$Q = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \frac{1}{\sqrt{10}} & -\frac{3}{\sqrt{10}} \\ 0 & \frac{3}{\sqrt{10}} & \frac{1}{\sqrt{10}} \end{pmatrix}.$$

As in the proof of 7.58, let $R$ be the 3-by-3 matrix whose entry in row $j$, column $k$ is $\langle v_k, e_j \rangle$, which gives

$$R = \begin{pmatrix} 1 & 2 & 1 \\ 0 & \sqrt{10} & \frac{\sqrt{10}}{5} \\ 0 & 0 & \frac{7\sqrt{10}}{5} \end{pmatrix}.$$

Note that $R$ is indeed an upper-triangular matrix with only positive numbers on the diagonal, as required by the QR factorization.

Now matrix multiplication can verify that $A = QR$ is the desired factorization of $A$:

$$QR = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \frac{1}{\sqrt{10}} & -\frac{3}{\sqrt{10}} \\ 0 & \frac{3}{\sqrt{10}} & \frac{1}{\sqrt{10}} \end{pmatrix} \begin{pmatrix} 1 & 2 & 1 \\ 0 & \sqrt{10} & \frac{\sqrt{10}}{5} \\ 0 & 0 & \frac{7\sqrt{10}}{5} \end{pmatrix} = \begin{pmatrix} 1 & 2 & 1 \\ 0 & 1 & -4 \\ 0 & 3 & 2 \end{pmatrix} = A.$$

Thus $A = QR$, as expected.

The QR factorization will be the major tool used in the proof of the Cholesky factorization (7.63) in the next subsection. For another nice application of the QR factorization, see the proof of Hadamard's inequality (9.66).

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If a QR factorization is available, then it can be used to solve a corresponding system of linear equations without using Gaussian elimination. Specifically, suppose $A$ is an $n$-by-$n$ square matrix with linearly independent columns. Suppose that $b \in \mathbf{F}^n$ and we want to solve the equation $Ax = b$ for $x = (x_1, \ldots, x_n) \in \mathbf{F}^n$ (as usual, we are identifying elements of $\mathbf{F}^n$ with $n$-by-1 column vectors).

Suppose $A = QR$, where $Q$ is unitary and $R$ is upper triangular with only positive numbers on its diagonal ($Q$ and $R$ are computable from $A$ using just the Gram–Schmidt procedure, as shown in the proof of 7.58). The equation $Ax = b$ is equivalent to the equation $QRx = b$. Multiplying both sides of this last equation by $Q^*$ on the left and using 7.57(d) gives the equation

$$Rx = Q^*b.$$

The matrix $Q^*$ is the conjugate transpose of the matrix $Q$. Thus computing $Q^*b$ is straightforward. Because $R$ is an upper-triangular matrix with positive numbers on its diagonal, the system of linear equations represented by the equation above can quickly be solved by first solving for $x_n$, then for $x_{n-1}$, and so on.

## *Cholesky Factorization*

We begin this subsection with a characterization of positive invertible operators in terms of inner products.

**7.61** *positive invertible operator*

A self-adjoint operator $T \in \mathcal{L}(V)$ is a positive invertible operator if and only if $\langle Tv, v\rangle > 0$ for every nonzero $v \in V$.

Proof First suppose $T$ is a positive invertible operator. If $v \in V$ and $v \neq 0$, then because $T$ is invertible we have $Tv \neq 0$. This implies that $\langle Tv, v\rangle \neq 0$ (by 7.43). Hence $\langle Tv, v\rangle > 0$.

To prove the implication in the other direction, suppose now that $\langle Tv, v\rangle > 0$ for every nonzero $v \in V$. Thus $Tv \neq 0$ for every nonzero $v \in V$. Hence $T$ is injective. Thus $T$ is invertible, as desired. ∎

The next definition transfers the result above to the language of matrices. Here we are using the usual Euclidean inner product on $\mathbf{F}^n$ and identifying elements of $\mathbf{F}^n$ with $n$-by-1 column vectors.

**7.62** definition: *positive definite*

A matrix $B \in \mathbf{F}^{n,n}$ is called *positive definite* if $B^* = B$ and

$$\langle Bx, x\rangle > 0$$

for every nonzero $x \in \mathbf{F}^n$.

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A matrix is upper triangular if and only if its conjugate transpose is lower triangular (meaning that all entries above the diagonal are 0). The factorization below, which has important consequences in computational linear algebra, writes a positive definite matrix as the product of a lower triangular matrix and its conjugate transpose.

Our next result is solely about matrices, although the proof makes use of the identification of results about operators with results about square matrices. In the result below, if the matrix $B$ is in $\mathbf{F}^{n,n}$, then the matrix $R$ is also in $\mathbf{F}^{n,n}$.

**7.63** *Cholesky factorization*

Suppose $B$ is a positive definite matrix. Then there exists a unique upper-triangular matrix $R$ with only positive numbers on its diagonal such that

$$B = R^*R.$$

Proof Because $B$ is positive definite, there exists an invertible square matrix $A$ of the same size as $B$ such that $B = A^*A$ [by the equivalence of (a) and (f) in 7.38].

Let $A = QR$ be the QR factorization of $A$ (see 7.58), where $Q$ is unitary and $R$ is upper triangular with only positive numbers on its diagonal. Then $A^* = R^*Q^*$.

Thus

$$B = A^*A = R^*Q^*QR = R^*R,$$

*André-Louis Cholesky (1875–1918) discovered this factorization, which was published posthumously in 1924.*

as desired.

To prove the uniqueness part of this result, suppose $S$ is an upper-triangular matrix with only positive numbers on its diagonal and $B = S^*S$. The matrix $S$ is invertible because $B$ is invertible (see Exercise 11 in Section 3D). Multiplying both sides of the equation $B = S^*S$ by $S^{-1}$ on the left gives the equation $BS^{-1} = S^*$.

Let $A$ be the matrix from the first paragraph of this proof. Then

$$\begin{aligned}(AS^{-1})^*(AS^{-1}) &= (S^*)^{-1}A^*AS^{-1}\\ &= (S^*)^{-1}BS^{-1}\\ &= (S^*)^{-1}S^*\\ &= I.\end{aligned}$$

Thus $AS^{-1}$ is unitary.

Hence $A = (AS^{-1})S$ is a factorization of $A$ as the product of a unitary matrix and an upper-triangular matrix with only positive numbers on its diagonal. The uniqueness of the QR factorization, as stated in 7.58, now implies that $S = R$. ∎

In the first paragraph of the proof above, we could have chosen $A$ to be the unique positive definite matrix that is a square root of $B$ (see 7.39). However, the proof was presented with the more general choice of $A$ because for specific positive definite matrices $B$, it may be easier to find a different choice of $A$.

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## *Exercises 7D*

**1** Suppose $\dim V \geq 2$ and $S \in \mathcal{L}(V, W)$. Prove that $S$ is an isometry if and only if $Se_1, Se_2$ is an orthonormal list in $W$ for every orthonormal list $e_1, e_2$ of length two in $V$.

**2** Suppose $T \in \mathcal{L}(V, W)$. Prove that $T$ is a scalar multiple of an isometry if and only if $T$ preserves orthogonality.

*The phrase "$T$ preserves orthogonality" means that $\langle Tu, Tv\rangle = 0$ for all $u, v \in V$ such that $\langle u, v\rangle = 0$.*

**3** (a) Show that the product of two unitary operators on $V$ is a unitary operator.
(b) Show that the inverse of a unitary operator on $V$ is a unitary operator.

*This exercise shows that the set of unitary operators on $V$ is a group, where the group operation is the usual product of two operators.*

**4** Suppose $\mathbf{F} = \mathbf{C}$ and $A, B \in \mathcal{L}(V)$ are self-adjoint. Show that $A + iB$ is unitary if and only if $AB = BA$ and $A^2 + B^2 = I$.

**5** Suppose $S \in \mathcal{L}(V)$. Prove that the following are equivalent.
(a) $S$ is a self-adjoint unitary operator.
(b) $S = 2P - I$ for some orthogonal projection $P$ on $V$.
(c) There exists a subspace $U$ of $V$ such that $Su = u$ for every $u \in U$ and $Sw = -w$ for every $w \in U^{\perp}$.

**6** Suppose $T_1, T_2$ are both normal operators on $\mathbf{F}^3$ with $2, 5, 7$ as eigenvalues. Prove that there exists a unitary operator $S \in \mathcal{L}(\mathbf{F}^3)$ such that $T_1 = S^* T_2 S$.

**7** Give an example of two self-adjoint operators $T_1, T_2 \in \mathcal{L}(\mathbf{F}^4)$ such that the eigenvalues of both operators are $2, 5, 7$ but there does not exist a unitary operator $S \in \mathcal{L}(\mathbf{F}^4)$ such that $T_1 = S^* T_2 S$. Be sure to explain why there is no unitary operator with the required property.

**8** Prove or give a counterexample: If $S \in \mathcal{L}(V)$ and there exists an orthonormal basis $e_1, \ldots, e_n$ of $V$ such that $\|Se_k\| = 1$ for each $e_k$, then $S$ is a unitary operator.

**9** Suppose $\mathbf{F} = \mathbf{C}$ and $T \in \mathcal{L}(V)$. Suppose every eigenvalue of $T$ has absolute value 1 and $\|Tv\| \leq \|v\|$ for every $v \in V$. Prove that $T$ is a unitary operator.

**10** Suppose $\mathbf{F} = \mathbf{C}$ and $T \in \mathcal{L}(V)$ is a self-adjoint operator such that $\|Tv\| \leq \|v\|$ for all $v \in V$.
(a) Show that $I - T^2$ is a positive operator.
(b) Show that $T + i\sqrt{I - T^2}$ is a unitary operator.

**11** Suppose $S \in \mathcal{L}(V)$. Prove that $S$ is a unitary operator if and only if

$$\{Sv : v \in V \text{ and } \|v\| \leq 1\} = \{v \in V : \|v\| \leq 1\}.$$

**12** Prove or give a counterexample: If $S \in \mathcal{L}(V)$ is invertible and $\|S^{-1}v\| = \|Sv\|$ for every $v \in V$, then $S$ is unitary.

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**13** Explain why the columns of a square matrix of complex numbers form an orthonormal list in $\mathbf{C}^n$ if and only if the rows of the matrix form an orthonormal list in $\mathbf{C}^n$.

**14** Suppose $v \in V$ with $\|v\| = 1$ and $b \in \mathbf{F}$. Also suppose $\dim V \geq 2$. Prove that there exists a unitary operator $S \in \mathcal{L}(V)$ such that $\langle Sv, v\rangle = b$ if and only if $|b| \leq 1$.

**15** Suppose $T$ is a unitary operator on $V$ such that $T - I$ is invertible.

(a) Prove that $(T + I)(T - I)^{-1}$ is a skew operator (meaning that it equals the negative of its adjoint).

(b) Prove that if $\mathbf{F} = \mathbf{C}$, then $i(T + I)(T - I)^{-1}$ is a self-adjoint operator.

*The function $z \mapsto i(z + 1)(z - 1)^{-1}$ maps the unit circle in $\mathbf{C}$ (except for the point 1) to $\mathbf{R}$. Thus (b) illustrates the analogy between the unitary operators and the unit circle in $\mathbf{C}$, along with the analogy between the self-adjoint operators and $\mathbf{R}$.*

**16** Suppose $\mathbf{F} = \mathbf{C}$ and $T \in \mathcal{L}(V)$ is self-adjoint. Prove that $(T + iI)(T - iI)^{-1}$ is a unitary operator and 1 is not an eigenvalue of this operator.

**17** Explain why the characterization of unitary matrices given by 7.57 holds.

**18** A square matrix $A$ is called *symmetric* if it equals its transpose. Prove that if $A$ is a symmetric matrix with real entries, then there exists a unitary matrix $Q$ with real entries such that $Q^* A Q$ is a diagonal matrix.

**19** Suppose $n$ is a positive integer. For this exercise, we adopt the notation that a typical element $z$ of $\mathbf{C}^n$ is denoted by $z = (z_0, z_1, \ldots, z_{n-1})$. Define linear functionals $\omega_0, \omega_1, \ldots, \omega_{n-1}$ on $\mathbf{C}^n$ by

$$\omega_j(z_0, z_1, \ldots, z_{n-1}) = \frac{1}{\sqrt{n}} \sum_{m=0}^{n-1} z_m e^{-2\pi i j m/n}.$$

The *discrete Fourier transform* is the operator $\mathcal{F} : \mathbf{C}^n \to \mathbf{C}^n$ defined by

$$\mathcal{F}z = (\omega_0(z), \omega_1(z), \ldots, \omega_{n-1}(z)).$$

(a) Show that $\mathcal{F}$ is a unitary operator on $\mathbf{C}^n$.

(b) Show that if $(z_0, \ldots, z_{n-1}) \in \mathbf{C}^n$ and $z_n$ is defined to equal $z_0$, then

$$\mathcal{F}^{-1}(z_0, z_1, \ldots, z_{n-1}) = \mathcal{F}(z_n, z_{n-1}, \ldots, z_1).$$

(c) Show that $\mathcal{F}^4 = I$.

*The discrete Fourier transform has many important applications in data analysis. The usual Fourier transform involves expressions of the form $\int_{-\infty}^{\infty} f(x) e^{-2\pi i t x} dx$ for complex-valued integrable functions $f$ defined on $\mathbf{R}$.*

**20** Suppose $A$ is a square matrix with linearly independent columns. Prove that there exist unique matrices $R$ and $Q$ such that $R$ is lower triangular with only positive numbers on its diagonal, $Q$ is unitary, and $A = RQ$.

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# 7E Singular Value Decomposition

## Singular Values

We will need the following result in this section.

**7.64** *properties of $T^*T$*

Suppose $T \in \mathcal{L}(V, W)$. Then

(a) $T^*T$ is a positive operator on $V$;

(b) $\operatorname{null} T^*T = \operatorname{null} T$;

(c) $\operatorname{range} T^*T = \operatorname{range} T^*$;

(d) $\dim \operatorname{range} T = \dim \operatorname{range} T^* = \dim \operatorname{range} T^*T$.

Proof

(a) We have

$$(T^*T)^* = T^*(T^*)^* = T^*T.$$

Thus $T^*T$ is self-adjoint.

If $v \in V$, then

$$\big\langle (T^*T)v, v \big\rangle = \langle T^*(Tv), v \rangle = \langle Tv, Tv \rangle = \|Tv\|^2 \geq 0.$$

Thus $T^*T$ is a positive operator.

(b) First suppose $v \in \operatorname{null} T^*T$. Then

$$\|Tv\|^2 = \langle Tv, Tv \rangle = \big\langle T^*Tv, v \big\rangle = \langle 0, v \rangle = 0.$$

Thus $Tv = 0$, proving that $\operatorname{null} T^*T \subseteq \operatorname{null} T$.

The inclusion in the other direction is clear, because if $v \in V$ and $Tv = 0$, then $T^*Tv = 0$.

Thus $\operatorname{null} T^*T = \operatorname{null} T$, completing the proof of (b).

(c) We already know from (a) that $T^*T$ is self-adjoint. Thus

$$\operatorname{range} T^*T = (\operatorname{null} T^*T)^\perp = (\operatorname{null} T)^\perp = \operatorname{range} T^*,$$

where the first and last equalities come from 7.6 and the second equality comes from (b).

(d) To verify the first equation in (d), note that

$$\dim \operatorname{range} T = \dim (\operatorname{null} T^*)^\perp = \dim W - \dim \operatorname{null} T^* = \dim \operatorname{range} T^*,$$

where the first equality comes from 7.6(d), the second equality comes from 6.51, and the last equality comes from the fundamental theorem of linear maps (3.21).

The equality $\dim \operatorname{range} T^* = \dim \operatorname{range} T^*T$ follows from (c).

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The eigenvalues of an operator tell us something about the behavior of the operator. Another collection of numbers, called the singular values, is also useful. Eigenspaces and the notation $E$ (used in the examples) were defined in 5.52.

**7.65 definition: *singular values***

Suppose $T \in \mathcal{L}(V, W)$. The *singular values* of $T$ are the nonnegative square roots of the eigenvalues of $T^*T$, listed in decreasing order, each included as many times as the dimension of the corresponding eigenspace of $T^*T$.

**7.66 example: *singular values of an operator on* $\mathbf{F}^4$**

Define $T \in \mathcal{L}(\mathbf{F}^4)$ by $T(z_1, z_2, z_3, z_4) = (0, 3z_1, 2z_2, -3z_4)$. A calculation shows that

$$T^*T(z_1, z_2, z_3, z_4) = (9z_1, 4z_2, 0, 9z_4),$$

as you should verify. Thus the standard basis of $\mathbf{F}^4$ diagonalizes $T^*T$, and we see that the eigenvalues of $T^*T$ are 9, 4, and 0. Also, the dimensions of the eigenspaces corresponding to the eigenvalues are

$$\dim E(9, T^*T) = 2 \quad \text{and} \quad \dim E(4, T^*T) = 1 \quad \text{and} \quad \dim E(0, T^*T) = 1.$$

Taking nonnegative square roots of these eigenvalues of $T^*T$ and using dimension information from above, we conclude that the singular values of $T$ are $3, 3, 2, 0$.

The only eigenvalues of $T$ are $-3$ and 0. Thus in this case, the collection of eigenvalues did not pick up the number 2 that appears in the definition (and hence the behavior) of $T$, but the list of singular values does include 2.

**7.67 example: *singular values of a linear map from* $\mathbf{F}^4$ *to* $\mathbf{F}^3$**

Suppose $T \in \mathcal{L}(\mathbf{F}^4, \mathbf{F}^3)$ has matrix (with respect to the standard bases)

$$\begin{pmatrix} 0 & 0 & 0 & -5 \\ 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \end{pmatrix}.$$

You can verify that the matrix of $T^*T$ is

$$\begin{pmatrix} 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 25 \end{pmatrix}$$

and that the eigenvalues of the operator $T^*T$ are 25, 2, 0, with $\dim E(25, T^*T) = 1$, $\dim E(2, T^*T) = 1$, and $\dim E(0, T^*T) = 2$. Thus the singular values of $T$ are $5, \sqrt{2}, 0, 0$.

See Exercise 2 for a characterization of the positive singular values.

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**7.68** *role of positive singular values*

Suppose that $T \in \mathcal{L}(V, W)$. Then

(a) $T$ is injective $\iff$ 0 is not a singular value of $T$;

(b) the number of positive singular values of $T$ equals dim range $T$;

(c) $T$ is surjective $\iff$ number of positive singular values of $T$ equals dim $W$.

Proof The linear map $T$ is injective if and only if null $T = \{0\}$, which happens if and only if null $T^*T = \{0\}$ [by 7.64(b)], which happens if and only if 0 is not an eigenvalue of $T^*T$, which happens if and only if 0 is not a singular value of $T$, completing the proof of (a).

The spectral theorem applied to $T^*T$ shows that dim range $T^*T$ equals the number of positive eigenvalues of $T^*T$ (counting repetitions). Thus 7.64(c) implies that dim range $T$ equals the number of positive singular values of $T$, proving (b).

Use (b) and 2.39 to show that (c) holds. ∎

The table below compares eigenvalues with singular values.

| **list of eigenvalues** | **list of singular values** |
|---|---|
| context: vector spaces | context: inner product spaces |
| defined only for linear maps from a vector space to itself | defined for linear maps from an inner product space to a possibly different inner product space |
| can be arbitrary real numbers (if $\mathbf{F} = \mathbf{R}$) or complex numbers (if $\mathbf{F} = \mathbf{C}$) | are nonnegative numbers |
| can be the empty list if $\mathbf{F} = \mathbf{R}$ | length of list equals dimension of domain |
| includes 0 $\iff$ operator is not invertible | includes 0 $\iff$ linear map is not injective |
| no standard order, especially if $\mathbf{F} = \mathbf{C}$ | always listed in decreasing order |

The next result nicely characterizes isometries in terms of singular values.

**7.69** *isometries characterized by having all singular values equal* 1

Suppose that $S \in \mathcal{L}(V, W)$. Then

$$S \text{ is an isometry} \iff \text{all singular values of } S \text{ equal } 1.$$

Proof We have

$$\begin{aligned} S \text{ is an isometry} &\iff S^*S = I \\ &\iff \text{all eigenvalues of } S^*S \text{ equal } 1 \\ &\iff \text{all singular values of } S \text{ equal } 1, \end{aligned}$$

where the first equivalence comes from 7.49 and the second equivalence comes from the spectral theorem (7.29 or 7.31) applied to the self-adjoint operator $S^*S$. ∎

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## *SVD for Linear Maps and for Matrices*

The next result shows that every linear map from $V$ to $W$ has a remarkably clean description in terms of its singular values and orthonormal lists in $V$ and $W$. In the next section we will see several important applications of the singular value decomposition (often called the SVD).

*The singular value decomposition is useful in computational linear algebra because good techniques exist for approximating eigenvalues and eigenvectors of positive operators such as $T^*T$, whose eigenvalues and eigenvectors lead to the singular value decomposition.*

**7.70 *singular value decomposition***

Suppose $T \in \mathcal{L}(V, W)$ and the positive singular values of $T$ are $s_1, \dots, s_m$. Then there exist orthonormal lists $e_1, \dots, e_m$ in $V$ and $f_1, \dots, f_m$ in $W$ such that

$$Tv = s_1\langle v, e_1\rangle f_1 + \cdots + s_m\langle v, e_m\rangle f_m \tag{7.71}$$

for every $v \in V$.

Proof Let $s_1, \dots, s_n$ denote the singular values of $T$ (thus $n = \dim V$). Because $T^*T$ is a positive operator [see 7.64(a)], the spectral theorem implies that there exists an orthonormal basis $e_1, \dots, e_n$ of $V$ with

$$T^*Te_k = s_k^2 e_k \tag{7.72}$$

for each $k = 1, \dots, n$.

For each $k = 1, \dots, m$, let

$$f_k = \frac{Te_k}{s_k}. \tag{7.73}$$

If $j, k \in \{1, \dots, m\}$, then

$$\langle f_j, f_k\rangle = \frac{1}{s_j s_k}\langle Te_j, Te_k\rangle = \frac{1}{s_j s_k}\langle e_j, T^*Te_k\rangle = \frac{s_k}{s_j}\langle e_j, e_k\rangle = \begin{cases} 0 & \text{if } j \neq k, \\ 1 & \text{if } j = k. \end{cases}$$

Thus $f_1, \dots, f_m$ is an orthonormal list in $W$.

If $k \in \{1, \dots, n\}$ and $k > m$, then $s_k = 0$ and hence $T^*Te_k = 0$ (by 7.72), which implies that $Te_k = 0$ [by 7.64(b)].

Suppose $v \in V$. Then

$$\begin{aligned} Tv &= T(\langle v, e_1\rangle e_1 + \cdots + \langle v, e_n\rangle e_n) \\ &= \langle v, e_1\rangle Te_1 + \cdots + \langle v, e_m\rangle Te_m \\ &= s_1\langle v, e_1\rangle f_1 + \cdots + s_m\langle v, e_m\rangle f_m, \end{aligned}$$

where the last index in the first line switched from $n$ to $m$ in the second line because $Te_k = 0$ if $k > m$ (as noted in the paragraph above) and the third line follows from 7.73. The equation above is our desired result. ∎

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Suppose $T \in \mathcal{L}(V, W)$, the positive singular values of $T$ are $s_1, \dots, s_m$, and $e_1, \dots, e_m$ and $f_1, \dots, f_m$ are as in the singular value decomposition 7.70. The orthonormal list $e_1, \dots, e_m$ can be extended to an orthonormal basis $e_1, \dots, e_{\dim V}$ of $V$ and the orthonormal list $f_1, \dots, f_m$ can be extended to an orthonormal basis $f_1, \dots, f_{\dim W}$ of $W$. The formula 7.71 shows that

$$Te_k = \begin{cases} s_k f_k & \text{if } 1 \le k \le m, \\ 0 & \text{if } m < k \le \dim V. \end{cases}$$

Thus the matrix of $T$ with respect to the orthonormal bases $(e_1, \dots, e_{\dim V})$ and $(f_1, \dots, f_{\dim W})$ has the simple form

$$\mathcal{M}(T, (e_1, \dots, e_{\dim V}), (f_1, \dots, f_{\dim W}))_{j,k} = \begin{cases} s_k & \text{if } 1 \le j = k \le m, \\ 0 & \text{otherwise.} \end{cases}$$

If $\dim V = \dim W$ (as happens, for example, if $W = V$), then the matrix described in the paragraph above is a diagonal matrix. If we extend the definition of diagonal matrix as follows to apply to matrices that are not necessarily square, then we have proved the wonderful result that every linear map from $V$ to $W$ has a diagonal matrix with respect to appropriate orthonormal bases.

**7.74 definition: *diagonal matrix***

An $M$-by-$N$ matrix $A$ is called a *diagonal matrix* if all entries of the matrix are 0 except possibly $A_{k,k}$ for $k = 1, \dots, \min\{M, N\}$.

The table below compares the spectral theorem (7.29 and 7.31) with the singular value decomposition (7.70).

| spectral theorem | singular value decomposition |
|---|---|
| describes only self-adjoint operators (when $\mathbf{F} = \mathbf{R}$) or normal operators (when $\mathbf{F} = \mathbf{C}$) | describes arbitrary linear maps from an inner product space to a possibly different inner product space |
| produces a single orthonormal basis | produces two orthonormal lists, one for domain space and one for range space, that are not necessarily the same even when range space equals domain space |
| different proofs depending on whether $\mathbf{F} = \mathbf{R}$ or $\mathbf{F} = \mathbf{C}$ | same proof works regardless of whether $\mathbf{F} = \mathbf{R}$ or $\mathbf{F} = \mathbf{C}$ |

The singular value decomposition gives us a new way to understand the adjoint and the inverse of a linear map. Specifically, the next result shows that given a singular value decomposition of a linear map $T \in \mathcal{L}(V, W)$, we can obtain the adjoint of $T$ simply by interchanging the roles of the $e$'s and the $f$'s (see 7.77). Similarly, we can obtain the pseudoinverse $T^\dagger$ (see 6.68) of $T$ by interchanging the roles of the $e$'s and the $f$'s and replacing each positive singular value $s_k$ of $T$ with $1/s_k$ (see 7.78).

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Recall that the pseudoinverse $T^\dagger$ in 7.78 below equals the inverse $T^{-1}$ if $T$ is invertible [see 6.69(a)].

**7.75** *singular value decomposition of adjoint and pseudoinverse*

Suppose $T \in \mathcal{L}(V, W)$ and the positive singular values of $T$ are $s_1, \dots, s_m$. Suppose $e_1, \dots, e_m$ and $f_1, \dots, f_m$ are orthonormal lists in $V$ and $W$ such that

$$Tv = s_1\langle v, e_1\rangle f_1 + \cdots + s_m\langle v, e_m\rangle f_m \tag{7.76}$$

for every $v \in V$. Then

$$T^*w = s_1\langle w, f_1\rangle e_1 + \cdots + s_m\langle w, f_m\rangle e_m \tag{7.77}$$

and

$$T^\dagger w = \frac{\langle w, f_1\rangle}{s_1}e_1 + \cdots + \frac{\langle w, f_m\rangle}{s_m}e_m \tag{7.78}$$

for every $w \in W$.

Proof If $v \in V$ and $w \in W$ then

$$\begin{aligned}\langle Tv, w\rangle &= \langle s_1\langle v, e_1\rangle f_1 + \cdots + s_m\langle v, e_m\rangle f_m, w\rangle \\ &= s_1\langle v, e_1\rangle\langle f_1, w\rangle + \cdots + s_m\langle v, e_m\rangle\langle f_m, w\rangle \\ &= \langle v, s_1\langle w, f_1\rangle e_1 + \cdots + s_m\langle w, f_m\rangle e_m\rangle.\end{aligned}$$

This implies that

$$T^*w = s_1\langle w, f_1\rangle e_1 + \cdots + s_m\langle w, f_m\rangle e_m,$$

proving 7.77.

To prove 7.78, suppose $w \in W$. Let

$$v = \frac{\langle w, f_1\rangle}{s_1}e_1 + \cdots + \frac{\langle w, f_m\rangle}{s_m}e_m.$$

Apply $T$ to both sides of the equation above, getting

$$\begin{aligned}Tv &= \frac{\langle w, f_1\rangle}{s_1}Te_1 + \cdots + \frac{\langle w, f_m\rangle}{s_m}Te_m \\ &= \langle w, f_1\rangle f_1 + \cdots + \langle w, f_m\rangle f_m \\ &= P_{\text{range } T}\, w,\end{aligned}$$

where the second line holds because 7.76 implies that $Te_k = s_k f_k$ if $k = 1, \dots, m$, and the last line above holds because 7.76 implies that $f_1, \dots, f_m$ spans range $T$ and thus is an orthonormal basis of range $T$ [and hence 6.57(i) applies]. The equation above, the observation that $v \in (\text{null } T)^\perp$ [see Exercise 8(b)], and the definition of $T^\dagger w$ (see 6.68) show that $v = T^\dagger w$, proving 7.78. ∎

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7.79 example: *finding a singular value decomposition*

Define $T \in \mathcal{L}(\mathbf{F}^4, \mathbf{F}^3)$ by $T(x_1, x_2, x_3, x_4) = (-5x_4, 0, x_1 + x_2)$. We want to find a singular value decomposition of $T$. The matrix of $T$ (with respect to the standard bases) is

$$\begin{pmatrix} 0 & 0 & 0 & -5 \\ 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \end{pmatrix}.$$

Thus, as discussed in Example 7.67, the matrix of $T^*T$ is

$$\begin{pmatrix} 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 25 \end{pmatrix},$$

and the positive eigenvalues of $T^*T$ are 25, 2, with $\dim E(25, T^*T) = 1$ and $\dim E(2, T^*T) = 1$. Hence the positive singular values of $T$ are 5, $\sqrt{2}$.

Thus to find a singular value decomposition of $T$, we must find an orthonormal list $e_1, e_2$ in $\mathbf{F}^4$ and an orthonormal list $f_1, f_2$ in $\mathbf{F}^3$ such that

$$Tv = 5\langle v, e_1\rangle f_1 + \sqrt{2}\langle v, e_2\rangle f_2$$

for all $v \in \mathbf{F}^4$.

An orthonormal basis of $E(25, T^*T)$ is the vector $(0, 0, 0, 1)$; an orthonormal basis of $E(2, T^*T)$ is the vector $\left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, 0, 0\right)$. Thus, following the proof of 7.70, we take

$$e_1 = (0, 0, 0, 1) \quad \text{and} \quad e_2 = \left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, 0, 0\right)$$

and

$$f_1 = \frac{Te_1}{5} = (-1, 0, 0) \quad \text{and} \quad f_2 = \frac{Te_2}{\sqrt{2}} = (0, 0, 1).$$

Then, as expected, we see that $e_1, e_2$ is an orthonormal list in $\mathbf{F}^4$ and $f_1, f_2$ is an orthonormal list in $\mathbf{F}^3$ and

$$Tv = 5\langle v, e_1\rangle f_1 + \sqrt{2}\langle v, e_2\rangle f_2$$

for all $v \in \mathbf{F}^4$. Thus we have found a singular value decomposition of $T$.

The next result translates the singular value decomposition from the context of linear maps to the context of matrices. Specifically, the following result gives a factorization of an arbitrary matrix as the product of three nice matrices. The proof gives an explicit construction of these three matrices in terms of the singular value decomposition.

In the next result, the phrase "orthogonal columns" should be interpreted to mean that the columns are orthogonal with respect to the standard Euclidean inner product.

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**7.80** *matrix version of SVD*

Suppose $A$ is an $M$-by-$n$ matrix of rank $m \geq 1$. Then there exist an $M$-by-$m$ matrix $B$ with orthonormal columns, an $m$-by-$m$ diagonal matrix $D$ with positive numbers on the diagonal, and an $n$-by-$m$ matrix $C$ with orthonormal columns such that

$$A = BDC^*.$$

Proof Let $T: \mathbf{F}^n \to \mathbf{F}^M$ be the linear map whose matrix with respect to the standard bases equals $A$. Then $\dim \operatorname{range} T = m$ (by 3.78). Let

7.81 $$Tv = s_1\langle v, e_1\rangle f_1 + \cdots + s_m\langle v, e_m\rangle f_m$$

be a singular value decomposition of $T$. Let

$$\begin{aligned} B &= \text{the } M\text{-by-}m \text{ matrix whose columns are } f_1, \ldots, f_m, \\ D &= \text{the } m\text{-by-}m \text{ diagonal matrix whose diagonal entries are } s_1, \ldots, s_m, \\ C &= \text{the } n\text{-by-}m \text{ matrix whose columns are } e_1, \ldots, e_m. \end{aligned}$$

Let $u_1, \ldots, u_m$ denote the standard basis of $\mathbf{F}^m$. If $k \in \{1, \ldots, m\}$ then

$$(AC - BD)u_k = Ae_k - B(s_k u_k) = s_k f_k - s_k f_k = 0.$$

Thus $AC = BD$.

Multiply both sides of this last equation by $C^*$ (the conjugate transpose of $C$) on the right to get

$$ACC^* = BDC^*.$$

Note that the rows of $C^*$ are the complex conjugates of $e_1, \ldots, e_m$. Thus if $k \in \{1, \ldots, m\}$, then the definition of matrix multiplication shows that $C^* e_k = u_k$; hence $CC^* e_k = e_k$. Thus $ACC^* v = Av$ for all $v \in \operatorname{span}(e_1, \ldots, e_m)$.

If $v \in (\operatorname{span}(e_1, \ldots, e_m))^\perp$, then $Av = 0$ (as follows from 7.81) and $C^* v = 0$ (as follows from the definition of matrix multiplication). Hence $ACC^* v = Av$ for all $v \in (\operatorname{span}(e_1, \ldots, e_m))^\perp$.

Because $ACC^*$ and $A$ agree on $\operatorname{span}(e_1, \ldots, e_m)$ and on $(\operatorname{span}(e_1, \ldots, e_m))^\perp$, we conclude that $ACC^* = A$. Thus the displayed equation above becomes

$$A = BDC^*,$$

as desired. ∎

Note that the matrix $A$ in the result above has $Mn$ entries. In comparison, the matrices $B$, $D$, and $C$ above have a total of

$$m(M + m + n)$$

entries. Thus if $M$ and $n$ are large numbers and the rank $m$ is considerably less than $M$ and $n$, then the number of entries that must be stored on a computer to represent $A$ is considerably less than $Mn$.

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## *Exercises 7E*

**1** Suppose $T \in \mathcal{L}(V, W)$. Show that $T = 0$ if and only if all singular values of $T$ are 0.

**2** Suppose $T \in \mathcal{L}(V, W)$ and $s > 0$. Prove that $s$ is a singular value of $T$ if and only if there exist nonzero vectors $v \in V$ and $w \in W$ such that

$$Tv = sw \quad \text{and} \quad T^*w = sv.$$

*The vectors $v, w$ satisfying both equations above are called a **Schmidt pair**. Erhard Schmidt introduced the concept of singular values in 1907.*

**3** Give an example of $T \in \mathcal{L}(\mathbf{C}^2)$ such that 0 is the only eigenvalue of $T$ and the singular values of $T$ are $5, 0$.

**4** Suppose that $T \in \mathcal{L}(V, W)$, $s_1$ is the largest singular value of $T$, and $s_n$ is the smallest singular value of $T$. Prove that

$$\{\|Tv\| : v \in V \text{ and } \|v\| = 1\} = [s_n, s_1].$$

**5** Suppose $T \in \mathcal{L}(\mathbf{C}^2)$ is defined by $T(x, y) = (-4y, x)$. Find the singular values of $T$.

**6** Find the singular values of the differentiation operator $D \in \mathcal{L}(\mathcal{P}_2(\mathbf{R}))$ defined by $Dp = p'$, where the inner product on $\mathcal{P}_2(\mathbf{R})$ is as in Example 6.34.

**7** Suppose that $T \in \mathcal{L}(V)$ is self-adjoint or that $\mathbf{F} = \mathbf{C}$ and $T \in \mathcal{L}(V)$ is normal. Let $\lambda_1, \ldots, \lambda_n$ be the eigenvalues of $T$, each included in this list as many times as the dimension of the corresponding eigenspace. Show that the singular values of $T$ are $|\lambda_1|, \ldots, |\lambda_n|$, after these numbers have been sorted into decreasing order.

**8** Suppose $T \in \mathcal{L}(V, W)$. Suppose $s_1 \geq s_2 \geq \cdots \geq s_m > 0$ and $e_1, \ldots, e_m$ is an orthonormal list in $V$ and $f_1, \ldots, f_m$ is an orthonormal list in $W$ such that

$$Tv = s_1\langle v, e_1\rangle f_1 + \cdots + s_m\langle v, e_m\rangle f_m$$

for every $v \in V$.

(a) Prove that $f_1, \ldots, f_m$ is an orthonormal basis of range $T$.
(b) Prove that $e_1, \ldots, e_m$ is an orthonormal basis of $(\operatorname{null} T)^\perp$.
(c) Prove that $s_1, \ldots, s_m$ are the positive singular values of $T$.
(d) Prove that if $k \in \{1, \ldots, m\}$, then $e_k$ is an eigenvector of $T^*T$ with corresponding eigenvalue $s_k^2$.
(e) Prove that

$$TT^*w = s_1^2\langle w, f_1\rangle f_1 + \cdots + s_m^2\langle w, f_m\rangle f_m$$

for all $w \in W$.

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**9** Suppose $T \in \mathcal{L}(V, W)$. Show that $T$ and $T^*$ have the same positive singular values.

**10** Suppose $T \in \mathcal{L}(V, W)$ has singular values $s_1, \ldots, s_n$. Prove that if $T$ is an invertible linear map, then $T^{-1}$ has singular values

$$\frac{1}{s_n}, \ldots, \frac{1}{s_1}.$$

**11** Suppose that $T \in \mathcal{L}(V, W)$ and $v_1, \ldots, v_n$ is an orthonormal basis of $V$. Let $s_1, \ldots, s_n$ denote the singular values of $T$.

(a) Prove that $\|Tv_1\|^2 + \cdots + \|Tv_n\|^2 = s_1^2 + \cdots + s_n^2$.

(b) Prove that if $W = V$ and $T$ is a positive operator, then

$$\langle Tv_1, v_1 \rangle + \cdots + \langle Tv_n, v_n \rangle = s_1 + \cdots + s_n.$$

*See the comment after Exercise 5 in Section 7A.*

**12** (a) Give an example of a finite-dimensional vector space and an operator $T$ on it such that the singular values of $T^2$ do not equal the squares of the singular values of $T$.

(b) Suppose $T \in \mathcal{L}(V)$ is normal. Prove that the singular values of $T^2$ equal the squares of the singular values of $T$.

**13** Suppose $T_1, T_2 \in \mathcal{L}(V)$. Prove that $T_1$ and $T_2$ have the same singular values if and only if there exist unitary operators $S_1, S_2 \in \mathcal{L}(V)$ such that $T_1 = S_1 T_2 S_2$.

**14** Suppose $T \in \mathcal{L}(V, W)$. Let $s_n$ denote the smallest singular value of $T$. Prove that $s_n \|v\| \le \|Tv\|$ for every $v \in V$.

**15** Suppose $T \in \mathcal{L}(V)$ and $s_1 \ge \cdots \ge s_n$ are the singular values of $T$. Prove that if $\lambda$ is an eigenvalue of $T$, then $s_1 \ge |\lambda| \ge s_n$.

**16** Suppose $T \in \mathcal{L}(V, W)$. Prove that $(T^*)^\dagger = (T^\dagger)^*$.

*Compare the result in this exercise to the analogous result for invertible linear maps* [*see 7.5*$(f)$].

**17** Suppose $T \in \mathcal{L}(V)$. Prove that $T$ is self-adjoint if and only if $T^\dagger$ is self-adjoint.

Matrices unfold
Singular values gleam like stars
Order in chaos shines

—written by ChatGPT with input *haiku about SVD*

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## 7F Consequences of Singular Value Decomposition

### Norms of Linear Maps

The singular value decomposition leads to the following upper bound for $\|Tv\|$.

**7.82 *upper bound for $\|Tv\|$***

Suppose $T \in \mathcal{L}(V, W)$. Let $s_1$ be the largest singular value of $T$. Then

$$\|Tv\| \le s_1\|v\|$$

for all $v \in V$.

*For a lower bound on $\|Tv\|$, look at Exercise 14 in Section 7E.*

Proof Let $s_1, \dots, s_m$ denote the positive singular values of $T$, and let $e_1, \dots, e_m$ be an orthonormal list in $V$ and $f_1, \dots, f_m$ be an orthonormal list in $W$ that provide a singular value decomposition of $T$. Thus

$$Tv = s_1\langle v, e_1\rangle f_1 + \cdots + s_m\langle v, e_m\rangle f_m \tag{7.83}$$

for all $v \in V$. Hence if $v \in V$ then

$$\begin{aligned}\|Tv\|^2 &= s_1^2\,|\langle v, e_1\rangle|^2 + \cdots + s_m^2\,|\langle v, e_m\rangle|^2\\ &\le s_1^2\big(|\langle v, e_1\rangle|^2 + \cdots + |\langle v, e_m\rangle|^2\big)\\ &\le s_1^2\,\|v\|^2,\end{aligned}$$

where the last inequality follows from Bessel's inequality (6.26). Taking square roots of both sides of the inequality above shows that $\|Tv\| \le s_1\|v\|$, as desired. ∎

Suppose $T \in \mathcal{L}(V, W)$ and $s_1$ is the largest singular value of $T$. The result above shows that

$$\|Tv\| \le s_1 \text{ for all } v \in V \text{ with } \|v\| \le 1. \tag{7.84}$$

Taking $v = e_1$ in 7.83 shows that $Te_1 = s_1 f_1$. Because $\|f_1\| = 1$, this implies that $\|Te_1\| = s_1$. Thus because $\|e_1\| = 1$, the inequality in 7.84 leads to the equation

$$\max\{\|Tv\| : v \in V \text{ and } \|v\| \le 1\} = s_1. \tag{7.85}$$

The equation above is the motivation for the following definition, which defines the norm of $T$ to be the left side of the equation above without needing to refer to singular values or the singular value decomposition.

**7.86 definition: *norm of a linear map, $\|\cdot\|$***

Suppose $T \in \mathcal{L}(V, W)$. Then the *norm* of $T$, denoted by $\|T\|$, is defined by

$$\|T\| = \max\{\|Tv\| : v \in V \text{ and } \|v\| \le 1\}.$$

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In general, the maximum of an infinite set of nonnegative numbers need not exist. However, the discussion before 7.86 shows that the maximum in the definition of the norm of a linear map $T$ from $V$ to $W$ does indeed exist (and equals the largest singular value of $T$).

We now have two different uses of the word *norm* and the notation $\|\cdot\|$. Our first use of this notation was in connection with an inner product on $V$, when we defined $\|v\| = \sqrt{\langle v, v\rangle}$ for each $v \in V$. Our second use of the norm notation and terminology is with the definition we just made of $\|T\|$ for $T \in \mathcal{L}(V, W)$. The norm $\|T\|$ for $T \in \mathcal{L}(V, W)$ does not usually come from taking an inner product of $T$ with itself (see Exercise 21). You should be able to tell from the context and from the symbols used which meaning of the norm is intended.

The properties of the norm on $\mathcal{L}(V, W)$ listed below look identical to properties of the norm on an inner product space (see 6.9 and 6.17). The inequality in (d) is called the *triangle inequality*, thus using the same terminology that we used for the norm on $V$. For the reverse triangle inequality, see Exercise 1.

**7.87** *basic properties of norms of linear maps*

Suppose $T \in \mathcal{L}(V, W)$. Then

(a) $\|T\| \geq 0$;

(b) $\|T\| = 0 \iff T = 0$;

(c) $\|\lambda T\| = |\lambda| \, \|T\|$ for all $\lambda \in \mathbf{F}$;

(d) $\|S + T\| \leq \|S\| + \|T\|$ for all $S \in \mathcal{L}(V, W)$.

Proof

(a) Because $\|Tv\| \geq 0$ for every $v \in V$, the definition of $\|T\|$ implies that $\|T\| \geq 0$.

(b) Suppose $\|T\| = 0$. Thus $Tv = 0$ for all $v \in V$ with $\|v\| \leq 1$. If $u \in V$ with $u \neq 0$, then

$$Tu = \|u\| \, T\left(\frac{u}{\|u\|}\right) = 0,$$

where the last equality holds because $u/\|u\|$ has norm 1. Because $Tu = 0$ for all $u \in V$, we have $T = 0$.

Conversely, if $T = 0$ then $Tv = 0$ for all $v \in V$ and hence $\|T\| = 0$.

(c) Suppose $\lambda \in \mathbf{F}$. Then

$$\begin{aligned}\|\lambda T\| &= \max\{\|\lambda Tv\| : v \in V \text{ and } \|v\| \leq 1\} \\ &= |\lambda| \max\{\|Tv\| : v \in V \text{ and } \|v\| \leq 1\} \\ &= |\lambda| \, \|T\|.\end{aligned}$$

(d) Suppose $S \in \mathcal{L}(V, W)$. The definition of $\|S + T\|$ implies that there exists $v \in V$ such that $\|v\| \leq 1$ and $\|S + T\| = \|(S + T)v\|$. Now

$$\|S + T\| = \|(S + T)v\| = \|Sv + Tv\| \leq \|Sv\| + \|Tv\| \leq \|S\| + \|T\|,$$

completing the proof of (d).

For $S, T \in \mathcal{L}(V, W)$, the quantity $\|S - T\|$ is often called the distance between $S$ and $T$. Informally, think of the condition that $\|S - T\|$ is a small number as meaning that $S$ and $T$ are close together. For example, Exercise 9 asserts that for every $T \in \mathcal{L}(V)$, there is an invertible operator as close to $T$ as we wish.

**7.88** *alternative formulas for $\|T\|$*

Suppose $T \in \mathcal{L}(V, W)$. Then

(a) $\|T\|$ = the largest singular value of $T$;

(b) $\|T\| = \max\{\|Tv\| : v \in V \text{ and } \|v\| = 1\}$;

(c) $\|T\|$ = the smallest number $c$ such that $\|Tv\| \le c\|v\|$ for all $v \in V$.

Proof

(a) See 7.85.

(b) Let $v \in V$ be such that $0 < \|v\| \le 1$. Let $u = v/\|v\|$. Then

$$\|u\| = \left\|\frac{v}{\|v\|}\right\| = 1 \quad \text{and} \quad \|Tu\| = \left\|T\Big(\frac{v}{\|v\|}\Big)\right\| = \frac{\|Tv\|}{\|v\|} \ge \|Tv\|.$$

Thus when finding the maximum of $\|Tv\|$ with $\|v\| \le 1$, we can restrict attention to vectors in $V$ with norm 1, proving (b).

(c) Suppose $v \in V$ and $v \ne 0$. Then the definition of $\|T\|$ implies that

$$\left\|T\Big(\frac{v}{\|v\|}\Big)\right\| \le \|T\|,$$

which implies that

7.89 $$\|Tv\| \le \|T\|\,\|v\|.$$

Now suppose $c \ge 0$ and $\|Tv\| \le c\|v\|$ for all $v \in V$. This implies that

$$\|Tv\| \le c$$

for all $v \in V$ with $\|v\| \le 1$. Taking the maximum of the left side of the inequality above over all $v \in V$ with $\|v\| \le 1$ shows that $\|T\| \le c$. Thus $\|T\|$ is the smallest number $c$ such that $\|Tv\| \le c\|v\|$ for all $v \in V$. ∎

When working with norms of linear maps, you will probably frequently use the inequality 7.89.

For computing an approximation of the norm of a linear map $T$ given the matrix of $T$ with respect to some orthonormal bases, 7.88(a) is likely to be most useful. The matrix of $T^*T$ is quickly computable from matrix multiplication. Then a computer can be asked to find an approximation for the largest eigenvalue of $T^*T$ (excellent numeric algorithms exist for this purpose). Then taking the square root and using 7.88(a) gives an approximation for the norm of $T$ (which usually cannot be computed exactly).

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You should verify all assertions in the example below.

7.90 example: *norms*

- If $I$ denotes the usual identity operator on $V$, then $\|I\| = 1$.
- If $T \in \mathcal{L}(\mathbf{F}^n)$ and the matrix of $T$ with respect to the standard basis of $\mathbf{F}^n$ consists of all 1's, then $\|T\| = n$.
- If $T \in \mathcal{L}(V)$ and $V$ has an orthonormal basis consisting of eigenvectors of $T$ with corresponding eigenvalues $\lambda_1, \ldots, \lambda_n$, then $\|T\|$ is the maximum of the numbers $|\lambda_1|, \ldots, |\lambda_n|$.
- Suppose $T \in \mathcal{L}(\mathbf{R}^5)$ is the operator whose matrix (with respect to the standard basis) is the 5-by-5 matrix whose entry in row $j$, column $k$ is $1/(j^2 + k)$. Standard mathematical software shows that the largest singular value of $T$ is approximately 0.8 and the smallest singular value of $T$ is approximately $10^{-6}$. Thus $\|T\| \approx 0.8$ and (using Exercise 10 in Section 7E) $\|T^{-1}\| \approx 10^6$. It is not possible to find exact formulas for these norms.

A linear map and its adjoint have the same norm, as shown by the next result.

7.91 *norm of the adjoint*

Suppose $T \in \mathcal{L}(V, W)$. Then $\|T^*\| = \|T\|$.

Proof Suppose $w \in W$. Then

$$\|T^*w\|^2 = \langle T^*w, T^*w\rangle = \langle TT^*w, w\rangle \le \|TT^*w\|\,\|w\| \le \|T\|\,\|T^*w\|\,\|w\|.$$

The inequality above implies that

$$\|T^*w\| \le \|T\|\,\|w\|,$$

which along with 7.88(c) implies that $\|T^*\| \le \|T\|$.

Replacing $T$ with $T^*$ in the inequality $\|T^*\| \le \|T\|$ and then using the equation $(T^*)^* = T$ shows that $\|T\| \le \|T^*\|$. Thus $\|T^*\| = \|T\|$, as desired. ∎

You may want to construct an alternative proof of the result above using Exercise 9 in Section 7E, which asserts that a linear map and its adjoint have the same positive singular values.

## *Approximation by Linear Maps with Lower-Dimensional Range*

The next result is a spectacular application of the singular value decomposition. It says that to best approximate a linear map by a linear map whose range has dimension at most $k$, chop off the singular value decomposition after the first $k$ terms. Specifically, the linear map $T_k$ in the next result has the property that $\dim \text{range}\, T_k = k$ and $T_k$ minimizes the distance to $T$ among all linear maps with range of dimension at most $k$. This result leads to algorithms for compressing huge matrices while preserving their most important information.

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**7.92** *best approximation by linear map whose range has dimension $\le k$*

Suppose $T \in \mathcal{L}(V, W)$ and $s_1 \ge \cdots \ge s_m$ are the positive singular values of $T$. Suppose $1 \le k < m$. Then

$$\min\{\|T - S\| : S \in \mathcal{L}(V, W) \text{ and } \dim \operatorname{range} S \le k\} = s_{k+1}.$$

Furthermore, if

$$Tv = s_1\langle v, e_1\rangle f_1 + \cdots + s_m\langle v, e_m\rangle f_m$$

is a singular value decomposition of $T$ and $T_k \in \mathcal{L}(V, W)$ is defined by

$$T_k v = s_1\langle v, e_1\rangle f_1 + \cdots + s_k\langle v, e_k\rangle f_k$$

for each $v \in V$, then $\dim \operatorname{range} T_k = k$ and $\|T - T_k\| = s_{k+1}$.

Proof If $v \in V$ then

$$\begin{aligned}
\|(T - T_k)v\|^2 &= \|s_{k+1}\langle v, e_{k+1}\rangle f_{k+1} + \cdots + s_m\langle v, e_m\rangle f_m\|^2 \\
&= s_{k+1}^2 |\langle v, e_{k+1}\rangle|^2 + \cdots + s_m^2 |\langle v, e_m\rangle|^2 \\
&\le s_{k+1}^2\left(|\langle v, e_{k+1}\rangle|^2 + \cdots + |\langle v, e_m\rangle|^2\right) \\
&\le s_{k+1}^2 \|v\|^2.
\end{aligned}$$

Thus $\|T - T_k\| \le s_{k+1}$. The equation $(T - T_k)e_{k+1} = s_{k+1}f_{k+1}$ now shows that $\|T - T_k\| = s_{k+1}$.

Suppose $S \in \mathcal{L}(V, W)$ and $\dim \operatorname{range} S \le k$. Thus $Se_1, \ldots, Se_{k+1}$, which is a list of length $k + 1$, is linearly dependent. Hence there exist $a_1, \ldots, a_{k+1} \in \mathbf{F}$, not all 0, such that

$$a_1 Se_1 + \cdots + a_{k+1}Se_{k+1} = 0.$$

Now $a_1e_1 + \cdots + a_{k+1}e_{k+1} \ne 0$ because $a_1, \ldots, a_{k+1}$ are not all 0. We have

$$\begin{aligned}
\|(T - S)(a_1e_1 + \cdots + a_{k+1}e_{k+1})\|^2 &= \|T(a_1e_1 + \cdots + a_{k+1}e_{k+1})\|^2 \\
&= \|s_1a_1f_1 + \cdots + s_{k+1}a_{k+1}f_{k+1}\|^2 \\
&= s_1^2|a_1|^2 + \cdots + s_{k+1}^2|a_{k+1}|^2 \\
&\ge s_{k+1}^2(|a_1|^2 + \cdots + |a_{k+1}|^2) \\
&= s_{k+1}^2\|a_1e_1 + \cdots + a_{k+1}e_{k+1}\|^2.
\end{aligned}$$

Because $a_1e_1 + \cdots + a_{k+1}e_{k+1} \ne 0$, the inequality above implies that

$$\|T - S\| \ge s_{k+1}.$$

Thus $S = T_k$ minimizes $\|T - S\|$ among $S \in \mathcal{L}(V, W)$ with $\dim \operatorname{range} S \le k$. ∎

For other examples of the use of the singular value decomposition in best approximation, see Exercise 22, which finds a subspace of given dimension on which the restriction of a linear map is as small as possible, and Exercise 27, which finds a unitary operator that is as close as possible to a given operator.

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## *Polar Decomposition*

Recall our discussion before 7.54 of the analogy between complex numbers $z$ with $|z| = 1$ and unitary operators. Continuing with this analogy, note that every complex number $z$ except 0 can be written in the form

$$z = \left(\frac{z}{|z|}\right)|z|$$

$$= \left(\frac{z}{|z|}\right)\sqrt{\bar{z}z},$$

where the first factor, namely, $z/|z|$, has absolute value 1.

Our analogy leads us to guess that every operator $T \in \mathcal{L}(V)$ can be written as a unitary operator times $\sqrt{T^*T}$. That guess is indeed correct. The corresponding result is called the polar decomposition, which gives a beautiful description of an arbitrary operator on $V$.

Note that if $T \in \mathcal{L}(V)$, then $T^*T$ is a positive operator [as was shown in 7.64(a)]. Thus the operator $\sqrt{T^*T}$ makes sense and is well defined as a positive operator on $V$.

The polar decomposition that we are about to state and prove says that every operator on $V$ is the product of a unitary operator and a positive operator. Thus we can write an arbitrary operator on $V$ as the product of two nice operators, each of which comes from a class that we can completely describe and that we understand reasonably well. The unitary operators are described by 7.55 if $\mathbf{F} = \mathbf{C}$; the positive operators are described by the real and complex spectral theorems (7.29 and 7.31).

Specifically, consider the case $\mathbf{F} = \mathbf{C}$, and suppose

$$T = S\sqrt{T^*T}$$

is a polar decomposition of an operator $T \in \mathcal{L}(V)$, where $S$ is a unitary operator. Then there is an orthonormal basis of $V$ with respect to which $S$ has a diagonal matrix, and there is an orthonormal basis of $V$ with respect to which $\sqrt{T^*T}$ has a diagonal matrix. **Warning:** There may not exist an orthonormal basis that simultaneously puts the matrices of both $S$ and $\sqrt{T^*T}$ into these nice diagonal forms—$S$ may require one orthonormal basis and $\sqrt{T^*T}$ may require a different orthonormal basis.

However (still assuming that $\mathbf{F} = \mathbf{C}$), if $T$ is normal, then an orthonormal basis of $V$ can be chosen such that both $S$ and $\sqrt{T^*T}$ have diagonal matrices with respect to this basis—see Exercise 31. The converse is also true: If $T \in \mathcal{L}(V)$ and $T = S\sqrt{T^*T}$ for some unitary operator $S \in \mathcal{L}(V)$ such that $S$ and $\sqrt{T^*T}$ both have diagonal matrices with respect to the same orthonormal basis of $V$, then $T$ is normal. This holds because $T$ then has a diagonal matrix with respect to this same orthonormal basis, which implies that $T$ is normal [by the equivalence of (c) and (a) in 7.31].

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The polar decomposition below is valid on both real and complex inner product spaces and for all operators on those spaces.

**7.93 *polar decomposition***

Suppose $T \in \mathcal{L}(V)$. Then there exists a unitary operator $S \in \mathcal{L}(V)$ such that

$$T = S\sqrt{T^*T}.$$

Proof Let $s_1, \dots, s_m$ be the positive singular values of $T$, and let $e_1, \dots, e_m$ and $f_1, \dots, f_m$ be orthonormal lists in $V$ such that

7.94 $$Tv = s_1\langle v, e_1\rangle f_1 + \cdots + s_m\langle v, e_m\rangle f_m$$

for every $v \in V$. Extend $e_1, \dots, e_m$ and $f_1, \dots, f_m$ to orthonormal bases $e_1, \dots, e_n$ and $f_1, \dots, f_n$ of $V$.

Define $S \in \mathcal{L}(V)$ by

$$Sv = \langle v, e_1\rangle f_1 + \cdots + \langle v, e_n\rangle f_n$$

for each $v \in V$. Then

$$\begin{aligned} \|Sv\|^2 &= \|\langle v, e_1\rangle f_1 + \cdots + \langle v, e_n\rangle f_n\|^2 \\ &= |\langle v, e_1\rangle|^2 + \cdots + |\langle v, e_n\rangle|^2 \\ &= \|v\|^2. \end{aligned}$$

Thus $S$ is a unitary operator.

Applying $T^*$ to both sides of 7.94 and then using the formula for $T^*$ given by 7.77 shows that

$$T^*Tv = s_1^2\langle v, e_1\rangle e_1 + \cdots + s_m^2\langle v, e_m\rangle e_m$$

for every $v \in V$. Thus if $v \in V$, then

$$\sqrt{T^*T}v = s_1\langle v, e_1\rangle e_1 + \cdots + s_m\langle v, e_m\rangle e_m$$

because the operator that sends $v$ to the right side of the equation above is a positive operator whose square equals $T^*T$. Now

$$\begin{aligned} S\sqrt{T^*T}v &= S(s_1\langle v, e_1\rangle e_1 + \cdots + s_m\langle v, e_m\rangle e_m) \\ &= s_1\langle v, e_1\rangle f_1 + \cdots + s_m\langle v, e_m\rangle f_m \\ &= Tv, \end{aligned}$$

where the last equation follows from 7.94. ∎

Exercise 27 shows that the unitary operator $S$ produced in the proof above is as close as a unitary operator can be to $T$.

Alternative proofs of the polar decomposition directly use the spectral theorem, avoiding the singular value decomposition. However, the proof above seems cleaner than those alternative proofs.

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## *Operators Applied to Ellipsoids and Parallelepipeds*

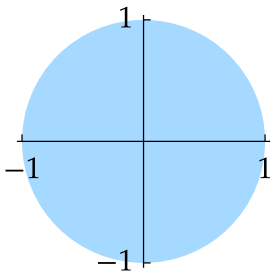
**7.95 definition: *ball, B***

The *ball* in $V$ of radius 1 centered at 0, denoted by $B$, is defined by

$$B = \{v \in V : \|v\| < 1\}.$$

If $\dim V = 2$, the word *disk* is sometimes used instead of *ball*. However, using *ball* in all dimensions is less confusing. Similarly, if $\dim V = 2$, then the word *ellipse* is sometimes used instead of the word *ellipsoid* that we are about to define. Again, using *ellipsoid* in all dimensions is less confusing.

You can think of the ellipsoid defined below as obtained by starting with the ball $B$ and then stretching by a factor of $s_k$ along each $f_k$ axis.




*The ball $B$ in $\mathbf{R}^2$.*

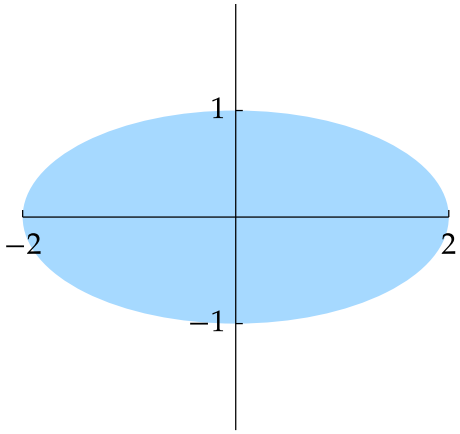
**7.96 definition: *ellipsoid, $E(s_1 f_1, \ldots, s_n f_n)$, principal axes***

Suppose that $f_1, \ldots, f_n$ is an orthonormal basis of $V$ and $s_1, \ldots, s_n$ are positive numbers. The *ellipsoid* $E(s_1 f_1, \ldots, s_n f_n)$ with *principal axes* $s_1 f_1, \ldots, s_n f_n$ is defined by

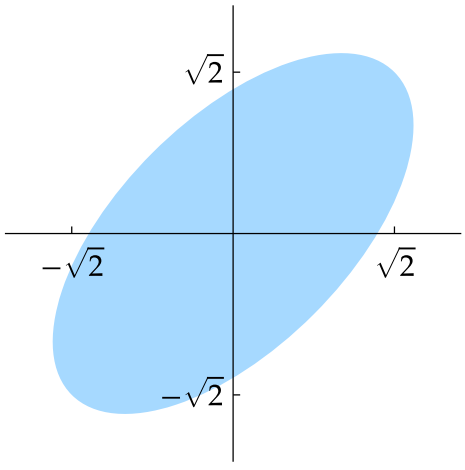
$$E(s_1 f_1, \ldots, s_n f_n) = \left\{v \in V : \frac{|\langle v, f_1\rangle|^2}{s_1^2} + \cdots + \frac{|\langle v, f_n\rangle|^2}{s_n^2} < 1\right\}.$$

The ellipsoid notation $E(s_1 f_1, \ldots, s_n f_n)$ does not explicitly include the inner product space $V$, even though the definition above depends on $V$. However, the inner product space $V$ should be clear from the context and also from the requirement that $f_1, \ldots, f_n$ be an orthonormal basis of $V$.

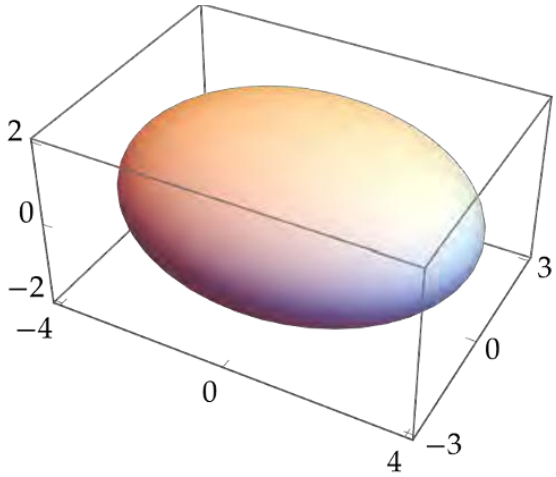
**7.97 example: *ellipsoids***




*The ellipsoid $E(2f_1, f_2)$ in $\mathbf{R}^2$, where $f_1, f_2$ is the standard basis of $\mathbf{R}^2$.*




*The ellipsoid $E(2f_1, f_2)$ in $\mathbf{R}^2$, where $f_1 = \left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right)$ and $f_2 = \left(-\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right)$.*




*The ellipsoid $E(4f_1, 3f_2, 2f_3)$ in $\mathbf{R}^3$, where $f_1, f_2, f_3$ is the standard basis of $\mathbf{R}^3$.*

The ellipsoid $E(f_1, ..., f_n)$ equals the ball $B$ in $V$ for every orthonormal basis $f_1, ..., f_n$ of $V$ [by Parseval's identity 6.30(b)].

**7.98 notation: $T(\Omega)$**

For $T$ a function defined on $V$ and $\Omega \subseteq V$, define $T(\Omega)$ by

$$T(\Omega) = \{Tv : v \in \Omega\}.$$

Thus if $T$ is a function defined on $V$, then $T(V) = \text{range } T$.

The next result states that every invertible operator $T \in \mathcal{L}(V)$ maps the ball $B$ in $V$ onto an ellipsoid in $V$. The proof shows that the principal axes of this ellipsoid come from the singular value decomposition of $T$.

**7.99 *invertible operator takes ball to ellipsoid***

Suppose $T \in \mathcal{L}(V)$ is invertible. Then $T$ maps the ball $B$ in $V$ onto an ellipsoid in $V$.

Proof Suppose $T$ has singular value decomposition

7.100 $$Tv = s_1\langle v, e_1\rangle f_1 + \cdots + s_n\langle v, e_n\rangle f_n$$

for all $v \in V$, where $s_1, ..., s_n$ are the singular values of $T$ and $e_1, ..., e_n$ and $f_1, ..., f_n$ are both orthonormal bases of $V$. We will show that $T(B) = E(s_1 f_1, ..., s_n f_n)$.

First suppose $v \in B$. Because $T$ is invertible, none of the singular values $s_1, ..., s_n$ equals 0 (see 7.68). Thus 7.100 implies that

$$\frac{|\langle Tv, f_1\rangle|^2}{s_1^2} + \cdots + \frac{|\langle Tv, f_n\rangle|^2}{s_n^2} = |\langle v, e_1\rangle|^2 + \cdots + |\langle v, e_n\rangle|^2 < 1.$$

Thus $Tv \in E(s_1 f_1, ..., s_n f_n)$. Hence $T(B) \subseteq E(s_1 f_1, ..., s_n f_n)$.

To prove inclusion in the other direction, now suppose $w \in E(s_1 f_1, ..., s_n f_n)$. Let

$$v = \frac{\langle w, f_1\rangle}{s_1} e_1 + \cdots + \frac{\langle w, f_n\rangle}{s_n} e_n.$$

Then $\|v\| < 1$ and 7.100 implies that $Tv = \langle w, f_1\rangle f_1 + \cdots + \langle w, f_n\rangle f_n = w$. Thus $T(B) \supseteq E(s_1 f_1, ..., s_n f_n)$.

We now use the previous result to show that invertible operators take all ellipsoids, not just the ball of radius 1, to ellipsoids.

**7.101** *invertible operator takes ellipsoids to ellipsoids*

Suppose $T \in \mathcal{L}(V)$ is invertible and $E$ is an ellipsoid in $V$. Then $T(E)$ is an ellipsoid in $V$.

Proof There exist orthonormal basis $f_1, \ldots, f_n$ of $V$ and positive numbers $s_1, \ldots, s_n$ such that $E = E(s_1 f_1, \ldots, s_n f_n)$. Define $S \in \mathcal{L}(V)$ by

$$S(a_1 f_1 + \cdots + a_n f_n) = a_1 s_1 f_1 + \cdots + a_n s_n f_n.$$

Then $S$ maps the ball $B$ of $V$ onto $E$, as you can verify. Thus

$$T(E) = T(S(B)) = (TS)(B).$$

The equation above and 7.99, applied to $TS$, show that $T(E)$ is an ellipsoid in $V$. ∎

Recall (see 3.95) that if $u \in V$ and $\Omega \subseteq V$ then $u + \Omega$ is defined by

$$u + \Omega = \{u + w : w \in \Omega\}.$$

Geometrically, the sets $\Omega$ and $u + \Omega$ look the same, but they are in different locations.

In the following definition, if $\dim V = 2$ then the word *parallelogram* is often used instead of *parallelepiped*.

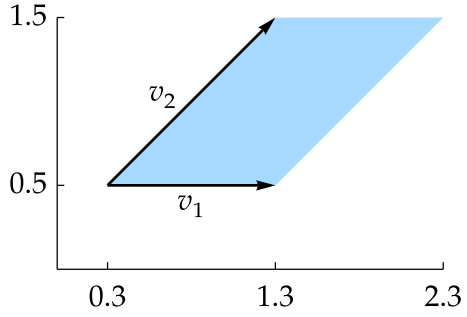
**7.102** definition: $P(v_1, \ldots, v_n)$, *parallelepiped*

Suppose $v_1, \ldots, v_n$ is a basis of $V$. Let

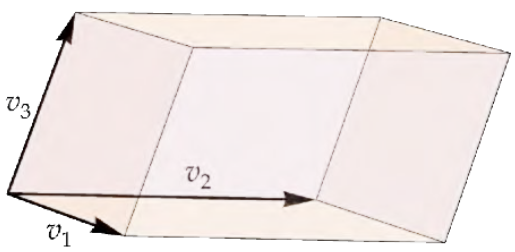
$$P(v_1, \ldots, v_n) = \{a_1 v_1 + \cdots + a_n v_n : a_1, \ldots, a_n \in (0, 1)\}.$$

A *parallelepiped* is a set of the form $u + P(v_1, \ldots, v_n)$ for some $u \in V$. The vectors $v_1, \ldots, v_n$ are called the *edges* of this parallelepiped.

**7.103** example: *parallelepipeds*




*The parallelepiped* $(0.3, 0.5) + P((1, 0), (1, 1))$ *in* $\mathbf{R}^2$.




*A parallelepiped in* $\mathbf{R}^3$.

**7.104** *invertible operator takes parallelepipeds to parallelepipeds*

Suppose $u \in V$ and $v_1, \dots, v_n$ is a basis of $V$. Suppose $T \in \mathcal{L}(V)$ is invertible. Then

$$T(u + P(v_1, \dots, v_n)) = Tu + P(Tv_1, \dots, Tv_n).$$

Proof Because $T$ is invertible, the list $Tv_1, \dots, Tv_n$ is a basis of $V$. The linearity of $T$ implies that

$$T(u + a_1v_1 + \cdots + a_nv_n) = Tu + a_1Tv_1 + \cdots + a_nTv_n$$

for all $a_1, \dots, a_n \in (0, 1)$. Thus $T(u + P(v_1, \dots, v_n)) = Tu + P(Tv_1, \dots, Tv_n)$. ∎

Just as the rectangles are distinguished among the parallelograms in $\mathbf{R}^2$, we give a special name to the parallelepipeds in $V$ whose defining edges are orthogonal to each other.

**7.105** definition: *box*

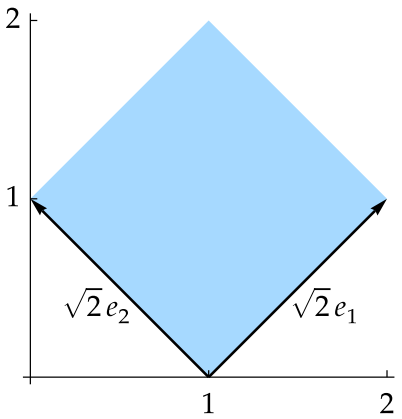
A *box* in $V$ is a set of the form

$$u + P(r_1e_1, \dots, r_ne_n),$$

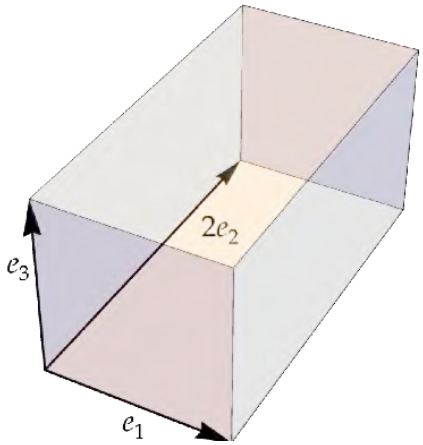
where $u \in V$ and $r_1, \dots, r_n$ are positive numbers and $e_1, \dots, e_n$ is an orthonormal basis of $V$.

Note that in the special case of $\mathbf{R}^2$ each box is a rectangle, but the terminology *box* can be used in all dimensions.

**7.106** example: *boxes*




*The box* $(1, 0) + P(\sqrt{2}e_1, \sqrt{2}e_2)$, *where* $e_1 = \left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right)$ *and* $e_2 = \left(-\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right)$.




*The box* $P(e_1, 2e_2, e_3)$, *where* $e_1, e_2, e_3$ *is the standard basis of* $\mathbf{R}^3$.

Suppose $T \in \mathcal{L}(V)$ is invertible. Then $T$ maps every parallelepiped in $V$ to a parallelepiped in $V$ (by 7.104). In particular, $T$ maps every box in $V$ to a parallelepiped in $V$. This raises the question of whether $T$ maps some boxes in $V$ to boxes in $V$. The following result answers this question, with the help of the singular value decomposition.

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**7.107** *every invertible operator takes some boxes to boxes*

Suppose $T \in \mathcal{L}(V)$ is invertible. Suppose $T$ has singular value decomposition

$$Tv = s_1\langle v, e_1\rangle f_1 + \cdots + s_n\langle v, e_n\rangle f_n,$$

where $s_1, \ldots, s_n$ are the singular values of $T$ and $e_1, \ldots, e_n$ and $f_1, \ldots, f_n$ are orthonormal bases of $V$ and the equation above holds for all $v \in V$. Then $T$ maps the box $u + P(r_1e_1, \ldots, r_ne_n)$ onto the box $Tu + P(r_1s_1f_1, \ldots, r_ns_nf_n)$ for all positive numbers $r_1, \ldots, r_n$ and all $u \in V$.

Proof If $a_1, \ldots, a_n \in (0, 1)$ and $r_1, \ldots, r_n$ are positive numbers and $u \in V$, then

$$T(u + a_1r_1e_1 + \cdots + a_nr_ne_n) = Tu + a_1r_1s_1f_1 + \cdots + a_nr_ns_nf_n.$$

Thus $T(u + P(r_1e_1, \ldots, r_ne_n)) = Tu + P(r_1s_1f_1, \ldots, r_ns_nf_n)$. ∎

## *Volume via Singular Values*

Our goal in this subsection is to understand how an operator changes the volume of subsets of its domain. Because notions of volume belong to analysis rather than to linear algebra, we will work only with an intuitive notion of volume. Our intuitive approach to volume can be converted into appropriate correct definitions, correct statements, and correct proofs using the machinery of analysis.

Our intuition about volume works best in real inner product spaces. Thus the assumption that $\mathbf{F} = \mathbf{R}$ will appear frequently in the rest of this subsection.

If $\dim V = n$, then by *volume* we will mean $n$-dimensional volume. You should be familiar with this concept in $\mathbf{R}^3$. When $n = 2$, this is usually called area instead of volume, but for consistency we use the word volume in all dimensions. The most fundamental intuition about volume is that the volume of a box (whose defining edges are by definition orthogonal to each other) is the product of the lengths of the defining edges. Thus we make the following definition.

**7.108** definition: *volume of a box*

Suppose $\mathbf{F} = \mathbf{R}$. If $u \in V$ and $r_1, \ldots, r_n$ are positive numbers and $e_1, \ldots, e_n$ is an orthonormal basis of $V$, then

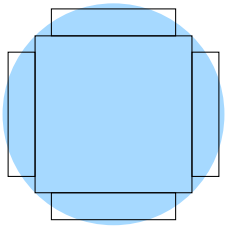
$$\text{volume}(u + P(r_1e_1, \ldots, r_ne_n)) = r_1 \times \cdots \times r_n.$$

The definition above agrees with the familiar formulas for the area (which we are calling the volume) of a rectangle in $\mathbf{R}^2$ and for the volume of a box in $\mathbf{R}^3$. For example, the first box in Example 7.106 has two-dimensional volume (or area) 2 because the defining edges of that box have length $\sqrt{2}$ and $\sqrt{2}$. The second box in Example 7.106 has three-dimensional volume 2 because the defining edges of that box have length 1, 2, and 1.

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To define the volume of a subset of $V$, approximate the subset by a finite collection of disjoint boxes, and then add up the volumes of the approximating collection of boxes. As we approximate a subset of $V$ more accurately by disjoint unions of more boxes, we get a better approximation to the volume.

These ideas should remind you of how the Riemann integral is defined by approximating the area under a curve by a disjoint collection of rectangles. This discussion leads to the following nonrigorous but intuitive definition.



*Volume of this ball ≈ sum of the volumes of the five boxes.*

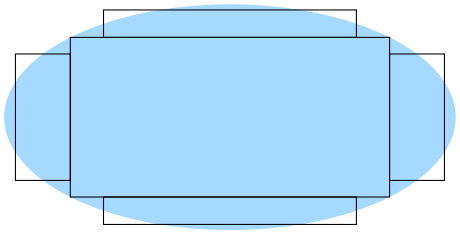
**7.109 definition: *volume***

Suppose $\mathbf{F} = \mathbf{R}$ and $\Omega \subseteq V$. Then the *volume* of $\Omega$, denoted by volume $\Omega$, is approximately the sum of the volumes of a collection of disjoint boxes that approximate $\Omega$.

We are ignoring many reasonable questions by taking an intuitive approach to volume. For example, if we approximate $\Omega$ by boxes with respect to one basis, do we get the same volume if we approximate $\Omega$ by boxes with respect to a different basis? If $\Omega_1$ and $\Omega_2$ are disjoint subsets of $V$, is volume$(\Omega_1 \cup \Omega_1) =$ volume $\Omega_1$ + volume $\Omega_2$? Provided that we consider only reasonably nice subsets of $V$, techniques of analysis show that both these questions have affirmative answers that agree with our intuition about volume.

**7.110 example: *volume change by a linear map***

Suppose that $T \in \mathcal{L}(\mathbf{R}^2)$ is defined by $Tv = 2\langle v, e_1\rangle e_1 + \langle v, e_2\rangle e_2$, where $e_1, e_2$ is the standard basis of $\mathbf{R}^2$. This linear map stretches by a factor of 2 along the $e_1$ axis. The ball approximated by five boxes above gets mapped by $T$ to the ellipsoid shown here. Each of the five boxes in the original figure gets mapped to a box of twice the width and the same height as in the original figure. Hence each box gets mapped to a box of twice the volume (area) as in the original figure. The sum of the volumes of the five new boxes approximates the volume of the ellipsoid. Thus $T$ changes the volume of the ball by a factor of 2.



*Each box here has twice the width and the same height as the boxes in the previous figure.*

In the example above, $T$ maps boxes with respect to the basis $e_1, e_2$ to boxes with respect to the same basis; thus we can see how $T$ changes volume. In general, an operator maps boxes to parallelepipeds that are not boxes. However, if we choose the right basis (coming from the singular value decomposition!), then boxes with respect to that basis get mapped to boxes with respect to a possibly different basis, as shown in 7.107. This observation leads to a natural proof of the following result.

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**7.111** *volume changes by a factor of the product of the singular values*

Suppose $\mathbf{F} = \mathbf{R}$, $T \in \mathcal{L}(V)$ is invertible, and $\Omega \subseteq V$. Then

$$\text{volume } T(\Omega) = (\text{product of singular values of } T)(\text{volume } \Omega).$$

Proof Suppose $T$ has singular value decomposition

$$Tv = s_1\langle v, e_1\rangle f_1 + \cdots + s_n\langle v, e_n\rangle f_n$$

for all $v \in V$, where $e_1, \ldots, e_n$ and $f_1, \ldots, f_n$ are orthonormal bases of $V$.

Approximate $\Omega$ by boxes of the form $u + P(r_1 e_1, \ldots, r_n e_n)$, which have volume $r_1 \times \cdots \times r_n$. The operator $T$ maps each box $u + P(r_1 e_1, \ldots, r_n e_n)$ onto the box $Tu + \mathcal{P}(r_1 s_1 f_1, \ldots, r_n s_n f_n)$, which has volume $(s_1 \times \cdots \times s_n)(r_1 \times \cdots \times r_n)$.

The operator $T$ maps a collection of boxes that approximate $\Omega$ onto a collection of boxes that approximate $T(\Omega)$. Because $T$ changes the volume of each box in a collection that approximates $\Omega$ by a factor of $s_1 \times \cdots \times s_n$, the linear map $T$ changes the volume of $\Omega$ by the same factor. ∎

Suppose $T \in \mathcal{L}(V)$. As we will see when we get to determinants, the product of the singular values of $T$ equals $|\det T|$; see 9.60 and 9.61.

## *Properties of an Operator as Determined by Its Eigenvalues*

We conclude this chapter by presenting the table below. The context of this table is a finite-dimensional complex inner product space. The first column of the table shows a property that a normal operator on such a space might have. The second column of the table shows a subset of $\mathbf{C}$ such that the operator has the corresponding property if and only if all eigenvalues of the operator lie in the specified subset. For example, the first row of the table states that a normal operator is invertible if and only if all its eigenvalues are nonzero (this first row is the only one in the table that does not need the hypothesis that the operator is normal).

Make sure you can explain why all results in the table hold. For example, the last row of the table holds because the norm of an operator equals its largest singular value (by 7.85) and the singular values of a normal operator, assuming $\mathbf{F} = \mathbf{C}$, equal the absolute values of the eigenvalues (by Exercise 7 in Section 7E).

| **properties of a normal operator** | **eigenvalues are contained in** |
|---|---|
| invertible | $\mathbf{C}\backslash\{0\}$ |
| self-adjoint | $\mathbf{R}$ |
| skew | $\{\lambda \in \mathbf{C} : \operatorname{Re}\lambda = 0\}$ |
| orthogonal projection | $\{0, 1\}$ |
| positive | $[0, \infty)$ |
| unitary | $\{\lambda \in \mathbf{C} : \lvert\lambda\rvert = 1\}$ |
| norm is less than 1 | $\{\lambda \in \mathbf{C} : \lvert\lambda\rvert < 1\}$ |

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## *Exercises 7F*

**1** Prove that if $S, T \in \mathcal{L}(V, W)$, then $\big|\, \|S\| - \|T\| \,\big| \le \|S - T\|$.

*The inequality above is called the* ***reverse triangle inequality****.*

**2** Suppose that $T \in \mathcal{L}(V)$ is self-adjoint or that $\mathbf{F} = \mathbf{C}$ and $T \in \mathcal{L}(V)$ is normal. Prove that

$$\|T\| = \max\{|\lambda| : \lambda \text{ is an eigenvalue of } T\}.$$

**3** Suppose $T \in \mathcal{L}(V, W)$ and $v \in V$. Prove that

$$\|Tv\| = \|T\| \, \|v\| \iff T^*Tv = \|T\|^2 v.$$

**4** Suppose $T \in \mathcal{L}(V, W)$, $v \in V$, and $\|Tv\| = \|T\| \, \|v\|$. Prove that if $u \in V$ and $\langle u, v\rangle = 0$, then $\langle Tu, Tv\rangle = 0$.

**5** Suppose $U$ is a finite-dimensional inner product space, $T \in \mathcal{L}(V, U)$, and $S \in \mathcal{L}(U, W)$. Prove that

$$\|ST\| \le \|S\| \, \|T\|.$$

**6** Prove or give a counterexample: If $S, T \in \mathcal{L}(V)$, then $\|ST\| = \|TS\|$.

**7** Show that defining $d(S, T) = \|S - T\|$ for $S, T \in \mathcal{L}(V, W)$ makes $d$ a metric on $\mathcal{L}(V, W)$.

*This exercise is intended for readers who are familiar with metric spaces.*

**8** (a) Prove that if $T \in \mathcal{L}(V)$ and $\|I - T\| < 1$, then $T$ is invertible.

(b) Suppose that $S \in \mathcal{L}(V)$ is invertible. Prove that if $T \in \mathcal{L}(V)$ and $\|S - T\| < 1/\|S^{-1}\|$, then $T$ is invertible.

*This exercise shows that the set of invertible operators in $\mathcal{L}(V)$ is an open subset of $\mathcal{L}(V)$, using the metric defined in Exercise 7.*

**9** Suppose $T \in \mathcal{L}(V)$. Prove that for every $\epsilon > 0$, there exists an invertible operator $S \in \mathcal{L}(V)$ such that $0 < \|T - S\| < \epsilon$.

**10** Suppose $\dim V > 1$ and $T \in \mathcal{L}(V)$ is not invertible. Prove that for every $\epsilon > 0$, there exists $S \in \mathcal{L}(V)$ such that $0 < \|T - S\| < \epsilon$ and $S$ is not invertible.

**11** Suppose $\mathbf{F} = \mathbf{C}$ and $T \in \mathcal{L}(V)$. Prove that for every $\epsilon > 0$ there exists a diagonalizable operator $S \in \mathcal{L}(V)$ such that $0 < \|T - S\| < \epsilon$.

**12** Suppose $T \in \mathcal{L}(V)$ is a positive operator. Show that $\left\|\sqrt{T}\right\| = \sqrt{\|T\|}$.

**13** Suppose $S, T \in \mathcal{L}(V)$ are positive operators. Show that

$$\|S - T\| \le \max\{\|S\|, \|T\|\} \le \|S + T\|.$$

**14** Suppose $U$ and $W$ are subspaces of $V$ such that $\|P_U - P_W\| < 1$. Prove that $\dim U = \dim W$.

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**15** Define $T \in \mathcal{L}(\mathbf{F}^3)$ by

$$T(z_1, z_2, z_3) = (z_3, 2z_1, 3z_2).$$

Find (explicitly) a unitary operator $S \in \mathcal{L}(\mathbf{F}^3)$ such that $T = S\sqrt{T^*T}$.

**16** Suppose $S \in \mathcal{L}(V)$ is a positive invertible operator. Prove that there exists $\delta > 0$ such that $T$ is a positive operator for every self-adjoint operator $T \in \mathcal{L}(V)$ with $\|S - T\| < \delta$.

**17** Prove that if $u \in V$ and $\varphi_u$ is the linear functional on $V$ defined by the equation $\varphi_u(v) = \langle v, u\rangle$, then $\|\varphi_u\| = \|u\|$.

*Here we are thinking of the scalar field* $\mathbf{F}$ *as an inner product space with* $\langle \alpha, \beta\rangle = \alpha\overline{\beta}$ *for all* $\alpha, \beta \in \mathbf{F}$. *Thus* $\|\varphi_u\|$ *means the norm of* $\varphi_u$ *as a linear map from* $V$ *to* $\mathbf{F}$.

**18** Suppose $e_1, \ldots, e_n$ is an orthonormal basis of $V$ and $T \in \mathcal{L}(V, W)$.

(a) Prove that $\max\{\|Te_1\|, \ldots, \|Te_n\|\} \le \|T\| \le (\|Te_1\|^2 + \cdots + \|Te_n\|^2)^{1/2}$.

(b) Prove that $\|T\| = (\|Te_1\|^2 + \cdots + \|Te_n\|^2)^{1/2}$ if and only if $\dim \operatorname{range} T \le 1$.

*Here* $e_1, \ldots, e_n$ *is an arbitrary orthonormal basis of* $V$, *not necessarily connected with a singular value decomposition of* $T$. *If* $s_1, \ldots, s_n$ *is the list of singular values of* $T$, *then the right side of the inequality above equals* $(s_1^2 + \cdots + s_n^2)^{1/2}$, *as was shown in Exercise 11(a) in Section 7E.*

**19** Prove that if $T \in \mathcal{L}(V, W)$, then $\|T^*T\| = \|T\|^2$.

*This formula for* $\|T^*T\|$ *leads to the important subject of* $C^*$*-algebras.*

**20** Suppose $T \in \mathcal{L}(V)$ is normal. Prove that $\|T^k\| = \|T\|^k$ for every positive integer $k$.

**21** Suppose $\dim V > 1$ and $\dim W > 1$. Prove that the norm on $\mathcal{L}(V, W)$ does not come from an inner product. In other words, prove that there does not exist an inner product on $\mathcal{L}(V, W)$ such that

$$\max\{\|Tv\| : v \in V \text{ and } \|v\| \le 1\} = \sqrt{\langle T, T\rangle}$$

for all $T \in \mathcal{L}(V, W)$.

**22** Suppose $T \in \mathcal{L}(V, W)$. Let $n = \dim V$ and let $s_1 \ge \cdots \ge s_n$ denote the singular values of $T$. Prove that if $1 \le k \le n$, then

$$\min\{\|T|_U\| : U \text{ is a subspace of } V \text{ with } \dim U = k\} = s_{n-k+1}.$$

**23** Suppose $T \in \mathcal{L}(V, W)$. Show that $T$ is uniformly continuous with respect to the metrics on $V$ and $W$ that arise from the norms on those spaces (see Exercise 23 in Section 6B).

**24** Suppose $T \in \mathcal{L}(V)$ is invertible. Prove that

$$\|T^{-1}\| = \|T\|^{-1} \iff \frac{T}{\|T\|} \text{ is a unitary operator.}$$

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**25** Fix $u, x \in V$ with $u \neq 0$. Define $T \in \mathcal{L}(V)$ by $Tv = \langle v, u\rangle x$ for every $v \in V$. Prove that

$$\sqrt{T^*T}v = \frac{\|x\|}{\|u\|}\langle v, u\rangle u$$

for every $v \in V$.

**26** Suppose $T \in \mathcal{L}(V)$. Prove that $T$ is invertible if and only if there exists a unique unitary operator $S \in \mathcal{L}(V)$ such that $T = S\sqrt{T^*T}$.

**27** Suppose $T \in \mathcal{L}(V)$ and $s_1, ..., s_n$ are the singular values of $T$. Let $e_1, ..., e_n$ and $f_1, ..., f_n$ be orthonormal bases of $V$ such that

$$Tv = s_1\langle v, e_1\rangle f_1 + \cdots + s_n\langle v, e_n\rangle f_n$$

for all $v \in V$. Define $S \in \mathcal{L}(V)$ by

$$Sv = \langle v, e_1\rangle f_1 + \cdots + \langle v, e_n\rangle f_n.$$

(a) Show that $S$ is unitary and $\|T - S\| = \max\{|s_1 - 1|, ..., |s_n - 1|\}$.
(b) Show that if $E \in \mathcal{L}(V)$ is unitary, then $\|T - E\| \geq \|T - S\|$.

*This exercise finds a unitary operator $S$ that is as close as possible (among the unitary operators) to a given operator $T$.*

**28** Suppose $T \in \mathcal{L}(V)$. Prove that there exists a unitary operator $S \in \mathcal{L}(V)$ such that $T = \sqrt{TT^*}\,S$.

**29** Suppose $T \in \mathcal{L}(V)$.

(a) Use the polar decomposition to show that there exists a unitary operator $S \in \mathcal{L}(V)$ such that $TT^* = ST^*TS^*$.
(b) Show how (a) implies that $T$ and $T^*$ have the same singular values.

**30** Suppose $T \in \mathcal{L}(V)$, $S \in \mathcal{L}(V)$ is a unitary operator, and $R \in \mathcal{L}(V)$ is a positive operator such that $T = SR$. Prove that $R = \sqrt{T^*T}$.

*This exercise shows that if we write $T$ as the product of a unitary operator and a positive operator (as in the polar decomposition 7.93), then the positive operator equals $\sqrt{T^*T}$.*

**31** Suppose $\mathbf{F} = \mathbf{C}$ and $T \in \mathcal{L}(V)$ is normal. Prove that there exists a unitary operator $S \in \mathcal{L}(V)$ such that $T = S\sqrt{T^*T}$ and such that $S$ and $\sqrt{T^*T}$ both have diagonal matrices with respect to the same orthonormal basis of $V$.

**32** Suppose that $T \in \mathcal{L}(V, W)$ and $T \neq 0$. Let $s_1, ..., s_m$ denote the positive singular values of $T$. Show that there exists an orthonormal basis $e_1, ..., e_m$ of $(\operatorname{null} T)^\perp$ such that

$$T\Big(E\Big(\frac{e_1}{s_1}, ..., \frac{e_m}{s_m}\Big)\Big)$$

equals the ball in range $T$ of radius 1 centered at 0.

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Chapter 8

# Operators on Complex Vector Spaces

In this chapter we delve deeper into the structure of operators, with most of the attention on complex vector spaces. Some of the results in this chapter apply to both real and complex vector spaces; thus we do not make a standing assumption that $\mathbf{F} = \mathbf{C}$. Also, an inner product does not help with this material, so we return to the general setting of a finite-dimensional vector space.

Even on a finite-dimensional complex vector space, an operator may not have enough eigenvectors to form a basis of the vector space. Thus we will consider the closely related objects called generalized eigenvectors. We will see that for each operator on a finite-dimensional complex vector space, there is a basis of the vector space consisting of generalized eigenvectors of the operator. The generalized eigenspace decomposition then provides a good description of arbitrary operators on a finite-dimensional complex vector space.

Nilpotent operators, which are operators that when raised to some power equal 0, have an important role in these investigations. Nilpotent operators provide a key tool in our proof that every invertible operator on a finite-dimensional complex vector space has a square root and in our approach to Jordan form.

This chapter concludes by defining the trace and proving its key properties.

*standing assumptions for this chapter*

- $\mathbf{F}$ denotes $\mathbf{R}$ or $\mathbf{C}$.
- $V$ denotes a finite-dimensional nonzero vector space over $\mathbf{F}$.



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*The Long Room of the Old Library at the University of Dublin, where William Hamilton (1805–1865) was a student and then a faculty member. Hamilton proved a special case of what we now call the Cayley–Hamilton theorem in 1853.*

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# 8A Generalized Eigenvectors and Nilpotent Operators

## Null Spaces of Powers of an Operator

We begin this chapter with a study of null spaces of powers of an operator.

**8.1 *sequence of increasing null spaces***

Suppose $T \in \mathcal{L}(V)$. Then

$$\{0\} = \operatorname{null} T^0 \subseteq \operatorname{null} T^1 \subseteq \cdots \subseteq \operatorname{null} T^k \subseteq \operatorname{null} T^{k+1} \subseteq \cdots.$$

Proof Suppose $k$ is a nonnegative integer and $v \in \operatorname{null} T^k$. Then $T^k v = 0$, which implies that $T^{k+1}v = T(T^k v) = T(0) = 0$. Thus $v \in \operatorname{null} T^{k+1}$. Hence $\operatorname{null} T^k \subseteq \operatorname{null} T^{k+1}$, as desired. ∎

The following result states that if two consecutive terms in the sequence of subspaces above are equal, then all later terms in the sequence are equal.

*For similar results about decreasing sequences of ranges, see Exercises 6, 7, and 8.*

**8.2 *equality in the sequence of null spaces***

Suppose $T \in \mathcal{L}(V)$ and $m$ is a nonnegative integer such that

$$\operatorname{null} T^m = \operatorname{null} T^{m+1}.$$

Then

$$\operatorname{null} T^m = \operatorname{null} T^{m+1} = \operatorname{null} T^{m+2} = \operatorname{null} T^{m+3} = \cdots.$$

Proof Let $k$ be a positive integer. We want to prove that

$$\operatorname{null} T^{m+k} = \operatorname{null} T^{m+k+1}.$$

We already know from 8.1 that $\operatorname{null} T^{m+k} \subseteq \operatorname{null} T^{m+k+1}$.

To prove the inclusion in the other direction, suppose $v \in \operatorname{null} T^{m+k+1}$. Then

$$T^{m+1}(T^k v) = T^{m+k+1} v = 0.$$

Hence

$$T^k v \in \operatorname{null} T^{m+1} = \operatorname{null} T^m.$$

Thus $T^{m+k}v = T^m(T^k v) = 0$, which means that $v \in \operatorname{null} T^{m+k}$. This implies that $\operatorname{null} T^{m+k+1} \subseteq \operatorname{null} T^{m+k}$, completing the proof. ∎

The result above raises the question of whether there exists a nonnegative integer $m$ such that $\operatorname{null} T^m = \operatorname{null} T^{m+1}$. The next result shows that this equality holds at least when $m$ equals the dimension of the vector space on which $T$ operates.

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**8.3 *null spaces stop growing***

Suppose $T \in \mathcal{L}(V)$. Then

$$\operatorname{null} T^{\dim V} = \operatorname{null} T^{\dim V+1} = \operatorname{null} T^{\dim V+2} = \cdots.$$

Proof We only need to prove that $\operatorname{null} T^{\dim V} = \operatorname{null} T^{\dim V+1}$ (by 8.2). Suppose this is not true. Then, by 8.1 and 8.2, we have

$$\{0\} = \operatorname{null} T^0 \subsetneq \operatorname{null} T^1 \subsetneq \cdots \subsetneq \operatorname{null} T^{\dim V} \subsetneq \operatorname{null} T^{\dim V+1},$$

where the symbol $\subsetneq$ means "contained in but not equal to". At each of the strict inclusions in the chain above, the dimension increases by at least 1. Thus $\dim \operatorname{null} T^{\dim V+1} \geq \dim V + 1$, a contradiction because a subspace of $V$ cannot have a larger dimension than $\dim V$. ∎

It is not true that $V = \operatorname{null} T \oplus \operatorname{range} T$ for every $T \in \mathcal{L}(V)$. However, the next result can be a useful substitute.

**8.4 *$V$ is the direct sum of* $\operatorname{null} T^{\dim V}$ *and* $\operatorname{range} T^{\dim V}$**

Suppose $T \in \mathcal{L}(V)$. Then

$$V = \operatorname{null} T^{\dim V} \oplus \operatorname{range} T^{\dim V}.$$

Proof Let $n = \dim V$. First we show that

$$(\operatorname{null} T^n) \cap (\operatorname{range} T^n) = \{0\}. \tag{8.5}$$

Suppose $v \in (\operatorname{null} T^n) \cap (\operatorname{range} T^n)$. Then $T^n v = 0$, and there exists $u \in V$ such that $v = T^n u$. Applying $T^n$ to both sides of the last equation shows that $T^n v = T^{2n} u$. Hence $T^{2n} u = 0$, which implies that $T^n u = 0$ (by 8.3). Thus $v = T^n u = 0$, completing the proof of 8.5.

Now 8.5 implies that $\operatorname{null} T^n + \operatorname{range} T^n$ is a direct sum (by 1.46). Also,

$$\dim(\operatorname{null} T^n \oplus \operatorname{range} T^n) = \dim \operatorname{null} T^n + \dim \operatorname{range} T^n = \dim V,$$

where the first equality above comes from 3.94 and the second equality comes from the fundamental theorem of linear maps (3.21). The equation above implies that $\operatorname{null} T^n \oplus \operatorname{range} T^n = V$ (see 2.39), as desired. ∎

For an improvement of the result above, see Exercise 19.

**8.6 example: $\mathbf{F}^3 = \operatorname{null} T^3 \oplus \operatorname{range} T^3$ *for* $T \in \mathcal{L}(\mathbf{F}^3)$**

Suppose $T \in \mathcal{L}(\mathbf{F}^3)$ is defined by

$$T(z_1, z_2, z_3) = (4z_2, 0, 5z_3).$$

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Then $\text{null } T = \{(z_1, 0, 0) : z_1 \in \mathbf{F}\}$ and $\text{range } T = \{(z_1, 0, z_3) : z_1, z_3 \in \mathbf{F}\}$. Thus $\text{null } T \cap \text{range } T \neq \{0\}$. Hence $\text{null } T + \text{range } T$ is not a direct sum. Also note that $\text{null } T + \text{range } T \neq \mathbf{F}^3$. However, we have $T^3(z_1, z_2, z_3) = (0, 0, 125z_3)$. Thus we see that

$$\text{null } T^3 = \{(z_1, z_2, 0) : z_1, z_2 \in \mathbf{F}\} \quad \text{and} \quad \text{range } T^3 = \{(0, 0, z_3) : z_3 \in \mathbf{F}\}.$$

Hence $\mathbf{F}^3 = \text{null } T^3 \oplus \text{range } T^3$, as expected by 8.4.

---

## *Generalized Eigenvectors*

Some operators do not have enough eigenvectors to lead to good descriptions of their behavior. Thus in this subsection we introduce the concept of generalized eigenvectors, which will play a major role in our description of the structure of an operator.

To understand why we need more than eigenvectors, let's examine the question of describing an operator by decomposing its domain into invariant subspaces. Fix $T \in \mathcal{L}(V)$. We seek to describe $T$ by finding a "nice" direct sum decomposition

$$V = V_1 \oplus \cdots \oplus V_m,$$

where each $V_k$ is a subspace of $V$ invariant under $T$. The simplest possible nonzero invariant subspaces are one-dimensional. A decomposition as above in which each $V_k$ is a one-dimensional subspace of $V$ invariant under $T$ is possible if and only if $V$ has a basis consisting of eigenvectors of $T$ (see 5.55). This happens if and only if $V$ has an eigenspace decomposition

8.7 $$V = E(\lambda_1, T) \oplus \cdots \oplus E(\lambda_m, T),$$

where $\lambda_1, \ldots, \lambda_m$ are the distinct eigenvalues of $T$ (see 5.55).

The spectral theorem in the previous chapter shows that if $V$ is an inner product space, then a decomposition of the form 8.7 holds for every self-adjoint operator if $\mathbf{F} = \mathbf{R}$ and for every normal operator if $\mathbf{F} = \mathbf{C}$ because operators of those types have enough eigenvectors to form a basis of $V$ (see 7.29 and 7.31).

However, a decomposition of the form 8.7 may not hold for more general operators, even on a complex vector space. An example was given by the operator in 5.57, which does not have enough eigenvectors for 8.7 to hold. Generalized eigenvectors and generalized eigenspaces, which we now introduce, will remedy this situation.

**8.8 definition: *generalized eigenvector***

Suppose $T \in \mathcal{L}(V)$ and $\lambda$ is an eigenvalue of $T$. A vector $v \in V$ is called a *generalized eigenvector* of $T$ corresponding to $\lambda$ if $v \neq 0$ and

$$(T - \lambda I)^k v = 0$$

for some positive integer $k$.

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A nonzero vector $v \in V$ is a generalized eigenvector of $T$ corresponding to $\lambda$ if and only if

$$(T - \lambda I)^{\dim V} v = 0,$$

as follows from applying 8.1 and 8.3 to the operator $T - \lambda I$.

*Generalized eigenvalues are not defined because doing so would not lead to anything new. Reason: if $(T - \lambda I)^k$ is not injective for some positive integer $k$, then $T - \lambda I$ is not injective, and hence $\lambda$ is an eigenvalue of $T$.*

As we know, an operator on a complex vector space may not have enough eigenvectors to form a basis of the domain. The next result shows that on a complex vector space there are enough generalized eigenvectors to do this.

**8.9** *a basis of generalized eigenvectors*

Suppose $\mathbf{F} = \mathbf{C}$ and $T \in \mathcal{L}(V)$. Then there is a basis of $V$ consisting of generalized eigenvectors of $T$.

Proof Let $n = \dim V$. We will use induction on $n$. To get started, note that the desired result holds if $n = 1$ because then every nonzero vector in $V$ is an eigenvector of $T$.

Now suppose $n > 1$ and the desired result holds for all smaller values of $\dim V$. Let $\lambda$ be an eigenvalue of $T$. Applying 8.4 to $T - \lambda I$ shows that

*This step is where we use the hypothesis that $\mathbf{F} = \mathbf{C}$, because if $\mathbf{F} = \mathbf{R}$ then $T$ may not have any eigenvalues.*

$$V = \operatorname{null}(T - \lambda I)^n \oplus \operatorname{range}(T - \lambda I)^n.$$

If $\operatorname{null}(T - \lambda I)^n = V$, then every nonzero vector in $V$ is a generalized eigenvector of $T$, and thus in this case there is a basis of $V$ consisting of generalized eigenvectors of $T$. Hence we can assume that $\operatorname{null}(T - \lambda I)^n \neq V$, which implies that $\operatorname{range}(T - \lambda I)^n \neq \{0\}$.

Also, $\operatorname{null}(T - \lambda I)^n \neq \{0\}$, because $\lambda$ is an eigenvalue of $T$. Thus we have

$$0 < \dim \operatorname{range}(T - \lambda I)^n < n.$$

Furthermore, $\operatorname{range}(T - \lambda I)^n$ is invariant under $T$ [by 5.18 with $p(z) = (z - \lambda)^n$]. Let $S \in \mathcal{L}(\operatorname{range}(T - \lambda I)^n)$ equal $T$ restricted to $\operatorname{range}(T - \lambda I)^n$. Our induction hypothesis applied to the operator $S$ implies that there is a basis of $\operatorname{range}(T - \lambda I)^n$ consisting of generalized eigenvectors of $S$, which of course are generalized eigenvectors of $T$. Adjoining that basis of $\operatorname{range}(T - \lambda I)^n$ to a basis of $\operatorname{null}(T - \lambda I)^n$ gives a basis of $V$ consisting of generalized eigenvectors of $T$. ∎

If $\mathbf{F} = \mathbf{R}$ and $\dim V > 1$, then some operators on $V$ have the property that there exists a basis of $V$ consisting of generalized eigenvectors of the operator, and (unlike what happens when $\mathbf{F} = \mathbf{C}$) other operators do not have this property. See Exercise 11 for a necessary and sufficient condition that determines whether an operator has this property.

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8.10 example: *generalized eigenvectors of an operator on* $\mathbf{C}^3$

Define $T \in \mathcal{L}(\mathbf{C}^3)$ by

$$T(z_1, z_2, z_3) = (4z_2, 0, 5z_3)$$

for each $(z_1, z_2, z_3) \in \mathbf{C}^3$. A routine use of the definition of eigenvalue shows that the eigenvalues of $T$ are 0 and 5. Furthermore, the eigenvectors corresponding to the eigenvalue 0 are the nonzero vectors of the form $(z_1, 0, 0)$, and the eigenvectors corresponding to the eigenvalue 5 are the nonzero vectors of the form $(0, 0, z_3)$. Hence this operator does not have enough eigenvectors to span its domain $\mathbf{C}^3$.

We compute that $T^3(z_1, z_2, z_3) = (0, 0, 125z_3)$. Thus 8.1 and 8.3 imply that the generalized eigenvectors of $T$ corresponding to the eigenvalue 0 are the nonzero vectors of the form $(z_1, z_2, 0)$.

We also have $(T - 5I)^3(z_1, z_2, z_3) = (-125z_1 + 300z_2, -125z_2, 0)$. Thus the generalized eigenvectors of $T$ corresponding to the eigenvalue 5 are the nonzero vectors of the form $(0, 0, z_3)$.

The paragraphs above show that each of the standard basis vectors of $\mathbf{C}^3$ is a generalized eigenvector of $T$. Thus $\mathbf{C}^3$ indeed has a basis consisting of generalized eigenvectors of $T$, as promised by 8.9.

---

If $v$ is an eigenvector of $T \in \mathcal{L}(V)$, then the corresponding eigenvalue $\lambda$ is uniquely determined by the equation $Tv = \lambda v$, which can be satisfied by only one $\lambda \in \mathbf{F}$ (because $v \neq 0$). However, if $v$ is a generalized eigenvector of $T$, then it is not obvious that the equation $(T - \lambda I)^{\dim V} v = 0$ can be satisfied by only one $\lambda \in \mathbf{F}$. Fortunately, the next result tells us that all is well on this issue.

8.11 *generalized eigenvector corresponds to a unique eigenvalue*

Suppose $T \in \mathcal{L}(V)$. Then each generalized eigenvector of $T$ corresponds to only one eigenvalue of $T$.

Proof Suppose $v \in V$ is a generalized eigenvector of $T$ corresponding to eigenvalues $\alpha$ and $\lambda$ of $T$. Let $m$ be the smallest positive integer such that $(T - \alpha I)^m v = 0$. Let $n = \dim V$. Then

$$\begin{aligned} 0 &= (T - \lambda I)^n v \\ &= \big((T - \alpha I) + (\alpha - \lambda)I\big)^n v \\ &= \sum_{k=0}^{n} b_k (\alpha - \lambda)^{n-k} (T - \alpha I)^k v, \end{aligned}$$

where $b_0 = 1$ and the values of the other binomial coefficients $b_k$ do not matter. Apply the operator $(T - \alpha I)^{m-1}$ to both sides of the equation above, getting

$$0 = (\alpha - \lambda)^n (T - \alpha I)^{m-1} v.$$

Because $(T - \alpha I)^{m-1} v \neq 0$, the equation above implies that $\alpha = \lambda$, as desired. ∎

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We saw earlier (5.11) that eigenvectors corresponding to distinct eigenvalues are linearly independent. Now we prove a similar result for generalized eigenvectors, with a proof that roughly follows the pattern of the proof of that earlier result.

**8.12** *linearly independent generalized eigenvectors*

Suppose that $T \in \mathcal{L}(V)$. Then every list of generalized eigenvectors of $T$ corresponding to distinct eigenvalues of $T$ is linearly independent.

Proof Suppose the desired result is false. Then there exists a smallest positive integer $m$ such that there exists a linearly dependent list $v_1, \ldots, v_m$ of generalized eigenvectors of $T$ corresponding to distinct eigenvalues $\lambda_1, \ldots, \lambda_m$ of $T$ (note that $m \geq 2$ because a generalized eigenvector is, by definition, nonzero). Thus there exist $a_1, \ldots, a_m \in \mathbf{F}$, none of which are 0 (because of the minimality of $m$), such that

$$a_1 v_1 + \cdots + a_m v_m = 0.$$

Let $n = \dim V$. Apply $(T - \lambda_m I)^n$ to both sides of the equation above, getting

8.13 $$a_1 (T - \lambda_m I)^n v_1 + \cdots + a_{m-1} (T - \lambda_m I)^n v_{m-1} = 0.$$

Suppose $k \in \{1, \ldots, m-1\}$. Then

$$(T - \lambda_m I)^n v_k \neq 0$$

because otherwise $v_k$ would be a generalized eigenvector of $T$ corresponding to the distinct eigenvalues $\lambda_k$ and $\lambda_m$, which would contradict 8.11. However,

$$(T - \lambda_k I)^n ((T - \lambda_m I)^n v_k) = (T - \lambda_m I)^n ((T - \lambda_k I)^n v_k) = 0.$$

Thus the last two displayed equations show that $(T - \lambda_m I)^n v_k$ is a generalized eigenvector of $T$ corresponding to the eigenvalue $\lambda_k$. Hence

$$(T - \lambda_m I)^n v_1, \ldots, (T - \lambda_m I)^n v_{m-1}$$

is a linearly dependent list (by 8.13) of $m-1$ generalized eigenvectors corresponding to distinct eigenvalues, contradicting the minimality of $m$. This contradiction completes the proof. ∎

## *Nilpotent Operators*

**8.14** definition: *nilpotent*

An operator is called *nilpotent* if some power of it equals 0.

Thus an operator on $V$ is nilpotent if every nonzero vector in $V$ is a generalized eigenvector of $T$ corresponding to the eigenvalue 0.

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8.15 example: *nilpotent operators*

(a) The operator $T \in \mathcal{L}(\mathbf{F}^4)$ defined by

$$T(z_1, z_2, z_3, z_4) = (0, 0, z_1, z_2)$$

is nilpotent because $T^2 = 0$.

(b) The operator on $\mathbf{F}^3$ whose matrix (with respect to the standard basis) is

$$\begin{pmatrix} -3 & 9 & 0 \\ -7 & 9 & 6 \\ 4 & 0 & -6 \end{pmatrix}$$

is nilpotent, as can be shown by cubing the matrix above to get the zero matrix.

(c) The operator of differentiation on $\mathcal{P}_m(\mathbf{R})$ is nilpotent because the $(m+1)^{\text{th}}$ derivative of every polynomial of degree at most $m$ equals 0. Note that on this space of dimension $m+1$, we need to raise the nilpotent operator to the power $m+1$ to get the 0 operator.

The next result shows that when raising a nilpotent operator to a power, we never need to use a power higher than the dimension of the space. For a slightly stronger result, see Exercise 18.

*The Latin word **nil** means nothing or zero; the Latin word **potens** means having power. Thus **nilpotent** literally means having a power that is zero.*

8.16 *nilpotent operator raised to dimension of domain is* 0

Suppose $T \in \mathcal{L}(V)$ is nilpotent. Then $T^{\dim V} = 0$.

Proof Because $T$ is nilpotent, there exists a positive integer $k$ such that $T^k = 0$. Thus $\operatorname{null} T^k = V$. Now 8.1 and 8.3 imply that $\operatorname{null} T^{\dim V} = V$. Thus $T^{\dim V} = 0$. ∎

8.17 *eigenvalues of nilpotent operator*

Suppose $T \in \mathcal{L}(V)$.

(a) If $T$ is nilpotent, then 0 is an eigenvalue of $T$ and $T$ has no other eigenvalues.

(b) If $\mathbf{F} = \mathbf{C}$ and 0 is the only eigenvalue of $T$, then $T$ is nilpotent.

Proof

(a) To prove (a), suppose $T$ is nilpotent. Hence there is a positive integer $m$ such that $T^m = 0$. This implies that $T$ is not injective. Thus 0 is an eigenvalue of $T$.

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To show that $T$ has no other eigenvalues, suppose $\lambda$ is an eigenvalue of $T$. Then there exists a nonzero vector $v \in V$ such that

$$\lambda v = Tv.$$

Repeatedly applying $T$ to both sides of this equation shows that

$$\lambda^m v = T^m v = 0.$$

Thus $\lambda = 0$, as desired.

(b) Suppose $\mathbf{F} = \mathbf{C}$ and 0 is the only eigenvalue of $T$. By 5.27(b), the minimal polynomial of $T$ equals $z^m$ for some positive integer $m$. Thus $T^m = 0$. Hence $T$ is nilpotent. ∎

Exercise 23 shows that the hypothesis that $\mathbf{F} = \mathbf{C}$ cannot be deleted in (b) of the result above.

Given an operator on $V$, we want to find a basis of $V$ such that the matrix of the operator with respect to this basis is as simple as possible, meaning that the matrix contains many 0's. The next result shows that if $T$ is nilpotent, then we can choose a basis of $V$ such that the matrix of $T$ with respect to this basis has more than half of its entries equal to 0. Later in this chapter we will do even better.

**8.18** *minimal polynomial and upper-triangular matrix of nilpotent operator*

Suppose $T \in \mathcal{L}(V)$. Then the following are equivalent.

(a) $T$ is nilpotent.

(b) The minimal polynomial of $T$ is $z^m$ for some positive integer $m$.

(c) There is a basis of $V$ with respect to which the matrix of $T$ has the form

$$\begin{pmatrix} 0 & & * \\ & \ddots & \\ 0 & & 0 \end{pmatrix},$$

where all entries on and below the diagonal equal 0.

Proof Suppose (a) holds, so $T$ is nilpotent. Thus there exists a positive integer $n$ such that $T^n = 0$. Now 5.29 implies that $z^n$ is a polynomial multiple of the minimal polynomial of $T$. Thus the minimal polynomial of $T$ is $z^m$ for some positive integer $m$, proving that (a) implies (b).

Now suppose (b) holds, so the minimal polynomial of $T$ is $z^m$ for some positive integer $m$. This implies, by 5.27(a), that 0 (which is the only zero of $z^m$) is the only eigenvalue of $T$. This further implies, by 5.44, that there is a basis of $V$ with respect to which the matrix of $T$ is upper triangular. This also implies, by 5.41, that all entries on the diagonal of this matrix are 0, proving that (b) implies (c).

Now suppose (c) holds. Then 5.40 implies that $T^{\dim V} = 0$. Thus $T$ is nilpotent, proving that (c) implies (a). ∎

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## *Exercises 8A*

**1** Suppose $T \in \mathcal{L}(V)$. Prove that if $\dim \operatorname{null} T^4 = 8$ and $\dim \operatorname{null} T^6 = 9$, then $\dim \operatorname{null} T^m = 9$ for all integers $m \geq 5$.

**2** Suppose $T \in \mathcal{L}(V)$, $m$ is a positive integer, $v \in V$, and $T^{m-1}v \neq 0$ but $T^m v = 0$. Prove that $v, Tv, T^2v, \ldots, T^{m-1}v$ is linearly independent.

*The result in this exercise is used in the proof of 8.45.*

**3** Suppose $T \in \mathcal{L}(V)$. Prove that

$$V = \operatorname{null} T \oplus \operatorname{range} T \iff \operatorname{null} T^2 = \operatorname{null} T.$$

**4** Suppose $T \in \mathcal{L}(V)$, $\lambda \in \mathbf{F}$, and $m$ is a positive integer such that the minimal polynomial of $T$ is a polynomial multiple of $(z - \lambda)^m$. Prove that

$$\dim \operatorname{null}(T - \lambda I)^m \geq m.$$

**5** Suppose $T \in \mathcal{L}(V)$ and $m$ is a positive integer. Prove that

$$\dim \operatorname{null} T^m \leq m \dim \operatorname{null} T.$$

*Hint: Exercise 21 in Section 3B may be useful.*

**6** Suppose $T \in \mathcal{L}(V)$. Show that

$$V = \operatorname{range} T^0 \supseteq \operatorname{range} T^1 \supseteq \cdots \supseteq \operatorname{range} T^k \supseteq \operatorname{range} T^{k+1} \supseteq \cdots.$$

**7** Suppose $T \in \mathcal{L}(V)$ and $m$ is a nonnegative integer such that

$$\operatorname{range} T^m = \operatorname{range} T^{m+1}.$$

Prove that $\operatorname{range} T^k = \operatorname{range} T^m$ for all $k > m$.

**8** Suppose $T \in \mathcal{L}(V)$. Prove that

$$\operatorname{range} T^{\dim V} = \operatorname{range} T^{\dim V + 1} = \operatorname{range} T^{\dim V + 2} = \cdots.$$

**9** Suppose $T \in \mathcal{L}(V)$ and $m$ is a nonnegative integer. Prove that

$$\operatorname{null} T^m = \operatorname{null} T^{m+1} \iff \operatorname{range} T^m = \operatorname{range} T^{m+1}.$$

**10** Define $T \in \mathcal{L}(\mathbf{C}^2)$ by $T(w, z) = (z, 0)$. Find all generalized eigenvectors of $T$.

**11** Suppose that $T \in \mathcal{L}(V)$. Prove that there is a basis of $V$ consisting of generalized eigenvectors of $T$ if and only if the minimal polynomial of $T$ equals $(z - \lambda_1) \cdots (z - \lambda_m)$ for some $\lambda_1, \ldots, \lambda_m \in \mathbf{F}$.

*Assume $\mathbf{F} = \mathbf{R}$ because the case $\mathbf{F} = \mathbf{C}$ follows from 5.27(b) and 8.9.*

*This exercise states that the condition for there to be a basis of $V$ consisting of generalized eigenvectors of $T$ is the same as the condition for there to be a basis with respect to which $T$ has an upper-triangular matrix (see 5.44).*

***Caution:** If $T$ has an upper-triangular matrix with respect to a basis $v_1, \ldots, v_n$ of $V$, then $v_1$ is an eigenvector of $T$ but it is not necessarily true that $v_2, \ldots, v_n$ are generalized eigenvectors of $T$.*

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**12** Suppose $T \in \mathcal{L}(V)$ is such that every vector in $V$ is a generalized eigenvector of $T$. Prove that there exists $\lambda \in \mathbf{F}$ such that $T - \lambda I$ is nilpotent.

**13** Suppose $S, T \in \mathcal{L}(V)$ and $ST$ is nilpotent. Prove that $TS$ is nilpotent.

**14** Suppose $T \in \mathcal{L}(V)$ is nilpotent and $T \neq 0$. Prove $T$ is not diagonalizable.

**15** Suppose $\mathbf{F} = \mathbf{C}$ and $T \in \mathcal{L}(V)$. Prove that $T$ is diagonalizable if and only if every generalized eigenvector of $T$ is an eigenvector of $T$.

*For $\mathbf{F} = \mathbf{C}$, this exercise adds another equivalence to the list of conditions for diagonalizability in 5.55.*

**16** (a) Give an example of nilpotent operators $S, T$ on the same vector space such that neither $S + T$ nor $ST$ is nilpotent.
(b) Suppose $S, T \in \mathcal{L}(V)$ are nilpotent and $ST = TS$. Prove that $S + T$ and $ST$ are nilpotent.

**17** Suppose $T \in \mathcal{L}(V)$ is nilpotent and $m$ is a positive integer such that $T^m = 0$.
(a) Prove that $I - T$ is invertible and that $(I - T)^{-1} = I + T + \cdots + T^{m-1}$.
(b) Explain how you would guess the formula above.

**18** Suppose $T \in \mathcal{L}(V)$ is nilpotent. Prove that $T^{1 + \dim \operatorname{range} T} = 0$.

*If $\dim \operatorname{range} T < \dim V - 1$, then this exercise improves 8.16.*

**19** Suppose $T \in \mathcal{L}(V)$ is not nilpotent. Show that

$$V = \operatorname{null} T^{\dim V - 1} \oplus \operatorname{range} T^{\dim V - 1}.$$

*For operators that are not nilpotent, this exercise improves 8.4.*

**20** Suppose $V$ is an inner product space and $T \in \mathcal{L}(V)$ is normal and nilpotent. Prove that $T = 0$.

**21** Suppose $T \in \mathcal{L}(V)$ is such that $\operatorname{null} T^{\dim V - 1} \neq \operatorname{null} T^{\dim V}$. Prove that $T$ is nilpotent and that $\dim \operatorname{null} T^k = k$ for every integer $k$ with $0 \leq k \leq \dim V$.

**22** Suppose $T \in \mathcal{L}(\mathbf{C}^5)$ is such that $\operatorname{range} T^4 \neq \operatorname{range} T^5$. Prove that $T$ is nilpotent.

**23** Give an example of an operator $T$ on a finite-dimensional real vector space such that 0 is the only eigenvalue of $T$ but $T$ is not nilpotent.

*This exercise shows that the implication $(b) \implies (a)$ in 8.17 does not hold without the hypothesis that $\mathbf{F} = \mathbf{C}$.*

**24** For each item in Example 8.15, find a basis of the domain vector space such that the matrix of the nilpotent operator with respect to that basis has the upper-triangular form promised by 8.18(c).

**25** Suppose that $V$ is an inner product space and $T \in \mathcal{L}(V)$ is nilpotent. Show that there is an orthonormal basis of $V$ with respect to which the matrix of $T$ has the upper-triangular form promised by 8.18(c).

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## 8B Generalized Eigenspace Decomposition

### Generalized Eigenspaces

**8.19 definition: *generalized eigenspace*, $G(\lambda, T)$**

Suppose $T \in \mathcal{L}(V)$ and $\lambda \in \mathbf{F}$. The *generalized eigenspace* of $T$ corresponding to $\lambda$, denoted by $G(\lambda, T)$, is defined by

$$G(\lambda, T) = \{v \in V : (T - \lambda I)^k v = 0 \text{ for some positive integer } k\}.$$

Thus $G(\lambda, T)$ is the set of generalized eigenvectors of $T$ corresponding to $\lambda$, along with the 0 vector.

Because every eigenvector of $T$ is a generalized eigenvector of $T$ (take $k = 1$ in the definition of generalized eigenvector), each eigenspace is contained in the corresponding generalized eigenspace. In other words, if $T \in \mathcal{L}(V)$ and $\lambda \in \mathbf{F}$, then $E(\lambda, T) \subseteq G(\lambda, T)$.

The next result implies that if $T \in \mathcal{L}(V)$ and $\lambda \in \mathbf{F}$, then the generalized eigenspace $G(\lambda, T)$ is a subspace of $V$ (because the null space of each linear map on $V$ is a subspace of $V$).

**8.20 *description of generalized eigenspaces***

Suppose $T \in \mathcal{L}(V)$ and $\lambda \in \mathbf{F}$. Then $G(\lambda, T) = \text{null}(T - \lambda I)^{\dim V}$.

Proof Suppose $v \in \text{null}(T - \lambda I)^{\dim V}$. The definitions imply $v \in G(\lambda, T)$. Thus $G(\lambda, T) \supseteq \text{null}(T - \lambda I)^{\dim V}$.

Conversely, suppose $v \in G(\lambda, T)$. Thus there is a positive integer $k$ such that $v \in \text{null}(T - \lambda I)^k$. From 8.1 and 8.3 (with $T - \lambda I$ replacing $T$), we get $v \in \text{null}(T - \lambda I)^{\dim V}$. Thus $G(\lambda, T) \subseteq \text{null}(T - \lambda I)^{\dim V}$, completing the proof. ∎

**8.21 example: *generalized eigenspaces of an operator on* $\mathbf{C}^3$**

Define $T \in \mathcal{L}(\mathbf{C}^3)$ by

$$T(z_1, z_2, z_3) = (4z_2, 0, 5z_3).$$

In Example 8.10, we saw that the eigenvalues of $T$ are 0 and 5, and we found the corresponding sets of generalized eigenvectors. Taking the union of those sets with $\{0\}$, we have

$$G(0, T) = \{(z_1, z_2, 0) : z_1, z_2 \in \mathbf{C}\} \quad \text{and} \quad G(5, T) = \{(0, 0, z_3) : z_3 \in \mathbf{C}\}.$$

Note that $\mathbf{C}^3 = G(0, T) \oplus G(5, T)$.

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In Example 8.21, the domain space $\mathbf{C}^3$ is the direct sum of the generalized eigenspaces of the operator $T$ in that example. Our next result shows that this behavior holds in general. Specifically, the following major result shows that if $\mathbf{F} = \mathbf{C}$ and $T \in \mathcal{L}(V)$, then $V$ is the direct sum of the generalized eigenspaces of $T$, each of which is invariant under $T$ and on which $T$ is a nilpotent operator plus a scalar multiple of the identity. Thus the next result achieves our goal of decomposing $V$ into invariant subspaces on which $T$ has a known behavior.

As we will see, the proof follows from putting together what we have learned about generalized eigenspaces and then using our result that for each operator $T \in \mathcal{L}(V)$, there exists a basis of $V$ consisting of generalized eigenvectors of $T$.

**8.22** *generalized eigenspace decomposition*

Suppose $\mathbf{F} = \mathbf{C}$ and $T \in \mathcal{L}(V)$. Let $\lambda_1, \ldots, \lambda_m$ be the distinct eigenvalues of $T$. Then

(a) $G(\lambda_k, T)$ is invariant under $T$ for each $k = 1, \ldots, m$;

(b) $(T - \lambda_k I)|_{G(\lambda_k, T)}$ is nilpotent for each $k = 1, \ldots, m$;

(c) $V = G(\lambda_1, T) \oplus \cdots \oplus G(\lambda_m, T)$.

Proof

(a) Suppose $k \in \{1, \ldots, m\}$. Then 8.20 shows that

$$G(\lambda_k, T) = \operatorname{null}(T - \lambda_k I)^{\dim V}.$$

Thus 5.18, with $p(z) = (z - \lambda_k)^{\dim V}$, implies that $G(\lambda_k, T)$ is invariant under $T$, proving (a).

(b) Suppose $k \in \{1, \ldots, m\}$. If $v \in G(\lambda_k, T)$, then $(T - \lambda_k I)^{\dim V} v = 0$ (by 8.20). Thus $\left((T - \lambda_k I)|_{G(\lambda_k, T)}\right)^{\dim V} = 0$. Hence $(T - \lambda_k I)|_{G(\lambda_k, T)}$ is nilpotent, proving (b).

(c) To show that $G(\lambda_1, T) + \cdots + G(\lambda_m, T)$ is a direct sum, suppose

$$v_1 + \cdots + v_m = 0,$$

where each $v_k$ is in $G(\lambda_k, T)$. Because generalized eigenvectors of $T$ corresponding to distinct eigenvalues are linearly independent (by 8.12), this implies that each $v_k$ equals 0. Thus $G(\lambda_1, T) + \cdots + G(\lambda_m, T)$ is a direct sum (by 1.45).

Finally, each vector in $V$ can be written as a finite sum of generalized eigenvectors of $T$ (by 8.9). Thus

$$V = G(\lambda_1, T) \oplus \cdots \oplus G(\lambda_m, T),$$

proving (c). ∎

For the analogous result when $\mathbf{F} = \mathbf{R}$, see Exercise 8.

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### *Multiplicity of an Eigenvalue*

If $V$ is a complex vector space and $T \in \mathcal{L}(V)$, then the decomposition of $V$ provided by the generalized eigenspace decomposition (8.22) can be a powerful tool. The dimensions of the subspaces involved in this decomposition are sufficiently important to get a name, which is given in the next definition.

**8.23 definition: *multiplicity***

- Suppose $T \in \mathcal{L}(V)$. The *multiplicity* of an eigenvalue $\lambda$ of $T$ is defined to be the dimension of the corresponding generalized eigenspace $G(\lambda, T)$.
- In other words, the multiplicity of an eigenvalue $\lambda$ of $T$ equals

$$\dim \operatorname{null}(T - \lambda I)^{\dim V}.$$

The second bullet point above holds because $G(\lambda, T) = \operatorname{null}(T - \lambda I)^{\dim V}$ (see 8.20).

**8.24 example: *multiplicity of each eigenvalue of an operator***

Suppose $T \in \mathcal{L}(\mathbf{C}^3)$ is defined by

$$T(z_1, z_2, z_3) = (6z_1 + 3z_2 + 4z_3, 6z_2 + 2z_3, 7z_3).$$

The matrix of $T$ (with respect to the standard basis) is

$$\begin{pmatrix} 6 & 3 & 4 \\ 0 & 6 & 2 \\ 0 & 0 & 7 \end{pmatrix}.$$

The eigenvalues of $T$ are the diagonal entries 6 and 7, as follows from 5.41. You can verify that the generalized eigenspaces of $T$ are as follows:

$$G(6, T) = \operatorname{span}((1, 0, 0), (0, 1, 0)) \quad \text{and} \quad G(7, T) = \operatorname{span}((10, 2, 1)).$$

Thus the eigenvalue 6 has multiplicity 2 and the eigenvalue 7 has multiplicity 1. The direct sum $\mathbf{C}^3 = G(6, T) \oplus G(7, T)$ is the generalized eigenspace decomposition promised by 8.22. A basis of $\mathbf{C}^3$ consisting of generalized eigenvectors of $T$, as promised by 8.9, is $(1, 0, 0), (0, 1, 0), (10, 2, 1)$. There does not exist a basis of $\mathbf{C}^3$ consisting of eigenvectors of this operator.

*In this example, the multiplicity of each eigenvalue equals the number of times that eigenvalue appears on the diagonal of an upper-triangular matrix representing the operator. This behavior always happens, as we will see in 8.31.*

In the example above, the sum of the multiplicities of the eigenvalues of $T$ equals 3, which is the dimension of the domain of $T$. The next result shows that this holds for all operators on finite-dimensional complex vector spaces.

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**8.25** *sum of the multiplicities equals* $\dim V$

Suppose $\mathbf{F} = \mathbf{C}$ and $T \in \mathcal{L}(V)$. Then the sum of the multiplicities of all eigenvalues of $T$ equals $\dim V$.

Proof The desired result follows from the generalized eigenspace decomposition (8.22) and the formula for the dimension of a direct sum (see 3.94). ∎

The terms *algebraic multiplicity* and *geometric multiplicity* are used in some books. In case you encounter this terminology, be aware that the algebraic multiplicity is the same as the multiplicity defined here and the geometric multiplicity is the dimension of the corresponding eigenspace. In other words, if $T \in \mathcal{L}(V)$ and $\lambda$ is an eigenvalue of $T$, then

$$\text{algebraic multiplicity of } \lambda = \dim \operatorname{null}(T - \lambda I)^{\dim V} = \dim G(\lambda, T),$$

$$\text{geometric multiplicity of } \lambda = \dim \operatorname{null}(T - \lambda I) = \dim E(\lambda, T).$$

Note that as defined above, the algebraic multiplicity also has a geometric meaning as the dimension of a certain null space. The definition of multiplicity given here is cleaner than the traditional definition that involves determinants; 9.62 implies that these definitions are equivalent.

If $V$ is an inner product space, $T \in \mathcal{L}(V)$ is normal, and $\lambda$ is an eigenvalue of $T$, then the algebraic multiplicity of $\lambda$ equals the geometric multiplicity of $\lambda$, as can be seen from applying Exercise 27 in Section 7A to the normal operator $T - \lambda I$. As a special case, the singular values of $S \in \mathcal{L}(V, W)$ (here $V$ and $W$ are both finite-dimensional inner product spaces) depend on the multiplicities (either algebraic or geometric) of the eigenvalues of the self-adjoint operator $S^*S$.

The next definition associates a monic polynomial with each operator on a finite-dimensional complex vector space.

**8.26** definition: *characteristic polynomial*

Suppose $\mathbf{F} = \mathbf{C}$ and $T \in \mathcal{L}(V)$. Let $\lambda_1, \ldots, \lambda_m$ denote the distinct eigenvalues of $T$, with multiplicities $d_1, \ldots, d_m$. The polynomial

$$(z - \lambda_1)^{d_1} \cdots (z - \lambda_m)^{d_m}$$

is called the *characteristic polynomial* of $T$.

**8.27** example: *the characteristic polynomial of an operator*

Suppose $T \in \mathcal{L}(\mathbf{C}^3)$ is defined as in Example 8.24. Because the eigenvalues of $T$ are 6, with multiplicity 2, and 7, with multiplicity 1, we see that the characteristic polynomial of $T$ is $(z - 6)^2(z - 7)$.

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**8.28** *degree and zeros of characteristic polynomial*

Suppose $\mathbf{F} = \mathbf{C}$ and $T \in \mathcal{L}(V)$. Then

(a) the characteristic polynomial of $T$ has degree $\dim V$;

(b) the zeros of the characteristic polynomial of $T$ are the eigenvalues of $T$.

Proof Our result about the sum of the multiplicities (8.25) implies (a). The definition of the characteristic polynomial implies (b). ∎

Most texts define the characteristic polynomial using determinants (the two definitions are equivalent by 9.62). The approach taken here, which is considerably simpler, leads to the following nice proof of the Cayley–Hamilton theorem.

**8.29** *Cayley–Hamilton theorem*

Suppose $\mathbf{F} = \mathbf{C}$, $T \in \mathcal{L}(V)$, and $q$ is the characteristic polynomial of $T$. Then $q(T) = 0$.

Proof Let $\lambda_1, \ldots, \lambda_m$ be the distinct eigenvalues of $T$, and let $d_k = \dim G(\lambda_k, T)$. For each $k \in \{1, \ldots, m\}$, we know that $(T - \lambda_k I)|_{G(\lambda_k, T)}$ is nilpotent. Thus we have

$$(T - \lambda_k I)^{d_k}|_{G(\lambda_k, T)} = 0$$

(by 8.16) for each $k \in \{1, \ldots, m\}$.

*Arthur Cayley (1821–1895) published three mathematics papers before completing his undergraduate degree.*

The generalized eigenspace decomposition (8.22) states that every vector in $V$ is a sum of vectors in $G(\lambda_1, T), \ldots, G(\lambda_m, T)$. Thus to prove that $q(T) = 0$, we only need to show that $q(T)|_{G(\lambda_k, T)} = 0$ for each $k$.

Fix $k \in \{1, \ldots, m\}$. We have

$$q(T) = (T - \lambda_1 I)^{d_1} \cdots (T - \lambda_m I)^{d_m}.$$

The operators on the right side of the equation above all commute, so we can move the factor $(T - \lambda_k I)^{d_k}$ to be the last term in the expression on the right. Because $(T - \lambda_k I)^{d_k}|_{G(\lambda_k, T)} = 0$, we have $q(T)|_{G(\lambda_k, T)} = 0$, as desired. ∎

The next result implies that if the minimal polynomial of an operator $T \in \mathcal{L}(V)$ has degree $\dim V$ (as happens almost always—see the paragraphs following 5.24), then the characteristic polynomial of $T$ equals the minimal polynomial of $T$.

**8.30** *characteristic polynomial is a multiple of minimal polynomial*

Suppose $\mathbf{F} = \mathbf{C}$ and $T \in \mathcal{L}(V)$. Then the characteristic polynomial of $T$ is a polynomial multiple of the minimal polynomial of $T$.

Proof The desired result follows immediately from the Cayley–Hamilton theorem (8.29) and 5.29. ∎

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Now we can prove that the result suggested by Example 8.24 holds for all operators on finite-dimensional complex vector spaces.

**8.31** *multiplicity of an eigenvalue equals number of times on diagonal*

Suppose $\mathbf{F} = \mathbf{C}$ and $T \in \mathcal{L}(V)$. Suppose $v_1, ..., v_n$ is a basis of $V$ such that $\mathcal{M}(T, (v_1, ..., v_n))$ is upper triangular. Then the number of times that each eigenvalue $\lambda$ of $T$ appears on the diagonal of $\mathcal{M}(T, (v_1, ..., v_n))$ equals the multiplicity of $\lambda$ as an eigenvalue of $T$.

Proof Let $A = \mathcal{M}(T, (v_1, ..., v_n))$. Thus $A$ is an upper-triangular matrix. Let $\lambda_1, ..., \lambda_n$ denote the entries on the diagonal of $A$. Thus for each $k \in \{1, ..., n\}$, we have

$$Tv_k = u_k + \lambda_k v_k$$

for some $u_k \in \text{span}(v_1, ..., v_{k-1})$. Hence if $k \in \{1, ..., n\}$ and $\lambda_k \neq 0$, then $Tv_k$ is not a linear combination of $Tv_1, ..., Tv_{k-1}$. The linear dependence lemma (2.19) now implies that the list of those $Tv_k$ such that $\lambda_k \neq 0$ is linearly independent.

Let $d$ denote the number of indices $k \in \{1, ..., n\}$ such that $\lambda_k = 0$. The conclusion of the previous paragraph implies that

$$\dim \text{range } T \geq n - d.$$

Because $n = \dim V = \dim \text{null } T + \dim \text{range } T$, the inequality above implies that

8.32 $$\dim \text{null } T \leq d.$$

The matrix of the operator $T^n$ with respect to the basis $v_1, ..., v_n$ is the upper-triangular matrix $A^n$, which has diagonal entries $\lambda_1^n, ..., \lambda_n^n$ [see Exercise 2(b) in Section 5C]. Because $\lambda_k^n = 0$ if and only if $\lambda_k = 0$, the number of times that 0 appears on the diagonal of $A^n$ equals $d$. Thus applying 8.32 with $T$ replaced with $T^n$, we have

8.33 $$\dim \text{null } T^n \leq d.$$

For $\lambda$ an eigenvalue of $T$, let $m_\lambda$ denote the multiplicity of $\lambda$ as an eigenvalue of $T$ and let $d_\lambda$ denote the number of times that $\lambda$ appears on the diagonal of $A$. Replacing $T$ in 8.33 with $T - \lambda I$, we see that

8.34 $$m_\lambda \leq d_\lambda$$

for each eigenvalue $\lambda$ of $T$. The sum of the multiplicities $m_\lambda$ over all eigenvalues $\lambda$ of $T$ equals $n$, the dimension of $V$ (by 8.25). The sum of the numbers $d_\lambda$ over all eigenvalues $\lambda$ of $T$ also equals $n$, because the diagonal of $A$ has length $n$.

Thus summing both sides of 8.34 over all eigenvalues $\lambda$ of $T$ produces an equality. Hence 8.34 must actually be an equality for each eigenvalue $\lambda$ of $T$. Thus the multiplicity of $\lambda$ as an eigenvalue of $T$ equals the number of times that $\lambda$ appears on the diagonal of $A$, as desired. ∎

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## *Block Diagonal Matrices*

*Often we can understand a matrix better by thinking of it as composed of smaller matrices.*

To interpret our results in matrix form, we make the following definition, generalizing the notion of a diagonal matrix. If each matrix $A_k$ in the definition below is a 1-by-1 matrix, then we actually have a diagonal matrix.

**8.35 definition:** *block diagonal matrix*

A *block diagonal matrix* is a square matrix of the form

$$\begin{pmatrix} A_1 & & 0 \\ & \ddots & \\ 0 & & A_m \end{pmatrix},$$

where $A_1, \ldots, A_m$ are square matrices lying along the diagonal and all other entries of the matrix equal 0.

**8.36 example:** *a block diagonal matrix*

The 5-by-5 matrix

$$A = \begin{pmatrix} \begin{pmatrix} 4 \end{pmatrix} & \begin{matrix} 0 & 0 \end{matrix} & \begin{matrix} 0 & 0 \end{matrix} \\ \begin{matrix} 0 \\ 0 \\ 0 \end{matrix} & \begin{pmatrix} 2 & -3 \\ 0 & 2 \end{pmatrix} & \begin{matrix} 0 & 0 \\ 0 & 0 \end{matrix} \\ \begin{matrix} 0 \\ 0 \end{matrix} & \begin{matrix} 0 & 0 \\ 0 & 0 \end{matrix} & \begin{pmatrix} 1 & 7 \\ 0 & 1 \end{pmatrix} \end{pmatrix}$$

is a block diagonal matrix with

$$A = \begin{pmatrix} A_1 & & 0 \\ & A_2 & \\ 0 & & A_3 \end{pmatrix},$$

where

$$A_1 = \begin{pmatrix} 4 \end{pmatrix}, \quad A_2 = \begin{pmatrix} 2 & -3 \\ 0 & 2 \end{pmatrix}, \quad A_3 = \begin{pmatrix} 1 & 7 \\ 0 & 1 \end{pmatrix}.$$

Here the inner matrices in the 5-by-5 matrix above are blocked off to show how we can think of it as a block diagonal matrix.

---

Note that in the example above, each of $A_1$, $A_2$, $A_3$ is an upper-triangular matrix whose diagonal entries are all equal. The next result shows that with respect to an appropriate basis, every operator on a finite-dimensional complex vector space has a matrix of this form. Note that this result gives us many more zeros in the matrix than are needed to make it upper triangular.

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**8.37** *block diagonal matrix with upper-triangular blocks*

Suppose $\mathbf{F} = \mathbf{C}$ and $T \in \mathcal{L}(V)$. Let $\lambda_1, \ldots, \lambda_m$ be the distinct eigenvalues of $T$, with multiplicities $d_1, \ldots, d_m$. Then there is a basis of $V$ with respect to which $T$ has a block diagonal matrix of the form

$$\begin{pmatrix} A_1 & & 0 \\ & \ddots & \\ 0 & & A_m \end{pmatrix},$$

where each $A_k$ is a $d_k$-by-$d_k$ upper-triangular matrix of the form

$$A_k = \begin{pmatrix} \lambda_k & & * \\ & \ddots & \\ 0 & & \lambda_k \end{pmatrix}.$$

Proof Each $(T - \lambda_k I)|_{G(\lambda_k, T)}$ is nilpotent (see 8.22). For each $k$, choose a basis of $G(\lambda_k, T)$, which is a vector space of dimension $d_k$, such that the matrix of $(T - \lambda_k I)|_{G(\lambda_k, T)}$ with respect to this basis is as in 8.18(c). Thus with respect to this basis, the matrix of $T|_{G(\lambda_k, T)}$, which equals $(T - \lambda_k I)|_{G(\lambda_k, T)} + \lambda_k I|_{G(\lambda_k, T)}$, looks like the desired form shown above for $A_k$.

The generalized eigenspace decomposition (8.22) shows that putting together the bases of the $G(\lambda_k, T)$'s chosen above gives a basis of $V$. The matrix of $T$ with respect to this basis has the desired form. ∎

**8.38** example: *block diagonal matrix via generalized eigenvectors*

Let $T \in \mathcal{L}(\mathbf{C}^3)$ be defined by $T(z_1, z_2, z_3) = (6z_1 + 3z_2 + 4z_3, 6z_2 + 2z_3, 7z_3)$. The matrix of $T$ (with respect to the standard basis) is

$$\begin{pmatrix} 6 & 3 & 4 \\ 0 & 6 & 2 \\ 0 & 0 & 7 \end{pmatrix},$$

which is an upper-triangular matrix but is not of the form promised by 8.37.

As we saw in Example 8.24, the eigenvalues of $T$ are 6 and 7, and

$$G(6, T) = \operatorname{span}((1, 0, 0), (0, 1, 0)) \quad \text{and} \quad G(7, T) = \operatorname{span}((10, 2, 1)).$$

We also saw that a basis of $\mathbf{C}^3$ consisting of generalized eigenvectors of $T$ is

$$(1, 0, 0), (0, 1, 0), (10, 2, 1).$$

The matrix of $T$ with respect to this basis is

$$\begin{pmatrix} \begin{pmatrix} 6 & 3 \\ 0 & 6 \end{pmatrix} & \begin{matrix} 0 \\ 0 \end{matrix} \\ \begin{matrix} 0 & 0 \end{matrix} & \begin{pmatrix} 7 \end{pmatrix} \end{pmatrix},$$

which is a matrix of the block diagonal form promised by 8.37.

## *Exercises 8B*

**1** Define $T \in \mathcal{L}(\mathbf{C}^2)$ by $T(w, z) = (-z, w)$. Find the generalized eigenspaces corresponding to the distinct eigenvalues of $T$.

**2** Suppose $T \in \mathcal{L}(V)$ is invertible. Prove that $G(\lambda, T) = G\left(\frac{1}{\lambda}, T^{-1}\right)$ for every $\lambda \in \mathbf{F}$ with $\lambda \neq 0$.

**3** Suppose $T \in \mathcal{L}(V)$. Suppose $S \in \mathcal{L}(V)$ is invertible. Prove that $T$ and $S^{-1}TS$ have the same eigenvalues with the same multiplicities.

**4** Suppose $\dim V \geq 2$ and $T \in \mathcal{L}(V)$ is such that $\operatorname{null} T^{\dim V - 2} \neq \operatorname{null} T^{\dim V - 1}$. Prove that $T$ has at most two distinct eigenvalues.

**5** Suppose $T \in \mathcal{L}(V)$ and 3 and 8 are eigenvalues of $T$. Let $n = \dim V$. Prove that $V = (\operatorname{null} T^{n-2}) \oplus (\operatorname{range} T^{n-2})$.

**6** Suppose $T \in \mathcal{L}(V)$ and $\lambda$ is an eigenvalue of $T$. Explain why the exponent of $z - \lambda$ in the factorization of the minimal polynomial of $T$ is the smallest positive integer $m$ such that $(T - \lambda I)^m|_{G(\lambda, T)} = 0$.

**7** Suppose $T \in \mathcal{L}(V)$ and $\lambda$ is an eigenvalue of $T$ with multiplicity $d$. Prove that $G(\lambda, T) = \operatorname{null}(T - \lambda I)^d$.

*If $d < \dim V$, then this exercise improves 8.20.*

**8** Suppose $T \in \mathcal{L}(V)$ and $\lambda_1, \ldots, \lambda_m$ are the distinct eigenvalues of $T$. Prove that

$$V = G(\lambda_1, T) \oplus \cdots \oplus G(\lambda_m, T)$$

if and only if the minimal polynomial of $T$ equals $(z - \lambda_1)^{k_1} \cdots (z - \lambda_m)^{k_m}$ for some positive integers $k_1, \ldots, k_m$.

*The case $\mathbf{F} = \mathbf{C}$ follows immediately from 5.27(b) and the generalized eigenspace decomposition (8.22); thus this exercise is interesting only when $\mathbf{F} = \mathbf{R}$.*

**9** Suppose $\mathbf{F} = \mathbf{C}$ and $T \in \mathcal{L}(V)$. Prove that there exist $D, N \in \mathcal{L}(V)$ such that $T = D + N$, the operator $D$ is diagonalizable, $N$ is nilpotent, and $DN = ND$.

**10** Suppose $V$ is a complex inner product space, $e_1, \ldots, e_n$ is an orthonormal basis of $V$, and $T \in \mathcal{L}(V)$. Let $\lambda_1, \ldots, \lambda_n$ be the eigenvalues of $T$, each included as many times as its multiplicity. Prove that

$$|\lambda_1|^2 + \cdots + |\lambda_n|^2 \leq \|Te_1\|^2 + \cdots + \|Te_n\|^2.$$

*See the comment after Exercise 5 in Section 7A.*

**11** Give an example of an operator on $\mathbf{C}^4$ whose characteristic polynomial equals $(z - 7)^2(z - 8)^2$.

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**12** Give an example of an operator on $\mathbf{C}^4$ whose characteristic polynomial equals $(z-1)(z-5)^3$ and whose minimal polynomial equals $(z-1)(z-5)^2$.

**13** Give an example of an operator on $\mathbf{C}^4$ whose characteristic and minimal polynomials both equal $z(z-1)^2(z-3)$.

**14** Give an example of an operator on $\mathbf{C}^4$ whose characteristic polynomial equals $z(z-1)^2(z-3)$ and whose minimal polynomial equals $z(z-1)(z-3)$.

**15** Let $T$ be the operator on $\mathbf{C}^4$ defined by $T(z_1, z_2, z_3, z_4) = (0, z_1, z_2, z_3)$. Find the characteristic polynomial and the minimal polynomial of $T$.

**16** Let $T$ be the operator on $\mathbf{C}^6$ defined by

$$T(z_1, z_2, z_3, z_4, z_5, z_6) = (0, z_1, z_2, 0, z_4, 0).$$

Find the characteristic polynomial and the minimal polynomial of $T$.

**17** Suppose $\mathbf{F} = \mathbf{C}$ and $P \in \mathcal{L}(V)$ is such that $P^2 = P$. Prove that the characteristic polynomial of $P$ is $z^m(z-1)^n$, where $m = \dim \operatorname{null} P$ and $n = \dim \operatorname{range} P$.

**18** Suppose $T \in \mathcal{L}(V)$ and $\lambda$ is an eigenvalue of $T$. Explain why the following four numbers equal each other.

(a) The exponent of $z - \lambda$ in the factorization of the minimal polynomial of $T$.

(b) The smallest positive integer $m$ such that $(T - \lambda I)^m|_{G(\lambda, T)} = 0$.

(c) The smallest positive integer $m$ such that

$$\operatorname{null}(T - \lambda I)^m = \operatorname{null}(T - \lambda I)^{m+1}.$$

(d) The smallest positive integer $m$ such that

$$\operatorname{range}(T - \lambda I)^m = \operatorname{range}(T - \lambda I)^{m+1}.$$

**19** Suppose $\mathbf{F} = \mathbf{C}$ and $S \in \mathcal{L}(V)$ is a unitary operator. Prove that the constant term in the characteristic polynomial of $S$ has absolute value 1.

**20** Suppose that $\mathbf{F} = \mathbf{C}$ and $V_1, \ldots, V_m$ are nonzero subspaces of $V$ such that

$$V = V_1 \oplus \cdots \oplus V_m.$$

Suppose $T \in \mathcal{L}(V)$ and each $V_k$ is invariant under $T$. For each $k$, let $p_k$ denote the characteristic polynomial of $T|_{V_k}$. Prove that the characteristic polynomial of $T$ equals $p_1 \cdots p_m$.

**21** Suppose $p, q \in \mathcal{P}(\mathbf{C})$ are monic polynomials with the same zeros and $q$ is a polynomial multiple of $p$. Prove that there exists $T \in \mathcal{L}(\mathbf{C}^{\deg q})$ such that the characteristic polynomial of $T$ is $q$ and the minimal polynomial of $T$ is $p$.

*This exercise implies that every monic polynomial is the characteristic polynomial of some operator.*

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**22** Suppose $A$ and $B$ are block diagonal matrices of the form

$$A = \begin{pmatrix} A_1 & & 0 \\ & \ddots & \\ 0 & & A_m \end{pmatrix}, \quad B = \begin{pmatrix} B_1 & & 0 \\ & \ddots & \\ 0 & & B_m \end{pmatrix},$$

where $A_k$ and $B_k$ are square matrices of the same size for each $k = 1, \ldots, m$. Show that $AB$ is a block diagonal matrix of the form

$$AB = \begin{pmatrix} A_1B_1 & & 0 \\ & \ddots & \\ 0 & & A_mB_m \end{pmatrix}.$$

**23** Suppose $\mathbf{F} = \mathbf{R}$, $T \in \mathcal{L}(V)$, and $\lambda \in \mathbf{C}$.

(a) Show that $u + iv \in G(\lambda, T_{\mathbf{C}})$ if and only if $u - iv \in G(\overline{\lambda}, T_{\mathbf{C}})$.

(b) Show that the multiplicity of $\lambda$ as an eigenvalue of $T_{\mathbf{C}}$ equals the multiplicity of $\overline{\lambda}$ as an eigenvalue of $T_{\mathbf{C}}$.

(c) Use (b) and the result about the sum of the multiplicities (8.25) to show that if $\dim V$ is an odd number, then $T_{\mathbf{C}}$ has a real eigenvalue.

(d) Use (c) and the result about real eigenvalues of $T_{\mathbf{C}}$ (Exercise 17 in Section 5A) to show that if $\dim V$ is an odd number, then $T$ has an eigenvalue (thus giving an alternative proof of 5.34).

*See Exercise 33 in Section 3B for the definition of the complexification $T_{\mathbf{C}}$.*

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# 8C Consequences of Generalized Eigenspace Decomposition

## Square Roots of Operators

Recall that a square root of an operator $T \in \mathcal{L}(V)$ is an operator $R \in \mathcal{L}(V)$ such that $R^2 = T$ (see 7.36). Every complex number has a square root, but not every operator on a complex vector space has a square root. For example, the operator on $\mathbf{C}^3$ defined by $T(z_1, z_2, z_3) = (z_2, z_3, 0)$ does not have a square root, as you are asked to show in Exercise 1. The noninvertibility of that operator is no accident, as we will soon see. We begin by showing that the identity plus any nilpotent operator has a square root.

**8.39** *identity plus nilpotent has a square root*

Suppose $T \in \mathcal{L}(V)$ is nilpotent. Then $I + T$ has a square root.

Proof Consider the Taylor series for the function $\sqrt{1+x}$:

8.40 $$\sqrt{1+x} = 1 + a_1 x + a_2 x^2 + \cdots.$$

We do not find an explicit formula for the coefficients or worry about whether the infinite sum converges because we use this equation only as motivation.

*Because $a_1 = \frac{1}{2}$, the formula above implies that $1 + \frac{x}{2}$ is a good estimate for $\sqrt{1+x}$ when $x$ is small.*

Because $T$ is nilpotent, $T^m = 0$ for some positive integer $m$. In 8.40, suppose we replace $x$ with $T$ and 1 with $I$. Then the infinite sum on the right side becomes a finite sum (because $T^k = 0$ for all $k \geq m$). Thus we guess that there is a square root of $I + T$ of the form

$$I + a_1 T + a_2 T^2 + \cdots + a_{m-1} T^{m-1}.$$

Having made this guess, we can try to choose $a_1, a_2, \ldots, a_{m-1}$ such that the operator above has its square equal to $I + T$. Now

$$\begin{aligned}(I + a_1 T + a_2 T^2 + a_3 T^3 + \cdots + a_{m-1} T^{m-1})^2 \\ = I + 2a_1 T + (2a_2 + a_1^2) T^2 + (2a_3 + 2a_1 a_2) T^3 + \cdots \\ + (2a_{m-1} + \text{terms involving } a_1, \ldots, a_{m-2}) T^{m-1}.\end{aligned}$$

We want the right side of the equation above to equal $I + T$. Hence choose $a_1$ such that $2a_1 = 1$ (thus $a_1 = 1/2$). Next, choose $a_2$ such that $2a_2 + a_1^2 = 0$ (thus $a_2 = -1/8$). Then choose $a_3$ such that the coefficient of $T^3$ on the right side of the equation above equals 0 (thus $a_3 = 1/16$). Continue in this fashion for each $k = 4, \ldots, m-1$, at each step solving for $a_k$ so that the coefficient of $T^k$ on the right side of the equation above equals 0. Actually we do not care about the explicit formula for the $a_k$'s. We only need to know that some choice of the $a_k$'s gives a square root of $I + T$. ∎

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The previous lemma is valid on real and complex vector spaces. However, the result below holds only on complex vector spaces. For example, the operator of multiplication by $-1$ on the one-dimensional real vector space $\mathbf{R}$ has no square root.

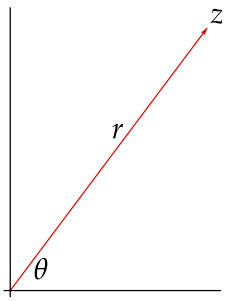
For the proof below, we need to know that every $z \in \mathbf{C}$ has a square root in $\mathbf{C}$. To show this, write

$$z = r(\cos\theta + i\sin\theta),$$

where $r$ is the length of the line segment in the complex plane from the origin to $z$ and $\theta$ is the angle of that line segment with the positive horizontal axis. Then

$$\sqrt{r}\left(\cos\tfrac{\theta}{2} + i\sin\tfrac{\theta}{2}\right)$$

is a square root of $z$, as you can verify by showing that the square of the complex number above equals $z$.




*Representation of a complex number with polar coordinates.*

**8.41** *over* $\mathbf{C}$*, invertible operators have square roots*

Suppose $V$ is a complex vector space and $T \in \mathcal{L}(V)$ is invertible. Then $T$ has a square root.

Proof Let $\lambda_1, \ldots, \lambda_m$ be the distinct eigenvalues of $T$. For each $k$, there exists a nilpotent operator $T_k \in \mathcal{L}(G(\lambda_k, T))$ such that $T|_{G(\lambda_k,T)} = \lambda_k I + T_k$ [see 8.22(c)]. Because $T$ is invertible, none of the $\lambda_k$'s equals 0, so we can write

$$T|_{G(\lambda_k,T)} = \lambda_k\left(I + \frac{T_k}{\lambda_k}\right)$$

for each $k$. Because $T_k/\lambda_k$ is nilpotent, $I + T_k/\lambda_k$ has a square root (by 8.39). Multiplying a square root of the complex number $\lambda_k$ by a square root of $I + T_k/\lambda_k$, we obtain a square root $R_k$ of $T|_{G(\lambda_k,T)}$.

By the generalized eigenspace decomposition (8.22), a typical vector $v \in V$ can be written uniquely in the form

$$v = u_1 + \cdots + u_m,$$

where each $u_k$ is in $G(\lambda_k, T)$. Using this decomposition, define an operator $R \in \mathcal{L}(V)$ by

$$Rv = R_1u_1 + \cdots + R_mu_m.$$

You should verify that this operator $R$ is a square root of $T$, completing the proof. ∎

By imitating the techniques in this subsection, you should be able to prove that if $V$ is a complex vector space and $T \in \mathcal{L}(V)$ is invertible, then $T$ has a $k^{\text{th}}$ root for every positive integer $k$.

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### *Jordan Form*

We know that if $V$ is a complex vector space, then for every $T \in \mathcal{L}(V)$ there is a basis of $V$ with respect to which $T$ has a nice upper-triangular matrix (see 8.37). In this subsection we will see that we can do even better—there is a basis of $V$ with respect to which the matrix of $T$ contains 0's everywhere except possibly on the diagonal and the line directly above the diagonal.

We begin by looking at two examples of nilpotent operators.

**8.42 example:** *nilpotent operator with nice matrix*

Let $T$ be the operator on $\mathbf{C}^4$ defined by

$$T(z_1, z_2, z_3, z_4) = (0, z_1, z_2, z_3).$$

Then $T^4 = 0$; thus $T$ is nilpotent. If $v = (1, 0, 0, 0)$, then $T^3v, T^2v, Tv, v$ is a basis of $\mathbf{C}^4$. The matrix of $T$ with respect to this basis is

$$\begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix}.$$

The next example of a nilpotent operator has more complicated behavior than the example above.

**8.43 example:** *nilpotent operator with slightly more complicated matrix*

Let $T$ be the operator on $\mathbf{C}^6$ defined by

$$T(z_1, z_2, z_3, z_4, z_5, z_6) = (0, z_1, z_2, 0, z_4, 0).$$

Then $T^3 = 0$; thus $T$ is nilpotent. In contraast to the nice behavior of the nilpotent operator of the previous example, for this nilpotent operator there does not exist a vector $v \in \mathbf{C}^6$ such that $T^5v, T^4v, T^3v, T^2v, Tv, v$ is a basis of $\mathbf{C}^6$. However, if we take $v_1 = (1, 0, 0, 0, 0, 0)$, $v_2 = (0, 0, 0, 1, 0, 0)$, and $v_3 = (0, 0, 0, 0, 0, 1)$, then $T^2v_1, Tv_1, v_1, Tv_2, v_2, v_3$ is a basis of $\mathbf{C}^6$. The matrix of $T$ with respect to this basis is

$$\begin{pmatrix} \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix} & \begin{matrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \end{matrix} & \begin{matrix} 0 \\ 0 \\ 0 \end{matrix} \\ \begin{matrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{matrix} & \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} & \begin{matrix} 0 \\ 0 \end{matrix} \\ \begin{matrix} 0 & 0 & 0 \end{matrix} & \begin{matrix} 0 & 0 \end{matrix} & \begin{pmatrix} 0 \end{pmatrix} \end{pmatrix}.$$

Here the inner matrices are blocked off to show that we can think of the 6-by-6 matrix above as a block diagonal matrix consisting of a 3-by-3 block with 1's on the line above the diagonal and 0's elsewhere, a 2-by-2 block with 1 above the diagonal and 0's elsewhere, and a 1-by-1 block containing 0.

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Our next goal is to show that every nilpotent operator $T \in \mathcal{L}(V)$ behaves similarly to the operator in the previous example. Specifically, there is a finite collection of vectors $v_1, \ldots, v_n \in V$ such that there is a basis of $V$ consisting of the vectors of the form $T^j v_k$, as $k$ varies from 1 to $n$ and $j$ varies (in reverse order) from 0 to the largest nonnegative integer $m_k$ such that $T^{m_k} v_k \neq 0$. With respect to this basis, the matrix of $T$ looks like the matrix in the previous example. More specifically, $T$ has a block diagonal matrix with respect to this basis, with each block a square matrix that is 0 everywhere except on the line above the diagonal.

In the next definition, the diagonal of each $A_k$ is filled with some eigenvalue $\lambda_k$ of $T$, the line directly above the diagonal of $A_k$ is filled with 1's, and all other entries in $A_k$ are 0 (to understand why each $\lambda_k$ is an eigenvalue of $T$, see 5.41). The $\lambda_k$'s need not be distinct. Also, $A_k$ may be a 1-by-1 matrix $(\lambda_k)$ containing just an eigenvalue of $T$. If each $\lambda_k$ is 0, then the next definition captures the behavior described in the paragraph above (recall that if $T$ is nilpotent, then 0 is the only eigenvalue of $T$).

**8.44 definition: *Jordan basis***

Suppose $T \in \mathcal{L}(V)$. A basis of $V$ is called a *Jordan basis* for $T$ if with respect to this basis $T$ has a block diagonal matrix

$$\begin{pmatrix} A_1 & & 0 \\ & \ddots & \\ 0 & & A_p \end{pmatrix}$$

in which each $A_k$ is an upper-triangular matrix of the form

$$A_k = \begin{pmatrix} \lambda_k & 1 & & 0 \\ & \ddots & \ddots & \\ & & \ddots & 1 \\ 0 & & & \lambda_k \end{pmatrix}.$$

Most of the work in proving that every operator on a finite-dimensional complex vector space has a Jordan basis occurs in proving the special case below of nilpotent operators. This special case holds on real vector spaces as well as complex vector spaces.

**8.45 *every nilpotent operator has a Jordan basis***

Suppose $T \in \mathcal{L}(V)$ is nilpotent. Then there is a basis of $V$ that is a Jordan basis for $T$.

Proof We will prove this result by induction on $\dim V$. To get started, note that the desired result holds if $\dim V = 1$ (because in that case, the only nilpotent operator is the 0 operator). Now assume that $\dim V > 1$ and that the desired result holds on all vector spaces of smaller dimension.

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Let $m$ be the smallest positive integer such that $T^m = 0$. Thus there exists $u \in V$ such that $T^{m-1}u \neq 0$. Let

$$U = \operatorname{span}(u, Tu, \ldots, T^{m-1}u).$$

The list $u, Tu, \ldots, T^{m-1}u$ is linearly independent (see Exercise 2 in Section 8A). If $U = V$, then writing this list in reverse order gives a Jordan basis for $T$ and we are done. Thus we can assume that $U \neq V$.

Note that $U$ is invariant under $T$. By our induction hypothesis, there is a basis of $U$ that is a Jordan basis for $T|_U$. The strategy of our proof is that we will find a subspace $W$ of $V$ such that $W$ is also invariant under $T$ and $V = U \oplus W$. Again by our induction hypothesis, there will be a basis of $W$ that is a Jordan basis for $T|_W$. Putting together the Jordan bases for $T|_U$ and $T|_W$, we will have a Jordan basis for $T$.

Let $\varphi \in V'$ be such that $\varphi(T^{m-1}u) \neq 0$. Let

$$W = \big\{v \in V : \varphi(T^k v) = 0 \text{ for each } k = 0, \ldots, m-1\big\}.$$

Then $W$ is a subspace of $V$ that is invariant under $T$ $\big($the invariance holds because if $v \in W$ then $\varphi(T^k(Tv)) = 0$ for $k = 0, \ldots, m-1$, where the case $k = m-1$ holds because $T^m = 0\big)$. We will show that $V = U \oplus W$, which by the previous paragraph will complete the proof.

To show that $U + W$ is a direct sum, suppose $v \in U \cap W$ with $v \neq 0$. Because $v \in U$, there exist $c_0, \ldots, c_{m-1} \in \mathbf{F}$ such that

$$v = c_0 u + c_1 Tu + \cdots + c_{m-1}T^{m-1}u.$$

Let $j$ be the smallest index such that $c_j \neq 0$. Apply $T^{m-j-1}$ to both sides of the equation above, getting

$$T^{m-j-1}v = c_j T^{m-1}u,$$

where we have used the equation $T^m = 0$. Now apply $\varphi$ to both sides of the equation above, getting

$$\varphi(T^{m-j-1}v) = c_j \varphi(T^{m-1}u) \neq 0.$$

The equation above shows that $v \notin W$. Hence we have proved that $U \cap W = \{0\}$, which implies that $U + W$ is a direct sum (see 1.46).

To show that $U \oplus W = V$, define $S : V \to \mathbf{F}^m$ by

$$Sv = \big(\varphi(v), \varphi(Tv), \ldots, \varphi(T^{m-1}v)\big).$$

Thus $\operatorname{null} S = W$. Hence

$$\dim W = \dim \operatorname{null} S = \dim V - \dim \operatorname{range} S \geq \dim V - m,$$

where the second equality comes from the fundamental theorem of linear maps (3.21). Using the inequality above, we have

$$\dim(U \oplus W) = \dim U + \dim W \geq m + (\dim V - m) = \dim V.$$

Thus $U \oplus W = V$ (by 2.39), completing the proof. ∎

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*Camille Jordan (1838–1922) published a proof of 8.46 in 1870.*

Now the generalized eigenspace decomposition allows us to extend the previous result to operators that may not be nilpotent. Doing this requires that we deal with complex vector spaces.

**8.46 *Jordan form***

Suppose $\mathbf{F} = \mathbf{C}$ and $T \in \mathcal{L}(V)$. Then there is a basis of $V$ that is a Jordan basis for $T$.

Proof Let $\lambda_1, \ldots, \lambda_m$ be the distinct eigenvalues of $T$. The generalized eigenspace decomposition states that

$$V = G(\lambda_1, T) \oplus \cdots \oplus G(\lambda_m, T),$$

where each $(T - \lambda_k I)|_{G(\lambda_k, T)}$ is nilpotent (see 8.22). Thus 8.45 implies that some basis of each $G(\lambda_k, T)$ is a Jordan basis for $(T - \lambda_k I)|_{G(\lambda_k, T)}$. Put these bases together to get a basis of $V$ that is a Jordan basis for $T$. ∎

## *Exercises 8C*

**1** Suppose $T \in \mathcal{L}(\mathbf{C}^3)$ is the operator defined by $T(z_1, z_2, z_3) = (z_2, z_3, 0)$. Prove that $T$ does not have a square root.

**2** Define $T \in \mathcal{L}(\mathbf{F}^5)$ by $T(x_1, x_2, x_3, x_4, x_5) = (2x_2, 3x_3, -x_4, 4x_5, 0)$.

(a) Show that $T$ is nilpotent.

(b) Find a square root of $I + T$.

**3** Suppose $V$ is a complex vector space. Prove that every invertible operator on $V$ has a cube root.

**4** Suppose $V$ is a real vector space. Prove that the operator $-I$ on $V$ has a square root if and only if $\dim V$ is an even number.

**5** Suppose $T \in \mathcal{L}(\mathbf{C}^2)$ is the operator defined by $T(w, z) = (-w - z, 9w + 5z)$. Find a Jordan basis for $T$.

**6** Find a basis of $\mathcal{P}_4(\mathbf{R})$ that is a Jordan basis for the differentiation operator $D$ on $\mathcal{P}_4(\mathbf{R})$ defined by $Dp = p'$.

**7** Suppose $T \in \mathcal{L}(V)$ is nilpotent and $v_1, \ldots, v_n$ is a Jordan basis for $T$. Prove that the minimal polynomial of $T$ is $z^{m+1}$, where $m$ is the length of the longest consecutive string of 1's that appears on the line directly above the diagonal in the matrix of $T$ with respect to $v_1, \ldots, v_n$.

**8** Suppose $T \in \mathcal{L}(V)$ and $v_1, \ldots, v_n$ is a basis of $V$ that is a Jordan basis for $T$. Describe the matrix of $T^2$ with respect to this basis.

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**9** Suppose $T \in \mathcal{L}(V)$ is nilpotent. Explain why there exist $v_1, ..., v_n \in V$ and nonnegative integers $m_1, ..., m_n$ such that (a) and (b) below both hold.
(a) $T^{m_1}v_1, ..., Tv_1, v_1, ..., T^{m_n}v_n, ..., Tv_n, v_n$ is a basis of $V$.
(b) $T^{m_1+1}v_1 = \cdots = T^{m_n+1}v_n = 0$.

**10** Suppose $T \in \mathcal{L}(V)$ and $v_1, ..., v_n$ is a basis of $V$ that is a Jordan basis for $T$. Describe the matrix of $T$ with respect to the basis $v_n, ..., v_1$ obtained by reversing the order of the $v$'s.

**11** Suppose $T \in \mathcal{L}(V)$. Explain why every vector in each Jordan basis for $T$ is a generalized eigenvector of $T$.

**12** Suppose $T \in \mathcal{L}(V)$ is diagonalizable. Show that $\mathcal{M}(T)$ is a diagonal matrix with respect to every Jordan basis for $T$.

**13** Suppose $T \in \mathcal{L}(V)$ is nilpotent. Prove that if $v_1, ..., v_n$ are vectors in $V$ and $m_1, ..., m_n$ are nonnegative integers such that

$$T^{m_1}v_1, ..., Tv_1, v_1, ..., T^{m_n}v_n, ..., Tv_n, v_n \text{ is a basis of } V$$

and

$$T^{m_1+1}v_1 = \cdots = T^{m_n+1}v_n = 0,$$

then $T^{m_1}v_1, ..., T^{m_n}v_n$ is a basis of $\operatorname{null} T$.

*This exercise shows that $n = \dim \operatorname{null} T$. Thus the positive integer $n$ that appears above depends only on $T$ and not on the specific Jordan basis chosen for $T$.*

**14** Suppose $\mathbf{F} = \mathbf{C}$ and $T \in \mathcal{L}(V)$. Prove that there does not exist a direct sum decomposition of $V$ into two nonzero subspaces invariant under $T$ if and only if the minimal polynomial of $T$ is of the form $(z - \lambda)^{\dim V}$ for some $\lambda \in \mathbf{C}$.

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## 8D Trace: A Connection Between Matrices and Operators

We begin this section by defining the trace of a square matrix. After developing some properties of the trace of a square matrix, we will use this concept to define the trace of an operator.

**8.47 definition: *trace of a matrix***

Suppose $A$ is a square matrix with entries in $\mathbf{F}$. The *trace* of $A$, denoted by $\operatorname{tr} A$, is defined to be the sum of the diagonal entries of $A$.

**8.48 example: *trace of a 3-by-3 matrix***

Suppose

$$A = \begin{pmatrix} 3 & -1 & -2 \\ 3 & 2 & -3 \\ 1 & 2 & 0 \end{pmatrix}.$$

The diagonal entries of $A$, which are shown in red above, are 3, 2, and 0. Thus $\operatorname{tr} A = 3 + 2 + 0 = 5$.

Matrix multiplication is not commutative, but the next result shows that the order of matrix multiplication does not matter to the trace.

**8.49 *trace of $AB$ equals trace of $BA$***

Suppose $A$ is an $m$-by-$n$ matrix and $B$ is an $n$-by-$m$ matrix. Then

$$\operatorname{tr}(AB) = \operatorname{tr}(BA).$$

Proof Suppose

$$A = \begin{pmatrix} A_{1,1} & \cdots & A_{1,n} \\ \vdots & & \vdots \\ A_{m,1} & \cdots & A_{m,n} \end{pmatrix}, \quad B = \begin{pmatrix} B_{1,1} & \cdots & B_{1,m} \\ \vdots & & \vdots \\ B_{n,1} & \cdots & B_{n,m} \end{pmatrix}.$$

The $j^{\text{th}}$ term on the diagonal of the $m$-by-$m$ matrix $AB$ equals $\sum_{k=1}^{n} A_{j,k}B_{k,j}$. Thus

$$\begin{aligned} \operatorname{tr}(AB) &= \sum_{j=1}^{m} \sum_{k=1}^{n} A_{j,k}B_{k,j} \\ &= \sum_{k=1}^{n} \sum_{j=1}^{m} B_{k,j}A_{j,k} \\ &= \sum_{k=1}^{n} (k^{\text{th}} \text{ term on diagonal of the } n\text{-by-}n \text{ matrix } BA) \\ &= \operatorname{tr}(BA), \end{aligned}$$

as desired.

We want to define the trace of an operator $T \in \mathcal{L}(V)$ to be the trace of the matrix of $T$ with respect to some basis of $V$. However, this definition should not depend on the choice of basis. The following result will make this possible.

**8.50** *trace of matrix of operator does not depend on basis*

Suppose $T \in \mathcal{L}(V)$. Suppose $u_1, \ldots, u_n$ and $v_1, \ldots, v_n$ are bases of $V$. Then

$$\operatorname{tr} \mathcal{M}(T, (u_1, \ldots, u_n)) = \operatorname{tr} \mathcal{M}(T, (v_1, \ldots, v_n)).$$

Proof Let $A = \mathcal{M}(T, (u_1, \ldots, u_n))$ and $B = \mathcal{M}(T, (v_1, \ldots, v_n))$. The change-of-basis formula tells us that there exists an invertible $n$-by-$n$ matrix $C$ such that $A = C^{-1}BC$ (see 3.84). Thus

$$\begin{aligned} \operatorname{tr} A &= \operatorname{tr}\big((C^{-1}B)C\big) \\ &= \operatorname{tr}\big(C(C^{-1}B)\big) \\ &= \operatorname{tr}\big((CC^{-1})B\big) \\ &= \operatorname{tr} B, \end{aligned}$$

where the second line comes from 8.49. ∎

Because of 8.50, the following definition now makes sense.

**8.51** definition: *trace of an operator*

Suppose $T \in \mathcal{L}(V)$. The *trace* of $T$, denote $\operatorname{tr} T$, is defined by

$$\operatorname{tr} T = \operatorname{tr} \mathcal{M}(T, (v_1, \ldots, v_n)),$$

where $v_1, \ldots, v_n$ is any basis of $V$.

Suppose $T \in \mathcal{L}(V)$ and $\lambda$ is an eigenvalue of $T$. Recall that we defined the multiplicity of $\lambda$ to be the dimension of the generalized eigenspace $G(\lambda, T)$ (see 8.23); we proved that this multiplicity equals $\dim \operatorname{null}(T - \lambda I)^{\dim V}$ (see 8.20). Recall also that if $V$ is a complex vector space, then the sum of the multiplicities of all eigenvalues of $T$ equals $\dim V$ (see 8.25).

In the definition below, the sum of the eigenvalues "with each eigenvalue included as many times as its multiplicity" means that if $\lambda_1, \ldots, \lambda_m$ are the distinct eigenvalues of $T$ with multiplicities $d_1, \ldots, d_m$, then the sum is

$$d_1\lambda_1 + \cdots + d_m\lambda_m.$$

Or if you prefer to work with a list of not-necessarily-distinct eigenvalues, with each eigenvalue included as many times as its multiplicity, then the eigenvalues could be denoted by $\lambda_1, \ldots, \lambda_n$ (where $n$ equals $\dim V$) and the sum is

$$\lambda_1 + \cdots + \lambda_n.$$

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**8.52** *on complex vector spaces, trace equals sum of eigenvalues*

Suppose $\mathbf{F} = \mathbf{C}$ and $T \in \mathcal{L}(V)$. Then $\operatorname{tr} T$ equals the sum of the eigenvalues of $T$, with each eigenvalue included as many times as its multiplicity.

Proof There is a basis of $V$ with respect to which $T$ has an upper-triangular matrix with the diagonal entries of the matrix consisting of the eigenvalues of $T$, with each eigenvalue included as many times as its multiplicity—see 8.37. Thus the definition of the trace of an operator along with 8.50, which allows us to use a basis of our choice, implies that $\operatorname{tr} T$ equals the sum of the eigenvalues of $T$, with each eigenvalue included as many times as its multiplicity. ∎

**8.53** example: *trace of an operator on* $\mathbf{C}^3$

Suppose $T \in \mathcal{L}(\mathbf{C}^3)$ is defined by

$$T(z_1, z_2, z_3) = (3z_1 - z_2 - 2z_3, 3z_1 + 2z_2 - 3z_3, z_1 + 2z_2).$$

Then the matrix of $T$ with respect to the standard basis of $\mathbf{C}^3$ is

$$\begin{pmatrix} 3 & -1 & -2 \\ 3 & 2 & -3 \\ 1 & 2 & 0 \end{pmatrix}.$$

Adding up the diagonal entries of this matrix, we see that $\operatorname{tr} T = 5$.

The eigenvalues of $T$ are 1, $2 + 3i$, and $2 - 3i$, each with multiplicity 1, as you can verify. The sum of these eigenvalues, each included as many times as its multiplicity, is $1 + (2 + 3i) + (2 - 3i)$, which equals 5, as expected by 8.52.

The trace has a close connection with the characteristic polynomial. Suppose $\mathbf{F} = \mathbf{C}$, $T \in \mathcal{L}(V)$, and $\lambda_1, \ldots, \lambda_n$ are the eigenvalues of $T$, with each eigenvalue included as many times as its multiplicity. Then by definition (see 8.26), the characteristic polynomial of $T$ equals

$$(z - \lambda_1) \cdots (z - \lambda_n).$$

Expanding the polynomial above, we can write the characteristic polynomial of $T$ in the form

$$z^n - (\lambda_1 + \cdots + \lambda_n) z^{n-1} + \cdots + (-1)^n (\lambda_1 \cdots \lambda_n).$$

The expression above immediately leads to the next result. Also see 9.65, which does not require the hypothesis that $\mathbf{F} = \mathbf{C}$.

**8.54** *trace and characteristic polynomial*

Suppose $\mathbf{F} = \mathbf{C}$ and $T \in \mathcal{L}(V)$. Let $n = \dim V$. Then $\operatorname{tr} T$ equals the negative of the coefficient of $z^{n-1}$ in the characteristic polynomial of $T$.

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The next result gives a nice formula for the trace of an operator on an inner product space.

**8.55** *trace on an inner product space*

Suppose $V$ is an inner product space, $T \in \mathcal{L}(V)$, and $e_1, \ldots, e_n$ is an orthonormal basis of $V$. Then

$$\operatorname{tr} T = \langle Te_1, e_1 \rangle + \cdots + \langle Te_n, e_n \rangle.$$

Proof The desired formula follows from the observation that the entry in row $k$, column $k$ of $\mathcal{M}(T, (e_1, \ldots, e_n))$ equals $\langle Te_k, e_k \rangle$ [use 6.30(a) with $v = Te_k$]. ∎

The algebraic properties of the trace as defined on square matrices translate to algebraic properties of the trace as defined on operators, as shown in the next result.

**8.56** *trace is linear*

The function $\operatorname{tr} : \mathcal{L}(V) \to \mathbf{F}$ is a linear functional on $\mathcal{L}(V)$ such that

$$\operatorname{tr}(ST) = \operatorname{tr}(TS)$$

for all $S, T \in \mathcal{L}(V)$.

Proof Choose a basis of $V$. All matrices of operators in this proof will be with respect to that basis. Suppose $S, T \in \mathcal{L}(V)$.

If $\lambda \in \mathbf{F}$, then

$$\operatorname{tr}(\lambda T) = \operatorname{tr} \mathcal{M}(\lambda T) = \operatorname{tr}(\lambda \mathcal{M}(T)) = \lambda \operatorname{tr} \mathcal{M}(T) = \lambda \operatorname{tr} T,$$

where the first and last equalities come from the definition of the trace of an operator, the second equality comes from 3.38, and the third equality follows from the definition of the trace of a square matrix.

Also,

$$\operatorname{tr}(S+T) = \operatorname{tr} \mathcal{M}(S+T) = \operatorname{tr}(\mathcal{M}(S) + \mathcal{M}(T)) = \operatorname{tr} \mathcal{M}(S) + \operatorname{tr} \mathcal{M}(T) = \operatorname{tr} S + \operatorname{tr} T,$$

where the first and last equalities come from the definition of the trace of an operator, the second equality comes from 3.35, and the third equality follows from the definition of the trace of a square matrix. The two paragraphs above show that $\operatorname{tr} : \mathcal{L}(V) \to \mathbf{F}$ is a linear functional on $\mathcal{L}(V)$.

Furthermore,

$$\operatorname{tr}(ST) = \operatorname{tr} \mathcal{M}(ST) = \operatorname{tr}(\mathcal{M}(S)\mathcal{M}(T)) = \operatorname{tr}(\mathcal{M}(T)\mathcal{M}(S)) = \operatorname{tr} \mathcal{M}(TS) = \operatorname{tr}(TS),$$

where the second and fourth equalities come from 3.43 and the crucial third equality comes from 8.49. ∎

The equations $\operatorname{tr}(ST) = \operatorname{tr}(TS)$ and $\operatorname{tr} I = \dim V$ uniquely characterize the trace among the linear functionals on $\mathcal{L}(V)$—see Exercise 10.

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The equation $\operatorname{tr}(ST) = \operatorname{tr}(TS)$ leads to our next result, which does not hold on infinite-dimensional vector spaces (see Exercise 13). However, additional hypotheses on $S$, $T$, and $V$ lead to an infinite-dimensional generalization of the result below, with important applications to quantum theory.

*The statement of the next result does not involve traces, but the short proof uses traces. When something like this happens in mathematics, then usually a good definition lurks in the background.*

**8.57** *identity operator is not the difference of $ST$ and $TS$*

There do not exist operators $S, T \in \mathcal{L}(V)$ such that $ST - TS = I$.

Proof Suppose $S, T \in \mathcal{L}(V)$. Then

$$\operatorname{tr}(ST - TS) = \operatorname{tr}(ST) - \operatorname{tr}(TS) = 0,$$

where both equalities come from 8.56. The trace of $I$ equals $\dim V$, which is not 0. Because $ST - TS$ and $I$ have different traces, they cannot be equal. ∎

## *Exercises 8D*

**1** Suppose $V$ is an inner product space and $v, w \in V$. Define an operator $T \in \mathcal{L}(V)$ by $Tu = \langle u, v\rangle w$. Find a formula for $\operatorname{tr} T$.

**2** Suppose $P \in \mathcal{L}(V)$ satisfies $P^2 = P$. Prove that

$$\operatorname{tr} P = \dim \operatorname{range} P.$$

**3** Suppose $T \in \mathcal{L}(V)$ and $T^5 = T$. Prove that the real and imaginary parts of $\operatorname{tr} T$ are both integers.

**4** Suppose $V$ is an inner product space and $T \in \mathcal{L}(V)$. Prove that

$$\operatorname{tr} T^* = \overline{\operatorname{tr} T}.$$

**5** Suppose $V$ is an inner product space. Suppose $T \in \mathcal{L}(V)$ is a positive operator and $\operatorname{tr} T = 0$. Prove that $T = 0$.

**6** Suppose $V$ is an inner product space and $P, Q \in \mathcal{L}(V)$ are orthogonal projections. Prove that $\operatorname{tr}(PQ) \geq 0$.

**7** Suppose $T \in \mathcal{L}(\mathbf{C}^3)$ is the operator whose matrix is

$$\begin{pmatrix} 51 & -12 & -21 \\ 60 & -40 & -28 \\ 57 & -68 & 1 \end{pmatrix}.$$

Someone tells you (accurately) that $-48$ and 24 are eigenvalues of $T$. Without using a computer or writing anything down, find the third eigenvalue of $T$.

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**8** Prove or give a counterexample: If $S, T \in \mathcal{L}(V)$, then $\operatorname{tr}(ST) = (\operatorname{tr} S)(\operatorname{tr} T)$.

**9** Suppose $T \in \mathcal{L}(V)$ is such that $\operatorname{tr}(ST) = 0$ for all $S \in \mathcal{L}(V)$. Prove that $T = 0$.

**10** Prove that the trace is the only linear functional $\tau\colon \mathcal{L}(V) \to \mathbf{F}$ such that

$$\tau(ST) = \tau(TS)$$

for all $S, T \in \mathcal{L}(V)$ and $\tau(I) = \dim V$.

*Hint: Suppose that $v_1, \dots, v_n$ is a basis of $V$. For $j, k \in \{1, \dots, n\}$, define $P_{j,k} \in \mathcal{L}(V)$ by $P_{j,k}(a_1 v_1 + \cdots + a_n v_n) = a_k v_j$. Prove that*

$$\tau(P_{j,k}) = \begin{cases} 1 & \text{if } j = k, \\ 0 & \text{if } j \neq k. \end{cases}$$

*Then for $T \in \mathcal{L}(V)$, use the equation $T = \sum_{k=1}^{n} \sum_{j=1}^{n} \mathcal{M}(T)_{j,k} P_{j,k}$ to show that $\tau(T) = \operatorname{tr} T$.*

**11** Suppose $V$ and $W$ are inner product spaces and $T \in \mathcal{L}(V, W)$. Prove that if $e_1, \dots, e_n$ is an orthonormal basis of $V$ and $f_1, \dots, f_m$ is an orthonormal basis of $W$, then

$$\operatorname{tr}(T^* T) = \sum_{k=1}^{n} \sum_{j=1}^{m} |\langle T e_k, f_j \rangle|^2.$$

*The numbers $\langle T e_k, f_j \rangle$ are the entries of the matrix of $T$ with respect to the orthonormal bases $e_1, \dots, e_n$ and $f_1, \dots, f_m$. These numbers depend on the bases, but $\operatorname{tr}(T^* T)$ does not depend on a choice of bases. Thus this exercise shows that the sum of the squares of the absolute values of the matrix entries does not depend on which orthonormal bases are used.*

**12** Suppose $V$ and $W$ are finite-dimensional inner product spaces.

(a) Prove that $\langle S, T \rangle = \operatorname{tr}(T^* S)$ defines an inner product on $\mathcal{L}(V, W)$.

(b) Suppose $e_1, \dots, e_n$ is an orthonormal basis of $V$ and $f_1, \dots, f_m$ is an orthonormal basis of $W$. Show that the inner product on $\mathcal{L}(V, W)$ from (a) is the same as the standard inner product on $\mathbf{F}^{mn}$, where we identify each element of $\mathcal{L}(V, W)$ with its matrix (with respect to the bases just mentioned) and then with an element of $\mathbf{F}^{mn}$.

*Caution: The norm of a linear map $T \in \mathcal{L}(V, W)$ as defined by 7.86 is not the same as the norm that comes from the inner product in (a) above. Unless explicitly stated otherwise, always assume that $\|T\|$ refers to the norm as defined by 7.86. The norm that comes from the inner product in (a) is called the **Frobenius norm** or the **Hilbert–Schmidt norm**.*

**13** Find $S, T \in \mathcal{L}(\mathcal{P}(\mathbf{F}))$ such that $ST - TS = I$.

*Hint: Make an appropriate modification of the operators in Example 3.9.*

*This exercise shows that additional hypotheses are needed on $S$ and $T$ to extend 8.57 to the setting of infinite-dimensional vector spaces.*

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Chapter 9

# Multilinear Algebra and Determinants

We begin this chapter by investigating bilinear forms and quadratic forms on a vector space. Then we will move on to multilinear forms. We will show that the vector space of alternating $n$-linear forms has dimension one on a vector space of dimension $n$. This result will allow us to give a clean basis-free definition of the determinant of an operator.

This approach to the determinant via alternating multilinear forms leads to straightforward proofs of key properties of determinants. For example, we will see that the determinant is multiplicative, meaning that $\det(ST) = (\det S)(\det T)$ for all operators $S$ and $T$ on the same vector space. We will also see that $T$ is invertible if and only if $\det T \neq 0$. Another important result states that the determinant of an operator on a complex vector space equals the product of the eigenvalues of the operator, with each eigenvalue included as many times as its multiplicity.

The chapter concludes with an introduction to tensor products.

*standing assumptions for this chapter*

- **F** denotes **R** or **C**.
- $V$ and $W$ denote finite-dimensional nonzero vector spaces over **F**.



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*The Mathematical Institute at the University of Göttingen. This building opened in 1930, when Emmy Noether (1882–1935) had already been a research mathematician and faculty member at the university for 15 years (the first eight years without salary). Noether was fired by the Nazi government in 1933. By then Noether and her collaborators had created many of the foundations of modern algebra, including an abstract algebra viewpoint that contributed to the development of linear algebra.*

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S. Axler, *Linear Algebra Done Right*, Undergraduate Texts in Mathematics,
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# 9A Bilinear Forms and Quadratic Forms

## Bilinear Forms

A bilinear form on $V$ is a function from $V \times V$ to $\mathbf{F}$ that is linear in each slot separately, meaning that if we hold either slot fixed then we have a linear function in the other slot. Here is the formal definition.

**9.1 definition: *bilinear form***

A *bilinear form* on $V$ is a function $\beta : V \times V \to \mathbf{F}$ such that

$$v \mapsto \beta(v, u) \quad \text{and} \quad v \mapsto \beta(u, v)$$

are both linear functionals on $V$ for every $u \in V$.

For example, if $V$ is a real inner product space, then the function that takes an ordered pair $(u, v) \in V \times V$ to $\langle u, v \rangle$ is a bilinear form on $V$. If $V$ is a nonzero complex inner product space, then this function is not a bilinear form because the inner product is not linear in the second slot (complex scalars come out of the second slot as their complex conjugates).

*Recall that the term **linear functional**, used in the definition above, means a linear function that maps into the scalar field $\mathbf{F}$. Thus the term **bilinear functional** would be more consistent terminology than **bilinear form**, which unfortunately has become standard.*

If $\mathbf{F} = \mathbf{R}$, then a bilinear form differs from an inner product in that an inner product requires symmetry [meaning that $\beta(v, w) = \beta(w, v)$ for all $v, w \in V$] and positive definiteness [meaning that $\beta(v, v) > 0$ for all $v \in V \backslash \{0\}$], but these properties are not required for a bilinear form.

**9.2 example: *bilinear forms***

- The function $\beta : \mathbf{F}^3 \times \mathbf{F}^3 \to \mathbf{F}$ defined by

$$\beta((x_1, x_2, x_3), (y_1, y_2, y_3)) = x_1 y_2 - 5 x_2 y_3 + 2 x_3 y_1$$

  is a bilinear form on $\mathbf{F}^3$.

- Suppose $A$ is an $n$-by-$n$ matrix with $A_{j,k} \in \mathbf{F}$ in row $j$, column $k$. Define a bilinear form $\beta_A$ on $\mathbf{F}^n$ by

$$\beta_A((x_1, \ldots, x_n), (y_1, \ldots, y_n)) = \sum_{k=1}^{n} \sum_{j=1}^{n} A_{j,k} x_j y_k.$$

  The first bullet point is a special case of this bullet point with $n = 3$ and

$$A = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & -5 \\ 2 & 0 & 0 \end{pmatrix}.$$

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- Suppose $V$ is a real inner product space and $T \in \mathcal{L}(V)$. Then the function $\beta : V \times V \to \mathbf{R}$ defined by
$$\beta(u, v) = \langle u, Tv \rangle$$
is a bilinear form on $V$.
- If $n$ is a positive integer, then the function $\beta : \mathcal{P}_n(\mathbf{R}) \times \mathcal{P}_n(\mathbf{R}) \to \mathbf{R}$ defined by
$$\beta(p, q) = p(2) \cdot q'(3)$$
is a bilinear form on $\mathcal{P}_n(\mathbf{R})$.
- Suppose $\varphi, \tau \in V'$. Then the function $\beta : V \times V \to \mathbf{F}$ defined by
$$\beta(u, v) = \varphi(u) \cdot \tau(v)$$
is a bilinear form on $V$.
- More generally, suppose that $\varphi_1, \ldots, \varphi_n, \tau_1, \ldots, \tau_n \in V'$. Then the function $\beta : V \times V \to \mathbf{F}$ defined by
$$\beta(u, v) = \varphi_1(u) \cdot \tau_1(v) + \cdots + \varphi_n(u) \cdot \tau_n(v)$$
is a bilinear form on $V$.

A bilinear form on $V$ is a function from $V \times V$ to $\mathbf{F}$. Because $V \times V$ is a vector space, this raises the question of whether a bilinear form can also be a linear map from $V \times V$ to $\mathbf{F}$. Note that none of the bilinear forms in 9.2 are linear maps except in some special cases in which the bilinear form is the zero map. Exercise 3 shows that a bilinear form $\beta$ on $V$ is a linear map on $V \times V$ only if $\beta = 0$.

**9.3 definition: $V^{(2)}$**

The set of bilinear forms on $V$ is denoted by $V^{(2)}$.

With the usual operations of addition and scalar multiplication of functions, $V^{(2)}$ is a vector space.

For $T$ an operator on an $n$-dimensional vector space $V$ and a basis $e_1, \ldots, e_n$ of $V$, we used an $n$-by-$n$ matrix to provide information about $T$. We now do the same thing for bilinear forms on $V$.

**9.4 definition: *matrix of a bilinear form*, $\mathcal{M}(\beta)$**

Suppose $\beta$ is a bilinear form on $V$ and $e_1, \ldots, e_n$ is a basis of $V$. The *matrix* of $\beta$ with respect to this basis is the $n$-by-$n$ matrix $\mathcal{M}(\beta)$ whose entry $\mathcal{M}(\beta)_{j,k}$ in row $j$, column $k$ is given by
$$\mathcal{M}(\beta)_{j,k} = \beta(e_j, e_k).$$
If the basis $e_1, \ldots, e_n$ is not clear from the context, then the notation $\mathcal{M}(\beta, (e_1, \ldots, e_n))$ is used.

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Recall that $\mathbf{F}^{n,n}$ denotes the vector space of $n$-by-$n$ matrices with entries in $\mathbf{F}$ and that $\dim \mathbf{F}^{n,n} = n^2$ (see 3.39 and 3.40).

**9.5 $\dim V^{(2)} = (\dim V)^2$**

Suppose $e_1, \ldots, e_n$ is a basis of $V$. Then the map $\beta \mapsto \mathcal{M}(\beta)$ is an isomorphism of $V^{(2)}$ onto $\mathbf{F}^{n,n}$. Furthermore, $\dim V^{(2)} = (\dim V)^2$.

Proof The map $\beta \mapsto \mathcal{M}(\beta)$ is clearly a linear map of $V^{(2)}$ into $\mathbf{F}^{n,n}$.

For $A \in \mathbf{F}^{n,n}$, define a bilinear form $\beta_A$ on $V$ by

$$\beta_A(x_1e_1 + \cdots + x_ne_n, y_1e_1 + \cdots + y_ne_n) = \sum_{k=1}^{n} \sum_{j=1}^{n} A_{j,k} x_j y_k$$

for $x_1, \ldots, x_n, y_1, \ldots, y_n \in \mathbf{F}$ (if $V = \mathbf{F}^n$ and $e_1, \ldots, e_n$ is the standard basis of $\mathbf{F}^n$, this $\beta_A$ is the same as the bilinear form $\beta_A$ in the second bullet point of Example 9.2).

The linear map $\beta \mapsto \mathcal{M}(\beta)$ from $V^{(2)}$ to $\mathbf{F}^{n,n}$ and the linear map $A \mapsto \beta_A$ from $\mathbf{F}^{n,n}$ to $V^{(2)}$ are inverses of each other because $\beta_{\mathcal{M}(\beta)} = \beta$ for all $\beta \in V^{(2)}$ and $\mathcal{M}(\beta_A) = A$ for all $A \in \mathbf{F}^{n,n}$, as you should verify.

Thus both maps are isomorphisms and the two spaces that they connect have the same dimension. Hence $\dim V^{(2)} = \dim \mathbf{F}^{n,n} = n^2 = (\dim V)^2$. ∎

Recall that $C^t$ denotes the transpose of a matrix $C$. The matrix $C^t$ is obtained by interchanging the rows and the columns of $C$.

**9.6 *composition of a bilinear form and an operator***

Suppose $\beta$ is a bilinear form on $V$ and $T \in \mathcal{L}(V)$. Define bilinear forms $\alpha$ and $\rho$ on $V$ by

$$\alpha(u, v) = \beta(u, Tv) \quad \text{and} \quad \rho(u, v) = \beta(Tu, v).$$

Let $e_1, \ldots, e_n$ be a basis of $V$. Then

$$\mathcal{M}(\alpha) = \mathcal{M}(\beta)\mathcal{M}(T) \quad \text{and} \quad \mathcal{M}(\rho) = \mathcal{M}(T)^t\mathcal{M}(\beta).$$

Proof If $j, k \in \{1, \ldots, n\}$, then

$$\begin{aligned}
\mathcal{M}(\alpha)_{j,k} &= \alpha(e_j, e_k) \\
&= \beta(e_j, Te_k) \\
&= \beta\Big(e_j, \sum_{m=1}^{n} \mathcal{M}(T)_{m,k} e_m\Big) \\
&= \sum_{m=1}^{n} \beta(e_j, e_m)\mathcal{M}(T)_{m,k} \\
&= (\mathcal{M}(\beta)\mathcal{M}(T))_{j,k}.
\end{aligned}$$

Thus $\mathcal{M}(\alpha) = \mathcal{M}(\beta)\mathcal{M}(T)$. The proof that $\mathcal{M}(\rho) = \mathcal{M}(T)^t\mathcal{M}(\beta)$ is similar. ∎

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The result below shows how the matrix of a bilinear form changes if we change the basis. The formula in the result below should be compared to the change-of-basis formula for the matrix of an operator (see 3.84). The two formulas are similar, except that the transpose $C^t$ appears in the formula below and the inverse $C^{-1}$ appears in the change-of-basis formula for the matrix of an operator.

**9.7** *change-of-basis formula*

Suppose $\beta \in V^{(2)}$. Suppose $e_1, \dots, e_n$ and $f_1, \dots, f_n$ are bases of $V$. Let

$$A = \mathcal{M}(\beta, (e_1, \dots, e_n)) \quad \text{and} \quad B = \mathcal{M}(\beta, (f_1, \dots, f_n))$$

and $C = \mathcal{M}(I, (e_1, \dots, e_n), (f_1, \dots, f_n))$. Then

$$A = C^t B C.$$

Proof The linear map lemma (3.4) tells us that there exists an operator $T \in \mathcal{L}(V)$ such that $Tf_k = e_k$ for each $k = 1, \dots, n$. The definition of the matrix of an operator with respect to a basis implies that

$$\mathcal{M}(T, (f_1, \dots, f_n)) = C.$$

Define bilinear forms $\alpha, \rho$ on $V$ by

$$\alpha(u, v) = \beta(u, Tv) \quad \text{and} \quad \rho(u, v) = \alpha(Tu, v) = \beta(Tu, Tv).$$

Then $\beta(e_j, e_k) = \beta(Tf_j, Tf_k) = \rho(f_j, f_k)$ for all $j, k \in \{1, \dots, n\}$. Thus

$$\begin{aligned} A &= \mathcal{M}(\rho, (f_1, \dots, f_n)) \\ &= C^t \mathcal{M}(\alpha, f_1, \dots, f_n)) \\ &= C^t B C, \end{aligned}$$

where the second and third lines each follow from 9.6. ∎

**9.8** example: *the matrix of a bilinear form on* $\mathcal{P}_2(\mathbf{R})$

Define a bilinear form $\beta$ on $\mathcal{P}_2(\mathbf{R})$ by $\beta(p, q) = p(2) \cdot q'(3)$. Let

$$A = \mathcal{M}(\beta, (1, x - 2, (x - 3)^2)) \quad \text{and} \quad B = \mathcal{M}(\beta, (1, x, x^2))$$

and $C = \mathcal{M}(I, (1, x - 2, (x - 3)^2), (1, x, x^2))$. Then

$$A = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix} \quad \text{and} \quad B = \begin{pmatrix} 0 & 1 & 6 \\ 0 & 2 & 12 \\ 0 & 4 & 24 \end{pmatrix} \quad \text{and} \quad C = \begin{pmatrix} 1 & -2 & 9 \\ 0 & 1 & -6 \\ 0 & 0 & 1 \end{pmatrix}.$$

Now the change-of-basis formula 9.7 asserts that $A = C^t B C$, which you can verify with matrix multiplication using the matrices above.

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## *Symmetric Bilinear Forms*

**9.9 definition: *symmetric bilinear form*, $V_{\text{sym}}^{(2)}$**

A bilinear form $\rho \in V^{(2)}$ is called *symmetric* if

$$\rho(u, w) = \rho(w, u)$$

for all $u, w \in V$. The set of symmetric bilinear forms on $V$ is denoted by $V_{\text{sym}}^{(2)}$.

**9.10 example: *symmetric bilinear forms***

- If $V$ is a real inner product space and $\rho \in V^{(2)}$ is defined by

$$\rho(u, w) = \langle u, w \rangle,$$

  then $\rho$ is a symmetric bilinear form on $V$.

- Suppose $V$ is a real inner product space and $T \in \mathcal{L}(V)$. Define $\rho \in V^{(2)}$ by

$$\rho(u, w) = \langle u, Tw \rangle.$$

  Then $\rho$ is a symmetric bilinear form on $V$ if and only if $T$ is a self-adjoint operator (the previous bullet point is the special case $T = I$).

- Suppose $\rho : \mathcal{L}(V) \times \mathcal{L}(V) \to \mathbf{F}$ is defined by

$$\rho(S, T) = \operatorname{tr}(ST).$$

  Then $\rho$ is a symmetric bilinear form on $\mathcal{L}(V)$ because trace is a linear functional on $\mathcal{L}(V)$ and $\operatorname{tr}(ST) = \operatorname{tr}(TS)$ for all $S, T \in \mathcal{L}(V)$; see 8.56.

**9.11 definition: *symmetric matrix***

A square matrix $A$ is called *symmetric* if it equals its transpose.

An operator on $V$ may have a symmetric matrix with respect to some but not all bases of $V$. In contrast, the next result shows that a bilinear form on $V$ has a symmetric matrix with respect to either all bases of $V$ or with respect to no bases of $V$.

**9.12 *symmetric bilinear forms are diagonalizable***

Suppose $\rho \in V^{(2)}$. Then the following are equivalent.

(a) $\rho$ is a symmetric bilinear form on $V$.

(b) $\mathcal{M}(\rho, (e_1, \ldots, e_n))$ is a symmetric matrix for every basis $e_1, \ldots, e_n$ of $V$.

(c) $\mathcal{M}(\rho, (e_1, \ldots, e_n))$ is a symmetric matrix for some basis $e_1, \ldots, e_n$ of $V$.

(d) $\mathcal{M}(\rho, (e_1, \ldots, e_n))$ is a diagonal matrix for some basis $e_1, \ldots, e_n$ of $V$.

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Proof First suppose (a) holds, so $\rho$ is a symmetric bilinear form. Suppose $e_1, ..., e_n$ is a basis of $V$ and $j, k \in \{1, ..., n\}$. Then $\rho(e_j, e_k) = \rho(e_k, e_j)$ because $\rho$ is symmetric. Thus $\mathcal{M}(\rho, (e_1, ..., e_n))$ is a symmetric matrix, showing that (a) implies (b).

Clearly (b) implies (c).

Now suppose (c) holds and $e_1, ..., e_n$ is a basis of $V$ such that $\mathcal{M}(\rho, (e_1, ..., e_n))$ is a symmetric matrix. Suppose $u, w \in V$. There exist $a_1, ..., a_n, b_1, ..., b_n \in \mathbf{F}$ such that $u = a_1e_1 + \cdots + a_ne_n$ and $w = b_1e_1 + \cdots + b_ne_n$. Now

$$\begin{aligned}
\rho(u, w) &= \rho\Big(\sum_{j=1}^{n} a_j e_j, \sum_{k=1}^{n} b_k e_k\Big) \\
&= \sum_{j=1}^{n} \sum_{k=1}^{n} a_j b_k \rho(e_j, e_k) \\
&= \sum_{j=1}^{n} \sum_{k=1}^{n} a_j b_k \rho(e_k, e_j) \\
&= \rho\Big(\sum_{k=1}^{n} b_k e_k, \sum_{j=1}^{n} a_j e_j\Big) \\
&= \rho(w, u),
\end{aligned}$$

where the third line holds because $\mathcal{M}(\rho)$ is a symmetric matrix. The equation above shows that $\rho$ is a symmetric bilinear form, proving that (c) implies (a).

At this point, we have proved that (a), (b), (c) are equivalent. Because every diagonal matrix is symmetric, (d) implies (c). To complete the proof, we will show that (a) implies (d) by induction on $n = \dim V$.

If $n = 1$, then (a) implies (d) because every 1-by-1 matrix is diagonal. Now suppose $n > 1$ and the implication (a) $\implies$ (d) holds for one less dimension. Suppose (a) holds, so $\rho$ is a symmetric bilinear form. If $\rho = 0$, then the matrix of $\rho$ with respect to every basis of $V$ is the zero matrix, which is a diagonal matrix. Hence we can assume that $\rho \neq 0$, which means there exist $u, w \in V$ such that $\rho(u, w) \neq 0$. Now

$$2\rho(u, w) = \rho(u + w, u + w) - \rho(u, u) - \rho(w, w).$$

Because the left side of the equation above is nonzero, the three terms on the right cannot all equal 0. Hence there exists $v \in V$ such that $\rho(v, v) \neq 0$.

Let $U = \{u \in V : \rho(u, v) = 0\}$. Thus $U$ is the null space of the linear functional $u \mapsto \rho(u, v)$ on $V$. This linear functional is not the zero linear functional because $v \notin U$. Thus $\dim U = n - 1$. By our induction hypothesis, there is a basis $e_1, ..., e_{n-1}$ of $U$ such that the symmetric bilinear form $\rho|_{U \times U}$ has a diagonal matrix with respect to this basis.

Because $v \notin U$, the list $e_1, ..., e_{n-1}, v$ is a basis of $V$. Suppose $k \in \{1, ..., n-1\}$. Then $\rho(e_k, v) = 0$ by the construction of $U$. Because $\rho$ is symmetric, we also have $\rho(v, e_k) = 0$. Thus the matrix of $\rho$ with respect to $e_1, ..., e_{n-1}, v$ is a diagonal matrix, completing the proof that (a) implies (d). ∎

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The previous result states that every symmetric bilinear form has a diagonal matrix with respect to some basis. If our vector space happens to be a real inner product space, then the next result shows that every symmetric bilinear form has a diagonal matrix with respect to some *orthonormal* basis. Note that the inner product here is unrelated to the bilinear form.

**9.13** *diagonalization of a symmetric bilinear form by an orthonormal basis*

Suppose $V$ is a real inner product space and $\rho$ is a symmetric bilinear form on $V$. Then $\rho$ has a diagonal matrix with respect to some orthonormal basis of $V$.

Proof Let $f_1, \dots, f_n$ be an orthonormal basis of $V$. Let $B = \mathcal{M}(\rho, (f_1, \dots, f_n))$. Then $B$ is a symmetric matrix (by 9.12). Let $T \in \mathcal{L}(V)$ be the operator such that $\mathcal{M}(T, (f_1, \dots, f_n)) = B$. Thus $T$ is self-adjoint.

The real spectral theorem (7.29) states that $T$ has a diagonal matrix with respect to some orthonormal basis $e_1, \dots, e_n$ of $V$. Let $C = \mathcal{M}(I, (e_1, \dots, e_n), (f_1, \dots, f_n))$. Thus $C^{-1}TC$ is the matrix of $T$ with respect to the basis $e_1, \dots, e_n$ (by 3.84). Hence $C^{-1}TC$ is a diagonal matrix. Now

$$M(\rho, (e_1, \dots, e_n)) = C^t TC = C^{-1}TC,$$

where the first equality holds by 9.7 and the second equality holds because $C$ is a unitary matrix with real entries (which implies that $C^{-1} = C^t$; see 7.57). ∎

Now we turn our attention to alternating bilinear forms. Alternating multilinear forms will play a major role in our approach to determinants later in this chapter.

**9.14** definition: *alternating bilinear form*, $V_{\text{alt}}^{(2)}$

A bilinear form $\alpha \in V^{(2)}$ is called *alternating* if

$$\alpha(v, v) = 0$$

for all $v \in V$. The set of alternating bilinear forms on $V$ is denoted by $V_{\text{alt}}^{(2)}$.

**9.15** example: *alternating bilinear forms*

- Suppose $n \geq 3$ and $\alpha : \mathbf{F}^n \times \mathbf{F}^n \to \mathbf{F}$ is defined by

$$\alpha((x_1, \dots, x_n), (y_1, \dots, y_n)) = x_1y_2 - x_2y_1 + x_1y_3 - x_3y_1.$$

  Then $\alpha$ is an alternating bilinear form on $\mathbf{F}^n$.

- Suppose $\varphi, \tau \in V'$. Then the bilinear form $\alpha$ on $V$ defined by

$$\alpha(u, w) = \varphi(u)\tau(w) - \varphi(w)\tau(u)$$

  is alternating.

The next result shows that a bilinear form is alternating if and only if switching the order of the two inputs multiplies the output by $-1$.

**9.16** *characterization of alternating bilinear forms*

A bilinear form $\alpha$ on $V$ is alternating if and only if

$$\alpha(u, w) = -\alpha(w, u)$$

for all $u, w \in V$.

Proof First suppose that $\alpha$ is alternating. If $u, w \in V$, then

$$\begin{aligned} 0 &= \alpha(u + w, u + w) \\ &= \alpha(u, u) + \alpha(u, w) + \alpha(w, u) + \alpha(w, w) \\ &= \alpha(u, w) + \alpha(w, u). \end{aligned}$$

Thus $\alpha(u, w) = -\alpha(w, u)$, as desired.

To prove the implication in the other direction, suppose $\alpha(u, w) = -\alpha(w, u)$ for all $u, w \in V$. Then $\alpha(v, v) = -\alpha(v, v)$ for all $v \in V$, which implies that $\alpha(v, v) = 0$ for all $v \in V$. Thus $\alpha$ is alternating. ∎

Now we show that the vector space of bilinear forms on $V$ is the direct sum of the symmetric bilinear forms on $V$ and the alternating bilinear forms on $V$.

**9.17** $V^{(2)} = V^{(2)}_{\text{sym}} \oplus V^{(2)}_{\text{alt}}$

The sets $V^{(2)}_{\text{sym}}$ and $V^{(2)}_{\text{alt}}$ are subspaces of $V^{(2)}$. Furthermore,

$$V^{(2)} = V^{(2)}_{\text{sym}} \oplus V^{(2)}_{\text{alt}}.$$

Proof The definition of symmetric bilinear form implies that the sum of any two symmetric bilinear forms on $V$ is a bilinear form on $V$, and any scalar multiple of any bilinear form on $V$ is a bilinear form on $V$. Thus $V^{(2)}_{\text{sym}}$ is a subspace of $V^{(2)}$. Similarly, the verification that $V^{(2)}_{\text{alt}}$ is a subspace of $V^{(2)}$ is straightforward.

Next, we want to show that $V^{(2)} = V^{(2)}_{\text{sym}} + V^{(2)}_{\text{alt}}$. To do this, suppose $\beta \in V^{(2)}$. Define $\rho, \alpha \in V^{(2)}$ by

$$\rho(u, w) = \frac{\beta(u, w) + \beta(w, u)}{2} \quad \text{and} \quad \alpha(u, w) = \frac{\beta(u, w) - \beta(w, u)}{2}.$$

Then $\rho \in V^{(2)}_{\text{sym}}$ and $\alpha \in V^{(2)}_{\text{alt}}$, and $\beta = \rho + \alpha$. Thus $V^{(2)} = V^{(2)}_{\text{sym}} + V^{(2)}_{\text{alt}}$.

Finally, to show that the intersection of the two subspaces under consideration equals $\{0\}$, suppose $\beta \in V^{(2)}_{\text{sym}} \cap V^{(2)}_{\text{alt}}$. Then 9.16 implies that

$$\beta(u, w) = -\beta(w, u) = -\beta(u, w)$$

for all $u, w \in V$, which implies that $\beta = 0$. Thus $V^{(2)} = V^{(2)}_{\text{sym}} \oplus V^{(2)}_{\text{alt}}$, as implied by 1.46. ∎

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## *Quadratic Forms*

**9.18 definition:** *quadratic form associated with a bilinear form, $q_\beta$*

For $\beta$ a bilinear form on $V$, define a function $q_\beta : V \to \mathbf{F}$ by $q_\beta(v) = \beta(v, v)$. A function $q : V \to \mathbf{F}$ is called a *quadratic form* on $V$ if there exists a bilinear form $\beta$ on $V$ such that $q = q_\beta$.

Note that if $\beta$ is a bilinear form, then $q_\beta = 0$ if and only if $\beta$ is alternating.

**9.19 example:** *quadratic form*

Suppose $\beta$ is the bilinear form on $\mathbf{R}^3$ defined by

$$\beta((x_1, x_2, x_3), (y_1, y_2, y_3)) = x_1y_1 - 4x_1y_2 + 8x_1y_3 - 3x_3y_3.$$

Then $q_\beta$ is the quadratic form on $\mathbf{R}^3$ given by the formula

$$q_\beta(x_1, x_2, x_3) = x_1^2 - 4x_1x_2 + 8x_1x_3 - 3x_3^2.$$

The quadratic form in the example above is typical of quadratic forms on $\mathbf{F}^n$, as shown in the next result.

**9.20** *quadratic forms on $\mathbf{F}^n$*

Suppose $n$ is a positive integer and $q$ is a function from $\mathbf{F}^n$ to $\mathbf{F}$. Then $q$ is a quadratic form on $\mathbf{F}^n$ if and only if there exist numbers $A_{j,k} \in \mathbf{F}$ for $j, k \in \{1, \ldots, n\}$ such that

$$q(x_1, \ldots, x_n) = \sum_{k=1}^{n} \sum_{j=1}^{n} A_{j,k} x_j x_k$$

for all $(x_1, \ldots, x_n) \in \mathbf{F}^n$.

Proof First suppose $q$ is a quadratic form on $\mathbf{F}^n$. Thus there exists a bilinear form $\beta$ on $\mathbf{F}^n$ such that $q = q_\beta$. Let $A$ be the matrix of $\beta$ with respect to the standard basis of $\mathbf{F}^n$. Then for all $(x_1, \ldots, x_n) \in \mathbf{F}^n$, we have the desired equation

$$q(x_1, \ldots, x_n) = \beta((x_1, \ldots, x_n), (x_1, \ldots, x_n)) = \sum_{k=1}^{n} \sum_{j=1}^{n} A_{j,k} x_j x_k.$$

Conversely, suppose there exist numbers $A_{j,k} \in \mathbf{F}$ for $j, k \in \{1, \ldots, n\}$ such that

$$q(x_1, \ldots, x_n) = \sum_{k=1}^{n} \sum_{j=1}^{k} A_{j,k} x_j x_k$$

for all $(x_1, \ldots, x_n) \in \mathbf{F}^n$. Define a bilinear form $\beta$ on $\mathbf{F}^n$ by

$$\beta((x_1, \ldots, x_n), (y_1, \ldots, y_n)) = \sum_{k=1}^{n} \sum_{j=1}^{k} A_{j,k} x_j y_k.$$

Then $q = q_\beta$, as desired.

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Although quadratic forms are defined in terms of an arbitrary bilinear form, the equivalence of (a) and (b) in the result below shows that a *symmetric* bilinear form can always be used.

**9.21** *characterization of quadratic forms*

Suppose $q : V \to \mathbf{F}$ is a function. The following are equivalent.

(a) $q$ is a quadratic form.

(b) There exists a unique symmetric bilinear form $\rho$ on $V$ such that $q = q_\rho$.

(c) $q(\lambda v) = \lambda^2 q(v)$ for all $\lambda \in \mathbf{F}$ and all $v \in V$, and the function

$$(u, w) \mapsto q(u + w) - q(u) - q(w)$$

is a symmetric bilinear form on $V$.

(d) $q(2v) = 4q(v)$ for all $v \in V$, and the function

$$(u, w) \mapsto q(u + w) - q(u) - q(w)$$

is a symmetric bilinear form on $V$.

Proof First suppose (a) holds, so $q$ is a quadratic form. Hence there exists a bilinear form $\beta$ such that $q = q_\beta$. By 9.17, there exist a symmetric bilinear form $\rho$ on $V$ and an alternating bilinear form $\alpha$ on $V$ such that $\beta = \rho + \alpha$. Now

$$q = q_\beta = q_\rho + q_\alpha = q_\rho.$$

If $\rho' \in V^{(2)}_{\text{sym}}$ also satisfies $q_{\rho'} = q$, then $q_{\rho' - \rho} = 0$; thus $\rho' - \rho \in V^{(2)}_{\text{sym}} \cap V^{(2)}_{\text{alt}}$, which implies that $\rho' = \rho$ (by 9.17). This completes the proof that (a) implies (b).

Now suppose (b) holds, so there exists a symmetric bilinear form $\rho$ on $V$ such that $q = q_\rho$. If $\lambda \in \mathbf{F}$ and $v \in V$ then

$$q(\lambda v) = \rho(\lambda v, \lambda v) = \lambda \rho(v, \lambda v) = \lambda^2 \rho(v, v) = \lambda^2 q(v),$$

showing that the first part of (c) holds.

If $u, w \in V$, then

$$q(u + w) - q(u) - q(w) = \rho(u + w, u + w) - \rho(u, u) - \rho(w, w) = 2\rho(u, w).$$

Thus the function $(u, w) \mapsto q(u+w)-q(u)-q(w)$ equals $2\rho$, which is a symmetric bilinear form on $V$, completing the proof that (b) implies (c).

Clearly (c) implies (d).

Now suppose (d) holds. Let $\rho$ be the symmetric bilinear form on $V$ defined by

$$\rho(u, w) = \frac{q(u + w) - q(u) - q(w)}{2}.$$

If $v \in V$, then

$$\rho(v, v) = \frac{q(2v) - q(v) - q(v)}{2} = \frac{4q(v) - 2q(v)}{2} = q(v).$$

Thus $q = q_\rho$, completing the proof that (d) implies (a).

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9.22 example: *symmetric bilinear form associated with a quadratic form*

Suppose $q$ is the quadratic form on $\mathbf{R}^3$ given by the formula

$$q(x_1, x_2, x_3) = x_1^2 - 4x_1x_2 + 8x_1x_3 - 3x_3^2.$$

A bilinear form $\beta$ on $\mathbf{R}^3$ such that $q = q_\beta$ is given by Example 9.19, but this bilinear form is not symmetric, as promised by 9.21(b). However, the bilinear form $\rho$ on $\mathbf{R}^3$ defined by

$$\rho((x_1, x_2, x_3), (y_1, y_2, y_3)) = x_1y_1 - 2x_1y_2 - 2x_2y_1 + 4x_1y_3 + 4x_3y_1 - 3x_3y_3$$

is symmetric and satisfies $q = q_\rho$.

The next result states that for each quadratic form we can choose a basis such that the quadratic form looks like a weighted sum of squares of the coordinates, meaning that there are no cross terms of the form $x_jx_k$ with $j \neq k$.

9.23 *diagonalization of quadratic form*

Suppose $q$ is a quadratic form on $V$.

(a) There exist a basis $e_1, \ldots, e_n$ of $V$ and $\lambda_1, \ldots, \lambda_n \in \mathbf{F}$ such that

$$q(x_1e_1 + \cdots + x_ne_n) = \lambda_1x_1^2 + \cdots + \lambda_nx_n^2$$

for all $x_1, \ldots, x_n \in \mathbf{F}$.

(b) If $\mathbf{F} = \mathbf{R}$ and $V$ is an inner product space, then the basis in (a) can be chosen to be an orthonormal basis of $V$.

Proof

(a) There exists a symmetric bilinear form $\rho$ on $V$ such that $q = q_\rho$ (by 9.21). Now there exists a basis $e_1, \ldots, e_n$ of $V$ such that $\mathcal{M}(\rho, (e_1, \ldots, e_n))$ is a diagonal matrix (by 9.12). Let $\lambda_1, \ldots, \lambda_n$ denote the entries on the diagonal of this matrix. Thus

$$\rho(e_j, e_k) = \begin{cases} \lambda_j & \text{if } j = k, \\ 0 & \text{if } j \neq k \end{cases}$$

for all $j, k \in \{1, \ldots, n\}$. If $x_1, \ldots, x_n \in \mathbf{F}$, then

$$\begin{aligned} q(x_1e_1 + \cdots + x_ne_n) &= \rho(x_1e_1 + \cdots + x_ne_n, x_1e_1 + \cdots + x_ne_n) \\ &= \sum_{k=1}^{n} \sum_{j=1}^{n} x_jx_k\rho(e_j, e_k) \\ &= \lambda_1x_1^2 + \cdots + \lambda_nx_n^2, \end{aligned}$$

as desired.

(b) Suppose $\mathbf{F} = \mathbf{R}$ and $V$ is an inner product space. Then 9.13 tells us that the basis in (a) can be chosen to be an orthonormal basis of $V$. ∎

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## *Exercises 9A*

**1** Prove that if $\beta$ is a bilinear form on $\mathbf{F}$, then there exists $c \in \mathbf{F}$ such that

$$\beta(x, y) = cxy$$

for all $x, y \in \mathbf{F}$.

**2** Let $n = \dim V$. Suppose $\beta$ is a bilinear form on $V$. Prove that there exist $\varphi_1, \ldots, \varphi_n, \tau_1, \ldots, \tau_n \in V'$ such that

$$\beta(u, v) = \varphi_1(u) \cdot \tau_1(v) + \cdots + \varphi_n(u) \cdot \tau_n(v)$$

for all $u, v \in V$.

*This exercise shows that if $n = \dim V$, then every bilinear form on $V$ is of the form given by the last bullet point of Example 9.2.*

**3** Suppose $\beta : V \times V \to \mathbf{F}$ is a bilinear form on $V$ and also is a linear functional on $V \times V$. Prove that $\beta = 0$.

**4** Suppose $V$ is a real inner product space and $\beta$ is a bilinear form on $V$. Show that there exists a unique operator $T \in \mathcal{L}(V)$ such that

$$\beta(u, v) = \langle u, Tv \rangle$$

for all $u, v \in V$.

*This exercise states that if $V$ is a real inner product space, then every bilinear form on $V$ is of the form given by the third bullet point in 9.2.*

**5** Suppose $\beta$ is a bilinear form on a real inner product space $V$ and $T$ is the unique operator on $V$ such that $\beta(u, v) = \langle u, Tv \rangle$ for all $u, v \in V$ (see Exercise 4). Show that $\beta$ is an inner product on $V$ if and only if $T$ is an invertible positive operator on $V$.

**6** Prove or give a counterexample: If $\rho$ is a symmetric bilinear form on $V$, then

$$\{v \in V : \rho(v, v) = 0\}$$

is a subspace of $V$.

**7** Explain why the proof of 9.13 (diagonalization of a symmetric bilinear form by an orthonormal basis on a real inner product space) fails if the hypothesis that $\mathbf{F} = \mathbf{R}$ is dropped.

**8** Find formulas for $\dim V^{(2)}_{\text{sym}}$ and $\dim V^{(2)}_{\text{alt}}$ in terms of $\dim V$.

**9** Suppose that $n$ is a positive integer and $V = \{p \in \mathcal{P}_n(\mathbf{R}) : p(0) = p(1)\}$. Define $\alpha : V \times V \to \mathbf{R}$ by

$$\alpha(p, q) = \int_0^1 pq'.$$

Show that $\alpha$ is an alternating bilinear form on $V$.

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**10** Suppose that $n$ is a positive integer and

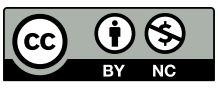
$$V = \{p \in \mathcal{P}_n(\mathbf{R}) : p(0) = p(1) \text{ and } p'(0) = p'(1)\}.$$

Define $\rho : V \times V \to \mathbf{R}$ by

$$\rho(p, q) = \int_0^1 pq''.$$

Show that $\rho$ is a symmetric bilinear form on $V$.

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# 9B Alternating Multilinear Forms

## Multilinear Forms

**9.24 definition: $V^m$**

For $m$ a positive integer, define $V^m$ by

$$V^m = \underbrace{V \times \cdots \times V}_{m \text{ times}}.$$

Now we can define $m$-linear forms as a generalization of the bilinear forms that we discussed in the previous section.

**9.25 definition: *$m$-linear form*, $V^{(m)}$, *multilinear form***

- For $m$ a positive integer, an *$m$-linear form* on $V$ is a function $\beta : V^m \to \mathbf{F}$ that is linear in each slot when the other slots are held fixed. This means that for each $k \in \{1, \ldots, m\}$ and all $u_1, \ldots, u_m \in V$, the function

  $$v \mapsto \beta(u_1, \ldots, u_{k-1}, v, u_{k+1}, \ldots, u_m)$$

  is a linear map from $V$ to $\mathbf{F}$.
- The set of $m$-linear forms on $V$ is denoted by $V^{(m)}$.
- A function $\beta$ is called a *multilinear form* on $V$ if it is an $m$-linear form on $V$ for some positive integer $m$.

In the definition above, the expression $\beta(u_1, \ldots, u_{k-1}, v, u_{k+1}, \ldots, u_m)$ means $\beta(v, u_2, \ldots, u_m)$ if $k = 1$ and means $\beta(u_1, \ldots, u_{m-1}, v)$ if $k = m$.

A 1-linear form on $V$ is a linear functional on $V$. A 2-linear form on $V$ is a bilinear form on $V$. You can verify that with the usual addition and scalar multiplication of functions, $V^{(m)}$ is a vector space.

**9.26 example: *$m$-linear forms***

- Suppose $\alpha, \rho \in V^{(2)}$. Define a function $\beta : V^4 \to \mathbf{F}$ by

  $$\beta(v_1, v_2, v_3, v_4) = \alpha(v_1, v_2)\, \rho(v_3, v_4).$$

  Then $\beta \in V^{(4)}$.
- Define $\beta : (\mathcal{L}(V))^m \to \mathbf{F}$ by

  $$\beta(T_1, \ldots, T_m) = \operatorname{tr}(T_1 \cdots T_m).$$

  Then $\beta$ is an $m$-linear form on $\mathcal{L}(V)$.

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Alternating multilinear forms, which we now define, play an important role as we head toward defining determinants.

**9.27 definition: *alternating forms*, $V_{\text{alt}}^{(m)}$**

Suppose $m$ is a positive integer.

- An $m$-linear form $\alpha$ on $V$ is called *alternating* if $\alpha(v_1, \dots, v_m) = 0$ whenever $v_1, \dots, v_m$ is a list of vectors in $V$ with $v_j = v_k$ for some two distinct values of $j$ and $k$ in $\{1, \dots, m\}$.
- $V_{\text{alt}}^{(m)} = \{\alpha \in V^{(m)} : \alpha \text{ is an alternating } m\text{-linear form on } V\}$.

You should verify that $V_{\text{alt}}^{(m)}$ is a subspace of $V^{(m)}$. See Example 9.15 for examples of alternating 2-linear forms. See Exercise 2 for an example of an alternating 3-linear form.

The next result tells us that if a linearly dependent list is input to an alternating multilinear form, then the output equals 0.

**9.28 *alternating multilinear forms and linear dependence***

Suppose $m$ is a positive integer and $\alpha$ is an alternating $m$-linear form on $V$. If $v_1, \dots, v_m$ is a linearly dependent list in $V$, then

$$\alpha(v_1, \dots, v_m) = 0.$$

Proof Suppose $v_1, \dots, v_m$ is a linearly dependent list in $V$. By the linear dependence lemma (2.19), some $v_k$ is a linear combination of $v_1, \dots, v_{k-1}$. Thus there exist $b_1, \dots, b_{k-1}$ such that $v_k = b_1 v_1 + \cdots + b_{k-1} v_{k-1}$. Now

$$\begin{aligned}
\alpha(v_1, \dots, v_m) &= \alpha\Big(v_1, \dots, v_{k-1}, \sum_{j=1}^{k-1} b_j v_j, v_{k+1}, \dots, v_m\Big) \\
&= \sum_{j=1}^{k-1} b_j \, \alpha(v_1, \dots, v_{k-1}, v_j, v_{k+1}, \dots, v_m) \\
&= 0.
\end{aligned}$$

The next result states that if $m > \dim V$, then there are no alternating $m$-linear forms on $V$ other than the function on $V^m$ that is identically 0.

**9.29 *no nonzero alternating $m$-linear forms for $m > \dim V$***

Suppose $m > \dim V$. Then 0 is the only alternating $m$-linear form on $V$.

Proof Suppose that $\alpha$ is an alternating $m$-linear form on $V$ and $v_1, \dots, v_m \in V$. Because $m > \dim V$, this list is not linearly independent (by 2.22). Thus 9.28 implies that $\alpha(v_1, \dots, v_m) = 0$. Hence $\alpha$ is the zero function from $V^m$ to $\mathbf{F}$.

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## *Alternating Multilinear Forms and Permutations*

**9.30** *swapping input vectors in an alternating multilinear form*

Suppose $m$ is a positive integer, $\alpha$ is an alternating $m$-linear form on $V$, and $v_1, \ldots, v_m$ is a list of vectors in $V$. Then swapping the vectors in any two slots of $\alpha(v_1, \ldots, v_m)$ changes the value of $\alpha$ by a factor of $-1$.

Proof Put $v_1 + v_2$ in both the first two slots, getting

$$0 = \alpha(v_1 + v_2, v_1 + v_2, v_3, \ldots, v_m).$$

Use the multilinear properties of $\alpha$ to expand the right side of the equation above (as in the proof of 9.16) to get

$$\alpha(v_2, v_1, v_3, \ldots, v_m) = -\alpha(v_1, v_2, v_3, \ldots, v_m).$$

Similarly, swapping the vectors in any two slots of $\alpha(v_1, \ldots, v_m)$ changes the value of $\alpha$ by a factor of $-1$. ∎

To see what can happen with multiple swaps, suppose $\alpha$ is an alternating 3-linear form on $V$ and $v_1, v_2, v_3 \in V$. To evaluate $\alpha(v_3, v_1, v_2)$ in terms of $\alpha(v_1, v_2, v_3)$, start with $\alpha(v_3, v_1, v_2)$ and swap the entries in the first and third slots, getting $\alpha(v_3, v_1, v_2) = -\alpha(v_2, v_1, v_3)$. Now in the last expression, swap the entries in the first and second slots, getting

$$\alpha(v_3, v_1, v_2) = -\alpha(v_2, v_1, v_3) = \alpha(v_1, v_2, v_3).$$

More generally, we see that if we do an odd number of swaps, then the value of $\alpha$ changes by a factor of $-1$, and if we do an even number of swaps, then the value of $\alpha$ does not change.

To deal with arbitrary multiple swaps, we need a bit of information about permutations.

**9.31** definition: *permutation,* perm $m$

Suppose $m$ is a positive integer.

- A *permutation* of $(1, \ldots, m)$ is a list $(j_1, \ldots, j_m)$ that contains each of the numbers $1, \ldots, m$ exactly once.
- The set of all permutations of $(1, \ldots, m)$ is denoted by perm $m$.

For example, $(2, 3, 4, 5, 1) \in \operatorname{perm} 5$. You should think of an element of perm $m$ as a rearrangement of the first $m$ positive integers.

The number of swaps used to change a permutation $(j_1, \ldots, j_m)$ to the standard order $(1, \ldots, m)$ can depend on the specific swaps selected. The following definition has the advantage of assigning a well-defined sign to every permutation.

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**9.32 definition: *sign of a permutation***

The *sign* of a permutation $(j_1, ..., j_m)$ is defined by

$$\text{sign}(j_1, ..., j_m) = (-1)^N,$$

where $N$ is the number of pairs of integers $(k, \ell)$ with $1 \le k < \ell \le m$ such that $k$ appears after $\ell$ in the list $(j_1, ..., j_m)$.

Hence the sign of a permutation equals 1 if the natural order has been changed an even number of times and equals $-1$ if the natural order has been changed an odd number of times.

**9.33 example: *signs***

- The permutation $(1, ..., m)$ [no changes in the natural order] has sign 1.
- The only pair of integers $(k, \ell)$ with $k < \ell$ such that $k$ appears after $\ell$ in the list $(2, 1, 3, 4)$ is $(1, 2)$. Thus the permutation $(2, 1, 3, 4)$ has sign $-1$.
- In the permutation $(2, 3, ..., m, 1)$, the only pairs $(k, \ell)$ with $k < \ell$ that appear with changed order are $(1, 2), (1, 3), ..., (1, m)$. Because we have $m - 1$ such pairs, the sign of this permutation equals $(-1)^{m-1}$.

**9.34 *swapping two entries in a permutation***

Swapping two entries in a permutation multiplies the sign of the permutation by $-1$.

Proof Suppose we have two permutations, where the second permutation is obtained from the first by swapping two entries. The two swapped entries were in their natural order in the first permutation if and only if they are not in their natural order in the second permutation. Thus we have a net change (so far) of 1 or $-1$ (both odd numbers) in the number of pairs not in their natural order.

Consider each entry between the two swapped entries. If an intermediate entry was originally in the natural order with respect to both swapped entries, then it is now in the natural order with respect to neither swapped entry. Similarly, if an intermediate entry was originally in the natural order with respect to neither of the swapped entries, then it is now in the natural order with respect to both swapped entries. If an intermediate entry was originally in the natural order with respect to exactly one of the swapped entries, then that is still true. Thus the net change (for each pair containing an entry between the two swapped entries) in the number of pairs not in their natural order is 2, $-2$, or 0 (all even numbers).

For all other pairs of entries, there is no change in whether or not they are in their natural order. Thus the total net change in the number of pairs not in their natural order is an odd number. Hence the sign of the second permutation equals $-1$ times the sign of the first permutation. ∎

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**9.35** *permutations and alternating multilinear forms*

Suppose $m$ is a positive integer and $\alpha \in V_{\text{alt}}^{(m)}$. Then

$$\alpha(v_{j_1}, \ldots, v_{j_m}) = (\text{sign}(j_1, \ldots, j_m))\alpha(v_1, \ldots, v_m)$$

for every list $v_1, \ldots, v_m$ of vectors in $V$ and all $(j_1, \ldots, j_m) \in \text{perm}\, m$.

Proof Suppose $v_1, \ldots, v_m \in V$ and $(j_1, \ldots, j_m) \in \text{perm}\, m$. We can get from $(j_1, \ldots, j_m)$ to $(1, \ldots, m)$ by a series of swaps of entries in different slots. Each such swap changes the value of $\alpha$ by a factor of $-1$ (by 9.30) and also changes the sign of the remaining permutation by a factor of $-1$ (by 9.34). After an appropriate number of swaps, we reach the permutation $1, \ldots, m$, which has sign 1. Thus the value of $\alpha$ changed signs an even number of times if $\text{sign}(j_1, \ldots, j_m) = 1$ and an odd number of times if $\text{sign}(j_1, \ldots, j_m) = -1$, which gives the desired result. ∎

Our use of permutations now leads in a natural way to the following beautiful formula for alternating $n$-linear forms on an $n$-dimensional vector space.

**9.36** *formula for* $(\dim V)$*-linear alternating forms on* $V$

Let $n = \dim V$. Suppose $e_1, \ldots, e_n$ is a basis of $V$ and $v_1, \ldots, v_n \in V$. For each $k \in \{1, \ldots, n\}$, let $b_{1,k}, \ldots, b_{n,k} \in \mathbf{F}$ be such that

$$v_k = \sum_{j=1}^{n} b_{j,k} e_j.$$

Then

$$\alpha(v_1, \ldots, v_n) = \alpha(e_1, \ldots, e_n) \sum_{(j_1, \ldots, j_n) \in \text{perm}\, n} (\text{sign}(j_1, \ldots, j_n)) b_{j_1,1} \cdots b_{j_n,n}$$

for every alternating $n$-linear form $\alpha$ on $V$.

Proof Suppose $\alpha$ is an alternating $n$-linear form $\alpha$ on $V$. Then

$$\begin{aligned}
\alpha(v_1, \ldots, v_n) &= \alpha\Big( \sum_{j_1=1}^{n} b_{j_1,1} e_{j_1}, \ldots, \sum_{j_n=1}^{n} b_{j_n,n} e_{j_n} \Big) \\
&= \sum_{j_1=1}^{n} \cdots \sum_{j_n=1}^{n} b_{j_1,1} \cdots b_{j_n,n}\, \alpha(e_{j_1}, \ldots, e_{j_n}) \\
&= \sum_{(j_1, \ldots, j_n) \in \text{perm}\, n} b_{j_1,1} \cdots b_{j_n,n}\, \alpha(e_{j_1}, \ldots, e_{j_n}) \\
&= \alpha(e_1, \ldots, e_n) \sum_{(j_1, \ldots, j_n) \in \text{perm}\, n} (\text{sign}(j_1, \ldots, j_n)) b_{j_1,1} \cdots b_{j_n,n},
\end{aligned}$$

where the third line holds because $\alpha(e_{j_1}, \ldots, e_{j_n}) = 0$ if $j_1, \ldots, j_n$ are not distinct integers, and the last line holds by 9.35. ∎

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The following result will be the key to our definition of the determinant in the next section.

**9.37** $\dim V_{\text{alt}}^{(\dim V)} = 1$

The vector space $V_{\text{alt}}^{(\dim V)}$ has dimension one.

Proof Let $n = \dim V$. Suppose $\alpha$ and $\alpha'$ are alternating $n$-linear forms on $V$ with $\alpha \neq 0$. Let $e_1, \ldots, e_n$ be such that $\alpha(e_1, \ldots, e_n) \neq 0$. There exists $c \in \mathbf{F}$ such that

$$\alpha'(e_1, \ldots, e_n) = c\alpha(e_1, \ldots, e_n).$$

Furthermore, 9.28 implies that $e_1, \ldots, e_n$ is linearly independent and thus is a basis of $V$.

Suppose $v_1, \ldots, v_n \in V$. Let $b_{j,k}$ be as in 9.36 for $j, k = 1, \ldots, n$. Then

$$\begin{aligned}
\alpha'(v_1, \ldots, v_n) &= \alpha'(e_1, \ldots, e_n) \sum_{(j_1, \ldots, j_n) \in \operatorname{perm} n} (\operatorname{sign}(j_1, \ldots, j_n)) b_{j_1,1} \cdots b_{j_n,n} \\
&= c\alpha(e_1, \ldots, e_n) \sum_{(j_1, \ldots, j_n) \in \operatorname{perm} n} (\operatorname{sign}(j_1, \ldots, j_n)) b_{j_1,1} \cdots b_{j_n,n} \\
&= c\alpha(v_1, \ldots, v_n),
\end{aligned}$$

where the first and last lines above come from 9.36. The equation above implies that $\alpha' = c\alpha$. Thus $\alpha', \alpha$ is not a linearly independent list, which implies that $\dim V_{\text{alt}}^{(n)} \leq 1$.

To complete the proof, we only need to show that there exists a nonzero alternating $n$-linear form $\alpha$ on $V$ (thus eliminating the possibility that $\dim V_{\text{alt}}^{(n)}$ equals 0). To do this, let $e_1, \ldots, e_n$ be any basis of $V$, and let $\varphi_1, \ldots, \varphi_n \in V'$ be the linear functionals on $V$ that allow us to express each element of $V$ as a linear combination of $e_1, \ldots, e_n$. In other words,

$$v = \sum_{j=1}^{n} \varphi_j(v) e_j$$

for every $v \in V$ (see 3.114). Now for $v_1, \ldots, v_n \in V$, define

$$\alpha(v_1, \ldots, v_n) = \sum_{(j_1, \ldots, j_n) \in \operatorname{perm} n} (\operatorname{sign}(j_1, \ldots, j_n)) \varphi_{j_1}(v_1) \cdots \varphi_{j_n}(v_n). \tag{9.38}$$

The verification that $\alpha$ is an $n$-linear form on $V$ is straightforward.

To see that $\alpha$ is alternating, suppose $v_1, \ldots, v_n \in V$ with $v_1 = v_2$. For each $(j_1, \ldots, j_n) \in \operatorname{perm} n$, the permutation $(j_2, j_1, j_3, \ldots, j_n)$ has the opposite sign. Because $v_1 = v_2$, the contributions from these two permutations to the sum in 9.38 cancel either other. Hence $\alpha(v_1, v_1, v_3, \ldots, v_n) = 0$. Similarly, $\alpha(v_1, \ldots, v_n) = 0$ if any two vectors in the list $v_1, \ldots, v_n$ are equal. Thus $\alpha$ is alternating.

Finally, consider 9.38 with each $v_k = e_k$. Because $\varphi_j(e_k)$ equals 0 if $j \neq k$ and equals 1 if $j = k$, only the permutation $(1, \ldots, n)$ makes a nonzero contribution to the right side of 9.38 in this case, giving the equation $\alpha(e_1, \ldots, e_n) = 1$. Thus we have produced a nonzero alternating $n$-linear form $\alpha$ on $V$, as desired. ∎

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*The formula 9.38 used in the last proof to construct a nonzero alternating $n$-linear form came from the formula in 9.36, and that formula arose naturally from the properties of an alternating multilinear form.*

Earlier we showed that the value of an alternating multilinear form applied to a linearly dependent list is 0; see 9.28. The next result provides a converse of 9.28 for $n$-linear multilinear forms when $n = \dim V$. In the following result, the statement that $\alpha$ is nonzero means (as usual for a function) that $\alpha$ is not the function on $V^n$ that is identically 0.

**9.39** *alternating $(\dim V)$-linear forms and linear independence*

Let $n = \dim V$. Suppose $\alpha$ is a nonzero alternating $n$-linear form on $V$ and $e_1, \ldots, e_n$ is a list of vectors in $V$. Then

$$\alpha(e_1, \ldots, e_n) \neq 0$$

if and only if $e_1, \ldots, e_n$ is linearly independent.

Proof First suppose $\alpha(e_1, \ldots, e_n) \neq 0$. Then 9.28 implies that $e_1, \ldots, e_n$ is linearly independent.

To prove the implication in the other direction, now suppose $e_1, \ldots, e_n$ is linearly independent. Because $n = \dim V$, this implies that $e_1, \ldots, e_n$ is a basis of $V$ (see 2.38).

Because $\alpha$ is not the zero $n$-linear form, there exist $v_1, \ldots, v_n \in V$ such that $\alpha(v_1, \ldots, v_n) \neq 0$. Now 9.36 implies that $\alpha(e_1, \ldots, e_n) \neq 0$. ∎

## *Exercises 9B*

**1** Suppose $m$ is a positive integer. Show that $\dim V^{(m)} = (\dim V)^m$.

**2** Suppose $n \geq 3$ and $\alpha : \mathbf{F}^n \times \mathbf{F}^n \times \mathbf{F}^n \to \mathbf{F}$ is defined by

$$\begin{aligned}&\alpha((x_1, \ldots, x_n), (y_1, \ldots, y_n), (z_1, \ldots, z_n))\\ &\quad = x_1y_2z_3 - x_2y_1z_3 - x_3y_2z_1 - x_1y_3z_2 + x_3y_1z_2 + x_2y_3z_1.\end{aligned}$$

Show that $\alpha$ is an alternating 3-linear form on $\mathbf{F}^n$.

**3** Suppose $m$ is a positive integer and $\alpha$ is an $m$-linear form on $V$ such that $\alpha(v_1, \ldots, v_m) = 0$ whenever $v_1, \ldots, v_m$ is a list of vectors in $V$ with $v_j = v_{j+1}$ for some $j \in \{1, \ldots, m-1\}$. Prove that $\alpha$ is an alternating $m$-linear form on $V$.

**4** Prove or give a counterexample: If $\alpha \in V^{(4)}_{\text{alt}}$, then

$$\{(v_1, v_2, v_3, v_4) \in V^4 : \alpha(v_1, v_2, v_3, v_4) = 0\}$$

is a subspace of $V^4$.

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**5** Suppose $m$ is a positive integer and $\beta$ is an $m$-linear form on $V$. Define an $m$-linear form $\alpha$ on $V$ by

$$\alpha(v_1, \dots, v_m) = \sum_{(j_1, \dots, j_m) \in \operatorname{perm} m} (\operatorname{sign}(j_1, \dots, j_m)) \beta(v_{j_1}, \dots, v_{j_m})$$

for $v_1, \dots, v_m \in V$. Explain why $\alpha \in V_{\text{alt}}^{(m)}$.

**6** Suppose $m$ is a positive integer and $\beta$ is an $m$-linear form on $V$. Define an $m$-linear form $\alpha$ on $V$ by

$$\alpha(v_1, \dots, v_m) = \sum_{(j_1, \dots, j_m) \in \operatorname{perm} m} \beta(v_{j_1}, \dots, v_{j_m})$$

for $v_1, \dots, v_m \in V$. Explain why

$$\alpha(v_{k_1}, \dots, v_{k_m}) = \alpha(v_1, \dots, v_m)$$

for all $v_1, \dots, v_m \in V$ and all $(k_1, \dots, k_m) \in \operatorname{perm} m$.

**7** Give an example of a nonzero alternating 2-linear form $\alpha$ on $\mathbf{R}^3$ and a linearly independent list $v_1, v_2$ in $\mathbf{R}^3$ such that $\alpha(v_1, v_2) = 0$.

*This exercise shows that 9.39 can fail if the hypothesis that $n = \dim V$ is deleted.*

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# 9C Determinants

## Defining the Determinant

The next definition will lead us to a clean, beautiful, basis-free definition of the determinant of an operator.

**9.40 definition: $\alpha_T$**

Suppose that $m$ is a positive integer and $T \in \mathcal{L}(V)$. For $\alpha \in V_{\text{alt}}^{(m)}$, define $\alpha_T \in V_{\text{alt}}^{(m)}$ by

$$\alpha_T(v_1, \dots, v_m) = \alpha(Tv_1, \dots, Tv_m)$$

for each list $v_1, \dots, v_m$ of vectors in $V$.

Suppose $T \in \mathcal{L}(V)$. If $\alpha \in V_{\text{alt}}^{(m)}$ and $v_1, \dots, v_m$ is a list of vectors in $V$ with $v_j = v_k$ for some $j \neq k$, then $Tv_j = Tv_k$, which implies that $\alpha_T(v_1, \dots, v_m) = \alpha(Tv_1, \dots, Tv_m) = 0$. Thus the function $\alpha \mapsto \alpha_T$ is a linear map of $V_{\text{alt}}^{(m)}$ to itself.

We know that $\dim V_{\text{alt}}^{(\dim V)} = 1$ (see 9.37). Every linear map from a one-dimensional vector space to itself is multiplication by some unique scalar. For the linear map $\alpha \mapsto \alpha_T$, we now define $\det T$ to be that scalar.

**9.41 definition: *determinant of an operator*, $\det T$**

Suppose $T \in \mathcal{L}(V)$. The *determinant* of $T$, denoted by $\det T$, is defined to be the unique number in $\mathbf{F}$ such that

$$\alpha_T = (\det T)\, \alpha$$

for all $\alpha \in V_{\text{alt}}^{(\dim V)}$.

**9.42 example: *determinants of operators***

Let $n = \dim V$.

- If $I$ is the identity operator on $V$, then $\alpha_I = \alpha$ for all $\alpha \in V_{\text{alt}}^{(n)}$. Thus $\det I = 1$.
- More generally, if $\lambda \in \mathbf{F}$, then $\alpha_{\lambda I} = \lambda^n \alpha$ for all $\alpha \in V_{\text{alt}}^{(n)}$. Thus $\det(\lambda I) = \lambda^n$.
- Still more generally, if $T \in \mathcal{L}(V)$ and $\lambda \in \mathbf{F}$, then $\alpha_{\lambda T} = \lambda^n \alpha_T = \lambda^n (\det T)\alpha$ for all $\alpha \in V_{\text{alt}}^{(n)}$. Thus $\det(\lambda T) = \lambda^n \det T$.
- Suppose $T \in \mathcal{L}(V)$ and there is a basis $e_1, \dots, e_n$ of $V$ consisting of eigenvectors of $T$, with corresponding eigenvalues $\lambda_1, \dots, \lambda_n$. If $\alpha \in V_{\text{alt}}^{(n)}$, then

$$\alpha_T(e_1, \dots, e_n) = \alpha(\lambda_1 e_1, \dots, \lambda_n e_n) = (\lambda_1 \cdots \lambda_n)\alpha(e_1, \dots, e_n).$$

If $\alpha \neq 0$, then 9.39 implies $\alpha(e_1, \dots, e_n) \neq 0$. Thus the equation above implies

$$\det T = \lambda_1 \cdots \lambda_n.$$

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Our next task is to define and give a formula for the determinant of a square matrix. To do this, we associate with each square matrix an operator and then define the determinant of the matrix to be the determinant of the associated operator.

**9.43 definition: *determinant of a matrix,* $\det A$**

Suppose that $n$ is a positive integer and $A$ is an $n$-by-$n$ square matrix with entries in $\mathbf{F}$. Let $T \in \mathcal{L}(\mathbf{F}^n)$ be the operator whose matrix with respect to the standard basis of $\mathbf{F}^n$ equals $A$. The *determinant* of $A$, denoted by $\det A$, is defined by $\det A = \det T$.

**9.44 example: *determinants of matrices***

- If $I$ is the $n$-by-$n$ identity matrix, then the corresponding operator on $\mathbf{F}^n$ is the identity operator $I$ on $\mathbf{F}^n$. Thus the first bullet point of 9.42 implies that the determinant of the identity matrix is 1.
- Suppose $A$ is a diagonal matrix with $\lambda_1, \dots, \lambda_n$ on the diagonal. Then the corresponding operator on $\mathbf{F}^n$ has the standard basis of $\mathbf{F}^n$ as eigenvectors, with eigenvalues $\lambda_1, \dots, \lambda_n$. Thus the last bullet point of 9.42 implies that $\det A = \lambda_1 \cdots \lambda_n$.

For the next result, think of each list $v_1, \dots, v_n$ of $n$ vectors in $\mathbf{F}^n$ as a list of $n$-by-1 column vectors. The notation $\begin{pmatrix} v_1 & \cdots & v_n \end{pmatrix}$ then denotes the $n$-by-$n$ square matrix whose $k^{\text{th}}$ column is $v_k$ for each $k = 1, \dots, n$.

**9.45 *determinant is an alternating multilinear form***

Suppose that $n$ is a positive integer. The map that takes a list $v_1, \dots, v_n$ of vectors in $\mathbf{F}^n$ to $\det \begin{pmatrix} v_1 & \cdots & v_n \end{pmatrix}$ is an alternating $n$-linear form on $\mathbf{F}^n$.

Proof Let $e_1, \dots, e_n$ be the standard basis of $\mathbf{F}^n$ and suppose $v_1, \dots, v_n$ is a list of vectors in $\mathbf{F}^n$. Let $T \in \mathcal{L}(\mathbf{F}^n)$ be the operator such that $Te_k = v_k$ for $k = 1, \dots, n$. Thus $T$ is the operator whose matrix with respect to $e_1, \dots, e_n$ is $\begin{pmatrix} v_1 & \cdots & v_n \end{pmatrix}$. Hence $\det \begin{pmatrix} v_1 & \cdots & v_n \end{pmatrix} = \det T$, by definition of the determinant of a matrix.

Let $\alpha$ be an alternating $n$-linear form on $\mathbf{F}^n$ such that $\alpha(e_1, \dots, e_n) = 1$. Then

$$\begin{aligned} \det \begin{pmatrix} v_1 & \cdots & v_n \end{pmatrix} &= \det T \\ &= (\det T)\, \alpha(e_1, \dots, e_n) \\ &= \alpha(Te_1, \dots, Te_n) \\ &= \alpha(v_1, \dots, v_n), \end{aligned}$$

where the third line follows from the definition of the determinant of an operator. The equation above shows that the map that takes a list of vectors $v_1, \dots, v_n$ in $\mathbf{F}^n$ to $\det \begin{pmatrix} v_1 & \cdots & v_n \end{pmatrix}$ is the alternating $n$-linear form $\alpha$ on $\mathbf{F}^n$.

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The previous result has several important consequences. For example, it immediately implies that a matrix with two identical columns has determinant 0. We will come back to other consequences later, but for now we want to give a formula for the determinant of a square matrix. Recall that if $A$ is a matrix, then $A_{j,k}$ denotes the entry in row $j$, column $k$ of $A$.

**9.46** *formula for determinant of a matrix*

Suppose that $n$ is a positive integer and $A$ is an $n$-by-$n$ square matrix. Then

$$\det A = \sum_{(j_1,\dots,j_n)\in \operatorname{perm} n} (\operatorname{sign}(j_1,\dots,j_n)) A_{j_1,1}\cdots A_{j_n,n}.$$

Proof Apply 9.36 with $V = \mathbf{F}^n$ and $e_1,\dots,e_n$ the standard basis of $\mathbf{F}^n$ and $\alpha$ the alternating $n$-linear form on $\mathbf{F}^n$ that takes $v_1,\dots,v_n$ to $\det \begin{pmatrix} v_1 & \cdots & v_n \end{pmatrix}$ [see 9.45]. If each $v_k$ is the $k^{\text{th}}$ column of $A$, then each $b_{j,k}$ in 9.36 equals $A_{j,k}$. Finally,

$$\alpha(e_1,\dots,e_n) = \det \begin{pmatrix} e_1 & \cdots & e_n \end{pmatrix} = \det I = 1.$$

Thus the formula in 9.36 becomes the formula stated in this result. ∎

**9.47** example: *explicit formula for determinant*

- If $A$ is a 2-by-2 matrix, then the formula in 9.46 becomes
$$\det A = A_{1,1}A_{2,2} - A_{2,1}A_{1,2}.$$
- If $A$ is a 3-by-3 matrix, then the formula in 9.46 becomes
$$\begin{aligned}\det A =& A_{1,1}A_{2,2}A_{3,3} - A_{2,1}A_{1,2}A_{3,3} - A_{3,1}A_{2,2}A_{1,3}\\ &- A_{1,1}A_{3,2}A_{2,3} + A_{3,1}A_{1,2}A_{2,3} + A_{2,1}A_{3,2}A_{1,3}.\end{aligned}$$

The sum in the formula in 9.46 contains $n!$ terms. Because $n!$ grows rapidly as $n$ increases, the formula in 9.46 is not a viable method to evaluate determinants even for moderately sized $n$. For example, 10! is over three million, and 100! is approximately $10^{158}$, leading to a sum that the fastest computer cannot evaluate. We will soon see some results that lead to faster evaluations of determinants than direct use of the sum in 9.46.

**9.48** *determinant of upper-triangular matrix*

Suppose that $A$ is an upper-triangular matrix with $\lambda_1,\dots,\lambda_n$ on the diagonal. Then $\det A = \lambda_1\cdots\lambda_n$.

Proof If $(j_1,\dots,j_n) \in \operatorname{perm} n$ with $(j_1,\dots,j_n) \neq (1,\dots,n)$, then $j_k > k$ for some $k \in \{1,\dots,n\}$, which implies that $A_{j_k,k} = 0$. Thus the only permutation that can make a nonzero contribution to the sum in 9.46 is the permutation $(1,\dots,n)$. Because $A_{k,k} = \lambda_k$ for each $k = 1,\dots,n$, this implies that $\det A = \lambda_1\cdots\lambda_n$. ∎

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## *Properties of Determinants*

Our definition of the determinant leads to the following magical proof that the determinant is multiplicative.

**9.49** *determinant is multiplicative*

(a) Suppose $S, T \in \mathcal{L}(V)$. Then $\det(ST) = (\det S)(\det T)$.

(b) Suppose $A$ and $B$ are square matrices of the same size. Then

$$\det(AB) = (\det A)(\det B)$$

Proof

(a) Let $n = \dim V$. Suppose $\alpha \in V_{\text{alt}}^{(n)}$ and $v_1, \dots, v_n \in V$. Then

$$\begin{aligned}\alpha_{ST}(v_1, \dots, v_n) &= \alpha(STv_1, \dots, STv_n)\\ &= (\det S)\alpha(Tv_1, \dots, Tv_n)\\ &= (\det S)(\det T)\alpha(v_1, \dots, v_n),\end{aligned}$$

where the first equation follows from the definition of $\alpha_{ST}$, the second equation follows from the definition of $\det S$, and the third equation follows from the definition of $\det T$. The equation above implies that $\det(ST) = (\det S)(\det T)$.

(b) Let $S, T \in \mathcal{L}(\mathbf{F}^n)$ be such that $\mathcal{M}(S) = A$ and $\mathcal{M}(T) = B$, where all matrices of operators in this proof are with respect to the standard basis of $\mathbf{F}^n$. Then $\mathcal{M}(ST) = \mathcal{M}(S)\mathcal{M}(T) = AB$ (see 3.43). Thus

$$\det(AB) = \det(ST) = (\det S)(\det T) = (\det A)(\det B),$$

where the second equality comes from the result in (a). ∎

The determinant of an operator determines whether the operator is invertible.

**9.50** *invertible ⟺ nonzero determinant*

An operator $T \in \mathcal{L}(V)$ is invertible if and only if $\det T \neq 0$. Furthermore, if $T$ is invertible, then $\det(T^{-1}) = \frac{1}{\det T}$.

Proof First suppose $T$ is invertible. Thus $TT^{-1} = I$. Now 9.49 implies that

$$1 = \det I = \det(TT^{-1}) = (\det T)\big(\det(T^{-1})\big).$$

Hence $\det T \neq 0$ and $\det(T^{-1})$ is the multiplicative inverse of $\det T$.

To prove the other direction, now suppose $\det T \neq 0$. Suppose $v \in V$ and $v \neq 0$. Let $v, e_2, \dots, e_n$ be a basis of $V$ and let $\alpha \in V_{\text{alt}}^{(n)}$ be such that $\alpha \neq 0$. Then $\alpha(v, e_2, \dots, e_n) \neq 0$ (by 9.39). Now

$$\alpha(Tv, Te_2, \dots, Te_n) = (\det T)\alpha(v, e_2, \dots, e_n) \neq 0,$$

Thus $Tv \neq 0$. Hence $T$ is invertible. ∎

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An $n$-by-$n$ matrix $A$ is invertible (see 3.80 for the definition of an invertible matrix) if and only if the operator on $\mathbf{F}^n$ associated with $A$ (via the standard basis of $\mathbf{F}^n$) is invertible. Thus the previous result shows that a square matrix $A$ is invertible if and only if $\det A \neq 0$.

**9.51** *eigenvalues and determinants*

Suppose $T \in \mathcal{L}(V)$ and $\lambda \in \mathbf{F}$. Then $\lambda$ is an eigenvalue of $T$ if and only if $\det(\lambda I - T) = 0$.

Proof The number $\lambda$ is an eigenvalue of $T$ if and only if $T - \lambda I$ is not invertible (see 5.7), which happens if and only if $\lambda I - T$ is not invertible, which happens if and only if $\det(\lambda I - T) = 0$ (by 9.50). ∎

Suppose $T \in \mathcal{L}(V)$ and $S \colon W \to V$ is an invertible linear map. To prove that $\det(S^{-1}TS) = \det T$, we could try to use 9.49 and 9.50, writing

$$\begin{aligned}\det(S^{-1}TS) &= (\det S^{-1})(\det T)(\det S)\\ &= \det T.\end{aligned}$$

That proof works if $W = V$, but if $W \neq V$ then it makes no sense because the determinant is defined only for linear maps from a vector space to itself, and $S$ maps $W$ to $V$, making $\det S$ undefined. The proof given below works around this issue and is valid when $W \neq V$.

**9.52** *determinant is a similarity invariant*

Suppose $T \in \mathcal{L}(V)$ and $S \colon W \to V$ is an invertible linear map. Then

$$\det(S^{-1}TS) = \det T.$$

Proof Let $n = \dim W = \dim V$. Suppose $\tau \in W^{(n)}_{\text{alt}}$. Define $\alpha \in V^{(n)}_{\text{alt}}$ by

$$\alpha(v_1, \ldots, v_n) = \tau(S^{-1}v_1, \ldots, S^{-1}v_n)$$

for $v_1, \ldots, v_n \in V$. Suppose $w_1, \ldots, w_n \in W$. Then

$$\begin{aligned}\tau_{S^{-1}TS}(w_1, \ldots, w_n) &= \tau(S^{-1}TSw_1, \ldots, S^{-1}TSw_n)\\ &= \alpha(TSw_1, \ldots, TSw_n)\\ &= \alpha_T(Sw_1, \ldots, Sw_n)\\ &= (\det T)\alpha(Sw_1, \ldots, Sw_n)\\ &= (\det T)\tau(w_1, \ldots, w_n).\end{aligned}$$

The equation above and the definition of the determinant of the operator $S^{-1}TS$ imply that $\det(S^{-1}TS) = \det T$. ∎

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For the special case in which $V = \mathbf{F}^n$ and $e_1, \ldots, e_n$ is the standard basis of $\mathbf{F}^n$, the next result is true by the definition of the determinant of a matrix. The left side of the equation in the next result does not depend on a choice of basis, which means that the right side is independent of the choice of basis.

**9.53** *determinant of operator equals determinant of its matrix*

Suppose $T \in \mathcal{L}(V)$ and $e_1, \ldots, e_n$ is a basis of $V$. Then

$$\det T = \det \mathcal{M}(T, (e_1, \ldots, e_n)).$$

Proof Let $f_1, \ldots, f_n$ be the standard basis of $\mathbf{F}^n$. Let $S : \mathbf{F}^n \to V$ be the linear map such that $Sf_k = e_k$ for each $k = 1, \ldots, n$. Thus $\mathcal{M}(S, (f_1, \ldots, f_n), (e_1, \ldots, e_n))$ and $\mathcal{M}(S^{-1}, (e_1, \ldots, e_n), (f_1, \ldots, f_n))$ both equal the $n$-by-$n$ identity matrix. Hence

$$\mathcal{M}(S^{-1}TS, (f_1, \ldots, f_n)) = \mathcal{M}(T, (e_1, \ldots, e_n)), \tag{9.54}$$

as follows from two applications of 3.43. Thus

$$\begin{aligned} \det T &= \det(S^{-1}TS) \\ &= \det \mathcal{M}(S^{-1}TS, (f_1, \ldots, f_n)) \\ &= \det \mathcal{M}(T, (e_1, \ldots, e_n)), \end{aligned}$$

where the first line comes from 9.52, the second line comes from the definition of the determinant of a matrix, and the third line follows from 9.54. ∎

The next result gives a more intuitive way to think about determinants than the definition or the formula in 9.46. We could make the characterization in the result below the definition of the determinant of an operator on a finite-dimensional complex vector space, with the current definition then becoming a consequence of that definition.

**9.55** *if $\mathbf{F} = \mathbf{C}$, then determinant equals product of eigenvalues*

Suppose $\mathbf{F} = \mathbf{C}$ and $T \in \mathcal{L}(V)$. Then $\det T$ equals the product of the eigenvalues of $T$, with each eigenvalue included as many times as its multiplicity.

Proof There is a basis of $V$ with respect to which $T$ has an upper-triangular matrix with the diagonal entries of the matrix consisting of the eigenvalues of $T$, with each eigenvalue included as many times as its multiplicity—see 8.37. Thus 9.53 and 9.48 imply that $\det T$ equals the product of the eigenvalues of $T$, with each eigenvalue included as many times as its multiplicity. ∎

As the next result shows, the determinant interacts nicely with the transpose of a square matrix, with the dual of an operator, and with the adjoint of an operator on an inner product space.

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**9.56** *determinant of transpose, dual, or adjoint*

(a) Suppose $A$ is a square matrix. Then $\det A^t = \det A$.

(b) Suppose $T \in \mathcal{L}(V)$. Then $\det T' = \det T$.

(c) Suppose $V$ is an inner product space and $T \in \mathcal{L}(V)$. Then

$$\det(T^*) = \overline{\det T}.$$

Proof

(a) Let $n$ be a positive integer. Define $\alpha : (\mathbf{F}^n)^n \to \mathbf{F}$ by

$$\alpha\Big(\begin{pmatrix} v_1 & \cdots & v_n \end{pmatrix}\Big) = \det\Big(\begin{pmatrix} v_1 & \cdots & v_n \end{pmatrix}^t\Big)$$

for all $v_1, \ldots, v_n \in \mathbf{F}^n$. The formula in 9.46 for the determinant of a matrix shows that $\alpha$ is an $n$-linear form on $\mathbf{F}^n$.

Suppose $v_1, \ldots, v_n \in \mathbf{F}^n$ and $v_j = v_k$ for some $j \neq k$. If $B$ is an $n$-by-$n$ matrix, then $\begin{pmatrix} v_1 & \cdots & v_n \end{pmatrix}^t B$ cannot equal the identity matrix because row $j$ and row $k$ of $\begin{pmatrix} v_1 & \cdots & v_n \end{pmatrix}^t B$ are equal. Thus $\begin{pmatrix} v_1 & \cdots & v_n \end{pmatrix}^t$ is not invertible, which implies that $\alpha\Big(\begin{pmatrix} v_1 & \cdots & v_n \end{pmatrix}\Big) = 0$. Hence $\alpha$ is an alternating $n$-linear form on $\mathbf{F}^n$.

Note that $\alpha$ applied to the standard basis of $\mathbf{F}^n$ equals 1. Because the vector space of alternating $n$-linear forms on $\mathbf{F}^n$ has dimension one (by 9.37), this implies that $\alpha$ is the determinant function. Thus (a) holds.

(b) The equation $\det T' = \det T$ follows from (a) and 9.53 and 3.132.

(c) Pick an orthonormal basis of $V$. The matrix of $T^*$ with respect to that basis is the conjugate transpose of the matrix of $T$ with respect to that basis (by 7.9). Thus 9.53, 9.46, and (a) imply that $\det(T^*) = \overline{\det T}$. ∎

**9.57** *helpful results in evaluating determinants*

(a) If either two columns or two rows of a square matrix are equal, then the determinant of the matrix equals 0.

(b) Suppose $A$ is a square matrix and $B$ is the matrix obtained from $A$ by swapping either two columns or two rows. Then $\det A = -\det B$.

(c) If one column or one row of a square matrix is multiplied by a scalar, then the value of the determinant is multiplied by the same scalar.

(d) If a scalar multiple of one column of a square matrix to added to another column, then the value of the determinant is unchanged.

(e) If a scalar multiple of one row of a square matrix to added to another row, then the value of the determinant is unchanged.

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Proof All the assertions in this result follow from the result that the maps $v_1, \dots, v_n \mapsto \det \begin{pmatrix} v_1 & \cdots & v_n \end{pmatrix}$ and $v_1, \dots, v_n \mapsto \det \begin{pmatrix} v_1 & \cdots & v_n \end{pmatrix}^t$ are both alternating $n$-linear forms on $\mathbf{F}^n$ [see 9.45 and 9.56(a)].

For example, to prove (d) suppose $v_1, \dots, v_n \in \mathbf{F}^n$ and $c \in \mathbf{F}$. Then

$$\begin{aligned}
&\det \begin{pmatrix} v_1 + cv_2 & v_2 & \cdots & v_n \end{pmatrix} \\
&\quad = \det \begin{pmatrix} v_1 & v_2 & \cdots & v_n \end{pmatrix} + c \det \begin{pmatrix} v_2 & v_2 & v_3 & \cdots & v_n \end{pmatrix} \\
&\quad = \det \begin{pmatrix} v_1 & v_2 & \cdots & v_n \end{pmatrix},
\end{aligned}$$

where the first equation follows from the multilinearity property and the second equation follows from the alternating property. The equation above shows that adding a multiple of the second column to the first column does not change the value of the determinant. The same conclusion holds for any two columns. Thus (d) holds.

The proof of (e) follows from (d) and from 9.56(a). The proofs of (a), (b), and (c) use similar tools and are left to the reader. ∎

For matrices whose entries are concrete numbers, the result above leads to a much faster way to evaluate the determinant than direct application of the formula in 9.46. Specifically, apply the Gaussian elimination procedure of swapping rows [by 9.48(b), this changes the determinant by a factor of $-1$], multiplying a row by a nonzero constant [by 9.48(c), this changes the determinant by the same constant], and adding a multiple of one row to another row [by 9.48(e), this does not change the determinant] to produce an upper-triangular matrix, whose determinant is the product of the diagonal entries (by 9.48). If your software keeps track of the number of row swaps and of the constants used when multiplying a row by a constant, then the determinant of the original matrix can be computed.

Because a number $\lambda \in \mathbf{F}$ is an eigenvalue of an operator $T \in \mathcal{L}(V)$ if and only if $\det(\lambda I - T) = 0$ (by 9.51), you may be tempted to think that one way to find eigenvalues quickly is to choose a basis of $V$, let $A = \mathcal{M}(T)$, evaluate $\det(\lambda I - A)$, and then solve the equation $\det(\lambda I - A) = 0$ for $\lambda$. However, that procedure is rarely efficient, except when $\dim V = 2$ (or when $\dim V$ equals 3 or 4 if you are willing to use the cubic or quartic formulas). One problem is that the procedure described in the paragraph above for evaluating a determinant does not work when the matrix includes a symbol (such as the $\lambda$ in $\lambda I - A$). This problem arises because decisions need to be made in the Gaussian elimination procedure about whether certain quantities equal 0, and those decisions become complicated in expressions involving a symbol $\lambda$.

Recall that an operator on a finite-dimensional inner product space is unitary if it preserves norms (see 7.51 and the paragraph following it). Every eigenvalue of a unitary operator has absolute value 1 (by 7.54). Thus the product of the eigenvalues of a unitary operator has absolute value 1. Hence (at least in the case $\mathbf{F} = \mathbf{C}$) the determinant of a unitary operator has absolute value 1 (by 9.55). The next result gives a proof that works without the assumption that $\mathbf{F} = \mathbf{C}$.

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**9.58** *every unitary operator has determinant with absolute value 1*

Suppose $V$ is an inner product space and $S \in \mathcal{L}(V)$ is a unitary operator. Then $|\det S| = 1$.

Proof Because $S$ is unitary, $I = S^*S$ (see 7.53). Thus

$$1 = \det(S^*S) = (\det S^*)(\det S) = \overline{(\det S)}(\det S) = |\det S|^2,$$

where the second equality comes from 9.49(a) and the third equality comes from 9.56(c). The equation above implies that $|\det S| = 1$. ∎

The determinant of a positive operator on an inner product space meshes well with the analogy that such operators correspond to the nonnegative real numbers.

**9.59** *every positive operator has nonnegative determinant*

Suppose $V$ is an inner product space and $T \in \mathcal{L}(V)$ is a positive operator. Then $\det T \geq 0$.

Proof By the spectral theorem (7.29 or 7.31), $V$ has an orthonormal basis consisting of eigenvectors of $T$. Thus by the last bullet point of 9.42, $\det T$ equals a product of the eigenvalues of $T$, possibly with repetitions. Each eigenvalue of $T$ is a nonnegative number (by 7.38). Thus we conclude that $\det T \geq 0$. ∎

Suppose $V$ is an inner product space and $T \in \mathcal{L}(V)$. Recall that the list of nonnegative square roots of the eigenvalues of $T^*T$ (each included as many times as its multiplicity) is called the list of singular values of $T$ (see Section 7E).

**9.60** $|\det T|$ = *product of singular values of* $T$

Suppose $V$ is an inner product space and $T \in \mathcal{L}(V)$. Then

$$|\det T| = \sqrt{\det(T^*T)} = \text{product of singular values of } T.$$

Proof We have

$$|\det T|^2 = \overline{(\det T)}(\det T) = \big(\det(T^*)\big)(\det T) = \det(T^*T),$$

where the middle equality comes from 9.56(c) and the last equality comes from 9.49(a). Taking square roots of both sides of the equation above shows that $|\det T| = \sqrt{\det(T^*T)}$.

Let $s_1, \ldots, s_n$ denote the list of singular values of $T$. Thus $s_1^2, \ldots, s_n^2$ is the list of eigenvalues of $T^*T$ (with appropriate repetitions), corresponding to an orthonormal basis of $V$ consisting of eigenvectors of $T^*T$. Hence the last bullet point of 9.42 implies that

$$\det(T^*T) = s_1^2 \cdots s_n^2.$$

Thus $|\det T| = s_1 \cdots s_n$, as desired. ∎

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An operator $T$ on a real inner product space changes volume by a factor of the product of the singular values (by 7.111). Thus the next result follows immediately from 7.111 and 9.60. This result explains why the absolute value of a determinant appears in the change of variables formula in multivariable calculus.

**9.61** *$T$ changes volume by factor of $|\det T|$*

Suppose $T \in \mathcal{L}(\mathbf{R}^n)$ and $\Omega \subseteq \mathbf{R}^n$. Then

$$\text{volume } T(\Omega) = |\det T|(\text{volume } \Omega).$$

For operators on finite-dimensional complex vector spaces, we now connect the determinant to a polynomial that we have previously seen.

**9.62** *if $\mathbf{F} = \mathbf{C}$, then characteristic polynomial of $T$ equals $\det(zI - T)$*

Suppose $\mathbf{F} = \mathbf{C}$ and $T \in \mathcal{L}(V)$. Let $\lambda_1, \ldots, \lambda_m$ denote the distinct eigenvalues of $T$, and let $d_1, \ldots, d_m$ denote their multiplicities. Then

$$\det(zI - T) = (z - \lambda_1)^{d_1} \cdots (z - \lambda_m)^{d_m}.$$

Proof There exists a basis of $V$ with respect to which $T$ has an upper-triangular matrix with each $\lambda_k$ appearing on the diagonal exactly $d_k$ times (by 8.37). With respect to this basis, $zI - T$ has an upper-triangular matrix with $z - \lambda_k$ appearing on the diagonal exactly $d_k$ times for each $k$. Thus 9.48 gives the desired equation. ∎

Suppose $\mathbf{F} = \mathbf{C}$ and $T \in \mathcal{L}(V)$. The characteristic polynomial of $T$ was defined in 8.26 as the polynomial on the right side of the equation in 9.62. We did not previously define the characteristic polynomial of an operator on a finite-dimensional real vector space because such operators may have no eigenvalues, making a definition using the right side of the equation in 9.62 inappropriate.

We now present a new definition of the characteristic polynomial, motivated by 9.62. This new definition is valid for both real and complex vector spaces. The equation in 9.62 shows that this new definition is equivalent to our previous definition when $\mathbf{F} = \mathbf{C}$ (8.26).

**9.63** definition: *characteristic polynomial*

Suppose $T \in \mathcal{L}(V)$. The polynomial defined by

$$z \mapsto \det(zI - T)$$

is called the *characteristic polynomial* of $T$.

The formula in 9.46 shows that the characteristic polynomial of an operator $T \in \mathcal{L}(V)$ is a monic polynomial of degree $\dim V$. The zeros in $\mathbf{F}$ of the characteristic polynomial of $T$ are exactly the eigenvalues of $T$ (by 9.51).

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Previously we proved the Cayley–Hamilton theorem (8.29) in the complex case. Now we can extend that result to operators on real vector spaces.

**9.64** *Cayley–Hamilton theorem*

Suppose $T \in \mathcal{L}(V)$ and $q$ is the characteristic polynomial of $T$. Then $q(T) = 0$.

Proof If $\mathbf{F} = \mathbf{C}$, then the equation $q(T) = 0$ follows from 9.62 and 8.29.

Now suppose $\mathbf{F} = \mathbf{R}$. Fix a basis of $V$, and let $A$ be the matrix of $T$ with respect to this basis. Let $S$ be the operator on $\mathbf{C}^{\dim V}$ such that the matrix of $S$ (with respect to the standard basis of $\mathbf{C}^{\dim V}$) is $A$. For all $z \in \mathbf{R}$ we have

$$q(z) = \det(zI - T) = \det(zI - A) = \det(zI - S).$$

Thus $q$ is the characteristic polynomial of $S$. The case $\mathbf{F} = \mathbf{C}$ (first sentence of this proof) now implies that $0 = q(S) = q(A) = q(T)$. ∎

The Cayley–Hamilton theorem (9.64) implies that the characteristic polynomial of an operator $T \in \mathcal{L}(V)$ is a polynomial multiple of the minimal polynomial of $T$ (by 5.29). Thus if the degree of the minimal polynomial of $T$ equals $\dim V$, then the characteristic polynomial of $T$ equals the minimal polynomial of $T$. This happens for a very large percentage of operators, including over 99.999% of 4-by-4 matrices with integer entries in $[-100, 100]$ (see the paragraph following 5.25).

The last sentence in our next result was previously proved in the complex case (see 8.54). Now we can give a proof that works on both real and complex vector spaces.

**9.65** *characteristic polynomial, trace, and determinant*

Suppose $T \in \mathcal{L}(V)$. Let $n = \dim V$. Then the characteristic polynomial of $T$ can be written as

$$z^n - (\operatorname{tr} T)z^{n-1} + \cdots + (-1)^n(\det T).$$

Proof The constant term of a polynomial function of $z$ is the value of the polynomial when $z = 0$. Thus the constant term of the characteristic polynomial of $T$ equals $\det(-T)$, which equals $(-1)^n \det T$ (by the third bullet point of 9.42).

Fix a basis of $V$, and let $A$ be the matrix of $T$ with respect to this basis. The matrix of $zI - T$ with respect to this basis is $zI - A$. The term coming from the identity permutation $\{1, \ldots, n\}$ in the formula 9.46 for $\det(zI - A)$ is

$$(z - A_{1,1}) \cdots (z - A_{n,n}).$$

The coefficient of $z^{n-1}$ in the expression above is $-(A_{1,1} + \cdots + A_{n,n})$, which equals $-\operatorname{tr} T$. The terms in the formula for $\det(zI - A)$ coming from other elements of perm $n$ contain at most $n-2$ factors of the form $z - A_{k,k}$ and thus do not contribute to the coefficient of $z^{n-1}$ in the characteristic polynomial of $T$. ∎

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*The next result was proved by Jacques Hadamard (1865–1963) in 1893.*

In the result below, think of the columns of the $n$-by-$n$ matrix $A$ as elements of $\mathbf{F}^n$. The norms appearing below then arise from the standard inner product on $\mathbf{F}^n$. Recall that the notation $R_{\cdot,k}$ in the proof below means the $k^{\text{th}}$ column of the matrix $R$ (as was defined in 3.44).

**9.66** *Hadamard's inequality*

Suppose $A$ is an $n$-by-$n$ matrix. Let $v_1, \ldots, v_n$ denote the columns of $A$. Then

$$|\det A| \le \prod_{k=1}^{n} \|v_k\|.$$

Proof If $A$ is not invertible, then $\det A = 0$ and hence the desired inequality holds in this case.

Thus assume that $A$ is invertible. The QR factorization (7.58) tells us that there exist a unitary matrix $Q$ and an upper-triangular matrix $R$ whose diagonal contains only positive numbers such that $A = QR$. We have

$$\begin{aligned}
|\det A| &= |\det Q| \, |\det R| \\
&= |\det R| \\
&= \prod_{k=1}^{n} R_{k,k} \\
&\le \prod_{k=1}^{n} \|R_{\cdot,k}\| \\
&= \prod_{k=1}^{n} \|QR_{\cdot,k}\| \\
&= \prod_{k=1}^{n} \|v_k\|,
\end{aligned}$$

where the first line comes from 9.49(b), the second line comes from 9.58, the third line comes from 9.48, and the fifth line holds because $Q$ is an isometry. ∎

To give a geometric interpretation to Hadamard's inequality, suppose $\mathbf{F} = \mathbf{R}$. Let $T \in \mathcal{L}(\mathbf{R}^n)$ be the operator such that $Te_k = v_k$ for each $k = 1, \ldots, n$, where $e_1, \ldots, e_n$ is the standard basis of $\mathbf{R}^n$. Then $T$ maps the box $P(e_1, \ldots, e_n)$ onto the parallelepiped $P(v_1, \ldots, v_n)$ [see 7.102 and 7.105 for a review of this notation and terminology]. Because the box $P(e_1, \ldots, e_n)$ has volume 1, this implies (by 9.61) that the parallelepiped $P(v_1, \ldots, v_n)$ has volume $|\det T|$, which equals $|\det A|$. Thus Hadamard's inequality above can be interpreted to say that among all parallelepipeds whose edges have lengths $\|v_1\|, \ldots, \|v_n\|$, the ones with largest volume have orthogonal edges $\left(\text{and thus have volume } \prod_{k=1}^{n} \|v_k\|\right)$.

For a necessary and sufficient condition for Hadamard's inequality to be an equality, see Exercise 18.

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The matrix in the next result is called the *Vandermonde matrix*. Vandermonde matrices have important applications in polynomial interpolation, the discrete Fourier transform, and other areas of mathematics. The proof of the next result is a nice illustration of the power of switching between matrices and linear maps.

**9.67** *determinant of Vandermonde matrix*

Suppose $n > 1$ and $\beta_1, \dots, \beta_n \in \mathbf{F}$. Then

$$\det \begin{pmatrix} 1 & \beta_1 & \beta_1^2 & \cdots & \beta_1^{n-1} \\ 1 & \beta_2 & \beta_2^2 & \cdots & \beta_2^{n-1} \\ & & \ddots & & \\ 1 & \beta_n & \beta_n^2 & \cdots & \beta_n^{n-1} \end{pmatrix} = \prod_{1 \le j < k \le n} (\beta_k - \beta_j).$$

Proof Let $1, z, \dots, z^{n-1}$ be the standard basis of $\mathcal{P}_{n-1}(\mathbf{F})$ and let $e_1, \dots, e_n$ denote the standard basis of $\mathbf{F}^n$. Define a linear map $S : \mathcal{P}_{n-1}(\mathbf{F}) \to \mathbf{F}^n$ by

$$Sp = (p(\beta_1), \dots, p(\beta_n)).$$

Let $A$ denote the Vandermonde matrix shown in the statement of this result. Note that

$$A = \mathcal{M}(S, (1, z, \dots, z^{n-1}), (e_1, \dots, e_n)).$$

Let $T : \mathcal{P}_{n-1}(\mathbf{F}) \to \mathcal{P}_{n-1}(\mathbf{F})$ be the operator on $\mathcal{P}_{n-1}(\mathbf{F})$ such that $T1 = 1$ and

$$Tz^k = (z - \beta_1)(z - \beta_2) \cdots (z - \beta_k)$$

for $k = 1, \dots, n-1$. Let $B = \mathcal{M}(T, (1, z, \dots, z^{n-1}), (1, z, \dots, z^{n-1}))$. Then $B$ is an upper-triangular matrix all of whose diagonal entries equal 1. Thus $\det B = 1$ (by 9.48).

Let $C = \mathcal{M}(ST, (1, z, \dots, z^{n-1}), (e_1, \dots, e_n))$. Thus $C = AB$ (by 3.81), which implies that

$$\det A = (\det A)(\det B) = \det C.$$

The definitions of $C$, $S$, and $T$ show that $C$ equals

$$\begin{pmatrix} 1 & 0 & 0 & \cdots & 0 \\ 1 & \beta_2 - \beta_1 & 0 & \cdots & 0 \\ 1 & \beta_2 - \beta_1 & (\beta_3 - \beta_1)(\beta_3 - \beta_2) & \cdots & 0 \\ & & & \ddots & \\ 1 & \beta_2 - \beta_1 & (\beta_3 - \beta_1)(\beta_3 - \beta_2) & \cdots & (\beta_n - \beta_1)(\beta_n - \beta_2) \cdots (\beta_n - \beta_{n-1}) \end{pmatrix}.$$

Now $\det A = \det C = \prod_{1 \le j < k \le n} (\beta_k - \beta_j)$, where we have used 9.56(a) and 9.48. ∎

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## *Exercises 9C*

**1** Prove or give a counterexample: $S, T \in \mathcal{L}(V) \implies \det(S+T) = \det S + \det T$.

**2** Suppose the first column of a square matrix $A$ consists of all zeros except possibly the first entry $A_{1,1}$. Let $B$ be the matrix obtained from $A$ by deleting the first row and the first column of $A$. Show that $\det A = A_{1,1} \det B$.

**3** Suppose $T \in \mathcal{L}(V)$ is nilpotent. Prove that $\det(I + T) = 1$.

**4** Suppose $S \in \mathcal{L}(V)$. Prove that $S$ is unitary if and only if $|\det S| = \|S\| = 1$.

**5** Suppose $A$ is a block upper-triangular matrix

$$A = \begin{pmatrix} A_1 & & * \\ & \ddots & \\ 0 & & A_m \end{pmatrix},$$

where each $A_k$ along the diagonal is a square matrix. Prove that

$$\det A = (\det A_1) \cdots (\det A_m).$$

**6** Suppose $A = \begin{pmatrix} v_1 & \cdots & v_n \end{pmatrix}$ is an $n$-by-$n$ matrix, with $v_k$ denoting the $k^{\text{th}}$ column of $A$. Show that if $(m_1, \ldots, m_n) \in \operatorname{perm} n$, then

$$\det \begin{pmatrix} v_{m_1} & \cdots & v_{m_n} \end{pmatrix} = (\operatorname{sign}(m_1, \ldots, m_n)) \det A.$$

**7** Suppose $T \in \mathcal{L}(V)$ is invertible. Let $p$ denote the characteristic polynomial of $T$ and let $q$ denote the characteristic polynomial of $T^{-1}$. Prove that

$$q(z) = \frac{1}{p(0)} z^{\dim V} p\left(\frac{1}{z}\right)$$

for all nonzero $z \in \mathbf{F}$.

**8** Suppose $T \in \mathcal{L}(V)$ is an operator with no eigenvalues (which implies that $\mathbf{F} = \mathbf{R}$). Prove that $\det T > 0$.

**9** Suppose that $V$ is a real vector space of even dimension, $T \in \mathcal{L}(V)$, and $\det T < 0$. Prove that $T$ has at least two distinct eigenvalues.

**10** Suppose $V$ is a real vector space of odd dimension and $T \in \mathcal{L}(V)$. Without using the minimal polynomial, prove that $T$ has an eigenvalue.

*This result was previously proved without using determinants or the characteristic polynomial—see 5.34.*

**11** Prove or give a counterexample: If $\mathbf{F} = \mathbf{R}$, $T \in \mathcal{L}(V)$, and $\det T > 0$, then $T$ has a square root.

*If $\mathbf{F} = \mathbf{C}$, $T \in \mathcal{L}(V)$, and $\det V \neq 0$, then $V$ has a square root (see 8.41).*

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**12** Suppose $S, T \in \mathcal{L}(V)$ and $S$ is invertible. Define $p\colon \mathbf{F} \to \mathbf{F}$ by

$$p(z) = \det(zS - T).$$

Prove that $p$ is a polynomial of degree $\dim V$ and that the coefficient of $z^{\dim V}$ in this polynomial is $\det S$.

**13** Suppose $\mathbf{F} = \mathbf{C}$, $T \in \mathcal{L}(V)$, and $n = \dim V > 2$. Let $\lambda_1, \ldots, \lambda_n$ denote the eigenvalues of $T$, with each eigenvalue included as many times as its multiplicity.

(a) Find a formula for the coefficient of $z^{n-2}$ in the characteristic polynomial of $T$ in terms of $\lambda_1, \ldots, \lambda_n$.

(b) Find a formula for the coefficient of $z$ in the characteristic polynomial of $T$ in terms of $\lambda_1, \ldots, \lambda_n$.

**14** Suppose $V$ is an inner product space and $T$ is a positive operator on $V$. Prove that

$$\det \sqrt{T} = \sqrt{\det T}.$$

**15** Suppose $V$ is an inner product space and $T \in \mathcal{L}(V)$. Use the polar decomposition to give a proof that

$$|\det T| = \sqrt{\det(T^*T)}$$

that is different from the proof given earlier (see 9.60).

**16** Suppose $T \in \mathcal{L}(V)$. Define $g\colon \mathbf{F} \to \mathbf{F}$ by $g(x) = \det(I + xT)$. Show that $g'(0) = \operatorname{tr} T$.

*Look for a clean solution to this exercise, without using the explicit but complicated formula for the determinant of a matrix.*

**17** Suppose $a, b, c$ are positive numbers. Find the volume of the ellipsoid

$$\left\{(x, y, z) \in \mathbf{R}^3 : \frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} < 1\right\}$$

by finding a set $\Omega \subseteq \mathbf{R}^3$ whose volume you know and an operator $T$ on $\mathbf{R}^3$ such that $T(\Omega)$ equals the ellipsoid above.

**18** Suppose that $A$ is an invertible square matrix. Prove that Hadamard's inequality (9.66) is an equality if and only if each column of $A$ is orthogonal to the other columns.

**19** Suppose $V$ is an inner product space, $e_1, \ldots, e_n$ is an orthonormal basis of $V$, and $T \in \mathcal{L}(V)$ is a positive operator.

(a) Prove that $\det T \le \prod_{k=1}^{n} \langle Te_k, e_k \rangle$.

(b) Prove that if $T$ is invertible, then the inequality in (a) is an equality if and only if $e_k$ is an eigenvector of $T$ for each $k = 1, \ldots, n$.

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**20** Suppose $A$ is an $n$-by-$n$ matrix, and suppose $c$ is such that $|A_{j,k}| \le c$ for all $j, k \in \{1, \dots, n\}$. Prove that

$$|\det A| \le c^n n^{n/2}.$$

*The formula for the determinant of a matrix (9.46) shows that* $|\det A| \le c^n n!$. *However, the estimate given by this exercise is much better. For example, if* $c = 1$ *and* $n = 100$, *then* $c^n n! \approx 10^{158}$, *but the estimate given by this exercise is the much smaller number* $10^{100}$. *If* $n$ *is an integer power of* 2, *then the inequality above is sharp and cannot be improved.*

**21** Suppose $n$ is a positive integer and $\delta : \mathbf{C}^{n,n} \to \mathbf{C}$ is a function such that

$$\delta(AB) = \delta(A) \cdot \delta(B)$$

for all $A, B \in \mathbf{C}^{n,n}$ and $\delta(A)$ equals the product of the diagonal entries of $A$ for each diagonal matrix $A \in \mathbf{C}^{n,n}$. Prove that

$$\delta(A) = \det A$$

for all $A \in \mathbf{C}^{n,n}$.

*Recall that* $\mathbf{C}^{n,n}$ *denotes set of* $n$*-by-*$n$ *matrices with entries in* $\mathbf{C}$. *This exercise shows that the determinant is the unique function defined on square matrices that is multiplicative and has the desired behavior on diagonal matrices. This result is analogous to Exercise 10 in Section 8D, which shows that the trace is uniquely determined by its algebraic properties.*

I find that in my own elementary lectures, I have, for pedagogical reasons, pushed determinants more and more into the background. Too often I have had the experience that, while the students acquired facility with the formulas, which are so useful in abbreviating long expressions, they often failed to gain familiarity with their *meaning*, and skill in manipulation prevented the student from going into all the details of the subject and so gaining a mastery.

—*Elementary Mathematics from an Advanced Standpoint: Geometry*, Felix Klein

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# 9D Tensor Products

## Tensor Product of Two Vector Spaces

The motivation for our next topic comes from wanting to form the product of a vector $v \in V$ and a vector $w \in W$. This product will be denoted by $v \otimes w$, pronounced "$v$ tensor $w$", and will be an element of some new vector space called $V \otimes W$ (also pronounced "$V$ tensor $W$").

We already have a vector space $V \times W$ (see Section 3E), called the product of $V$ and $W$. However, $V \times W$ will not serve our purposes here because it does not provide a natural way to multiply an element of $V$ by an element of $W$. We would like our tensor product to satisfy some of the usual properties of multiplication. For example, we would like the distributive property to be satisfied, meaning that if $v_1, v_2, v \in V$ and $w_1, w_2, w \in W$, then

$$(v_1 + v_2) \otimes w = v_1 \otimes w + v_2 \otimes w \quad \text{and} \quad v \otimes (w_1 + w_2) = v \otimes w_1 + v \otimes w_2.$$

We would also like scalar multiplication to interact well with this new multiplication, meaning that

*To produce $\otimes$ in TEX, type* `\otimes`.

$$\lambda(v \otimes w) = (\lambda v) \otimes w = v \otimes (\lambda w)$$

for all $\lambda \in \mathbf{F}$, $v \in V$, and $w \in W$.

Furthermore, it would be nice if each basis of $V$ when combined with each basis of $W$ produced a basis of $V \otimes W$. Specifically, if $e_1, \ldots, e_m$ is a basis of $V$ and $f_1, \ldots, f_n$ is a basis of $W$, then we would like a list (in any order) consisting of $e_j \otimes f_k$, as $j$ ranges from 1 to $m$ and $k$ ranges from 1 to $n$, to be a basis of $V \otimes W$. This implies that $\dim(V \otimes W)$ should equal $(\dim V)(\dim W)$. Recall that $\dim(V \times W) = \dim V + \dim W$ (see 3.92), which shows that the product $V \times W$ will not serve our purposes here.

To produce a vector space whose dimension is $(\dim V)(\dim W)$ in a natural fashion from $V$ and $W$, we look at the vector space of bilinear functionals, as defined below.

**9.68 definition: *bilinear functional on $V \times W$, the vector space $\mathcal{B}(V, W)$***

- A *bilinear functional* on $V \times W$ is a function $\beta : V \times W \to \mathbf{F}$ such that $v \mapsto \beta(v, w)$ is a linear functional on $V$ for each $w \in W$ and $w \mapsto \beta(v, w)$ is a linear functional on $W$ for each $v \in V$.
- The vector space of bilinear functionals on $V \times W$ is denoted by $\mathcal{B}(V, W)$.

If $W = V$, then a bilinear functional on $V \times W$ is a bilinear form; see 9.1.

The operations of addition and scalar multiplication on $\mathcal{B}(V, W)$ are defined to be the usual operations of addition and scalar multiplication of functions. As you can verify, these operations make $\mathcal{B}(V, W)$ into a vector space whose additive identity is the zero function from $V \times W$ to $\mathbf{F}$.

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9.69 example: *bilinear functionals*

- Suppose $\varphi \in V'$ and $\tau \in W'$. Define $\beta : V \times W \to \mathbf{F}$ by $\beta(v, w) = \varphi(v)\tau(w)$. Then $\beta$ is a bilinear functional on $V \times W$.
- Suppose $v \in V$ and $w \in W$. Define $\beta : V' \times W' \to \mathbf{F}$ by $\beta(\varphi, \tau) = \varphi(v)\tau(w)$. Then $\beta$ is a bilinear functional on $V' \times W'$.
- Define $\beta : V \times V' \to \mathbf{F}$ by $\beta(v, \varphi) = \varphi(v)$. Then $\beta$ is a bilinear functional on $V \times V'$.
- Suppose $\varphi \in V'$. Define $\beta : V \times \mathcal{L}(V) \to \mathbf{F}$ by $\beta(v, T) = \varphi(Tv)$. Then $\beta$ is a bilinear functional on $V \times \mathcal{L}(V)$.
- Suppose $m$ and $n$ are positive integers. Define $\beta : \mathbf{F}^{m,n} \times \mathbf{F}^{n,m} \to \mathbf{F}$ by $\beta(A, B) = \operatorname{tr}(AB)$. Then $\beta$ is a bilinear functional on $\mathbf{F}^{m,n} \times \mathbf{F}^{n,m}$.

9.70 *dimension of the vector space of bilinear functionals*

$$\dim \mathcal{B}(V, W) = (\dim V)(\dim W).$$

Proof Let $e_1, \ldots, e_m$ be a basis of $V$ and $f_1, \ldots, f_n$ be a basis of $W$. For a bilinear functional $\beta \in \mathcal{B}(V, W)$, let $\mathcal{M}(\beta)$ be the $m$-by-$n$ matrix whose entry in row $j$, column $k$ is $\beta(e_j, f_k)$. The map $\beta \mapsto \mathcal{M}(\beta)$ is a linear map of $\mathcal{B}(V, W)$ into $\mathbf{F}^{m,n}$.

For a matrix $C \in \mathbf{F}^{m,n}$, define a bilinear functional $\beta_C$ on $V \times W$ by

$$\beta_C(a_1 e_1 + \cdots + a_m e_m, b_1 f_1 + \cdots + b_n f_n) = \sum_{k=1}^{n} \sum_{j=1}^{m} C_{j,k} a_j b_k$$

for $a_1, \ldots, a_m, b_1, \ldots, b_n \in \mathbf{F}$.

The linear map $\beta \mapsto \mathcal{M}(\beta)$ from $\mathcal{B}(V, W)$ to $\mathbf{F}^{m,n}$ and the linear map $C \mapsto \beta_C$ from $\mathbf{F}^{m,n}$ to $\mathcal{B}(V, W)$ are inverses of each other because $\beta_{\mathcal{M}(\beta)} = \beta$ for all $\beta \in \mathcal{B}(V, W)$ and $\mathcal{M}(\beta_C) = C$ for all $C \in \mathbf{F}^{m,n}$, as you should verify.

Thus both maps are isomorphisms and the two spaces that they connect have the same dimension. Hence $\dim \mathcal{B}(V, W) = \dim \mathbf{F}^{m,n} = mn = (\dim V)(\dim W)$. ∎

Several different definitions of $V \otimes W$ appear in the mathematical literature. These definitions are equivalent to each other, at least in the finite-dimensional context, because any two vector spaces of the same dimension are isomorphic.

The result above states that $\mathcal{B}(V, W)$ has the dimension that we seek, as do $\mathcal{L}(V, W)$ and $\mathbf{F}^{\dim V, \dim W}$. Thus it may be tempting to define $V \otimes W$ to be $\mathcal{B}(V, W)$ or $\mathcal{L}(V, W)$ or $\mathbf{F}^{\dim V, \dim W}$. However, none of those definitions would lead to a basis-free definition of $v \otimes w$ for $v \in V$ and $w \in W$.

The following definition, while it may seem a bit strange and abstract at first, has the huge advantage that it defines $v \otimes w$ in a basis-free fashion. We define $V \otimes W$ to be the vector space of bilinear functionals on $V' \times W'$ instead of the more tempting choice of the vector space of bilinear functionals on $V \times W$.

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**9.71 definition: *tensor product, $V \otimes W$, $v \otimes w$***

- The *tensor product* $V \otimes W$ is defined to be $\mathcal{B}(V', W')$.
- For $v \in V$ and $w \in W$, the *tensor product* $v \otimes w$ is the element of $V \otimes W$ defined by
$$(v \otimes w)(\varphi, \tau) = \varphi(v)\tau(w)$$
for all $(\varphi, \tau) \in V' \times W'$.

We can quickly prove that the definition of $V \otimes W$ gives it the desired dimension.

**9.72 *dimension of the tensor product of two vector spaces***

$$\dim(V \otimes W) = (\dim V)(\dim W).$$

Proof Because a vector space and its dual have the same dimension (by 3.111), we have $\dim V' = \dim V$ and $\dim W' = \dim W$. Thus 9.70 tells us that the dimension of $\mathcal{B}(V', W')$ equals $(\dim V)(\dim W)$. ∎

To understand the definition of the tensor product $v \otimes w$ of two vectors $v \in V$ and $w \in W$, focus on the kind of object it is. An element of $V \otimes W$ is a bilinear functional on $V' \times W'$, and in particular it is a function from $V' \times W'$ to $\mathbf{F}$. Thus for each element of $V' \times W'$, it should produce an element of $\mathbf{F}$. The definition above has this behavior, because $v \otimes w$ applied to a typical element $(\varphi, \tau)$ of $V' \times W'$ produces the number $\varphi(v)\tau(w)$.

The somewhat abstract nature of $v \otimes w$ should not matter. The important point is the behavior of these objects. The next result shows that tensor products of vectors have the desired bilinearity properties.

**9.73 *bilinearity of tensor product***

Suppose $v, v_1, v_2 \in V$ and $w, w_1, w_2 \in W$ and $\lambda \in \mathbf{F}$. Then

$$(v_1 + v_2) \otimes w = v_1 \otimes w + v_2 \otimes w \quad \text{and} \quad v \otimes (w_1 + w_2) = v \otimes w_1 + v \otimes w_2$$

and

$$\lambda(v \otimes w) = (\lambda v) \otimes w = v \otimes (\lambda w).$$

Proof Suppose $(\varphi, \tau) \in V' \times W'$. Then

$$\begin{aligned}((v_1 + v_2) \otimes w)(\varphi, \tau) &= \varphi(v_1 + v_2)\tau(w) \\ &= \varphi(v_1)\tau(w) + \varphi(v_2)\tau(w) \\ &= (v_1 \otimes w)(\varphi, \tau) + (v_2 \otimes w)(\varphi, \tau) \\ &= (v_1 \otimes w + v_2 \otimes w)(\varphi, \tau).\end{aligned}$$

Thus $(v_1 + v_2) \otimes w = v_1 \otimes w + v_2 \otimes w$.

The other two equalities are proved similarly. ∎

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Lists are, by definition, ordered. The order matters when, for example, we form the matrix of an operator with respect to a basis. For lists in this section with two indices, such as $\{e_j \otimes f_k\}_{j=1,\dots,m;k=1,\dots,n}$ in the next result, the ordering does not matter and we do not specify it—just choose any convenient ordering.

The linear independence of elements of $V \otimes W$ in (a) of the result below captures the idea that there are no relationships among vectors in $V \otimes W$ other than the relationships that come from bilinearity of the tensor product (see 9.73) and the relationships that may be present due to linear dependence of a list of vectors in $V$ or a list of vectors in $W$.

**9.74** *basis of $V \otimes W$*

Suppose $e_1, \dots, e_m$ is a list of vectors in $V$ and $f_1, \dots, f_n$ is a list of vectors in $W$.

(a) If $e_1, \dots, e_m$ and $f_1, \dots, f_n$ are both linearly independent lists, then

$$\{e_j \otimes f_k\}_{j=1,\dots,m;k=1,\dots,n}$$

is a linearly independent list in $V \otimes W$.

(b) If $e_1, \dots, e_m$ is a basis of $V$ and $f_1, \dots, f_n$ is a basis of $W$, then the list $\{e_j \otimes f_k\}_{j=1,\dots,m;k=1,\dots,n}$ is a basis of $V \otimes W$.

Proof To prove (a), suppose $e_1, \dots, e_m$ and $f_1, \dots, f_n$ are both linearly independent lists. This linear independence and the linear map lemma (3.4) imply that there exist $\varphi_1, \dots, \varphi_m \in V'$ and $\tau_1, \dots, \tau_n \in W'$ such that

$$\varphi_j(e_k) = \begin{cases} 1 & \text{if } j = k, \\ 0 & \text{if } j \neq k \end{cases} \quad \text{and} \quad \tau_j(f_k) = \begin{cases} 1 & \text{if } j = k, \\ 0 & \text{if } j \neq k, \end{cases}$$

where $j, k \in \{1, \dots, m\}$ in the first equation and $j, k \in \{1, \dots, n\}$ in the second equation.

Suppose $\{a_{j,k}\}_{j=1,\dots,m;k=1,\dots,n}$ is a list of scalars such that

$$\text{9.75} \qquad \sum_{k=1}^{n} \sum_{j=1}^{m} a_{j,k}(e_j \otimes f_k) = 0.$$

Note that $(e_j \otimes f_k)(\varphi_M, \tau_N)$ equals 1 if $j = M$ and $k = N$, and equals 0 otherwise. Thus applying both sides of 9.75 to $(\varphi_M, \tau_N)$ shows that $a_{M,N} = 0$, proving that $\{e_j \otimes f_k\}_{j=1,\dots,m;k=1,\dots,n}$ is linearly independent.

Now (b) follows from (a), the equation $\dim V \otimes W = (\dim V)(\dim W)$ [see 9.72], and the result that a linearly independent list of the right length is a basis (see 2.38). ∎

Every element of $V \times W$ is a finite sum of elements of the form $v \otimes w$, where $v \in V$ and $w \in W$, as implied by (b) in the result above. However, if $\dim V > 1$ and $\dim W > 1$, then Exercise 4 shows that

$$\{v \otimes w : (v, w) \in V \times W\} \neq V \otimes W.$$

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9.76 example: *tensor product of element of* $\mathbf{F}^m$ *with element of* $\mathbf{F}^n$

Suppose $m$ and $n$ are positive integers. Let $e_1, \dots, e_m$ denote the standard basis of $\mathbf{F}^m$ and let $f_1, \dots, f_n$ denote the standard basis of $\mathbf{F}^n$. Suppose

$$v = (v_1, \dots, v_m) \in \mathbf{F}^m \quad \text{and} \quad w = (w_1, \dots, w_n) \in \mathbf{F}^n.$$

Then

$$\begin{aligned} v \otimes w &= \Big(\sum_{j=1}^{m} v_j e_j\Big) \otimes \Big(\sum_{k=1}^{n} w_k f_k\Big) \\ &= \sum_{k=1}^{n} \sum_{j=1}^{m} (v_j w_k)(e_j \otimes f_k). \end{aligned}$$

Thus with respect to the basis $\{e_j \otimes f_k\}_{j=1,\dots,m;\,k=1,\dots,n}$ of $\mathbf{F}^m \otimes \mathbf{F}^n$ provided by 9.74(b), the coefficients of $v \otimes w$ are the numbers $\{v_j w_k\}_{j=1,\dots,m;\,k=1,\dots,n}$. If instead of writing these numbers in a list, we write them in an $m$-by-$n$ matrix with $v_j w_k$ in row $j$, column $k$, then we can identify $v \otimes w$ with the $m$-by-$n$ matrix

$$\begin{pmatrix} v_1 w_1 & \cdots & v_1 w_n \\ & \ddots & \\ v_m w_1 & \cdots & v_m w_n \end{pmatrix}.$$

See Exercises 5 and 6 for practice in using the identification from the example above.

We now define bilinear maps, which differ from bilinear functionals in that the target space can be an arbitrary vector space rather than just the scalar field.

9.77 definition: *bilinear map*

A *bilinear map* from $V \times W$ to a vector space $U$ is a function $\Gamma : V \times W \to U$ such that $v \mapsto \Gamma(v, w)$ is a linear map from $V$ to $U$ for each $w \in W$ and $w \mapsto \Gamma(v, w)$ is a linear map from $W$ to $U$ for each $v \in V$.

9.78 example: *bilinear maps*

- Every bilinear functional on $V \times W$ is a bilinear map from $V \times W$ to $\mathbf{F}$.
- The function $\Gamma : V \times W \to V \otimes W$ defined by $\Gamma(v, w) = v \otimes w$ is a bilinear map from $V \times W$ to $V \otimes W$ (by 9.73).
- The function $\Gamma : \mathcal{L}(V) \times \mathcal{L}(V) \to \mathcal{L}(V)$ defined by $\Gamma(S, T) = ST$ is a bilinear map from $\mathcal{L}(V) \times \mathcal{L}(V)$ to $\mathcal{L}(V)$.
- The function $\Gamma : V \times \mathcal{L}(V, W) \to W$ defined by $\Gamma(v, T) = Tv$ is a bilinear map from $V \times \mathcal{L}(V, W)$ to $W$.

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Tensor products allow us to convert bilinear maps on $V \times W$ into linear maps on $V \otimes W$ (and vice versa), as shown by the next result. In the mathematical literature, (a) of the result below is called the "universal property" of tensor products.

**9.79** *converting bilinear maps to linear maps*

Suppose $U$ is a vector space.

(a) Suppose $\Gamma : V \times W \to U$ is a bilinear map. Then there exists a unique linear map $\hat{\Gamma} : V \otimes W \to U$ such that

$$\hat{\Gamma}(v \otimes w) = \Gamma(v, w)$$

for all $(v, w) \in V \times W$.

(b) Conversely, suppose $T : V \otimes W \to U$ is a linear map. There there exists a unique bilinear map $T^{\#} : V \times W \to U$ such that

$$T^{\#}(v, w) = T(v \otimes w)$$

for all $(v, w) \in V \times W$.

Proof Let $e_1, \ldots, e_m$ be a basis of $V$ and let $f_1, \ldots, f_n$ be a basis of $W$. By the linear map lemma (3.4) and 9.74(b), there exists a unique linear map $\hat{\Gamma} : V \otimes W \to U$ such that

$$\hat{\Gamma}(e_j \otimes f_k) = \Gamma(e_j, f_k)$$

for all $j \in \{1, \ldots, m\}$ and $k \in \{1, \ldots, n\}$.

Now suppose $(v, w) \in V \times W$. There exist $a_1, \ldots, a_m, b_1, \ldots, b_n \in \mathbf{F}$ such that $v = a_1 e_1 + \cdots + a_m e_m$ and $w = b_1 f_1 + \cdots + b_n f_n$. Thus

$$\begin{aligned}
\hat{\Gamma}(v \otimes w) &= \hat{\Gamma}\Big( \sum_{k=1}^{n} \sum_{j=1}^{m} (a_j b_k)(e_j \otimes f_k) \Big) \\
&= \sum_{k=1}^{n} \sum_{j=1}^{m} a_j b_k \hat{\Gamma}(e_j \otimes f_k) \\
&= \sum_{k=1}^{n} \sum_{j=1}^{m} a_j b_k \Gamma(e_j, f_k) \\
&= \Gamma(v, w),
\end{aligned}$$

as desired, where the second line holds because $\hat{\Gamma}$ is linear, the third line holds by the definition of $\hat{\Gamma}$, and the fourth line holds because $\Gamma$ is bilinear.

The uniqueness of the linear map $\hat{\Gamma}$ satisfying $\hat{\Gamma}(v \otimes w) = \Gamma(v, w)$ follows from 9.74(b), completing the proof of (a).

To prove (b), define a function $T^{\#} : V \times W \to U$ by $T^{\#}(v, w) = T(v \otimes w)$ for all $(v, w) \in V \times W$. The bilinearity of the tensor product (see 9.73) and the linearity of $T$ imply that $T^{\#}$ is bilinear.

Clearly the choice of $T^{\#}$ that satisfies the conditions is unique.

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To prove 9.79(a), we could not just define $\hat{\Gamma}(v \otimes w) = \Gamma(v, w)$ for all $v \in V$ and $w \in W$ (and then extend $\hat{\Gamma}$ linearly to all of $V \otimes W$) because elements of $V \otimes W$ do not have unique representations as finite sums of elements of the form $v \otimes w$. Our proof used a basis of $V$ and a basis of $W$ to get around this problem.

Although our construction of $\hat{\Gamma}$ in the proof of 9.79(a) depended on a basis of $V$ and a basis of $W$, the equation $\hat{\Gamma}(v \otimes w) = \Gamma(v, w)$ that holds for all $v \in V$ and $w \in W$ shows that $\hat{\Gamma}$ does not depend on the choice of bases for $V$ and $W$.

## *Tensor Product of Inner Product Spaces*

The result below features three inner products—one on $V \otimes W$, one on $V$, and one on $W$, although we use the same symbol $\langle \cdot, \cdot \rangle$ for all three inner products.

**9.80** *inner product on tensor product of two inner product spaces*

Suppose $V$ and $W$ are inner product spaces. Then there is a unique inner product on $V \otimes W$ such that

$$\langle v \otimes w, u \otimes x \rangle = \langle v, u \rangle \langle w, x \rangle$$

for all $v, u \in V$ and $w, x \in W$.

Proof Suppose $e_1, \ldots, e_m$ is an orthonormal basis of $V$ and $f_1, \ldots, f_n$ is an orthonormal basis of $W$. Define an inner product on $V \otimes W$ by

$$\text{9.81} \qquad \left\langle \sum_{k=1}^{n} \sum_{j=1}^{m} b_{j,k} e_j \otimes f_k, \sum_{k=1}^{n} \sum_{j=1}^{m} c_{j,k} e_j \otimes f_k \right\rangle = \sum_{k=1}^{n} \sum_{j=1}^{m} b_{j,k} \overline{c_{j,k}}.$$

The straightforward verification that 9.81 defines an inner product on $V \times W$ is left to the reader [use 9.74(b)].

Suppose that $v, u \in V$ and $w, x \in W$. Let $v_1, \ldots, v_m \in \mathbf{F}$ be such that $v = v_1 e_1 + \cdots + v_m e_m$, with similar expressions for $u$, $w$, and $x$. Then

$$\begin{aligned}
\langle v \otimes w, u \otimes x \rangle &= \left\langle \sum_{j=1}^{m} v_j e_j \otimes \sum_{k=1}^{n} w_k f_k, \sum_{j=1}^{m} u_j e_j \otimes \sum_{k=1}^{n} x_k f_k \right\rangle \\
&= \left\langle \sum_{k=1}^{n} \sum_{j=1}^{m} v_j w_k e_j \otimes f_k, \sum_{k=1}^{n} \sum_{j=1}^{m} u_j x_k e_j \otimes f_k \right\rangle \\
&= \sum_{k=1}^{n} \sum_{j=1}^{m} v_j \overline{u_j} w_k \overline{x_k} \\
&= \left( \sum_{j=1}^{m} v_j \overline{u_j} \right) \left( \sum_{k=1}^{n} w_k \overline{x_k} \right) \\
&= \langle v, u \rangle \langle w, x \rangle.
\end{aligned}$$

There is only one inner product on $V \otimes W$ such that $\langle v \otimes w, u \otimes x \rangle = \langle v, u \rangle \langle w, x \rangle$ for all $v, u \in V$ and $w, x \in W$ because every element of $V \otimes W$ can be written as a linear combination of elements of the form $v \otimes w$ [by 9.74(b)]. ∎

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The definition below of a natural inner product on $V \otimes W$ is now justified by 9.80. We could not have simply defined $\langle v \otimes w, u \otimes x\rangle$ to be $\langle v, u\rangle\langle w, x\rangle$ (and then used additivity in each slot separately to extend the definition to $V \otimes W$) without some proof because elements of $V \otimes W$ do not have unique representations as finite sums of elements of the form $v \otimes w$.

**9.82 definition:** *inner product on tensor product of two inner product spaces*

Suppose $V$ and $W$ are inner product spaces. The inner product on $V \otimes W$ is the unique function $\langle \cdot, \cdot \rangle$ from $(V \otimes W) \times (V \otimes W)$ to $\mathbf{F}$ such that

$$\langle v \otimes w, u \otimes x\rangle = \langle v, u\rangle\langle w, x\rangle$$

for all $v, u \in V$ and $w, x \in W$.

Take $u = v$ and $x = w$ in the equation above and then take square roots to show that

$$\|v \otimes w\| = \|v\| \, \|w\|$$

for all $v \in V$ and all $w \in W$.

The construction of the inner product in the proof of 9.80 depended on an orthonormal basis $e_1, \ldots, e_m$ of $V$ and an orthonormal basis $f_1, \ldots, f_n$ of $W$. Formula 9.81 implies that $\{e_j \otimes f_k\}_{j=1,\ldots,m;k=1,\ldots,n}$ is a doubly indexed orthonormal list in $V \otimes W$ and hence is an orthonormal basis of $V \otimes W$ [by 9.74(b)]. The importance of the next result arises because the orthonormal bases used there can be different from the orthonormal bases used to define the inner product in 9.80. Although the notation for the bases is the same in the proof of 9.80 and in the result below, think of them as two different sets of orthonormal bases.

**9.83** *orthonormal basis of $V \otimes W$*

Suppose $V$ and $W$ are inner product spaces, and $e_1, \ldots, e_m$ is an orthonormal basis of $V$ and $f_1, \ldots, f_n$ is an orthonormal basis of $W$. Then

$$\{e_j \otimes f_k\}_{j=1,\ldots,m;k=1,\ldots,n}$$

is an orthonormal basis of $V \otimes W$.

Proof We know that $\{e_j \otimes f_k\}_{j=1,\ldots,m;k=1,\ldots,n}$ is a basis of $V \otimes W$ [by 9.74(b)]. Thus we only need to verify orthonormality. To do this, suppose $j, M \in \{1, \ldots, m\}$ and $k, N \in \{1, \ldots, n\}$. Then

$$\langle e_j \otimes f_k, e_N \otimes f_M\rangle = \langle e_j, e_N\rangle\langle f_k, f_M\rangle = \begin{cases} 1 & \text{if } j = N \text{ and } k = M, \\ 0 & \text{otherwise.} \end{cases}$$

Hence the doubly indexed list $\{e_j \otimes f_k\}_{j=1,\ldots,m;k=1,\ldots,n}$ is indeed an orthonormal basis of $V \otimes W$. ∎

See Exercise 11 for an example of how the inner product structure on $V \otimes W$ interacts with operators on $V$ and $W$.

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## *Tensor Product of Multiple Vector Spaces*

We have been discussing properties of the tensor product of two finite-dimensional vector spaces. Now we turn our attention to the tensor product of multiple finite-dimensional vector spaces. This generalization requires no new ideas, only some slightly more complicated notation. Readers with a good understanding of the tensor product of two vector spaces should be able to make the extension to the tensor product of more than two vector spaces.

Thus in this subsection, no proofs will be provided. The definitions and the statements of results that will be provided should be enough information to enable readers to fill in the details, using what has already been learned about the tensor product of two vector spaces.

We begin with the following notational assumption.

**9.84 notation: $V_1, \ldots, V_m$**

For the rest of this subsection, $m$ denotes an integer greater than 1 and $V_1, \ldots, V_m$ denote finite-dimensional vector spaces.

The notion of an $m$-linear functional, which we are about to define, generalizes the notion of a bilinear functional (see 9.68). Recall that the use of the word "functional" indicates that we are mapping into the scalar field $\mathbf{F}$. Recall also that the terminology "$m$-linear form" is used in the special case $V_1 = \cdots = V_m$ (see 9.25). The notation $\mathcal{B}(V_1, \ldots, V_m)$ generalizes our previous notation $\mathcal{B}(V, W)$.

**9.85 definition: *$m$-linear functional, the vector space $\mathcal{B}(V_1, \ldots, V_m)$***

- An *$m$-linear functional* on $V_1 \times \cdots \times V_m$ is a function $\beta : V_1 \times \cdots \times V_m \to \mathbf{F}$ that is a linear functional in each slot when the other slots are held fixed.
- The vector space of $m$-linear functionals on $V_1 \times \cdots \times V_m$ is denoted by $\mathcal{B}(V_1, \ldots, V_m)$.

**9.86 example: *$m$-linear functional***

Suppose $\varphi_k \in (V_k)'$ for each $k \in \{1, \ldots, m\}$. Define $\beta : V_1 \times \cdots \times V_m \to \mathbf{F}$ by

$$\beta(v_1, \ldots, v_m) = \varphi_1(v_1) \times \cdots \times \varphi_m(v_m).$$

Then $\beta$ is an $m$-linear functional on $V_1 \times \cdots \times V_m$.

The next result can be proved by imitating the proof of 9.70.

**9.87 *dimension of the vector space of $m$-linear functionals***

$$\dim \mathcal{B}(V_1, \ldots, V_m) = (\dim V_1) \times \cdots \times (\dim V_m).$$

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Now we can define the tensor product of multiple vector spaces and the tensor product of elements of those vector spaces. The following definition is completely analogous to our previous definition (9.71) in the case $m = 2$.

**9.88 definition: *tensor product*, $V_1 \otimes \cdots \otimes V_m$, $v_1 \otimes \cdots \otimes v_m$**

- The *tensor product* $V_1 \otimes \cdots \otimes V_m$ is defined to be $\mathcal{B}(V_1', \ldots, V_m')$.
- For $v_1 \in V_1, \ldots, v_m \in V_m$, the *tensor product* $v_1 \otimes \cdots \otimes v_m$ is the element of $V_1 \otimes \cdots \otimes V_m$ defined by

$$(v_1 \otimes \cdots \otimes v_m)(\varphi_1, \ldots, \varphi_m) = \varphi_1(v_1) \cdots \varphi_m(v_m)$$

for all $(\varphi_1 \ldots, \varphi_m) \in V_1' \times \cdots \times V_m'$.

The next result can be proved by following the pattern of the proof of the analogous result when $m = 2$ (see 9.72).

**9.89 *dimension of the tensor product***

$$\dim(V_1 \otimes \cdots \otimes V_m) = (\dim V_1) \cdots (\dim V_m).$$

Our next result generalizes 9.74.

**9.90 *basis of $V_1 \otimes \cdots \otimes V_m$***

Suppose $\dim V_k = n_k$ and $e_1^k, \ldots, e_{n_k}^k$ is a basis of $V_k$ for $k = 1, \ldots, m$. Then

$$\{e_{j_1}^1 \otimes \cdots \otimes e_{j_m}^m\}_{j_1 = 1, \ldots, n_1; \cdots; j_m = 1, \ldots, n_m}$$

is a basis of $V_1 \otimes \cdots \otimes V_m$.

Suppose $m = 2$ and $e_1^1, \ldots, e_{n_1}^1$ is a basis of $V_1$ and $e_1^2, \ldots, e_{n_2}^2$ is a basis of $V_2$. Then with respect to the basis $\{e_{j_1}^1 \otimes e_{j_2}^2\}_{j_1 = 1, \ldots, n_1; j_2 = 1, \ldots, n_2}$ in the result above, the coefficients of an element of $V_1 \otimes V_2$ can be represented by an $n_1$-by-$n_2$ matrix that contains the coefficient of $e_{j_1}^1 \otimes e_{j_2}^2$ in row $j_1$, column $j_2$. Thus we need a matrix, which is an array specified by two indices, to represent an element of $V_1 \otimes V_2$.

If $m > 2$, then the result above shows that we need an array specified by $m$ indices to represent an arbitrary element of $V_1 \otimes \cdots \otimes V_m$. Thus tensor products may appear when we deal with objects specified by arrays with multiple indices.

The next definition generalizes the notion of a bilinear map (see 9.77). As with bilinear maps, the target space can be an arbitrary vector space.

**9.91 definition: *$m$-linear map***

An *$m$-linear map* from $V_1 \times \cdots \times V_m$ to a vector space $U$ is a function $\Gamma : V_1 \times \cdots \times V_m \to U$ that is a linear map in each slot when the other slots are held fixed.

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The next result can be proved by following the pattern of the proof of 9.79.

**9.92** *converting m-linear maps to linear maps*

Suppose $U$ is a vector space.

(a) Suppose that $\Gamma\colon V_1 \times \cdots \times V_m \to U$ is an $m$-linear map. Then there exists a unique linear map $\hat{\Gamma}\colon V_1 \otimes \cdots \otimes V_m \to U$ such that

$$\hat{\Gamma}(v_1 \otimes \cdots \otimes v_m) = \Gamma(v_1, \ldots, v_m)$$

for all $(v_1, \ldots, v_m) \in V_1 \times \cdots \times V_m$.

(b) Conversely, suppose $T\colon V_1 \otimes \cdots \otimes V_m \to U$ is a linear map. There there exists a unique $m$-linear map $T^{\#}\colon V_1 \times \cdots \times V_m \to U$ such that

$$T^{\#}(v_1, \ldots, v_m) = T(v_1 \otimes \cdots \otimes v_m)$$

for all $(v_1, \ldots, v_m) \in V_1 \times \cdots \times V_m$.

See Exercises 12 and 13 for tensor products of multiple inner product spaces.

## *Exercises 9D*

**1** Suppose $v \in V$ and $w \in W$. Prove that $v \otimes w = 0$ if and only if $v = 0$ or $w = 0$.

**2** Give an example of six distinct vectors $v_1, v_2, v_3, w_1, w_2, w_3$ in $\mathbf{R}^3$ such that

$$v_1 \otimes w_1 + v_2 \otimes w_2 + v_3 \otimes w_3 = 0$$

but none of $v_1 \otimes w_1, v_2 \otimes w_2, v_3 \otimes w_3$ is a scalar multiple of another element of this list.

**3** Suppose that $v_1, \ldots, v_m$ is a linearly independent list in $V$. Suppose also that $w_1, \ldots, w_m$ is a list in $W$ such that

$$v_1 \otimes w_1 + \cdots + v_m \otimes w_m = 0.$$

Prove that $w_1 = \cdots = w_m = 0$.

**4** Suppose $\dim V > 1$ and $\dim W > 1$. Prove that

$$\{v \otimes w : (v, w) \in V \times W\}$$

is not a subspace of $V \otimes W$.

*This exercise implies that if* $\dim V > 1$ *and* $\dim W > 1$, *then*

$$\{v \otimes w : (v, w) \in V \times W\} \neq V \otimes W.$$

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**5** Suppose $m$ and $n$ are positive integers. For $v \in \mathbf{F}^m$ and $w \in \mathbf{F}^n$, identify $v \otimes w$ with an $m$-by-$n$ matrix as in Example 9.76. With that identification, show that the set

$$\{v \otimes w : v \in \mathbf{F}^m \text{ and } w \in \mathbf{F}^n\}$$

is the set of $m$-by-$n$ matrices (with entries in $\mathbf{F}$) that have rank at most one.

**6** Suppose $m$ and $n$ are positive integers. Give a description, analogous to Exercise 5, of the set of $m$-by-$n$ matrices (with entries in $\mathbf{F}$) that have rank at most two.

**7** Suppose $\dim V > 2$ and $\dim W > 2$. Prove that

$$\{v_1 \otimes w_1 + v_2 \otimes w_2 : v_1, v_2 \in V \text{ and } w_1, w_2 \in W\} \neq V \otimes W.$$

**8** Suppose $v_1, \ldots, v_m \in V$ and $w_1, \ldots, w_m \in W$ are such that

$$v_1 \otimes w_1 + \cdots + v_m \otimes w_m = 0.$$

Suppose that $U$ is a vector space and $\Gamma : V \times W \to U$ is a bilinear map. Show that

$$\Gamma(v_1, w_1) + \cdots + \Gamma(v_m, w_m) = 0.$$

**9** Suppose $S \in \mathcal{L}(V)$ and $T \in \mathcal{L}(W)$. Prove that there exists a unique operator on $V \otimes W$ that takes $v \otimes w$ to $Sv \otimes Tw$ for all $v \in V$ and $w \in W$.

*In an abuse of notation, the operator on $V \otimes W$ given by this exercise is often called $S \otimes T$.*

**10** Suppose $S \in \mathcal{L}(V)$ and $T \in \mathcal{L}(W)$. Prove that $S \otimes T$ is an invertible operator on $V \otimes W$ if and only if both $S$ and $T$ are invertible operators. Also, prove that if both $S$ and $T$ are invertible operators, then $(S \otimes T)^{-1} = S^{-1} \otimes T^{-1}$, where we are using the notation from the comment after Exercise 9.

**11** Suppose $V$ and $W$ are inner product spaces. Prove that if $S \in \mathcal{L}(V)$ and $T \in \mathcal{L}(W)$, then $(S \otimes T)^* = S^* \otimes T^*$, where we are using the notation from the comment after Exercise 9.

**12** Suppose that $V_1, \ldots, V_m$ are finite-dimensional inner product spaces. Prove that there is a unique inner product on $V_1 \otimes \cdots \otimes V_m$ such that

$$\langle v_1 \otimes \cdots \otimes v_m, u_1 \otimes \cdots \otimes u_m \rangle = \langle v_1, u_1 \rangle \cdots \langle v_m, u_m \rangle$$

for all $(v_1, \ldots, v_m)$ and $(u_1, \ldots, u_m)$ in $V_1 \times \cdots \times V_m$.

*Note that the equation above implies that*

$$\|v_1 \otimes \cdots \otimes v_m\| = \|v_1\| \times \cdots \times \|v_m\|$$

*for all $(v_1, \ldots, v_m) \in V_1 \times \cdots \times V_m$.*

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**13** Suppose that $V_1, ..., V_m$ are finite-dimensional inner product spaces and $V_1 \otimes \cdots \otimes V_m$ is made into an inner product space using the inner product from Exercise 12. Suppose $e_1^k, ..., e_{n_k}^k$ is an orthonormal basis of $V_k$ for each $k = 1, ..., m$. Show that the list

$$\{e_{j_1}^1 \otimes \cdots \otimes e_{j_m}^m\}_{j_1 = 1, ..., n_1; \cdots; j_m = 1, ..., n_m}$$

is an orthonormal basis of $V_1 \otimes \cdots \otimes V_m$.

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# *Colophon: Notes on Typesetting*

- This book was typeset in LuaLaTeX by the author, who wrote the LaTeX code to implement the book's design.
- The LaTeX software used for this book was written by Leslie Lamport. The TeX software, which forms the base for LaTeX, was written by Donald Knuth.
- The main text font in this book is the Open Type Format version of TeX Gyre Termes, a font based on Times, which was designed by Stanley Morison and Victor Lardent for the British newspaper *The Times* in 1931.
- The main math font in this book is the Open Type Format version of TeX Gyre Pagella Math, a font based on Palatino, which was designed by Hermann Zapf.
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- The LuaLaTeX packages fontspec and unicode-math, both written by Will Robertson, were used to manage fonts.
- The LaTeX package fontsize, written by Ivan Valbusa, was used to gracefully change the main fonts to 10.5 point size.
- The figures in the book were produced by *Mathematica*, using *Mathematica* code written by the author. *Mathematica* was created by Stephen Wolfram. The *Mathematica* package MaTeX, written by Szabolcs Horvát, was used to place LaTeX-generated labels in the *Mathematica* figures.
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# Linear Algebra Done Right
# Solutions Manual

Sheldon Axler

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This solutions manual provides answers to all the exercises in my book *Linear Algebra Done Right*. It is distributed (without charge) only to instructors who are using *Linear Algebra Done Right* as a textbook. If you are such an instructor, please contact me (preferably via e-mail) to obtain the appropriate version of the solutions manual. This version of the solutions manual is intended to accompany the second edition (1997) of *Linear Algebra Done Right*.

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This solutions manual has not been subjected to the same amount of scrutiny as the book, so errors are more likely. I would be grateful for information about any errors that you notice. If you know nicer solutions to any of the exercises than the solutions given here, please let me know so that I can improve future versions of this solutions manual.

Please check my web site for errata and other information about *Linear Algebra Done Right*. I welcome comments about either the book or the solutions manual.

Have fun!

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CHAPTER 1

# Vector Spaces

1. Suppose $a$ and $b$ are real numbers, not both 0. Find real numbers $c$ and $d$ such that

$$1/(a+bi) = c+di.$$

SOLUTION: Multiplying both the numerator and the denominator of the left side of the equation above by $a-bi$ gives

$$\frac{a-bi}{a^2+b^2} = c+di.$$

Thus we must have

$$c = \frac{a}{a^2+b^2} \quad \text{and} \quad d = \frac{-b}{a^2+b^2};$$

because $a$ and $b$ are not both 0, we are not dividing by 0.

COMMENT: Note that these formulas for $c$ and $d$ are derived under the assumption that $a+bi$ has a multiplicative inverse. However, we can forget about the derivation and verify (using the definition of complex multiplication) that

$$(a+bi)\Big(\frac{a}{a^2+b^2} - \frac{b}{a^2+b^2}i\Big) = 1,$$

which shows that every nonzero complex number does indeed have a multiplicative inverse.



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2. Show that

$$\frac{-1+\sqrt{3}i}{2}$$

is a cube root of 1 (meaning that its cube equals 1).

SOLUTION: Using the definition of complex multiplication, we have

$$\left(\frac{-1+\sqrt{3}i}{2}\right)^2 = \frac{-1-\sqrt{3}i}{2}.$$

Thus

$$\begin{aligned}\left(\frac{-1+\sqrt{3}i}{2}\right)^3 &= \left(\frac{-1-\sqrt{3}i}{2}\right)\left(\frac{-1+\sqrt{3}i}{2}\right)\\ &= 1.\end{aligned}$$

3. Prove that $-(-v) = v$ for every $v \in V$.

SOLUTION: Let $v \in V$. By the definition of additive inverse, we have

$$v + (-v) = 0.$$

The additive inverse of $-v$, which by definition is $-(-v)$, is the unique vector that when added to $-v$ gives 0. The equation above shows that $v$ has this property. Thus $-(-v) = v$.

COMMENT: Using 1.6 twice leads to another proof that $-(-v) = v$. However, the proof given above uses only the additive structure of $V$, whereas a proof using 1.6 also uses the multiplicative structure.

4. Prove that if $a \in \mathbf{F}$, $v \in V$, and $av = 0$, then $a = 0$ or $v = 0$.

SOLUTION: Suppose that $a \in \mathbf{F}$, $v \in V$, and

$$av = 0.$$

We want to prove that $a = 0$ or $v = 0$. If $a = 0$, then we are done. So suppose that $a \neq 0$. Multiplying both sides of the equation above by $1/a$ gives

$$\frac{1}{a}(av) = \frac{1}{a}0.$$

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The associative property shows that the left side of the equation above equals $1v$, which equals $v$. The right side of the equation above equals 0 (by 1.5). Thus $v = 0$, completing the proof.

5. For each of the following subsets of $\mathbf{F}^3$, determine whether it is a subspace of $\mathbf{F}^3$:

(a) $\{(x_1, x_2, x_3) \in \mathbf{F}^3 : x_1 + 2x_2 + 3x_3 = 0\}$;

(b) $\{(x_1, x_2, x_3) \in \mathbf{F}^3 : x_1 + 2x_2 + 3x_3 = 4\}$;

(c) $\{(x_1, x_2, x_3) \in \mathbf{F}^3 : x_1 x_2 x_3 = 0\}$;

(d) $\{(x_1, x_2, x_3) \in \mathbf{F}^3 : x_1 = 5x_3\}$.

SOLUTION: (a) Let

$$U = \{(x_1, x_2, x_3) \in \mathbf{F}^3 : x_1 + 2x_2 + 3x_3 = 0\}.$$

To show that $U$ is a subspace of $\mathbf{F}^3$, first note that $(0, 0, 0) \in U$, so $U$ is nonempty.

Next, suppose that $(x_1, x_2, x_3) \in U$ and $(y_1, y_2, y_3) \in U$. Then

$$x_1 + 2x_2 + 3x_3 = 0$$
$$y_1 + 2y_2 + 3y_3 = 0.$$

Adding these equations, we have

$$(x_1 + y_1) + 2(x_2 + y_2) + 3(x_3 + y_3) = 0,$$

which means that $(x_1 + y_1, x_2 + y_2, x_3 + y_3) \in U$. Thus $U$ is closed under addition.

Next, suppose that $(x_1, x_2, x_3) \in U$ and $a \in \mathbf{F}$. Then

$$x_1 + 2x_2 + 3x_3 = 0.$$

Multiplying this equation by $a$, we have

$$(ax_1) + 2(ax_2) + 3(ax_3) = 0,$$

which means that $(ax_1, ax_2, ax_3) \in U$. Thus $U$ is closed under scalar multiplication.

Because $U$ is a nonempty subset of $\mathbf{F}^3$ that is closed under addition and scalar multiplication, $U$ is a subspace of $\mathbf{F}^3$.

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(b) Let

$$U = \{(x_1, x_2, x_3) \in \mathbf{F}^3 : x_1 + 2x_2 + 3x_3 = 4\}.$$

Then $(4, 0, 0) \in U$ but $0(4, 0, 0)$, which equals $(0, 0, 0)$, is not in $U$. Thus $U$ is not closed under scalar multiplication. Thus $U$ is not a subspace of $\mathbf{F}^3$.

(c) Let

$$U = \{(x_1, x_2, x_3) \in \mathbf{F}^3 : x_1 x_2 x_3 = 0\}.$$

Then $(1, 1, 0) \in U$ and $(0, 0, 1) \in U$, but the sum of these two vectors, which equals $(1, 1, 1)$, is not in $U$. Thus $U$ is not closed under addition. Thus $U$ is not a subspace of $\mathbf{F}^3$.

(d) Let

$$U = \{(x_1, x_2, x_3) \in \mathbf{F}^3 : x_1 = 5x_3\}.$$

To show that $U$ is a subspace of $\mathbf{F}^3$, first note that $(0, 0, 0) \in U$, so $U$ is nonempty.

Next, suppose that $(x_1, x_2, x_3) \in U$ and $(y_1, y_2, y_3) \in U$. Then

$$x_1 = 5x_3$$
$$y_1 = 5y_3.$$

Adding these equations, we have

$$x_1 + y_1 = 5(x_3 + y_3),$$

which means that $(x_1 + y_1, x_2 + y_2, x_3 + y_3) \in U$. Thus $U$ is closed under addition.

Next, suppose that $(x_1, x_2, x_3) \in U$ and $a \in \mathbf{F}$. Then

$$x_1 = 5x_3.$$

Multiplying this equation by $a$, we have

$$ax_1 = 5(ax_3),$$

which means that $(ax_1, ax_2, ax_3) \in U$. Thus $U$ is closed under scalar multiplication.

Because $U$ is a nonempty subset of $\mathbf{F}^3$ that is closed under addition and scalar multiplication, $U$ is a subspace of $\mathbf{F}^3$.

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6. Give an example of a nonempty subset $U$ of $\mathbf{R}^2$ such that $U$ is closed under addition and under taking additive inverses (meaning $-u \in U$ whenever $u \in U$), but $U$ is not a subspace of $\mathbf{R}^2$.

SOLUTION: Let $U = \{(m, n) : m \text{ and } n \text{ are integers}\}$. Then clearly $U$ is closed under addition and under taking additive inverses. However, $(1, 1) \in U$ but $\frac{1}{2}(1, 1)$, which equals $(\frac{1}{2}, \frac{1}{2})$, is not in $U$, so $U$ is not closed under scalar multiplication. Thus $U$ is not a subspace of $\mathbf{R}^2$.

Of course there are also many other examples.

7. Give an example of a nonempty subset $U$ of $\mathbf{R}^2$ such that $U$ is closed under scalar multiplication, but $U$ is not a subspace of $\mathbf{R}^2$.

SOLUTION: Let $U$ be the union of the two coordinate axes in $\mathbf{R}^2$. More precisely, let

$$U = \{(x, 0) : x \in \mathbf{R}\} \cup \{(0, y) : y \in \mathbf{R}\}.$$

Then clearly $U$ is closed under scalar multiplication. However, $(1, 0)$ and $(0, 1)$ are in $U$ but their sum, which equals $(1, 1)$ is not in $U$, so $U$ is not closed under addition. Thus $U$ is not a subspace of $\mathbf{R}^2$.

Of course there are also many other examples.

8. Prove that the intersection of any collection of subspaces of $V$ is a subspace of $V$.

SOLUTION: Suppose $\{U_\alpha\}_{\alpha \in \Gamma}$ is a collection of subspaces of $V$; here $\Gamma$ is an arbitrary index set. We need to prove that $\bigcap_{\alpha \in \Gamma} U_\alpha$, which equals the set of vectors that are in $U_\alpha$ for every $\alpha \in \Gamma$, is a subspace of $V$.

The additive identity 0 is in $U_\alpha$ for every $\alpha \in \Gamma$ (because each $U_\alpha$ is a subspace of $V$). Thus $0 \in \bigcap_{\alpha \in \Gamma} U_\alpha$. In particular, $\bigcap_{\alpha \in \Gamma} U_\alpha$ is a nonempty subset of $V$.

Suppose $u, v \in \bigcap_{\alpha \in \Gamma} U_\alpha$. Then $u, v. \in U_\alpha$ for every $\alpha \in \Gamma$. Thus $u + v \in U_\alpha$ for every $\alpha \in \Gamma$ (because each $U_\alpha$ is a subspace of $V$). Thus $u + v \in \bigcap_{\alpha \in \Gamma} U_\alpha$. Thus $\bigcap_{\alpha \in \Gamma} U_\alpha$ is closed under addition.

Suppose $u \in \bigcap_{\alpha \in \Gamma} U_\alpha$ and $a \in \mathbf{F}$. Then $u \in U_\alpha$ for every $\alpha \in \Gamma$. Thus $au \in U_\alpha$ for every $\alpha \in \Gamma$ (because each $U_\alpha$ is a subspace of $V$). Thus $au \in \bigcap_{\alpha \in \Gamma} U_\alpha$. Thus $\bigcap_{\alpha \in \Gamma} U_\alpha$ is closed under scalar multiplication.

Because $\bigcap_{\alpha \in \Gamma} U_\alpha$ is a nonempty subset of $V$ that is closed under addition and scalar multiplication, $\bigcap_{\alpha \in \Gamma} U_\alpha$ is a subspace of $V$.

COMMENT: For many students, the hardest part of this exercise is understanding the meaning of an arbitrary intersection of sets. Instructors who

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do not want to deal with this issue should change the exercise to "Prove that the intersection of any finite collection of subspaces of $V$ is a subspace of $V$." Many students will then prove that the intersection of two subspaces of $V$ is a subspace of $V$ and use induction to get the result for finite collections of subspaces.

9. Prove that the union of two subspaces of $V$ is a subspace of $V$ if and only if one of the subspaces is contained in the other.

SOLUTION: Suppose $U$ and $W$ are subspaces of $V$ such that $U \cup W$ is a subspace of $V$. We will use proof by contradiction to show that $U \subset W$ or $W \subset U$. Suppose that our desired result is false. Then $U \not\subset W$ and $W \not\subset U$. This means that there exists $u \in U$ such that $u \notin W$ and there exists $w \in W$ such that $w \notin U$. Because $u$ and $w$ are both in $U \cup W$, which is a subspace of $V$, we can conclude that $u + w \in U \cup W$. Thus $u + w \in U$ or $u + w \in W$.

First consider the possibility that $u + w \in U$. In this case $w$, which equals $(u + w) + (-u)$, would be in the sum of two elements of $U$ and hence we would have $w \in U$, contradicting our assumption that $w \notin U$.

Now consider the possibility that $u + w \in W$. In this case $u$, which equals $(u + w) + (-w)$, would be in the sum of two elements of $W$ and hence we would have $u \in W$, contradicting our assumption that $u \notin W$.

The two paragraphs above show that $u + w \notin U$ and $u + w \notin W$, contradicting the final sentence of the first paragraph of this solution. This contradiction completes our proof that $U \subset W$ or $W \subset U$.

The other direction of this exercise is trivial: if we have two subspaces of $V$, one of which is contained in the other, then the union of these two subspaces equals the larger of them, which is a subspace of $V$.

10. Suppose that $U$ is a subspace of $V$. What is $U + U$?

SOLUTION: By definition, $U+U = \{u+v : u, v \in U\}$. Clearly $U \subset U+U$ because if $u \in U$, then $u$ equals $u + 0$, which expresses $u$ as a sum of two elements of $U$. Conversely, $U+U \subset U$ because the sum of two elements of $U$ is an element of $U$ (because $U$ is a subspace of $V$). Conclusion: $U + U = U$.

11. Is the operation of addition on the subspaces of $V$ commutative? Associative? (In other words, if $U_1, U_2, U_3$ are subspaces of $V$, is $U_1 + U_2 = U_2 + U_1$? Is $(U_1 + U_2) + U_3 = U_1 + (U_2 + U_3)$?)

SOLUTION: Suppose $U_1, U_2, U_3$ are subspaces of $V$.

A typical element of $U_1 + U_2$ is a vector of the form $u_1 + u_2$, where $u_1 \in U_1$ and $u_2 \in U_2$. Because addition of vectors is commutative, $u_1 + u_2$ equals

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$u_2 + u_1$, which is a typical element of $U_2 + U_1$. Thus $U_1 + U_2 = U_2 + U_1$. In other words, the operation of addition on the subspaces of $V$ is commutative.

A typical element of $(U_1 + U_2) + U_3$ is a vector of the form $(u_1 + u_2) + u_3$, where $u_1 \in U_1$, $u_2 \in U_2$, and $u_3 \in U_3$. Because addition of vectors is associative, $(u_1 + u_2) + u_3$ equals $u_1 + (u_2 + u_3)$, which is a typical element of $U_1 + (U_2 + U_3)$. Thus $(U_1 + U_2) + U_3 = U_1 + (U_2 + U_3)$. In other words, the operation of addition on the subspaces of $V$ is associative.

12. Does the operation of addition on the subspaces of $V$ have an additive identity? Which subspaces have additive inverses?

SOLUTION: The subspace $\{0\}$ is an additive identity for the operation of addition on the subspaces of $V$. More precisely, if $U$ is a subspace of $V$, then $U + \{0\} = \{0\} + U = U$.

For a subspace $U$ of $V$ to have an additive inverse, there would have to be another subspace $W$ of $V$ such that $U + W = \{0\}$. Because both $U$ and $W$ are contained in $U + W$, this is possible only if $U = W = \{0\}$. Thus $\{0\}$ is the only subspace of $V$ that has an additive inverse.

13. Prove or give a counterexample: if $U_1, U_2, W$ are subspaces of $V$ such that

$$U_1 + W = U_2 + W,$$

then $U_1 = U_2$.

SOLUTION: To construct a counterexample for the assertion above, choose $V$ to be any nonzero vector space. Let $U_1 = \{0\}$, $U_2 = V$, and $W = V$. Then $U_1 + W$ and $U_2 + W$ are both equal to $V$, but $U_1 \neq U_2$.

Of course there are also many other examples.

14. Suppose $U$ is the subspace of $\mathcal{P}(\mathbf{F})$ consisting of all polynomials $p$ of the form

$$p(z) = az^2 + bz^5,$$

where $a, b \in \mathbf{F}$. Find a subspace $W$ of $\mathcal{P}(\mathbf{F})$ such that $\mathcal{P}(\mathbf{F}) = U \oplus W$.

SOLUTION: Let $W$ be the set of all polynomials (with coefficients in $\mathbf{F}$) whose $z^2$-coefficient and $z^5$-coefficient both equal 0. Then every polynomial in $\mathcal{P}(\mathbf{F})$ can be written uniquely in the form $p + q$, where $p \in U$ and $q \in W$. Thus $\mathcal{P}(\mathbf{F}) = U \oplus W$.

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COMMENT: There are other possible choices for $W$ that give a correct solution to this exercise, but the choice for $W$ made above is certainly the most natural one.

15. Prove or give a counterexample: if $U_1, U_2, W$ are subspaces of $V$ such that

$$V = U_1 \oplus W \quad \text{and} \quad V = U_2 \oplus W,$$

then $U_1 = U_2$.

SOLUTION: To construct a counterexample for the assertion above, let $V = \mathbf{F}^2$, let $U_1 = \{(x, 0) : x \in \mathbf{F}\}$, let $U_2 = \{(0, y) : y \in \mathbf{F}\}$, and let $W = \{(z, z) : z \in \mathbf{F}\}$. Then

$$\mathbf{F}^2 = U_1 \oplus W \quad \text{and} \quad \mathbf{F}^2 = U_2 \oplus W,$$

as is easy to verify, but $U_1 \neq U_2$.

Of course there are also many other examples.

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CHAPTER 2

# Finite-Dimensional Vector Spaces

1. Prove that if $(v_1, \ldots, v_n)$ spans $V$, then so does the list

$$(v_1 - v_2, v_2 - v_3, \ldots, v_{n-1} - v_n, v_n)$$

obtained by subtracting from each vector (except the last one) the following vector.

SOLUTION: Suppose $(v_1, \ldots, v_n)$ spans $V$. Let $v \in V$. To show that $v \in \text{span}(v_1 - v_2, v_2 - v_3, \ldots, v_{n-1} - v_n, v_n)$, we need to find $a_1, \ldots, a_n \in \mathbf{F}$ such that

$$v = a_1(v_1 - v_2) + a_2(v_2 - v_3) + \ldots a_{n-1}(v_{n-1} - v_n) + a_n v_n.$$

Rearranging terms of the equation above, we see that we need to find $a_1, \ldots, a_n \in \mathbf{F}$ such that

(a) $$v = a_1 v_1 + (a_2 - a_1)v_2 + (a_3 - a_2)v_3 + \cdots + (a_n - a_{n-1})v_n.$$

Because $(v_1, \ldots, v_n)$ spans $V$, there exist $b_1, \ldots, b_n \in \mathbf{F}$ such that

(b) $$v = b_1 v_1 + b_2 v_2 + b_3 v_3 + \cdots + b_n v_n.$$

Comparing equations (a) and (b), we see that (a) will be satisfied if we choose $a_1$ to equal $b_1$ and then choose $a_2$ to equal $b_2 + a_1$ and then choose $a_3$ to equal $b_3 + a_2$, and so on.



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2. Prove that if $(v_1, \ldots, v_n)$ is linearly independent in $V$, then so is the list

$$(v_1 - v_2, v_2 - v_3, \ldots, v_{n-1} - v_n, v_n)$$

obtained by subtracting from each vector (except the last one) the following vector.

SOLUTION: Suppose $(v_1, \ldots, v_n)$ is linearly independent in $V$. To prove that the list displayed above is linearly independent, suppose $a_1, \ldots, a_n \in \mathbf{F}$ are such that

$$a_1(v_1 - v_2) + a_2(v_2 - v_3) + \cdots + a_{n-1}(v_{n-1} - v_n) + a_n v_n = 0.$$

Rearranging terms, the equation above can be rewritten as

$$a_1 v_1 + (a_2 - a_1)v_2 + (a_3 - a_2)v_3 + \cdots + (a_n - a_{n-1})v_n = 0.$$

Because $(v_1, \ldots, v_n)$ is linearly independent, the equation above implies that

$$\begin{aligned} a_1 &= 0 \\ a_2 - a_1 &= 0 \\ a_3 - a_2 &= 0 \\ &\vdots \\ a_n - a_{n-1} &= 0. \end{aligned}$$

The first equation above tells us that $a_1 = 0$. That information, combined with the second equation, tells us that $a_2 = 0$. That information, combined with the third equation, tells us that $a_3 = 0$. Continue in this fashion, getting $a_1 = \cdots = a_n = 0$. Thus $(v_1 - v_2, v_2 - v_3, \ldots, v_{n-1} - v_n, v_n)$ is linearly independent.

3. Suppose $(v_1, \ldots, v_n)$ is linearly independent in $V$ and $w \in V$. Prove that if $(v_1 + w, \ldots, v_n + w)$ is linearly dependent, then $w \in \operatorname{span}(v_1, \ldots, v_n)$.

SOLUTION: Suppose $(v_1 + w, \ldots, v_n + w)$ is linearly dependent. Then there exist scalars $a_1, \ldots, a_n$, not all 0, such that

$$a_1(v_1 + w) + \cdots + a_n(v_n + w) = 0.$$

Rearranging this equation, we have

$$a_1 v_1 + \cdots + a_n v_n = -(a_1 + \cdots + a_n)w.$$

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If $a_1 + \cdots + a_n$ were 0, then the equation above would contradict the linear independence of $(v_1, \ldots, v_n)$. Thus $a_1 + \cdots + a_n \neq 0$. Hence we can divide both sides of the equation above by $-(a_1 + \cdots + a_n)$, showing that $w \in \text{span}(v_1, \ldots, v_n)$.

4. Suppose $m$ is a positive integer. Is the set consisting of 0 and all polynomials with coefficients in $\mathbf{F}$ and with degree equal to $m$ a subspace of $\mathcal{P}(\mathbf{F})$?

SOLUTION: The set consisting of 0 and all polynomials with coefficients in $\mathbf{F}$ and with degree equal to $m$ is not a subspace of $\mathcal{P}(\mathbf{F})$ because it is not closed under addition. Specifically, the sum of two polynomials of degree $m$ may be a polynomial with degree less than $m$. For example, suppose $m = 2$. Then $7 + 4z + 5z^2$ and $1 + 2z - 5z^2$ are both polynomials of degree 2 but their sum, which equals $8 + 6z$, is a polynomial of degree 1.

5. Prove that $\mathbf{F}^\infty$ is infinite dimensional.

SOLUTION: For each positive integer $m$, let $e_m$ be the element of $\mathbf{F}^\infty$ whose $m^{\text{th}}$ coordinate equals 1 and whose other coordinates equal 0:

$$e_m = (0, \ldots, 0, \underset{\substack{\uparrow \\ m^{\text{th}} \text{ coordinate}}}{1}, 0, \ldots).$$

Then $(e_1, \ldots, e_m)$ is a linearly independent list of vectors in $\mathbf{F}^\infty$, as is easy to verify. This implies, by the marginal comment attached to 2.6, that $\mathbf{F}^\infty$ is infinite dimensional.

6. Prove that the real vector space consisting of all continuous real-valued functions on the interval $[0, 1]$ is infinite dimensional.

SOLUTION: Let $V$ denote the real vector space of all continuous real-valued functions on the interval $[0, 1]$. For each positive integer $m$, the list $(1, x, \ldots, x^m)$ is linearly independent in $V$ (because if $a_0, \ldots, a_m \in \mathbf{R}$ are such that

$$a_0 + a_1 x + \cdots + a_m x^m = 0$$

for every $x \in [0, 1]$, then the polynomial above has infinitely many roots and hence all its coefficients must equal 0). This implies, by the marginal comment attached to 2.6, that $V$ is infinite dimensional.

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7. Prove that $V$ is infinite dimensional if and only if there is a sequence $v_1, v_2, \ldots$ of vectors in $V$ such that $(v_1, \ldots, v_n)$ is linearly independent for every positive integer $n$.

SOLUTION: First suppose that $V$ is infinite dimensional. Choose $v_1$ to be any nonzero vector in $V$. Choose $v_2, v_3, \ldots$ by the following inductive process: suppose that $v_1, \ldots, v_{n-1}$ have been chosen; choose any vector $v_n \in V$ such that $v_n \notin \operatorname{span}(v_1, \ldots, v_{n-1})$—because $V$ is not finite dimensional, $\operatorname{span}(v_1, \ldots, v_{n-1})$ cannot equal $V$ so choosing $v_n$ in this fashion is possible. The linear dependence lemma (2.4) implies that $(v_1, \ldots, v_n)$ is linearly independent for every positive integer $n$, as desired.

Conversely, suppose there is a sequence $v_1, v_2, \ldots$ of vectors in $V$ such that $(v_1, \ldots, v_n)$ is linearly independent for every positive integer $n$. This implies, by the marginal comment attached to 2.6, that $V$ is infinite dimensional.

8. Let $U$ be the subspace of $\mathbf{R}^5$ defined by

$$U = \{(x_1, x_2, x_3, x_4, x_5) \in \mathbf{R}^5 : x_1 = 3x_2 \text{ and } x_3 = 7x_4\}.$$

Find a basis of $U$.

SOLUTION: Obviously

$$U = \{(3x_2, x_2, 7x_4, x_4, x_5) : x_2, x_4, x_5 \in \mathbf{R}\}.$$

From this representation of $U$, we see easily that

$$\big((3,1,0,0,0), (0,0,7,1,0), (0,0,0,0,1)\big)$$

is a basis of $U$.

Of course there are also other possible choices of bases of $U$.

9. Prove or disprove: there exists a basis $(p_0, p_1, p_2, p_3)$ of $\mathcal{P}_3(\mathbf{F})$ such that none of the polynomials $p_0, p_1, p_2, p_3$ has degree 2.

SOLUTION: Define $p_0, p_1, p_2, p_3 \in \mathcal{P}_3(\mathbf{F})$ by

$$\begin{aligned} p_0(z) &= 1, \\ p_1(z) &= z, \\ p_2(z) &= z^2 + z^3, \\ p_3(z) &= z^3. \end{aligned}$$

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None of the polynomials $p_0, p_1, p_2, p_3$ has degree 2, but $(p_0, p_1, p_2, p_3)$ is a basis of $\mathcal{P}_3(\mathbf{F})$, as is easy to verify.

Of course there are also other possible choices of bases of $\mathcal{P}_3(\mathbf{F})$ without using polynomials of degree 2.

10. Suppose that $V$ is finite dimensional, with $\dim V = n$. Prove that there exist one-dimensional subspaces $U_1, \ldots, U_n$ of $V$ such that

$$V = U_1 \oplus \cdots \oplus U_n.$$

SOLUTION: Let $(v_1, \ldots, v_n)$ be a basis of $V$. For each $j$, let $U_j$ equal $\operatorname{span}(v_j)$; in other words, $U_j = \{av_j : a \in \mathbf{F}\}$. Because $(v_1, \ldots, v_n)$ is a basis of $V$, each vector in $V$ can be written uniquely in the form

$$a_1 v_1 + \cdots + a_n v_n,$$

where $a_1, \ldots, a_n \in \mathbf{F}$ (see 2.8). By definition of direct sum, this means that $V = U_1 \oplus \cdots \oplus U_n$.

11. Suppose that $V$ is finite dimensional and $U$ is a subspace of $V$ such that $\dim U = \dim V$. Prove that $U = V$.

SOLUTION: Let $(u_1, \ldots, u_n)$ be a basis of $U$. Thus $n = \dim U$, and by hypothesis we also have $n = \dim V$. Thus $(u_1, \ldots, u_n)$ is a linearly independent (because it is a basis of $U$) list of vectors in $V$ with length $\dim V$. From 2.17, we see that $(u_1, \ldots, u_n)$ is a basis of $V$. In particular every vector in $V$ is a linear combination of $(u_1, \ldots, u_n)$. Because each $u_j \in U$, this implies that $U = V$.

12. Suppose that $p_0, p_1, \ldots, p_m$ are polynomials in $\mathcal{P}_m(\mathbf{F})$ such that $p_j(2) = 0$ for each $j$. Prove that $(p_0, p_1, \ldots, p_m)$ is not linearly independent in $\mathcal{P}_m(\mathbf{F})$.

SOLUTION: Because $p_j(2) = 0$ for each $j$, the constant polynomial 1 is not in $\operatorname{span}(p_0, \ldots, p_m)$. Thus $(p_0, \ldots, p_m)$ is not a basis of $\mathcal{P}_m(\mathbf{F})$. Because $(p_0, \ldots, p_m)$ is a list of length $m + 1$ and $\mathcal{P}_m(\mathbf{F})$ has dimension $m + 1$, this implies (by 2.17) that $(p_0, \ldots, p_m)$ is not linearly independent.

13. Suppose $U$ and $W$ are subspaces of $\mathbf{R}^8$ such that $\dim U = 3$, $\dim W = 5$, and $U + W = \mathbf{R}^8$. Prove that $U \cap W = \{0\}$.

SOLUTION: We know (from 2.18) that

$$\dim(U + W) = \dim U + \dim W - \dim(U \cap W).$$

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Because $\dim(U+W) = 8$, $\dim U = 3$, and $\dim W = 5$, this implies that $\dim(U \cap W) = 0$. Thus $U \cap W = \{0\}$.

14. Suppose that $U$ and $W$ are both five-dimensional subspaces of $\mathbf{R}^9$. Prove that $U \cap W \neq \{0\}$.

SOLUTION: Using 2.18 we have

$$\begin{aligned} 9 &\geq \dim(U+W) \\ &= \dim U + \dim W - \dim(U \cap W) \\ &= 10 - \dim(U \cap W). \end{aligned}$$

Thus $\dim(U \cap W) \geq 1$. In particular, $U \cap W \neq \{0\}$.

15. You might guess, by analogy with the formula for the number of elements in the union of three subsets of a finite set, that if $U_1, U_2, U_3$ are subspaces of a finite-dimensional vector space, then

$$\begin{aligned} &\dim(U_1 + U_2 + U_3) \\ &\qquad = \dim U_1 + \dim U_2 + \dim U_3 \\ &\qquad\quad - \dim(U_1 \cap U_2) - \dim(U_1 \cap U_3) - \dim(U_2 \cap U_3) \\ &\qquad\quad + \dim(U_1 \cap U_2 \cap U_3). \end{aligned}$$

Prove this or give a counterexample.

SOLUTION: To give a counterexample, let $V = \mathbf{R}^2$, and let

$$\begin{aligned} U_1 &= \{(x,0) : x \in \mathbf{R}\}, \\ U_2 &= \{(0,y) : y \in \mathbf{R}\}, \\ U_3 &= \{(x,x) : x \in \mathbf{R}\}. \end{aligned}$$

Then $U_1 + U_2 + U_3 = \mathbf{R}^2$, so $\dim(U_1 + U_2 + U_3) = 2$. However,

$$\dim U_1 = \dim U_2 = \dim U_3 = 1$$

and

$$\dim(U_1 \cap U_2) = \dim(U_1 \cap U_3) = \dim(U_2 \cap U_3) = \dim(U_1 \cap U_2 \cap U_3) = 0.$$

Thus in this case our guess would reduce to the formula $2 = 3$, which obviously is false.

Of course there are also many other examples.

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16. Prove that if $V$ is finite dimensional and $U_1, \ldots, U_m$ are subspaces of $V$, then

$$\dim(U_1 + \cdots + U_m) \leq \dim U_1 + \cdots + \dim U_m.$$

SOLUTION: For each $j = 1, \ldots m$, choose a basis for $U_j$. Put these bases together to form a single list of vectors in $V$. Clearly this list spans $U_1 + \cdots + U_m$. Hence the dimension of $U_1 + \cdots + U_m$ is less than or equal to the number of vectors in this list (by 2.10), which equals $\dim U_1 + \cdots + \dim U_m$. In other words,

$$\dim(U_1 + \cdots + U_m) \leq \dim U_1 + \cdots + \dim U_m.$$

17. Suppose $V$ is finite dimensional. Prove that if $U_1, \ldots, U_m$ are subspaces of $V$ such that $V = U_1 \oplus \cdots \oplus U_m$, then

$$\dim V = \dim U_1 + \cdots + \dim U_m.$$

COMMENT: This exercise deepens the analogy between direct sums of subspaces and disjoint unions of subsets. Specifically, compare this exercise to the following obvious statement: if a finite set is written as a disjoint union of subsets, then the number of elements in the set equals the sum of the number of elements in the disjoint subsets.

SOLUTION: Suppose that $U_1, \ldots, U_m$ are subspaces of $V$ such that $V = U_1 \oplus \cdots \oplus U_m$. For each $j = 1, \ldots m$, choose a basis for $U_j$. Put these bases together to form a single list $B$ of vectors in $V$. Clearly $B$ spans $U_1 + \cdots + U_m$, which equals $V$. If we show that $B$ is also linearly independent, then it will be a basis of $V$. Thus the dimension of $V$ will equal the number of vectors $B$. In other words, we will have

$$\dim V = \dim U_1 + \cdots + \dim U_m,$$

as desired.

We still need to show that $B$ is linearly independent. To do this, suppose that some linear combination of $B$ equals 0. Write this linear combination as $u_1 + \cdots + u_m$, where we have grouped together the terms that come from the basis vectors of $U_1$ and called their sum $u_1$, and similarly up to $u_m$. Thus we have

$$u_1 + \cdots + u_m = 0,$$

where each $u_j \in U_j$. Because $V = U_1 \oplus \cdots \oplus U_m$, this implies that each $u_j$ equals 0. Because each $u_j$ is a linear combination of our basis of $U_j$, all the

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coefficients in the linear combination defining $u_j$ must equal 0. Thus all the coefficients in our original linear combination of $B$ must equal 0. In other words, $B$ is linearly independent, completing our proof.

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CHAPTER 3

# Linear Maps

1. Show that every linear map from a one-dimensional vector space to itself is multiplication by some scalar. More precisely, prove that if $\dim V = 1$ and $T \in \mathcal{L}(V, V)$, then there exists $a \in \mathbf{F}$ such that $Tv = av$ for all $v \in V$.

   SOLUTION: Suppose $\dim V = 1$ and $T \in \mathcal{L}(V, V)$. Let $u$ be any nonzero vector in $V$. Then every vector in $V$ is a scalar multiple of $u$. In particular, $Tu = au$ for some $a \in \mathbf{F}$.

   Now consider a typical vector $v \in V$. There exists $b \in \mathbf{F}$ such that $v = bu$. Thus

$$\begin{aligned} Tv &= T(bu) \\ &= bT(u) \\ &= b(au) \\ &= a(bu) \\ &= av. \end{aligned}$$

2. Give an example of a function $f: \mathbf{R}^2 \to \mathbf{R}$ such that

$$f(av) = af(v)$$

   for all $a \in \mathbf{R}$ and all $v \in \mathbf{R}^2$ but $f$ is not linear.

   SOLUTION: Define $f: \mathbf{R}^2 \to \mathbf{R}$ by

$$f(x, y) = (x^3 + y^3)^{1/3}.$$

   Then $f(av) = af(v)$ for all $a \in \mathbf{R}$ and all $v \in \mathbf{R}^2$. However, $f$ is not linear because $f(1, 0) = 1$ and $f(0, 1) = 1$ but



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$$\begin{aligned} f\big((1,0)+(0,1)\big) &= f(1,1) \\ &= 2^{1/3} \\ &\neq f(1,0)+f(0,1). \end{aligned}$$

Of course there are also many other examples.

COMMENT: This exercise shows that homogeneity alone is not enough to imply that a function is a linear map. Additivity alone is also not enough to imply that a function is a linear map, although the proof of this involves advanced tools that are beyond the scope of this book.

3. Suppose that $V$ is finite dimensional. Prove that any linear map on a subspace of $V$ can be extended to a linear map on $V$. In other words, show that if $U$ is a subspace of $V$ and $S \in \mathcal{L}(U,W)$, then there exists $T \in \mathcal{L}(V,W)$ such that $Tu = Su$ for all $u \in U$.

SOLUTION: Suppose $U$ is a subspace of $V$ and $S \in \mathcal{L}(U,W)$. Let $(u_1,\dots,u_m)$ be a basis of $U$. Then $(u_1,\dots,u_m)$ is a linearly independent list of vectors in $V$, and so can be extended to a basis $(u_1,\dots,u_m,v_1,\dots,v_n)$ of $V$ (by 2.12). Define $T \in \mathcal{L}(V,W)$ by

$$T(a_1u_1+\dots a_mu_m+b_1v_1+\dots b_nv_n) = a_1Su_1+\cdots+a_mSu_m.$$

Then $Tu = Su$ for all $u \in U$.

COMMENT: Defining $T\colon V \to W$ by

$$Tv = \begin{cases} Sv & \text{if } v \in U; \\ 0 & \text{if } v \notin U. \end{cases}$$

does not work because this map is not linear.

4. Suppose that $T$ is a linear map from $V$ to $\mathbf{F}$. Prove that if $u \in V$ is not in $\operatorname{null} T$, then

$$V = \operatorname{null} T \oplus \{au : a \in \mathbf{F}\}.$$

SOLUTION: Suppose $u \in V$ is not in $\operatorname{null} T$. If $a \in \mathbf{F}$ and $au \in \operatorname{null} T$, then $0 = T(au) = aTu$, which implies that $a = 0$ (because $Tu \neq 0$). Thus

$$\operatorname{null} T \cap \{au : a \in \mathbf{F}\} = \{0\}.$$

If $v \in V$, then

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$$v = \left(v - \frac{Tv}{Tu}u\right) + \frac{Tv}{Tu}u.$$

Note that $T\left(v - \frac{Tv}{Tu}u\right) = Tv - \frac{Tv}{Tu}Tu = 0$. Thus the equation above expresses an arbitrary vector $v \in V$ as the sum of a vector in $\operatorname{null} T$ and a scalar multiple of $u$. Hence $V = \operatorname{null} T + \{au : a \in \mathbf{F}\}$. Using 1.9, we conclude that $V = \operatorname{null} T \oplus \{au : a \in \mathbf{F}\}$.

5. Suppose that $T \in \mathcal{L}(V, W)$ is injective and $(v_1, \ldots, v_n)$ is linearly independent in $V$. Prove that $(Tv_1, \ldots, Tv_n)$ is linearly independent in $W$.

SOLUTION: To show that $(Tv_1, \ldots, Tv_n)$ is linearly independent, suppose $a_1, \ldots, a_n \in \mathbf{F}$ are such that

$$a_1Tv_1 + \cdots + a_nTv_n = 0.$$

Because $T$ is a linear map, this equation can be rewritten as

$$T(a_1v_1 + \cdots + a_nv_n) = 0.$$

Because $T$ is injective, this implies that

$$a_1v_1 + \cdots + a_nv_n = 0.$$

Because $(v_1, \ldots, v_n)$ is linearly independent, the equation above implies that $a_1 = \cdots = a_n = 0$. Thus $(Tv_1, \ldots, Tv_n)$ is linearly independent.

6. Prove that if $S_1, \ldots, S_n$ are injective linear maps such that $S_1 \ldots S_n$ makes sense, then $S_1 \ldots S_n$ is injective.

SOLUTION: Suppose that $S_1, \ldots, S_n$ are injective linear maps such that $S_1 \ldots S_n$ makes sense (which means that the domains of $S_1, \ldots, S_n$ are such that $S_1 \ldots S_n$ is well defined). Suppose $v$ is a vector in the domain of $S_1 \ldots S_n$ (which equals the domain of $S_n$) such that

$$(S_1 \ldots S_n)v = 0.$$

To show that $S_1 \ldots S_n$ is injective, we need to show that $v = 0$ (see 3.2). To do this, rewrite the equation above as

$$S_1\big((S_2 \ldots S_n)v\big) = 0.$$

Because $S_1$ is injective, this implies that

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$$(S_2 \dots S_n)v = 0.$$

The same argument, now applied to the equation above, shows that

$$(S_3 \dots S_n)v = 0.$$

Repeat this process until reaching the equation $S_n v = 0$, which implies (because $S_n$ is injective) that $v = 0$, as desired.

7. Prove that if $(v_1, \dots, v_n)$ spans $V$ and $T \in \mathcal{L}(V, W)$ is surjective, then $(Tv_1, \dots, Tv_n)$ spans $W$.

SOLUTION: Suppose that $(v_1, \dots, v_n)$ spans $V$ and $T \in \mathcal{L}(V, W)$ is surjective. Let $w \in W$. Because $T$ is surjective, there exists $v \in V$ such that $Tv = w$. Because $(v_1, \dots, v_n)$ spans $V$, there exist $a_1, \dots, a_n \in \mathbf{F}$ such that

$$v = a_1 v_1 + \cdots + a_n v_n.$$

Applying $T$ to both sides of this equation, we get

$$Tv = a_1 Tv_1 + \cdots + a_n Tv_n.$$

Because $Tv = w$, the equation above implies that $w \in \operatorname{span}(Tv_1, \dots, Tv_n)$. Because $w$ was an arbitrary vector in $W$, this implies that $(Tv_1, \dots, Tv_n)$ spans $W$.

8. Suppose that $V$ is finite dimensional and that $T \in \mathcal{L}(V, W)$. Prove that there exists a subspace $U$ of $V$ such that $U \cap \operatorname{null} T = \{0\}$ and $\operatorname{range} T = \{Tu : u \in U\}$.

SOLUTION: There exists a subspace $U$ of $V$ such that

$$V = \operatorname{null} T \oplus U;$$

this follows from 2.13 (with $\operatorname{null} T$ playing the role of $U$ and $U$ playing the role of $W$).

From the definition of direct sum, we have $U \cap \operatorname{null} T = \{0\}$.

Obviously $\operatorname{range} T \supset \{Tu : u \in U\}$. To prove the inclusion in the other direction, suppose $v \in V$. Then there exist $w \in \operatorname{null} T$ and $u' \in U$ such that

$$v = w + u'.$$

Applying $T$ to both sides of this equation, we have $Tv = Tw + Tu' = Tu'$. Thus $Tv \in \{Tu : u \in U\}$. Because $v$ was an arbitrary vector in $V$ (and thus $Tv$ is an arbitrary vector in $\operatorname{range} T$), this implies that

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$$\text{range}\, T \subset \{Tu : u \in U\}.$$

Thus $\text{range}\, T = \{Tu : u \in U\}$, as desired.

9. Prove that if $T$ is a linear map from $\mathbf{F}^4$ to $\mathbf{F}^2$ such that

$$\text{null}\, T = \{(x_1, x_2, x_3, x_4) \in \mathbf{F}^4 : x_1 = 5x_2 \text{ and } x_3 = 7x_4\},$$

then $T$ is surjective.

SOLUTION: Suppose $T \in \mathcal{L}(F4, \mathbf{F}^2)$ is such that $\text{null}\, T$ is as above. Then $\big((5,1,0,0),(0,0,7,1)\big)$ is a basis of $\text{null}\, T$, and hence $\dim \text{null}\, T = 2$. From 3.4 we have

$$\begin{aligned}\dim \text{range}\, T &= \dim \mathbf{F}^4 - \dim \text{null}\, T \\ &= 4 - 2 \\ &= 2.\end{aligned}$$

Because $\text{range}\, T$ is a two-dimensional subspace of $\mathbf{R}^2$, we have $\text{range}\, T = \mathbf{R}^2$. In other words, $T$ is surjective.

10. Prove that there does not exist a linear map from $\mathbf{F}^5$ to $\mathbf{F}^2$ whose null space equals

$$\{(x_1, x_2, x_3, x_4, x_5) \in \mathbf{F}^5 : x_1 = 3x_2 \text{ and } x_3 = x_4 = x_5\}.$$

SOLUTION: Suppose $U$ is the subspace of $\mathbf{F}^5$ displayed above. Then $\big((3,1,0,0,0),(0,0,1,1,1)\big)$ is a basis of $U$, and hence $\dim U = 2$.

If $T \in \mathcal{L}(\mathbf{F}^5, \mathbf{F}^2)$ then from 3.4 we have

$$\begin{aligned}\dim \text{null}\, T &= \dim \mathbf{F}^5 - \dim \text{range}\, T \\ &= 5 - \dim \text{range}\, T \\ &\geq 3 \\ &> \dim U,\end{aligned}$$

where the first inequality holds because $\text{range}\, T \subset \mathbf{F}^2$. The inequality above shows that if $T \in \mathcal{L}(\mathbf{F}^5, \mathbf{F}^2)$, then $\text{null}\, T \neq U$, as desired.

11. Prove that if there exists a linear map on $V$ whose null space and range are both finite dimensional, then $V$ is finite dimensional.

SOLUTION: Suppose there exists a linear map $T$ from $V$ into some vector space such that $\text{null}\, T$ and $\text{range}\, T$ are both finite dimensional. Thus

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there exist vectors $u_1, \dots, u_m \in V$ and $w_1, \dots, w_n \in \text{range}\, T$ such that $(u_1, \dots, u_m)$ spans $\text{null}\, T$ and $(w_1, \dots, w_n)$ spans $\text{range}\, T$. Because each $w_j \in \text{range}\, T$, there exists $v_j \in V$ such that $w_j = Tv_j$.

Suppose $v \in V$. Then $Tv \in \text{range}\, T$, so there exist $b_1, \dots, b_n \in \mathbf{F}$ such that

$$\begin{aligned} Tv &= b_1 w_1 + \cdots + b_n w_n \\ &= b_1 Tv_1 + \cdots + b_n Tv_n \\ &= T(b_1 v_1 + \dots b_n v_n). \end{aligned}$$

The equation above implies that $T(v - b_1 v_1 - \cdots - b_n v_n) = 0$. In other words, $v - b_1 v_1 - \cdots - b_n v_n \in \text{null}\, T$. Thus there exist $a_1, \dots, a_m \in \mathbf{F}$ such that

$$v - b_1 v_1 - \cdots - b_n v_n = a_1 u_1 + \cdots + a_m u_m.$$

The equation above can be rewritten as

$$v = a_1 u_1 + \cdots + a_m u_m + b_1 v_1 + \cdots + b_n v_n.$$

The equation above shows that an arbitrary vector $v \in V$ is a linear combination of $(u_1, \dots, u_m, v_1, \dots, v_n)$. In other words, $(u_1, \dots, u_m, v_1, \dots, v_n)$ spans $V$. Thus $V$ is finite dimensional.

COMMENT: The hypothesis of 3.4 is that $V$ is finite dimensional (which is what we are trying to prove in this exercise), so 3.4 cannot be used in this exercise.

12. Suppose that $V$ and $W$ are both finite dimensional. Prove that there exists a surjective linear map from $V$ onto $W$ if and only if $\dim W \leq \dim V$.

SOLUTION: First suppose that there exists a surjective linear map $T$ from $V$ onto $W$. Then

$$\begin{aligned} \dim W &= \dim \text{range}\, T \\ &= \dim V - \dim \text{null}\, T \\ &\leq \dim V, \end{aligned}$$

where the second equality comes from 3.4.

To prove the other direction, now suppose that $\dim W \leq \dim V$. Let $(w_1, \dots, w_m)$ be a basis of $W$ and let $(v_1, \dots, v_n)$ be a basis of $V$. For $a_1, \dots, a_n \in \mathbf{F}$ define $T(a_1 v_1 + \cdots + a_n v_n)$ by

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$$T(a_1v_1 + \cdots + a_nv_n) = a_1w_1 + \cdots + a_mw_m.$$

Because $\dim W \leq \dim V$, we have $m \leq n$ and so $a_m$ on the right side of the equation above makes sense. Clearly $T$ is a surjective linear map from $V$ onto $W$.

13. Suppose that $V$ and $W$ are finite dimensional and that $U$ is a subspace of $V$. Prove that there exists $T \in \mathcal{L}(V, W)$ such that $\operatorname{null} T = U$ if and only if $\dim U \geq \dim V - \dim W$.

SOLUTION: First suppose that there exists $T \in \mathcal{L}(V, W)$ such that $\operatorname{null} T = U$. Then

$$\begin{aligned} \dim U &= \dim \operatorname{null} T \\ &= \dim V - \dim \operatorname{range} T \\ &\geq \dim V - \dim W, \end{aligned}$$

where the second equality comes from 3.4.

To prove the other direction, now suppose that $\dim U \geq \dim V - \dim W$. Let $(u_1, \ldots, u_m)$ be a basis of $U$. Extend to a basis $(u_1, \ldots, u_m, v_1, \ldots, v_n)$ of $V$. Let $(w_1, \ldots, w_p)$ be a basis of $W$. For $a_1, \ldots, a_m, b_1, \ldots, b_n \in \mathbf{F}$ define $T(a_1u_1 + \cdots + a_mu_m + b_1v_1 + \cdots + b_nv_n)$ by

$$T(a_1u_1 + \cdots + a_mu_m + b_1v_1 + \cdots + b_nv_n) = b_1w_1 + \cdots + b_nw_n.$$

Because $\dim W \geq \dim V - \dim U$, we have $p \geq n$ and so $w_n$ on the right side of the equation above makes sense. Clearly $T \in \mathcal{L}(V, W)$ and $\operatorname{null} T = U$.

14. Suppose that $W$ is finite dimensional and $T \in \mathcal{L}(V, W)$. Prove that $T$ is injective if and only if there exists $S \in \mathcal{L}(W, V)$ such that $ST$ is the identity map on $V$.

SOLUTION: First suppose that $T$ is injective. Define $S' \colon \operatorname{range} T \to V$ by

$$S'(Tv) = v;$$

because $T$ is injective, each element of $\operatorname{range} T$ can be represented in the form $Tv$ in only one way, so $T$ is well defined. As can be easily checked, $S'$ is a linear map on $\operatorname{range} T$. By Exercise 3 of this chapter, $S'$ can be extended to a linear map $S \in \mathcal{L}(W, V)$. If $v \in V$, then $(ST)v = S(Tv) = S'(Tv) = v$. Thus $ST$ is the identity map on $V$, as desired.

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To prove the implication in the other direction, now suppose that there exists $S \in \mathcal{L}(W, V)$ such that $ST$ is the identity map on $V$. If $u, v \in V$ are such that $Tu = Tv$, then

$$u = (ST)(u) = S(Tu) = S(Tv) = (ST)v = v$$

and hence $u = v$. Thus $T$ is injective, as desired.

15. Suppose that $V$ is finite dimensional and $T \in \mathcal{L}(V, W)$. Prove that $T$ is surjective if and only if there exists $S \in \mathcal{L}(W, V)$ such that $TS$ is the identity map on $W$.

SOLUTION: First suppose that $T$ is surjective. Thus $W$, which equals range $T$, is finite dimensional (by 3.4). Let $(w_1, \ldots, w_m)$ be a basis of $W$. Because $T$ is surjective, for each $j$ there exists $v_j \in V$ such that $w_j = Tv_j$. Define $S \in \mathcal{L}(W, V)$ by

$$S(a_1 w_1 + \cdots + a_m w_m) = a_1 v_1 + \cdots + a_m v_m.$$

Then

$$\begin{aligned}(TS)(a_1 w_1 + \cdots + a_m w_m) &= T(a_1 v_1 + \cdots + a_m v_m) \\ &= a_1 T v_1 + \cdots + a_m T v_m \\ &= a_1 w_1 + \cdots + a_m w_m.\end{aligned}$$

Thus $TS$ is the identity map on $W$.

To prove the implication in the other direction, now suppose that there exists $S \in \mathcal{L}(W, V)$ such that $TS$ is the identity map on $W$. If $w \in W$, then $w = T(Sw)$, and hence $w \in$ range $T$. Thus range $T = W$. In other words, $T$ is surjective, as desired.

16. Suppose that $U$ and $V$ are finite-dimensional vector spaces and that $S \in \mathcal{L}(V, W)$, $T \in \mathcal{L}(U, V)$. Prove that

$$\dim \operatorname{null} ST \leq \dim \operatorname{null} S + \dim \operatorname{null} T.$$

SOLUTION: Define a linear map $T' \colon \operatorname{null} ST \to V$ by $T'u = Tu$. If $u \in \operatorname{null} ST$, then $S(Tu) = 0$, which means that $Tu \in \operatorname{null} S$. In other words, range $T' \subset \operatorname{null} S$. Now

$$\begin{aligned}\dim \operatorname{null} ST &= \dim \operatorname{null} T' + \dim \operatorname{range} T' \\ &\leq \dim \operatorname{null} T' + \dim \operatorname{null} S \\ &\leq \dim \operatorname{null} T + \dim \operatorname{null} S,\end{aligned}$$

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where the first line follows from 3.4 (applied to $T'$), the second line holds because range $T' \subset$ null $S$, and the third line holds because of the obvious inclusion null $T' \subset$ null $T$.

17. Prove that the distributive property holds for matrix addition and matrix multiplication. In other words, suppose $A$, $B$, and $C$ are matrices whose sizes are such that $A(B+C)$ makes sense. Prove that $AB+AC$ makes sense and that $A(B+C) = AB + AC$.

SOLUTION: Because $A(B+C)$ makes sense, $B$ and $C$ must have the same size. Furthermore, the number of columns of $A$ (let's call this number $n$) must equal the number of rows of $B$ and $C$. All this means that $AB + AC$ makes sense.

To prove that $A(B + C) = AB + AC$, just use the definition of matrix addition, the definition of matrix multiplication, and the usual distributive property for elements of $\mathbf{F}$. Specifically, let $a_{j,k}$, $b_{j,k}$, and $c_{j,k}$ denote the entries in row $j$, column $k$ of $A$, $B$, and $C$, respectively. The entry in row $j$, column $k$ of $B + C$ is $b_{j,k} + c_{j,k}$. Thus the entry in row $j$, column $k$ of $A(B + C)$ is

$$\sum_{r=1}^{n} a_{j,r}(b_{r,k} + c_{r,k}),$$

which equals

$$\sum_{r=1}^{n} a_{j,r}b_{r,k} + \sum_{r=1}^{n} a_{j,r}c_{r,k},$$

which equals the entry in row $j$, column $k$ of $AB + AC$, as desired.

18. Prove that matrix multiplication is associative. In other words, suppose $A$, $B$, and $C$ are matrices whose sizes are such that $(AB)C$ makes sense. Prove that $A(BC)$ makes sense and that $(AB)C = A(BC)$.

SOLUTION: This exercise can be done by a brute force calculation, in the style of the solution to the previous exercise. Here is a solution that uses only the associativity of the product of linear maps (which is easy to verify because composition of functions is clearly associative) and the nice property that the matrix of the product of two linear maps equals the product of the matrices of the two linear maps (see 3.11).

Suppose $A$ is an $m$-by-$n$ matrix, $B$ is an $n$-by-$p$ matrix, and $C$ is a $p$-by-$q$ matrix; the sizes much match up like this in order for $(AB)C$ to make sense.

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Let $R \in \mathcal{L}(\mathbf{F}^n, \mathbf{F}^m), S \in \mathcal{L}(\mathbf{F}^p, \mathbf{F}^n), T \in \mathcal{L}(\mathbf{F}^q, \mathbf{F}^p)$ be such that, with respect to the standard bases, $\mathcal{M}(R) = A, \mathcal{M}(S) = B, \mathcal{M}(T) = C$; 3.19 insures that such linear maps exist. Now

$$\begin{aligned}(AB)C &= \big(\mathcal{M}(R)\mathcal{M}(S)\big)\mathcal{M}(T)\\ &= \mathcal{M}(RS)\mathcal{M}(T)\\ &= \mathcal{M}\big((RS)T\big)\\ &= \mathcal{M}\big(R(ST)\big)\\ &= \mathcal{M}(R)\mathcal{M}(ST)\\ &= \mathcal{M}(R)\big(\mathcal{M}(S)\mathcal{M}(T)\big)\\ &= A(BC).\end{aligned}$$

19. Suppose $T \in \mathcal{L}(\mathbf{F}^n, \mathbf{F}^m)$ and that

$$\mathcal{M}(T) = \begin{bmatrix} a_{1,1} & \dots & a_{1,n} \\ \vdots & & \vdots \\ a_{m,1} & \dots & a_{m,n} \end{bmatrix},$$

where we are using the standard bases. Prove that

$$T(x_1, \dots, x_n) = (a_{1,1}x_1 + \cdots + a_{1,n}x_n, \dots, a_{m,1}x_1 + \cdots + a_{m,n}x_n)$$

for every $(x_1, \dots, x_n) \in \mathbf{F}^n$.

COMMENT: This exercise shows $T$ has the form promised on page 39.

SOLUTION: Let $x = (x_1, \dots, x_n) \in \mathbf{F}^n$. Using the standard bases, we then have

$$\begin{aligned}\mathcal{M}(Tx) &= \mathcal{M}(T)\mathcal{M}(x)\\ &= \begin{bmatrix} a_{1,1} & \dots & a_{1,n} \\ \vdots & & \vdots \\ a_{m,1} & \dots & a_{m,n} \end{bmatrix} \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}\\ &= \begin{bmatrix} a_{1,1}x_1 + \cdots + a_{1,n}x_n \\ \vdots \\ a_{m,1}x_1 + \cdots + a_{m,n}x_n \end{bmatrix},\end{aligned}$$

where the first equality comes from 3.14. The last equation implies that

$$Tx = (a_{1,1}x_1 + \cdots + a_{1,n}x_n, \ldots, a_{m,1}x_1 + \cdots + a_{m,n}x_n),$$

as desired.

20. Suppose $(v_1, \ldots, v_n)$ is a basis of $V$. Prove that the function $T: V \to \mathrm{Mat}(n, 1, \mathbf{F})$ defined by

$$Tv = \mathcal{M}(v)$$

is an invertible linear map of $V$ onto $\mathrm{Mat}(n, 1, \mathbf{F})$; here $\mathcal{M}(v)$ is the matrix of $v \in V$ with respect to the basis $(v_1, \ldots, v_n)$.

SOLUTION: Suppose $u, w \in V$. We can write

$$u = a_1v_1 + \cdots + a_nv_n \quad \text{and} \quad w = b_1v_1 + \cdots + b_nv_n$$

for some $a_1, \ldots, a_n, b_1, \ldots, b_n \in \mathbf{F}$. Thus

$$u + w = (a_1 + b_1)v_1 + \cdots + (a_n + b_n)v_n.$$

Hence

$$\begin{aligned} T(u + w) &= \mathcal{M}(u + w) \\ &= \begin{bmatrix} a_1 + b_1 \\ \vdots \\ a_n + b_n \end{bmatrix} \\ &= \begin{bmatrix} a_1 \\ \vdots \\ a_n \end{bmatrix} + \begin{bmatrix} b_1 \\ \vdots \\ b_n \end{bmatrix} \\ &= \mathcal{M}(u) + \mathcal{M}(w) \\ &= Tu + Tw, \end{aligned}$$

which shows that $T$ satisfies that additivity property required for linearity.

If $c \in \mathbf{F}$, then

$$cu = ca_1v_1 + \cdots + ca_nv_n.$$

Hence

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$$\begin{aligned} T(cu) &= \mathcal{M}(cu) \\ &= \begin{bmatrix} ca_1 \\ \vdots \\ ca_n \end{bmatrix} \\ &= c \begin{bmatrix} a_1 \\ \vdots \\ a_n \end{bmatrix} \\ &= c\mathcal{M}(u) \\ &= cTu, \end{aligned}$$

which shows that $T$ satisfies the homogeneity property required for linearity. Thus $T$ is linear.

If $Tu = 0$, then $a_1 = \cdots = a_n = 0$, which implies that $u = 0$. Thus $T$ is injective.

If $c_1, \ldots, c_n \in \mathbf{F}$, then

$$T(c_1 v_1 + \cdots + c_n v_n) = \begin{bmatrix} c_1 \\ \vdots \\ c_n \end{bmatrix},$$

which implies that $T$ is surjective.

Because the linear map $T$ is injective and surjective, it is invertible (see 3.17).

21. Prove that every linear map from $\mathrm{Mat}(n, 1, \mathbf{F})$ to $\mathrm{Mat}(m, 1, \mathbf{F})$ is given by a matrix multiplication. In other words, prove that if

$$T \in \mathcal{L}(\mathrm{Mat}(n, 1, \mathbf{F}), \mathrm{Mat}(m, 1, \mathbf{F})),$$

then there exists an $m$-by-$n$ matrix $A$ such that $TB = AB$ for every $B \in \mathrm{Mat}(n, 1, \mathbf{F})$.

SOLUTION: The vector spaces $\mathrm{Mat}(n, 1, \mathbf{F})$ and $\mathrm{Mat}(m, 1, \mathbf{F})$ have obvious bases (consisting of matrices that have 0 in all entries except for a 1 in one entry). Let $A$ be the matrix of $T$ with respect to these bases. Note that if $B \in \mathrm{Mat}(n, 1, \mathbf{F})$, then $\mathcal{M}(B) = B$ and $\mathcal{M}(TB) = TB$. Thus

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$$\begin{aligned} TB &= \mathcal{M}(TB) \\ &= \mathcal{M}(T)\mathcal{M}(B) \\ &= AB, \end{aligned}$$

where the second equality comes from 3.14.

22. Suppose that $V$ is finite dimensional and $S, T \in \mathcal{L}(V)$. Prove that $ST$ is invertible if and only if both $S$ and $T$ are invertible.

SOLUTION: First suppose $ST$ is invertible. Thus there exists $R \in \mathcal{L}(V)$ such that $R(ST) = (ST)R = I$. If $v \in V$ is such that $Tv = 0$, then

$$\begin{aligned} v &= Iv \\ &= R(ST)v \\ &= 0. \end{aligned}$$

Because $v$ was an arbitrary vector in $\operatorname{null} T$, this shows that $\operatorname{null} T = \{0\}$. Thus $T$ is injective (by 3.2), and hence $T$ is invertible (by 3.21), as desired.

If $u \in V$, then

$$\begin{aligned} u &= Iu \\ &= (ST)Ru \\ &= S(TRu), \end{aligned}$$

which shows that $u \in \operatorname{range} S$. Because $u$ was an arbitrary vector in $V$, this implies that $\operatorname{range} S = V$. Thus $V$ is surjective, and hence $V$ is invertible (by 3.21), as desired.

To prove the implication in the other direction, now suppose that both $S$ and $T$ are invertible. Then

$$\begin{aligned} (ST)(T^{-1}S^{-1}) &= S(TT^{-1})S^{-1} \\ &= SS^{-1} \\ &= I \end{aligned}$$

and

$$\begin{aligned} (T^{-1}S^{-1})(ST) &= T^{-1}(S^{-1}S)T \\ &= T^{-1}T \\ &= I. \end{aligned}$$

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Thus $T^{-1}S^{-1}$ satisfies the properties required for an inverse of $ST$. Thus $ST$ is invertible and $(ST)^{-1} = T^{-1}S^{-1}$.

23. Suppose that $V$ is finite dimensional and $S, T \in \mathcal{L}(V)$. Prove that $ST = I$ if and only if $TS = I$.

SOLUTION: First suppose that

$$ST = I.$$

Because $I$ is invertible, the previous exercise implies that $S$ and $T$ are both invertible. Multiply both sides of the equation above by $T^{-1}$ on the right, getting

$$S = T^{-1}.$$

Now multiply both sides of the equation above by $T$ on the left, getting

$$TS = I,$$

as desired.

To prove the implication in the other direction, simply reverse the roles of $S$ and $T$ in the direction we have already proved, showing that if $TS = I$, then $ST = I$.

24. Suppose that $V$ is finite dimensional and $T \in \mathcal{L}(V)$. Prove that $T$ is a scalar multiple of the identity if and only if $ST = TS$ for every $S \in \mathcal{L}(V)$.

SOLUTION: First suppose that $T = aI$ for some $a \in \mathbf{F}$. Let $S \in \mathcal{L}(V)$. Then

$$\begin{aligned} ST &= S(aI) \\ &= aS \\ &= (aI)S \\ &= TS. \end{aligned}$$

To prove the implication in the other direction, suppose now that $ST = TS$ for all $S \in \mathcal{L}(V)$. We begin by proving that $(v, Tv)$ is linearly dependent for every $v \in V$. To do this, fix $v \in V$, and suppose that $(v, Tv)$ is linearly independent. Then $(v, Tv)$ can be extended to a basis $(v, Tv, u_1, \ldots, u_n)$ of $V$. Define $S \in \mathcal{L}(V)$ by

$$S(av + bTv + c_1u_1 + \cdots + c_nu_n) = bv.$$

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Thus $S(Tv) = v$ and $Sv = 0$. Thus the equation $S(Tv) = T(Sv)$ becomes the equation $v = 0$, a contradiction because $(v, Tv)$ was assumed to be linearly independent. This contradiction shows that $(v, Tv)$ is linearly dependent for every $v \in V$. This implies that for each $v \in V \setminus \{0\}$, there exists $a_v \in \mathbf{F}$ such that

$$Tv = a_v v.$$

To show that $T$ is a scalar multiple of the identity, we must show that $a_v$ is independent of $v$. To do this, suppose $v, w \in V \setminus \{0\}$. We want to show that $a_v = a_w$. First consider the case where $(v, w)$ is linearly dependent. Then there exists $b \in \mathbf{F}$ such that $w = bv$. We have

$$\begin{aligned} a_w w &= Tw \\ &= T(bv) \\ &= bTv \\ &= b(a_v v) \\ &= a_v w, \end{aligned}$$

which shows that $a_v = a_w$, as desired.

Finally, consider the case where $(v, w)$ is linearly independent. We have

$$\begin{aligned} a_{v+w}(v + w) &= T(v + w) \\ &= Tv + Tw \\ &= a_v v + a_w w, \end{aligned}$$

which implies that

$$(a_{v+w} - a_v)v + (a_{v+w} - a_w)w = 0.$$

Because $(v, w)$ is linearly independent, this implies that $a_{v+w} = a_v$ and $a_{v+w} = a_w$, so again we have $a_v = a_w$, as desired.

25. Prove that if $V$ is finite dimensional with $\dim V > 1$, then the set of non-invertible operators on $V$ is not a subspace of $\mathcal{L}(V)$.

SOLUTION: Suppose that $V$ is finite dimensional with $\dim V > 1$. Let $n = \dim V$ and let $(v_1, \ldots, v_n)$ be a basis of $V$. Define $S, T \in \mathcal{L}(V)$ by

$$S(a_1 v_1 + \cdots + a_n v_n) = a_1 v_1$$

and

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$$T(a_1v_1 + \cdots + a_nv_n) = a_2v_2 + \cdots + a_nv_n.$$

Then $S$ is not injective because $Sv_2 = 0$ (this is where we use the hypothesis that $\dim V > 1$), and $T$ is not injective because $Tv_1 = 0$. Thus both $S$ and $T$ are not invertible. However, $S+T$ equals $I$, which is invertible. Thus the set of noninvertible operators on $V$ is not closed under addition, and hence it is not a subspace of $\mathcal{L}(V)$.

COMMENT: If $\dim V = 1$, then the set of noninvertible operators on $V$ equals $\{0\}$, which is a subspace of $\mathcal{L}(V)$.

26. Suppose $n$ is a positive integer and $a_{i,j} \in \mathbf{F}$ for $i, j = 1, \ldots, n$. Prove that the following are equivalent:

(a) The trivial solution $x_1 = \cdots = x_n = 0$ is the only solution to the homogeneous system of equations

$$\sum_{k=1}^{n} a_{1,k}x_k = 0$$
$$\vdots$$
$$\sum_{k=1}^{n} a_{n,k}x_k = 0.$$

(b) For every $c_1, \ldots, c_n \in \mathbf{F}$, there exists a solution to the system of equations

$$\sum_{k=1}^{n} a_{1,k}x_k = c_1$$
$$\vdots$$
$$\sum_{k=1}^{n} a_{n,k}x_k = c_n.$$

Note that here we have the same number of equations as variables.

SOLUTION: Define $T \in \mathcal{L}(\mathbf{F}^n)$ by

$$T(x_1, \ldots, x_n) = \Big(\sum_{k=1}^{n} a_{1,k}x_k, \ldots, \sum_{k=1}^{n} a_{n,k}x_k\Big).$$

Then (a) above is the assertion that $T$ is injective, and (b) above is the assertion that $T$ is surjective. By 3.21, these two assertions are equivalent.

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CHAPTER 4

# Polynomials

1. Suppose $m$ and $n$ are positive integers with $m \leq n$. Prove that there exists a polynomial $p \in \mathcal{P}_n(\mathbf{F})$ with exactly $m$ distinct roots.

   SOLUTION: Define $p \in \mathcal{P}_n(\mathbf{F})$ by

   $$p(z) = (z-1)^{n-m+1}(z-2)(z-3)\ldots(z-m).$$

   Then $p$ is a polynomial of degree $n$ with exactly $m$ distinct roots (which are $1, \ldots, m$).

2. Suppose that $z_1, \ldots, z_{m+1}$ are distinct elements of $\mathbf{F}$ and $w_1, \ldots, w_{m+1} \in \mathbf{F}$. Prove that there exists a unique polynomial $p \in \mathcal{P}_m(\mathbf{F})$ such that

   $$p(z_j) = w_j$$

   for $j = 1, \ldots, m+1$.

   SOLUTION: Define $T\colon \mathcal{P}_m(\mathbf{F}) \to \mathbf{F}^{m+1}$ by

   $$Tp = \big(p(z_1), \ldots, p(z_{m+1})\big).$$

   We need to prove that $T$ is injective (which implies that at most one polynomial $p$ satisfies the condition required by the exercise) and surjective (which implies that at least one polynomial $p$ satisfies the condition required by the exercise).

   Clearly $T$ is a linear map. If $p \in \operatorname{null} T$, then

   $$p(z_1) = \cdots = p(z_{m+1}) = 0,$$

   which means that $p$ is a polynomial of degree $m$ with at least $m+1$ distinct roots, which means that $p = 0$ (by 4.3). Thus $p$ is injective, as desired.



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Now

$$\begin{aligned}\dim \text{range}\, T &= \dim \mathcal{P}_m(\mathbf{F}) - \dim \text{null}\, T \\ &= (m+1) - 0 \\ &= \dim \mathbf{F}^{m+1},\end{aligned}$$

where the first equality comes from 3.4 and the second equality holds because $\text{null}\, T = \{0\}$. The last equality above implies that $\text{range}\, T = \mathbf{F}^{m+1}$. Thus $T$ is surjective, as desired.

COMMENT: Surjectivity of $T$ can also be proved by using an explicit construction. But linear algebra, specifically 3.4, gives us surjectivity easily once we get injectivity.

3. Prove that if $p, q \in \mathcal{P}(\mathbf{F})$, with $p \neq 0$, then there exist unique polynomials $s, r \in \mathcal{P}(\mathbf{F})$ such that

$$q = sp + r$$

and $\deg r < \deg p$. In other words, add a uniqueness statement to the division algorithm (4.5).

SOLUTION: Suppose $p, q \in \mathcal{P}(\mathbf{F})$, with $p \neq 0$. We know from the division algorithm (4.5) that there exist $s, r \in \mathcal{P}(\mathbf{F})$, with $\deg r < \deg p$, such that

$$q = sp + r.$$

To prove that $s$ and $r$ are unique, suppose that $\tilde{s}, \tilde{r}$ are in $\mathcal{P}(\mathbf{F})$, with $\deg \tilde{r} < \deg p$ and

$$q = \tilde{s}p + \tilde{r}.$$

Subtracting the last two equations are reârranging, we have

$$(\tilde{s} - s)p = r - \tilde{r}.$$

The right side of the equation above is a polynomial whose degree is less than $\deg p$. If $\tilde{s}$ were not equal to $s$, then the left side of the equation above would be a polynomial whose degree is at least $\deg p$. Thus we must have $\tilde{s} = s$, which, from the equation above, implies that $\tilde{r} = r$. Thus the choices of $s$ and $r$ were indeed unique.

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4. Suppose $p \in \mathcal{P}(\mathbf{C})$ has degree $m$. Prove that $p$ has $m$ distinct roots if and only if $p$ and its derivative $p'$ have no roots in common.

SOLUTION: First suppose that $p$ has $m$ distinct roots. Because $p$ has degree $m$, this implies that $p$ can be written in the form

$$p(z) = c(z - \lambda_1) \dots (z - \lambda_m),$$

where $\lambda_1, \dots, \lambda_m$ are distinct. To prove that $p$ and $p'$ have no roots in common, we must show that $p'(\lambda_j) \neq 0$ for each $j$. To do this, fix $j$. The expression above for $p$ shows that we can write $p$ in the form

$$p(z) = (z - \lambda_j)q(z),$$

where $q$ is a polynomial such that $q(\lambda_j) \neq 0$. Differentiating both sides of this equation, we have

$$p'(z) = (z - \lambda_j)q'(z) + q(z).$$

Thus

$$\begin{aligned} p'(\lambda_j) &= q(\lambda_j) \\ &\neq 0, \end{aligned}$$

as desired.

To prove the other direction, we will proved the contrapositive, meaning that we will prove that if $p$ has less than $m$ distinct roots, then $p$ and $p'$ have at least one root in common. To do this, suppose that $p$ has less than $m$ distinct roots. Then for some root $\lambda$ of $p$, we can write $p$ in the form

$$p(z) = (z - \lambda)^n q(z),$$

where $n \geq 2$ and $q$ is a polynomial. Differentiating both sides of this equation, we have

$$p'(z) = (z - \lambda)^n q'(z) + n(z - \lambda)^{n-1} q(z).$$

Thus $p'(\lambda) = 0$, and so $\lambda$ is a common root of $p$ and $p'$, as desired.

5. Prove that every polynomial with odd degree and real coefficients has a real root.

SOLUTION: Suppose that $p$ is a polynomial with odd degree and real coefficients. By 4.14, $p$ is a constant times the product of factors of the form

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$x - \lambda$ and/or $x^2 + \alpha x + \beta$, where $\lambda, \alpha, \beta \in \mathbf{R}$. Not all the factors can be of the form $x^2 + \alpha x + \beta$, because otherwise $p$ would have even degree. Thus at least one factor must be of the form $x - \lambda$. Any such $\lambda$ is a real root of $p$.

COMMENT: Here is another proof, using calculus but not using 4.14. Suppose $p$ is a polynomial with odd degree $m$. We can write $p$ in the form

$$p(x) = a_0 + a_1 x + \cdots + a_m x^m,$$

where $a_0, \ldots, a_m \in \mathbf{R}$ and $a_m \neq 0$. Replacing $p$ with $-p$ if necessary, we can assume that $a_m > 0$. Now

$$p(x) = x^m\left(\frac{a_0}{x^m} + \frac{a_1}{x^{m-1}} + \cdots + \frac{a_{m-1}}{x} + a_m\right).$$

This implies that

$$\lim_{x \to -\infty} p(x) = -\infty \quad \text{and} \quad \lim_{x \to \infty} p(x) = \infty.$$

The intermediate value theorem now implies that there is a real number $\lambda$ such that $p(\lambda) = 0$. In other words, $p$ has a real root.

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CHAPTER 5

# Eigenvalues and Eigenvectors

1. Suppose $T \in \mathcal{L}(V)$. Prove that if $U_1, \ldots, U_m$ are subspaces of $V$ invariant under $T$, then $U_1 + \cdots + U_m$ is invariant under $T$.

   SOLUTION: Suppose $U_1, \ldots, U_m$ are subspaces of $V$ invariant under $T$. Consider a vector $u \in U_1 + \cdots + U_m$. There exist $u_1 \in U_1, \ldots, u_m \in U_m$ such that

   $$u = u_1 + \cdots + u_m.$$

   Applying $T$ to both sides of this equation, we get

   $$Tu = Tu_1 + \cdots + Tu_m.$$

   Because each $U_j$ is invariant under $T$, we have $Tu_1 \in U_1, \ldots, Tu_m \in U_m$. Thus the equation above shows that $Tu \in U_1 + \cdots + U_m$, which implies that $U_1 + \cdots + U_m$ is invariant under $T$.

2. Suppose $T \in \mathcal{L}(V)$. Prove that the intersection of any collection of subspaces of $V$ invariant under $T$ is invariant under $T$.

   SOLUTION: Suppose $\{U_\alpha\}_{\alpha \in \Gamma}$ is a collection of subspaces of $V$ invariant under $T$; here $\Gamma$ is an arbitrary index set. We need to prove that $\bigcap_{\alpha \in \Gamma} U_\alpha$, which equals the set of vectors that are in $U_\alpha$ for every $\alpha \in \Gamma$, is invariant under $T$. To do this, suppose $u \in \bigcap_{\alpha \in \Gamma} U_\alpha$. Then $u \in U_\alpha$ for every $\alpha \in \Gamma$. Thus $Tu \in U_\alpha$ for every $\alpha \in \Gamma$ (because every $U_\alpha$ is invariant under $T$). Thus $Tu \in \bigcap_{\alpha \in \Gamma} U_\alpha$, which implies that $\bigcap_{\alpha \in \Gamma} U_\alpha$ is invariant under $T$.

3. Prove or give a counterexample: if $U$ is a subspace of $V$ that is invariant under every operator on $V$, then $U = \{0\}$ or $U = V$.



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SOLUTION: We will prove that if $U$ is a subspace of $V$ that is invariant under every operator on $V$, then $U = \{0\}$ or $U = V$. Actually we will prove the (logically equivalent) contrapositive, meaning that we will prove that if $U$ is a subspace of $V$ such that $U \neq \{0\}$ and $U \neq V$, then there exists $T \in \mathcal{L}(V)$ such that $U$ is not invariant under $T$. To do this, suppose $U$ is a subspace of $V$ such that $U \neq \{0\}$ and $U \neq V$. Choose $u \in U \setminus \{0\}$ (this is possible because $U \neq \{0\}$) and $w \in V \setminus U$ (this is possible because $U \neq V$). Extend the list $(u)$, which is linearly independent because $u \neq 0$, to a basis $(u, v_1, \ldots, v_n)$ of $V$. Define $T \in \mathcal{L}(V)$ by

$$T(au + b_1 v_1 + \ldots b_n v_n) = aw.$$

Thus $Tu = w$. Because $u \in U$ but $w \notin U$, this shows that $U$ is not invariant under $T$, as desired.

4. Suppose that $S, T \in \mathcal{L}(V)$ are such that $ST = TS$. Prove that $\text{null}(T - \lambda I)$ is invariant under $S$ for every $\lambda \in \mathbf{F}$.

SOLUTION: Fix $\lambda \in \mathbf{F}$. Suppose $v \in \text{null}(T - \lambda I)$. Then

$$\begin{aligned}(T - \lambda I)(Sv) &= TSv - \lambda Sv \\ &= STv - \lambda Sv \\ &= S(Tv - \lambda v) \\ &= 0.\end{aligned}$$

Thus $Sv \in \text{null}(T - \lambda I)$. Hence $\text{null}(T - \lambda I)$ is invariant under $S$.

5. Define $T \in \mathcal{L}(\mathbf{F}^2)$ by

$$T(w, z) = (z, w).$$

Find all eigenvalues and eigenvectors of $T$.

SOLUTION: Suppose $\lambda$ is an eigenvalue of $T$. For this particular operator, the eigenvalue-eigenvector equation $T(w, z) = \lambda(w, z)$ becomes the system of equations

$$\begin{aligned}z &= \lambda w \\ w &= \lambda z.\end{aligned}$$

Substituting the value for $z$ from the first equation into the second equation gives $w = \lambda^2 w$. Thus $1 = \lambda^2$ (we can ignore the possibility that $w = 0$

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because if $w = 0$, then the first equation above implies that $z = 0$). Thus $\lambda = 1$ or $\lambda = -1$. The set of eigenvectors corresponding to the eigenvalue 1 is

$$\{(w, w) : w \in \mathbf{F}\};$$

The set of eigenvectors corresponding to the eigenvalue $-1$ is

$$\{(w, -w) : w \in \mathbf{F}\}.$$

6. Define $T \in \mathcal{L}(\mathbf{F}^3)$ by

$$T(z_1, z_2, z_3) = (2z_2, 0, 5z_3).$$

Find all eigenvalues and eigenvectors of $T$.

SOLUTION: Suppose $\lambda$ is an eigenvalue of $T$. For this particular operator, the eigenvalue-eigenvector equation $T(z_1, z_2, z_3) = \lambda(z_1, z_2, z_3)$ becomes the system of equations

$$\begin{aligned} 2z_2 &= \lambda z_1 \\ 0 &= \lambda z_2 \\ 5z_3 &= \lambda z_3. \end{aligned}$$

If $\lambda \neq 0$, then the second equation implies that $z_2 = 0$, and the first equation then implies that $z_1 = 0$. Because an eigenvalue must have a nonzero eigenvector, there must be a solution to the system above with $z_3 \neq 0$. The third equation then shows that $\lambda = 5$. In other words, 5 is the only nonzero eigenvalue of $T$. The set of eigenvectors corresponding to the eigenvalue 5 is

$$\{(0, 0, z_3) : z_3 \in \mathbf{F}\}.$$

If $\lambda = 0$, the first and third equations above show that $z_2 = 0$ and $z_3 = 0$. With these values for $z_2, z_3$, the equations above are satisfied for all values of $z_1$. Thus 0 is an eigenvalue of $T$. The set of eigenvectors corresponding to the eigenvalue 0 is

$$\{(z_1, 0, 0) : z_1 \in \mathbf{F}\}.$$

7. Suppose $n$ is a positive integer and $T \in \mathcal{L}(\mathbf{F}^n)$ is defined by

$$T(x_1, \ldots, x_n) = (x_1 + \cdots + x_n, \ldots, x_1 + \cdots + x_n);$$

in other words, $T$ is the operator whose matrix (with respect to the standard basis) consists of all 1's. Find all eigenvalues and eigenvectors of $T$.

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SOLUTION: Suppose $\lambda$ is an eigenvalue of $T$. For this particular operator, the eigenvalue-eigenvector equation $Tx = \lambda x$ becomes the system of equations

$$\begin{aligned} x_1 + \cdots + x_n &= \lambda x_1 \\ &\vdots \\ x_1 + \cdots + x_n &= \lambda x_n. \end{aligned}$$

Thus

$$\lambda x_1 = \cdots = \lambda x_n.$$

Hence either $\lambda = 0$ or $x_1 = \cdots = x_n$.

Consider first the possibility that $\lambda = 0$. In this case all the equations in the eigenvector-eigenvalue system of equations above become the equation $x_1 + \cdots + x_n = 0$. Thus we see that 0 is an eigenvalue of $T$ and that the corresponding set of eigenvectors equals

$$\{(x_1, \ldots, x_n) \in \mathbf{F}^n : x_1 + \cdots + x_n = 0\}.$$

Now consider the possibility that $x_1 = \cdots = x_n$; let $t$ denote the common value of $x_1, \ldots, x_n$. In this case all the equations in the eigenvector-eigenvalue system of equations above become the equation $nt = \lambda t$. Hence $\lambda$ must equal $n$ (an eigenvalue must have a nonzero eigenvector, so we can take $t \neq 0$). Thus we see that $n$ is an eigenvalue of $T$ and that the corresponding set of eigenvectors equals

$$\{(x_1, \ldots, x_n) \in \mathbf{F}^n : x_1 = \cdots = x_n\}.$$

Because the eigenvector-eigenvalue system of equations above implies that $\lambda = 0$ or $x_1 = \cdots = x_n$, we see that $T$ has no eigenvalues other than 0 and $n$.

8. Find all eigenvalues and eigenvectors of the backward shift operator $T \in \mathcal{L}(\mathbf{F}^\infty)$ defined by

$$T(z_1, z_2, z_3, \ldots) = (z_2, z_3, \ldots).$$

SOLUTION: Suppose $\lambda$ is an eigenvalue of $T$. For this particular operator, the eigenvalue-eigenvector equation $Tz = \lambda z$ becomes the system of equations

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$$\begin{aligned} z_2 &= \lambda z_1 \\ z_3 &= \lambda z_2 \\ z_4 &= \lambda z_3 \\ &\vdots \end{aligned}$$

From this we see that we can choose $z_1$ arbitrarily and then solve for the other coordinates:

$$\begin{aligned} z_2 &= \lambda z_1 \\ z_3 &= \lambda z_2 = \lambda^2 z_1 \\ z_4 &= \lambda z_3 = \lambda^3 z_1 \\ &\vdots \end{aligned}$$

Thus each $\lambda \in \mathbf{F}$ is an eigenvalue of $T$ and the set of corresponding eigenvectors is

$$\{(w, \lambda w, \lambda^2 w, \lambda^3 w \ldots) : w \in \mathbf{F}\}.$$

9. Suppose $T \in \mathcal{L}(V)$ and $\dim \operatorname{range} T = k$. Prove that $T$ has at most $k+1$ distinct eigenvalues.

SOLUTION: Let $\lambda_1, \ldots, \lambda_m$ be the distinct eigenvalues of $T$, and let $v_1, \ldots, v_m$ be corresponding nonzero eigenvectors. If $\lambda_j \neq 0$, then

$$T(v_j/\lambda_j) = v_j.$$

Because at most one of $\lambda_1, \ldots, \lambda_m$ equals 0, this implies that at least $m-1$ of the vectors $v_1, \ldots, v_m$ are in $\operatorname{range} T$. These vectors are linearly independent (by 5.6), which implies that

$$m - 1 \leq \dim \operatorname{range} T = k.$$

Thus $m \leq k+1$, as desired.

10. Suppose $T \in \mathcal{L}(V)$ is invertible and $\lambda \in \mathbf{F} \backslash \{0\}$. Prove that $\lambda$ is an eigenvalue of $T$ if and only if $\frac{1}{\lambda}$ is an eigenvalue of $T^{-1}$.

SOLUTION: First suppose that $\lambda$ is an eigenvalue of $T$. Thus there exists a nonzero vector $v \in V$ such that

$$Tv = \lambda v.$$

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Applying $T^{-1}$ to both sides of the equation above, we get $v = \lambda T^{-1}v$, which is equivalent to the equation $T^{-1}v = \frac{1}{\lambda}v$. Thus $\frac{1}{\lambda}$ is an eigenvalue of $T^{-1}$.

To prove the implication in the other direction, replace $T$ by $T^{-1}$ and $\lambda$ by $\frac{1}{\lambda}$ and then apply the result from the paragraph above.

11. Suppose $S, T \in \mathcal{L}(V)$. Prove that $ST$ and $TS$ have the same eigenvalues.

SOLUTION: Suppose that $\lambda \in \mathbf{F}$ is an eigenvalue of $ST$. We want to prove that $\lambda$ is an eigenvalue of $TS$. Because $\lambda$ is an eigenvalue of $ST$, there exists a nonzero vector $v \in V$ such that

$$(ST)v = \lambda v.$$

Now

$$\begin{aligned}(TS)(Tv) &= T(STv)\\ &= T(\lambda v)\\ &= \lambda Tv.\end{aligned}$$

If $Tv \neq 0$, then the equation above shows that $\lambda$ is an eigenvalue of $TS$, as desired.

If $Tv = 0$, then $\lambda = 0$ (because $S(Tv) = \lambda v$) and furthermore $T$ is not invertible, which implies that $TS$ is not invertible (by Exercise 22 in Chapter 3), which implies that $\lambda$ (which equals 0) is an eigenvalue of $TS$.

Regardless of whether or not $Tv = 0$, we have shown that $\lambda$ is an eigenvalue of $TS$. Because $\lambda$ was an arbitrary eigenvalue of $ST$, we have shown that every eigenvalue of $ST$ is an eigenvalue of $TS$.

Reversing the roles of $S$ and $T$, we conclude that every eigenvalue of $TS$ is also an eigenvalue of $ST$. Thus $ST$ and $TS$ have the same eigenvalues.

12. Suppose $T \in \mathcal{L}(V)$ is such that every vector in $V$ is an eigenvector of $T$. Prove that $T$ is a scalar multiple of the identity operator.

SOLUTION: For each $v \in V$, there exists $a_v \in \mathbf{F}$ such that

$$Tv = a_v v.$$

Because $T0 = 0$, we can choose $a_0$ to be any number in $\mathbf{F}$, but for $v \in V \setminus \{0\}$ the value of $a_v$ is uniquely determined by the equation above.

To show that $T$ is a scalar multiple of the identity, we must show that $a_v$ is independent of $v$ for $v \in V \setminus \{0\}$. To do this, suppose $v, w \in V \setminus \{0\}$. We want to show that $a_v = a_w$. First consider the case where $(v, w)$ is linearly dependent. Then there exists $b \in \mathbf{F}$ such that $w = bv$. We have

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$$\begin{aligned} a_w w &= Tw \\ &= T(bv) \\ &= bTv \\ &= b(a_v v) \\ &= a_v w, \end{aligned}$$

which shows that $a_v = a_w$, as desired.

Finally, consider the case where $(v, w)$ is linearly independent. We have

$$\begin{aligned} a_{v+w}(v+w) &= T(v+w) \\ &= Tv + Tw \\ &= a_v v + a_w w, \end{aligned}$$

which implies that

$$(a_{v+w} - a_v)v + (a_{v+w} - a_w)w = 0.$$

Because $(v, w)$ is linearly independent, this implies that $a_{v+w} = a_v$ and $a_{v+w} = a_w$, so again we have $a_v = a_w$, as desired.

13. Suppose $T \in \mathcal{L}(V)$ is such that every subspace of $V$ with dimension

$$\dim V - 1$$

is invariant under $T$. Prove that $T$ is a scalar multiple of the identity operator.

SOLUTION: Suppose that $T$ is not a scalar multiple of the identity operator. By the previous exercise, there exists $u \in V$ such that $u$ is not an eigenvector of $T$. Thus $(u, Tu)$ is linearly independent. Extend $(u, Tu)$ to a basis $(u, Tu, v_1, \ldots, v_n)$ of $V$. Let

$$U = \operatorname{span}(u, v_1, \ldots, v_n).$$

Then $U$ is a subspace of $V$ and $\dim U \doteq \dim V - 1$. However, $U$ is not invariant under $T$ because $u \in U$ but $Tu \notin U$. This contradiction to our hypothesis about $T$ shows that our assumption that $T$ is not a scalar multiple of the identity must have been false.

14. Suppose $S, T \in \mathcal{L}(V)$ and $S$ is invertible. Prove that if $p \in \mathcal{P}(\mathbf{F})$ is a polynomial, then

$$p(STS^{-1}) = Sp(T)S^{-1}.$$

SOLUTION: First suppose $m$ is a positive integer. Then

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$$\begin{aligned}(STS^{-1})^m &= (STS^{-1})(STS^{-1})\dots(STS^{-1}) \\ &= ST(S^{-1}S)T(S^{-1}S)\dots(S^{-1}S)TS^{-1} \\ &= ST^mS^{-1},\end{aligned}$$

which is our desired equation in the special case when $p(z) = z^m$. Multiplying both sides of the equation above by a scalar and then summing a finite number of equations of the resulting form shows that $p(STS^{-1}) = Sp(T)S^{-1}$ for every polynomial $p \in \mathcal{P}(\mathbf{F})$.

15. Suppose $\mathbf{F} = \mathbf{C}$, $T \in \mathcal{L}(V)$, $p \in \mathcal{P}(\mathbf{C})$, and $a \in \mathbf{C}$. Prove that $a$ is an eigenvalue of $p(T)$ if and only if $a = p(\lambda)$ for some eigenvalue $\lambda$ of $T$.

SOLUTION: First suppose that $a$ is an eigenvalue of $p(T)$. Thus $p(T) - aI$ is not injective. Write the polynomial $p(z) - a$ in factored form:

$$p(z) - a = c(z - \lambda_1)\dots(z - \lambda_m),$$

where $c, \lambda_1, \dots, \lambda_m \in \mathbf{C}$. We can assume that $c \neq 0$ (otherwise $p$ is a constant polynomial, in which case the desired result clearly holds). The equation above implies that

$$p(T) - aI = c(T - \lambda_1 I)\dots(T - \lambda_m I).$$

Because $p(T) - aI$ is not injective, this implies that $T - \lambda_j I$ is not injective for some $j$. In other words, some $\lambda_j$ is an eigenvalue of $T$. The formula above for $p(z) - a$ shows that $p(\lambda_j) - a = 0$. Hence $a = p(\lambda_j)$, as desired.

For the other direction, now suppose that $a = p(\lambda)$ for some eigenvalue $\lambda$ of $T$. Thus there exists a nonzero vector $v \in V$ such that

$$Tv = \lambda v.$$

Repeatedly applying $T$ to both sides of this equation shows that $T^k v = \lambda^k v$ for every positive integer $k$. Thus

$$\begin{aligned}p(T)v &= p(\lambda)v \\ &= av.\end{aligned}$$

Thus $a$ is an eigenvalue of $p(T)$.

16. Show that the result in the previous exercise does not hold if $\mathbf{C}$ is replaced with $\mathbf{R}$.

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SOLUTION: Define $T \in \mathcal{L}(\mathbf{R}^2)$ by $T(x,y) = (-y,x)$. Define $p \in \mathcal{P}(\mathbf{R})$ by $p(x) = x^2$. Then $p(T) = T^2 = -I$, and hence $-1$ is an eigenvalue of $p(T)$. However, $T$ has no eigenvalues (as we saw on page 78 of the textbook; the point here is that eigenvalues are required to be real because we are working on a real vector space), so there does not exist an eigenvalue $\lambda$ of $T$ such that $-1 = p(\lambda)$.

Of course there are also many other examples.

17. Suppose $V$ is a complex vector space and $T \in \mathcal{L}(V)$. Prove that $T$ has an invariant subspace of dimension $j$ for each $j = 1, \dots, \dim V$.

SOLUTION: There is a basis $(v_1, \dots, v_{\dim V})$ with respect to which $T$ has an upper-triangular matrix (see 5.13). For each $j = 1, \dots, \dim V$, the span of $(v_1, \dots, v_j)$ is a $j$-dimensional subspace of $V$ that is invariant under $T$ (by 5.12).

18. Give an example of an operator whose matrix with respect to some basis contains only 0's on the diagonal, but the operator is invertible.

SOLUTION: Let $T \in \mathcal{L}(\mathbf{F}^2)$ be the operator whose matrix (with respect to the standard basis) is

$$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}.$$

Obviously this matrix has only 0's on the diagonal, but $T$ is invertible (because $TT = I$, as is clear from squaring the matrix above).

Of course there are also many other examples.

COMMENT: This exercise and the next one show that 5.16 fails without the hypothesis that an upper-triangular matrix is under consideration.

19. Give an example of an operator whose matrix with respect to some basis contains only nonzero numbers on the diagonal, but the operator is not invertible.

SOLUTION: Define $T \in \mathcal{L}(\mathbf{F}^2)$ to be the operator whose matrix (with respect to the standard basis) is

$$\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}.$$

Then $T(1,0) = T(0,1) = (1,1)$, so $T$ is not injective, so $T$ is not invertible, even though the diagonal of the matrix above contains only nonzero numbers.

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Of course there are also many other examples.

20. Suppose that $T \in \mathcal{L}(V)$ has $\dim V$ distinct eigenvalues and that $S \in \mathcal{L}(V)$ has the same eigenvectors as $T$ (not necessarily with the same eigenvalues). Prove that $ST = TS$.

SOLUTION: Let $n = \dim V$. There is a basis $(v_1, \ldots, v_n)$ of $V$ consisting of eigenvectors of $T$ (see 5.20 and 5.21). Letting $\lambda_1, \ldots, \lambda_n$ be the corresponding eigenvalues, we have

$$Tv_j = \lambda_j v_j$$

for each $j$. Each $v_j$ is also an eigenvector of $S$, so

$$Sv_j = \alpha_j v_j$$

for some $\alpha_j \in \mathbf{F}$.

For each $j$, we have

$$(ST)v_j = S(Tv_j) = \lambda_j Sv_j = \alpha_j \lambda_j v_j$$

and

$$(TS)v_j = T(Sv_j) = \alpha_j Tv_j = \alpha_j \lambda_j v_j.$$

Because the operators $ST$ and $TS$ agree on a basis, they are equal.

21. Suppose $P \in \mathcal{L}(V)$ and $P^2 = P$. Prove that $V = \operatorname{null} P \oplus \operatorname{range} P$.

SOLUTION: First suppose $u \in \operatorname{null} P \cap \operatorname{range} P$. Then $Pu = 0$, and there exists $w \in V$ such that $u = Pw$. Applying $P$ to both sides of the last equation, we have $Pu = P^2 w = Pw$. But $Pu = 0$, so this implies that $Pw = 0$. Because $u = Pw$, this implies that $u = 0$. Because $u$ was an arbitrary vector in $\operatorname{null} P \cap \operatorname{range} P$, this implies that $\operatorname{null} P \cap \operatorname{range} P = \{0\}$.

Now suppose $v \in V$. Then obviously

$$v = (v - Pv) + Pv.$$

Note that $P(v - Pv) = Pv - P^2 v = 0$, so $(v - Pv) \in \operatorname{null} P$. Clearly $Pv \in \operatorname{range} P$. Thus the equation above shows that $v \in \operatorname{null} P + \operatorname{range} P$. Because $v$ was an arbitrary vector in $V$, this implies that $V = \operatorname{null} P + \operatorname{range} P$.

We have shown that $\operatorname{null} P \cap \operatorname{range} P = \{0\}$ and $V = \operatorname{null} P + \operatorname{range} P$. Thus $V = \operatorname{null} P \oplus \operatorname{range} P$ (by 1.9).

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22. Suppose $V = U \oplus W$, where $U$ and $W$ are nonzero subspaces of $V$. Find all eigenvalues and eigenvectors of $P_{U,W}$.

SOLUTION: Because $V = U \oplus W$, each vector $v \in V$ can be written uniquely in the form

$$v = u + w,$$

where $u \in U$ and $w \in W$. Recall that if $v$ is represented as above, then $P_{U,W}v = u$.

Suppose $\lambda \in \mathbf{F}$ is an eigenvalue of $P_{U,W}$. Then there exists a nonzero vector $v \in V$ such that $P_{U,W}v = \lambda v$. Writing this equation using the representation of $v$ given above, we have $u = \lambda(u + w)$. Thus

$$(1 - \lambda)u - \lambda w = 0.$$

Because $V = U \oplus W$, if 0 is written as the sum of a vector in $U$ and a vector in $W$, then both vectors must be 0. Thus the equation above implies that $(1 - \lambda)u = \lambda w = 0$. Because $u$ and $w$ are not both 0 (because $v \neq 0$), this implies that $\lambda = 1$ or $\lambda = 0$.

For $v \in V$ with representation as above, the equation $P_{U,W}v = 0$ is equivalent to the equation $u = 0$, which is equivalent to the equation $v = w$, which is equivalent to the statement that $v \in W$. This means that 0 is an eigenvalue of $P_{U,W}$ (because $W$ is a nonzero subspace of $V$) and that $W$ equals the set of eigenvectors corresponding to the eigenvalue 0.

For $v \in V$ with representation as above, the equation $P_{U,W}v = v$ is equivalent to the equation $v = u$, which is equivalent to the statement that $v \in U$. This means that 1 is an eigenvalue of $P_{U,W}$ (because $U$ is a nonzero subspace of $V$) and that $U$ equals the set of eigenvectors corresponding to the eigenvalue 1.

23. Give an example of an operator $T \in \mathcal{L}(\mathbf{R}^4)$ such that $T$ has no (real) eigenvalues.

SOLUTION: Define $T \in \mathcal{L}(\mathbf{R}^4)$ by

$$T(x_1, x_2, x_3, x_4) = (-x_2, x_1, -x_4, x_3).$$

Suppose $\lambda \in \mathbf{R}$. For this particular operator, the eigenvalue-eigenvector equation $T(x_1, x_2, x_3, x_4) = \lambda(x_1, x_2, x_3, x_4)$ becomes the system of equations

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$$\begin{aligned} -x_2 &= \lambda x_1 \\ x_1 &= \lambda x_2 \\ -x_4 &= \lambda x_3 \\ x_3 &= \lambda x_4. \end{aligned}$$

Multiplying together the first two equations and also multiplying together the last two equations gives $-x_1x_2 = \lambda^2 x_1x_2$ and $-x_3x_4 = \lambda^2 x_3x_4$. If either $x_1$ or $x_2$ does not equal 0, then the first two equations show that neither of $x_1, x_2$ equals 0. Similarly, if either $x_3$ or $x_4$ does not equal 0, then the last two equations show that neither of $x_3, x_4$ equals 0. Thus if $\lambda$ is an eigenvalue of $T$, then there is a solution to the system of equations above with $x_1x_2 \neq 0$ or $x_3x_4 \neq 0$. Either way, we conclude that $-1 = \lambda^2$, which is impossible for any real number $\lambda$. Thus $T$ has no real eigenvalues.

24. Suppose $V$ is a real vector space and $T \in \mathcal{L}(V)$ has no eigenvalues. Prove that every subspace of $V$ invariant under $T$ has even dimension.

SOLUTION: Suppose $U$ is a subspace of $V$ that is invariant under $T$. Thus $T|_U \in \mathcal{L}(U)$. If $\dim U$ were odd, then $T|_U$ would have an eigenvalue $\lambda \in \mathbf{R}$ (by 5.26), so there would exist a nonzero vector $u \in U$ such that

$$T|_U u = \lambda u.$$

Obviously this would imply that $Tu = \lambda u$, which would imply that $\lambda$ is an eigenvalue of $T$. But $T$ has no eigenvalues, so $\dim U$ must be even, as desired.

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CHAPTER 6

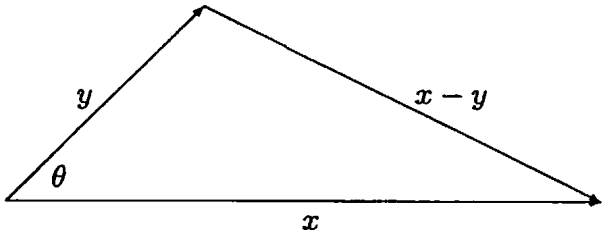
# Inner-Product Spaces

1. Prove that if $x, y$ are nonzero vectors in $\mathbf{R}^2$, then

$$\langle x, y \rangle = \|x\| \|y\| \cos\theta,$$

where $\theta$ is the angle between $x$ and $y$ (thinking of $x$ and $y$ as arrows with initial point at the origin). *Hint:* draw the triangle formed by $x$, $y$, and $x-y$; then use the law of cosines.

SOLUTION: Suppose that $x, y$ are nonzero vectors in $\mathbf{R}^2$ and $\theta$ is the angle between $x$ and $y$. Consider the triangle formed by $x$, $y$, and $x - y$:




The law of cosines states that

$$\|x - y\|^2 = \|x\|^2 + \|y\|^2 - 2\|x\| \|y\| \cos\theta.$$

As usual, we compute the norm of a vector squared by taking the inner product of the vector with itself:

$$\begin{aligned} \|x - y\|^2 &= \langle x - y, x - y \rangle \\ &= \langle x, x \rangle - \langle x, y \rangle - \langle y, x \rangle + \langle y, y \rangle \\ &= \|x\|^2 + \|y\|^2 - 2\langle x, y \rangle. \end{aligned}$$



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Substitute the last expression for $\|x-y\|^2$ into the left side of the law of cosines, obtaining

$$\|x\|^2 + \|y\|^2 - 2\langle x, y\rangle = \|x\|^2 + \|y\|^2 - 2\|x\|\|y\|\cos\theta.$$

Now subtract $\|x\|^2 + \|y\|^2$ from both sides of the equation above, and then divide both sides by $-2$, obtaining

$$\langle x, y\rangle = \|x\|\|y\|\cos\theta.$$

2. Suppose $u, v \in V$. Prove that $\langle u, v\rangle = 0$ if and only if

$$\|u\| \le \|u + av\|$$

for all $a \in \mathbf{F}$.

SOLUTION: First suppose that $\langle u, v\rangle = 0$. Let $a \in \mathbf{F}$. Then $u, av$ are orthogonal. The Pythagorean theorem thus implies that

$$\begin{aligned}\|u + av\|^2 &= \|u\|^2 + \|av\|^2 \\ &\ge \|u\|^2.\end{aligned}$$

Taking square roots gives $\|u\| \le \|u + av\|$, as desired.

To prove the implication in the other direction, now suppose that $\|u\| \le \|u + av\|$ for all $a \in \mathbf{F}$. Squaring this inequality, we get

$$\begin{aligned}\|u\|^2 &\le \|u + av\|^2 \\ &= \langle u + av, u + av\rangle \\ &= \langle u, u\rangle + \langle u, av\rangle + \langle av, u\rangle + \langle av, av\rangle \\ &= \|u\|^2 + \bar{a}\langle u, v\rangle + a\overline{\langle u, v\rangle} + |a|^2\|v\|^2 \\ &= \|u\|^2 + 2\operatorname{Re}\bar{a}\langle u, v\rangle + |a|^2\|v\|^2\end{aligned}$$

for all $a \in \mathbf{F}$. Thus

$$-2\operatorname{Re}\bar{a}\langle u, v\rangle \le |a|^2\|v\|^2$$

for all $a \in \mathbf{F}$. In particular, we can let $a$ equal $-t\langle u, v\rangle$ for $t > 0$. Substituting this value for $a$ into the inequality above gives

$$2t|\langle u, v\rangle|^2 \le t^2|\langle u, v\rangle|^2\|v\|^2$$

for all $t > 0$. Divide both sides of the inequality above by $t$, getting

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$$2|\langle u, v\rangle|^2 \le t|\langle u, v\rangle|^2\|v\|^2$$

for all $t > 0$. If $v = 0$, then $\langle u, v\rangle = 0$, as desired. If $v \neq 0$, set $t$ equal to $1/\|v\|^2$ in the inequality above, getting

$$2|\langle u, v\rangle|^2 \le |\langle u, v\rangle|^2,$$

which implies that $\langle u, v\rangle = 0$, as desired.

3. Prove that

$$\Big(\sum_{j=1}^{n} a_j b_j\Big)^2 \le \Big(\sum_{j=1}^{n} j a_j{}^2\Big)\Big(\sum_{j=1}^{n} \frac{b_j{}^2}{j}\Big)$$

for all real numbers $a_1, \ldots, a_n$ and $b_1, \ldots, b_n$.

SOLUTION: Suppose $a_1, \ldots, a_n, b_1, \ldots, b_n \in \mathbf{R}$. Using the usual inner product on $\mathbf{R}^n$, we have

$$\begin{aligned}
\Big(\sum_{j=1}^{n} a_j b_j\Big)^2 &= \Big(\sum_{j=1}^{n} (\sqrt{j}a_j)(b_j/\sqrt{j})\Big)^2 \\
&= \langle (a_1, \sqrt{2}a_2, \ldots, \sqrt{n}a_n), (b_1, b_2/\sqrt{2}, \ldots, b_n/\sqrt{n})\rangle^2 \\
&\le \|(a_1, \sqrt{2}a_2, \ldots, \sqrt{n}a_n)\|^2 \|(b_1, b_2/\sqrt{2}, \ldots, b_n/\sqrt{n})\|^2 \\
&= \Big(\sum_{j=1}^{n} j a_j{}^2\Big)\Big(\sum_{j=1}^{n} b_j{}^2/j\Big),
\end{aligned}$$

where the inequality above comes from the Cauchy-Schwarz inequality.

4. Suppose $u, v \in V$ are such that

$$\|u\| = 3, \quad \|u + v\| = 4, \quad \|u - v\| = 6.$$

What number must $\|v\|$ equal?

SOLUTION: From the parallelogram equality, we have

$$\begin{aligned}
\|v\|^2 &= \frac{\|u + v\|^2 + \|u - v\|^2 - 2\|u\|^2}{2} \\
&= \frac{16 + 36 - 18}{2} \\
&= 17.
\end{aligned}$$

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Thus $\|v\| = \sqrt{17}$.

5. Prove or disprove: there is an inner product on $\mathbf{R}^2$ such that the associated norm is given by

$$\|(x_1, x_2)\| = |x_1| + |x_2|$$

for all $(x_1, x_2) \in \mathbf{R}^2$.

SOLUTION: We will show that there does not exist an inner product on $\mathbf{R}^2$ such that the associated norm is given by the formula above by showing that the parallelogram equality is violated. Let

$$u = (3, 2) \quad \text{and} \quad v = (1, 3).$$

Then

$$u + v = (4, 5) \quad \text{and} \quad u - v = (2, -1).$$

Using the formula above, we then have

$$\begin{aligned} \|u + v\|^2 + \|u - v\|^2 &= 81 + 9 \\ &= 90 \end{aligned}$$

and

$$\begin{aligned} 2(\|u\|^2 + \|v\|^2) &= 2(25 + 16) \\ &= 82. \end{aligned}$$

Thus the parallelogram equality fails, as desired.

6. Prove that if $V$ is a real inner-product space, then

$$\langle u, v \rangle = \frac{\|u + v\|^2 - \|u - v\|^2}{4}$$

for all $u, v \in V$.

SOLUTION: Suppose $V$ is a real inner-product space and $u, v \in V$. Then

$$\begin{aligned} \frac{\|u + v\|^2 - \|u - v\|^2}{4} &= \frac{\langle u + v, u + v \rangle - \langle u - v, u - v \rangle}{4} \\ &= \frac{\|u\|^2 + 2\langle u, v \rangle + \|v\|^2 - (\|u\|^2 - 2\langle u, v \rangle + \|v\|^2)}{4} \\ &= \frac{4\langle u, v \rangle}{4} \\ &= \langle u, v \rangle, \end{aligned}$$

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as desired.

7. Prove that if $V$ is a complex inner-product space, then

$$\langle u, v\rangle = \frac{\|u+v\|^2 - \|u-v\|^2 + \|u+iv\|^2 i - \|u-iv\|^2 i}{4}$$

for all $u, v \in V$.

SOLUTION: Suppose $V$ is a complex inner-product space and $u, v \in V$. Then

$$\begin{aligned}\|u+v\|^2 &= \langle u+v, u+v\rangle \\ &= \|u\|^2 + \langle u, v\rangle + \langle v, u\rangle + \|v\|^2\end{aligned}$$

and

$$\begin{aligned}-\|u-v\|^2 &= -\langle u-v, u-v\rangle \\ &= -\|u\|^2 + \langle u, v\rangle + \langle v, u\rangle - \|v\|^2\end{aligned}$$

and

$$\begin{aligned}i\|u+iv\|^2 &= i\langle u+iv, u+iv\rangle \\ &= i\|u\|^2 + \langle u, v\rangle - \langle v, u\rangle + i\|v\|^2\end{aligned}$$

and

$$\begin{aligned}-i\|u-iv\|^2 &= -i\langle u-iv, u-iv\rangle \\ &= -i\|u\|^2 + \langle u, v\rangle - \langle v, u\rangle - i\|v\|^2.\end{aligned}$$

Adding the four equations, we have

$$\|u+v\|^2 - \|u-v\|^2 + i\|u+iv\|^2 - i\|u-iv\|^2 = 4\langle u, v\rangle,$$

as desired.

8. A norm on a vector space $U$ is a function $\| \ \| : U \to [0, \infty)$ such that $\|u\| = 0$ if and only if $u = 0$, $\|\alpha u\| = |\alpha|\|u\|$ for all $\alpha \in \mathbf{F}$ and all $u \in U$, and $\|u+v\| \leq \|u\| + \|v\|$ for all $u, v \in U$. Prove that a norm satisfying the parallelogram equality comes from an inner product (in other words, show that if $\| \ \|$ is a norm on $U$ satisfying the parallelogram equality, then there is an inner product $\langle \ , \ \rangle$ on $U$ such that $\|u\| = \langle u, u\rangle^{1/2}$ for all $u \in U$).

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COMMENT: This is among the hardest exercises in the book. Instructors may want to simplify this exercise slightly by allowing students to consider only the case where $\mathbf{F} = \mathbf{R}$.

SOLUTION: Suppose that $U$ is a vector space and $\| \ \|$ is a norm on $U$ satisfying the parallelogram equality. We want to find an inner product $\langle \ , \ \rangle$ on $U$ such that $\|u\| = \langle u, u\rangle^{1/2}$ for all $u \in U$.

First consider the case where $\mathbf{F} = \mathbf{R}$. For $u, v \in U$, define $\langle u, v\rangle$ by

$$\langle u, v\rangle = \frac{\|u+v\|^2 - \|u-v\|^2}{4}.$$

This definition is motivated by Exercise 6 of this chapter, which gives a formula for the inner product in terms of norms.

For $u \in U$ we have

$$\begin{aligned}\langle u, u\rangle &= \frac{\|u+u\|^2 - \|u-u\|^2}{4} \\ &= \frac{\|2u\|^2 - \|0\|^2}{4} \\ &= \|u\|^2.\end{aligned}$$

Thus $\|u\| = \langle u, u\rangle^{1/2}$, as desired. However, we still must show that $\langle \ , \ \rangle$ satisfies the properties required of an inner product.

Because $\langle u, u\rangle = \|u\|^2$ (as shown above), we have $\langle u, u\rangle \geq 0$ for all $u \in U$, with equality if and only if $u = 0$; these properties follow from the properties of a norm. Thus $\langle \ , \ \rangle$ satisfies the positivity and definiteness properties required of an inner product.

To prove that $\langle \ , \ \rangle$ is additive in the first slot, let $u, v, w \in U$. Then

$$\begin{aligned}&4(\langle u+v, w\rangle - \langle u, w\rangle - \langle v, w\rangle) \\ &\quad = \|u+v+w\|^2 - \|u+v-w\|^2 - \|u+w\|^2 \\ &\qquad + \|u-w\|^2 - \|v+w\|^2 + \|v-w\|^2 \\ &\quad = \|u+v+w\|^2 + (\|u-w\|^2 + \|v-w\|^2) \\ &\qquad - \|u+v-w\|^2 - (\|u+w\|^2 + \|v+w\|^2),\end{aligned}$$

where the first equality comes from the definition of $\langle \ , \ \rangle$. In the last equality above, the parentheses indicate groupings to which we will apply

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the parallelogram equality, which asserts that the sum of the norms squared of two vectors $x, y \in U$ can be computed from the formula

$$\|x\|^2 + \|y\|^2 = \frac{\|x+y\|^2}{2} + \frac{\|x-y\|^2}{2}.$$

Applying the parallelogram equality to the two terms in parentheses above (take $x = u - w, y = v - w$ for the first sum in parentheses, then $x = u + w, y = v + w$ for the second term in parentheses) gives

$$\begin{aligned}
&4(\langle u+v, w\rangle - \langle u, w\rangle - \langle v, w\rangle) \\
&= \|u+v+w\|^2 + \frac{\|u+v-2w\|^2}{2} + \frac{\|u-v\|^2}{2} \\
&\quad - \|u+v-w\|^2 - \frac{\|u+v+2w\|^2}{2} - \frac{\|u-v\|^2}{2} \\
&= (\|u+v+w\|^2 + \|w\|^2) + \frac{\|u+v-2w\|^2}{2} \\
&\quad - (\|u+v-w\|^2 + \|w\|^2) - \frac{\|u+v+2w\|^2}{2}.
\end{aligned}$$

Applying the parallelogram equality to the two terms in parentheses above (take $x = u + v + w, y = w$ for the first sum in parentheses, then $x = u + v - w, y = w$ for the second term in parentheses) gives

$$\begin{aligned}
&4(\langle u+v, w\rangle - \langle u, w\rangle - \langle v, w\rangle) \\
&= \frac{\|u+v+2w\|^2}{2} + \frac{\|u+v\|^2}{2} + \frac{\|u+v-2w\|^2}{2} \\
&\quad - \frac{\|u+v\|^2}{2} - \frac{\|u+v-2w\|^2}{2} - \frac{\|u+v+2w\|^2}{2} \\
&= 0.
\end{aligned}$$

Thus

$$\langle u+v, w\rangle = \langle u, w\rangle + \langle v, w\rangle,$$

completing the proof that $\langle\ ,\ \rangle$ is additive in the first slot.

To prove that $\langle\ ,\ \rangle$ is homogeneous in the first slot, let $u, v \in U$. If $n$ is a positive integer, then

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$$\begin{aligned}\langle nu, v\rangle &= \langle \underbrace{u + \cdots + u}_{n \text{ times}}, v\rangle \\ &= \underbrace{\langle u, v\rangle + \cdots + \langle u, v\rangle}_{n \text{ times}} \\ &= n\langle u, v\rangle,\end{aligned}$$

where the second equality comes from additivity in the first slot, which we have already verified. Replacing $u$ with $u/n$ in the equality above gives $\langle u, v\rangle = n\langle u/n, v\rangle$, which implies that

$$\langle \frac{u}{n}, v\rangle = \frac{1}{n}\langle u, v\rangle.$$

Let $m$ be another positive integer, and replace $u$ with $mu$ in the equality above, getting

$$\begin{aligned}\langle \frac{m}{n}u, v\rangle &= \frac{1}{n}\langle mu, v\rangle \\ &= \frac{m}{n}\langle u, v\rangle,\end{aligned}$$

where the second equality holds because we have already shown that $\langle\ ,\ \rangle$ is homogeneous in the first slot with respect to positive integers. We have now shown that $\langle\ ,\ \rangle$ is homogeneous in the first slot with respect to positive rational numbers.

From the definition of $\langle\ ,\ \rangle$, we have

$$\begin{aligned}\langle -u, v\rangle &= \frac{\|-u+v\|^2 - \|-u-v\|^2}{4} \\ &= -\frac{\|u+v\|^2 - \|u-v\|^2}{4} \\ &= -\langle u, v\rangle.\end{aligned}$$

Combining this with the result from the previous paragraph, we can now conclude that $\langle\ ,\ \rangle$ is homogeneous in the first slot with respect to all rational numbers.

Now suppose that $\lambda \in \mathbf{R}$. There exists a sequence $r_1, r_2, \ldots$ of rational numbers such that $\lim_{n\to\infty} r_n = \lambda$. Thus

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$$\begin{aligned}\lambda\langle u,v\rangle &= \lim_{n\to\infty} r_n\langle u,v\rangle \\ &= \lim_{n\to\infty} \langle r_n u,v\rangle \\ &= \lim_{n\to\infty} \frac{\|r_n u+v\|^2-\|r_n u-v\|^2}{4}.\end{aligned}$$

In the next paragraph we will show that $\lim_{n\to\infty}\|r_n u+v\| = \|\lambda u+v\|$ and $\lim_{n\to\infty}\|r_n u-v\| = \|\lambda u-v\|$. Combining this with the last equation above we can conclude that

$$\begin{aligned}\lambda\langle u,v\rangle &= \frac{\|\lambda u+v\|^2-\|\lambda u-v\|^2}{4} \\ &= \langle \lambda u,v\rangle,\end{aligned}$$

which will complete the proof that $\langle\ ,\ \rangle$ is homogeneous in the first slot.

If $x, y \in U$, then

$$\begin{aligned}\|x\| &= \|y+(x-y)\| \\ &\le \|y\|+\|x-y\|\end{aligned}$$

and thus

$$\|x\|-\|y\| \le \|x-y\|.$$

Interchanging the roles of $x$ and $y$, we get

$$\|y\|-\|x\| \le \|x-y\|.$$

Because $\big|\|x\|-\|y\|\big|$ equals $\|x\|-\|y\|$ or $\|y\|-\|x\|$, we can now conclude that

$$\big|\|x\|-\|y\|\big| \le \|x-y\|.$$

With $x = r_n u+v$ and $y = \lambda u+v$, this inequality gives

$$\begin{aligned}\big|\|r_n u+v\|-\|\lambda u+v\|\big| &\le \|r_n u-\lambda u\| \\ &= |r_n-\lambda|\|u\|.\end{aligned}$$

Because $\lim_{n\to\infty} r_n = \lambda$, this shows that

$$\lim_{n\to\infty}\|r_n u+v\| = \|\lambda u+v\|.$$

Replacing $v$ with $-v$, we have

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$$\lim_{n\to\infty} \|r_n u - v\| = \|\lambda u - v\|.$$

The last two equations are the promised ingredients that were needed for the proof that $\langle\ ,\ \rangle$ is homogeneous in the first slot.

Finally, we must show that $\langle u, v\rangle = \langle v, u\rangle$ (recall that we are considering the case where $\mathbf{F} = \mathbf{R}$). This last step is easy:

$$\begin{aligned}\langle u, v\rangle &= \frac{\|u+v\|^2 - \|u-v\|^2}{4}\\ &= \frac{\|v+u\|^2 - \|v-u\|^2}{4}\\ &= \langle v, u\rangle.\end{aligned}$$

This completes the proof that $\langle\ ,\ \rangle$ is an inner product when $\mathbf{F} = \mathbf{R}$. Whew!

Now consider the case where $\mathbf{F} = \mathbf{C}$. For $u, v \in U$, define $\langle u, v\rangle$ by

$$\langle u, v\rangle = \frac{\|u+v\|^2 - \|u-v\|^2 + \|u+iv\|^2 i - \|u-iv\|^2 i}{4}.$$

This definition is motivated by Exercise 7 of this chapter, which gives a formula for the inner product in terms of norms.

For $u \in U$ we have

$$\begin{aligned}\langle u, u\rangle &= \frac{\|u+u\|^2 - \|u-u\|^2 + \|u+iu\|^2 i - \|u-iu\|^2 i}{4}\\ &= \frac{\|2u\|^2 + |1+i|^2\|u\|^2 i - |1-i|^2\|u\|^2 i}{4}\\ &= \frac{4\|u\|^2 + 2\|u\|^2 i - 2\|u\|^2 i}{4}\\ &= \|u\|^2.\end{aligned}$$

Thus $\|u\| = \langle u, u\rangle^{1/2}$, as desired. However, we still must show that $\langle\ ,\ \rangle$ satisfies the properties required of an inner product.

Because $\langle u, u\rangle = \|u\|^2$ (as shown above), we have $\langle u, u\rangle \geq 0$ for all $u \in U$, with equality if and only if $u = 0$; these properties follow from the properties of a norm. Thus $\langle\ ,\ \rangle$ satisfies the positivity and definiteness properties required of an inner product.

For convenience, let's define $\langle\ ,\ \rangle_{\mathbf{R}}$ by

$$\langle u, v\rangle_{\mathbf{R}} = \frac{\|u+v\|^2 - \|u-v\|^2}{4}.$$

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Here the subscript $\mathbf{R}$ reminds us that $\langle\ ,\ \rangle_{\mathbf{R}}$ was the inner product we defined when considering the case $\mathbf{F} = \mathbf{R}$. Now we are assuming that $\mathbf{F} = \mathbf{C}$, but $\langle\ ,\ \rangle_{\mathbf{R}}$ is still well defined. Note that

$$\langle u, v\rangle = \langle u, v\rangle_{\mathbf{R}} + \langle u, iv\rangle_{\mathbf{R}} i.$$

We have already proved that $\langle\ ,\ \rangle_{\mathbf{R}}$ is additive in the first slot, and now we use that information. Let $u, v, w \in U$. Then

$$\begin{aligned}\langle u+v, w\rangle &= \langle u+v, w\rangle_{\mathbf{R}} + \langle u+v, iw\rangle_{\mathbf{R}} i \\ &= \langle u, w\rangle_{\mathbf{R}} + \langle v, w\rangle_{\mathbf{R}} + \langle u, iw\rangle_{\mathbf{R}} i + \langle v, iw\rangle_{\mathbf{R}} i \\ &= (\langle u, w\rangle_{\mathbf{R}} + \langle u, iw\rangle_{\mathbf{R}} i) + (\langle v, w\rangle_{\mathbf{R}} + \langle v, iw\rangle_{\mathbf{R}} i) \\ &= \langle u, w\rangle + \langle v, w\rangle.\end{aligned}$$

Thus $\langle\ ,\ \rangle$ is additive in the first slot.

To prove that $\langle\ ,\ \rangle$ is homogeneous in the first slot, let $u, v \in U$. If $\lambda \in \mathbf{R}$, then

$$\begin{aligned}\langle \lambda u, v\rangle &= \langle \lambda u, v\rangle_{\mathbf{R}} + \langle \lambda u, iv\rangle_{\mathbf{R}} i \\ &= \lambda\langle u, v\rangle_{\mathbf{R}} + \lambda\langle u, iv\rangle_{\mathbf{R}} i \\ &= \lambda\langle u, v\rangle,\end{aligned}$$

where we have used the homogeneity of $\langle\ ,\ \rangle_{\mathbf{R}}$ in the first slot. The last equation above shows that $\langle\ ,\ \rangle$ is homogeneous in the first slot with respect to all real numbers. We must still extend this result to complex numbers.

Note that

$$\begin{aligned}\langle iu, v\rangle &= \frac{\|iu+v\|^2 - \|iu-v\|^2 + \|iu+iv\|^2 i - \|iu-iv\|^2 i}{4} \\ &= \frac{\|i(u+v)\|^2 i - \|i(u-v)\|^2 i - \|i(u+iv)\|^2 + \|i(u-iv)\|^2}{4} \\ &= \frac{\|u+v\|^2 i - \|u-v\|^2 i - \|u+iv\|^2 + \|u-iv\|^2}{4} \\ &= i\langle u, v\rangle.\end{aligned}$$

Combining this result with additivity and homogeneity with respect to real numbers, we get that

$$\langle (a+bi)u, v\rangle = (a+bi)\langle u, v\rangle$$

for all $a, b \in \mathbf{R}$. In other words, $\langle\ ,\ \rangle$ is homogeneous in the first slot with respect to all complex numbers.

Finally, we must show that $\langle u, v\rangle = \overline{\langle v, u\rangle}$. This last step is easy:

$$\begin{aligned}
\langle u, v\rangle &= \frac{\|u+v\|^2 - \|u-v\|^2 + \|u+iv\|^2 i - \|u-iv\|^2 i}{4} \\
&= \frac{\|v+u\|^2 - \|v-u\|^2 + \|i(-iu+v)\|^2 i - \|(-i)(iu+v)\|^2 i}{4} \\
&= \overline{\frac{\|v+u\|^2 - \|v-u\|^2 + \|v+iu\|^2 i - \|v-iu\|^2 i}{4}} \\
&= \overline{\langle v, u\rangle}.
\end{aligned}$$

This completes the proof that $\langle\ ,\ \rangle$ is an inner product when $\mathbf{F} = \mathbf{C}$.

9. Suppose $n$ is a positive integer. Prove that

$$\Big(\frac{1}{\sqrt{2\pi}}, \frac{\sin x}{\sqrt{\pi}}, \frac{\sin 2x}{\sqrt{\pi}}, \ldots, \frac{\sin nx}{\sqrt{\pi}}, \frac{\cos x}{\sqrt{\pi}}, \frac{\cos 2x}{\sqrt{\pi}}, \ldots, \frac{\cos nx}{\sqrt{\pi}}\Big)$$

is an orthonormal list of vectors in $C[-\pi, \pi]$, the vector space of continuous real-valued functions on $[-\pi, \pi]$ with inner product

$$\langle f, g\rangle = \int_{-\pi}^{\pi} f(x)g(x)\,dx.$$

COMMENT: This orthonormal list is often used for modeling periodic phenomena such as tides.

SOLUTION: First we need to show that each element of the list above has norm 1. This follows easily from the following formulas:

$$\int (\sin jt)^2\,dt = \frac{2jt - \sin 2jt}{4j}$$

$$\int (\cos jt)^2\,dt = \frac{2jt + \sin 2jt}{4j}.$$

Next we need to show that any two distinct elements of the list above are orthogonal. This follows easily from the following formulas, valid when $j \neq k$:

$$\int (\sin jt)(\sin kt)\,dt =$$
$$\frac{j\sin(j-k)t + k\sin(j-k)t - j\sin(j+k)t + k\sin(j+k)t}{2(j-k)(j+k)}$$

$$\int (\sin jt)(\cos kt)\,dt =$$
$$\frac{j\cos(j-k)t + k\cos(j-k)t + j\cos(j+k)t - k\cos(j+k)t}{2(k-j)(j+k)}$$

$$\int (\cos jt)(\cos kt)\,dt =$$
$$\frac{j\sin(j-k)t + k\sin(j-k)t + j\sin(j+k)t - k\sin(j+k)t}{2(j-k)(j+k)}$$

$$\int (\sin jt)(\cos jt)\,dt = -\frac{(\cos jt)^2}{2j}.$$

10. On $\mathcal{P}_2(\mathbf{R})$, consider the inner product given by

$$\langle p, q\rangle = \int_0^1 p(x)q(x)\,dx.$$

Apply the Gram-Schmidt procedure to the basis $(1, x, x^2)$ to produce an orthonormal basis of $\mathcal{P}_2(\mathbf{R})$.

SOLUTION: Applying the Gram-Schmidt procedure to $(1, x, x^2)$ produces (using elementary calculus and some arithmetic) the following orthonormal basis of $\mathcal{P}_2(\mathbf{R})$:

$$\big(1, \sqrt{3}(-1+2x), \sqrt{5}(1-6x+6x^2)\big).$$

11. What happens if the Gram-Schmidt procedure is applied to a list of vectors that is not linearly independent?

SOLUTION: Suppose $(v_1, \ldots, v_m)$ is a linearly dependent list of vectors in $V$.

If $v_1 = 0$, then at the first step of applying the Gram-Schmidt procedure to $(v_1, \ldots, v_m)$ we will be dividing by 0 when trying to set $e_1 = v_1/\|v_1\|$.

If $v_1 \neq 0$, then by the linear dependence lemma (2.4), some $v_j$ is in $\operatorname{span}(v_1, \ldots, v_{j-1})$; here we choose $j$ to be the smallest positive integer with this property. If we apply the Gram-Schmidt procedure to produce $(e_1, \ldots, e_{j-1})$ at the end of step $j$, then

$$\operatorname{span}(v_1, \ldots, v_{j-1}) = \operatorname{span}(e_1, \ldots, e_{j-1}).$$

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Thus $v_j \in \text{span}(e_1, \ldots, e_{j-1})$. By 6.17, this implies that

$$v_j = \langle v_j, e_1 \rangle e_1 + \cdots + \langle v_j, e_{j-1} \rangle e_{j-1}.$$

Thus the Gram-Schmidt formula 6.23 for $e_j$ includes a division by 0, which is not allowed.

12. Suppose $V$ is a real inner-product space and $(v_1, \ldots, v_m)$ is a linearly independent list of vectors in $V$. Prove that there exist exactly $2^m$ orthonormal lists $(e_1, \ldots, e_m)$ of vectors in $V$ such that

$$\text{span}(v_1, \ldots, v_j) = \text{span}(e_1, \ldots, e_j)$$

for all $j \in \{1, \ldots, m\}$.

SOLUTION: For $j = 1$, the condition above states that

$$\text{span}(v_1) = \text{span}(e_1).$$

Because there are only two vectors in $\text{span}(v_1)$ with norm 1 (these two vectors are $v_1/\|v_1\|$ and $-v_1/\|v_1\|$), we have only these two choices for $e_1$.

Now suppose that $j > 1$ and that an orthonormal list $(e_1, \ldots, e_{j-1})$ has been chosen such that

$$\text{span}(v_1, \ldots, v_{j-1}) = \text{span}(e_1, \ldots, e_{j-1}).$$

The Gram-Schmidt procedure produces $e_j \in V$ such that $(e_1, \ldots, e_j)$ is an orthonormal list and

$$\text{span}(v_1, \ldots, v_j) = \text{span}(e_1, \ldots, e_j).$$

Suppose $e_j' \in V$ is another vector with these properties, meaning that $(e_1, \ldots, e_{j-1}, e_j')$ is an orthonormal list and

$$\text{span}(v_1, \ldots, v_j) = \text{span}(e_1, \ldots, e_{j-1}, e_j').$$

The last two equations show that $\text{span}(e_1, \ldots, e_{j-1}, e_j') = \text{span}(e_1, \ldots, e_j)$. In particular, $e_j' \in \text{span}(e_1, \ldots, e_j)$, which implies that

$$\begin{aligned} e_j' &= \langle e_j', e_1 \rangle e_1 + \cdots + \langle e_j', e_j \rangle e_j \\ &= \langle e_j', e_j \rangle e_j, \end{aligned}$$

where the first equality comes from 6.17 (with $\text{span}(e_1, \ldots, e_j)$ replacing $V$ and $j$ replacing $n$) and the second equality holds because $(e_1, \ldots, e_{j-1}, e_j')$

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is an orthonormal list. Taking norms of both sides of the last equation, and recalling that $e_j$ and $e_j'$ both have norm 1, we see that $|\langle e_j', e_j\rangle| = 1$. Thus $\langle e_j', e_j\rangle = 1$ or $\langle e_j', e_j\rangle = -1$. Hence the last equation above implies that $e_j' = e_j$ or $e_j' = -e_j$.

We have shown that there are exactly two possible choices for each $e_j$. As $j$ ranges from 1 to $m$, this gives us exactly $2^m$ possible choices for $(e_1, \dots, e_m)$.

13. Suppose $(e_1, \dots, e_m)$ is an orthonormal list of vectors in $V$. Let $v \in V$. Prove that

$$\|v\|^2 = |\langle v, e_1\rangle|^2 + \cdots + |\langle v, e_m\rangle|^2$$

if and only if $v \in \operatorname{span}(e_1, \dots, e_m)$.

SOLUTION: Extend $(e_1, \dots, e_m)$ to an orthonormal basis $(e_1, \dots, e_n)$ of $V$. Then

$$v = \langle v, e_1\rangle e_1 + \cdots + \langle v, e_n\rangle e_n$$

and

$$\|v\|^2 = |\langle v, e_1\rangle|^2 + \cdots + |\langle v, e_n\rangle|^2;$$

see 6.17. From the last equation, we see that

$$\|v\|^2 = |\langle v, e_1\rangle|^2 + \cdots + |\langle v, e_m\rangle|^2$$

if and only if $\langle v, e_{m+1}\rangle = \cdots = \langle v, e_n\rangle = 0$. From the first equation above, this happens if and only if

$$v = \langle v, e_1\rangle e_1 + \cdots + \langle v, e_m\rangle e_m,$$

which happens if and only if $v \in \operatorname{span}(e_1, \dots, e_m)$.

14. Find an orthonormal basis of $\mathcal{P}_2(\mathbf{R})$ (with inner product as in Exercise 10) such that the differentiation operator (the operator that takes $p$ to $p'$) on $\mathcal{P}_2(\mathbf{R})$ has an upper-triangular matrix with respect to this basis.

SOLUTION: Because $1' = 0$, $x' = 1$, and $(x^2)' = 2x$, the differentiation operator on $\mathcal{P}_2(\mathbf{R})$ has an upper-triangular matrix with respect to the basis $(1, x, x^2)$. However, $(1, x, x^2)$ is not an orthonormal basis. But, as can be seen from the proof of 6.27, if the Gram-Schmidt procedure is applied to this

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basis, we will get an orthonormal basis with respect to which the differentiation operator has an upper-triangular matrix. As we saw in Exercise 10 of this chapter, the Gram-Schmidt procedure applied to $(1, x, x^2)$ gives

$$(1, \sqrt{3}(-1+2x), \sqrt{5}(1-6x+6x^2)).$$

which is our desired orthonormal basis of $\mathcal{P}_2(\mathbf{R})$.

15. Suppose $U$ is a subspace of $V$. Prove that

$$\dim U^\perp = \dim V - \dim U.$$

SOLUTION: From 6.29, we know that

$$V = U \oplus U^\perp.$$

Thus by Exercise 17 in Chapter 2, we have

$$\dim V = \dim U + \dim U^\perp,$$

which implies that $\dim U^\perp = \dim V - \dim U$.

16. Suppose $U$ is a subspace of $V$. Prove that $U^\perp = \{0\}$ if and only if $U = V$.

SOLUTION: From 6.29, we know that

$$V = U \oplus U^\perp.$$

This clearly implies that $U^\perp = \{0\}$ if and only if $U = V$.

17. Prove that if $P \in \mathcal{L}(V)$ is such that $P^2 = P$ and every vector in $\operatorname{null} P$ is orthogonal to every vector in $\operatorname{range} P$, then $P$ is an orthogonal projection.

SOLUTION: Suppose $P \in \mathcal{L}(V)$ is such that $P^2 = P$ and every vector in $\operatorname{null} P$ is orthogonal to every vector in $\operatorname{range} P$. Let $U = \operatorname{range} P$. We will show that $P$ equals the orthogonal projection $P_U$. To do this, suppose $v \in V$. Then

$$v = Pv + (v - Pv).$$

Clearly $Pv \in \operatorname{range} P = U$. Also, $P(v - Pv) = Pv - P^2v = 0$, which means that $v - Pv \in \operatorname{null} P$. Thus $v - Pv$ is orthogonal to every vector in $U$. In other words, $v - Pv \in U^\perp$. Thus the equation above writes $v$ as the sum of a vector in $U$ and a vector in $U^\perp$. In this decomposition, the vector in $U$ equals, by definition, $P_U v$. Hence $Pv = P_U v$, as desired.

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18. Prove that if $P \in \mathcal{L}(V)$ is such that $P^2 = P$ and

$$\|Pv\| \leq \|v\|$$

for every $v \in V$, then $P$ is an orthogonal projection.

SOLUTION: Suppose $u \in \operatorname{range} P$ and $w \in \operatorname{null} P$. If we can show that $\langle u, w \rangle = 0$, then by the previous exercise we can conclude that $P$ is an orthogonal projection.

Because $u \in \operatorname{range} P$, there exists $u' \in V$ such that

$$u = Pu'.$$

Applying $P$ to both sides of this equation, we have

$$\begin{aligned} Pu &= P^2u' \\ &= Pu' \\ &= u. \end{aligned}$$

Because $w \in \operatorname{null} P$, this implies that

$$P(u + aw) = u$$

for every $a \in \mathbf{F}$. Thus

$$\begin{aligned} \|u\|^2 &= \|P(u + aw)\|^2 \\ &\leq \|u + aw\|^2 \end{aligned}$$

for every $a \in \mathbf{F}$, where the second line follows from our hypothesis that $\|Pv\| \leq \|v\|$ for every $v \in V$. The inequality above implies (see Exercise 2 of this chapter) that $\langle u, w \rangle = 0$, as desired.

19. Suppose $T \in \mathcal{L}(V)$ and $U$ is a subspace of $V$. Prove that $U$ is invariant under $T$ if and only if $P_U T P_U = T P_U$.

SOLUTION: First suppose that $U$ is invariant under $T$. Let $v \in V$. Then $P_U v \in U$ and hence $T(P_U v) \in U$ (because $U$ is invariant under $T$). Thus $P_U\big(T(P_U v)\big) = T(P_U v)$. Because $v$ was an arbitrary vector in $V$, this implies that $P_U T P_U = T P_U$, as desired.

To prove the implication in the other direction, now suppose that

$$P_U T P_U = T P_U.$$

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Suppose $u \in U$. Then $P_U u = u$, so applying both sides of the equation above to $u$ gives $P_U(Tu) = Tu$, which implies that $Tu \in U$. Because $u$ was an arbitrary vector in $U$, this implies that $T$ is invariant under $U$, as desired.

20. Suppose $T \in \mathcal{L}(V)$ and $U$ is a subspace of $V$. Prove that $U$ and $U^\perp$ are both invariant under $T$ if and only if $P_U T = T P_U$.

SOLUTION: First suppose that $U$ and $U^\perp$ are both invariant under $T$. By the previous exercise, this implies that

$$P_U T P_U = T P_U$$

and

$$P_{U^\perp} T P_{U^\perp} = T P_{U^\perp}.$$

But $P_{U^\perp} = I - P_U$, so the last equation becomes

$$(I - P_U)T(I - P_U) = T(I - P_U).$$

Expanding both sides of the equation above and rearranging terms, we get

$$P_U T P_U = P_U T.$$

Combining this with the first equation above, we get $P_U T = T P_U$, as desired.

To prove the implication in the other direction, suppose now that

$$P_U T = T P_U.$$

Then

$$\begin{aligned} P_U T P_U &= (P_U T) P_U \\ &= (T P_U) P_U \\ &= T {P_U}^2 \\ &= T P_U, \end{aligned}$$

which implies (by the previous exercise) that $U$ is invariant under $T$, as desired. Also,

$$\begin{aligned} P_{U^\perp} T P_{U^\perp} &= \big((I - P_U)T\big) P_{U^\perp} \\ &= (T - P_U T) P_{U^\perp} \\ &= (T - T P_U) P_{U^\perp} \\ &= T(1 - P_U) P_{U^\perp} \\ &= T {P_{U^\perp}}^2 \\ &= T P_{U^\perp}, \end{aligned}$$

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which implies (by the previous exercise) that $U^\perp$ is invariant under $T$, as desired.

21. In $\mathbf{R}^4$, let

$$U = \operatorname{span}\big((1,1,0,0),(1,1,1,2)\big).$$

Find $u \in U$ such that $\|u - (1,2,3,4)\|$ is as small as possible.

SOLUTION: First we find an orthonormal basis of $U$ by applying the Gram-Schmidt procedure to $\big((1,1,0,0),(1,1,1,2)\big)$, getting

$$e_1 = \Big(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, 0, 0\Big)$$

$$e_2 = \Big(0, 0, \frac{1}{\sqrt{5}}, \frac{2}{\sqrt{5}}\Big).$$

Thus with $e_1, e_2$ as above, $(e_1, e_2)$ is an orthonormal basis of $U$. By 6.36 and 6.35, the closest point $u \in U$ to $(1,2,3,4)$ is

$$\langle (1,2,3,4), e_1\rangle e_1 + \langle (1,2,3,4), e_2\rangle e_2,$$

which equals

$$\Big(\frac{3}{2}, \frac{3}{2}, \frac{11}{5}, \frac{22}{5}\Big).$$

22. Find $p \in \mathcal{P}_3(\mathbf{R})$ such that $p(0) = 0$, $p'(0) = 0$, and

$$\int_0^1 |2 + 3x - p(x)|^2 \, dx$$

is as small as possible.

SOLUTION: Define an inner product on $\mathcal{P}_3(\mathbf{R})$ by

$$\langle f, g\rangle = \int_0^1 f(x)g(x)\, dx.$$

Let $q(x) = 2 + 3x$, and let

$$U = \{p \in \mathcal{P}_3(\mathbf{R}) : p(0) = 0, p'(0) = 0\}.$$

With this notation, our problem is to find the closest point $p \in U$ to $q$. To do this, first we find an orthonormal basis of $U$.

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A polynomial $p$ satisfying $p(0) = 0$, $p'(0) = 0$ has constant term 0 and first degree term also equal to 0. Thus a basis of $U$ is

$$(x^2, x^3).$$

Apply the Gram-Schmidt procedure to this basis, getting

$$e_1 = \sqrt{5}x^2$$

$$e_2 = \sqrt{7}(-5x^2 + 6x^3).$$

Thus with $e_1, e_2$ as above, $(e_1, e_2)$ is an orthonormal basis of $U$. By 6.36 and 6.35, the closest point $p \in U$ to $q$ is given by the formula

$$p = \langle q, e_1 \rangle e_1 + \langle q, e_2 \rangle e_2.$$

A short computation now shows that

$$p(x) = 24x^2 - \frac{203}{10}x^3.$$

23. Find $p \in \mathcal{P}_5(\mathbf{R})$ that makes

$$\int_{-\pi}^{\pi} |\sin x - p(x)|^2 \, dx$$

as small as possible. (The polynomial 6.40 is an excellent approximation to the answer to this exercise, but here you are asked to find the exact solution, which involves powers of $\pi$. A computer that can perform symbolic integration will be useful.)

SOLUTION: Let $C[-\pi, \pi]$ denote the real vector space of continuous real-valued functions on $[-\pi, \pi]$ with inner product

$$\langle f, g \rangle = \int_{-\pi}^{\pi} f(x)g(x) \, dx.$$

Let $v \in C[-\pi, \pi]$ be the function defined by $v(x) = \sin x$. Let $U$ denote the subspace of $C[-\pi, \pi]$ consisting of the polynomials with real coefficients and degree at most 5. We need to find $p \in U$ such that $\|v - p\|$ is as small as possible.

First find an orthonormal basis of $U$ by applying the Gram-Schmidt procedure (using the inner product above) to the basis $(1, x, x^2, x^3, x^4, x^5)$ of $U$, producing the orthonormal basis $(e_1, e_2, e_3, e_4, e_5, e_6)$, where

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$$e_1 = \frac{1}{\sqrt{2\pi}},$$

$$e_2 = \frac{\sqrt{\frac{3}{2}}x}{\pi^{3/2}},$$

$$e_3 = -\frac{\sqrt{\frac{5}{2}}(\pi^2 - 3x^2)}{2\pi^{5/2}},$$

$$e_4 = -\frac{\sqrt{\frac{7}{2}}(3\pi^2 x - 5x^3)}{2\pi^{7/2}},$$

$$e_5 = \frac{3(3\pi^4 - 30\pi^2 x^2 + 35x^4)}{8\sqrt{2}\pi^{9/2}},$$

$$e_6 = -\frac{\sqrt{\frac{11}{2}}(15\pi^4 x - 70\pi^2 x^3 + 63x^5)}{8\pi^{11/2}}.$$

Now compute $P_U v$ using 6.35 (with $m = 6$), getting

$$P_U v = \frac{105(1485 - 153\pi^2 + \pi^4)}{8\pi^6}x - \frac{315(1155 - 125\pi^2 + \pi^4)}{4\pi^8}x^3 + \frac{693(945 - 105\pi^2 + \pi^4)}{8\pi^{10}}x^5.$$

Finally, 6.36 and the discussion following 6.42 show that the function above is the one we seek.

24. Find a polynomial $q \in \mathcal{P}_2(\mathbf{R})$ such that

$$p(\frac{1}{2}) = \int_0^1 p(x)q(x)\,dx$$

for every $p \in \mathcal{P}_2(\mathbf{R})$.

SOLUTION: We will need an orthonormal basis of $\mathcal{P}_2(\mathbf{R})$, where the inner product of two polynomials in $\mathcal{P}_2(\mathbf{R})$ is defined to be the integral from 0 to 1 of the product of the two polynomials. An orthonormal basis of $\mathcal{P}_2(\mathbf{R})$ was already computed in Exercise 10 of this chapter. Specifically, let

$$e_1(x) = 1$$
$$e_2(x) = \sqrt{3}(-1 + 2x)$$
$$e_3(x) = \sqrt{5}(1 - 6x + 6x^2).$$

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Then $(e_1, e_2, e_3)$ is an orthonormal basis of $\mathcal{P}_2(\mathbf{R})$.

Define a linear functional $\varphi$ on $\mathcal{P}_2(\mathbf{R})$ by

$$\varphi(p) = p(\tfrac{1}{2}).$$

We seek $q \in \mathcal{P}_2(\mathbf{R})$ such that $\varphi(p) = \langle p, q\rangle$ for every $p \in \mathcal{P}_2(\mathbf{R})$. By the formula given in the proof of 6.45, we have

$$q = \varphi(e_1)e_1 + \varphi(e_2)e_2 + \varphi(e_3)e_3.$$

Evaluate the right side of the equation above to get

$$q(x) = -\frac{3}{2} + 15x - 15x^2.$$

25. Find a polynomial $q \in \mathcal{P}_2(\mathbf{R})$ such that

$$\int_0^1 p(x)(\cos \pi x)\, dx = \int_0^1 p(x)q(x)\, dx$$

for every $p \in \mathcal{P}_2(\mathbf{R})$.

SOLUTION: Define a linear functional $\varphi$ on $\mathcal{P}_2(\mathbf{R})$ by

$$\varphi(p) = \int_0^1 p(x)(\cos \pi x)\, dx.$$

We seek $q \in \mathcal{P}_2(\mathbf{R})$ such that $\varphi(p) = \langle p, q\rangle$ for every $p \in \mathcal{P}_2(\mathbf{R})$, where the inner product on $\mathcal{P}_2(\mathbf{R})$ is defined as in the previous exercise. Letting $e_1, e_2, e_3$ be as in the previous exercise, but using our new definition of $\varphi$, we again have

$$q = \varphi(e_1)e_1 + \varphi(e_2)e_2 + \varphi(e_3)e_3.$$

Evaluate the right side of the equation above to get

$$q(x) = \frac{12 - 24x}{\pi^2}.$$

26. Fix a vector $v \in V$ and define $T \in \mathcal{L}(V, \mathbf{F})$ by $Tu = \langle u, v\rangle$. For $a \in \mathbf{F}$, find a formula for $T^*a$.

SOLUTION: Because $T \in \mathcal{L}(V, \mathbf{F})$, we know that $T^* \in \mathcal{L}(\mathbf{F}, V)$. Fix $a \in \mathbf{F}$. Then $T^*a$ is the unique vector in $V$ such that

(a) $$\langle Tu, a\rangle = \langle u, T^*a\rangle$$

for all $u \in U$. The inner product on the right is the inner product in $V$, but the inner product on the left is the usual inner product on $\mathbf{F}$: the product of the entry in the first slot with the complex conjugate of the entry in the second slot. Thus

$$\begin{aligned}\langle Tu, a\rangle &= (Tu)\bar{a} \\ &= \langle u, v\rangle\bar{a} \\ &= \langle u, av\rangle. \qquad \text{(b)}\end{aligned}$$

Comparing (a) with (b) gives

$$\langle u, T^*a\rangle = \langle u, av\rangle$$

for all $u \in U$. Thus $T^*a = av$.

27. Suppose $n$ is a positive integer. Define $T \in \mathcal{L}(\mathbf{F}^n)$ by

$$T(z_1, \ldots, z_n) = (0, z_1, \ldots, z_{n-1}).$$

Find a formula for $T^*(z_1, \ldots, z_n)$.

SOLUTION: Fix $(z_1, \ldots, z_n) \in \mathbf{F}^n$. Then for every $(w_1, \ldots, w_n) \in \mathbf{F}^n$, we have

$$\begin{aligned}\langle (w_1, \ldots, w_n), T^*(z_1, \ldots, z_n)\rangle &= \langle T(w_1, \ldots, w_n), (z_1, \ldots, z_n)\rangle \\ &= \langle (0, w_1, \ldots, w_{n-1}), (z_1, \ldots, z_n)\rangle \\ &= w_1\overline{z_2} + \cdots + w_{n-1}\overline{z_n} \\ &= \langle (w_1, \ldots, w_n), (z_2, \ldots, z_n, 0)\rangle.\end{aligned}$$

Thus

$$T^*(z_1, \ldots, z_n) = (z_2, \ldots, z_n, 0).$$

28. Suppose $T \in \mathcal{L}(V)$ and $\lambda \in \mathbf{F}$. Prove that $\lambda$ is an eigenvalue of $T$ if and only if $\bar{\lambda}$ is an eigenvalue of $T^*$.

SOLUTION: We have

$$\begin{aligned}
\lambda \text{ is an not eigenvalue of } T &\iff T - \lambda I \text{ is invertible} \\
&\iff S(T - \lambda I) = (T - \lambda I)S = I \\
&\qquad \text{for some } S \in \mathcal{L}(V) \\
&\iff (T - \lambda I)^* S^* = S^*(T - \lambda I)^* = I \\
&\qquad \text{for some } S \in \mathcal{L}(V) \\
&\iff (T - \lambda I)^* \text{ is invertible} \\
&\iff T^* - \bar{\lambda} I \text{ is invertible} \\
&\iff \bar{\lambda} \text{ is not an eigenvalue of } T^*.
\end{aligned}$$

Thus $\lambda$ is an eigenvalue of $T$ if and only if $\bar{\lambda}$ is an eigenvalue of $T^*$.

29. Suppose $T \in \mathcal{L}(V)$ and $U$ is a subspace of $V$. Prove that $U$ is invariant under $T$ if and only if $U^\perp$ is invariant under $T^*$.

SOLUTION: First suppose that $U$ is invariant under $T$. To prove that $U^\perp$ is invariant under $T^*$, let $v \in U^\perp$. We need to show that $T^*v \in U^\perp$. But

$$\begin{aligned}
\langle u, T^*v \rangle &= \langle Tu, v \rangle \\
&= 0
\end{aligned}$$

for every $u \in U$ (because if $u \in U$, then $Tu \in U$ and hence $Tu$ is orthogonal to $v$, an element of $U^\perp$). Thus $T^*v \in U^\perp$, and hence $U^\perp$ is invariant under $T^*$, as desired.

To prove the other direction, now suppose that $U^\perp$ is invariant under $T^*$. Then by the first direction, we know that $(U^\perp)^\perp$ is invariant under $(T^*)^*$. But $(U^\perp)^\perp = U$ (by 6.33) and $(T^*)^* = T$, so $U$ is invariant under $T$, completing the proof.

30. Suppose $T \in \mathcal{L}(V, W)$. Prove that

(a) $T$ is injective if and only if $T^*$ is surjective;

(b) $T$ is surjective if and only if $T^*$ is injective.

SOLUTION: First we prove (a):

$$\begin{aligned}
T \text{ is injective} &\iff \operatorname{null} T = \{0\} \\
&\iff (\operatorname{range} T^*)^\perp = \{0\} \\
&\iff \operatorname{range} T^* = W \\
&\iff T^* \text{ is surjective},
\end{aligned}$$

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where the second line comes from 6.46(c).

Now that (a) has been proved, (b) follows immediately by replacing $T$ with $T^*$ in (a).

31. Prove that

$$\dim \operatorname{null} T^* = \dim \operatorname{null} T + \dim W - \dim V$$

and

$$\dim \operatorname{range} T^* = \dim \operatorname{range} T$$

for every $T \in \mathcal{L}(V, W)$.

SOLUTION: Let $T \in \mathcal{L}(V, W)$. Then

$$\begin{aligned} \dim \operatorname{null} T^* &= \dim(\operatorname{range} T)^\perp \\ &= \dim W - \dim \operatorname{range} T \\ &= \dim \operatorname{null} T + \dim W - \dim V, \end{aligned}$$

where the first equality comes from 6.46(a), the second equality comes from Exercise 15 of this chapter, and the third equality comes from 3.4. This proves the first equality that we seek.

To prove the second equality, note that

$$\begin{aligned} \dim \operatorname{range} T^* &= \dim W - \dim \operatorname{null} T^* \\ &= \dim V - \dim \operatorname{null} T \\ &= \dim \operatorname{range} T, \end{aligned}$$

where the first and third equalities come from 3.4 and the second equality comes from the first part of this exercise. This proves the second equality that we seek.

32. Suppose $A$ is an $m$-by-$n$ matrix of real numbers. Prove that the dimension of the span of the columns of $A$ (in $\mathbf{R}^m$) equals the dimension of the span of the rows of $A$ (in $\mathbf{R}^n$).

SOLUTION: Let $T \in \mathcal{L}(\mathbf{R}^n, \mathbf{R}^m)$ be such that the matrix of $T$ (with respect to the standard bases) equals $A$. Then $\operatorname{range} T$ equals the span of the columns of $A$. Thus

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dimension of the span of the columns of $A$

$$\begin{aligned}
&= \dim \text{range}\, T \\
&= \dim \text{range}\, T^* \\
&= \text{dimension of the span of the columns of } \mathcal{M}(T^*) \\
&= \text{dimension of the span of the columns of the transpose of } A \\
&= \text{dimension of the span of the rows of } A,
\end{aligned}$$

where the second equality comes from the previous exercise and the fourth equality comes from 6.47.

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CHAPTER 7

# Operators on Inner-Product Spaces

1. Make $\mathcal{P}_2(\mathbf{R})$ into an inner-product space by defining

$$\langle p, q \rangle = \int_0^1 p(x)q(x)\,dx.$$

Define $T \in \mathcal{L}(\mathcal{P}_2(\mathbf{R}))$ by $T(a_0 + a_1x + a_2x^2) = a_1x$.

(a) Show that $T$ is not self-adjoint.

(b) The matrix of $T$ with respect to the basis $(1, x, x^2)$ is

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

This matrix equals its conjugate transpose, even though $T$ is not self-adjoint. Explain why this is not a contradiction.

SOLUTION: (a): Note that

$$\begin{aligned} \langle T1, x \rangle &= \langle 0, x \rangle \\ &= 0 \end{aligned}$$

but

$$\begin{aligned} \langle 1, Tx \rangle &= \langle 1, x \rangle \\ &= \frac{1}{2}. \end{aligned}$$



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Thus $\langle T1, x\rangle \neq \langle 1, Tx\rangle$, which shows that $T$ is not self-adjoint.

(b): The result stating that the matrix of $T^*$ is the conjugate transpose of the matrix of $T$ has as a hypothesis that we are working with orthonormal bases (see 6.47). Because $(1, x, x^2)$ is not an orthonormal basis of $\mathcal{P}_2(\mathbf{R})$, we cannot compute the matrix of $T^*$ with respect to this basis by taking the conjugate transpose of the matrix of $T$.

2. Prove or give a counterexample: the product of any two self-adjoint operators on a finite-dimensional inner-product space is self-adjoint.

SOLUTION: Let $S, T \in \mathcal{L}(\mathbf{F}^2)$ be the operators whose matrices (with respect to the standard basis) are given by

$$\mathcal{M}(S) = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix} \quad \text{and} \quad \mathcal{M}(T) = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}.$$

Each of these matrices obviously equals its conjugate transpose, and hence $S, T$ are self-adjoint. Now

$$\mathcal{M}(ST) = \mathcal{M}(S)\mathcal{M}(T) = \begin{bmatrix} 0 & 1 \\ 2 & 0 \end{bmatrix}.$$

Because $\mathcal{M}(ST)$ does not equal its conjugate transpose, $ST$ is not self-adjoint. Thus we have an example of two self-adjoint operators whose product is not self-adjoint.

Of course there are also many other examples.

COMMENT: Suppose $S, T \in \mathcal{L}(V)$ are self-adjoint. Then $ST$ is self-adjoint if and only if $ST = TS$ (as is easy to see).

3. (a) Show that if $V$ is a real inner-product space, then the set of self-adjoint operators on $V$ is a subspace of $\mathcal{L}(V)$.

(b) Show that if $V$ is a complex inner-product space, then the set of self-adjoint operators on $V$ is not a subspace of $\mathcal{L}(V)$.

SOLUTION: (a): Suppose $V$ is a real inner-product space. Obviously the zero operator is self-adjoint. Furthermore, if $S, T \in \mathcal{L}(V)$ are self-adjoint, then

$$\begin{aligned}(S+T)^* &= S^* + T^* \\ &= S + T,\end{aligned}$$

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and thus $S+T$ is self-adjoint. Finally, if $T \in \mathcal{L}(V)$ is self-adjoint and $a \in \mathbf{R}$, then

$$\begin{aligned}(aT)^* &= aT^* \\ &= aT,\end{aligned}$$

and thus $aT$ is self-adjoint. We have shown that the set of self-adjoint operators on $V$ contains the zero operator and that it is closed under addition and scalar multiplication. Thus the set of self-adjoint operators on $V$ is a subspace of $\mathcal{L}(V)$.

(b): Suppose now that $V$ is a complex vector space. The identity operator $I$ is self-adjoint, but $(iI)^* = -iI$ so $iI$ is not self-adjoint. Thus the set of self-adjoint operators on $V$ is not closed under scalar multiplication and hence it is not a subspace of $\mathcal{L}(V)$.

4. Suppose $P \in \mathcal{L}(V)$ is such that $P^2 = P$. Prove that $P$ is an orthogonal projection if and only if $P$ is self-adjoint.

SOLUTION: First suppose that $P$ is an orthogonal projection. Thus there is a subspace $U$ of $V$ such that $P = P_U$. Suppose $v_1, v_2 \in V$. Write

$$v_1 = u_1 + w_1, \quad v_2 = u_2 + w_2,$$

where $u_1, u_2 \in U$ and $w_1, w_2 \in U^\perp$ (see 6.29). Now

$$\begin{aligned}\langle Pv_1, v_2\rangle &= \langle u_1, u_2 + w_2\rangle \\ &= \langle u_1, u_2\rangle + \langle u_1, w_2\rangle \\ &= \langle u_1, u_2\rangle \\ &= \langle u_1, u_2\rangle + \langle w_1, u_2\rangle \\ &= \langle u_1 + w_1, u_2\rangle \\ &= \langle v_1, Pv_2\rangle.\end{aligned}$$

Thus $P = P^*$, and hence $P$ is self-adjoint.

To prove the implication in the other direction, now suppose that $P$ is self-adjoint. Let $v \in V$. Because $P(v - Pv) = Pv - P^2v = 0$, we have

$$v - Pv \in \operatorname{null} P = (\operatorname{range} P^*)^\perp = (\operatorname{range} P)^\perp,$$

where the first equality comes from 6.46(c). Writing

$$v = Pv + (v - Pv),$$

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we have $Pv \in \operatorname{range} P$ and $(v - Pv) \in (\operatorname{range} P)^{\perp}$. Thus $Pv = P_{\operatorname{range} P} v$. Because this holds for all $v \in V$, we have $P = P_{\operatorname{range} P}$, which shows that $P$ is an orthogonal projection.

5. Show that if $\dim V \geq 2$, then the set of normal operators on $V$ is not a subspace of $\mathcal{L}(V)$.

SOLUTION: Suppose $\dim V \geq 2$. Let $(e_1, \ldots, e_n)$ be an orthonormal basis of $V$. Define $S, T \in \mathcal{L}(V)$ by

$$S(a_1 e_1 + \cdots + a_n e_n) = a_2 e_1 - a_1 e_2$$

and

$$T(a_1 e_1 + \cdots + a_n e_n) = a_2 e_1 + a_1 e_2.$$

A simple calculation verifies that

$$S^*(a_1 e_1 + \cdots + a_n e_n) = -a_2 e_1 + a_1 e_2.$$

From this formula, another simple calculation shows that $SS^* = S^*S$. Yet another simple calculation shows that $T$ is self-adjoint. Thus both $S$ and $T$ are normal. However, $S + T$ is given by the formula

$$(S + T)(a_1 e_1 + \cdots + a_n e_n) = 2a_2 e_1.$$

A simple calculation verifies that

$$(S + T)^*(a_1 e_1 + \cdots + a_n e_n) = 2a_1 e_2.$$

A final simple calculation shows that $(S + T)(S + T)^* \neq (S + T)^*(S + T)$. In other words, $S + T$ is not normal. Thus the set of normal operators on $V$ is not closed under addition and hence is not a subspace of $\mathcal{L}(V)$.

6. Prove that if $T \in \mathcal{L}(V)$ is normal, then

$$\operatorname{range} T = \operatorname{range} T^*.$$

SOLUTION: Suppose $T$ is normal. Then

$$\begin{aligned} \operatorname{range} T &= (\operatorname{null} T^*)^{\perp} \\ &= (\operatorname{null} T)^{\perp} \\ &= \operatorname{range} T^*, \end{aligned}$$

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where the first equality comes from 6.46(d), the second equality comes from 7.6 (see especially the marginal comment at 7.6), and the third equality comes from 6.46(b).

7. Prove that if $T \in \mathcal{L}(V)$ is normal, then

$$\operatorname{null} T^k = \operatorname{null} T \quad \text{and} \quad \operatorname{range} T^k = \operatorname{range} T$$

for every positive integer $k$.

SOLUTION: Suppose $T \in \mathcal{L}(V)$ is normal and that $k$ is a positive integer. Obviously we can assume that $k \geq 2$.

First we will prove that $\operatorname{null} T^k = \operatorname{null} T$. If $v \in \operatorname{null} T$, then

$$\begin{aligned} T^k v &= T^{k-1}(Tv) \\ &= T^{k-1}0 \\ &= 0, \end{aligned}$$

and so $v \in \operatorname{null} T^k$. Thus $\operatorname{null} T \subset \operatorname{null} T^k$.

To prove an inclusion in the other direction, suppose now that $v \in \operatorname{null} T^k$. Then

$$\begin{aligned} \langle T^*T^{k-1}v, T^*T^{k-1}v\rangle &= \langle TT^*T^{k-1}v, T^{k-1}v\rangle \\ &= \langle T^*T^k v, T^{k-1}v\rangle \\ &= \langle 0, T^{k-1}v\rangle \\ &= 0, \end{aligned}$$

where the second equality holds because $T^*T = TT^*$. The last equality above implies that $T^*T^{k-1}v = 0$. Thus

$$\begin{aligned} 0 &= \langle T^*T^{k-1}v, T^{k-2}v\rangle \\ &= \langle T^{k-1}v, T^{k-1}v\rangle. \end{aligned}$$

Hence $T^{k-1}v = 0$. In other words, $v \in \operatorname{null} T^{k-1}$. The same argument, with $k$ replaced with $k-1$, shows that $v \in \operatorname{null} T^{k-2}$. Repeat this process until reaching the conclusion that $v \in \operatorname{null} T$. This shows that $\operatorname{null} T^k \subset \operatorname{null} T$, completing that proof that $\operatorname{null} T^k = \operatorname{null} T$.

Now we will show that $\operatorname{range} T^k = \operatorname{range} T$. If $v \in \operatorname{range} T^k$, then there exists $u \in V$ such that $v = T^k u = T(T^{k-1})u$, which implies that $v \in \operatorname{range} T$. Thus $\operatorname{range} T^k \subset \operatorname{range} T$. Note that

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$$\begin{aligned}\dim \operatorname{range} T^k &= \dim V - \dim \operatorname{null} T^k \\ &= \dim V - \dim \operatorname{null} T \\ &= \dim \operatorname{range} T,\end{aligned}$$

where the first and third equalities come from 3.4 and the second equality comes from the first part of this exercise. Because $\operatorname{range} T^k$ and $\operatorname{range} T$ have the same dimension and one of them is contained in the other, these two subspaces of $V$ must be equal, completing the proof.

8. Prove that there does not exist a self-adjoint operator $T \in \mathcal{L}(\mathbf{R}^3)$ such that $T(1,2,3) = (0,0,0)$ and $T(2,5,7) = (2,5,7)$.

SOLUTION: Suppose $T \in \mathcal{L}(\mathbf{R}^3)$ is such that $T(1,2,3) = (0,0,0)$ and $T(2,5,7) = (2,5,7)$. Obviously $(1,2,3)$ is an eigenvector of $T$ with eigenvalue 0 and $(2,5,7)$ is an eigenvector of $T$ with eigenvalue 1. If $T$ were self-adjoint, then eigenvectors corresponding to distinct eigenvalues would be orthogonal (see 7.8). Because $(1,2,3)$ and $(2,5,7)$ are not orthogonal, $T$ cannot be self-adjoint.

9. Prove that a normal operator on a complex inner-product space is self-adjoint if and only if all its eigenvalues are real.

COMMENT: This exercise strengthens the analogy (for normal operators) between self-adjoint operators and real numbers.

SOLUTION: Suppose $V$ is a complex inner product space and $T \in \mathcal{L}(V)$ is normal.

If $T$ is self-adjoint, then by 7.1 all its eigenvalues are real.

Conversely, suppose that all the eigenvalues of $T$ are real. By the complex spectral theorem (7.9), there is an orthonormal basis $(e_1, \ldots, e_n)$ of $V$ consisting of eigenvectors of $T$. Thus there exist real numbers $\lambda_1, \ldots, \lambda_n$ such that $Te_j = \lambda_j e_j$ for $j = 1, \ldots, n$. .The matrix of $T$ with respect to the basis $(e_1, \ldots, e_n)$ is the diagonal matrix with $\lambda_1, \ldots, \lambda_n$ on the diagonal. This matrix equals its conjugate transpose. Thus $T = T^*$. In other words, $T$ is self-adjoint, as desired.

10. Suppose $V$ is a complex inner-product space and $T \in \mathcal{L}(V)$ is a normal operator such that $T^9 = T^8$. Prove that $T$ is self-adjoint and $T^2 = T$.

SOLUTION: By the complex spectral theorem (7.9), there is an orthonormal basis $(e_1, \ldots, e_n)$ of $V$ consisting of eigenvectors of $T$. Let $\lambda_1, \ldots, \lambda_n$ be the corresponding eigenvalues. Thus

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$$Te_j = \lambda_j e_j$$

for $j = 1, \ldots, n$. Applying $T$ repeatedly to both sides of the equation above, we get $T^9 e_j = {\lambda_j}^9 e_j$ and $T^8 e_j = {\lambda_j}^8 e_j$. Thus ${\lambda_j}^9 = {\lambda_j}^8$, which implies that $\lambda_j$ equals 0 or 1. In particular, all the eigenvalues of $T$ are real. This implies (by the previous exercise) that $T$ is self-adjoint.

Applying $T$ to both sides of the equation above, we get

$$\begin{aligned} T^2 e_j &= {\lambda_j}^2 e_j \\ &= \lambda_j e_j \\ &= T e_j, \end{aligned}$$

where the second equality holds because $\lambda_j$ equals 0 or 1. Because $T^2$ and $T$ agree on a basis, they must be equal.

11. Suppose $V$ is a complex inner-product space. Prove that every normal operator on $V$ has a square root. (An operator $S \in \mathcal{L}(V)$ is called a *square root* of $T \in \mathcal{L}(V)$ if $S^2 = T$.)

SOLUTION: Suppose $T \in \mathcal{L}(V)$ is normal. By the complex spectral theorem (7.9), there is an orthonormal basis $(e_1, \ldots, e_n)$ of $V$ consisting of eigenvectors of $T$. Thus there exist complex numbers $\lambda_1, \ldots, \lambda_n$ such that $Te_j = \lambda_j e_j$ for $j = 1, \ldots, n$. Define $S$ to be the operator on $V$ such that $Se_j = {\lambda_j}^{1/2} e_j$ for $j = 1, \ldots, n$; here ${\lambda_j}^{1/2}$ denotes a complex square root of $\lambda_j$ (every nonzero complex number has two square roots—it does not matter which one is chosen). Then, as is easy to verify, $S^2 = T$. Thus $S$ is a square root of $T$.

12. Give an example of a real inner-product space $V$ and $T \in \mathcal{L}(V)$ and real numbers $\alpha, \beta$ with $\alpha^2 < 4\beta$ such that $T^2 + \alpha T + \beta I$ is not invertible.

COMMENT: This exercise shows that the hypothesis that $T$ is self-adjoint is needed in 7.11, even for real vector spaces.

SOLUTION: Let $T \in \mathcal{L}(\mathbf{R}^2)$ be the counterclockwise rotation on $\mathbf{R}^2$; so $T(x, y) = (-y, x)$ for $(x, y) \in \mathbf{R}^2$. Thus $T^2 = -I$. Taking $\alpha = 0$ and $\beta = 1$, we have $\alpha^2 < 4\beta$ and

$$\begin{aligned} T^2 + \alpha T + \beta I &= T^2 + I \\ &= 0. \end{aligned}$$

In particular, $T^2 + \alpha T + \beta I$ is not invertible.

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13. Prove or give a counterexample: every self-adjoint operator on $V$ has a cube root. (An operator $S \in \mathcal{L}(V)$ is called a *cube root* of $T \in \mathcal{L}(V)$ if $S^3 = T$.)

SOLUTION: Suppose $T \in \mathcal{L}(V)$ is self-adjoint. By the spectral theorem (7.13), there is an orthonormal basis $(e_1, \ldots, e_n)$ of $V$ consisting of eigenvectors of $T$. The corresponding eigenvalues must be real (by 7.1). Thus there exist real numbers $\lambda_1, \ldots, \lambda_n$ such that $Te_j = \lambda_j e_j$ for $j = 1, \ldots, n$. Define $S$ to be the operator on $V$ such that $Se_j = \lambda_j{}^{1/3} e_j$ for $j = 1, \ldots, n$. Then, as is easy to verify, $S^3 = T$. Thus $S$ is a cube root of $T$, completing the proof that every self-adjoint operator on $V$ has a cube root.

14. Suppose $T \in \mathcal{L}(V)$ is self-adjoint, $\lambda \in \mathbf{F}$, and $\epsilon > 0$. Prove that if there exists $v \in V$ such that $\|v\| = 1$ and

$$\|Tv - \lambda v\| < \epsilon,$$

then $T$ has an eigenvalue $\lambda'$ such that $|\lambda - \lambda'| < \epsilon$.

SOLUTION: By the spectral theorem (7.13), there is an orthonormal basis $(e_1, \ldots, e_n)$ of $V$ consisting of eigenvectors of $T$. Let $\lambda_1, \ldots, \lambda_n$ be the corresponding eigenvalues.

Suppose $v \in V$ is such that $\|v\| = 1$ and $\|Tv - \lambda v\| < \epsilon$. From 6.17 we have

$$v = \langle v, e_1 \rangle e_1 + \cdots + \langle v, e_n \rangle e_n,$$

and so

$$Tv = \lambda_1 \langle v, e_1 \rangle e_1 + \cdots + \lambda_n \langle v, e_n \rangle e_n.$$

Thus

$$\begin{aligned}
\epsilon^2 &> \|Tv - \lambda v\|^2 \\
&= \|(\lambda_1 - \lambda)\langle v, e_1 \rangle e_1 + \cdots + (\lambda_n - \lambda)\langle v, e_n \rangle e_n\|^2 \\
&= |\lambda_1 - \lambda|^2 |\langle v, e_1 \rangle|^2 + \cdots + |\lambda_n - \lambda|^2 |\langle v, e_n \rangle|^2 \\
&\geq \left(\min\{|\lambda_1 - \lambda|^2, \ldots, |\lambda_n - \lambda|^2\}\right) \left(|\langle v, e_1 \rangle|^2 + \cdots + |\langle v, e_n \rangle|^2\right) \\
&= \min\{|\lambda_1 - \lambda|^2, \ldots, |\lambda_n - \lambda|^2\}
\end{aligned}$$

Thus $\epsilon > |\lambda_j - \lambda|$ for some $j$. In other words, there is an eigenvalue whose distance from $\lambda$ is less than $\epsilon$, as desired.

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15. Suppose $U$ is a finite-dimensional real vector space and $T \in \mathcal{L}(U)$. Prove that $U$ has a basis consisting of eigenvectors of $T$ if and only if there is an inner product on $U$ that makes $T$ into a self-adjoint operator.

SOLUTION: First suppose that $U$ has a basis $(e_1, \dots, e_n)$ of eigenvectors of $T$. Because $(e_1, \dots, e_n)$ is a basis of $U$, every element of $U$ can be uniquely written as a linear combination of $(e_1, \dots, e_n)$. Thus we can define an inner product on $U$ by

$$\langle a_1 e_1 + \cdots + a_n e_n, b_1 e_1 + \cdots + b_n e_n \rangle = a_1 b_1 + \cdots + a_n b_n.$$

It is easy to verify that this is indeed an inner product on $U$ and that $(e_1, \dots, e_n)$ is on orthonormal basis of $U$ with respect to this inner product. Because each $e_j$ is an eigenvector of $T$, the operator $T$ has a diagonal matrix with respect to the orthonormal basis $(e_1, \dots, e_n)$. Thus $T$ is self-adjoint.

Conversely, now suppose that there is an inner product on $U$ that makes $T$ into a self-adjoint operator. Then by the spectral theorem (7.13), $U$ has a basis consisting of eigenvectors of $T$.

16. Give an example of an operator $T$ on an inner product space such that $T$ has an invariant subspace whose orthogonal complement is not invariant under $T$.

COMMENT: This exercise shows that 7.18 can fail without the hypothesis that $T$ is normal.

SOLUTION: Define $T \in \mathcal{L}(\mathbf{F}^2)$ by $T(w, z) = (z, 0)$. Then $T(w, 0) = (0, 0)$ for all $w \in \mathbf{F}$. Thus the subspace $U$ defined by $U = \{(w, 0) : w \in \mathbf{F}\}$ is invariant under $T$. However, $U^\perp = \{(0, z) : \in \mathbf{F}\}$, which is not invariant under $T$ because $(0, 1) \in U^\perp$ but $T(0, 1) = (1, 0) \notin U^\perp$.

Of course there are also many other examples.

17. Prove that the sum of any two positive operators on $V$ is positive.

SOLUTION: Suppose $S$ and $T$ are positive operators on $V$. Because $S$ and $T$ are self-adjoint, so is $S + T$. Furthermore,

$$\begin{aligned} \langle (S + T)v, v \rangle &= \langle Sv, v \rangle + \langle Tv, v \rangle \\ &\geq 0. \end{aligned}$$

Thus $S + T$ is a positive operator, as desired.

18. Prove that if $T \in \mathcal{L}(V)$ is positive, then so is $T^k$ for every positive integer $k$.

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SOLUTION: Suppose $T \in \mathcal{L}(V)$ is positive and $k$ is a positive integer. Then $T^k$ is self-adjoint (because $T$ is self-adjoint).

First consider the case where $k$ is an even integer. Then we can write $k = 2m$ for some positive integer $m$. Now

$$\begin{aligned}\langle T^k v, v\rangle &= \langle T^{2m} v, v\rangle \\ &= \langle T^m v, T^m v\rangle \\ &\geq 0\end{aligned}$$

for every $v \in V$, where the second equality holds because $T$ is self-adjoint. The inequality above shows that $T^k$ is positive, as desired.

Now consider the case where $k$ is an odd integer. Then we can write $k = 2m + 1$ for some nonnegative integer $m$. Now

$$\begin{aligned}\langle T^k v, v\rangle &= \langle T^{2m+1} v, v\rangle \\ &= \langle T(T^m v), T^m v\rangle \\ &\geq 0\end{aligned}$$

for every $v \in V$, where the second equality holds because $T$ is self-adjoint and the inequality holds because $T$ is positive. The inequality above shows that $T^k$ is positive, as desired.

19. Suppose that $T$ is a positive operator on $V$. Prove that $T$ is invertible if and only if

$$\langle Tv, v\rangle > 0$$

for every $v \in V \setminus \{0\}$.

SOLUTION: First suppose that $T$ is invertible. By 7.27, there exists an operator $S \in \mathcal{L}(V)$ such that $T = S^*S$. Suppose $v \in V \setminus \{0\}$. Then $Sv \neq 0$ because otherwise we would have $Tv = S^*Sv = 0$, which would contradict the invertibility of $T$. Now

$$\begin{aligned}\langle Tv, v\rangle &= \langle S^*Sv, v\rangle \\ &= \langle Sv, Sv\rangle \\ &> 0,\end{aligned}$$

as desired.

Now suppose that $\langle Tv, v\rangle > 0$ for every $v \in V \setminus \{0\}$. In particular, this means that $Tv \neq 0$ for every $v \in V \setminus \{0\}$. Thus $T$ is injective, and hence $T$ is invertible (see 3.21), as desired.

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20. Prove or disprove: the identity operator on $\mathbf{F}^2$ has infinitely many self-adjoint square roots.

SOLUTION: For each $t \in [-1, 1]$, the operator whose matrix (with respect to the standard basis) equals

$$\begin{bmatrix} t & \sqrt{1-t^2} \\ \sqrt{1-t^2} & -t \end{bmatrix}$$

is self-adjoint and a square root of the identity operator, as can be verified by squaring the matrix above. Thus the identity operator has infinitely many self-adjoint square roots.

21. Prove or give a counterexample: if $S \in \mathcal{L}(V)$ and there exists an orthonormal basis $(e_1, \ldots, e_n)$ of $V$ such that $\|Se_j\| = 1$ for each $e_j$, then $S$ is an isometry.

SOLUTION: Define $S \in \mathcal{L}(\mathbf{F}^2)$ by

$$S(w, z) = (w + z, 0).$$

With the usual inner product on $\mathbf{F}^2$, the standard basis $\big((1,0),(0,1)\big)$ is an orthonormal basis of $\mathbf{F}^2$. Note that $\|S(1,0)\| = \|S(0,1)\| = 1$. However, $S$ is not an isometry because $\|S(1,-1)\| = 0$.

Of course there are also many other examples.

22. Prove that if $S \in \mathcal{L}(\mathbf{R}^3)$ is an isometry, then there exists a nonzero vector $x \in \mathbf{R}^3$ such that $S^2x = x$.

SOLUTION: Suppose $S \in \mathcal{L}(\mathbf{R}^3)$ is an isometry. Then there is a basis of $\mathbf{R}^3$ with respect to which $S$ has a block diagonal matrix, where each block on the diagonal is a 1-by-1 matrix containing 1 or $-1$ or is a 2-by-2 matrix (see 7.38). Because $\mathbf{R}^3$ has odd dimension, at least one of these blocks must be a 1-by-1 matrix. In other words, either 1 or $-1$ must be an eigenvalue of $S$. Thus there is a nonzero vector $x \in \mathbf{R}^3$ such that $Sx = \lambda x$, where $\lambda = \pm 1$. Hence

$$S^2x = S(Sx) = S(\lambda x) = \lambda Sx = \lambda^2 x = x.$$

23. Define $T \in \mathcal{L}(\mathbf{F}^3)$ by

$$T(z_1, z_2, z_3) = (z_3, 2z_1, 3z_2).$$

Find (explicitly) an isometry $S \in \mathcal{L}(\mathbf{F}^3)$ such that $T = S\sqrt{T^*T}$.

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SOLUTION: With respect to the standard basis of $\mathbf{F}^3$, we have

$$\mathcal{M}(T) = \begin{bmatrix} 0 & 0 & 1 \\ 2 & 0 & 0 \\ 0 & 3 & 0 \end{bmatrix}.$$

Thus

$$\mathcal{M}(T^*) = \begin{bmatrix} 0 & 2 & 0 \\ 0 & 0 & 3 \\ 1 & 0 & 0 \end{bmatrix}.$$

Computing the product $\mathcal{M}(T^*)\mathcal{M}(T)$, which equals $\mathcal{M}(T^*T)$, we get

$$\mathcal{M}(T^*T) = \begin{bmatrix} 4 & 0 & 0 \\ 0 & 9 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

From the matrix above, we see that $(T^*T)(z_1, z_2, z_3) = (4z_1, 9z_2, z_3)$. Thus $\sqrt{T^*T}(z_1, z_2, z_3) = (2z_1, 3z_2, z_3)$. Hence if we define $S \in \mathcal{L}(\mathbf{F}^3)$ by

$$S(z_1, z_2, z_3) = (z_3, z_1, z_2),$$

then $S$ is an isometry and $T = S\sqrt{T^*T}$.

24. Suppose $T \in \mathcal{L}(V)$, $S \in \mathcal{L}(V)$ is an isometry, and $R \in \mathcal{L}(V)$ is a positive operator such that $T = SR$. Prove that $R = \sqrt{T^*T}$.

COMMENT: This exercise shows that if we write $T$ as the product of an isometry and a positive operator (as in the polar decomposition), then the positive operator must equal $\sqrt{T^*T}$.

SOLUTION: Taking adjoints of both sides of the equation $T = SR$, we have

$$\begin{aligned} T^* &= R^*S^* \\ &= RS^*, \end{aligned}$$

where the last equation holds because $R$ is positive (and hence self-adjoint). Multiplying together our formulas for $T^*$ and $T$, we get

$$\begin{aligned} T^*T &= RS^*SR \\ &= R^2, \end{aligned}$$

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where the last equation holds because $S$ is an isometry (and hence $S^*S = I$ by 7.36). The equation above asserts that $R$ is a square root of $T^*T$; because $R$ is positive, this implies that $R = \sqrt{T^*T}$.

25. Suppose $T \in \mathcal{L}(V)$. Prove that $T$ is invertible if and only if there exists a unique isometry $S \in \mathcal{L}(V)$ such that $T = S\sqrt{T^*T}$.

SOLUTION: First suppose that $T$ is invertible. The polar decomposition (7.41) states that there exists an isometry $S \in \mathcal{L}(V)$ such that

$$T = S\sqrt{T^*T}.$$

Because $T$ is invertible, this implies that $\sqrt{T^*T}$ is invertible (see Exercise 22 in Chapter 3). Thus the equation above implies that $S = T(\sqrt{T^*T})^{-1}$. Because $S$ must be given by this formula, we see that there is a unique operator $S \in \mathcal{L}(V)$ such that $T = S\sqrt{T^*T}$, as desired.

Now suppose that there exists a unique isometry $S \in \mathcal{L}(V)$ such that $T = S\sqrt{T^*T}$. This means that the linear map $S_2$ in the proof of the polar decomposition (7.41) must be 0 because otherwise we could replace $S_2$ with $-S_2$ and get another choice for $S$. But $\operatorname{range} S_2$ equals $(\operatorname{range} T)^\perp$, and hence $(\operatorname{range} T)^\perp = \{0\}$. This implies that $\operatorname{range} T = V$, which implies that $T$ is invertible (by 3.21), as desired.

26. Prove that if $T \in \mathcal{L}(V)$ is self-adjoint, then the singular values of $T$ equal the absolute values of the eigenvalues of $T$ (repeated appropriately).

SOLUTION: Suppose $T \in \mathcal{L}(V)$ is self-adjoint. There exists an orthonormal basis $(e_1, \dots, e_n)$ of $V$ consisting of eigenvectors of $T$. Thus

$$Te_j = \lambda_j e_j$$

for each $j$, where $\lambda_1, \dots, \lambda_n \in \mathbf{R}$ are the eigenvalues of $T$. Thus

$$\begin{aligned} T^*Te_j &= T^2 e_j \\ &= (\lambda_j)^2 e_j \end{aligned}$$

for each $j$. The equation above implies that $\sqrt{T^*T}e_j = |\lambda_j|e_j$ for each $j$. Thus the singular values of $T$ are $|\lambda_1|, \dots, |\lambda_n|$, as desired.

27. Prove or give a counterexample: if $T \in \mathcal{L}(V)$, then the singular values of $T^2$ equal the squares of the singular values of $T$.

SOLUTION: Define $T \in \mathcal{L}(\mathbf{F}^2)$ by

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$$T(z_1, z_2) = (z_2, 0).$$

Then $T^*T(z_1, z_2) = (0, z_2)$ and hence $\sqrt{T^*T}(z_1, z_2) = (0, z_2)$. Thus the eigenvalues of $\sqrt{T^*T}$ are $0, 1$. Hence the singular values of $T$ are $0, 1$.

However, $T^2 = 0$, so the singular values of $T^2$ are $0, 0$. Thus for this operator $T$, the singular values of $T^2$ do not equal the squares of the singular values of $T$.

Of course there are also many other examples.

28. Suppose $T \in \mathcal{L}(V)$. Prove that $T$ is invertible if and only if 0 is not a singular value of $T$.

SOLUTION: If $S \in \mathcal{L}(V)$ and $ST = TS = I$, then taking adjoints we get $T^*S^* = S^*T^* = I$. Thus if $T$ is invertible, then so is $T^*$.

Now

$$\begin{aligned} T \text{ is invertible} &\iff T \text{ and } T^* \text{ are invertible} \\ &\iff T^*T \text{ is invertible} \\ &\iff \sqrt{T^*T}\sqrt{T^*T} \text{ is invertible} \\ &\iff \sqrt{T^*T} \text{ is invertible} \\ &\iff 0 \text{ is not an eigenvalue of } \sqrt{T^*T} \\ &\iff 0 \text{ is not a singular value of } T, \end{aligned}$$

where the second and fourth equivalences follow from Exercise 22 in Chapter 2.

29. Suppose $T \in \mathcal{L}(V)$. Prove that $\dim \operatorname{range} T$ equals the number of nonzero singular values of $T$.

SOLUTION: By the singular value decomposition (7.46), there exist orthonormal bases $(u_1, \dots, u_n)$ and $(w_1, \dots, w_n)$ of $V$ such that

$$Tv = s_1\langle v, u_1\rangle w_1 + \cdots + s_n\langle v, u_n\rangle w_n$$

for every $v \in V$, where $s_1, \dots, s_n$ are the singular values of $T$. For each $j$, we have $Tu_j = s_j w_j$. Thus each $w_j$ corresponding to a nonzero $s_j$ is in $\operatorname{range} T$. The equation above also shows that the $w_j$'s corresponding to nonzero $s_j$'s span $\operatorname{range} T$. Thus $\dim \operatorname{range} T$ equals the number of nonzero singular values of $T$.

30. Suppose $S \in \mathcal{L}(V)$. Prove that $S$ is an isometry if and only if all the singular values of $S$ equal 1.

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SOLUTION: We have

$$\begin{aligned} S \text{ is an isometry} &\iff S^*S = I \\ &\iff \sqrt{S^*S} = I \\ &\iff \text{all the eigenvalues of } \sqrt{S^*S} \text{ equal } 1 \\ &\iff \text{all the singular values of } S \text{ equal } 1, \end{aligned}$$

where the first equivalence comes from 7.36 and the third equivalence comes from the spectral theorem (7.9 or 7.13) applied to the self-adjoint operator $\sqrt{S^*S}$.

31. Suppose $T_1, T_2 \in \mathcal{L}(V)$. Prove that $T_1$ and $T_2$ have the same singular values if and only if there exist isometries $S_1, S_2 \in \mathcal{L}(V)$ such that $T_1 = S_1 T_2 S_2$.

SOLUTION: First suppose that $T_1$ and $T_2$ have the same singular values $s_1, \dots, s_n$. By the singular-value decomposition (7.46), there exist orthonormal bases $(e_1, \dots, e_n), (f_1, \dots, f_n), (e'_1, \dots, e'_n), (f'_1, \dots, f'_n)$ of $V$ such that

$$\begin{aligned} T_1 v &= s_1\langle v, e_1\rangle f_1 + \cdots + s_n\langle v, e_n\rangle f_n, \\ T_2 v &= s_1\langle v, e'_1\rangle f'_1 + \cdots + s_n\langle v, e'_n\rangle f'_n \end{aligned}$$

for every $v \in V$. Define $S_1, S_2 \in \mathcal{L}(V)$ by

$$\begin{aligned} S_1(a_1 f'_1 + \cdots + a_n f'_n) &= a_1 f_1 + \cdots + a_n f_n, \\ S_2(a_1 e_1 + \cdots + a_n e_n) &= a_1 e'_1 + \cdots + a_n e'_n. \end{aligned}$$

Then

$$\begin{aligned} \|S_1(a_1 f'_1 + \cdots + a_n f'_n)\|^2 &= \|a_1 f_1 + \cdots + a_n f_n\|^2 \\ &= |a_1|^2 + \cdots + |a_n|^2 \\ &= \|a_1 f'_1 + \cdots + a_n f'_n\|^2, \end{aligned}$$

and thus $S_1$ is an isometry. Similarly, $S_2$ is an isometry. This implies that $S_2^* = S_2^{-1}$ (see 7.36). In particular, $S_2^* e'_j = e_j$. Now for $v \in V$ we have

$$\begin{aligned} T_2(S_2 v) &= s_1\langle S_2 v, e'_1\rangle f'_1 + \cdots + s_n\langle S_2 v, e'_n\rangle f'_n \\ &= s_1\langle v, S_2^* e'_1\rangle f'_1 + \cdots + s_n\langle v, S_2^* e'_n\rangle f'_n \\ &= s_1\langle v, e_1\rangle f'_1 + \cdots + s_n\langle v, e_n\rangle f'_n. \end{aligned}$$

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Thus

$$\begin{aligned} S_1(T_2S_2v) &= s_1\langle v, e_1\rangle S_1f_1' + \cdots + s_n\langle v, e_n\rangle S_1f_n' \\ &= s_1\langle v, e_1\rangle f_1 + \cdots + s_n\langle v, e_n\rangle f_n \\ &= T_1v \end{aligned}$$

for every $v \in V$. Hence $S_1T_2S_2 = T_1$, as desired.

To prove the implication in the other direction, now suppose that there exist isometries $S_1, S_2 \in \mathcal{L}(V)$ such that $T_1 = S_1T_2S_2$. Using 7.36, we have

$$\begin{aligned} T_1{}^*T_1 &= S_2{}^*T_2{}^*S_1{}^*S_1T_2S_2 \\ &= S_2{}^{-1}T_2{}^*T_2S_2. \end{aligned}$$

This implies that $T_1{}^*T_1$ and $T_2{}^*T_2$ have the same eigenvalues (and that the corresponding spaces of eigenvectors have the same dimensions). Thus $T_1$ and $T_2$ have the same singular values.

32. Suppose $T \in \mathcal{L}(V)$ has singular-value decomposition given by

$$Tv = s_1\langle v, e_1\rangle f_1 + \cdots + s_n\langle v, e_n\rangle f_n$$

for every $v \in V$, where $s_1, \ldots, s_n$ are the singular values of $T$ and $(e_1, \ldots, e_n)$ and $(f_1, \ldots, f_n)$ are orthonormal bases of $V$.

(a) Prove that

$$T^*v = s_1\langle v, f_1\rangle e_1 + \cdots + s_n\langle v, f_n\rangle e_n$$

for every $v \in V$.

(b) Prove that if $T$ is invertible, then

$$T^{-1}v = \frac{\langle v, f_1\rangle e_1}{s_1} + \cdots + \frac{\langle v, f_n\rangle e_n}{s_n}$$

for every $v \in V$.

SOLUTION: (a): Fix $v \in V$. Then

$$\begin{aligned} \langle w, T^*v\rangle &= \langle Tw, v\rangle \\ &= \langle s_1\langle w, e_1\rangle f_1 + \cdots + s_n\langle w, e_n\rangle f_n, v\rangle \\ &= s_1\langle w, e_1\rangle\langle f_1, v\rangle + \cdots + s_n\langle w, e_n\rangle\langle f_n, v\rangle \\ &= \langle w, s_1\langle v, f_1\rangle e_1 + \cdots + s_n\langle v, f_n\rangle e_n\rangle \end{aligned}$$

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for all $w \in V$. This implies that

$$T^*v = s_1\langle v, f_1\rangle e_1 + \cdots + s_n\langle v, f_n\rangle e_n,$$

as desired.

(b): Suppose $T$ is invertible. Let $v \in V$ and let

$$w = \frac{\langle v, f_1\rangle e_1}{s_1} + \cdots + \frac{\langle v, f_n\rangle e_n}{s_n};$$

none of the singular values $s_1, \ldots, s_n$ equals 0 (see Exercise 28 of this chapter), so this makes sense. Now

$$\begin{aligned} Tw &= \frac{\langle v, f_1\rangle Te_1}{s_1} + \cdots + \frac{\langle v, f_n\rangle Te_n}{s_n} \\ &= \frac{\langle v, f_1\rangle s_1 f_1}{s_1} + \cdots + \frac{\langle v, f_n\rangle s_n f_n}{s_n} \\ &= \langle v, f_1\rangle f_1 + \cdots + \langle v, f_n\rangle f_n \\ &= v. \end{aligned}$$

Thus $w = T^{-1}v$, as desired.

33. Suppose $T \in \mathcal{L}(V)$. Let $\hat{s}$ denote the smallest singular value of $T$, and let $s$ denote the largest singular value of $T$. Prove that

$$\hat{s}\|v\| \le \|Tv\| \le s\|v\|$$

for every $v \in V$.

SOLUTION: Let $v \in V$. By the singular value decomposition (7.46), there exist orthonormal bases $(u_1, \ldots, u_n)$ and $(w_1, \ldots, w_n)$ of $V$ such that

$$Tv = s_1\langle v, u_1\rangle w_1 + \cdots + s_n\langle v, u_n\rangle w_n,$$

where $s_1, \ldots, s_n$ are the singular values of $T$. Because $(u_1, \ldots, u_n)$ and $(w_1, \ldots, w_n)$ are both orthonormal bases of $V$, we have

$$\begin{aligned} \hat{s}^2\|v\|^2 &= \hat{s}^2(|\langle v, u_1\rangle|^2 + \cdots + |\langle v, u_n\rangle|^2) \\ &\le {s_1}^2|\langle v, u_1\rangle|^2 + \cdots + {s_n}^2|\langle v, u_n\rangle|^2 \\ &= \|Tv\|^2, \end{aligned}$$

giving the first desired inequality. Also,

$$\begin{aligned}\|Tv\|^2 &= {s_1}^2|\langle v, u_1\rangle|^2 + \cdots + {s_n}^2|\langle v, u_n\rangle|^2 \\ &\leq s^2(|\langle v, u_1\rangle|^2 + \cdots + |\langle v, u_n\rangle|^2) \\ &= s^2\|v\|^2,\end{aligned}$$

giving the second desired inequality.

34. Suppose $T', T'' \in \mathcal{L}(V)$. Let $s'$ denote the largest singular value of $T'$, let $s''$ denote the largest singular value of $T''$, and let $s$ denote the largest singular value of $T' + T''$. Prove that $s \leq s' + s''$.

SOLUTION: Let $T = T' + T''$. Because $s$ is a singular value of $T$, we know that $s$ is an eigenvalue of $\sqrt{T^*T}$. Thus there exists a vector $v \in V$ such that $\|v\| = 1$ and $\sqrt{T^*T}v = sv$. Now

$$\begin{aligned}s &= \|sv\| \\ &= \|\sqrt{T^*T}v\| \\ &= \|Tv\| \\ &= \|T'v + T''v\| \\ &\leq \|T'v\| + \|T''v\| \\ &\leq s'\|v\| + s''\|v\| \\ &= s' + s'',\end{aligned}$$

where the third line above comes from 7.42 and the sixth line above comes from the previous exercise.

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CHAPTER 8

# Operators on Complex Vector Spaces

1. Define $T \in \mathcal{L}(\mathbf{C}^2)$ by

$$T(w, z) = (z, 0).$$

Find all generalized eigenvectors of $T$.

SOLUTION: Suppose $\lambda$ is an eigenvalue of $T$. For this particular operator, the eigenvalue-eigenvector equation $T(w, z) = \lambda(w, z)$ becomes the system of equations

$$z = \lambda w$$
$$0 = \lambda z.$$

If $\lambda \neq 0$, then the second equation implies that $z = 0$, and the first equation then implies that $w = 0$. Because an eigenvalue must have a nonzero eigenvector, this shows that 0 is the only possible eigenvalue of $T$. For $\lambda = 0$, the equations above show that $z$ must equal 0, but $w$ can be arbitrary. Thus 0 is indeed an eigenvalue of $T$, and the set of eigenvectors corresponding to this eigenvalue is

$$\{(w, 0) : w \in \mathbf{F}\}.$$

Note that $T^2 = 0$. Thus every vector in $\mathbf{C}^2$ is a generalized eigenvector of $T$ (corresponding to the eigenvalue 0).



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2. Define $T \in \mathcal{L}(\mathbf{C}^2)$ by

$$T(w, z) = (-z, w).$$

Find all generalized eigenvectors of $T$.

SOLUTION: On page 78 of the textbook we saw that the eigenvalues of $T$ are $i$ and $-i$. Note that $T^2 = -I$.

The set of generalized eigenvectors of $T$ corresponding to the eigenvalue $i$ equals $\text{null}(T - iI)^2$ (by 8.7). To compute this, note that

$$\begin{aligned}(T - iI)^2 &= T^2 - 2iT - I \\ &= -2I - 2iT \\ &= -2i(T - iI).\end{aligned}$$

Thus $\text{null}(T - iI)^2$ equals the set of eigenvectors of $T$ corresponding to the eigenvalue $i$. On page 78 of the textbook we noted that this equals $\{(a, -ia) : a \in \mathbf{C}\}$.

The set of generalized eigenvectors of $T$ corresponding to the eigenvalue $-i$ equals $\text{null}(T + iI)^2$ (by 8.7). To compute this, note that

$$\begin{aligned}(T + iI)^2 &= T^2 + 2iT - I \\ &= -2I + 2iT \\ &= 2i(T + iI).\end{aligned}$$

Thus $\text{null}(T + iI)^2$ equals the set of eigenvectors of $T$ corresponding to the eigenvalue $-i$. On page 78 of the textbook we noted that this equals $\{(a, ia) : a \in \mathbf{C}\}$.

3. Suppose $T \in \mathcal{L}(V)$, $m$ is a positive integer, and $v \in V$ is such that $T^{m-1}v \neq 0$ but $T^m v = 0$. Prove that

$$(v, Tv, T^2v, \ldots, T^{m-1}v)$$

is linearly independent.

SOLUTION: Suppose $a_0, a_1, a_2, \ldots, a_{m-1} \in \mathbf{F}$ are such that

$$a_0 v + a_1 Tv + a_2 T^2 v + \cdots + a_{m-1} T^{m-1} v = 0.$$

Because $T^m v = 0$, if we apply $T^{m-1}$ to both sides of the equation above, we get $a_0 T^{m-1} v = 0$. Because $T^{m-1} v \neq 0$, this implies that $a_0 = 0$. Thus the equation above can be rewritten as

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$$a_1Tv + a_2T^2v + \cdots + a_{m-1}T^{m-1}v = 0.$$

Applying $T^{m-2}$ to both sides of this equation, we get $a_1T^{m-1}v = 0$. Thus $a_1 = 0$. Continuing in this fashion, we have $a_0 = a_1 = a_2 = \cdots = a_{m-1} = 0$, which means that $(v, Tv, T^2v, \ldots, T^{m-1}v)$ is linearly independent.

4. Suppose $T \in \mathcal{L}(\mathbf{C}^3)$ is defined by $T(z_1, z_2, z_3) = (z_2, z_3, 0)$. Prove that $T$ has no square root. More precisely, prove that there does not exist $S \in \mathcal{L}(\mathbf{C}^3)$ such that $S^2 = T$.

SOLUTION: Note that $T^3 = 0$. Suppose there exists $S \in \mathcal{L}(\mathbf{C}^3)$ such that $S^2 = T$. Then $S^6 = T^3 = 0$, so $S$ is nilpotent. By 8.8, this implies that $S^3 = 0$. Thus

$$\begin{aligned} T^2 &= S^4 \\ &= SS^3 \\ &= 0. \end{aligned}$$

But $T^2(z_1, z_2, z_3) = (z_3, 0, 0)$, so $T^2$ is not the 0 operator, contradicting the equation above. This contradiction shows that our supposition that there exists $S \in \mathcal{L}(\mathbf{C}^3)$ such that $S^2 = T$ must have been false.

5. Suppose $S, T \in \mathcal{L}(V)$. Prove that if $ST$ is nilpotent, then $TS$ is nilpotent.

SOLUTION: Suppose $ST$ is nilpotent. Thus there exists a positive integer $n$ such that $(ST)^n = 0$. Now

$$\begin{aligned} (TS)^{n+1} &= (TS)(TS)\ldots(TS) \\ &= T(ST)(ST)\ldots(ST)S \\ &= T(ST)^nS \\ &= (T)(0)(S) \\ &= 0, \end{aligned}$$

and thus $TS$ is nilpotent.

6. Suppose $N \in \mathcal{L}(V)$ is nilpotent. Prove (without using 8.26) that 0 is the only eigenvalue of $N$.

SOLUTION: There is a positive integer $m$ such that $N^m = 0$. This implies that $N$ is not injective, so 0 is an eigenvalue of $N$.

Conversely, suppose $\lambda$ is an eigenvalue of $N$. Then there exists a nonzero vector $v \in V$ such that

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$$\lambda v = Nv.$$

Repeatedly applying $N$ to both sides of this equation shows that

$$\begin{aligned}\lambda^m v &= N^m v\\ &= 0.\end{aligned}$$

Thus $\lambda = 0$, as desired.

7. Suppose $V$ is an inner-product space. Prove that if $N \in \mathcal{L}(V)$ is self-adjoint and nilpotent, then $N = 0$.

SOLUTION: Suppose $N \in \mathcal{L}(V)$ is self-adjoint and nilpotent. Because $N$ is self-adjoint, there is an orthonormal basis $(e_1, \ldots, e_n)$ of $V$ consisting of eigenvectors of $N$ (by the spectral theorem). Because $N$ is nilpotent, 0 is the only eigenvalue of $N$ (see Exercise 6 of this chapter). Thus the eigenvalue corresponding to each $e_j$ must equal 0. In other words, $Ne_j = 0$ for each $j$. Because $(e_1, \ldots, e_n)$ is a basis of $V$, this implies that $N = 0$.

8. Suppose $N \in \mathcal{L}(V)$ is such that $\operatorname{null} N^{\dim V - 1} \neq \operatorname{null} N^{\dim V}$. Prove that $N$ is nilpotent and that

$$\dim \operatorname{null} N^j = j$$

for every integer $j$ with $0 \leq j \leq \dim V$.

SOLUTION: Because $\operatorname{null} N^{\dim V - 1} \neq \operatorname{null} N^{\dim V}$, we know (by 8.5) that $\operatorname{null} N^{j-1} \neq \operatorname{null} N^j$ whenever $0 \leq j \leq \dim V$. Thus

$$\{0\} = \operatorname{null} N^0 \subsetneq \operatorname{null} N^1 \subsetneq \ldots \subsetneq \operatorname{null} N^{\dim V - 1} \subsetneq \operatorname{null} N^{\dim V}.$$

At each of the strict inclusions in the chain above, the dimension must increase by at least 1. However, if the dimension increases by more than 1 at any step, we would end up with $\dim \operatorname{null} N^{\dim V} > \dim V$, a contradiction because a subspace of $V$ cannot have dimension larger than $\dim V$. Thus the dimension increases by exactly one at each step. In other words, $\dim \operatorname{null} N^j = j$ for every integer $j$ with $0 \leq j \leq \dim V$. In particular, taking $j = \dim V$, we have $\dim \operatorname{null} N^{\dim V} = \dim V$. This means that $\operatorname{null} N^{\dim V} = V$. Thus $N^{\dim V} = 0$, and so $N$ is nilpotent.

9. Suppose $T \in \mathcal{L}(V)$ and $m$ is a nonnegative integer such that

$$\operatorname{range} T^m = \operatorname{range} T^{m+1}.$$

Prove that $\operatorname{range} T^k = \operatorname{range} T^m$ for all $k > m$.

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SOLUTION: Suppose $u \in \text{range}\, T^{m+1}$. Thus there exists a vector $v$ in $V$ (the domain of $T$) such that $u = T^{m+1}v$. Now $T^m v$ is in $\text{range}\, T^m$, which by our hypothesis equals $\text{range}\, T^{m+1}$. Thus there exists $w \in V$ such that $T^m v = T^{m+1}w$. Putting all this together, we have

$$\begin{aligned} u &= T^{m+1}v \\ &= T(T^m v) \\ &= T(T^{m+1}w) \\ &= T^{m+2}w. \end{aligned}$$

Thus $u \in \text{range}\, T^{m+2}$. Because $u$ was an arbitrary vector in $\text{range}\, T^{m+1}$, we have shown that $\text{range}\, T^{m+1} \subset \text{range}\, T^{m+2}$. We also have an easy inclusion in the other direction, so we conclude that $\text{range}\, T^{m+1} = \text{range}\, T^{m+2}$.

In the paragraph above, we showed that $\text{range}\, T^m = \text{range}\, T^{m+1}$ implies $\text{range}\, T^{m+1} = \text{range}\, T^{m+2}$. Apply that result, with $m$ replaced with $m+1$, to conclude that $\text{range}\, T^{m+2} = \text{range}\, T^{m+3}$. Continuing in this fashion, we see that

$$\text{range}\, T^m = \text{range}\, T^{m+1} = \text{range}\, T^{m+2} = \ldots,$$

as desired.

10. Prove or give a counterexample: if $T \in \mathcal{L}(V)$, then

$$V = \text{null}\, T \oplus \text{range}\, T.$$

SOLUTION: Define $T \in \mathcal{L}(\mathbf{F}^2)$ by $T(w,z) = (z,0)$. Thus

$$\text{null}\, T = \text{range}\, T = \{(w,0) : w \in \mathbf{F}\},$$

which clearly implies that $\mathbf{F}^2$ is not the direct sum of $\text{null}\, T$ and $\text{range}\, T$.

Of course there are also many other examples.

11. Prove that if $T \in \mathcal{L}(V)$, then

$$V = \text{null}\, T^n \oplus \text{range}\, T^n,$$

where $n = \dim V$.

SOLUTION: Let $T \in \mathcal{L}(V)$. First we show that

$$V = \text{null}\, T^n + \text{range}\, T^n.$$

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To do this, let $v \in V$. Then

$$v = (v - T^n u) + T^n u$$

for any vector $u \in V$. Obviously $T^n u \in \operatorname{range} T^n$. Thus looking at the equation above, we see that we need to show that there exists $u \in V$ such that $v - T^n u \in \operatorname{null} T^n$. In other words, we want a vector $u \in V$ such that $T^n(v - T^n u) = 0$, which is equivalent to $T^n v = T^{2n} u$. But $T^n v \in \operatorname{range} T^n$, and $\operatorname{range} T^n = \operatorname{range} T^{2n}$ (by 8.9), so $T^n v \in \operatorname{range} T^{2n}$. Thus there indeed exists $u \in V$ such that $T^n v = T^{2n} u$, completing our proof $v \in \operatorname{null} T^n + \operatorname{range} T^n$. Because $v$ was an arbitrary vector in $V$, this implies that $V = \operatorname{null} T^n + \operatorname{range} T^n$.

For any linear map (and in particular for $T^n$), the dimension of the domain equals the sum of the dimensions of the null space and range (by 3.4). In other words,

$$\dim V = \dim \operatorname{null} T^n + \dim \operatorname{range} T^n.$$

This equation, along with the equation $V = \operatorname{null} T^n + \operatorname{range} T^n$, implies that $V = \operatorname{null} T^n \oplus \operatorname{range} T^n$ (by 2.19).

12. Suppose $V$ is a complex vector space, $N \in \mathcal{L}(V)$, and 0 is the only eigenvalue of $N$. Prove that $N$ is nilpotent. Give an example to show that this is not necessarily true on a real vector space.

SOLUTION: Because 0 is the only eigenvalue of $N$, 8.23(a) implies that every vector in $V$ is a generalized eigenvector of $T$ corresponding to the eigenvalue 0. This implies that $N$ is nilpotent.

Define $T \in \mathcal{L}(\mathbf{R}^3)$ by

$$T(x, y, z) = (-y, x, 0).$$

Then 0 is an eigenvalue of $T$ because $T(0, 0, 1) = (0, 0, 0)$. As can be verified from the definition of eigenvalue, $T$ has no other eigenvalues (which must be in $\mathbf{R}$, because $T$ is an operator on a real vector space). However, $T^3(x, y, z) = (y, -x, 0)$. In particular, $T^3 \neq 0$. Thus $T$ is not nilpotent.

Of course there are also many other examples.

13. Suppose that $V$ is a complex vector space with $\dim V = n$ and $T \in \mathcal{L}(V)$ is such that

$$\operatorname{null} T^{n-2} \neq \operatorname{null} T^{n-1}.$$

Prove that $T$ has at most two distinct eigenvalues.

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SOLUTION: Because $\operatorname{null} T^{n-2} \neq \operatorname{null} T^{n-1}$, we see that $\dim \operatorname{null} T^j$ is at least 1 more than $\dim \operatorname{null} T^{j-1}$ for $j = 1, \ldots, n-1$ (by 8.5). Thus $\dim \operatorname{null} T^{n-1} \geq n-1$. In particular, 0 is an eigenvalue of $T$ with multiplicity at least $n-1$. Because the sum of the multiplicities of all the eigenvalues of $T$ equals $n$ (by 8.18), this implies that $T$ can have at most one additional eigenvalue.

14. Give an example of an operator on $\mathbf{C}^4$ whose characteristic polynomial equals $(z-7)^2(z-8)^2$.

SOLUTION: Define $T \in \mathcal{L}(\mathbf{C}^4)$ by

$$T(z_1, z_2, z_3, z_4) = (7z_1, 7z_2, 8z_3, 8z_4).$$

Then $\operatorname{null}(T - 7I)$ is the two-dimensional subspace

$$\{(z_1, z_2, 0, 0) : z_1, z_2 \in \mathbf{C}\}$$

and $\operatorname{null}(T - 8I)$ is the two-dimensional subspace

$$\{(0, 0, z_3, z_4) : z_3, z_4 \in \mathbf{C}\}.$$

Thus 7 is an eigenvalue of $T$ with multiplicity at least 2 and 8 is an eigenvalue of $T$ with multiplicity at least 2. Because $2+2 = 4 = \dim \mathbf{C}^4$, there can be no other eigenvalues of $T$ and the eigenvalues 7 and 8 must have multiplicity 2 (by 8.18). Thus the characteristic polynomial of $T$ equals $(z-7)^2(z-8)^2$.

Of course there are also many other examples.

15. Suppose $V$ is a complex vector space. Suppose $T \in \mathcal{L}(V)$ is such that 5 and 6 are eigenvalues of $T$ and that $T$ has no other eigenvalues. Prove that

$$(T - 5I)^{n-1}(T - 6I)^{n-1} = 0,$$

where $n = \dim V$.

SOLUTION: Because 5 and 6 are eigenvalues of $T$ and $T$ has no other eigenvalues, the characteristic polynomial of $T$ must be of the form

$$(z-5)^{d_1}(z-6)^{d_2},$$

where $1 \leq d_1$ and $1 \leq d_2$. Because $d_1 + d_2 = n$, we must also have $d_1 \leq n-1$ and $d_2 \leq n-1$. The Cayley-Hamilton theorem implies that

$$(T - 5I)^{d_1}(T - 6I)^{d_2} = 0.$$

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Because $d_1 \le n-1$ and $d_2 \le n-1$, we can multiply the equation above by appropriate powers of $T-5I$ and $T-6I$ to get $(T-5I)^{n-1}(T-6I)^{n-1}=0$.

16. Suppose $V$ is a complex vector space and $T \in \mathcal{L}(V)$. Prove that $V$ has a basis consisting of eigenvectors of $T$ if and only if every generalized eigenvector of $T$ is an eigenvector of $T$.

COMMENT: For complex vector spaces, this exercise adds another equivalence to the list given by 5.21.

SOLUTION: First suppose that $V$ has a basis consisting of eigenvectors of $T$. Thus there exists a basis $(v_1,\dots,v_n)$ of $V$ and $\lambda_1,\dots,\lambda_n \in \mathbf{C}$ such that $Tv_j=\lambda_j v_j$ for each $j$. Suppose $v \in V$ is a generalized eigenvector of $T$ corresponding to an eigenvalue $\lambda$. Because $(v_1,\dots,v_n)$ is a basis of $V$, there exist $a_1,\dots,a_n \in \mathbf{C}$ such that

$$v = a_1v_1 + \cdots + a_nv_n.$$

Thus

$$(T-\lambda I)v = (\lambda_1-\lambda)a_1v_1 + \cdots + (\lambda_n-\lambda)a_nv_n.$$

Applying $T-\lambda I$ repeatedly to both sides of this equation, we get

$$(T-\lambda I)^n v = (\lambda_1-\lambda)^n a_1v_1 + \cdots + (\lambda_n-\lambda)^n a_nv_n.$$

The left side of the equation above equals 0 (because $v$ is a generalized eigenvector of $T$ corresponding to the eigenvalue $\lambda$). Thus the right side of the equation above equals 0. Thus $(\lambda_j-\lambda)^n a_j = 0$ for each $j$. This implies that the indices $j$ such that $a_j \ne 0$ must all have all satisfy $\lambda_j=\lambda$, which means that $v$ is a linear combination of the $v_j$'s that correspond to eigenvalue $\lambda$, which means that $v$ is itself an eigenvector corresponding to eigenvalue $\lambda$, as desired.

To prove the other direction, now suppose that every generalized eigenvector of $T$ is an eigenvector of $T$. By 8.25, there exists a basis of $V$ consisting of generalized eigenvectors of $T$. Because every generalized eigenvector of $T$ is an eigenvector of $T$, this gives a basis of $V$ consisting of eigenvectors of $T$, as desired.

17. Suppose $V$ is an inner-product space and $N \in \mathcal{L}(V)$ is nilpotent. Prove that there exists an orthonormal basis of $V$ with respect to which $N$ has an upper-triangular matrix.

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SOLUTION: By 8.26, there is a basis of $V$ with respect to which $N$ has an upper-triangular matrix. By 6.27, this basis can be chosen to be orthonormal. (If $V$ is a complex inner-product space, then we can just use 6.28 directly, so the hypothesis that $N$ is nilpotent is not needed.)

18. Define $N \in \mathcal{L}(\mathbf{F}^5)$ by

$$N(x_1, x_2, x_3, x_4, x_5) = (2x_2, 3x_3, -x_4, 4x_5, 0).$$

Find a square root of $I + N$.

SOLUTION: Note that

$$\begin{aligned}
N^2(x_1, x_2, x_3, x_4, x_5) &= (6x_3, -3x_4, -4x_5, 0, 0) \\
N^3(x_1, x_2, x_3, x_4, x_5) &= (-6x_4, -12x_5, 0, 0, 0) \\
N^4(x_1, x_2, x_3, x_4, x_5) &= (-24x_5, 0, 0, 0, 0) \\
N^5(x_1, x_2, x_3, x_4, x_5) &= (0, 0, 0, 0, 0).
\end{aligned}$$

Because $N^5 = 0$, the proof of 8.30 shows that

$$I + \frac{1}{2}N - \frac{1}{8}N^2 + \frac{1}{16}N^3 - \frac{5}{128}N^4$$

is a square root of $I + N$. Using the formulas above, we calculate that the operator $S \in \mathcal{L}(\mathbf{F}^5)$ defined by

$$\begin{aligned}
&S(x_1, x_2, x_3, x_4, x_5) = \\
&\Big(x_1 + x_2 - \frac{3x_3}{4} - \frac{3x_4}{8} + \frac{15x_5}{16},\ x_2 + \frac{3x_3}{2} + \frac{3x_4}{8} - \frac{3x_5}{4}, \\
&\ x_3 - \frac{x_4}{2} + \frac{x_5}{2},\ x_4 + 2x_5,\ x_5\Big)
\end{aligned}$$

is a square root of $I + N$.

19. Prove that if $V$ is a complex vector space, then every invertible operator on $V$ has a cube root.

SOLUTION: First suppose that $N \in \mathcal{L}(V)$ is nilpotent. We will show that $I + N$ has a cube root by imitating the proof of 8.30. Specifically, we guess that there is a cube root of $I + N$ of the form

$$I + a_1 N + a_2 N^2 + \cdots + a_{m-1} N^{m-1},$$

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where $m$ is such that $N^m = 0$. Having made this guess, we can try to choose $a_1, a_2, \ldots, a_{m-1}$ so that the operator above has its cube equal to $I+N$. Now

$$\begin{aligned}(I + a_1N + a_2N^2 + a_3N^3 + \cdots + a_{m-1}N^{m-1})^3 \\ = I + 3a_1N + (3a_2 + 3{a_1}^2)N^2 + (3a_3 + 6a_1a_2 + {a_1}^3)N^3 + \cdots \\ + (3a_{m-1} + \text{terms involving } a_1, \ldots, a_{m-2})N^{m-1}.\end{aligned}$$

We want the right side of the equation above to equal $I + N$. Hence choose $a_1$ so that $3a_1 = 1$ (thus $a_1 = 1/3$). Next, choose $a_2$ so that $3a_2 + 3{a_1}^2 = 0$ (thus $a_2 = -1/9$). Then choose $a_3$ so that the coefficient of $N^3$ on the right side of the equation above equals 0 (thus $a_3 = 5/81$). Continue in this fashion for $j = 4, \ldots, m-1$, at each step solving for $a_j$ so that the coefficient of $N^j$ on the right side of the equation above equals 0. Actually we don't care about the explicit formula for the $a_j$'s. We need only know that some choice of the $a_j$'s gives a cube root of $I + N$.

Having shown that the identity plus a nilpotent always has a cube root, we now look at the proof that every invertible operator on a complex vector space has a square root (see 8.32). In that proof, if we replace the words "square root" with "cube root", we get a proof that every invertible operator on a complex vector space has a cube root.

20. Suppose $T \in \mathcal{L}(V)$ is invertible. Prove that there exists a polynomial $p \in \mathcal{P}(\mathbf{F})$ such that $T^{-1} = p(T)$.

SOLUTION: Let $a_0 + a_1z + \cdots + a_{m-1}z^{m-1} + z^m$ denote the minimal polynomial of $T$. This is the monic polynomial of smallest degree such that

$$a_0I + a_1T + \cdots + a_{m-1}T^{m-1} + T^m = 0.$$

If $a_0$ were equal to 0, then we could multiply both sides of the equation above by $T^{-1}$ to get

$$a_1I + a_2T + \cdots + a_{m-1}T^{m-2} + T^{m-1} = 0,$$

which would give a monic polynomial $q$ of smaller degree such that $q(T) = 0$, contradicting the definition of the minimal polynomial. Thus $a_0 \neq 0$.

Because $a_0 \neq 0$, we can solve the first equation above for the identity operator $I$, getting

$$I = -\frac{a_1}{a_0}T - \cdots - \frac{a_{m-1}}{a_0}T^{m-1} - \frac{1}{a_0}T^m.$$

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Now multiply both sides of the equation above by $T^{-1}$, getting

$$T^{-1} = -\frac{a_1}{a_0}I - \frac{a_2}{a_0}T - \cdots - \frac{a_{m-1}}{a_0}T^{m-2} - \frac{1}{a_0}T^{m-1}.$$

Setting

$$p(z) = -\frac{a_1}{a_0} - \frac{a_2}{a_0}z - \cdots - \frac{a_{m-1}}{a_0}z^{m-2} - \frac{1}{a_0}z^{m-1},$$

we thus have $T^{-1} = p(T)$.

21. Give an example of an operator on $\mathbf{C}^3$ whose minimal polynomial equals $z^2$.

SOLUTION: Define $T \in \mathcal{L}(\mathbf{C}^3)$ by

$$T(w_1, w_2, w_3) = (w_3, 0, 0).$$

Clearly $T^2 = 0$. In other words, the polynomial $z^2$ when applied to $T$ gives 0. Thus the minimal polynomial of $T$ is a divisor of $z^2$ (by 8.34). But the only monic polynomials that divide $z^2$ are 1, $z$, and $z^2$. The polynomial 1 applied to $T$ gives the identity operator, which is not 0, and the polynomial $z$ applied to $T$ gives $T$, which is also not 0. Thus the minimal polynomial of $T$ must be $z^2$.

Of course there are also many other examples.

22. Give an example of an operator on $\mathbf{C}^4$ whose minimal polynomial equals $z(z-1)^2$.

SOLUTION: Define $T \in \mathcal{L}(\mathbf{C}^4)$ by

$$T(w_1, w_2, w_3, w_4) = (0, w_2 + w_4, w_3, w_4).$$

Then

$$(T - I)(w_1, w_2, w_3, w_4) = (-w_1, w_4, 0, 0),$$

which implies that

$$(T - I)^2(w_1, w_2, w_3, w_4) = (w_1, 0, 0, 0),$$

which implies that

$$T(T - I)^2 = 0.$$

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In other words, the polynomial $z(z-1)^2$ when applied to $T$ gives 0. Thus the minimal polynomial of $T$ is a divisor of $z(z-1)^2$ (by 8.34).

Note that 0 is an eigenvalue of $T$ because $T(1,0,0,0) = (0,0,0,0)$ and 1 is an eigenvalue of $T$ because $T(0,1,0,0) = (0,1,0,0)$. Thus 0 and 1 must both be roots of the minimal polynomial of $T$ (by 8.36).

The only monic polynomials that divide $z(z-1)^2$ and have $0, 1$ as roots are $z(z-1)$ and $z(z-1)^2$. Because $T(T-I) \neq 0$, as is easy to check, this implies that $z(z-1)^2$ is the minimal polynomial of $T$.

Of course there are also many other examples.

23. Suppose $V$ is a complex vector space and $T \in \mathcal{L}(V)$. Prove that $V$ has a basis consisting of eigenvectors of $T$ if and only if the minimal polynomial of $T$ has no repeated roots.

COMMENT: For complex vector spaces, this exercise adds another equivalence to the list given by 5.21.

SOLUTION: First suppose that there is a basis $(v_1, \ldots, v_n)$ of $V$ consisting of eigenvectors of $T$. Let $\lambda_1, \ldots, \lambda_m$ be the distinct eigenvalues of $T$. Then for each $v_j$, there exists $\lambda_k$ with $(T - \lambda_k I)v_j = 0$. Thus

$$(T - \lambda_1 I) \ldots (T - \lambda_m I)v_j = 0$$

for each $j$ (because all the operators in sight commute, for each $j$ the appropriate $T - \lambda_k I$ can be moved to the last position in the product above). An operator that sends each vector in a basis to the 0 vector is the 0 operator, so

$$(T - \lambda_1 I) \ldots (T - \lambda_m I) = 0.$$

Thus the polynomial $(z - \lambda_1) \ldots (z - \lambda_m)$ when applied to $T$ gives 0. Thus the minimal polynomial of $T$ is a divisor of $(z - \lambda_1) \ldots (z - \lambda_m)$ (by 8.34). Because $(z - \lambda_1) \ldots (z - \lambda_m)$ has no repeated roots, this implies that the minimal polynomial of $T$ has no repeated roots, as desired. (Because each eigenvalue of $T$ must be a root of the minimal polynomial of $T$, the minimal polynomial of $T$ actually equals $(z - \lambda_1) \ldots (z - \lambda_m)$.)

To prove the implication in the other direction, now suppose that the minimal polynomial of $T$ has no repeated roots. Letting $\lambda_1, \ldots, \lambda_m$ denote the distinct eigenvalues of $T$, this means that the minimal polynomial of $T$ equals

$$(z - \lambda_1) \ldots (z - \lambda_m).$$

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Thus

$$(T - \lambda_1 I) \dots (T - \lambda_m I) = 0.$$

Let $U_m$ be the subspace of generalized eigenvectors corresponding to the eigenvalue $\lambda_m$. Recall that $U_m$ is invariant under $T$ (see 8.23(b)). Suppose $v \in U_m$. Let $u = (T - \lambda_m I)v$, so $u \in U_m$. The equation above implies that

$$\begin{aligned}(T|_{U_m} - \lambda_1 I) \dots (T|_{U_m} - \lambda_{m-1} I)u &= (T - \lambda_1 I) \dots (T - \lambda_m I)v \\ &= 0.\end{aligned}$$

Because $(T - \lambda_m I)|_{U_m}$ is nilpotent (see sec 8.23(c)), 0 is the only eigenvalue of $(T - \lambda_m I)|_{U_m}$ (this follows from 8.26 and 5.18). Thus $T|_{U_m} - \lambda_j I$ is invertible (as an operator on $U_m$) for $j = 1, \dots, m-1$. The equation above thus implies that $u = 0$. In other words, $v$ is an eigenvector of $T$.

We have shown that every generalized eigenvector of $T$ corresponding to the eigenvalue $\lambda_m$ is an eigenvector of $T$. There is nothing special about the eigenvalue $\lambda_m$—we could have relabeled the eigenvalues so that any of them was called $\lambda_m$. Thus we can conclude that every generalized eigenvector of $T$ is actually an eigenvector of $T$. Because there is a basis of $V$ consisting of generalized eigenvectors of $T$ (see 8.25), this means that there is a basis of $V$ consisting of eigenvectors of $T$, as desired.

24. Suppose $V$ is an inner-product space. Prove that if $T \in \mathcal{L}(V)$ is normal, then the minimal polynomial of $T$ has no repeated roots.

SOLUTION: Suppose $T \in \mathcal{L}(V)$ is normal. Let $p$ denote the minimal polynomial of $T$. Suppose $\lambda \in \mathbf{F}$ is an eigenvalue of $T$. We can write

$$p(z) = (z - \lambda)^m q(z)$$

where $m$ is a positive integer and $q$ is a monic polynomial such that $q(\lambda) \neq 0$. Our goal is to prove that $m = 1$, which implies that $p$ has no repeated roots.

Because $p(T) = 0$, we know that $(T - \lambda I)^m q(T) = 0$. This is equivalent to the statement that

$$\operatorname{range} q(T) \subset \operatorname{null}(T - \lambda I)^m.$$

Because $T$ is normal, so is $T - \lambda I$, and thus $\operatorname{null}(T - \lambda I)^m = \operatorname{null}(T - \lambda I)$ (by Exercise 7 in Chapter 7). Hence the set inclusion above becomes

$$\operatorname{range} q(T) \subset \operatorname{null}(T - \lambda I),$$

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which implies that $(T-\lambda I)q(T) = 0$. Letting $p_1(z) = (z-\lambda)q(z)$, this means that $p_1$ is a monic polynomial with the property that $p_1(T) = 0$. If $m > 1$, then the degree of $p_1$ would be less than the degree of $p$, contradicting the definition of minimal polynomial. Thus we can conclude that $m = 1$, as desired.

COMMENT: The proof given above works on both real and complex vector spaces. If $V$ is a complex vector space, then this exercise can be done by using the complex spectral theorem (7.9) and Exercise 23 of this chapter.

25. Suppose $T \in \mathcal{L}(V)$ and $v \in V$. Let $p$ be the monic polynomial of smallest degree such that

$$p(T)v = 0.$$

Prove that $p$ divides the minimal polynomial of $T$.

SOLUTION: Let $q$ denote the minimal polynomial of $T$. By the division algorithm (4.5), there exist polynomials $s, r \in \mathcal{P}(\mathbf{F})$ such that

$$q = sp + r$$

and $\deg r < \deg p$. Thus

$$q(T)v = s(T)p(T)v + r(T)v.$$

Because $q(T) = 0$ and $p(T)v = 0$, the equation above shows that $r(T)v = 0$. This implies that $r = 0$ (otherwise we could multiply $r$ by a scalar to get a monic polynomial with degree smaller than $\deg p$ that when applied to $T$ gives an operator having $v$ in its null space, which would contract the definition of $p$ as the monic polynomial of smallest degree with this property).

Using the information that $r = 0$, rewrite the formula above for $q$ as

$$q = sp.$$

Thus $p$ divides $q$, the minimal polynomial of $T$.

26. Give an example of an operator on $\mathbf{C}^4$ whose characteristic and minimal polynomials both equal $z(z-1)^2(z-3)$.

SOLUTION: Define $T \in \mathcal{L}(\mathbf{C}^4)$ by

$$T(w_1, w_2, w_3, w_4) = (0, w_2 + w_3, w_3, 3w_4).$$

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An easy computation shows that $T(T-I)^2(T-3I) = 0$. Thus the minimal polynomial of $T$ is a divisor of $z(z-1)^2(z-3)$ (by 8.34).

Note that 0 is an eigenvalue of $T$ because $T(1,0,0,0) = (0,0,0,0)$ and 1 is an eigenvalue of $T$ because $T(0,1,0,0) = (0,1,0,0)$ and 3 is an eigenvalue of $T$ because $T(0,0,0,1) = (0,0,0,3)$. Thus 0, 1, and 3 must be roots of the minimal polynomial of $T$ (by 8.36).

The only monic polynomials that divide $z(z-1)^2(z-3)$ and have $0, 1, 3$ as roots are $z(z-1)(z-3)$ and $z(z-1)^2(z-3)$. Because $T(T-1)(T-3) \neq 0$, as is easy to check, this implies that $z(z-1)^2(z-3)$ is the minimal polynomial of $T$, as desired.

Because $T$ is an operator on a four-dimensional complex vector space and the minimal polynomial of $T$ has degree 4, the characteristic polynomial of $T$ (which is a monic polynomial of degree 4 that is divisible by the minimal polynomial of $T$) must equal the minimal polynomial of $T$.

Of course there are also many other examples.

27. Give an example of an operator on $\mathbf{C}^4$ whose characteristic polynomial equals $z(z-1)^2(z-3)$ and whose minimal polynomial equals $z(z-1)(z-3)$.

SOLUTION: Define $T \in \mathcal{L}(\mathbf{C}^4)$ by

$$T(w_1, w_2, w_3, w_4) = (0, w_2, w_3, 3w_4).$$

An easy computation shows that $T(T-I)(T-3I) = 0$. Thus the minimal polynomial of $T$ is a divisor of $z(z-1)(z-3)$ (by 8.34).

Note that 0 is an eigenvalue of $T$ because $T(1,0,0,0) = (0,0,0,0)$ and 1 is an eigenvalue of $T$ because $T(0,1,0,0) = (0,1,0,0)$ and 3 is an eigenvalue of $T$ because $T(0,0,0,1) = (0,0,0,3)$. Thus 0, 1, and 3 must be roots of the minimal polynomial of $T$ (by 8.36).

The only monic polynomial that is a divisor of $z(z-1)(z-3)$ and has $0, 1, 3$ as roots is $z(z-1)(z-3)$. Thus $z(z-1)(z-3)$ is the minimal polynomial of $T$, as desired.

Note that every vector in $\{(0, w_2, w_3, 0) : w_2, w_3 \in \mathbf{C}\}$ is an eigenvector of $T$ corresponding to the eigenvalue 1. Thus the eigenvalue 1 of $T$ has multiplicity at least 2. The eigenvalues 0 and 3 of $T$ have multiplicity at least 1.

Because $T$ is an operator on a four-dimensional complex vector space, the sum of the multiplicities of all the eigenvalues equals 4 (by 8.18). Thus the each use of the phrase "at least" in the previous paragraph can be replaced by "equal" because if any of the eigenvalues had larger multiplicity, the sum of the multiplicities of all the eigenvalues would exceed 4.

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Because 0 and 3 are eigenvalues of $T$ with multiplicity 1 and 1 is an eigenvalue of $T$ with multiplicity 2 and $T$ has no other eigenvalues (the multiplicities of the eigenvalues mentioned already sum to 4), the characteristic polynomial of $T$ equals $z(z-1)^2(z-3)$, as desired.

Of course there are also many other examples.

28. Suppose $a_0, \dots, a_{n-1} \in \mathbf{C}$. Find the minimal and characteristic polynomials of the operator on $\mathbf{C}^n$ whose matrix (with respect to the standard basis) is

$$\begin{bmatrix} 0 & & & & -a_0 \\ 1 & 0 & & & -a_1 \\ & 1 & \ddots & & -a_2 \\ & & \ddots & & \vdots \\ & & & 0 & -a_{n-2} \\ & & & 1 & -a_{n-1} \end{bmatrix}.$$

COMMENT: This exercise shows that every monic polynomial is the characteristic polynomial of some operator.

SOLUTION: Suppose that $T \in \mathcal{L}(\mathbf{C}^n)$ has matrix as above with respect to the standard basis $(e_1, \dots, e_n)$ of $\mathbf{C}^n$. Thus

$$\begin{aligned} Te_1 &= e_2 \\ T^2e_1 &= Te_2 = e_3 \\ &\vdots \\ T^{n-1}e_1 &= Te_{n-1} = e_n \\ T^ne_1 &= Te_n = -a_0e_1 - a_1e_2 - \cdots - a_{n-1}e_n. \end{aligned}$$

Thus

$$(e_1, Te_1, T^2e_1, \dots, T^{n-1}e_1) \doteq (e_1, e_2, e_3, \dots, e_n).$$

In particular, $(e_1, Te_1, T^2e_1, \dots, T^{n-1}e_1)$ is linearly independent. Thus if $p$ is a monic polynomial with degree less than $n$, then $p(T)e_1 \neq 0$. Thus the minimal polynomial of $T$ must have degree $n$.

Writing $T^ne_1$ as a linear combination of $(e_1, Te_1, T^2e_1, \dots, T^{n-1}e_1)$ is possible in only one way:

$$T^ne_1 = -a_0e_1 - a_1Te_1 - \cdots - a_{n-1}T^{n-1}e_1.$$

Thus setting

$$p(z) = a_0 + a_1 z + \cdots + a_{n-1} z^{n-1} + z^n,$$

we see that $p$ is the only monic polynomial of degree $n$ such that $p(T)e_1 = 0$. Hence $p$ must equal the minimal polynomial of $T$.

Because $T$ is an operator on an $n$-dimensional complex vector space and the minimal polynomial of $T$ has degree $n$, the characteristic polynomial of $T$ (which is a monic polynomial of degree $n$ that is divisible by the minimal polynomial of $T$) must equal the minimal polynomial of $T$. Thus the characteristic polynomial of $T$ also equals $p$.

29. Suppose $N \in \mathcal{L}(V)$ is nilpotent. Prove that the minimal polynomial of $N$ is $z^{m+1}$, where $m$ is the length of the longest consecutive string of 1's that appears on the line directly above the diagonal in the matrix of $N$ with respect to any Jordan basis for $N$.

SOLUTION: Suppose $(v_1, \ldots, v_n)$ is a Jordan basis for $N$ and that $m$ equals the length of the longest consecutive string of 1's that appears on the line directly above the diagonal of $\mathcal{M}\big(N, (v_1, \ldots, v_n)\big)$. The diagonal of $\mathcal{M}\big(N, (v_1, \ldots, v_n)\big)$ contains only 0's (by Exercise 6 of this chapter). Thus $\mathcal{M}\big(N, (v_1, \ldots, v_n)\big)$ is a block diagonal matrix whose blocks have the form

$$\begin{bmatrix} 0 & 1 & & 0 \\ & \ddots & \ddots & \\ & & \ddots & 1 \\ 0 & & & 0 \end{bmatrix},$$

and the largest such block in an $(m+1)$-by-$(m+1)$ matrix. For each $v_j$, we see that $N^{m+1} v_j = 0$. Because $N^{m+1}$ equals 0 on a basis of $V$, we conclude that $N^{m+1} = 0$. Thus the minimal polynomial of $N$ must divide $z^{m+1}$ (by 8.34) and hence must be of the form $z^k$ for some $k \le m+1$. But there is a basis vector $v_j$ such that $N^m v_j = v_{j-m} \ne 0$. Thus $N^m \ne 0$, which implies that the minimal polynomial of $N$ equals $z^{m+1}$.

30. Suppose $V$ is a complex vector space and $T \in \mathcal{L}(V)$. Prove that there does not exist a direct sum decomposition of $V$ into two proper subspaces invariant under $T$ if and only if the minimal polynomial of $T$ is of the form $(z - \lambda)^{\dim V}$ for some $\lambda \in \mathbf{C}$.

SOLUTION: First suppose that there does not exist a direct sum decomposition of $V$ into two proper subspaces invariant under $T$. Thus $T$ has only one eigenvalue (by 8.23), which we will call $\lambda$.

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There is a Jordan basis of $T$ (by 8.47), meaning that with respect to this basis $T$ has a block diagonal matrix

$$\begin{bmatrix} A_1 & & 0 \\ & \ddots & \\ 0 & & A_m \end{bmatrix},$$

where each $A_j$ is an upper-triangular matrix of the form

$$A_j = \begin{bmatrix} \lambda & 1 & & 0 \\ & \ddots & \ddots & \\ & & \ddots & 1 \\ 0 & & & \lambda \end{bmatrix}.$$

If $m > 1$, then we could let $U$ denote the span of the basis vectors corresponding to $A_1$ and let $W$ denote the span of the basis vectors corresponding to $U_2, \ldots, U_m$; we would have $V = U \oplus W$, where $U$ and $W$ would be proper subspaces of $V$ invariant under $T$. This contradiction shows that $m = 1$. Thus the matrix of $T$ with respect to our Jordan basis is just $A_1$. In other words, there is a basis $(v_1, \ldots, v_{\dim V})$ of $V$ such that the matrix of $T - \lambda I$ with respect to this basis is

$$\begin{bmatrix} 0 & 1 & & 0 \\ & \ddots & \ddots & \\ & & \ddots & 1 \\ 0 & & & 0 \end{bmatrix}.$$

The previous problem now implies that the minimal polynomial of $T - \lambda I$ equals $z^{\dim V}$. This clearly implies that the minimal polynomial of $T$ equals $(z - \lambda)^{\dim V}$, as desired.

To prove the implication in the other direction, suppose that the minimal polynomial of $T$ equals $(z - \lambda)^{\dim V}$. Suppose that there exist two proper subspaces $U_1, U_2$ of $V$, each invariant under $T$, such that $V = U_1 \oplus U_2$.

Let $p_1$ denote the minimal polynomial of $T|_{U_1}$ and $p_2$ denote the minimal polynomial of $T|_{U_2}$. If $u_1 \in U_1$, then

$$\begin{aligned}(p_1 p_2)(T)u_1 &= p_2(T)p_1(T)u_1 \\ &= 0.\end{aligned}$$

Similarly, if $u_2 \in U_2$, then

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$$\begin{aligned}(p_1p_2)(T)u_2 &= p_1(T)p_2(T)u_2\\ &= 0.\end{aligned}$$

Because every vector in $V$ can be written in the form $u_1+u_2$, where $u_1 \in U_1$ and $u_2 \in U_2$, the equations above imply that $(p_1p_2)(T) = 0$. Thus the minimal polynomial of $T$, which equals $(z-\lambda)^{\dim V}$, is a divisor of $p_1p_2$ (by 8.34).

The degree of the monic polynomial $p_1p_2$ equals the degree of $p_1$ plus the degree of $p_2$, which is less than or equal to $\dim U_1 + \dim U_2$, which equals $\dim V$. Because $(z-\lambda)^{\dim V}$ is a divisor of $p_1p_2$, this implies that

$$p_1(z)p_2(z) = (z-\lambda)^{\dim V}.$$

This implies that

$$p_1(z) = (z-\lambda)^{\dim U_1} \quad \text{and} \quad p_2(z) = (z-\lambda)^{\dim U_2}.$$

Let $m = \max\{\dim U_1, \dim U_2\}$. Let $p(z) = (z-\lambda)^m$. The equations above show that $p(T)|_{U_1} = 0$ and $p(T)|_{U_2} = 0$. Thus $p(T) = 0$. Because the degree of $p$ is less than $\dim V$, this contradicts our hypothesis that $(z-\lambda)^{\dim V}$ is the minimal polynomial of $T$. This contradiction means that our assumption that there exists a direct sum decomposition of $V$ into two proper subspaces invariant under $T$ must have been false, as desired.

31. Suppose $T \in \mathcal{L}(V)$ and $(v_1, \dots, v_n)$ is a basis of $V$ that is a Jordan basis for $T$. Describe the matrix of $T$ with respect to the basis $(v_n, \dots, v_1)$ obtained by reversing the order of the $v$'s.

SOLUTION: The matrix of $T$ with respect to $(v_1, \dots, v_n)$ is a block diagonal matrix

$$\begin{bmatrix} A_1 & & 0 \\ & \ddots & \\ 0 & & A_m \end{bmatrix},$$

where each $A_j$ is an upper-triangular matrix of the form

$$A_j = \begin{bmatrix} \lambda_j & 1 & & 0 \\ & \ddots & \ddots & \\ & & \ddots & 1 \\ 0 & & & \lambda_j \end{bmatrix}.$$

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Temporarily fix $j$, and let $(u_1, \dots, u_k)$ be the part of $(v_1, \dots, v_n)$ corresponding to the block $A_j$. Thus

$$\begin{aligned} Tu_1 &= \lambda_j u_1 \\ Tu_2 &= u_1 + \lambda_j u_2 \\ Tu_3 &= u_2 + \lambda_j u_3 \\ &\vdots \\ Tu_k &= u_{k-1} + \lambda_j u_k. \end{aligned}$$

Write the equations above in the form

$$\begin{aligned} Tu_k &= \lambda_j u_k + u_{k-1} \\ Tu_{k-1} &= \lambda_j u_{k-1} + u_{k-2} \\ &\vdots \\ Tu_2 &= \lambda_j u_2 + u_1 \\ Tu_1 &= \lambda_j u_1. \end{aligned}$$

Thus the matrix of $T|_{\text{span}(u_1,\dots,u_k)}$ with respect to $(u_k, \dots, u_1)$ is

$$B_j = \begin{bmatrix} \lambda_j & & & 0 \\ 1 & \ddots & & \\ & \ddots & \ddots & \\ 0 & & 1 & \lambda_j \end{bmatrix}.$$

In other words, $B_j$ is obtained from $A_j$ by reflection across the diagonal, so in $B_j$ the 1's lie below the diagonal instead of above it. Now we see that the matrix of $T$ with respect to $(v_n, \dots, v_1)$ is the block diagonal matrix

$$\begin{bmatrix} B_m & & 0 \\ & \ddots & \\ 0 & & B_1 \end{bmatrix}.$$

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CHAPTER 9

# Operators on Real Vector Spaces

1. Prove that 1 is an eigenvalue of every square matrix with the property that the sum of the entries in each row equals 1.

SOLUTION: Suppose that $A$ is an $n$-by-$n$ matrix such that the sum of the entries in each row of $A$ equals 1. Let $x$ be the $n$-by-1 matrix all of whose entries equal 1. Then from the definition of matrix multiplication we see that the entry in row $j$, column 1 of $Ax$ equals that sum of the entries in row $j$ of $A$, which equals 1. Thus $Ax = x$, which implies that 1 is an eigenvalue of $A$.

2. Consider a 2-by-2 matrix of real numbers

$$A = \begin{bmatrix} a & c \\ b & d \end{bmatrix}.$$

Prove that $A$ has an eigenvalue (in $\mathbf{R}$) if and only if

$$(a-d)^2 + 4bc \geq 0.$$

SOLUTION: A number $\lambda \in \mathbf{R}$ is an eigenvalue of $A$ if and only if there exist numbers $x, y \in \mathbf{R}$, not both 0, such that

$$\begin{bmatrix} a & c \\ b & d \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \lambda \begin{bmatrix} x \\ y \end{bmatrix}.$$

The left side of this equation equals $\begin{bmatrix} ax+cy \\ bx+dy \end{bmatrix}$, so the equation above is equivalent to the system of equations



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$$\begin{aligned}(a-\lambda)x + \quad cy &= 0\\ bx + (d-\lambda)y &= 0.\end{aligned}$$

It is easy to see that this system of equations has a solution other than $x = y = 0$ if and only if

$$(a-\lambda)(d-\lambda) = bc,$$

which is equivalent to the equation

$$\lambda^2 - (a+d)\lambda + (ad-bc) = 0.$$

There is a real number $\lambda$ satisfying the equation above if and only if

$$(a+d)^2 - 4(ad-bc) \geq 0.$$

The left side of the inequality above equals $(a-d)^2 + 4bc$, and thus we conclude that $A$ has a real eigenvalue if and only if

$$(a-d)^2 + 4bc \geq 0.$$

3. Suppose $A$ is a block diagonal matrix

$$A = \begin{bmatrix} A_1 & & 0 \\ & \ddots & \\ 0 & & A_m \end{bmatrix},$$

where each $A_j$ is a square matrix. Prove that the set of eigenvalues of $A$ equals the union of the eigenvalues of $A_1, \ldots, A_m$.

SOLUTION: Suppose that $A_j$ has size $n_j$-by-$n_j$. Let

$$x = \begin{bmatrix} x_1 \\ \vdots \\ x_m \end{bmatrix},$$

where each $x_j$ is an $n_j$-by-1 matrix. Then

$$Ax = \begin{bmatrix} A_1x_1 \\ \vdots \\ A_mx_m \end{bmatrix}.$$

Thus the equation $Ax = \lambda x$ is equivalent to the system of equations

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$$A_1 x_1 = \lambda x_1$$
$$\vdots$$
$$A_m x_m = \lambda x_m.$$

Suppose that $\lambda$ is an eigenvalue of $A$. Then there is a nonzero vector $x$ satisfying the system of equations above. Because $x$ is nonzero, there exists $k$ such that $x_k$ is nonzero. Because $A_k x_k = \lambda x_k$, this implies that $\lambda$ is an eigenvalue of $A_k$. Thus $\lambda$ is in the union of the eigenvalues of $A_1, \ldots, A_m$, as desired.

To prove the implication in the other direction, suppose now that $\lambda$ is in the union of the eigenvalues of $A_1, \ldots, A_m$. Then there exists $k$ such that $\lambda$ is an eigenvalue of $A_k$. Thus there exists a nonzero $n_k$-by-1 vector $x_k$ such that $A_k x_k = \lambda x_k$. For $j \neq k$, define $x_j$ to be the $n_j$-by-1 matrix all of whose entries equal 0, and define $x$ to be the matrix determined by $x_1, \ldots, x_m$ as above. Then $x$ is nonzero (because $x_k$ is nonzero) and $Ax = \lambda x$, which shows that $\lambda$ is an eigenvalue of $A$, as desired.

4. Suppose $A$ is a block upper-triangular matrix

$$A = \begin{bmatrix} A_1 & & * \\ & \ddots & \\ 0 & & A_m \end{bmatrix},$$

where each $A_j$ is a square matrix. Prove that the set of eigenvalues of $A$ equals the union of the eigenvalues of $A_1, \ldots, A_m$.

COMMENT: Clearly Exercise 4 is a stronger statement than Exercise 3. Even so, students may want to do Exercise 3 first because it is easier than Exercise 4.

SOLUTION: We will prove that 0 is an eigenvalue of $A$ if and only if 0 is an eigenvalue of at least one of the $A_k$'s. This is all we need to do, because for arbitrary $\lambda \in \mathbf{F}$, we can replace $A$ with $A - \lambda I$ and each $A_k$ with $A_k - \lambda I$, concluding that $\lambda$ is an eigenvalue of $A$ if and only if $\lambda$ is an eigenvalue of at least one of the $A_k$'s. This last statement implies that the set of eigenvalues of $A$ equals the union of the eigenvalues of $A_1, \ldots, A_m$.

Suppose that $A$ has size $n$-by-$n$ and each $A_j$ has size $n_j$-by-$n_j$. We can write a typical $n$-by-1 matrix $x$ in the form

$$x = \begin{bmatrix} x_1 \\ \vdots \\ x_m \end{bmatrix},$$

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where each $x_j$ is an $n_j$-by-1 matrix. The product $Ax$ can be computed by multiplying together the block matrices of $A$ and $x$ given above, using the same formula as one would use when multiplying matrices of numbers (this follows from the definition of matrix multiplication).

First suppose that 0 is an eigenvalue of $A$. Thus there exists a nonzero $n$-by-1 matrix $x$ such that $Ax = 0$. Write $x$ in the form above, and let $k$ be the largest index with nonzero $x_k$; thus we can write

$$x = \begin{bmatrix} x_1 \\ \vdots \\ x_k \\ 0 \\ \vdots \\ 0 \end{bmatrix}.$$

(If $k = m$, then the 0's shown above at the tail of $x$ do not appear.) If we break $Ax$ into blocks of the same size as was done for $x$, then the $k^{\text{th}}$ block of $Ax$ will equal $A_k x_k$; this follows from the block upper-triangular form of $A$ and the 0's that appear in $x$ after the $k^{\text{th}}$ block. But $Ax = 0$, so the $k^{\text{th}}$ block of $Ax$ equals 0, so $A_k x_k = 0$. Because $x_k \neq 0$, this implies that 0 is an eigenvalue of $A_k$, as desired.

To prove the implication in the other direction, suppose now that 0 is an eigenvalue of some $A_k$. This means that the operator on $\text{Mat}(n, 1, \mathbf{F})$ (the vector space of $n$-by-1 matrices) that sends $x_k \in \text{Mat}(n, 1, \mathbf{F})$ to $A_k x_k$ is not injective. Thus this operator is not surjective (by 3.21). Thus the operator on $\text{Mat}(n_1 + \cdots + n_k, 1, \mathbf{F})$ that sends

$$\begin{bmatrix} x_1 \\ \vdots \\ x_k \end{bmatrix}$$

to

$$\begin{bmatrix} A_1 & & * \\ & \ddots & \\ 0 & & A_k \end{bmatrix} \begin{bmatrix} x_1 \\ \vdots \\ x_k \end{bmatrix}$$

is not surjective (because the last block in the product above will be $A_k x_k$, which cannot be an arbitrary $n_k$-by-1 matrix). Again using 3.21, this means that the last operator is not injective. In other words, there exists a nonzero vector

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$$\begin{bmatrix} x_1 \\ \vdots \\ x_k \end{bmatrix} \in \mathrm{Mat}(n_1 + \cdots + n_k, 1, \mathbf{F})$$

such that

$$\begin{bmatrix} A_1 & & * \\ & \ddots & \\ 0 & & A_k \end{bmatrix} \begin{bmatrix} x_1 \\ \vdots \\ x_k \end{bmatrix} = 0.$$

Adjoining an appropriate number of 0's, this implies that

$$\begin{bmatrix} A_1 & & & & * \\ & \ddots & & & \\ & & A_k & & \\ & & & A_{k+1} & \\ & & & & \ddots \\ 0 & & & & A_m \end{bmatrix} \begin{bmatrix} x_1 \\ \vdots \\ x_k \\ 0 \\ \vdots \\ 0 \end{bmatrix} = 0.$$

In other words, 0 is an eigenvalue of $A$, as desired.

5. Suppose $V$ is a real vector space and $T \in \mathcal{L}(V)$. Suppose $\alpha, \beta \in \mathbf{R}$ are such that $T^2 + \alpha T + \beta I = 0$. Prove that $T$ has an eigenvalue if and only if $\alpha^2 \geq 4\beta$.

SOLUTION: First suppose that $T$ has an eigenvalue $\lambda \in \mathbf{R}$. Thus there exists a nonzero vector $v \in V$ such that $Tv = \lambda v$. Applying $T$ to both sides of the last equation, we get $T^2 v = \lambda^2 v$. Thus

$$\begin{aligned} 0 &= (T^2 + \alpha T + \beta I)v \\ &= \lambda^2 v + \alpha\lambda v + \beta v \\ &= (\lambda^2 + \alpha\lambda + \beta)v. \end{aligned}$$

Because $v \neq 0$, the last equation implies that

$$\lambda^2 + \alpha\lambda + \beta = 0,$$

which implies (recall that $\lambda$, $\alpha$, and $\beta$ are all real) that $\alpha^2 \geq 4\beta$, as desired.

To prove the implication in the opposite direction, suppose now that $\alpha^2 \geq 4\beta$. Then the polynomial $x^2 + \alpha x + \beta$ has two real roots, which means that we can write

$$x^2 + \alpha x + \beta = (x - r_1)(x - r_2)$$

for some $r_1, r_2 \in \mathbf{R}$. Thus

$$\begin{aligned} 0 &= T^2 + \alpha T + \beta I \\ &= (T - r_1 I)(T - r_2 I). \end{aligned}$$

In particular, $(T - r_1 I)(T - r_2 I)$ is not injective, which implies that at least one of $T - r_1 I$ and $T - r_2 I$ is not injective. In other words, at least one of $r_1, r_2$ must be an eigenvalue of $T$. Thus $T$ has an eigenvalue, as desired.

6. Suppose $V$ is a real inner-product space and $T \in \mathcal{L}(V)$. Prove that there is an orthonormal basis of $V$ with respect to which $T$ has a block upper-triangular matrix

$$\begin{bmatrix} A_1 & & * \\ & \ddots & \\ 0 & & A_m \end{bmatrix},$$

where each $A_j$ is a 1-by-1 matrix or a 2-by-2 matrix with no eigenvalues.

SOLUTION: We know that there is a basis $(v_1, \ldots, v_n)$ of $V$ with respect to which the matrix of $T$ has the block upper-triangular form above, where each $A_j$ is a 1-by-1 matrix or a 2-by-2 matrix with no eigenvalues (see 9.4).

Apply the Gram-Schmidt procedure to $(v_1, \ldots, v_n)$, getting an orthonormal basis $(e_1, \ldots, e_n)$ of $V$ such that

$$\operatorname{span}(v_1, \ldots, v_j) = \operatorname{span}(e_1, \ldots, e_j)$$

for $j = 1, \ldots, n$ (see 6.20). This condition on the spans implies that the matrix of $T$ with respect to $(e_1, \ldots, e_n)$ is also a block upper-triangular of the form above, where each $A_j$ is a 1-by-1 matrix or a 2-by-2 matrix (these $A_j$'s may differ from the $A_j$'s corresponding to the matrix of $T$ with respect to $(v_1, \ldots, v_n)$). All that remains is to show that, if necessary, we can modify our orthonormal basis so that none of the 2-by-2 blocks on the diagonal have eigenvalues.

Suppose that $A_j$ is a 2-by-2 block on the diagonal of the matrix of $T$ with respect to $(e_1, \ldots, e_n)$ and that $A_j$ has an eigenvalue $\lambda$. Thus there exist $x, y \in \mathbf{R}$, not both 0, such that

$$A_j \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} x \\ y \end{bmatrix}.$$

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Let $e_k, e_{k+1}$ denote the basis vectors corresponding to $A_j$. Then, as is easy to verify,

$$T(xe_k + ye_{k+1}) = u + \lambda(xe_k + ye_{k+1})$$

for some $u \in \text{span}(e_1, \ldots, e_{k-1})$. Let

$$f_k = \frac{xe_k + ye_{k+1}}{\|xe_k + ye_{k+1}\|}$$

and choose $f_{k+1} \in V$ such that $(f_k, f_{k+1})$ is orthonormal and

$$\text{span}(f_k, f_{k+1}) = \text{span}(e_k, e_{k+1}).$$

Then the matrix of $T$ with respect to the orthonormal basis

$$(e_1, \ldots, e_{k-1}, f_k, f_{k+1}, e_{k+2}, \ldots, e_n)$$

will be a block upper-triangular matrix with the same entries on the diagonal as previously, except that $A_j$ will be replaced by the 2-by-2 matrix

$$\begin{bmatrix} \lambda & * \\ 0 & * \end{bmatrix}.$$

In other words, where we had $A_j$ on the diagonal, we can now think of having two 1-by-1 matrices on the diagonal (and we still have a block upper-triangular matrix because of the 0 in the lower-left entry of the matrix above).

Repeating, when necessary, the procedure described above, we obtain an orthonormal basis of $V$ with respect to which $T$ has a block upper-triangular matrix

$$\begin{bmatrix} A_1 & & * \\ & \ddots & \\ 0 & & A_m \end{bmatrix},$$

where each $A_j$ is a 1-by-1 matrix or a 2-by-2 matrix with no eigenvalues.

7. Prove that if $T \in \mathcal{L}(V)$ and $j$ is a positive integer such that $j \leq \dim V$, then $T$ has an invariant subspace whose dimension equals $j - 1$ or $j$.

SOLUTION: Suppose $T \in \mathcal{L}(V)$ and $j$ is a positive integer such that $j \leq \dim V$. If $V$ is a complex vector space, then $T$ has an invariant subspace

whose dimension equals $j$ (by Exercise 17 in Chapter 5). So we can assume that $V$ is a real vector space.

By 9.4, there is a basis $(v_1, \dots, v_n)$ of $V$ with respect to which $T$ has a block upper-triangular matrix

$$\begin{bmatrix} A_1 & & * \\ & \ddots & \\ 0 & & A_m \end{bmatrix},$$

where each $A_k$ is a 1-by-1 matrix or a 2-by-2 matrix. Either $v_{j-1}$ or $v_j$ must be the last vector in a block of vectors corresponding to some $A_k$. Thus either $\operatorname{span}(v_1, \dots, v_{j-1})$ or $\operatorname{span}(v_1, \dots, v_j)$ must be invariant under $T$.

8. Prove that there does not exist an operator $T \in \mathcal{L}(\mathbf{R}^7)$ such that $T^2 + T + I$ is nilpotent.

SOLUTION: Suppose $T \in \mathcal{L}(\mathbf{R}^7)$. From part (b) of 9.9 (combined with 9.4, which insures that $T$ has a matrix of the form 9.10 with respect to some basis of $\mathbf{R}^7$), we see that

$$\dim \operatorname{null}(T^2 + T + I)^7$$

must be an even integer. In particular, $\dim \operatorname{null}(T^2 + T + I)^7 \neq 7$, which implies that $\operatorname{null}(T^2 + T + I)^7 \neq \mathbf{R}^7$, which implies that $(T^2 + T + I)^7 \neq 0$, which implies that $T$ is not nilpotent (by 8.8).

9. Give an example of an operator $T \in \mathcal{L}(\mathbf{C}^7)$ such that $T^2 + T + I$ is nilpotent.

SOLUTION: Let $\lambda = (-1 + 3i)/2$. Define $T \in \mathcal{L}(\mathbf{C}^7)$ by $T = \lambda I$. Then

$$\begin{aligned} T^2 + T + I &= (\lambda^2 + \lambda + 1)I \\ &= 0. \end{aligned}$$

In particular, $T^2 + T + I$ is nilpotent.

10. Suppose $V$ is a real vector space and $T \in \mathcal{L}(V)$. Suppose $\alpha, \beta \in \mathbf{R}$ are such that $\alpha^2 < 4\beta$. Prove that

$$\operatorname{null}(T^2 + \alpha T + \beta I)^k$$

has even dimension for every positive integer $k$.

SOLUTION: Let $k$ be a positive integer, and let $U = \operatorname{null}(T^2 + \alpha T + \beta I)^k$. We need to prove that $\dim U$ is even.

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Because $U$ is invariant under $T$ (by 8.22, with $p(x) = (x^2 + \alpha x + \beta)^k$), we can define $S \in \mathcal{L}(U)$ by $S = T|_U$. Clearly $(S^2 + \alpha S + \beta I)^k$ is the 0 operator on $U$. Thus $S^2 + \alpha S + \beta I$ is a nilpotent operator on $U$, which implies that $\operatorname{null}(S^2 + \alpha S + \beta I)^{\dim U} = U$ (by 8.8). Now part (b) of 9.9, applied to $S$ and $U$ instead of $T$ and $V$, shows that $\dim U$ is an even integer, as desired.

11. Suppose $V$ is a real vector space and $T \in \mathcal{L}(V)$. Suppose $\alpha, \beta \in \mathbf{R}$ are such that $\alpha^2 < 4\beta$ and $T^2 + \alpha T + \beta I$ is nilpotent. Prove that $\dim V$ is even and

$$(T^2 + \alpha T + \beta I)^{\dim V/2} = 0.$$

SOLUTION: Let $S = T^2 + \alpha T + \beta I$. Because $S$ is nilpotent, there is a smallest positive integer $m$ such that $S^m = 0$. Thus

$$\{0\} = \operatorname{null} S^0 \subsetneq \operatorname{null} S \subsetneq \operatorname{null} S^2 \subsetneq \ldots \subsetneq \operatorname{null} S^m = V,$$

where the proper inclusions come from 8.5. The previous exercise states that each $\operatorname{null} S^k$ has even dimension (in particular $V$, which equals $\operatorname{null} S^m$, has even dimension). Hence the dimension must increase by at least 2 in all the proper inclusions above. Thus $\dim S^m \geq 2m$, which clearly implies that $m \leq (\dim V)/2$. Because $S^m = 0$, this implies that $S^{\dim V/2} = 0$, as desired.

12. Prove that if $T \in \mathcal{L}(\mathbf{R}^3)$ and $5, 7$ are eigenvalues of $T$, then $T$ has no eigenpairs.

SOLUTION: Suppose that $T \in \mathcal{L}(\mathbf{R}^3)$ and $5, 7$ are eigenvalues of $T$. Of course each of these two eigenvalues must have multiplicity at least 1. By 9.17, the sum of the multiplicities of all the eigenvalues of $T$ plus twice the sum of the multiplicities of all the eigenpairs of $T$ equals 3. Because the sum of the multiplicities of all the eigenvalues of $T$ is at least 2, there is no room for an eigenpair, which would add at least 2 more to the sum (because we take twice the sum of the multiplicities of all the eigenpairs). Thus $T$ has no eigenpairs.

13. Suppose $V$ is a real vector space with $\dim V = n$ and $T \in \mathcal{L}(V)$ is such that

$$\operatorname{null} T^{n-2} \neq \operatorname{null} T^{n-1}.$$

Prove that $T$ has at most two distinct eigenvalues and that $T$ has no eigenpairs.

SOLUTION: Because $\operatorname{null} T^{n-2} \neq \operatorname{null} T^{n-1}$, we see that $\dim \operatorname{null} T^j$ is at least 1 more than $\dim \operatorname{null} T^{j-1}$ for $j = 1, \ldots, n-1$ (by 8.5). Thus

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$\dim \operatorname{null} T^{n-1} \geq n-1$. In particular, 0 is an eigenvalue of $T$ with multiplicity at least $n-1$. Because the sum of the multiplicities of all the eigenvalues of $T$ plus the sum of twice the multiplicities of all the eigenpairs of $T$ equals $n$ (by 9.17), this implies that $T$ can have no eigenpairs and at most one additional eigenvalue.

14. Suppose $V$ is a vector space with dimension 2 and $T \in \mathcal{L}(V)$. Prove that if

$$\begin{bmatrix} a & c \\ b & d \end{bmatrix}$$

is the matrix of $T$ with respect to some basis of $V$, then the characteristic polynomial of $T$ equals $(z-a)(z-d)-bc$.

COMMENT: As usual unless otherwise specified, here $V$ may be a real or complex vector space.

SOLUTION: Let $q(z) = (z-a)(z-d)-bc$. If $(v_1, v_2)$ is the basis of $V$ with respect to which $T$ has the matrix above, then

$$Tv_1 = av_1 + bv_2 \quad \text{and} \quad Tv_2 = cv_1 + dv_2.$$

From these equations you can easily verify that $q(T)v_1 = 0$ and $q(T)v_2 = 0$. Because $q(T)$ is 0 on a basis of $V$, we conclude that $q(T) = 0$.

Because $q$ is a monic polynomial of degree 2 and $q(T) = 0$, we conclude that the minimal polynomial of $T$ has degree 1 or 2.

Suppose first that the minimal polynomial of $T$ has degree 2. Because the minimal polynomial of $T$ is a divisor of $q$ (by 8.34), and because a monic polynomial of degree 2 can be a divisor of another monic polynomial of degree 2 only if the two polynomials are equal, we conclude that $q$ is the minimal polynomial of $T$. The Cayley-Hamilton theorem now implies that $q$ is a divisor of the characteristic polynomial of $T$, which is also a monic polynomial of degree 2. This implies that $q$ is the characteristic polynomial of $T$, as desired.

Now consider the only remaining possible case, which is that the minimal polynomial of $T$ has degree 1, meaning that it equals $z-\lambda$ for some $\lambda \in \mathbf{F}$. This implies that $T = \lambda I$, which implies that the characteristic polynomial of $T$ equals $(z-\lambda)^2$. Because $T = \lambda I$, we must have $a = d = \lambda$ and $b = c = 0$. Thus $q(z) = (z-\lambda)^2$. In particular, $q$ is the characteristic polynomial of $T$, as desired.

COMMENT: Note that we did not need to find the eigenvalues of $T$ to do this exercise.

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15. Suppose $V$ is a real inner-product space and $S \in \mathcal{L}(V)$ is an isometry. Prove that if $(\alpha, \beta)$ is an eigenpair of $S$, then $\beta = 1$.

SOLUTION: There is a basis of $V$ with respect to which $S$ has a block diagonal matrix, where each block on the diagonal is a 1-by-1 matrix or a 2-by-2 matrix of the form

$$\begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix},$$

with $\theta \in (0, \pi)$ (see 7.38). The characteristic polynomial of the matrix above is $(x - \cos\theta)^2 + \sin^2\theta$, which equals $x^2 - 2(\cos\theta)x + 1$.

If $(\alpha, \beta)$ is an eigenpair of $S$, then

$$\dim \operatorname{null}(S^2 + \alpha S + \beta I)^{\dim V} > 0,$$

which implies that $x^2 + \alpha x + \beta$ is the characteristic polynomial of a 2-by-2 matrix of the form displayed above (see 9.9). Thus $\beta = 1$.

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CHAPTER 10

# Trace and Determinant

1. Suppose that $T \in \mathcal{L}(V)$ and $(v_1, \ldots, v_n)$ is a basis of $V$. Prove that $\mathcal{M}(T, (v_1, \ldots, v_n))$ is invertible if and only if $T$ is invertible.

SOLUTION: First suppose that $\mathcal{M}(T)$ is an invertible matrix (because the only basis is sight is $(v_1, \ldots, v_n)$, we can leave the basis out of the notation). Thus there exists an $n$-by-$n$ matrix $B$ such that

$$\mathcal{M}(T)B = B\mathcal{M}(T) = I.$$

There exists an operator $S \in \mathcal{L}(V)$ such that $\mathcal{M}(S) = B$ (see 3.19). Thus the equation above becomes

$$\mathcal{M}(T)\mathcal{M}(S) = \mathcal{M}(S)\mathcal{M}(T) = I,$$

which we can rewrite as

$$\mathcal{M}(TS) = \mathcal{M}(ST) = \mathcal{M}(I),$$

which implies that

$$TS = ST = I.$$

Thus $T$ is invertible, as desired, with inverse $S$.

To prove the implication in the other direction, suppose now that $T$ is invertible. Thus there exists $S \in \mathcal{L}(V)$ such that

$$TS = ST = I.$$

This implies that



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$$\mathcal{M}(TS) = \mathcal{M}(ST) = \mathcal{M}(I),$$

which implies that

$$\mathcal{M}(T)\mathcal{M}(S) = \mathcal{M}(S)\mathcal{M}(T) = I.$$

Thus $\mathcal{M}(T)$ is invertible, as desired, with inverse $\mathcal{M}(S)$.

2. Prove that if $A$ and $B$ are square matrices of the same size and $AB = I$, then $BA = I$.

SOLUTION: Suppose that $A$ and $B$ are $n$-by-$n$ matrices and $AB = I$. There exist $S, T \in \mathcal{L}(\mathbf{F}^n)$ such that

$$\mathcal{M}(S) = A \quad \text{and} \quad \mathcal{M}(T) = B;$$

here we are using the standard basis of $\mathbf{F}^n$ (the existence of $S, T \in \mathcal{L}(\mathbf{F}^n)$ satisfying the equations above follows from 3.19). Because $AB = I$, we have $\mathcal{M}(S)\mathcal{M}(T) = I$, which implies that $\mathcal{M}(ST) = \mathcal{M}(I)$, which implies that $ST = I$, which implies that $TS = I$ (by Exercise 23 in Chapter 3). Thus

$$\begin{aligned} BA &= \mathcal{M}(T)\mathcal{M}(S) \\ &= \mathcal{M}(TS) \\ &= \mathcal{M}(I) \\ &= I. \end{aligned}$$

3. Suppose $T \in \mathcal{L}(V)$ has the same matrix with respect to every basis of $V$. Prove that $T$ is a scalar multiple of the identity operator.

SOLUTION: We begin by proving that $(v, Tv)$ is linearly dependent for every $v \in V$. To do this, fix $v \in V$, and suppose that $(v, Tv)$ is linearly independent. Then $(v, Tv)$ can be extended to a basis $(v, Tv, u_1, \ldots, u_n)$ of $V$. The first column of the matrix of $T$ with respect to this basis is

$$\begin{bmatrix} 0 \\ 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}.$$

Clearly $(2v, Tv, u_1, \ldots, u_n)$ is also a basis of $V$. The first column of the matrix of $T$ with respect to this basis is

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$$\begin{bmatrix} 0 \\ 2 \\ 0 \\ \vdots \\ 0 \end{bmatrix}.$$

Thus $T$ has different matrices with respect to the two bases we have considered. This contradiction shows that $(v, Tv)$ is linearly dependent for every $v \in V$. This implies that for every vector in $V$ is an eigenvector of $T$. This implies that $T$ is a scalar multiple of the identity operator (by Exercise 12 in Chapter 5).

4. Suppose that $(u_1, \ldots, u_n)$ and $(v_1, \ldots, v_n)$ are bases of $V$. Let $T \in \mathcal{L}(V)$ be the operator such that $Tv_k = u_k$ for $k = 1, \ldots, n$. Prove that

$$\mathcal{M}\big(T, (v_1, \ldots, v_n)\big) = \mathcal{M}\big(I, (u_1, \ldots, u_n), (v_1, \ldots, v_n)\big).$$

SOLUTION: Fix $k$. Write

$$u_k = a_1 v_1 + \cdots + a_n v_n,$$

where $a_1, \ldots, a_n \in \mathbf{F}$. Because $Tv_k = u_k$, the $k^{\text{th}}$ column of the matrix $\mathcal{M}\big(T, (v_1, \ldots, v_n)\big)$ consists of the numbers $a_1, \ldots, a_n$. Because $Iu_k = u_k$, the $k^{\text{th}}$ column of $\mathcal{M}\big(I, (u_1, \ldots, u_n), (v_1, \ldots, v_n)\big)$ also consists of the numbers $a_1, \ldots, a_n$.

Because $\mathcal{M}\big(T, (v_1, \ldots, v_n)\big)$ and $\mathcal{M}\big(I, (u_1, \ldots, u_n), (v_1, \ldots, v_n)\big)$ have the same columns, these two matrices must be equal.

5. Prove that if $B$ is a square matrix with complex entries, then there exists an invertible square matrix $A$ with complex entries such that $A^{-1}BA$ is an upper-triangular matrix.

SOLUTION: Suppose $B$ is an $n$-by-$n$ matrix with complex entries. Let $(e_1, \ldots, e_n)$ denote the standard basis of $\mathbf{C}^n$. There exists $T \in \mathcal{L}(\mathbf{C}^n)$ such that $\mathcal{M}\big(T, (e_1, \ldots, e_n)\big) = B$ (see 3.19).

There is a basis $(v_1, \ldots, v_n)$ of $V$ such that $\mathcal{M}\big(T, (v_1, \ldots, v_n)\big)$ is an upper-triangular matrix (see 5.13). Let $A = \mathcal{M}\big((v_1, \ldots, v_n), (e_1, \ldots, e_n)\big)$. Then $A$ is invertible (by 10.2) and

$$\begin{aligned} A^{-1}BA &= A^{-1}\mathcal{M}\big(T, (e_1, \ldots, e_n)\big)A \\ &= \mathcal{M}\big(T, (v_1, \ldots, v_n)\big), \end{aligned}$$

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where the second equality comes from 10.3. Thus $A^{-1}BA$ is an upper-triangular matrix.

6. Give an example of a real vector space $V$ and $T \in \mathcal{L}(V)$ such that

$$\text{trace}(T^2) < 0.$$

SOLUTION: Define $T \in \mathcal{L}(\mathbf{R}^2)$ by

$$T(x, y) = (-y, x).$$

Then $T^2 = -I$, so $\text{trace}(T^2) = -2$.

7. Suppose $V$ is a real vector space, $T \in \mathcal{L}(V)$, and $V$ has a basis consisting of eigenvectors of $T$. Prove that $\text{trace}(T^2) \geq 0$.

SOLUTION: Let $(v_1, \ldots, v_n)$ be a basis of $V$ consisting of eigenvectors of $T$. Thus there exist $\lambda_1, \ldots, \lambda_n \in \mathbf{R}$ such that $Tv_j = \lambda_j v_j$ for each $j$. Clearly the matrix of $T^2$ with respect to the basis $(v_1, \ldots, v_n)$ is the diagonal matrix

$$\begin{bmatrix} {\lambda_1}^2 & & 0 \\ & \ddots & \\ 0 & & {\lambda_n}^2 \end{bmatrix}.$$

Thus $\text{trace}\, T^2 = {\lambda_1}^2 + \cdots + {\lambda_n}^2 \geq 0$.

8. Suppose $V$ is an inner-product space and $v, w \in \mathcal{L}(V)$. Define $T \in \mathcal{L}(V)$ by $Tu = \langle u, v \rangle w$. Find a formula for $\text{trace}\, T$.

SOLUTION: First suppose that $v \neq 0$. Extend $(\frac{v}{\|v\|})$ to an orthonormal basis $(\frac{v}{\|v\|}, e_1, \ldots, e_n)$ of $V$. Note that for each $j$, we have $Te_j = 0$ (because $\langle e_j, v \rangle = 0$). The trace of $T$ equals the sum of the diagonal entries in the matrix of $T$ with respect to the basis $(\frac{v}{\|v\|}, e_1, \ldots, e_n)$. Thus

$$\begin{aligned} \text{trace}\, T &= \langle T(\frac{v}{\|v\|}), \frac{v}{\|v\|} \rangle + \langle Te_1, e_1 \rangle + \cdots + \langle Te_n, e_n \rangle \\ &= \langle \langle \frac{v}{\|v\|}, v \rangle w, \frac{v}{\|v\|} \rangle \\ &= \langle w, v \rangle. \end{aligned}$$

If $v = 0$, then $T = 0$ and so $\text{trace}\, T = 0 = \langle w, v \rangle$. Thus we have the formula

$$\operatorname{trace} T = \langle w, v \rangle$$

regardless of whether or not $v = 0$.

9. Prove that if $P \in \mathcal{L}(V)$ satisfies $P^2 = P$, then $\operatorname{trace} P$ is a nonnegative integer.

SOLUTION: Suppose that $T \in \mathcal{L}(V)$ satisfies $P^2 = P$. Let $(u_1, \ldots, u_m)$ be a basis of $\operatorname{range} P$ and let $(v_1, \ldots, v_n)$ be a basis of $\operatorname{null} P$. Then

$$(u_1, \ldots, u_m, v_1, \ldots, v_n)$$

is a basis of $V$ (this holds because $V = \operatorname{range} T \oplus \operatorname{null} T$; see Exercise 21 in Chapter 5). For each $u_j$ we have $Pu_j = u_j$ and for each $v_k$ we have $Pv_k = 0$. Thus the matrix of $P$ with respect to the basis above of $V$ is a diagonal matrix whose diagonal contains $m$ 1's followed by $n$ 0's. Thus $\operatorname{trace} P = m$, which is a nonnegative integer, as desired. In fact, we have shown that

$$\operatorname{trace} P = \dim \operatorname{range} P.$$

10. Prove that if $V$ is an inner-product space and $T \in \mathcal{L}(V)$, then

$$\operatorname{trace} T^* = \overline{\operatorname{trace} T}.$$

SOLUTION: Suppose that $V$ is an inner-product space and $T \in \mathcal{L}(V)$. Let $(e_1, \ldots, e_n)$ be an orthonormal basis of $V$. The trace of any operator on $V$ equals the sum of the diagonal entries on the matrix of the operator with respect to this basis. Thus

$$\begin{aligned}
\operatorname{trace} T^* &= \langle T^* e_1, e_1 \rangle + \cdots + \langle T^* e_n, e_n \rangle \\
&= \langle e_1, Te_1 \rangle + \cdots + \langle e_n, Te_n \rangle \\
&= \overline{\langle Te_1, e_1 \rangle} + \cdots + \overline{\langle Te_n, e_n \rangle} \\
&= \overline{\langle Te_1, e_1 \rangle + \cdots + \langle Te_n, e_n \rangle} \\
&= \overline{\operatorname{trace} T}.
\end{aligned}$$

11. Suppose $V$ is an inner-product space. Prove that if $T \in \mathcal{L}(V)$ is a positive operator and $\operatorname{trace} T = 0$, then $T = 0$.

SOLUTION: Suppose $T \in \mathcal{L}(V)$ is a positive operator and $\operatorname{trace} T = 0$. There exists an operator $S \in \mathcal{L}(V)$ such that $T = S^* S$ (by 7.27). Let $(e_1, \ldots, e_n)$ be an orthonormal basis of $V$. Then

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$$\begin{aligned} 0 &= \operatorname{trace} T \\ &= \langle Te_1, e_1\rangle + \cdots + \langle Te_n, e_n\rangle \\ &= \langle S^*Se_1, e_1\rangle + \cdots + \langle S^*Se_n, e_n\rangle \\ &= \|Se_1\|^2 + \cdots + \|Se_n\|^2. \end{aligned}$$

The equation above implies that $Se_j = 0$ for each $j$. Because $S$ is 0 on a basis of $V$, we have $S = 0$. Because $T = S^*S$, this implies that $T = 0$.

12. Suppose $T \in \mathcal{L}(\mathbf{C}^3)$ is the operator whose matrix is

$$\begin{bmatrix} 51 & -12 & -21 \\ 60 & -40 & -28 \\ 57 & -68 & 1 \end{bmatrix}.$$

Someone tells you (accurately) that $-48$ and 24 are eigenvalues of $T$. Without using a computer or writing anything down, find the third eigenvalue of $T$.

SOLUTION: The sum of the eigenvalues of $T$ equals the sum of the diagonal terms of the matrix above (both quantities equal $\operatorname{trace} T$). The sum of the diagonal terms of the matrix above equals 12. The sum of two of the eigenvalues of $T$, $-48$ and 24, equals $-24$. Because the sum of all three eigenvalues of $T$ must equal 12, the third eigenvalue of $T$ must be 36.

13. Prove or give a counterexample: if $T \in \mathcal{L}(V)$ and $c \in \mathbf{F}$, then $\operatorname{trace}(cT) = c \operatorname{trace} T$.

SOLUTION: Suppose $T \in \mathcal{L}(V)$ and $c \in \mathbf{F}$. To prove that $\operatorname{trace}(cT) = c \operatorname{trace} T$, consider a basis of $V$. Then $\operatorname{trace} T$ equals the sum of the diagonal terms of the matrix of $T$ with respect to this basis. The matrix of $cT$, with respect to the same basis, equals $c$ times the matrix of $T$. Thus the sum of the diagonal terms of the matrix of $cT$ equals $c$ times the sum of the diagonal terms of the matrix of $T$. In other words, $\operatorname{trace}(cT) = c \operatorname{trace} T$.

14. Prove or give a counterexample: if $S, T \in \mathcal{L}(V)$, then

$$\operatorname{trace}(ST) = (\operatorname{trace} S)(\operatorname{trace} T).$$

SOLUTION: Define $S, T \in \mathcal{L}(\mathbf{F}^2)$ by $S(x, y) = T(x, y) = (-y, x)$. Then with respect to the standard bases the matrix of $S$ (which of course equals the matrix of $T$) is

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$$\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}.$$

Thus $\operatorname{trace} S = \operatorname{trace} T = 0$. However, $ST = -I$, so $\operatorname{trace} ST = -2$. Thus for this choice of $S$ and $T$, we have $\operatorname{trace}(ST) \neq (\operatorname{trace} S)(\operatorname{trace} T)$.

Of course there are also many other examples.

15. Suppose $T \in \mathcal{L}(V)$. Prove that if $\operatorname{trace}(ST) = 0$ for all $S \in \mathcal{L}(V)$, then $T = 0$.

SOLUTION: Suppose that $\operatorname{trace}(ST) = 0$ for all $S \in \mathcal{L}(V)$. Then $\operatorname{trace}(TS) = 0$ for all $S \in \mathcal{L}(V)$ (by 10.12). Suppose that there exists $v \in V$ such that $Tv \neq 0$. Then $(Tv)$ can be extended to a basis $(Tv, u_1, \ldots, u_n)$ of $V$. Define $S \in \mathcal{L}(V)$ by

$$S(aTv + b_1u_1 + \cdots + b_nu_n) = av.$$

Thus $S(Tv) = v$ and $Su_j = 0$ for each $j$. Hence $(TS)(Tv) = T(S(Tv)) = Tv$ and $(TS)(u_j) = 0$ for each $j$. This implies that with respect to the basis $(Tv, u_1, \ldots, u_n)$, the matrix of $TS$ consists of all 0's except for a 1 in the upper-left corner. Thus $\operatorname{trace}(TS) = 1$. This contradiction shows that our assumption that $Tv \neq 0$ must have been false. Thus $Tv = 0$ for every $v \in V$, which means that $T = 0$.

16. Suppose $V$ is an inner-product space and $T \in \mathcal{L}(V)$. Prove that if $(e_1, \ldots, e_n)$ is an orthonormal basis of $V$, then

$$\operatorname{trace}(T^*T) = \|Te_1\|^2 + \cdots + \|Te_n\|^2.$$

Conclude that the right side of the equation above is independent of which orthonormal basis $(e_1, \ldots, e_n)$ is chosen for $V$.

SOLUTION: Suppose that $(e_1, \ldots, e_n)$ is an orthonormal basis of $V$. Then

$$\begin{aligned} \operatorname{trace} T^*T &= \langle T^*Te_1, e_1\rangle + \cdots + \langle T^*Te_n, e_n\rangle \\ &= \langle Te_1, Te_1\rangle + \cdots + \langle Te_n, Te_n\rangle \\ &= \|Te_1\|^2 + \cdots + \|Te_n\|^2. \end{aligned}$$

Because $\operatorname{trace} T^*T$ does not depend upon the choice of a basis of $V$, the formula above shows that $\|Te_1\|^2 + \cdots + \|Te_n\|^2$ is independent of the orthonormal basis $(e_1, \ldots, e_n)$.

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17. Suppose $V$ is a complex inner-product space and $T \in \mathcal{L}(V)$. Let $\lambda_1, \ldots, \lambda_n$ be the eigenvalues of $T$, repeated according to multiplicity. Suppose

$$\begin{bmatrix} a_{1,1} & \cdots & a_{1,n} \\ \vdots & & \vdots \\ a_{n,1} & \cdots & a_{n,n} \end{bmatrix}$$

is the matrix of $T$ with respect to some orthonormal basis of $V$. Prove that

$$|\lambda_1|^2 + \cdots + |\lambda_n|^2 \leq \sum_{k=1}^{n} \sum_{j=1}^{n} |a_{j,k}|^2.$$

SOLUTION: Suppose that $(e_1, \ldots, e_n)$ is the orthonormal basis with respect to which $T$ has the matrix above. Thus for each $k$, we have

$$Te_k = a_{1,k}e_1 + \cdots + a_{n,k}e_n,$$

which implies that

$$\|Te_k\|^2 = |a_{1,k}|^2 + \cdots + |a_{n,k}|^2.$$

Thus

$$\|Te_1\|^2 + \cdots + \|Te_n\|^2 = \sum_{k=1}^{n} \sum_{j=1}^{n} |a_{j,k}|^2.$$

By the previous exercise, the left side of this equation equals $\text{trace}(T^*T)$. This reduces the exercise at hand to proving that

$$|\lambda_1|^2 + \cdots + |\lambda_n|^2 \leq \text{trace}(T^*T).$$

There is an orthonormal basis $(f_1, \ldots, f_n)$ with respect to which $T$ has an upper-triangular matrix (by 6.28). The diagonal entries of the matrix of $T$ with respect to $(f_1, \ldots, f_n)$ are precisely $\lambda_1, \ldots, \lambda_n$ (by 8.10), where we can relabel the eigenvalues of $T$ so that they appear in the order $\lambda_1, \ldots, \lambda_n$ along the diagonal. In other words,

$$\mathcal{M}(T, (f_1, \ldots, f_n)) = \begin{bmatrix} \lambda_1 & & * \\ & \ddots & \\ 0 & & \lambda_n \end{bmatrix}.$$

From the matrix above, we see that

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$$|\lambda_k|^2 \le \|Tf_k\|^2$$

for each $k$. Thus

$$\begin{aligned}|\lambda_1|^2 + \cdots + |\lambda_n|^2 &\le \|Tf_1\|^2 + \cdots + \|Tf_n\|^2 \\ &= \operatorname{trace}(T^*T),\end{aligned}$$

as desired (here the last equality comes from the previous exercise).

18. Suppose $V$ is an inner-product space. Prove that

$$\langle S, T\rangle = \operatorname{trace}(ST^*)$$

defines an inner product on $\mathcal{L}(V)$.

SOLUTION: Suppose that $\langle\cdot,\cdot\rangle$ is defined as above and $R, S, T \in \mathcal{L}(V)$. Then $\langle T, T\rangle = \operatorname{trace}(TT^*)$, and thus $\langle T, T\rangle \ge 0$ (by the formula given in Exercise 16 of this chapter, with $T$ replaced with $T^*$). Because $TT^*$ is a positive operator (see 7.27), we also see that $\langle T, T\rangle = 0$ if and only if $T = 0$ (by Exercise 11 of this chapter).

Now

$$\begin{aligned}\langle R+S, T\rangle &= \operatorname{trace}\big((R+S)T^*\big) \\ &= \operatorname{trace}(RT^* + ST^*) \\ &= \operatorname{trace}(RT^*) + \operatorname{trace}(ST^*) \\ &= \langle R, T\rangle + \langle S, T\rangle,\end{aligned}$$

where the third equality comes from 10.12.

For $c \in \mathbf{F}$, we have

$$\begin{aligned}\langle cS, T\rangle &= \operatorname{trace}(cST^*) \\ &= c\operatorname{trace}(ST^*) \\ &= c\langle S, T\rangle,\end{aligned}$$

where the second equality comes from Exercise 13 of this chapter.

Finally,

$$\begin{aligned}\langle S, T\rangle &= \operatorname{trace}(ST^*) \\ &= \overline{\operatorname{trace}\big((ST^*)^*\big)} \\ &= \overline{\operatorname{trace}(TS^*)} \\ &= \overline{\langle S, T\rangle},\end{aligned}$$

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where the second equality comes from Exercise 10 of this chapter.

We have shown that $\langle \cdot, \cdot \rangle$ satisfies all the properties required of an inner product.

COMMENT: Suppose $(e_1, \ldots, e_n)$ is an orthonormal basis of $V$ and

$$\begin{bmatrix} a_{1,1} & \ldots & a_{1,n} \\ \vdots & & \vdots \\ a_{n,1} & \ldots & a_{n,n} \end{bmatrix}$$

is the matrix of $T$ with respect to this basis. Then

$$\begin{aligned} \langle T, T \rangle &= \operatorname{trace}(TT^*) \\ &= \operatorname{trace}(T^*T) \\ &= \sum_{k=1}^{n} \sum_{j=1}^{n} |a_{j,k}|^2, \end{aligned}$$

where the second equality comes from 10.12 and the third equality comes from Exercise 16 of this chapter. Thus the norm on $\mathcal{L}(V)$ induced by $\langle \cdot, \cdot \rangle$ is the same as the standard norm on $\mathbf{F}^{n^2}$ (here we are identifying each operator with its matrix, which has $n^2$ entries). Because norms determine the inner product (see Exercises 6 and 7 in Chapter 6), this means that the inner product $\langle \cdot, \cdot \rangle$ is the same as the standard inner product of $\mathbf{F}^{n^2}$ (again using the identification via matrices with respect to the orthonormal basis $(e_1, \ldots, e_n)$).

19. Suppose $V$ is an inner-product space and $T \in \mathcal{L}(V)$. Prove that if

$$\|T^*v\| \le \|Tv\|$$

for every $v \in V$, then $T$ is normal.

SOLUTION: Suppose that $\|T^*v\| \le \|Tv\|$ for every $v \in V$. Suppose $u \in V$ with $\|u\| = 1$. Extend $(u)$ to an orthonormal basis $(u, e_1, \ldots, e_n)$ of $V$. Then

$$\begin{aligned} \operatorname{trace}(TT^*) &= \|T^*u\|^2 + \|T^*e_1\|^2 + \cdots + \|T^*e_n\|^2 \\ &\le \|Tu\|^2 + \|Te_1\|^2 + \cdots + \|Te_n\|^2 \\ &= \operatorname{trace}(T^*T) \\ &= \operatorname{trace}(TT^*), \end{aligned}$$

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where the first and third lines come from Exercise 16 of this chapter and the last line comes from 10.12. Because the first and last lines above are equal, we must have equality throughout. Thus $\|T^*u\| = \|Tu\|$. This clearly implies that

$$\|T^*(au)\| = \|T(au)\|$$

for every $a \in \mathbf{F}$. Because every vector in $V$ can be written in the form $au$ for some $a \in \mathbf{F}$ and some $u \in V$ with $\|u\| = 1$, this implies that $\|T^*v\| = \|Tv\|$ for every $v \in V$. This implies that $T$ is normal (by 7.6).

COMMENT: This exercise fails on infinite-dimensional inner-product spaces, leading to what are called hyponormal operators, which have a well-developed theory.

20. Prove or give a counterexample: if $T \in \mathcal{L}(V)$ and $c \in \mathbf{F}$, then $\det(cT) = c^{\dim V} \det T$.

SOLUTION: Let $n = \dim V$. If $A$ is an $n$-by-$n$ matrix, then

$$\det(cA) = c^n \det A$$

for every $c \in \mathbf{F}$ (this follows immediately from the definition 10.25). Now suppose that $T \in \mathcal{L}(V)$. Because $\det T$ equals the determinant of the matrix of $T$ with respect to any basis (see 10.33), the equation above implies that for every $c \in \mathbf{F}$ we have

$$\begin{aligned} \det(cT) &= \det \mathcal{M}(cT) \\ &= \det(c\mathcal{M}(T)) \\ &= c^n \det \mathcal{M}(T) \\ &= c^n \det T, \end{aligned}$$

as desired.

21. Prove or give a counterexample: if $S, T \in \mathcal{L}(V)$, then

$$\det(S + T) = \det S + \det T.$$

SOLUTION: Define $S, T \in \mathcal{L}(\mathbf{F}^2)$ by

$$S(x, y) = (x, 0) \quad \text{and} \quad T(x, y) = (0, y).$$

Then $S$ and $T$ are both not invertible. Thus $\det S = \det T = 0$ (by 10.14). However, $S + T = I$ and $\det I = 1$, so $\det(S + T) \neq \det S + \det T$.

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Of course there are also many other examples.

22. Suppose $A$ is a block upper-triangular matrix

$$A = \begin{bmatrix} A_1 & & * \\ & \ddots & \\ 0 & & A_m \end{bmatrix},$$

where each $A_j$ along the diagonal is a square matrix. Prove that

$$\det A = (\det A_1) \dots (\det A_m).$$

SOLUTION: First consider the case where $m = 2$, so we can write $A$ in the form

$$A = \begin{bmatrix} B & * \\ 0 & C \end{bmatrix},$$

where $B$ is an $n$-by-$n$ matrix

$$B = \begin{bmatrix} b_{1,1} & \dots & b_{1,n} \\ \vdots & & \vdots \\ b_{n,1} & \dots & b_{n,n} \end{bmatrix}$$

and $C$ is a $p$-by-$p$ matrix

$$C = \begin{bmatrix} c_{1,1} & \dots & c_{1,p} \\ \vdots & & \vdots \\ c_{p,1} & \dots & c_{p,p} \end{bmatrix}.$$

Let $a_{j,k}$ denote the entry in row $j$, column $k$ of $A$. Note that $a_{j,k} = 0$ if $j > n$ and $k \leq n$ (this follows from the block upper-triangular form of $A$). Because $A$ is an $(n+p)$-by-$(n+p)$ matrix, to compute $\det A$ we need to consider a typical permutation $(m_1, \dots, m_{n+p}) \in \text{perm}(n+p)$. If any of $m_1, \dots, m_n$ is greater than $n$, then

$$a_{m_1,1} \dots a_{m_{n+p},n+p} = 0.$$

Thus in computing $\det A$ we need only consider permutations in which the first $n$ coordinates all come from $\{1, \dots, n\}$, which means that the last $p$ coordinates all come from $\{n+1, \dots, n+p\}$. We can break any such permutation $(m_1, \dots, m_{n+p})$ into two pieces: a permutation of $\{1, \dots, n\}$ and (relabeling $n+1, \dots, n+p$ as $1, \dots, p$ to correspond to the labeling of the

entries of $C$) a permutation of $\{1,\dots,p\}$. Clearly the sign of $(m_1,\dots,m_{n+p})$ will equal the product of the signs of these two permutations. Putting all this together, we have

$$\begin{aligned}
\det A &= \sum_{(m_1,\dots,m_{n+p})\in \mathrm{perm}(n+p)} \big(\mathrm{sign}(m_1,\dots,m_{n+p})\big) a_{m_1,1}\dots a_{m_{n+p},n+p} \\
&= \sum_{\substack{(j_1,\dots,j_n)\in \mathrm{perm}\, n \\ (k_1,\dots,k_p)\in \mathrm{perm}\, p}} \big[\big(\mathrm{sign}(j_1,\dots,j_n)\big)\big(\mathrm{sign}(k_1,\dots,k_p)\big)\cdot \\
&\qquad\qquad b_{j_1,1}\dots b_{j_n,n} c_{k_1,1}\dots c_{k_p,p}\big] \\
&= \sum_{(j_1,\dots,j_n)\in \mathrm{perm}\, n} \big(\mathrm{sign}(j_1,\dots,j_n)\big) b_{j_1,1}\dots b_{j_n,n}\cdot \\
&\qquad\qquad \sum_{(k_1,\dots,k_p)\in \mathrm{perm}\, p} \big(\mathrm{sign}(k_1,\dots,k_p)\big) c_{k_1,1}\dots c_{k_p,p} \\
&= (\det B)(\det C),
\end{aligned}$$

completing the proof when $m = 2$.

Suppose now that $m > 2$ and

$$A = \begin{bmatrix} A_1 & & * \\ & \ddots & \\ 0 & & A_m \end{bmatrix}.$$

Writing

$$B = \begin{bmatrix} A_1 & & * \\ & \ddots & \\ 0 & & A_{m-1} \end{bmatrix},$$

we have

$$A = \begin{bmatrix} B & * \\ 0 & A_m \end{bmatrix}.$$

Thus

$$\begin{aligned}
\det A &= (\det B)(\det A_m) \\
&= (\det A_1)\dots(\det A_{m-1})(\det A_m),
\end{aligned}$$

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where the first equality holds by the $m = 2$ case proved above and the second equality comes from induction on $m$ (meaning that we can assume the desired result holds when $m$ is replaced with $m - 1$).

23. Suppose $A$ is an $n$-by-$n$ matrix with real entries. Let $S \in \mathcal{L}(\mathbf{C}^n)$ denote the operator on $\mathbf{C}^n$ whose matrix equals $A$, and let $T \in \mathcal{L}(\mathbf{R}^n)$ denote the operator on $\mathbf{R}^n$ whose matrix equals $A$. Prove that $\operatorname{trace} S = \operatorname{trace} T$ and $\det S = \det T$.

SOLUTION: The formulas defining the trace and determinant of a matrix do not depend upon whether we think of the matrix entries as real or complex numbers. We have $\operatorname{trace} A = \operatorname{trace} S$ and $\operatorname{trace} A = \operatorname{trace} T$ (by 10.11). Thus $\operatorname{trace} S = \operatorname{trace} T$. Similarly, we have $\det A = \det S$ and $\det A = \det T$ (by 10.33). Thus $\det S = \det T$.

24. Suppose $V$ is an inner-product space and $T \in \mathcal{L}(V)$. Prove that

$$\det T^* = \overline{\det T}.$$

Use this to prove that $|\det T| = \det \sqrt{T^* T}$, giving a different proof than was given in 10.37.

SOLUTION: Let $n = \dim V$. If $\lambda \in \mathbf{F}$, then $\big((T - \lambda I)^n\big)^* = (T^* - \bar{\lambda} I)^n$, which implies that

$$\dim \operatorname{null}(T - \lambda I)^n = \dim \operatorname{null}(T^* - \bar{\lambda} I)^n,$$

where we have used Exercise 31 in Chapter 7. The equation above shows that the eigenvalues of $T^*$ are precisely the complex conjugates of the eigenvalues of $T$, with the same multiplicities. If $\mathbf{F} = \mathbf{C}$, the determinant equals the product of the eigenvalues, counting multiplicity, so we have $\det T^* = \overline{\det T}$ (so far just on complex vector spaces).

Now suppose $\mathbf{F} = \mathbf{R}$, so we must consider eigenpairs. If $\alpha, \beta \in \mathbf{R}$, then

$$\dim \operatorname{null}(T^2 + \alpha T + \beta I)^n = \dim \operatorname{null}((T^*)^2 + \alpha T^* + \beta I)^n,$$

again by Exercise 31 in Chapter 7. Thus $T$ and $T^*$ have the same eigenpairs with the same multiplicities. From the paragraph above, $T$ and $T^*$ have the same eigenvalues with the same multiplicities. Because the determinant equals the product of the eigenvalues (counting multiplicity) times the product of the second coordinates of the eigenpairs (counting multiplicity), this implies that $\det T^* = \det T = \overline{\det T}$.

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At this point, we know that $\det T^* = \overline{\det T}$ regardless of whether $\mathbf{F} = \mathbf{C}$ or $\mathbf{F} = \mathbf{R}$. Thus

$$\begin{aligned}(\det\sqrt{T^*T})^2 &= (\det\sqrt{T^*T})(\det\sqrt{T^*T})\\ &= \det(T^*T)\\ &= (\det T^*)(\det T)\\ &= (\overline{\det T})(\det T)\\ &= |\det T|^2.\end{aligned}$$

Taking square roots (and recalling that the positive operator $\sqrt{T^*T}$ has a nonnegative determinant), we have $\det\sqrt{T^*T} = |\det T|$, as desired.

25. Let $a, b, c$ be positive numbers. Find the volume of the ellipsoid

$$\left\{(x,y,z) \in \mathbf{R}^3 : \frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} < 1\right\}$$

by finding a set $\Omega \subset \mathbf{R}^3$ whose volume you know and an operator $T \in \mathcal{L}(\mathbf{R}^3)$ such that $T(\Omega)$ equals the ellipsoid above.

SOLUTION: Let $E$ denote the ellipsoid defined above and let $\Omega$ be the ball with radius 1 defined by

$$\Omega = \{(x,y,z) \in \mathbf{R}^3 : x^2 + y^2 + z^2 < 1\}.$$

Of course volume $\Omega = \frac{4}{3}\pi$. Define $T \in \mathcal{L}(\mathbf{R}^3)$ by

$$T(x,y,z) = (ax, by, cz).$$

If $(x,y,z) \in \Omega$, then

$$\begin{aligned}\frac{(ax)^2}{a^2} + \frac{(by)^2}{b^2} + \frac{(cz)^2}{c^2} &= x^2 + y^2 + z^2\\ &< 1,\end{aligned}$$

which shows that $T(x,y,z) \in E$. Thus $T(\Omega) \subset E$.

Conversely, if $(x,y,z) \in E$, then

$$\begin{aligned}\left(\frac{x}{a}\right)^2 + \left(\frac{y}{b}\right)^2 + \left(\frac{z}{c}\right)^2 &= \frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2}\\ &< 1,\end{aligned}$$

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which shows that $(\frac{x}{a}, \frac{y}{b}, \frac{z}{c}) \in \Omega$. Because $T(\frac{x}{a}, \frac{y}{b}, \frac{z}{c}) = (x, y, z)$, this implies that $E \subset T(\Omega)$.

The last two paragraphs show that $E = T(\Omega)$. Thus

$$\begin{aligned} \text{volume}\, E &= \text{volume}\, T(\Omega) \\ &= |\det T|(\text{volume}\, \Omega) \\ &= \frac{4\pi abc}{3}, \end{aligned}$$

where the second equality comes from 10.38.

# Linear Algebra Done Right Solutions

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# Chapter 1. Vector Spaces

## 1.A. $\mathbf{R}^n$ and $\mathbf{C}^n$

**Exercise 1.A.1.** Show that $\alpha + \beta = \beta + \alpha$ for all $\alpha, \beta \in \mathbf{C}$.

**Solution.** If $\alpha = x + yi$ and $\beta = u + vi$, then

$$\alpha + \beta = (x + u) + (y + v)i = (u + x) + (v + y)i = \beta + \alpha,$$

where we have used the commutativity of addition in $\mathbf{R}$.

**Exercise 1.A.2.** Show that $(\alpha + \beta) + \lambda = \alpha + (\beta + \lambda)$ for all $\alpha, \beta \in \mathbf{C}$.

**Solution.** If $\alpha = x + yi, \beta = u + vi$, and $\lambda = s + ti$, then

$$\begin{aligned}(\alpha + \beta) + \lambda &= ((x + u) + (y + v))i + \lambda = ((x + u) + s) + ((y + v) + t)i \\ &= (x + (u + s)) + (y + (v + t))i = \alpha + ((u + s) + (v + t)i) = \alpha + (\beta + \lambda),\end{aligned}$$

where we have used the associativity of addition in $\mathbf{R}$.

**Exercise 1.A.3.** Show that $(\alpha\beta)\lambda = \alpha(\beta\lambda)$ for all $\alpha, \beta, \lambda \in \mathbf{C}$.

**Solution.** If $\alpha = x + yi, \beta = u + vi$, and $\lambda = s + ti$, then

$$\begin{aligned}(\alpha\beta)\lambda &= [(xu - yv) + (xv + yu)i]\lambda \\ &= [(xu - yv)s - (xv + yu)t] + [(xu - yv)t + (xv + yu)s]i \\ &= [(xu)s - (yv)s - (xv)t - (yu)t] + [(xu)t - (yv)t + (xv)s + (yu)s]i \\ &= [x(us) - x(vt) - y(ut) - y(vs)] + [x(ut) + x(vs) + y(us) - y(vt)]i \\ &= [x(us - vt) - y(ut + vs)] + [x(ut + vs) + y(us - vt)]i \\ &= \alpha[(us - vt) + (ut + vs)i] \\ &= \alpha(\beta\lambda),\end{aligned}$$

where we have used several algebraic properties of $\mathbf{R}$.

**Exercise 1.A.4.** Show that $\lambda(\alpha + \beta) = \lambda\alpha + \lambda\beta$ for all $\lambda, \alpha, \beta \in \mathbf{C}$.

**Solution.** If $\alpha = x + yi, \beta = u + vi$, and $\lambda = s + ti$, then



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$$\begin{aligned}\lambda(\alpha+\beta) &= [s(x+u)-t(y+v)]+[s(y+v)+t(x+u)i]\\ &= (sx+su-ty-tv)+(sy+sv+tx+tu)i\\ &= [(sx-ty)+(sy+tx)i]+[(su-tv)+(sv+tu)i]\\ &= \lambda\alpha+\lambda\beta,\end{aligned}$$

where we have used distributivity in $\mathbf{R}$.

**Exercise 1.A.5.** Show that for every $\alpha \in \mathbf{C}$, there exists a unique $\beta \in \mathbf{C}$ such that $\alpha+\beta=0$.

**Solution.** Suppose that $\alpha = x+yi$. Let $\beta = -x-yi$ and observe that

$$\alpha+\beta = (x-x)+(y-y)i = 0+0i = 0.$$

To see that $\beta$ is unique, suppose that $\beta'$ also satisfies $\alpha+\beta'=0$ and notice that

$$\beta = \beta = 0 = \beta+(\alpha+\beta') = (\alpha+\beta)+\beta' = 0+\beta' = \beta',$$

where we have used the associativity of addition in $\mathbf{C}$ (Exercise 1.A.2) and the commutativity of addition in $\mathbf{C}$ (Exercise 1.A.1).

**Exercise 1.A.6.** Show that for every $\alpha \in \mathbf{C}$ with $\alpha \neq 0$, there exists a unique $\beta \in \mathbf{C}$ such that $\alpha\beta = 1$.

**Solution.** Suppose that $\alpha = x+yi$. Since $\alpha \neq 0$, it must be the case that $x$ and $y$ are not both zero, so that $x^2+y^2 \neq 0$. Let $\beta = \frac{x}{x^2+y^2} - \frac{y}{x^2+y^2}i$ and observe that

$$\alpha\beta = (x+yi)\left(\frac{x}{x^2+y^2}-\frac{y}{x^2+y^2}i\right) = \frac{x^2+y^2}{x^2+y^2}+\frac{xy-xy}{x^2+y^2}i = 1+0i = 1.$$

To see that $\beta$ is unique, suppose $\beta'$ also satisfies $\alpha\beta'=1$ and notice that

$$\beta = \beta 1 = \beta(\alpha\beta') = (\alpha\beta)\beta' = 1\beta' = \beta',$$

where we have used the associativity of multiplication in $\mathbf{C}$ (Exercise 1.A.3) and the commutativity of multiplication in $\mathbf{C}$ (1.4).

**Exercise 1.A.7.** Show that

$$\frac{-1+\sqrt{3}i}{2}$$

is a cube root of 1 (meaning that its cube equals 1).

**Solution.** Let $z = \frac{-1+\sqrt{3}i}{2}$, so that $2z = -1+\sqrt{3}i$. Observe that

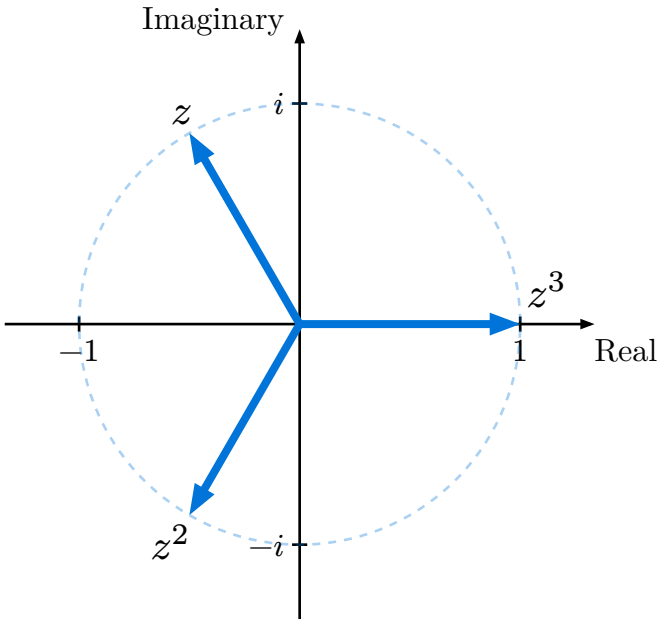


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$$(2z)^2 = 4z^2 = \left(-1+\sqrt{3}i\right)^2 = 1 - 2\sqrt{3}i - 3 = -2 - 2\sqrt{3}i$$

$$\Rightarrow (2z)^3 = (4z^2)(2z) = \left(-2-2\sqrt{3}i\right)\left(-1+\sqrt{3}i\right) = 2 - 2\sqrt{3}i + 2\sqrt{3}i + 6 = 8,$$

i.e. $8z^3 = 8$. It follows that $z^3 = 1$.

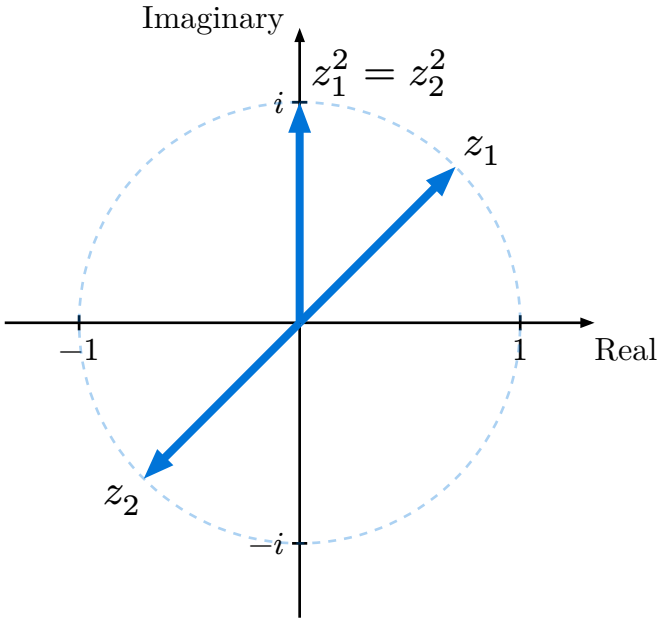



**Exercise 1.A.8.** Find two distinct square roots of $i$.

**Solution.** Let $z_1 = \frac{1+i}{\sqrt{2}}$ and $z_2 = -z_1$ ($z_1$ and $z_2$ are distinct since $z_1 \neq 0$) and observe that

$$2z_1^2 = (1+i)^2 = 2i \quad \Rightarrow \quad z_1^2 = i,$$

i.e. $z_1$ is a square root of $i$. Furthermore, $z_2^2 = (-z_1)^2 = z_1^2 = i$, so that $z_2$ is a square root of $i$ also.






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**Exercise 1.A.9.** Find $x \in \mathbf{R}^4$ such that

$$(4, -3, 1, 7) + 2x = (5, 9, -6, 8).$$

**Solution.** The unique solution is $x = (\frac{1}{2}, 6, -\frac{7}{2}, \frac{1}{2})$.

**Exercise 1.A.10.** Explain why there does not exist $\lambda \in \mathbf{C}$ such that

$$\lambda(2 - 3i, 5 + 4i, -6 + 7i) = (12 - 5i, 7 + 22i, -32 - 9i).$$

**Solution.** If there was such a $\lambda$, then

$$\lambda(2 - 3i) = 12 - 5i \quad \Rightarrow \quad \lambda = \frac{12 - 5i}{2 - 3i} = 3 + 2i.$$

However,

$$(3 + 2i)(-6 + 7i) = -32 + 9i \neq -32 - 9i.$$

**Exercise 1.A.11.** Show that $(x + y) + z = x + (y + z)$ for all $x, y, z \in \mathbf{F}^n$.

**Solution.** If $x = (x_1, ..., x_n), y = (y_1, ..., y_n)$, and $z = (z_1, ..., z_n)$, then

$$\begin{aligned}(x + y) + z &= (x_1 + y_1, ..., x_n + y_n) + z = ((x_1 + y_1) + z_1, ..., (x_n + y_n) + z_n) \\ &= (x_1 + (y_1 + z_1), ..., x_n + (y_n + z_n)) = x + (y_1 + z_1, ..., y_n + z_n) = x + (y + z),\end{aligned}$$

where we have used the associativity of addition in $\mathbf{F}$ (we proved this for $\mathbf{C}$ in Exercise 1.A.2).

**Exercise 1.A.12.** Show that $(ab)x = a(bx)$ for all $x \in \mathbf{F}^n$ and all $a, b \in \mathbf{F}$.

**Solution.** If $x = (x_1, ..., x_n)$, then

$$(ab)x = ((ab)x_1, ..., (ab)x_n) = (a(bx_1), ..., a(bx_n)) = a(bx_1, ..., bx_n) = a(bx),$$

where we have used the associativity of multiplication in $\mathbf{F}$ (we proved this for $\mathbf{C}$ in Exercise 1.A.3).

**Exercise 1.A.13.** Show that $1x = x$ for all $x \in \mathbf{F}^n$.

**Solution.** If $x = (x_1, ..., x_n)$, then

$$1x = (1x_1, ..., 1x_n) = (x_1, ..., x_n) = x,$$

where we have used that $1x_j = x_j$ for any $x_j \in \mathbf{F}$.

**Exercise 1.A.14.** Show that $\lambda(x + y) = \lambda x + \lambda y$ for all $\lambda \in \mathbf{F}$ and all $x, y \in \mathbf{F}^n$.



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**Solution.** If $x = (x_1, ..., x_n)$ and $y = (y_1, ..., y_n)$, then

$$\begin{aligned}
\lambda(x + y) &= \lambda(x_1 + y_1, ..., x_n + y_n) \\
&= (\lambda(x_1 + y_1), ..., \lambda(x_n + y_n)) \\
&= (\lambda x_1 + \lambda y_1, ..., \lambda x_n + \lambda y_n) \\
&= (\lambda x_1, ..., \lambda x_n) + (\lambda y_1, ..., \lambda y_n) \\
&= \lambda(x_1, ..., x_n) + \lambda(y_1, ..., y_n) \\
&= \lambda x + \lambda y,
\end{aligned}$$

where we have used distributivity in $\mathbf{F}$ (we proved this for $\mathbf{C}$ in Exercise 1.A.4).

**Exercise 1.A.15.** Show that $(a + b)x = ax + bx$ for all $a, b \in \mathbf{F}$ and all $x \in \mathbf{F}^n$.

**Solution.** If $x = (x_1, ..., x_n)$, then

$$\begin{aligned}
(a + b)x &= (a + b)(x_1, ..., x_n) \\
&= ((a + b)x_1, ..., (a + b)x_n) \\
&= (ax_1 + bx_1, ..., ax_n + bx_n) \\
&= (ax_1, ..., ax_n) + (bx_1, ..., bx_n) \\
&= a(x_1, ..., x_n) + b(x_1, ..., x_n) \\
&= ax + bx,
\end{aligned}$$

where we have used distributivity in $\mathbf{F}$ (we proved this for $\mathbf{C}$ in Exercise 1.A.4).



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## 1.B. Definition of Vector Space

**Exercise 1.B.1.** Show that $-(-v) = v$ for every $v \in V$.

**Solution.** Since $v + (-v) = 0$ and the additive inverse of a vector is unique (1.27), it must be the case that $-(-v) = v$.

**Exercise 1.B.2.** Suppose $a \in \mathbf{F}, v \in V$, and $av = 0$. Prove that $a = 0$ or $v = 0$.

**Solution.** It will suffice to show that if $av = 0$ and $a \neq 0$, so that $a^{-1}$ exists, then $v = 0$. Indeed,

$$av = 0 \quad \Rightarrow \quad a^{-1}(av) = 0 \quad \Rightarrow \quad (a^{-1}a)v = 0 \quad \Rightarrow \quad 1v = 0 \quad \Rightarrow \quad v = 0.$$

**Exercise 1.B.3.** Suppose $v, w \in V$. Explain why there exists a unique $x \in V$ such that $v + 3x = w$.

**Solution.** For $v, w, x \in V$, notice that

$$v + 3x = w \quad \Leftrightarrow \quad 3x = w - v \quad \Leftrightarrow \quad x = \tfrac{1}{3}(w - v).$$

**Exercise 1.B.4.** The empty set is not a vector space. The empty set fails to satisfy only one of the requirements listed in the definition of a vector space (1.20). Which one?

**Solution.** The empty set does not contain an additive identity.

**Exercise 1.B.5.** Show that in the definition of a vector space (1.20), the additive inverse condition can be replaced with the condition that

$$0v = 0 \text{ for all } v \in V.$$

Here the 0 on the left side is the number 0, and the 0 on the right side is the additive identity of $V$.

> *The phrase a "condition can be replaced" in a definition means that the collection of objects satisfying the definition is unchanged if the original condition is replaced with the new condition.*

**Solution.** If $V$ satisfies all of the conditions in 1.20, then as shown in 1.30 we have $0v = 0$ for all $v \in V$. Suppose that $V$ satisfies all of the conditions in 1.20, except we have replaced the additive inverse condition with the condition that $0v = 0$ for all $v \in V$. We want to show that for each $v \in V$, there exists an element $w \in V$ such that $v + w = 0$. Indeed, for $v \in V$, let $w = (-1)v$ and observe that



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$$v + w = 1v + (-1)v = (1 - 1)v = 0v = 0.$$

**Exercise 1.B.6.** Let $\infty$ and $-\infty$ denote two distinct objects, neither of which is in $\mathbf{R}$. Define an addition and scalar multiplication on $\mathbf{R} \cup \{\infty, -\infty\}$ as you could guess from the notation. Specifically, the sum and product of two real numbers is as usual, and for $t \in \mathbf{R}$ define

$$t\infty = \begin{cases} -\infty & \text{if } t < 0, \\ 0 & \text{if } t = 0, \\ \infty & \text{if } t > 0, \end{cases} \qquad t(-\infty) = \begin{cases} \infty & \text{if } t < 0, \\ 0 & \text{if } t = 0, \\ -\infty & \text{if } t > 0, \end{cases}$$

and

$$t + \infty = \infty + t = \infty + \infty = \infty,$$
$$t + (-\infty) = (-\infty) + t = (-\infty) + (-\infty) = -\infty,$$
$$\infty + (-\infty) = (-\infty) + \infty = 0.$$

With these operations of addition and scalar multiplication, is $\mathbf{R} \cup \{\infty, -\infty\}$ a vector space over $\mathbf{R}$? Explain.

**Solution.** This is not a vector space over $\mathbf{R}$, since addition is not associative:

$$(1 + \infty) + (-\infty) = \infty + (-\infty) = 0 \neq 1 = 1 + 0 = 1 + (\infty + (-\infty)).$$

**Exercise 1.B.7.** Suppose $S$ is a nonempty set. Let $V^S$ denote the set of functions from $S$ to $V$. Define a natural addition and scalar multiplication on $V^S$, and show that $V^S$ is a vector space with these definitions.

**Solution.** We define addition and scalar multiplication on $V^S$ as in 1.24, i.e. for $f, g \in V^S$ the sum $f + g \in V^S$ is the function

$$\begin{aligned} f + g \ : \ S &\to V \\ x &\mapsto f(x) + g(x); \end{aligned}$$

the addition $f(x) + g(x)$ is vector addition in $V$. Similarly, for $\lambda \in \mathbf{F}$ and $f \in V^S$, the product $\lambda f \in V^S$ is the function

$$\begin{aligned} \lambda f \ : \ S &\to V \\ x &\mapsto \lambda f(x); \end{aligned}$$

the product $\lambda f(x)$ is scalar multiplication in $V$. We now show that $V^S$ with these definitions satisfies each condition in definition 1.20.

**Commutativity.** Let $f, g \in V^S$ and $x \in S$ be given. Observe that

$$(f + g)(x) = f(x) + g(x) = g(x) + f(x) = (g + f)(x),$$



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where we have used the commutativity of addition in $V$ for the second equality. It follows that $f + g = g + f$.

**Associativity.** Let $f, g, h \in V^S$ and $x \in S$ be given. Observe that

$$\begin{aligned}((f + g) + h)(x) &= (f + g)(x) + h(x) = (f(x) + g(x)) + h(x) \\ &= f(x) + (g(x) + h(x)) = f(x) + (g + h)(x) = (f + (g + h))(x),\end{aligned}$$

where we have used the associativity of addition in $V$ for the third equality. It follows that $(f + g) + h = f + (g + h)$. Similarly, let $f \in V^S$ and $a, b \in \mathbf{F}$ be given. Observe that, for any $x \in S$,

$$((ab)f)(x) = (ab)f(x) = a(bf(x)) = a((bf)(x)) = (a(bf))(x),$$

where we have used the associativity of scalar multiplication in $V$ for the second equality. It follows that $(ab)f = a(bf)$.

**Additive identity.** We claim that the additive identity in $V^S$ is the function $0 : S \to V$ given by $0(x) = 0$ for any $x \in S$; the 0 on the right-hand side is the additive identity in $V$. Indeed, for any $f \in V^S$ and $x \in S$ we have

$$(f + 0)(x) = f(x) + 0(x) = f(x) + 0 = f(x).$$

It follows that $f + 0 = f$.

**Additive inverse.** For $f \in V^S$, define $g : S \to V$ by $g(x) = -f(x)$ for $x \in S$, where $-f(x)$ is the additive inverse in $V$ of $f(x)$. We claim that $g$ is the additive inverse of $f$. To see this, let $x \in S$ be given and observe that

$$(f + g)(x) = f(x) + g(x) = f(x) + (-f(x)) = 0 = 0(x);$$

it follows that $f + g = 0$.

**Multiplicative identity.** Let $f \in V^S$ and $x \in S$ be given. Observe that

$$(1f)(x) = 1f(x) = f(x),$$

where we have used that $1v = v$ for any $v \in V$. It follows that $1f = f$.

**Distributive properties.** Let $a \in \mathbf{F}$ and $f, g \in V^S$ be given. Observe that, for any $x \in S$,

$$\begin{aligned}(a(f + g))(x) &= a(f + g)(x) = a((f(x) + g(x)) \\ &= af(x) + ag(x) = (af)(x) + (ag)(x) = (af + ag)(x),\end{aligned}$$

where we have used the first distributive property in $V$ for the third equality. It follows that $a(f + g) = af + ag$. Similarly, let $a, b \in \mathbf{F}$ and $f \in V^S$ be given. For any $x \in S$, observe that

$$((a + b)f)(x) = (a + b)f(x) = af(x) + bf(x) = (af)(x) + (bf)(x) = (af + bf)(x),$$

where we have used the second distributive property in $V$ for the second equality. It follows that $(a + b)f = af + bf$.

We may conclude that $V^S$ is a vector space over $\mathbf{F}$.



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**Exercise 1.B.8.** Suppose $V$ is a real vector space.

- The *complexification* of $V$, denoted by $V_{\mathbf{C}}$, equals $V \times V$. An element of $V_{\mathbf{C}}$ is an ordered pair $(u, v)$, where $u, v \in V$, but we write this as $u + iv$.
- Addition on $V_{\mathbf{C}}$ is defined by

$$(u_1 + iv_1) + (u_2 + iv_2) = (u_1 + u_2) + i(v_1 + v_2)$$

  for all $u_1, v_1, u_2, v_2 \in V$.
- Complex scalar multiplication on $V_{\mathbf{C}}$ is defined by

$$(a + bi)(u + iv) = (au - bv) + i(av + bu)$$

  for all $a, b \in \mathbf{R}$ and all $u, v \in V$.

Prove that with the definitions of addition and scalar multiplication as above, $V_{\mathbf{C}}$ is a complex vector space.

> *Think of $V$ as a subset of $V_{\mathbf{C}}$ by identifying $u \in V$ with $u + i0$. The construction of $V_{\mathbf{C}}$ from $V$ can then be thought of as generalizing the construction of $\mathbf{C}^n$ from $\mathbf{R}^n$.*

**Solution.** We need to verify each condition in definition 1.20. The algebraic manipulations required to show that commutativity, associativity, and the first distributive property hold for $V_{\mathbf{C}}$ are essentially the same algebraic manipulations we performed in Exercise 1.A.1, Exercise 1.A.2, Exercise 1.A.3, and Exercise 1.A.4, except instead of using the algebraic properties of $\mathbf{R}$, we use the algebraic properties of $V$ (i.e. the properties listed in 1.20); we will avoid repeating ourselves and instead verify the remaining conditions.

**Additive identity.** We claim that the additive identity in $V_{\mathbf{C}}$ is $0 + i0$, where $0$ is the additive identity in $V$. Indeed, for any $u + iv \in V_{\mathbf{C}}$ we have

$$(u + iv) + (0 + i0) = (u + 0) + i(v + 0) = u + iv.$$

**Additive inverse.** We claim that the additive inverse of an element $u + iv \in V_{\mathbf{C}}$ is the element $(-u) + i(-v)$, where $-u$ is the additive inverse of $u$ in $V$. Indeed,

$$(u + iv) + ((-u) + i(-v)) = (u + (-u)) + i(v + (-v)) = 0 + i0.$$

**Multiplicative identity.** For any $u + iv \in V_{\mathbf{C}}$, we have

$$(1 + 0i)(u + iv) = (1u - 0v) + i(1v + 0u) = u + iv.$$

**Distributive properties.** For the second distributive property, let $a + bi, c + di \in \mathbf{C}$ and $u + iv \in V_{\mathbf{C}}$ be given. Observe that

$$\begin{aligned}((a + bi) + (c + di))(u + iv) &= ((a + c) + (b + d)i)(u + iv) \\ &= ((a + c)u - (b + d)v) + i((a + c)v + (b + d)u) \\ &= (au + cu - bv - dv) + i(av + cv + bu + du)\end{aligned}$$



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$$= ((au - bv) + i(av + bu)) + ((cu - dv) + i(cv + du))$$
$$= (a + bi)(u + iv) + (c + di)(u + iv),$$

where we have used the second distributive property for $V$ for the third equality.



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## 1.C. Subspaces

**Exercise 1.C.1.** For each of the following subsets of $\mathbf{F}^3$, determine whether it is a subspace of $\mathbf{F}^3$.

(a) $\{(x_1, x_2, x_3) \in \mathbf{F}^3 : x_1 + 2x_2 + 3x_3 = 0\}$

(b) $\{(x_1, x_2, x_3) \in \mathbf{F}^3 : x_1 + 2x_2 + 3x_3 = 4\}$

(c) $\{(x_1, x_2, x_3) \in \mathbf{F}^3 : x_1 x_2 x_3 = 0\}$

(d) $\{(x_1, x_2, x_3) \in \mathbf{F}^3 : x_1 = 5x_3\}$

**Solution.** Let $U$ denote the set in each part of this question.

(a) This is a subspace of $\mathbf{F}^3$. Certainly the zero vector belongs to $U$. Suppose that $x = (x_1, x_2, x_3), y = (y_1, y_2, y_3) \in U$ and $\alpha \in \mathbf{F}$ and observe that

$$(x_1 + y_1) + 2(x_2 + y_2) + 3(x_3 + y_3) = (x_1 + 2x_2 + 3x_3) + (y_1 + 2y_2 + 3y_3) = 0 + 0 = 0,$$

$$\alpha x_1 + 2(\alpha x_2) + 3(\alpha x_3) = \alpha(x_1 + 2x_2 + 3x_3) = \alpha 0 = 0.$$

Thus $x + y$ and $\alpha x$ also belong to $U$. It follows from 1.34 that $U$ is a subspace of $V$.

(b) This is not a subspace of $\mathbf{F}^3$ because it does not contain the zero vector.

(c) This is not a subspace of $\mathbf{F}^3$ because it is not closed under addition: $(1, 1, 0)$ and $(0, 0, 1)$ belong to $U$, but $(1, 1, 0) + (0, 0, 1) = (1, 1, 1)$ does not belong to $U$.

(d) This is a subspace of $\mathbf{F}^3$. Note that $U = \{(x_1, x_2, x_3) \in \mathbf{F}^3 : x_1 - 5x_3 = 0\}$. Certainly the zero vector belongs to $U$. Suppose that $x = (x_1, x_2, x_3), y = (y_1, y_2, y_3) \in U$ and $\alpha \in \mathbf{F}$ and observe that

$$(x_1 + y_1) - 5(x_3 + y_3) = (x_1 - 5x_3) + (y_1 - 5y_3) = 0 + 0 = 0,$$

$$\alpha x_1 - 5(\alpha x_3) = \alpha(x_1 - 5x_3) = \alpha 0 = 0.$$

Thus $x + y$ and $\alpha x$ also belong to $U$. It follows from 1.34 that $U$ is a subspace of $V$.

**Exercise 1.C.2.** Verify all assertions about subspaces in Example 1.35.

**Solution.**

(a) The assertion is that if $b \in \mathbf{F}$, then

$$U = \{(x_1, x_2, x_3, x_4) \in \mathbf{F}^4 : x_3 = 5x_4 + b\} = \{(x_1, x_2, x_3, x_4) \in \mathbf{F}^4 : x_3 - 5x_4 = b\}$$

is a subspace of $\mathbf{F}^4$ if and only if $b = 0$. Indeed, if $b \neq 0$ then $U$ is not a subspace of $\mathbf{F}^4$ because the zero vector does not belong to $U$, and if $b = 0$ then we may argue as in Exercise 1.C.1 (d) to see that $U$ is a subspace of $\mathbf{F}^4$.



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(b) The assertion is that the set of continuous real-valued functions on the interval $[0,1]$ is a subspace of $\mathbf{R}^{[0,1]}$, i.e.

$$U = \{f : [0,1] \to \mathbf{R}, f \text{ continuous}\}$$

is a subspace of $\mathbf{R}^{[0,1]}$. Certainly the zero function $x \mapsto 0$ on $[0,1]$ is continuous and hence belongs to $U$, and it is well-known from elementary real analysis that sums and constant multiples of continuous functions are again continuous. It follows from 1.34 that $U$ is a subspace of $\mathbf{R}^{[0,1]}$.

(c) The assertion is that the set of differentiable real-valued functions on $\mathbf{R}$ is a subspace of $\mathbf{R}^{\mathbf{R}}$, i.e.

$$U = \{f : \mathbf{R} \to \mathbf{R}, f \text{ differentiable}\}$$

is a subspace of $\mathbf{R}^{\mathbf{R}}$. Certainly the zero function $x \mapsto 0$ on $\mathbf{R}$ is differentiable and hence belongs to $U$, and it is well-known from elementary real analysis that sums and constant multiples of differentiable functions are again differentiable. It follows from 1.34 that $U$ is a subspace of $\mathbf{R}^{\mathbf{R}}$.

(d) The assertion is that the set $U$ of differentiable real-valued functions $f$ on the interval $(0,3)$ such that $f'(2) = b$ is a subspace of $\mathbf{R}^{(0,3)}$ if and only if $b = 0$. If $b \neq 0$, then the zero function $x \mapsto 0$ on $(0,3)$, which has derivative $0 \neq b$ at $x = 2$, does not belong to $U$ and thus $U$ is not a subspace of $\mathbf{R}^{(0,3)}$.

Suppose that $b = 0$ and note that the zero function now belongs to $U$. If $f, g \in U$ and $\alpha \in \mathbf{R}$, then

$$(f+g)'(2) = f'(2) + g'(2) = 0 + 0 = 0 \quad \text{and} \quad (\alpha f)'(2) = \alpha f'(2) = \alpha 0 = 0.$$

Thus $f + g$ and $\alpha f$ belong to $U$. It follows from 1.34 that $U$ is a subspace of $\mathbf{R}^{(0,3)}$.

(e) The assertion is that the set $U$ of all sequences of complex numbers with limit 0 is a subspace of $\mathbf{C}^{\infty}$. Certainly the zero sequence $(0, 0, 0, ...)$ has limit 0 and hence belongs to $U$. Suppose that $x = (x_n)_{n=1}^{\infty}$ and $y = (y_n)_{n=1}^{\infty}$ belong to $U$ and $\alpha \in \mathbf{C}$. Using basic results about limits, observe that

$$\lim_{n\to\infty}(x_n + y_n) = \lim_{n\to\infty} x_n + \lim_{n\to\infty} y_n = 0 + 0 = 0$$
$$\text{and} \quad \lim_{n\to\infty}(\alpha x_n) = \alpha \lim_{n\to\infty} x_n = \alpha 0 = 0.$$

Thus $x + y$ and $\alpha x$ belong to $U$. It follows from 1.34 that $U$ is a subspace of $\mathbf{C}^{(0,3)}$.

**Exercise 1.C.3.** Show that the set of differentiable real-valued functions $f$ on the interval $(-4,4)$ such that $f'(-1) = 3f(2)$ is a subspace of $\mathbf{R}^{(-4,4)}$.

**Solution.** Let $U$ be the set in question; it is straightforward to verify that the zero function belongs to $U$. Suppose that $f, g \in U$ and $\alpha \in \mathbf{R}$. Observe that

$$(f+g)'(-1) = f'(-1) + g'(-1) = 3f(2) + 3g(2) = 3(f(2) + g(2)) = 3(f+g)(2)$$



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$$\text{and} \quad (\alpha f)'(-1) = \alpha f'(-1) = \alpha(3f(2)) = 3(\alpha f(2)) = 3(\alpha f)(2).$$

Thus $f + g$ and $\alpha f$ belong to $U$. It follows from 1.34 that $U$ is a subspace of $\mathbf{R}^{(-4,4)}$.

**Exercise 1.C.4.** Suppose $b \in \mathbf{R}$. Show that the set of continuous real-valued functions $f$ on the interval $[0, 1]$ such that $\int_0^1 f = b$ is a subspace of $\mathbf{R}^{[0,1]}$ if and only if $b = 0$.

**Solution.** Let $U$ be the set in question. If $b \neq 0$ then the zero function $x \mapsto 0$ on $[0, 1]$, which has integral $0 \neq b$ over $[0, 1]$, does not belong to $U$ and thus $U$ is not a subspace of $\mathbf{R}^{[0,1]}$.

Suppose that $b = 0$ and note that the zero function now belongs to $U$. If $f, g \in U$ and $\alpha \in \mathbf{R}$, then using basic properties of integration we have

$$\int_0^1 (f + g) = \int_0^1 f + \int_0^1 g = 0 + 0 = 0 \quad \text{and} \quad \int_0^1 (\alpha f) = \alpha \int_0^1 f = \alpha 0 = 0.$$

Thus $f + g$ and $\alpha f$ belong to $U$. It follows from 1.34 that $U$ is a subspace of $\mathbf{R}^{[0,1]}$.

**Exercise 1.C.5.** Is $\mathbf{R}^2$ a subspace of the complex vector space $\mathbf{C}^2$?

**Solution.** The question is whether the subset

$$\mathbf{R}^2 = \{(x, y) : x, y \in \mathbf{R}\} \subseteq \{(z, w) : z, w \in \mathbf{C}\} = \mathbf{C}^2$$

is a subspace, where we are taking complex scalars in $\mathbf{C}^2$. This is not a subspace because it is not closed under scalar multiplication: $(1, 0) \in \mathbf{R}^2$ but $i(1, 0) = (i, 0) \notin \mathbf{R}^2$.

**Exercise 1.C.6.**

(a) Is $\{(a, b, c) \in \mathbf{R}^3 : a^3 = b^3\}$ a subspace of $\mathbf{R}^3$?

(b) Is $\{(a, b, c) \in \mathbf{C}^3 : a^3 = b^3\}$ a subspace of $\mathbf{C}^3$?

**Solution.**

(a) Let $U$ be the set in question. For $a, b \in \mathbf{R}$ we have $a^3 = b^3$ if and only if $a = b$ and thus the set $U$ can be expressed as

$$U = \{(a, a, c) \in \mathbf{R}^3 : a, c \in \mathbf{R}\}.$$

Certainly $(0, 0, 0) \in U$. If $(a, a, c), (x, x, y) \in U$ and $\lambda \in \mathbf{R}$, then observe that

$$(a, a, c) + (x, x, y) = (a + x, a + x, c + y) \in U \quad \text{and} \quad \lambda(a, a, c) = (\lambda a, \lambda a, \lambda c) \in U.$$

It follows from 1.34 that $U$ is a subspace of $\mathbf{R}^3$.

(b) Let $U$ be the set in question. Observe that

$$\left(\frac{-1 + \sqrt{3}i}{2}\right)^3 = \left(\frac{-1 - \sqrt{3}i}{2}\right)^3 = 1.$$



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It follows that $u := \left(\frac{-1+\sqrt{3}i}{2}, 1, 0\right)$ and $v := \left(\frac{-1-\sqrt{3}i}{2}, 1, 0\right)$ belong to $U$. However,

$$u + v = (-1, 2, 0) \notin U.$$

Thus $U$ is not a subspace of $\mathbf{C}^3$ because it is not closed under addition.

**Exercise 1.C.7.** Prove or give a counterexample: If $U$ is a nonempty subset of $\mathbf{R}^2$ such that $U$ is closed under addition and under taking additive inverses (meaning $-u \in U$ whenever $u \in U$), then $U$ is a subspace of $\mathbf{R}^2$.

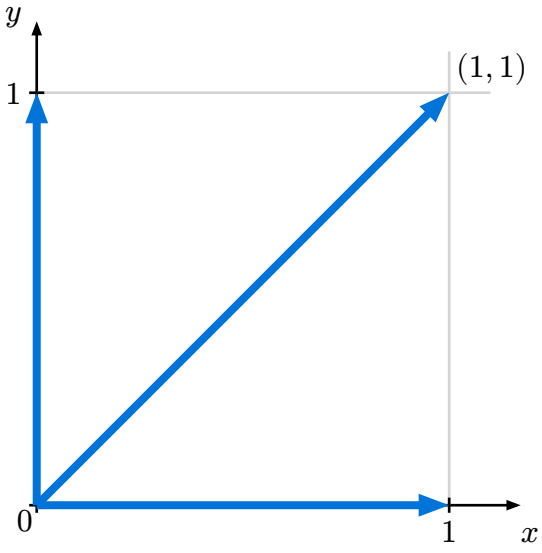
**Solution.** For a counterexample, consider $U = \{(a, b) : a, b \in \mathbf{Q}\} \subseteq \mathbf{R}^2$, which satisfies the required conditions since the sum of two rational numbers is a rational number and the additive inverse of a rational number is a rational number. However, $U$ is not a subspace of $\mathbf{R}^2$ because it is not closed under scalar multiplication: $(1, 0) \in U$ but $\sqrt{2}(1, 0) = \left(\sqrt{2}, 0\right) \notin U$.

**Exercise 1.C.8.** Give an example of a nonempty subset $U$ of $\mathbf{R}^2$ such that $U$ is closed under scalar multiplication, but $U$ is not a subspace of $\mathbf{R}^2$.

**Solution.** Let $U$ be the union of the $x$- and $y$-axes, i.e.

$$U = \{(x, 0) : x \in \mathbf{R}\} \cup \{(0, y) : y \in \mathbf{R}\}.$$

It is straightforward to verify that $U$ is closed under scalar multiplication. However, $U$ is not a subspace of $\mathbf{R}^2$ because it is not closed under addition: $(1, 0)$ and $(0, 1)$ belong to $U$, but $(1, 0) + (0, 1) = (1, 1)$ does not.




**Exercise 1.C.9.** A function $f : \mathbf{R} \to \mathbf{R}$ is called *periodic* if there exists a positive number $p$ such that $f(x) = f(x + p)$ for all $x \in \mathbf{R}$. Is the set of periodic functions from $\mathbf{R}$ to $\mathbf{R}$ a subspace of $\mathbf{R}^{\mathbf{R}}$? Explain.

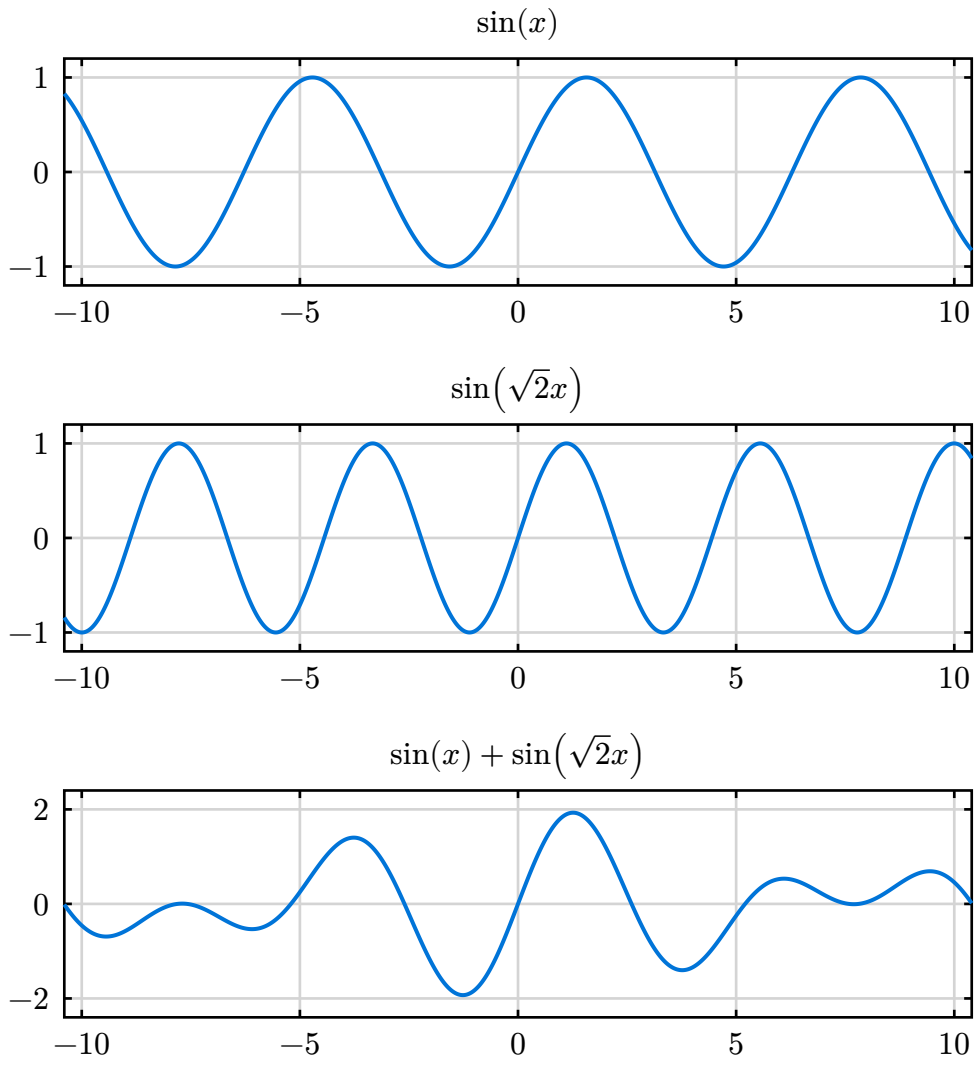
**Solution.** Consider the periodic functions $\sin(x)$ and $\sin\left(\sqrt{2}x\right)$ and let



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$$f(x) = \sin(x) + \sin\left(\sqrt{2}x\right).$$

We will show that $f$ is not periodic.




Suppose there was a positive real number $p$ such that $f(x) = f(x+p)$ for all $x \in \mathbf{R}$, i.e.

$$\sin(x) + \sin\left(\sqrt{2}x\right) = \sin(x+p) + \sin\left(\sqrt{2}x + \sqrt{2}p\right) \text{ for all } x \in \mathbf{R}. \tag{1}$$

By differentiating this equation twice, we see that

$$\sin(x) + 2\sin\left(\sqrt{2}x\right) = \sin(x+p) + 2\sin\left(\sqrt{2}x + \sqrt{2}p\right) \text{ for all } x \in \mathbf{R}. \tag{2}$$

Subtracting equation (1) from equation (2) gives us

$$\sin\left(\sqrt{2}x\right) = \sin\left(\sqrt{2}x + \sqrt{2}p\right) \text{ for all } x \in \mathbf{R}, \tag{3}$$

which together with equation (1) implies that

$$\sin(x) = \sin(x+p) \text{ for all } x \in \mathbf{R}. \tag{4}$$

By taking $x = 0$ in equation (4) we see that $0 = \sin(p)$, which is the case if and only if $p = n\pi$ for some positive integer $n$ ($p$ was assumed to be positive). Substituting this value of $p$ and $x = 0$ into equation (3) gives $0 = \sin\left(n\sqrt{2}\pi\right)$, which is the case if and only if $n\sqrt{2}\pi = m\pi$ for some integer $m$, which must be positive since $n$ is positive. It follows that $\sqrt{2} = \frac{m}{n}$, contradicting the irrationality of $\sqrt{2}$.



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Thus $f$ is not periodic and we may conclude that the set of periodic functions from $\mathbf{R}$ to $\mathbf{R}$ is not a subspace of $\mathbf{R}^{\mathbf{R}}$ because it is not closed under addition.

**Exercise 1.C.10.** Suppose $V_1$ and $V_2$ are subspaces of $V$. Prove that the intersection $V_1 \cap V_2$ is a subspace of $V$.

**Solution.** Because $V_1$ and $V_2$ are subspaces of $V$, we have $0 \in V_1$ and $0 \in V_2$ and thus $0 \in V_1 \cap V_2$. Suppose $u, v \in V_1 \cap V_2$ and $\lambda \in \mathbf{F}$. Since $u, v \in V_1$ and $V_1$ is a subspace of $V$, we have $u + v \in V_1$ and $\lambda u \in V_1$. Similarly, $u + v \in V_2$ and $\lambda u \in V_2$. Thus $u + v \in V_1 \cap V_2$ and $\lambda u \in V_1 \cap V_2$. We may use 1.34 to conclude that $V_1 \cap V_2$ is a subspace of $V$.

**Exercise 1.C.11.** Prove that the intersection of every collection of subspaces of $V$ is a subspace of $V$.

**Solution.** Let $\mathcal{U}$ be an arbitrary collection of subspaces of $V$. We will show that $\bigcap \mathcal{U}$ is a subspace of $V$. The zero vector belongs to $\bigcap \mathcal{U}$ because each $U \in \mathcal{U}$ is a subspace of $V$ and hence contains the zero vector. If $u, v \in \bigcap \mathcal{U}, \lambda \in \mathbf{F}$, and $U \in \mathcal{U}$, then $u, v \in U$ and thus $u + v \in U$ and $\lambda u \in U$ since $U$ is a subspace of $V$. Because $U \in \mathcal{U}$ was arbitrary, it follows that $u + v \in \bigcap \mathcal{U}$ and $\lambda u \in \bigcap \mathcal{U}$. We may use 1.34 to conclude that $\bigcap \mathcal{U}$ is a subspace of $V$.

**Exercise 1.C.12.** Prove that the union of two subspaces of $V$ is a subspace of $V$ if and only if one of the subspaces is contained in the other.

**Solution.** Suppose that $U$ and $W$ are subspaces of $V$. We want to show that $U \cup W$ is a subspace of $V$ if and only if $U \subseteq W$ or $W \subseteq U$. If one of $U$ or $W$ is contained in the other then either $U \cup W = U$ or $U \cup W = W$; in either case, $U \cup W$ is then a subspace of $V$ by assumption.

For the converse, it will suffice to show that if $U \cup W$ is a subspace of $V$ and $U \nsubseteq W$, then $W \subseteq U$. Since $U \nsubseteq W$, there is some $u \in U$ such that $u \notin W$. Let $w \in W$ be given and note that, because $U \cup W$ is a subspace of $V$ and $u, w \in U \cup W$, we must have $u + w \in U \cup W$. It cannot be the case that $u + w \in W$, since then $u + w - w = u \in W$, so it must be the case that $u + w \in U$. It follows that $u + w - u = w \in U$ and hence that $W \subseteq U$, as desired.

**Exercise 1.C.13.** Prove that the union of three subspaces of $V$ is a subspace of $V$ if and only if one of the subspaces contains the other two.

*This exercise is surprisingly harder than Exercise 12, possibly because this exercise is not true if we replace* $\mathbf{F}$ *with a field containing only two elements.*

**Solution.** Let $U_1, U_2$, and $U_3$ be subspaces of $V$. We want to show that $U = U_1 \cup U_2 \cup U_3$ is a subspace of $V$ if and only if some $U_j$ contains the other two. If some $U_j$ contains the other two, then $U = U_j$ is a subspace of $V$ by assumption.



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Suppose that $U$ is a subspace of $V$. If any $U_j$ is contained in the union of the other two, say $U_1 \subseteq U_2 \cup U_3$, then $U = U_2 \cup U_3$ and we may apply Exercise 1.C.12 to see that either $U_2 \subseteq U_3$ or $U_3 \subseteq U_2$; in either case, one $U_j$ contains the other two. Suppose therefore that no $U_j$ is contained in the union of the other two. Seeking a contradiction, suppose further that no $U_j$ contains the other two, so that

$$U_1 \nsubseteq (U_2 \cup U_3) \quad \text{and} \quad (U_2 \cup U_3) \nsubseteq U_1.$$

It follows that there exists some $u \in U_1$ such that $u \notin U_2 \cup U_3$ and some $v \in U_2 \cup U_3$ such that $v \notin U_1$. Let $W = \{v + \lambda u : \lambda \in \mathbf{F}\} \subseteq U$ and observe that no element of $W$ belongs to $U_1$, for if $v + \lambda u \in U_1$ then $v + \lambda u - \lambda u = v \in U_1$—but $v \notin U_1$. Thus

$$W \cap U_1 = \emptyset \quad \text{and} \quad W \subseteq (U_1 \cup U_2 \cup U_3) \quad \Rightarrow \quad W \subseteq (U_2 \cup U_3).$$

Because $W$ contains infinitely many elements, there must be some $i \in \{2, 3\}$ such that $U_i$ contains infinitely many elements of $W$. There then exist $\lambda, \mu \in \mathbf{F}$ such that $\lambda \neq \mu$ and such that $v + \lambda u$ and $v + \mu u$ both belong to $U_i$, which implies that $(\lambda - \mu)u \in U_i$ since $U_i$ is a subspace of $V$. This gives $u \in U_i$ since $\lambda - \mu \neq 0$, contradicting that $u \notin U_2 \cup U_3$. We may conclude that some $U_j$ contains the other two.

**Exercise 1.C.14.** Suppose

$$U = \{(x, -x, 2x) \in \mathbf{F}^3 : x \in \mathbf{F}\} \quad \text{and} \quad W = \{(x, x, 2x) \in \mathbf{F}^3 : x \in \mathbf{F}\}.$$

Describe $U + W$ using symbols, and also give a description of $U + W$ that uses no symbols.

**Solution.** We claim that $U + W$ is the subspace

$$E = \{(x, y, 2x) \in \mathbf{F}^3 : x, y \in \mathbf{F}\}.$$

To see this, let $(x, -x, 2x) \in U$ and $(y, y, 2y) \in W$ be given and notice that

$$(x, -x, 2x) + (y, y, 2y) = (x + y, -x + y, 2(x + y)) \in E.$$

Thus $U + W \subseteq E$. For the reverse inclusion, let $(x, y, 2x) \in E$ be given and observe that

$$(x, y, 2x) = \left(\frac{x - y}{2}, \frac{y - x}{2}, x - y\right) + \left(\frac{x + y}{2}, \frac{x + y}{2}, x + y\right) \in U + W.$$

Thus $U + W = E$, as claimed. In words, $U + W$ is the subspace of $\mathbf{F}^3$ consisting of those vectors whose third coordinate is twice their first coordinate.

**Exercise 1.C.15.** Suppose $U$ is a subspace of $V$. What is $U + U$?

**Solution.** For $u + v \in U + U$ we have $u + v \in U$ since $U$ is a subspace of $V$, and for $u \in U$ we have $u = u + 0 \in U + U$. Thus $U + U = U$.



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**Exercise 1.C.16.** Is the operation of addition on the subspaces of $V$ commutative? In other words, if $U$ and $W$ are subspaces of $V$, is $U + W = W + U$?

**Solution.** The operation is commutative, since addition of vectors in $V$ is commutative. If $u + w \in U + W$, then $u + w = w + u \in W + U$, so that $U + W \subseteq W + U$. Similarly, $W + U \subseteq U + W$.

**Exercise 1.C.17.** Is the operation of addition on the subspaces of $V$ associative? In other words, if $V_1, V_2, V_3$ are subspaces of $V$, is

$$(V_1 + V_2) + V_3 = V_1 + (V_2 + V_3)?$$

**Solution.** The operation is associative, since addition of vectors in $V$ is associative. If $(u_1 + u_2) + u_3 \in (U_1 + U_2) + U_3$, then

$$(u_1 + u_2) + u_3 = u_1 + (u_2 + u_3) \in U_1 + (U_2 + U_3),$$

so that $(U_1 + U_2) + U_3 \subseteq U_1 + (U_2 + U_3)$. Similarly, $U_1 + (U_2 + U_3) \subseteq (U_1 + U_2) + U_3$.

**Exercise 1.C.18.** Does the operation of addition on the subspaces of $V$ have an additive identity? Which subspaces have additive inverses?

**Solution.** The subspace $\{0\}$ is the additive identity for the operation. If $U$ is a subspace of $V$ then $u + 0 = u$ for any $u \in U$; it follows that $U + \{0\} = U$.

Since $\{0\} + \{0\} = \{0\}$, the subspace $\{0\}$ is its own additive inverse. We claim that no other subspace of $V$ has an additive inverse, i.e. if $U$ is a subspace of $V$ with $U \neq \{0\}$, then there does not exist a subspace $W$ satisfying $U + W = \{0\}$. Indeed, simply observe that $U \subseteq U + W$ for any subspace $W$, so that $U + W \neq \{0\}$.

**Exercise 1.C.19.** Prove or give a counterexample: If $V_1, V_2, U$ are subspaces of $V$ such that

$$V_1 + U = V_2 + U,$$

then $V_1 = V_2$.

**Solution.** This is false. For a counterexample, consider the real vector space $\mathbf{R}$ and observe that

$$\{0\} + \mathbf{R} = \mathbf{R} + \mathbf{R} = \mathbf{R},$$

but $\{0\} \neq \mathbf{R}$.



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**Exercise 1.C.20.** Suppose

$$U = \{(x, x, y, y) \in \mathbf{F}^4 : x, y \in \mathbf{F}\}.$$

Find a subspace $W$ of $\mathbf{F}^4$ such that $\mathbf{F}^4 = U \oplus W$.

**Solution.** Let

$$W = \{(0, a, 0, b) \in \mathbf{F}^4 : a, b \in \mathbf{F}\};$$

it is straightforward to verify that $W$ is a subspace of $\mathbf{F}^4$. If $v \in U \cap W$, then

$$v \in W \quad \Rightarrow \quad v = (0, a, 0, b) \text{ for some } a, b \in \mathbf{F},$$
$$v \in U \quad \Rightarrow \quad a = b = 0 \quad \Rightarrow \quad v = 0.$$

Thus $U \cap W = \{0\}$ and it follows from 1.46 that the sum $U + W$ is direct.

Let $(v_1, v_2, v_3, v_4) \in \mathbf{F}^4$ be given and observe that

$$(v_1, v_2, v_3, v_4) = (v_1, v_1, v_3, v_3) + (0, v_2 - v_1, 0, v_4 - v_3) \in U \oplus W.$$

Thus $\mathbf{F}^4 = U \oplus W$.

**Exercise 1.C.21.** Suppose

$$U = \{(x, y, x + y, x - y, 2x) \in \mathbf{F}^5 : x, y \in \mathbf{F}\}.$$

Find a subspace $W$ of $\mathbf{F}^5$ such that $\mathbf{F}^5 = U \oplus W$.

**Solution.** Let

$$W = \{(0, 0, a, b, c) \in \mathbf{F}^5 : a, b, c \in \mathbf{F}\};$$

it is straightforward to verify that $W$ is a subspace of $\mathbf{F}^5$. If $v \in U \cap W$, then

$$v \in U \quad \Rightarrow \quad v = (x, y, x + y, x - y, 2x) \text{ for some } x, y \in \mathbf{F},$$
$$v \in W \quad \Rightarrow \quad x = y = 0 \quad \Rightarrow \quad v = 0.$$

Thus $U \cap W = \{0\}$ and it follows from 1.46 that the sum $U + W$ is direct.

Let $v = (v_1, v_2, v_3, v_4, v_5) \in \mathbf{F}^5$ be given and observe that

$$\begin{aligned}(v_1, v_2, v_3, v_4, v_5) = (v_1, v_2, v_1 + v_2, v_1 - v_2, 2v_1) \\ + (0, 0, v_3 - (v_1 + v_2), v_4 - (v_1 - v_2), v_5 - 2v_1) \in U \oplus W.\end{aligned}$$

Thus $\mathbf{F}^5 = U \oplus W$.



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**Exercise 1.C.22.** Suppose

$$U = \{(x, y, x + y, x - y, 2x) \in \mathbf{F}^5 : x, y \in \mathbf{F}\}.$$

Find three subspaces $W_1, W_2, W_3$ of $\mathbf{F}^5$, none of which equals $\{0\}$, such that $\mathbf{F}^5 = U \oplus W_1 \oplus W_2 \oplus W_3$.

**Solution.** Let

$$W_1 = \{(0, 0, a, 0, 0) \in \mathbf{F}^5 : a \in \mathbf{F}\}, \quad W_2 = \{(0, 0, 0, b, 0) \in \mathbf{F}^5 : b \in \mathbf{F}\},$$
$$W_3 = \{(0, 0, 0, 0, c) \in \mathbf{F}^5 : c \in \mathbf{F}\};$$

it is straightforward to verify that $W_1, W_2$, and $W_3$ are subspaces of $\mathbf{F}^5$. Suppose that

$$u = (x, y, x + y, x - y, 2x) \in U, \qquad w_1 = (0, 0, a, 0, 0) \in W_1,$$
$$w_2 = (0, 0, 0, b, 0) \in W_2, \quad \text{and} \quad w_3 = (0, 0, 0, 0, c) \in W_3$$

are such that $u + w_1 + w_2 + w_3 = 0$. That is,

$$(x, y, x + y + a, x - y + b, 2x + c) = (0, 0, 0, 0, 0),$$

from which it follows that $x = y = a = b = c = 0$. Thus the only way to express the zero vector as a sum $u + w_1 + w_2 + w_3 \in U + W_1 + W_2 + W_3$ is to take $u = w_1 = w_2 = w_3 = 0$ and so it follows from 1.45 that the sum $U + W_1 + W_2 + W_3$ is direct.

Let $(v_1, v_2, v_3, v_4, v_5) \in \mathbf{F}^5$ be given and observe that

$$\begin{aligned}(v_1, v_2, v_3, v_4, v_5) &= (v_1, v_2, v_1 + v_2, v_1 - v_2, 2v_1) + (0, 0, v_3 - (v_1 + v_2), 0, 0) \\ &\quad + (0, 0, 0, v_4 - (v_1 - v_2), 0) + (0, 0, 0, 0, v_5 - 2v_1) \in U \oplus W_1 \oplus W_2 \oplus W_3.\end{aligned}$$

Thus $\mathbf{F}^5 = U \oplus W_1 \oplus W_2 \oplus W_3$.

**Exercise 1.C.23.** Prove or give a counterexample: If $V_1, V_2, U$ are subspaces of $V$ such that

$$V = V_1 \oplus U \quad \text{and} \quad V = V_2 \oplus U,$$

then $V_1 = V_2$.

> *Hint: When trying to discover whether a conjecture in linear algebra is true or false, it is often useful to start by experimenting in* $\mathbf{F}^2$.

**Solution.** This is false. For a counterexample, consider $V = \mathbf{R}^2$,

$$U = \{(x, 0) \in \mathbf{R}^2 : x \in \mathbf{R}\}, \quad V_1 = \{(0, y) \in \mathbf{R}^2 : y \in \mathbf{R}\}, \quad V_2 = \{(y, y) \in \mathbf{R}^2 : y \in \mathbf{R}\}.$$

It is straightforward to verify that $U \cap V_1 = U \cap V_2 = \{0\}$, so that $U + V_1$ and $U + V_2$ are both direct sums (1.46), and that $U \oplus V_1 = U \oplus V_2 = \mathbf{R}^2$. However, $V_1 \neq V_2$ since $(1, 1) \in V_2$ but $(1, 1) \notin V_1$.



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**Exercise 1.C.24.** A function $f : \mathbf{R} \to \mathbf{R}$ is called *even* if

$$f(-x) = f(x)$$

for all $x \in \mathbf{R}$. A function $f : \mathbf{R} \to \mathbf{R}$ is called *odd* if

$$f(-x) = -f(x)$$

for all $x \in \mathbf{R}$. Let $V_{\text{e}}$ denote the set of real-valued even functions on $\mathbf{R}$ and let $V_{\text{o}}$ denote the set of real-valued odd functions on $\mathbf{R}$. Show that $\mathbf{R}^{\mathbf{R}} = V_{\text{e}} \oplus V_{\text{o}}$.

**Solution.** Suppose that $f \in V_{\text{e}} \cap V_{\text{o}}$, so that $f(x) = -f(x)$ for all $x \in \mathbf{R}$. This implies that $f(x) = 0$ for all $x \in \mathbf{R}$, i.e. $f = 0$. Thus $V_{\text{e}} \cap V_{\text{o}} = \{0\}$ and it follows from 1.46 that the sum $V_{\text{e}} + V_{\text{o}}$ is direct. For $f : \mathbf{R} \to \mathbf{R}$, define $f_{\text{e}} : \mathbf{R} \to \mathbf{R}$ and $f_{\text{o}} : \mathbf{R} \to \mathbf{R}$ by

$$f_{\text{e}}(x) = \frac{f(x) + f(-x)}{2} \quad \text{and} \quad f_{\text{o}}(x) = \frac{f(x) - f(-x)}{2}.$$

It is straightforward to verify that $f_{\text{e}}$ is an even function, $f_{\text{o}}$ is an odd function, and $f = f_{\text{e}} + f_{\text{o}}$. We may conclude that $\mathbf{R}^{\mathbf{R}} = V_{\text{e}} \oplus V_{\text{o}}$.



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# Chapter 2. Finite-Dimensional Vector Spaces

## 2.A. Span and Linear Independence

**Exercise 2.A.1.** Find a list of four distinct vectors in $\mathbf{F}^3$ whose span equals

$$\{(x, y, z) \in \mathbf{F}^3 : x + y + z = 0\}.$$

**Solution.** Let $W$ be the subspace in question and consider the list

$$v_1 = (1, 0, -1), \quad v_2 = (0, 1, -1), \quad v_3 = (1, 1, -2), \quad v_4 = (1, -1, 0).$$

We claim that $\text{span}(v_1, v_2, v_3, v_4) = W$. If $a_1, a_2, a_3, a_4 \in \mathbf{F}$, then

$$a_1v_1 + a_2v_2 + a_3v_3 + a_4v_4 = (a_1 + a_3 + a_4, a_2 + a_3 - a_4, -a_1 - a_2 - 2a_3) \in W$$
$$\text{since} \quad (a_1 + a_3 + a_4) + (a_2 + a_3 - a_4) + (-a_1 - a_2 - 2a_3) = 0.$$

Thus $\text{span}(v_1, v_2, v_3, v_4) \subseteq W$. Now suppose that $(x, y, z) \in W$ and observe that $z = -x - y$. It follows that

$$(x, y, z) = (x, y, -x - y) = xv_1 + yv_2 \in \text{span}(v_1, v_2, v_3, v_4).$$

Thus $W \subseteq \text{span}(v_1, v_2, v_3, v_4)$ and we may conclude that $\text{span}(v_1, v_2, v_3, v_4) = W$, as claimed.

**Exercise 2.A.2.** Prove or give a counterexample: If $v_1, v_2, v_3, v_4$ spans $V$, then the list

$$v_1 - v_2, v_2 - v_3, v_3 - v_4, v_4$$

also spans $V$.

**Solution.** This is true. Let $v \in V$ be given. Since $V = \text{span}(v_1, v_2, v_3, v_4)$, there are scalars $a_1, a_2, a_3, a_4$ such that $v = a_1v_1 + a_2v_2 + a_3v_3 + a_4v_4$. Observe that

$$a_1(v_1 - v_2) + (a_1 + a_2)(v_2 - v_3) + (a_1 + a_2 + a_3)(v_3 - v_4) + (a_1 + a_2 + a_3 + a_4)v_4$$
$$= a_1v_1 + a_2v_2 + a_3v_3 + a_4v_4 = v.$$

Thus $v \in \text{span}(v_1 - v_2, v_2 - v_3, v_3 - v_4, v_4)$. It follows that

$$V = \text{span}(v_1 - v_2, v_2 - v_3, v_3 - v_4, v_4).$$

**Exercise 2.A.3.** Suppose $v_1, ..., v_m$ is a list of vectors in $V$. For $k \in \{1, ..., m\}$, let

$$w_k = v_1 + \cdots + v_k.$$

Show that $\text{span}(v_1, ..., v_m) = \text{span}(w_1, ..., w_m)$.



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**Solution.** For any scalars $a_1, ..., a_m \in \mathbf{F}$, observe that

$$a_1v_1 + a_2(v_1 + v_2) + \cdots + a_m(v_1 + \cdots + v_m) \\ = (a_1 + \cdots + a_m)v_1 + (a_2 + \cdots + a_m)v_2 + \cdots + a_mv_m.$$

It follows that $\operatorname{span}(w_1, ..., w_m) \subseteq \operatorname{span}(v_1, ..., v_m)$. Similarly, for any scalars $a_1, ..., a_m \in \mathbf{F}$, notice that

$$a_1v_1 + a_2v_2 + \cdots + a_mv_m = (a_1 - a_2)v_1 + (a_2 - a_3)(v_1 + v_2) \\ + \cdots + (a_{m-1} - a_m)(v_1 + \cdots + v_{m-1}) + a_m(v_1 + \cdots + v_m).$$

Thus $\operatorname{span}(v_1, ..., v_m) \subseteq \operatorname{span}(w_1, ..., w_m)$.

**Exercise 2.A.4.**

(a) Show that a list of length one in a vector space is linearly independent if and only if the vector in the list is not 0.

(b) Show that a list of length two in a vector space is linearly independent if and only if neither of the two vectors in the list is a scalar multiple of the other.

**Solution.**

(a) Suppose the list consists of the single vector $v \in V$. If $v \neq 0$ and $a \in \mathbf{F}$ is such that $av = 0$, then Exercise 1.B.2 shows that we must have $a = 0$; it follows that the list $v$ is linearly independent. If $v = 0$ then simply observe that $1v = 0$, demonstrating that the list $v$ is linearly dependent.

(b) Suppose that the list consists of the vectors $u, v \in V$. If one of these vectors is a scalar multiple of the other, say $v = \lambda u$ for some $\lambda \in \mathbf{F}$, then observe that $v - \lambda u = 0$. Because the coefficient of $v$ in this linear combination is non-zero, we see that the list $u, v$ is linearly dependent.

Conversely, suppose that the list $u, v$ is linearly dependent, so that $\mu v + \lambda u = 0$ with at least one of the coefficients $\mu, \lambda$ non-zero, say $\mu \neq 0$; it follows that $v = -\frac{\lambda}{\mu}u$.

**Exercise 2.A.5.** Find a number $t$ such that

$$(3, 1, 4), (2, -3, 5), (5, 9, t)$$

is not linearly independent in $\mathbf{R}^3$.

**Solution.** Let $t = 2$ and observe that

$$3(3, 1, 4) - 2(2, -3, 5) - (5, 9, 2) = (0, 0, 0).$$

**Exercise 2.A.6.** Show that the list $(2, 3, 1), (1, -1, 2), (7, 3, c)$ is linearly dependent in $\mathbf{F}^3$ if and only if $c = 8$.



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**Solution.** That the list is linearly dependent if $c = 8$ was shown in the first bullet point of (2.20). Conversely, suppose that the list is linearly dependent. Since $(1, -1, 2)$ is evidently not a scalar multiple of $(2, 3, 1)$, the linear dependence lemma (2.19) implies that $(7, 3, c)$ lies in the span of $(2, 3, 1)$ and $(1, -1, 2)$, i.e. there are scalars $x$ and $y$ such that

$$x(2, 3, 1) + y(1, -1, 2) = (7, 3, c).$$

Solving the equations $2x + y = 7$ and $3x - y = 3$ gives $x = 2$ and $y = 3$, whence $c = x + 2y = 8$.

**Exercise 2.A.7.**

(a) Show that if we think of $\mathbf{C}$ as a vector space over $\mathbf{R}$, then the list $1 + i, 1 - i$ is linearly independent.

(b) Show that if we think of $\mathbf{C}$ as a vector space over $\mathbf{C}$, then the list $1 + i, 1 - i$ is linearly dependent.

**Solution.**

(a) Suppose that $x$ and $y$ are real numbers such that

$$x(1 + i) + y(1 - i) = (x + y) + (x - y)i = 0.$$

Since a complex number is zero if and only if its real and imaginary parts are zero, we must have

$$x + y = 0 \text{ and } x - y = 0 \quad \Leftrightarrow \quad x = y = 0.$$

Thus the list $1 + i, 1 - i$ is linearly independent.

(b) Observe that $i(1 - i) = 1 + i$, so that $1 + i$ is a scalar multiple of $1 - i$. It follows from Exercise 2.A.4 (b) that the list $1 + i, 1 - i$ is linearly dependent.

**Exercise 2.A.8.** Suppose $v_1, v_2, v_3, v_4$ is linearly independent in $V$. Prove that the list

$$v_1 - v_2, v_2 - v_3, v_3 - v_4, v_4$$

is also linearly independent.

**Solution.** Suppose that $a_1, a_2, a_3, a_4$ are scalars such that

$$a_1(v_1 - v_2) + a_2(v_2 - v_3) + a_3(v_3 - v_4) + a_4 v_4 = 0$$
$$\Leftrightarrow \quad a_1 v_1 + (a_2 - a_1)v_2 + (a_3 - a_2)v_3 + (a_4 - a_3)v_4 = 0.$$

Since the list $v_1, v_2, v_3, v_4$ is linearly independent, we must have

$$a_1 = a_2 - a_1 = a_3 - a_2 = a_4 - a_3 = 0,$$

which implies that $a_1 = a_2 = a_3 = a_4 = 0$. It follows that the list $v_1 - v_2, v_2 - v_3, v_3 - v_4, v_4$ is linearly independent.



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**Exercise 2.A.9.** Prove or give a counterexample: If $v_1, v_2, ..., v_m$ is a linearly independent list of vectors in $V$, then

$$5v_1 - 4v_2, v_2, v_3, ..., v_m$$

is linearly independent.

**Solution.** Suppose that $a_1, a_2, ..., a_m$ are scalars such that

$$a_1(5v_1 - 4v_2) + a_2v_2 + a_3v_3 + \cdots + a_mv_m = 0$$
$$\Leftrightarrow \quad 5a_1v_1 + (a_2 - 4a_1)v_2 + a_3v_3 + \cdots + a_mv_m = 0.$$

Since the list $v_1, v_2, ..., v_m$ is linearly independent, we must have

$$5a_1 = a_2 - 4a_1 = a_3 = \cdots = a_m = 0,$$

which implies that $a_1 = a_2 = a_3 = \cdots = a_m = 0$. It follows that the list $5v_1 - 4v_2, v_2, v_3, ..., v_m$ is linearly independent.

**Exercise 2.A.10.** Prove or give a counterexample: If $v_1, v_2, ..., v_m$ is a linearly independent list of vectors in $V$ and $\lambda \in \mathbf{F}$ with $\lambda \neq 0$, then $\lambda v_1, \lambda v_2, ..., \lambda v_m$ is linearly independent.

**Solution.** Suppose that $a_1, a_2, ..., a_m$ are scalars such that

$$a_1\lambda v_1 + a_2\lambda v_2 + \cdots + a_m\lambda v_m = 0.$$

Since $\lambda \neq 0$, we may multiply both sides of this equation by $\lambda^{-1}$ to obtain the equation

$$a_1v_1 + a_2v_2 + \cdots + a_mv_m = 0.$$

Since the list $v_1, v_2, ..., v_m$ is linearly independent, this implies that $a_1 = a_2 = \cdots = a_m = 0$. It follows that the list $\lambda v_1, \lambda v_2, ..., \lambda v_m$ is linearly independent.

**Exercise 2.A.11.** Prove or give a counterexample: If $v_1, ..., v_m$ and $w_1, ..., w_m$ are linearly independent lists of vectors in $V$, then the list $v_1 + w_1, ..., v_m + w_m$ is linearly independent.

**Solution.** This is false. Consider $\mathbf{R}$ as a vector space over itself. We have two linearly independent lists 1 and $-1$, but the list $1 + (-1) = 0$ is linearly dependent.

**Exercise 2.A.12.** Suppose $v_1, ..., v_m$ is linearly independent in $V$ and $w \in V$. Prove that if $v_1 + w, ..., v_m + w$ is linearly dependent, then $w \in \operatorname{span}(v_1, ..., v_m)$.

**Solution.** By the linear dependence lemma (2.19), there is a $j \in \{1, 2, ..., m\}$ such that $v_j + w \in \operatorname{span}\big(v_1 + w, ..., v_{j-1} + w\big)$. If $j = 1$ then $v_1 + w = 0$, i.e. $w = -v_1$. It follows that $w \in \operatorname{span}(v_1, ..., v_m)$.



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If $j \geq 2$, then there are scalars $a_1, ..., a_{j-1}$ such that

$$v_j + w = a_1(v_1 + w) + \cdots + a_{j-1}\big(v_{j-1} + w\big) \quad \Leftrightarrow \quad v_j + \lambda w = a_1 v_1 + \cdots + a_{j-1} v_{j-1},$$

where $\lambda = 1 - \big(a_1 + \cdots + a_{j-1}\big)$. Note that $\lambda$ must be non-zero: if this were not the case, then $v_j$ would lie in the span of $v_1, ..., v_{j-1}$, which cannot happen since the list $v_1, ..., v_j$ is linearly independent. It follows that

$$w = \lambda^{-1}\big(a_1 v_1 + \cdots + a_{j-1} v_{j-1} - v_j\big),$$

so that $w \in \operatorname{span}(v_1, ..., v_m)$.

**Exercise 2.A.13.** Suppose $v_1, ..., v_m$ is linearly independent in $V$ and $w \in V$. Show that

$$v_1, ..., v_m, w \text{ is linearly independent} \quad \Leftrightarrow \quad w \notin \operatorname{span}(v_1, ..., v_m).$$

**Solution.** If $w \in \operatorname{span}(v_1, ..., v_m)$ then the list $v_1, ..., v_m, w$ is linearly dependent by the third bullet point of 2.18. Conversely, suppose that the list $v_1, ..., v_m, w$ is linearly dependent. By the linear dependence lemma (2.19), one of the vectors in the list must be in the span of the previous vectors. It cannot be the case that some $v_j$ belongs to $\operatorname{span}\big(v_1, ..., v_{j-1}\big)$ since this would contradict the linear independence of the list $v_1, ..., v_m$, so it must be the case that $w \in \operatorname{span}(v_1, ..., v_m)$.

**Exercise 2.A.14.** Suppose $v_1, ..., v_m$ is a list of vectors in $V$. For $k \in \{1, ..., m\}$, let

$$w_k = v_1 + \cdots + v_k.$$

Show that the list $v_1, ..., v_m$ is linearly independent if and only if the list $w_1, ..., w_m$ is linearly independent.

**Solution.** Let $W = \operatorname{span}(w_1, ..., w_m)$; by Exercise 2.A.3 we also have $W = \operatorname{span}(v_1, ..., v_m)$. If the list $w_1, ..., w_m$ is linearly dependent, then using the linear dependence lemma (2.19) we may remove some $w_j$ from the list $w_1, ..., w_m$ to obtain a spanning list for $W$ of length $m - 1$. It follows from 2.22 that the list $v_1, ..., v_m$, which spans $W$, must be linearly dependent. A similar argument shows that the list $w_1, ..., w_m$ must be linearly dependent if the list $v_1, ..., v_m$ is linearly dependent.

**Exercise 2.A.15.** Explain why there does not exist a list of six polynomials that is linearly independent in $\mathcal{P}_4(\mathbf{F})$.

**Solution.** As noted in the textbook, $\mathcal{P}_4(\mathbf{F})$ is spanned by the list $1, z, z^2, z^3, z^4$, which has length 5. It follows from 2.22 that any linearly independent list in $\mathcal{P}_4(\mathbf{F})$ can have length at most 5.

**Exercise 2.A.16.** Explain why no list of four polynomials spans $\mathcal{P}_4(\mathbf{F})$.



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**Solution.** As shown in (2.16) (b), the list $1, z, z^2, z^3, z^4$, which has length 5, is linearly independent in $\mathcal{P}_4(\mathbf{F})$. It follows from 2.22 that any spanning list for $\mathcal{P}_4(\mathbf{F})$ must have length at least 5.

**Exercise 2.A.17.** Prove that $V$ is infinite-dimensional if and only if there is a sequence $v_1, v_2, \ldots$ of vectors in $V$ such that $v_1, \ldots, v_m$ is linearly independent for every positive integer $m$.

**Solution.** First suppose that $V$ is finite-dimensional, so that it is spanned by some list $w_1, \ldots, w_m$, and let $v_1, v_2, \ldots$ be any sequence of vectors in $V$; by 2.22, the list $v_1, v_2, \ldots, v_{m+1}$ must be linearly dependent.

Now suppose that $V$ is infinite-dimensional, so that no list of vectors in $V$ is a spanning list. Certainly $V \neq \{0\}$, so pick any $v_1 \neq 0$ in $V$ and note that the list $v_1$ is linearly independent. Suppose that after $m$ steps we have chosen a linearly independent list $v_1, \ldots, v_m$. By assumption $V \neq \operatorname{span}(v_1, \ldots, v_m)$, so pick any $v_{m+1} \notin \operatorname{span}(v_1, \ldots, v_m)$ and note that, by Exercise 2.A.13, the list $v_1, \ldots, v_m, v_{m+1}$ is linearly independent. This process recursively defines a sequence of vectors $v_1, v_2, \ldots$ such that $v_1, \ldots, v_m$ is linearly independent for each positive integer $m$.

**Exercise 2.A.18.** Prove that $\mathbf{F}^\infty$ is infinite-dimensional.

**Solution.** Consider the sequence of vectors $v_1, v_2, \ldots$, where $v_j \in \mathbf{F}^\infty$ is the sequence with a 1 in the $j^{\text{th}}$ position and 0's elsewhere. For each positive integer $m$, it is straightforward to verify that the list $v_1, \ldots, v_m$ is linearly independent; it follows from Exercise 2.A.17 that $\mathbf{F}^\infty$ is infinite-dimensional.

**Exercise 2.A.19.** Prove that the real vector space of all continuous real-valued functions on the interval $[0, 1]$ is infinite-dimensional.

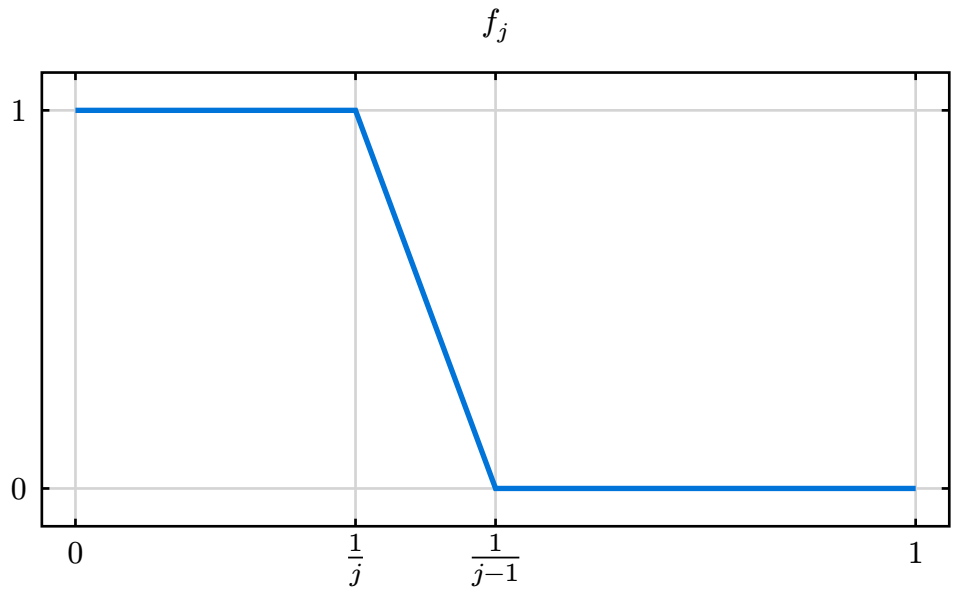
**Solution.** Consider the sequence of continuous functions $f_1, f_2, \ldots$ on the interval $[0, 1]$, where $f_1(x) = 1$ for all $x \in [0, 1]$ and, for $j \geq 2$,

$$f_j(x) = \begin{cases} 1 & \text{if } 0 \leq x \leq \frac{1}{j}, \\ 0 & \text{if } \frac{1}{j-1} \leq x \leq 1. \end{cases}$$

On the interval $\left(\frac{1}{j}, \frac{1}{j-1}\right)$, take $f_j$ to be the line segment joining the points $\left(\frac{1}{j}, 1\right)$ and $\left(\frac{1}{j-1}, 0\right)$, so that $f_j$ is a continuous function on $[0, 1]$.



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Let $m$ be a positive integer and suppose we have real numbers $a_1, ..., a_m$ such that

$$a_1 f_1(x) + a_2 f_2(x) + \cdots + a_m f_m(x) = 0$$

for all $x \in [0, 1]$. Taking $x = 1$ gives us

$$a_1 f_1(1) + a_2 f_2(1) + \cdots + a_m f_m(1) = a_1 = 0.$$

Similarly, taking $x = \frac{1}{2}$ gives us

$$a_2 f_2\left(\tfrac{1}{2}\right) + \cdots + a_m f_m\left(\tfrac{1}{2}\right) = a_2 = 0.$$

By continuing in this fashion, taking $x = \frac{1}{j}$ for each $j \in \{1, 2, ..., m\}$, we see that $a_1 = a_2 = \cdots = a_m = 0$. It follows that the list $f_1, f_2, ..., f_m$ is linearly independent for each positive integer $m$ and thus by Exercise 2.A.17 the real vector space of all continuous real-valued functions on the interval $[0, 1]$ is infinite-dimensional.

**Exercise 2.A.20.** Suppose $p_0, p_1, ..., p_m$ are polynomials in $\mathcal{P}_m(\mathbf{F})$ such that $p_k(2) = 0$ for each $k \in \{0, ..., m\}$. Prove that $p_0, p_1, ..., p_m$ is not linearly independent in $\mathcal{P}_m(\mathbf{F})$.

**Solution.** Since $\mathcal{P}_m(\mathbf{F})$ is spanned by the list $1, x, ..., x^m$ of length $m + 1$, 2.22 implies that the list $p_0, p_1, ..., p_m, x$ of length $m + 2$ is linearly dependent. The linear dependence lemma (2.19) implies that one of the vectors from this list belongs to the span of the previous vectors. Notice that for any scalars $a_0, ..., a_m$,

$$x = a_0 p_0(x) + \cdots + a_m p_m(x) \text{ for all } x \in \mathbf{F} \quad \Rightarrow \quad 2 = a_0 p_0(2) + \cdots + a_m p_m(2) = 0,$$

which is a contradiction; it follows that $x \notin \operatorname{span}(p_0, p_1, ..., p_m)$ and thus there must be some $j \in \{0, ..., m\}$ such that $p_j \in \operatorname{span}\left(p_0, p_1, ..., p_{j-1}\right)$. The third bullet point of 2.18 then implies that the list $p_0, p_1, ..., p_m$ is linearly dependent.



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## 2.B. Bases

**Exercise 2.B.1.** Find all vector spaces that have exactly one basis.

**Solution.** We will consider only finite-dimensional vector spaces over $\mathbf{R}$ or $\mathbf{C}$.

First consider the trivial vector space $\{0\}$. There are two possible lists of vectors: the empty list and the list 0. Since any list containing the zero vector is linearly dependent, the list 0 cannot be a basis of $\{0\}$. By definition the empty list is linearly independent and has span $\{0\}$; it follows that the empty list is a basis of $\{0\}$. Thus the trivial vector space has exactly one basis.

Now suppose that $V \neq \{0\}$. By 2.31, $V$ has a basis $v_1, ..., v_m$. Since $V \neq \{0\}$, this basis is not the empty list, so $v_1$ exists and is non-zero. It follows that $B = 2v_1, ..., 2v_m$ is distinct from $v_1, ..., v_m$. By Exercise 2.A.10, $B$ is linearly independent. Furthermore, we claim that $\operatorname{span} B = V$. Let $v \in V$ be given. Since $v_1, ..., v_m$ is a basis, there are scalars $a_1, ..., a_m$ such that $v = \sum_{j=1}^{m} a_j v_j$. This is equivalent to

$$v = \sum_{j=1}^{m} \left(\tfrac{1}{2} a_j\right)(2v_j);$$

it follows that $v \in \operatorname{span} B$ and hence that $\operatorname{span} B = V$, as claimed. Thus $B$ is a basis of $V$, distinct from the original basis $v_1, ..., v_m$. We may conclude that the trivial vector space is the only vector space which has exactly one basis.

**Exercise 2.B.2.** Verify all assertions in Example 2.27.

**Solution.**

(a) The assertion is that the list $B = (1, 0, ..., 0), (0, 1, 0, ..., 0), ..., (0, ..., 0, 1)$ is a basis of $\mathbf{F}^n$. Since any $(x_1, x_2, ..., x_n) \in \mathbf{F}^n$ can be expressed as

$$x_1(1, 0, ..., 0) + x_2(0, 1, 0, ..., 0) + \cdots + x_n(0, ..., 0, 1),$$

we see that $\operatorname{span} B = \mathbf{F}^n$. Setting the above expression equal to $(0, 0, 0, ..., 0)$ immediately gives us $x_1 = x_2 = \cdots = x_n = 0$, so that the list $B$ is linearly independent. Thus $B$ is a basis of $\mathbf{F}^n$.

(b) The assertion is that the list $B = (1, 2), (3, 5)$ is a basis of $\mathbf{F}^2$. Since neither of these vectors is a scalar multiple of the other, Exercise 2.A.4 (b) shows that $B$ is linearly independent. If $(a, b) \in \mathbf{F}^2$, then observe that

$$(-5a + 3b)(1, 2) + (2a - b)(3, 5) = (a, b).$$

Thus $\operatorname{span} B = V$ and we may conclude that $B$ is a basis of $\mathbf{F}^2$.



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(c) The assertion is that the list $B = (1, 2, -4), (7, -5, 6)$ is linearly independent in $\mathbf{F}^3$ but is not a basis of $\mathbf{F}^3$ because it does not span $\mathbf{F}^3$. Since neither of these vectors is a scalar multiple of the other, Exercise 2.A.4 (b) shows that $B$ is linearly independent. However, since the list $(1, 0, 0), (0, 1, 0), (0, 0, 1)$ of length 3 is linearly independent in $\mathbf{F}^3$ (see (a)), 2.22 implies that $B$ cannot span $\mathbf{F}^3$.

(d) The assertion is that the list $B = (1, 2), (3, 5), (4, 13)$ spans $\mathbf{F}^2$ but is not a basis of $\mathbf{F}^2$ because it is not linearly independent. Indeed, part (b) shows that $B$ spans $\mathbf{F}^2$ and that $(4, 13)$ lies in the span of $(1, 2)$ and $(3, 5)$, so that $B$ is linearly dependent.

(e) The assertion is that the list $B = (1, 1, 0), (0, 0, 1)$ is a basis of

$$U = \{(x, x, y) \in \mathbf{F}^3 : x, y \in \mathbf{F}\}.$$

Indeed, $\operatorname{span} B = U$ since $x(1, 1, 0) + y(0, 0, 1) = (x, x, y)$ for any scalars $x, y$, and $B$ is linearly independent since $(x, x, y) = (0, 0, 0)$ forces $x = y = 0$.

(f) The assertion is that the list $B = (1, -1, 0), (1, 0, -1)$ is a basis of

$$U = \{(x, y, z) \in \mathbf{F}^3 : x + y + z = 0\}.$$

Observe that $B$ is linearly independent since

$$x(1, -1, 0) + y(1, 0, -1) = (x + y, -x, -y) = (0, 0, 0)$$

gives us $x = y = 0$, and $B$ spans $U$ by Exercise 2.A.1 (using the notation of that exercise, we have $B = v_4, v_1$).

(g) The assertion is that the list $B = 1, z, ..., z^m$ is a basis of $\mathcal{P}_m(\mathbf{F})$. The fact that $\operatorname{span} B = \mathcal{P}_m(\mathbf{F})$ was noted on p. 31 of the textbook, and the linear independence of $B$ was shown in 2.16(b).

**Exercise 2.B.3.**

(a) Let $U$ be the subspace of $\mathbf{R}^5$ defined by

$$U = \{(x_1, x_2, x_3, x_4, x_5) \in \mathbf{R}^5 : x_1 = 3x_2 \text{ and } x_3 = 7x_4\}.$$

Find a basis of $U$.

(b) Extend the basis in (a) to a basis of $\mathbf{R}^5$.

(c) Find a subspace $W$ of $\mathbf{R}^5$ such that $\mathbf{R}^5 = U \oplus W$.

**Solution.**

(a) Note that

$$U = \{(3x_1, x_1, 7x_2, x_2, x_3) \in \mathbf{R}^5 : x_1, x_2, x_3 \in \mathbf{R}\}.$$

Let $u_1 = (3, 1, 0, 0, 0), u_2 = (0, 0, 7, 1, 0), u_3 = (0, 0, 0, 0, 1)$ and $B = u_1, u_2, u_3$. Since

$$x_1u_1 + x_2u_2 + x_3u_3 = (3x_1, x_1, 7x_2, x_2, x_3)$$



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for scalars $x_1, x_2, x_3$, we see that $\operatorname{span} B = U$. Setting the above expression equal to $(0,0,0,0,0)$, it is immediate that $x_1 = x_2 = x_3 = 0$, so that $B$ is linearly independent. Thus $B$ is a basis of $U$.

(b) Denote the $j^{\text{th}}$ standard basis vector of $\mathbf{R}^5$ by $e_j$. Following the procedure outlined in 2.30 and 2.32, we adjoin the five standard basis vectors to $B$ to obtain the spanning list

$$u_1, u_2, u_3, e_1, e_2, e_3, e_4, e_5.$$

- $e_1$ does not belong to $\operatorname{span}(u_1, u_2, u_3)$, so we do not delete it.
- Note that $e_2 = u_1 - 3e_1$, so we delete $e_2$ from the list.
- $e_3$ does not belong to $\operatorname{span}(u_1, u_2, u_3, e_1)$, so we do not delete it.
- Note that $e_4 = u_2 - 7e_3$, so we delete $e_4$ from the list.
- Since $e_5 = u_3$, we delete $e_5$ from the list.

We are left with the list $u_1, u_2, u_3, e_1, e_3$; as the proof of (2.32) shows, this must be a basis of $\mathbf{R}^5$.

(c) As shown in the proof of (2.33), if we let

$$W = \operatorname{span}(e_1, e_3) = \{(x_1, 0, x_3, 0, 0) \in \mathbf{R}^5 : x_1, x_3 \in \mathbf{R}\},$$

then $\mathbf{R}^5 = U \oplus W$.

**Exercise 2.B.4.**

(a) Let $U$ be the subspace of $\mathbf{C}^5$ defined by

$$U = \{(z_1, z_2, z_3, z_4, z_5) \in \mathbf{C}^5 : 6z_1 = z_2 \text{ and } z_3 + 2z_4 + 3z_5 = 0\}.$$

Find a basis of $U$.

(b) Extend the basis in (a) to a basis of $\mathbf{C}^5$.

(c) Find a subspace $W$ of $\mathbf{C}^5$ such that $\mathbf{C}^5 = U \oplus W$.

**Solution.**

(a) Note that

$$U = \{(z_1, 6z_1, -2z_2 - 3z_3, z_2, z_3) \in \mathbf{C}^5 : z_1, z_2, z_3 \in \mathbf{C}\}.$$

Let $u_1 = (1,6,0,0,0), u_2 = (0,0,-2,1,0), u_3 = (0,0,-3,0,1)$, and $B = u_1, u_2, u_3$. Since

$$z_1u_1 + z_2u_2 + z_3u_3 = (z_1, 6z_1, -2z_2 - 3z_3, z_2, z_3)$$

for scalars $z_1, z_2, z_3$, we see that $\operatorname{span} B = U$. Setting the above expression equal to $(0,0,0,0,0)$, it is immediate that $z_1 = z_2 = z_3 = 0$, so that $B$ is linearly independent. Thus $B$ is a basis of $U$.

(b) Denote the $j^{\text{th}}$ standard basis vector of $\mathbf{C}^5$ by $e_j$. Following the procedure outlined in 2.30 and 2.32, we adjoin the five standard basis vectors to $B$ to obtain the spanning list



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$$u_1, u_2, u_3, e_1, e_2, e_3, e_4, e_5.$$

- $e_1$ does not belong to $\operatorname{span}(u_1, u_2, u_3)$, so we do not delete it.
- Note that $e_2 = \frac{1}{6}(u_1 - e_1)$, so we delete $e_2$ from the list.
- $e_3$ does not belong to $\operatorname{span}(u_1, u_2, u_3, e_1)$, so we do not delete it.
- Note that $e_4 = u_2 + 2e_3$, so we delete $e_4$ from the list.
- Since $e_5 = u_3 + 3e_3$, we delete $e_5$ from the list.

We are left with the list $u_1, u_2, u_3, e_1, e_3$; as the proof of 2.32 shows, this must be a basis of $\mathbf{C}^5$.

(c) As shown in the proof of (2.33), if we let

$$W = \operatorname{span}(e_1, e_3) = \{(z_1, 0, z_3, 0, 0) \in \mathbf{C}^5 : x_1, x_3 \in \mathbf{C}\},$$

then $\mathbf{C}^5 = U \oplus W$.

**Exercise 2.B.5.** Suppose $V$ is finite-dimensional and $U, W$ are subspaces of $V$ such that $V = U + W$. Prove that there exists a basis of $V$ consisting of vectors in $U \cup W$.

**Solution.** Let $u_1, ..., u_m$ be a basis of $U$ and let $w_1, ..., w_n$ be a basis of $W$; these bases exist by 2.25 and 2.31. Since

$$U = \operatorname{span}(u_1, ..., u_m), \quad W = \operatorname{span}(w_1, ..., w_n), \quad \text{and} \quad V = U + W,$$

we see that $V = \operatorname{span}(u_1, ..., u_m, w_1, ..., w_n)$. Thus, using the procedure of 2.30, we can reduce the list $u_1, ..., u_m, w_1, ..., w_n$ to a basis of $V$ consisting of vectors in $U \cup W$.

**Exercise 2.B.6.** Prove or give a counterexample: If $p_0, p_1, p_2, p_3$ is a list in $\mathcal{P}_3(\mathbf{F})$ such that none of the polynomials $p_0, p_1, p_2, p_3$ has degree 2, then $p_0, p_1, p_2, p_3$ is not a basis of $\mathcal{P}_3(\mathbf{F})$.

**Solution.** For a counterexample, consider $B = 1, x, x^2 + x^3, x^3$; none of the polynomials in this list has degree 2. Suppose $a_0, a_1, a_2, a_3$ are scalars such that

$$a_0 + a_1 x + a_2(x^2 + x^3) + a_3 x^3 = a_0 + a_1 x + a_2 x^2 + (a_2 + a_3)x^3 = 0$$

for all $x \in \mathbf{F}$. This implies that $a_0 = a_1 = a_2 = a_2 + a_3 = 0$ (we will prove this in 4.8), which in turn gives $a_3 = 0$. It follows that $B$ is linearly independent. Now suppose that $p = a_0 + a_1 x + a_2 x^2 + a_3 x^3 \in \mathcal{P}_3(\mathbf{F})$ is given and observe that

$$a_0 + a_1 x + a_2(x^2 + x^3) + (a_3 - a_2)x^3 = p,$$

so that $p \in \operatorname{span} B$. It follows that $\mathcal{P}_3(\mathbf{F}) = \operatorname{span} B$ and hence that $B$ is a basis of $\mathcal{P}_3(\mathbf{F})$.



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**Exercise 2.B.7.** Suppose $v_1, v_2, v_3, v_4$ is a basis of $V$. Prove that

$$v_1 + v_2, v_2 + v_3, v_3 + v_4, v_4$$

is also a basis of $V$.

**Solution.** Let $B = v_1 + v_2, v_2 + v_3, v_3 + v_4, v_4$. Suppose there are scalars $a_1, a_2, a_3, a_4$ such that

$$\begin{aligned} &a_1(v_1 + v_2) + a_2(v_2 + v_3) + a_3(v_3 + v_4) + a_4 v_4 \\ &\qquad = a_1 v_1 + (a_1 + a_2)v_2 + (a_2 + a_3)v_3 + (a_3 + a_4)v_4 = 0. \end{aligned}$$

Since $v_1, v_2, v_3, v_4$ is a basis, this implies that

$$a_1 = a_1 + a_2 = a_2 + a_3 = a_3 + a_4 = 0 \quad \Rightarrow \quad a_1 = a_2 = a_3 = a_4 = 0.$$

Thus the list $B$ is linearly independent. Let $v \in V$ be given. Since $v_1, v_2, v_3, v_4$ is a basis of $V$, there are scalars $a_1, a_2, a_3, a_4$ such that $v = a_1 v_1 + a_2 v_2 + a_3 v_3 + a_4 v_4$. Observe that

$$\begin{aligned} &a_1(v_1 + v_2) + (a_2 - a_1)(v_2 + v_3) + (a_3 - a_2 + a_1)(v_3 + v_4) \\ &\qquad + (a_4 - a_3 + a_2 - a_1)v_4 = a_1 v_1 + a_2 v_2 + a_3 v_3 + a_4 v_4 = v. \end{aligned}$$

It follows that $\operatorname{span} B = V$ and hence that $B$ is a basis of $V$.

**Exercise 2.B.8.** Prove or give a counterexample: If $v_1, v_2, v_3, v_4$ is a basis of $V$ and $U$ is a subspace of $V$ such that $v_1, v_2 \in U$ and $v_3 \notin U$ and $v_4 \notin U$, then $v_1, v_2$ is a basis of $U$.

**Solution.** For a counterexample, consider $V = \mathbf{R}^4$ and let $e_j$ be the $j^{\text{th}}$ standard basis vector of $\mathbf{R}^4$. It is straightforward to verify that the list

$$v_1 = e_1, \quad v_2 = e_2, \quad v_3 = e_3 + e_4, \quad v_4 = e_1 + e_4$$

is a basis of $\mathbf{R}^4$. Let $U = \operatorname{span}(e_1, e_2, e_3)$ and note that $v_1, v_2 \in U$. Note further that, since each vector $(a_1, a_2, a_3, a_4) \in U$ must satisfy $a_4 = 0$, we have $v_3, v_4 \notin U$. However, $v_1, v_2$ is not a basis for $U$: since $e_1, e_2, e_3$ is linearly independent, any spanning list for $U$ must contain at least three vectors.

**Exercise 2.B.9.** Suppose $v_1, ..., v_m$ is a list of vectors in $V$. For $k \in \{1, ..., m\}$, let

$$w_k = v_1 + \cdots + v_k.$$

Show that $v_1, ..., v_m$ is a basis of $V$ if and only if $w_1, ..., w_m$ is a basis of $V$.

**Solution.** This is immediate from Exercise 2.A.3 and Exercise 2.A.14.



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**Exercise 2.B.10.** Suppose $U$ and $W$ are subspaces of $V$ such that $V = U \oplus W$. Suppose also that $u_1, ..., u_m$ is a basis of $U$ and $w_1, ..., w_n$ is a basis of $W$. Prove that

$$u_1, ..., u_m, w_1, ..., w_n$$

is a basis of $V$.

**Solution.** Let $v \in V$ be given. Since the sum $V = U \oplus W$ is direct, there are unique vectors $u \in U$ and $w \in W$ such that $v = u + w$. Because $u_1, ..., u_m$ is a basis of $U$, 2.28 implies that there are unique scalars $a_1, ..., a_m$ such that $u = a_1 u_1 + \cdots + a_m u_m$. Similarly, there are unique scalars $b_1, ..., b_n$ such that $w = b_1 w_1 + \cdots + b_n w_n$. It follows that $v$ can be uniquely represented as

$$v = a_1 u_1 + \cdots + a_m u_m + b_1 w_1 + \cdots + b_n w_n.$$

Thus, by 2.28, $u_1, ..., u_m, w_1, ..., w_n$ is a basis of $V$.

**Exercise 2.B.11.** Suppose $V$ is a real vector space. Show that if $v_1, ..., v_n$ is a basis of $V$ (as a real vector space), then $v_1, ..., v_n$ is also a basis of the complexification $V_{\mathbf{C}}$ (as a complex vector space).

*See Exercise 8 in Section 1B for the definition of the complexification $V_{\mathbf{C}}$.*

**Solution.** Let $u + iv \in V_{\mathbf{C}}$ be given. By 2.28, there are unique real scalars $a_1, ..., a_n, b_1, ..., b_n$ such that

$$u = a_1 v_1 + \cdots + a_n v_n \quad \text{and} \quad v = b_1 v_1 + \cdots + b_n v_n.$$

Using the definitions of vector addition and complex scalar multiplication in $V_{\mathbf{C}}$ given in Exercise 1.B.8, observe that

$$\sum_{k=1}^{n} (a_k + b_k i) v_k = \left( \sum_{k=1}^{n} a_k v_k \right) + i \left( \sum_{k=1}^{n} b_k v_k \right) = u + iv. \tag{1}$$

Because two ordered pairs $(u, v)$ and $(w, x)$ are equal if and only if $u = w$ and $v = x$, and the real scalars $a_1, ..., a_n, b_1, ..., b_n$ are unique, we see that the complex scalars $a_1 + b_1 i, ..., a_n + b_n i$ in the linear combination on the left-hand side of equation (1) are unique. It follows from 2.28 that $v_1, ..., v_n$ is a basis of $V_{\mathbf{C}}$.

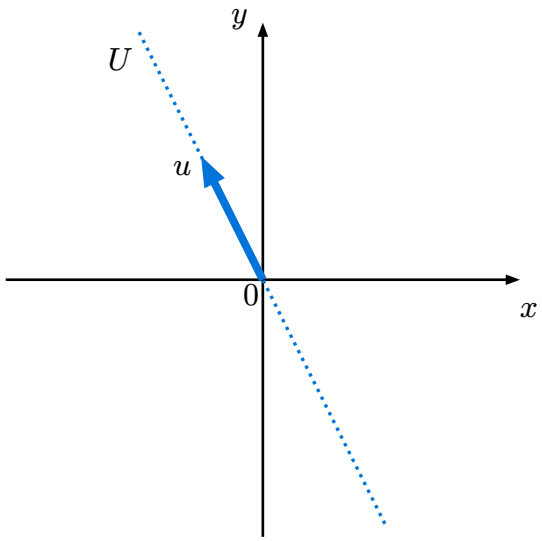


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## 2.C. Dimension

**Exercise 2.C.1.** Show that the subspaces of $\mathbf{R}^2$ are precisely $\{0\}$, all lines in $\mathbf{R}^2$ containing the origin, and $\mathbf{R}^2$.

**Solution.** It is easily verified that $\{0\}$, all lines in $\mathbf{R}^2$ through the origin, and $\mathbf{R}^2$ are indeed subspaces of $\mathbf{R}^2$. To see that these are the only subspaces of $\mathbf{R}^2$, suppose that $U$ is a subspace of $\mathbf{R}^2$ and note that by 2.37 we must have $\dim U \in \{0, 1, 2\}$. If $\dim U = 0$ then $U = \{0\}$, if $\dim U = 2$ then $U = \mathbf{R}^2$ by 2.39, and if $\dim U = 1$ then there exists a basis $u \neq 0$ of $U$, so that $U = \operatorname{span}(u) = \{\lambda u : \lambda \in \mathbf{R}\}$, i.e. $U$ is a line through the origin with direction vector $u$.




**Exercise 2.C.2.** Show that the subspaces of $\mathbf{R}^3$ are precisely $\{0\}$, all lines in $\mathbf{R}^3$ containing the origin, all planes in $\mathbf{R}^3$ containing the origin, and $\mathbf{R}^3$.

**Solution.** It is easily verified that $\{0\}$, all lines in $\mathbf{R}^3$ through the origin, all planes in $\mathbf{R}^3$ through the origin, and $\mathbf{R}^3$ are indeed subspaces of $\mathbf{R}^3$. To see that these are the only subspaces of $\mathbf{R}^3$, suppose that $U$ is a subspace of $\mathbf{R}^3$ and note that by 2.37 we must have $\dim U \in \{0, 1, 2, 3\}$. If $\dim U = 0$ then $U = \{0\}$, if $\dim U = 3$ then $U = \mathbf{R}^3$ by 2.39, and if $\dim U = 1$ then there exists a basis $u \neq 0$ of $U$, so that $U = \operatorname{span}(u) = \{\lambda u : \lambda \in \mathbf{R}\}$, i.e. $U$ is a line through the origin with direction vector $u$. If $\dim U = 2$ then there is a basis $u_1, u_2$ of $U$, so that $U = \operatorname{span}(u_1, u_2) = \{\lambda_1 u_1 + \lambda_2 u_2 : \lambda_1, \lambda_2 \in \mathbf{R}\}$. Because neither of $u_1, u_2$ is a scalar multiple of the other, i.e. $u_1$ and $u_2$ are not collinear, this describes a plane through the origin in $\mathbf{R}^3$.



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**Exercise 2.C.3.**

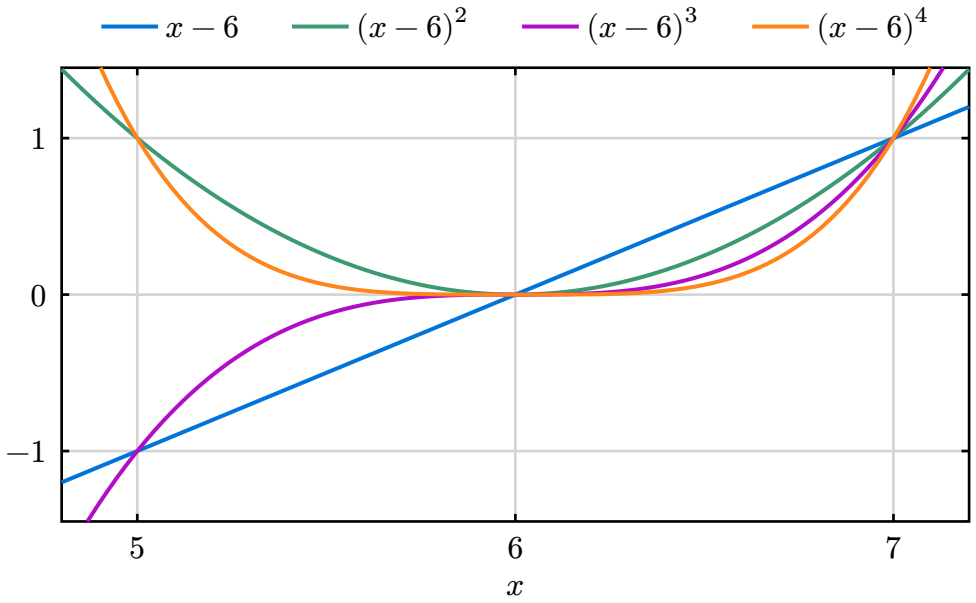
(a) Let $U = \{p \in \mathcal{P}_4(\mathbf{F}) : p(6) = 0\}$. Find a basis of $U$.

(b) Extend the basis in (a) to a basis of $\mathcal{P}_4(\mathbf{F})$.

(c) Find a subspace $W$ of $\mathcal{P}_4(\mathbf{F})$ such that $\mathcal{P}_4(\mathbf{F}) = U \oplus W$.

**Solution.**

(a) Let $B = x - 6, (x-6)^2, (x-6)^3, (x-6)^4$; certainly each of these polynomials belongs to $U$.




Suppose we have scalars $a_1, a_2, a_3, a_4$ such that

$$a_1(x-6) + a_2(x-6)^2 + a_3(x-6)^3 + a_4(x-6)^4 = 0$$

for all $x \in \mathbf{F}$. Using the reasoning of 2.41, we see that this equation implies that $a_1 = a_2 = a_3 = a_4 = 0$. It follows that $B$ is linearly independent and thus by 2.22 we have $\dim U \geq 4$. Using 2.37, we also find that $\dim U \leq \dim \mathcal{P}_4(\mathbf{F}) = 5$. However, notice that $U \neq \mathcal{P}_4(\mathbf{F})$ because the non-zero constant polynomials do not belong to $U$; it follows from 2.39 that $\dim U$ must be strictly less than $\dim \mathcal{P}_4(\mathbf{F}) = 5$. Thus $\dim U = 4$ and using 2.38 we may conclude that $B$ is a basis of $U$.

(b) Certainly the constant polynomial 1 does not belong to $U = \operatorname{span} B$. It then follows from Exercise 2.A.13 that the list $B' = 1, x-6, (x-6)^2, (x-6)^3, (x-6)^4$ is linearly independent. Since $\dim \mathcal{P}_4(\mathbf{F}) = 5$, 2.38 allows us to conclude that $B'$ is a basis of $\mathcal{P}_4(\mathbf{F})$.

(c) Let $W = \operatorname{span}(1)$, i.e. the subspace of all constant polynomials. As the proof of 2.33 shows, we then have $\mathcal{P}_4(\mathbf{F}) = U \oplus W$.



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**Exercise 2.C.4.**

(a) Let $U = \{p \in \mathcal{P}_4(\mathbf{R}) : p''(6) = 0\}$. Find a basis of $U$.

(b) Extend the basis in (a) to a basis of $\mathcal{P}_4(\mathbf{R})$.

(c) Find a subspace $W$ of $\mathcal{P}_4(\mathbf{R})$ such that $\mathcal{P}_4(\mathbf{R}) = U \oplus W$.

**Solution.**

(a) Let $B = 1, x, (x-6)^3, (x-6)^4$; it is straightforward to verify that each of these polynomials belongs to $U$. Suppose we have scalars $a_0, a_1, a_3, a_4$ such that

$$a_0 + a_1 x + a_3(x-6)^3 + a_4(x-6)^4 = 0$$

for all $x \in \mathbf{R}$. Using the reasoning of 2.41, we see that this equation implies that $a_0 = a_1 = a_3 = a_4 = 0$. It follows that $B$ is linearly independent and thus by 2.22 we have $\dim U \geq 4$. Using 2.37, we also find that $\dim U \leq \dim \mathcal{P}_4(\mathbf{R}) = 5$. However, notice that $U \neq \mathcal{P}_4(\mathbf{R})$ because $x^2 \notin U$; it follows from 2.39 that $\dim U$ must be strictly less than $\dim \mathcal{P}_4(\mathbf{R}) = 5$. Thus $\dim U = 4$ and using 2.38 we may conclude that $B$ is a basis of $U$.

(b) As noted in part (a), $x^2 \notin U = \operatorname{span} B$. It then follows from Exercise 2.A.13 that the list $B' = 1, x, x^2, (x-6)^3, (x-6)^4$ is linearly independent. Since $\dim \mathcal{P}_4(\mathbf{R}) = 5$, (2.38) allows us to conclude that $B'$ is a basis of $\mathcal{P}_4(\mathbf{R})$.

(c) Let $W = \operatorname{span}(x^2)$. As the proof of 2.33 shows, we then have $\mathcal{P}_4(\mathbf{R}) = U \oplus W$.

**Exercise 2.C.5.**

(a) Let $U = \{p \in \mathcal{P}_4(\mathbf{F}) : p(2) = p(5)\}$. Find a basis of $U$.

(b) Extend the basis in (a) to a basis of $\mathcal{P}_4(\mathbf{F})$.

(c) Find a subspace $W$ of $\mathcal{P}_4(\mathbf{F})$ such that $\mathcal{P}_4(\mathbf{F}) = U \oplus W$.

**Solution.**

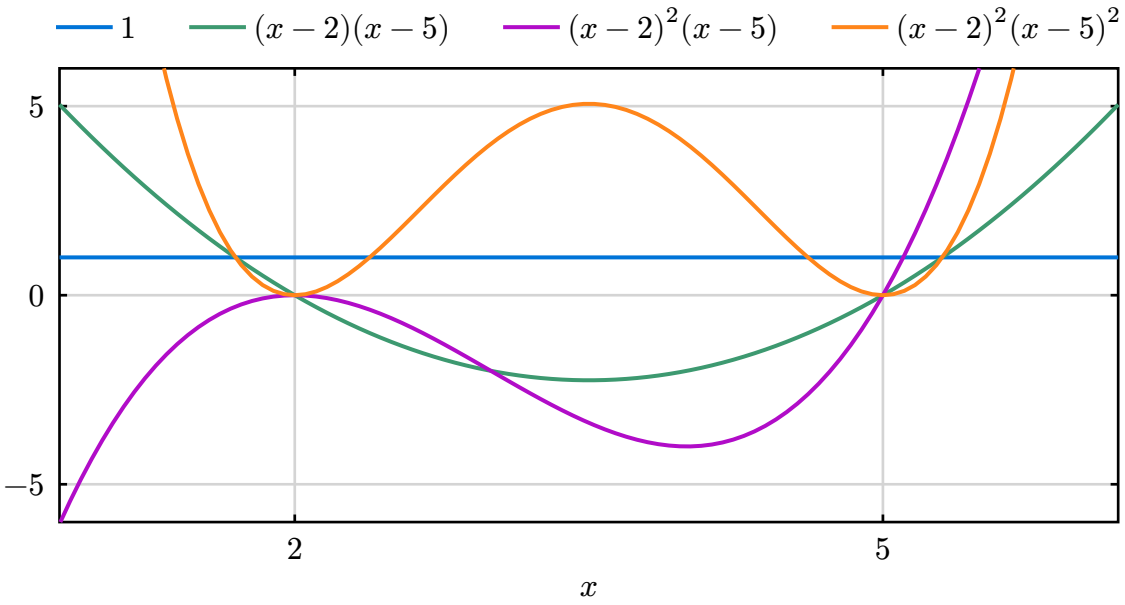
(a) Let $B = 1, (x-2)(x-5), (x-2)^2(x-5), (x-2)^2(x-5)^2$; it is straightforward to verify that each of these polynomials belongs to $U$. Suppose we have scalars $a_0, a_2, a_3, a_4$ such that

$$a_0 + a_2(x-2)(x-5) + a_3(x-2)^2(x-5) + a_4(x-2)^2(x-5)^2 = 0$$

for all $x \in \mathbf{F}$. Using the reasoning of 2.41, we see that this equation implies that $a_0 = a_2 = a_3 = a_4 = 0$. It follows that $B$ is linearly independent and thus by 2.22 we have $\dim U \geq 4$. Using 2.37, we also find that $\dim U \leq \dim \mathcal{P}_4(\mathbf{F}) = 5$. However, notice that $U \neq \mathcal{P}_4(\mathbf{F})$ because $x \notin U$; it follows from 2.39 that $\dim U$ must be strictly less than $\dim \mathcal{P}_4(\mathbf{F}) = 5$. Thus $\dim U = 4$ and using 2.38 we may conclude that $B$ is a basis of $U$.



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(b) As noted in part (a), $x \notin U = \operatorname{span} B$. It then follows from Exercise 2.A.13 that the list $B' = 1, x, (x-2)(x-5), (x-2)^2(x-5), (x-2)^2(x-5)^2$ is linearly independent. Since $\dim \mathcal{P}_4(\mathbf{F}) = 5$, 2.38 allows us to conclude that $B'$ is a basis of $\mathcal{P}_4(\mathbf{F})$.

(c) Let $W = \operatorname{span}(x)$. As the proof of 2.33 shows, we then have $\mathcal{P}_4(\mathbf{F}) = U \oplus W$.

**Exercise 2.C.6.**

(a) Let $U = \{p \in \mathcal{P}_4(\mathbf{F}) : p(2) = p(5) = p(6)\}$. Find a basis of $U$.

(b) Extend the basis in (a) to a basis of $\mathcal{P}_4(\mathbf{F})$.

(c) Find a subspace $W$ of $\mathcal{P}_4(\mathbf{F})$ such that $\mathcal{P}_4(\mathbf{F}) = U \oplus W$.

**Solution.**

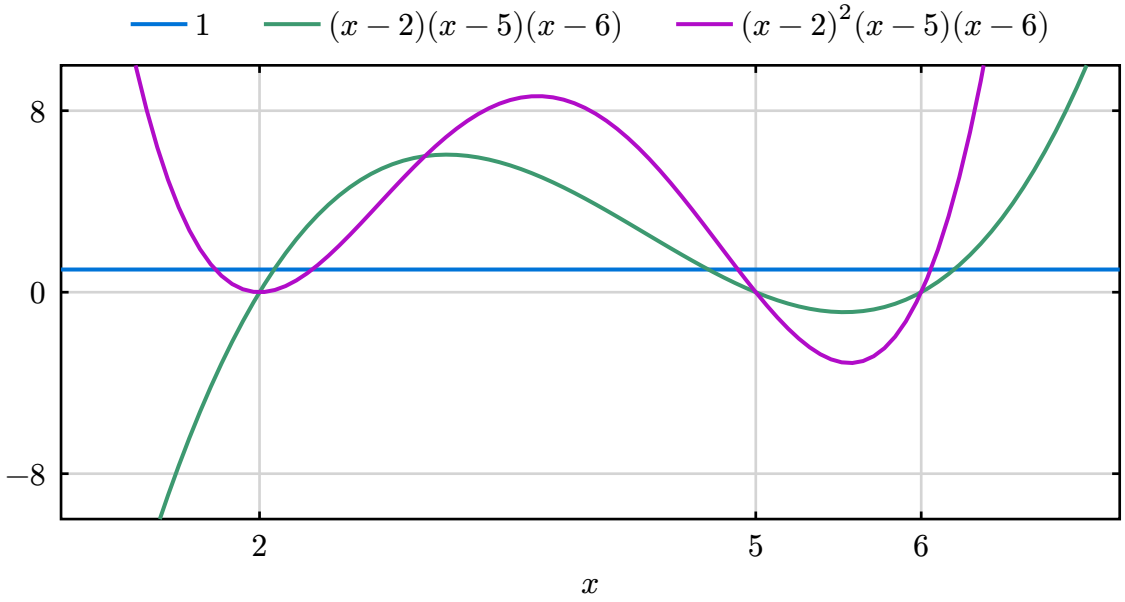
(a) Let $B = 1, (x-2)(x-5)(x-6), (x-2)^2(x-5)(x-6)$; it is straightforward to verify that each of these polynomials belongs to $U$. Suppose we have scalars $a_0, a_3, a_4$ such that

$$a_0 + a_3(x-2)(x-5)(x-6) + a_4(x-2)^2(x-5)(x-6) = 0$$

for all $x \in \mathbf{F}$. Using the reasoning of 2.41, we see that this equation implies that $a_0 = a_3 = a_4 = 0$. It follows that $B$ is linearly independent and thus by 2.22 we have $\dim U \geq 3$. Let $Y$ denote the subspace from Exercise 2.C.5 and notice that $U$ is a subspace of $Y$. Using 2.37, we then find that $\dim U \leq \dim Y = 4$. However, notice that $U \neq Y$ because $(x-2)(x-5) \in Y$ but $(x-2)(x-5) \notin U$; it follows from 2.39 that $\dim U$ must be strictly less than $\dim Y = 4$. Thus $\dim U = 3$ and using 2.38 we may conclude that $B$ is a basis of $U$.



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(b) As noted in part (a), $(x-2)(x-5) \in Y$ but $(x-2)(x-5) \notin U = \operatorname{span} B$. It then follows from Exercise 2.A.13 that the list

$$B' = 1, (x-2)(x-5), (x-2)(x-5)(x-6), (x-2)^2(x-5)(x-6)$$

is linearly independent. Since $\dim Y = 4$, 2.38 shows that $B'$ is a basis of $Y$. We can now argue as in Exercise 2.C.5 (b) to conclude that the list

$$1, x, (x-2)(x-5), (x-2)(x-5)(x-6), (x-2)^2(x-5)(x-6)$$

is a basis of $\mathcal{P}_4(\mathbf{F})$.

(c) Let $W = \operatorname{span}(x, (x-2)(x-5))$. As the proof of 2.33 shows, we then have $\mathcal{P}_4(\mathbf{F}) = U \oplus W$.

**Exercise 2.C.7.**

(a) Let $U = \left\{p \in \mathcal{P}_4(\mathbf{R}) : \int_{-1}^{1} p = 0\right\}$. Find a basis of $U$.

(b) Extend the basis in (a) to a basis of $\mathcal{P}_4(\mathbf{R})$.

(c) Find a subspace $W$ of $\mathcal{P}_4(\mathbf{R})$ such that $\mathcal{P}_4(\mathbf{R}) = U \oplus W$.

**Solution.**

(a) Let $B = x, x^2 - \frac{1}{3}, x^3, x^4 - \frac{1}{5}$; it is straightforward to verify that each of these polynomials belongs to $U$. Suppose we have scalars $a_1, a_2, a_3, a_4$ such that

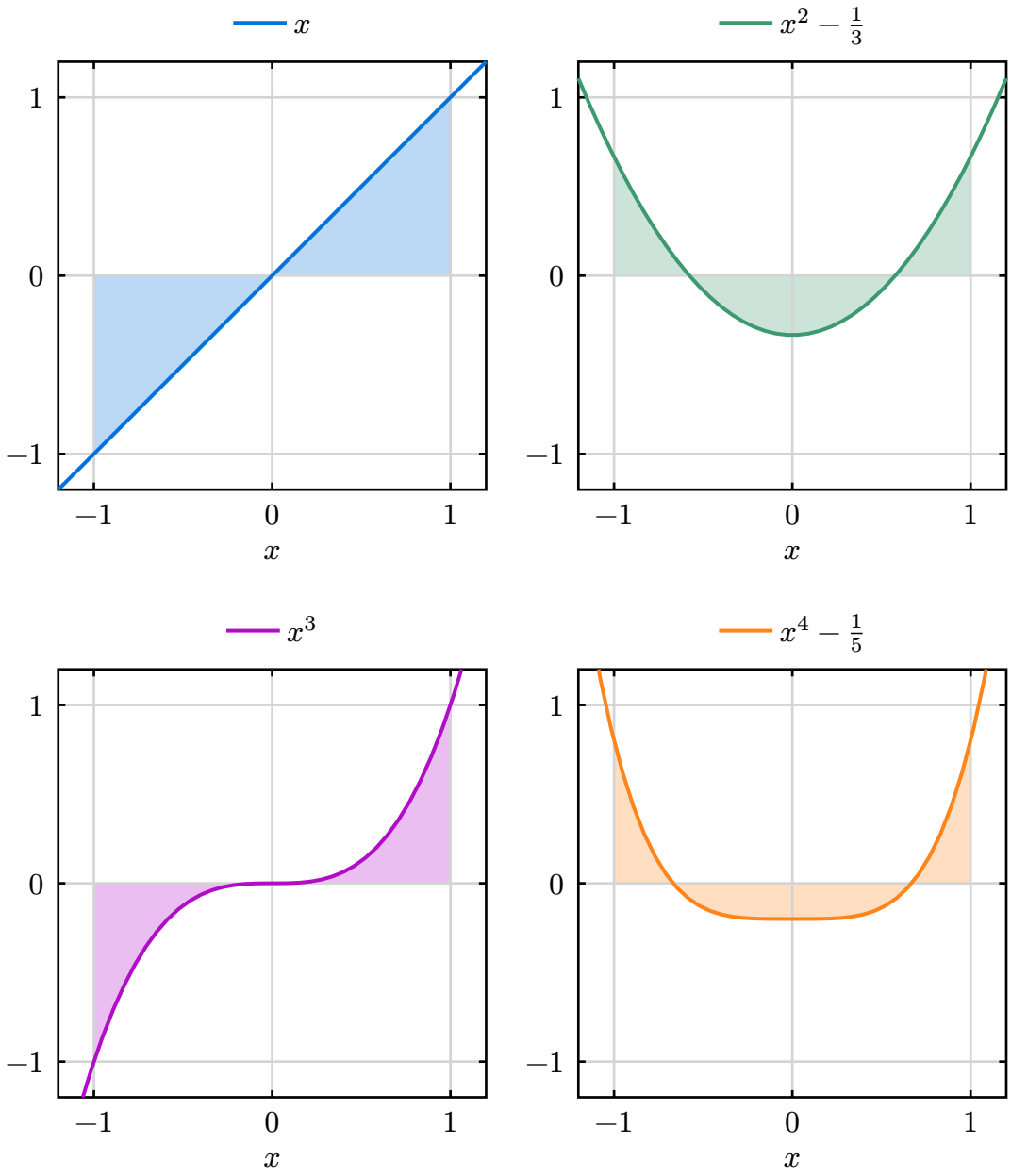
$$a_1 x + a_2\left(x^2 - \tfrac{1}{3}\right) + a_3 x^3 + a_4\left(x^4 - \tfrac{1}{5}\right) = 0$$

for all $x \in \mathbf{R}$. Using the reasoning of 2.41, we see that this equation implies that $a_1 = a_2 = a_3 = a_4 = 0$. It follows that $B$ is linearly independent and thus by 2.22 we have $\dim U \geq 4$. Using 2.37, we also find that $\dim U \leq \dim \mathcal{P}_4(\mathbf{R}) = 5$. However, notice that $U \neq \mathcal{P}_4(\mathbf{R})$ because $1 \notin U$; it follows from 2.39 that $\dim U$ must be strictly less



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than $\dim \mathcal{P}_4(\mathbf{R}) = 5$. Thus $\dim U = 4$ and using 2.38 we may conclude that $B$ is a basis of $U$.




(b) As noted in part (a), $1 \notin U = \operatorname{span} B$. It then follows from Exercise 2.A.13 that the list $B' = 1, x, x^2 - \frac{1}{3}, x^3, x^4 - \frac{1}{5}$ is linearly independent. Since $\dim \mathcal{P}_4(\mathbf{R}) = 5$, 2.38 allows us to conclude that $B'$ is a basis of $\mathcal{P}_4(\mathbf{R})$.

(c) Let $W = \operatorname{span}(1)$. As the proof of 2.33 shows, we then have $\mathcal{P}_4(\mathbf{R}) = U \oplus W$.

**Exercise 2.C.8.** Suppose $v_1, ..., v_m$ is linearly independent in $V$ and $w \in V$. Prove that

$$\dim \operatorname{span}(v_1 + w, ..., v_m + w) \geq m - 1.$$

**Solution.** If $m = 1$ then certainly $\dim \operatorname{span}(v_1 + w) \geq 0$, so suppose that $m \geq 2$. Because $v_1, ..., v_m$ is linearly independent, notice that:



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- $-v_1 \notin \operatorname{span}(v_2, ..., v_m)$;
- $v_2, ..., v_m$ is linearly independent.

It then follows from the contrapositive of Exercise 2.A.12 that the list $B = v_2 - v_1, ..., v_m - v_1$ is linearly independent. Now observe that

$$v_j - v_1 = (v_j + w) - (v_1 + w) \in \operatorname{span}(v_1 + w, ..., v_m + w)$$

for any $2 \leq j \leq m$. Thus $B$ is a linearly independent list of length $m - 1$ contained in $\operatorname{span}(v_1 + w, ..., v_m + w)$ and we may use 2.22 to conclude that

$$\dim \operatorname{span}(v_1 + w, ..., v_m + w) \geq m - 1.$$

**Exercise 2.C.9.** Suppose $m$ is a positive integer and $p_0, p_1, ..., p_m \in \mathcal{P}(\mathbf{F})$ are such that each $p_k$ has degree $k$. Prove that $p_0, p_1, ..., p_m$ is a basis of $\mathcal{P}_m(\mathbf{F})$.

**Solution.** Suppose we have scalars $a_0, a_1, ..., a_m$ such that

$$a_0 p_0(x) + a_1 p_1(x) + \cdots + a_m p_m(x) = 0 \tag{$*$}$$

for all $x \in \mathbf{F}$. Let $c$ be the coefficient of $x^m$ in the polynomial $p_m$ and note that $c \neq 0$ since $p_m$ has degree $m$. Because each $p_k$ has degree $k$, the coefficient of $x^m$ in the polynomial $p_k$ must be zero for $k < m$. Thus the left-hand side of $(*)$ has an $a_m c x^m$ term whereas the right-hand side has no $x^m$ term. It follows that $a_m c = 0$ and hence that $a_m = 0$, since $c \neq 0$. Repeating this argument for the lower degree terms, we find that $a_0 = a_1 = \cdots = a_m = 0$. Thus $p_0, p_1, ..., p_m$ is a linearly independent list of length $m + 1$ contained in $\mathcal{P}_m(\mathbf{F})$. Since $\dim \mathcal{P}_m(\mathbf{F}) = m + 1$, 2.38 allows us to conclude that $p_0, p_1, ..., p_m$ is a basis of $\mathcal{P}_m(\mathbf{F})$.

**Exercise 2.C.10.** Suppose $m$ is a positive integer. For $0 \leq k \leq m$, let

$$p_k(x) = x^k (1 - x)^{m-k}.$$

Show that $p_0, ..., p_m$ is a basis of $\mathcal{P}_m(\mathbf{F})$.

> *The basis in this exercise leads to what are called* ***Bernstein polynomials****. You can do a web search to learn how Bernstein polynomials are used to approximate continuous functions on* $[0, 1]$.

**Solution.** To remind us that these polynomials depend on $m$, let us use the notation

$$p_{k,m}(x) = x^k (1 - x)^{m-k}.$$

For a positive integer $m$, let $S(m)$ be the statement that the list $p_{0,m}, ..., p_{m,m}$ is linearly independent. We will use induction (twice) to show that $S(m)$ holds for all positive integers. First, we show the truth of $S(1)$ and $S(2)$.

For $S(1)$, suppose that $a_0, a_1$ are scalars such that

$$a_0 p_{0,1}(x) + a_1 p_{1,1}(x) = a_0(1 - x) + a_1 x = 0$$



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for all $x \in \mathbf{F}$. Taking $x = 0$ and $x = 1$ immediately gives us $a_0 = a_1 = 0$.

For $S(2)$, suppose that $a_0, a_1, a_2$ are scalars such that

$$a_0 p_{0,2}(x) + a_1 p_{1,2}(x) + a_2 p_{2,2}(x) = a_0(1-x)^2 + a_1 x(1-x) + a_2 x^2 = 0$$

for all $x \in \mathbf{F}$. Taking $x = 0$ and $x = 1$ immediately gives us $a_0 = a_2 = 0$, and then taking any $x \notin \{0, 1\}$ gives us $a_2 = 0$.

Now suppose that $S(m)$ holds for some positive integer $m$ and let $a_0, ..., a_{m+2}$ be scalars such that

$$\sum_{k=0}^{m+2} a_k p_{k,m+2}(x) = \sum_{k=0}^{m+2} a_k x^k (1-x)^{m+2-k} = 0$$

for all $x \in \mathbf{F}$. Taking $x = 0$ and $x = 1$ immediately gives us $a_0 = a_{m+2} = 0$, so that we now have the equation

$$\sum_{k=1}^{m+1} a_k x^k (1-x)^{m+2-k} = 0$$

for all $x \in \mathbf{F}$. Observe that

$$\begin{aligned}
\sum_{k=1}^{m+1} a_k x^k (1-x)^{m+2-k} &= x(1-x) \sum_{k=1}^{m+1} a_k x^{k-1} (1-x)^{m+1-k} \\
&= x(1-x) \sum_{k=0}^{m} a_{k+1} x^k (1-x)^{m-k} \\
&= x(1-x) \sum_{k=0}^{m} a_{k+1} p_{k,m}(x).
\end{aligned}$$

Thus $x(1-x)\sum_{k=0}^{m} a_{k+1} p_{k,m}(x) = 0$ for all $x \in \mathbf{F}$, which implies that $\sum_{k=0}^{m} a_{k+1} p_{k,m}(x) = 0$ for all $x \neq 0, 1$. Because the only polynomial with infinitely many roots is the zero polynomial (we will prove this in 4.8), in fact we must have

$$\sum_{k=0}^{m} a_{k+1} p_{k,m}(x) = 0$$

for all $x \in \mathbf{F}$. The induction hypothesis now implies that $a_1 = \cdots = a_m = 0$ and thus the list $p_0, ..., p_{m+2}$ is linearly independent, i.e. $S(m+2)$ holds. This completes the induction step.

We have now shown that $S(m) \Rightarrow S(m+2)$ for a positive integer $m$. Since $S(1)$ holds, an application of induction shows that $S(1), S(3), S(5), ...$ all hold. Similarly, since $S(2)$ holds, another application of induction shows that $S(2), S(4), S(6), ...$ all hold. Thus $S(m)$ holds for all positive integers $m$.

To complete the exercise, let $m$ be a positive integer. As we just showed, $p_0, ..., p_m$ is a linearly independent list of length $m+1$ contained in $\mathcal{P}_m(\mathbf{F})$. Since $\dim \mathcal{P}_m(\mathbf{F}) = m+1$, 2.38 allows us to conclude that $p_0, ..., p_m$ is a basis of $\mathcal{P}_m(\mathbf{F})$.



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**Exercise 2.C.11.** Suppose $U$ and $W$ are both four-dimensional subspaces of $\mathbf{C}^6$. Prove that there exist two vectors in $U \cap W$ such that neither of these vectors is a scalar multiple of the other.

**Solution.** Notice that

$$6 = \dim \mathbf{C}^6 \geq \dim(U + W) = \dim U + \dim W - \dim(U \cap W) = 8 - \dim(U \cap W),$$

where we have used 2.37 and 2.43. It follows that $\dim(U \cap W) \geq 2$ and thus we can find a linearly independent list $v_1, v_2$ in $U \cap W$; by Exercise 2.A.4 (b), neither of these vectors is a scalar multiple of the other.

**Exercise 2.C.12.** Suppose that $U$ and $W$ are subspaces of $\mathbf{R}^8$ such that $\dim U = 3$, $\dim W = 5$, and $U + W = \mathbf{R}^8$. Prove that $\mathbf{R}^8 = U \oplus W$.

**Solution.** By 2.43 we have

$$8 = \dim \mathbf{R}^8 = \dim(U + W) = \dim U + \dim W - \dim(U \cap W) = 8 - \dim(U \cap W).$$

It follows that $\dim(U \cap W) = 0$ and hence that $U \cap W = \{0\}$. Thus, by 1.46, the sum $\mathbf{R}^8 = U \oplus W$ is direct.

**Exercise 2.C.13.** Suppose $U$ and $W$ are both five-dimensional subspaces of $\mathbf{R}^9$. Prove that $U \cap W \neq \{0\}$.

**Solution.** By 2.43 we have

$$9 = \dim \mathbf{R}^9 \geq \dim(U + W) = \dim U + \dim W - \dim(U \cap W) = 10 - \dim(U \cap W).$$

It follows that $\dim(U \cap W) \geq 1$ and hence that $U \cap W \neq \{0\}$.

**Exercise 2.C.14.** Suppose $V$ is a ten-dimensional vector space and $V_1, V_2, V_3$ are subspaces of $V$ with $\dim V_1 = \dim V_2 = \dim V_3 = 7$. Prove that $V_1 \cap V_2 \cap V_3 \neq \{0\}$.

**Solution.** By 2.43 we have

$$\dim((V_1 \cap V_2) + V_3) = \dim(V_1 \cap V_2) + 7 - \dim(V_1 \cap V_2 \cap V_3), \tag{1}$$

$$\dim(V_1 + V_2) = 14 - \dim(V_1 \cap V_2). \tag{2}$$

Combining equations (1) and (2) gives us

$$\dim(V_1 \cap V_2 \cap V_3) = 21 - \dim(V_1 + V_2) - \dim((V_1 \cap V_2) + V_3).$$

Now we use the above equation and 2.37:

$$\dim(V_1 \cap V_2 \cap V_3) \geq 21 - 2 \dim V = 1.$$

Thus $V_1 \cap V_2 \cap V_3 \neq \{0\}$.



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**Exercise 2.C.15.** Suppose $V$ is finite-dimensional and $V_1, V_2, V_3$ are subspaces of $V$ with $\dim V_1 + \dim V_2 + \dim V_3 > 2 \dim V$. Prove that $V_1 \cap V_2 \cap V_3 \neq \{0\}$.

**Solution.** By 2.43 we have

$$\dim((V_1 \cap V_2) + V_3) = \dim(V_1 \cap V_2) + \dim V_3 - \dim(V_1 \cap V_2 \cap V_3), \tag{1}$$

$$\dim(V_1 + V_2) = \dim V_1 + \dim V_2 - \dim(V_1 \cap V_2). \tag{2}$$

Combining equations (1) and (2) gives us

$$\dim(V_1 \cap V_2 \cap V_3) = \dim V_1 + \dim V_2 + \dim V_3 - \dim(V_1 + V_2) - \dim((V_1 \cap V_2) + V_3).$$

Now we use the above equation and 2.37:

$$\dim(V_1 \cap V_2 \cap V_3) \geq \dim V_1 + \dim V_2 + \dim V_3 - 2 \dim V > 0.$$

Thus $V_1 \cap V_2 \cap V_3 \neq \{0\}$.

**Exercise 2.C.16.** Suppose $V$ is finite-dimensional and $U$ is a subspace of $V$ with $U \neq V$. Let $n = \dim V$ and $m = \dim U$. Prove that there exist $n - m$ subspaces of $V$, each of dimension $n - 1$, whose intersection equals $U$.

**Solution.** Let $u_1, ..., u_m$ be a basis of $U$ and, using 2.32, extend this to a basis $B = u_1, ..., u_m, v_1, ..., v_k$ of $V$; note that $k \geq 1$ since $U \neq V$ and that $n - m = k$. For each $j \in \{1, ..., k\}$, let $B_j$ be the list of vectors obtained by removing $v_j$ from $B$, i.e.

$$B_j = u_1, ..., u_m, v_1, ..., v_{j-1}, v_{j+1}, ..., v_k.$$

Now let $U_j = \operatorname{span} B_j$. Observe that $B_j$ is linearly independent since $B$ is linearly independent and thus $B_j$ is a basis of $U_j$, so that $\dim U_j = m + k - 1 = n - 1$. Furthermore, for $i \neq j$ we have $v_j \in U_i$ but $v_j \notin U_j$ by the linear independence of $B$; it follows that $U_i \neq U_j$. Thus the collection $\{U_j : 1 \leq j \leq k\}$ consists of $k = n - m$ distinct subspaces of $V$ each of dimension $n - 1$.

We now show that $U = U_1 \cap \cdots \cap U_k$. If $k = 1$ then $U_1 = U$ and the equality is clear, so suppose that $k \geq 2$. Certainly $U \subseteq U_j$ for each $j \in \{1, ..., k\}$ and thus $U \subseteq U_1 \cap \cdots \cap U_k$. Let $u \in U_1 \cap \cdots \cap U_k$ be given. In particular $u \in U_1$, so there are scalars $a_1, ..., a_m, c_2, ..., c_k$ such that

$$u = a_1 u_1 + \cdots + a_m u_m + c_2 v_2 + \cdots + c_k v_k. \tag{$*$}$$

For each $j \in \{2, ..., k\}$ we have $u \in U_j$ and thus $u$ can also be expressed as a linear combination of the list $B_j$. The coefficient of $v_j$ in this linear combination is zero and, because $B$ is a basis of $V$, it then follows from unique representation 2.28 that the coefficient $c_j$ in the linear combination $(*)$ is also zero. Thus $u = a_1 u_1 + \cdots + a_m u_m \in U$, so that $U_1 \cap \cdots \cap U_m \subseteq U$. We may conclude that $U = U_1 \cap \cdots \cap U_k$.



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**Exercise 2.C.17.** Suppose that $V_1, ..., V_m$ are finite-dimensional subspaces of $V$. Prove that $V_1 + \cdots + V_m$ is finite-dimensional and

$$\dim(V_1 + \cdots + V_m) \leq \dim V_1 + \cdots + \dim V_m.$$

*The inequality above is an equality if and only if $V_1 + \cdots + V_m$ is a direct sum, as will be shown in 3.94.*

**Solution.** Each $V_j$ has a basis $B_j$ by 2.31, so that $V_j = \operatorname{span} B_j$. Let $B$ be the list $B_1, B_2, ..., B_m$ (removing duplicate vectors if necessary) and, letting $|B|$ denote the length of the list $B$, notice that $|B| \leq |B_1| + \cdots + |B_m|$. Notice further that $V_1 + \cdots + V_m = \operatorname{span} B$. It follows that $V_1 + \cdots + V_m$ is finite-dimensional and furthermore, by 2.22,

$$\dim(V_1 + \cdots + V_m) \leq |B| \leq |B_1| + \cdots + |B_m| = \dim V_1 + \cdots + \dim V_m.$$

**Exercise 2.C.18.** Suppose $V$ is finite-dimensional, with $\dim V = n \geq 1$. Prove that there exist one-dimensional subspaces $V_1, ..., V_n$ of $V$ such that

$$V = V_1 \oplus \cdots \oplus V_n.$$

**Solution.** $V$ has a non-empty (since $n \geq 1$) basis $v_1, ..., v_n$. For each $j \in \{1, ..., n\}$ let $V_j = \operatorname{span}(v_j)$ and note that $\dim V_j = 1$ because $v_j \neq 0$. By 2.28 each vector in $V$ is a linear combination of the basis vectors $v_1, ..., v_n$, so that $V = V_1 + \cdots + V_n$, and furthermore this linear combination is unique, so that the sum $V = V_1 \oplus \cdots + \oplus V_n$ is direct.

**Exercise 2.C.19.** Explain why you might guess, motivated by analogy with the formula for the number of elements in the union of three finite sets, that if $V_1, V_2, V_3$ are subspaces of a finite-dimensional vector space, then

$$\begin{aligned}\dim(V_1 + V_2 + V_3)&\\ &= \dim V_1 + \dim V_2 + \dim V_3\\ &\quad - \dim(V_1 \cap V_2) - \dim(V_1 \cap V_3) - \dim(V_2 \cap V_3)\\ &\quad + \dim(V_1 \cap V_2 \cap V_3).\end{aligned}$$

Then either prove the formula above or give a counterexample.

**Solution.** If $S_1, S_2, S_3$ are finite sets and $|S|$ denotes the number of elements in a finite set $S$, then the inclusion-exclusion principle gives us the formula

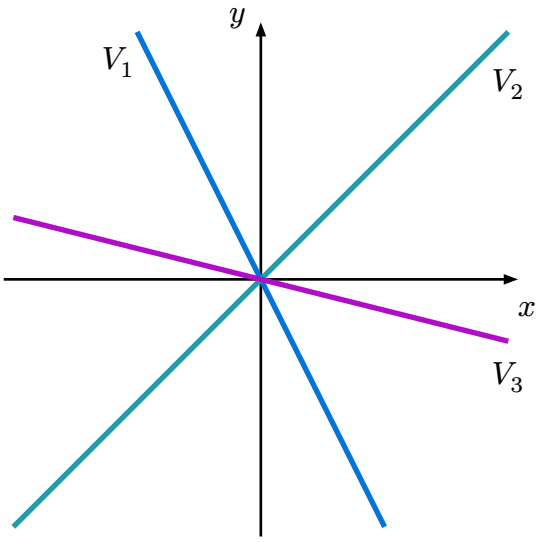
$$|S_1 \cup S_2 \cup S_3| = |S_1| + |S_2| + |S_3| - |S_1 \cap S_2| - |S_1 \cap S_3| - |S_2 \cap S_3| + |S_1 \cap S_2 \cap S_3|.$$

However, the analogous formula for the dimensions of finite-dimensional subspaces does not hold, as the following counterexample shows. Consider $\mathbf{R}^2$ and let $V_1, V_2, V_3$ be three distinct lines through the origin, say



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$$V_1 = \operatorname{span}((-1,2)), \quad V_2 = \operatorname{span}((1,1)), \quad \text{and} \quad V_3 = \operatorname{span}((4,-1)).$$




It is straightforward to verify that $V_1 + V_2 + V_3 = \mathbf{R}^2$ and that

$$V_1 \cap V_2 = V_1 \cap V_3 = V_2 \cap V_3 = V_1 \cap V_2 \cap V_3 = \{0\}.$$

Thus $\dim(V_1 + V_2 + V_3) = \dim \mathbf{R}^2 = 2$, whereas the right-hand side of the proposed formula is

$$1 + 1 + 1 - 0 - 0 - 0 + 0 = 3.$$

**Exercise 2.C.20.** Prove that if $V_1, V_2$, and $V_3$ are subspaces of a finite-dimensional vector space, then

$$\begin{aligned}
&\dim(V_1 + V_2 + V_3) \\
&\quad = \dim V_1 + \dim V_2 + \dim V_3 \\
&\qquad - \frac{\dim(V_1 \cap V_2) + \dim(V_1 \cap V_3) + \dim(V_2 \cap V_3)}{3} \\
&\qquad - \frac{\dim((V_1 + V_2) \cap V_3) + \dim((V_1 + V_3) \cap V_2) + \dim((V_2 + V_3) \cap V_1)}{3}.
\end{aligned}$$

*The formula above may seem strange because the right side does not look like an integer.*

**Solution.** Using 2.43 twice, observe that

$$\begin{aligned}
\dim(V_1 + V_2 + V_3) &= \dim(V_1 + V_2) + \dim V_3 - \dim((V_1 + V_2) \cap V_3) \\
&= \dim V_1 + \dim V_2 + \dim V_3 - \dim(V_1 \cap V_2) - \dim((V_1 + V_2) \cap V_3).
\end{aligned}$$

Similarly, we find that

$$\dim(V_1 + V_2 + V_3) = \dim V_1 + \dim V_2 + \dim V_3 - \dim(V_1 \cap V_3) - \dim((V_1 + V_3) \cap V_2),$$

$$\dim(V_1 + V_2 + V_3) = \dim V_1 + \dim V_2 + \dim V_3 - \dim(V_2 \cap V_3) - \dim((V_2 + V_3) \cap V_1).$$



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Adding these three formulas together and then dividing through by **3** gives us the desired formula.



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# Chapter 3. Linear Maps

## 3.A. Vector Space of Linear Maps

**Exercise 3.A.1.** Suppose $b, c \in \mathbf{R}$. Define $T : \mathbf{R}^3 \to \mathbf{R}^2$ by

$$T(x, y, z) = (2x - 4y + 3z + b, 6x + cxyz).$$

Show that $T$ is linear if and only if $b = c = 0$.

**Solution.** First suppose that $b = c = 0$, so that $T$ is the map

$$T(x, y, z) = (2x - 4y + 3z, 6x).$$

Let $(x_1, y_1, z_1), (x_2, y_2, z_2) \in \mathbf{R}^3$ and $\lambda \in \mathbf{R}$ be given. Observe that

$$\begin{aligned} T(x_1 + x_2, y_1 + y_2, z_1 + z_2) &= (2(x_1 + x_2) - 4(y_1 + y_2) + 3(z_1 + z_2), 6(x_1 + x_2)) \\ &= (2x_1 + 2x_2 - 4y_1 - 4y_2 + 3z_1 + 3z_2, 6x_1 + 6x_2) \\ &= (2x_1 - 4y_1 + 3z_1, 6x_1) + (2x_2 - 4y_2 + 3z_2, 6x_2) \\ &= T(x_1, y_1, z_1) + T(x_2, y_2, z_2) \end{aligned}$$

$$\begin{aligned} T(\lambda x_1, \lambda y_1, \lambda z_1) &= (2\lambda x_1 - 4\lambda y_1 + 3\lambda z_1, 6\lambda x_1) \\ &= (\lambda(2x_1 - 4y_1 + 3z_1), \lambda(6x_1)) \\ &= \lambda(2x_1 - 4y_1 + 3z_1, 6x_1) \\ &= \lambda T(x_1, y_1, z_1). \end{aligned}$$

Thus $T$ is linear.

Now suppose that $b \neq 0$ and notice that $T(0, 0, 0) = (b, 0) \neq (0, 0)$; it follows from 3.10 that $T$ is not linear. If $c \neq 0$ then note that

$$T(1, 1, 1) = (1 + b, 6 + c) \quad \text{and} \quad T(2, 2, 2) = (2 + b, 12 + 8c).$$

Since $2(6 + c) = 12 + 2c \neq 12 + 8c$ for $c \neq 0$, we see that $2T(1, 1, 1) \neq T(2, 2, 2)$ and thus $T$ is not linear.



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**Exercise 3.A.2.** Suppose $b, c \in \mathbf{R}$. Define $T : \mathcal{P}(\mathbf{R}) \to \mathbf{R}^2$ by

$$Tp = \left(3p(4) + 5p'(6) + bp(1)p(2), \int_{-1}^{2} x^3 p(x)\,\mathrm{d}x + c\sin p(0)\right).$$

Show that $T$ is linear if and only if $b = c = 0$.

**Solution.** First suppose that $b = c = 0$, so that $T$ is the map

$$Tp = \left(3p(4) + 5p'(6), \int_{-1}^{2} x^3 p(x)\,\mathrm{d}x\right).$$

Let $p, q \in \mathcal{P}(\mathbf{R})$ and $\lambda \in \mathbf{R}$ be given. Observe that

$$\begin{aligned}
T(p+q) &= \left(3(p+q)(4) + 5(p+q)'(6), \int_{-1}^{2} x^3 (p+q)(x)\,\mathrm{d}x\right) \\
&= \left(3(p(4)+q(4)) + 5(p'(6)+q'(6)), \int_{-1}^{2} x^3 (p(x)+q(x))\,\mathrm{d}x\right) \\
&= \left(3p(4) + 3q(4) + 5p'(6) + 5q'(6), \int_{-1}^{2} x^3 p(x)\,\mathrm{d}x + \int_{-1}^{2} x^3 q(x)\,\mathrm{d}x\right) \\
&= \left(3p(4) + 5p'(6), \int_{-1}^{2} x^3 p(x)\,\mathrm{d}x\right) + \left(3q(4) + 5q'(6), \int_{-1}^{2} x^3 q(x)\,\mathrm{d}x\right) \\
&= Tp + Tq.
\end{aligned}$$

$$\begin{aligned}
T(\lambda p) &= \left(3(\lambda p)(4) + 5(\lambda p)'(6), \int_{-1}^{2} x^3 (\lambda p)(x)\,\mathrm{d}x\right) \\
&= \left(3(\lambda p(4)) + 5(\lambda p'(6)), \int_{-1}^{2} x^3 (\lambda p(x))\,\mathrm{d}x\right) \\
&= \left(\lambda(3p(4) + 5p'(6)), \lambda \int_{-1}^{2} x^3 p(x)\,\mathrm{d}x\right) \\
&= \lambda\left(3p(4) + 5p'(6), \int_{-1}^{2} x^3 p(x)\,\mathrm{d}x\right) \\
&= \lambda Tp.
\end{aligned}$$

Thus $T$ is linear.

Now suppose that $T$ is linear and observe that

$$2T(\pi) = \left(6\pi + 2b\pi^2, \tfrac{15}{2}\pi + 2c\right) \quad \text{and} \quad T(2\pi) = \left(6\pi + 4b\pi^2, \tfrac{15}{2}\pi\right).$$



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Since $T$ is linear we must have $2T(\pi) = T(2\pi)$:

$$\left(6\pi + 2b\pi^2, \tfrac{15}{2}\pi + 2c\right) = \left(6\pi + 4b\pi^2, \tfrac{15}{2}\pi\right) \quad \Leftrightarrow \quad (2b\pi^2, 2c) = (4b\pi^2, 0) \quad \Leftrightarrow \quad b = c = 0.$$

**Exercise 3.A.3.** Suppose that $T \in \mathcal{L}(\mathbf{F}^n, \mathbf{F}^m)$. Show that there exist scalars $A_{j,k} \in \mathbf{F}$ for $j = 1, ..., m$ and $k = 1, ..., n$ such that

$$T(x_1, ..., x_n) = \left(A_{1,1}x_1 + \cdots + A_{1,n}x_n, ..., A_{m,1}x_1 + \cdots + A_{m,n}x_n\right)$$

for every $(x_1, ..., x_n) \in \mathbf{F}^n$.

*This exercise shows that the linear map $T$ has the form promised in the second to last item of Example 3.3.*

**Solution.** Let $e_1, ..., e_n$ be the standard basis of $\mathbf{F}^n$ and let $f_1, ..., f_m$ be the standard basis of $\mathbf{F}^m$. For any $k \in \{1, ..., n\}$, there are scalars $A_{1,k}, ..., A_{m,k}$ such that

$$Te_k = \sum_{j=1}^{m} A_{j,k} f_j.$$

Let $x = (x_1, ..., x_n) = \sum_{k=1}^{n} x_k e_k$ be given and observe that by linearity,

$$Tx = T\left(\sum_{k=1}^{n} x_k e_k\right) = \sum_{k=1}^{n} x_k Te_k = \sum_{k=1}^{n} x_k \sum_{j=1}^{m} A_{j,k} f_j = \sum_{j=1}^{m} \left(\sum_{k=1}^{n} A_{j,k} x_k\right) f_j$$

$$= \left(\sum_{k=1}^{n} A_{1,k}x_k, ..., \sum_{k=1}^{n} A_{m,k}x_k\right) = \left(A_{1,1}x_1 + \cdots + A_{1,n}x_n, ..., A_{m,1}x_1 + \cdots + A_{m,n}x_n\right).$$

**Exercise 3.A.4.** Suppose $T \in \mathcal{L}(V, W)$ and $v_1, ..., v_m$ is a list of vectors in $V$ such that $Tv_1, ..., Tv_m$ is a linearly independent list in $W$. Prove that $v_1, ..., v_m$ is linearly independent.

**Solution.** Suppose we have scalars $a_1, ..., a_m$ such that $a_1v_1 + \cdots + a_mv_m = 0$. Applying $T$ to both sides of this equation and using linearity and 3.10, we obtain

$$T(a_1v_1 + \cdots + a_mv_m) = T(0) \quad \Leftrightarrow \quad a_1Tv_1 + \cdots + a_mTv_m = 0.$$

Since the list $Tv_1, ..., Tv_m$ is linearly independent, this equation implies that $a_1 = \cdots = a_m = 0$. Thus the list $v_1, ..., v_m$ is linearly independent.

**Exercise 3.A.5.** Prove that $\mathcal{L}(V, W)$ is a vector space, as was asserted in 3.6.

**Solution.** First let us show that $\mathcal{L}(V, W)$ is closed under addition and scalar multiplication. Let $S, T \in \mathcal{L}(V, W), u, v \in V$, and $\lambda, \alpha \in \mathbf{F}$ be given. Observe that

$$(S + T)(u + v) = S(u + v) + T(u + v) = Su + Sv + Tu + Tv$$

$$= Su + Tu + Sv + Tv = (S + T)(u) + (S + T)(v),$$



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$$(S+T)(\alpha v) = S(\alpha v) + T(\alpha v) = \alpha Sv + \alpha Tv = \alpha(Sv+Tv) + \alpha(S+T)(v).$$

Thus $S+T \in \mathcal{L}(V,W)$. Similarly,

$$(\lambda S)(u+v) = \lambda S(u+v) = \lambda(Su+Sv) = \lambda Su + \lambda Sv = (\lambda S)(u) + (\lambda S)(v),$$

$$(\lambda S)(\alpha v) = \lambda S(\alpha v) = \lambda(\alpha Sv) = \alpha(\lambda Sv) = \alpha(\lambda S)(v).$$

Thus $\lambda S \in \mathcal{L}(V,W)$. We now verify each requirement in definition 1.20.

**Commutativity.** Suppose $S, T \in \mathcal{L}(V,W)$ and $v \in V$. Observe that

$$(S+T)(v) = Sv + Tv = Tv + Sv = (T+S)(v).$$

Thus $S+T = T+S$.

**Associativity.** Suppose $R, S, T \in \mathcal{L}(V,W), a, b \in \mathbf{F}$, and $v \in V$. Observe that

$$\begin{aligned}((R+S)+T)(v) = (R+S)(v) = Tv = (Rv+Sv) = Tv \\ = Rv + (Sv+Tv) = Rv + (S+T)(v) = (R+(S+T))(v),\end{aligned}$$

$$((ab)R)(v) = (ab)Rv = a(bRv) = a((bR)(v)) = (a(bR))(v).$$

Thus $(R+S)+T = R+(S+T)$ and $(ab)R = a(bR)$.

**Additive identity.** Certainly the map $0 : V \to W$ given by $v \mapsto 0$ belongs to $\mathcal{L}(V,W)$; we claim that this map is the additive identity in $\mathcal{L}(V,W)$. Indeed, let $S \in \mathcal{L}(V,W)$ and $v \in V$ be given and observe that

$$(S+0)(v) = Sv + 0v = Sv + 0 = Sv.$$

Thus $S+0 = S$.

**Additive inverse.** For $S \in \mathcal{L}(V,W)$, define $T : V \to W$ by $Tv = -Sv$; it is straightforward to verify that $T$ is linear. We claim that $T$ is the additive inverse to $S$. Indeed, for any $v \in V$,

$$(S+T)(v) = Sv + Tv = Sv + (-Sv) = 0.$$

Thus $S+T = 0$.

**Multiplicative identity.** Let $S \in \mathcal{L}(V,W)$ and $v \in V$ be given and observe that

$$(1S)(v) = 1Sv = Sv.$$

Thus $1S = S$.

**Distributive properties.** Let $S, T \in \mathcal{L}(V,W), a, b \in \mathbf{F}$, and $v \in V$ be given. Observe that

$$(a(S+T))(v) = a(S+T)(v) = a(Sv+Tv) = aSv + aTv = (aS)(v) + (aT)(v),$$

$$((a+b)S)(v) = (a+b)Sv = aSv + bSv = (aS)(v) + (bS)(v).$$

Thus $a(S+T) = aS + aT$ and $(a+b)S = aS + bS$.



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**Exercise 3.A.6.** Prove that multiplication of linear maps has the associative, identity, and distributive properties asserted in 3.8.

**Solution.** The associative property is immediate from the associativity of composition of functions.

For the identity property, let $I_V$ be the identity map on $V$ and let $I_W$ be the identity map on $W$. For $T \in \mathcal{L}(V,W)$ and $v \in V$, observe that

$$(TI_V)(v) = T(I_V v) = Tv \quad \text{and} \quad (I_W T)(v) = I_W(Tv) = Tv.$$

Thus $TI_V = I_W T = T$.

For the distributive properties, let $T, T_1, T_2 \in \mathcal{L}(U,V)$ and $S, S_1, S_2 \in \mathcal{L}(V,W)$ be given. For any $u \in U$, observe that

$$((S_1 + S_2)T)(u) = (S_1 + S_2)(Tu) = S_1(Tu) + S_2(Tu) = (S_1T)(u) = (S_2T)(u),$$

$$\begin{aligned}(S(T_1 + T_2))(u) = S((T_1 + T_2)(u)) &= S(T_1u + T_2u) \\ &= S(T_1u) + S(T_2u) = (ST_1)(u) + (ST_2)(u).\end{aligned}$$

Thus $(S_1 + S_2)T = S_1T + S_2T$ and $S(T_1 + T_2) = ST_1 + ST_2$.

**Exercise 3.A.7.** Show that every linear map from a one-dimensional vector space to itself is multiplication by some scalar. More precisely, prove that if $\dim V = 1$ and $T \in \mathcal{L}(V)$, then there exists $\lambda \in \mathbf{F}$ such that $Tv = \lambda v$ for all $v \in V$.

**Solution.** Since $\dim V = 1$ there is a basis $u$ of $V$ and thus $Tu = \lambda u$ for some $\lambda \in \mathbf{F}$. Let $v = \alpha u \in V$ be given and observe that

$$Tv = T(\alpha u) = \alpha Tu = \alpha(\lambda u) = \lambda(\alpha u) = \lambda v.$$

**Exercise 3.A.8.** Give an example of a function $\varphi : \mathbf{R}^2 \to \mathbf{R}$ such that

$$\varphi(av) = a\varphi(v)$$

for all $a \in \mathbf{R}$ and all $v \in \mathbf{R}^2$ but $\varphi$ is not linear.

*This exercise and the next exercise show that neither homogeneity nor additivity alone is enough to imply that a function is a linear map.*

**Solution.** Let $\varphi : \mathbf{R}^2 \to \mathbf{R}$ be given by $\varphi(x,y) = \left(x^3 + y^3\right)^{1/3}$ and observe that for any $a \in \mathbf{R}$ and $(x,y) \in \mathbf{R}^2$,

$$\varphi(ax, ay) = \left((ax)^3 + (ay)^3\right)^{1/3} = \left(a^3\right)^{1/3}\left(x^3 + y^3\right)^{1/3} = a\left(x^3 + y^3\right)^{1/3} = a\varphi(x,y).$$

However, notice that



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$$\varphi(1,0)+\varphi(0,1)=1+1=2\neq 2^{1/3}=\varphi(1,1).$$

Thus $\varphi$ is not linear.

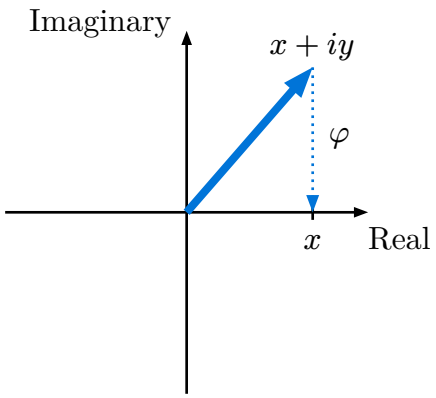
**Exercise 3.A.9.** Give an example of a function $\varphi : \mathbf{C} \to \mathbf{C}$ such that

$$\varphi(w+z)=\varphi(w)+\varphi(z)$$

for all $w, z \in \mathbf{C}$ but $\varphi$ is not linear. (Here $\mathbf{C}$ is thought of as a complex vector space.)

> *There also exists a function $\varphi : \mathbf{R} \to \mathbf{R}$ such that $\varphi$ satisfies the additivity condition above but $\varphi$ is not linear. However, showing the existence of such a function involves considerably more advanced tools.*

**Solution.** Let $\varphi : \mathbf{C} \to \mathbf{C}$ be given by $\varphi(x+iy)=x$, i.e. $\varphi$ takes a complex number to its real part.




Observe that

$$\varphi((x+iy)+(u+iv))=\varphi((x+u)+i(y+v))=x+u=\varphi(x+iy)+\varphi(u+iv).$$

However, $\varphi(i)=0$ but $\varphi(i^2)=\varphi(-1)=-1\neq i\varphi(i)$. Thus $\varphi$ is not linear.

**Exercise 3.A.10.** Prove or give a counterexample: If $q \in \mathcal{P}(\mathbf{R})$ and $T : \mathcal{P}(\mathbf{R}) \to \mathcal{P}(\mathbf{R})$ is defined by $Tp = q \circ p$, then $T$ is linear map.

> *The function $T$ defined here differs from the function $T$ defined in the last bullet point of 3.3 by the order of the functions in the compositions.*

**Solution.** This does not necessarily define a linear map. For example, consider $q(x)=1$ and observe that $(T0)(x)=q(0(x))=q(0)=1\neq 0$. It follows from 3.10 that $T$ is not linear.

**Exercise 3.A.11.** Suppose $V$ is finite-dimensional and $T \in \mathcal{L}(V)$. Prove that $T$ is a scalar multiple of the identity if and only if $ST=TS$ for every $S \in \mathcal{L}(V)$.

**Solution.** Suppose that $T$ is a scalar multiple of the identity, say $T=\lambda I$ for some $\lambda \in \mathbf{F}$, and let $S \in \mathcal{L}(V)$ and $v \in V$ be given. Observe that



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$$S(Tv) = S((\lambda I)(v)) = S(\lambda v) = \lambda(Sv) = (\lambda I)(Sv) = T(Sv).$$

Thus $ST = TS$.

Now suppose that $ST = TS$ for every $S \in \mathcal{L}(V)$. If $V = \{0\}$ then $T = 0I$ and we are done. Otherwise, let $v_1, ..., v_m$ be a basis of $V$ and, using the linear map lemma (3.4), define a linear map $\varphi : V \to \mathbf{F}$ satisfying $\varphi(v_1) = \cdots = \varphi(v_m) = 1$. Let $\lambda = \varphi(Tv_1) \in \mathbf{F}$. For a fixed $v \in V$, define $S_v \in \mathcal{L}(V)$ by $S_v u = \varphi(u)v$; the linearity of $S$ follows from the linearity of $\varphi$. Now observe that

$$Tv = T(\varphi(v_1)v) = T(S_v v_1) = S_v(Tv_1) = \varphi(Tv_1)v = \lambda v,$$

where we have used that $T$ commutes with every $S \in \mathcal{L}(V)$ for the third equality. Because $v \in V$ was arbitrary, we may conclude that $T = \lambda I$.

**Exercise 3.A.12.** Suppose $U$ is a subspace of $V$ with $U \neq V$. Suppose $S \in \mathcal{L}(U, W)$ and $S \neq 0$ (which means that $Su \neq 0$ for some $u \in U$). Define $T : V \to W$ by

$$Tv = \begin{cases} Sv & \text{if } v \in U, \\ 0 & \text{if } v \in V \text{ and } v \notin U. \end{cases}$$

Prove that $T$ is not a linear map on $V$.

**Solution.** There is some $u \in U$ such that $Su \neq 0$ and since $U \neq V$ there is some $v \in V$ such that $v \notin U$. This implies that $u - v \notin U$, otherwise $v = -(u - v) + u \in U$. Observe that

$$Tv + T(u - v) = 0 + 0 = 0 \neq Su = Tu = T(v + u - v).$$

Thus $T$ is not linear.

**Exercise 3.A.13.** Suppose $V$ is finite-dimensional. Prove that every linear map on a subspace of $V$ can be extended to a linear map on $V$. In other words, show that if $U$ is a subspace of $V$ and $S \in \mathcal{L}(U, W)$, then there exists $T \in \mathcal{L}(V, W)$ such that $Tu = Su$ for all $u \in U$.

*The result in this exercise is used in the proof of 3.125.*

**Solution.** Let $u_1, ..., u_m$ be a basis of $U$ and extend this to a basis $u_1, ..., u_m, v_1, ..., v_n$ of $V$. If we use the linear map lemma (3.4) to define $T \in \mathcal{L}(V, W)$ by $Tu_k = Su_k$ and $Tv_k = 0$, then for any $u = a_1u_1 + \cdots + a_mu_m \in U$ we have

$$Tu = a_1Tu_1 + \cdots + a_mTu_m = a_1Su_1 + \cdots + a_mSu_m = Su.$$

Thus $T$ extends $S$.

**Exercise 3.A.14.** Suppose $V$ is finite-dimensional with $\dim V > 0$, and suppose $W$ is infinite-dimensional. Prove that $\mathcal{L}(V, W)$ is infinite-dimensional.



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**Solution.** By Exercise 2.A.17 there is a sequence $w_1, w_2, w_3, \ldots$ in $W$ such that $w_1, \ldots, w_m$ is linearly independent for each positive integer $m$. Let $v_1, \ldots, v_n$ be a basis of $V$ and note that $n \geq 1$. For each positive integer $k$, use the linear map lemma (3.4) to define a linear map $T_k \in \mathcal{L}(V, W)$ satisfying

$$T_k v_j = \begin{cases} w_k & \text{if } j = 1, \\ 0 & \text{otherwise.} \end{cases}$$

Let $m$ be a positive integer and suppose we have scalars $a_1, \ldots, a_m$ such that $a_1 T_1 + \cdots + a_m T_m = 0$. In particular,

$$0 = (a_1 T_1 + \cdots + a_m T_m)(v_1) = a_1 T_1 v_1 + \cdots + a_m T_m v_1 = a_1 w_1 + \cdots + a_m w_m.$$

This implies that $a_1 = \cdots = a_m = 0$ since the list $w_1, \ldots, w_m$ is linearly independent and it follows that the list $T_1, \ldots, T_m$ is linearly independent. We may use Exercise 2.A.17 to conclude that $\mathcal{L}(V, W)$ is infinite-dimensional.

**Exercise 3.A.15.** Suppose $v_1, \ldots, v_m$ is a linearly dependent list of vectors in $V$. Suppose also that $W \neq \{0\}$. Prove that there exist $w_1, \ldots, w_m \in W$ such that no $T \in \mathcal{L}(V, W)$ satisfies $Tv_k = w_k$ for each $k = 1, \ldots, m$.

**Solution.** (We will use complex conjugation for this solution; complex conjugation is defined and its properties are studied in Chapter 4 of the textbook.)

We will prove the contrapositive statement. That is, assuming that for all lists $w_1, \ldots, w_m \in W$ there is a linear map $T \in \mathcal{L}(V, W)$ such that $Tv_k = w_k$ for each $k \in \{1, \ldots, m\}$, we will prove that the list $v_1, \ldots, v_m$ is linearly independent. Indeed, suppose we have scalars $a_1, \ldots, a_m$ such that $a_1 v_1 + \cdots + a_m v_m = 0$. There is some non-zero $w \in W$; by assumption there is a linear map $T \in \mathcal{L}(V, W)$ such that $Tv_k = \overline{a_k} w$ for each $k \in \{1, \ldots, m\}$. It follows that

$$0 = T\left(\sum_{k=1}^{m} a_k v_k\right) = \sum_{k=1}^{m} a_k T v_k = \sum_{k=1}^{m} a_k \overline{a_k} w = \left(\sum_{k=1}^{m} |a_k|^2\right) w.$$

Since $w \neq 0$, Exercise 1.B.2 implies that $\sum_{k=1}^{m} |a_k|^2 = 0$, which is the case if and only if $a_k = 0$ for all $k \in \{1, \ldots, m\}$. Thus the list $v_1, \ldots, v_m$ is linearly independent.

**Exercise 3.A.16.** Suppose $V$ is finite-dimensional with $\dim V > 1$. Prove that there exist $S, T \in \mathcal{L}(V)$ such that $ST \neq TS$.

**Solution.** There is a basis $v_1, v_2, \ldots, v_n$ for $V$ with $n \geq 2$. Using the linear map lemma (3.4), define linear maps $S, T \in \mathcal{L}(V)$ satisfying

$$Sv_k = \begin{cases} v_2 & \text{if } k = 1, \\ v_1 & \text{if } k = 2, \\ 0 & \text{otherwise,} \end{cases} \qquad Tv_k = \begin{cases} 2v_2 & \text{if } k = 1, \\ v_1 & \text{if } k = 2, \\ 0 & \text{otherwise.} \end{cases}$$



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Observe that

$$(ST - TS)(v_1) = S(Tv_1) - T(Sv_1) = S(2v_2) - Tv_2 = 2v_1 - v_1 = v_1 \neq 0.$$

Thus $ST \neq TS$.

**Exercise 3.A.17.** Suppose $V$ is finite-dimensional. Show that the only two-sided ideals of $\mathcal{L}(V)$ are $\{0\}$ and $\mathcal{L}(V)$.

*A subspace $\mathcal{E}$ of $\mathcal{L}(V)$ is called a **two-sided ideal** of $\mathcal{L}(V)$ if $TE \in \mathcal{E}$ and $ET \in \mathcal{E}$ for all $E \in \mathcal{E}$ and all $T \in \mathcal{L}(V)$.*

**Solution.** Certainly $\{0\}$ is a two-sided ideal of $\mathcal{L}(V)$, so suppose that $\mathcal{E}$ is a two-sided ideal of $\mathcal{L}(V)$ containing some non-zero linear map $T \in \mathcal{L}(V)$; we must show that $\mathcal{E} = \mathcal{L}(V)$.

Let $v_1, ..., v_m$ be a basis of $V$. For each pair $(i, j)$, use the linear map lemma (3.4) to define a linear map $E_{i,j} \in \mathcal{L}(V)$ which sends $v_i$ to $v_j$ and each other basis vector to 0. Since $T \neq 0$, there must be some $\ell \in \{1, ..., m\}$ such that

$$Tv_\ell = a_1 v_1 + \cdots + a_m v_m \neq 0;$$

there must then be some non-zero coefficient in this linear combination, say $a_n$. For each $i \in \{1, ..., m\}$, observe that

$$E_{n,i} T E_{i,\ell} v_i = E_{n,i} T v_\ell = E_{n,i}(a_1 v_1 + \cdots + a_m v_m) = a_n v_i,$$

$$\text{and for } k \neq i, \qquad E_{n,i} T E_{i,\ell} v_k = E_{n,i} T(0) = 0.$$

Thus, letting $L_i$ be the linear map $E_{n,i} T E_{i,\ell}$, we have $L_i v_i = a_n v_i$ and $L_i v_k = 0$ for $k \neq i$; it follows that $L_1 + \cdots + L_m = a_n I$, where $I$ is the identity map on $V$. Now observe that

$$T \in \mathcal{E} \quad \Rightarrow \quad E_{n,i} T E_{i,\ell} = L_i \in \mathcal{E} \text{ for each } i \in \{1, ..., m\}$$

$$\Rightarrow \quad L_1 + \cdots + L_m = a_n I \in \mathcal{E} \quad \Rightarrow \quad a_n^{-1} a_n I = I \in \mathcal{E};$$

each of these implications is justified because $\mathcal{E}$ is a two-sided ideal of $\mathcal{L}(V)$. It follows that $SI = S \in \mathcal{E}$ for any $S \in \mathcal{L}(V)$ and we may conclude that $\mathcal{E} = \mathcal{L}(V)$.



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## 3.B. Null Spaces and Ranges

**Exercise 3.B.1.** Give an example of a linear map $T$ with $\dim \operatorname{null} T = 3$ and $\dim \operatorname{range} T = 2$.

**Solution.** Let $T : \mathbf{R}^5 \to \mathbf{R}^2$ be given by

$$T(x_1, x_2, x_3, x_4, x_5) = (x_1, x_2)$$

and observe that

$$\operatorname{null} T = \{(0, 0, x_3, x_4, x_5) \in \mathbf{R}^5 : x_3, x_4, x_5 \in \mathbf{R}\} \quad \text{and} \quad \operatorname{range} T = \mathbf{R}^2.$$

Thus $\dim \operatorname{null} T = 3$ and $\dim \operatorname{range} T = 2$.

**Exercise 3.B.2.** Suppose $S, T \in \mathcal{L}(V)$ are such that $\operatorname{range} S \subseteq \operatorname{null} T$. Prove that $(ST)^2 = 0$.

**Solution.** For any $v \in V$ we have $S(Tv) \in \operatorname{range} S \subseteq \operatorname{null} T$, so that $T(S(Tv)) = 0$. It follows that

$$(ST)^2(v) = S(T(S(Tv))) = S(0) = 0.$$

Thus $(ST)^2 = 0$.

**Exercise 3.B.3.** Suppose $v_1, ..., v_m$ is a list of vectors in $V$. Define $T \in \mathcal{L}(\mathbf{F}^m, V)$ by

$$T(z_1, ..., z_m) = z_1 v_1 + \cdots + z_m v_m.$$

(a) What property of $T$ corresponds to $v_1, ..., v_m$ spanning $V$?

(b) What property of $T$ corresponds to the list $v_1, ..., v_m$ being linearly independent?

**Solution.**

(a) The surjectivity of $T$ corresponds to $v_1, ..., v_m$ spanning $V$. Indeed, observe that $T$ is surjective if and only if for every $v \in V$ there exists $(z_1, ..., z_m) \in \mathbf{F}^m$ such that $T(z_1, ..., z_m) = z_1 v_1 + \cdots + z_m v_m = v$, which is the case if and only if $V = \operatorname{span}(v_1, ..., v_m)$.

(b) The injectivity of $T$ corresponds to $v_1, ..., v_m$ being linearly independent. By 3.15, $T$ is injective if and only if $\operatorname{null} T = \{0\}$, i.e. if and only if the only choice of $(z_1, ..., z_m) \in \mathbf{F}^m$ which gives $z_1 v_1 + \cdots + z_m v_m = 0$ is $(0, ..., 0)$; this is the definition of linear independence.



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**Exercise 3.B.4.** Show that $\{T \in \mathcal{L}(\mathbf{R}^5, \mathbf{R}^4) : \dim \operatorname{null} T > 2\}$ is not a subspace of $\mathcal{L}(\mathbf{R}^5, \mathbf{R}^4)$.

**Solution.** Let $W = \{T \in \mathcal{L}(\mathbf{R}^5, \mathbf{R}^4) : \dim \operatorname{null} T > 2\}$. Define $S, T \in \mathcal{L}(\mathbf{R}^5, \mathbf{R}^4)$ by

$$S(x_1, x_2, x_3, x_4, x_5) = (x_1, x_2, 0, 0) \quad \text{and} \quad T(x_1, x_2, x_3, x_4, x_5) = (0, 0, x_3, x_4)$$

and observe that

$$\operatorname{null} S = \{(0, 0, x_3, x_4, x_5) \in \mathbf{R}^5 : x_3, x_4, x_5 \in \mathbf{R}\}$$
$$\text{and} \quad \operatorname{null} T = \{(x_1, x_2, 0, 0, x_5) \in \mathbf{R}^5 : x_1, x_2, x_5 \in \mathbf{R}\},$$

so that $\dim \operatorname{null} S = \dim \operatorname{null} T = 3$. Thus $S, T \in W$. However, note that

$$(S + T)(x_1, x_2, x_3, x_4, x_5) = (x_1, x_2, x_3, x_4)$$
$$\Rightarrow \quad \operatorname{null}(S + T) = \{(0, 0, 0, 0, x_5) \in \mathbf{R}^5 : x_5 \in \mathbf{R}\} \quad \Rightarrow \quad \dim \operatorname{null}(S + T) = 1.$$

Thus $S + T \notin W$. It follows that $W$ is not closed under addition and hence that $W$ is not a subspace of $\mathcal{L}(\mathbf{R}^5, \mathbf{R}^4)$.

**Exercise 3.B.5.** Give an example of $T \in \mathcal{L}(\mathbf{R}^4)$ such that $\operatorname{range} T = \operatorname{null} T$.

**Solution.** Let $T \in \mathcal{L}(\mathbf{R}^4)$ be given by

$$T(x_1, x_2, x_3, x_4) = (x_3, x_4, 0, 0),$$

which satisfies $\operatorname{range} T = \operatorname{null} T = \{(a, b, 0, 0) \in \mathbf{R}^4 : a, b \in \mathbf{R}\}$.

**Exercise 3.B.6.** Prove that there does not exist $T \in \mathcal{L}(\mathbf{R}^5)$ such that $\operatorname{range} T = \operatorname{null} T$.

**Solution.** If $T \in \mathcal{L}(V)$ for some finite-dimensional vector space $V$ and $\operatorname{range} T = \operatorname{null} T$, then the fundamental theorem of linear maps (3.21) implies that

$$\dim V = \dim \operatorname{null} T + \dim \operatorname{range} T = 2 \dim \operatorname{null} T.$$

Thus the dimension of $V$ must be a non-negative even integer, which 5 is not.

**Exercise 3.B.7.** Suppose $V$ and $W$ are finite-dimensional with $2 \leq \dim V \leq \dim W$. Show that $\{T \in \mathcal{L}(V, W) : T \text{ is not injective}\}$ is not a subspace of $\mathcal{L}(V, W)$.

**Solution.** Let $X = \{T \in \mathcal{L}(V, W) : T \text{ is not injective}\}$ and note that by 3.15 we have

$$X = \{T \in \mathcal{L}(V, W) : \operatorname{null} T \neq \{0\}\}.$$

Let $v_1, ..., v_m$ be a basis of $V$ and let $w_1, ..., w_n$ be a basis of $W$; by assumption we have $2 \leq m \leq n$. Define $S, T \in \mathcal{L}(V, W)$ by



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$$Sv_k = \begin{cases} 0 & \text{if } k = 1, \\ w_2 & \text{if } k = 2, \\ \frac{1}{2}w_k & \text{otherwise,} \end{cases} \qquad Tv_k = \begin{cases} w_1 & \text{if } k = 1, \\ 0 & \text{if } k = 2, \\ \frac{1}{2}w_k & \text{otherwise.} \end{cases}$$

Notice that $S, T \in X$ since $v_1, v_2$ are non-zero, $v_1 \in \operatorname{null} S$, and $v_2 \in \operatorname{null} T$. Notice further that

$$(S+T)(v_k) = w_k \text{ for all } k \in \{1, ..., m\} \quad \Rightarrow \quad \operatorname{range}(S+T) = \operatorname{span}(w_1, ..., w_m).$$

The linear independence of the list $w_1, ..., w_n$ then implies that $w_1, ..., w_m$ is a basis of $\operatorname{range}(S+T)$ and thus $\dim \operatorname{range}(S+T) = \dim \operatorname{span}(w_1, ..., w_m) = m$. Because $\dim V = m$, it follows from the fundamental theorem of linear maps (3.21) that $\dim \operatorname{null}(S+T) = 0$, hence $\operatorname{null}(S+T) = \{0\}$, hence $S+T \notin X$. We may conclude that $X$ is not closed under addition and hence is not a subspace of $\mathcal{L}(V, W)$.

**Exercise 3.B.8.** Suppose $V$ and $W$ are finite-dimensional with $\dim V \geq \dim W \geq 2$. Show that $\{T \in \mathcal{L}(V, W) : T \text{ is not surjective}\}$ is not a subspace of $\mathcal{L}(V, W)$.

**Solution.** Let $X = \{T \in \mathcal{L}(V, W) : T \text{ is not surjective}\}$, let $v_1, ..., v_m$ be a basis of $V$, and let $w_1, ..., w_n$ be a basis of $W$; by assumption we have $m \geq n \geq 2$. Define $S, T \in \mathcal{L}(V, W)$ by

$$Sv_k = \begin{cases} 0 & \text{if } k = 1 \text{ or } k > n, \\ w_2 & \text{if } k = 2, \\ \frac{1}{2}w_k & \text{otherwise,} \end{cases} \qquad Tv_k = \begin{cases} w_1 & \text{if } k = 1, \\ 0 & \text{if } k = 2 \text{ or } k > n, \\ \frac{1}{2}w_k & \text{otherwise.} \end{cases}$$

It is straightforward to verify that $\operatorname{range} S = \operatorname{span}(w_2, ..., w_n)$. The linear independence of the list $w_1, ..., w_n$ then implies that $w_1 \notin \operatorname{range} S$ and thus $S$ is not surjective, i.e. $S \in X$. Similarly, we find that $w_2 \notin \operatorname{range} T$ and thus $T \in X$. Now observe that

$$(S+T)(v_k) = \begin{cases} w_k & \text{if } k \leq n, \\ 0 & \text{if } k > n, \end{cases} \quad \Rightarrow \quad \operatorname{range}(S+T) = \operatorname{span}(w_1, ..., w_n) = W.$$

Thus $S+T$ is surjective, i.e. $S+T \notin X$. It follows that $X$ is not closed under addition and hence is not a subspace of $\mathcal{L}(V, W)$.

**Exercise 3.B.9.** Suppose $T \in \mathcal{L}(V, W)$ is injective and $v_1, ..., v_n$ is linearly independent in $V$. Prove that $Tv_1, ..., Tv_n$ is linearly independent in $W$.

**Solution.** Suppose we have scalars $a_1, ..., a_n$ such that $a_1 Tv_1 + \cdots + a_n Tv_n = 0$. By linearity, this is equivalent to $T(a_1 v_1 + \cdots + a_n v_n) = 0$, i.e. $a_1 v_1 + \cdots + a_n v_n \in \operatorname{null} T$. Because $T$ is injective, it follows from 3.15 that $a_1 v_1 + \cdots + a_n v_n = 0$. The linear independence of $v_1, ..., v_n$ then implies that $a_1 = \cdots = a_n = 0$. Thus $Tv_1, ..., Tv_n$ is linearly independent.



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**Exercise 3.B.10.** Suppose $v_1, ..., v_n$ spans $V$ and $T \in \mathcal{L}(V, W)$. Show that $Tv_1, ..., Tv_n$ spans $\operatorname{range} T$.

**Solution.** Let $w \in \operatorname{range} T$ be given, so that $w = Tv$ for some $v \in V$. Since $v_1, ..., v_n$ spans $V$, there are scalars $a_1, ..., a_n$ such that $v = a_1 + \cdots + a_n v_n$. It follows from the linearity of $T$ that

$$w = Tv = T(a_1 v_1 + \cdots + a_n v_n) = a_1 Tv_1 + \cdots + a_n Tv_n \in \operatorname{span}(Tv_1, ..., Tv_n).$$

Thus $\operatorname{range} T \subseteq \operatorname{span}(Tv_1, ..., Tv_n)$. Now let $a_1 Tv_1 + \cdots + a_n Tv_n \in \operatorname{span}(Tv_1, ..., Tv_n)$ be given. The linearity of $T$ gives us

$$a_1 Tv_1 + \cdots + a_n Tv_n = T(a_1 v_1 + \cdots + a_n v_n) \in \operatorname{range} T.$$

Thus $\operatorname{span}(Tv_1, ..., Tv_n) \subseteq \operatorname{range} T$ and we may conclude that $\operatorname{range} T = \operatorname{span}(Tv_1, ..., Tv_n)$.

**Exercise 3.B.11.** Suppose that $V$ is finite-dimensional and that $T \in \mathcal{L}(V, W)$. Prove that there exists a subspace $U$ of $V$ such that

$$U \cap \operatorname{null} T = \{0\} \quad \text{and} \quad \operatorname{range} T = \{Tu : u \in U\}.$$

**Solution.** By 2.33 there is a subspace $U$ of $V$ such that $V = U \oplus \operatorname{null} T$ and 1.46 then gives us $U \cap \operatorname{null} T = \{0\}$. Suppose that $w \in \operatorname{range} T$, so that $w = Tv$ for some $v \in V$. Since $V = U \oplus \operatorname{null} T$, there are vectors $u \in U$ and $x \in \operatorname{null} T$ such that $v = u + x$ and it follows that

$$w = Tv = T(u + x) = Tu + Tx = Tu.$$

Thus $\operatorname{range} T \subseteq \{Tu : u \in U\}$; the reverse inclusion $\{Tu : u \in U\} \subseteq \operatorname{range} T$ is clear.

**Exercise 3.B.12.** Suppose $T$ is a linear map from $\mathbf{F}^4$ to $\mathbf{F}^2$ such that

$$\operatorname{null} T = \{(x_1, x_2, x_3, x_4) \in \mathbf{F}^4 : x_1 = 5x_2 \text{ and } x_3 = 7x_4\}.$$

Prove that $T$ is surjective.

**Solution.** It is straightforward to verify that $(5, 1, 0, 0), (0, 0, 7, 1)$ is a basis of $\operatorname{null} T$, so that $\dim \operatorname{null} T = 2$. The fundamental theorem of linear maps (3.21) then gives

$$\dim \operatorname{range} T = \dim \mathbf{F}^4 - \dim \operatorname{null} T = 2.$$

Thus $\dim \operatorname{range} T = 2 = \dim \mathbf{F}^2$ and it follows from 2.39 that $\operatorname{range} T = \mathbf{F}^2$, i.e. $T$ is surjective.

**Exercise 3.B.13.** Suppose $U$ is a three-dimensional subspace of $\mathbf{R}^8$ and that $T$ is a linear map from $\mathbf{R}^8$ to $\mathbf{R}^5$ such that $\operatorname{null} T = U$. Prove that $T$ is surjective.



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**Solution.** The fundamental theorem of linear maps (3.21) gives us

$$\dim \operatorname{range} T = \dim \mathbf{R}^8 - \dim \operatorname{null} T = 8 - \dim U = 5.$$

Thus $\dim \operatorname{range} T = 5 = \dim \mathbf{R}^5$ and it follows from (2.39) that $\operatorname{range} T = \mathbf{R}^5$, i.e. $T$ is surjective.

**Exercise 3.B.14.** Prove that there does not exist a linear map from $\mathbf{F}^5$ to $\mathbf{F}^2$ whose null space equals $\{(x_1, x_2, x_3, x_4, x_5) \in \mathbf{F}^5 : x_1 = 3x_2 \text{ and } x_3 = x_4 = x_5\}$.

**Solution.** Let $U = \{(x_1, x_2, x_3, x_4, x_5) \in \mathbf{F}^5 : x_1 = 3x_2 \text{ and } x_3 = x_4 = x_5\}$. It is straightforward to verify that $(3, 1, 0, 0, 0), (0, 0, 1, 1, 1)$ is a basis of $U$, so that $\dim U = 2$. For any $T \in \mathcal{L}(\mathbf{F}^5, \mathbf{F}^2)$, the fundamental theorem of linear maps (3.21) implies that

$$\dim \operatorname{null} T = \dim \mathbf{F}^5 - \dim \operatorname{range} T \geq 5 - \dim \mathbf{F}^2 = 3.$$

It follows that $U$ cannot be the null space of $T$.

**Exercise 3.B.15.** Suppose there exists a linear map on $V$ whose null space and range are both finite-dimensional. Prove that $V$ is finite-dimensional.

**Solution.** Let $T$ be the linear map in question, let $Tv_1, ..., Tv_m$ be a basis of $\operatorname{range} T$, and let $w_1, ..., w_n$ be a basis of $\operatorname{null} T$. For any $v \in V$ there are scalars $a_1, ..., a_m$ such that

$$Tv = a_1 Tv_1 + \cdots + a_m Tv_m = T(a_1 v_1 + \cdots + a_m v_m)$$
$$\Rightarrow \quad T(v - (a_1 v_1 + \cdots + a_m v_m)) = 0 \quad \Rightarrow \quad v - (a_1 v_1 + \cdots + a_m v_m) \in \operatorname{null} T.$$

It follows that there are scalars $b_1, ..., b_n$ such that

$$v - (a_1 v_1 + \cdots + a_m v_m) = b_1 w_1 + \cdots + b_n w_n$$
$$\Rightarrow \quad v = a_1 v_1 + \cdots + a_m v_m + b_1 w_1 + \cdots + b_n w_n.$$

Thus the list $v_1, ..., v_m, w_1, ..., w_n$ spans $V$. We may conclude that $V$ is finite-dimensional.

**Exercise 3.B.16.** Suppose $V$ and $W$ are finite-dimensional. Prove that there exists an injective linear map from $V$ to $W$ if and only if $\dim V \leq \dim W$.

**Solution.** If $\dim V > \dim W$ then 3.22 guarantees that no linear map from $V$ to $W$ is injective. Suppose therefore that $\dim V = m \leq n = \dim W$, let $v_1, ..., v_m$ be a basis of $V$, and let $w_1, ..., w_n$ be a basis of $W$. Define $T \in \mathcal{L}(V, W)$ by $Tv_k = w_k$. The linear independence of the list $w_1, ..., w_m$ and Exercise 3.B.10 imply that $w_1, ..., w_m$ is a basis of $\operatorname{range} T$, so that $\dim \operatorname{range} T = m = \dim V$. It then follows from the fundamental theorem of linear maps (3.21) that $\dim \operatorname{null} T = 0$, i.e. $T$ is injective.



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**Exercise 3.B.17.** Suppose $V$ and $W$ are both finite-dimensional. Prove that there exists a surjective linear map from $V$ onto $W$ if and only if $\dim V \geq \dim W$.

**Solution.** If $\dim V < \dim W$ then 3.24 guarantees that no linear map from $V$ to $W$ is surjective. Suppose therefore that $\dim V = m \geq n = \dim W$, let $v_1, ..., v_m$ be a basis of $V$, and let $w_1, ..., w_n$ be a basis of $W$. Define $T \in \mathcal{L}(V, W)$ by

$$Tv_k = \begin{cases} w_k & \text{if } k \leq n, \\ 0 & \text{otherwise.} \end{cases}$$

It follows from Exercise 3.B.10 that

$$\operatorname{range} T = \operatorname{span}(Tv_1, ..., Tv_m) = \operatorname{span}(w_1, ..., w_n) = W.$$

Thus $T$ is surjective.

**Exercise 3.B.18.** Suppose $V$ and $W$ are finite-dimensional and that $U$ is a subspace of $V$. Prove that there exists $T \in \mathcal{L}(V, W)$ such that $\operatorname{null} T = U$ if and only if $\dim U \geq \dim V - \dim W$.

**Solution.** If there exists such a $T$ then the fundamental theorem of linear maps (3.21) and 2.37 give us

$$\dim U = \dim \operatorname{null} T = \dim V - \dim \operatorname{range} T \geq \dim V - \dim W.$$

Conversely, suppose that $\dim U \geq \dim V - \dim W$. Let $u_1, ..., u_m$ be a basis of $U$, extend this to a basis $u_1, ..., u_m, v_1, ..., v_n$ of $V$, and let $X = \operatorname{span}(v_1, ..., v_n)$, so that $V = U \oplus X$ and $\dim V = \dim U + \dim X$. Combining this with our hypothesis $\dim U \geq \dim V - \dim W$ gives us $\dim X \leq \dim W$ and so we may invoke Exercise 3.B.16 to obtain an injective linear map $S : X \to W$. Extend this to a linear map $T : V \to W$ satisfying $T(u + x) = Sx$ as in Exercise 3.A.13 ($X$ is playing the role of $U$ here). Now observe that

$$T(u + x) = 0 \quad \Leftrightarrow \quad Sx = 0 \quad \Leftrightarrow \quad x = 0 \quad \Leftrightarrow \quad u + x \in U,$$

where we have used the injectivity of $S$ for the second equivalence. It follows that $\operatorname{null} T = U$.

**Exercise 3.B.19.** Suppose $W$ is finite-dimensional and $T \in \mathcal{L}(V, W)$. Prove that $T$ is injective if and only if there exists $S \in \mathcal{L}(W, V)$ such that $ST$ is the identity operator on $V$.

**Solution.** Suppose there exists such an $S$ and let $v \in \operatorname{null} T$ be given. Observe that

$$v = (ST)(v) = S(0) = 0.$$

Thus $\operatorname{null} T = \{0\}$, which implies that $T$ is injective (by 3.15).



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Now suppose that $T$ is injective. Let $Tv_1, ..., Tv_m$ be a basis of $\operatorname{range} T$ and extend this to a basis $Tv_1, ..., Tv_m, w_1, ..., w_n$ of $W$. Define $S \in \mathcal{L}(W, V)$ by $S(Tv_k) = v_k$ and $Sw_k = 0$. For any $v \in V$ we have $Tv = a_1 Tv_1 + \cdots + a_m Tv_m$ for some scalars $a_1, ..., a_m$, which implies

$$T(v - (a_1 v_1 + \cdots + a_m v_m)) = 0.$$

Because $T$ is injective it must then be the case that $v = a_1 v_1 + \cdots + a_m v_m$. Now observe that

$$S(Tv) = a_1 S(Tv_1) + \cdots + a_m S(Tv_m) = a_1 v_1 + \cdots + a_m v_m = v.$$

Thus $ST$ is the identity map on $V$.

**Exercise 3.B.20.** Suppose $W$ is finite-dimensional and $T \in \mathcal{L}(V, W)$. Prove that $T$ is surjective if and only if there exists $S \in \mathcal{L}(W, V)$ such that $TS$ is the identity operator on $W$.

**Solution.** Suppose there exists such a map $S$ and let $w \in W$ be given. Observe that

$$w = T(Sw) \in \operatorname{range} T.$$

It follows that $W = \operatorname{range} T$, i.e. $T$ is surjective.

Now suppose that $T$ is surjective, i.e. $W = \operatorname{range} T$. Let $Tv_1, ..., Tv_m$ be a basis of $\operatorname{range} T = W$ and define $S \in \mathcal{L}(W, V)$ by $S(Tv_k) = v_k$ for $k \in \{1, ..., m\}$. For any $w = a_1 Tv_1 + \cdots + a_m Tv_m \in W$, observe that

$$Sw = a_1 S(Tv_1) + \cdots + a_m S(Tv_m) = a_1 v_1 + \cdots + a_m v_m$$
$$\Rightarrow \quad T(Sw) = a_1 Tv_1 + \cdots + a_m Tv_m = w.$$

Thus $TS$ is the identity map on $W$.

**Exercise 3.B.21.** Suppose $V$ is finite-dimensional, $T \in \mathcal{L}(V, W)$, and $U$ is a subspace of $W$. Prove that $\{v \in V : Tv \in U\}$ is a subspace of $V$ and

$$\dim\{v \in V : Tv \in U\} = \dim \operatorname{null} T + \dim(U \cap \operatorname{range} T).$$

**Solution.** Let $X = \{v \in V : Tv \in U\}$ and note that $0 \in X$ since $T(0) = 0 \in U$. Suppose $v_1, v_2 \in X$, so that $Tv_1, Tv_2 \in U$, and $\lambda \in \mathbf{F}$. Because $T$ is linear and $U$ is a subspace, we then have

$$Tv_1, Tv_2 \in U \quad \Rightarrow \quad T(v_1) + T(v_2) = T(v_1 + v_2) \in U \quad \text{and} \quad \lambda Tv_1 = T(\lambda v_1) \in U.$$

Thus $v_1 + v_2$ and $\lambda v_1$ belong to $X$. It follows from 1.34 that $X$ is a subspace of $V$.

Let $S$ be the restriction of $T$ to $X$, i.e. $S : X \to W$ is given by $Sv = Tv$, and notice that $S$ is linear because $T$ is linear. Notice further that

$$v \in \operatorname{null} T \quad \Rightarrow \quad Tv = 0 \in U \quad \Rightarrow \quad v \in X \quad \Rightarrow \quad Sv = Tv = 0 \quad \Rightarrow \quad v \in \operatorname{null} S,$$

$$v \in \operatorname{null} S \quad \Rightarrow \quad Sv = Tv = 0 \quad \Rightarrow \quad v \in \operatorname{null} T.$$



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Thus $\operatorname{null} S = \operatorname{null} T$. Similarly,

$$Tv \in U \cap \operatorname{range} T \text{ for some } v \in V \quad \Rightarrow \quad v \in X \quad \Rightarrow \quad Tv = Sv \in \operatorname{range} S,$$

$$Sv \in \operatorname{range} S \text{ for some } v \in X \quad \Rightarrow \quad Sv = Tv \in U \cap \operatorname{range} T.$$

Thus $\operatorname{range} S = U \cap \operatorname{range} T$. The fundamental theorem of linear maps (3.21) now implies that

$$\dim X = \dim \operatorname{null} S + \dim \operatorname{range} S = \dim \operatorname{null} T + \dim(U \cap \operatorname{range} T).$$

**Exercise 3.B.22.** Suppose $U$ and $V$ are finite-dimensional vector spaces and $S \in \mathcal{L}(V, W)$ and $T \in \mathcal{L}(U, V)$. Prove that

$$\dim \operatorname{null} ST \leq \dim \operatorname{null} S + \dim \operatorname{null} T.$$

**Solution.** For any $u \in U$ observe that

$$u \in \operatorname{null} ST \quad \Leftrightarrow \quad S(Tu) = 0 \quad \Leftrightarrow \quad Tu \in \operatorname{null} S.$$

Thus $\operatorname{null} ST = \{u \in U : Tu \in \operatorname{null} S\}$. It follows from Exercise 3.B.21 that

$$\dim \operatorname{null} ST = \dim \operatorname{null} T + \dim(\operatorname{null} S \cap \operatorname{range} T) \leq \dim \operatorname{null} T + \dim \operatorname{null} S.$$

**Exercise 3.B.23.** Suppose $U$ and $V$ are finite-dimensional vector spaces and $S \in \mathcal{L}(V, W)$ and $T \in \mathcal{L}(U, V)$. Prove that

$$\dim \operatorname{range} ST \leq \min\{\dim \operatorname{range} S, \dim \operatorname{range} T\}.$$

**Solution.** Certainly $\operatorname{range} ST \subseteq \operatorname{range} S$ and thus $\dim \operatorname{range} ST \leq \dim \operatorname{range} S$. Let $R \in \mathcal{L}(\operatorname{range} T, W)$ be the restriction of $S$ to $\operatorname{range} T$, so that $\operatorname{range} R = \operatorname{range} ST$. The fundamental theorem of linear maps (3.21) then implies that

$$\dim \operatorname{range} ST = \dim \operatorname{range} R = \dim \operatorname{range} T - \dim \operatorname{null} R \leq \dim \operatorname{range} T.$$

Thus $\dim \operatorname{range} ST \leq \dim \operatorname{range} S$ and $\dim \operatorname{range} ST \leq \dim \operatorname{range} T$; it follows that

$$\dim \operatorname{range} ST \leq \min\{\dim \operatorname{range} S, \dim \operatorname{range} T\}.$$

**Exercise 3.B.24.**

(a) Suppose $\dim V = 5$ and $S, T \in \mathcal{L}(V)$ are such that $ST = 0$. Prove that $\dim \operatorname{range} TS \leq 2$.

(b) Give an example of $S, T \in \mathcal{L}(\mathbf{F}^5)$ with $ST = 0$ and $\dim \operatorname{range} TS = 2$.

**Solution.**

(a) The fundamental theorem of linear maps (3.21) and Exercise 3.B.23 give us

$$\dim \operatorname{null} S = 5 - \dim \operatorname{range} S \leq 5 - \dim \operatorname{range} TS. \tag{1}$$



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Note that $ST = 0$ implies $\operatorname{range} T \subseteq \operatorname{null} S$, so that $\dim \operatorname{range} T \leq \dim \operatorname{null} S$. It then follows from Exercise 3.B.23 and equation (1) that

$$\dim \operatorname{range} TS \leq \dim \operatorname{range} T \leq \dim \operatorname{null} S \leq 5 - \dim \operatorname{range} TS$$
$$\Rightarrow \quad 2 \dim \operatorname{range} TS \leq 5 \quad \Rightarrow \quad \dim \operatorname{range} TS \leq \frac{5}{2}.$$

Thus $\dim \operatorname{range} TS \leq 2$, since the dimension of a vector space must be an integer.

(b) Let $S, T \in \mathcal{L}(\mathbf{F}^5)$ be given by

$$S(x_1, x_2, x_3, x_4, x_5) = (0, 0, 0, x_4, x_5) \quad \text{and} \quad T(x_1, x_2, x_3, x_4, x_5) = (x_3, x_4, x_5, 0, 0).$$

Observe that

$$(ST)(x_1, x_2, x_3, x_4, x_5) = (0, 0, 0, 0, 0)$$
$$\text{and} \quad (TS)(x_1, x_2, x_3, x_4, x_5) = (0, x_4, x_5, 0, 0).$$

Thus $ST = 0$ and $\dim \operatorname{range} TS = 2$.

**Exercise 3.B.25.** Suppose that $W$ is finite-dimensional and $S, T \in \mathcal{L}(V, W)$. Prove that $\operatorname{null} S \subseteq \operatorname{null} T$ if and only if there exists $E \in \mathcal{L}(W)$ such that $T = ES$.

**Solution.** If there exists such a map $E$ then for any $v \in \operatorname{null} S$ we have

$$Tv = E(Sv) = E(0) = 0 \quad \Rightarrow \quad v \in \operatorname{null} T.$$

Thus $\operatorname{null} S \subseteq \operatorname{null} T$.

Now suppose that $\operatorname{null} S \subseteq \operatorname{null} T$. Let $Sv_1, ..., Sv_m$ be a basis of $\operatorname{range} S$ and extend this to a basis $Sv_1, ..., Sv_m, w_1, ..., w_n$ of $W$. Define $E \in \mathcal{L}(W)$ by $E(Sv_k) = Tv_k$ and $Ew_k = 0$. For any $v \in V$ we have $Sv = a_1 Sv_1 + \cdots + a_m Sv_m$ for some scalars $a_1, ..., a_m$, which implies

$$S(v - (a_1 v_1 + \cdots + a_m v_m)) = 0 \quad \Rightarrow \quad v - (a_1 v_1 + \cdots + a_m v_m) \in \operatorname{null} S$$
$$\Rightarrow \quad v - (a_1 v_1 + \cdots + a_m v_m) \in \operatorname{null} T \quad \Rightarrow \quad T(v - (a_1 v_1 + \cdots + a_m v_m)) = 0.$$

Thus $Tv = a_1 Tv_1 + \cdots + a_m Tv_m$. It follows that

$$E(Sv) = a_1 E(Sv_1) + \cdots + a_m E(Sv_m) = a_1 Tv_1 + \cdots + a_m Tv_m = Tv.$$

Hence $T = ES$.

**Exercise 3.B.26.** Suppose that $V$ is finite-dimensional and $S, T \in \mathcal{L}(V, W)$. Prove that $\operatorname{range} S \subseteq \operatorname{range} T$ if and only if there exists $E \in \mathcal{L}(V)$ such that $S = TE$.

**Solution.** If there exists such a map $E$ then for any $Sv \in \operatorname{range} S$ we have $Sv = T(Ev) \in \operatorname{range} T$ also. Thus $\operatorname{range} S \subseteq \operatorname{range} T$.

Now suppose that $\operatorname{range} S \subseteq \operatorname{range} T$ and let $v_1, ..., v_m$ be a basis of $V$. Since $\operatorname{range} S \subseteq \operatorname{range} T$, for each $k \in \{1, ..., m\}$ we have $Sv_k = Tu_k$ for some $u_k \in V$. Define $E \in \mathcal{L}(V)$ by $Ev_k = u_k$ and observe that for any $v = a_1 v_1 + \cdots + a_m v_m \in V$ we have



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$$(TE)(v) = (TE)(a_1v_1 + \cdots + a_mv_m) = a_1T(E(v_1)) + \cdots + a_mT(E(v_m))$$
$$= a_1Tu_1 + \cdots + a_mTu_m = a_1Sv_1 + \cdots + a_mSv_m = S(a_1v_1 + \cdots + a_mv_m) = Sv.$$

Thus $S = TE$.

**Exercise 3.B.27.** Suppose $P \in \mathcal{L}(V)$ and $P^2 = P$. Prove that $V = \operatorname{null} P \oplus \operatorname{range} P$.

**Solution.** For any $v \in V$ note that

$$P^2v = Pv \quad \Rightarrow \quad P(Pv) - Pv = 0 \quad \Rightarrow \quad P(Pv - v) = 0$$
$$\Rightarrow \quad Pv - v = u \in \operatorname{null} P \quad \Rightarrow \quad v = u + Pv \in \operatorname{null} P + \operatorname{range} P.$$

Thus $V = \operatorname{null} P + \operatorname{range} P$. Suppose that $v = Pu \in \operatorname{null} P \cap \operatorname{range} P$. It follows that

$$0 = Pv = P^2u = Pu = v.$$

Thus $\operatorname{null} P \cap \operatorname{range} P = \{0\}$ and it follows from 1.46 that the sum $V = \operatorname{null} P \oplus \operatorname{range} P$ is direct.

**Exercise 3.B.28.** Suppose $D \in \mathcal{L}(\mathcal{P}(\mathbf{R}))$ is such that $\deg Dp = (\deg p) - 1$ for every non-constant polynomial $p \in \mathcal{P}(\mathbf{R})$. Prove that $D$ is surjective.

*The notation $D$ is used above to remind you of the differentiation map that sends a polynomial $p$ to $p'$.*

**Solution.** First, we will recursively define a sequence of polynomials $(p_k)_{k=0}^{\infty}$ such that $Dp_k = x^k$. By assumption $\deg Dx = (\deg x) - 1 = 0$, so that $Dx = b$ for some non-zero $b \in \mathbf{F}$. Define $p_0 = b^{-1}x$ and, using the linearity of $D$, observe that

$$Dp_0 = D(b^{-1}x) = b^{-1}Dx = b^{-1}b = 1.$$

Now suppose that we have defined polynomials $p_0, ..., p_n$ such that $Dp_k = x^k$ for each $k \in \{0, ..., n\}$. By assumption $D(x^{n+2})$ must have degree $n + 1$, i.e. must be of the form

$$D(x^{n+2}) = b_{n+1}x^{n+1} + b_nx^n + \cdots + b_1x + b_0$$

where $b_{n+1} \neq 0$. Because $Dp_k = x^k$ and $D$ is linear, it follows that

$$b_{n+1}^{-1}D(x^{n+2}) = x^{n+1} + b_{n+1}^{-1}(b_nDp_n + \cdots + b_1Dp_1 + b_0Dp_0)$$
$$\Rightarrow \quad x^{n+1} = D(b_{n+1}^{-1}(x^{n+2} - b_np_n - \cdots - b_1p_1 - b_0p_0)).$$

Thus, defining $p_{n+1} = b_{n+1}^{-1}(x^{n+2} - b_np_n - \cdots - b_1p_1 - b_0p_0)$, we have $Dp_{n+1} = x^{n+1}$. We now obtain the desired sequence $(p_k)_{k=0}^{\infty}$ of polynomials by recursion.

We can now show that $D$ is surjective. Let $p = \sum_{k=0}^{\deg p} a_kx^k \in \mathcal{P}(\mathbf{R})$ be given. Because $D$ is linear, it follows that



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$$D\left(\sum_{k=0}^{\deg p} a_k p_k\right) = \sum_{k=0}^{\deg p} a_k Dp_k = \sum_{k=0}^{\deg p} a_k x^k = p.$$

Thus $D$ is surjective.

**Exercise 3.B.29.** Suppose $p \in \mathcal{P}(\mathbf{R})$. Prove that there exists a polynomial $q \in \mathcal{P}(\mathbf{R})$ such that $5q'' + 3q' = p$.

*This exercise can be done without linear algebra, but it's more fun to do it using linear algebra.*

**Solution.** Define a map $D : \mathcal{P}(\mathbf{R}) \to \mathcal{P}(\mathbf{R})$ by $Dq = 5q'' + 3q'$; it will suffice to show that $D$ is surjective. The linearity of $D$ follows from the linearity of differentiation. Suppose $q \in \mathcal{P}(\mathbf{R})$ is a non-constant polynomial of degree $n \geq 1$, so that $q = \sum_{k=0}^{n} a_k x^k$ with $a_n \neq 0$. Some calculations reveals that

$$Dq = \begin{cases} 3a_n & \text{if } n = 1, \\ 3na_n x^{n-1} + \sum_{k=0}^{n-2} (k+1)\big[3a_{k+1} + 5(k+2)a_{k+2}\big]x^k & \text{if } n \geq 2. \end{cases}$$

In either case, because $a_n \neq 0$, the polynomial $Dq$ has degree $n-1$. Thus $D$ satisfies the hypotheses of Exercise 3.B.26 and hence must be surjective.

**Exercise 3.B.30.** Suppose $\varphi \in \mathcal{L}(V, \mathbf{F})$ and $\varphi \neq 0$. Suppose $u \in V$ is not in $\operatorname{null} \varphi$. Prove that

$$V = \operatorname{null} \varphi \oplus \{au : a \in \mathbf{F}\}.$$

**Solution.** For any $v \in V$, the linearity of $\varphi$ gives us

$$1 = \varphi\left(\frac{1}{\varphi(u)}u\right) \quad \Rightarrow \quad \varphi(v) = \varphi\left(\frac{\varphi(v)}{\varphi(u)}u\right) \quad \Rightarrow \quad \varphi\left(v - \frac{\varphi(v)}{\varphi(u)}u\right) = 0$$

$$\Rightarrow \quad v - \frac{\varphi(v)}{\varphi(u)}u = w \text{ for some } w \in \operatorname{null} \varphi \quad \Rightarrow \quad v = w + \frac{\varphi(v)}{\varphi(u)}u \in \operatorname{null} \varphi + \{au : a \in \mathbf{F}\}.$$

Thus $V = \operatorname{null} \varphi + \{au : a \in \mathbf{F}\}$. Suppose that $v \in \operatorname{null} \varphi \cap \{au : a \in \mathbf{F}\}$, so that $v = au$ for some $a \in \mathbf{F}$. Observe that

$$0 = \varphi(v) = \varphi(au) = a\varphi(u).$$

Since $\varphi(u) \neq 0$, Exercise 1.B.2 implies that $a = 0$ and hence that $v = 0$. Thus

$$\operatorname{null} \varphi \cap \{au : a \in \mathbf{F}\} = \{0\}$$

and it follows from 1.46 that the sum $V = \operatorname{null} \varphi \oplus \{au : a \in \mathbf{F}\}$ is direct.



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**Exercise 3.B.31.** Suppose $V$ is finite-dimensional, $X$ is a subspace of $V$, and $Y$ is a finite-dimensional subspace of $W$. Prove that there exists $T \in \mathcal{L}(V, W)$ such that $\operatorname{null} T = X$ and $\operatorname{range} T = Y$ if and only if $\dim X + \dim Y = \dim V$.

**Solution.** If there exists such a map $T$ then the equality $\dim X + \dim Y = \dim V$ is immediate from the fundamental theorem of linear maps (3.21).

Suppose that $\dim X + \dim Y = \dim V$. Let $x_1, ..., x_\ell$ be a basis of $X$, which we extend to a basis $x_1, ..., x_\ell, v_1, ..., v_m$, and let $y_1, ..., y_n$ be a basis of $Y$. By assumption we have

$$\dim X + \dim Y = \dim V \quad \Leftrightarrow \quad \ell + n = \ell + m \quad \Leftrightarrow \quad n = m,$$

so the map $T \in \mathcal{L}(V, W)$ given by $Tx_k = 0$ and $Tv_k = y_k$ is well-defined (i.e. there are enough $y_k$'s to define this map). Suppose $v = a_1 x_1 + \cdots + a_\ell x_\ell + b_1 v_1 + \cdots + b_m v_m \in V$ is such that $Tv = 0$. Observe that

$$0 = Tv = a_1 Tx_1 + \cdots + a_\ell Tx_\ell + b_1 Tv_1 + \cdots + b_m Tv_m = b_1 y_1 + \cdots + b_m y_m.$$

The linear independence of $y_1, ..., y_n$ then implies that $b_1 = \cdots = b_m = 0$ and thus $v \in X$, so that $\operatorname{null} T \subseteq X$. Certainly $X \subseteq \operatorname{null} T$ and it follows that $\operatorname{null} T = X$. Furthermore, Exercise 3.B.10 implies that

$$\operatorname{range} T = \operatorname{span}(Tx_1, ..., Tx_\ell, Tv_1, ..., Tv_m) = \operatorname{span}(y_1, ..., y_m) = \operatorname{span}(y_1, ..., y_n) = Y,$$

where we have used $m = n$ for the third equality.

**Exercise 3.B.32.** Suppose $V$ is finite-dimensional with $\dim V > 1$. Show that if $\varphi : \mathcal{L}(V) \to \mathbf{F}$ is a linear map such that $\varphi(ST) = \varphi(S)\varphi(T)$ for all $S, T \in \mathcal{L}(V)$, then $\varphi = 0$.

*Hint: The description of the two-sided ideals of $\mathcal{L}(V)$ given by Exercise 17 in Section 3A might be useful.*

**Solution.** First note that by Exercise 3.A.16 there exist $S, T \in \mathcal{L}(V)$ such that $ST - TS \neq 0$. Note further that

$$\varphi(ST - TS) = \varphi(ST) - \varphi(TS) = \varphi(S)\varphi(T) - \varphi(T)\varphi(S) = 0,$$

where we have used that multiplication in $\mathbf{F}$ is commutative. It follows that $ST - TS \in \operatorname{null} \varphi$ and hence that $\operatorname{null} \varphi \neq \{0\}$.

Now suppose that $E \in \operatorname{null} \varphi$ and $T \in \mathcal{L}(V)$. Observe that

$$\varphi(ET) = \varphi(E)\varphi(T) = 0 \cdot \varphi(T) = 0 \quad \text{and} \quad \varphi(TE) = \varphi(T)\varphi(E) = \varphi(T) \cdot 0 = 0.$$

Thus $ET$ and $TE$ also belong to $\operatorname{null} \varphi$, so that $\operatorname{null} \varphi$ is a two-sided ideal of $\mathcal{L}(V)$. As we showed in Exercise 3.A.17, any non-zero two-sided ideal of $\mathcal{L}(V)$ must be $\mathcal{L}(V)$ itself, i.e. $\operatorname{null} \varphi = \mathcal{L}(V)$. It follows that $\varphi = 0$.



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**Exercise 3.B.33.** Suppose that $V$ and $W$ are real vector spaces and $T \in \mathcal{L}(V, W)$. Define $T_{\mathbf{C}} : V_{\mathbf{C}} \to W_{\mathbf{C}}$ by

$$T_{\mathbf{C}}(u + iv) = Tu + iTv$$

for all $u, v \in V$.

(a) Show that $T_{\mathbf{C}}$ is a (complex) linear map from $V_{\mathbf{C}}$ to $W_{\mathbf{C}}$.

(b) Show that $T_{\mathbf{C}}$ is injective if and only if $T$ is injective.

(c) Show that $\operatorname{range} T_{\mathbf{C}} = W_{\mathbf{C}}$ if and only if $\operatorname{range} T = W$.

*See Exercise 8 in Section 1B for the definition of the complexification $V_{\mathbf{C}}$. The linear map $T_{\mathbf{C}}$ is called the **complexification** of the linear map $T$.*

**Solution.**

(a) Let $u_1 + iv_1$ and $u_2 + iv_2 \in V_{\mathbf{C}}$ be given. Using the linearity of $T$, observe that

$$\begin{aligned} T_{\mathbf{C}}((u_1 + iv_1) + (u_2 + iv_2)) &= T_{\mathbf{C}}((u_1 + u_2) + i(v_1 + v_2)) \\ &= T(u_1 + u_2) + iT(v_1 + v_2) \\ &= Tu_1 + Tu_2 + iTv_1 + iTv_2 \\ &= Tu_1 + iTv_1 + Tu_2 + iTv_2 \\ &= T_{\mathbf{C}}(u_1 + iv_1) + T_{\mathbf{C}}(u_2 + iv_2). \end{aligned}$$

Similarly, let $u + iv \in V_{\mathbf{C}}$ and $a + bi \in \mathbf{C}$ be given. The linearity of $T$ gives us

$$\begin{aligned} T_{\mathbf{C}}((a + bi)(u + iv)) &= T_{\mathbf{C}}((au - bv) + i(av + bu)) \\ &= T(au - bv) + iT(av + bu) \\ &= (aTu - bTv) + i(aTv + bTu) \\ &= (a + bi)(Tu + iTv) \\ &= (a + bi)T_{\mathbf{C}}(u + iv). \end{aligned}$$

Thus $T_{\mathbf{C}}$ is a $\mathbf{C}$-linear map from $V_{\mathbf{C}}$ to $W_{\mathbf{C}}$.

(b) Suppose that $T$ is injective and observe that for any $u + iv \in V_{\mathbf{C}}$,

$$\begin{aligned} T_{\mathbf{C}}(u + iv) = 0 \quad &\Leftrightarrow \quad Tu + iTv = 0 \quad \Leftrightarrow \quad Tu = 0 \text{ and } Tv = 0 \\ &\Leftrightarrow \quad u = 0 \text{ and } v = 0 \quad \Leftrightarrow \quad u + iv = 0, \end{aligned}$$

where we have used the injectivity of $T$ for the third equivalence. Thus $\operatorname{null} T_{\mathbf{C}} = \{0\}$, i.e. $T_{\mathbf{C}}$ is injective.

Now suppose that $T_{\mathbf{C}}$ is injective and observe that for any $u \in V$,

$$Tu = 0 \quad \Leftrightarrow \quad T_{\mathbf{C}}(u + i0) = 0 \quad \Leftrightarrow \quad u + i0 = 0 \quad \Leftrightarrow \quad u = 0,$$



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where we have used the injectivity of $T_{\mathbf{C}}$ for the second equivalence. It follows that $\operatorname{null} T = \{0\}$, i.e. $T$ is injective.

(c) Suppose that $\operatorname{range} T = W$ and let $w + ix \in W_{\mathbf{C}}$ be given. There exist $u, v \in V$ such that $Tu = w$ and $Tv = x$. It follows that

$$T_{\mathbf{C}}(u + iv) = Tu + iTv = w + ix.$$

Thus $\operatorname{range} T_{\mathbf{C}} = W_{\mathbf{C}}$.

Now suppose that $\operatorname{range} T_{\mathbf{C}} = W_{\mathbf{C}}$ and let $w \in W$ be given. There exists $u + iv \in V_{\mathbf{C}}$ such that $T_{\mathbf{C}}(u + iv) = w + i0$. It follows that

$$T_{\mathbf{C}}(u + iv) = Tu + iTv = w + i0 \quad \Rightarrow \quad Tu = w.$$

Thus $\operatorname{range} T = W$.



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## 3.C. Matrices

**Exercise 3.C.1.** Suppose $T \in \mathcal{L}(V, W)$. Show that with respect to each choice of bases of $V$ and $W$, the matrix of $T$ has at least $\dim \operatorname{range} T$ nonzero entries.

**Solution.** Let $v_1, ..., v_n$ be a basis of $V$ and $w_1, ..., w_m$ be a basis of $W$, so that the matrix of $T$ with respect to these bases is the $m$-by-$n$ matrix $\mathcal{M}(T)$ whose entries $A_{j,k}$ are defined by

$$Tv_k = A_{1,k}w_1 + \cdots + A_{m,k}w_m.$$

Let $p = \dim \operatorname{null} T$ and $q = \dim \operatorname{range} T$, so that $p + q = n$. Because the list $v_1, ..., v_n$ is linearly independent, at most $p$ of the $v_k$'s can belong to $\operatorname{null} T$. Equivalently, at least $n - p = q$ of the $v_k$'s do not belong to $\operatorname{null} T$. Letting $v_k$ be such a vector, we have

$$Tv_k = A_{1,k}w_1 + \cdots + A_{m,k}w_m \neq 0.$$

This implies that at least one of the scalars $A_{j,k}$ is non-zero, i.e. column $k$ of $\mathcal{M}(T)$ has at least one non-zero entry. Since there are at least $q$ choices of $k$ resulting in a non-zero column $k$ of $\mathcal{M}(T)$, we see that $\mathcal{M}(T)$ has at least $q = \dim \operatorname{range} T$ non-zero entries.

**Exercise 3.C.2.** Suppose $V$ and $W$ are finite-dimensional and $T \in \mathcal{L}(V, W)$. Prove that $\dim \operatorname{range} T = 1$ if and only if there exist a basis of $V$ and a basis of $W$ such that with respect to these bases, all entries of $\mathcal{M}(T)$ equal 1.

**Solution.** Suppose there exists a basis $v_1, ..., v_n$ of $V$ and a basis $w_1, ..., w_m$ of $W$ such that with respect to these bases all entries of $\mathcal{M}(T)$ equal 1. That is,

$$Tv_1 = \cdots = Tv_n = w_1 + \cdots + w_m.$$

It follows from Exercise 3.B.10 that

$$\operatorname{range} T = \operatorname{span}(Tv_1, ..., Tv_n) = \operatorname{span}(w_1 + \cdots + w_m).$$

Because $w_1, ..., w_m$ is linearly independent, we must have $w_1 + \cdots + w_m \neq 0$ and thus $w_1, ..., w_m$ is a basis of $\operatorname{range} T$. It follows that $\dim \operatorname{range} T = 1$.

To prove the converse, let us first prove the following lemmas.



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**Lemma L.1.** If $W$ is finite-dimensional with $\dim W = m$ and $w \in W$ is non-zero, then there exists a basis $w_1, ..., w_m$ of $W$ such that $w = w_1 + \cdots + w_m$.

*Proof.* Note that $w$ being non-zero implies $m \geq 1$. If $m = 1$ then take $w_1 = w$; otherwise, extend the list $w$ to a basis $w, w_1, ..., w_{m-1}$ of $W$ and define

$$w_m = w - w_1 - \cdots - w_{m-1},$$

so that $w = w_1 + \cdots + w_m$. Observe that each vector in the basis $w, w_1, ..., w_{m-1}$ can be expressed as a linear combination of vectors from the list $w_1, ..., w_m$. It follows that $W = \text{span}(w_1 ..., w_m)$ and hence, by 2.42, $w_1, ..., w_m$ is a basis of $W$. □

**Lemma L.2.** If $V$ is finite-dimensional with $\dim V = n$ and $U$ is a subspace of $V$ with $U \neq V$, then there exists a basis $v_1, ..., v_n$ of $V$ such that $v_k \notin U$ for each $k \in \{1, ..., n\}$.

*Proof.* Note that $U \neq V$ implies $n \geq 1$. We will construct the required basis $v_1, ..., v_n$ via the following process.

**Step 1.** Because $U \neq V$, there exists some $v_1 \in V$ such that $v_1 \notin U$, which implies that $v_1 \neq 0$. If $\text{span}(v_1) = V$ then the process stops and $v_1$ is the required basis. Otherwise, move to step 2.

**Step $k$.** Suppose we have chosen linearly independent vectors $v_1, ..., v_{k-1}$, none of which belong to $U$, such that $\text{span}(v_1, ..., v_{k-1}) \neq V$. Observe that

$$\text{span}(v_1, ..., v_{k-1}) \cup U = V \quad \Rightarrow \quad U \subseteq \text{span}(v_1, ..., v_{k-1})$$
$$\Rightarrow \quad \text{span}(v_1, ..., v_{k-1}) = V,$$

where we have used Exercise 1.C.12 and the fact that $\text{span}(v_1, ..., v_{k-1})$ is not contained in $U$ (since $v_1 \notin U$) for the first implication. Given that $\text{span}(v_1, ..., v_{k-1}) \neq V$, it must be the case that $\text{span}(v_1, ..., v_{k-1}) \cup U \neq V$ and thus there exists some $v_k \in V$ such that

$$v_k \notin \text{span}(v_1, ..., v_{k-1}) \quad \text{and} \quad v_k \notin U.$$

It follows from Exercise 2.A.13 that the list $v_1, ..., v_k$ is linearly independent. If $\text{span}(v_1, ..., v_k) = V$ then the process stops and $v_1, ..., v_k$ is the required basis. Otherwise, move to step $k + 1$.

Because $V$ is finite-dimensional, this process must stop after a finite number of steps (indeed, it stops after $n$ steps). □

Returning to the exercise, suppose that $\dim \text{range}\, T = 1$, so that $\text{range}\, T$ has a basis $w$. By Lemma L.1 there is a basis $w_1, ..., w_m$ of $W$ such that $w = w_1 + \cdots + w_m$, and by Lemma L.2 there is a basis $u_1, ..., u_n$ of $V$ such that each $u_k \notin \text{null}\, T$. For any $k \in \{1, ..., n\}$ we have $Tu_k \in \text{range}\, T = \text{span}(w)$ and thus $Tu_k = \lambda_k w$ for some scalar $\lambda_k$; this scalar must be non-



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zero since $Tu_k \neq 0$. Let $v_k = \lambda_k^{-1} u_k$ and observe that, because each $\lambda_k^{-1}$ is non-zero, $v_1, ..., v_n$ is a basis of $V$. It follows that

$$Tv_k = w = w_1 + \cdots + w_m$$

for each $k \in \{1, ..., n\}$. Thus with respect to the bases $v_1, ..., v_n$ and $w_1, ..., w_m$ all entries of $\mathcal{M}(T)$ equal 1.

**Exercise 3.C.3.** Suppose $v_1, ..., v_n$ is a basis of $V$ and $w_1, ..., w_m$ is a basis of $W$.

(a) Show that if $S, T \in \mathcal{L}(V, W)$, then $\mathcal{M}(S + T) = \mathcal{M}(S) + \mathcal{M}(T)$.

(b) Show that if $\lambda \in \mathbf{F}$ and $T \in \mathcal{L}(V, W)$, then $\mathcal{M}(\lambda T) = \lambda \mathcal{M}(T)$.

*This exercise asks you to verify 3.35 and 3.38.*

**Solution.**

(a) Suppose $\mathcal{M}(S)$ has entries $A_{j,k}$ and $\mathcal{M}(T)$ has entries $B_{j,k}$, i.e.

$$Sv_k = A_{1,k} w_1 + \cdots + A_{m,k} w_m \quad \text{and} \quad Tv_k = B_{1,k} w_1 + \cdots + B_{m,k} w_m.$$

It follows that

$$(S + T)(v_k) = Sv_k + Tv_k = \big(A_{1,k} + B_{1,k}\big) w_1 + \cdots + \big(A_{m,k} + B_{m,k}\big) w_m.$$

Thus $\mathcal{M}(S + T)$ has entries $A_{j,k} + B_{j,k}$. That is, $\mathcal{M}(S + T) = \mathcal{M}(S) + \mathcal{M}(T)$.

(b) Suppose $\mathcal{M}(T)$ has entries $A_{j,k}$, i.e.

$$Tv_k = A_{1,k} w_1 + \cdots + A_{m,k} w_m.$$

It follows that

$$(\lambda T)(v_k) = \lambda T v_k = \big(\lambda A_{1,k}\big) w_1 + \cdots + \big(\lambda A_{m,k}\big) w_m.$$

Thus $\mathcal{M}(S + T)$ has entries $\lambda A_{j,k}$. That is, $\mathcal{M}(\lambda T) = \lambda \mathcal{M}(T)$.

**Exercise 3.C.4.** Suppose that $D \in \mathcal{L}(\mathcal{P}_3(\mathbf{R}), \mathcal{P}_2(\mathbf{R}))$ is the differentiation map defined by $Dp = p'$. Find a basis of $\mathcal{P}_3(\mathbf{R})$ and a basis of $\mathcal{P}_2(\mathbf{R})$ such that the matrix of $D$ with respect to these bases is

$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix}.$$

*Compare with Example 3.33. The next exercise generalizes this exercise.*

**Solution.** Take $x^3, x^2, x, 1$ as a basis of $\mathcal{P}_3(\mathbf{R})$ and $3x^2, 2x, 1$ as a basis of $\mathcal{P}_2(\mathbf{R})$ and observe that

$$D(x^3) = 3x^2, \quad D(x^2) = 2x, \quad D(x) = 1, \quad \text{and} \quad D(1) = 0.$$

Thus the matrix of $D$ with respect to these bases is



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$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix}.$$

**Exercise 3.C.5.** Suppose $V$ and $W$ are finite-dimensional and $T \in \mathcal{L}(V, W)$. Prove that there exist a basis of $V$ and a basis of $W$ such that with respect to these bases, all entries of $\mathcal{M}(T)$ are 0 except that the entries in row $k$, column $k$, equal 1 if $1 \leq k \leq \dim \operatorname{range} T$.

**Solution.** Let $v_1, ..., v_m$ be a basis of $\operatorname{null} T$ and extend this to a basis $u_1, ..., u_\ell, v_1, ..., v_m$ of $V$ (note the ordering). As the proof of 2.33 shows, we then have $U \cap \operatorname{null} T = \{0\}$, where $U = \operatorname{span}(u_1, ..., u_\ell)$. The restriction of $T$ to $U$ is a linear map in its own right. Moreover, this restriction is injective:

$$u \in U \text{ and } Tu = 0 \quad \Rightarrow \quad u \in U \cap \operatorname{null} T = \{0\} \quad \Rightarrow \quad u = 0.$$

It then follows from Exercise 3.B.9, Exercise 3.B.10, and Exercise 3.B.11 that the list $Tu_1, ..., Tu_\ell$ is a basis of $\{Tu : u \in U\} = \operatorname{range} T$. Thus $\dim \operatorname{range} T = \ell$. Extend the list $Tu_1, ..., Tu_\ell$ to a basis $Tu_1, ..., Tu_\ell, w_1, ..., w_n$ of $W$ and let $\mathcal{M}(T)$ be the matrix of $T$ with respect to the bases $u_1, ..., u_\ell, v_1, ..., v_m$ and $Tu_1, ..., Tu_\ell, w_1, ..., w_n$. If $\mathcal{M}(T)$ has entries $A_{j,k}$, then notice that $A_{j,k} = 1$ if $j = k$ and $1 \leq k \leq \ell = \dim \operatorname{range} T$, and $A_{j,k} = 0$ otherwise.

**Exercise 3.C.6.** Suppose $v_1, ..., v_m$ is a basis of $V$ and $W$ is finite-dimensional. Suppose $T \in \mathcal{L}(V, W)$. Prove that there exists a basis $w_1, ..., w_n$ of $W$ such that all entries in the first column of $\mathcal{M}(T)$ [with respect to the bases $v_1, ..., v_m$ and $w_1, ..., w_n$] are 0 except for possibly a 1 in the first row, first column.

> *In this exercise, unlike Exercise 5, you are given the basis of $V$ instead of being able to choose a basis of $V$.*

**Solution.** If $v_1 \in \operatorname{null} T$ then let $w_1, ..., w_n$ be any basis of $W$. Since $Tv_1 = 0$, it follows that the first column of $\mathcal{M}(T)$ is zero.

Suppose that $v_1 \notin \operatorname{null} T$, so that $Tv_1 \neq 0$. Let $w_1 = Tv_1$ and extend this to a basis $w_1, ..., w_n$ of $W$. It follows that the entries in the first column of $\mathcal{M}(T)$ are all 0 except for a 1 in the first row.

**Exercise 3.C.7.** Suppose $w_1, ..., w_n$ is a basis of $W$ and $V$ is finite-dimensional. Suppose $T \in \mathcal{L}(V, W)$. Prove that there exists a basis $v_1, ..., v_m$ of $V$ such that all entries in the first row of $\mathcal{M}(T)$ [with respect to the bases $v_1, ..., v_m$ and $w_1, ..., w_n$] are 0 except for possibly a 1 in the first row, first column.

> *In this exercise, unlike Exercise 5, you are given the basis of $W$ instead of being able to choose a basis of $W$.*



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**Solution.** Let $u_1, ..., u_m$ be any basis of $V$ and suppose $\mathcal{M}(T, (u_1, ..., u_m), (w_1, ..., w_n))$ has entries $A_{j,k}$. If the first row of this matrix is zero then $u_1, ..., u_m$ is the desired basis. Otherwise, there exists some $i \in \{1, ..., m\}$ such that $A_{1,i}$ is non-zero. Let $\lambda = A_{1,i}^{-1}$ and define

$$v_1 = \lambda u_i, \quad v_i = u_1 - \lambda A_{1,1} u_i, \quad \text{and} \quad v_k = u_k - \lambda A_{1,k} u_i \text{ for } 2 \le k \le m \text{ and } k \ne i.$$

Because each $u_k$ belongs to $\operatorname{span}(v_1, ..., v_m)$, the list $v_1, ..., v_m$ spans $V$. It follows from 2.42 that $v_1, ..., v_m$ is a basis of $V$. Now observe that

$$Tv_1 = \lambda T u_i = \lambda\big(A_{1,i} w_1 + \cdots + A_{n,i} w_n\big) = 1 w_1 + \cdots + \lambda A_{n,i} w_n,$$

$$\begin{aligned} Tv_i = Tu_1 - A_{1,1}(\lambda T u_i) &= A_{1,1} w_1 + \cdots + A_{n,1} w_n - A_{1,1}\big(w_1 + \cdots + \lambda A_{n,i} w_n\big) \\ &= 0 w_1 + \cdots + \big(A_{n,1} - \lambda A_{1,1} A_{n,i}\big) w_n. \end{aligned}$$

For $2 \le k \le m$ and $k \ne i$,

$$\begin{aligned} Tv_k = Tu_k - A_{1,k}(\lambda T u_i) &= A_{1,k} w_1 + \cdots + A_{n,k} w_n - A_{1,k}\big(w_1 + \cdots + \lambda A_{n,i} w_n\big) \\ &= 0 w_1 + \cdots + \big(A_{n,k} - \lambda A_{1,k} A_{n,i}\big) w_n. \end{aligned}$$

Thus the entries in the first row of the matrix of $T$ with respect to the bases $v_1, ..., v_m$ and $w_1, ..., w_n$ are 0, except for a 1 in the first column.

**Exercise 3.C.8.** Suppose $A$ is an $m$-by-$n$ matrix and $B$ is an $n$-by-$p$ matrix. Prove that

$$(AB)_{j,\cdot} = A_{j,\cdot} B$$

for each $1 \le j \le m$. In other words, show that row $j$ of $AB$ equals (row $j$ of $A$) times $B$.

*This exercise gives the row version of 3.48.*

**Solution.** $(AB)_{j,\cdot}$ is a 1-by-$p$ matrix whose entry in the $k^{\text{th}}$ column is

$$\big((AB)_{j,\cdot}\big)_{1,k} = (AB)_{j,k} = \sum_{r=1}^{n} A_{j,r} B_{r,k}.$$

$A_{j,\cdot}$ is a 1-by-$n$ matrix and so $A_{j,\cdot} B$ is a 1-by-$p$ matrix whose entry in the $k^{\text{th}}$ column is

$$\big(A_{j,\cdot} B\big)_{1,k} = \sum_{r=1}^{n} \big(A_{j,\cdot}\big)_{1,r} B_{r,k} = \sum_{r=1}^{n} A_{j,r} B_{r,k}.$$

Thus $(AB)_{j,\cdot} = A_{j,\cdot} B$.



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**Exercise 3.C.9.** Suppose $a = \begin{pmatrix} a_1 & \cdots & a_n \end{pmatrix}$ is a 1-by-$n$ matrix and $B$ is an $n$-by-$p$ matrix. Prove that

$$aB = a_1 B_{1,\cdot} + \cdots + a_n B_{n,\cdot}.$$

In other words, show that $aB$ is a linear combination of the rows of $B$, with the scalars that multiply the rows coming from $a$.

*This exercise gives the row version of 3.50.*

**Solution.** $aB$ is a 1-by-$p$ matrix whose entry in the $k^{\text{th}}$ column is

$$(aB)_{1,k} = \sum_{r=1}^{n} a_r B_{r,k} = a_1 B_{1,k} + \cdots + a_n B_{n,k}.$$

Thus

$$aB = a_1 B_{1,\cdot} + \cdots + a_n B_{n,\cdot}.$$

**Exercise 3.C.10.** Give an example of 2-by-2 matrices $A$ and $B$ such that $AB \neq BA$.

**Solution.** Let

$$A = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \quad \text{and} \quad B = \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix}$$

and observe that

$$AB = \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix} \neq \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} = BA.$$

**Exercise 3.C.11.** Prove that the distributive property holds for matrix addition and matrix multiplication. In other words, suppose $A, B, C, D, E$, and $F$ are matrices whose sizes are such that $A(B+C)$ and $(D+E)F$ make sense. Explain why $AB + AC$ and $DF + EF$ both make sense and prove that

$$A(B+C) = AB + AC \quad \text{and} \quad (D+E)F = DF + EF.$$

**Solution.** For $B + C$ to make sense, $B$ and $C$ must have the same sizes; suppose they are both $n$-by-$p$ matrices. For $A(B+C)$ to make sense, $A$ must then be an $m$-by-$n$ matrix for some $m$. Given this, both $AB$ and $AC$ are $m$-by-$p$ matrices and thus $AB + AC$ makes sense.

Similarly, suppose $D$ and $E$ are both $m$-by-$n$ matrices. For $(D+E)F$ to make sense, $F$ must be an $n$-by-$p$ matrix for some $p$. Given this, both $DF$ and $EF$ are $m$-by-$p$ matrices and thus $DF + EF$ makes sense.

The entry of $A(B+C)$ in row $j$, column $k$ is given by



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$$\sum_{r=1}^{n} A_{j,r}(B+C)_{r,k} = \sum_{r=1}^{n} A_{j,r}\big(B_{r,k}+C_{r,k}\big) = \sum_{r=1}^{n} A_{j,r}B_{r,k} + \sum_{r=1}^{n} A_{j,r}C_{r,k},$$

where we have used distributivity in $\mathbf{F}$. The expression on the right-hand side gives the entry of $AB+AC$ in row $j$, column $k$. Thus $A(B+C) = AB+AC$. A similar argument shows that $(D+E)F = DF+EF$.

**Exercise 3.C.12.** Prove that matrix multiplication is associative. In other words, suppose $A, B$, and $C$ are matrices whose sizes are such that $(AB)C$ makes sense. Explain why $A(BC)$ makes sense and prove that

$$(AB)C = A(BC).$$

*Try to find a clean proof that illustrates the following quote from Emil Artin: "It is my experience that proofs involving matrices can be shortened by 50% if one throws the matrices out."*

**Solution.** If $A$ is an $m$-by-$n$ matrix then for $AB$ to make sense, $B$ must be an $n$-by-$p$ matrix for some $p$, so that $AB$ is an $m$-by-$p$ matrix. For $(AB)C$ to make sense, $C$ must then be a $p$ -by-$q$ matrix for some $q$, so that $(AB)C$ is an $m$-by-$q$ matrix. Thus $BC$ is an $n$-by-$q$ matrix and $A(BC)$ is an $m$-by-$q$ matrix.

Let $e_1, ..., e_n$ be the standard basis of $\mathbf{F}^n$, let $f_1, ..., f_m$ be the standard basis of $\mathbf{F}^m$, and define $R \in \mathcal{L}(\mathbf{F}^n, \mathbf{F}^m)$ by $Re_k = A_{1,k}f_1 + \cdots + A_{m,k}f_m$. Thus, with respect to the standard bases, $\mathcal{M}(R) = A$. Define $S \in \mathcal{L}(\mathbf{F}^p, \mathbf{F}^n)$ and $T \in \mathcal{L}(\mathbf{F}^q, \mathbf{F}^p)$ similarly, so that $\mathcal{M}(S) = B$ and $\mathcal{M}(T) = C$, with respect to the standard bases. Now observe that

$$\begin{aligned}(AB)C = (\mathcal{M}(R)\mathcal{M}(S))\mathcal{M}(T) &\overset{(3.43)}{=} \mathcal{M}(RS)\mathcal{M}(T) \overset{(3.43)}{=} \mathcal{M}((RS)T) \\ &\overset{(3.8)}{=} \mathcal{M}(R(ST)) \overset{(3.43)}{=} \mathcal{M}(R)\mathcal{M}(ST) \overset{(3.43)}{=} \mathcal{M}(R)(\mathcal{M}(S)\mathcal{M}(T)) = A(BC);\end{aligned}$$

the number above the equals sign is the textbook reference justifying the equality.

**Exercise 3.C.13.** Suppose $A$ is an $n$-by-$n$ matrix and $1 \le j, k \le n$. Show that the entry in row $j$, column $k$, of $A^3$ (which is defined to mean $AAA$) is

$$\sum_{p=1}^{n}\sum_{r=1}^{n} A_{j,p}A_{p,r}A_{r,k}.$$

**Solution.** By the definition of matrix multiplication, we have

$$(A^3)_{j,k} = (A^2A)_{j,k} = \sum_{r=1}^{n}(A^2)_{j,r}A_{r,k} = \sum_{r=1}^{n}\left(\sum_{p=1}^{n} A_{j,p}A_{p,r}\right)A_{r,k} = \sum_{p=1}^{n}\sum_{r=1}^{n} A_{j,p}A_{p,r}A_{r,k}.$$



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**Exercise 3.C.14.** Suppose $m$ and $n$ are positive integers. Prove that the function $A \mapsto A^{\mathrm{t}}$ is a linear map from $\mathbf{F}^{m,n}$ to $\mathbf{F}^{n,m}$.

**Solution.** Let $A, B \in \mathbf{F}^{m,n}$ and $\lambda \in \mathbf{F}$ be given. Observe that

$$\big((A+B)^{\mathrm{t}}\big)_{k,j} = (A+B)_{j,k} = A_{j,k} + B_{j,k} = (A^{\mathrm{t}})_{k,j} + (B^{\mathrm{t}})_{k,j},$$

$$\big((\lambda A)^{\mathrm{t}}\big)_{k,j} = (\lambda A)_{j,k} = \lambda A_{j,k} = \lambda (A^{\mathrm{t}})_{k,j}.$$

Thus $(A+B)^{\mathrm{t}} = A^{\mathrm{t}} + B^{\mathrm{t}}$ and $(\lambda A)^{\mathrm{t}} = \lambda A^{\mathrm{t}}$.

**Exercise 3.C.15.** Prove that if $A$ is an $m$-by-$n$ matrix and $C$ is an $n$-by-$p$ matrix, then

$$(AC)^{\mathrm{t}} = C^{\mathrm{t}} A^{\mathrm{t}}.$$

*This exercise shows that the transpose of the product of two matrices is the product of the transposes in the opposite order.*

**Solution.** Observe that

$$\big((AC)^{\mathrm{t}}\big)_{k,j} = (AC)_{j,k} = \sum_{r=1}^{n} A_{j,r} C_{r,k} = \sum_{r=1}^{n} (C^{\mathrm{t}})_{k,r} (A^{\mathrm{t}})_{r,j} = (C^{\mathrm{t}} A^{\mathrm{t}})_{k,j}.$$

Thus $(AC)^{\mathrm{t}} = C^{\mathrm{t}} A^{\mathrm{t}}$.

**Exercise 3.C.16.** Suppose $A$ is an $m$-by-$n$ matrix with $A \neq 0$. Prove that the rank of $A$ is 1 if and only if there exist $(c_1, ..., c_m) \in \mathbf{F}^m$ and $(d_1, ..., d_n) \in \mathbf{F}^n$ such that $A_{j,k} = c_j d_k$ for every $j = 1, ..., m$ and every $k = 1, ..., n$.

**Solution.** If the rank of $A$ is 1 then by 3.56 there is an $m$-by-1 matrix $C$ and a 1-by-$n$ matrix $R$ such that $A = CR$; take $c_j$ to be the $j^{\text{th}}$ entry of $C$ and take $d_k$ to be the $k^{\text{th}}$ entry of $R$.

Now suppose there exist $(c_1, ..., c_m) \in \mathbf{F}^m$ and $(d_1, ..., d_n) \in \mathbf{F}^n$ such that $A_{j,k} = c_j d_k$ for every $j = 1, ..., m$ and every $k = 1, ..., n$. If we define

$$C = \begin{pmatrix} c_1 \\ \vdots \\ c_m \end{pmatrix} \in \mathbf{F}^{m,1} \quad \text{and} \quad D = \begin{pmatrix} d_1 & \cdots & d_n \end{pmatrix} \in \mathbf{F}^{1,n},$$

then $A = CD$. By (3.51) (a), every column of $A$ is a scalar multiple of $C$. It follows that $\operatorname{rank} A \leq 1$, and since $A \neq 0$ we must have $\operatorname{rank} A \geq 1$. Thus $\operatorname{rank} A = 1$.



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**Exercise 3.C.17.** Suppose $T \in \mathcal{L}(V)$, and $u_1, ..., u_n$ and $v_1, ..., v_n$ are bases of $V$. Prove that the following are equivalent.

(a) $T$ is injective.

(b) The columns of $\mathcal{M}(T)$ are linearly independent in $\mathbf{F}^{n,1}$.

(c) The columns of $\mathcal{M}(T)$ span $\mathbf{F}^{n,1}$.

(d) The rows of $\mathcal{M}(T)$ span $\mathbf{F}^{1,n}$.

(e) The rows of $\mathcal{M}(T)$ are linearly independent in $\mathbf{F}^{1,n}$.

Here $\mathcal{M}(T)$ means $\mathcal{M}(T, (u_1, ..., u_n), (v_1, ..., v_n))$.

**Solution.** Suppose that $\mathcal{M}(T)$ has entries $A_{j,k}$ and suppose that (a) holds. Let $b_1, ..., b_n$ be scalars such that

$$b_1 \begin{pmatrix} A_{1,1} \\ \vdots \\ A_{n,1} \end{pmatrix} + \cdots + b_n \begin{pmatrix} A_{1,n} \\ \vdots \\ A_{n,n} \end{pmatrix} = \begin{pmatrix} b_1 A_{1,1} + \cdots + b_n A_{1,n} \\ \vdots \\ b_1 A_{n,1} + \cdots + b_n A_{n,n} \end{pmatrix} = \begin{pmatrix} 0 \\ \vdots \\ 0 \end{pmatrix}.$$

Let $u = b_1 u_1 + \cdots + b_n u_n$ and observe that

$$\begin{aligned} Tu &= b_1 Tu_1 + \cdots + b_n Tu_n \\ &= b_1 (A_{1,1} v_1 + \cdots + A_{n,1} v_n) + \cdots + b_n (A_{1,n} v_1 + \cdots + A_{n,n} v_n) \\ &= (b_1 A_{1,1} + \cdots + b_n A_{1,n}) v_1 + \cdots + (b_1 A_{n,1} + \cdots + b_n A_{n,n}) v_n \\ &= 0 v_1 + \cdots + 0 v_n = 0. \end{aligned}$$

Thus $u \in \operatorname{null} T$. Since $T$ is injective, this implies that $u = 0$. The linear independence of the basis $u_1, ..., u_n$ then gives us $b_1 = \cdots = b_n = 0$. Thus the columns of $\mathcal{M}(T)$ are linearly independent, i.e. (b) holds.

Now suppose that (b) holds and let $u = b_1 u_1 + \cdots + b_n u_n$ be such that $Tu = 0$. As in the previous paragraph, this is equivalent to

$$(b_1 A_{1,1} + \cdots + b_n A_{1,n}) v_1 + \cdots + (b_1 A_{n,1} + \cdots + b_n A_{n,n}) v_n = 0.$$

The linear independence of the basis $v_1, ..., v_n$ then implies that

$$b_1 A_{1,1} + \cdots + b_n A_{1,n} = \cdots = b_1 A_{n,1} + \cdots + b_n A_{n,n} = 0,$$

which in turn gives us

$$\begin{pmatrix} b_1 A_{1,1} + \cdots + b_n A_{1,n} \\ \vdots \\ b_1 A_{n,1} + \cdots + b_n A_{n,n} \end{pmatrix} = b_1 \begin{pmatrix} A_{1,1} \\ \vdots \\ A_{n,1} \end{pmatrix} + \cdots + b_n \begin{pmatrix} A_{1,n} \\ \vdots \\ A_{n,n} \end{pmatrix} = \begin{pmatrix} 0 \\ \vdots \\ 0 \end{pmatrix}.$$

It follows from the linear independence of the columns of $\mathcal{M}(T)$ that $b_1 = \cdots = b_n = 0$, so that $u = 0$. Thus $T$ is injective, i.e. (a) holds. This gives us the equivalence of (a) and (b).



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For the equivalence of (b) and (c), note that $\dim \mathbf{F}^{n,1} = n$ by 3.40 and thus, by 2.38 and 2.40, a list of $n$ vectors in $\mathbf{F}^{n,1}$ is linearly independent if and only if it spans $\mathbf{F}^{n,1}$. The same argument gives us the equivalence of (d) and (e), since we also have $\dim \mathbf{F}^{1,n} = n$ by 3.40.

Finally, the equivalence of (c) and (d) is given by 3.57.



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## 3.D. Invertibility and Isomorphisms

**Exercise 3.D.1.** Suppose $T \in \mathcal{L}(V, W)$ is invertible. Show that $T^{-1}$ is invertible and

$$(T^{-1})^{-1} = T.$$

**Solution.** By the definition of invertibility of $T$ we have $TT^{-1} = I$ and $T^{-1}T = I$. These equations show that $T^{-1}$ is invertible and its inverse is $T$.

**Exercise 3.D.2.** Suppose $T \in \mathcal{L}(U, V)$ and $S \in \mathcal{L}(V, W)$ are both invertible linear maps. Prove that $ST \in \mathcal{L}(U, W)$ is invertible and that $(ST)^{-1} = T^{-1}S^{-1}$.

**Solution.** Using the algebraic properties of 3.8, observe that

$$T^{-1}S^{-1}ST = T^{-1}IT = T^{-1}T = I \quad \text{and} \quad STT^{-1}S^{-1} = SIS^{-1} = SS^{-1} = I.$$

Thus $T^{-1}S^{-1}ST$ is the identity map on $U$ and $STT^{-1}S^{-1}$ is the identity map on $W$. It follows that $ST$ is invertible and that $(ST)^{-1} = T^{-1}S^{-1}$.

**Exercise 3.D.3.** Suppose $V$ is finite-dimensional and $T \in \mathcal{L}(V)$. Prove that the following are equivalent.

(a) $T$ is invertible.

(b) $Tv_1, ..., Tv_n$ is a basis of $V$ for every basis $v_1, ..., v_n$ of $V$.

(c) $Tv_1, ..., Tv_n$ is a basis of $V$ for some basis $v_1, ..., v_n$ of $V$.

**Solution.** Suppose that (a) holds and let $v_1, ..., v_n$ be a basis of $V$. Since $T$ is invertible, it is injective. It follows from Exercise 3.B.9 that $Tv_1, ..., Tv_n$ is linearly independent. Thus by 2.38, $Tv_1, ..., Tv_n$ is a basis of $V$. Hence (b) holds.

Suppose that (b) holds. By 2.31 there is a basis $v_1, ..., v_n$ of $V$. By assumption $Tv_1, ..., Tv_n$ is a basis of $V$. Thus (c) holds.

Suppose that (c) holds, so that there is a basis $v_1, ..., v_n$ of $V$ such that $Tv_1, ..., Tv_n$ is a basis of $V$. Let $v = a_1v_1 + \cdots + a_nv_n$ be such that $Tv = 0$ and observe that

$$0 = Tv = T(a_1v_1 + \cdots + a_nv_n) = a_1Tv_1 + \cdots + a_nTv_n.$$

The linear independence of $Tv_1, ..., Tv_n$ then implies that $a_1 = \cdots = a_n = 0$ and thus $v = 0$. It follows that $\operatorname{null} T = \{0\}$ and hence that $T$ is injective. Thus by 3.65, $T$ is invertible.

**Exercise 3.D.4.** Suppose $V$ is finite-dimensional and $\dim V > 1$. Prove that the set of noninvertible linear maps from $V$ to itself is not a subspace of $\mathcal{L}(V)$.



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**Solution.** By 3.65 this is equivalent to showing that the set of linear maps from $V$ to itself which are not injective is not a subspace of $\mathcal{L}(V)$. We showed this in Exercise 3.B.7.

**Exercise 3.D.5.** Suppose $V$ is finite-dimensional, $U$ is a subspace of $V$, and $S \in \mathcal{L}(U, V)$. Prove that there exists an invertible linear map $T$ from $V$ to itself such that $Tu = Su$ for every $u \in U$ if and only if $S$ is injective.

**Solution.** If there is such a map $T$, then

$$\operatorname{null} S = \operatorname{null} T \cap U = \{0\} \cap U = \{0\}.$$

Thus $S$ is injective.

Suppose that $S$ is injective. Let $u_1, ..., u_m$ be a basis of $U$, which we extend to a basis $u_1, ..., u_m, x_1, ..., x_n$ of $V$. The injectivity of $S$ and Exercise 3.B.9 imply that $Su_1, ..., Su_m$ is linearly independent and thus can be extended to a basis $Su_1, ..., Su_m, y_1, ..., y_n$ of $V$. Define $T \in \mathcal{L}(V)$ by $Tu_k = Su_k$ and $Tx_k = y_k$. Certainly $T$ extends $S$. Furthermore, since $T$ maps a basis of $V$ to a basis of $V$, $T$ must be invertible by Exercise 3.B.3.

**Exercise 3.D.6.** Suppose that $W$ is finite-dimensional and $S, T \in \mathcal{L}(V, W)$. Prove that $\operatorname{null} S = \operatorname{null} T$ if and only if there exists an invertible $E \in \mathcal{L}(W)$ such that $S = ET$.

**Solution.** Suppose there exists an invertible $E \in \mathcal{L}(W)$ such that $S = ET$, which implies $T = E^{-1}S$. It follows from Exercise 3.B.25 that $\operatorname{null} T \subseteq \operatorname{null} S$ and $\operatorname{null} S \subseteq \operatorname{null} T$. Thus $\operatorname{null} S = \operatorname{null} T$.

Now suppose that $\operatorname{null} S = \operatorname{null} T$. It follows from Exercise 3.B.25 that there are maps $R, R' \in \mathcal{L}(W)$ such that $T = RS$ and $S = R'T$ and thus by Exercise 3.B.23 we have $\dim \operatorname{range} T = \dim \operatorname{range} S$. Let $Tv_1, ..., Tv_m$ be a basis of $\operatorname{range} T$ and notice that this linearly independent list is equal to $R(Sv_1), ..., R(Sv_m)$. It follows from Exercise 3.A.4 that the list $Sv_1, ..., Sv_m$ is linearly independent and hence is a basis of $\operatorname{range} S$, since $\dim \operatorname{range} S = m$. Extend these lists to bases

$$Tv_1, ..., Tv_m, x_1, ..., x_n \quad \text{and} \quad Sv_1, ..., Sv_m, y_1, ..., y_n$$

of $W$, and define $E \in \mathcal{L}(W)$ by $E(Tv_k) = Sv_k$ and $Ex_k = y_k$. Notice that $E$ maps a basis of $W$ to a basis of $W$; it follows from Exercise 3.D.3 that $E$ is invertible. For any $v \in V$ we have $Tv = a_1 Tv_1 + \cdots + a_m Tv_m$ for some scalars $a_1, ..., a_m$. As we showed in Exercise 3.B.25, this implies $Sv = a_1 Sv_1 + \cdots + a_m Sv_m$ since $\operatorname{null} T = \operatorname{null} S$. It follows that

$$E(Tv) = a_1 E(Tv_1) + \cdots + a_m E(Tv_m) = a_1 Sv_1 + \cdots + a_m Sv_m = Sv.$$

Thus $S = ET$.

**Exercise 3.D.7.** Suppose that $V$ is finite-dimensional and $S, T \in \mathcal{L}(V, W)$. Prove that $\operatorname{range} S = \operatorname{range} T$ if and only if there exists an invertible $E \in \mathcal{L}(V)$ such that $S = TE$.



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**Solution.** Suppose there exists an invertible $E \in \mathcal{L}(V)$ such that $S = TE$, which implies $T = SE^{-1}$. It follows from Exercise 3.B.26 that $\operatorname{range} T \subseteq \operatorname{range} S$ and $\operatorname{range} S \subseteq \operatorname{range} T$. Thus $\operatorname{range} S = \operatorname{range} T$.

Now suppose that $\operatorname{range} S = \operatorname{range} T$. Let $u_1, ..., u_m$ be a basis of $\operatorname{null} S$ and extend this to a basis of $u_1, ..., u_m, x_1, ..., x_n$ of $V$; as the proof of the fundamental theorem of linear maps (3.21) shows, $Sx_1, ..., Sx_n$ is a basis of $\operatorname{range} S$. Our assumption $\operatorname{range} S = \operatorname{range} T$ implies that $Sx_k = Ty_k$ for some $y_1, ..., y_n \in V$, and also that $\dim \operatorname{null} T = \dim \operatorname{null} S = m$. Let $v_1, ..., v_m$ be a basis of $\operatorname{null} T$ and suppose we have scalars $a_1, ..., a_m, b_1, ..., b_n$ such that

$$a_1 v_1 + \cdots + a_m v_m + b_1 y_1 + \cdots + b_n y_n = 0.$$

Applying $T$ to both sides of this equation gives us

$$b_1 Ty_1 + \cdots + b_n Ty_n = b_1 Sx_1 + \cdots + b_n Sx_n = 0.$$

The linear independence of $Sx_1, ..., Sx_n$ implies that $b_1 = \cdots = b_n = 0$ and the linear independence of $v_1, ..., v_m$ then gives us $a_1 = \cdots = a_m = 0$. Thus the list $v_1, ..., v_m, y_1, ..., y_n$ is linearly independent and hence is a basis of $V$ by 2.38. Define $E \in \mathcal{L}(V)$ by $Eu_k = v_k$ and $Ex_k = y_k$. Because $E$ maps a basis of $V$ to a basis of $V$, Exercise 3.D.3 shows that $E$ is invertible. For any $v = a_1 u_1 + \cdots + a_m u_m + b_1 x_1 + \cdots + b_n x_n \in V$, observe that

$$\begin{aligned} T(Ev) &= T(a_1 v_1 + \cdots + a_m v_m + b_1 y_1 + \cdots + b_n y_n) \\ &= b_1 Ty_1 + \cdots + b_n Ty_n = b_1 Sx_1 + \cdots + b_n Sx_n = Sv. \end{aligned}$$

Thus $S = TE$.

**Exercise 3.D.8.** Suppose $V$ and $W$ are finite-dimensional and $S, T \in \mathcal{L}(V, W)$. Prove that there exist invertible $E_1 \in \mathcal{L}(V)$ and $E_2 \in \mathcal{L}(W)$ such that $S = E_2 T E_1$ if and only if $\dim \operatorname{null} S = \dim \operatorname{null} T$.

**Solution.** Suppose there exist such maps $E_1, E_2$. It follows from Exercise 3.B.22 that

$$\dim \operatorname{null} S \leq \dim \operatorname{null} E_2 + \dim \operatorname{null} T + \dim \operatorname{null} E_1 = \dim \operatorname{null} T.$$

Notice that $T = E_2^{-1} S E_1^{-1}$; repeating the previous argument gives us $\dim \operatorname{null} T \leq \dim \operatorname{null} S$ and thus $\dim \operatorname{null} S = \dim \operatorname{null} T$.

Now suppose that $\dim \operatorname{null} S = \dim \operatorname{null} T$. Let $u_1, ..., u_m$ be a basis of $\operatorname{null} S$ and let $v_1, ..., v_m$ be a basis of $\operatorname{null} T$. Extend these to bases

$$u_1, ..., u_m, x_1, ..., x_n \quad \text{and} \quad v_1, ..., v_m, y_1, ..., y_n$$

of $V$, and define $E_1 \in \mathcal{L}(V)$ by $E_1 u_k = v_k$ and $E_1 x_k = y_k$. Because $E_1$ maps a basis to a basis, Exercise 3.D.3 shows that $E_1$ is invertible. It is straightforward to verify that $\operatorname{null} S = \operatorname{null} TE_1$ and thus by Exercise 3.D.6 there is an invertible $E_2 \in \mathcal{L}(W)$ such that $S = E_2 T E_1$.



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**Exercise 3.D.9.** Suppose $V$ is finite-dimensional and $T : V \to W$ is a surjective linear map of $V$ onto $W$. Prove that there is a subspace $U$ of $V$ such that $T|_U$ is an isomorphism of $U$ onto $W$.

*Here $T|_U$ means the function $T$ restricted to $U$. Thus $T|_U$ is the function whose domain is $U$, with $T|_U$ defined by $T|_U\,(u) = Tu$ for every $u \in U$.*

**Solution.** By Exercise 3.B.11 there is a subspace $U$ of $V$ such that

$$U \cap \operatorname{null} T = \{0\} \quad \text{and} \quad W = \operatorname{range} T = \{Tu : u \in U\}.$$

The equation $U \cap \operatorname{null} T = \{0\}$ shows that $T|_U$ is injective and the equation $W = \{Tu : u \in U\}$ shows that $T|_U$ is surjective. Thus $T|_U$ is an isomorphism of $U$ onto $W$.

**Exercise 3.D.10.** Suppose $V$ and $W$ are finite-dimensional and $U$ is a subspace of $V$. Let

$$\mathcal{E} = \{T \in \mathcal{L}(V, W) : U \subseteq \operatorname{null} T\}.$$

(a) Show that $\mathcal{E}$ is a subspace of $\mathcal{L}(V, W)$.

(b) Find a formula for $\dim \mathcal{E}$ in terms of $\dim V$, $\dim W$, and $\dim U$.

*Hint: Define $\Phi : \mathcal{L}(V, W) \to \mathcal{L}(U, W)$ by $\Phi(T) = T|_U$. What is $\operatorname{null} \Phi$? What is $\operatorname{range} \Phi$?*

**Solution.**

(a) Because the null space of the zero map is all of $V$, we certainly have $0 \in \mathcal{E}$. Suppose that $S, T \in \mathcal{E}$ and $\lambda \in \mathbf{F}$. For any $u \in U$, observe that

$$(S + T)(u) = Su + Tu = 0 \quad \text{and} \quad (\lambda T)(u) = \lambda Tu = 0,$$

where we have used that $u \in \operatorname{null} S$ and $u \in \operatorname{null} T$. It follows that $U \subseteq \operatorname{null}(S + T)$ and $U \subseteq \operatorname{null}(\lambda T)$, so that $S + T \in \mathcal{E}$ and $\lambda T \in \mathcal{E}$. Thus $\mathcal{E}$ is a subspace of $\mathcal{L}(V, W)$.

(b) Following the hint, define $\Phi : \mathcal{L}(V, W) \to \mathcal{L}(U, W)$ by $\Phi(T) = T|_U$; it is straightforward to verify that $\Phi$ is linear. Note that

$$\Phi(T) = 0 \quad \Leftrightarrow \quad T|_U = 0 \quad \Leftrightarrow \quad Tu = 0 \text{ for all } u \in U \quad \Leftrightarrow \quad U \subseteq \operatorname{null} T.$$

Thus $\operatorname{null} \Phi = \mathcal{E}$. For any $S \in \mathcal{L}(U, W)$ we can use Exercise 3.B.13 to extend $S$ to a linear map $T \in \mathcal{L}(V, W)$; it follows that $\Phi(T) = S$. Thus $\Phi$ is surjective, i.e. $\operatorname{range} \Phi = \mathcal{L}(U, W)$. It now follows from the fundamental theorem of linear maps (3.21) and 3.72 that

$$\dim \mathcal{L}(V, W) = \dim \operatorname{null} \Phi + \dim \operatorname{range} \Phi \quad \Rightarrow \quad \dim \mathcal{E} = \dim W(\dim V - \dim U).$$



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**Exercise 3.D.11.** Suppose $V$ is finite-dimensional and $S, T \in \mathcal{L}(V)$. Prove that

$$ST \text{ is invertible} \quad \Leftrightarrow \quad S \text{ and } T \text{ are invertible.}$$

**Solution.** If $S$ and $T$ are invertible then Exercise 3.D.2 shows that $ST$ is invertible. If $S$ is not invertible then $S$ is not surjective by 3.65, so that $\dim \operatorname{range} S < \dim V$. It follows from Exercise 3.B.23 that

$$\dim \operatorname{range} ST \leq \dim \operatorname{range} S < \dim V.$$

Thus $ST$ is not surjective and hence not invertible. A similar argument shows that if $T$ is not invertible then $ST$ is not invertible.

**Exercise 3.D.12.** Suppose $V$ is finite-dimensional and $S, T, U \in \mathcal{L}(V)$ and $STU = I$. Show that $T$ is invertible and that $T^{-1} = US$.

**Solution.** It follows from 3.68 that $S$ commutes with $TU$ and that $ST$ commutes with $U$. Thus

$$STU = I \quad \Rightarrow \quad TUS = I \text{ and } UST = I.$$

Hence $T$ is invertible and $T^{-1} = US$.

**Exercise 3.D.13.** Show that the result in Exercise 12 can fail without the hypothesis that $V$ is finite-dimensional.

**Solution.** Consider $V = \mathbf{F}^\infty$. Let $S$ be the backward shift operator, let $T$ be the forward shift operator, and let $U$ be the identity on $\mathbf{F}^\infty$. For any $(x_1, x_2, x_3, ...) \in \mathbf{F}^\infty$, observe that

$$(STU)(x_1, x_2, x_3, ...) = S(T(x_1, x_2, x_3, ...)) = S(0, x_1, x_2, x_3, ...) = (x_1, x_2, x_3, ...).$$

Thus $STU = I$. However, $T$ is not invertible because $T$ is not surjective: $(1, 0, 0, ...) \notin \operatorname{range} T$.

**Exercise 3.D.14.** Prove or give a counterexample: If $V$ is a finite-dimensional vector space and $R, S, T \in \mathcal{L}(V)$ are such that $RST$ is surjective, then $S$ is injective.

**Solution.** This is true. $RST$ must be invertible by 3.65 and thus by Exercise 3.D.11 $S$ is invertible and hence injective.

**Exercise 3.D.15.** Suppose $T \in \mathcal{L}(V)$ and $v_1, ..., v_m$ is a list in $V$ such that $Tv_1, ..., Tv_m$ spans $V$. Prove that $v_1, ..., v_m$ spans $V$.

**Solution.** Using 2.30 we can reduce the list $Tv_1, ..., Tv_m$ to a basis $Tv_{k_1}, ..., Tv_{k_n}$ for some indices $1 \leq k_1 < \cdots < k_n \leq m$. It follows from Exercise 3.A.4 and 2.38 that $v_{k_1}, ..., v_{k_n}$ is a basis of $V$ and thus



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$$V = \operatorname{span}\left(v_{k_1}, ..., v_{k_n}\right) \subseteq \operatorname{span}(v_1, ..., v_m) \quad \Rightarrow \quad V = \operatorname{span}(v_1, ..., v_m).$$

**Exercise 3.D.16.** Prove that every linear map from $\mathbf{F}^{n,1}$ to $\mathbf{F}^{m,1}$ is given by a matrix multiplication. In other words, prove that if $T \in \mathcal{L}(\mathbf{F}^{n,1}, \mathbf{F}^{m,1})$, then there exists an $m$ -by-$n$ matrix $A$ such that $Tx = Ax$ for every $x \in \mathbf{F}^{n,1}$.

**Solution.** Let $A$ be the matrix of $T$ with respect to the standard bases of $\mathbf{F}^{n,1}$ and $\mathbf{F}^{m,1}$. With respect to these standard bases we have $\mathcal{M}(x) = x$ and $\mathcal{M}(y) = y$ for any $x \in \mathbf{F}^{n,1}$ and $y \in \mathbf{F}^{m,1}$. It then follows from 3.76 that

$$Tx = \mathcal{M}(Tx) = \mathcal{M}(T)\mathcal{M}(x) = Ax$$

for every $x \in \mathbf{F}^{n,1}$.

**Exercise 3.D.17.** Suppose $V$ is finite-dimensional and $S \in \mathcal{L}(V)$. Define $\mathcal{A} \in \mathcal{L}(\mathcal{L}(V))$ by

$$\mathcal{A}(T) = ST$$

for $T \in \mathcal{L}(V)$.

(a) Show that $\dim \operatorname{null} \mathcal{A} = (\dim V)(\dim \operatorname{null} S)$.

(b) Show that $\dim \operatorname{range} \mathcal{A} = (\dim V)(\dim \operatorname{range} S)$.

**Solution.**

(a) For $T \in \mathcal{L}(V)$, note that $ST = 0$ if and only if $\operatorname{range} T \subseteq \operatorname{null} S$. Thus we can identify $\operatorname{null} \mathcal{A}$ with $\mathcal{L}(V, \operatorname{null} S)$ and it follows from 3.72 that

$$\dim \operatorname{null} \mathcal{A} = (\dim V)(\dim \operatorname{null} S).$$

(b) For $R \in \mathcal{L}(V)$, Exercise 3.B.26 implies that $R = ST$ for some $T \in \mathcal{L}(V)$ if and only if $\operatorname{range} R \subseteq \operatorname{range} S$. Thus we can identify $\operatorname{range} \mathcal{A}$ with $\mathcal{L}(V, \operatorname{range} S)$ and it follows from 3.72 that

$$\dim \operatorname{range} \mathcal{A} = (\dim V)(\dim \operatorname{range} S).$$

**Exercise 3.D.18.** Show that $V$ and $\mathcal{L}(\mathbf{F}, V)$ are isomorphic vector spaces.

**Solution.** Define a map $\Phi : \mathcal{L}(\mathbf{F}, V) \to V$ by

$$\Phi(T) = T(1).$$

It is straightforward to verify that $\Phi$ is linear. Now define a map $\Psi : V \to \mathcal{L}(\mathbf{F}, V)$ by

$$[\Psi(v)](x) = xv.$$

It is straightforward to check that $\Psi(v)$ indeed belongs to $\mathcal{L}(\mathbf{F}, V)$ for any $v \in V$. For any $T \in \mathcal{L}(\mathbf{F}, V)$, observe that



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$$[\Psi(\Phi(T))](x) = [\Psi(T(1))](x) = xT(1) = T(x).$$

Thus $\Psi(\Phi(T)) = T$, i.e. $\Psi \circ \Phi$ is the identity map on $\mathcal{L}(\mathbf{F}, V)$. Now let $v \in V$ be given and observe that

$$\Phi(\Psi(v)) = [\Psi(v)](1) = v.$$

Thus $\Phi \circ \Psi$ is the identity map on $V$. As the proof of 3.63 shows, it now follows that $\Psi$ is a linear map. Thus $\Phi$ is an isomorphism from $\mathcal{L}(\mathbf{F}, V)$ to $V$ and its inverse is $\Psi$.

**Exercise 3.D.19.** Suppose $V$ is a finite-dimensional and $T \in \mathcal{L}(V)$. Prove that $T$ has the same matrix with respect to every basis of $V$ if and only if $T$ is a scalar multiple of the identity operator.

**Solution.** If $T = \lambda I$ for some $\lambda \in \mathbf{F}$ then the matrix of $T$ with respect to any basis of $V$ must be $\lambda I$ (here $I$ is the identity matrix).

Suppose that $T$ has the same matrix with respect to every basis of $V$, i.e. there is some matrix $A$ with entries $A_{j,k}$ such that

$$Tu_k = A_{1,k}u_1 + \cdots + A_{m,k}u_m$$

for any basis $u_1, ..., u_m$ of $V$. Let $v_1, ..., v_m$ be a fixed basis of $V$ and let $k \in \{1, ..., m\}$ be given; it is straightforward to verify that $v_1, ..., \frac{1}{2}v_k, ..., v_m$ is also a basis of $V$. By assumption we must then have

$$T\left(\tfrac{1}{2}v_k\right) = A_{1,k}v_1 + \cdots + A_{k,k}\left(\tfrac{1}{2}v_k\right) + \cdots + A_{m,k}v_m$$

$$\Rightarrow \quad Tv_k = 2A_{1,k}v_1 + \cdots + A_{k,k}v_k + \cdots + 2A_{m,k}v_m.$$

On the other hand we must have

$$Tv_k = A_{1,k}v_1 + \cdots + A_{k,k}v_k + \cdots + A_{m,k}v_m.$$

Hence by unique representation we must have $A_{j,k} = 2A_{j,k}$, so that $A_{j,k} = 0$, for all $j \neq k$; it follows that $Tu_k = A_{k,k}u_k$ for any basis $u_1, ..., u_m$ of $V$. Let $1 \leq j < k \leq m$ be given and consider the basis $v_1, ..., v_k, ..., v_j, ..., v_m$ of $V$, i.e. the basis obtained by swapping the basis vectors $v_j$ and $v_k$. This gives us the two equations

$$Tv_k = A_{k,k}v_k \quad \text{and} \quad Tv_k = A_{j,j}v_k.$$

It follows from unique representation that $A_{j,j} = A_{k,k}$ and thus, letting $\lambda = A_{1,1}$, we have $Tv_k = \lambda v_k$ for all $k \in \{1, ..., m\}$. Thus $T = \lambda I$.



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**Exercise 3.D.20.** Suppose $q \in \mathcal{P}(\mathbf{R})$. Prove that there exists a polynomial $p \in \mathcal{P}(\mathbf{R})$ such that

$$q(x) = (x^2 + x)p''(x) + 2xp'(x) + p(3)$$

for all $x \in \mathbf{R}$.

**Solution.** There is a non-negative integer $m$ such that $q \in \mathcal{P}_m(\mathbf{R})$; either $m = \deg q$ if $q \neq 0$ or $m = 0$ if $q = 0$. Define $T : \mathcal{P}_m(\mathbf{R}) \to \mathcal{P}_m(\mathbf{R})$ by

$$Tp = (x^2 + x)p''(x) + 2xp'(x) + p(3).$$

It is straightforward to check that $T$ is linear and some calculations reveal that $\deg Tp = \deg p$ for any $p \in \mathcal{P}_m(\mathbf{R})$. It follows that if $p \in \mathcal{P}_m(\mathbf{R})$ is such that $Tp = 0$, so that $\deg Tp = -\infty$, then $\deg p = -\infty$, i.e. $p = 0$. Thus $T$ is injective. By 3.65 this implies that $T$ is surjective and so there must exist some $p \in \mathcal{P}_m(\mathbf{R})$ such that $Tp = q$.

**Exercise 3.D.21.** Suppose $n$ is a positive integer and $A_{j,k} \in \mathbf{F}$ for all $j, k = 1, ..., n$. Prove that the following are equivalent (note that in both parts below, the number of equations equals the number of variables).

(a) The trivial solution $x_1 = \cdots = x_n = 0$ is the only solution to the homogeneous system of equations

$$\sum_{k=1}^{n} A_{1,k}x_k = 0$$

$$\vdots$$

$$\sum_{k=1}^{n} A_{n,k}x_k = 0.$$

(b) For every $c_1, ..., c_n \in \mathbf{F}$, there exists a solution to the system of equations

$$\sum_{k=1}^{n} A_{1,k}x_k = c_1$$

$$\vdots$$

$$\sum_{k=1}^{n} A_{n,k}x_k = c_n.$$

**Solution.** Define $T \in \mathcal{L}(\mathbf{F}^n)$ by

$$T(x_1, ..., x_n) = \left( \sum_{k=1}^{n} A_{1,k}x_k, ..., \sum_{k=1}^{n} A_{n,k}x_k \right)$$



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and notice that (a) is equivalent to the injectivity of $T$ and (b) is equivalent to the surjectivity of (b). It then follows from 3.65 that (a) and (b) are equivalent.

**Exercise 3.D.22.** Suppose $T \in \mathcal{L}(V)$ and $v_1, ..., v_n$ is a basis of $V$. Prove that

$$\mathcal{M}(T, (v_1, ..., v_n)) \text{ is invertible} \quad \Leftrightarrow \quad T \text{ is invertible.}$$

**Solution.** In what follows, all matrices of linear maps and vectors are understood to be with respect to the basis $v_1, ..., v_n$ of $V$.

Suppose that $\mathcal{M}(T)$ is invertible and let $B$ be its inverse. Define $S \in \mathcal{L}(V)$ by

$$Sv_k = B_{1,k}v_1 + \cdots + B_{n,k}v_n,$$

so that $\mathcal{M}(S) = B$. Note that for any $u, v \in V$ we have $u = v$ if and only if $\mathcal{M}(u) = \mathcal{M}(v)$, by unique representation in the basis $v_1, ..., v_n$. Let $v \in V$ be given and observe that

$$\mathcal{M}((ST)(v)) \overset{(3.76)}{=} \mathcal{M}(ST)\mathcal{M}(v) \overset{(3.43)}{=} \mathcal{M}(S)\mathcal{M}(T)\mathcal{M}(v) = I\mathcal{M}(v) = \mathcal{M}(v).$$

The number above the equals sign is the textbook reference justifying the equality; the third equality is justified as $\mathcal{M}(S) = B$ is the inverse of $\mathcal{M}(T)$. Thus $(ST)(v) = v$ for all $v \in V$, so that $ST$ is the identity map on $V$. It follows from 3.68 that $TS$ is also the identity map on $V$ and we may conclude that $T$ is invertible with inverse $S$.

The converse statement is the content of 3.86, which we now prove. Suppose that $T$ is invertible. Using 3.43, observe that

$$\mathcal{M}(T)\mathcal{M}(T^{-1}) = \mathcal{M}(TT^{-1}) = \mathcal{M}(I) = I = \mathcal{M}(I) = \mathcal{M}(T^{-1}T) = \mathcal{M}(T^{-1})\mathcal{M}(T).$$

Thus $(\mathcal{M}(T))^{-1} = \mathcal{M}(T^{-1})$.

**Exercise 3.D.23.** Suppose that $u_1, ..., u_n$ and $v_1, ..., v_n$ are bases of $V$. Let $T \in \mathcal{L}(V)$ be such that $Tv_k = u_k$ for each $k = 1, ..., n$. Prove that

$$\mathcal{M}(T, (v_1, ..., v_n)) = \mathcal{M}(I, (u_1, ..., u_n), (v_1, ..., v_n)).$$

**Solution.** For ease of notation, let us write

$$\mathcal{M}(T, u, v) = \mathcal{M}(T, (u_1, ..., u_n), (v_1, ..., v_n)) \quad \text{and} \quad \mathcal{M}(T, v) = \mathcal{M}(T, (v_1, ..., v_n)).$$

Note that, by 3.81,

$$\mathcal{M}(T, u)\mathcal{M}(I, v, u) = \mathcal{M}(TI, v, u) = \mathcal{M}(T, v, u) = I, \tag{$*$}$$

where the last equality follows from the definition of $T$. Using 3.84, 3.82, and $(*)$, observe that

$$\mathcal{M}(T, v) = (\mathcal{M}(I, v, u))^{-1}\mathcal{M}(T, u)\mathcal{M}(I, v, u) = \mathcal{M}(I, u, v)\mathcal{M}(T, u)\mathcal{M}(I, v, u) = \mathcal{M}(I, u, v).$$



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**Exercise 3.D.24.** Suppose $A$ and $B$ are square matrices of the same size and $AB = I$. Prove that $BA = I$.

**Solution.** Suppose that $A$ and $B$ are $n$-by-$n$ matrices and let $e_1, ..., e_n$ be the standard basis of $\mathbf{F}^n$. In what follows, all matrices of linear maps are understood to be with respect to this standard basis.

Let $S, T \in \mathcal{L}(\mathbf{F}^n)$ be given by

$$Se_k = \sum_{j=1}^{n} A_{j,k} e_k \quad \text{and} \quad Te_k = \sum_{j=1}^{n} B_{j,k} e_k,$$

so that $\mathcal{M}(S) = A$ and $\mathcal{M}(T) = B$. Using 3.43, we then have

$$\mathcal{M}(ST) = \mathcal{M}(S)\mathcal{M}(T) = AB = I = \mathcal{M}(I).$$

Thus, by the uniqueness part of the linear map lemma (3.4), we have $ST = I$, which implies $TS = I$ by 3.68. It follows from 3.43 that

$$BA = \mathcal{M}(T)\mathcal{M}(S) = \mathcal{M}(TS) = \mathcal{M}(I) = I.$$



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## 3.E. Products and Quotients of Vector Spaces

**Exercise 3.E.1.** Suppose $T$ is a function from $V$ to $W$. The *graph* of $T$ is the subset of $V \times W$ defined by

$$\text{graph of } T = \{(v, Tv) \in V \times W : v \in V\}.$$

Prove that $T$ is a linear map if and only if the graph of $T$ is a subspace of $V \times W$.

> *Formally, a function $T$ from $V$ to $W$ is a subset $T$ of $V \times W$ such that for each $v \in V$, there exists exactly one element $(v, w) \in T$. In other words, formally a function is what is called above its graph. We do not usually think of functions in this formal manner. However, if we do become formal, then this exercise could be rephrased as follows: Prove that a function $T$ from $V$ to $W$ is a linear map if and only if $T$ is a subspace of $V \times W$.*

**Solution.** Let $\mathsf{graph}(T)$ be the graph of $T$. First suppose that $T$ is linear, so that $T(0) = 0$; it follows that $(0, 0) = (0, T(0)) \in \mathsf{graph}(T)$. For any $(u, Tu), (v, Tv) \in \mathsf{graph}(T)$ and $\lambda \in \mathbf{F}$, the linearity of $T$ implies that

$$(u, Tu) + (v, Tv) = (u + v, Tu + Tv) = (u + v, T(u + v)) \in \mathsf{graph}(T),$$

$$\lambda(v, Tv) = (\lambda v, \lambda Tv) = (\lambda v, T(\lambda v)) \in \mathsf{graph}(T).$$

Thus $\mathsf{graph}(T)$ is a subspace of $V \times W$.

Now suppose that $\mathsf{graph}(T)$ is a subspace of $V \times W$. Let $u, v \in V$ and $\lambda \in \mathbf{F}$ be given. Observe that

$$(u, Tu), (v, Tv) \in \mathsf{graph}(T) \quad \Rightarrow \quad (u, Tu) + (v, Tv) = (u + v, Tu + Tv) \in \mathsf{graph}(T).$$

Because the second component of an element of $\mathsf{graph}(T)$ must be $T$ applied to the first component, and $(u + v, Tu + Tv)$ belongs to $\mathsf{graph}(T)$, it must be that $T(u + v) = Tu + Tv$. Similarly,

$$(v, Tv) \in \mathsf{graph}(T) \quad \Rightarrow \quad \lambda(v, Tv) = (\lambda v, \lambda Tv) \in \mathsf{graph}(T) \quad \Rightarrow \quad T(\lambda v) = \lambda Tv.$$

Thus $T$ is linear.

**Exercise 3.E.2.** Suppose that $V_1, ..., V_m$ are vector spaces such that $V_1 \times \cdots \times V_m$ is finite-dimensional. Prove that $V_k$ is finite-dimensional for each $k = 1, ..., m$.

**Solution.** For any $k \in \{1, ..., m\}$, let $p_k : V_1 \times \cdots \times V_m \to V_k$ be given by $p_k(v_1, ..., v_m) = v_k$. It is straightforward to verify that $p_k$ is a surjective linear map and thus, by the fundamental theorem of linear maps (3.21), $\mathsf{range}\, p_k = V_k$ is finite-dimensional.



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**Exercise 3.E.3.** Suppose $V_1, ..., V_m$ are vector spaces. Prove that $\mathcal{L}(V_1 \times \cdots \times V_m, W)$ and $\mathcal{L}(V_1, W) \times \cdots \times \mathcal{L}(V_m, W)$ are isomorphic vector spaces.

**Solution.** Define a map

$$\Phi : \mathcal{L}(V_1, W) \times \cdots \times \mathcal{L}(V_m, W) \to \mathcal{L}(V_1 \times \cdots \times V_m, W),$$

where $\Phi(T_1, ..., T_m)$ is the map $V_1 \times \cdots \times V_m \to W$ given by

$$(v_1, ..., v_m) \mapsto T_1 v_1 + \cdots + T_m v_m.$$

It is straightforward to verify that $\Phi(T_1, ..., T_m)$ is indeed a linear map for any $(T_1, ..., T_m)$, and that $\Phi$ itself is linear.

For $k \in \{1, ..., m\}$, define $\iota_k : V_k \to V_1 \times \cdots \times V_m$ by $\iota_k(v) = (0, ..., v, ..., 0)$, where the $v$ is in the $k^{\text{th}}$ position; it is straightforward to check that each $\iota_k$ is a linear map. Define a map

$$\Psi : \mathcal{L}(V_1 \times \cdots \times V_m, W) \to \mathcal{L}(V_1, W) \times \cdots \times \mathcal{L}(V_m, W),$$

where $\Psi(T)$ is given by $(T \circ \iota_1, ..., T \circ \iota_m)$. The linearity of each $T \circ \iota_k$ follows from the linearity of $T$ and the linearity of $\iota_k$. Let $(T_1, ..., T_m) \in \mathcal{L}(V_1, W) \times \cdots \times \mathcal{L}(V_m, W)$ be given and observe that

$$\Psi(\Phi(T_1, ..., T_m)) = (\Phi(T_1, ..., T_m) \circ \iota_1, ..., \Phi(T_1, ..., T_m) \circ \iota_m).$$

For any $k \in \{1, ..., m\}$ and $v \in V_k$ we have

$$[\Phi(T_1, ..., T_m)](\iota_k(v)) = [\Phi(T_1, ..., T_m)](0, ..., v, ..., 0) = T_1(0) + \cdots + T_k v + \cdots + T_m(0) = T_k v.$$

Thus $\Phi(T_1, ..., T_m) \circ \iota_k = T_k$ and it follows that $\Psi(\Phi(T_1, ..., T_m)) = (T_1, ..., T_m)$, i.e. $\Psi \circ \Phi$ is the identity map on $\mathcal{L}(V_1, W) \times \cdots \times \mathcal{L}(V_m, W)$. Now let $T \in \mathcal{L}(V_1 \times \cdots \times V_m, W)$ be given and observe that

$$\begin{aligned}
[\Phi(\Psi(T))](v_1, ..., v_m) &= [\Phi(T \circ \iota_1, ..., T \circ \iota_m)](v_1, ..., v_m) \\
&= (T \circ \iota_1)(v_1) + \cdots + (T \circ \iota_m)(v_m) \\
&= T(v_1, ..., 0) + \cdots + T(0, ..., v_m) \\
&= T(v_1, ..., v_m).
\end{aligned}$$

Thus $\Phi(\Psi(T)) = T$, i.e. $\Phi \circ \Psi$ is the identity map on $\mathcal{L}(V_1 \times \cdots \times V_m, W)$. As the proof of 3.63 shows, it now follows that $\Psi$ is a linear map. Thus $\Phi$ is an isomorphism from $\mathcal{L}(V_1, W) \times \cdots \times \mathcal{L}(V_m, W)$ to $\mathcal{L}(V_1 \times \cdots \times V_m, W)$ and its inverse is $\Psi$.

**Exercise 3.E.4.** Suppose $W_1, ..., W_m$ are vector spaces. Prove that $\mathcal{L}(V, W_1 \times \cdots \times W_m)$ and $\mathcal{L}(V, W_1) \times \cdots \times \mathcal{L}(V, W_m)$ are isomorphic vector spaces.

**Solution.** Define a map

$$\Phi : \mathcal{L}(V, W_1) \times \cdots \times \mathcal{L}(V, W_m) \to \mathcal{L}(V, W_1 \times \cdots \times W_m),$$



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where $\Phi(T_1, ..., T_m)$ is the map $V \to W_1 \times \cdots \times W_m$ given by

$$v \mapsto (T_1 v, ..., T_m v).$$

It is straightforward to verify that $\Phi(T_1, ..., T_m)$ is indeed a linear map for any $(T_1, ..., T_m)$, and that $\Phi$ itself is linear.

For each $k \in \{1, ..., m\}$, define $p_k : W_1 \times \cdots \times W_m \to W_k$ by $p_k(w_1, ..., w_m) = w_k$; it is straightforward to check that each $p_k$ is a linear map. Define a map

$$\Psi : \mathcal{L}(V, W_1 \times \cdots \times W_m) \to \mathcal{L}(V, W_1) \times \cdots \times \mathcal{L}(V, W_m),$$

where $\Psi(T)$ is given by $(p_1 \circ T, ..., p_m \circ T)$. The linearity of each $p_k \circ T$ is given by the linearity of $p_k$ and the linearity of $T$. Let $(T_1, ..., T_m) \in \mathcal{L}(V, W_1) \times \cdots \times \mathcal{L}(V, W_m)$ be given and observe that

$$\Psi(\Phi(T_1, ..., T_m)) = (p_1 \circ \Phi(T_1, ..., T_m), ..., p_m \circ \Phi(T_1, ..., T_m)).$$

For any $k \in \{1, ..., m\}$ and $v \in V$ we have

$$p_k([\Phi(T_1, ..., T_m)](v)) = p_k(T_1 v, ..., T_m v) = T_k v.$$

Thus $p_k \circ \Phi(T_1, ..., T_m) = T_k$ and it follows that $\Psi(\Phi(T_1, ..., T_m)) = (T_1, ..., T_m)$, i.e. $\Psi \circ \Phi$ is the identity map on $\mathcal{L}(V, W_1) \times \cdots \times \mathcal{L}(V, W_m)$. Now let $T \in \mathcal{L}(V, W_1 \times \cdots \times W_m)$ be given and observe that

$$[\Phi(\Psi(T))](v) = [\Phi(p_1 \circ T, ..., p_m \circ T)](v) = (p_1(Tv), ..., p_m(Tv)) = Tv.$$

Thus $\Phi(\Psi(T)) = T$, i.e. $\Phi \circ \Psi$ is the identity map on $\mathcal{L}(V, W_1 \times \cdots \times W_m)$. As the proof of 3.63 shows, it now follows that $\Psi$ is a linear map. Thus $\Phi$ is an isomorphism from $\mathcal{L}(V, W_1) \times \cdots \times \mathcal{L}(V, W_m)$ to $\mathcal{L}(V, W_1 \times \cdots \times W_m)$ and its inverse is $\Psi$.

**Exercise 3.E.5.** For $m$ a positive integer, define $V^m$ by

$$V^m = \underbrace{V \times \cdots \times V}_{m \text{ times}}.$$

Prove that $V^m$ and $\mathcal{L}(\mathbf{F}^m, V)$ are isomorphic vector spaces.

**Solution.** Define a map $\Phi : \mathcal{L}(\mathbf{F}^m, V) \to V^m$ by

$$\Phi(T) = (Te_1, ..., Te_m),$$

where $e_1, ..., e_m$ is the standard basis of $\mathbf{F}^m$. It is straightforward to verify that $\Phi$ is linear. Now define a map $\Psi : V^m \to \mathcal{L}(\mathbf{F}^m, V)$ by

$$[\Psi(v_1, ..., v_m)](x_1, ..., x_m) = x_1 v_1 + \cdots + x_m v_m.$$

It is straightforward to check that $\Psi(v_1, ..., v_m)$ indeed belongs to $\mathcal{L}(\mathbf{F}^m, V)$ for any $(v_1, ..., v_m) \in V^m$. For any $T \in \mathcal{L}(\mathbf{F}^m, V)$, observe that



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$$\begin{aligned}[\Psi(\Phi(T))](x_1,...,x_m) &= [\Psi(Te_1,...,Te_m)](x_1,...,x_m)\\ &= x_1Te_1+\cdots+x_mTe_m\\ &= T(x_1e_1+\cdots+x_me_m)\\ &= T(x_1,...,x_m).\end{aligned}$$

Thus $\Psi(\Phi(T))=T$, i.e. $\Psi\circ\Phi$ is the identity map on $\mathcal{L}(\mathbf{F}^m,V)$. Now let $(v_1,...,v_m)\in V^m$ be given and observe that

$$\Phi(\Psi(v_1,...,v_m))=([\Psi(v_1,...,v_m)](e_1),...,[\Psi(v_1,...,v_m)](e_m))=(v_1,...,v_m).$$

Thus $\Phi\circ\Psi$ is the identity map on $V^m$. As the proof of 3.63 shows, it now follows that $\Psi$ is a linear map. Thus $\Phi$ is an isomorphism from $\mathcal{L}(\mathbf{F}^m,V)$ to $V^m$ and its inverse is $\Psi$.

**Exercise 3.E.6.** Suppose that $v,x$ are vectors in $V$ and that $U,W$ are subspaces of $V$ such that $v+U=x+W$. Prove that $U=W$.

**Solution.** Since $v\in v+U=x+W$, there is some $w\in W$ such that $v=x+w$, which implies that $x-v=-w\in W$. For any $u\in U$ we have $v+u=x+w$ for some $w\in W$, so that $u=x-v+w\in W$. Thus $U\subseteq W$. A similar argument shows that $W\subseteq U$ and it follows that $U=W$.

**Exercise 3.E.7.** Let $U=\{(x,y,z)\in\mathbf{R}^3:2x+3y+5z=0\}$. Suppose $A\subseteq\mathbf{R}^3$. Prove that $A$ is a translate of $U$ if and only if there exists $c\in\mathbf{R}$ such that

$$A=\{(x,y,z)\in\mathbf{R}^3:2x+3y+5z=c\}.$$

**Solution.** Suppose that $A$ is a translate of $U$, i.e. there is some $(a_1,a_2,a_3)\in\mathbf{R}^3$ such that $A=(a_1,a_2,a_3)+U$. Let $c=2a_1+3a_2+5a_3$ and let

$$W=\{(x_1,x_2,x_3)\in\mathbf{R}^3:2x_1+3x_2+5x_3=c\}.$$

We need to show that $A=W$. Suppose that $(x_1,x_2,x_3)\in A$, so that

$$(x_1,x_2,x_3)=(a_1,a_2,a_3)+(u_1,u_2,u_3)$$

for some $(u_1,u_2,u_3)\in U$. It follows that

$$2x_1+3x_2+5x_3=2u_1+3u_2+5u_3+2a_1+3a_2+5a_3=0+c=c.$$

Thus $A\subseteq W$. If $(x_1,x_2,x_3)\in W$ then note that

$$2x_1+3x_2+5x_3=c=2a_1+3a_2+5a_3\quad\Rightarrow\quad(x_1-a_1,x_2-a_2,x_3-a_3)\in U.$$

Thus $(x_1,x_2,x_3)=(a_1,a_2,a_3)+(x_1-a_1,x_2-a_2,x_3-a_3)\in(a_1,a_2,a_3)+U=A$ and it follows that $W\subseteq A$. Hence $A=W$.

Now suppose that there is some $c\in\mathbf{R}$ such that



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$$A = \{(x_1, x_2, x_3) \in \mathbf{R}^3 : 2x_1 + 3x_2 + 5x_3 = c\}.$$

We claim that $A = \left(\frac{c}{6}, \frac{c}{9}, \frac{c}{15}\right) + U$. If $(x_1, x_2, x_3) \in A$ then observe that

$$(x_1, x_2, x_3) = \left(\tfrac{c}{6}, \tfrac{c}{9}, \tfrac{c}{15}\right) + \left(x_1 - \tfrac{c}{6}, x_2 - \tfrac{c}{9}, x_3 - \tfrac{c}{15}\right) \in \left(\tfrac{c}{6}, \tfrac{c}{9}, \tfrac{c}{15}\right) + U,$$

where $\left(x_1 - \frac{c}{6}, x_2 - \frac{c}{9}, x_3 - \frac{c}{15}\right) \in U$ since

$$2\left(x_1 - \tfrac{c}{6}\right) + 3\left(x_2 - \tfrac{c}{9}\right) + 5\left(x_3 - \tfrac{c}{15}\right) = 2x_1 + 3x_2 + 5x_3 - \left(\tfrac{c}{3} + \tfrac{c}{3} + \tfrac{c}{3}\right) = c - c = 0.$$

Thus $A \subseteq \left(\frac{c}{6}, \frac{c}{9}, \frac{c}{15}\right) + U$. If $\left(\frac{c}{6}, \frac{c}{9}, \frac{c}{15}\right) + (u_1, u_2, u_3) \in \left(\frac{c}{6}, \frac{c}{9}, \frac{c}{15}\right) + U$ then

$$2\left(\tfrac{c}{6} + u_1\right) + 3\left(\tfrac{c}{9} + u_2\right) + 5\left(\tfrac{c}{15} + u_3\right) = \left(\tfrac{c}{3} + \tfrac{c}{3} + \tfrac{c}{3}\right) + 2u_1 + 3u_2 + 5u_3 = c + 0 = c.$$

Thus $\left(\frac{c}{6}, \frac{c}{9}, \frac{c}{15}\right) + (u_1, u_2, u_3) \in A$ and we may conclude that $A = \left(\frac{c}{6}, \frac{c}{9}, \frac{c}{15}\right) + U$.

**Exercise 3.E.8.**

(a) Suppose $T \in \mathcal{L}(V, W)$ and $c \in W$. Prove that $\{x \in V : Tx = c\}$ is either the empty set or is a translate of $\operatorname{null} T$.

(b) Explain why the set of solutions to a system of linear equations such as 3.27 is either the empty set or is a translate of some subspace of $\mathbf{F}^n$.

**Solution.**

(a) If $c \notin \operatorname{range} T$ then $\{x \in V : Tx = c\}$ must be empty. Suppose that $c \in \operatorname{range} T$, so that $c = Tu$ for some $u \in V$. We claim that $\{x \in V : Tx = c\} = u + \operatorname{null} T$. Indeed, for $x = u + v \in u + \operatorname{null} T$ we have

$$Tx = Tu + Tv = c \quad \Rightarrow \quad x \in \{x \in V : Tx = c\}.$$

Thus $u + \operatorname{null} T \subseteq \{x \in V : Tx = c\}$. If $x \in V$ is such that $Tx = c$ then observe that

$$Tx = c = Tu \quad \Rightarrow \quad T(x - u) = 0 \quad \Rightarrow \quad x - u \in \operatorname{null} T$$
$$\Rightarrow \quad x = u + v \text{ for some } v \in \operatorname{null} T.$$

Thus $\{x \in V : Tx = c\} \subseteq u + \operatorname{null} T$ and our claim follows.

(b) Consider a system of linear equations

$$\sum_{k=1}^{n} A_{1,k} x_k = c_1$$
$$\vdots$$
$$\sum_{k=1}^{n} A_{m,k} x_k = c_m.$$

Define $T \in \mathcal{L}(\mathbf{F}^n, \mathbf{F}^m)$ by



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$$T(x_1, ..., x_n) = \left( \sum_{k=1}^{n} A_{1,k} x_k, ..., \sum_{k=1}^{n} A_{m,k} x_k \right)$$

and let $c = (c_1, ..., c_m) \in \mathbf{F}^m$. The solution set of the system of linear equations is then precisely the set

$$\{(x_1, ..., x_n) \in \mathbf{F}^n : T(x_1, ..., x_n) = c\}.$$

As we showed in part (a), this set is either empty or is a translate of the subspace $\operatorname{null} T$.

**Exercise 3.E.9.** Prove that a nonempty subset $A$ of $V$ is a translate of some subspace of $V$ if and only if $\lambda v + (1 - \lambda)w \in A$ for all $v, w \in A$ and all $\lambda \in \mathbf{F}$.

**Solution.** Suppose that $A = x + U$ for some $x \in V$ and some subspace $U$ of $V$. Let $v, w \in A$ and $\lambda \in \mathbf{F}$ be given and note that $v = x + t$ and $w = x + u$ for some $t, u \in U$. It follows that

$$\lambda v + (1 - \lambda)w = \lambda(x + t) + (1 - \lambda)(x + u) = x + (\lambda t + (1 - \lambda)u) \in x + U = A.$$

Now suppose that $\lambda v + (1 - \lambda)w \in A$ for all $v, w \in A$ and all $\lambda \in \mathbf{F}$. Because $A$ is non-empty, there is some $x \in A$. We claim that $-x + A$ is a subspace of $V$. Certainly $0 = -x + x \in -x + A$. Suppose that $-x + v, -x + w \in -x + A$ and $\lambda \in \mathbf{F}$, and observe that

$$x, v \in A \quad \Rightarrow \quad -x + 2v \in A \quad \text{and} \quad x, w \in A \quad \Rightarrow \quad -x + 2w \in A.$$

It follows that

$$\left(-\tfrac{1}{2}x + v\right) + \left(-\tfrac{1}{2}x + w\right) = -x + v + w \in A \quad \Rightarrow \quad -2x + v + w \in -x + A.$$

Thus $(-x + v) + (-x + w) \in -x + A$, so that $-x + A$ is closed under vector addition. Furthermore, $-x + A$ is closed under scalar multiplication:

$$x, v \in A \quad \Rightarrow \quad \lambda(-x + v) = -x + (\lambda v + (1 - \lambda)x) \in -x + A.$$

Thus $-x + A$ is a subspace of $V$. It follows that $A = x + (-x + A)$ is a translate of the subspace $-x + A$.

**Exercise 3.E.10.** Suppose $A_1 = v + U_1$ and $A_2 = w + U_2$ for some $v, w \in V$ and some subspaces $U_1, U_2$ of $V$. Prove that the intersection $A_1 \cap A_2$ is either a translate of some subspace of $V$ or is the empty set.

**Solution.** If $A_1 \cap A_2$ is non-empty, so that there is some $x \in A_1 \cap A_2$, then by 3.101 we have $A_1 = x + U_1$ and $A_2 = x + U_2$. We claim that $A_1 \cap A_2 = x + (U_1 \cap U_2)$. If $y \in A_1 \cap A_2$ then $y = x + u_1 = x + u_2$ for some $u_1 \in U_1$ and some $u_2 \in U_2$. It follows that $u_1 = u_2 \in U_1 \cap U_2$ and thus $y \in x + (U_1 \cap U_2)$. This gives us the inclusion $A_1 \cap A_2 \subseteq x + (U_1 \cap U_2)$. If $y = x + u \in x + (U_1 \cap U_2)$ then $y = x + u \in x + U_1 = A_1$ and $y = x + u \in x + U_2 = A_2$, so that $y \in A_1 \cap A_2$. Thus $x + (U_1 \cap U_2) \subseteq A_1 \cap A_2$ and our claim follows.



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**Exercise 3.E.11.** Suppose $U = \{(x_1, x_2, ...) \in \mathbf{F}^\infty : x_k \neq 0 \text{ for only finitely many } k\}$.

(a) Show that $U$ is a subspace of $\mathbf{F}^\infty$.

(b) Prove that $\mathbf{F}^\infty / U$ is infinite-dimensional.

**Solution.**

(a) Notice that

$$U = \{(x_1, x_2, ...) \in \mathbf{F}^\infty : \text{there exists } K \text{ such that } x_k = 0 \text{ for all } k \geq K\}.$$

Certainly $(0, 0, 0, ...) \in U$. Suppose that $(x_1, x_2, ...), (y_1, y_2, ...) \in U$ and $\lambda \in \mathbf{F}$. There are positive integers $K, L$ such that $x_k = 0$ for all $k \geq K$ and $y_k = 0$ for all $k \geq L$. It follows that $x_k + y_k = 0$ for all $k \geq \max\{K, L\}$ and $\lambda x_k = 0$ for all $k \geq K$. Thus

$$(x_1, x_2, ...) + (y_1, y_2, ...) \in U \quad \text{and} \quad \lambda(x_1, x_2, ...) \in U.$$

It follows that $U$ is a subspace of $\mathbf{F}^\infty$.

(b) For $x \in \mathbf{F}^\infty$ we will use the notation $x(k)$ to denote the $k^{\text{th}}$ term of $x$. Let $e_n \in \mathbf{F}^\infty$ be the sequence given by

$$e_n(k) = \begin{cases} 1 & \text{if } k \text{ is divisible by } 2^n, \\ 0 & \text{otherwise.} \end{cases}$$

$$e_1 = (0, 1, 0, 1, 0, 1, 0, 1, ...),$$
$$e_2 = (0, 0, 0, 1, 0, 0, 0, 1, ...),$$
$$e_3 = (0, 0, 0, 0, 0, 0, 0, 1, ...), \text{ etc.}$$

Let $m \in \mathbf{N}$ be given. We claim that the list $e_1 + U, ..., e_m + U$ is linearly independent. Suppose $a_1, ..., a_m$ are such that

$$a_1(e_1 + U) + \cdots + a_m(e_m + U) = (a_1 e_1 + \cdots + a_m e_m) + U = 0.$$

This is the case if and only if $e \in U$, where $e = a_1 e_1 + \cdots + a_m e_m$, which implies that there is a positive integer $K$ such that $e(k) = 0$ for all $k \geq K$. Let $N \in \mathbf{N}$ be such that $2^{mN} \geq K$ and note that $2^{mN} + 2$ is divisible by 2 but not by $2^2, ..., 2^m$. It follows that

$$0 = e(2^{mN} + 2) = a_1 e_1(2^{mN} + 2) + a_2 e_2(2^{mN} + 2) + \cdots + a_m e_m(2^{mN} + 2) = a_1.$$

Similarly, $2^{mN} + 2^2$ is divisible by $2^2$ but not by $2^3, ..., 2^m$ and thus

$$0 = e(2^{mN} + 2^2) = a_2 e_2(2^{mN} + 2^2) + a_3 e_3(2^{mN} + 2^2) + \cdots + a_m e_m(2^{mN} + 2^2) = a_2.$$

Continuing in this manner, we find that $a_1 = \cdots = a_m = 0$ and our claim follows.

We can now use Exercise 2.A.17 to conclude that $\mathbf{F}^\infty / U$ is infinite-dimensional.



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**Exercise 3.E.12.** Suppose $v_1, ..., v_m \in V$. Let

$$A = \{\lambda_1 v_1 + \cdots + \lambda_m v_m : \lambda_1, ..., \lambda_m \in \mathbf{F} \text{ and } \lambda_1 + \cdots + \lambda_m = 1\}.$$

(a) Prove that $A$ is a translate of some subspace of $V$.

(b) Prove that if $B$ is a translate of some subspace of $V$ and $\{v_1, ..., v_m\} \subseteq B$, then $A \subseteq B$.

(c) Prove that $A$ is a translate of some subspace of $V$ of dimension less than $m$.

**Solution.**

(a) Suppose $v = \sum_{k=1}^m \lambda_k v_k$ and $w = \sum_{k=1}^m \mu_k v_k$ belong to $A$ and $\gamma \in \mathbf{F}$. Observe that

$$\sum_{k=1}^m [\gamma\lambda_k + (1-\gamma)\mu_k] = \gamma \sum_{k=1}^m \lambda_k + (1-\gamma) \sum_{k=1}^m \mu_k = \gamma + (1-\gamma) = 1.$$

It follows that

$$\gamma v + (1-\gamma)w = \gamma \sum_{k=1}^m \lambda_k v_k + (1-\gamma) \sum_{k=1}^m \mu_k v_k = \sum_{k=1}^m [\gamma\lambda_k + (1-\gamma)\mu_k] v_k$$

belongs to $A$. Thus $A$ is a translate of some subspace of $V$ by Exercise 3.E.9.

(b) Suppose that $B = v + U$ for some $v \in U$ and some subspace $U$ of $V$, and suppose that $\{v_1, ..., v_m\} \subseteq B$, so that each $v_k = v + u_k$ for some $u_k \in U$. Let $\sum_{k=1}^m \lambda_k v_k \in A$ be given and observe that

$$\sum_{k=1}^m \lambda_k v_k = \sum_{k=1}^m \lambda_k (v + u_k) = \left( \sum_{k=1}^m \lambda_k \right) v + \sum_{k=1}^m \lambda_k u_k = v + \sum_{k=1}^m \lambda_k u_k \in v + U = B.$$

Thus $A \subseteq B$.

(c) If $m = 1$ then $A = \{v_1\} = v_1 + \{0\}$. If $m \geq 2$ then let $U = \operatorname{span}(v_2 - v_1, ..., v_m - v_1)$ and note that $\dim U \leq m - 1$. Note further that $v_1 + U$ is a translate of $U$ containing $\{v_1, ..., v_m\}$; it follows from part (b) that $A \subseteq v_1 + U$. Let $v_1 + \sum_{k=2}^m a_k (v_k - v_1)$ in $v_1 + U$ be given and observe that

$$v_1 + \sum_{k=2}^m a_k (v_k - v_1) = \left( 1 - \sum_{k=2}^m a_k \right) v_1 + \sum_{k=2}^m a_k v_k \in A.$$

Thus $v_1 + U \subseteq A$ and we may conclude that $A = v_1 + U$, where $\dim U \leq m - 1$.

**Exercise 3.E.13.** Suppose $U$ is a subspace of $V$ such that $V/U$ is finite-dimensional. Prove that $V$ is isomorphic to $U \times (V/U)$.

**Solution.** Let $v_1 + U, ..., v_m + U$ be a basis of $V/U$. Define $T \in \mathcal{L}(V/U, V)$ by $T(v_k + U) = v_k$ and notice that $\pi \circ T$ is the identity map on $V/U$. Now define



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$S : U \times (V/U) \to V$ by $S(u, v + U) = u + T(v + U)$; the linearity of $S$ follows from the linearity of $T$.

Suppose $(u, v + U)$ is such that

$$S(u, v + U) = u + T(v + U) = 0.$$

Using $\operatorname{null} \pi = U$ and $\pi \circ T = I$, applying $\pi$ to the equation above shows that $v + U = 0$. It follows that $T(v + U) = 0$ and hence that $u = 0$. Thus $S$ is injective.

For any $v \in V$ there are scalars $a_1, ..., a_m$ such that $v + U = a_1 v_1 + \cdots + a_m v_m + U$ and thus by 3.101 we have $v = u + a_1 v_1 + \cdots + a_m v_m$ for some $u \in U$. Observe that

$$\begin{aligned} S(u, a_1 v_1 + \cdots + a_m v_m + U) &= u + T(a_1 v_1 + \cdots + a_m v_m + U) \\ &= u + a_1 v_1 + \cdots + a_m v_m = v. \end{aligned}$$

Thus $S$ is surjective and we may conclude that $S$ is an isomorphism from $U \times (V/U)$ to $V$.

**Exercise 3.E.14.** Suppose $U$ and $W$ are subspaces of $V$ and $V = U \oplus W$. Suppose $w_1, ..., w_m$ is a basis of $W$. Prove that $w_1 + U, ..., w_m + U$ is a basis of $V/U$.

**Solution.** Suppose $a_1, ..., a_m$ are scalars such that $a_1 w_1 + \cdots + a_m w_m + U = 0$, which by 3.101 is the case if and only if $a_1 w_1 + \cdots + a_m w_m \in U$. Because the sum $U \oplus W$ is direct, 1.46 shows that $a_1 w_1 + \cdots + a_m w_m = 0$ and it follows from the linear independence of $w_1, ..., w_m$ that $a_1 = \cdots = a_m = 0$. Thus $w_1 + U, ..., w_m + U$ is linearly independent.

For any $v + U \in V/U$, we have $v = u + a_1 w_1 + \cdots + a_m w_m$ for some $u \in U$ and some scalars $a_1, ..., a_m$. It follows that

$$v + U = \pi(v) = a_1 w_1 + \cdots + a_m w_m + U = a_1(w_1 + U) + \cdots + a_m(w_m + U).$$

Thus $w_1 + U, ..., w_m + U$ spans $V/U$ and we may conclude that $w_1 + U, ..., w_m + U$ is a basis of $V/U$.

**Exercise 3.E.15.** Suppose $U$ is a subspace of $V$ and $v_1 + U, ..., v_m + U$ is a basis of $V/U$ and $u_1, ..., u_n$ is a basis of $U$. Prove that $v_1, ..., v_m, u_1, ..., u_n$ is a basis of $V$.

**Solution.** Suppose there are scalars $a_1, ..., a_m, b_1, ..., b_n$ such that

$$a_1 v_1 + \cdots + a_m v_m + b_1 u_1 + \cdots + b_n u_n = 0.$$

Applying $\pi$ to both sides of this equation shows that $a_1(v_1 + U) + \cdots + a_m(v_m + U) = 0$. The linear independence of $v_1 + U, ..., v_m + U$ then implies that $a_1 = \cdots = a_m = 0$, and the linear independence of $u_1, ..., u_n$ then gives us $b_1 = \cdots = b_n = 0$. Thus $v_1, ..., v_m, u_1, ..., u_n$ is linearly independent.

Let $v \in V$ be given. There are scalars $a_1, ..., a_m$ such that $v + U = a_1 v_1 + \cdots + a_m v_m + U$. It follows from 3.101 that $v = a_1 v_1 + \cdots + a_m v_m + u$ for some $u \in U$, so that there are scalars $b_1, ..., b_n$ such that



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$$v = a_1 v_1 + \cdots + a_m v_m + b_1 u_1 + \cdots + b_n u_n.$$

Thus $v_1, ..., v_m, u_1, ..., u_n$ spans $V$ and we may conclude that $v_1, ..., v_m, u_1, ..., u_n$ is a basis of $V$.

**Exercise 3.E.16.** Suppose $\varphi \in \mathcal{L}(V, \mathbf{F})$ and $\varphi \neq 0$. Prove that $\dim V/(\operatorname{null} \varphi) = 1$.

**Solution.** There is some $u \in V$ such that $\varphi(u) \neq 0$; Exercise 3.B.30 then implies that

$$V = \operatorname{null} \varphi \oplus \{au : a \in \mathbf{F}\}.$$

Notice that $u$ is a basis of $\{au : a \in \mathbf{F}\}$. It follows from Exercise 3.E.14 that $u + \operatorname{null} \varphi$ is a basis of $V/(\operatorname{null} \varphi)$. Thus $\dim V/(\operatorname{null} \varphi) = 1$.

**Exercise 3.E.17.** Suppose $U$ is a subspace of $V$ such that $\dim V/U = 1$. Prove that there exists $\varphi \in \mathcal{L}(V, \mathbf{F})$ such that $\operatorname{null} \varphi = U$.

**Solution.** Let $w + U$ be a basis of $V/U$. For any $v \in V$ there is a unique $a \in \mathbf{F}$ such that $v + U = aw + U$. Given this uniqueness, the map $\varphi : V \to \mathbf{F}$ defined by $\varphi(v) = a$ is well-defined. Moreover, $\varphi$ is linear. For any $v_1, v_2 \in V$ there are unique scalars $a_1, a_2$ such that $v_1 + U = a_1 w + U$ and $v_2 + U = a_2 w + U$. Let $\lambda \in \mathbf{F}$ be given. Since

$$(v_1 + v_2) + U = (a_1 + a_2)w + U \quad \text{and} \quad \lambda v_1 + U = (\lambda a_1)w + U,$$

$a_1 + a_2$ must be the unique coefficient of $w$ in the linear combination representing $(v_1 + v_2) + U$ in the basis $w + U$ of $V/U$; similarly, $\lambda a_1$ must be the unique coefficient of $w$ in the representation of $\lambda v_1 + U$. It follows that

$$\varphi(v_1 + v_2) = a_1 + a_2 = \varphi(v_1) + \varphi(v_2) \quad \text{and} \quad \varphi(\lambda v_1) = \lambda a_1 = \lambda \varphi(v_1).$$

Thus $\varphi$ is linear.

For any $u \in U$ we have $u + U = 0w + U$ and thus $\varphi(u) = 0$, so that $U \subseteq \operatorname{null} \varphi$. Conversely, if $v \in V$ is such that $\varphi(v) = 0$ then $v + U = 0w + U$; it follows from 3.101 that $v \in U$. Thus $\operatorname{null} \varphi = U$.

**Exercise 3.E.18.** Suppose that $U$ is a subspace of $V$ such that $V/U$ is finite-dimensional.

(a) Show that if $W$ is a finite-dimensional subspace of $V$ and $V = U + W$, then $\dim W \geq \dim V/U$.

(b) Prove that there exists a finite-dimensional subspace $W$ of $V$ such that $\dim W = \dim V/U$ and $V = U \oplus W$.

**Solution.**

(a) Let $w_1, ..., w_m$ be a basis of $W$. For any $v + U \in V/U$ we have



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$$v = u + a_1 w_1 + \cdots + a_m w_m$$

for some $u \in U$ and some scalars $a_1, ..., a_m$, so that

$$v + U = a_1(w_1 + U) + \cdots + a_m(w_m + U).$$

Thus $w_1 + U, ..., w_m + U$ spans $V/U$. It follows that $\dim V/U \leq m = \dim W$.

(b) Let $v_1 + U, ..., v_m + U$ be a basis of $V/U$ and define $T \in \mathcal{L}(V/U, V)$ by $T(v_k + U) = v_k$. Notice that

$$\pi \circ T = I \quad \text{and} \quad \operatorname{range} T = \operatorname{span}(v_1, ..., v_m).$$

It follows from Exercise 3.B.19 that $T$ is injective; letting $W = \operatorname{range} T$, we see that $T$ is an isomorphism between $V/U$ and $W$. Thus $\dim W = \dim V/U$.

For any $v \in V$ we have $v + U = a_1 v_1 + \cdots + a_m v_m + U$ for some scalars $a_1, ..., a_m$. It follows from 3.101 that $v = u + a_1 v_1 + \cdots + a_m v_m$ for some $u \in U$. Thus $V = U + W$. Suppose that $v \in U \cap W$, so that $v + U = 0$ and $v = a_1 v_1 + \cdots + a_m v_m$ for some scalars $a_1, ..., a_m$. It follows that

$$0 = v + U = a_1(v_1 + U) + \cdots + a_m(v_m + U).$$

The linear independence of $v_1 + U, ..., v_m + U$ then implies that $a_1 = \cdots = a_m = 0$, whence $v = 0$. Thus $U \cap W = \{0\}$ and we may use 1.46 to conclude that the sum $V = U \oplus W$ is direct.

**Exercise 3.E.19.** Suppose $T \in \mathcal{L}(V, W)$ and $U$ is a subspace of $V$. Let $\pi$ denote the quotient map from $V$ onto $V/U$. Prove that there exists $S \in \mathcal{L}(V/U, W)$ such that $T = S \circ \pi$ if and only if $U \subseteq \operatorname{null} T$.

**Solution.** If there exists such a map $S$ and $u \in U$, then

$$Tu = S(\pi(u)) = S(0) = 0.$$

Thus $U \subseteq \operatorname{null} T$.

Now suppose $U \subseteq \operatorname{null} T$. Define $S : V/U \to W$ by $S(v + U) = Tv$. This map is well-defined:

$$v_1 + U = v_2 + U \quad \Leftrightarrow \quad v_1 - v_2 \in U \quad \Rightarrow \quad v_1 - v_2 \in \operatorname{null} T \quad \Rightarrow \quad Tv_1 = Tv_2.$$

The linearity of $S$ follows from the linearity of $T$ and the definition of $S$ makes it clear that $T = S \circ \pi$.



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## 3.F. Duality

**Exercise 3.F.1.** Explain why each linear functional is surjective or is the zero map.

**Solution.** Suppose $\varphi \in V'$ is non-zero, so that there is some $v \in V$ such that $\varphi(v) \neq 0$, and notice that

$$\varphi\left(\frac{\lambda}{\varphi(v)}v\right) = \lambda$$

for any $\lambda \in \mathbf{F}$. Thus $\varphi$ is surjective.

**Exercise 3.F.2.** Give three distinct examples of linear functionals on $\mathbf{R}^{[0,1]}$.

**Solution.** For $k \in \{0, 1, 2\}$, define $\varphi_k : \mathbf{R}^{[0,1]} \to \mathbf{R}$ by $\varphi_k(f) = f\left(\frac{k}{2}\right)$. It is straightforward to check that each $\varphi_k$ is a linear functional on $\mathbf{R}^{[0,1]}$.

**Exercise 3.F.3.** Suppose $V$ is finite-dimensional and $v \in V$ with $v \neq 0$. Prove that there exists $\varphi \in V'$ such that $\varphi(v) = 1$.

**Solution.** Let $v_1 = v$ and extend this to a basis $v_1, ..., v_n$ of $V$. Now define $\varphi \in V'$ by $\varphi(v_k) = 1$ for each $k \in \{1, ..., n\}$.

**Exercise 3.F.4.** Suppose $V$ is finite-dimensional and $U$ is a subspace of $V$ such that $U \neq V$. Prove that there exists $\varphi \in V'$ such that $\varphi(u) = 0$ for every $u \in U$ but $\varphi \neq 0$.

**Solution.** Let $u_1, ..., u_m$ be a basis of $U$ and extend this to a basis $u_1, ..., u_m, v_1, ..., v_n$ of $V$; there must be at least one $v_k$ since $U \neq V$. Define $\varphi \in V'$ by

$$\varphi(u_1) = \cdots = \varphi(u_m) = 0 \quad \text{and} \quad \varphi(v_1) = \cdots = \varphi(v_n) = 1.$$

It follows that $\varphi(u) = 0$ for all $u \in U$ but $\varphi \neq 0$.

**Exercise 3.F.5.** Suppose $T \in \mathcal{L}(V, W)$ and $w_1, ..., w_m$ is a basis of $\operatorname{range} T$. Hence for each $v \in V$, there exist unique numbers $\varphi_1(v), ..., \varphi_m(v)$ such that

$$Tv = \varphi_1(v)w_1 + \cdots + \varphi_m(v)w_m,$$

thus defining functions $\varphi_1, ..., \varphi_m$ from $V$ to $\mathbf{F}$. Show that each of the functions $\varphi_1, ..., \varphi_m$ is a linear functional on $V$.

**Solution.** Let us think of $T$ as a linear map $V \to \operatorname{range} T$, so that the dual map $T'$ is a linear map $(\operatorname{range} T)' \to V'$. Let $\psi_1, ..., \psi_m$ be the dual basis to $w_1, ..., w_m$. For any $v \in V$, the definition of each $\varphi_k$ and 3.114 show that



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$$\varphi_1(v)w_1 + \cdots + \varphi_m(v)w_m = Tv = \psi_1(Tv)w_1 + \cdots + \psi_m(Tv)w_m.$$

It follows from unique representation that $\varphi_k(v) = \psi_k(Tv)$ for all $v \in V$, i.e.

$$\varphi_k = \psi_k \circ T = T'(\psi_k) \in V'.$$

**Exercise 3.F.6.** Suppose $\varphi, \beta \in V'$. Prove that $\operatorname{null}\varphi \subseteq \operatorname{null}\beta$ if and only if there exists $c \in \mathbf{F}$ such that $\beta = c\varphi$.

**Solution.** By Exercise 3.B.25 we have $\operatorname{null}\varphi \subseteq \operatorname{null}\beta$ if and only if there exists $E \in \mathcal{L}(\mathbf{F})$ such that $\beta = E \circ \varphi$. If there exists such an $E$, then let $c = E(1)$ and observe that, for any $v \in V$,

$$\beta(v) = E(\varphi(v)) = E(1)\varphi(v) = c\varphi(v) \quad \Rightarrow \quad \beta = c\varphi.$$

Conversely, if there exists such a $c \in \mathbf{F}$ then define $E \in \mathcal{L}(\mathbf{F})$ by $E(x) = cx$ and observe that, for any $v \in V$,

$$\beta(v) = c\varphi(v) = E(\varphi(v)) \quad \Rightarrow \quad \beta = E \circ \varphi.$$

**Exercise 3.F.7.** Suppose that $V_1, ..., V_m$ are vector spaces. Prove that $(V_1 \times \cdots \times V_m)'$ and $V_1' \times \cdots \times V_m'$ are isomorphic vector spaces.

**Solution.** This is immediate from Exercise 3.E.3, taking $W = \mathbf{F}$.

**Exercise 3.F.8.** Suppose $v_1, ..., v_n$ is a basis of $V$ and $\varphi_1, ..., \varphi_n$ is the dual basis of $V'$. Define $\Gamma : V \to \mathbf{F}^n$ and $\Lambda : \mathbf{F}^n \to V$ by

$$\Gamma(v) = (\varphi_1(v), ..., \varphi_n(v)) \quad \text{and} \quad \Lambda(a_1, ..., a_n) = a_1v_1 + \cdots + a_nv_n.$$

Explain why $\Gamma$ and $\Lambda$ are inverses of each other.

**Solution.** It is straightforward to verify that $\Gamma$ and $\Lambda$ are linear. For any $(a_1, ..., a_n) \in \mathbf{F}^n$, 3.114 shows that

$$\begin{aligned}\Gamma(\Lambda(a_1, ..., a_n)) &= \Gamma(a_1v_1 + \cdots + a_nv_n) \\ &= (\varphi_1(a_1v_1 + \cdots + a_nv_n), ..., \varphi_n(a_1v_1 + \cdots + a_nv_n)) = (a_1, ..., a_n).\end{aligned}$$

Thus $\Gamma\Lambda$ is the identity on $\mathbf{F}^n$. For any $v \in V$, 3.114 gives us

$$\Lambda(\Gamma(v)) = \Lambda(\varphi_1(v), ..., \varphi_n(v)) = \varphi_1(v)v_1 + \cdots + \varphi_n(v)v_n = v.$$

Thus $\Lambda\Gamma$ is the identity on $V$ and we may conclude that $\Gamma$ and $\Lambda$ are inverses of each other.



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**Exercise 3.F.9.** Suppose $m$ is a positive integer. Show that the dual basis of the basis $1, x, ..., x^m$ of $\mathcal{P}_m(\mathbf{R})$ is $\varphi_0, \varphi_1, ..., \varphi_m$, where

$$\varphi_k(p) = \frac{p^{(k)}(0)}{k!}.$$

*Here $p^{(k)}$ denotes the $k^{th}$ derivative of $p$, with the understanding that the $0^{th}$ derivative of $p$ is $p$.*

**Solution.** Let $k \in \{0, ..., m\}$ be given and observe that, for $j \neq k$,

$$\varphi_k(x^k) = \frac{(x^k)^{(k)}(0)}{k!} = \frac{k!}{k!} = 1 \quad \text{and} \quad \varphi_k(x^j) = \frac{(x^j)^{(k)}(0)}{k!} = \frac{0}{k!} = 0.$$

Thus the linear functionals $\varphi_k$ and $p \mapsto \frac{p^{(k)}(0)}{k!}$ agree on the basis $1, x, ..., x^m$; it follows that they are equal as functions.

**Exercise 3.F.10.** Suppose $m$ is a positive integer.

(a) Show that $1, x-5, ..., (x-5)^m$ is a basis of $\mathcal{P}_m(\mathbf{R})$.

(b) What is the dual basis of the basis in (a)?

**Solution.**

(a) This is immediate from Exercise 2.C.9.

(b) We can argue as we did in Exercise 3.F.9 to see that the dual basis is $\varphi_0, ..., \varphi_m$, where

$$\varphi_k(p) = \frac{p^{(k)}(5)}{k!}.$$

**Exercise 3.F.11.** Suppose $v_1, ..., v_n$ is a basis of $V$ and $\varphi_1, ..., \varphi_n$ is the corresponding dual basis of $V'$. Suppose $\psi \in V'$. Prove that

$$\psi = \psi(v_1)\varphi_1 + \cdots + \psi(v_n)\varphi_n.$$

**Solution.** Let $v = a_1v_1 + \cdots + a_nv_n \in V$ be given and note that, by 3.114, we have $\varphi_k(a_1v_1 + \cdots + a_nv_n) = a_k$. It follows that

$$\begin{aligned}
[\psi(v_1)\varphi_1 + \cdots + \psi(v_n)\varphi_n](a_1v_1 + \cdots + a_nv_n) &= \psi(v_1)\varphi_1(a_1v_1 + \cdots + a_nv_n) \\
&\quad + \cdots + \psi(v_n)\varphi_n(a_1v_1 + \cdots + a_nv_n) \\
&= \psi(v_1)a_1 + \cdots + \psi(v_n)a_n \\
&= \psi(a_1v_1 + \cdots + a_nv_n).
\end{aligned}$$

Thus $\psi(v_1)\varphi_1 + \cdots + \psi(v_n)\varphi_n = \psi$.



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**Exercise 3.F.12.** Suppose $S, T \in \mathcal{L}(V, W)$.

(a) Prove that $(S+T)' = S' + T'$.

(b) Prove that $(\lambda T)' = \lambda T'$ for all $\lambda \in \mathbf{F}$.

*This exercise asks you to verify (a) and (b) in 3.120.*

**Solution.**

(a) For any $\psi \in W'$ and $v \in V$, observe that

$$\begin{aligned}\big[(S+T)'(\psi)\big](v) &= \psi((S+T)(v)) = \psi(Sv+Tv) = \psi(Sv)+\psi(Tv)\\ &= [S'(\psi)](v) + [T'(\psi)](v) = [S'(\psi)+T'(\psi)](v) = [(S'+T')(\psi)](v).\end{aligned}$$

Thus $(S+T)' = S' + T'$.

(b) For any $\psi \in W'$ and $v \in V$, observe that

$$\begin{aligned}\big[(\lambda T)'(\psi)\big](v) &= \psi((\lambda T)(v)) = \psi(\lambda Tv) = \lambda\psi(Tv)\\ &= \lambda[T'(\psi)](v) = [\lambda T'(\psi)](v) = [(\lambda T')(\psi)](v).\end{aligned}$$

Thus $(\lambda T)' = \lambda T'$.

**Exercise 3.F.13.** Show that the dual map of the identity operator on $V$ is the identity operator on $V'$.

**Solution.** For any $\varphi \in V'$ and $v \in V$, observe that

$$[I'(\varphi)](v) = \varphi(Iv) = \varphi(v).$$

Thus $I'(\varphi) = \varphi$, i.e. $I'$ is the identity operator on $V'$.

**Exercise 3.F.14.** Define $T : \mathbf{R}^3 \to \mathbf{R}^2$ by

$$T(x, y, z) = (4x + 5y + 6z, 7x + 8y + 9z).$$

Suppose $\varphi_1, \varphi_2$ denotes the dual basis of the standard basis of $\mathbf{R}^2$ and $\psi_1, \psi_2, \psi_3$ denotes the dual basis of the standard basis of $\mathbf{R}^3$.

(a) Describe the linear functionals $T'(\varphi_1)$ and $T'(\varphi_2)$.

(b) Write $T'(\varphi_1)$ and $T'(\varphi_2)$ as linear combinations of $\psi_1, \psi_2, \psi_3$.

**Solution.**

(a) By the definition of the dual map, we have

$$\begin{aligned}[T'(\varphi_1)](x, y, z) &= \varphi_1(T(x, y, z)) = \varphi_1(4x + 5y + 6z, 7x + 8y + 9z) = 4x + 5y + 6z,\\ [T'(\varphi_2)](x, y, z) &= \varphi_2(T(x, y, z)) = \varphi_2(4x + 5y + 6z, 7x + 8y + 9z) = 7x + 8y + 9z.\end{aligned}$$



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(b) Note that

$$\psi_1(x,y,z) = x, \quad \psi_2(x,y,z) = y, \quad \text{and} \quad \psi_3(x,y,z) = z.$$

Thus

$$T'(\varphi_1) = 4\psi_1 + 5\psi_2 + 6\psi_3 \quad \text{and} \quad T'(\varphi_2) = 6\psi_1 + 7\psi_2 + 8\psi_3.$$

**Exercise 3.F.15.** Define $T : \mathcal{P}(\mathbf{R}) \to \mathcal{P}(\mathbf{R})$ by

$$(Tp)(x) = x^2 p(x) + p''(x)$$

for each $x \in \mathbf{R}$.

(a) Suppose $\varphi \in \mathcal{P}(\mathbf{R})'$ is defined by $\varphi(p) = p'(4)$. Describe the linear functional $T'(\varphi)$ on $\mathcal{P}(\mathbf{R})$.

(b) Suppose $\varphi \in \mathcal{P}(\mathbf{R})'$ is defined by $\varphi(p) = \int_0^1 p$. Evaluate $(T'(\varphi))(x^3)$.

**Solution.**

(a) We have

$$\begin{aligned}[T'(\varphi)](p) = \varphi(Tp) = \varphi(x^2p + p'') &= (x^2p + p'')'(4) \\ &= (2xp + x^2p' + p''')(4) = 8p(4) + 16p'(4) + p'''(4).\end{aligned}$$

(b) We have

$$[T'(\varphi)](x^3) = \varphi(Tx^3) = \varphi(x^5 + 6x) = \int_0^1 x^5 + 6x \, \mathrm{d}x = \frac{19}{6}.$$

**Exercise 3.F.16.** Suppose $W$ is finite-dimensional and $T \in \mathcal{L}(V,W)$. Prove that

$$T' = 0 \quad \Leftrightarrow \quad T = 0.$$

**Solution.** If $T = 0$ and $\psi \in W'$, then

$$T'(\psi) = \psi \circ T = \psi \circ 0 = 0.$$

Thus $T' = 0$.

Now suppose that $T' = 0$, so that $\operatorname{null} T' = W'$. It follows from 3.128(a) that $(\operatorname{range} T)^0 = W'$, which by 3.127(b) is equivalent to $\operatorname{range} T = \{0\}$. Thus $T = 0$.

**Exercise 3.F.17.** Suppose $V$ and $W$ are finite-dimensional and $T \in \mathcal{L}(V,W)$. Prove that $T$ is invertible if and only if $T' \in \mathcal{L}(W',V')$ is invertible.

**Solution.** If either of $T, T'$ is invertible then, using 3.111, it must be the case that

$$\dim V' = \dim V = \dim W = \dim W'.$$

Thus, by 3.65 and 3.129,



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$$T \text{ is invertible} \quad \Leftrightarrow \quad T \text{ is surjective} \quad \Leftrightarrow \quad T' \text{ is injective} \quad \Leftrightarrow \quad T' \text{ is invertible.}$$

**Exercise 3.F.18.** Suppose $V$ and $W$ are finite-dimensional. Prove that the map that takes $T \in \mathcal{L}(V,W)$ to $T' \in \mathcal{L}(W',V')$ is an isomorphism of $\mathcal{L}(V,W)$ onto $\mathcal{L}(W',V')$.

**Solution.** Let $\Phi : \mathcal{L}(V,W) \to \mathcal{L}(W',V')$ be the map in question, i.e. $\Phi(T) = T'$. Exercise 3.F.12 shows that $\Phi$ is linear and Exercise 3.F.16 shows that $\Phi$ is injective. Note that, by 3.72 and 3.111,

$$\dim \mathcal{L}(V,W) = (\dim V)(\dim W) = (\dim V')(\dim W') = \dim \mathcal{L}(W',V').$$

3.65 allows us to conclude that $\Phi$ is an isomorphism.

**Exercise 3.F.19.** Suppose $U \subseteq V$. Explain why

$$U^0 = \{\varphi \in V' : U \subseteq \operatorname{null} \varphi\}.$$

**Solution.** This is immediate from the equivalence

$$\varphi(u) = 0 \text{ for all } u \in U \quad \Leftrightarrow \quad U \subseteq \operatorname{null} \varphi.$$

**Exercise 3.F.20.** Suppose $V$ is finite-dimensional and $U$ is a subspace of $V$. Show that

$$U = \{v \in V : \varphi(v) = 0 \text{ for every } \varphi \in U^0\}.$$

**Solution.** If $v \in U$ then certainly $\varphi(v) = 0$ for every $\varphi \in U^0$. Suppose $v \notin U$ and let $u_1, ..., u_m$ be a basis of $U$. Let $v_1 = v$ and note that the list $u_1, ..., u_m, v_1$ is linearly independent since $v_1 \notin U = \operatorname{span}(u_1, ..., u_m)$. Extend this list to a basis $u_1, ..., u_m, v_1, ..., v_n$ of $V$ and define $\varphi \in V'$ by

$$\varphi(u_1) = \cdots = \varphi(u_m) = 0 \quad \text{and} \quad \varphi(v_1) = \cdots = \varphi(v_n) = 1.$$

It follows that $\varphi \in U^0$ and $\varphi(v) \neq 0$. Thus

$$v \in U \quad \Leftrightarrow \quad v \in \{v \in V : \varphi(v) = 0 \text{ for every } \varphi \in U^0\}.$$

**Exercise 3.F.21.** Suppose $V$ is finite-dimensional and $U$ and $W$ are subspaces of $V$.

(a) Prove that $W^0 \subseteq U^0$ if and only if $U \subseteq W$.

(b) Prove that $W^0 = U^0$ if and only if $U = W$.

**Solution.** For a subspace $U$ of $V$, let

$$A_U = \{v \in V : \varphi(v) = 0 \text{ for every } \varphi \in U^0\}.$$



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(a) If $W^0 \subseteq U^0$ and $v \in A_U$ then in particular $\varphi(v) = 0$ for every $\varphi \in W^0$, i.e. $v \in A_W$. Thus $A_U \subseteq A_W$, which by Exercise 3.F.21 is equivalent to $U \subseteq W$. If $U \subseteq W$ and $\varphi \in W^0$ then in particular $\varphi(v) = 0$ for all $v \in U$, i.e. $\varphi \in U^0$. Thus $W^0 \subseteq U^0$.

(b) This follows from part (a):

$$W^0 = U^0 \quad \Leftrightarrow \quad W^0 \subseteq U^0 \text{ and } U^0 \subseteq W^0 \quad \Leftrightarrow \quad U \subseteq W \text{ and } W \subseteq U \quad \Leftrightarrow \quad U = W.$$

**Exercise 3.F.22.** Suppose $V$ is finite-dimensional and $U$ and $W$ are subspaces of $V$.

(a) Show that $(U + W)^0 = U^0 \cap W^0$.

(b) Show that $(U \cap W)^0 = U^0 + W^0$.

**Solution.**

(a) Suppose $\varphi \in (U + W)^0$. For any $u \in U$ and $w \in W$ observe that $u, w \in U + W$, so that $\varphi(u) = \varphi(w) = 0$. Thus $\varphi \in U^0 \cap W^0$ and it follows that $(U + W)^0 \subseteq U^0 \cap W^0$.

Suppose $\varphi \in U^0 \cap W^0$ and observe that, for any $u + w \in U + W$, we have

$$\varphi(u + w) = \varphi(u) + \varphi(w) = 0 \quad \Rightarrow \quad \varphi \in (U + W)^0.$$

It follows that $U^0 \cap W^0 \subseteq (U + W)^0$ and thus $(U + W)^0 = U^0 \cap W^0$.

(b) Suppose $\varphi \in (U \cap W)^0$. There are subspaces $X, Y$ of $V$ such that $V = U \oplus X$ and $V = W \oplus Y$. Define $\psi, \beta \in V'$ by

$$\psi(u + x) = \tfrac{1}{2}\varphi(x) \quad \text{and} \quad \beta(w + y) = \tfrac{1}{2}\varphi(y).$$

It is straightforward to verify that $\psi \in U^0$ and $\beta \in W^0$. Let $v = u + x = w + y \in V$ be given and note that, because $\varphi \in (U \cap W)^0$, $\varphi(v) = \varphi(x) = \varphi(y)$. It follows that

$$\varphi(v) = \tfrac{1}{2}\varphi(v) + \tfrac{1}{2}\varphi(v) = \tfrac{1}{2}\varphi(x) + \tfrac{1}{2}\varphi(y) = \psi(v) + \beta(v).$$

Thus $\varphi \in U^0 + W^0$, whence $(U \cap W)^0 \subseteq U^0 + W^0$.

Now suppose that $\varphi \in U^0 + W^0$, so that $\varphi = \psi + \beta$ for some $\psi \in U^0$ and some $\beta \in W^0$. Let $v \in U \cap W$ be given and observe that

$$\varphi(v) = \psi(v) + \beta(v) = 0 \quad \Rightarrow \quad \varphi \in (U \cap W)^0.$$

It follows that $U^0 + W^0 \subseteq (U \cap W)^0$ and thus $(U \cap W)^0 = U^0 + W^0$.

**Exercise 3.F.23.** Suppose $V$ is finite-dimensional and $\varphi_1, \ldots, \varphi_m \in V'$. Prove that the following three sets are equal to each other.

(a) $\operatorname{span}(\varphi_1, \ldots, \varphi_m)$

(b) $((\operatorname{null} \varphi_1) \cap \cdots \cap (\operatorname{null} \varphi_m))^0$

(c) $\{\varphi \in V' : (\operatorname{null} \varphi_1) \cap \cdots \cap (\operatorname{null} \varphi_m) \subseteq \operatorname{null} \varphi\}$



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**Solution.** (b) and (c) are equal by Exercise 3.F.19.

To show that (a) and (b) are equal, let us prove the following lemma.

**Lemma L.3.** If $\varphi \in V'$ then $\operatorname{span}(\varphi) \subseteq (\operatorname{null}\varphi)^0$; if $V$ is finite-dimensional then this containment is an equality.

*Proof.* For any $a \in \mathbf{F}$ and $v \in \operatorname{null}\varphi$ we have $a\varphi(v) = 0$. Thus $\operatorname{span}(\varphi) \subseteq (\operatorname{null}\varphi)^0$. Now observe that $\dim \operatorname{span}(\varphi) = \dim \operatorname{range}\varphi$, since

$$\begin{aligned} \varphi = 0 \quad &\Rightarrow \quad \dim \operatorname{span}(\varphi) = 0 = \dim \operatorname{range}\varphi, \\ \varphi \neq 0 \quad &\Rightarrow \quad \dim \operatorname{span}(\varphi) = 1 = \dim \operatorname{range}\varphi, \end{aligned}$$

where we have used Exercise 3.F.1 for the second implication. Assuming that $V$ is finite-dimensional, we can use (3.21) and (3.125) to obtain the equality

$$\dim \operatorname{span}(\varphi) = \dim \operatorname{range}\varphi = \dim (\operatorname{null}\varphi)^0.$$

2.39 allows us to conclude that $\operatorname{span}(\varphi) = (\operatorname{null}\varphi)^0$. □

Using Exercise 3.F.22 and Lemma L.3, observe that

$$\begin{aligned} ((\operatorname{null}\varphi_1) \cap \cdots \cap (\operatorname{null}\varphi_m))^0 &= (\operatorname{null}\varphi_1)^0 + \cdots + (\operatorname{null}\varphi_m)^0 \\ &= \operatorname{span}(\varphi_1) + \cdots + \operatorname{span}(\varphi_m) = \operatorname{span}(\varphi_1, ..., \varphi_m). \end{aligned}$$

Thus (a) and (b) are equal.

**Exercise 3.F.24.** Suppose $V$ is finite-dimensional and $v_1, ..., v_m \in V$. Define a linear map $\Gamma : V' \to \mathbf{F}^m$ by $\Gamma(\varphi) = (\varphi(v_1), ..., \varphi(v_m))$.

(a) Prove that $v_1, ..., v_m$ spans $V$ if and only if $\Gamma$ is injective.

(b) Prove that $v_1, ..., v_m$ is linearly independent if and only if $\Gamma$ is surjective.

**Solution.** Let $e_1, ..., e_m$ be the standard basis of $\mathbf{F}^m$ and let $\psi_1, ..., \psi_m$ be the corresponding dual basis of $(\mathbf{F}^m)'$, so that the map $\Psi : \mathbf{F}^m \to (\mathbf{F}^m)'$ given by $\Psi(e_k) = \psi_k$ is an isomorphism. Define $T \in \mathcal{L}(\mathbf{F}^m, V)$ by $Te_k = v_k$, i.e.

$$T(x_1, ..., x_m) = x_1 v_1 + \cdots + x_m v_m.$$

For any $\varphi \in V'$ and $k \in \{1, ..., m\}$, observe that

$$\begin{aligned} [T'(\varphi)](e_k) = \varphi(Te_k) = \varphi(v_k) &= \sum_{j=1}^{m} \varphi(v_j)\psi_j(e_k) \\ &= [\Psi(\varphi(v_1), ..., \varphi(v_m))](e_k) = [\Psi(\Gamma(\varphi))](e_k). \end{aligned}$$

Thus $T' = \Psi \circ \Gamma$. Because $\Psi$ is a bijection, it follows that the injectivity of $\Gamma$ is equivalent to the injectivity of $T'$ and the surjectivity of $\Gamma$ is equivalent to the surjectivity of $T'$.



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(a) Observe that

$$\operatorname{span}(v_1, ..., v_m) = V \quad \Leftrightarrow \quad T \text{ is surjective} \quad \Leftrightarrow \quad T' \text{ is injective} \quad \Leftrightarrow \quad \Gamma \text{ is injective},$$

where the first equivalence follows from Exercise 3.B.3, the second equivalence follows from 3.129, and the third equivalence follows from our previous discussion.

(b) Observe that

$$v_1, ..., v_m \text{ is linearly independent} \quad \Leftrightarrow \quad T \text{ is injective}$$

$$\Leftrightarrow \quad T' \text{ is surjective} \quad \Leftrightarrow \quad \Gamma \text{ is surjective},$$

where the first equivalence follows from Exercise 3.B.3, the second equivalence follows from 3.131, and the third equivalence follows from our previous discussion.

**Exercise 3.F.25.** Suppose $V$ is finite-dimensional and $\varphi_1, ..., \varphi_m \in V'$. Define a linear map $\Gamma : V \to \mathbf{F}^m$ by $\Gamma(v) = (\varphi_1(v), ..., \varphi_m(v))$.

(a) Prove that $\varphi_1, ..., \varphi_m$ spans $V'$ if and only if $\Gamma$ is injective.

(b) Prove that $\varphi_1, ..., \varphi_m$ is linearly independent if and only if $\Gamma$ is surjective.

**Solution.** Let $e_1, ..., e_m$ be the standard basis of $\mathbf{F}^m$ and let $\psi_1, ..., \psi_m$ be the corresponding dual basis of $(\mathbf{F}^m)'$, so that the map $\Psi : \mathbf{F}^m \to (\mathbf{F}^m)'$ given by $\Psi(e_k) = \psi_k$ is an isomorphism. For any $(x_1, ..., x_m) \in \mathbf{F}^m$ and $v \in V$, observe that

$$\begin{aligned}
[\Gamma'(\Psi(x_1, ..., x_m))](v) &= [\Gamma'(x_1\psi_1 + \cdots + x_m\psi_m)](v) \\
&= [x_1\psi_1 + \cdots + x_m\psi_m](\Gamma(v)) \\
&= [x_1\psi_1 + \cdots + x_m\psi_m](\varphi_1(v), ..., \varphi_m(v)) \\
&= x_1\varphi_1(v) + \cdots + x_m\varphi_m(v) \\
&= [x_1\varphi_1 + \cdots + x_m\varphi_m](v).
\end{aligned}$$

It follows that $\Gamma' \circ \Psi : \mathbf{F}^m \to V'$ is given by

$$\Gamma'(\Psi(x_1, ..., x_m)) = x_1\varphi_1 + \cdots + x_m\varphi_m.$$

Because $\Psi$ is a bijection, the injectivity of $\Gamma'$ is equivalent to the injectivity of $\Gamma' \circ \Psi$ and the surjectivity of $\Gamma'$ is equivalent to the surjectivity of $\Gamma' \circ \Psi$.

(a) Observe that

$$\operatorname{span}(\varphi_1, ..., \varphi_m) = V' \quad \Leftrightarrow \quad \Gamma' \circ \Psi \text{ is surjective}$$

$$\Leftrightarrow \quad \Gamma' \text{ is surjective} \quad \Leftrightarrow \quad \Gamma \text{ is injective},$$

where the first equivalence follows from Exercise 3.B.3, the second equivalence follows from our previous discussion, and the third equivalence follows from 3.131.

(b) Observe that



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$$\varphi_1, ..., \varphi_m \text{ is linearly independent} \quad \Leftrightarrow \quad \Gamma' \circ \Psi \text{ is injective}$$

$$\Leftrightarrow \quad \Gamma' \text{ is injective} \quad \Leftrightarrow \quad \Gamma \text{ is surjective,}$$

where the first equivalence follows from Exercise 3.B.3, the second equivalence follows from our previous discussion, and third equivalence follows from 3.129.

**Exercise 3.F.26.** Suppose $V$ is finite-dimensional and $\Omega$ is a subspace of $V'$. Prove that

$$\Omega = \{v \in V : \varphi(v) = 0 \text{ for every } \varphi \in \Omega\}^0.$$

**Solution.** Let $U = \{v \in V : \varphi(v) = 0 \text{ for every } \varphi \in \Omega\}$ and let $\varphi_1, ..., \varphi_m$ be a basis of $\Omega$. Certainly $U \subseteq (\operatorname{null} \varphi_1) \cap \cdots \cap (\operatorname{null} \varphi_m)$. Suppose $v \in (\operatorname{null} \varphi_1) \cap \cdots \cap (\operatorname{null} \varphi_m)$ and let $\varphi \in \Omega$ be given. There are scalars $a_1, ..., a_m$ such that $\varphi = a_1\varphi_1 + \cdots + a_m\varphi_m$, which gives us

$$\varphi(v) = (a_1\varphi_1 + \cdots + a_m\varphi_m)(v) = a_1\varphi_1(v) + \cdots + a_m\varphi_m(v) = 0.$$

Thus $v \in U$ and it follows that $U = (\operatorname{null} \varphi_1) \cap \cdots \cap (\operatorname{null} \varphi_m)$. We may now apply Exercise 3.F.23 to see that

$$U^0 = ((\operatorname{null} \varphi_1) \cap \cdots \cap (\operatorname{null} \varphi_m))^0 = \operatorname{span}(\varphi_1, ..., \varphi_m) = \Omega.$$

**Exercise 3.F.27.** Suppose $T \in \mathcal{L}(\mathcal{P}_5(\mathbf{R}))$ and $\operatorname{null} T' = \operatorname{span}(\varphi)$, where $\varphi$ is the linear functional on $\mathcal{P}_5(\mathbf{R})$ defined by $\varphi(p) = p(8)$. Prove that

$$\operatorname{range} T = \{p \in \mathcal{P}_5(\mathbf{R}) : p(8) = 0\}.$$

**Solution.** Observe that

$$\begin{aligned}
\operatorname{range} T &= \left\{p \in \mathcal{P}_5(\mathbf{R}) : \psi(p) = 0 \text{ for every } \psi \in (\operatorname{range} T)^0\right\} && \text{(Exercise 3.F.20)} \\
&= \{p \in \mathcal{P}_5(\mathbf{R}) : \psi(p) = 0 \text{ for every } \psi \in \operatorname{null} T'\} && (3.128) \\
&= \{p \in \mathcal{P}_5(\mathbf{R}) : \psi(p) = 0 \text{ for every } \psi \in \operatorname{span}(\varphi)\} \\
&= \{p \in \mathcal{P}_5(\mathbf{R}) : \lambda\varphi(p) = 0 \text{ for every } \lambda \in \mathbf{R}\} \\
&= \{p \in \mathcal{P}_5(\mathbf{R}) : \lambda p(8) = 0 \text{ for every } \lambda \in \mathbf{R}\} \\
&= \{p \in \mathcal{P}_5(\mathbf{R}) : p(8) = 0\}.
\end{aligned}$$

**Exercise 3.F.28.** Suppose $V$ is finite-dimensional and $\varphi_1, ..., \varphi_m$ is a linearly independent list in $V'$. Prove that

$$\dim((\operatorname{null} \varphi_1) \cap \cdots \cap (\operatorname{null} \varphi_m)) = (\dim V) - m.$$

**Solution.** By Exercise 3.F.23 and the linear independence of the list $\varphi_1, ..., \varphi_m$ we have



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$$\dim\left((\operatorname{null}\varphi_1)\cap\cdots\cap(\operatorname{null}\varphi_m)\right)^0=\dim\operatorname{span}(\varphi_1,\ldots,\varphi_m)=m.$$

It then follows from 3.125 that

$$\dim((\operatorname{null}\varphi_1)\cap\cdots\cap(\operatorname{null}\varphi_m))=(\dim V)-m.$$

**Exercise 3.F.29.** Suppose $V$ and $W$ are finite-dimensional and $T\in\mathcal{L}(V,W)$.

(a) Prove that if $\varphi\in W'$ and $\operatorname{null}T'=\operatorname{span}(\varphi)$, then $\operatorname{range}T=\operatorname{null}\varphi$.

(b) Prove that if $\psi\in V'$ and $\operatorname{range}T'=\operatorname{span}(\psi)$, then $\operatorname{null}T=\operatorname{null}\psi$.

**Solution.**

(a) Observe that

$$\begin{aligned}
\operatorname{range}T&=\left\{w\in W:\psi(w)=0\text{ for every }\psi\in(\operatorname{range}T)^0\right\} && \text{(Exercise 3.F.20)}\\
&=\{w\in W:\psi(w)=0\text{ for every }\psi\in\operatorname{null}T'\} && (3.128)\\
&=\{w\in W:\psi(w)=0\text{ for every }\psi\in\operatorname{span}(\varphi)\}\\
&=\{w\in W:\lambda\varphi(w)=0\text{ for every }\lambda\in\mathbf{F}\}\\
&=\{w\in W:\varphi(w)=0\}\\
&=\operatorname{null}\varphi.
\end{aligned}$$

(b) Observe that

$$\begin{aligned}
\operatorname{null}T&=\left\{v\in V:\varphi(v)=0\text{ for every }\varphi\in(\operatorname{null}T)^0\right\} && \text{(Exercise 3.F.20)}\\
&=\{v\in V:\varphi(v)=0\text{ for every }\varphi\in\operatorname{range}T'\} && (3.130)\\
&=\{v\in V:\varphi(v)=0\text{ for every }\varphi\in\operatorname{span}(\psi)\}\\
&=\{v\in V:\lambda\psi(v)=0\text{ for every }\lambda\in\mathbf{F}\}\\
&=\{v\in V:\psi(v)=0\}\\
&=\operatorname{null}\psi.
\end{aligned}$$

**Exercise 3.F.30.** Suppose $V$ is finite-dimensional and $\varphi_1,\ldots,\varphi_n$ is a basis of $V'$. Show that there exists a basis of $V$ whose dual basis is $\varphi_1,\ldots,\varphi_n$.

**Solution.** Let $k\in\{1,\ldots,n\}$ be given and define

$$U_k=\bigcap\left\{\operatorname{null}\varphi_j:j\in\{1,\ldots,n\}\setminus\{k\}\right\}.$$

By 3.111 and Exercise 3.F.28 we have $\dim U_k=1$ and thus there is some non-zero $u_k\in V$ such that $U_k=\operatorname{span}(u_k)$. Note that Exercise 3.F.28 also implies that



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$$(\operatorname{null}\varphi_1)\cap\cdots\cap(\operatorname{null}\varphi_n)=\{0\}.$$

Since $u_k\neq 0$ it must then be the case that $u_k\notin\operatorname{null}\varphi_k$. Thus we can define $v_k=(\varphi_k(u_k))^{-1}u_k$. Notice that $\varphi_k(v_k)=1$ and, for $j\neq k$,

$$u_k\in\operatorname{null}\varphi_j\quad\Rightarrow\quad\varphi_j(v_k)=0.$$

If $a_1,\ldots,a_n$ are scalars such that $a_1v_1+\cdots+a_nv_n=0$ then, for each $j\in\{1,\ldots,n\}$, applying $\varphi_j$ to both sides of this equation shows that $a_j=0$. Thus $v_1,\ldots,v_n$ is a linearly independent list of length $n=\dim V$; it follows that $v_1,\ldots,v_n$ is a basis of $V$. Because

$$\varphi_j(v_k)=\begin{cases}1 & \text{if } j=k,\\ 0 & \text{if } j\neq k,\end{cases}$$

the uniqueness part of the linear map lemma (3.4) shows that $\varphi_1,\ldots,\varphi_n$ is the dual basis of $v_1,\ldots,v_n$.

**Exercise 3.F.31.** Suppose $U$ is a subspace of $V$. Let $i:U\to V$ be the inclusion map defined by $i(u)=u$. Thus $i'\in\mathcal{L}(V',U')$.

(a) Show that $\operatorname{null}i'=U^0$.

(b) Prove that if $V$ is finite-dimensional, then $\operatorname{range}i'=U'$.

(c) Prove that if $V$ is finite-dimensional, then $\tilde{i'}$ is an isomorphism from $V'/U^0$ onto $U'$.

*The isomorphism in (c) is natural in that it does not depend on a choice of basis in either vector space.*

**Solution.**

(a) For $\varphi\in V'$ the map $i'(\varphi)=\varphi\circ i$ is simply the restriction of $\varphi$ to $U$. Thus

$$i'(\varphi)=0\quad\Leftrightarrow\quad\varphi(u)=0\text{ for all }u\in U.$$

It follows that $\operatorname{null}i'=U^0$.

(b) Let $\psi\in U'$ be given. By Exercise 3.A.13 we can extend $\psi$ to a linear functional $\varphi\in V'$ such that the restriction of $\varphi$ to $U$ is equal to $\psi$. Thus $i'(\varphi)=\psi$ and it follows that $i'$ is surjective.

(c) By 3.107 and parts (a) and (b), $\tilde{i'}$ is an isomorphism from $V'/\operatorname{null}i'=V'/U^0$ onto $\operatorname{range}i'=U'$.



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**Exercise 3.F.32.** The *double dual space* of $V$, denoted by $V''$, is defined to be the dual space of $V'$. In other words, $V'' = (V')'$. Define $\Lambda : V \to V''$ by

$$(\Lambda v)(\varphi) = \varphi(v)$$

for each $v \in V$ and each $\varphi \in V'$.

(a) Show that $\Lambda$ is a linear map from $V$ to $V''$.

(b) Show that if $T \in \mathcal{L}(V)$, then $T'' \circ \Lambda = \Lambda \circ T$, where $T'' = (T')'$.

(c) Show that if $V$ is finite-dimensional, then $\Lambda$ is an isomorphism from $V$ onto $V''$.

*Suppose $V$ is finite-dimensional. Then $V$ and $V'$ are isomorphic, but finding an isomorphism from $V$ onto $V'$ generally requires choosing a basis of $V$. In contrast, the isomorphism $\Lambda$ from $V$ onto $V''$ does not require a choice of basis and thus is considered more natural.*

**Solution.**

(a) Suppose $u, v \in V$ and $\mu \in \mathbf{F}$. For any $\varphi \in V'$ we have

$$(\Lambda(u+v))(\varphi) = \varphi(u+v) = \varphi(u) + \varphi(v) = (\Lambda u)(\varphi) + (\Lambda v)(\varphi) = (\Lambda u + \Lambda v)(\varphi).$$

Thus $\Lambda(u+v) = \Lambda u + \Lambda v$. Similarly, for any $\varphi \in V'$,

$$(\Lambda(\mu v))(\varphi) = \varphi(\mu v) = \mu\varphi(v) = \mu(\Lambda v)(\varphi) = (\mu \Lambda v)(\varphi).$$

Thus $\Lambda(\mu v) = \mu \Lambda v$. It follows that $\Lambda$ is linear.

(b) Let $v \in V$ be given and observe that

$$(T''(\Lambda v))(\varphi) = (\Lambda v)(T'(\varphi)) = (\Lambda v)(\varphi \circ T) = \varphi(Tv) \quad \text{and} \quad (\Lambda(Tv))(\varphi) = \varphi(Tv).$$

Thus $T'' \circ \Lambda = \Lambda \circ T$.

(c) Let $v_1, ..., v_n$ be a basis of $V$ and let $\varphi_1, ..., \varphi_n$ be the corresponding dual basis of $V'$. Suppose $v = a_1 v_1 + \cdots + a_n v_n$ is such that $\Lambda v = 0$, i.e. $\varphi(v) = 0$ for every $\varphi \in V'$. For each $k \in \{1, ..., n\}$ it follows that

$$0 = \varphi_k(v) = \varphi_k(a_1 v_1 + \cdots + a_n v_n) = a_k.$$

Thus $v = 0$, so that $\operatorname{null} \Lambda = \{0\}$, i.e. $\Lambda$ is injective. By 3.111 we have

$$\dim V'' = \dim V' = \dim V$$

and so, by 3.65, we may conclude that $\Lambda$ is an isomorphism.



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**Exercise 3.F.33.** Suppose $U$ is a subspace of $V$. Let $\pi : V \to V/U$ be the usual quotient map. Thus $\pi' \in \mathcal{L}\big((V/U)', V'\big)$.

(a) Show that $\pi'$ is injective.

(b) Show that $\operatorname{range} \pi' = U^0$.

(c) Conclude that $\pi'$ is an isomorphism from $(V/U)'$ onto $U^0$.

*The isomorphism in (c) is natural in that it does not depend on a choice of basis in either vector space. In fact, there is no assumption here that any of these vector spaces are finite-dimensional.*

**Solution.**

(a) Suppose $\varphi \in (V/U)'$ is such that $\pi'(\varphi) = 0$, i.e. $\varphi(v + U) = 0$ for all $v + U \in V/U$. Thus $\varphi = 0$, so that $\operatorname{null} \pi' = \{0\}$. It follows that $\pi'$ is injective.

(b) Note that $U^0 = \{\varphi \in V' : U \subseteq \operatorname{null} \varphi\}$ by Exercise 3.F.19. Taking $W = \mathbf{F}$ in Exercise 3.E.19 then shows that $\operatorname{range} \pi' = U^0$.

(c) This is immediate from parts (a) and (b).



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# Chapter 4. Polynomials

**Exercise 4.1.** Suppose $w, z \in \mathbf{C}$. Verify the following equalities and inequalities.

(a) $z + \overline{z} = 2 \operatorname{Re} z$

(b) $z - \overline{z} = 2(\operatorname{Im} z)i$

(c) $z\overline{z} = |z|^2$

(d) $\overline{w + z} = \overline{w} + \overline{z}$ and $\overline{wz} = \overline{w}\,\overline{z}$

(e) $\overline{\overline{z}} = z$

(f) $|\operatorname{Re} z| \leq |z|$ and $|\operatorname{Im} z| \leq |z|$

(g) $|\overline{z}| = |z|$

(h) $|wz| = |w||z|$

*The results above are the parts of 4.4 that were left to the reader.*

**Solution.** Suppose $w = a + bi$ and $z = x + yi$.

(a) Observe that

$$z + \overline{z} = (x + yi) + (x - yi) = 2x = 2 \operatorname{Re} z.$$

(b) Observe that

$$z - \overline{z} = (x + yi) - (x - yi) = 2yi = 2(\operatorname{Im} z)i.$$

(c) Observe that

$$z\overline{z} = (x + yi)(x - yi) = x^2 + y^2 = (\operatorname{Re} z)^2 + (\operatorname{Im} z)^2 = |z|^2.$$

(d) Observe that

$$\overline{w + z} = (a + x) - (b + y)i = (a - bi) + (x - yi) = \overline{w} + \overline{z},$$

$$\overline{wz} = (ax - by) - (ay + bx)i = (a - bi)(x - yi) = \overline{w}\,\overline{z}.$$

(e) Observe that

$$\overline{\overline{z}} = \overline{x - yi} = x + yi = z.$$

(f) Since each quantity involved is positive, it will suffice to show that $|\operatorname{Re} z|^2 \leq |z|^2$ and $|\operatorname{Im}(z)|^2 \leq |z|^2$; these inequalities are immediate from the equation $|z|^2 = |\operatorname{Re} z|^2 + |\operatorname{Im} z|^2$.

(g) Observe that

$$|\overline{z}| = (\operatorname{Re} \overline{z})^2 + (\operatorname{Im} \overline{z})^2 = (\operatorname{Re} z)^2 + (-\operatorname{Im} z)^2 = (\operatorname{Re} z)^2 + (\operatorname{Im} z)^2 = |z|.$$



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(h) Since both sides are positive, it will suffice to show that $|wz|^2 = |w|^2|z|^2$. Indeed, using parts (c) and (d),

$$|wz|^2 = wz\overline{wz} = wz\overline{w}\,\overline{z} = w\overline{w}z\overline{z} = |w|^2|z|^2.$$

**Exercise 4.2.** Prove that if $w, z \in \mathbf{C}$, then $||w| - |z|| \leq |w - z|$.

*The inequality above is called the* ***reverse triangle inequality****.*

**Solution.** Notice that

$$|w| = |w - z + z| \leq |w - z| + |z| \quad \Rightarrow \quad |w| - |z| \leq |w - z|,$$
$$|z| = |z - w + w| \leq |z - w| + |w| \quad \Rightarrow \quad |z| - |w| \leq |w - z|.$$

Thus $||w| - |z|| \leq |w - z|$.

**Exercise 4.3.** Suppose $V$ is a complex vector space and $\varphi \in V'$. Define $\sigma : V \to \mathbf{R}$ by $\sigma(v) = \operatorname{Re}\varphi(v)$ for each $v \in V$. Show that

$$\varphi(v) = \sigma(v) - i\sigma(iv)$$

for all $v \in V$.

**Solution.** For any $z = x + iy \in \mathbf{C}$, note that

$$\operatorname{Re}(iz) = \operatorname{Re}(-y + ix) = -y = -\operatorname{Im} z.$$

For any $v \in V$ it follows that

$$\sigma(v) - i\sigma(iv) = \operatorname{Re}\varphi(v) - i\operatorname{Re}(\varphi(iv)) = \operatorname{Re}\varphi(v) - i\operatorname{Re}(i\varphi(v)) = \operatorname{Re}\varphi(v) + \operatorname{Im}\varphi(v) = \varphi(v).$$

**Exercise 4.4.** Suppose $m$ is a positive integer. Is the set

$$\{0\} \cup \{p \in \mathcal{P}(\mathbf{F}) : \deg p = m\}$$

a subspace of $\mathcal{P}(\mathbf{F})$?

**Solution.** Let $U$ be the set in question and observe that $x^m, 1 - x^m \in U$ but

$$x^m + 1 - x^m = 1 \notin U.$$

Thus $U$ is not a subspace of $\mathcal{P}(\mathbf{F})$.

**Exercise 4.5.** Is the set

$$\{0\} \cup \{p \in \mathcal{P}(\mathbf{F}) : \deg p \text{ is even}\}$$

a subspace of $\mathcal{P}(\mathbf{F})$?

**Solution.** Let $U$ be the set in question and observe that $x^2, x - x^2 \in U$ but



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$$x^2 + x - x^2 = x \notin U.$$

Thus $U$ is not a subspace of $\mathcal{P}(\mathbf{F})$.

**Exercise 4.6.** Suppose that $m$ and $n$ are positive integers with $m \leq n$, and suppose $\lambda_1, ..., \lambda_m \in \mathbf{F}$. Prove that there exists a polynomial $p \in \mathcal{P}(\mathbf{F})$ with $\deg p = n$ such that $0 = p(\lambda_1) = \cdots = p(\lambda_m)$ and such that $p$ has no other zeros.

**Solution.** Let $p(z) = (z - \lambda_1)(z - \lambda_2) \cdots (z - \lambda_m)^{n-m+1}$ and note that $\deg p = n$ and that each $\lambda_k$ is a zero of $p$. The uniqueness parts of 4.13 and 4.16, together with 4.6, shows that $p$ can have no other zeros.

**Exercise 4.7.** Suppose that $m$ is a nonnegative integer, $z_1, ..., z_{m+1}$ are distinct elements of $\mathbf{F}$, and $w_1, ..., w_{m+1} \in \mathbf{F}$. Prove that there exists a unique polynomial $p \in \mathcal{P}_m(\mathbf{F})$ such that

$$p(z_k) = w_k$$

for each $k = 1, ..., m + 1$.

> *This result can be proved without using linear algebra. However, try to find the clearer, shorter proof that uses some linear algebra.*

**Solution.** Note that the list $p_0, ..., p_{m+1} \in \mathcal{P}_m(\mathbf{F})$ given by

$$\begin{aligned} p_0 &= 1, \\ p_1 &= z - z_1, \\ p_2 &= (z - z_1)(z - z_2), \\ &\vdots \\ p_m &= (z - z_1)(z - z_2) \cdots (z - z_m) \end{aligned}$$

is a basis of $\mathcal{P}_m(\mathbf{F})$ by Exercise 2.C.9. Define a map $T \in \mathcal{L}(\mathcal{P}_m(\mathbf{F}), \mathbf{F}^{m+1})$ by

$$Tp = (p(z_1), ..., p(z_{m+1})).$$

Notice that the matrix of $T$ with respect to the basis $p_0, ..., p_m$ of $\mathcal{P}_m(\mathbf{F})$ and the standard basis of $\mathbf{F}^{m+1}$ is

$$\begin{pmatrix} 1 & 0 & 0 & \cdots & 0 \\ 1 & A_{1,1} & 0 & \cdots & 0 \\ 1 & A_{2,1} & A_{2,2} & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & A_{m,1} & A_{m,2} & \cdots & A_{m,m} \end{pmatrix}, \quad \text{where } A_{j,k} = \prod_{i=1}^{k} (z_{j+1} - z_i) \text{ for } j \geq k.$$

Notice further that each $A_{j,k}$ is non-zero because the elements $z_1, ..., z_{m+1}$ are distinct. A straightforward calculation then shows that the rows of this matrix are linearly



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independent; it follows from Exercise 3.C.17 that $T$ is injective and hence, because $\dim \mathcal{P}_m(\mathbf{F}) = \dim \mathbf{F}^{m+1}$, invertible. Thus there exists a unique $p \in \mathcal{P}_m(\mathbf{F})$ such that

$$Tp = (p(z_1), ..., p(z_{m+1})) = (w_1, w_2, ..., w_{m+1}).$$

**Exercise 4.8.** Suppose $p \in \mathcal{P}(\mathbf{C})$ has degree $m$. Prove that $p$ has $m$ distinct zeros if and only if $p$ and its derivative $p'$ have no zeros in common.

**Solution.** The cases $m = 0$ and $m = 1$ are straightforward to check. For $m \geq 2$, we will prove the equivalent statement

$$p \text{ has strictly less than } m \text{ distinct zeros} \quad \Leftrightarrow \quad p \text{ and } p' \text{ have a zero in common.}$$

If $p$ has strictly less than $m$ distinct zeros then it must be the case that $p$ has a zero $\lambda \in \mathbf{C}$ such that $p(z) = (z - \lambda)^k q(z)$ for some positive integer $k \geq 2$ and some $q \in \mathcal{P}(\mathbf{C})$. It follows that

$$p'(z) = k(z - \lambda)^{k-1} q(z) + (z - \lambda)^k q'(z)$$

and hence that $p'(\lambda) = 0$, since $k \geq 2$. Thus $p$ and $p'$ have the zero $\lambda$ in common.

Now suppose that $p$ and $p'$ have a zero in common, say $\lambda \in \mathbf{F}$, so that

$$p(z) = (z - \lambda)q(z) \quad \text{and} \quad p'(z) = (z - \lambda)r(z)$$

for some $q, r \in \mathcal{P}(\mathbf{C})$. The product rule gives us

$$q(z) + (z - \lambda)q'(z) = p'(z) = (z - \lambda)r(z).$$

Evaluating this expression at $z = \lambda$ shows that $q(\lambda) = 0$, so that $z - \lambda$ is a factor of $q$. It follows that $p$ is of the form $p(z) = (z - \lambda)^2 t(z)$ for some $t \in \mathcal{P}(\mathbf{C})$ satisfying $\deg t = m - 2$. Thus $p$ has strictly less than $m$ zeros.

**Exercise 4.9.** Prove that every polynomial of odd degree with real coefficients has a real zero.

**Solution.** Let $p \in \mathcal{P}(\mathbf{R})$ be a polynomial of odd degree. By 4.16, $p$ is of the form

$$p(x) = c(x - \lambda_1) \cdots (x - \lambda_m)(x^2 + b_1 x + c_1) \cdots (x^2 + b_M x + c_M),$$

where $c, \lambda_1, ..., \lambda_m, b_1, ..., b_M, c_1, ..., c_M \in \mathbf{R}$, $c > 0$, and $b_k^2 < 4c_k$ for each $k$. This implies that $\deg p = m + 2M$; since $\deg p$ is odd, it must be the case that $m > 0$. Thus $p$ has at least one real zero.



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**Exercise 4.10.** For $p \in \mathcal{P}(\mathbf{R})$, define $Tp : \mathbf{R} \to \mathbf{R}$ by

$$(Tp)(x) = \begin{cases} \dfrac{p(x) - p(3)}{x - 3} & \text{if } x \neq 3, \\ p'(3) & \text{if } x = 3 \end{cases}$$

for each $x \in \mathbf{R}$. Show that $Tp \in \mathcal{P}(\mathbf{R})$ for every polynomial $p \in \mathcal{P}(\mathbf{R})$ and also show that $T : \mathcal{P}(\mathbf{R}) \to \mathcal{P}(\mathbf{R})$ is a linear map.

**Solution.** Let $p \in \mathcal{P}(\mathbf{R})$ be given and notice that $p(x) - p(3)$ has a zero at $x = 3$, so that

$$p(x) - p(3) = (x - 3)q(x)$$

for some unique $q \in \mathcal{P}(\mathbf{R})$. It follows that for any $x \neq 3$ we have

$$q(x) = \frac{p(x) - p(3)}{x - 3}.$$

Differentiating the equality $p(x) - p(3) = (x - 3)q(x)$ gives us $p'(x) = q(x) + (x - 3)q'(x)$, whence $p'(3) = q(3)$. Thus $Tp = q \in \mathcal{P}(\mathbf{R})$.

Let $p_1, p_2, \in \mathcal{P}(\mathbf{R})$ and $\lambda \in \mathbf{R}$ be given. There are unique polynomials $q_1, q_2 \in \mathcal{P}(\mathbf{R})$ such that

$$p_1(x) - p_1(3) = (x - 3)q_1(x) \quad \text{and} \quad p_2(x) - p_2(3) = (x - 3)q_2(x).$$

As we showed above, it follows that $Tp_1 = q_1$ and $Tp_2 = q_2$. Notice that

$$(p_1 + p_2)(x) - (p_1 + p_2)(3) = (x - 3)(q_1 + q_2)(x)$$

$$\text{and} \quad (\lambda p_1)(x) - (\lambda p_1)(3) = (x - 3)(\lambda q_1)(x).$$

By uniqueness we must have $T(p_1 + p_2) = q_1 + q_2 = Tp_1 + Tp_2$ and $T(\lambda p_1) = \lambda q_1 = \lambda T p_1$. Thus $T$ is linear.

**Exercise 4.11.** Suppose $p \in \mathcal{P}(\mathbf{C})$. Define $q : \mathbf{C} \to \mathbf{C}$ by

$$q(z) = p(z)\overline{p(\overline{z})}.$$

Prove that $q$ is a polynomial with real coefficients.

**Solution.** Suppose $p(z) = a_0 + a_1 z + \cdots + a_m z^m$ for some non-negative integer $m$. Observe that $\overline{p(\overline{z})} = \overline{a_0} + \overline{a_1} z + \cdots + \overline{a_m} z^m$. It follows that

$$q(z) = p(z)\overline{p(\overline{z})} = \sum_{k=0}^{2m} \left( \sum_{i+j=k} a_i \overline{a_j} \right) z^k.$$

For any $k \in \{0, ..., 2m\}$ note that



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$$\overline{\sum_{i+j=k} a_i\overline{a_j}} = \sum_{i+j=k} \overline{a_i\overline{a_j}} = \sum_{i+j=k} a_j\overline{a_i} = \sum_{i+j=k} a_i\overline{a_j},$$

where the last equality follows by reindexing. Thus $\sum_{i+j=k} a_i\overline{a_j} \in \mathbf{R}$ for each $k \in \{0, ..., 2m\}$, i.e. $q$ has real coefficients.

**Exercise 4.12.** Suppose $m$ is a nonnegative integer and $p \in \mathcal{P}_m(\mathbf{C})$ is such that there are distinct real numbers $x_0, x_1, ..., x_m$ with $p(x_k) \in \mathbf{R}$ for each $k = 0, 1, ..., m$. Prove that all coefficients of $p$ are real.

**Solution.** By Exercise 4.7 there is a unique polynomial $q \in \mathcal{P}_m(\mathbf{R})$ such that $q(x_k) = p(x_k)$ for each $k \in \{0, ..., m\}$. It follows that the polynomial $p - q \in \mathcal{P}_m(\mathbf{C})$ has $m + 1$ distinct zeros and thus, by (4.8), $p = q \in \mathcal{P}_m(\mathbf{R})$.

**Exercise 4.13.** Suppose $p \in \mathcal{P}(\mathbf{F})$ with $p \neq 0$. Let $U = \{pq : q \in \mathcal{P}(\mathbf{F})\}$.

(a) Show that $\dim \mathcal{P}(\mathbf{F})/U = \deg p$.

(b) Find a basis of $\mathcal{P}(\mathbf{F})/U$.

**Solution.**

(a) Let $m = \deg p$. If $m = 0$, i.e. $p$ is a non-zero constant polynomial, then $U = \mathcal{P}(\mathbf{F})$ and thus $\mathcal{P}(\mathbf{F})/U = \{0\}$, so that $\dim \mathcal{P}(\mathbf{F})/U = 0 = \deg p$. Suppose that $m \geq 1$. For any $s \in \mathcal{P}(\mathbf{F})$, the division algorithm for polynomials (4.9) implies that there are unique polynomials $q, r \in \mathcal{P}(\mathbf{F})$ such that $s = pq + r$ and $\deg r < \deg p$. Thus the map $T : \mathcal{P}(\mathbf{F}) \to \mathcal{P}_{m-1}(\mathbf{F})$ given by $Ts = r$ is well-defined. Let $s_1, s_2 \in \mathcal{P}(\mathbf{F})$ and $\lambda \in \mathbf{F}$ be given. There are unique polynomials $q_1, q_2, r_1, r_2 \in \mathcal{P}(\mathbf{F})$ such that

$$s_1 = pq_1 + r_1, \quad s_2 = pq_2 + r_2, \quad \deg r_1 < \deg p, \quad \text{and} \quad \deg r_2 < \deg p.$$

Thus $Ts_1 = r_1$ and $Ts_2 = r_2$. Observe that

$$s_1 + s_2 = p(q_1 + q_2) + (r_1 + r_2), \quad \deg(r_1 + r_2) \leq \max\{\deg r_1, \deg r_2\} < \deg p,$$
$$\lambda s_1 = p(\lambda q_1) + (\lambda r_1), \quad \deg(\lambda r_1) \leq \deg r_1 < \deg p.$$

It follows from the uniqueness part of the division algorithm that

$$T(s_1 + s_2) = r_1 + r_2 = Ts_1 + Ts_2 \quad \text{and} \quad T(\lambda s_1) = \lambda r_1 = \lambda T s_1.$$

Thus $T$ is linear.

For any $r \in \mathcal{P}_{m-1}(\mathbf{F})$ we have $\deg r < \deg p$ and thus $Tr = r$. Hence $T$ is surjective. Notice that $pq \in U$ has remainder zero upon division by $p$; it follows that $T(pq) = 0$. Conversely, if $s = pq + r$ is such that $Ts = r = 0$ then $s = pq \in U$. Thus $\operatorname{null} T = U$. It now follows from 3.107 that $\mathcal{P}(\mathbf{F})/U$ is isomorphic to $\mathcal{P}_{m-1}(\mathbf{F})$ via the map $\tilde{T} : \mathcal{P}(\mathbf{F})/U \to \mathcal{P}_{m-1}(\mathbf{F})$ given by $\tilde{T}(s + U) = Ts$. Thus



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$$\dim \mathcal{P}(\mathbf{F})/U = \dim \mathcal{P}_{m-1}(\mathbf{F}) = m = \deg p.$$

(b) Notice that $\tilde{T}(1+U), \tilde{T}(z+U), ..., \tilde{T}(z^{m-1}+U)$ is the list $1, z, ..., z^{m-1}$, i.e. the standard basis of $\mathcal{P}_{m-1}(\mathbf{F})$. Because $\tilde{T}^{-1}$ is an isomorphism, it follows that

$$\tilde{T}^{-1}(1), \tilde{T}^{-1}(z), ..., \tilde{T}^{-1}(z^{m-1}) = 1+U, z+U, ..., z^{m-1}+U$$

is a basis of $\mathcal{P}(\mathbf{F})/U$.

**Exercise 4.14.** Suppose $p, q \in \mathcal{P}(\mathbf{C})$ are nonconstant polynomials with no zeros in common. Let $m = \deg p$ and $n = \deg q$. Use linear algebra as outlined below in (a)-(c) to prove that there exist $r \in \mathcal{P}_{n-1}(\mathbf{C})$ and $s \in \mathcal{P}_{m-1}(\mathbf{C})$ such that

$$rp + sq = 1.$$

(a) Define $T : \mathcal{P}_{n-1}(\mathbf{C}) \times \mathcal{P}_{m-1}(\mathbf{C}) \to \mathcal{P}_{m+n-1}(\mathbf{C})$ by

$$T(r, s) = rp + sq.$$

Show that the linear map $T$ is injective.

(b) Show that the linear map $T$ in (a) is surjective.

(c) Use (b) to conclude that there exist $r \in \mathcal{P}_{n-1}(\mathbf{C})$ and $s \in \mathcal{P}_{m-1}(\mathbf{C})$ such that $rp + sq = 1$.

**Solution.**

(a) Note that $m, n \geq 1$ since $p, q$ are non-constant. Let $\lambda_1, ..., \lambda_m$ be the zeros of $p$ and let $\mu_1, ..., \mu_n$ be the zeros of $q$; by assumption we have $p(\mu_k) \neq 0$ and $q(\lambda_k) \neq 0$ for all $k$. Suppose that $r \in \mathcal{P}_{n-1}(\mathbf{C})$ and $s \in \mathcal{P}_{m-1}(\mathbf{C})$ are such that $rp + sq = 0$. In particular, for each $k \in \{1, ..., n\}$,

$$r(\mu_k)p(\mu_k) + s(\mu_k)q(\mu_k) = r(\mu_k)p(\mu_k) = 0 \quad \Rightarrow \quad r(\mu_k) = 0,$$

where we have used that $q(\mu_k) = 0$ and $p(\mu_k) \neq 0$. Thus $r$ is a polynomial of degree at most $n-1$ with $n$ zeros; it follows from 4.8 that $r = 0$. A similar argument with the $\lambda_k$'s shows that $s = 0$. Thus $\operatorname{null} T = \{0\}$, i.e. $T$ is injective.

(b) Notice that

$$\begin{aligned}\dim(\mathcal{P}_{n-1}(\mathbf{C}) \times \mathcal{P}_{m-1}(\mathbf{C})) &= \dim \mathcal{P}_{n-1}(\mathbf{C}) + \dim \mathcal{P}_{m-1}(\mathbf{C}) \\ &= n + m = \dim \mathcal{P}_{m+n-1}(\mathbf{C}).\end{aligned}$$

Since $T$ is injective, it follows from 3.65 that $T$ is also surjective.

(c) This is immediate from the surjectivity of $T$.



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# Chapter 5. Eigenvalues and Eigenvectors

## 5.A. Invariant Subspaces

**Exercise 5.A.1.** Suppose $T \in \mathcal{L}(V)$ and $U$ is a subspace of $V$.

(a) Prove that if $U \subseteq \operatorname{null} T$, then $U$ is invariant under $T$.

(b) Prove that if $\operatorname{range} T \subseteq U$, then $U$ is invariant under $T$.

**Solution.**

(a) Let $u \in U \subseteq \operatorname{null} T$ be given and observe that $Tu = 0 \in U$. Thus $U$ is invariant under $T$.

(b) Let $u \in U$ be given and observe that $Tu \in \operatorname{range} T \subseteq U$. Thus $U$ is invariant under $T$.

**Exercise 5.A.2.** Suppose that $T \in \mathcal{L}(V)$ and $V_1, ..., V_m$ are subspaces of $V$ invariant under $T$. Prove that $V_1 + \cdots + V_m$ is invariant under $T$.

**Solution.** Let $v = a_1 v_1 + \cdots + a_m v_m \in V_1 + \cdots + V_m$ be given. By assumption each $Tv_k \in V_k$ and thus

$$Tv = a_1 Tv_1 + \cdots + a_m Tv_k \in V_1 + \cdots + V_m.$$

Hence $V_1 + \cdots + V_m$ is invariant under $T$.

**Exercise 5.A.3.** Suppose $T \in \mathcal{L}(V)$. Prove that the intersection of every collection of subspaces of $V$ invariant under $T$ is invariant under $T$.

**Solution.** Let $\mathcal{U}$ be a collection of subspaces of $V$ invariant under $T$. For any $u \in \bigcap \mathcal{U}$ and any $U \in \mathcal{U}$, we have $u \in U$ and thus $Tu \in U$. It follows that $Tu \in \bigcap \mathcal{U}$ and hence that $\bigcap \mathcal{U}$ is invariant under $T$.

**Exercise 5.A.4.** Prove or give a counterexample: If $V$ is finite-dimensional and $U$ is a subspace of $V$ that is invariant under every operator on $V$, then $U = \{0\}$ or $U = V$.

**Solution.** This is true. It will suffice to show that if $U \neq \{0\}$ then $U = V$. Suppose therefore that there exists some $v_1 \in U$ with $v_1 \neq 0$ and extend this to a basis $v_1, ..., v_m$ of $V$. For each $k \in \{1, ..., m\}$ define an operator $T_k \in \mathcal{L}(V)$ by $T_k v_1 = v_k$ and $T_k v_j = 0$ for $j \neq 1$. By assumption $U$ is invariant under $T_k$ and thus $T_k v_1 = v_k \in U$. It follows that $U$ contains the basis $v_1, ..., v_m$ of $V$ and hence that $U = V$.



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**Exercise 5.A.5.** Suppose $T \in \mathcal{L}(\mathbf{R}^2)$ is defined by $T(x, y) = (-3y, x)$. Find the eigenvalues of $T$.

**Solution.** Geometrically, we can observe that $T$ is a counterclockwise rotation by 90° about the origin followed by a dilation of the $x$-axis by a factor of 3. We can now argue as in 5.9(a) to see that $T$ has no eigenvalues.

Algebraically, for $\lambda \in \mathbf{R}$ we can try to solve the equation $T(x, y) = (-3y, x) = (\lambda x, \lambda y)$. Substituting $x = \lambda y$ into $-3y = \lambda x$ gives us $-3y = \lambda^2 y$. Because $y = 0$ implies $x = 0$, and eigenvectors are non-zero, we may assume that $y \neq 0$ and thus obtain the equation $\lambda^2 + 3 = 0$. Since this equation has no real solutions, we see that $T$ has no eigenvalues.

**Exercise 5.A.6.** Define $T \in \mathcal{L}(\mathbf{F}^2)$ by $T(w, z) = (z, w)$. Find all eigenvalues and eigenvectors of $T$.

**Solution.** $T$ is a reflection in the line $w = z$. An appeal to our geometric intuition suggests that 1 is an eigenvalue with corresponding eigenvector $(1, 1)$ and that $-1$ is an eigenvalue with corresponding eigenvector $(-1, 1)$. To see this algebraically, suppose $\lambda \in \mathbf{F}$ and $(w, z) \neq (0, 0)$ are such that $T(w, z) = (z, w) = (\lambda w, \lambda z)$. Substituting $z = \lambda w$ into $w = \lambda z$ gives us $w = \lambda^2 w$. Since $w = 0$ implies $z = 0$, and eigenvectors are non-zero, we may assume that $w \neq 0$ and thus obtain the equation $\lambda^2 - 1 = 0$, which has solutions $\lambda = \pm 1$. These are both indeed eigenvalues, since

$$T(1, 1) = (1, 1) \quad \text{and} \quad T(-1, 1) = (1, -1) = -(-1, 1).$$

Since $\dim \mathbf{F}^2 = 2$, it follows from 5.11 and 5.12 that there are no other eigenvalues of $T$ and no other eigenvectors of $T$ linearly independent from the two given above. We may conclude that the eigenvalues and eigenvectors of $T$ are precisely:

| eigenvalue | corresponding eigenvectors |
|---|---|
| 1 | $(w, w)$ for $w \in \mathbf{F} \setminus \{0\}$ |
| $-1$ | $(-w, w)$ for $w \in \mathbf{F} \setminus \{0\}$ |

**Exercise 5.A.7.** Define $T \in \mathcal{L}(\mathbf{F}^3)$ by $T(z_1, z_2, z_3) = (2z_2, 0, 5z_3)$. Find all eigenvalues and eigenvectors of $T$.

**Solution.** $T$ can be thought of as the composition of the following transformations:

- a projection onto the $z_2 z_3$-plane;
- a clockwise rotation of 90° around the $z_3$-axis; after the projection onto the $z_2 z_3$-plane, this is equivalent to a reflection in the plane $z_1 = z_2$;
- a dilation of the $z_1$-axis by a factor of 2;



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- a dilation of the $z_3$-axis by a factor of 5.

In other words, $T$ maps $(z_1, z_2, z_3) \in \mathbf{F}^3$ like so:

$$(z_1, z_2, z_3) \mapsto (0, z_2, z_3) \mapsto (z_2, 0, z_3) \mapsto (2z_2, 0, z_3) \mapsto (2z_2, 0, 5z_3).$$

An appeal to our geometric intuition suggests that 5 is an eigenvalue with corresponding eigenvector $(0, 0, 1)$ and that 0 is an eigenvector with corresponding eigenvector $(1, 0, 0)$. To prove this algebraically, suppose that $\lambda \in \mathbf{F}$ and $(z_1, z_2, z_3) \neq (0, 0, 0)$ are such that

$$T(z_1, z_2, z_3) = (2z_2, 0, 5z_3) = (\lambda z_1, \lambda z_2, \lambda z_3).$$

If $\lambda \neq 0$ then the equation $\lambda z_2 = 0$ implies that $z_2 = 0$ and thus the equation $2z_2 = \lambda z_1$ gives us $z_1 = 0$. Since eigenvectors are non-zero, it must be the case that $z_3 \neq 0$ and so the equation $5z_3 = \lambda z_3$ implies that $\lambda = 5$. So the only possible eigenvalues are 0 and 5, which are indeed eigenvalues since

$$T(0, 0, 1) = (0, 0, 5) = 5(0, 0, 1) \quad \text{and} \quad T(1, 0, 0) = (0, 0, 0) = 0(1, 0, 0).$$

We claim that there are no other eigenvectors of $T$ linearly independent from these two. As we just showed, any eigenvector of $T$ corresponding to the eigenvalue 5 must satisfy $z_1 = z_2 = 0$ and thus each such eigenvector is a scalar multiple of $(0, 0, 1)$. If $(z_1, z_2, z_3)$ is an eigenvector corresponding to the eigenvalue 0, i.e. $(z_1, z_2, z_3) \in \operatorname{null} T$, then

$$T(z_1, z_2, z_3) = (2z_2, 0, 5z_3) = (0, 0, 0) \quad \Rightarrow \quad z_2 = z_3 = 0 \quad \Rightarrow \quad (z_1, z_2, z_3) = z_1(1, 0, 0).$$

We may conclude that the eigenvalues and eigenvectors of $T$ are precisely:

| eigenvalue | corresponding eigenvectors |
|---|---|
| 5 | $(0, 0, w)$ for $w \in \mathbf{F} \setminus \{0\}$ |
| 0 | $(w, 0, 0)$ for $w \in \mathbf{F} \setminus \{0\}$ |

**Exercise 5.A.8.** Suppose $P \in \mathcal{L}(V)$ is such that $P^2 = P$. Prove that if $\lambda$ is an eigenvalue of $P$, then $\lambda = 0$ or $\lambda = 1$.

**Solution.** Suppose that $\lambda$ is an eigenvalue of $P$, i.e. there is some $v \neq 0$ such that $Pv = \lambda v$. Notice that

$$\lambda v = Pv = P(Pv) = P(\lambda v) = \lambda Pv = \lambda^2 v.$$

Because $v \neq 0$, this implies that $\lambda = \lambda^2$. Thus $\lambda = 0$ or $\lambda = 1$.

**Exercise 5.A.9.** Define $T : \mathcal{P}(\mathbf{R}) \to \mathcal{P}(\mathbf{R})$ by $Tp = p'$. Find all eigenvalues and eigenvectors of $T$.



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**Solution.** Note that $T(1) = 0 = 0(1)$, so that 0 is an eigenvalue of $T$. Note further that the only polynomials whose derivative is zero are the constant polynomials. Thus the eigenvectors corresponding to the eigenvalue 0 of $T$ are precisely the non-zero constant polynomials.

Suppose $p \in \mathcal{P}(\mathbf{R})$ satisfies $\deg p \geq 1$. If $\lambda \neq 0$ then $\deg(\lambda p) = \deg p$, whereas $\deg p' = \deg p - 1$. Thus it cannot be the case that $Tp = p' = \lambda p$ and we may conclude that the eigenvalues and eigenvectors of $T$ are precisely:

| eigenvalue | corresponding eigenvectors |
|---|---|
| 0 | non-zero constant polynomials in $\mathcal{P}(\mathbf{R})$ |

**Exercise 5.A.10.** Define $T \in \mathcal{L}(\mathcal{P}_4(\mathbf{R}))$ by $(Tp)(x) = xp'(x)$ for all $x \in \mathbf{R}$. Find all eigenvalues and eigenvectors of $T$.

**Solution.** Letting $p_k \in \mathcal{P}_4(\mathbf{R})$ be given by $p_k(x) = x^k$ for $k \in \{0, ..., 4\}$, notice that

$$(Tp_k)(x) = kx^k = kp_k(x).$$

By 5.11 and 5.12 we may conclude that the eigenvalues and eigenvectors of $T$ are precisely:

| eigenvalue | corresponding eigenvectors |
|---|---|
| $k \in \{0, ..., 4\}$ | $\alpha p_k$ for $\alpha \in \mathbf{R} \setminus \{0\}$ |

**Exercise 5.A.11.** Suppose $V$ is finite-dimensional, $T \in \mathcal{L}(V)$, and $\alpha \in \mathbf{F}$. Prove that there exists $\delta > 0$ such that $T - \lambda I$ is invertible for all $\lambda \in \mathbf{F}$ such that $0 < |\alpha - \lambda| < \delta$.

**Solution.** If $T$ has no eigenvalues then 5.7 shows that $T - \lambda I$ is invertible for all $\lambda \in \mathbf{F}$, so that any $\delta > 0$ will suffice, say $\delta = 1$. Suppose therefore that $T$ has at least one eigenvalue and let $\lambda_1, ..., \lambda_n$ be the distinct eigenvalues of $T$; there are only finitely many eigenvalues of $T$ by 5.12. Let

$$\delta = \min\{|\alpha - \lambda_k| : k \in \{1, ..., n\} \text{ and } \lambda_k \neq \alpha\}.$$

It follows from this definition that $\delta$ is positive and furthermore that $\lambda \neq \lambda_k$ for any $k \in \{1, ..., n\}$ and any $\lambda \in \mathbf{F}$ such that $0 < |\alpha - \lambda| < \delta$. That is, $\lambda$ is not an eigenvalue of $T$ for any $\lambda \in \mathbf{F}$ such that $0 < |\alpha - \lambda| < \delta$. By 5.7 this is equivalent to saying that $T - \lambda I$ is invertible for any $\lambda \in \mathbf{F}$ such that $0 < |\alpha - \lambda| < \delta$.

**Exercise 5.A.12.** Suppose $V = U \oplus W$, where $U$ and $W$ are nonzero subspaces of $V$. Define $P \in \mathcal{L}(V)$ by $P(u + w) = u$ for each $u \in U$ and each $w \in W$. Find all eigenvalues and eigenvectors of $P$.



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**Solution.** Notice that $P^2 = P$. It follows from Exercise 5.A.8 that the only possible eigenvalues of $P$ are 1 and 0. Because $U$ and $W$ are non-zero, there is some non-zero $u \in U$ and some non-zero $w \in W$. Observe that

$$Pu = 1u \quad \text{and} \quad Pw = 0w.$$

Thus 1 and 0 are indeed eigenvalues of $P$. The above equations show that any non-zero elements of $U$ are eigenvectors of $T$ corresponding to the eigenvalue 1 and any non-zero elements of $W$ are eigenvectors of $T$ corresponding to the eigenvalue 0. If $v = u + w$ is an eigenvector of $T$ corresponding to the eigenvalue 1, then observe that

$$u + w = P(u + w) = u \quad \Rightarrow \quad w = 0 \quad \Rightarrow \quad v \in U \setminus \{0\}.$$

Similarly, if $v = u + w$ is an eigenvector of $T$ corresponding to the eigenvalue 0, then observe that

$$0 = P(u + w) = u \quad \Rightarrow \quad v \in W \setminus \{0\}.$$

We may conclude that the eigenvalues and eigenvectors of $P$ are precisely:

| eigenvalue | corresponding eigenvectors |
|---|---|
| 1 | $u \in U \setminus \{0\}$ |
| 0 | $w \in W \setminus \{0\}$ |

**Exercise 5.A.13.** Suppose $T \in \mathcal{L}(V)$. Suppose $S \in \mathcal{L}(V)$ is invertible.

(a) Prove that $T$ and $S^{-1}TS$ have the same eigenvalues.

(b) What is the relationship between the eigenvectors of $T$ and the eigenvectors of $S^{-1}TS$?

**Solution.**

(a) Suppose $\lambda \in \mathbf{F}$ is an eigenvalue of $T$ a corresponding eigenvector $v \in V$. Because $S$ is surjective there is some $u \in V$ such that $v = Su$; notice that $u \neq 0$ since $v \neq 0$. It follows that

$$Tv = \lambda v \quad \Leftrightarrow \quad (TS)(u) = \lambda Su \quad \Leftrightarrow \quad (S^{-1}TS)(u) = \lambda u.$$

Thus $\lambda$ is an eigenvalue of $S^{-1}TS$ with a corresponding eigenvector $u$.

Similarly, suppose $\lambda \in \mathbf{F}$ is an eigenvalue of $S^{-1}TS$ with a corresponding eigenvector $u \in V$. Because $S^{-1}$ is surjective there is some $v \in V$ such that $u = S^{-1}v$; notice that $v \neq 0$ since $u \neq 0$. It follows that

$$(S^{-1}TS)(u) = \lambda u \quad \Leftrightarrow \quad (S^{-1}T)(v) = \lambda S^{-1}v \quad \Leftrightarrow \quad Tv = \lambda v.$$

Thus $\lambda$ is an eigenvalue of $T$ with a corresponding eigenvector $v$.



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(b) Let $\lambda \in \mathbf{F}$ be an eigenvalue of $T$. As we showed in part (a), this is the case if and only if $\lambda$ is an eigenvalue of $S^{-1}TS$. Define

$$E(\lambda, T) = \{v \in V : v \neq 0 \text{ and } Tv = \lambda v\}$$

$$\text{and} \quad E(\lambda, S^{-1}TS) = \{u \in V : u \neq 0 \text{ and } (S^{-1}TS)(u) = \lambda u\}.$$

That is, $E(\lambda, T)$ is the collection of eigenvectors of $T$ corresponding to the eigenvalue $\lambda$ and $E(\lambda, S^{-1}TS)$ is the collection of eigenvectors of $S^{-1}TS$ corresponding to the eigenvalue $\lambda$. Our calculations in part (a) show that

$$E(\lambda, T) = \{Su : u \in E(\lambda, S^{-1}TS)\} \quad \text{and} \quad E(\lambda, S^{-1}TS) = \{S^{-1}v : v \in E(\lambda, T)\}.$$

**Exercise 5.A.14.** Give an example of an operator on $\mathbf{R}^4$ that has no (real) eigenvalues.

**Solution.** Define $T \in \mathcal{L}(\mathbf{R}^4)$ by $T(x, y, z, t) = (y, z, t, -x)$ and suppose $\lambda \in \mathbf{R}$ is such that

$$T(x, y, z, t) = (y, z, t, -x) = \lambda(x, y, z, t).$$

We then have $-x = \lambda t = \lambda^2 z = \lambda^3 y = \lambda^4 x$. Notice that $x = 0$ implies $y = z = t = 0$. Since we are looking for eigenvectors, we can assume that $x \neq 0$ and thus arrive at the equation $\lambda^4 + 1 = 0$, which has no real solutions. It follows that $T$ has no real eigenvalues.

**Exercise 5.A.15.** Suppose $V$ is finite-dimensional, $T \in \mathcal{L}(V)$, and $\lambda \in \mathbf{F}$. Show that $\lambda$ is an eigenvalue of $T$ if and only if $\lambda$ is an eigenvalue of the dual operator $T' \in \mathcal{L}(V')$.

**Solution.** Observe that

$$\begin{aligned}
\lambda \text{ is an eigenvalue of } T &\iff T - \lambda I \text{ is not injective} && (5.7)\\
&\iff (T - \lambda I)' \text{ is not surjective} && (3.131)\\
&\iff T' - \lambda I' \text{ is not surjective} && (\text{Exercise 3.F.12})\\
&\iff T' - \lambda I \text{ is not surjective} && (\text{Exercise 3.F.13})\\
&\iff \lambda \text{ is an eigenvalue of } T'. && (5.7)
\end{aligned}$$

**Exercise 5.A.16.** Suppose $v_1, \ldots, v_n$ is a basis of $V$ and $T \in \mathcal{L}(V)$. Prove that if $\lambda$ is an eigenvalue of $T$, then

$$|\lambda| \leq n \max\Big\{\big|\mathcal{M}(T)_{j,k}\big| : 1 \leq j, k \leq n\Big\},$$

where $\mathcal{M}(T)_{j,k}$ denotes the entry in row $j$, column $k$ of the matrix of $T$ with respect to the basis $v_1, \ldots, v_n$.

*See Exercise 19 in Section 6A for a different bound on $|\lambda|$.*



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**Solution.** Let $A_{j,k} = \mathcal{M}(T)_{j,k}$ and suppose that $v = b_1 v_1 + \cdots + b_n v_n$ is an eigenvector of $T$ corresponding to the eigenvalue $\lambda$. Notice that

$$\begin{aligned}\lambda b_1 v_1 + \cdots + \lambda b_n v_n &= b_1 T v_1 + \cdots + b_n T v_n \\ &= b_1 \left( \sum_{j=1}^{n} A_{j,1} v_j \right) + \cdots + b_n \left( \sum_{j=1}^{n} A_{j,n} v_j \right) = \sum_{j=1}^{n} \left( \sum_{k=1}^{n} A_{j,k} b_k \right) v_j.\end{aligned}$$

It follows from unique representation that

$$\lambda b_j = \sum_{k=1}^{n} A_{j,k} b_k \quad \Rightarrow \quad |\lambda||b_j| \leq \sum_{k=1}^{n} |A_{j,k}||b_k|$$

for each $j \in \{1, ..., n\}$. Let $|b_i|$ be the largest amongst the values $|b_1|, ..., |b_n|$ and notice that $|b_i| > 0$, since $|b_i| = 0$ implies $b_1 = \cdots = b_n = 0$, so that $v$ is zero—but $v$ is an eigenvector and thus non-zero. Let $M = \max\left\{ \left| \mathcal{M}(T)_{j,k} \right| : 1 \leq j, k \leq n \right\}$ and observe that

$$|\lambda||b_i| \leq \sum_{k=1}^{n} |A_{i,k}||b_k| \quad \Rightarrow \quad |\lambda| \leq \sum_{k=1}^{n} |A_{i,k}| \frac{|b_k|}{|b_i|} \leq \sum_{k=1}^{n} |A_{i,k}| \leq \sum_{k=1}^{n} M = nM.$$

**Exercise 5.A.17.** Suppose $\mathbf{F} = \mathbf{R}, T \in \mathcal{L}(V)$, and $\lambda \in \mathbf{R}$. Prove that $\lambda$ is an eigenvalue of $T$ if and only if $\lambda$ is an eigenvalue of the complexification $T_{\mathbf{C}}$.

*See Exercise 33 in Section 3B for the definition of $T_{\mathbf{C}}$.*

**Solution.** Suppose that $\lambda$ is an eigenvalue of $T$ with a corresponding eigenvector $v \in V$. Notice that $v + i0$ is non-zero since $v$ is non-zero, and notice further that

$$T_{\mathbf{C}}(v + 0i) = Tv + iT(0) = \lambda v + 0i = \lambda(v + 0i).$$

Thus $\lambda$ is an eigenvalue of $T_{\mathbf{C}}$ with a corresponding eigenvector $v + 0i$.

Now suppose that $\lambda$ is an eigenvalue of $T_{\mathbf{C}}$ with a corresponding eigenvector $u + iv$. It follows that

$$T_{\mathbf{C}}(u + iv) = Tu + iTv = \lambda(u + iv) = \lambda u + i(\lambda v) \quad \Rightarrow \quad Tu = \lambda u \text{ and } Tv = \lambda v.$$

Since $u + iv \neq 0$, at least one of $u, v$ is non-zero. Thus $\lambda$ is an eigenvalue of $T$ with $u$ or $v$ (or both) as a corresponding eigenvector.

**Exercise 5.A.18.** Suppose $\mathbf{F} = \mathbf{R}, T \in \mathcal{L}(V)$, and $\lambda \in \mathbf{C}$. Prove that $\lambda$ is an eigenvalue of the complexification $T_{\mathbf{C}}$ if and only if $\overline{\lambda}$ is an eigenvalue of $T_{\mathbf{C}}$.

**Solution.** Suppose that $\lambda = a + bi$ and $u + iv$ is an eigenvector of $T_{\mathbf{C}}$ corresponding to $\lambda$. Observe that

$$\lambda(u + iv) = T_{\mathbf{C}}(u + iv) \quad \Leftrightarrow \quad (a + bi)(u + iv) = (au - bv) + (av + bu)i = Tu + iTv.$$

Thus $Tu = au - bv$ and $Tv = av + bu$. It follows that



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$$T_{\mathbf{C}}(u - iv) = Tu - iTv = (au - bv) - (av + bu) = (a + bi)(u - iv) = \overline{\lambda}(u - iv).$$

Furthermore, $u - iv$ is non-zero since $u + iv$ is non-zero. Thus $\overline{\lambda}$ is an eigenvalue of $T_{\mathbf{C}}$ with a corresponding eigenvector $u - iv$.

We have now shown that

$$\lambda \in \mathbf{C} \text{ is an eigenvalue of } T_{\mathbf{C}} \quad \Rightarrow \quad \overline{\lambda} \text{ is an eigenvalue of } T_{\mathbf{C}}.$$

The converse can be obtained by replacing $\lambda$ with $\overline{\lambda}$ in the implication above and using that $\overline{\overline{\lambda}} = \lambda$.

**Exercise 5.A.19.** Show that the forward shift operator $T \in \mathcal{L}(\mathbf{F}^\infty)$ defined by

$$T(z_1, z_2, ...) = (0, z_1, z_2, ...)$$

has no eigenvalues.

**Solution.** We are looking for solutions to the equation

$$(0, z_1, z_2, ...) = (\lambda z_1, \lambda z_2, \lambda z_3, ...),$$

where $(z_1, z_2, ...) \neq 0$ and $\lambda \in \mathbf{F}$. Notice that $\lambda = 0$ implies that each $z_k = 0$. If $\lambda \neq 0$ then the equation $0 = \lambda z_1$ implies that $z_1 = 0$, which gives us the equation $0 = \lambda z_2$, which implies that $z_2 = 0$, and so on. Thus both assumptions $\lambda = 0$ and $\lambda \neq 0$ imply that $(z_1, z_2, ...) = 0$. We may conclude that $T$ has no eigenvalues.

**Exercise 5.A.20.** Define the backward shift operator $S \in \mathcal{L}(\mathbf{F}^\infty)$ by

$$S(z_1, z_2, z_3, ...) = (z_2, z_3, ...).$$

(a) Show that every element of $\mathbf{F}$ is an eigenvalue of $S$.

(b) Find all eigenvectors of $S$.

**Solution.**

(a) Observe that for any $\lambda \in \mathbf{F}$ and any $\alpha \in \mathbf{F} \setminus \{0\}$ we have $\alpha(1, \lambda, \lambda^2, ...) \neq 0$ and

$$S(\alpha(1, \lambda, \lambda^2, ...)) = \alpha(\lambda, \lambda^2, \lambda^3, ...) = \lambda\alpha(1, \lambda, \lambda^2, ...).$$

Thus each $\lambda \in \mathbf{F}$ is an eigenvalue of $S$.

(b) Let $\lambda \in \mathbf{F}$ be given and suppose that $(z_1, z_2, z_3, ...)$ is an eigenvector of $S$ corresponding to $\lambda$, i.e.

$$(z_2, z_3, z_4, ...) = (\lambda z_1, \lambda z_2, \lambda z_3, ...).$$

This equation implies that $z_2 = \lambda z_1$, which gives us $z_3 = \lambda z_2 = \lambda^2 z_1$, and so on. Thus

$$(z_1, z_2, z_3, ...) = (z_1, \lambda z_1, \lambda^2 z_1, ...) = z_1(1, \lambda, \lambda^2, ...);$$



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this implies $z_1 \neq 0$ since eigenvectors are non-zero. Conversely, any vector of the form $\alpha(1, \lambda, \lambda^2, ...)$ for $\alpha \in \mathbf{F} \setminus \{0\}$ is an eigenvector of $S$ corresponding to $\lambda$, as we showed in part (a). We may conclude that the eigenvalues and eigenvectors of $S$ are precisely:

| eigenvalue | corresponding eigenvectors |
|---|---|
| $\lambda \in \mathbf{F}$ | $\alpha(1, \lambda, \lambda^2, ...)$ for $\alpha \in \mathbf{F} \setminus \{0\}$ |

**Exercise 5.A.21.** Suppose $T \in \mathcal{L}(V)$ is invertible.

(a) Suppose $\lambda \in \mathbf{F}$ with $\lambda \neq 0$. Prove that $\lambda$ is an eigenvalue of $T$ if and only if $\frac{1}{\lambda}$ is an eigenvalue of $T^{-1}$.

(b) Prove that $T$ and $T^{-1}$ have the same eigenvectors.

**Solution.**

(a) For $\lambda \neq 0$ and $v \neq 0$, observe that

$$Tv = \lambda v \quad \Leftrightarrow \quad v = \lambda T^{-1} v \quad \Leftrightarrow \quad \lambda^{-1} v = T^{-1} v.$$

(b) Notice that $T$ and $T^{-1}$ are both injective and so neither has 0 as an eigenvalue. Thus any eigenvector of $T$ or $T^{-1}$ must correspond to a non-zero eigenvalue. It follows from part (a) that

$$v \text{ is an eigenvector of } T \quad \Leftrightarrow \quad v \text{ is an eigenvector of } T^{-1}.$$

**Exercise 5.A.22.** Suppose $T \in \mathcal{L}(V)$ and there exist nonzero vectors $u$ and $w$ in $V$ such that

$$Tu = 3w \quad \text{and} \quad Tw = 3u.$$

Prove that 3 or $-3$ is an eigenvalue of $T$.

**Solution.** Applying $T$ to both sides of the equation $Tu = 3w$ shows that $T^2 u = 9u$ or equivalently $(T^2 - 9I)(u) = 0$. Because $u$ is non-zero, this demonstrates that the operator $T^2 - 9I = (T - 3I)(T + 3I)$ is not injective. It must then be the case that at least one of the operators $T - 3I, T + 3I$ is not injective and thus 3 or $-3$ is an eigenvalue of $T$.

**Exercise 5.A.23.** Suppose $V$ is finite-dimensional and $S, T \in \mathcal{L}(V)$. Prove that $ST$ and $TS$ have the same eigenvalues.

**Solution.** Exercise 3.D.11 shows that $ST$ is invertible if and only if $TS$ is invertible; by 5.7, this is equivalent to $ST$ having 0 as an eigenvalue if and only if $TS$ has 0 as an eigenvalue.

Suppose that $\lambda \neq 0$ is an eigenvalue of $ST$ with corresponding eigenvector $v \in V$, i.e. $S(Tv) = \lambda v$, and note that $Tv \neq 0$ since $\lambda \neq 0$ and $v \neq 0$. Note further that

$$(TS)(Tv) = T(S(Tv)) = T(\lambda v) = \lambda T v.$$



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Thus $\lambda$ is an eigenvalue of $TS$ with a corresponding eigenvector $Tv$. Swapping the roles of $S$ and $T$ in this argument shows that any non-zero eigenvalue of $TS$ must also be an eigenvalue of $ST$.

**Exercise 5.A.24.** Suppose $A$ is an $n$-by-$n$ matrix with entries in $\mathbf{F}$. Define $T \in \mathcal{L}(\mathbf{F}^n)$ by $Tx = Ax$, where elements of $\mathbf{F}^n$ are thought of as $n$-by-1 column vectors.

(a) Suppose the sum of the entries in each row of $A$ equals 1. Prove that 1 is an eigenvalue of $T$.

(b) Suppose the sum of the entries in each column of $A$ equals 1. Prove that 1 is an eigenvalue of $T$.

**Solution.**

(a) Let $A_{j,k}$ be the entries of $A$; our assumption is that $\sum_{k=1}^{n} A_{j,k} = 1$ for each $j \in \{1, ..., n\}$. Observe that

$$T(1, ..., 1) = \begin{pmatrix} A_{1,1} & \cdots & A_{1,n} \\ \vdots & \ddots & \vdots \\ A_{n,1} & \cdots & A_{n,n} \end{pmatrix} \begin{pmatrix} 1 \\ \vdots \\ 1 \end{pmatrix} = \begin{pmatrix} \sum_{k=1}^{n} A_{1,k} \\ \vdots \\ \sum_{k=1}^{n} A_{n,k} \end{pmatrix} = \begin{pmatrix} 1 \\ \vdots \\ 1 \end{pmatrix}.$$

Thus 1 is an eigenvalue of $T$.

(b) Here are two arguments.

(1) Let $e_1, ..., e_n$ be the standard basis of $\mathbf{F}^n$ and let $\psi \in (\mathbf{F}^n)'$ be the linear functional given by $\psi(e_k) = 1$. Certainly the matrix of $T$ with respect to $e_1, ..., e_n$ is $A$. It follows that

$$(\psi \circ (T - I))(e_k) = \psi\left(\left(\sum_{j=1}^{n} A_{j,k} e_j\right) - e_k\right) = \left(\sum_{j=1}^{n} A_{j,k}\right) - 1 = 0,$$

where we have used our assumption that the sum of the entries in each column of $A$ equals 1. Thus $\psi \circ (T - I) \in (\mathbf{F}^n)'$ is the zero map. If the operator $T - I$ were invertible then it would have to be the case that $\psi = 0$; because $\psi$ is non-zero, we see that $T - I$ is not invertible. That is, 1 is an eigenvalue of $T$.

(2) Define $S \in \mathcal{L}(\mathbf{F}^n)$ by $Sx = A^{\mathrm{t}}x$. Because the rows of $A^{\mathrm{t}}$ are the columns of $A$, part (a) shows that 1 is an eigenvalue of $S$. Let $e_1, ..., e_n$ be the standard basis of $\mathbf{F}^n$, let $\varphi_1, ..., \varphi_n$ be the corresponding dual basis, and define an isomorphism $\Phi \in \mathcal{L}\big(\mathbf{F}^n, (\mathbf{F}^n)'\big)$ by $\Phi e_k = \varphi_k$. Certainly the matrix of $T$ with respect to $e_1, ..., e_n$ is $A$ and thus, by 3.132, the matrix of $T'$ with respect to $\varphi_1, ..., \varphi_n$ is $A^{\mathrm{t}}$. It follows that

$$\begin{aligned}(\Phi^{-1}T'\Phi)(e_k) = (\Phi^{-1}T')(\varphi_k) = \Phi^{-1}\big(A_{k,1}\psi_1 + \cdots + A_{k,n}\psi_n\big) \\ = A_{k,1}e_1 + \cdots + A_{k,n}e_n = A^{\mathrm{t}}e_k = Se_k.\end{aligned}$$



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Thus $\Phi^{-1}T'\Phi = S$, so that $\Phi^{-1}T'\Phi$ has 1 as an eigenvalue. A small modification of the argument given in Exercise 5.A.13 (a) shows that $\Phi^{-1}T'\Phi$ and $T'$ have the same eigenvalues. It follows that 1 is an eigenvalue of $T'$ and hence, by Exercise 5.A.15, 1 is an eigenvalue of $T$.

**Exercise 5.A.25.** Suppose $T \in \mathcal{L}(V)$ and $u, w$ are eigenvectors of $T$ such that $u + w$ is also an eigenvector of $T$. Prove that $u$ and $w$ are eigenvectors of $T$ corresponding to the same eigenvalue.

**Solution.** Suppose that the eigenvectors $u, w, u + w$ correspond to the eigenvalues $\alpha, \beta, \lambda$, respectively. Observe that

$$\alpha u + \beta w = Tu + Tw = T(u + w) = \lambda(u + w) = \lambda u + \lambda w.$$

It follows that $(\alpha - \lambda)u + (\beta - \lambda)w = 0$. If we suppose that $\alpha \neq \beta$ then 5.11 shows that $u$ and $w$ are linearly independent and the equation $(\alpha - \lambda)u + (\beta - \lambda)w = 0$ then implies that $\alpha = \beta = \lambda$, contradicting our assumption. Thus $\alpha = \beta$.

**Exercise 5.A.26.** Suppose $T \in \mathcal{L}(V)$ is such that every nonzero vector in $V$ is an eigenvector of $T$. Prove that $T$ is a scalar multiple of the identity operator.

**Solution.** The case where $V = \{0\}$ is easily handled, so assume that $V \neq \{0\}$ and fix some non-zero $u \in V$; by assumption we have $Tu = \lambda u$ for some $\lambda \in \mathbf{F}$. Suppose that $v \in V$ is non-zero. If $u + v \neq 0$ then by assumption $v$ and $u + v$ are both eigenvectors of $T$. It then follows from Exercise 5.A.25 that $u$ and $v$ correspond to the same eigenvalue $\lambda$, so that $Tv = \lambda v$. If $u + v = 0$ then $Tv = -Tu = -\lambda u = \lambda v$. Thus we have $Tv = \lambda v$ for all $v \in V$, i.e. $T = \lambda I$.

**Exercise 5.A.27.** Suppose that $V$ is finite-dimensional and $k \in \{1, ..., \dim V - 1\}$. Suppose $T \in \mathcal{L}(V)$ is such that every subspace of $V$ of dimension $k$ is invariant under $T$. Prove that $T$ is a scalar multiple of the identity operator.

**Solution.** If $\dim V = 0$ then $T = 0I$ and if $\dim V = 1$ then Exercise 3.A.7 shows that $T = \lambda I$ for some $\lambda \in \mathbf{F}$. Suppose that $\dim V \geq 2$. For $k \in \{1, ..., \dim V - 1\}$, let $P(k)$ be the statement that if every subspace of $V$ of dimension $k$ is invariant under $T$, then $T$ is a scalar multiple of the identity operator. We will prove that $P(1)$ holds and that $P(k)$ implies $P(k + 1)$, provided $k + 1 \in \{1, ..., \dim V - 1\}$.

Suppose that every one-dimensional subspace of $V$ is invariant under $T$ and let $v \in V$ be non-zero. By assumption the subspace $\operatorname{span}(v)$ is invariant under $T$ and thus $Tv = \lambda v$ for some $\lambda \in \mathbf{F}$, i.e. $v$ is an eigenvector of $T$. It follows from Exercise 5.A.26 that $T$ is a scalar multiple of the identity operator. Thus $P(1)$ holds.

Now suppose that $P(k)$ holds for some $k$ such that $k + 1 \in \{1, ..., \dim V - 1\}$ and suppose that every subspace of dimension $k + 1$ is invariant under $T$. Let $U$ be a subspace of dimen-



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sion $k$. Because $k \leq \dim V - 2$, we can find linearly independent vectors $v, w \in V$ such that $v, w \notin U$. Let $u \in U$ be given. By assumption the subspaces $U \oplus \operatorname{span}(v)$ and $U \oplus \operatorname{span}(w)$, which have dimension $k+1$, are invariant under $T$. Since $u$ belongs to each of these subspaces, it follows that

$$Tu = a_1 u_1 + bv \quad \text{and} \quad Tu = a_2 u_2 + cw$$

for some $u_1, u_2 \in U$ and some $a_1, a_2, b, c \in \mathbf{F}$. This implies that $a_1 u_1 - a_2 u_2 = cw - bv$, so that $cw - bv \in U \cap \operatorname{span}(v, w) = \{0\}$. Thus $cw - bv = 0$ and the linear independence of $v, w$ then gives us $b = c = 0$, so that $Tu \in U$. That is, $U$ is invariant under $T$. Because we assumed that $P(k)$ holds, it now follows that $T$ is a scalar multiple of the identity, i.e. $P(k+1)$ holds.

Thus, by induction, $P(k)$ holds for each $k \in \{1, ..., \dim V - 1\}$.

**Exercise 5.A.28.** Suppose $V$ is finite-dimensional and $T \in \mathcal{L}(V)$. Prove that $T$ has at most $1 + \dim \operatorname{range} T$ distinct eigenvalues.

**Solution.** Suppose $\lambda_1, ..., \lambda_n$ are $n$ distinct eigenvalues of $T$ with corresponding eigenvectors $v_1, ..., v_n$ and note that each $Tv_k = \lambda_k v_k \in \operatorname{range} T$. The list $v_1, ..., v_n$ is linearly independent by 5.11 and thus the list $\lambda_1 v_1, ..., \lambda_n v_n$ is also linearly independent, provided each $\lambda_k$ is non-zero; if some $\lambda_k = 0$ (since the eigenvalues are distinct there can be at most one such $\lambda_k$), we can discard $\lambda_k v_k$ from the list and be left with a linearly independent list of $n-1$ vectors. In either case, there are at least $n-1$ linearly independent vectors in $\operatorname{range} T$ and thus $n \leq 1 + \dim \operatorname{range} T$.

**Exercise 5.A.29.** Suppose $T \in \mathcal{L}(\mathbf{R}^3)$ and $-4, 5$, and $\sqrt{7}$ are eigenvalues of $T$. Prove that there exists $x \in \mathbf{R}^3$ such that $Tx - 9x = \left(-4, 5, \sqrt{7}\right)$.

**Solution.** Because $T$ has $3 = \dim \mathbf{R}^3$ distinct eigenvalues, 5.12 shows that 9 cannot be an eigenvalue of $T$. By 5.7 this is equivalent to the operator $T - 9I$ being invertible. Thus the desired $x \in \mathbf{R}^3$ is $(T - 9I)^{-1}\left(-4, 5, \sqrt{7}\right)$.

**Exercise 5.A.30.** Suppose $T \in \mathcal{L}(V)$ and $(T - 2I)(T - 3I)(T - 4I) = 0$. Suppose $\lambda$ is an eigenvalue of $T$. Prove that $\lambda = 2$ or $\lambda = 3$ or $\lambda = 4$.

**Solution.** We have $Tv = \lambda v$ for some $v \neq 0$. Observe that

$$0 = (T - 2I)(T - 3I)(T - 4I)v = (\lambda - 2)(\lambda - 3)(\lambda - 4)v.$$

Since $v$ is non-zero, this equation implies that $\lambda \in \{2, 3, 4\}$.

**Exercise 5.A.31.** Give an example of $T \in \mathcal{L}(\mathbf{R}^2)$ such that $T^4 = -I$.

**Solution.** Define $T \in \mathcal{L}(\mathbf{R}^2)$ by



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$$T(1,0) = \left(\cos\tfrac{\pi}{4}, \sin\tfrac{\pi}{4}\right) = \tfrac{1}{\sqrt{2}}(1,1) \quad \text{and} \quad T(0,1) = \left(\cos\tfrac{3\pi}{4}, \sin\tfrac{3\pi}{4}\right) = \tfrac{1}{\sqrt{2}}(-1,1).$$

Note that $T$ is a counterclockwise rotation about the origin by 45°. It follows that $T^4$ is a counterclockwise rotation about the origin by 180°, i.e.

$$T^4(1,0) = (\cos\pi, \sin\pi) = (-1,0) \quad \text{and} \quad T(0,1) = \left(\cos\tfrac{3\pi}{2}, \sin\tfrac{3\pi}{2}\right) = (0,-1).$$

Thus $T^4 = -I$.

**Exercise 5.A.32.** Suppose $T \in \mathcal{L}(V)$ has no eigenvalues and $T^4 = I$. Prove that $T^2 = -I$.

**Solution.** Observe that $T^4 = I$ is equivalent to $(T^2 - I)(T^2 + I) = 0$. Because $T$ has no eigenvalues, the operators $T - I$ and $T + I$ must be injective and thus their product $(T - I)(T + I) = T^2 - I$ is also injective. It follows that for any $v \neq 0$,

$$0 = (T^2 - I)(T^2 + I)v \quad \Leftrightarrow \quad 0 = (T^2 + I)v \quad \Leftrightarrow \quad T^2 v = -v.$$

Thus $T^2 = -I$.

**Exercise 5.A.33.** Suppose $T \in \mathcal{L}(V)$ and $m$ is a positive integer.

(a) Prove that $T$ is injective if and only if $T^m$ is injective.

(b) Prove that $T$ is surjective if and only if $T^m$ is surjective.

**Solution.** Certainly these results are true if $m = 1$, so suppose that $m \geq 2$.

(a) The composition of injective functions is again injective, so $T^m$ is injective if $T$ is injective. Suppose that $T^m$ is injective and let $v \in \operatorname{null} T$ be given. It follows that

$$Tv = 0 \quad \Rightarrow \quad T^m v = T^{m-1}(0) = 0 \quad \Rightarrow \quad v \in \operatorname{null} T^m = \{0\} \quad \Rightarrow \quad v = 0.$$

Thus $\operatorname{null} T = \{0\}$, i.e. $T$ is injective.

(b) Suppose that $T^m$ is surjective and let $w \in V$ be given. There exists some $v \in V$ such that $T^m v = w$, which is equivalent to $T(T^{m-1}v) = w$. Thus $T$ is surjective.

Suppose that $T$ is surjective and let $w \in V$ be given. There exist vectors $v_1, ..., v_m$ such that:

$$Tv_1 = w \quad \text{and} \quad Tv_k = v_{k-1} \text{ for } k \geq 2.$$

It follows that $T^m v_m = w$ and hence that $T^m$ is surjective.

**Exercise 5.A.34.** Suppose $V$ is finite-dimensional and $v_1, ..., v_m \in V$. Prove that the list $v_1, ..., v_m$ is linearly independent if and only if there exists $T \in \mathcal{L}(V)$ such that $v_1, ..., v_m$ are eigenvectors of $T$ corresponding to distinct eigenvalues.



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**Solution.** Suppose $v_1, ..., v_m$ is linearly independent and extend this to a basis $v_1, ..., v_m, w_1, ..., w_n$ of $V$. Define $T \in \mathcal{L}(V)$ by

$$Tv_k = kv_k \text{ for } k \in \{1, ..., m\} \quad \text{and} \quad Tw_k = 0 \text{ for } k \in \{1, ..., n\}.$$

It follows that $v_1, ..., v_m$ are eigenvectors of $T$ corresponding to distinct eigenvalues.

The converse implication is the content of 5.11.

**Exercise 5.A.35.** Suppose that $\lambda_1, ..., \lambda_n$ is a list of distinct real numbers. Prove that the list $e^{\lambda_1 x}, ..., e^{\lambda_n x}$ is linearly independent in the vector space of real-valued functions on $\mathbf{R}$.

*Hint: Let $V = \operatorname{span}\left(e^{\lambda_1 x}, ..., e^{\lambda_n x}\right)$, and define an operator $D \in \mathcal{L}(V)$ by $Df = f'$. Find eigenvalues and eigenvectors of $D$.*

**Solution.** Let $V = \operatorname{span}\left(e^{\lambda_1 x}, ..., e^{\lambda_n x}\right)$, and define an operator $D \in \mathcal{L}(V)$ by $Df = f'$. For each $k \in \{1, ..., n\}$,

$$D\left(e^{\lambda_k x}\right) = \left(e^{\lambda_k x}\right)' = \lambda_k e^{\lambda_k x}.$$

This demonstrates that $D$ indeed maps $V$ into $V$ and also that each $\lambda_k$ is an eigenvalue of $D$ with a corresponding eigenvector $e^{\lambda_k x}$. Because the eigenvalues $\lambda_1, ..., \lambda_n$ are distinct, 5.11 shows that the corresponding eigenvectors $e^{\lambda_1 x}, ..., e^{\lambda_n x}$ are linearly independent.

**Exercise 5.A.36.** Suppose that $\lambda_1, ..., \lambda_n$ is a list of distinct positive numbers. Prove that the list $\cos(\lambda_1 x), ..., \cos(\lambda_n x)$ is linearly independent in the vector space of real-valued functions on $\mathbf{R}$.

**Solution.** Let $V = \operatorname{span}(\cos(\lambda_1 x), ..., \cos(\lambda_n x))$ and define $D \in \mathcal{L}(V)$ by $Df = \frac{\mathrm{d}^4}{\mathrm{d}x^4} f$. For each $k \in \{1, ..., n\}$,

$$D(\cos(\lambda_k x)) = \frac{\mathrm{d}^4}{\mathrm{d}x^4} \cos(\lambda_k x) = \lambda_k^4 \cos(\lambda_k x).$$

This demonstrates that $D$ indeed maps $V$ into $V$ and also that each $\lambda_k^4$ is an eigenvalue of $D$ with a corresponding eigenvector $\cos(\lambda_k x)$. Because $\lambda_1, ..., \lambda_n$ are distinct and positive, the real numbers $\lambda_1^4, ..., \lambda_n^4$ are also distinct. It then follows from 5.11 that the list $\cos(\lambda_1 x), ..., \cos(\lambda_n x)$ is linearly independent.

**Exercise 5.A.37.** Suppose $V$ is finite-dimensional and $T \in \mathcal{L}(V)$. Define $\mathcal{A} \in \mathcal{L}(\mathcal{L}(V))$ by

$$\mathcal{A}(S) = TS$$

for each $S \in \mathcal{L}(V)$. Prove that the set of eigenvalues of $T$ equals the set of eigenvalues of $\mathcal{A}$.



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**Solution.** Suppose that $\lambda \in \mathbf{F}$ is an eigenvalue of $T$, so that there is some non-zero $v_1 \in V$ such that $Tv_1 = \lambda v_1$. Extend $v_1$ to a basis $v_1, ..., v_m$ of $V$ and define $S \in \mathcal{L}(V)$ by

$$Sv_1 = v_1 \quad \text{and} \quad Sv_k = 0 \text{ for } k \geq 2.$$

Notice that $S \neq 0$ and that

$$(T - \lambda I)Sv_1 = (T - \lambda I)v_1 = 0 \quad \text{and} \quad (T - \lambda I)Sv_k = 0 \text{ for } k \geq 2.$$

Thus $(T - \lambda I)S = 0$, i.e. $TS = \lambda S$. It follows that $\lambda$ is an eigenvalue of $\mathcal{A}$ with a corresponding eigenvector $S$.

Now suppose that $\lambda \in \mathbf{F}$ is an eigenvalue of $\mathcal{A}$, i.e. there is some non-zero $S \in \mathcal{L}(V)$ such that $TS = \lambda S$, or equivalently $(T - \lambda I)S = 0$. If $T - \lambda I$ were injective then the equation $(T - \lambda I)S = 0$ would imply that $S = 0$. Given that $S$ is non-zero, it must be the case that $T - \lambda I$ is not injective and thus $\lambda$ is an eigenvalue of $T$.

**Exercise 5.A.38.** Suppose $V$ is finite-dimensional, $T \in \mathcal{L}(V)$, and $U$ is a subspace of $V$ invariant under $T$. The *quotient operator* $T/U \in \mathcal{L}(V/U)$ is defined by

$$(T/U)(v + U) = Tv + U$$

for each $v \in V$.

(a) Show that the definition of $T/U$ makes sense (which requires using the condition that $U$ is invariant under $T$) and show that $T/U$ is an operator on $V/U$.

(b) Show that each eigenvalue of $T/U$ is an eigenvalue of $T$.

**Solution.**

(a) Suppose that $v, w \in V$ are such that $v + U = w + U$, which by 3.101 is equivalent to $v - w \in U$. Because $U$ is invariant under $T$ it follows that $Tv - Tw \in U$ and another application of 3.101 gives us $Tv + U = Tw + U$. Thus the definition of $T/U$ makes sense.

Certainly $T/U$ maps $V/U$ into $V/U$. Let $v + U, w + U \in V/U$ and $\lambda \in \mathbf{F}$ be given. Observe that

$$\begin{aligned}(T/U)((v + U) + (w + U)) &= (T/U)((v + w) + U) = T(v + w) + U \\ &= (Tv + Tw) + U = (Tv + U) + (Tw + U) = (T/U)(v + U) + (T/U)(w + U),\end{aligned}$$

$$\begin{aligned}(T/U)(\lambda(v + U)) &= (T/U)(\lambda v + U) = T(\lambda v) + U \\ &= \lambda Tv + U = \lambda(Tv + U) = \lambda(T/U)(v + U).\end{aligned}$$

Thus $T/U$ is a linear operator on $V/U$.

(b) Suppose $\lambda \in \mathbf{F}$ is an eigenvalue of $T/U$, i.e. there exists a non-zero $v + U \in V/U$ such that

$$(T/U)(v + U) = Tv + U = \lambda(v + U) = \lambda v + U.$$



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Note that $U$ is invariant under $T - \lambda I$: for any $u \in U$ we have $Tu - \lambda u \in U$ since $U$ is invariant under $T$. Here are two ways to proceed; notice that in each argument we only require that $U$ is finite-dimensional.

(1) Consider the restriction operator $R \coloneqq (T - \lambda I)|_U \in \mathcal{L}(U)$. There are two cases.

Suppose that $R$ is not surjective; by 3.65 this is equivalent to $\operatorname{null} R \neq \{0\}$, so that there exists some non-zero $u \in U$ such that $Ru = 0$, i.e. $Tu = \lambda u$. Thus $\lambda$ is an eigenvalue of $T$.

Now suppose that $R$ is surjective. Because $Tv + U = \lambda v + U$, we have $Tv = \lambda v + w$ for some $w \in U$. The surjectivity of $R$ implies that there exists some $u \in U$ such that $-w = Ru = Tu - \lambda u$. Observe that

$$T(v + u) = Tv + Tu = \lambda v + w + Tu = \lambda v + \lambda u = \lambda(v + u).$$

Note that $v + u$ must be non-zero, otherwise $v$ would belong to $U$, contradicting that $v + U$ is non-zero. Thus $\lambda$ is an eigenvalue of $T$.

(2) Let $u_1, ..., u_n$ be a basis of $U$. Because $Tv + U = \lambda v + U$ and $U$ is invariant under $T - \lambda I$, the list

$$(T - \lambda I)v, (T - \lambda I)u_1, ..., (T - \lambda I)u_n$$

is contained in $U$. This is a list of $n + 1$ vectors in an $n$-dimensional space and hence must be linearly dependent, i.e. there are scalars $a_0, a_1, ..., a_n$, not all zero, such that $w \coloneqq a_0 v + a_1 u_1 + \cdots + a_n u_n$ satisfies $(T - \lambda I)w = 0$. Note that $w$ must be non-zero: if $w = 0$ and $a_0 \neq 0$ then $v \in U$, contradicting $v + U \neq 0$, and if $w = 0$ and $a_0 = 0$ then $a_1 u_1 + \cdots + a_n u_n$ is a non-trivial linear combination, contradicting the linear independence of $u_1, ..., u_n$. Thus $\lambda$ is an eigenvalue of $T$.

**Exercise 5.A.39.** Suppose $V$ is finite-dimensional and $T \in \mathcal{L}(V)$. Prove that $T$ has an eigenvalue if and only if there exists a subspace of $V$ of dimension $\dim V - 1$ that is invariant under $T$.

**Solution.** Suppose that $T$ has an eigenvalue $\lambda \in \mathbf{F}$ with a corresponding eigenvector $v \in V$ it follows that $\dim \operatorname{null}(T - \lambda I) \geq 1$ and hence that $\dim \operatorname{range}(T - \lambda I) \leq \dim V - 1$. By taking a basis of $\operatorname{range}(T - \lambda I)$ and, if necessary, extending it to a linearly independent list of length $\dim V - 1$, we can obtain a subspace $U$ of $V$ satisfying $\operatorname{range}(T - \lambda I) \subseteq U$ and $\dim U = \dim V - 1$. Exercise 5.A.39 shows that $U$ is invariant under $T - \lambda I$, which implies that $U$ is invariant under $T$:

$$u \in U \quad \Rightarrow \quad Tu - \lambda u \in U \quad \Rightarrow \quad Tu \in U.$$

Now suppose that there exists a subspace $U$ of $V$ such that $U$ is invariant under $T$ and $\dim U = \dim V - 1$, and consider the quotient operator $T/U \in \mathcal{L}(V/U)$. Since $\dim V/U = 1$, Exercise 3.A.7 shows that $T/U = \lambda I$ for some $\lambda \in \mathbf{F}$; it follows that $\lambda$ is an eigenvalue of $T/U$ and thus, by Exercise 5.A.38, $\lambda$ is an eigenvalue of $T$.



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**Exercise 5.A.40.** Suppose $S, T \in \mathcal{L}(V)$ and $S$ is invertible. Suppose $p \in \mathcal{P}(\mathbf{F})$ is a polynomial. Prove that

$$p(STS^{-1}) = Sp(T)S^{-1}.$$

**Solution.** Notice that $(STS^{-1})^0 = I = SS^{-1} = SIS^{-1} = ST^0S^{-1}$. For a non-negative integer $k$, suppose that $(STS^{-1})^k = ST^kS^{-1}$ and observe that

$$(STS^{-1})^{k+1} = (STS^{-1})^k STS^{-1} = ST^kS^{-1}STS^{-1} = ST^kITS^{-1} = ST^{k+1}S^{-1}.$$

Thus $(STS^{-1})^k = ST^kS^{-1}$ for all non-negative integers $k$. Suppose $p = \sum_{k=0}^n a_k z^k$ and observe that

$$p(STS^{-1}) = \sum_{k=0}^n a_k (STS^{-1})^k = \sum_{k=0}^n a_k ST^kS^{-1} = S\left(\sum_{k=0}^n a_k T^k\right)S^{-1} = Sp(T)S^{-1}.$$

**Exercise 5.A.41.** Suppose $T \in \mathcal{L}(V)$ and $U$ is a subspace of $V$ invariant under $T$. Prove that $U$ is invariant under $p(T)$ for every polynomial $p \in \mathcal{P}(\mathbf{F})$.

**Solution.** Certainly $U$ is invariant under $T^k$ for any non-negative integer $k$. Suppose $p = \sum_{k=0}^n a_k z^k$ and observe that

$$p(T)u = \left(\sum_{k=0}^n a_k T^k\right)u = \sum_{k=0}^n a_k T^k u;$$

this belongs to $U$ since each $T^k u \in U$ and $U$ is closed under vector addition and scalar multiplication.

**Exercise 5.A.42.** Define $T \in \mathcal{L}(\mathbf{F}^n)$ by $T(x_1, x_2, x_3, ..., x_n) = (x_1, 2x_2, 3x_3, ..., nx_n)$.

(a) Find all eigenvalues and eigenvectors of $T$.

(b) Find all subspaces of $\mathbf{F}^n$ that are invariant under $T$.

**Solution.**

(a) Let $e_1, ..., e_n$ be the standard basis of $\mathbf{F}^n$ and notice that $Te_k = ke_k$ for each $k \in \{1, ..., n\}$. Thus $k$ is an eigenvalue of $T$ with a corresponding eigenvector $e_k$. By 5.11 and 5.12 we can conclude that the eigenvalues and eigenvectors of $T$ are precisely:

| eigenvalue | corresponding eigenvectors |
|---|---|
| $k \in \{1, ..., n\}$ | $\text{span}(e_k) \setminus \{0\}$ |

(b) First, let us prove some useful results.



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**Lemma L.4.** Suppose $T \in \mathcal{L}(V)$ and $U$ is a subspace of $V$ invariant under $T$. If $\lambda_1, ..., \lambda_m$ are distinct eigenvalues of $T$ with corresponding eigenvectors $v_1, ..., v_m$, then

$$v_1 + \cdots + v_m \in U \quad \Leftrightarrow \quad v_k \in U \text{ for each } k \in \{1, ..., m\}.$$

*Proof.* If $v_k \in U$ for each $k \in \{1, ..., m\}$ then $v_1 + \cdots + v_m \in U$ since $U$ is a subspace of $V$.

For the converse, we will proceed by induction on $m$. The base case $m = 1$ is clear, so suppose that the result holds for some positive integer $m$ and let $\lambda_1, ..., \lambda_{m+1}$ be distinct eigenvalues of $T$ with corresponding eigenvectors $v_1, ..., v_{m+1}$. Suppose that $v = v_1 + \cdots + v_{m+1} \in U$. Because $U$ is invariant under $T$ we have

$$Tv = \lambda_1 v_1 + \cdots + \lambda_{m+1} v_{m+1} \in U$$

$$\Rightarrow \quad Tv - \lambda_{m+1} v = (\lambda_1 - \lambda_{m+1})v_1 + \cdots + (\lambda_m - \lambda_{m+1})v_m \in U$$

Let $k \in \{1, ..., m\}$ be given. By assumption the eigenvalues $\lambda_1, ..., \lambda_{m+1}$ are distinct and thus $\lambda_k - \lambda_{m+1} \neq 0$. It follows that $(\lambda_k - \lambda_{m+1})v_k$ is an eigenvector of $T$ corresponding to the eigenvalue $\lambda_k$. Our induction hypothesis then guarantees that $(\lambda_k - \lambda_{m+1})v_k$ belongs to $U$. Thus $v_k$ belongs to $U$ for each $k \in \{1, ..., m\}$, which gives us

$$v_{m+1} = v - v_1 - \cdots - v_m \in U.$$

This completes the induction step and the proof. □



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**Lemma L.5.** Suppose $T \in \mathcal{L}(V), \dim V = n$, and $\lambda_1, ..., \lambda_n$ are distinct eigenvalues of $T$ with corresponding eigenvectors $v_1, ..., v_n$. If $U$ is a subspace of $V$ invariant under $T$, then

$$U = (U \cap E_1) \oplus \cdots \oplus (U \cap E_n) \quad \text{where} \quad E_k = \operatorname{span}(v_k).$$

*Proof.* By 5.11 the eigenvectors $v_1, ..., v_n$ are linearly independent and hence form a basis of $V$, so that $V = E_1 \oplus \cdots \oplus E_n$. For any $u \in U$ we have $u = e_1 + \cdots + e_n$, where each $e_k \in E_k$. If any $e_k = 0$ then certainly $e_k \in U$; otherwise, $e_k$ is an eigenvector of $T$ corresponding to the eigenvalue $\lambda_k$ and thus, by Lemma L.4, the non-zero $e_k$'s belong to $U$ also. It follows that $u \in (U \cap E_1) + \cdots + (U \cap E_n)$ and hence that

$$U = (U \cap E_1) + \cdots + (U \cap E_n).$$

The directness of this sum follows immediately from the directness of the sum $V = E_1 \oplus \cdots \oplus E_n$. □

**Lemma L.6.** If $T \in \mathcal{L}(V), \dim V = n \geq 1$, and $\lambda_1, ..., \lambda_n$ are distinct eigenvalues of $T$ with corresponding eigenvectors $v_1, ..., v_n$, then the non-zero subspaces of $V$ which are invariant under $T$ are precisely those of the form

$$\operatorname{span}(v_{k_1}, ..., v_{k_m})$$

for some choice of integers $1 \leq k_1 < \cdots < k_m \leq n$ with $1 \leq m \leq n$.

*Proof.* It is straightforward to verify that each $\operatorname{span}(v_{k_1}, ..., v_{k_m})$ is indeed a subspace of $V$ invariant under $T$. For $k \in \{1, ..., n\}$ let $E_k = \operatorname{span}(v_k)$ and suppose $U$ is a non-zero subspace of $V$ invariant under $T$. By Lemma L.5 we have

$$U = (U \cap E_1) \oplus \cdots \oplus (U \cap E_n).$$

For each $k$, since $\dim E_k = 1$, we can either have $U \cap E_k = \{0\}$ or $U \cap E_k = E_k$. Because $U$ is non-zero, there must be at least one $k$ such that $U \cap E_k = E_k$; let $1 \leq k_1 < \cdots < k_m \leq n$ be those indices for which $U \cap E_k = E_k$. It follows that

$$U = (U \cap E_1) \oplus \cdots \oplus (U \cap E_n) = E_{k_1} \oplus \cdots \oplus E_{k_m} = \operatorname{span}(v_{k_1}, ..., v_{k_m}).$$ □

Now let us return to the exercise. As we showed in part (a), the eigenvalues of $T$ are $1, ..., n$ with corresponding eigenvectors $e_1, ..., e_n$. It then follows from Lemma L.6 that the non-zero subspaces of $\mathbf{F}^n$ which are invariant under $T$ are precisely those of the form

$$\operatorname{span}(e_{k_1}, ..., e_{k_m})$$

for some choice of integers $1 \leq k_1 < \cdots < k_m \leq n$ with $1 \leq m \leq n$.



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**Exercise 5.A.43.** Suppose that $V$ is finite-dimensional, $\dim V > 1$, and $T \in \mathcal{L}(V)$. Prove that $\{p(T) : p \in \mathcal{P}(\mathbf{F})\} \neq \mathcal{L}(V)$.

**Solution.** If every operator in $\mathcal{L}(V)$ could be realized as $p(T)$ for some $p \in \mathcal{P}(\mathbf{F})$, then each pair of operators in $\mathcal{L}(V)$ would commute with each other by 5.17. However, because $\dim V > 1$, Exercise 3.A.16 shows that there exist two operators in $\mathcal{L}(V)$ which do not commute with each other.



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## 5.B. The Minimal Polynomial

**Exercise 5.B.1.** Suppose $T \in \mathcal{L}(V)$. Prove that 9 is an eigenvalue of $T^2$ if and only if 3 or $-3$ is an eigenvalue of $T$.

**Solution.** We showed in the solution to Exercise 5.A.22 that if 9 is an eigenvalue of $T^2$ then 3 or $-3$ is an eigenvalue of $T$. Conversely, suppose there is some non-zero $v \in V$ such that $Tv = \pm 3v$. It follows that $T^2 v = (\pm 3)^2 v = 9v$ and thus 9 is an eigenvalue of $T^2$.

**Exercise 5.B.2.** Suppose $V$ is a complex vector space and $T \in \mathcal{L}(V)$ has no eigenvalues. Prove that every subspace of $V$ invariant under $T$ is either $\{0\}$ or infinite-dimensional.

**Solution.** Suppose $U$ is a non-zero subspace of $V$ invariant under $T$ and consider the restriction operator $T|_U$. If $U$ were finite-dimensional then 5.19 would imply that $T|_U$ has an eigenvalue, which would also be an eigenvalue of $T$. Since $T$ has no eigenvalues it must be the case that $U$ is infinite-dimensional.

**Exercise 5.B.3.** Suppose $n$ is a positive integer and $T \in \mathcal{L}(\mathbf{F}^n)$ is defined by

$$T(x_1, ..., x_n) = (x_1 + \cdots + x_n, ..., x_1 + \cdots + x_n).$$

(a) Find all eigenvalues and eigenvectors of $T$.

(b) Find the minimal polynomial of $T$.

*The matrix of $T$ with respect to the standard basis of $\mathbf{F}^n$ consists of all 1's.*

**Solution.**

(a) If $n = 1$ then $T$ is the identity operator on $\mathbf{F}$, whose only eigenvalue is 1 with corresponding eigenvectors $x \in \mathbf{F} \setminus \{0\}$.

Suppose that $n \geq 2$. Some straightforward calculations reveal that

$$\operatorname{null} T = \{(-x_2 - \cdots - x_n, x_2, ..., x_n) \in \mathbf{F}^n : x_2, ..., x_n \in \mathbf{F}\}$$

$$\text{and} \quad \operatorname{range} T = \operatorname{span}((1, ..., 1)).$$

Note that $\dim \operatorname{null} T = n - 1$ and $\dim \operatorname{range} T = 1$. Since $\dim \operatorname{null} T \geq 1$, it follows that 0 is an eigenvalue of $T$. Notice that $n$ is also an eigenvalue of $T$, since

$$T(1, ..., 1) = (n, ..., n) = n(1, ..., 1).$$

We claim that these are the only eigenvalues of $T$. Indeed, if $x \neq 0$ and $\lambda \neq 0$ are such that $Tx = \lambda x$, then since $\operatorname{range} T = \operatorname{span}((1, ..., 1))$ there must exist some $\alpha \in \mathbf{F}$ such that

$$Tx = \lambda x = \alpha(1, ..., 1) \quad \Rightarrow \quad x = \lambda^{-1}\alpha(1, ..., 1).$$



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Thus the eigenvector $x$, which corresponds to the eigenvalue $\lambda$, and the eigenvector $(1, ..., 1)$, which corresponds to the eigenvalue $n$, are linearly dependent. It follows from the contrapositive of 5.11 that $\lambda = n$.

Certainly $\operatorname{null} T \setminus \{0\}$ is the collection of eigenvectors of $T$ corresponding to the eigenvalue 0. Because $\operatorname{range} T = \operatorname{span}((1, ..., 1))$, the collection of eigenvectors of $T$ corresponding to the eigenvalue $n$ must be $\operatorname{span}((1, ..., 1)) \setminus \{0\}$.

(b) If $n = 1$ then $T$ is the identity operator on $\mathbf{F}$ and it is then clear that the minimal polynomial of $T$ is $p(z) = z - 1$.

Suppose that $n \geq 2$ and let $p \in \mathcal{P}(\mathbf{F})$ be the minimal polynomial of $T$. As we showed in part (a) the eigenvalues of $T$ are 0 and $n$, which must be the zeros of $p$ by 5.27(a), i.e. $p$ is divisible by $z$ and $z - n$. This implies that $\deg p \geq 2$. A straightforward calculation shows that $T(T - nI) = 0$. The uniqueness of $p$ and the minimality of its degree allow us to conclude that $p(z) = z(z - n)$.

**Exercise 5.B.4.** Suppose $\mathbf{F} = \mathbf{C}, T \in \mathcal{L}(V), p \in \mathcal{P}(\mathbf{C})$, and $\alpha \in \mathbf{C}$. Prove that $\alpha$ is an eigenvalue of $p(T)$ if and only if $\alpha = p(\lambda)$ for some eigenvalue $\lambda$ of $T$.

**Solution.** Suppose that $\alpha = p(\lambda)$ for some eigenvalue $\lambda$ of $T$. There is some non-zero $v \in V$ such that $Tv = \lambda v$; as shown in the proof of 5.27 we then have $p(T)v = p(\lambda)v = \alpha v$ and thus $\alpha$ is an eigenvalue of $p(T)$.

For the converse, we must assume that $p$ is non-constant (see the errata for the third edition of LADR), or that $V$ is finite-dimensional. To demonstrate this, let $V = \mathbf{C}^\infty, p(z) = \alpha \in \mathbf{C}$, and let $T \in \mathcal{L}(\mathbf{C}^\infty)$ be the forward shift operator. Certainly $\alpha$ is an eigenvalue of $p(T) = \alpha I$, but we may not express $\alpha$ as $p(\lambda)$ for some eigenvalue $\lambda$ of $T$ because $T$ has no eigenvalues, as shown in Exercise 5.A.19. Since any operator on a non-zero complex vector space has an eigenvalue (5.19), we will not encounter this issue if we assume that $V$ is finite-dimensional. The result as stated is true if we assume that $p$ is non-constant, as we now show.

Suppose that $\deg p \geq 1$ and that $\alpha$ is an eigenvalue of $p(T)$, i.e. there is some non-zero $v \in V$ such that $p(T)v = \alpha v$. Let $q \in \mathcal{P}(\mathbf{C})$ be given by $q(z) = p(z) - \alpha$. Because $q$ is a polynomial over $\mathbf{C}$, 4.13 shows that there is a factorization

$$q(z) = c(z - \lambda_1) \cdots (z - \lambda_m)$$

for some $c, \lambda_1, ..., \lambda_m \in \mathbf{C}$. Since $\deg q = \deg p \geq 1$, it must be the case that $c \neq 0$ and $m \geq 1$. Note that $q(T)v = 0$ since $p(T)v = \alpha v$ and thus

$$0 = q(T)v = c(T - \lambda_1 I) \cdots (T - \lambda_m I)v.$$

Because $c \neq 0$ and $v \neq 0$, the equation above implies that there is some $k \in \{1, ..., m\}$ such that $T - \lambda_k I$ is not injective, i.e. $\lambda_k$ is an eigenvalue of $T$. Furthermore, $p(\lambda_k) = q(\lambda_k) + \alpha = \alpha$ since $\lambda_k$ is a zero of $q$.



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**Exercise 5.B.5.** Give an example of an operator on $\mathbf{R}^2$ that shows the result in Exercise 4 does not hold if $\mathbf{C}$ is replaced with $\mathbf{R}$.

**Solution.** Let $T \in \mathcal{L}(\mathbf{R}^2)$ be given by $T(x, y) = (-y, x)$, i.e. a counterclockwise rotation about the origin by 90°. Let $p(t) = t^2$ and notice that $p(T) = T^2 = -I$, since $T^2$ is a counterclockwise rotation about the origin by 180°. Thus $p(T)$ has the eigenvalue $-1$. However, we cannot possibly express $-1$ as $p(\lambda)$ for some eigenvalue $\lambda$ of $T$ because $T$ has no eigenvalues, as shown in 5.9(a).

**Exercise 5.B.6.** Suppose $T \in \mathcal{L}(\mathbf{F}^2)$ is defined by $T(w, z) = (-z, w)$. Find the minimal polynomial of $T$.

**Solution.** Let $e_1, e_2$ be the standard basis of $\mathbf{F}^2$ and note that $Te_1 = e_2$ and $T^2e_1 = -e_1$. Note further that the system

$$c_0e_1 + c_1Te_1 = -T^2e_1, \quad \text{i.e.} \quad c_0e_1 + c_1e_2 = e_1,$$

has the unique solution $c_0 = 1$ and $c_1 = 0$. As shown in the textbook (see the discussion after 5.24), this implies that the minimal polynomial of $T$ is $p(t) = 1 + t^2$.

**Exercise 5.B.7.**

(a) Give an example of $S, T \in \mathcal{L}(\mathbf{F}^2)$ such that the minimal polynomial of $ST$ does not equal the minimal polynomial of $TS$.

(b) Suppose $V$ is finite-dimensional and $S, T \in \mathcal{L}(V)$. Prove that if at least one of $S, T$ is invertible, then the minimal polynomial of $ST$ equals the minimal polynomial of $TS$.

*Hint: Show that if $S$ is invertible and $p \in \mathcal{P}(\mathbf{F})$, then $p(TS) = S^{-1}p(ST)S$.*

**Solution.**

(a) Let $S, T \in \mathcal{L}(\mathbf{F}^2)$ be the operators whose matrices with respect to the standard basis of $\mathbf{F}^2$ are

$$\mathcal{M}(S) = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \quad \text{and} \quad \mathcal{M}(T) = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}.$$

A simple computation shows that

$$\mathcal{M}(ST) = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \quad \text{and} \quad \mathcal{M}(TS) = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}.$$

It follows that the minimal polynomial of $TS$ is $p(z) = z$. Since $p(ST) = ST \neq 0$, it cannot be the case that $p$ is the minimal polynomial of $ST$.



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(b) Suppose at least one of $S, T$ is invertible, say $S$ (the case where $T$ is invertible is handled similarly). Notice that $(TS)^0 = I = S^{-1}IS = S^{-1}(ST)^0 S$. Suppose that $(TS)^k = S^{-1}(ST)^k S$ for some non-negative integer $k$ and observe that

$$(TS)^{k+1} = (TS)^k TS = S^{-1}(ST)^k STS = S^{-1}(ST)^{k+1} S.$$

It follows by induction that $(TS)^k = S^{-1}(ST)^k S$ for all non-negative integers $k$. Now suppose $p(z) = \sum_{k=0}^{n} a_k z^k$ is some polynomial in $\mathcal{P}(\mathbf{F})$ and observe that

$$p(TS) = \sum_{k=0}^{n} a_k (TS)^k = \sum_{k=0}^{n} a_k S^{-1}(ST)^k S = S^{-1}\left(\sum_{k=0}^{n} a_k (ST)^k\right) S = S^{-1} p(ST) S.$$

We can now show that the minimal polynomial of $ST$ equals the minimal polynomial of $TS$. Let $p$ be the minimal polynomial of $ST$ and let $q$ be the minimal polynomial of $TS$. Notice that

$$p(TS) = S^{-1}p(ST)S = 0 \quad \text{and} \quad 0 = q(TS) = S^{-1}q(ST)S \quad \Rightarrow \quad q(ST) = 0.$$

It follows from 5.29 that $p$ is a polynomial multiple of $q$ and $q$ is a polynomial multiple of $p$. This implies $p = cq$ for some $c \in \mathbf{F}$ (as the next lemma shows); because $p$ and $q$ are monic we must have $c = 1$ and thus $p = q$.

**Lemma L.7.** Suppose $p, q \in \mathcal{P}(\mathbf{F})$ are non-zero. If $p$ is a polynomial multiple of $q$ and $q$ is a polynomial multiple of $p$ then $p = cq$ for some $c \in \mathbf{F} \setminus \{0\}$.

*Proof.* There are polynomials $r, s \in \mathcal{P}(\mathbf{F})$ such that $p = rq$ and $q = sp$, which gives us $p = rsp$. This implies that $r(x)s(x) = 1$ for all $x \in \mathbf{F}$ such that $p(x) \neq 0$; there are finitely many such $x$ because $p$ is non-zero. Thus $r(x)s(x) = 1$ holds for infinitely many $x$ and so must hold for all $x \in \mathbf{F}$ (otherwise $rs - 1$ is a non-zero polynomial with infinitely many roots). This equation forces $\deg r = 1$, so that $r = c$ for some $c \in \mathbf{F} \setminus \{0\}$. Thus $p = cq$. □

**Exercise 5.B.8.** Suppose $T \in \mathcal{L}(\mathbf{R}^2)$ is the operator of counterclockwise rotation by 1°. Find the minimal polynomial of $T$.

*Because* $\dim \mathbf{R}^2 = 2$*, the degree of the minimal polynomial of* $T$ *is at most* 2*. Thus the minimal polynomial of* $T$ *is not the tempting polynomial* $x^{180} + 1$*, even though* $T^{180} = -I$*.*

**Solution.** Let $e_1, e_2$ be the standard basis of $\mathbf{R}^2$ and observe that

$$Te_1 = \cos\left(\tfrac{\pi}{180}\right)e_1 + \sin\left(\tfrac{\pi}{180}\right)e_2 \quad \text{and} \quad T^2 e_1 = \cos\left(\tfrac{\pi}{90}\right)e_1 + \sin\left(\tfrac{\pi}{90}\right)e_2.$$

Thus, solving the system of equations $c_0 e_1 + c_1 Te_1 = -Te_2$ amounts to solving the system



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$$\begin{pmatrix} 1 & \cos\left(\frac{\pi}{180}\right) \\ 0 & \sin\left(\frac{\pi}{180}\right) \end{pmatrix} \begin{pmatrix} c_0 \\ c_1 \end{pmatrix} = \begin{pmatrix} -\cos\left(\frac{\pi}{90}\right) \\ -\sin\left(\frac{\pi}{90}\right) \end{pmatrix}.$$

This system of equations has the unique solution $c_0 = 1$ and $c_1 = -2\cos\left(\frac{\pi}{180}\right)$. As shown in the textbook (see the discussion after 5.24), this implies that the minimal polynomial of $T$ is $p(x) = 1 - 2\cos\left(\frac{\pi}{180}\right)x + x^2$.

**Exercise 5.B.9.** Suppose $T \in \mathcal{L}(V)$ is such that with respect to some basis of $V$, all entries of the matrix of $T$ are rational numbers. Explain why all coefficients of the minimal polynomial of $T$ are rational numbers.

**Solution.** By 5.22, there exists a minimal positive integer $m \leq \dim V$ such that the equation

$$c_0 I + c_1 T + \cdots + c_{m-1} T^{m-1} = -T^m \tag{1}$$

has a unique solution $c_0, c_1, ..., c_{m-1} \in \mathbf{F}$. Moreover, the numbers $c_0, c_1, ..., c_{m-1}, 1$ are the coefficients of the minimal polynomial of $T$. Thus it will suffice to show that each $c_i$ is a rational number.

By assumption there is a basis $v_1, ..., v_n$ of $V$ such that the entries of the matrix of $T$ with respect to this basis are rational numbers. Let $A$ denote this matrix and consider the matrix equation

$$x_0 I + x_1 A + \cdots + x_{m-1} A^{m-1} = -A^m.$$

As noted in the textbook (see the discussion after 5.24), this equation can be thought of as a system of $n^2$ equations in the $m$ unknowns $x_0, x_1, ..., x_{m-1}$. That is, letting $(A^i)_{j,k}$ be the entry in the $j^{\text{th}}$ row and $k^{\text{th}}$ column of $A^i$, for each $j, k \in \{1, ..., n\}$ we have a linear equation

$$\sum_{i=0}^{m-1} (A^i)_{j,k}\, x_i = (-A^m)_{j,k}.$$

Because the entries of $A$ are rational, it follows from the definition of matrix multiplication that each $(A^i)_{j,k}$ is also rational. Thus each coefficient in this system of equations is a rational number. If this system has a solution $(x_0, x_1, ..., x_{m-1})$, then Gaussian elimination (or some other method) shows that each $x_i$ is a rational function of the coefficients of the system; it follows that each $x_i$ is rational. Now observe that equation (1) implies that $(c_0, c_1, ..., c_{m-1})$ is a solution of this system of equations. Thus each $c_i$ is rational.

**Exercise 5.B.10.** Suppose $V$ is finite-dimensional, $T \in \mathcal{L}(V)$, and $v \in V$. Prove that

$$\operatorname{span}(v, Tv, ..., T^m v) = \operatorname{span}\left(v, Tv, ..., T^{\dim V - 1} v\right)$$

for all integers $m \geq \dim V - 1$.



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**Solution.** Let $n = \dim V$ and let $U_m = \text{span}(v, Tv, ..., T^m v)$ for a non-negative integer $m$; define $U_{-1} = \{0\}$. Our goal is to show that $U_m = U_{n-1}$ for all $m \geq n-1$.

First we will use induction to show that if $T^{k+1}v \in U_k$ for some $k \geq -1$ then $T^m v \in U_k$ for all $m \geq k+1$. The base case $m = k+1$ is clear, so suppose that $T^m v \in U_k$ for some $m \geq k+1$, i.e.

$$T^m v = a_0 v + a_1 Tv + \cdots + a_{k-1} T^{k-1} v + a_k T^k v.$$

Now observe that

$$T^{m+1} v = a_0 Tv + a_1 T^2 v + \cdots + a_{k-1} T^k v + a_k T^{k+1} v.$$

Certainly $a_0 Tv + a_1 T^2 v + \cdots + a_{k-1} T^k v \in U_k$, and since $T^{k+1}v$ belongs to $U_k$ by assumption, it follows that $T^{m+1} v \in U_k$. This completes the induction step.

The previous result implies that if $T^{k+1}v \in U_k$ for some $k$ then $U_m = U_k$ for all $m \geq k$. For a non-negative integer $m$, note that if $T^{k+1}v \notin U_k$ for all $k \in \{-1, ..., m-1\}$ then the linear dependence lemma (2.19) shows that the list $v, Tv, ..., T^m v$ is linearly independent. Because the list $v, Tv, ..., T^n v$ is linearly dependent (it has length $n+1$ and $\dim V = n$), it follows that there exists some $k \in \{-1, ..., n-1\}$ such that $T^{k+1}v \in U_k$, which implies that $U_m = U_k$ for all $m \geq k$. In particular, $U_m = U_{n-1}$ for all $m \geq n-1$.

**Exercise 5.B.11.** Suppose $V$ is a two-dimensional vector space, $T \in \mathcal{L}(V)$, and the matrix of $T$ with respect to some basis of $V$ is $\begin{pmatrix} a & c \\ b & d \end{pmatrix}$.

(a) Show that $T^2 - (a+d)T + (ad-bc)I = 0$.

(b) Show that the minimal polynomial of $T$ equals

$$\begin{cases} z - a & \text{if } b = c = 0 \text{ and } a = d, \\ z^2 - (a+d)z + (ad-bc) & \text{otherwise.} \end{cases}$$

**Solution.**

(a) Letting $A = \begin{pmatrix} a & c \\ b & d \end{pmatrix}$, a straightforward calculation shows that

$$A^2 - (a+d)A + (ad-bc)I = 0.$$

It follows that $T^2 - (a+d)T + (ad-bc)I = 0$.

(b) If $b = c = 0$ and $a = d$ then $T = aI$ and it is then clear that the minimal polynomial of $T$ is $z - a$. If $b \neq 0$, or $c \neq 0$, or $a \neq d$, then $T$ is not a scalar multiple of the identity. It follows that the equation $xI = -T$ has no solution for $x \in \mathbf{F}$ and thus the degree of the minimal polynomial of $T$ must be at least 2. Since the degree of the minimal polynomial of $T$ can be at most $\dim V = 2$, we see that the degree of the minimal polynomial of $T$ must equal 2. Thus, because $p(z) = z^2 - (a+d)z + (ad-bc)$ is monic, satisfies $p(T) = 0$, and has degree equal to the degree of the minimal polynomial of $T$, it follows from the uniqueness of the minimal polynomial of an operator that $p$ is the minimal polynomial of $T$.



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**Exercise 5.B.12.** Define $T \in \mathcal{L}(\mathbf{F}^n)$ by $T(x_1, x_2, x_3, ..., x_n) = (x_1, 2x_2, 3x_3, ..., nx_n)$. Find the minimal polynomial of $T$.

**Solution.** Let $p \in \mathcal{P}(\mathbf{F})$ be the minimal polynomial of $T$. As we showed in Exercise 5.A.42 (a), each $k \in \{1, ..., n\}$ is an eigenvalue of $T$. It follows from 5.27 that each $k \in \{1, ..., n\}$ is a zero of $p$, which implies $\deg p \geq n$. Since $\deg p \leq n$ by 5.22, it must be the case that

$$p(z) = (z-1)(z-2)(z-3)\cdots(z-n).$$

**Exercise 5.B.13.** Suppose $T \in \mathcal{L}(V)$ and $p \in \mathcal{P}(\mathbf{F})$. Prove that there exists a unique $r \in \mathcal{P}(\mathbf{F})$ such that $p(T) = r(T)$ and $\deg r$ is less than the degree of the minimal polynomial of $T$.

**Solution.** Let $q \in \mathcal{P}(\mathbf{F})$ be the minimal polynomial of $T$, which must be non-zero. The division algorithm for polynomials (4.9) shows that there exist unique polynomials $s, r \in \mathcal{P}(\mathbf{F})$ such that $p = sq + r$ and $\deg r < \deg q$. Because $q(T) = 0$, it follows that $p(T) = r(T)$.

**Exercise 5.B.14.** Suppose $V$ is finite-dimensional and $T \in \mathcal{L}(V)$ has minimal polynomial $4 + 5z - 6z^2 - 7z^3 + 2z^4 + z^5$. Find the minimal polynomial of $T^{-1}$.

**Solution.** We will use the following useful lemma.



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**Lemma L.8.** Suppose $V$ is finite-dimensional and $T \in \mathcal{L}(V)$ is invertible. If the minimal polynomial of $T$ is $p \in \mathcal{P}(\mathbf{F})$ with $p = \sum_{k=0}^{m} c_k z^k$, then the minimal polynomial of $T^{-1}$ is

$$\frac{1}{c_0} \sum_{k=0}^{m} c_{m-k} z^k.$$

(Note that $c_0 \neq 0$ since $T$ is invertible.)

*Proof.* Let $q = \frac{1}{c_0} \sum_{k=0}^{m} c_{m-k} z^k$ and note that $q$ is monic and $\deg q = m$. Note further that

$$0 = T^{-m} p(T) = T^{-m} \sum_{k=0}^{m} c_k T^k = \sum_{k=0}^{m} c_k T^{k-m}$$

$$= \sum_{k=0}^{m} c_k (T^{-1})^{m-k} = \sum_{k=0}^{m} c_{m-k} (T^{-1})^k = q(T^{-1}).$$

It follows that the degree of the minimal polynomial of $T^{-1}$ is at most $\deg q = m$. By replacing $T$ with $T^{-1}$ in the previous argument and using that $(T^{-1})^{-1} = T$, we see that the minimal polynomials of $T$ and $T^{-1}$ must have the same degree. Because $q$ is monic, satisfies $q(T^{-1}) = 0$, and has the same degree as the minimal polynomial of $T^{-1}$, it follows from the uniqueness of the minimal polynomial of an operator that $q$ is the minimal polynomial of $T^{-1}$. □

It is now immediate from Lemma L.8 that the minimal polynomial of $T^{-1}$ is

$$\frac{1}{4} + \frac{1}{2} z - \frac{7}{4} z^2 - \frac{3}{2} z^3 + \frac{5}{4} z^4 + z^5.$$

**Exercise 5.B.15.** Suppose $V$ is a finite-dimensional complex vector space with $\dim V > 0$ and $T \in \mathcal{L}(V)$. Define $f : \mathbf{C} \to \mathbf{R}$ by

$$f(\lambda) = \dim \operatorname{range}(T - \lambda I).$$

Prove that $f$ is not a continuous function.

**Solution.** Let $m = \dim V$ and note that, by 5.19, there exists an eigenvalue $\lambda \in \mathbf{C}$ of $T$. It follows from 5.7 that $T - \lambda I$ is not surjective, so that $f(\lambda) < m$. Consider the sequence $(\lambda_n)_{n=1}^{\infty}$ of distinct complex numbers given by $\lambda_n = \lambda + \frac{1}{n}$, which satisfies $\lim_{n\to\infty} \lambda_n = \lambda$. Because $T$ can have at most $m$ distinct eigenvalues 5.13, we may choose a subsequence $(\lambda_{n_k})_{k=1}^{\infty}$ such that each $\lambda_{n_k}$ is not an eigenvalue of $T$. By 5.7 each operator $T - \lambda_{n_k} I$ must be surjective. It follows that $f(\lambda_{n_k}) = m$ for each positive integer $k$, which implies

$$\lim_{k\to\infty} f(\lambda_{n_k}) = m > f(\lambda) = f\Big(\lim_{k\to\infty} \lambda_{n_k}\Big).$$



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Thus $f$ is not continuous at $\lambda$.

**Exercise 5.B.16.** Suppose $a_0, ..., a_{n-1} \in \mathbf{F}$. Let $T$ be the operator on $\mathbf{F}^n$ whose matrix (with respect to the standard basis) is

$$\begin{pmatrix} 0 & & & & -a_0 \\ 1 & 0 & & & -a_1 \\ & 1 & \ddots & & -a_2 \\ & & \ddots & & \vdots \\ & & & 0 & -a_{n-2} \\ & & & 1 & -a_{n-1} \end{pmatrix},$$

Here all entries of the matrix are 0 except for all 1's on the line under the diagonal and the entries in the last column (some of which might also be 0). Show that the minimal polynomial of $T$ is the polynomial

$$a_0 + a_1 z + \cdots + a_{n-1} z^{n-1} + z^n.$$

*The matrix above is called the **companion matrix** of the polynomial above. This exercise shows that every monic polynomial is the minimal polynomial of some operator. Hence a formula or an algorithm that could produce exact eigenvalues for each operator on each $\mathbf{F}^n$ could then produce exact zeros for each polynomial [by 5.27(a)]. Thus there is no such formula or algorithm. However, efficient numeric methods exist for obtaining very good approximations for the eigenvalues of an operator.*

**Solution.** Let $e_0, ..., e_{n-1}$ be the standard basis of $\mathbf{F}^n$ and observe that

$$Te_0 = e_1, \quad T^2 e_0 = e_2, \quad ..., \quad T^{n-1} e_0 = e_{n-1}, \quad T^n e_0 = -(a_0 e_0 + \cdots + a_{n-1} e_{n-1}).$$

It follows that the equation $c_0 e_0 + c_1 T e_0 + c_2 T^2 e_0 + \cdots + c_{n-1} T^{n-1} e_0 = -T^n e_0$ is equivalent to

$$c_0 e_0 + c_1 e_1 + c_2 e_2 + \cdots + c_{n-1} e_{n-1} = a_0 e_0 + a_1 e_1 + a_2 e_2 + \cdots + a_{n-1} e_{n-1}.$$

By unique representation, this equation has the unique solution $(c_0, ..., c_{n-1}) = (a_0, ..., a_{n-1})$. It follows (see the discussion in the textbook after 5.24) that the minimal polynomial of $T$ is $a_0 + a_1 z + \cdots + a_{n-1} z^{n-1} + z^n$.

**Exercise 5.B.17.** Suppose $V$ is finite-dimensional, $T \in \mathcal{L}(V)$, and $p$ is the minimal polynomial of $T$. Suppose $\lambda \in \mathbf{F}$. Show that the minimal polynomial of $T - \lambda I$ is the polynomial $q$ defined by $q(z) = p(z + \lambda)$.

**Solution.** Let $s$ be the minimal polynomial of $T - \lambda I$. Notice that $q$ is monic and satisfies $\deg q = \deg p$. Notice further that

$$q(T - \lambda I) = p(T - \lambda I + \lambda I) = p(T) = 0.$$



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It follows that $\deg s \le \deg q = \deg p$. Now define a polynomial $r$ by $r(z) = s(z - \lambda)$ and observe that $\deg r = \deg s$ and that

$$r(T) = s(T - \lambda I) = 0.$$

It follows that $\deg p \le \deg r = \deg s$. Thus $\deg s = \deg p$. Because $q$ is a monic polynomial satisfying $q(T - \lambda I) = 0$, and the degree of $q$ equals the degree of the minimal polynomial of $T - \lambda I$, it follows from the uniqueness of the minimal polynomial of an operator that $q$ is the minimal polynomial of $T - \lambda I$.

**Exercise 5.B.18.** Suppose $V$ is finite-dimensional, $T \in \mathcal{L}(V)$, and $p$ is the minimal polynomial of $T$. Suppose $\lambda \in \mathbf{F} \setminus \{0\}$. Show that the minimal polynomial of $\lambda T$ is the polynomial $q$ defined by $q(z) = \lambda^{\deg p} p\left(\frac{z}{\lambda}\right)$.

**Solution.** Let $s$ be the minimal polynomial of $\lambda T$. Observe that $\deg q = \deg p$ and that

$$q(\lambda T) = \lambda^{\deg p} p(T) = 0.$$

Thus $\deg s \le \deg q$. Let $r$ be given by $r(z) = s(\lambda z)$ and notice that $\deg r = \deg s$ since $\lambda \neq 0$. Furthermore, $r(T) = s(\lambda T) = 0$, which implies $\deg p \le \deg r$, i.e. $\deg q \le \deg s$. Thus $\deg s = \deg q$. Because $p$ is monic, the highest degree term of $p\left(\frac{z}{\lambda}\right)$ is $\lambda^{-\deg p} z^{\deg p}$ and it follows that $q$ is monic. Thus $q$ is a monic polynomial satisfying $q(\lambda T) = 0$ whose degree equals the degree of the minimal polynomial of $\lambda T$. The uniqueness of the minimal polynomial of an operator then implies that $q$ is the minimal polynomial of $\lambda T$.

**Exercise 5.B.19.** Suppose $V$ is finite-dimensional and $T \in \mathcal{L}(V)$. Let $\mathcal{E}$ be the subspace of $\mathcal{L}(V)$ defined by

$$\mathcal{E} = \{q(T) : q \in \mathcal{P}(\mathbf{F})\}.$$

Prove that $\dim \mathcal{E}$ equals the degree of the minimal polynomial of $T$.

**Solution.** Let $p$ be the minimal polynomial of $T$. Define $\Phi \in \mathcal{L}(\mathcal{P}(\mathbf{F}), \mathcal{L}(V))$ by $\Phi q = q(T)$ and notice that $\operatorname{range} \Phi = \mathcal{E}$. Notice further that, by 5.29, $\operatorname{null} \Phi = \{pq : q \in \mathcal{P}(\mathbf{F})\}$. Exercise 4.13 shows that $\dim \mathcal{P}(\mathbf{F})/(\operatorname{null} \Phi) = \deg p$ and 3.107 shows that $\Phi$ is an isomorphism from $\mathcal{P}(\mathbf{F})/(\operatorname{null} \Phi)$ onto $\mathcal{E}$. Thus $\dim \mathcal{E} = \deg p$.

**Exercise 5.B.20.** Suppose $T \in \mathcal{L}(\mathbf{F}^4)$ is such that the eigenvalues of $T$ are $3, 5, 8$. Prove that $(T - 3I)^2 (T - 5I)^2 (T - 8I)^2 = 0$.

**Solution.** Let $p$ be the minimal polynomial of $T$ and let $q(z) = (z - 3)^2 (z - 5)^2 (z - 8)^2$. 5.27 shows that $3, 5, 8$ are zeros of $p$, so that $p$ is of the form

$$p(z) = s(z)(z - 3)(z - 5)(z - 8)$$



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for some polynomial $s$. Because $\deg p \leq \dim \mathbf{F}^4 = 4$ and $3, 5, 8$ are the only zeros of $p$, we must have $s \in \{1, z-3, z-5, z-8\}$. It follows that $q$ is a polynomial multiple of $p$ and thus, by 5.29, $q(T) = 0$.

**Exercise 5.B.21.** Suppose $V$ is finite-dimensional and $T \in \mathcal{L}(V)$. Prove that the minimal polynomial of $T$ has degree at most $1 + \dim \operatorname{range} T$.

*If $\dim \operatorname{range} T < \dim V - 1$, then this exercise gives a better upper bound than 5.22 for the degree of the minimal polynomial of $T$.*

**Solution.** Let $p \in \mathcal{P}(\mathbf{F})$ be the minimal polynomial of $T$ and let $q \in \mathcal{P}(\mathbf{F})$ be the minimal polynomial of $T|_{\operatorname{range} T}$. For any $v \in V$ observe that

$$q(T)Tv = q\big(T|_{\operatorname{range} T}\big)Tv = 0.$$

Thus $q(T)T = 0$. It follows from the minimality of $\deg p$ that

$$\deg p \leq \deg(xq(x)) = 1 + \deg q \leq 1 + \dim \operatorname{range} T.$$

**Exercise 5.B.22.** Suppose $V$ is finite-dimensional and $T \in \mathcal{L}(V)$. Prove that $T$ is invertible if and only if $I \in \operatorname{span}\big(T, T^2, ..., T^{\dim V}\big)$.

**Solution.** Let $p(z) = c_0 + c_1 z + \cdots + c_{m-1} z^{m-1} + z^m$ be the minimal polynomial of $T$. If $T$ is invertible then 5.32 shows that $c_0 \neq 0$ and thus

$$c_0 I + c_1 T + \cdots + c_{m-1} T^{m-1} + T^m = 0$$

$$\Rightarrow \quad I = -c_0^{-1}\big(c_1 T + \cdots + c_{m-1} T^{m-1} + T^m\big) \in \operatorname{span}\big(T, ..., T^{\dim V}\big).$$

Now suppose that $I \in \operatorname{span}(T, T^2, ..., T^n)$, where $n = \dim V$, so that

$$I = a_1 T + a_2 T^2 + \cdots + a_n T^n$$

for some $a_1, a_2, ..., a_n \in \mathbf{F}$. Let $q \in \mathcal{P}(\mathbf{F})$ be given by $q(z) = -1 + a_1 z + a_2 z^2 + \cdots + a_n z^n$ and note that $q(T) = 0$. It follows from 5.29 that $q$ is a polynomial multiple of $p$. Because 0 is not a root of $q$, it must be that 0 is not a root of $p$ either, i.e. the constant term of $p$ is not zero. 5.32 allows us to conclude that $T$ is invertible.

**Exercise 5.B.23.** Suppose $V$ is finite-dimensional and $T \in \mathcal{L}(V)$. Let $n = \dim V$. Prove that if $v \in V$, then $\operatorname{span}(v, Tv, ..., T^{n-1}v)$ is invariant under $T$.

**Solution.** It will suffice to show that $T^n v \in \operatorname{span}(v, Tv, ..., T^{n-1}v)$. This is immediate from Exercise 5.B.10.



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**Exercise 5.B.24.** Suppose $V$ is a finite-dimensional complex vector space. Suppose $T \in \mathcal{L}(V)$ is such that 5 and 6 are eigenvalues of $T$ and that $T$ has no other eigenvalues. Prove that $(T-5I)^{\dim V-1}(T-6I)^{\dim V-1}=0$.

**Solution.** Let $n=\dim V$ and note that because $T$ has 2 distinct eigenvalues, 5.11 implies $n \geq 2$. Let $q(z)=(z-5)^{n-1}(z-6)^{n-1}$, and let $p$ be the minimal polynomial of $T$. Since 5 and 6 are the only eigenvalues of $T$, 5.27(b) shows that $p$ is of the form $p(z)=(z-5)^k(z-6)^\ell$ for some positive integers $k,\ell$. Thus $q$ is a polynomial multiple of $p$ and it then follows from 5.29 that $q(T)=0$.

**Exercise 5.B.25.** Suppose $V$ is finite-dimensional, $T \in \mathcal{L}(V)$, and $U$ is a subspace of $V$ that is invariant under $T$.

(a) Prove that the minimal polynomial of $T$ is a polynomial multiple of the minimal polynomial of the quotient operator $T/U$.

(b) Prove that

$$(\text{minimal polynomial of } T|_U) \times (\text{minimal polynomial of } T/U)$$

is a polynomial multiple of the minimal polynomial of $T$.

*The quotient operator $T/U$ was defined in Exercise 38 in Section 5A.*

**Solution.**

(a) We will use the following lemma.

**Lemma L.9.** Suppose $T \in \mathcal{L}(V), U$ is a subspace of $V$ invariant under $T$, and $\pi : V \to V/U$ is the quotient map. If $p \in \mathcal{P}(\mathbf{F})$ then $\pi p(T) = p(T/U)\pi$.

*Proof.* Suppose $p=\sum_{k=0}^m c_k z^k$ and let $\pi : V \to V/U$ be the quotient map. For any non-negative integer $k$ and any $v \in V$, the definition of the quotient operator implies that $(T/U)^k(v+U)=T^kv+U$. Thus, for any $v \in V$,

$$\pi p(T)v = \sum_{k=0}^m c_k \pi(T^k v) = \sum_{k=0}^m c_k (T^k v + U)$$
$$= \sum_{k=0}^m c_k (T/U)^k (v+U) = p(T/U)(\pi(v)).$$

It follows that $\pi p(T) = p(T/U)\pi$. □

Let $p=\sum_{k=0}^m c_k z^k$ be the minimal polynomial of $T$ and let $\pi : V \to V/U$ be the quotient map. For any $v+U \in V/U$, it follows from Lemma L.9 that

$$p(T/U)(v+U) = p(T/U)(\pi(v)) = \pi p(T)v = \pi(0) = 0.$$



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Thus $p(T/U) = 0$. It then follows from 5.29 that $p$ is a multiple of the minimal polynomial of $T/U$.

(b) Let $r$ and $s$ be the minimal polynomials of $T|_U$ and $T/U$, and let $\pi : V \to V/U$ be the quotient map. By 2.33 there is a subspace $W$ of $V$ such that $V = U \oplus W$. For any $u \in U$, notice that $r(T)u = r(T|_U)u = 0$. It then follows from Lemma L.9 that, for any $w \in W$,

$$\pi(s(T)w) = s(T/U)(w + U) = 0 \quad \Rightarrow \quad s(T)w \in U \quad \Rightarrow \quad r(T)s(T)w = 0.$$

Let $v = u + w \in V$ be given and observe that

$$r(T)s(T)v = r(T)s(T)u + r(T)s(T)w = s(T)r(T)u = 0.$$

Thus $r(T)s(T) = 0$ and it then follows from 5.29 that $rs$ is a polynomial multiple of the minimal polynomial of $T$.

**Exercise 5.B.26.** Suppose $V$ is finite-dimensional, $T \in \mathcal{L}(V)$, and $U$ is a subspace of $V$ that is invariant under $T$. Prove that the set of eigenvalues of $T$ equals the union of the set of eigenvalues of $T|_U$ and the set of eigenvalues of $T/U$.

**Solution.** Let $p, r$, and $s$ be the minimal polynomials of $T, T|_U$, and $T/U$, and let $Z_p, Z_r$, and $Z_s$ be the collection of zeros of $p, r$, and $s$. By 5.27(a), it will suffice to show that $Z_p = Z_r \cup Z_s$. It follows from Exercise 5.B.25 and 5.31 that there exist polynomials $q, a, b$ such that:

(1) $r(x)s(x) = p(x)q(x)$;

(2) $p(x) = a(x)r(x)$;

(3) $p(x) = b(x)s(x)$.

Equation (1) shows that if $\lambda \in \mathbf{F}$ is such that $p(\lambda) = 0$ then $r(\lambda) = 0$ or $s(\lambda) = 0$. That is, $Z_p \subseteq Z_r \cup Z_s$. Equations (2) and (3) show that if $\lambda \in \mathbf{F}$ is such that $r(\lambda) = 0$ or $s(\lambda) = 0$ then $p(\lambda) = 0$. That is, $Z_r \cup Z_s \subseteq Z_p$. Thus $Z_p = Z_r \cup Z_s$.

**Exercise 5.B.27.** Suppose $\mathbf{F} = \mathbf{R}, V$ is finite-dimensional, and $T \in \mathcal{L}(V)$. Prove that the minimal polynomial of $T_{\mathbf{C}}$ equals the minimal polynomial of $T$.

*The complexification $T_{\mathbf{C}}$ was defined in Exercise 33 of Section 3B.*

**Solution.** Let $p \in \mathcal{P}(\mathbf{R})$ be the minimal polynomial of $T$ and let $q \in \mathcal{P}(\mathbf{C})$ be the minimal polynomial of $T_{\mathbf{C}}$. Because we can identify $a \in \mathbf{R}$ with $a + 0i \in \mathbf{C}$, we can think of $p$ as a polynomial with complex coefficients. Thus it makes sense to consider $p(T_{\mathbf{C}})$. From the definitions of $T_{\mathbf{C}}$ and of scalar multiplication in $V_{\mathbf{C}}$, note that, for any non-negative integer $k$, any $a \in \mathbf{R}$, and any $u + iv \in V_{\mathbf{C}}$,

$$T_{\mathbf{C}}^k(u + iv) = \big(T^k u\big) + i\big(T^k v\big) \quad \text{and} \quad aT_{\mathbf{C}}(u + iv) = (aTu) + i(aTv).$$



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Combining this with the definition of vector addition in $V_{\mathbf{C}}$, for all $u + iv \in V_{\mathbf{C}}$ it follows that

$$p(T_{\mathbf{C}})(u + iv) = (p(T)u) + i(p(T)v) = 0 + 0i.$$

Thus $p(T_{\mathbf{C}}) = 0$ and hence, by 5.29, $p$ is a polynomial multiple of $q$, where we think of $p$ as an element of $\mathcal{P}(\mathbf{C})$.

Let $\mathcal{B} := v_1, ..., v_n$ be a basis of $V$ and let $\mathcal{B}_{\mathbf{C}} := v_1 + 0i, ..., v_n + 0i$; it follows from Exercise 2.B.11 that $\mathcal{B}_{\mathbf{C}}$ is a basis of $V_{\mathbf{C}}$. Because $T_{\mathbf{C}}(v_k + 0i) = Tv_k + 0i$, the matrix of $T_{\mathbf{C}}$ with respect to $\mathcal{B}_{\mathbf{C}}$ must be equal to the matrix of $T$ with respect to $\mathcal{B}$, where we think of $\mathcal{M}(T, \mathcal{B})$ as a matrix with complex entries. Letting $A$ denote this matrix, it follows that each entry of $A$ is a real number and hence that $\overline{A} = A$, where $\overline{A}$ is the matrix obtained by taking the complex conjugate of each entry of $A$. Note that, for any non-negative integer $k$, each entry of $A^k$ must also be a real number and thus $\overline{A^k} = A^k$. Suppose that $q = \sum_{k=0}^{m} a_k z^k$ and observe that

$$0 = q(A) \quad \Rightarrow \quad 0 = \overline{q(A)} = \overline{\sum_{k=0}^{m} a_k A^k} = \sum_{k=0}^{m} \overline{a_k A^k} = \sum_{k=0}^{m} \overline{a_k}\,\overline{A^k} = \sum_{k=0}^{m} \overline{a_k} A^k;$$

the algebraic properties of "matrix complex conjugation" used here follow quickly from 4.4. Thus, letting $\overline{q} = \sum_{k=0}^{m} \overline{a_k} z^k$, we have $\overline{q}(A) = 0$, which implies that $(q - \overline{q})(A) = 0$ and hence that $(q - \overline{q})(T_{\mathbf{C}}) = 0$. Note that, because $q$ and $\overline{q}$ are monic, we have $\deg(q - \overline{q}) < \deg q$. It must then be the case that $q - \overline{q}$ is the zero polynomial, since $q$ is the minimal polynomial of $T_{\mathbf{C}}$. This gives us $a_k = \overline{a_k}$, i.e. $a_k \in \mathbf{R}$, for each $k$ and thus $q$ can be thought of as a polynomial with real coefficients, so that $q(T)$ makes sense. Because $A$ is also the matrix of $T$ with respect to $\mathcal{B}$, the equation $0 = q(A)$ shows that $0 = q(T)$ and thus, by 5.29, $q$ must be a polynomial multiple of $p$. We may now appeal to Lemma L.7 and the fact that $p$ and $q$ are both monic to conclude that $p = q$.

**Exercise 5.B.28.** Suppose $V$ is finite-dimensional and $T \in \mathcal{L}(V)$. Prove that the minimal polynomial of $T' \in \mathcal{L}(V')$ equals the minimal polynomial of $T$.

*The dual map $V'$ was defined in Section 3F.*

**Solution.** Let $p$ be the minimal polynomial of $T$ and let $q$ be the minimal polynomial of $T'$. For any $s \in \mathcal{P}(\mathbf{F})$, 3.120 shows that $s(T)' = s(T')$, and Exercise 3.F.16 shows that $0' = 0$. Thus

$$p(T) = 0 \quad \Rightarrow \quad p(T)' = 0' \quad \Leftrightarrow \quad p(T') = 0.$$

It follows from 5.29 that $p$ is a polynomial multiple of $q$. Let $\Lambda : V \to V''$ be the isomorphism defined in Exercise 3.F.32 and observe that

$$0 = q(T') = q(T)' \quad \Rightarrow \quad q(T)'' = 0 \quad \Rightarrow \quad q(T)'' \circ \Lambda = 0 \quad \Rightarrow \quad \Lambda \circ q(T) = 0 \quad \Rightarrow \quad q(T) = 0,$$



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where we have used Exercise 3.F.32 (b) for the third implication and the injectivity of $\Lambda$ for the last implication. Thus, by 5.29, $q$ must be a polynomial multiple of $p$. We may now appeal to Lemma L.7 and the fact that $p$ and $q$ are both monic to conclude that $p = q$.

**Exercise 5.B.29.** Show that every operator on a finite-dimensional vector space of dimension at least two has an invariant subspace of dimension two.

*Exercise 6 in Section 5C will give an improvement of this result when* $\mathbf{F} = \mathbf{C}$.

**Solution.** For an integer $k \geq 2$, let $P(k)$ be the statement that any operator on a vector space of dimension $k$ has an invariant subspace of dimension two. We will proceed by induction on $k$. For the base case $P(2)$, we can take the invariant subspace to be the vector space itself.

Now suppose that $P(k)$ holds for some $k \geq 2$, let $T$ be an operator on some vector space $V$ satisfying $\dim V = k + 1$, and let $p \in \mathcal{P}(\mathbf{F})$ be the minimal polynomial of $T$; note that $\deg p \geq 1$ since $V \neq \{0\}$.

If $p$ has a linear factor, i.e. if $p$ has a zero, then 5.27 shows that $T$ has an eigenvalue. It then follows from Exercise 5.A.39 that there exists a subspace $U$ of $V$ which is invariant under $T$ and satisfies $\dim U = k$. Our induction hypothesis guarantees that there is a subspace $W$ of $U$ which is invariant under $T|_U$ and such that $\dim W = 2$. It follows that $W$ is a two-dimensional subspace of $V$ which is invariant under $T$.

If $p$ has no linear factor then note that $T$ has no eigenvalues by 5.27(a). Note further that, by the fundamental theorem of algebra (4.12/4.13), we must have $\mathbf{F} = \mathbf{R}$. It then follows from 4.16 that $p$ has a factorization $p = f_1 \cdots f_m$ where each $f_j \in \mathcal{P}(\mathbf{R})$ is quadratic. Because $0 = p(T) = f_1(T) \cdots f_m(T)$, there must exist some $j \in \{1, ..., m\}$ such that $f_j(T)$ is not injective, i.e. there exists some non-zero $v \in V$ such that $f_j(T)v = 0$. Since $f_j$ is quadratic we have $f_j(x) = ax^2 + bx + c$ for some $a, b, c \in \mathbf{R}$ such that $a \neq 0$. Thus

$$0 = f_j(T)v = aT^2v + bTv + cv \quad \Rightarrow \quad T^2v \in \operatorname{span}(v, Tv).$$

It follows that $\operatorname{span}(v, Tv)$ is invariant under $T$. Furthermore, because $v \neq 0$ and $T$ has no eigenvalues, we must have $\dim \operatorname{span}(v, Tv) = 2$. Thus $V$ has a two-dimensional subspace which is invariant under $T$.

In either case, $P(k + 1)$ holds. This completes the induction step and the proof.



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## 5.C. Upper-Triangular Matrices

**Exercise 5.C.1.** Prove or give a counterexample: If $T \in \mathcal{L}(V)$ and $T^2$ has an upper-triangular matrix with respect to some basis of $V$, then $T$ has an upper triangular matrix with respect to some basis of $V$.

**Solution.** This is false. For a counterexample, take $T \in \mathcal{L}(\mathbf{R}^2)$ to be a counterclockwise rotation about the origin by 90°, i.e. $T(x, y) = (-y, x)$. Then $T^2 = -I$, so that the matrix of $T^2$ with respect to the standard basis of $\mathbf{R}^2$ is the upper-triangular matrix

$$-\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}.$$

However, note that if $T$ had an upper-triangular matrix with respect to some basis of $\mathbf{R}^2$ then $T$ would have an eigenvalue—but $T$ has no eigenvalues, as shown in 5.9(a).

**Exercise 5.C.2.** Suppose $A$ and $B$ are upper-triangular matrices of the same size, with $\alpha_1, ..., \alpha_n$ on the diagonal of $A$ and $\beta_1, ..., \beta_n$ on the diagonal of $B$.

(a) Show that $A + B$ is an upper-triangular matrix with $\alpha_1 + \beta_1, ..., \alpha_n + \beta_n$ on the diagonal.

(b) Show that $AB$ is an upper-triangular matrix with $\alpha_1\beta_1, ..., \alpha_n\beta_n$ on the diagonal.

*The results in this exercise are used in the proof of 5.81.*

**Solution.** Note that $A$ and $B$ satisfy

$$A_{j,k} = \begin{cases} \alpha_j & \text{if } j = k, \\ 0 & \text{if } j > k, \end{cases} \qquad B_{j,k} = \begin{cases} \beta_j & \text{if } j = k, \\ 0 & \text{if } j > k. \end{cases}$$

(a) From the definition of matrix addition, we have

$$(A + B)_{j,k} = A_{j,k} + B_{j,k} = \begin{cases} \alpha_j + \beta_j & \text{if } j = k, \\ 0 & \text{if } j > k. \end{cases}$$

Thus $A + B$ is upper-triangular with $\alpha_1 + \beta_1, ..., \alpha_n + \beta_n$ on the diagonal.

(b) By the definition of matrix multiplication, we have

$$(AB)_{j,j} = \sum_{r=1}^{n} A_{j,r}B_{r,j} = A_{j,j}B_{j,j} + \sum_{r \neq j} A_{j,r}B_{r,j} = \alpha_j\beta_j + \sum_{r \neq j} A_{j,r}B_{r,j}.$$

For $r \neq j$ we either have $j > r$, in which case $A_{j,r} = 0$, or we have $r > j$, in which case $B_{r,j} = 0$. Thus $\sum_{r \neq j} A_{j,r}B_{r,j} = 0$ and it follows that $(AB)_{j,j} = \alpha_j\beta_j$.

For $j > k$, observe that



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$$(AB)_{j,k} = \sum_{r=1}^{n} A_{j,r}B_{r,k} = \sum_{r=1}^{j-1} A_{j,r}B_{r,k} + \sum_{r=j}^{n} A_{j,r}B_{r,k}.$$

For $1 \le r \le j-1$ we have $j > r$ and thus $A_{j,r} = 0$; it follows that $\sum_{r=1}^{j-1} A_{j,r}B_{r,k} = 0$. For $j \le r \le n$ we have $r \ge j > k$ and thus $B_{r,k} = 0$; it follows that $\sum_{r=j}^{n} A_{j,r}B_{r,k} = 0$. Hence $(AB)_{j,k} = 0$ and we may conclude that $AB$ is upper-triangular with $\alpha_1\beta_1, ..., \alpha_n\beta_n$ on the diagonal.

**Exercise 5.C.3.** Suppose $T \in \mathcal{L}(V)$ is invertible and $v_1, ..., v_n$ is a basis of $V$ with respect to which the matrix of $T$ is upper triangular, with $\lambda_1, ..., \lambda_n$ on the diagonal. Show that the matrix of $T^{-1}$ is also upper triangular with respect to the basis $v_1, ..., v_n$, with

$$\frac{1}{\lambda_1}, ..., \frac{1}{\lambda_n}$$

on the diagonal.

**Solution.** Because the matrix of $T$ with respect to $v_1, ..., v_n$ is upper-triangular with $\lambda_1, ..., \lambda_n$ on the diagonal, we have $Tv_1 = \lambda_1 v_1$ and, for each $k \ge 2, Tv_k = u_k + \lambda_k v_k$ for some $u_k \in \text{span}(v_1, ..., v_{k-1})$. Observe that

$$Tv_1 = \lambda_1 v_1 \quad \Rightarrow \quad T^{-1}v_1 = \lambda_1^{-1} v_1.$$

This shows that $\text{span}(v_1)$ is invariant under $T^{-1}$ and that the first diagonal entry of $\mathcal{M}(T^{-1})$ is $\lambda_1^{-1}$. Now observe that

$$Tv_2 = u_2 + \lambda_2 v_2 \quad \Rightarrow \quad T^{-1}v_2 = \lambda_2^{-1}T^{-1}u_2 + \lambda_2^{-1}v_2;$$

note that $\lambda_2^{-1}T^{-1}u_2 \in \text{span}(v_1)$ because $u_2 \in \text{span}(v_1)$ and $\text{span}(v_1)$ is invariant under $T^{-1}$. Thus $T^{-1}v_2 \in \text{span}(v_1, v_2)$, so that $\text{span}(v_1, v_2)$ is invariant under $T^{-1}$, and the second diagonal entry of $\mathcal{M}(T^{-1})$ is $\lambda_2^{-1}$. Continuing in this manner, we see that each $\text{span}(v_1, ..., v_k)$ is invariant under $T^{-1}$, whence $\mathcal{M}(T^{-1})$ is upper-triangular, and that the diagonal entries of $\mathcal{M}(T^{-1})$ are $\lambda_1^{-1}, ..., \lambda_n^{-1}$.

**Exercise 5.C.4.** Give an example of an operator whose matrix with respect to some basis contains only 0's on the diagonal, but the operator is invertible.

*This exercise and the exercise below show that 5.41 fails without the hypothesis that an upper-triangular matrix is under consideration.*

**Solution.** Let $T \in \mathcal{L}(\mathbf{R}^2)$ be given by $T(x, y) = (y, x)$. The matrix of $T$ with respect to the standard basis of $\mathbf{R}^2$ is

$$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

Furthermore, $T$ is invertible since $T^2 = I$.



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**Exercise 5.C.5.** Give an example of an operator whose matrix with respect to some basis contains only nonzero numbers on the diagonal, but the operator is not invertible.

**Solution.** Let $T \in \mathcal{L}(\mathbf{R}^2)$ be given by $T(x, y) = (x + y, x + y)$. The matrix of $T$ with respect to the standard basis of $\mathbf{R}^2$ is

$$\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}.$$

Furthermore, $T$ is not invertible because $T$ is not injective: $T(1, -1) = (0, 0)$.

**Exercise 5.C.6.** Suppose $\mathbf{F} = \mathbf{C}$, $V$ is finite-dimensional, and $T \in \mathcal{L}(V)$. Prove that if $k \in \{1, ..., \dim V\}$, then $V$ has a $k$-dimensional subspace invariant under $T$.

**Solution.** By 5.47 there is a basis $v_1, ..., v_n$ of $V$ such that the matrix of $T$ with respect to $v_1, ..., v_n$ is upper-triangular. It follows from 5.39 that for each $k \in \{1, ..., n\}$, the subspace $\operatorname{span}(v_1, ..., v_k)$ is $k$-dimensional and invariant under $T$.

**Exercise 5.C.7.** Suppose $V$ is finite-dimensional, $T \in \mathcal{L}(V)$, and $v \in V$.

(a) Prove that there exists a unique monic polynomial $p_v$ of smallest degree such that $p_v(T)v = 0$.

(b) Prove that the minimal polynomial of $T$ is a polynomial multiple of $p_v$.

**Solution.**

(a) If $v = 0$ then we can take $p_v = 1$, so suppose that $v \neq 0$. The list $v, Tv, ..., T^{\dim V}v$ has length $1 + \dim V$ and hence must be linearly dependent. By the linear dependence lemma (2.19), there exists a least integer $k \in \{1, ..., \dim V\}$ such that $T^k v \in U \coloneqq \operatorname{span}\big(v, Tv, ..., T^{k-1}v\big)$ and such that $v, Tv, ..., T^{k-1}v$ is linearly independent, so that $\dim U = k$. Note that $U$ is invariant under $T$ because $T^k v \in U$. Let $p_v$ be the minimal polynomial of $T|_U$ and observe that, since $v \in U$,

$$p_v(T)v = p_v(T|_U)v = 0.$$

If $q$ is a polynomial of degree $\ell < \deg p_v \leq \dim U = k$ such that $q(T)v = 0$ then $T^\ell v \in \operatorname{span}\big(v, Tv, ..., T^{\ell-1}v\big)$, contradicting the minimality of $k$. Thus the degree of $p_v$ is minimal. If $r$ is a monic polynomial of degree $\deg p_v$ satisfying $r(T)v = 0$ then $p_v - r$ satisfies $(p_v - r)(T)v = 0$ and $\deg(p_v - r) < \deg p_v$. If $p_v - r$ were not zero then we could divide by the leading coefficient to obtain a monic polynomial $s$ satisfying $s(T)v = 0$ and $\deg s < \deg p_v$, which contradicts the minimality of $\deg p_v$. Thus $p_v$ is unique.

(b) This is immediate from part (a) and 5.31.



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**Exercise 5.C.8.** Suppose $V$ is finite-dimensional, $T \in \mathcal{L}(V)$, and there exists a nonzero vector $v \in V$ such that $T^2v + 2Tv = -2v$.

(a) Prove that if $\mathbf{F} = \mathbf{R}$, then there does not exist a basis of $V$ with respect to which $T$ has an upper triangular matrix.

(b) Prove that if $\mathbf{F} = \mathbf{C}$ and $A$ is an upper-triangular matrix that equals the matrix of $T$ with respect to some basis of $V$, then $-1 + i$ or $-1 - i$ appears on the diagonal of $A$.

**Solution.**

(a) Let $q(x) = x^2 + 2x + 2$, so that $q(T)v = 0$, and let $p_v$ be defined as in Exercise 5.C.7. Note that $\deg p_v \geq 1$ since $v \neq 0$. A straightforward modification of 5.29 shows that $q$ must be a polynomial multiple of $p_v$. It follows that $\deg p_v \geq 2$ since $q$ has no real roots. The minimality of $\deg p_v$ then implies that $p_v = q$ and thus, by Exercise 5.C.7 (b), the minimal polynomial of $T$ must be a polynomial multiple of $q$. Thus the minimal polynomial of $T$ does not split into linear factors and it then follows from 5.44 that there is no basis of $V$ with respect to which $T$ has an upper-triangular matrix.

(b) Let $q(z) = z^2 + 2z + 2 = (z + 1 - i)(z + 1 + i)$, so that $q(T)v = 0$, and let $p_v$ be defined as in Exercise 5.C.7. Note that $\deg p_v \geq 1$ since $v \neq 0$. A straightforward modification of 5.29 shows that $q$ must be a polynomial multiple of $p_v$. There are then three possibilities:

$$p_v = z + 1 - i, \quad p_v = z + 1 + i, \quad \text{or} \quad p_v = q.$$

In any case, at least one of $-1 + i, -1 - i$ is a root of $p_v$. It follows from Exercise 5.C.7 (b) that at least one of $-1 + i, -1 - i$ is a root of the minimal polynomial of $T$ and hence, by 5.27, at least one of $-1 + i, -1 - i$ is an eigenvalue of $T$. 5.41 allows us to conclude that at least one of $-1 + i, -1 - i$ appears on the diagonal of $A$.

**Exercise 5.C.9.** Suppose $B$ is a square matrix with complex entries. Prove that there exists an invertible square matrix $A$ with complex entries such that $A^{-1}BA$ is an upper-triangular matrix.

**Solution.** Suppose $B$ is an $n$-by-$n$ matrix and let $T \in \mathcal{L}(\mathbf{C}^n)$ be given by $Tx = Bx$, where we think of elements of $\mathbf{C}^n$ as column vectors. Evidently, the matrix of $T$ with respect to the standard basis $e_1, ..., e_n$ of $\mathbf{C}^n$ is $B$. By 5.47 there is a basis $v_1, ..., v_n$ of $\mathbf{C}^n$ such that $\mathcal{M}(T, (v_1, ..., v_n))$ is upper-triangular. Let $A = \mathcal{M}(I, (v_1, ..., v_n), (e_1, ..., e_n))$; as 3.84 shows, it follows that $A^{-1}BA$ equals the upper-triangular matrix $\mathcal{M}(T, (v_1, ..., v_n))$.



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**Exercise 5.C.10.** Suppose $T \in \mathcal{L}(V)$ and $v_1, ..., v_n$ is a basis of $B$. Show that the following are equivalent.

(a) The matrix of $T$ with respect to $v_1, ..., v_n$ is lower triangular.

(b) $\text{span}(v_k, ..., v_n)$ is invariant under $T$ for each $k = 1, ..., n$.

(c) $Tv_k \in \text{span}(v_k, ..., v_n)$ for each $k = 1, ..., n$.

*A square matrix is called **lower triangular** if all entries above the diagonal are* 0.

**Solution.** Suppose that (a) holds and let $k \in \{1, ..., n\}$ be given. For any $j \in \{k, ..., n\}$ we have $Tv_j \in \text{span}(v_j, ..., v_n)$ since the matrix of $T$ with respect to $v_1, ..., v_n$ is lower-triangular. Because $\text{span}(v_j, ..., v_n) \subseteq \text{span}(v_k, ..., v_n)$ for $j \geq k$, it follows that $Tv_j \in \text{span}(v_k, ..., v_n)$ for each $j \in \{k, ...n\}$. Thus $\text{span}(v_k, ..., v_n)$ is invariant under $T$, i.e. (b) holds.

Now suppose that (b) holds. For any $k \in \{1, ..., n\}$ we have $v_k \in \text{span}(v_k, ..., v_n)$, which is invariant under $T$ by assumption. Thus $Tv_k \in \text{span}(v_k, ..., v_n)$, i.e. (c) holds.

Suppose that (c) holds, so that each $Tv_k$ can be written as a linear combination of the basis vectors $v_k, ..., v_n$ only. It follows that each entry above the diagonal of $\mathcal{M}(T)$ is zero, i.e. $\mathcal{M}(T)$ is lower-triangular. Thus (a) holds.

**Exercise 5.C.11.** Suppose $\mathbf{F} = \mathbf{C}$ and $V$ is finite-dimensional. Prove that if $T \in \mathcal{L}(V)$, then there exists a basis of $V$ with respect to which $T$ has a lower-triangular matrix.

**Solution.** By 5.47 there is a basis $v_1, ..., v_n$ of $V$ with respect to which the matrix of $T$ is upper-triangular. For each $k \in \{1, ..., n\}$ define $u_k = v_{n-k+1}$ and observe that, using 5.39,

$$Tu_k = Tv_{n-k+1} \in \text{span}(v_1, v_2, ..., v_{n-k}, v_{n-k+1}) = \text{span}(u_n, u_{n-1}, ..., u_{k+1}, u_k).$$

It follows from Exercise 5.C.10 that the matrix of $T$ with respect to $u_1, ..., u_n$ is lower-triangular.

**Exercise 5.C.12.** Suppose $V$ is finite-dimensional, $T \in \mathcal{L}(V)$ has an upper-triangular matrix with respect to some basis of $V$, and $U$ is a subspace of $V$ that is invariant under $T$.

(a) Prove that $T|_U$ has an upper-triangular matrix with respect to some basis of $U$.

(b) Prove that the quotient operator $T/U$ has an upper-triangular matrix with respect to some basis of $V/U$.

*The quotient operator $T/U$ was defined in Exercise 38 in Section 5A.*

**Solution.**

(a) Let $p$ be the minimal polynomial of $T$ and let $q$ be the minimal polynomial of $T|_U$. By 5.31 and 5.44, $p$ is a product of linear factors and also a polynomial multiple of



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$q$. It follows that $q$ is a product of linear factors and thus, by 5.44, $T|_U$ has an upper-triangular matrix with respect to some basis of $U$.

(b) Similarly to part (a), let $p$ be the minimal polynomial of $T$ and let $q$ be the minimal polynomial of $T/U$. By Exercise 5.B.25 (a) and 5.44, $p$ is a product of linear factors and also a polynomial multiple of $q$. It follows that $q$ is a product of linear factors and thus, by 5.44, $T/U$ has an upper-triangular matrix with respect to some basis of $V/U$.

**Exercise 5.C.13.** Suppose $V$ is finite-dimensional and $T \in \mathcal{L}(V)$. Suppose there exists a subspace $U$ of $V$ that is invariant under $T$ such that $T|_U$ has an upper-triangular matrix with respect to some basis of $U$ and also $T/U$ has an upper-triangular matrix with respect to some basis of $V/U$. Prove that $T$ has an upper-triangular matrix with respect to some basis of $V$.

**Solution.** Let $p, q, r$ be the minimal polynomials of $T, T|_U$, and $T/U$. By Exercise 5.B.25 (b) and 5.44, $q$ and $r$ are products of linear factors and $qr$ is a polynomial multiple of $p$. It follows that $p$ is a product of linear factors and thus, by 5.44, $T$ has an upper-triangular matrix with respect to some basis of $V$.

**Exercise 5.C.14.** Suppose $V$ is finite-dimensional and $T \in \mathcal{L}(V)$. Prove that $T$ has an upper-triangular matrix with respect to some basis of $V$ if and only if the dual operator $T'$ has an upper-triangular matrix with respect to some basis of the dual space $V'$.

**Solution.** Exercise 5.B.28 shows that $T$ and $T'$ have the same minimal polynomial. The desired equivalence now follows from 5.44.



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## 5.D. Diagonalizable Operators

**Exercise 5.D.1.** Suppose $V$ is a finite-dimensional complex vector space and $T \in \mathcal{L}(V)$.

(a) Prove that if $T^4 = I$, then $T$ is diagonalizable.

(b) Prove that if $T^4 = T$, then $T$ is diagonalizable.

(c) Give an example of an operator $T \in \mathcal{L}(\mathbf{C}^2)$ such that $T^4 = T^2$ and $T$ is not diagonalizable.

**Solution.**

(a) Let $q(z) = z^4 - 1 = (z-1)(z+1)(z-i)(z+i)$, let $p$ be the minimal polynomial of $T$, and note that $q(T) = 0$. It follows from 5.29 that $q$ is a polynomial multiple of $p$ and so $p$ must be a product of distinct linear factors. Thus, by 5.62, $T$ is diagonalizable.

(b) Let $q(z) = z^4 - z = z(z-1)\left(z + \frac{1}{2} - \frac{\sqrt{3}}{2}i\right)\left(z + \frac{1}{2} + \frac{\sqrt{3}}{2}i\right)$, let $p$ be the minimal polynomial of $T$, and note that $q(T) = 0$. It follows from 5.29 that $q$ is a polynomial multiple of $p$ and so $p$ must be a product of distinct linear factors. Thus, by 5.62, $T$ is diagonalizable.

(c) Let $T \in \mathcal{L}(\mathbf{C}^2)$ be given by $T(w, z) = (z, 0)$ and notice that $T^2 = T^4 = 0$. The matrix of $T$ with respect to the standard basis of $\mathbf{C}^2$ is

$$\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}.$$

Thus, by 5.41, the only eigenvalue of $T$ is 0. Letting $p$ be the minimal polynomial of $T$, 5.27 shows that $p$ is of the form $z^k$ for some $k \in \{1, 2\}$. Since $T \neq 0$, it must be that $p(z) = z^2$. It follows from 5.62 that $T$ is not diagonalizable.

**Exercise 5.D.2.** Suppose $T \in \mathcal{L}(V)$ has a diagonal matrix $A$ with respect to some basis of $V$. Prove that if $\lambda \in \mathbf{F}$, then $\lambda$ appears on the diagonal of $A$ precisely $\dim E(\lambda, T)$ times.

**Solution.** Suppose $v_1, ..., v_n$ is a basis of $V$ such that $A \coloneqq \mathcal{M}(T, (v_1, ..., v_n))$ is diagonal and let $\lambda_1, ..., \lambda_n$ denote the diagonal entries of $A$. Note that the list of those $Tv_k = \lambda_k v_k$ such that $\lambda_k \neq 0$ is linearly independent since $v_1, ..., v_n$ is linearly independent. Thus, letting $d$ be the number of indices $k \in \{1, ..., n\}$ such that $\lambda_k = 0$, we have $\dim \operatorname{range} T \geq n - d$. It follows that

$$\dim \operatorname{null} T \leq d. \tag{1}$$

For $\lambda \in \mathbf{F}$, let $d_\lambda$ be the number of times $\lambda$ appears on the diagonal of $A$. By replacing $T$ with $T - \lambda I$ in (1), we find that $\dim E(\lambda, T) \leq d_\lambda$. Now observe that, by 5.55(d),



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$$\sum_{\lambda \in \mathbf{F}} \dim E(\lambda, T) = n = \sum_{\lambda \in \mathbf{F}} d_\lambda. \tag{2}$$

(Both of these are finite sums since $\dim E(\lambda, T) = d_\lambda = 0$ for all but finitely many $\lambda \in \mathbf{F}$.) It follows that each inequality $\dim E(\lambda, T) \leq d_\lambda$ must in fact be an equality, otherwise the left-hand side of (2) would be strictly less than the right-hand side.

**Exercise 5.D.3.** Suppose $V$ is finite-dimensional and $T \in \mathcal{L}(V)$. Prove that if the operator $T$ is diagonalizable, then $V = \operatorname{null} T \oplus \operatorname{range} T$.

**Solution.** Let $\lambda_1, ..., \lambda_m$ be the distinct non-zero eigenvalues of $T$ (this list may be empty). By 5.55, we have

$$V = E(0, T) \oplus E(\lambda_1, T) \oplus \cdots \oplus E(\lambda_m, T) = \operatorname{null} T \oplus W,$$

where $W = E(\lambda_1, T) \oplus \cdots \oplus E(\lambda_m, T)$; if the list $\lambda_1, ..., \lambda_m$ is empty then take $W = \{0\}$. Let $Tv \in \operatorname{range} T$ be given. The direct sum expression above shows that $v$ is of the form

$$v = u + w_1 + \cdots + w_m \in \operatorname{null} T \oplus W \quad \Rightarrow \quad Tv = \lambda_1 w_1 + \cdots + \lambda_m w_m \in W.$$

Thus $\operatorname{range} T \subseteq W$. Now let $w_1 + \cdots + w_m \in W$ be given. Because each $\lambda_k \neq 0$, it follows that

$$w_1 + \cdots + w_m = T(\lambda_1^{-1} w_1 + \cdots + \lambda_m^{-1} w_m) \in \operatorname{range} T.$$

Thus $W = \operatorname{range} T$ and we may conclude that $V = \operatorname{null} T \oplus \operatorname{range} T$.

**Exercise 5.D.4.** Suppose $V$ is finite-dimensional and $T \in \mathcal{L}(V)$. Prove that the following are equivalent.

(a) $V = \operatorname{null} T \oplus \operatorname{range} T$.

(b) $V = \operatorname{null} T + \operatorname{range} T$.

(c) $\operatorname{null} T \cap \operatorname{range} T = \{0\}$.

**Solution.** Certainly (a) implies (b). Suppose that (b) holds. By 2.43, we have

$$\dim(\operatorname{null} T \cap \operatorname{range} T) = \dim \operatorname{null} T + \dim \operatorname{range} T - \dim(\operatorname{null} T + \operatorname{range} T).$$

By assumption $\dim(\operatorname{null} T + \operatorname{range} T) = \dim V$ and the fundamental theorem of linear maps (3.21) shows that $\dim \operatorname{null} T + \dim \operatorname{range} T = \dim V$ also. Thus $\dim(\operatorname{null} T \cap \operatorname{range} T) = 0$, so that $\operatorname{null} T \cap \operatorname{range} T = \{0\}$, i.e. (c) holds.

Suppose that (c) holds. It follows from 1.46 that the sum $\operatorname{null} T \oplus \operatorname{range} T$ is direct. Furthermore,

$$\dim(\operatorname{null} T \oplus \operatorname{range} T) = \dim \operatorname{null} T + \dim \operatorname{range} T = \dim V$$

by 1.46 and 2.43. Thus, by 2.39, $V = \operatorname{null} T \oplus \operatorname{range} T$, i.e. (a) holds.



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**Exercise 5.D.5.** Suppose $V$ is a finite-dimensional complex vector space and $T \in \mathcal{L}(V)$. Prove that $T$ is diagonalizable if and only if

$$V = \text{null}(T - \lambda I) \oplus \text{range}(T - \lambda I)$$

for every $\lambda \in \mathbf{C}$.

**Solution.** Suppose that $T$ is diagonalizable, so that there is some basis of $V$ with respect to which the matrix of $T$ is diagonal:

$$\begin{pmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \end{pmatrix}.$$

For any $\lambda \in \mathbf{C}$, the matrix of the operator $T - \lambda I$ with respect to this same basis is also diagonal:

$$\begin{pmatrix} \lambda_1 - \lambda & & 0 \\ & \ddots & \\ 0 & & \lambda_n - \lambda \end{pmatrix}.$$

So $T - \lambda I$ is also diagonalizable and thus by Exercise 5.D.3 we have

$$V = \text{null}(T - \lambda I) \oplus \text{range}(T - \lambda I).$$

Here are two proofs of the converse.

(1) By contrapositive: suppose that $T$ is not diagonalizable. Let $\lambda_1, ..., \lambda_m$ be the distinct eigenvalues of $T$ and let $p$ be the minimal polynomial of $T$; by 5.27(b) we have

$$p(z) = (z - \lambda_1)^{n_1} \cdots (z - \lambda_m)^{n_m}$$

for some positive integers $n_1, ..., n_m$. Because $T$ is not diagonalizable, 5.62 shows that there must be some $k \in \{1, ..., m\}$ such that $n_k \geq 2$. Let $q$ be the polynomial given by

$$q(z) = (z - \lambda_k)^{n_k - 1} \prod_{j \neq k} (z - \lambda_j)^{n_j}.$$

Notice that $p(z) = (z - \lambda_k)q(z)$, so that $\deg q < \deg p$, and, for any $v \in V$,

$$0 = p(T)v = (T - \lambda_k I)q(T)v \quad \Rightarrow \quad q(T)v \in \text{null}(T - \lambda_k I),$$

$$n_k - 1 \geq 1 \quad \Rightarrow \quad q(T)v \in \text{range}(T - \lambda_k I).$$

Thus $q(T)v \in \text{null}(T - \lambda_k I) \cap \text{range}(T - \lambda_k I)$ for all $v \in V$. Since $\deg q < \deg p$ it must be the case that $q(T) \neq 0$, i.e. there exists some $v \in V$ such that $q(T)v \neq 0$. It follows that

$$\text{null}(T - \lambda_k I) \cap \text{range}(T - \lambda_k I) \neq \{0\}$$

and hence, by Exercise 5.D.4, $V \neq \text{null}(T - \lambda_k I) \oplus \text{range}(T - \lambda_k I)$.



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(2) By strong induction on $\dim V$. Let $P(n)$ be the following statement: if $V$ is an $n$-dimensional complex vector space, $T \in \mathcal{L}(V)$, and

$$V = \operatorname{null}(T - \lambda I) \oplus \operatorname{range}(T - \lambda I)$$

for all $\lambda \in \mathbf{C}$, then $T$ is diagonalizable. The truth of $P(0)$ is clear, so suppose that $P(0), ..., P(n)$ all hold for some $n \geq 1$. Let $V$ be an $(n+1)$-dimensional vector space and suppose that $T \in \mathcal{L}(V)$ satisfies

$$V = \operatorname{null}(T - \lambda I) \oplus \operatorname{range}(T - \lambda I)$$

for all $\lambda \in \mathbf{C}$. By 5.19 there exists an eigenvalue $\lambda_0 \in \mathbf{C}$ of $T$; let $U = \operatorname{range}(T - \lambda_0 I)$ and note that $U$ is invariant under $T$ by (5.18). Note further that, for any $\lambda \in \mathbf{C}$,

$$\operatorname{null}(T|_U - \lambda I|_U) \cap \operatorname{range}(T|_U - \lambda I|_U) \subseteq \operatorname{null}(T - \lambda I) \cap \operatorname{range}(T - \lambda I) = \{0\}.$$

It follows that $U = \operatorname{null}(T|_U - \lambda I|_U) \oplus \operatorname{range}(T|_U - \lambda I|_U)$ for every $\lambda \in \mathbf{C}$, where we have used the equivalence of (a) and (c) in Exercise 5.D.4. By assumption $V = E(\lambda_0, T) \oplus U$; since $\lambda_0$ is an eigenvalue of $T$ we have $\dim E(\lambda_0, T) \geq 1$ and thus $\dim U < \dim V$. Our induction hypothesis now guarantees that there is a basis of $U$ consisting of eigenvectors of $T|_U$, which must also be eigenvectors of $T$. Combining this basis with a basis of $E(\lambda_0, T)$, we obtain a basis of $V$ consisting of eigenvectors of $T$. It follows from 5.55 that $T$ is diagonalizable. This completes the induction step and the proof.

**Exercise 5.D.6.** Suppose $T \in \mathcal{L}(\mathbf{F}^5)$ and $\dim E(8, T) = 4$. Prove that $T - 2I$ or $T - 6I$ is invertible.

**Solution.** We will prove the contrapositive statement. Suppose that neither $T - 2I$ nor $T - 6I$ is invertible, so that $\dim E(2, T) \geq 1$ and $\dim E(6, T) \geq 1$. If 8 is an eigenvalue of $T$ then 5.54 shows that $\dim E(8, T) + \dim E(2, T) + \dim E(6, T) \leq \dim \mathbf{F}^5$, and if 8 is not an eigenvalue of $T$ then 5.54 together with $\dim E(8, T) = 0$ gives us the same inequality. In either case,

$$\dim E(8, T) + \dim E(2, T) + \dim E(6, T) \leq \dim \mathbf{F}^5 = 5$$
$$\Rightarrow \quad \dim E(8, T) \leq 5 - \dim E(2, T) - \dim E(6, T) \leq 3 < 4.$$

Thus $\dim E(8, T) \neq 4$.

**Exercise 5.D.7.** Suppose $T \in \mathcal{L}(V)$ is invertible. Prove that

$$E(\lambda, T) = E\left(\frac{1}{\lambda}, T^{-1}\right)$$

for every $\lambda \in \mathbf{F}$ with $\lambda \neq 0$.

**Solution.** This follows from the equivalence, for $\lambda \neq 0$ and any $v \in V$,



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$$Tv = \lambda v \quad \Leftrightarrow \quad T^{-1}v = \lambda^{-1}v.$$

**Exercise 5.D.8.** Suppose $V$ is finite-dimensional and $T \in \mathcal{L}(V)$. Let $\lambda_1, ..., \lambda_m$ denote the distinct nonzero eigenvalues of $T$. Prove that

$$\dim E(\lambda_1, T) + \cdots + \dim E(\lambda_m, T) \leq \dim \operatorname{range} T.$$

**Solution.** Note that

$$\dim E(0, T) + \dim E(\lambda_1, T) + \cdots + \dim E(\lambda_m, T) \leq \dim V \tag{1}$$

follows from 5.54: if 0 is an eigenvalue of $T$ then the inequality is immediate from 5.54, and if 0 is not an eigenvalue of $T$ then $\dim E(0, T) = 0$ and thus we can add $\dim E(0, T)$ to the left-hand side of the inequality in 5.54. The fundamental theorem of linear maps (3.21) shows that

$$\dim V = \dim \operatorname{null} T + \dim \operatorname{range} T = \dim E(0, T) + \dim \operatorname{range} T.$$

Combining this with inequality (1) gives us the desired inequality.

**Exercise 5.D.9.** Suppose $R, T \in \mathcal{L}(\mathbf{F}^3)$ each have $2, 6, 7$ as eigenvalues. Prove that there exists invertible operator $S \in \mathcal{L}(\mathbf{F}^3)$ such that $R = S^{-1}TS$.

**Solution.** Since $R$ and $T$ both have $3 = \dim \mathbf{F}^3$ distinct eigenvalues, 5.58 shows that they are both diagonalizable, i.e. there exists a basis $u_1, u_2, u_3$ and a basis $v_1, v_2, v_3$ of $V$ such that

$$Ru_1 = 2u_1, \quad Ru_2 = 6u_2, \quad Ru_3 = 7u_3, \quad Tv_1 = 2v_1, \quad Tv_2 = 6v_2, \quad Tv_3 = 7v_3.$$

Define $S \in \mathcal{L}(\mathbf{F}^3)$ by $Su_k = v_k$ and note that $S$ is invertible since it maps a basis to a basis. Furthermore,

$$S^{-1}TSu_1 = S^{-1}Tv_1 = 2S^{-1}v_1 = 2u_1 = Ru_1.$$

Similarly, $S^{-1}TSu_k = Ru_k$ for $k \in \{2, 3\}$. Thus $S^{-1}TS = R$.

**Exercise 5.D.10.** Find $R, T \in \mathcal{L}(\mathbf{F}^4)$ such that $R$ and $T$ each have $2, 6, 7$ as eigenvalues, $R$ and $T$ have no other eigenvalues, and there does not exist an invertible operator $S \in \mathcal{L}(\mathbf{F}^4)$ such that $R = S^{-1}TS$.

**Solution.** Let $R$ and $T$ be the operators which have the matrices

$$\mathcal{M}(R) = \begin{pmatrix} 2 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 6 & 0 \\ 0 & 0 & 0 & 7 \end{pmatrix} \quad \text{and} \quad \mathcal{M}(T) = \begin{pmatrix} 2 & 0 & 0 & 0 \\ 0 & 6 & 0 & 0 \\ 0 & 0 & 6 & 0 \\ 0 & 0 & 0 & 7 \end{pmatrix}$$

with respect to the standard basis $e_1, e_2, e_3, e_4$ of $\mathbf{F}^4$. Since these matrices are upper-triangular, 5.41 shows that the eigenvalues of $R$ and $T$ are precisely $2, 6, 7$. To disprove the existence of an invertible operator $S \in \mathcal{L}(\mathbf{F}^4)$ such that $R = S^{-1}TS$, let $S \in \mathcal{L}(\mathbf{F}^4)$ be any



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invertible operator. By Exercise 5.A.13, $S^{-1}TS$ also has 2 as an eigenvalue. Furthermore, the eigenspace $E(2,T)$ is the image under $S$ of the eigenspace $E(2,S^{-1}TS)$. A restriction of $S$ thus provides us with an isomorphism between $E(2,T)$ and $E(2,S^{-1}TS)$; in particular, these eigenspaces must have the same dimension. However, note that

$$\dim E(2,T) = \dim \operatorname{span}(e_1) = 1 \neq 2 = \dim \operatorname{span}(e_1,e_2) = \dim E(2,R).$$

Thus there cannot exist an invertible $S \in \mathcal{L}(\mathbf{F}^4)$ such that $R = S^{-1}TS$.

**Exercise 5.D.11.** Find $T \in \mathcal{L}(\mathbf{C}^3)$ such that 6 and 7 are eigenvalues of $T$ and such that $T$ does not have a diagonal matrix with respect to any basis of $\mathbf{C}^3$.

**Solution.** Let $T$ be the operator which has the matrix

$$\mathcal{M}(T) = \begin{pmatrix} 6 & 1 & 0 \\ 0 & 6 & 0 \\ 0 & 0 & 7 \end{pmatrix}$$

with respect to the standard basis $e_1, e_2, e_3$ of $\mathbf{C}^3$. Since this matrix is upper-triangular, 5.41 shows that the eigenvalues of $T$ are precisely 6 and 7. Some routine calculations reveal that $E(6,T) = \operatorname{span}(e_1)$ and $E(7,T) = \operatorname{span}(e_3)$. It follows that

$$\dim E(6,T) + \dim E(7,T) = 2 \neq 3 = \dim \mathbf{C}^3.$$

Thus, by 5.55, $T$ is not diagonalizable.

**Exercise 5.D.12.** Suppose $T \in \mathcal{L}(\mathbf{C}^3)$ is such that 6 and 7 are eigenvalues of $T$. Furthermore, suppose $T$ does not have a diagonal matrix with respect to any basis of $\mathbf{C}^3$. Prove that there exists $(z_1,z_2,z_3) \in \mathbf{C}^3$ such that

$$T(z_1,z_2,z_3) = (6+8z_1, 7+8z_2, 13+8z_3).$$

**Solution.** Since $\dim \mathbf{C}^3 = 3$, it must be the case that 6 and 7 are the only eigenvalues of $T$; if $T$ had another distinct eigenvalue then, by 5.58, $T$ would be diagonalizable. It follows from 5.7 that $T - 8I$ is surjective and thus there exists $(z_1,z_2,z_3) \in \mathbf{C}^3$ such that

$$(T-8I)(z_1,z_2,z_3) = (6,7,13) \quad \Leftrightarrow \quad T(z_1,z_2,z_3) = (6+8z_1, 7+8z_2, 13+8z_3).$$

**Exercise 5.D.13.** Suppose $A$ is a diagonal matrix with distinct entries on the diagonal and $B$ is a matrix of the same size as $A$. Show that $AB = BA$ if and only if $B$ is a diagonal matrix.

**Solution.** Suppose $A$ has diagonal entries $a_1, \ldots, a_n$. If $B$ is also diagonal with diagonal entries $b_1, \ldots, b_n$, then



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$$AB = \begin{pmatrix} a_1 & & 0 \\ & \ddots & \\ 0 & & a_n \end{pmatrix} \begin{pmatrix} b_1 & & 0 \\ & \ddots & \\ 0 & & b_n \end{pmatrix} = \begin{pmatrix} a_1 b_1 & & 0 \\ & \ddots & \\ 0 & & a_n b_n \end{pmatrix} = \begin{pmatrix} b_1 & & 0 \\ & \ddots & \\ 0 & & b_n \end{pmatrix} \begin{pmatrix} a_1 & & 0 \\ & \ddots & \\ 0 & & a_n \end{pmatrix} = BA.$$

Now suppose that $B$ is not diagonal, i.e. there exist $j, k \in \{1, ..., n\}$ such that $j \neq k$ and $B_{j,k} \neq 0$. Observe that

$$(AB)_{j,k} = \sum_{r=1}^{n} A_{j,r} B_{r,k} = a_j B_{j,k} \neq B_{j,k} a_k = \sum_{r=1}^{n} B_{j,r} A_{r,k} = (BA)_{j,k},$$

where $a_j B_{j,k} \neq B_{j,k} a_k$ follows since $B_{j,k} \neq 0$ and $a_j \neq a_k$. Thus $AB \neq BA$.

**Exercise 5.D.14.**

(a) Give an example of a finite-dimensional complex vector space and an operator $T$ on that vector space such that $T^2$ is diagonalizable but $T$ is not diagonalizable.

(b) Suppose $\mathbf{F} = \mathbf{C}$, $k$ is a positive integer, and $T \in \mathcal{L}(V)$ is invertible. Prove that $T$ is diagonalizable if and only if $T^k$ is diagonalizable.

**Solution.**

(a) Define $T$ as in Exercise 5.D.1 (c). As we showed there, $T$ is not diagonalizable. However, $T^2 = 0$ is certainly diagonalizable.

(b) Suppose that $T$ is diagonalizable, so that there exists a basis $v_1, ..., v_n$ of $V$ with respect to which the matrix of $T$ is diagonal, say

$$\mathcal{M}(T) = \begin{pmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \end{pmatrix}$$

for some $\lambda_1, ..., \lambda_n \in \mathbf{C}$. It follows from 3.43 and direct calculation that

$$\mathcal{M}\left(T^k\right) = [\mathcal{M}(T)]^k = \begin{pmatrix} \lambda_1^k & & 0 \\ & \ddots & \\ 0 & & \lambda_n^k \end{pmatrix}.$$

Thus $T^k$ is diagonalizable.

Now suppose that $T^k$ is diagonalizable and let $p$ be the minimal polynomial of $T^k$. By 5.27 and 5.62 we have $p(z) = (z - \lambda_1) \cdots (z - \lambda_m)$, where $\lambda_1, ..., \lambda_m$ are the distinct eigenvalues of $T^k$. Note that $T^k$ is invertible since $T$ is invertible; it follows that each $\lambda_j$ is non-zero. It can be shown that any non-zero complex number has exactly $k$ distinct $k^{\text{th}}$ roots. For each $j \in \{1, ..., m\}$, let $\mu_{1,j}, ..., \mu_{k,j}$ be the $k$ distinct solutions to $z^k = \lambda_j$. Observe that, for any $a, i \in \{1, ..., k\}$,

$$b \neq j \quad \Rightarrow \quad \lambda_b \neq \lambda_j \quad \Leftrightarrow \quad \mu_{a,b}^k \neq \mu_{i,j}^k \quad \Rightarrow \quad \mu_{a,b} \neq \mu_{i,j}.$$

Thus, if we let $q$ be the polynomial given by



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$$q(z) = \big(z^k - \lambda_1\big) \cdots \big(z^k - \lambda_m\big) = \prod_{j=1}^{m} \prod_{i=1}^{k} (z - \mu_{i,j}),$$

then $q$ is a product of distinct linear factors. Notice that $q(T) = p\big(T^k\big) = 0$; it follows from 5.29 that $q$ is a polynomial multiple of the minimal polynomial of $T$. Thus the minimal polynomial of $T$ is a product of distinct linear factors. 5.62 allows us to conclude that $T$ is diagonalizable.

**Exercise 5.D.15.** Suppose $V$ is a finite-dimensional complex vector space, $T \in \mathcal{L}(V)$, and $p$ is the minimal polynomial of $T$. Prove that the following are equivalent.

(a) $T$ is diagonalizable.

(b) There does not exist $\lambda \in \mathbf{C}$ such that $p$ is a polynomial multiple of $(z - \lambda)^2$.

(c) $p$ and its derivative $p'$ have no zeros in common.

(d) The greatest common divisor of $p$ and $p'$ is the constant polynomial 1.

*The **greatest common divisor** of $p$ and $p'$ is the monic polynomial $q$ of largest degree such that $p$ and $p'$ are both polynomial multiples of $q$. The Euclidean algorithm for polynomials (look it up) can quickly determine the greatest common divisor of two polynomials, without requiring any information about the zeros of the polynomials. Thus the equivalence of (a) and (d) above shows that we can determine whether $T$ is diagonalizable without knowing anything about the zeros of $p$.*

**Solution.** By 5.27(b), $p$ is of the form $p(z) = (z - \lambda_1)^{n_1} \cdots (z - \lambda_m)^{n_m}$, where $\lambda_1, ..., \lambda_m$ is a list of the distinct eigenvalues of $T$ and each $n_k$ is a positive integer.

Note that (b) is equivalent to $n_1 = \cdots = n_m = 1$, which by 5.62 is equivalent to (a). Thus (a) and (b) are equivalent.

Certainly $p$ has $m$ distinct zeros, and observe that $n_1 = \cdots = n_m = 1$ if and only if $\deg p = m$. Exercise 4.8 now shows that (b) and (c) are equivalent.

Suppose the negation of (b) holds, so that $p(z) = (z - \lambda)^2 s(z)$ for some polynomial $s$; the product rule gives us $p'(z) = (z - \lambda)(2s(z) + (z - \lambda)s'(z))$. Thus the greatest common divisor of $p$ and $p'$ must have degree at least 1, so that (d) does not hold. The contrapositive of this and the equivalence of (b) and (c) shows that (d) implies (c).

Let $q$ be the greatest common divisor of $p$ and $p'$ and suppose the negation of (d) holds, so that $\deg q \geq 1$. The fundamental theorem of algebra shows that $q$ has some zero, which must also be a zero of $p$ and $p'$, i.e. the negation of (c) holds. Thus (c) and (d) are equivalent.



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**Exercise 5.D.16.** Suppose that $T \in \mathcal{L}(V)$ is diagonalizable. Let $\lambda_1, ..., \lambda_m$ denote the distinct eigenvalues of $T$. Prove that a subspace $U$ of $V$ is invariant under $T$ if and only if there exist subspaces $U_1, ..., U_m$ of $V$ such that $U_k \subseteq E(\lambda_k, T)$ for each $k$ and $U = U_1 \oplus \cdots \oplus U_m$.

**Solution.** Suppose there exist such subspaces and let $u = u_1 + \cdots + u_m \in U$ be given. Observe that

$$Tu = Tu_1 + \cdots + Tu_m = \lambda_1 u_1 + \cdots + \lambda_m u_m \in U_1 \oplus \cdots \oplus U_m = U.$$

Thus $U$ is invariant under $T$.

Now suppose that $U$ is invariant under $T$. Because $T$ is diagonalizable, (5.55) shows that

$$V = E(\lambda_1, T) \oplus \cdots \oplus E(\lambda_m, T).$$

For each $k \in \{1, ..., m\}$, let $U_k = U \cap E(\lambda_k, T) \subseteq E(\lambda_k, T)$. The directness of the sum $U_1 \oplus \cdots \oplus U_m$ is immediate from the directness of the sum $E(\lambda_1, T) \oplus \cdots \oplus E(\lambda_m, T)$, and since each $U_k \subseteq U$ we have $U_1 \oplus \cdots \oplus U_m \subseteq U$. For any $u \in U$ we have $u = v_1 + \cdots + v_m$, where each $v_k \in E(\lambda_k, T)$. Lemma L.4 shows that each $v_k \in U$ and thus $u$ belongs to $U_1 \oplus \cdots \oplus U_m$, so that $U \subseteq U_1 \oplus \cdots \oplus U_m$. We may conclude that $U = U_1 \oplus \cdots \oplus U_m$.

**Exercise 5.D.17.** Suppose $V$ is finite-dimensional. Prove that $\mathcal{L}(V)$ has a basis consisting of diagonalizable operators.

**Solution.** Let $v_1, ..., v_n$ be a basis of $V$; in what follows, all matrices of operators are with respect to this basis. For $i, j \in \{1, ..., n\}$ such that $i \neq j$, define $T_{i,j} \in \mathcal{L}(V)$ by

$$T_{i,j} v_k = \begin{cases} k v_k & \text{if } k \neq j, \\ k v_k + v_i & \text{if } k = j. \end{cases}$$

Thus the matrix of $T_{i,j}$ has diagonal entries $1, ..., n$, a 1 in the $i^{\text{th}}$ row and $j^{\text{th}}$ column, and 0's elsewhere. Notice that this matrix is either upper- or lower-triangular. It follows that the eigenvalues of $T_{i,j}$ are precisely $1, ..., n$ and hence $T_{i,j}$ is diagonalizable by 5.58. For each $j \in \{1, ..., n\}$, define $S_j \in \mathcal{L}(V)$ by

$$S_j v_k = \begin{cases} v_k & \text{if } k = j, \\ 0 & \text{if } k \neq j. \end{cases}$$

Notice that the matrix of $S_j$ has a 1 in the $j^{\text{th}}$ row and $j^{\text{th}}$ column, and 0's elsewhere. Thus each $S_j$ is diagonalizable.

Suppose we have a linear combination of the list

$$\mathcal{B} := T_{1,2}, ..., T_{1,n}, ..., T_{n,1}, ..., T_{n,n-1}, S_1, ..., S_n$$



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which equals zero. For $i \neq j$, only the operator $T_{i,j}$ has a non-zero entry in the $i^{\text{th}}$ row and $j^{\text{th}}$ column of its matrix. It follows that the coefficient of $T_{i,j}$ in the linear combination must be zero and we are left with a linear combination $a_1 S_1 + \cdots + a_n S_n = 0$, which clearly implies $a_1 = \cdots = a_n = 0$. Thus $\mathcal{B}$ is linearly independent. A straightforward counting argument shows that $\mathcal{B}$ has length $n^2$ and so we may conclude that $\mathcal{B}$ is a basis of $\mathcal{L}(V)$ consisting of diagonalizable operators.

**Exercise 5.D.18.** Suppose that $T \in \mathcal{L}(V)$ is diagonalizable and $U$ is a subspace of $V$ that is invariant under $T$. Prove that the quotient operator $T/U$ is a diagonalizable operator on $V/U$.

*The quotient operator $T/U$ was defined in Exercise 38 in Section 5A.*

**Solution.** By 5.62 and Exercise 5.B.25 (a), the minimal polynomial of $T$ is a product of distinct linear factors and also a polynomial multiple of the minimal polynomial of $T/U$; it follows that the minimal polynomial of $T/U$ is a product of distinct linear factors and 5.62 allows us to conclude that $T/U$ is diagonalizable.

**Exercise 5.D.19.** Prove or give a counterexample: If $T \in \mathcal{L}(V)$ and there exists a subspace $U$ of $V$ that is invariant under $T$ such that $T|_U$ and $T/U$ are both diagonalizable, then $T$ is diagonalizable.

*See Exercise 13 in Section 5C for an analogous statement about upper-triangular matrices.*

**Solution.** This is false. For a counterexample, consider the operator $T \in \mathcal{L}(\mathbf{F}^2)$ given by $Te_1 = 0$ and $Te_2 = e_1$, where $e_1, e_2$ is the standard basis of $\mathbf{F}^2$. It is straightforward to verify that the only eigenvalue of $T$ is 0 and that $E(0, T) = \operatorname{span}(e_1)$. It then follows from 5.55 that $T$ is not diagonalizable. However, if we let $U$ be the $T$-invariant subspace $E(0, T)$, then $T|_U$ and $T/U$ are both operators on 1-dimensional vector spaces and hence are diagonalizable.

**Exercise 5.D.20.** Suppose $V$ is finite-dimensional and $T \in \mathcal{L}(V)$. Prove that $T$ is diagonalizable if and only if the dual operator $T'$ is diagonalizable.

**Solution.** By Exercise 5.B.28, $T$ and $T'$ have the same minimal polynomial. The desired equivalence is then immediate from 5.62.



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**Exercise 5.D.21.** The *Fibonacci sequence* $F_0, F_1, F_2, \ldots$ is defined by

$$F_0 = 0,\ F_1 = 1, \text{ and } F_n = F_{n-2} + F_{n-1} \text{ for } n \geq 2.$$

Define $T \in \mathcal{L}(\mathbf{R}^2)$ by $T(x, y) = (y, x + y)$.

(a) Show that $T^n(0, 1) = (F_n, F_{n+1})$ for each nonnegative integer $n$.

(b) Find the eigenvalues of $T$.

(c) Find a basis of $\mathbf{R}^2$ consisting of eigenvectors of $T$.

(d) Use the solution to (c) to compute $T^n(0, 1)$. Conclude that

$$F_n = \frac{1}{\sqrt{5}}\left[\left(\frac{1+\sqrt{5}}{2}\right)^n - \left(\frac{1-\sqrt{5}}{2}\right)^n\right]$$

for each nonnegative integer $n$.

(e) Use (d) to conclude that if $n$ is a nonnegative integer, then the Fibonacci number $F_n$ is the integer that is closest to

$$\frac{1}{\sqrt{5}}\left(\frac{1+\sqrt{5}}{2}\right)^n.$$

*Each $F_n$ is a nonnegative integer, even though the right side of the formula in (d) does not look like an integer. The number*

$$\frac{1+\sqrt{5}}{2}$$

*is called the **golden ratio**.*

**Solution.**

(a) We will proceed by induction. The base case $n = 0$ is clear, so suppose that $T^n(0, 1) = (F_n, F_{n+1})$ holds for some non-negative integer $n$ and observe that

$$T^{n+1}(0, 1) = T(T^n(0, 1)) = T(F_n, F_{n+1}) = (F_{n+1}, F_n + F_{n+1}) = (F_{n+1}, F_{n+2}).$$

This completes the induction step and the proof.

(b) We are looking for solutions $(x, y) \neq (0, 0)$ and $\lambda \in \mathbf{R}$ of the equation

$$T(x, y) = (y, x + y) = (\lambda x, \lambda y).$$

From the equation $y = \lambda x$ we see that $x = 0$ implies $y = 0$, so we may assume that $x$ is non-zero. Substituting $y = \lambda x$ into the equation $x + y = \lambda y$ and cancelling $x$ gives us the equation $\lambda^2 - \lambda - 1 = 0$, which has two distinct real solutions:

$$\lambda_1 = \frac{1+\sqrt{5}}{2} \quad \text{and} \quad \lambda_2 = \frac{1-\sqrt{5}}{2}.$$



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These are indeed eigenvalues, since

$$T(1,\lambda_1) = (\lambda_1, \lambda_1 + 1) = (\lambda_1, \lambda_1^2) = \lambda_1(1,\lambda_1),$$

where we have used that $\lambda_1$ satisfies the equation $\lambda_1^2 - \lambda_1 - 1 = 0$ for the second equality. Similarly, $T(1,\lambda_2) = \lambda_2(1,\lambda_2)$. Since $\dim \mathbf{R}^2 = 2$, 5.12 allows us to conclude that the eigenvalues of $T$ are precisely $\lambda_1$ and $\lambda_2$.

(c) Since $\lambda_1 \neq \lambda_2$, the eigenvectors $v_1 = (1,\lambda_1)$ and $v_2 = (1,\lambda_2)$ found in part (b) are linearly independent by 5.11 and thus form a basis of $\mathbf{R}^2$.

(d) Observe that

$$v_1 - v_2 = (0, \lambda_1 - \lambda_2) = \left(0, \sqrt{5}\right).$$

Thus $(0,1) = \frac{1}{\sqrt{5}}(v_1 - v_2)$. For any positive integer $n$, it follows that

$$T^n(0,1) = \frac{1}{\sqrt{5}}(T^n v_1 - T^n v_2) = \frac{1}{\sqrt{5}}(\lambda_1^n v_1 - \lambda_2^n v_2) = \frac{1}{\sqrt{5}}\left(\lambda_1^n - \lambda_2^n, \lambda_1^{n+1} - \lambda_2^{n+1}\right).$$

Given the result of part (a), we may conclude that

$$F_n = \frac{1}{\sqrt{5}}(\lambda_1^n - \lambda_2^n) = \frac{1}{\sqrt{5}}\left[\left(\frac{1+\sqrt{5}}{2}\right)^n - \left(\frac{1-\sqrt{5}}{2}\right)^n\right].$$

(e) Certainly $F_0 = 0$ is the closest integer to $\frac{1}{\sqrt{5}}$. For any positive integer $n$, observe that

$$\begin{aligned}
2 < \sqrt{5} < 3 \quad &\Rightarrow \quad -1 < \frac{1-\sqrt{5}}{2} < -\frac{1}{2} \\
&\Rightarrow \quad -1 < \left(\frac{1-\sqrt{5}}{2}\right)^n < 1 \\
&\Rightarrow \quad -\frac{1}{\sqrt{5}} < -\frac{1}{\sqrt{5}}\left(\frac{1-\sqrt{5}}{2}\right)^n < \frac{1}{\sqrt{5}} \\
&\Rightarrow \quad -\frac{1}{2} < -\frac{1}{\sqrt{5}}\left(\frac{1-\sqrt{5}}{2}\right)^n < \frac{1}{2}.
\end{aligned}$$

It then follows from part (d) that

$$\frac{1}{\sqrt{5}}\lambda_1^n - \frac{1}{2} < F_n < \frac{1}{\sqrt{5}}\lambda_1^n + \frac{1}{2},$$

i.e. $F_n$ is an integer belonging to the open interval $\left(\frac{1}{\sqrt{5}}\lambda_1^n - \frac{1}{2}, \frac{1}{\sqrt{5}}\lambda_1^n + \frac{1}{2}\right)$, which has length 1. We may conclude that $F_n$ is the integer closest to $\frac{1}{\sqrt{5}}\lambda_1^n$.



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**Exercise 5.D.22.** Suppose $T \in \mathcal{L}(V)$ and $A$ is an $n$-by-$n$ matrix that is the matrix of $T$ with respect to some basis of $V$. Prove that if

$$|A_{j,j}| > \sum_{\substack{k=1 \\ k \neq j}}^{n} |A_{j,k}|$$

for each $j \in \{1, ..., n\}$, then $T$ is invertible.

*This exercise states that if the diagonal entries of the matrix of $T$ are large compared to the nondiagonal entries, then $T$ is invertible.*

**Solution.** If $T$ has no eigenvalues then certainly 0 is not an eigenvalue of $T$ and thus $T$ is invertible. Otherwise, let $\lambda \in \mathbf{F}$ be an eigenvalue of $T$. 5.67 shows that there exists a $j \in \{1, ..., n\}$ such that

$$|\lambda - A_{j,j}| \leq \sum_{\substack{k=1 \\ k \neq j}}^{n} |A_{j,k}| < |A_{j,j}| \quad \Rightarrow \quad |\lambda| > 0 \quad \Rightarrow \quad \lambda \neq 0.$$

Thus 0 is not an eigenvalue of $T$ and it follows that $T$ is invertible.

**Exercise 5.D.23.** Suppose the definition of the Gershgorin disks is changed so that the radius of the $k^{\text{th}}$ disk is the sum of the absolute values of the entries in column (instead of row) $k$ of $A$, excluding the diagonal entry. Show that the Gershgorin disk theorem (5.67) still holds with this changed definition.

**Solution.** Suppose $T \in \mathcal{L}(V)$ and $v_1, ..., v_n$ is a basis of $V$. Let $A$ be the matrix of $T$ with respect to $v_1, ..., v_n$; it follows from 3.132 that $A^{\text{t}}$ is the matrix of $T'$ with respect to the dual basis $\varphi_1, ..., \varphi_n$ of $V'$. Let $\lambda \in \mathbf{F}$ be an eigenvalue of $T$. Exercise 5.A.15 shows that $\lambda$ is also an eigenvalue of $T'$ and thus, by 5.67, there exists a $k \in \{1, ..., n\}$ such that

$$|\lambda - A_{k,k}| \leq \sum_{\substack{j=1 \\ j \neq k}}^{n} \left|A^{\text{t}}_{k,j}\right| = \sum_{\substack{j=1 \\ j \neq k}}^{n} |A_{j,k}|.$$

Thus $\lambda$ is contained in the $k^{\text{th}}$ Gershgorin "column-disk" of $T$ with respect to $v_1, ..., v_n$.



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## 5.E. Commuting Operators

**Exercise 5.E.1.** Give an example of two commuting operators $S, T$ on $\mathbf{F}^4$ such that there is a subspace of $\mathbf{F}^4$ that is invariant under $S$ but not under $T$ and there is a subspace of $\mathbf{F}^4$ that is invariant under $T$ but not under $S$.

**Solution.** Let $S, T \in \mathcal{L}(\mathbf{F}^4)$ be given by

$$S(x_1, x_2, x_3, x_4) = (x_2, x_1, 0, 0) \quad \text{and} \quad T(x_1, x_2, x_3, x_4) = (0, 0, x_4, x_3).$$

Notice that $S$ and $T$ commute, since for any $x \in \mathbf{F}^4$ we have $STx = TSx = 0$. Notice further that, for any $\lambda \in \mathbf{F}$,

$$S(\lambda, 0, 0, 0) = (0, \lambda, 0, 0), \quad T(\lambda, 0, 0, 0) = 0, \quad S(0, 0, \lambda, 0) = 0, \quad T(0, 0, \lambda, 0) = (0, 0, 0, \lambda).$$

It follows that $\operatorname{span}((1, 0, 0, 0))$ is invariant under $T$ but not under $S$, and $\operatorname{span}((0, 0, 1, 0))$ is invariant under $S$ but not under $T$.

**Exercise 5.E.2.** Suppose $\mathcal{E}$ is a subset of $\mathcal{L}(V)$ and every element of $\mathcal{E}$ is diagonalizable. Prove that there exists a basis of $V$ with respect to which every element of $\mathcal{E}$ has a diagonal matrix if and only if every pair of elements of $\mathcal{E}$ commutes.

> *This exercise extends 5.76, which considers the case in which $\mathcal{E}$ contains only two elements. For this exercise, $\mathcal{E}$ may contain any number of elements, and $\mathcal{E}$ may even be an infinite set.*

**Solution.** Suppose there exists such a basis $v_1, ..., v_n$ and let $S, T \in \mathcal{E}$ be given. The matrices of $S$ and $T$ with respect to $v_1, ..., v_n$ are diagonal and hence commute. It follows from 5.74 that $S$ and $T$ commute.

Suppose that every pair of elements of $\mathcal{E}$ commutes and suppose that $\dim V = n$. Because $\dim \mathcal{L}(V) = n^2$, there must exist a subset $\mathcal{F} \subseteq \mathcal{E}$ of cardinality at most $n^2$ such that every operator in $\mathcal{E}$ is a linear combination of operators in $\mathcal{F}$. Suppose that $\mathcal{F} = \{T_1, ..., T_m\}$ for some $m \leq n^2$. Since $T_1$ is diagonalizable, 5.55 shows that

$$V = \bigoplus_{\lambda_1 \in \mathbf{F}} E(\lambda_1, T_1);$$

note that, since $T_1$ has at most $n$ distinct eigenvalues, all but finitely many of the summands $E(\lambda_1, T_1)$ are equal to $\{0\}$, so that this direct sum is finite. Let $\lambda_1 \in \mathbf{F}$ be given. Because $T_1$ and $T_2$ commute, 5.75 shows that $E(\lambda_1, T_1)$ is invariant under $T_2$. It then follows from 5.65 that $T_2|_{E(\lambda_1, T_1)}$ is diagonalizable and thus, by 5.55,

$$E(\lambda_1, T_1) = \bigoplus_{\lambda_2 \in \mathbf{F}} E\big(\lambda_2, T_2|_{E(\lambda_1, T_1)}\big);$$

again, this direct sum is finite since $T_2$ has at most $n$ distinct eigenvalues. Notice that



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$$E\big(\lambda_2, T_2|_{E(\lambda_1,T_1)}\big) = \{v \in E(\lambda_1, T_1) : T_2 v = \lambda_2 v\}$$
$$= \{v \in V : T_1 v = \lambda_1 v \text{ and } T_2 v = \lambda_2 v\} = E(\lambda_1, T_1) \cap E(\lambda_2, T_2).$$

Combining this with $V = \bigoplus_{\lambda_1 \in \mathbf{F}} E(\lambda_1, T_1)$ and $E(\lambda_1, T_1) = \bigoplus_{\lambda_2 \in \mathbf{F}} E\big(\lambda_2, T_2|_{E(\lambda_1,T_1)}\big)$, we see that

$$V = \bigoplus_{\lambda_1, \lambda_2 \in \mathbf{F}} (E(\lambda_1, T_1) \cap E(\lambda_2, T_2)).$$

If we continue this process, we find that

$$V = \bigoplus_{\lambda_1, \ldots, \lambda_m \in \mathbf{F}} (E(\lambda_1, T_1) \cap \cdots \cap E(\lambda_m, T_m)),$$

where this direct sum is finite because each $T_k$ has at most $n$ distinct eigenvalues. If we take a basis for each non-zero summand $E(\lambda_1, T_1) \cap \cdots \cap E(\lambda_m, T_m)$ and combine these bases, we obtain a basis $v_1, \ldots, v_n$ of $V$ such that each basis vector is an eigenvector of each $T_k$. Thus the matrix of each $T_k$ with respect to $v_1, \ldots, v_n$ is diagonal. Because each $T \in \mathcal{E}$ is a linear combination of $T_1, \ldots, T_m$, and a linear combination of diagonal matrices is a diagonal matrix, we see that the matrix of each $T \in \mathcal{E}$ is diagonal with respect to $v_1, .., v_n$.

**Exercise 5.E.3.** Suppose $S, T \in \mathcal{L}(V)$ are such that $ST = TS$. Suppose $p \in \mathcal{P}(\mathbf{F})$.

(a) Prove that $\operatorname{null} p(S)$ is invariant under $T$.

(b) Prove that $\operatorname{range} p(S)$ is invariant under $T$.

*See 5.18 for the special case $S = T$.*

**Solution.** Suppose $p = \sum_{k=0}^{m} c_k z^k$.

(a) Let $v \in \operatorname{null} p(S)$ be given and observe that

$$p(S)(Tv) = \sum_{k=0}^{m} c_k S^k (Tv) = T\left(\sum_{k=0}^{m} c_k S^k v\right) = T(p(S)v) = T(0) = 0,$$

where we have used that $T$ is linear and that $T$ commutes with $S$ for the second equality. Thus $\operatorname{null} p(S)$ is invariant under $T$.

(b) Let $v \in \operatorname{range} p(S)$ be given, so that $v = p(S)w$ for some $w \in V$, and observe that

$$Tv = T\left(\sum_{k=0}^{m} c_k S^k w\right) = \sum_{k=0}^{m} c_k S^k (Tw) = p(S)(Tw) \in \operatorname{range} p(S),$$

where we have used that $T$ is linear and that $T$ commutes with $S$ for the second equality. Thus $\operatorname{range} p(S)$ is invariant under $T$.

**Exercise 5.E.4.** Prove or give a counterexample: If $A$ is a diagonal matrix and $B$ is an upper-triangular matrix of the same size as $A$, then $A$ and $B$ commute.



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**Solution.** This is false:

$$\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \neq \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}.$$

**Exercise 5.E.5.** Prove that a pair of operators on a finite-dimensional vector space commute if and only if their dual operators commute.

*See 3.118 for the definition of the dual of an operator.*

**Solution.** Suppose $V$ is a finite-dimensional vector space and $S, T \in \mathcal{L}(V)$. Let $v_1, ..., v_n$ be a basis of $V$, let $\varphi_1, ..., \varphi_n$ be the corresponding dual basis of $V'$, and let

$$A = \mathcal{M}(S, (v_1, ..., v_n)) \quad \text{and} \quad B = \mathcal{M}(T, (v_1, ..., v_n)).$$

It follows from 3.132 that the matrices of $S'$ and $T'$ with respect to $\varphi_1, ..., \varphi_n$ are $A^t$ and $B^t$. To show that $S$ and $T$ commute if and only if $S'$ and $T'$ commute, it will suffice, by 5.74, to show that $A$ and $B$ commute if and only if $A^t$ and $B^t$ commute. Indeed, using Exercise 3.C.15,

$$AB = BA \quad \Leftrightarrow \quad (AB)^t = (BA)^t \quad \Leftrightarrow \quad B^t A^t = A^t B^t.$$

**Exercise 5.E.6.** Suppose $V$ is a finite-dimensional complex vector space and $S, T \in \mathcal{L}(V)$ commute. Prove that there exist $\alpha, \lambda \in \mathbf{C}$ such that

$$\operatorname{range}(S - \alpha I) + \operatorname{range}(T - \lambda I) \neq V.$$

**Solution.** By 5.80 there is a basis $v_1, ..., v_n$ such that the matrices $\mathcal{M}(S)$ and $\mathcal{M}(T)$ are both upper-triangular, say

$$\mathcal{M}(S) = \begin{pmatrix} * & \cdots & * \\ \vdots & \ddots & \vdots \\ 0 & \cdots & \alpha \end{pmatrix} \quad \text{and} \quad \mathcal{M}(T) = \begin{pmatrix} * & \cdots & * \\ \vdots & \ddots & \vdots \\ 0 & \cdots & \lambda \end{pmatrix}$$

for some $\alpha, \lambda \in \mathbf{C}$. It follows that

$$\mathcal{M}(S - \alpha I) = \begin{pmatrix} * & \cdots & * \\ \vdots & \ddots & \vdots \\ 0 & \cdots & 0 \end{pmatrix} \quad \text{and} \quad \mathcal{M}(T - \lambda I) = \begin{pmatrix} * & \cdots & * \\ \vdots & \ddots & \vdots \\ 0 & \cdots & 0 \end{pmatrix}$$

and hence that $\operatorname{range}(S - \alpha I)$ and $\operatorname{range}(T - \lambda I)$ are both contained in $\operatorname{span}(v_1, ..., v_{n-1})$. Thus $\operatorname{range}(S - \alpha I) + \operatorname{range}(T - \lambda I) \subseteq \operatorname{span}(v_1, ..., v_{n-1}) \neq V$.

**Exercise 5.E.7.** Suppose $V$ is a complex vector space, $S \in \mathcal{L}(V)$ is diagonalizable, and $T \in \mathcal{L}(V)$ commutes with $S$. Prove that there is a basis of $V$ such that $S$ has a diagonal matrix with respect to this basis and $T$ has an upper-triangular matrix with respect to this basis.



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**Solution.** We will proceed by induction on $\dim V$. Certainly the result is true for $\dim V = 1$, since all 1-by-1 matrices are diagonal. Let $V$ be a complex vector space of dimension $n > 1$, suppose the result holds for all complex vector spaces of smaller dimension, let $S \in \mathcal{L}(V)$ be diagonalizable, and suppose $T \in \mathcal{L}(V)$ commutes with $S$. By 5.19 there exists an eigenvalue $\lambda \in \mathbf{C}$ of $S$ and it then follows from Exercise 5.D.5 that $V = U \oplus W$, where

$$U = \operatorname{null}(S - \lambda I) \quad \text{and} \quad W = \operatorname{range}(S - \lambda I).$$

If $S = \lambda I$ then the matrix of $S$ with respect to any basis of $V$ is diagonal and thus the desired basis of $V$ is given by 5.47. If $S \neq \lambda I$ then $1 \leq \dim U < n$ and $1 \leq \dim W < n$. Furthermore, Exercise 5.E.3 shows that $U$ and $W$ are invariant under both $S$ and $T$. Because $S$ and $T$ commute, their restrictions to any subspace of $V$ will also commute and thus we can apply our induction hypothesis to both $U$ and $W$ to obtain a basis $v_1, ..., v_m$ of $U$ and a basis $v_{m+1}, ..., v_n$ of $W$ such that each $v_k$ is an eigenvector of $S$ and such that

$$Tv_k \in \operatorname{span}(v_1, ..., v_k) \text{ for } k \in \{1, ..., m\},$$

$$\text{and} \quad Tv_k \in \operatorname{span}(v_{m+1}, ..., v_k) \subseteq \operatorname{span}(v_1, ..., v_k) \text{ for } k \in \{m+1, ..., n\}.$$

Thus $v_1, ..., v_n$ is a basis of $V$ such that $\mathcal{M}(S, (v_1, ..., v_n))$ is diagonal and such that $\mathcal{M}(T, (v_1, ..., v_n))$ is upper-triangular. This completes the induction step and the proof.

**Exercise 5.E.8.** Suppose $m = 3$ in Example 5.72 and $D_x, D_y$ are the commuting partial differentiation operators on $\mathcal{P}_3(\mathbf{R}^2)$ from that example. Find a basis of $\mathcal{P}_3(\mathbf{R}^2)$ with respect to which $D_x$ and $D_y$ each have an upper-triangular matrix.

**Solution.** Consider the list

$$B := 1, x, y, x^2, xy, y^2, x^3, x^2y, xy^2, y^3.$$

From the definition of $\mathcal{P}_3(\mathbf{R}^2)$, it is clear that $B$ spans $\mathcal{P}_3(\mathbf{R}^2)$. Suppose we have a linear combination

$$a_{0,0} + a_{1,0}x + a_{0,1}y + a_{2,0}x^2 + a_{1,1}xy + a_{0,2}y^2 + a_{3,0}x^3 + a_{2,1}x^2y + a_{1,2}xy^2 + a_{0,3}y^3 = 0.$$

Taking $y = 0$ shows that $a_{0,0} = a_{1,0} = a_{2,0} = a_{3,0} = 0$, given the linear independence of $1, x, x^2, x^3$ in $\mathcal{P}(\mathbf{R})$, and similarly taking $x = 0$ gives us $a_{0,1} = a_{0,2} = a_{0,3} = 0$. Thus we are left with the linear combination

$$a_{1,1}xy + a_{2,1}x^2y + a_{1,2}xy^2 = 0.$$

Taking $(x, y) \in \{(1, 1), (2, 1), (1, 2)\}$ gives us the system of linear equations

$$\begin{pmatrix} 1 & 1 & 1 \\ 2 & 4 & 2 \\ 2 & 2 & 4 \end{pmatrix} \begin{pmatrix} a_{1,1} \\ a_{2,1} \\ a_{1,2} \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix},$$



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which has the unique solution $a_{1,1} = a_{2,1} = a_{1,2} = 0$. Thus $B$ is linearly independent and hence forms a basis of $\mathcal{P}_3(\mathbf{R}^2)$. Now observe that applying $D_x$ to each vector in $B$ gives us the list

$$0, 1, 0, 2x, y, 0, 3x^2, 2xy, y^2, 0.$$

It follows that the matrix of $D_x$ with respect to $B$ is upper-triangular:

$$\begin{pmatrix} 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 2 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 3 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 2 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

Similarly, we find that the matrix of $D_y$ with respect to $B$ is upper-triangular.

**Exercise 5.E.9.** Suppose $V$ is a finite-dimensional nonzero complex vector space. Suppose that $\mathcal{E} \subseteq \mathcal{L}(V)$ is such that $S$ and $T$ commute for all $S, T \in \mathcal{E}$.

(a) Prove that there is a vector in $V$ that is an eigenvector for every element of $\mathcal{E}$.

(b) Prove that there is a basis of $V$ with respect to which every element of $\mathcal{E}$ has an upper-triangular matrix.

*This exercise extends 5.78 and 5.80, which consider the case in which $\mathcal{E}$ contains only two elements. For this exercise, $\mathcal{E}$ may contain any number of elements, and $\mathcal{E}$ may even be an infinite set.*

**Solution.**

(a) Suppose that $\dim V = n$. Because $\dim \mathcal{L}(V) = n^2$, there must exist a subset $\mathcal{F} \subseteq \mathcal{E}$ of cardinality at most $n^2$ such that every operator in $\mathcal{E}$ is a linear combination of operators in $\mathcal{F}$. Suppose that $\mathcal{F} = \{T_1, ..., T_m\}$ for some $m \leq n^2$. By 5.19 $T_1$ has an eigenvalue $\lambda_1$, and because $T_1$ and $T_2$ commute, 5.75 shows that $E(\lambda_1, T_1)$ is invariant under $T_2$. Another application of 5.19 shows that $T_2|_{E(\lambda_1, T_1)}$ has an eigenvector, which must also be an eigenvector of $T_1$. Thus $E(\lambda_1, T_1) \cap E(\lambda_2, T_2) \neq \{0\}$. Since $T_3$ commutes with both $T_1$ and $T_2$, 5.75 shows that $E(\lambda_1, T_1)$ and $E(\lambda_2, T_2)$ are both invariant under $T_3$ and thus, by Exercise 5.A.3, the intersection $E(\lambda_1, T_1) \cap E(\lambda_2, T_2)$ is also invariant under $T_3$. It then follows from 5.19 that $T_3$ restricted to $E(\lambda_1, T_1) \cap E(\lambda_2, T_2)$ has an eigenvector, which must also be an eigenvector of $T_1$ and $T_2$. By continuing in this manner, we obtain a $v \in V$ that is an eigenvector of each $T_k$, say $T_k v = \lambda_k v$ for some $\lambda_k \in \mathbf{C}$.



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Let $T \in \mathcal{E}$ be given. As noted above, $T$ must be a linear combination of operators in $\mathcal{F}$, say $T = \sum_{k=1}^{m} c_k T_k$. It follows that

$$Tv = \left(\sum_{k=1}^{m} c_k T_k\right) v = \sum_{k=1}^{m} c_k \lambda_k v = \left(\sum_{k=1}^{m} c_k \lambda_k\right) v.$$

Thus $v$ is an eigenvector of $T$.

(b) Let us first consider the special case where $\mathcal{E}$ is finite. Our proof here is a generalization of the proof of 5.80. For a positive integer $n$, let $P(n)$ be the statement that if $V$ is an $n$-dimensional complex vector space and $\{T_1, ..., T_m\}$ is a collection of pairwise commuting operators on $V$ for some $m \geq 2$, then there is a basis of $V$ with respect to which each $T_k$ has an upper-triangular matrix.

The truth of $P(1)$ is clear. For some $n > 1$, suppose that $P(n-1)$ holds, let $V$ be an $n$-dimensional complex vector space, and let $\{T_1, ..., T_m\}$ be a collection of pairwise commuting operators on $V$ for some $m \geq 2$. By part (a) there exists a $v_1 \in V$ which is an eigenvector of each $T_k$, so that $T_k v_1 \in \operatorname{span}(v_1)$. Using 2.33, let $W$ be such that $V = \operatorname{span}(v_1) \oplus W$ and define $P \in \mathcal{L}(V, W)$ by $P(av_1 + w) = w$. For each $k \in \{1, ..., m\}$, define $\hat{T}_k \in \mathcal{L}(W)$ by $\hat{T}_k w = P(T_k w)$. Because each pair of operators in $\{T_1, ..., T_m\}$ commutes, the proof of 5.80 shows that each pair of operators in $\{\hat{T}_1, ..., \hat{T}_m\}$ also commutes. We can now apply our induction hypothesis to obtain a basis $v_2, ..., v_n$ of $W$ with respect to which the matrix of each $\hat{T}_k$ is upper-triangular. The list $v_1, ..., v_n$ is a basis of $V$. For each $j \in \{2, ..., n\}$ and each $k \in \{1, ..., m\}$, there exists $a_{j,k} \in \mathbf{C}$ such that

$$T_k v_j = a_{j,k} v_1 + \hat{T}_k v_j.$$

Because $\hat{T}_k v_j \in \operatorname{span}(v_2, ..., v_j)$, this equation implies that $T_k v_j \in \operatorname{span}(v_1, ..., v_j)$. Thus the matrix of each $T_k$ with respect to the basis $v_1, ..., v_n$ is upper triangular. This completes the induction step and the proof.

Now let us consider the general case where $\mathcal{E}$ may be infinite. Because $\dim \mathcal{L}(V) = n^2$, there must exist a subset $\mathcal{F} \subseteq \mathcal{E}$ of cardinality at most $n^2$ such that every operator in $\mathcal{E}$ is a linear combination of operators in $\mathcal{F}$. Suppose that $\mathcal{F} = \{T_1, ..., T_m\}$ for some $m \leq n^2$. The special case we just proved implies that there is a basis $v_1, ..., v_n$ of $V$ with respect to which each $T_k$ has an upper-triangular matrix. Because a linear combination of upper-triangular matrices is again an upper-triangular matrix and each $T \in \mathcal{E}$ is a linear combination of the operators $\{T_1, ..., T_m\}$, we see that the matrix of each $T \in \mathcal{E}$ with respect to $v_1, ..., v_n$ is upper-triangular.



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**Exercise 5.E.10.** Give an example of two commuting operators $S, T$ on a finite-dimensional real vector space such that $S + T$ has an eigenvalue that does not equal an eigenvalue of $S$ plus an eigenvalue of $T$ and $ST$ has an eigenvalue that does not equal an eigenvalue of $S$ times an eigenvalue of $T$.

*This exercise shows that 5.81 does not hold on real vector spaces.*

**Solution.** Let $S, T \in \mathcal{L}(\mathbf{R}^2)$ be given by $S(x, y) = (-y, x)$ and $T = -S$, i.e. $S$ is a counterclockwise rotation about the origin by 90° and $T$ is a clockwise rotation about the origin by 90°. It follows that $S + T = 0$ and $ST = I$, so that 0 is an eigenvalue of $S + T$ and 1 is an eigenvalue of $ST$. However, we may not express either of these eigenvalues as a sum or product of eigenvalues of $S$ and $T$, because $S$ and $T$ do not have eigenvalues (see 5.9(a)).



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# Chapter 6. Inner Product Spaces

## 6.A. Inner Products and Norms

**Exercise 6.A.1.** Prove or give a counterexample: If $v_1, ..., v_m \in V$, then

$$\sum_{j=1}^{m}\sum_{k=1}^{m}\langle v_j, v_k\rangle \geq 0.$$

**Solution.** Suppose we have a sequence of vectors $v_1, v_2, v_3, ...$ in $V$. We will use induction on $m$ to prove that

$$\|v_1 + \cdots + v_m\|^2 = \sum_{j=1}^{m}\sum_{k=1}^{m}\langle v_j, v_k\rangle$$

for each positive integer $m$. The base case is clear, so suppose that the result holds for some positive integer $m$ and observe that

$$\begin{aligned}
\|v_1 + \cdots + v_m + v_{m+1}\|^2 &= \langle v_1 + \cdots + v_m + v_{m+1}, v_1 + \cdots + v_m + v_{m+1}\rangle \\
&= \langle v_1 + \cdots + v_m, v_1 + \cdots + v_m\rangle + \langle v_1 + \cdots + v_m, v_{m+1}\rangle \\
&\quad + \langle v_{m+1}, v_1 + \cdots + v_m\rangle + \langle v_{m+1}, v_{m+1}\rangle \\
&= \sum_{j=1}^{m}\sum_{k=1}^{m}\langle v_j, v_k\rangle + \sum_{j=1}^{m}\langle v_j, v_{m+1}\rangle + \sum_{k=1}^{m}\langle v_{m+1}, v_k\rangle + \langle v_{m+1}, v_{m+1}\rangle \\
&= \sum_{j=1}^{m+1}\sum_{k=1}^{m+1}\langle v_j, v_k\rangle.
\end{aligned}$$

This completes the induction step. The desired inequality is now immediate:

$$\sum_{j=1}^{m}\sum_{k=1}^{m}\langle v_j, v_k\rangle = \|v_1 + \cdots + v_m\|^2 \geq 0.$$

**Exercise 6.A.2.** Suppose $S \in \mathcal{L}(V)$. Define $\langle \cdot, \cdot \rangle_1$ by

$$\langle u, v\rangle_1 = \langle Su, Sv\rangle$$

for all $u, v \in V$. Show that $\langle \cdot, \cdot \rangle_1$ is an inner product on $V$ if and only if $S$ is injective.

**Solution.** If $S$ is not injective then there exists some non-zero $v \in V$ such that $Sv = 0$. It follows that



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$$\langle v, v\rangle_1 = \langle Sv, Sv\rangle = \langle 0, 0\rangle = 0.$$

Thus $\langle \cdot, \cdot \rangle_1$ fails to have the definiteness property required by 6.2 and hence is not an inner product on $V$.

Now suppose that $S$ is injective. We verify each property required by 6.2.

**Positivity.** We have $\langle v, v\rangle_1 = \langle Sv, Sv\rangle \geq 0$ for all $v \in V$ by the positivity of $\langle \cdot, \cdot \rangle$.

**Definiteness.** We have $\langle v, v\rangle_1 = \langle Sv, Sv\rangle = 0$ if and only if $Sv = 0$ by the definiteness of $\langle \cdot, \cdot \rangle$, and $Sv = 0$ if and only if $v = 0$ by the injectivity of $S$. Thus $\langle v, v\rangle_1 = 0$ if and only if $v = 0$.

**Additivity in the first slot.** Let $u, v, w \in V$ be given and observe that

$$\langle u + v, w\rangle_1 = \langle S(u + v), Sw\rangle = \langle Su + Sv, Sw\rangle = \langle Su, Sw\rangle + \langle Sv, Sw\rangle = \langle u, w\rangle_1 + \langle v, w\rangle_1,$$

where we have used the linearity of $S$ and the additivity in the first slot of $\langle \cdot, \cdot \rangle$.

**Homogeneity in the first slot.** Let $\lambda \in \mathbf{F}$ and $u, v \in V$ be given and observe that

$$\langle \lambda u, v\rangle_1 = \langle S(\lambda u), Sv\rangle = \langle \lambda Su, Sv\rangle = \lambda \langle Su, Sv\rangle = \lambda \langle u, v\rangle_1,$$

where we have used the linearity of $S$ and the homogeneity in the first slot of $\langle \cdot, \cdot \rangle$.

**Conjugate symmetry.** Let $u, v \in V$ be given and observe that

$$\overline{\langle v, u\rangle_1} = \overline{\langle Sv, Su\rangle} = \langle Su, Sv\rangle = \langle u, v\rangle_1,$$

where we have used the conjugate symmetry of $\langle \cdot, \cdot \rangle$.

**Exercise 6.A.3.**

(a) Show that the function taking an ordered pair $((x_1, x_2), (y_1, y_2))$ of elements of $\mathbf{R}^2$ to $|x_1 y_1| + |x_2 y_2|$ is not an inner product on $\mathbf{R}^2$.

(b) Show that the function taking an ordered pair $((x_1, x_2, x_3), (y_1, y_2, y_3))$ of elements of $\mathbf{R}^3$ to $x_1 y_1 + x_3 y_3$ is not an inner product on $\mathbf{R}^3$.

**Solution.**

(a) Let $f$ be the function in question, i.e. $f : \mathbf{R}^2 \times \mathbf{R}^2 \to \mathbf{R}$ is given by

$$f((x_1, x_2), (y_1, y_2)) = |x_1 y_1| + |x_2 y_2|,$$

and notice that

$$f((-1, 0), (1, 0)) = 1 \neq -1 = -f((1, 0), (1, 0)).$$

Thus $f$ is not homogeneous in the first slot and hence is not an inner product on $\mathbf{R}^2$.

(b) Let $f$ be the function in question, i.e. $f : \mathbf{R}^3 \times \mathbf{R}^3 \to \mathbf{R}$ is given by

$$f((x_1, x_2, x_3), (y_1, y_2, y_3)) = x_1 y_1 + x_3 y_3,$$

and notice that



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$$f((0,1,0),(0,1,0)) = 0.$$

Thus $f$ fails to have the definiteness property required by 6.2 and hence is not an inner product on $\mathbf{R}^3$.

**Exercise 6.A.4.** Suppose $T \in \mathcal{L}(V)$ is such that $\|Tv\| \leq \|v\|$ for every $v \in V$. Prove that $T - \sqrt{2}I$ is injective.

**Solution.** We will prove the contrapositive statement. If $T - \sqrt{2}I$ is not injective then there is some non-zero $v \in V$ such that $Tv = \sqrt{2}v$. Because $v \neq 0$ we have $\|v\| \neq 0$ and thus

$$\|Tv\| = \left\|\sqrt{2}v\right\| = \sqrt{2}\|v\| > \|v\|.$$

**Exercise 6.A.5.** Suppose $V$ is a real inner product space.

(a) Show that $\langle u+v, u-v\rangle = \|u\|^2 - \|v\|^2$ for every $u, v \in V$.

(b) Show that if $u, v \in V$ have the same norm, then $u+v$ is orthogonal to $u-v$.

(c) Use (b) to show that the diagonals of a rhombus are perpendicular to each other.

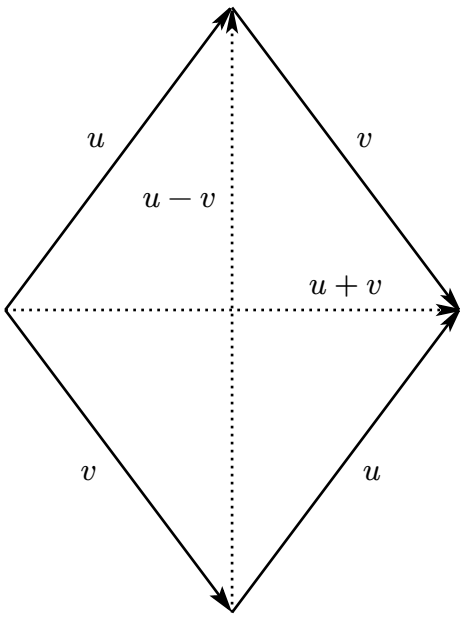
**Solution.**

(a) For any $u, v \in V$ we have

$$\langle u+v, u-v\rangle = \langle u,u\rangle - \langle u,v\rangle + \langle v,u\rangle - \langle v,v\rangle = \|u\|^2 - \|v\|^2.$$

(b) This is immediate from part (a).

(c) In plane geometry, a rhombus is a quadrilateral whose four sides have the same length. Letting $u$ and $v$ denote the two non-parallel sides, the diagonals are given by $u+v$ and $u-v$. Since $\|u\| = \|v\|$, part (b) shows that $u+v$ and $u-v$ are perpendicular to each other.






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**Exercise 6.A.6.** Suppose $u, v \in V$. Prove that $\langle u, v\rangle = 0 \Leftrightarrow \|u\| \leq \|u + av\|$ for all $a \in \mathbf{F}$.

**Solution.** Suppose that $\langle u, v\rangle = 0$ and let $a \in \mathbf{F}$ be given. Observe that

$$\langle u, av\rangle = \overline{a}\langle u, v\rangle = 0.$$

Thus $u$ and $av$ are orthogonal. It follows from the Pythagorean Theorem (6.12) that

$$\|u + av\|^2 = \|u\|^2 + \|av\|^2 \geq \|u\|^2.$$

Taking square roots gives the desired inequality.

Now suppose that $\langle u, v\rangle \neq 0$. By 6.11 it must be the case that $v \neq 0$. Thus we can define $c$ and $w$ as in 6.13, so that $\langle w, v\rangle = 0$ and $u = cv + w$. Since $w$ and $v$ are orthogonal, the Pythagorean Theorem (6.12) shows that

$$\|u\|^2 = \|cv + w\|^2 = |c|^2\|v\|^2 + \|w\|^2 > \|w\|^2 = \|u - cv\|^2;$$

the inequality is strict here because $c \neq 0$ and $v \neq 0$. Taking square roots gives us $\|u\| > \|u - cv\|$ and thus a choice of $a = -c$ gives us the desired result.

**Exercise 6.A.7.** Suppose $u, v \in V$. Prove that $\|au + bv\| = \|bu + av\|$ for all $a, b \in \mathbf{R}$ if and only if $\|u\| = \|v\|$.

**Solution.** For any $a, b \in \mathbf{R}$, note that

$$\|au + bv\| = \|bu + av\| \quad \Leftrightarrow \quad \|au + bv\|^2 = \|bu + av\|^2.$$

Note further that

$$\|au + bv\|^2 = a^2\|u\|^2 + 2ab\langle u, v\rangle + b^2\|v\|^2$$

$$\text{and} \quad \|bu + av\|^2 = b^2\|u\|^2 + 2ab\langle u, v\rangle + a^2\|v\|^2.$$

Thus $\|au + bv\|^2 = \|bu + av\|^2$ holds if and only if

$$(a^2 - b^2)\left(\|u\|^2 - \|v\|^2\right) = 0.$$

Given this, it will suffice to show that $(a^2 - b^2)\left(\|u\|^2 - \|v\|^2\right) = 0$ for all $a, b \in \mathbf{R}$ if and only if $\|u\| = \|v\|$. The reverse implication is clear; for the forward implication, simply take $a = 1$ and $b = 0$.

**Exercise 6.A.8.** Suppose $a, b, c, x, y \in \mathbf{R}$ and $a^2 + b^2 + c^2 + x^2 + y^2 \leq 1$. Prove that $a + b + c + 4x + 9y \leq 10$.

**Solution.** Let $u, v \in \mathbf{R}^5$ be given by $u = (a, b, c, x, y)$ and $v = (1, 1, 1, 4, 9)$. The Cauchy-Schwarz inequality (6.14) shows that



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$$a+b+c+4x+9y \le |a+b+c+4x+9y| = |\langle u, v\rangle|$$
$$\le \|u\|\|v\| = \sqrt{a^2+b^2+c^2+x^2+y^2}\sqrt{1+1+1+16+81} \le 10.$$

**Exercise 6.A.9.** Suppose $u, v \in V$ and $\|u\| = \|v\| = 1$ and $\langle u, v\rangle = 1$. Prove that $u = v$.

**Solution.** Observe that

$$\langle u-v, u-v\rangle = \langle u,u\rangle - \langle u,v\rangle - \langle v,u\rangle + \langle v,v\rangle = \|u\|^2 - 2\operatorname{Re}\langle u,v\rangle + \|v\|^2 = 0.$$

It follows from definiteness that $u - v = 0$.

**Exercise 6.A.10.** Suppose $u, v \in V$ and $\|u\| \le 1$ and $\|v\| \le 1$. Prove that

$$\sqrt{1-\|u\|^2}\sqrt{1-\|v\|^2} \le 1 - |\langle u, v\rangle|.$$

**Solution.** Observe that

$$\begin{aligned}
0 \le (\|u\| - \|v\|)^2 &\Leftrightarrow 0 \le \|u\|^2 - 2\|u\|\|v\| + \|v\|^2 \\
&\Leftrightarrow -\|u\|^2 - \|v\|^2 \le -2\|u\|\|v\| \\
&\Leftrightarrow 1 - \|u\|^2 - \|v\|^2 + \|u\|^2\|v\|^2 \le 1 - 2\|u\|\|v\| + \|u\|^2\|v\|^2 \\
&\Leftrightarrow \left(1-\|u\|^2\right)\left(1-\|v\|^2\right) \le (1 - \|u\|\|v\|)^2.
\end{aligned}$$

Since $\|u\| \le 1$ and $\|v\| \le 1$, the quantities $1-\|u\|^2, 1-\|v\|^2$, and $1-\|u\|\|v\|$ are non-negative. Thus we may take square roots to obtain the inequality

$$\sqrt{1-\|u\|^2}\sqrt{1-\|v\|^2} \le 1 - \|u\|\|v\|.$$

The Cauchy-Schwarz inequality (6.14) shows that $1 - \|u\|\|v\| \le 1 - |\langle u, v\rangle|$ and thus

$$\sqrt{1-\|u\|^2}\sqrt{1-\|v\|^2} \le 1 - |\langle u, v\rangle|.$$

**Exercise 6.A.11.** Find vectors $u, v \in \mathbf{R}^2$ such that $u$ is a scalar multiple of $(1,3)$, $v$ is orthogonal to $(1,3)$, and $(1,2) = u + v$.

**Solution.** Let $x = (1,2), y = (1,3)$, and let

$$c = \frac{\langle x, y\rangle}{\|y\|^2} = \frac{7}{10}, \quad u = cy, \quad \text{and} \quad v = x - cy.$$

Then $u$ is a scalar multiple of $y$ and, as 6.13 shows, $\langle v, y\rangle = 0$ and $x = u + v$.



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**Exercise 6.A.12.** Suppose $a, b, c, d$ are positive numbers.

(a) Prove that $(a+b+c+d)\left(\frac{1}{a}+\frac{1}{b}+\frac{1}{c}+\frac{1}{d}\right) \geq 16$.

(b) For which positive numbers $a, b, c, d$ is the inequality above an equality?

**Solution.**

(a) If we let

$$u = \left(\sqrt{a}, \sqrt{b}, \sqrt{c}, \sqrt{d}\right) \quad \text{and} \quad v = \left(\frac{1}{\sqrt{a}}, \frac{1}{\sqrt{b}}, \frac{1}{\sqrt{c}}, \frac{1}{\sqrt{d}}\right),$$

then

$$\langle u, v\rangle = 4, \quad \|u\| = \sqrt{a+b+c+d}, \quad \text{and} \quad \|v\| = \sqrt{\frac{1}{a}+\frac{1}{b}+\frac{1}{c}+\frac{1}{d}}.$$

Squaring both sides of the Cauchy-Schwarz inequality (6.14) gives the desired inequality.

(b) For positive numbers $a, b, c, d$, we claim that

$$(a+b+c+d)\left(\frac{1}{a}+\frac{1}{b}+\frac{1}{c}+\frac{1}{d}\right) = 16 \quad \Leftrightarrow \quad a = b = c = d.$$

The reverse implication is straightforward to check. For the forward implication, define $u$ and $v$ as in part (a) and note that the Cauchy-Schwarz inequality is an equality if and only if one of $u, v$ is a scalar multiple of the other. If $u = \lambda v$ for some $\lambda \in \mathbf{R}$, then necessarily $\lambda > 0$ since $a > 0$ and

$$\sqrt{a} = \frac{\lambda}{\sqrt{a}} \quad \Rightarrow \quad a = \lambda.$$

Similarly we find that $b = c = d = \lambda$. If $v = \lambda u$ for some $\lambda \in \mathbf{R}$ then again $\lambda$ must be positive and

$$\frac{1}{\sqrt{a}} = \lambda\sqrt{a} \quad \Rightarrow \quad a = \frac{1}{\lambda}.$$

Similarly we find that $b = c = d = \frac{1}{\lambda}$. In either case we have $a = b = c = d$.

**Exercise 6.A.13.** Show that the square of an average is less than or equal to the average of the squares. More precisely, show that if $a_1, ..., a_n \in \mathbf{R}$, then the square of the average of $a_1, ..., a_n$ is less than or equal to the average of $a_1^2, ..., a_n^2$.

**Solution.** For a positive integer $n$ and real numbers $a_1, ..., a_n$, let

$$u = (a_1, ..., a_n) \in \mathbf{R}^n \quad \text{and} \quad v = \left(\frac{1}{n}, ..., \frac{1}{n}\right) \in \mathbf{R}^n.$$

Observe that



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$$\langle u, v\rangle^2 = \left(\frac{a_1 + \cdots + a_n}{n}\right)^2, \quad \|u\|^2 = a_1^2 + \cdots + a_n^2, \quad \text{and} \quad \|v\|^2 = \frac{1}{n}.$$

Squaring both sides of the Cauchy-Schwarz inequality (6.14) gives us the desired inequality.

**Exercise 6.A.14.** Suppose $v \in V$ and $v \neq 0$. Prove that $v/\|v\|$ is the unique closest element on the unit sphere of $V$ to $v$. More precisely, prove that if $u \in V$ and $\|u\| = 1$, then

$$\left\| v - \frac{v}{\|v\|} \right\| \leq \|v - u\|,$$

with equality only if $u = v/\|v\|$.

**Solution.** Some routine calculations show that

$$\left\| v - \frac{v}{\|v\|} \right\|^2 = \|v\|^2 + 1 - 2\|v\| \quad \text{and} \quad \|v - u\|^2 = \|v\|^2 + 1 - 2\,\mathrm{Re}\langle v, u\rangle.$$

Thus

$$\left\| v - \frac{v}{\|v\|} \right\| \leq \|v - u\| \quad \Leftrightarrow \quad \left\| v - \frac{v}{\|v\|} \right\|^2 \leq \|v - u\|^2 \quad \Leftrightarrow \quad \mathrm{Re}\langle v, u\rangle \leq \|v\|.$$

Indeed, using the Cauchy-Schwarz inequality,

$$\mathrm{Re}\langle v, u\rangle \leq |\langle v, u\rangle| \leq \|v\|\|u\| = \|v\|.$$

As the proof of 6.17 shows, we have equality here if and only if one of $u, v$ is a non-negative real multiple of the other. If $u = \lambda v$ for some $\lambda \geq 0$, then

$$1 = \|u\| = |\lambda|\|v\| = \lambda\|v\| \quad \Rightarrow \quad \lambda = \frac{1}{\|v\|} \quad \Rightarrow \quad u = \frac{v}{\|v\|},$$

and if $v = \lambda u$ for some $\lambda \geq 0$ then

$$\|v\| = |\lambda|\|u\| = \lambda \quad \Rightarrow \quad u = \frac{v}{\|v\|}.$$

**Exercise 6.A.15.** Suppose $u, v$ are nonzero vectors in $\mathbf{R}^2$. Prove that

$$\langle u, v\rangle = \|u\|\|v\| \cos\theta,$$

where $\theta$ is the angle between $u$ and $v$ (thinking of $u$ and $v$ as arrows with initial point at the origin).

*Hint: Use the law of cosines on the triangle formed by $u, v$, and $u - v$.*

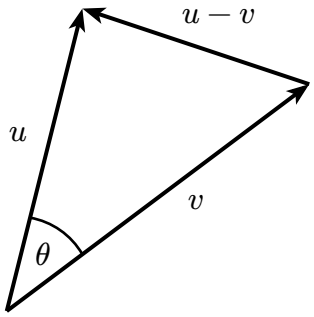
**Solution.** The law of cosines applied to the triangle formed by $u, v$, and $u - v$ states that

$$\|u - v\|^2 = \|u\|^2 + \|v\|^2 - 2\|u\|\|v\| \cos\theta.$$



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This expression together with the identity $\|u-v\|^2 = \|u\|^2 + \|v\|^2 - 2\langle u, v\rangle$ gives us the desired equality.




**Exercise 6.A.16.** The angle between two vectors (thought of as arrows with initial point at the origin) in $\mathbf{R}^2$ or $\mathbf{R}^3$ can be defined geometrically. However, geometry is not as clear in $\mathbf{R}^n$ for $n > 3$. Thus the angle between two nonzero vectors $x, y \in \mathbf{R}^n$ is defined to be

$$\arccos \frac{\langle x, y\rangle}{\|x\|\|y\|},$$

where the motivation for this definition comes from Exercise 15. Explain why the Cauchy-Schwarz inequality is needed to show that this definition makes sense.

**Solution.** The arccos function is only defined on the interval $[-1, 1]$; for the definition in question to make sense, we must have

$$\frac{\langle x, y\rangle}{\|x\|\|y\|} \in [-1, 1] \quad \Leftrightarrow \quad \frac{|\langle x, y\rangle|}{\|x\|\|y\|} \in [0, 1]$$

for any non-zero $x, y \in \mathbf{R}^n$. The Cauchy-Schwarz inequality ensures this.

**Exercise 6.A.17.** Prove that

$$\left(\sum_{k=1}^{n} a_k b_k\right)^2 \leq \left(\sum_{k=1}^{n} k a_k^2\right)\left(\sum_{k=1}^{n} \frac{b_k^2}{k}\right)$$

for all real numbers $a_1, ..., a_n$ and $b_1, ..., b_n$.

**Solution.** Let $n$ be a positive integer and let $a_1, ..., a_n, b_1, ..., b_n$ be real numbers. Define

$$u = \left(a_1, \sqrt{2}a_2, \sqrt{3}a_3, ..., \sqrt{n}a_n\right) \quad \text{and} \quad v = \left(b_1, \frac{b_2}{\sqrt{2}}, \frac{b_3}{\sqrt{3}}, ..., \frac{b_n}{\sqrt{n}}\right).$$

Then

$$\langle u, v\rangle = \sum_{k=1}^{n} a_k b_k, \quad \|u\| = \left(\sum_{k=1}^{n} k a_k^2\right)^{1/2}, \quad \text{and} \quad \|v\| = \left(\sum_{k=1}^{n} \frac{b_k^2}{k}\right)^{1/2}.$$



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Squaring both sides of the Cauchy-Schwarz inequality gives us the desired inequality.

**Exercise 6.A.18.**

(a) Suppose $f : [1, \infty) \to [0, \infty)$ is continuous. Show that

$$\left( \int_1^\infty f \right)^2 \leq \int_1^\infty x^2 (f(x))^2 \, dx.$$

(b) For which continuous functions $f : [1, \infty) \to [0, \infty)$ is the inequality in (a) an equality with both sides finite?

**Solution.**

(a) For $t \geq 1$ consider the vector space of continuous real-valued functions on the interval $[1, t]$ equipped with the inner product

$$\langle g, h \rangle = \int_1^t g(x) h(x) \, \mathrm{d}x.$$

The Cauchy-Schwarz inequality shows that

$$\left( \int_1^t f(x) \, \mathrm{d}x \right)^2 = \left( \int_1^t \frac{x}{x} f(x) \, \mathrm{d}x \right)^2 \leq \left( \int_1^t x^2 (f(x))^2 \, \mathrm{d}x \right) \left( \int_1^t \frac{1}{x^2} \, \mathrm{d}x \right)$$
$$= \left( \int_1^t x^2 (f(x))^2 \, \mathrm{d}x \right) \left( 1 - \frac{1}{t} \right).$$

Because $f$ is non-negative, both integrals $\int_1^\infty f(x) \, \mathrm{d}x$ and $\int_1^\infty x^2 (f(x))^2 \, \mathrm{d}x$ either converge or diverge to infinity. If $\int_1^\infty x^2 (f(x))^2 \, \mathrm{d}x = \infty$ then the desired inequality certainly holds, and if $\int_1^\infty x^2 (f(x))^2 \, \mathrm{d}x$ converges then the inequality

$$\left( \int_1^t f(x) \, \mathrm{d}x \right)^2 \leq \left( \int_1^t x^2 (f(x))^2 \, \mathrm{d}x \right) \left( 1 - \frac{1}{t} \right)$$

shows that $\int_1^\infty f(x) \, \mathrm{d}x$ also converges and furthermore that

$$\left( \int_1^\infty f(x) \, \mathrm{d}x \right)^2 \leq \int_1^\infty x^2 (f(x))^2 \, \mathrm{d}x.$$

(b) The Cauchy-Schwarz inequality used in part (a) is an equality if and only if $xf(x)$ and $x^{-1}$ are linearly dependent as functions on $[1, \infty)$, i.e. if and only if $f(x) = \lambda x^{-2}$ for all $x \geq 1$ and some $\lambda \geq 0$. In this case we obtain

$$\left( \int_1^\infty f(x) \, \mathrm{d}x \right)^2 = \int_1^\infty x^2 (f(x))^2 \, \mathrm{d}x = \lambda^2.$$



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**Exercise 6.A.19.** Suppose $v_1, ..., v_n$ is a basis of $V$ and $T \in \mathcal{L}(V)$. Prove that if $\lambda$ is an eigenvalue of $T$, then

$$|\lambda|^2 \leq \sum_{j=1}^{n} \sum_{k=1}^{n} |\mathcal{M}(T)_{j,k}|^2,$$

where $\mathcal{M}(T)_{j,k}$ denotes the entry in row $j$, column $k$ of the matrix of $T$ with respect to the basis $v_1, ..., v_n$.

**Solution.** It is straightforward to verify that

$$\langle a_1 v_1 + \cdots + a_n v_n, b_1 v_1 + \cdots + b_n v_n \rangle = a_1 \overline{b_1} + \cdots + a_n \overline{b_n}$$

is an inner product on $V$ (this is essentially the Euclidean inner product after identifying $V$ with $\mathbf{F}^n$). Because $\lambda$ is an eigenvalue of $T$, there is a non-zero $v \in V$ such that $Tv = \lambda v$; by replacing $v$ with $v/\|v\|$ if necessary, we may assume that $\|v\| = 1$. Suppose that $v = a_1 v_1 + \cdots + a_n v_n$ and observe that

$$Tv = \sum_{k=1}^{n} a_k T v_k = \sum_{k=1}^{n} a_k \sum_{j=1}^{n} \mathcal{M}(T)_{j,k} v_j = \sum_{j=1}^{n} \left( \sum_{k=1}^{n} a_k \mathcal{M}(T)_{j,k} \right) v_j$$

$$\Rightarrow \quad \|Tv\|^2 = \sum_{j=1}^{n} \left| \sum_{k=1}^{n} a_k \mathcal{M}(T)_{j,k} \right|^2.$$

For each $j \in \{1, ..., n\}$, applying the Cauchy-Schwarz inequality to the vectors

$$(a_1, ..., a_n) \quad \text{and} \quad \left( \overline{\mathcal{M}(T)_{j,1}}, ..., \overline{\mathcal{M}(T)_{j,n}} \right)$$

in $\mathbf{F}^n$ with the Euclidean inner product shows that

$$\left| \sum_{k=1}^{n} a_k \mathcal{M}(T)_{j,k} \right|^2 \leq \left( \sum_{k=1}^{n} |a_k|^2 \right) \left( \sum_{k=1}^{n} |\mathcal{M}(T)_{j,k}|^2 \right)$$

$$= \|v\|^2 \left( \sum_{k=1}^{n} |\mathcal{M}(T)_{j,k}|^2 \right) = \sum_{k=1}^{n} |\mathcal{M}(T)_{j,k}|^2.$$

Thus

$$|\lambda|^2 = \|\lambda v\|^2 = \|Tv\|^2 = \sum_{j=1}^{n} \left| \sum_{k=1}^{n} a_k \mathcal{M}(T)_{j,k} \right|^2 \leq \sum_{j=1}^{n} \sum_{k=1}^{n} |\mathcal{M}(T)_{j,k}|^2.$$

**Exercise 6.A.20.** Prove that if $u, v \in V$ then $|\|u\| - \|v\|| \leq \|u - v\|$.

*The inequality above is called the* ***reverse triangle inequality****. For the reverse triangle inequality when $V = \mathbf{C}$, see Exercise 2 in Chapter 4.*

**Solution.** Notice that



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$$\|u\| = \|u - v + v\| \leq \|u - v\| + \|v\| \quad \Rightarrow \quad \|u\| - \|v\| \leq \|u - v\|,$$
$$\|v\| = \|v - u + u\| \leq \|u - v\| + \|u\| \quad \Rightarrow \quad \|v\| - \|u\| \leq \|u - v\|.$$

Thus $|\|u\| - \|v\|| \leq \|u - v\|$.

**Exercise 6.A.21.** Suppose $u, v \in V$ are such that

$$\|u\| = 3, \quad \|u + v\| = 4, \quad \|u - v\| = 6.$$

What number does $\|v\|$ equal?

**Solution.** Rearranging the parallelogram equality (6.21) for $\|v\|$ gives

$$\|v\| = \left( \frac{\|u + v\|^2 + \|u - v\|^2}{2} - \|u\|^2 \right)^{1/2}.$$

Substituting the given values, we find $\|v\| = \sqrt{17}$.

**Exercise 6.A.22.** Show that if $u, v \in V$, then

$$\|u + v\|\|u - v\| \leq \|u\|^2 + \|v\|^2.$$

**Solution.** Notice that

$$0 \leq (\|u + v\| - \|u - v\|)^2 \quad \Leftrightarrow \quad 4\|u + v\|\|u - v\| \leq (\|u + v\| + \|u - v\|)^2$$
$$\Leftrightarrow \quad \|u + v\|\|u - v\| \leq \tfrac{1}{2}(\|u + v\| + \|u - v\|)^2 - \|u + v\|\|u - v\| = \|u\|^2 + \|v\|^2,$$

where the last equality is the parallelogram equality (6.21).

**Exercise 6.A.23.** Suppose $v_1, ..., v_m \in V$ are such that $\|v_k\| \leq 1$ for each $k = 1, ..., m$. Show that there exist $a_1, ..., a_m \in \{1, -1\}$ such that

$$\|a_1 v_1 + \cdots + a_m v_m\| \leq \sqrt{m}.$$

**Solution.** We will inductively define the integers $a_1, ..., a_m$. To begin, simply take $a_1 = 1$. For $k \in \{1, ..., m - 1\}$, suppose we have chosen $a_1, ..., a_k \in \{1, -1\}$ such that

$$\|u\| \leq \sqrt{k}, \quad \text{where} \quad u = a_1 v_1 + \cdots + a_k v_k.$$

It follows from Exercise 6.A.22 that

$$\|u + v_{k+1}\|\|u - v_{k+1}\| \leq \|u\|^2 + \|v_{k+1}\|^2 \leq k + 1.$$

Thus at least one of $\|u + v_{k+1}\|, \|u - v_{k+1}\|$ is less than or equal to $\sqrt{k+1}$. Let $a_{k+1} = 1$ if $\|u + v_{k+1}\| \leq \sqrt{k+1}$ and let $a_{k+1} = -1$ otherwise. By repeating this process until $k = m - 1$, we obtain the desired integers $a_1, ..., a_m$.



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**Exercise 6.A.24.** Prove or give a counterexample: If $\|\cdot\|$ is the norm associated with an inner product on $\mathbf{R}^2$, then there exists $(x, y) \in \mathbf{R}^2$ such that $\|(x, y)\| \neq \max\{|x|, |y|\}$.

**Solution.** Let $f : \mathbf{R}^2 \to \mathbf{R}$ be given by $f(x, y) = \max\{|x|, |y|\}$, let $u = (1, 0)$ and $v = (1, 1)$, and observe that

$$[f(u + v)]^2 + [f(u - v)]^2 = 5 \neq 4 = 2\Big([f(u)]^2 + [f(v)]^2\Big).$$

Since a norm associated with an inner product must satisfy the parallelogram equality (6.21), it follows that any norm $\|\cdot\|$ associated with an inner product on $\mathbf{R}^2$ cannot be given by $f$. That is, there must exist some $(x, y) \in \mathbf{R}^2$ such that $\|(x, y)\| \neq f(x, y)$.

**Exercise 6.A.25.** Suppose $p > 0$. Prove that there is an inner product on $\mathbf{R}^2$ such that the associated norm is given by

$$\|(x, y)\| = \left(|x|^p + |y|^p\right)^{1/p}$$

for all $(x, y) \in \mathbf{R}^2$ if and only if $p = 2$.

**Solution.** Let $f : \mathbf{R}^2 \to \mathbf{R}$ be given by $f(x, y) = \left(|x|^p + |y|^p\right)^{1/p}$, let $u = (1, 0)$ and $v = (0, 1)$, and observe that

$$[f(u + v)]^2 + [f(u - v)]^2 = 2^{1+2/p} \quad \text{and} \quad 2\Big([f(u)]^2 + [f(v)]^2\Big) = 4.$$

If $f$ was indeed a norm arising from an inner product then $f$ would satisfy the parallelogram equality (6.21). Since the quantities above are equal if and only if $2^{1+2/p} = 4$, i.e. if and only if $p = 2$, the only possible value for $p$ is 2, which indeed gives the norm associated with the Euclidean inner product on $\mathbf{R}^2$, as 6.8(a) shows.

**Exercise 6.A.26.** Suppose $V$ is a real inner product space. Prove that

$$\langle u, v\rangle = \frac{\|u + v\|^2 - \|u - v\|^2}{4}$$

for all $u, v \in V$.

**Solution.** For any $u, v \in V$, observe that

$$\begin{aligned}\|u + v\|^2 - \|u - v\|^2 &= \langle u + v, u + v\rangle - \langle u - v, u - v\rangle \\ &= \langle u, u\rangle + 2\langle u, v\rangle + \langle v, v\rangle - \langle u, u\rangle + 2\langle u, v\rangle - \langle v, v\rangle \\ &= 4\langle u, v\rangle.\end{aligned}$$



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**Exercise 6.A.27.** Suppose $V$ is a complex inner product space. Prove that

$$\langle u, v\rangle = \frac{\|u+v\|^2 - \|u-v\|^2 + \|u+iv\|^2 i - \|u-iv\|^2 i}{4}$$

for all $u, v \in V$.

**Solution.** For any $u, v \in V$, observe that

$$\begin{aligned}
\|u+v\|^2 - \|u-v\|^2 &= \langle u+v, u+v\rangle - \langle u-v, u-v\rangle \\
&= \langle u,u\rangle + \langle u,v\rangle + \overline{\langle u,v\rangle} + \langle v,v\rangle \\
&\quad - \langle u,u\rangle + \langle u,v\rangle + \overline{\langle u,v\rangle} - \langle v,v\rangle \\
&= 2\Big(\langle u,v\rangle + \overline{\langle u,v\rangle}\Big) \\
&= 4\,\mathrm{Re}\langle u,v\rangle.
\end{aligned}$$

Furthermore,

$$\begin{aligned}
i\|u+iv\|^2 - i\|u-iv\|^2 &= i\langle u+iv, u+iv\rangle - i\langle u-iv, u-iv\rangle \\
&= i\langle u,u\rangle - i^2\langle u,v\rangle + i^2\overline{\langle u,v\rangle} + i\langle v,v\rangle \\
&\quad - i\langle u,u\rangle - i^2\langle u,v\rangle + i^2\overline{\langle u,v\rangle} - i\langle v,v\rangle \\
&= 2\Big(\langle u,v\rangle - \overline{\langle u,v\rangle}\Big) \\
&= 4\,\mathrm{Im}\langle u,v\rangle.
\end{aligned}$$

It follows that

$$\|u+v\|^2 - \|u-v\|^2 + \|u+iv\|^2 i - \|u-iv\|^2 i = 4\langle u,v\rangle.$$

**Exercise 6.A.28.** A norm on a vector space $U$ is a function

$$\|\cdot\| : U \to [0,\infty)$$

such that $\|u\| = 0$ if and only if $u = 0, \|\alpha u\| = |\alpha|\|u\|$ for all $\alpha \in \mathbf{F}$ and all $u \in U$, and $\|u+v\| \le \|u\| + \|v\|$ for all $u, v \in U$. Prove that a norm satisfying the parallelogram equality comes from an inner product (in other words, show that if $\|\cdot\|$ is a norm on $U$ satisfying the parallelogram equality, then there is an inner product $\langle\cdot,\cdot\rangle$ on $U$ such that $\|u\| = \langle u,u\rangle^{1/2}$ for all $u \in U$).

**Solution.** Let us first consider the case where $U$ is a real vector space. Define $\langle\cdot,\cdot\rangle : U \times U \to \mathbf{R}$ by

$$\langle u,v\rangle = \frac{\|u+v\|^2 - \|u-v\|^2}{4}.$$



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For any $u \in U$ we have

$$\langle u, u \rangle = \frac{\|2u\|^2}{4} = \|u\|^2.$$

Thus the norm is given by $\|u\| = \langle u, u \rangle^{1/2}$. We now show that $\langle \cdot, \cdot \rangle$ is an inner product on $U$.

**Positive-definiteness.** Combining the identity $\langle u, u \rangle = \|u\|^2$ with the properties of the norm $\|\cdot\|$ shows that $\langle \cdot, \cdot \rangle$ is positive-definite.

**Symmetry.** For any $u, v \in V$, observe that $\|v - u\| = |-1|\|u - v\| = \|u - v\|$; it follows that

$$\langle u, v \rangle = \frac{\|u + v\|^2 - \|u - v\|^2}{4} = \frac{\|v + u\|^2 - \|v - u\|^2}{4} = \langle v, u \rangle.$$

**Additivity in the first slot.** Let $u, v, w \in U$ be given. Since $\|\cdot\|$ satisfies the parallelogram equality, we have

$$\|v + 2w\|^2 + \|v\|^2 = 2\|v + w\|^2 + 2\|w\|^2,$$
$$\|v - 2w\|^2 + \|v\|^2 = 2\|v - w\|^2 + 2\|w\|^2.$$

Subtracting the latter of these equations from the former gives us

$$\|v + 2w\|^2 - \|v - 2w\|^2 = 2\|v + w\|^2 - 2\|v - w\|^2. \tag{1}$$

Now we use the parallelogram equality two more times:

$$2\|u + v + w\|^2 + 2\|u - w\|^2 = \|v + 2u\|^2 + \|v + 2w\|^2,$$
$$2\|u + v - w\|^2 + 2\|u + w\|^2 = \|v + 2u\|^2 + \|v - 2w\|^2.$$

Subtracting the latter of these equations from the former gives us

$$2\Big(\|u + v + w\|^2 + \|u - w\|^2\Big) - 2\Big(\|u + v - w\|^2 + \|u + w\|^2\Big) = \|v + 2w\|^2 - \|v - 2w\|^2.$$

Combining this with equation (1), we see that

$$2\Big(\|u + v + w\|^2 + \|u - w\|^2\Big) - 2\Big(\|u + v - w\|^2 + \|u + w\|^2\Big) = 2\|v + w\|^2 - 2\|v - w\|^2.$$

Equivalently,

$$\frac{\|u + v + w\|^2 - \|u + v - w\|^2}{4} = \frac{\|u + w\|^2 - \|u - w\|^2 + \|v + w\|^2 - \|v - w\|^2}{4},$$

which is exactly the statement $\langle u + v, w \rangle = \langle u, w \rangle + \langle v, w \rangle$.

**Homogeneity in the first slot.** Suppose $u, v \in U$. First, we will use induction to show that $\langle nu, v \rangle = n\langle u, v \rangle$ for all positive integers $n$. The base case $n = 1$ is clear, so suppose that the result holds for some positive integer $n$ and observe that

$$\langle (n + 1)u, v \rangle = \langle nu + u, v \rangle = \langle nu, v \rangle + \langle u, v \rangle = n\langle u, v \rangle + \langle u, v \rangle = (n + 1)\langle u, v \rangle,$$

where we have used additivity in the first slot and the induction hypothesis. This completes the induction step and thus $\langle nu, v \rangle = n\langle u, v \rangle$ for all positive integers $n$. Certainly $\langle 0, v \rangle = 0$



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and so we may extend this result to all non-negative integers. If $n$ is a positive integer then observe that

$$\langle -nu, v\rangle + n\langle u, v\rangle = \langle -nu, v\rangle + \langle nu, v\rangle = \langle 0, v\rangle = 0,$$

where we have used additivity in the first slot and homogeneity in the first slot for positive integers. It follows that $\langle -nu, v\rangle = -n\langle u, v\rangle$ and thus we have homogeneity in the first slot for all integers.

To extend homogeneity in the first slot to rational numbers, let $n$ be a positive integer. By additivity in the first slot, we have

$$n\langle n^{-1}u, v\rangle = \textstyle\sum_{j=1}^{n}\langle n^{-1}u, v\rangle = \left\langle \sum_{j=1}^{n} n^{-1}u, v\right\rangle = \langle u, v\rangle,$$

which implies that $\langle n^{-1}u, v\rangle = n^{-1}\langle u, v\rangle$. Combining this with homogeneity in the first slot for integers allows us to extend homogeneity in the first slot to rational numbers.

Finally, to obtain homogeneity in the first slot for all real numbers, let $\lambda \in \mathbf{R}$ be given. There exists a sequence $(r_n)$ of rational numbers satisfying $\lim_{n\to\infty} r_n = \lambda$. The reverse triangle inequality (Exercise 6.A.20) shows that the function $U \to \mathbf{R}$ given by $u \mapsto \|u\|$ is continuous. Combining this with standard results on compositions and linear combinations of continuous functions, we have

$$\begin{aligned}
\lambda\langle u, v\rangle &= \lim_{n\to\infty} r_n\langle u, v\rangle \\
&= \lim_{n\to\infty} \langle r_n u, v\rangle \\
&= \lim_{n\to\infty} \frac{\|r_n u + v\|^2 - \|r_n u - v\|^2}{4} \\
&= \frac{\|\lambda u + v\|^2 - \|\lambda u - v\|^2}{4} \\
&= \langle \lambda u, v\rangle.
\end{aligned}$$

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Now let us consider the case where $U$ is a complex vector space. Define $B : U \times U \to \mathbf{R}$ by

$$B(u, v) = \frac{\|u + v\|^2 - \|u - v\|^2}{4}$$

and define $\langle \cdot, \cdot\rangle : U \times U \to \mathbf{C}$ by

$$\langle u, v\rangle = B(u, v) + iB(u, iv) = \frac{\|u + v\|^2 - \|u - v\|^2}{4} + i\frac{\|u + iv\|^2 - \|u - iv\|^2}{4}.$$

Observe that for any $u \in U$ we have $B(u, u) = \|u\|^2$ and

$$B(u, iu) = \frac{\|u + iu\|^2 - \|u - iu\|^2}{4} = \frac{|1 + i|^2\|u\|^2 - |1 - i|^2\|u\|^2}{4} = 0.$$

Thus $\|u\| = \langle u, u\rangle^{1/2}$. We now show that $\langle \cdot, \cdot\rangle$ is an inner product on $U$.



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**Positive-definiteness.** Combining the identity $\langle u, u\rangle = \|u\|^2$ with the properties of the norm $\|\cdot\|$ shows that $\langle \cdot, \cdot \rangle$ is positive-definite.

**Conjugate symmetry.** For any $u, v \in U$, note that

$$\overline{\langle v, u\rangle} = \overline{B(v,u) + iB(v,iu)} = B(u,v) - iB(v,iu),$$

where we have used that $B$ is real-valued and symmetric (we showed this in the case where $U$ is a real vector space.) Given the expression above, to verify conjugate symmetry of $\langle \cdot, \cdot \rangle$ it will suffice to show that $B(u, iv) = -B(v, iu)$. Indeed,

$$\begin{aligned} -4B(v,iu) &= \|v - iu\|^2 - \|v + iu\|^2 = \|-i(u+iv)\|^2 - \|i(u-iv)\|^2 \\ &= |-i|^2\|u+iv\|^2 - \|i\|^2\|u-iv\|^2 = \|u+iv\|^2 - \|u-iv\|^2 = 4B(u,iv). \end{aligned}$$

**Additivity in the first slot.** Let $u, v, w \in U$ be given. The proof of additivity in the first slot we gave for the case where $U$ is a real vector space equally shows that $B$ is additive in the first slot. It follows that

$$\begin{aligned} \langle u+v, w\rangle &= B(u+v,w) + iB(u+v,iw) \\ &= B(u,w) + iB(u,iw) + B(v,w) + iB(v,iw) = \langle u, w\rangle + \langle v, w\rangle. \end{aligned}$$

**Homogeneity in the first slot.** Let $u, v \in U$ be given. The proof of homogeneity in the first slot we gave for the case where $U$ is a real vector space equally shows that $B$ is homogeneous in the first slot with respect to real numbers. It follows that for any $\lambda \in \mathbf{R}$ we have $\langle \lambda u, v\rangle = \lambda\langle u, v\rangle$. Now observe that

$$\begin{aligned} 4\langle iu, v\rangle &= \|iu + v\|^2 - \|iu - v\|^2 + i\big(\|iu + iv\|^2 - \|iu - iv\|^2\big) \\ &= \|i(u - iv)\|^2 - \|i(u + iv)\|^2 + i\big(\|i(u+v)\|^2 - \|i(u-v)\|^2\big) \\ &= \|u - iv\|^2 - \|u + iv\|^2 + i\big(\|u+v\|^2 - \|u-v\|^2\big) \\ &= i\big(\|u+v\|^2 - \|u-v\|^2 + i\big(\|u+iv\|^2 - \|u-iv\|^2\big)\big) \\ &= 4i\langle u, v\rangle. \end{aligned}$$

Thus $\langle iu, v\rangle = i\langle u, v\rangle$. It follows that, for any $x + iy \in \mathbf{C}$,

$$\langle (x+iy)u, v\rangle = \langle xu + iyu, v\rangle = \langle xu, v\rangle + \langle iyu, v\rangle = x\langle u, v\rangle + iy\langle u, v\rangle = (x+iy)\langle u, v\rangle.$$

**Exercise 6.A.29.** Suppose $V_1, ..., V_m$ are inner product spaces. Show that the equation

$$\langle (u_1, ..., u_m), (v_1, ..., v_m)\rangle = \langle u_1, v_1\rangle + \cdots + \langle u_m, v_m\rangle$$

defines an inner product on $V_1 \times \cdots \times V_m$.

*In the expression above on the right, for each $k = 1, ..., m$, the inner product $\langle u_k, v_k\rangle$ denotes the inner product on $V_k$. Each of the spaces $V_1, ..., V_m$ may have a different inner product, even though the same notation is used here.*



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**Solution.** For convenience, let $\mathbf{V} = V_1 \times \cdots \times V_m$. We verify each property in definition 6.2.

**Positivity.** Let $(v_1, ..., v_m) \in \mathbf{V}$ be given and observe that

$$\langle (v_1, ..., v_m), (v_1, ..., v_m) \rangle = \langle v_1, v_1 \rangle + \cdots + \langle v_m, v_m \rangle.$$

Since each $\langle v_k, v_k \rangle$ is non-negative, it follows that $\langle (v_1, ..., v_m), (v_1, ..., v_m) \rangle$ is non-negative.

**Definiteness.** For $(v_1, ..., v_m) \in \mathbf{V}$, note that $(v_1, ..., v_m) = 0$ if and only if each $v_k = 0$. By the definiteness of the inner product on each $V_k$, this is the case if and only if each $\langle v_k, v_k \rangle = 0$. From the non-negativity of the expression

$$\langle (v_1, ..., v_m), (v_1, ..., v_m) \rangle = \langle v_1, v_1 \rangle + \cdots + \langle v_m, v_m \rangle,$$

we see that each $\langle v_k, v_k \rangle = 0$ if and only if $\langle (v_1, ..., v_m), (v_1, ..., v_m) \rangle = 0$.

**Additivity in the first slot.** Let $(u_1, ..., u_m), (v_1, ..., v_m), (w_1, ..., w_m) \in \mathbf{V}$ be given and observe that

$$\begin{aligned}
&\langle (u_1, ..., u_m) + (v_1, ..., v_m), (w_1, ..., w_m) \rangle \\
&= \langle (u_1 + v_1, ..., u_m + v_m), (w_1, ..., w_m) \rangle \\
&= \langle u_1 + v_1, w_1 \rangle + \cdots + \langle u_m + v_m, w_m \rangle \\
&= \langle u_1, w_1 \rangle + \langle v_1, w_1 \rangle + \cdots + \langle u_m, w_m \rangle + \langle v_m, w_m \rangle \\
&= \langle u_1, w_1 \rangle + \cdots + \langle u_m, w_m \rangle + \langle v_1, w_1 \rangle + \cdots + \langle v_m, w_m \rangle \\
&= \langle (u_1, ..., u_m), (w_1, ..., w_m) \rangle + \langle (v_1, ..., v_m), (w_1, ..., w_m) \rangle,
\end{aligned}$$

where we have used the additivity in the first slot of the inner product on each $V_k$.

**Homogeneity in the first slot.** Let $\lambda \in \mathbf{F}$ and $(u_1, ..., u_m), (v_1, ..., v_m) \in \mathbf{V}$ be given, and observe that

$$\begin{aligned}
\langle \lambda(u_1, ..., u_m), (v_1, ..., v_m) \rangle &= \langle (\lambda u_1, ..., \lambda u_m), (v_1, ..., v_m) \rangle \\
&= \langle \lambda u_1, v_1 \rangle + \cdots + \langle \lambda u_m, v_m \rangle \\
&= \lambda \langle u_1, v_1 \rangle + \cdots + \lambda \langle u_m, v_m \rangle \\
&= \lambda (\langle u_1, v_1 \rangle + \cdots + \langle u_m, v_m \rangle) \\
&= \lambda \langle (u_1, ..., u_m), (v_1, ..., v_m) \rangle,
\end{aligned}$$

where we have used homogeneity in the first slot of the inner product on each $V_k$.

**Conjugate symmetry.** Let $(u_1, ..., u_m), (v_1, ..., v_m) \in \mathbf{V}$ be given and observe that

$$\begin{aligned}
&\overline{\langle (v_1, ..., v_m), (u_1, ..., u_m) \rangle} = \overline{\langle v_1, u_1 \rangle + \cdots + \langle v_m, u_m \rangle} \\
&= \overline{\langle v_1, u_1 \rangle} + \cdots + \overline{\langle v_m, u_m \rangle} = \langle u_1, v_1 \rangle + \cdots + \langle u_m, v_m \rangle = \langle (u_1, ..., u_m), (v_1, ..., v_m) \rangle,
\end{aligned}$$

where we have used the conjugate symmetry of the inner product on each $V_k$.



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**Exercise 6.A.30.** Suppose $V$ is a real inner product space. For $u, v, w, x \in V$, define

$$\langle u + iv, w + ix \rangle_{\mathbf{C}} = \langle u, w \rangle + \langle v, x \rangle + (\langle v, w \rangle - \langle u, x \rangle)i.$$

(a) Show that $\langle \cdot, \cdot \rangle_{\mathbf{C}}$ makes $V_{\mathbf{C}}$ into a complex inner product space.

(b) Show that if $u, v \in V$, then

$$\langle u, v \rangle_{\mathbf{C}} = \langle u, v \rangle \quad \text{and} \quad \|u + iv\|_{\mathbf{C}}^2 = \|u\|^2 + \|v\|^2.$$

*See Exercise 8 in Section 1B for the definition of the complexification $V_{\mathbf{C}}$.*

**Solution.**

(a) We verify each property in definition 6.2.

**Positive-definiteness.** For any $u + iv \in V_{\mathbf{C}}$,

$$\langle u + iv, u + iv \rangle_{\mathbf{C}} = \langle u, u \rangle + \langle v, v \rangle + (\langle v, u \rangle - \langle u, v \rangle)i = \langle u, u \rangle + \langle v, v \rangle,$$

where we have used the symmetry of $\langle \cdot, \cdot \rangle$. The positivity of $\langle \cdot, \cdot \rangle$ and the expression above gives us the positivity of $\langle \cdot, \cdot \rangle_{\mathbf{C}}$. Moreover,

$$\langle u + iv, u + iv \rangle_{\mathbf{C}} = 0 \quad \Leftrightarrow \quad \langle u, u \rangle = 0 \text{ and } \langle v, v \rangle = 0 \quad \Leftrightarrow \quad u = v = 0 \quad \Leftrightarrow \quad u + iv = 0.$$

**Conjugate symmetry.** For any $u + iv, w + ix \in V_{\mathbf{C}}$, observe that

$$\begin{aligned} \overline{\langle w + ix, u + iv \rangle_{\mathbf{C}}} &= \overline{\langle w, u \rangle + \langle x, v \rangle + (\langle x, u \rangle - \langle w, v \rangle)i} \\ &= \langle u, w \rangle + \langle v, x \rangle + (\langle v, w \rangle - \langle u, x \rangle)i = \langle u + iv, w + ix \rangle_{\mathbf{C}}, \end{aligned}$$

where we have used the symmetry of $\langle \cdot, \cdot \rangle$.

**Additivity in the first slot.** Let $u + iv, w + ix, y + iz \in V_{\mathbf{C}}$ be given and observe that

$$\begin{aligned} \langle (u + iv) + (w + ix), y + iz \rangle_{\mathbf{C}} &= \langle (u + w) + i(v + x), y + iz \rangle_{\mathbf{C}} \\ &= \langle u + w, y \rangle + \langle v + x, z \rangle + (\langle v + x, y \rangle - \langle u + w, z \rangle)i \\ &= \langle u, y \rangle + \langle w, y \rangle + \langle v, z \rangle + \langle x, z \rangle \\ &\quad + (\langle v, y \rangle + \langle x, y \rangle - \langle u, z \rangle - \langle w, z \rangle)i \\ &= \langle u, y \rangle + \langle v, z \rangle + (\langle v, y \rangle - \langle u, z \rangle)i \\ &\quad + \langle w, y \rangle + \langle x, z \rangle + (\langle x, y \rangle - \langle w, z \rangle)i \\ &= \langle u + iv, y + iz \rangle_{\mathbf{C}} + \langle w + ix, y + iz \rangle_{\mathbf{C}}, \end{aligned}$$

where we have used the additivity in the first slot of $\langle \cdot, \cdot \rangle$.

**Homogeneity in the first slot.** Let $u + iv, w + ix \in V_{\mathbf{C}}$ and $a + bi \in \mathbf{C}$ be given and observe that



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$$\begin{aligned}
\langle (a+bi)(u+iv), w+ix\rangle_{\mathbf{C}} &= \langle (au-bv)+i(av+bu), w+ix\rangle_{\mathbf{C}} \\
&= \langle au-bv, w\rangle + \langle av+bu, x\rangle \\
&\quad + (\langle av+bu, w\rangle - \langle au-bv, x\rangle)i \\
&= a\langle u, w\rangle - b\langle v, w\rangle + a\langle v, x\rangle + b\langle u, x\rangle \\
&\quad + (a\langle v, w\rangle + b\langle u, w\rangle - a\langle u, x\rangle + b\langle v, x\rangle)i \\
&= [a(\langle u, w\rangle + \langle v, x\rangle) - b(\langle v, w\rangle - \langle u, x\rangle)] \\
&\quad + [a(\langle v, w\rangle - \langle u, x\rangle) + b(\langle u, w\rangle + \langle v, x\rangle)]i \\
&= (a+bi)[\langle u, w\rangle + \langle v, x\rangle + (\langle v, w\rangle - \langle u, x\rangle)i] \\
&= (a+bi)\langle u+iv, w+ix\rangle_{\mathbf{C}},
\end{aligned}$$

where we have used the homogeneity in the first slot of $\langle \cdot, \cdot \rangle$.

(b) For $u, v \in V$ we have

$$\langle u, v\rangle_{\mathbf{C}} = \langle u, v\rangle + \langle 0, 0\rangle + (\langle 0, v\rangle - \langle u, 0\rangle)i = \langle u, v\rangle.$$

Furthermore, as we showed in part (a) when we verified the positive-definiteness of $\langle \cdot, \cdot \rangle_{\mathbf{C}}$,

$$\|u+iv\|_{\mathbf{C}}^2 = \langle u+iv, u+iv\rangle_{\mathbf{C}} = \langle u, u\rangle + \langle v, v\rangle = \|u\|^2 + \|v\|^2.$$

**Exercise 6.A.31.** Suppose $u, v, w \in V$. Prove that

$$\left\|w - \tfrac{1}{2}(u+v)\right\|^2 = \frac{\|w-u\|^2 + \|w-v\|^2}{2} - \frac{\|u-v\|^2}{4}.$$

**Solution.** It will suffice to prove that

$$4\left\|w - \tfrac{1}{2}(u+v)\right\|^2 = 2\Big(\|w-u\|^2 + \|w-v\|^2\Big) - \|u-v\|^2,$$

which is equivalent to

$$\|2w-u-v\|^2 + \|u-v\|^2 = 2\Big(\|w-u\|^2 + \|w-v\|^2\Big),$$

which follows immediately from the parallelogram equality (6.21).

**Exercise 6.A.32.** Suppose that $E$ is a subset of $V$ with the property that $u, v \in E$ implies $\frac{1}{2}(u+v) \in E$. Let $w \in V$. Show that there is at most one point in $E$ that is closest to $w$. In other words, show that there is at most one $u \in E$ such that

$$\|w-u\| \le \|w-x\|$$

for all $x \in E$.



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**Solution.** Suppose $u, v \in E$ are both closest to $w$, so that $\|w-u\| = \|w-v\|$. It follows from Exercise 6.A.31 that

$$\frac{\|u-v\|^2}{4} = \|w-u\|^2 - \left\|w - \tfrac{1}{2}(u+v)\right\|^2.$$

Since $u$ and $v$ belong to $E$ we must have $\frac{1}{2}(u+v) \in E$, and then since $u$ is a point in $E$ closest to $w$ we must have $\|w-u\| \le \left\|w - \frac{1}{2}(u+v)\right\|$. It follows that $\|u-v\|^2 \le 0$, which is the case if and only if $u = v$.

**Exercise 6.A.33.** Suppose $f, g$ are differentiable functions from $\mathbf{R}$ to $\mathbf{R}^n$.

(a) Show that

$$\langle f(t), g(t)\rangle' = \langle f'(t), g(t)\rangle + \langle f(t), g'(t)\rangle.$$

(b) Suppose $c$ is a positive number and $\|f(t)\| = c$ for every $t \in \mathbf{R}$. Show that $\langle f'(t), f(t)\rangle = 0$ for every $t \in \mathbf{R}$.

(c) Interpret the result in (b) geometrically in terms of the tangent vector to a curve lying on a sphere in $\mathbf{R}^n$ centered at the origin.

*A function $f : \mathbf{R} \to \mathbf{R}^n$ is called differentiable if there exist differentiable functions $f_1, ..., f_n$ from $\mathbf{R}$ to $\mathbf{R}$ such that $f(t) = (f_1(t), ..., f_n(t))$ for each $t \in \mathbf{R}$. Furthermore, for each $t \in \mathbf{R}$, the derivative $f'(t) \in \mathbf{R}^n$ is defined by $f'(t) = (f_1'(t), ..., f_n'(t))$.*

**Solution.**

(a) Suppose that $f(t) = (f_1(t), ..., f_n(t))$ and $g(t) = (g_1(t), ..., g_n(t))$ for some differentiable functions $f_1, ..., f_n, g_1, ..., g_n : \mathbf{R} \to \mathbf{R}$. By the usual rules of differentiation we have

$$\begin{aligned}
\langle f(t), g(t)\rangle' &= (f_1(t)g_1(t) + \cdots + f_n(t)g_n(t))' \\
&= f_1'(t)g_1(t) + f_1(t)g_1'(t) + \cdots + f_n'(t)g_n(t) + f_n(t)g_n'(t) \\
&= f_1'(t)g_1(t) + \cdots + f_n'(t)g_n(t) + f_1(t)g_1'(t) + \cdots + f_n(t)g_n'(t) \\
&= \langle f'(t), g(t)\rangle + \langle f(t), g'(t)\rangle.
\end{aligned}$$

(b) By part (a) we have

$$0 = (c^2)' = \left(\|f(t)\|^2\right)' = \langle f(t), f(t)\rangle' = 2\langle f'(t), f(t)\rangle.$$

Thus $\langle f'(t), f(t)\rangle = 0$.

(c) Suppose $c > 0$. A differentiable function $f : \mathbf{R} \to \mathbf{R}^n$ satisfying $\|f(t)\| = c$ for every $t \in \mathbf{R}$ traces out a curve which lies on an $(n-1)$-sphere of radius $c$ centered at the origin in $\mathbf{R}^n$. The tangent vector to this curve is given by $f'$; the result of part (b) states that this tangent vector is always orthogonal to the curve $f$.

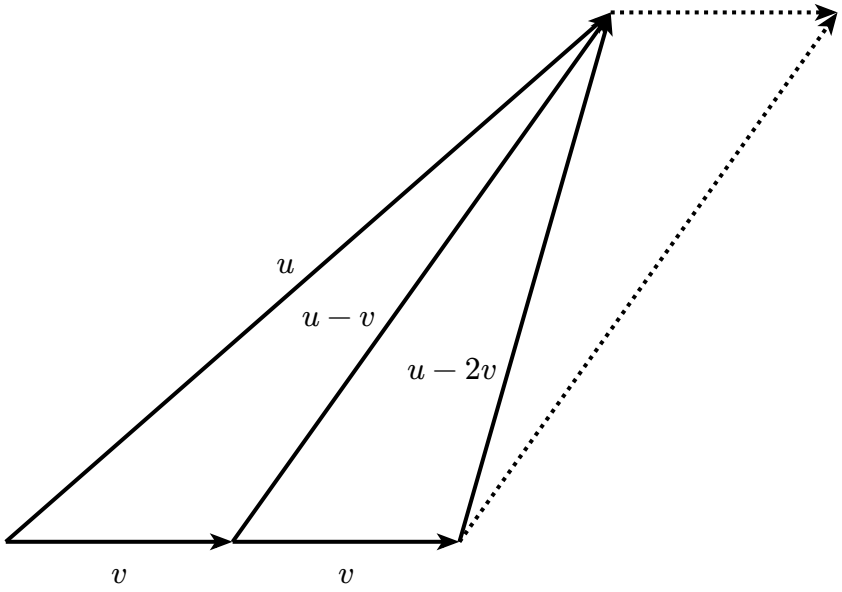


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**Exercise 6.A.34.** Use inner products to prove Apollonius's identity: In a triangle with sides of length $a, b$, and $c$, let $d$ be the length of the line segment from the midpoint of the side of length $c$ to the opposite vertex. Then

$$a^2 + b^2 = \tfrac{1}{2}c^2 + 2d^2.$$

**Solution.** Set up the triangle as follows.




Thus

$$\|u\| = a, \quad \|v\| = \tfrac{1}{2}c, \quad \|u - v\| = d, \quad \|u - 2v\| = b.$$

Consider the parallelogram formed by the vectors $u - v$ and $v$. The parallelogram equality states that

$$\|u\|^2 = \|u - 2v\|^2 = 2\|v\|^2 + 2\|u - v\|^2.$$

Substituting the given side lengths, we obtain

$$a^2 + b^2 = \tfrac{1}{2}c^2 + 2d^2.$$



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**Exercise 6.A.35.** Fix a positive integer $n$. The *Laplacian* $\Delta p$ of a twice differentiable real-valued function $p$ on $\mathbf{R}^n$ is the function on $\mathbf{R}^n$ defined by

$$\Delta p = \frac{\partial^2 p}{\partial x_1^2} + \cdots + \frac{\partial^2 p}{\partial x_n^2}.$$

The function $p$ is called *harmonic* if $\Delta p = 0$.

A *polynomial* on $\mathbf{R}^n$ is a linear combination (with coefficients in $\mathbf{R}$) of functions of the form $x_1^{m_1} \cdots x_n^{m_n}$, where $m_1, ..., m_n$ are nonnegative integers.

Suppose $q$ is a polynomial on $\mathbf{R}^n$. Prove that there exists a harmonic polynomial $p$ on $\mathbf{R}^n$ such that $p(x) = q(x)$ for every $x \in \mathbf{R}^n$ with $\|x\| = 1$.

> *The only fact about harmonic functions that you need for this exercise is that if $p$ is a harmonic function on $\mathbf{R}^n$ and $p(x) = 0$ for all $x \in \mathbf{R}^n$ with $\|x\| = 1$, then $p = 0$.*
>
> *Hint: A reasonable guess is that the desired harmonic polynomial $p$ is of the form $q + \big(1 - \|x\|^2\big)r$ for some polynomial $r$. Prove that there is a polynomial $r$ on $\mathbf{R}^n$ such that $q + \big(1 - \|x\|^2\big)r$ is harmonic by defining an operator $T$ on a suitable vector space by*
>
> $$Tr = \Delta\Big(\big(1 - \|x\|^2\big)r\Big)$$
>
> *and then showing that $T$ is injective and hence surjective.*

**Solution.** Let us first provide a few definitions. A monomial on $\mathbf{R}^n$ is a polynomial on $\mathbf{R}^n$ of the form $x_1^{m_1} \cdots x_n^{m_n}$, where $m_1, ..., m_n$ are non-negative integers. The degree of such a monomial is the sum $m_1 + \cdots + m_n$. The degree of a non-zero polynomial $p$ on $\mathbf{R}^n$ is the greatest degree amongst its monomial terms $x_1^{m_1} \cdots x_n^{m_n}$ and the degree of the zero polynomial is defined to be $-\infty$.

Define $\mathcal{P}_m^n(\mathbf{R})$ to be the collection of all polynomials on $\mathbf{R}^n$ of degree at most $m$ and note that $\mathcal{P}_m^n(\mathbf{R})$ is a subset of the vector space $\mathbf{R}^{\mathbf{R}^n}$; in fact, size the zero polynomial is simply the zero function, and addition and scalar multiplication of polynomials of degree at most $m$ will not result in a polynomial of degree greater than $m$, $\mathcal{P}_m^n(\mathbf{R})$ is a vector subspace of $\mathbf{R}^{\mathbf{R}^n}$.

It is straightforward to verify that the collection of all monomials of degree at most $m$ forms a basis of $\mathcal{P}_m^n(\mathbf{R})$. This collection is finite and thus $\mathcal{P}_m^n(\mathbf{R})$ is a finite-dimensional vector space.

Clearly, the Laplacian $\Delta p$ of a polynomial $p$ on $\mathbf{R}^n$ is itself a polynomial and furthermore satisfies either $\deg \Delta p = -\infty$ or $\deg \Delta p = \deg p - 2$. Thus the Laplacian defines an operator $\Delta \in \mathcal{L}(\mathcal{P}_m^n(\mathbf{R}))$; the linearity of $\Delta$ follows from the linearity of partial differentiation.

The function

$$x = (x_1, ..., x_n) \in \mathbf{R}^n \mapsto \|x\|^2 = x_1^2 + \cdots + x_n^2$$



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is a polynomial on $\mathbf{R}^n$ of degree 2. Given this, if $r$ is a non-zero polynomial on $\mathbf{R}^n$, then $\big(1-\|x\|^2\big)r$ is also a polynomial on $\mathbf{R}^n$ of degree $r+2$; if $r$ is the zero polynomial then so is $\big(1-\|x\|^2\big)r$. It follows that $\Delta\big(1-\|x\|^2\big)r$ is a polynomial on $\mathbf{R}^n$ of degree at most $\deg r$.

Let $q$ be a polynomial on $\mathbf{R}^n$ and let $m = \deg q$. By our previous discussion, the operator $T \in \mathcal{L}(\mathcal{P}_m^n(\mathbf{R}))$ given by

$$T(r) = \Delta\Big(\big(1-\|x\|^2\big)r\Big)$$

is well-defined; the linearity of $T$ follows from the linearity of $\Delta$ and distributivity on $\mathbf{R}$.

We claim that $T$ is injective. If $T(r) = 0$ for some $r \in \mathcal{P}_m^n(\mathbf{R})$ then $\big(1-\|x\|^2\big)r$ is a harmonic polynomial on $\mathbf{R}^n$ which satisfies $\big(1-\|x\|^2\big)r = 0$ for all $x \in \mathbf{R}^n$ such that $\|x\| = 1$; the fact about harmonic functions given in the exercise then implies that $\big(1-\|x\|^2\big)r = 0$. It follows that $r$ is identically zero on the open set $\{x \in \mathbf{R}^n : \|x\| = 1\}^{\mathrm{c}}$ and hence that $r$ is the zero polynomial. Thus $\operatorname{null} T = \{0\}$, i.e. $T$ is injective.

By 3.65 $T$ must be surjective. Hence there exists some $r \in \mathcal{P}_m^n(\mathbf{R})$ such that $T(r) = \Delta(-q)$, which by linearity is equivalent to

$$\Delta\Big(q + \big(1-\|x\|^2\big)r\Big) = 0.$$

Thus $p = q + \big(1-\|x\|^2\big)r$ is a harmonic polynomial on $\mathbf{R}^n$ which satisfies $p(x) = q(x)$ for all $x \in \mathbf{R}^n$ with $\|x\| = 1$.



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## 6.B. Orthonormal Bases

**Exercise 6.B.1.** Suppose $e_1, ..., e_m$ is a list of vectors in $V$ such that

$$\|a_1 e_1 + \cdots + a_m e_m\|^2 = |a_1|^2 + \cdots + |a_m|^2$$

for all $a_1, ..., a_m \in \mathbf{F}$. Show that $e_1, ..., e_m$ is an orthonormal list.

*This exercise provides a converse to 6.24.*

**Solution.** For each $k \in \{1, ..., m\}$, taking each of $a_1, ..., a_m$ to be 0 except $a_k = 1$ shows that $\|e_k\| = 1$. Suppose $j, k \in \{1, ..., m\}$ are such that $j \neq k$ and let $a \in \mathbf{F}$ be given. Observe that

$$\|e_j\|^2 = 1 \leq 1 + |a|^2 = \|e_j + a e_k\|^2 \quad \Rightarrow \quad \|e_j\| \leq \|e_j + a e_k\|.$$

It follows from Exercise 6.A.6 that $\langle e_j, e_k \rangle = 0$. Thus $e_1, ..., e_m$ is an orthonormal list.

**Exercise 6.B.2.**

(a) Suppose $\theta \in \mathbf{R}$. Show that both

$$(\cos\theta, \sin\theta), (-\sin\theta, \cos\theta) \quad \text{and} \quad (\cos\theta, \sin\theta), (\sin\theta, -\cos\theta)$$

are orthonormal bases of $\mathbf{R}^2$.

(b) Show that each orthonormal basis of $\mathbf{R}^2$ is of the form given by one of the two possibilities in (a).

**Solution.**

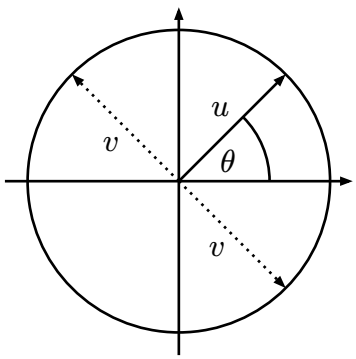
(a) Observe that

$$\langle (\cos\theta, \sin\theta), (\cos\theta, \sin\theta) \rangle = \langle (-\sin\theta, \cos\theta), (-\sin\theta, \cos\theta) \rangle = \cos^2\theta + \sin^2\theta = 1,$$

$$\langle (\cos\theta, \sin\theta), (-\sin\theta, \cos\theta) \rangle = \cos\theta \sin\theta - \cos\theta \sin\theta = 0.$$

Thus $(\cos\theta, \sin\theta), (-\sin\theta, \cos\theta)$ is an orthonormal basis of $\mathbf{R}^2$. A similar calculation shows that $(\cos\theta, \sin\theta), (\sin\theta, -\cos\theta)$ is also an orthonormal basis of $\mathbf{R}^2$.

(b) Suppose $u, v$ is an orthonormal basis of $\mathbf{R}^2$. Let $\theta$ be the angle that $u$ makes with the positive $x$-axis, as shown below.






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Note that $\|u\| = 1$, so that $u$ lies on the circle of radius 1 centered at the origin in $\mathbf{R}^2$. It follows that $u = (\cos\theta, \sin\theta)$.

Since $u$ and $v$ are orthogonal, plane geometry tells us that $v$ either makes an angle of $\theta + \frac{\pi}{2}$ or $\theta - \frac{\pi}{2}$ with the positive $x$-axis, and since $\|v\| = 1$ we know that $v$ also lies on the circle of radius 1 centered at the origin. It follows that

$$v = \big(\cos\big(\theta + \tfrac{\pi}{2}\big), \sin\big(\theta + \tfrac{\pi}{2}\big)\big) = (-\sin\theta, \cos\theta)$$
$$\text{or} \quad v = \big(\cos\big(\theta - \tfrac{\pi}{2}\big), \sin\big(\theta - \tfrac{\pi}{2}\big)\big) = (\sin\theta, -\cos\theta).$$

**Exercise 6.B.3.** Suppose $e_1, ..., e_m$ is an orthonormal list in $V$ and $v \in V$. Prove that

$$\|v\|^2 = |\langle v, e_1\rangle|^2 + \cdots + |\langle v, e_m\rangle|^2 \quad \Leftrightarrow \quad v \in \operatorname{span}(e_1, ..., e_m).$$

**Solution.** Suppose $v \in \operatorname{span}(e_1, ..., e_m)$. Note that $e_1, ..., e_m$ is linearly independent by 6.25 and hence is a basis of $\operatorname{span}(e_1, ..., e_m)$. It follows from 6.30 that

$$\|v\|^2 = |\langle v, e_1\rangle|^2 + \cdots + |\langle v, e_m\rangle|^2.$$

Now suppose that $\|v\|^2 = |\langle v, e_1\rangle|^2 + \cdots + |\langle v, e_m\rangle|^2$, let $u = \langle v, e_1\rangle e_1 + \cdots + \langle v, e_m\rangle e_m$, and note that $\|u\|^2 = \|v\|^2$ by 6.24. Observe that

$$\langle u, v\rangle = \Big\langle \textstyle\sum_{k=1}^m \langle v, e_k\rangle e_k, v\Big\rangle = \displaystyle\sum_{k=1}^m \langle\langle v, e_k\rangle e_k, v\rangle = \sum_{k=1}^m |\langle v, e_k\rangle|^2 = \|v\|^2.$$

Thus

$$\|u - v\|^2 = \|u\|^2 + \|v\|^2 - 2\operatorname{Re}\langle u, v\rangle = 2\|v\|^2 - 2\|v\|^2 = 0,$$

from which it follows that $v = u \in \operatorname{span}(e_1, ..., e_m)$.



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**Exercise 6.B.4.** Suppose $n$ is a positive integer. Prove that

$$\frac{1}{\sqrt{2\pi}}, \frac{\cos x}{\sqrt{\pi}}, \frac{\cos 2x}{\sqrt{\pi}}, ..., \frac{\cos nx}{\sqrt{\pi}}, \frac{\sin x}{\sqrt{\pi}}, \frac{\sin 2x}{\sqrt{\pi}}, ..., \frac{\sin nx}{\sqrt{\pi}}$$

is an orthonormal list of vectors in $C[-\pi, \pi]$, the vector space of continuous real-valued functions on $[-\pi, \pi]$ with inner product

$$\langle f, g \rangle = \int_{-\pi}^{\pi} fg.$$

*Hint: The following formulas should help.*

$$(\sin x)(\cos y) = \frac{\sin(x-y) + \sin(x+y)}{2}$$

$$(\sin x)(\sin y) = \frac{\cos(x-y) - \cos(x+y)}{2}$$

$$(\cos x)(\cos y) = \frac{\cos(x-y) + \cos(x+y)}{2}$$

**Solution.** We calculate, for $k \in \{1, ..., n\}$,

$$\left\langle \frac{1}{\sqrt{2\pi}}, \frac{1}{\sqrt{2\pi}} \right\rangle = \int_{-\pi}^{\pi} \left( \frac{1}{\sqrt{2\pi}} \right)^2 \mathrm{d}x = \frac{1}{2\pi} \int_{-\pi}^{\pi} \mathrm{d}x = 1;$$

$$\left\langle \frac{\cos kx}{\sqrt{\pi}}, \frac{\cos kx}{\sqrt{\pi}} \right\rangle = \int_{-\pi}^{\pi} \left( \frac{\cos kx}{\sqrt{\pi}} \right)^2 \mathrm{d}x = \frac{2}{k\pi} \int_{0}^{k\pi} \cos^2 y \, \mathrm{d}y = \frac{1}{k\pi} [y + \sin y \cos y]_{y=0}^{y=k\pi} = 1;$$

$$\left\langle \frac{\sin kx}{\sqrt{\pi}}, \frac{\sin kx}{\sqrt{\pi}} \right\rangle = \int_{-\pi}^{\pi} \left( \frac{\sin kx}{\sqrt{\pi}} \right)^2 \mathrm{d}x = \frac{2}{k\pi} \int_{0}^{k\pi} \sin^2 y \, \mathrm{d}y = \frac{1}{k\pi} [y - \sin y \cos y]_{y=0}^{y=k\pi} = 1;$$

$$\left\langle \frac{1}{\sqrt{2\pi}}, \frac{\cos kx}{\sqrt{\pi}} \right\rangle = \int_{-\pi}^{\pi} \frac{1}{\sqrt{2\pi}} \frac{\cos kx}{\sqrt{\pi}} \, \mathrm{d}x = \frac{\sqrt{2}}{k\pi} \int_{0}^{k\pi} \cos y \, \mathrm{d}y = \frac{\sqrt{2}}{k\pi} [\sin y]_{y=0}^{y=k\pi} = 0;$$

$$\left\langle \frac{1}{\sqrt{2\pi}}, \frac{\sin kx}{\sqrt{\pi}} \right\rangle = \int_{-\pi}^{\pi} \frac{1}{\sqrt{2\pi}} \frac{\sin kx}{\sqrt{\pi}} \, \mathrm{d}x = 0,$$

where we have used that $\sin kx$ is an odd function for the last equality. For $j, k \in \{1, ..., n\}$ such that $j \neq k$, we have

$$\left\langle \frac{\cos jx}{\sqrt{\pi}}, \frac{\cos kx}{\sqrt{\pi}} \right\rangle = \int_{-\pi}^{\pi} \frac{\cos jx}{\sqrt{\pi}} \frac{\cos kx}{\sqrt{\pi}} \, \mathrm{d}x = \frac{1}{\pi} \int_{0}^{\pi} \cos((j-k)x) + \cos((j+k)x) \, \mathrm{d}x$$

$$= \frac{1}{\pi} \left[ \frac{\sin((j-k)x)}{j-k} \right]_{x=0}^{x=\pi} + \frac{1}{\pi} \left[ \frac{\sin((j+k)x)}{j+k} \right]_{x=0}^{x=\pi} = 0;$$



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$$\left\langle \frac{\sin jx}{\sqrt{\pi}}, \frac{\sin kx}{\sqrt{\pi}} \right\rangle = \int_{-\pi}^{\pi} \frac{\sin jx}{\sqrt{\pi}} \frac{\sin kx}{\sqrt{\pi}} \, \mathrm{d}x = \frac{1}{\pi} \int_{0}^{\pi} \cos((j-k)x) - \cos((j+k)x) \, \mathrm{d}x$$

$$= \frac{1}{\pi} \left[ \frac{\sin((j-k)x)}{j-k} \right]_{x=0}^{x=\pi} - \frac{1}{\pi} \left[ \frac{\sin((j+k)x)}{j+k} \right]_{x=0}^{x=\pi} = 0.$$

Finally, for any $j, k \in \{1, ..., n\}$ we have

$$\left\langle \frac{\cos jx}{\sqrt{\pi}}, \frac{\sin kx}{\sqrt{\pi}} \right\rangle = \int_{-\pi}^{\pi} \frac{\cos jx}{\sqrt{\pi}} \frac{\sin kx}{\sqrt{\pi}} \, \mathrm{d}x = 0,$$

where we have used that $\cos jx \sin kx$ is an odd function for the last equality.

**Exercise 6.B.5.** Suppose $f : [-\pi, \pi] \to \mathbf{R}$ is continuous. For each nonnegative integer $k$, define

$$a_k = \frac{1}{\sqrt{\pi}} \int_{-\pi}^{\pi} f(x) \cos(kx) \, dx \quad \text{and} \quad b_k = \frac{1}{\sqrt{\pi}} \int_{-\pi}^{\pi} f(x) \sin(kx) \, dx.$$

Prove that

$$\frac{a_0^2}{2} + \sum_{k=1}^{\infty} (a_k^2 + b_k^2) \le \int_{-\pi}^{\pi} f^2.$$

*The inequality above is actually an equality for all continuous functions $f : [-\pi, \pi] \to \mathbf{R}$. However, proving that this inequality is an equality involves Fourier series techniques beyond the scope of this book.*

**Solution.** Consider $C[-\pi, \pi]$, the vector space of continuous real-valued functions on $[-\pi, \pi]$, with inner product

$$\langle f, g \rangle = \int_{-\pi}^{\pi} fg.$$

As we showed in Exercise 6.B.5,

$$\frac{1}{\sqrt{2\pi}}, \frac{\cos x}{\sqrt{\pi}}, \frac{\cos 2x}{\sqrt{\pi}}, ..., \frac{\cos nx}{\sqrt{\pi}}, \frac{\sin x}{\sqrt{\pi}}, \frac{\sin 2x}{\sqrt{\pi}}, ..., \frac{\sin nx}{\sqrt{\pi}}$$

is an orthonormal list of vectors in $C[-\pi, \pi]$ for any $n \ge 1$. Observe that, for $k \ge 1$,

$$a_0 = \frac{1}{\sqrt{\pi}} \int_{-\pi}^{\pi} f(x) \, \mathrm{d}x = \sqrt{2} \left\langle f, \frac{1}{\sqrt{2\pi}} \right\rangle;$$

$$a_k = \frac{1}{\sqrt{\pi}} \int_{-\pi}^{\pi} f(x) \cos(kx) \, \mathrm{d}x = \left\langle f, \frac{\cos kx}{\sqrt{\pi}} \right\rangle;$$

$$b_k = \frac{1}{\sqrt{\pi}} \int_{-\pi}^{\pi} f(x) \sin(kx) \, \mathrm{d}x = \left\langle f, \frac{\sin kx}{\sqrt{\pi}} \right\rangle.$$



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Thus, by Bessel's inequality (6.26),

$$\frac{a_0^2}{2} + \sum_{k=1}^{n}(a_k^2 + b_k^2) \leq \|f\|^2 = \int_{-\pi}^{\pi} f^2.$$

for all $n \geq 1$. It then follows from the monotone convergence theorem that $\sum_{k=1}^{\infty}(a_k^2 + b_k^2)$ is a convergent series. Furthermore,

$$\frac{a_0^2}{2} + \sum_{k=1}^{\infty}(a_k^2 + b_k^2) \leq \int_{-\pi}^{\pi} f^2.$$

**Exercise 6.B.6.** Suppose $e_1, ..., e_n$ is an orthonormal basis of $V$.

(a) Prove that if $v_1, ..., v_n$ are vectors in $V$ such that

$$\|e_k - v_k\| < \frac{1}{\sqrt{n}}$$

for each $k$, then $v_1, ..., v_n$ is a basis of $V$.

(b) Show that there exist $v_1, ..., v_n \in V$ such that

$$\|e_k - v_k\| \leq \frac{1}{\sqrt{n}}$$

for each $k$, but $v_1, ..., v_n$ is not linearly independent.

*This exercise states in (a) that an appropriately small perturbation of an orthonormal basis is a basis. Then (b) shows that the number $1/\sqrt{n}$ on the right side of the inequality in (a) cannot be improved upon.*

**Solution.**

(a) It will suffice to show that $v_1, ..., v_n$ is linearly independent, so suppose that $\sum_{k=1}^{n} a_k v_k = 0$ and observe that

$$\begin{aligned}
\textstyle\sum_{k=1}^{n} |a_k|^2 &= \left\|\textstyle\sum_{k=1}^{n} a_k e_k\right\|^2 && (6.24)\\
&= \left\|\textstyle\sum_{k=1}^{n} a_k (e_k - v_k)\right\|^2 && \left(\textstyle\sum_{k=1}^{n} a_k v_k = 0\right)\\
&\leq \left(\textstyle\sum_{k=1}^{n} |a_k| \|e_k - v_k\|\right)^2 && \text{(triangle inequality)}\\
&\leq \left(\textstyle\sum_{k=1}^{n} |a_k|^2\right)\left(\textstyle\sum_{k=1}^{n} \|e_k - v_k\|^2\right). && \text{(Cauchy-Schwarz inequality)}
\end{aligned}$$

By assumption we have $\sum_{k=1}^{n} \|e_k - v_k\|^2 < 1$. It follows that $\sum_{k=1}^{n} |a_k|^2 = 0$, which is the case if and only if each $a_k = 0$. Thus $v_1, ..., v_n$ is linearly independent.

(b) For each $k \in \{1, ..., n\}$ let

$$v_k = e_k - \frac{e_1 + \cdots + e_n}{n} = \left(-\frac{1}{n}\right)e_1 + \cdots + \left(1 - \frac{1}{n}\right)e_k + \cdots + \left(-\frac{1}{n}\right)e_n.$$



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Notice that $v_1 + \cdots + v_n = 0$, so that $v_1, ..., v_n$ is linearly dependent. Furthermore, for each $k \in \{1, ..., n\}$, 6.24 shows that

$$\|e_k - v_k\| = \left\| \frac{1}{n} e_1 + \cdots + \frac{1}{n} e_k + \cdots + \frac{1}{n} e_n \right\| = \frac{1}{n} \|e_1 + \cdots + e_n\| = \frac{1}{n} \cdot \sqrt{n} = \frac{1}{\sqrt{n}}.$$

**Exercise 6.B.7.** Suppose $T \in \mathcal{L}(\mathbf{R}^3)$ has an upper-triangular matrix with respect to the basis $(1, 0, 0), (1, 1, 1), (1, 1, 2)$. Find an orthonormal basis of $\mathbf{R}^3$ with respect to which $T$ has an upper-triangular matrix.

**Solution.** Performing the Gram-Schmidt procedure on the basis $(1, 0, 0), (1, 1, 1), (1, 1, 2)$ yields the orthonormal basis

$$(1, 0, 0), \quad \left(0, \frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right), \quad \left(0, -\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right).$$

As the proof of 6.37 shows, the matrix of $T$ with respect to this orthonormal basis must also be upper-triangular.

**Exercise 6.B.8.** Make $\mathcal{P}_2(\mathbf{R})$ into an inner product space by defining $\langle p, q \rangle = \int_0^1 pq$ for all $p, q \in \mathcal{P}_2(\mathbf{R})$.

(a) Apply the Gram-Schmidt procedure to the basis $1, x, x^2$ to produce an orthonormal basis of $\mathcal{P}_2(\mathbf{R})$.

(b) The differentiation operator (the operator that takes $p$ to $p'$) on $\mathcal{P}_2(\mathbf{R})$ has an upper-triangular matrix with respect to the basis $1, x, x^2$, which is not an orthonormal basis. Find the matrix of the differentiation operator on $\mathcal{P}_2(\mathbf{R})$ with respect to the orthonormal basis produced in (a) and verify that this matrix is upper triangular, as expected from the proof of 6.37.

**Solution.**

(a) By applying the Gram-Schmidt procedure we obtain the orthonormal basis

$$1, \quad 2\sqrt{3}(x - \tfrac{1}{2}), \quad 6\sqrt{5}(x^2 - x + \tfrac{1}{6}).$$

(b) Some routine calculations reveal that the matrix of $T$ with respect to the orthonormal basis found in part (a) is

$$\begin{pmatrix} 0 & 0 & 0 \\ 0 & 2\sqrt{3} & 0 \\ 0 & 0 & 6\sqrt{\frac{5}{3}} \end{pmatrix},$$

which is indeed upper-triangular.



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**Exercise 6.B.9.** Suppose $e_1, ..., e_m$ is the result of applying the Gram-Schmidt procedure to a linearly independent list $v_1, ..., v_m \in V$. Prove that $\langle v_k, e_k \rangle > 0$ for each $k = 1, ..., m$.

**Solution.** Let $u_1, ..., u_m$ be a list in $V$ and let $f_1, ..., f_m$ be an orthonormal list in $V$ with the property that $\text{span}(f_1, ..., f_k) = \text{span}(u_1, ..., u_k)$ for each $k \in \{1, ..., m\}$. We will prove that

$$\langle u_k, f_k \rangle = 0 \text{ for some } k \in \{1, ..., m\} \quad \Rightarrow \quad u_1, ..., u_m \text{ is linearly dependent.}$$

The contrapositive of this implication will give us the desired result.

Suppose that there exists some $k \in \{1, ..., m\}$ such that $\langle u_k, f_k \rangle = 0$. Notice that $u_k \in \text{span}(u_1, ..., u_k) = \text{span}(f_1, ..., f_k)$; it follows from Exercise 6.B.3 that

$$\|u_k\|^2 = |\langle u_k, f_1 \rangle|^2 + \cdots + |\langle u_k, f_{k-1} \rangle|^2 + |\langle u_k, f_k \rangle|^2 = |\langle u_k, f_1 \rangle|^2 + \cdots + |\langle u_k, f_{k-1} \rangle|^2.$$

Another application of Exercise 6.B.3 shows that $u_k \in \text{span}(f_1, ..., f_{k-1}) = \text{span}(u_1, ..., u_{k-1})$. Thus $u_1, ..., u_m$ is linearly dependent.

**Exercise 6.B.10.** Suppose $v_1, ..., v_m$ is a linearly independent list in $V$. Explain why the orthonormal list produced by the formulas of the Gram-Schmidt procedure (6.32) is the only orthonormal list $e_1, ..., e_m$ in $V$ such that $\langle v_k, e_k \rangle > 0$ and $\text{span}(v_1, ..., v_k) = \text{span}(e_1, ..., e_k)$ for each $k = 1, ..., m$.

*The result in this exercise is used in the proof of 7.58.*

**Solution.** Here is a useful lemma.



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**Lemma L.10.** Suppose $v_1, ..., v_m$ is a linearly independent list in $V$ and let $e_1, ..., e_m$ be the orthonormal list obtained by applying the Gram-Schmidt procedure to $v_1, ..., v_m$. Let $S = \{\lambda \in \mathbf{F} : |\lambda| = 1\}$ (if $\mathbf{F} = \mathbf{R}$ then $S = \{-1, 1\}$ and if $\mathbf{F} = \mathbf{C}$ then $S$ is the unit circle in the complex plane) and let $S^m$ be the collection of functions $\{1, ..., m\} \to S$. The orthonormal lists $u_1, ..., u_m$ satisfying $\text{span}(u_1, ..., u_k) = \text{span}(v_1, ..., v_k)$ for each $k \in \{1, ..., m\}$ are exactly those of the form $f(1)e_1, ..., f(m)e_m$ for some $f \in S^m$.

*Proof.* Let $f \in S^m$ be given and suppose $j, k \in \{1, ..., m\}$ are such that $j \neq k$. Observe that

$$\|f(k)e_k\| = |f(k)|\|e_k\| = 1 \quad \text{and} \quad \langle f(j)e_j, f(k)e_k\rangle = f(j)\overline{f(k)}\langle e_j, e_k\rangle = 0.$$

Furthermore, since $0 \notin S$ we have for each $k \in \{1, ..., m\}$,

$$\text{span}(f(1)e_1, ..., f(k)e_k) = \text{span}(e_1, ..., e_k) = \text{span}(v_1, ..., v_k).$$

Thus $f(1)e_1, ..., f(m)e_m$ is an orthonormal list satisfying

$$\text{span}(f(1)e_1, ..., f(k)e_k) = \text{span}(v_1, ..., v_k)$$

for each $k \in \{1, ..., m\}$.

Now suppose that $u_1, ..., u_m$ is an orthonormal list satisfying

$$\text{span}(u_1, ..., u_k) = \text{span}(v_1, ..., v_k) = \text{span}(e_1, ..., e_k)$$

for each $k \in \{1, ..., m\}$. In particular $\text{span}(e_1) = \text{span}(u_1)$, from which it follows that $u_1 = \lambda_1 e_1$ for some $\lambda_1 \in \mathbf{F}$. Because $\|u_1\| = \|e_1\| = 1$ we see that $|\lambda_1| = 1$, so that $\lambda_1 \in S$; let $f(1) = \lambda_1$.

Given that $\text{span}(e_1, e_2) = \text{span}(u_1, u_2)$, notice that $e_1, e_2$ is an orthonormal basis of $\text{span}(e_1, e_2)$ and that $u_2 \in \text{span}(e_1, e_2)$. It follows from 6.30 that

$$u_2 = \langle u_2, e_1\rangle e_1 + \langle u_2, e_2\rangle e_2.$$

The orthonormality of the list $u_1, u_2$ shows that

$$0 = \langle u_1, u_2\rangle = \langle f(1)e_1, u_2\rangle = f(1)\langle e_1, u_2\rangle,$$

which implies $\langle e_1, u_2\rangle = 0$ since $f(1) \neq 0$. Thus $u_2 = \lambda_2 e_2$ for some $\lambda \in \mathbf{F}$. Because $\|u_2\| = \|e_2\| = 1$ we see that $|\lambda_2| = 1$, so that $\lambda_2 \in S$; let $f(2) = \lambda_2$.

By continuing in this manner we obtain an $f \in S^m$ such that $u_k = f(k)e_k$ for each $k \in \{1, ..., m\}$. □

Returning to the exercise, 6.32 and Exercise 6.B.9 show that the orthonormal list $e_1, ..., e_m$ produced by the Gram-Schmidt procedure indeed satisfies $\langle v_k, e_k\rangle > 0$ and

$$\text{span}(v_1, ..., v_k) = \text{span}(e_1, ..., e_k)$$

for each $k \in \{1, ..., m\}$.



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Conversely, suppose that $u_1, ..., u_m$ is an orthonormal list in $V$ such that $\langle v_k, u_k \rangle > 0$ and

$$\operatorname{span}(v_1, ..., v_k) = \operatorname{span}(u_1, ..., u_k)$$

for each $k \in \{1, ..., m\}$. Lemma L.10 shows that $u_1, ..., u_m$ is of the form $f(1)e_1, ..., f(m)e_m$ for some $f : \{1, ..., m\} \to \mathbf{F}$ satisfying $|f(k)| = 1$ for each $k \in \{1, ..., m\}$. For such a $k$, observe that

$$\langle v_k, u_k \rangle > 0 \quad \Leftrightarrow \quad f(k)\langle v_k, e_k \rangle > 0.$$

Since $\langle v_k, e_k \rangle > 0$ by Exercise 6.B.9, $f(k)$ must be a positive real number. Combining this with $|f(k)| = 1$, we see that $f(k) = 1$ for each $k \in \{1, ..., m\}$. Thus $u_1, ..., u_m$ is nothing but $e_1, ..., e_m$.

**Exercise 6.B.11.** Find a polynomial $q \in \mathcal{P}_2(\mathbf{R})$ such that $p\left(\frac{1}{2}\right) = \int_0^1 pq$ for every $p \in \mathcal{P}_2(\mathbf{R})$.

**Solution.** Equip $\mathcal{P}_2(\mathbf{R})$ with the inner product $\langle p, q \rangle = \int_0^1 pq$ and define $\varphi \in (\mathcal{P}_2(\mathbf{R}))'$ by $\varphi(p) = p\left(\frac{1}{2}\right)$. As the proof of 6.42 shows, if we take

$$q = \varphi(e_1)e_1 + \varphi(e_2)e_2 + \varphi(e_3)e_3 = -15x^2 + 15x - \tfrac{3}{2},$$

where $e_1, e_2, e_3$ is the orthonormal basis of $\mathcal{P}_2(\mathbf{R})$ found in Exercise 6.B.8 (a), then

$$\varphi(p) = p\left(\tfrac{1}{2}\right) = \langle p, q \rangle = \int_0^1 pq$$

for every $p \in \mathcal{P}_2(\mathbf{R})$.

**Exercise 6.B.12.** Find a polynomial $q \in \mathcal{P}_2(\mathbf{R})$ such that

$$\int_0^1 p(x)\cos(\pi x)\, dx = \int_0^1 pq$$

for every $p \in \mathcal{P}_2(\mathbf{R})$.

**Solution.** Equip $\mathcal{P}_2(\mathbf{R})$ with the inner product $\langle p, q \rangle = \int_0^1 pq$ and define $\varphi \in (\mathcal{P}_2(\mathbf{R}))'$ by $\varphi(p) = \int_0^1 p(x)\cos(\pi x)\, \mathrm{d}x$. As the proof of 6.42 shows, if we take

$$q = \varphi(e_1)e_1 + \varphi(e_2)e_2 + \varphi(e_3)e_3 = -\frac{24}{\pi^2}\left(x - \tfrac{1}{2}\right),$$

where $e_1, e_2, e_3$ is the orthonormal basis of $\mathcal{P}_2(\mathbf{R})$ found in Exercise 6.B.8 (a), then

$$\varphi(p) = \int_0^1 p(x)\cos(\pi x)\, \mathrm{d}x = \langle p, q \rangle = \int_0^1 pq$$

for every $p \in \mathcal{P}_2(\mathbf{R})$.



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**Exercise 6.B.13.** Show that a list $v_1, ..., v_m$ of vectors in $V$ is linearly dependent if and only if the Gram-Schmidt formula in 6.32 produces $f_k = 0$ for some $k \in \{1, ..., m\}$.

*This exercise gives an alternative to Gaussian elimination techniques for determining whether a list of vectors in an inner product space is linearly dependent.*

**Solution.** If $v_1, ..., v_m$ is linearly independent then each $f_k$ must be non-zero, as the proof of 6.32 shows.

Suppose that $v_1, ..., v_m$ is linearly dependent. If $v_1 = 0$ then $f_1 = 0$; otherwise, let $k \in \{2, ..., m\}$ be the least integer such that $v_k \in \text{span}(v_1, ..., v_{k-1})$ and note that $v_1, ..., v_{k-1}$ is linearly independent. This linear independence allows us to construct $f_1, ..., f_{k-1}$ as in 6.32 so that:

- each $f_i \neq 0$;
- $\text{span}(v_1, ..., v_i) = \text{span}(f_1, ..., f_i)$ for each $i \in \{1, ..., k-1\}$;
- $f_1, ..., f_{i-1}$ is pairwise orthogonal.

It follows that $v_k \in \text{span}(f_1, ..., f_{k-1})$, say $v_k = a_1 f_1 + \cdots + a_{k-1} f_{k-1}$. Notice that, for each $i \in \{1, ..., k-1\}$,

$$\langle v_k, f_i \rangle = \langle a_i f_i, f_i \rangle = a_i \|f_i\|^2 \quad \Rightarrow \quad a_i = \frac{\langle v_k, f_i \rangle}{\|f_i\|^2}.$$

Thus, using the formula for $f_k$ in 6.32,

$$f_k = v_k - \sum_{i=1}^{k-1} \frac{\langle v_k, f_i \rangle}{\|f_i\|^2} f_i = v_k - \sum_{i=1}^{k-1} a_i f_i = v_k - v_k = 0.$$

**Exercise 6.B.14.** Suppose $V$ is a real inner product space and $v_1, ..., v_m$ is a linearly independent list of vectors in $V$. Prove that there exist exactly $2^m$ orthonormal lists $e_1, ..., e_m$ of vectors in $V$ such that

$$\text{span}(v_1, ..., v_k) = \text{span}(e_1, ..., e_k)$$

for each $k \in \{1, ..., m\}$.

**Solution.** Let $e_1, ..., e_m$ be the orthonormal list obtained by applying the Gram-Schmidt procedure to $v_1, ..., v_m$. Lemma L.10 shows that the orthonormal lists $u_1, ..., u_m$ satisfying $\text{span}(v_1, ..., v_k) = \text{span}(u_1, ..., u_k)$ for each $k \in \{1, ..., m\}$ are precisely those of the form

$$f(1)e_1, ..., f(m)e_m$$

for some $f : \{1, ..., m\} \to \{-1, 1\}$. It is straightforward to verify that there are $2^m$ such functions, and that each $f : \{1, ..., m\} \to \{-1, 1\}$ gives a distinct orthonormal list $f(1)e_1, ..., f(m)e_m$. Thus there $2^m$ orthonormal lists $u_1, ..., u_m$ of vectors in $V$ such that

$$\text{span}(v_1, ..., v_k) = \text{span}(u_1, ..., u_k)$$



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for each $k \in \{1, ..., m\}$.

**Exercise 6.B.15.** Suppose $\langle \cdot, \cdot \rangle_1$ and $\langle \cdot, \cdot \rangle_2$ are inner products on $V$ such that $\langle u, v \rangle_1 = 0$ if and only if $\langle u, v \rangle_2 = 0$. Prove that there is a positive number $c$ such that $\langle u, v \rangle_1 = \langle u, v \rangle_2$ for every $u, v \in V$.

*This exercise shows that if two inner products have the same pairs of orthogonal vectors, then each of the inner products is a scalar multiple of the other inner product.*

**Solution.** If $V = \{0\}$ then we may take any $c > 0$ we like, since the only inner product on $V$ is the map $(0, 0) \mapsto 0$. Suppose therefore that $V \neq \{0\}$ and for each non-zero $v \in V$ define

$$c_v = \frac{\langle v, v \rangle_1}{\langle v, v \rangle_2};$$

notice that $c_v$ is positive. Suppose $u, v \in V$ are non-zero. Using orthogonal decomposition 6.13, we have

$$\left\langle u - \frac{\langle u, v \rangle_2}{\langle v, v \rangle_2} v, v \right\rangle_2 = 0.$$

Our assumption is that orthogonality with respect to $\langle \cdot, \cdot \rangle_2$ is equivalent to orthogonality with respect to $\langle \cdot, \cdot \rangle_1$ and thus

$$\left\langle u - \frac{\langle u, v \rangle_2}{\langle v, v \rangle_2} v, v \right\rangle_1 = 0 \quad \Leftrightarrow \quad \langle u, v \rangle_1 - \frac{\langle u, v \rangle_2}{\langle v, v \rangle_2} \langle v, v \rangle_1 = 0 \quad \Leftrightarrow \quad \langle u, v \rangle_1 = c_v \langle u, v \rangle_2.$$

Reversing the roles of $u$ and $v$ shows that $\langle v, u \rangle_1 = c_u \langle v, u \rangle_2$ and combining this with conjugate symmetry gives us

$$c_v \langle u, v \rangle_2 = \langle u, v \rangle_1 = \overline{\langle v, u \rangle_1} = \overline{c_u \langle v, u \rangle_2} = c_u \langle u, v \rangle_2.$$

Thus, for all non-zero $u, v \in V$, we have

(1) $\langle u, v \rangle_1 = c_v \langle u, v \rangle_2$;

(2) $c_v \langle u, v \rangle_2 = c_u \langle u, v \rangle_2$.

Given non-zero $u, v \in V$, there exists a non-zero $w \in V$ such that $\langle w, u \rangle_2 \neq 0$ and $\langle v, w \rangle_2 \neq 0$: if $\langle u, v \rangle_2 \neq 0$ then take $w = u$ and if $\langle u, v \rangle_2 = 0$ then take $w = u + v$. Using (2), it follows that

$$c_u \langle w, u \rangle_2 = c_w \langle w, u \rangle_2 \quad \text{and} \quad c_w \langle v, w \rangle_2 = c_v \langle v, w \rangle_2.$$

Since $\langle w, u \rangle_2 \neq 0$ and $\langle v, w \rangle_2 \neq 0$, these two equations imply that $c_u = c_w = c_v$. If we denote this common value by $c$ (noting that $c > 0$), then we have shown that $c_v = c$ for all non-zero $v \in V$. It follows from (1) that $\langle u, v \rangle_1 = c \langle u, v \rangle_2$ for all non-zero $u, v \in V$. Certainly this equation also holds if $u = 0$ or $v = 0$ and thus $\langle u, v \rangle_1 = c \langle u, v \rangle_2$ for all $u, v \in V$.



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**Exercise 6.B.16.** Suppose $V$ is finite-dimensional. Suppose $\langle \cdot, \cdot \rangle_1, \langle \cdot, \cdot \rangle_2$ are inner products on $V$ with corresponding norms $\| \cdot \|_1$ and $\| \cdot \|_2$. Prove that there exists a positive number $c$ such that $\|v\|_1 \leq c\|v\|_2$ for every $v \in V$.

**Solution.** By 6.35 there exists an orthonormal basis $e_1, ..., e_n$ of $V$ with respect to $\langle \cdot, \cdot \rangle_2$. Let $v \in V$ be given, so that $v = a_1 e_1 + \cdots + a_n e_n$ for some scalars $a_1, ..., a_n$. Observe that

$$|a_1| + \cdots + |a_n| \leq n \max\{|a_1|, ..., |a_n|\} \leq n\sqrt{|a_1|^2 + \cdots + |a_n|^2} = n\|v\|_2, \tag{1}$$

where the last equality follows from 6.24. If we let $M = \max\left\{\|e_1\|_1, ..., \|e_n\|_1\right\}$, which is positive since each $e_k \neq 0$, then it follows from (1) and the triangle inequality that

$$\|v\|_1 \leq |a_1|\|e_1\|_1 + \cdots + |a_n|\|e_n\|_1 \leq M(|a_1| + \cdots + |a_n|) \leq nM\|v\|_2.$$

Thus the desired positive constant is $c = nM$.

**Exercise 6.B.17.** Suppose $\mathbf{F} = \mathbf{C}$ and $V$ is finite-dimensional. Prove that if $T$ is an operator on $V$ such that 1 is the only eigenvalue of $T$ and $\|Tv\| \leq \|v\|$ for all $v \in V$, then $T$ is the identity operator.

**Solution.** By Schur's theorem (6.38), there is an orthonormal basis $e_1, ..., e_n$ of $V$ with respect to which the matrix of $T$ is upper-triangular; because the only eigenvalue of $T$ is 1, the diagonal entries of this matrix must equal 1. Thus, for each $k \in \{1, ..., n\}$,

$$Te_k = A_{1,k}e_1 + \cdots + A_{1,k-1}e_{k-1} + e_k \quad \Rightarrow \quad \|Te_k\|^2 = |A_{1,k}|^2 + \cdots + |A_{1,k-1}|^2 + 1,$$

where we have used 6.24. By assumption $\|Te_k\|^2 \leq \|e_k\|^2 = 1$ and thus

$$|A_{1,k}|^2 + \cdots + |A_{1,k-1}|^2 \leq 0 \quad \Rightarrow \quad A_{1,k} = \cdots = A_{1,k-1} = 0.$$

It follows that the matrix of $T$ with respect to $e_1, ..., e_n$ is diagonal. Since each diagonal entry is equal to 1, we may conclude that $T$ is the identity operator.

**Exercise 6.B.18.** Suppose $u_1, ..., u_m$ is a linearly independent list in $V$. Show that there exists $v \in V$ such that $\langle u_k, v \rangle = 1$ for all $k \in \{1, ..., m\}$.

**Solution.** Let $U = \text{span}(u_1, ..., u_m)$, so that $u_1, ..., u_m$ is a basis of $U$, let $\varphi_1, ..., \varphi_m$ be the dual basis of $U'$, and let $\varphi = \varphi_1 + \cdots + \varphi_m$. The Riesz representation theorem (6.42) shows that there is a unique $v \in U$ such that $\varphi(u) = \langle u, v \rangle$ for each $u \in U$. In particular, $\langle u_k, v \rangle = \varphi(u_k) = 1$ for each $k \in \{1, ..., m\}$.



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**Exercise 6.B.19.** Suppose $v_1, ..., v_n$ is a basis of $V$. Prove that there exists a basis $u_1, ..., u_n$ of $V$ such that

$$\langle v_j, u_k \rangle = \begin{cases} 0 & \text{if } j \neq k, \\ 1 & \text{if } j = k. \end{cases}$$

**Solution.** Let $\varphi_1, ..., \varphi_n$ be the dual basis of $v_1, ..., v_n$. For each $k \in \{1, ..., n\}$, the Riesz representation theorem (6.42) shows that there is some $u_k \in V$ such that $\varphi_k(v) = \langle v, u_k \rangle$ for every $v \in V$. It follows that

$$\langle v_j, u_k \rangle = \varphi_k(v_j) = \begin{cases} 0 & \text{if } j \neq k, \\ 1 & \text{if } j = k. \end{cases}$$

Suppose $a_1, ..., a_n$ are scalars such that $a_1 u_1 + \cdots + a_n u_n = 0$. For each $k \in \{1, ..., n\}$, observe that

$$0 = \langle a_1 u_1 + \cdots + a_n u_n, v_k \rangle = a_1 \langle u_1, v_k \rangle + \cdots + a_n \langle u_n, v_k \rangle = a_k.$$

It follows that $u_1, ..., u_n$ is linearly independent and hence forms a basis of $V$.

**Exercise 6.B.20.** Suppose $\mathbf{F} = \mathbf{C}, V$ is finite-dimensional, and $\mathcal{E} \subseteq \mathcal{L}(V)$ is such that

$$ST = TS$$

for all $S, T \in \mathcal{E}$. Prove that there is an orthonormal basis of $V$ with respect to which every element of $\mathcal{E}$ has an upper-triangular matrix.

> *This exercise strengthens Exercise 9(b) in Section 5E (in the context of inner product spaces) by asserting that the basis in that exercise can be chosen to be orthonormal.*

**Solution.** By Exercise 5.E.9 (b) there is a basis $v_1, ..., v_n$ of $V$ with respect to which every element of $\mathcal{E}$ has an upper-triangular matrix, i.e. such that

$$Te_k \in \operatorname{span}(v_1, ..., v_k)$$

for every $k \in \{1, ..., m\}$ and every $T \in \mathcal{E}$. Let $e_1, ..., e_n$ be the orthonormal basis obtained by applying the Gram-Schmidt procedure (6.32) to $v_1, ..., v_n$ and note that

$$Te_k \in \operatorname{span}(v_1, ..., v_k) = \operatorname{span}(e_1, ..., e_k)$$

for every $k \in \{1, ..., m\}$ and every $T \in \mathcal{E}$. Thus $e_1, ..., e_n$ is an orthonormal basis of $V$ with respect to which every element of $\mathcal{E}$ has an upper-triangular matrix.

**Exercise 6.B.21.** Suppose $\mathbf{F} = \mathbf{C}, V$ is finite-dimensional, $T \in \mathcal{L}(V)$, and all eigenvalues of $T$ have absolute value less than 1. Let $\varepsilon > 0$. Prove that there exists a positive integer $m$ such that $\|T^m v\| \leq \varepsilon \|v\|$ for every $v \in V$.



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**Solution.** We will first translate the problem into a statement involving column vectors in $\mathbf{C}^{n,1}$ and matrices in $\mathbf{C}^{n,n}$, where $n = \dim V$. We shall use the following notation, which differs from the notation of the question.

- $\langle \cdot, \cdot \rangle_V : V \times V \to \mathbf{C}$. This is the given inner product on $V$.
- $\| \cdot \|_V : V \to [0, \infty)$. This is the norm on $V$ arising from $\langle \cdot, \cdot \rangle_V$, i.e.
$$\|v\|_V = \sqrt{\langle v, v \rangle_V}.$$
- $\langle \cdot, \cdot \rangle : \mathbf{C}^{n,1} \times \mathbf{C}^{n,1} \to \mathbf{C}$. This is the Euclidean inner product on $\mathbf{C}^{n,1}$, i.e.
$$\langle x, y \rangle = \sum_{k=1}^{n} x_k \overline{y_k}, \quad \text{where } x = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} \text{ and } y = \begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix}.$$
- $\| \cdot \| : \mathbf{C}^{n,1} \to [0, \infty)$. This is the Euclidean norm on $\mathbf{C}^{n,1}$ arising from $\langle \cdot, \cdot \rangle$, i.e.
$$\|x\| = \sqrt{\langle x, x \rangle} = \left( \sum_{k=1}^{n} |x_k|^2 \right)^{1/2}, \quad \text{where } x = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}.$$

Schur's theorem (6.38) implies that there is an orthonormal (with respect to $\langle \cdot, \cdot \rangle_V$) basis $e_1, ..., e_n$ of $V$ such that the matrix $A \in \mathbf{C}^{n,n}$ of $T$ with respect to $e_1, ..., e_n$ is upper-triangular. Given $v = x_1 e_1 + \cdots + x_n e_n \in V$, observe that

$$\|v\|_V = \left( \sum_{k=1}^{n} |x_k|^2 \right)^{1/2} = \|x\|, \quad \text{where } x = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}.$$

Thus it will suffice to show that there exists a positive integer $m$ such that

$$\|A^m x\| \leq \varepsilon \|x\| \text{ for all } x \in \mathbf{C}^{n,1}.$$

In what follows, by a strictly upper-triangular matrix we mean a matrix that is upper-triangular and whose diagonal entries are zero. We shall use the following two easily verified facts about strictly upper-triangular matrices:

(i) if $D \in \mathbf{C}^{n,n}$ is diagonal and $N \in \mathbf{C}^{n,n}$ is strictly upper-triangular then $DN$ and $ND$ are both strictly upper-triangular;

(ii) if $N \in \mathbf{C}^{n,n}$ is strictly upper-triangular then $N^n = 0$.

Let $D$ be the diagonal matrix whose diagonal entries are exactly those of $U$ and let $N$ be the strictly upper-triangular matrix whose entries above the diagonal are exactly those of $U$, so that $U = D + N$. Let

$$\rho = \max\{|D_{1,1}|, ..., |D_{n,n}|\} = \max\{|U_{1,1}|, ..., |U_{n,n}|\}$$

and note that $0 \leq \rho < 1$ since the diagonal elements of $D$ and $U$ are precisely the eigenvalues of $T$. Note further that

$$\|Dx\| = \left( |D_{1,1}|^2 |x_1|^2 + \cdots + |D_{n,n}|^2 |x_n|^2 \right)^{1/2} \leq \rho \|x\|$$



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for any $x \in \mathbf{C}^{n,1}$. Let

$$C = \sum_{j=1}^{n} \sum_{k=1}^{n} |N_{j,k}|^2;$$

a calculation similar to the one given in Exercise 6.A.19 shows that $\|Nx\| \leq C\|x\|$ for all $x \in \mathbf{C}^{n,1}$. Putting everything together, for any integer $m \geq n$ we have the inequality

$$\|U^m x\| = \|(D+N)^m x\| \leq \sum_{k=0}^{n-1} \binom{m}{k} \rho^{m-k} C^k \|x\|.$$

This can be shown using induction, but is best illustrated by example. If $m = 3$ and $n = 2$ then (i) and (ii) show that $DNN = NDN = NND = NNN = 0$. Thus

$$\begin{aligned}
\|U^m x\| &= \|(D+N)^m x\| \\
&\leq \|(DDD + DDN + DND + NDD + DNN + NDN + NND + NNN)x\| \\
&= \|(DDD + DDN + DND + NDD)x\| \\
&\leq \|DDDx\| + \|DDNx\| + \|DNDx\| + \|NDDx\| \\
&\leq \rho(\rho(\rho\|x\|)) + \rho(\rho(C\|x\|)) + \rho(C(\rho\|x\|)) + C(\rho(\rho\|x\|)) \\
&= \sum_{k=0}^{1} \binom{3}{k} \rho^{3-k} C^k \|x\|.
\end{aligned}$$

For any $0 \leq k \leq n-1$ we have $\binom{m}{k} \leq m^{n-1}$ and $\rho^{m-k} \leq \rho^{m-n+1}$, since $0 \leq \rho < 1$. Thus, letting $\mu = \max\{1, C, ..., C^{n-1}\}$, we have the inequality

$$\|U^m x\| \leq \sum_{k=0}^{n-1} \binom{m}{k} \rho^{m-k} C^k \|x\| \leq \mu n m^{n-1} \rho^{m-n+1} \|x\|$$

for any $x \in \mathbf{C}^{n,1}$. Because $0 \leq \rho < 1$ we have $\lim_{m\to\infty} m^{n-1}\rho^{m-n+1} = 0$. Thus there exists a positive integer $m$ such that $\|U^m x\| \leq \varepsilon\|x\|$ for every $x \in \mathbf{C}^{n,1}$.



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**Exercise 6.B.22.** Suppose $C[-1,1]$ is the vector space of continuous real-valued functions on the interval $[-1,1]$ with inner product given by

$$\langle f, g\rangle = \int_{-1}^{1} fg$$

for all $f, g \in C[-1,1]$. Let $\varphi$ be the linear functional on $C[-1,1]$ defined by $\varphi(f) = f(0)$. Show that there does not exist $g \in C[-1,1]$ such that

$$\varphi(f) = \langle f, g\rangle$$

for every $f \in C[-1,1]$.

*This exercise shows that the Riesz representation theorem (6.42) does not hold on infinite-dimensional vector spaces without additional hypotheses on $V$ and $\varphi$.*

**Solution.** Suppose such a $g$ exists and define $h \in C[-1,1]$ by $h(x) = x^2 g(x)$. Observe that

$$0 = h(0) = \varphi(h) = \langle h, g\rangle = \int_{-1}^{1} [xg(x)]^2 \, \mathrm{d}x.$$

Because the integrand is non-negative and continuous, we have $\int_{-1}^{1} [xg(x)]^2 \, \mathrm{d}x = 0$ if and only if $xg(x) = 0$ for all $x \in [-1,1]$. This implies that $g(x) = 0$ for all $x \in [-1,1] \setminus \{0\}$; the continuity of $g$ then implies that $g(0) = 0$ also. Thus

$$f(0) = \int_{-1}^{1} f(x)g(x) \, \mathrm{d}x = \int_{-1}^{1} 0 \, \mathrm{d}x = 0$$

for every $f \in C[-1,1]$, which is certainly not true. We may conclude that no such $g$ exists.

**Exercise 6.B.23.** For all $u, v \in V$, define $d(u,v) = \|u - v\|$.

(a) Show that $d$ is a metric on $V$.

(b) Show that if $V$ is finite-dimensional, then $d$ is a complete metric on $V$ (meaning that every Cauchy sequence converges).

(c) Show that every finite-dimensional subspace of $V$ is a closed subset of $V$ (with respect to the metric $d$).

*This exercise requires familiarity with metric spaces.*

**Solution.**

(a) Certainly $d$ is non-negative. The equivalence of $d(u,v) = 0$ and $u = v$ follows from 6.9(a). The symmetry of $d$ follows from 6.9(b), and the triangle inequality for $d$ is immediate from 6.17.

(b) Let $v_1, ..., v_n$ be a basis of $V$ and consider the 1-norm with respect to this basis:

$$\|v\|_1 = |a_1| + \cdots + |a_n|,$$



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where $v = a_1 v_1 + \cdots + a_n v_n$. By Exercise 6.B.16, it will suffice to show that $V$ is complete with respect to $\|\cdot\|_1$.

Let $(v_m)_{m=1}^{\infty}$ be a Cauchy sequence in $\left(V, \|\cdot\|_1\right)$, where $v_m = a_{m,1} v_1 + \cdots + a_{m,n} v_n$. For any $j \in \{1, ..., n\}$ and any positive integers $k$ and $m$, we have the inequality

$$\left|a_{m,j} - a_{k,j}\right| \leq \|v_m - v_k\|_1.$$

It follows that $\left(a_{m,j}\right)_{m=1}^{\infty}$ is a Cauchy sequence in the complete metric space $\mathbf{F}$ and thus there exists some $a_j \in \mathbf{F}$ such that $\lim_{m\to\infty} a_{m,j} = a_j$. Define $v = a_1 v_1 + \cdots + a_n v_n$ and observe that

$$\|v_m - v\|_1 = \left|a_{m,1} - a_1\right| + \cdots + \left|a_{m,n} - a_n\right| \to 0 \text{ as } m \to \infty.$$

Thus $(v_m)_{m=1}^{\infty}$ is convergent. We may conclude that $V$ is complete with respect to $\|\cdot\|_1$.

(c) Suppose $U$ is a finite-dimensional subspace of $V$ and $(u_m)_{m=1}^{\infty}$ is a sequence contained in $U$ satisfying $\lim_{m\to\infty}\|u_m - v\|$ for some $v \in V$. We need to show that $v \in U$. The norm $\|\cdot\|$ restricts to a norm on $U$; by part (b), this normed space $(U, \|\cdot\|)$ must be complete since $U$ is finite-dimensional. Because convergent sequences are necessarily Cauchy, completeness implies that the sequence $(u_m)_{m=1}^{\infty}$ converges to some $u \in U$. Since limits of sequences are unique it follows that $v = u \in U$, as desired.



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## 6.C. Orthogonal Complements and Minimization Problems

**Exercise 6.C.1.** Suppose $v_1, ..., v_m \in V$. Prove that

$$\{v_1, ..., v_m\}^\perp = (\text{span}(v_1, ..., v_m))^\perp.$$

**Solution.** Suppose that $v \in (\text{span}(v_1, ..., v_m))^\perp$. In particular, for every $k \in \{1, ..., m\}$,

$$v \in \text{span}(v_1, ..., v_m) \quad \Rightarrow \quad \langle v, v_k \rangle = 0.$$

It follows that $v \in \{v_1, ..., v_m\}^\perp$ and hence that $(\text{span}(v_1, ..., v_m))^\perp \subseteq \{v_1, ..., v_m\}^\perp$.

Now suppose that $v \in \{v_1, ..., v_m\}^\perp$ and let $a_1 v_1 + \cdots + a_m v_m \in \text{span}(v_1, ..., v_m)$ be given. Observe that

$$\langle v, a_1 v_1 + \cdots + a_m v_m \rangle = \overline{a_1} \langle v, v_1 \rangle + \cdots + \overline{a_m} \langle v, v_m \rangle = 0.$$

It follows that $v \in (\text{span}(v_1, ..., v_m))^\perp$ and hence that $\{v_1, ..., v_m\}^\perp \subseteq (\text{span}(v_1, ..., v_m))^\perp$. We may conclude that

$$\{v_1, ..., v_m\}^\perp = (\text{span}(v_1, ..., v_m))^\perp.$$

**Exercise 6.C.2.** Suppose $U$ is a subspace of $V$ with basis $u_1, ..., u_m$ and

$$u_1, ..., u_m, v_1, ..., v_n$$

is a basis of $V$. Prove that if the Gram-Schmidt procedure is applied to the basis of $V$ above, producing a list $e_1, ..., e_m, f_1, ..., f_n$, then $e_1, ..., e_m$ is an orthonormal basis of $U$ and $f_1, ..., f_n$ is an orthonormal basis of $U^\perp$.

**Solution.** The Gram-Schmidt procedure guarantees that

$$\text{span}(e_1, ..., e_m) = \text{span}(u_1, ..., u_m) = U,$$

and 6.25 shows that $e_1, ..., e_m$ is linearly independent. Thus $e_1, ..., e_m$ is an orthonormal basis of $U$.

The Gram-Schmidt procedure also guarantees that for any $k \in \{1, ..., n\}$ the vector $f_k$ is orthogonal to each vector in the list $e_1, ..., e_n$. By Exercise 6.C.1, this implies that

$$f_k \in (\text{span}(e_1, ..., e_m))^\perp = U^\perp.$$

Note that $f_1, ..., f_n$ is linearly independent by 6.25 and $\dim U^\perp = \dim V - \dim U = n$ by 6.51. Thus, by 2.38, $f_1, ..., f_n$ is an orthonormal basis of $U^\perp$.



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**Exercise 6.C.3.** Suppose $U$ is the subspace of $\mathbf{R}^4$ defined by

$$U = \operatorname{span}((1,2,3,-4),(-5,4,3,2)).$$

Find an orthonormal basis of $U$ and an orthonormal basis of $U^\perp$.

**Solution.** It is straightforward to verify that

$$u_1 = (1,2,3,-4), \quad u_2 = (-5,4,3,2), \quad v_1 = (1,0,0,0), \quad v_2 = (0,1,0,0)$$

is a basis of $\mathbf{R}^4$. Certainly $u_1, u_2$ is a basis of $U$. Performing the Gram-Schmidt procedure on this list yields the orthonormal list

$$e_1 = \frac{1}{\sqrt{30}}(1,2,3,-4), \quad e_2 = \frac{1}{\sqrt{12030}}(-77,56,39,38),$$

$$f_1 = \frac{1}{\sqrt{76190}}(190,117,60,151), \quad f_2 = \frac{1}{9\sqrt{190}}(0,81,-90,27).$$

As we showed in Exercise 6.C.3, $e_1, e_2$ must be an orthonormal basis of $U$ and $f_1, f_2$ must be an orthonormal basis of $U^\perp$.

**Exercise 6.C.4.** Suppose $e_1, ..., e_n$ is a list of vectors in $V$ with $\|e_k\| = 1$ for each $k = 1, ..., n$ and

$$\|v\|^2 = |\langle v, e_1\rangle|^2 + \cdots + |\langle v, e_n\rangle|^2$$

for all $v \in V$. Prove that $e_1, ..., e_n$ is an orthonormal basis of $V$.

*This exercise provides a converse to 6.30(b).*

**Solution.** Let $k \in \{1, ..., n\}$ be given and observe that

$$\|e_k\|^2 = |\langle e_k, e_k\rangle|^2 + \sum_{j=1,\, j\neq k}^{n} |\langle e_k, e_j\rangle|^2 \quad \Leftrightarrow \quad 1 = 1 + \sum_{j=1,\, j\neq k}^{n} |\langle e_k, e_j\rangle|^2$$

$$\Rightarrow \quad \sum_{j=1,\, j\neq k}^{n} |\langle e_k, e_j\rangle|^2 = 0.$$

Thus $\langle e_k, e_j\rangle = 0$ for each $j \neq k$. It follows that $e_1, ..., e_n$ is an orthonormal list and hence, by 6.25, $e_1, ..., e_n$ is linearly independent. Suppose that $v \in \{e_1, ..., e_n\}^\perp$ and observe that

$$\|v\|^2 = |\langle v, e_1\rangle|^2 + \cdots + |\langle v, e_n\rangle|^2 = 0 \quad \Leftrightarrow \quad v = 0.$$

Thus $\{e_1, ..., e_n\}^\perp = \{0\}$. It follows from Exercise 6.C.1 and 6.54 that $\operatorname{span}(e_1, ..., e_n) = V$. We may conclude that $e_1, ..., e_n$ is an orthonormal basis of $V$.

**Exercise 6.C.5.** Suppose that $V$ is finite-dimensional and $U$ is a subspace of $V$. Show that $P_{U^\perp} = I - P_U$, where $I$ is the identity operator on $V$.



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**Solution.** By 6.49 and 6.52, for any $v \in V$ we can write $v = u + w$ for unique vectors $u \in U = (U^\perp)^\perp$ and $w \in U^\perp$. It follows that

$$P_U(v) + P_{U^\perp}(v) = u + w = v.$$

Thus $P_U + P_{U^\perp} = I$.

**Exercise 6.C.6.** Suppose $V$ is finite-dimensional and $T \in \mathcal{L}(V, W)$. Show that

$$T = TP_{(\operatorname{null} T)^\perp} = P_{\operatorname{range} T}T.$$

**Solution.** By 6.49 we have $V = \operatorname{null} T \oplus (\operatorname{null} T)^\perp$. Thus, for any $v \in V$, we can write $v = x + y$, where $x \in \operatorname{null} T$ and $y \in (\operatorname{null} T)^\perp$. It follows that

$$Tv = Ty = TP_{(\operatorname{null} T)^\perp}(v)$$

and hence that $T = TP_{(\operatorname{null} T)^\perp}$.

Because $\operatorname{range} T$ is finite-dimensional, 6.49 allows us to write $W = \operatorname{range} T \oplus (\operatorname{range} T)^\perp$. For any $v \in V$, observe that $Tv = Tv + 0 \in \operatorname{range} T \oplus (\operatorname{range} T)^\perp$. It follows that

$$Tv = P_{\operatorname{range} T}Tv$$

and hence that $T = P_{\operatorname{range} T}T$.

**Exercise 6.C.7.** Suppose that $X$ and $Y$ are finite-dimensional subspaces of $V$. Prove that $P_X P_Y = 0$ if and only if $\langle x, y\rangle = 0$ for all $x \in X$ and all $y \in Y$.

**Solution.** Suppose that $\langle x, y\rangle = 0$ for all $x \in X$ and $y \in Y$. For $v \in V$ write $v = y + z$, where $y \in Y$ and $z \in Y^\perp$, so that $P_Y v = y$. Our hypothesis ensures that $y \in X^\perp$ and thus

$$P_X P_Y v = P_X v = 0$$

by 6.57(c). Hence $P_X P_Y = 0$.

For the converse, suppose that $P_X P_Y = 0$ and let $x \in X$ and $y \in Y$ be given. Using 6.57(b) and 6.57(e), observe that

$$P_X y = P_X P_Y y = 0 \quad \Rightarrow \quad y \in \operatorname{null} P_X \quad \Rightarrow \quad y \in X^\perp \quad \Rightarrow \quad \langle x, y\rangle = 0.$$



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**Exercise 6.C.8.** Suppose $U$ is a finite-dimensional subspace of $V$ and $v \in V$. Define a linear functional $\varphi : U \to \mathbf{F}$ by

$$\varphi(u) = \langle u, v \rangle$$

for all $u \in U$. By the Riesz representation theorem (6.42) as applied to the inner product space $U$, there exists a unique vector $w \in U$ such that

$$\varphi(u) = \langle u, w \rangle$$

for all $u \in U$. Show that $w = P_U v$.

**Solution.** We have $\langle u, v \rangle = \langle u, w \rangle$ for every $u \in U$. Equivalently, $\langle u, v - w \rangle = 0$ for every $u \in U$, so that $v - w \in U^\perp$. It follows that $v = w + v - w$, where $w \in U$ and $v - w \in U^\perp$. Thus $P_U v = w$.

**Exercise 6.C.9.** Suppose $V$ is finite-dimensional. Suppose $P \in \mathcal{L}(V)$ is such that $P^2 = P$ and every vector in $\operatorname{null} P$ is orthogonal to every vector in $\operatorname{range} P$. Prove that there exists a subspace $U$ of $V$ such that $P = P_U$.

**Solution.** By Exercise 3.B.27 and 6.49 we have the decompositions

$$V = \operatorname{range} P \oplus \operatorname{null} P \quad \text{and} \quad V = \operatorname{range} P \oplus (\operatorname{range} P)^\perp,$$

which implies $\dim \operatorname{null} P = \dim (\operatorname{range} P)^\perp$. Combining this with the hypothesis $\operatorname{null} P \subseteq (\operatorname{range} P)^\perp$, we see that $\operatorname{null} P = (\operatorname{range} P)^\perp$. Let $U = \operatorname{range} P$; we claim that $P = P_U$. Let $v = Px + w \in V$ be given, where $Px \in \operatorname{range} P$ and $w \in (\operatorname{range} P)^\perp = \operatorname{null} P$. Observe that

$$P_U v = Px = P(Px + w) = Pv,$$

where we have used $P^2 = P$ and $w \in \operatorname{null} P$ for the second equality. Thus $P = P_U$.

**Exercise 6.C.10.** Suppose $V$ is finite-dimensional and $P \in \mathcal{L}(V)$ is such that $P^2 = P$ and

$$\|Pv\| \le \|v\|$$

for every $v \in V$. Prove that there exists a subspace $U$ of $V$ such that $P = P_U$.

**Solution.** Suppose $w \in \operatorname{null} P$ and $Px \in \operatorname{range} P$. Our hypothesis gives us the inequality

$$\|Px\| = \|P(Px + \lambda w)\| \le \|Px + \lambda w\|$$

for any $\lambda \in \mathbf{F}$. It follows from Exercise 6.A.6 that $\langle w, Px \rangle = 0$ and hence that $\operatorname{null} P$ is contained in $(\operatorname{range} P)^\perp$. We can now let $U = \operatorname{range} P$ and proceed as in Exercise 6.C.9 to see that $P = P_U$.



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**Exercise 6.C.11.** Suppose $T \in \mathcal{L}(V)$ and $U$ is a finite-dimensional subspace of $V$. Prove that

$$U \text{ is invariant under } T \quad \Leftrightarrow \quad P_U T P_U = T P_U.$$

**Solution.** Suppose that $U$ is invariant under $T$ and let $v \in V$ be given. Observe that

$$P_U v \in U \quad \Rightarrow \quad T P_U v \in U \quad \Rightarrow \quad P_U T P_U v = T P_U v,$$

where the last implication follows from 6.57(b). Now suppose that $U$ is not invariant under $T$, i.e. there is some $u \in U$ such that $Tu \notin U$. Note that

$$T P_U u = Tu \notin U \quad \text{and} \quad P_U T P_U u \in U,$$

where we have used 6.57(b) and 6.57(d). It follows that

$$P_U T P_U u \neq T P_U u \quad \Rightarrow \quad P_U T P_U \neq P_U T.$$

**Exercise 6.C.12.** Suppose $V$ is finite-dimensional, $T \in \mathcal{L}(V)$, and $U$ is a subspace of $V$. Prove that

$$U \text{ and } U^\perp \text{ are both invariant under } T \quad \Leftrightarrow \quad P_U T = T P_U.$$

**Solution.** Suppose that $U$ and $U^\perp$ are both invariant under $T$ and let $v = u + w \in V$ be given, where $u \in U$ and $w \in U^\perp$. By assumption we have $Tu \in U$ and $Tw \in U^\perp$; using 6.57, it follows that

$$P_U T v = P_U(Tu + Tw) = Tu = T P_U u = T P_U(u + w) = T P_U v.$$

If $U$ is not invariant under $T$ then there exists some $u \in U$ such that $Tu \notin U$. It follows that

$$T P_U u = Tu \notin U \quad \text{and} \quad P_U T u \in U,$$

so that $T P_U \neq P_U T$. Similarly, if $U^\perp$ is not invariant under $T$ then there exists some $w \in U^\perp$ such that $Tw \notin U^\perp$. It follows that

$$T P_U w = T(0) = 0 \quad \text{and} \quad P_U T w \neq 0,$$

so that $T P_U \neq P_U T$.



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**Exercise 6.C.13.** Suppose $\mathbf{F} = \mathbf{R}$ and $V$ is finite-dimensional. For each $v \in V$, let $\varphi_v$ denote the linear functional on $V$ defined by

$$\varphi_v(u) = \langle u, v \rangle$$

for all $u \in V$.

(a) Show that $v \mapsto \varphi_v$ is an injective linear map from $V$ to $V'$.

(b) Use (a) and a dimension-counting argument to show that $v \mapsto \varphi_v$ is an isomorphism from $V$ onto $V'$.

*The purpose of this exercise is to give an alternative proof of the Riesz representation theorem (6.42 and 6.58) when $\mathbf{F} = \mathbf{R}$. Thus you should not use the Riesz representation theorem as a tool in your solution.*

**Solution.**

(a) Let $u, v, w \in V$ be given and note that

$$\varphi_{v+w}(u) = \langle u, v + w \rangle = \langle u, v \rangle + \langle u, w \rangle = \varphi_v(u) + \varphi_w(u).$$

Similarly, for any $u, v \in V$ and any $\lambda \in \mathbf{R}$,

$$\varphi_{\lambda v}(u) = \langle u, \lambda v \rangle = \lambda \langle u, v \rangle = \lambda \varphi_v(u).$$

Thus $v \mapsto \varphi_v$ is linear. Suppose that $v \in V$ is such that $\varphi_v = 0$, i.e. $\langle u, v \rangle = 0$ for every $u \in U$. It follows that $v \in V^\perp$ and hence, by 6.48(c), $v = 0$. Thus $v \mapsto \varphi_v$ is injective.

(b) By 3.111, 3.65, and part (a), the map $v \mapsto \varphi_v$ must be an isomorphism.

**Exercise 6.C.14.** Suppose that $e_1, ..., e_n$ is an orthonormal basis of $V$. Explain why the dual basis (see 3.112) of $e_1, ..., e_n$ is $e_1, ..., e_n$ under the identification of $V'$ with $V$ provided by the Riesz representation theorem (6.58).

**Solution.** Let $\varphi_1, ..., \varphi_n$ be the dual basis of $e_1, ..., e_n$. Using the notation of 6.58, for any $j, k \in \{1, ..., n\}$ we have

$$\varphi_{e_k}(e_j) = \langle e_j, e_k \rangle = \begin{cases} 1 & \text{if } j = k, \\ 0 & \text{if } j \neq k. \end{cases}$$

Thus $\varphi_{e_k} = \varphi_k$ for each $k \in \{1, ..., n\}$, i.e. we may identify $e_k$ with $\varphi_k$.

**Exercise 6.C.15.** In $\mathbf{R}^4$, let

$$U = \operatorname{span}((1, 1, 0, 0), (1, 1, 1, 2)).$$

Find $u \in U$ such that $\|u - (1, 2, 3, 4)\|$ is as small as possible.



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**Solution.** Let $u_1 = (1, 1, 0, 0)$ and $u_2 = (1, 1, 1, 2)$, so that $u_1, u_2$ is a basis of $U$. Performing the Gram-Schmidt procedure on the list $u_1, u_2$ yields the list

$$e_1 = \frac{1}{\sqrt{2}}(1, 1, 0, 0), \quad e_2 = \frac{1}{\sqrt{5}}(0, 0, 1, 2),$$

which is an orthonormal basis of $U$. Let $v = (1, 2, 3, 4)$. According to 6.61, to minimize $\|u - v\|$ we should take $u = P_U v$. This can be calculated using 6.57(i):

$$P_U v = \langle v, e_1 \rangle e_1 + \langle v, e_2 \rangle e_2 = \left(\frac{3}{2}, \frac{3}{2}, \frac{11}{5}, \frac{22}{5}\right).$$

**Exercise 6.C.16.** Suppose $C[-1, 1]$ is the vector space of continuous real-valued functions on the interval $[-1, 1]$ with inner product given by

$$\langle f, g \rangle = \int_{-1}^{1} fg$$

for all $f, g \in C[-1, 1]$. Let $U$ be the subspace of $C[-1, 1]$ defined by

$$U = \{f \in C[-1, 1] : f(0) = 0\}.$$

(a) Show that $U^\perp = \{0\}$.

(b) Show that 6.49 and 6.52 do not hold without the finite-dimensional hypothesis.

**Solution.**

(a) Certainly $0 \in U^\perp$. Suppose that $g \in U^\perp$. Let $f : [-1, 1] \to \mathbf{R}$ be given by $f(x) = x^2 g(x)$ and note that $f \in U$, so that

$$0 = \langle f, g \rangle = \int_{-1}^{1} [xg(x)]^2 \, \mathrm{d}x.$$

Because the integrand $[xg(x)]^2$ is continuous and non-negative we must have $xg(x) = 0$ for every $x \in [-1, 1]$, which implies $g(x) = 0$ for all non-zero $x \in [-1, 1]$. The continuity of $g$ gives us $g(0) = 0$ also and thus $g = 0$. We may conclude that $U^\perp = \{0\}$.

(b) From part (a) we have $U \oplus U^\perp = U \neq C[-1, 1]$ and thus 6.49 does not hold. Part (a) and 6.48(b) give us

$$(U^\perp)^\perp = \{0\}^\perp = C[-1, 1] \neq U,$$

so that 6.52 does not hold.

**Exercise 6.C.17.** Find $p \in \mathcal{P}_3(\mathbf{R})$ such that $p(0) = 0$, $p'(0) = 0$, and

$$\int_0^1 |2 + 3x - p(x)|^2 \, dx$$

is as small as possible.



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**Solution.** Equip $\mathcal{P}_3(\mathbf{R})$ with the inner product

$$\langle p, q\rangle = \int_0^1 p(x)q(x)\,\mathrm{d}x$$

and let $U = \{p \in \mathcal{P}_3(\mathbf{R}) : p(0) = p'(0) = 0\}$. It is straightforward to verify that $U$ is a subspace of $\mathcal{P}_3(\mathbf{R})$ and that $x^2, x^3$ is a basis of $U$. Performing the Gram-Schmidt procedure on this basis yields the orthonormal basis $e_1, e_2$ of $U$, where

$$e_1 = \sqrt{5}x^2, \quad e_2 = 6\sqrt{7}\big(x^3 - \tfrac{5}{6}x^2\big).$$

Let $q(x) = 2 + 3x$. According to 6.61, to minimize $\|q - p\|^2 = \int_0^1 |2 + 3x - p(x)|^2\,\mathrm{d}x$ we should take $p = P_U q$. This can be calculated using 6.57(i):

$$P_U q = \langle q, e_1\rangle e_1 + \langle q, e_2\rangle e_2 = 24x^2 - \frac{203}{10}x^3.$$

**Exercise 6.C.18.** Find $p \in \mathcal{P}_5(\mathbf{R})$ that makes $\int_{-\pi}^{\pi} |\sin x - p(x)|^2\,dx$ as small as possible.

> *The polynomial 6.65 is an excellent approximation to the answer to this exercise, but here you are asked to find the exact solution, which involves powers of $\pi$. A computer that can perform symbolic integration should help.*

**Solution.** Equip $C[-\pi, \pi]$ with the inner product

$$\langle p, q\rangle = \int_{-\pi}^{\pi} p(x)q(x)\,\mathrm{d}x$$

and let $U = \mathcal{P}_5(\mathbf{R})$. Performing the Gram-Schmidt procedure on the basis $1, x, x^2, x^3, x^4, x^5$ of $U$ gives us the orthonormal basis

$$e_1 = \frac{1}{\sqrt{2\pi}}, \quad e_2 = \sqrt{\frac{3}{2\pi^3}}x, \quad e_3 = -\frac{1}{2}\sqrt{\frac{5}{2\pi^5}}(\pi^2 - 3x^2),$$

$$e_4 = -\frac{1}{2}\sqrt{\frac{7}{2\pi^7}}(3\pi^2 x - 5x^3), \quad e_5 = \frac{3}{8\sqrt{2\pi^9}}(3\pi^4 - 30\pi^2 x^2 + 35x^4),$$

$$e_6 = -\frac{1}{8}\sqrt{\frac{11}{2\pi^{11}}}(15\pi^4 x - 70\pi^2 x^3 + 63x^5).$$

According to 6.61, to minimize $\|\sin x - p\|^2 = \int_{-\pi}^{\pi} |\sin x - p(x)|^2\,\mathrm{d}x$ we should take $p = P_U(\sin x)$. This can be calculated using 6.57(i):

$$P_U(\sin x) = \frac{105(1465 - 153\pi^2 + \pi^4)}{8\pi^6}x - \frac{315(1155 - 125\pi^2 + \pi^4)}{4\pi^8}x^3 + \frac{693(945 - 105\pi^2 + \pi^4)}{8\pi^{10}}x^5.$$



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**Exercise 6.C.19.** Suppose $V$ is finite-dimensional and $P \in \mathcal{L}(V)$ is an orthogonal projection of $V$ onto some subspace of $V$. Prove that $P^\dagger = P$.

**Solution.** Suppose $U$ is the subspace of $V$ such that $P = P_U$. Using 6.57(e) and 6.52, observe that

$$(\operatorname{null} P)^\perp = (\operatorname{null} P_U)^\perp = (U^\perp)^\perp = U.$$

Thus $P|_{(\operatorname{null} P)^\perp} = P|_U$. Since $P = P_U$, 6.57(b) shows that $P|_{(\operatorname{null} P)^\perp}$ is simply the identity operator on $(\operatorname{null} P)^\perp = U$. Combining this with 6.57(d), we have

$$P^\dagger = \left(P|_{(\operatorname{null} P)^\perp}\right)^{-1} P_{\operatorname{range} P} = P_U = P.$$

**Exercise 6.C.20.** Suppose $V$ is finite-dimensional and $T \in \mathcal{L}(V, W)$. Show that

$$\operatorname{null} T^\dagger = (\operatorname{range} T)^\perp \quad \text{and} \quad \operatorname{range} T^\dagger = (\operatorname{null} T)^\perp.$$

**Solution.** Because $\left(T|_{(\operatorname{null} T)^\perp}\right)^{-1}$ is injective (by 6.67), we have

$$\operatorname{null} T^\dagger = \operatorname{null} P_{\operatorname{range} T} = (\operatorname{range} T)^\perp,$$

where we have used 6.57(e) for the last equality.

6.57(d) shows that $P_{\operatorname{range} T}$ is a surjection from $W$ onto $\operatorname{range} T$, and 6.67 shows that $\left(T|_{(\operatorname{null} T)^\perp}\right)^{-1}$ is a surjection from $\operatorname{range} T$ onto $(\operatorname{null} T)^\perp$. Thus $\operatorname{range} T^\dagger = (\operatorname{null} T)^\perp$.

$$W \xrightarrow{P_{\operatorname{range} T}} \operatorname{range} T \xrightarrow{\left(T|_{(\operatorname{null} T)^\perp}\right)^{-1}} (\operatorname{null} T)^\perp$$

**Exercise 6.C.21.** Suppose $T \in \mathcal{L}(\mathbf{F}^3, \mathbf{F}^2)$ is defined by

$$T(a, b, c) = (a + b + c, 2b + 3c).$$

(a) For $(x, y) \in \mathbf{F}^2$, find a formula for $T^\dagger(x, y)$.

(b) Verify that the equation $TT^\dagger = P_{\operatorname{range} T}$ from 6.69(b) holds with the formula for $T^\dagger$ obtained in (a).

(c) Verify that the equation $T^\dagger T = P_{(\operatorname{null} T)^\perp}$ from 6.69(c) holds with the formula for $T^\dagger$ obtained in (a).

**Solution.**

(a) We proceed as in example 6.71. Note that $T$ is surjective, so that $P_{\operatorname{range} T}$ is the identity operator on $W$. Note further that

$$\operatorname{null} T = \{(a, b, c) \in \mathbf{F}^3 : a + b + c = 0, 2b + 3c = 0\} = \operatorname{span}((1, -3, 2)).$$

For $(x, y) \in \mathbf{F}^2$, it follows that



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$$T^\dagger(x,y) = (T|_{(\operatorname{null} T)^\perp})^{-1} P_{\operatorname{range} T}(x,y) = (T|_{(\operatorname{null} T)^\perp})^{-1}(x,y).$$

If $(T|_{(\operatorname{null} T)^\perp})^{-1}(x,y) = (a,b,c) \in \mathbf{F}^3$ then $(a,b,c)$ must satisfy $T(a,b,c) = (x,y)$ and $\langle (a,b,c), (1,-3,2)\rangle = 0$. In other words, $(a,b,c)$ must satisfy the following equations

$$a + b + c = x,$$

$$2b + 3c = y,$$

$$a - 3b + 2c = 0.$$

Solving this system of equations yields the solution

$$a = \tfrac{1}{14}(13x - 5y), \quad b = \tfrac{1}{14}(3x + y), \quad c = \tfrac{1}{7}(-x + 2y).$$

Thus

$$T^\dagger(x,y) = \tfrac{1}{14}(13x - 5y, 3x + y, -2x + 4y).$$

(b) As noted in part (a), $P_{\operatorname{range} T}$ is the identity operator on $W$. Observe that

$$TT^\dagger(x,y) = \tfrac{1}{14} T(13x - 5y, 3x + y, -2x + 4y) = \tfrac{1}{14}(14x, 14y) = (x,y).$$

Thus $TT^\dagger = P_{\operatorname{range} T}$.

(c) As noted in part (a), $\operatorname{null} T = \operatorname{span}((1,-3,2))$. Thus

$$(\operatorname{null} T)^\perp = \{(a,b,c) \in \mathbf{F}^3 : a - 3b + 2c = 0\}.$$

It is straightforward to verify that $(1,1,1), (0,2,3)$ is a basis of $(\operatorname{null} T)^\perp$. Performing the Gram-Schmidt procedure on this basis gives us the orthonormal basis

$$\frac{1}{\sqrt{3}}(1,1,1), \quad \frac{1}{\sqrt{42}}(-5,1,4).$$

A tedious calculation using the formula for $T^\dagger$ found in part (a) and the formula for $P_{(\operatorname{null} T)^\perp}$ given by 6.57(i) shows that

$$T^\dagger T(a,b,c) = P_{(\operatorname{null} T)^\perp}(a,b,c) = \tfrac{1}{14}(13a + 3b - 2c, 3a + 5b + 6c, -2a + 6b + 10c).$$

**Exercise 6.C.22.** Suppose $V$ is finite-dimensional and $T \in \mathcal{L}(V,W)$. Prove that

$$TT^\dagger T = T \quad \text{and} \quad T^\dagger T T^\dagger = T^\dagger.$$

*Both formulas above clearly hold if $T$ is invertible because in that case we can replace $T^\dagger$ with $T^{-1}$.*

**Solution.** By 6.69(b) we have $TT^\dagger = P_{\operatorname{range} T}$. It follows that $TT^\dagger T = P_{\operatorname{range} T} T = T$, since $P_{\operatorname{range} T}$ is the identity operator on $\operatorname{range} T$ by 6.57(b).

By 6.69(c) we have $T^\dagger T = P_{(\operatorname{null} T)^\perp}$ and thus $T^\dagger T T^\dagger = P_{(\operatorname{null} T)^\perp} T^\dagger = T^\dagger$, since $T^\dagger$ maps into $(\operatorname{null} T)^\perp$ and $P_{(\operatorname{null} T)^\perp}$ is the identity operator on $(\operatorname{null} T)^\perp$.



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**Exercise 6.C.23.** Suppose $V$ and $W$ are finite-dimensional and $T \in \mathcal{L}(V, W)$. Prove that

$$\left(T^\dagger\right)^\dagger = T.$$

*The equation above is analogous to the equation $\left(T^{-1}\right)^{-1} = T$ that holds if $T$ is invertible.*

**Solution.** By Exercise 6.C.20 and 6.52 we have

$$\left(T^\dagger\right)^\dagger = \left(T^\dagger|_{(\operatorname{null} T^\dagger)^\perp}\right)^{-1} P_{\operatorname{range} T^\dagger} = \left(T^\dagger|_{((\operatorname{range} T)^\perp)^\perp}\right)^{-1} P_{(\operatorname{null} T)^\perp} = \left(T^\dagger|_{\operatorname{range} T}\right)^{-1} P_{(\operatorname{null} T)^\perp}.$$

Because $P_{\operatorname{range} T}$ is the identity operator on $\operatorname{range} T$ we have $T^\dagger|_{\operatorname{range} T} = \left(T|_{(\operatorname{null} T)^\perp}\right)^{-1}$. Thus

$$\left(T^\dagger\right)^\dagger = \left(\left(T|_{(\operatorname{null} T)^\perp}\right)^{-1}\right)^{-1} P_{(\operatorname{null} T)^\perp} = T|_{(\operatorname{null} T)^\perp}\, P_{(\operatorname{null} T)^\perp} = T P_{(\operatorname{null} T)^\perp}.$$

For any $v \in V$ we have $v = u + w$, where $u \in \operatorname{null} T$ and $w \in (\operatorname{null} T)^\perp$. It follows that

$$\left(T^\dagger\right)^\dagger(v) = T P_{(\operatorname{null} T)^\perp}(v) = Tw = Tv.$$

Thus $\left(T^\dagger\right)^\dagger = T$.



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# Chapter 7. Operators on Inner Product Spaces

## 7.A. Self-Adjoint and Normal Operators

**Exercise 7.A.1.** Suppose $n$ is a positive integer. Define $T \in \mathcal{L}(\mathbf{F}^n)$ by

$$T(z_1, ..., z_n) = (0, z_1, ..., z_{n-1}).$$

Find a formula for $T^*(z_1, ..., z_n)$.

**Solution.** Observe that

$$\begin{aligned}\langle (w_1, ..., w_n), T^*(z_1, ..., z_n)\rangle &= \langle T(w_1, ..., w_n), (z_1, ..., z_n)\rangle \\ &= \langle (0, w_1, ..., w_{n-1}), (z_1, ..., z_n)\rangle \\ &= w_1 z_2 + \cdots + w_{n-1} z_n \\ &= \langle (w_1, ..., w_n), (z_2, ..., z_n, 0)\rangle.\end{aligned}$$

Thus $T^*(z_1, ..., z_n) = (z_2, ..., z_n, 0)$.

**Exercise 7.A.2.** Suppose $T \in \mathcal{L}(V, W)$. Prove that

$$T = 0 \quad \Leftrightarrow \quad T^* = 0 \quad \Leftrightarrow \quad T^*T = 0 \quad \Leftrightarrow \quad TT^* = 0.$$

**Solution.** Suppose $T = 0$, fix $w \in W$, and observe that

$$\langle v, T^*w\rangle = \langle Tv, w\rangle = \langle 0, w\rangle = 0$$

for any $v \in V$. Thus $T^*w \in V^\perp$. It follows from 6.48(c) that $T^*w = 0$ and hence that $T^* = 0$. Combining this with 7.5(c) shows that $T = 0$ if and only if $T^* = 0$.

That $T = 0$ implies $T^*T = 0$ is clear. Suppose that $T^*T = 0$, let $v \in V$ be given, and observe that

$$0 = \langle 0, v\rangle = \langle T^*Tv, v\rangle = \langle Tv, Tv\rangle = \|Tv\|^2 \quad \Rightarrow \quad Tv = 0.$$

Thus $T = 0$, so that $T = 0$ if and only if $T^*T = 0$. Replacing $T$ with $T^*$ in this result and using 7.5(c) shows that $T^* = 0$ if and only if $TT^* = 0$.

**Exercise 7.A.3.** Suppose $T \in \mathcal{L}(V)$ and $\lambda \in \mathbf{F}$. Prove that

$$\lambda \text{ is an eigenvalue of } T \quad \Leftrightarrow \quad \overline{\lambda} \text{ is an eigenvalue of } T^*.$$

**Solution.** Observe that



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$$
\begin{aligned}
\lambda \text{ is an eigenvalue of } T &\Leftrightarrow T-\lambda I \text{ is not surjective} && (5.7(c)) \\
&\Leftrightarrow \operatorname{range}(T-\lambda I) \neq V \\
&\Leftrightarrow (\operatorname{range}(T-\lambda I))^{\perp} \neq \{0\} && (7.48(c)) \\
&\Leftrightarrow \operatorname{null}(T-\lambda I)^{*} \neq \{0\} && (7.6(a)) \\
&\Leftrightarrow \operatorname{null}(T^{*}-\overline{\lambda} I) \neq \{0\} && (7.5(a), (b), (e)) \\
&\Leftrightarrow T^{*}-\overline{\lambda} I \text{ is not injective} \\
&\Leftrightarrow \overline{\lambda} \text{ is an eigenvalue of } T^{*}. && (5.7(b))
\end{aligned}
$$

**Exercise 7.A.4.** Suppose $T \in \mathcal{L}(V)$ and $U$ is a subspace of $V$. Prove that

$$U \text{ is invariant under } T \quad \Leftrightarrow \quad U^{\perp} \text{ is invariant under } T^{*}.$$

**Solution.** Suppose that $U$ is invariant under $T$ and let $v \in U^{\perp}$ be given. Observe that

$$\langle u, T^{*}v\rangle = \langle Tu, v\rangle = 0$$

for any $u \in U$, where the last equality follows since $Tu \in U$ and $v \in U^{\perp}$. Thus $T^{*}v \in U^{\perp}$ and it follows that $U^{\perp}$ is invariant under $T^{*}$.

Now suppose that $U^{\perp}$ is invariant under $T^{*}$. The previous paragraph shows that $(U^{\perp})^{\perp}$ is invariant under $(T^{*})^{*}$, which by 6.52 and 7.5(c) is exactly the statement that $U$ is invariant under $T$.

**Exercise 7.A.5.** Suppose $T \in \mathcal{L}(V, W)$. Suppose $e_1, ..., e_n$ is an orthonormal basis of $V$ and $f_1, ..., f_m$ is an orthonormal basis of $W$. Prove that

$$\|Te_1\|^2 + \cdots + \|Te_n\|^2 = \|T^{*}f_1\|^2 + \cdots + \|T^{*}f_m\|^2.$$

*The numbers $\|Te_1\|^2, ..., \|Te_n\|^2$ in the equation above depend on the orthonormal basis $e_1, ..., e_n$, but the right side of the equation does not depend on $e_1, ..., e_n$. Thus the equation above shows that the sum on the left side does not depend on which orthonormal basis $e_1, ..., e_n$ is used.*

**Solution.** Using 6.30(b), observe that

$$
\begin{aligned}
\sum_{j=1}^{n}\|Te_j\|^2 = \sum_{j=1}^{n}\sum_{k=1}^{n}\left|\langle Te_j, f_k\rangle\right|^2 &= \sum_{j=1}^{n}\sum_{k=1}^{n}\left|\langle e_j, T^{*}f_k\rangle\right|^2 \\
&= \sum_{k=1}^{n}\sum_{j=1}^{n}\left|\langle T^{*}f_k, e_j\rangle\right|^2 = \sum_{k=1}^{n}\|T^{*}f_k\|^2.
\end{aligned}
$$



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**Exercise 7.A.6.** Suppose $T \in \mathcal{L}(V, W)$. Prove that

(a) $T$ is injective $\Leftrightarrow$ $T^*$ is surjective;

(b) $T$ is surjective $\Leftrightarrow$ $T^*$ is injective.

**Solution.**

(a) Observe that

$$\begin{aligned} T \text{ is injective} &\Leftrightarrow \operatorname{null} T = \{0\} \\ &\Leftrightarrow (\operatorname{range} T^*)^\perp = \{0\} && (7.6(c)) \\ &\Leftrightarrow \operatorname{range} T^* = V && (7.48(c)) \\ &\Leftrightarrow T^* \text{ is surjective.} \end{aligned}$$

(b) Part (a) shows that $T^*$ is injective if and only if $(T^*)^*$ is surjective, which by 7.5(c) is exactly the statement that $T^*$ is injective if and only if $T$ is surjective.

**Exercise 7.A.7.** Prove that if $T \in \mathcal{L}(V, W)$, then

(a) $\dim \operatorname{null} T^* = \dim \operatorname{null} T + \dim W - \dim V$;

(b) $\dim \operatorname{range} T^* = \dim \operatorname{range} T$.

**Solution.**

(a) We have

$$\begin{aligned} \dim \operatorname{null} T^* = \dim (\operatorname{range} T)^\perp = \dim W - \dim \operatorname{range} T \\ = \dim \operatorname{null} T + \dim W - \dim V, \end{aligned}$$

where the first equality is 7.6(a), the second equality is 6.51, and the last equality follows from the fundamental theorem of linear maps (3.21).

(b) We have

$$\dim \operatorname{range} T^* = \dim (\operatorname{null} T)^\perp = \dim V - \dim \operatorname{null} T = \dim \operatorname{range} T,$$

where the first equality is 7.6(b), the second equality is 6.51, and the last equality follows from the fundamental theorem of linear maps (3.21).

**Exercise 7.A.8.** Suppose $A$ is an $m$-by-$n$ matrix with entries in $\mathbf{F}$. Use (b) in Exercise 7 to prove that the row rank of $A$ equals the column rank of $A$.

*This exercise asks for yet another alternative proof of a result that was previously proved in 3.57 and 3.133.*



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**Solution.** For a column vector $v$ with entries in $\mathbf{F}$, let $\overline{v}$ denote the column vector obtained by taking the complex conjugate of each entry of $v$, e.g.

$$v = \begin{pmatrix} 1-i \\ \pi + 9i \\ \sqrt{2} \end{pmatrix} \quad \Rightarrow \quad \overline{v} = \begin{pmatrix} 1+i \\ \pi - 9i \\ \sqrt{2} \end{pmatrix}.$$

For a matrix $M$ with entries in $\mathbf{F}$, let $\overline{M}$ be the matrix obtained by taking the complex conjugate of each entry of $M$. Suppose that $v_1, ..., v_\ell$ is a basis of the span of the columns of $M$. We claim that the list $\overline{v_1}, ..., \overline{v_\ell}$ is linearly independent. Indeed, for scalars $a_1, ..., a_\ell \in \mathbf{F}$, observe that

$$a_1\overline{v_1} + \cdots + a_\ell\overline{v_\ell} = 0 \quad \Rightarrow \quad \overline{a_1}v_1 + \cdots + \overline{a_\ell}v_\ell = 0$$
$$\Rightarrow \quad \overline{a_1} = \cdots = \overline{a_\ell} = 0 \quad \Rightarrow \quad a_1 = \cdots = a_\ell = 0.$$

It follows that the column rank of $\overline{M}$ is greater than or equal to the column rank of $M$. By replacing $M$ with $\overline{M}$ in this result, we see that the column rank of $\overline{M}$ must equal the column rank of $M$.

Let $T \in \mathcal{L}(\mathbf{F}^n, \mathbf{F}^m)$ be such that the matrix of $T$ with respect to the standard orthonormal bases of $\mathbf{F}^n$ and $\mathbf{F}^m$ is $A$; 7.9 shows that the matrix of $T^*$ with respect to the standard orthonormal bases of $\mathbf{F}^m$ and $\mathbf{F}^n$ is $A^*$. Using Exercise 7.A.7 and our previous discussion, it follows that

$$\text{column rank of } A = \dim \operatorname{range} T = \dim \operatorname{range} T^*$$
$$= \text{column rank of } A^* = \text{column rank of } A^{\mathrm{t}} = \text{row rank of } A.$$

**Exercise 7.A.9.** Prove that the product of two self-adjoint operators on $V$ is self-adjoint if and only if the two operators commute.

**Solution.** Suppose $S, T \in \mathcal{L}(V)$ are self-adjoint and observe that, by 7.5(d),

$$(ST)^* = T^*S^* = TS.$$

Thus $(ST)^* = ST$ if and only if $TS = ST$. That is, $ST$ is self-adjoint if and only if $S$ and $T$ commute.

**Exercise 7.A.10.** Suppose $\mathbf{F} = \mathbf{C}$ and $T \in \mathcal{L}(V)$. Prove that $T$ is self-adjoint if and only if

$$\langle Tv, v\rangle = \langle T^*v, v\rangle$$

for all $v \in V$.

**Solution.** Note that

$$\langle Tv, v\rangle = \langle T^*v, v\rangle \text{ for all } v \in V \quad \Leftrightarrow \quad \langle (T - T^*)v, v\rangle = 0 \text{ for all } v \in V.$$



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The desired equivalence then follows from 7.13.

**Exercise 7.A.11.** Define an operator $S : \mathbf{F}^2 \to \mathbf{F}^2$ by $S(w, z) = (-z, w)$.

(a) Find a formula for $S^*$.

(b) Show that $S$ is normal but not self-adjoint.

(c) Find all eigenvalues of $S$.

*If $\mathbf{F} = \mathbf{R}$, then $S$ is the operator on $\mathbf{R}^2$ of counterclockwise rotation by $90°$.*

**Solution.**

(a) Observe that

$$\langle S(w, z), (x, y)\rangle = \langle (-z, w), (x, y)\rangle = -zx + wy = \langle (w, z), (y, -x)\rangle.$$

Thus $S^*(x, y) = (y, -x)$.

(b) Certainly $S^* \neq S$, but notice that $S^* = S^{-1}$. It follows that $SS^* = I = S^*S$, so that $S$ is normal.

(c) As shown in Example 5.9, $S$ has no eigenvalues if $\mathbf{F} = \mathbf{R}$ and $S$ has $\pm i$ as eigenvalues if $\mathbf{F} = \mathbf{C}$.

**Exercise 7.A.12.** An operator $B \in \mathcal{L}(V)$ is called *skew* if

$$B^* = -B.$$

Suppose that $T \in \mathcal{L}(V)$. Prove that $T$ is normal if and only if there exist commuting operators $A$ and $B$ such that $A$ is self-adjoint, $B$ is a skew operator, and $T = A + B$.

**Solution.** Suppose there exist such operators $A$ and $B$. Observe that

$$\begin{aligned} TT^* - T^*T &= (A + B)(A^* + B^*) - (A^* + B^*)(A + B) \\ &= (A + B)(A - B) - (A - B)(A + B) = A^2 - B^2 - A^2 + B^2 = 0, \end{aligned}$$

where we have used 7.5(a) for the first equality, that $A$ is self-adjoint and $B$ is skew for the second equality, and that $A$ and $B$ commute for the third equality. Thus $T$ is normal.

Suppose that $T$ is normal and define

$$A = \frac{T + T^*}{2} \quad \text{and} \quad B = \frac{T - T^*}{2}.$$

Certainly $A + B = T$, and 7.5(a), (b), and (c) show that $A$ is self-adjoint and $B$ is skew. Observe that

$$(T + T^*)(T - T^*) - (T - T^*)(T + T^*) = T^2 - (T^*)^2 - T^2 + (T^*)^2 = 0,$$

where we have used that $T$ and $T^*$ commute for the first equality. It follows that $4AB = 4BA$ and hence that $A$ and $B$ commute.



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**Exercise 7.A.13.** Suppose $\mathbf{F} = \mathbf{R}$. Define $\mathcal{A} \in \mathcal{L}(\mathcal{L}(V))$ by $\mathcal{A}T = T^*$ for all $T \in \mathcal{L}(V)$.

(a) Find all eigenvalues of $\mathcal{A}$.

(b) Find the minimal polynomial of $\mathcal{A}$.

**Solution.**

(a) We are looking for $T \neq 0$ and $\lambda \in \mathbf{R}$ such that $T^* = \lambda T$. Taking the adjoint of both sides of this equation and using 7.5 shows that $T = \lambda T^*$ and thus $T^* = \lambda^2 T^*$. Exercise 7.A.2 shows that $T^* \neq 0$ since $T \neq 0$ and thus $\lambda^2 = 1$, so that $\pm 1$ are the only possible eigenvalues of $\mathcal{A}$. These are indeed eigenvalues of $\mathcal{A}$, since

$$I^* = I \quad \text{and} \quad (-I)^* = -I^* = -I,$$

where we have used 7.5.

(b) By part (a) and 5.27(a), the minimal polynomial of $\mathcal{A}$ has two distinct zeros and hence must have degree at least two. Using 7.5(c), observe that

$$(\mathcal{A}^2 - I)(T) = (T^*)^* - T = 0$$

for any $T \in \mathcal{L}(V)$. Thus $z^2 - 1$ is the minimal polynomial of $\mathcal{A}$.

**Exercise 7.A.14.** Define an inner product on $\mathcal{P}_2(\mathbf{R})$ by $\langle p, q \rangle = \int_0^1 pq$. Define an operator $T \in \mathcal{L}(\mathcal{P}_2(\mathbf{R}))$ by

$$T(ax^2 + bx + c) = bx.$$

(a) Show that with this inner product, the operator $T$ is not self-adjoint.

(b) The matrix of $T$ with respect to the basis $1, x, x^2$ is

$$\begin{pmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

This matrix equals its conjugate transpose, even though $T$ is not self-adjoint. Explain why this is not a contradiction.

**Solution.**

(a) Let $p, q \in \mathcal{P}_2(\mathbf{R})$ be given by $p = 2x$ and $q = 1$, so that $Tp = p$ and $Tq = 0$. Observe that

$$\langle Tp, q \rangle = \int_0^1 2x \, dx = 1 \neq 0 = \langle p, Tq \rangle.$$

Thus $T$ is not self-adjoint.

(b) The result in 7.9 requires that the basis of $\mathcal{P}_2(\mathbf{R})$ is orthonormal, but $1, x, x^2$ is not an orthonormal basis:



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$$\langle 1, x\rangle = \int_0^1 x \, dx = \tfrac{1}{2} \neq 0.$$

**Exercise 7.A.15.** Suppose $T \in \mathcal{L}(V)$ is invertible. Prove that

(a) $T$ is self-adjoint $\Leftrightarrow$ $T^{-1}$ is self-adjoint;

(b) $T$ is normal $\Leftrightarrow$ $T^{-1}$ is normal.

**Solution.**

(a) Suppose that $T$ is self-adjoint. Using 7.5(f), observe that

$$(T^{-1})^* = (T^*)^{-1} = T^{-1}.$$

Thus $T^{-1}$ is self-adjoint. Replacing $T$ with $T^{-1}$ in the previous result and using that $(T^{-1})^{-1} = T$ gives us the desired equivalence.

(b) Suppose that $T$ is normal. Using 7.5(f), observe that

$$T^{-1}(T^{-1})^* = T^{-1}(T^*)^{-1} = (T^*T)^{-1} = (TT^*)^{-1} = (T^*)^{-1}T^{-1} = (T^{-1})^*T^{-1}.$$

Thus $T^{-1}$ is normal. Replacing $T$ with $T^{-1}$ in the previous result and using that $(T^{-1})^{-1} = T$ gives us the desired equivalence.

**Exercise 7.A.16.** Suppose $\mathbf{F} = \mathbf{R}$.

(a) Show that the set of self-adjoint operators on $V$ is a subspace of $\mathcal{L}(V)$.

(b) What is the dimension of the subspace of $\mathcal{L}(V)$ in (a) [in terms of $\dim V$]?

**Solution.**

(a) The zero operator is self-adjoint by Exercise 7.A.2; closure under operator addition and closure under scalar multiplication follow from 7.5(a) and 7.5(b).

(b) Suppose $\dim V = n$. By 3.71 and 7.9, it will suffice to find the dimension of the subspace $\mathcal{E}$ of $\mathbf{R}^{n,n}$ consisting of those matrices $A$ such that $A = A^t$. For $k \in \{1, ..., n\}$ let $E_{k,k}$ be the matrix with a 1 in the $k^{\text{th}}$ diagonal entry and zeros elsewhere, and for $j, k \in \{1, ..., n\}$ with $j < k$ let $E_{j,k}$ be the matrix with a 1 in the $j^{\text{th}}$ row and $k^{\text{th}}$ column, a 1 in the $k^{\text{th}}$ row and $j^{\text{th}}$ column, and zeros elsewhere. Let $\mathcal{B}$ be the list consisting of the matrices $E_{j,k}$ with $j \leq k$. It is straightforward to verify that $\mathcal{B}$ is linearly independent. Since any $A \in \mathcal{E}$ satisfies $A = A^t$, the entry in row $j$ and column $k$ of $A$ must equal the entry in row $k$ and column $j$ of $A$. It follows that $\mathcal{B}$ spans $\mathcal{E}$ and hence that $\mathcal{B}$ is a basis of $\mathcal{E}$. A simple counting argument shows that $\mathcal{B}$ has length $n(n+1)/2$ and thus

$$\dim \mathcal{E} = \frac{n(n+1)}{2}.$$



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**Exercise 7.A.17.** Suppose $\mathbf{F} = \mathbf{C}$. Show that the set of self-adjoint operators on $V$ is not a subspace of $\mathcal{L}(V)$.

**Solution.** 7.5(e) shows that the identity operator $I$ is self-adjoint. Let $v \in V$ be non-zero and observe that

$$(iI)(v) = iv \neq -iv = \big(\bar{i}I\big)(v) = (iI)^*(v),$$

where we have used 7.5(b). It follows that $iI$ is not self-adjoint, hence that the set of self-adjoint operators on $V$ is not closed under scalar multiplication, and hence that this set is not a subspace of $V$.

**Exercise 7.A.18.** Suppose $\dim V \geq 2$. Show that the set of normal operators on $V$ is not a subspace of $\mathcal{L}(V)$.

**Solution.** Let $e_1, e_2, ..., e_n$ be an orthonormal basis of $V$ and let $S, T \in \mathcal{L}(V)$ be the operators whose matrices with respect to this basis are

$$A = \begin{pmatrix} 1 & 0 & 0 & \cdots & 0 \\ 0 & 0 & 0 & \cdots & 0 \\ 0 & 0 & 0 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 0 \end{pmatrix} \quad \text{and} \quad B = \begin{pmatrix} 0 & 1 & 0 & \cdots & 0 \\ -1 & 0 & 0 & \cdots & 0 \\ 0 & 0 & 0 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 0 \end{pmatrix}.$$

Note that $S$ is self-adjoint and hence normal. Note further that $T$ satisfies $T^* = -T$, so that $TT^* = T^*T = -T^2$; it follows that $T$ is also normal. However, some calculations reveal that

$$(A+B)(A+B)^* = \begin{pmatrix} 2 & -1 & 0 & \cdots & 0 \\ -1 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 0 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 0 \end{pmatrix} \neq (A+B)^*(A+B) = \begin{pmatrix} 2 & 1 & 0 & \cdots & 0 \\ 1 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 0 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 0 \end{pmatrix}.$$

Thus $S+T$ is not normal. It follows that the set of normal operators on $V$ is not closed under addition and hence cannot be a subspace of $\mathcal{L}(V)$.

**Exercise 7.A.19.** Suppose $T \in \mathcal{L}(V)$ and $\|T^*v\| \leq \|Tv\|$ for every $v \in V$. Prove that $T$ is normal.

> *This exercise fails on infinite-dimensional inner product spaces, leading to what are called hyponormal operators, which have a well-developed theory.*

**Solution.** Let $e_1, ..., e_n$ be an orthonormal basis of $V$. It follows from Exercise 7.A.5 that

$$\|Te_1\|^2 + \cdots + \|Te_n\|^2 = \|T^*e_1\|^2 + \cdots + \|T^*e_n\|^2. \qquad (*)$$



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By assumption we have $\|T^*e_k\| \leq \|Te_k\|$ for each $k \in \{1, ..., n\}$. In fact, each of these inequalities must be an equality, otherwise the right-hand side of $(*)$ would be strictly less than the left-hand side. Because $e_1, ..., e_n$ was an arbitrary orthonormal basis of $V$, we have now shown that $\|T^*e_1\| = \|Te_1\|$ for any $e_1 \in V$ such that $\|e_1\| = 1$. Thus for any non-zero $v \in V$ we have

$$\left\|T^*\left(\frac{v}{\|v\|}\right)\right\| = \left\|T\left(\frac{v}{\|v\|}\right)\right\| \quad \Rightarrow \quad \|T^*v\| = \|Tv\|.$$

7.20 allows us to conclude that $T$ is normal.

**Exercise 7.A.20.** Suppose $P \in \mathcal{L}(V)$ is such that $P^2 = P$. Prove that the following are equivalent.

(a) $P$ is self-adjoint.

(b) $P$ is normal.

(c) There is a subspace $U$ of $V$ such that $P = P_U$.

**Solution.** Certainly (a) implies (b).

Suppose (b) holds and note that

$$\operatorname{null} P = \operatorname{null} P^* = (\operatorname{range} P)^\perp,$$

where the first equality is 7.21(a) and the second equality is 7.6(a). It follows from Exercise 6.C.9 that (c) holds.

Suppose that (c) holds, let $v = u_1 + x_1$ and $w = u_2 + x_2$ be given, where $u_1, u_2 \in U$ and $x_1, x_2 \in U^\perp$, and observe that

$$\langle P_U v, w\rangle = \langle u_1, u_2 + x_2\rangle = \langle u_1, u_2\rangle = \langle u_1 + x_1, u_2\rangle = \langle v, P_U w\rangle.$$

Thus $P_U = P$ is self-adjoint, i.e. (a) holds.

**Exercise 7.A.21.** Suppose $D : \mathcal{P}_8(\mathbf{R}) \to \mathcal{P}_8(\mathbf{R})$ is the differentiation operator defined by $Dp = p'$. Prove that there does not exist an inner product on $\mathcal{P}_8(\mathbf{R})$ that makes $D$ a normal operator.

**Solution.** If $T \in \mathcal{L}(V)$ is normal then $\operatorname{null} T^2 = \operatorname{null} T$ (we will prove a stronger result in Exercise 7.A.27), since if $v \in \operatorname{null} T^2$ then $v \in \operatorname{null} T^*T$ by 7.20 and thus

$$\langle T^*Tv, v\rangle = 0 \quad \Leftrightarrow \quad \langle Tv, Tv\rangle = 0 \quad \Leftrightarrow \quad Tv = 0.$$

Notice that $\operatorname{null} D^2 \neq \operatorname{null} D$ since $D^2(x) = 0$ but $D(x) = 1$. It follows from our previous discussion that there does not exist an inner product on $\mathcal{P}_8(\mathbf{R})$ that makes $D$ a normal operator.



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**Exercise 7.A.22.** Give an example of an operator $T \in \mathcal{L}(\mathbf{R}^3)$ such that $T$ is normal but not self-adjoint.

**Solution.** Let $T \in \mathcal{L}(\mathbf{R}^3)$ be the operator whose matrix with respect to the standard orthonormal basis of $\mathbf{R}^3$ is

$$\begin{pmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

It follows from 7.9 that $T^* = -T$ and hence that $TT^* = T^*T = -T^2$. Thus $T$ is normal but not self-adjoint (since $T \neq 0$).

**Exercise 7.A.23.** Suppose $T$ is a normal operator on $V$. Suppose also that $v, w \in V$ satisfy the equations

$$\|v\| = \|w\| = 2, \quad Tv = 3v, \quad Tw = 4w.$$

Show that $\|T(v + w)\| = 10$.

**Solution.** Because $v$ and $w$ are eigenvectors of $T$ corresponding to distinct eigenvalues, they must be orthogonal by 7.22. The Pythagorean theorem then implies that

$$\|T(v + w)\|^2 = \|3v + 4w\|^2 = \|3v\|^2 + \|4w\|^2 = 36 + 64 = 100.$$

Thus $\|T(v + w)\| = 10$.

**Exercise 7.A.24.** Suppose $T \in \mathcal{L}(V)$ and

$$a_0 + a_1 z + a_2 z^2 + \cdots + a_{m-1} z^{m-1} + z^m$$

is the minimal polynomial of $T$. Prove that the minimal polynomial of $T^*$ is

$$\overline{a_0} + \overline{a_1} z + \overline{a_2} z^2 + \cdots + \overline{a_{m-1}} z^{m-1} + z^m.$$

*This exercise shows that the minimal polynomial of $T^*$ equals the minimal polynomial of $T$ if $\mathbf{F} = \mathbf{R}$.*

**Solution.** For $p \in \mathcal{P}(\mathbf{F})$, let $\overline{p} \in \mathcal{P}(\mathbf{F})$ be the polynomial whose coefficients are the complex conjugates of the coefficients of $p$; notice that $\deg \overline{p} = \deg p$. Letting $p \in \mathcal{P}(\mathbf{F})$ be the minimal polynomial of $T$, i.e.

$$p(z) = a_0 + a_1 z + a_2 z^2 + \cdots + a_{m-1} z^{m-1} + z^m,$$

our aim is to show that $\overline{p}$ is the minimal polynomial of $T^*$. Notice that, by 7.5 and Exercise 7.A.2,

$$p(T) = 0 \quad \Rightarrow \quad [p(T)]^* = 0 \quad \Rightarrow \quad \overline{p}(T^*) = 0.$$



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Suppose $s \in \mathcal{P}(\mathbf{F})$ satisfies $\deg s < \deg p$. It follows that $\deg \overline{s} < \deg p$ and hence that

$$\overline{s}(T) \neq 0 \quad \Rightarrow \quad [\overline{s}(T)]^* \neq 0 \quad \Rightarrow \quad s(T^*) \neq 0.$$

Thus the minimal polynomial of $T^*$ must have degree at least $\deg p$. Since $\overline{p}$ is monic, $\deg \overline{p} = \deg p$, and $\overline{p}(T^*) = 0$, we may conclude that $\overline{p}$ is the minimal polynomial of $T^*$.

**Exercise 7.A.25.** Suppose $T \in \mathcal{L}(V)$. Prove that $T$ is diagonalizable if and only if $T^*$ is diagonalizable.

**Solution.** Let $p \in \mathcal{P}(\mathbf{F})$ be the minimal polynomial of $T$; using the notation of Exercise 7.A.24, $\overline{p}$ is the minimal polynomial of $T^*$. For $\alpha \in \mathbf{F}$ notice that

$$p(\alpha) = 0 \quad \Leftrightarrow \quad \overline{p(\alpha)} = 0 \quad \Leftrightarrow \quad \overline{p}(\overline{\alpha}) = 0.$$

It follows that if $p$ is of the form $p(z) = (z - \alpha_1) \cdots (z - \alpha_m)$ for some distinct $\alpha_1, ..., \alpha_m \in \mathbf{F}$ then $\overline{p}$ is of the form $\overline{p}(z) = (z - \overline{\alpha_1}) \cdots (z - \overline{\alpha_m})$, where $\overline{\alpha_1}, ..., \overline{\alpha_m}$ are distinct. By 5.62 this is exactly the statement that $T^*$ is diagonalizable if $T$ is diagonalizable. Replacing $T$ with $T^*$ in this implication and using that $(T^*)^* = T$ gives us the desired equivalence.

**Exercise 7.A.26.** Fix $u, x \in V$. Define $T \in \mathcal{L}(V)$ by $Tv = \langle v, u \rangle x$ for every $v \in V$.

(a) Prove that if $V$ is a real vector space, then $T$ is self-adjoint if and only if the list $u, x$ is linearly dependent.

(b) Prove that $T$ is normal if and only if the list $u, x$ is linearly dependent.

**Solution.** Note that example 7.3 gives us the formula

$$T^*v = \langle v, x \rangle u.$$

(a) Suppose that $u, x$ is linearly dependent, say $x = \lambda u$ for some $\lambda \in \mathbf{R}$, and observe that

$$Tv = \langle v, u \rangle x = \lambda \langle v, u \rangle u = \langle v, \lambda u \rangle u = \langle v, x \rangle u = T^*v$$

for any $v \in V$. Thus $T$ is self-adjoint.

Now suppose that $T$ is self-adjoint. If $u = 0$ then we are done, so suppose that $u \neq 0$. Since $T$ is self-adjoint we must have

$$Tv = \langle v, u \rangle x = \langle v, x \rangle u = T^*v$$

for every $v \in V$. In particular,

$$\langle u, u \rangle x = \langle u, x \rangle u \quad \Rightarrow \quad x = \frac{\langle u, x \rangle}{\langle u, u \rangle} u,$$

demonstrating that $u, x$ is linearly dependent.

(b) Note that

$$(TT^* - T^*T)v = \langle v, x \rangle \langle u, u \rangle x - \langle v, u \rangle \langle x, x \rangle u$$



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for any $v \in V$. Suppose that $u, x$ is linearly dependent, say $x = \lambda u$ for some $\lambda \in \mathbf{F}$, and observe that

$$\begin{aligned}(TT^* - T^*T)v &= \langle v, x\rangle\langle u, u\rangle x - \langle v, u\rangle\langle x, x\rangle u \\ &= |\lambda|^2\langle v, u\rangle\langle u, u\rangle u - |\lambda|^2\langle v, u\rangle\langle u, u\rangle u = 0\end{aligned}$$

for every $v \in V$. Thus $T$ is normal.

Conversely, suppose that $T$ is normal. If $u = 0$ then we are done, so suppose that $u \neq 0$ and observe that

$$(TT^* - T^*T)x = \langle x, x\rangle\langle u, u\rangle x - \langle x, u\rangle\langle x, x\rangle u = 0 \quad \Rightarrow \quad x = \frac{\langle x, u\rangle}{\langle u, u\rangle}u,$$

demonstrating that $u, x$ is linearly dependent.

**Exercise 7.A.27.** Suppose $T \in \mathcal{L}(V)$ is normal. Prove that

$$\operatorname{null} T^k = \operatorname{null} T \quad \text{and} \quad \operatorname{range} T^k = \operatorname{range} T$$

for every positive integer $k$.

**Solution.** We will use induction to prove that $\operatorname{null} T^k = \operatorname{null} T$ for every positive integer $k$. The base case $k = 1$ is clear, so suppose that the result holds for some positive integer $k$. Certainly $\operatorname{null} T^k \subseteq \operatorname{null} T^{k+1}$, so suppose that $v \in \operatorname{null} T^{k+1}$. It follows from 7.20 that $v \in \operatorname{null} T^*T^k$ and hence that

$$\langle T^*T^k v, T^{k-1}v\rangle = 0 \quad \Leftrightarrow \quad \langle T^k v, T^k v\rangle = 0 \quad \Leftrightarrow \quad T^k v = 0.$$

Thus $\operatorname{null} T^{k+1} = \operatorname{null} T^k = \operatorname{null} T$, where the last equality is our induction hypothesis. This completes the induction step and the proof.

If $T$ is normal then $T^*$ is normal, and 7.5(d) shows that $\left(T^k\right)^* = \left(T^*\right)^k$ for every positive integer $k$. It then follows from the previous result that

$$\begin{aligned}\operatorname{range} T^k &= \left(\operatorname{null}\left(T^k\right)^*\right)^\perp && \text{(7.6(d))} \\ &= \left(\operatorname{null}\left(T^*\right)^k\right)^\perp \\ &= \left(\operatorname{null} T^*\right)^\perp && (T^* \text{ is normal}) \\ &= \operatorname{range} T && \text{(7.6(d))}\end{aligned}$$

for any positive integer $k$.

**Exercise 7.A.28.** Suppose $T \in \mathcal{L}(V)$ is normal. Prove that if $\lambda \in \mathbf{F}$, then the minimal polynomial of $T$ is not a polynomial multiple of $(x - \lambda)^2$.



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**Solution.** We will prove the contrapositive. Suppose $T \in \mathcal{L}(V)$ has minimal polynomial $p \in \mathcal{P}(\mathbf{F})$ of the form $p(x) = (x-\lambda)^2 q(x)$ for some $\lambda \in \mathbf{F}$ and some $q \in \mathcal{P}(\mathbf{F})$. Because the polynomial $(x-\lambda)q(x)$ has degree strictly less than $p$, there must exist some $v \in V$ such that $q(T)v \notin \operatorname{null}(T-\lambda I)$. Since $p(T) = 0$ we have $q(T)v \in \operatorname{null}(T-\lambda I)^2$ and thus $\operatorname{null}(T-\lambda I)^2 \neq \operatorname{null}(T-\lambda I)$. It follows from Exercise 7.A.27 that $T - \lambda I$ is not normal. The contrapositive of 7.21(d) allows us to conclude that $T$ is not normal.

**Exercise 7.A.29.** Prove or give a counterexample: If $T \in \mathcal{L}(V)$ and there is an orthonormal basis $e_1, \ldots, e_n$ of $V$ such that $\|Te_k\| = \|T^* e_k\|$ for each $k = 1, \ldots, n$, then $T$ is normal.

**Solution.** This is false. Let $T$ be the operator on $\mathbf{F}^2$ whose matrix with respect to the standard orthonormal basis $e_1, e_2$ of $\mathbf{F}^2$ is

$$\begin{pmatrix} 1 & 1 \\ -1 & 0 \end{pmatrix};$$

As we showed in Exercise 7.A.18, $T$ is not normal. However,

$$\|Te_1\| = \|T^* e_1\| = \sqrt{2} \quad \text{and} \quad \|Te_2\| = \|T^* e_2\| = 1.$$

**Exercise 7.A.30.** Suppose that $T \in \mathcal{L}(\mathbf{F}^3)$ is normal and $T(1,1,1) = (2,2,2)$. Suppose $(z_1, z_2, z_3) \in \operatorname{null} T$. Prove that $z_1 + z_2 + z_3 = 0$.

**Solution.** If $u := (z_1, z_2, z_3) = 0$ then we are done, so suppose that $u \neq 0$. It follows that $u$ is an eigenvector of $T$ corresponding to the eigenvalue 0. Note that $v := (1,1,1)$ is an eigenvector of $T$ corresponding to the eigenvalue 2. Since these are eigenvectors of a normal operator corresponding to distinct eigenvalues, they must be orthogonal by 7.22. That is,

$$\langle u, v \rangle = z_1 + z_2 + z_3 = 0.$$

**Exercise 7.A.31.** Fix a positive integer $n$. In the inner product space of continuous real-valued functions on $[-\pi, \pi]$ with inner product $\langle f, g \rangle = \int_{-\pi}^{\pi} fg$, let

$$V = \operatorname{span}(1, \cos x, \cos 2x, \ldots, \cos nx, \sin x, \sin 2x, \ldots, \sin nx).$$

(a) Define $D \in \mathcal{L}(V)$ by $Df = f'$. Show that $D^* = -D$. Conclude that $D$ is normal but not self-adjoint.

(b) Define $T \in \mathcal{L}(V)$ by $Tf = f''$. Show that $T$ is self-adjoint.

**Solution.**

(a) For each $k \in \{1, \ldots, n\}$, let

$$v = \frac{1}{\sqrt{2\pi}}, \quad e_k = \frac{\cos kx}{\sqrt{\pi}}, \quad \text{and} \quad f_k = \frac{\sin kx}{\sqrt{\pi}}.$$



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If we let $\mathcal{B} = v, e_1, ..., e_n, f_1, ..., f_n$, then $\mathcal{B}$ is an orthonormal basis of $V$, as shown in Exercise 6.B.4. Observe that $Dv = 0, De_k = -kf_k$, and $Df_k = ke_k$ for each $k \in \{1, ..., n\}$. It follows from 7.9 that

$$D^*v = 0 = -Dv, \quad D^*e_k = kf_k = -De_k, \quad \text{and} \quad D^*f_k = -ke_k = -Df_k$$

for each $k \in \{1, ..., n\}$. Thus $D^* = -D$, so that $D^*$ is normal ($D^*D = DD^* = -D^2$) but not self-adjoint (since $V \neq 0$).

(b) Notice that $T = D^2$. It follows from 7.5 that

$$T^* = \left(D^2\right)^* = \left(D^*\right)^2 = (-D)^2 = D^2 = T.$$

Thus $T$ is self-adjoint.

**Exercise 7.A.32.** Suppose $T : V \to W$ is a linear map. Show that under the standard identification of $V$ with $V'$ (see 6.58) and the corresponding identification of $W$ with $W'$, the adjoint map $T^* : W \to V$ corresponds to the dual map $T' : W' \to V'$. More precisely, show that

$$T'(\varphi_w) = \varphi_{T^*w}$$

for all $w \in W$, where $\varphi_w$ and $\varphi_{T^*w}$ are defined as in 6.58.

**Solution.** For any $v \in V$ and $w \in W$, observe that

$$[T'(\varphi_w)]v = \varphi_w(Tv) = \langle Tv, w\rangle = \langle v, T^*w\rangle = \varphi_{T^*w}(v).$$



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## 7.B. Spectral Theorem

**Exercise 7.B.1.** Prove that a normal operator on a complex inner product space is self-adjoint if and only if all its eigenvalues are real.

*This exercise strengthens the analogy (for normal operators) between self-adjoint operators and real numbers.*

**Solution.** Suppose $V$ is a complex inner product space and $T \in \mathcal{L}(V)$ is normal. If $T$ is self-adjoint then 7.12 shows that every eigenvalue of $T$ is real. Suppose that every eigenvalue of $T$ is real. The complex spectral theorem (7.31) implies that there is an orthonormal basis of $V$ with respect to which the matrix $\mathcal{M}(T)$ is diagonal. Because each eigenvalue of $T$ is real, the diagonal entries of $\mathcal{M}(T)$ must be real. It follows that $\mathcal{M}(T)$ equals its own conjugate transpose and hence that $T$ is self-adjoint.

**Exercise 7.B.2.** Suppose $\mathbf{F} = \mathbf{C}$. Suppose $T \in \mathcal{L}(V)$ is normal and has only one eigenvalue. Prove that $T$ is a scalar multiple of the identity operator.

**Solution.** Suppose that $\lambda \in \mathbf{C}$ is the sole eigenvalue of $T$. The complex spectral theorem (7.31) implies that there is an orthonormal basis of $V$ with respect to which $\mathcal{M}(T)$ is diagonal. Because $\lambda$ is the only eigenvalue of $T$, each diagonal entry of $\mathcal{M}(T)$ must be equal to $\lambda$. Thus $T = \lambda I$.

**Exercise 7.B.3.** Suppose $\mathbf{F} = \mathbf{C}$ and $T \in \mathcal{L}(V)$ is normal. Prove that the set of eigenvalues of $T$ is contained in $\{0, 1\}$ if and only if there is a subspace $U$ of $V$ such that $T = P_U$.

**Solution.** If there exists a subspace $U$ of $V$ such that $T = P_U$ then $T^2 = T$ and it follows from Exercise 5.A.8 that the set of eigenvalues of $T$ is contained in $\{0, 1\}$.

Suppose that the set of eigenvalues of $T$ is contained in $\{0, 1\}$. The complex spectral theorem (7.31) implies that there is an orthonormal basis $\mathcal{B}$ of $V$ consisting of eigenvectors of $T$. Each basis vector in $\mathcal{B}$ must correspond either to the eigenvalue 0 or the eigenvalue 1. Let $u_1, ..., u_m$ be those basis vectors in $\mathcal{B}$ corresponding to the eigenvalue 1 and let $v_1, ..., v_n$ be those basis vectors in $\mathcal{B}$ corresponding to the eigenvalue 0; either of these lists may be empty. Let $U = \operatorname{span}(u_1, ..., u_m)$. Because $\mathcal{B}$ is an orthonormal basis of $V$, it follows that $U^\perp = \operatorname{span}(v_1, ..., v_n)$ and hence that

$$P_U u_k = u_k = T u_k \quad \text{and} \quad P_U v_k = 0 = T v_k.$$

Thus $T = P_U$.



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**Exercise 7.B.4.** Prove that a normal operator on a complex inner product space is skew (meaning that it equals the negative of its adjoint) if and only if all its eigenvalues are purely imaginary (meaning that they have real part equal to 0).

**Solution.** Suppose $V$ is a complex inner product space and $T \in \mathcal{L}(V)$ is normal. By the complex spectral theorem (7.31) there is an orthonormal basis of $V$ with respect to which $\mathcal{M}(T)$ is of the form

$$\begin{pmatrix} \lambda_1 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & \lambda_n \end{pmatrix},$$

where $\lambda_1, ..., \lambda_n \in \mathbf{C}$ are the eigenvalues of $T$. Observe that

$$T \text{ is skew} \quad \Leftrightarrow \quad \mathcal{M}(T) = -[\mathcal{M}(T)]^* \quad \Leftrightarrow \quad \text{each } \lambda_k = -\overline{\lambda_k}$$
$$\Leftrightarrow \quad \text{each } \lambda_k \text{ is purely imaginary} \quad \Leftrightarrow \quad \text{each eigenvalue of } T \text{ is purely imaginary.}$$

**Exercise 7.B.5.** Prove or give a counterexample: If $T \in \mathcal{L}(\mathbf{C}^3)$ is a diagonalizable operator, then $T$ is normal (with respect to the usual inner product).

**Solution.** This is false. Consider the basis

$$v_1 = (1, 0, 0), \quad v_2 = (0, 1, 0), \quad v_3 = (1, 0, 1)$$

of $\mathbf{C}^3$ and define $T \in \mathcal{L}(\mathbf{C}^3)$ by

$$Tv_1 = v_1, \quad Tv_2 = v_2, \quad Tv_3 = 2v_3.$$

Observe that $T$ is diagonalizable since $\mathbf{C}^3$ has a basis $v_1, v_2, v_3$ consisting of eigenvectors of $T$. Observe further that $v_1, v_3$ are eigenvectors of $T$ corresponding to distinct eigenvalues, and that $v_1$ and $v_3$ are not orthogonal: $\langle v_1, v_3 \rangle = 1$. It follows from 7.22 that $T$ is not normal.

**Exercise 7.B.6.** Suppose $V$ is a complex inner product space and $T \in \mathcal{L}(V)$ is a normal operator such that $T^9 = T^8$. Prove that $T$ is self-adjoint and $T^2 = T$.

**Solution.** The complex spectral theorem (7.31) implies that there is an orthonormal basis $e_1, ..., e_n$ of $V$ consisting of eigenvectors of $T$, so that $Te_k = \lambda_k e_k$ for each $k \in \{1, ..., n\}$, where $\lambda_1, ..., \lambda_n$ are the eigenvalues of $T$. By assumption we have

$$T^9 e_k = T^8 e_k \quad \Leftrightarrow \quad \lambda_k^9 e_k = \lambda_k^8 e_k \quad \Leftrightarrow \quad \lambda_k^9 = \lambda_k^8 \quad \Leftrightarrow \quad \lambda_k \in \{0, 1\}.$$

It follows from Exercise 7.B.3 that $T = P_U$ for some subspace $U$ of $V$, so that $T^2 = T$, and Exercise 7.B.1 (or Exercise 7.A.20) shows that $T$ is self-adjoint.



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**Exercise 7.B.7.** Give an example of an operator $T$ on a complex vector space such that $T^9 = T^8$ but $T^2 \neq T$.

**Solution.** Let $T$ be the operator on $\mathbf{C}^2$ whose matrix with respect to the standard basis of $\mathbf{C}^2$ is

$$\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}.$$

It follows that $T^9 = T^8 = T^2 = 0$ but $T \neq 0$.

**Exercise 7.B.8.** Suppose $\mathbf{F} = \mathbf{C}$ and $T \in \mathcal{L}(V)$. Prove that $T$ is normal if and only if every eigenvector of $T$ is also an eigenvector of $T^*$.

**Solution.** If $T$ is normal then every eigenvector of $T$ is an eigenvector of $T^*$ by 7.21(e).

Suppose that every eigenvector of $T$ is also an eigenvector of $T^*$. By Schur's theorem (6.38) there is an orthonormal basis $e_1, ..., e_n$ of $V$ with respect to which $A := \mathcal{M}(T)$ is upper-triangular. It follows that $Te_1 = A_{1,1}e_1$, so that $e_1$ is an eigenvector of $T$. Our assumption implies that $e_1$ is also an eigenvector of $T^*$, say $T^*e_1 = \mu_1 e_1$. On the other hand, by 7.9,

$$T^*e_1 = \overline{A_{1,1}}e_1 + \overline{A_{1,2}}e_2 + \cdots + \overline{A_{1,n}}e_n.$$

It follows from unique representation that $A_{1,2} = \cdots = A_{1,n} = 0$. Thus $Te_2 = A_{2,2}e_2$, so that $e_2$ is an eigenvector of $T$. Our assumption implies that $e_2$ is also an eigenvector of $T^*$, say $T^*e_2 = \mu_2 e_2$. On the other hand, by 7.9,

$$T^*e_2 = \overline{A_{2,2}}e_2 + \overline{A_{2,3}}e_3 + \cdots + \overline{A_{2,n}}e_n.$$

It follows from unique representation that $A_{2,3} = \cdots = A_{2,n} = 0$. Continuing in this manner, we see that $A$ is a diagonal matrix. The complex spectral theorem allows us to conclude that $T$ is normal.

**Exercise 7.B.9.** Suppose $\mathbf{F} = \mathbf{C}$ and $T \in \mathcal{L}(V)$. Prove that $T$ is normal if and only if there exists a polynomial $p \in \mathcal{P}(\mathbf{C})$ such that $T^* = p(T)$.

**Solution.** If there exists such a polynomial $p$ then $T$ commutes with $T^* = p(T)$ by 5.17(b). Thus $T$ is normal.

Suppose that $T$ is normal. By the complex spectral theorem (7.31), there is an orthonormal basis $e_1, ..., e_n$ of $V$ such that each $e_k$ is an eigenvector of $T$. Let $\lambda_1, ..., \lambda_m$ be the distinct eigenvalues of $T$. Exercise 4.7 shows that there is a polynomial $p \in \mathcal{P}(\mathbf{C})$ satisfying $p(\lambda_j) = \overline{\lambda_j}$ for each $j \in \{1, ..., m\}$. For any $k \in \{1, ..., n\}$ we have $Te_k = \lambda_j e_k$ for some $j \in \{1, ..., m\}$. It follows that

$$p(T)e_k = p(\lambda_j)e_k = \overline{\lambda_j}e_k = T^*e_k,$$



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where we have used 7.21(e) for the last equality. Thus $p(T) = T^*$.

**Exercise 7.B.10.** Suppose $V$ is a complex inner product space. Prove that every normal operator on $V$ has a square root.

*An operator $S \in \mathcal{L}(V)$ is called a **square root** of $T \in \mathcal{L}(V)$ if $S^2 = T$. We will discuss more about square roots of operators in Sections 7C and 8C.*

**Solution.** Let $T \in \mathcal{L}(V)$ be normal. The complex spectral theorem (7.31) implies that there is an orthonormal basis $e_1, ..., e_n$ of $V$ such that $Te_k = \lambda_k e_k$ for each $k \in \{1, ..., n\}$, where $\lambda_1, ..., \lambda_n \in \mathbf{C}$ are the eigenvalues of $T$. Because any complex number has a square root, for each $k$ there exists some $\mu_k \in \mathbf{C}$ such that $\mu_k^2 = \lambda_k$. Define $S \in \mathcal{L}(V)$ by $Se_k = \mu_k e_k$ and observe that

$$S^2 e_k = \mu_k^2 e_k = \lambda_k e_k = Te_k.$$

Thus $S^2 = T$.

**Exercise 7.B.11.** Prove that every self-adjoint operator on $V$ has a cube root.

*An operator $S \in \mathcal{L}(V)$ is called a **cube root** of $T \in \mathcal{L}(V)$ if $S^3 = T$.*

**Solution.** Let $T \in \mathcal{L}(V)$ be self-adjoint. The relevant spectral theorem (7.29 or 7.31) implies that there is an orthonormal basis $e_1, ..., e_n$ of $V$ such that $Te_k = \lambda_k e_k$ for each $k \in \{1, ..., n\}$, where $\lambda_1, ..., \lambda_n \in \mathbf{F}$ are the eigenvalues of $T$. Note that each $\lambda_k$ must be real by 7.12. Because any real number has a cube root, for each $k$ there exists some $\mu_k \in \mathbf{R}$ such that $\mu_k^3 = \lambda_k$. Define $S \in \mathcal{L}(V)$ by $Se_k = \mu_k e_k$ and observe that

$$S^3 e_k = \mu_k^3 e_k = \lambda_k e_k = Te_k.$$

Thus $S^3 = T$.

**Exercise 7.B.12.** Suppose $V$ is a complex vector space and $T \in \mathcal{L}(V)$ is normal. Prove that if $S$ is an operator on $V$ that commutes with $T$, then $S$ commutes with $T^*$.

*The result in this exercise is called Fuglede's theorem.*

**Solution.** By Exercise 7.B.9 there is a polynomial $p = \sum_{k=0}^m a_k z^k \in \mathcal{P}(\mathbf{C})$ such that $T^* = p(T)$. Since $ST = TS$, a straightforward induction argument shows that $ST^k = T^k S$ for any non-negative integer $k$. It follows that

$$ST^* = Sp(T) = S\left(\sum_{k=0}^m a_k T^k\right) = \sum_{k=0}^m a_k ST^k = \sum_{k=0}^m a_k T^k S = \left(\sum_{k=0}^m a_k T^k\right) S = p(T)S = T^*S.$$



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**Exercise 7.B.13.** Without using the complex spectral theorem, use the version of Schur's theorem that applies to two commuting operators (take $\mathcal{E} = \{T, T^*\}$ in Exercise 20 in Section 6B) to give a different proof that if $\mathbf{F} = \mathbf{C}$ and $T \in \mathcal{L}(V)$ is normal, then $T$ has a diagonal matrix with respect to some orthonormal basis of $V$.

**Solution.** Since $T$ and $T^*$ commute, Exercise 6.B.20 implies that there is an orthonormal basis $e_1, ..., e_n$ of $V$ such that $\mathcal{M}(T)$ and $\mathcal{M}(T^*)$ are upper-triangular. Because $\mathcal{M}(T^*)$ is the conjugate transpose of $\mathcal{M}(T)$ (by 7.9), it must be that $\mathcal{M}(T)$ is also lower-triangular. Thus $\mathcal{M}(T)$ is diagonal.

**Exercise 7.B.14.** Suppose $\mathbf{F} = \mathbf{R}$ and $T \in \mathcal{L}(V)$. Prove that $T$ is self-adjoint if and only if all pairs of eigenvectors corresponding to distinct eigenvalues of $T$ are orthogonal and $V = E(\lambda_1, T) \oplus \cdots \oplus E(\lambda_m, T)$, where $\lambda_1, ..., \lambda_m$ denote the distinct eigenvalues of $T$.

**Solution.** If $T$ is self-adjoint then 7.22 shows that all pairs of eigenvectors corresponding to distinct eigenvalues of $T$ are orthogonal, and the real spectral theorem (7.29) shows that $T$ is diagonalizable, which by 5.55 is equivalent to

$$V = E(\lambda_1, T) \oplus \cdots \oplus E(\lambda_m, T),$$

where $\lambda_1, ..., \lambda_m$ are the distinct eigenvalues of $T$.

Now suppose that all pairs of eigenvectors corresponding to distinct eigenvalues of $T$ are orthogonal and

$$V = E(\lambda_1, T) \oplus \cdots \oplus E(\lambda_m, T),$$

where $\lambda_1, ..., \lambda_m$ are the distinct eigenvalues of $T$. Choose an orthonormal basis for each eigenspace $E(\lambda_k, T)$. Our hypotheses ensure that the list obtained by concatenating these orthonormal bases is an orthonormal basis of $V$ consisting of eigenvectors of $T$. It follows from the real spectral theorem (7.29) that $T$ is self-adjoint.

**Exercise 7.B.15.** Suppose $\mathbf{F} = \mathbf{C}$ and $T \in \mathcal{L}(V)$. Prove that $T$ is normal if and only if all pairs of eigenvectors corresponding to distinct eigenvalues of $T$ are orthogonal and $V = E(\lambda_1, T) \oplus \cdots \oplus E(\lambda_m, T)$, where $\lambda_1, ..., \lambda_m$ denote the distinct eigenvalues of $T$.

**Solution.** If $T$ is normal then 7.22 shows that all pairs of eigenvectors corresponding to distinct eigenvalues of $T$ are orthogonal, and the complex spectral theorem (7.31) shows that $T$ is diagonalizable, which by 5.55 is equivalent to

$$V = E(\lambda_1, T) \oplus \cdots \oplus E(\lambda_m, T),$$

where $\lambda_1, ..., \lambda_m$ are the distinct eigenvalues of $T$.



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Now suppose that all pairs of eigenvectors corresponding to distinct eigenvalues of $T$ are orthogonal and

$$V = E(\lambda_1, T) \oplus \cdots \oplus E(\lambda_m, T),$$

where $\lambda_1, ..., \lambda_m$ are the distinct eigenvalues of $T$. Choose an orthonormal basis for each eigenspace $E(\lambda_k, T)$. Our hypotheses ensure that the list obtained by concatenating these orthonormal bases is an orthonormal basis of $V$ consisting of eigenvectors of $T$. It follows from the complex spectral theorem (7.31) that $T$ is normal.

**Exercise 7.B.16.** Suppose $\mathbf{F} = \mathbf{C}$ and $\mathcal{E} \subseteq \mathcal{L}(V)$. Prove that there is an orthonormal basis of $V$ with respect to which every element of $\mathcal{E}$ has a diagonal matrix if and only if $S$ and $T$ are commuting normal operators for all $S, T \in \mathcal{E}$.

*This exercise extends the complex spectral theorem to the context of a collection of commuting normal operators.*

**Solution.** If there exists such an orthonormal basis of $V$ then each $T \in \mathcal{E}$ is normal by the complex spectral theorem (7.31) and each pair $S, T \in \mathcal{E}$ commutes by 5.74 (since diagonal matrices always commute).

Suppose $S$ and $T$ are commuting normal operators for all $S, T \in \mathcal{E}$. By Exercise 6.B.20, there is an orthonormal basis of $V$ with respect to which the matrix of each $T \in \mathcal{E}$ is upper-triangular. Because $T$ is normal this matrix must actually be diagonal, as the proof of the complex spectral theorem (7.31) shows.

**Exercise 7.B.17.** Suppose $\mathbf{F} = \mathbf{R}$ and $\mathcal{E} \subseteq \mathcal{L}(V)$. Prove that there is an orthonormal basis of $V$ with respect to which every element of $\mathcal{E}$ has a diagonal matrix if and only if $S$ and $T$ are commuting self-adjoint operators for all $S, T \in \mathcal{E}$.

*This exercise extends the real spectral theorem to the context of a collection of commuting self-adjoint operators.*

**Solution.** If there exists such an orthonormal basis of $V$ then each $T \in \mathcal{E}$ is self-adjoint by the real spectral theorem (7.29) and each pair $S, T \in \mathcal{E}$ commutes by 5.74 (since diagonal matrices always commute).

Suppose $S$ and $T$ are commuting self-adjoint operators for all $S, T \in \mathcal{E}$. The real spectral theorem (7.29) shows that each $T \in \mathcal{E}$ is diagonalizable and thus by Exercise 5.E.2 there is a basis $v_1, ..., v_n$ of $V$ with respect to which the matrix of each $T \in \mathcal{E}$ is diagonal. Perform the Gram-Schmidt procedure on $v_1, ..., v_n$ to obtain an orthonormal basis $e_1, ..., e_n$ of $V$ such that

$$\operatorname{span}(e_1, ..., e_k) = \operatorname{span}(v_1, ..., v_k)$$

for each $k \in \{1, ..., n\}$. Let $T \in \mathcal{E}$ be given and note that $\operatorname{span}(v_1, ..., v_k) = \operatorname{span}(e_1, ..., e_k)$ is invariant under $T$ for each $k \in \{1, ..., n\}$ since $\mathcal{M}(T, (v_1, ..., v_n))$ is diagonal. Thus, by 5.39,



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$\mathcal{M}(T, (e_1, ..., e_n))$ is upper-triangular. Because $T$ is self-adjoint this matrix must actually be diagonal, as the proof of the real spectral theorem (7.29) shows.

**Exercise 7.B.18.** Give an example of a real inner product space $V$, an operator $T \in \mathcal{L}(V)$, and real numbers $b^2 < 4c$ such that

$$T^2 + bT + cI$$

is not invertible.

*This exercise shows that the hypothesis that $T$ is self-adjoint cannot be deleted in 7.26, even for real vector spaces.*

**Solution.** Let $V = \mathbf{R}^2$ with the usual inner product and let $T \in \mathcal{L}(\mathbf{R}^2)$ be a counterclockwise rotation about the origin by 90°, so that the matrix of $T$ with respect to the standard orthonormal basis of $\mathbf{R}^2$ is

$$\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}.$$

Observe that $T^2 + I$ (i.e. taking $b = 0$ and $c = 1$) is zero and hence not invertible.

**Exercise 7.B.19.** Suppose $T \in \mathcal{L}(V)$ is self-adjoint and $U$ is a subspace of $V$ that is invariant under $T$.

(a) Prove that $U^\perp$ is invariant under $T$.

(b) Prove that $T|_U \in \mathcal{L}(U)$ is self-adjoint.

(c) Prove that $T|_{U^\perp} \in \mathcal{L}(U^\perp)$ is self-adjoint.

**Solution.**

(a) This is immediate from Exercise 7.A.4 and the fact that $T^* = T$.

(b) Because $T$ is self-adjoint we have $\langle Tx, y\rangle = \langle x, Ty\rangle$ for all $x, y \in V$. In particular this equality holds for all $x, y \in U$. It follows that $T|_U$ is self-adjoint.

(c) This follows by replacing $U$ with $U^\perp$ in part (b), which is valid by part (a).



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**Exercise 7.B.20.** Suppose $T \in \mathcal{L}(V)$ is normal and $U$ is a subspace of $V$ that is invariant under $T$.

(a) Prove that $U^\perp$ is invariant under $T$.

(b) Prove that $U$ is invariant under $T^*$.

(c) Prove that $(T|_U)^* = (T^*)|_U$.

(d) Prove that $T|_U \in \mathcal{L}(U)$ and $T|_{U^\perp} \in \mathcal{L}(U^\perp)$ are normal operators.

*This exercise can be used to give yet another proof of the complex spectral theorem (use induction on* $\dim V$ *and the result that* $T$ *has an eigenvector).*

**Solution.**

(a) Let $e_1, ..., e_m$ be an orthonormal basis of $U$ and let $e_{m+1}, ..., e_n$ be an orthonormal basis of $U^\perp$, so that $e_1, ..., e_n$ is an orthonormal basis of $V$. Because $U$ is invariant under $T$, the matrix of $T$ with respect to $e_1, ..., e_n$ is of the form

$$\begin{pmatrix} A_{1,1} & \cdots & A_{1,m} & A_{1,m+1} & \cdots & A_{1,n} \\ \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ A_{m,1} & \cdots & A_{m,m} & A_{m,m+1} & \cdots & A_{m,n} \\ 0 & \cdots & 0 & A_{m+1,m+1} & \cdots & A_{m+1,n} \\ \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ 0 & \cdots & 0 & A_{n,m+1} & \cdots & A_{n,n} \end{pmatrix}.$$

It follows from 7.9 and the matrix above that

$$\sum_{k=1}^{m} \|Te_k\|^2 = \sum_{k=1}^{m}\sum_{j=1}^{m} |A_{j,k}|^2 \quad \text{and} \quad \sum_{k=1}^{m} \|T^*e_k\|^2 = \sum_{k=1}^{m}\sum_{j=1}^{n} |A_{k,j}|^2 = \sum_{k=1}^{n}\sum_{j=1}^{m} |A_{j,k}|^2,$$

where we have swapped the indices $j$ and $k$ for the last equality. For each $k \in \{1, ..., m\}$ we have $\|Te_k\|^2 = \|T^*e_k\|^2$ and thus

$$0 = \sum_{k=1}^{m} \|T^*e_k\|^2 - \sum_{k=1}^{m} \|Te_k\|^2 = \sum_{k=1}^{n}\sum_{j=1}^{m} |A_{j,k}|^2 - \sum_{k=1}^{m}\sum_{j=1}^{m} |A_{j,k}|^2 = \sum_{k=m+1}^{n}\sum_{j=1}^{m} |A_{j,k}|^2.$$

It follows that $A_{j,k} = 0$ for each $j \in \{1, ..., m\}$ and each $k \in \{m+1, ..., n\}$, so that the matrix of $T$ with respect to $e_1, ..., e_n$ is of the form

$$\begin{pmatrix} A_{1,1} & \cdots & A_{1,m} & 0 & \cdots & 0 \\ \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ A_{m,1} & \cdots & A_{m,m} & 0 & \cdots & 0 \\ 0 & \cdots & 0 & A_{m+1,m+1} & \cdots & A_{m+1,n} \\ \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ 0 & \cdots & 0 & A_{n,m+1} & \cdots & A_{n,n} \end{pmatrix}.$$

Thus $U^\perp$ is invariant under $T$.



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(b) With respect to the orthonormal basis $e_1, ..., e_n$ of $V$ from part (a), 7.9 shows that the matrix of $T^*$ is

$$\begin{pmatrix} \overline{A_{1,1}} & \cdots & \overline{A_{m,1}} & 0 & \cdots & 0 \\ \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ \overline{A_{1,m}} & \cdots & \overline{A_{m,m}} & 0 & \cdots & 0 \\ 0 & \cdots & 0 & \overline{A_{m+1,m+1}} & \cdots & \overline{A_{n,m+1}} \\ \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ 0 & \cdots & 0 & \overline{A_{m+1,n}} & \cdots & \overline{A_{n,n}} \end{pmatrix}.$$

Thus $U$ is invariant under $T^*$.

(c) With respect to the orthonormal basis $e_1, ..., e_m$ of $U$ from part (a) we have

$$\mathcal{M}(T|_U) = \begin{pmatrix} A_{1,1} & \cdots & A_{1,m} \\ \vdots & \ddots & \vdots \\ A_{m,1} & \cdots & A_{m,m} \end{pmatrix} \quad \Rightarrow \quad \mathcal{M}\big((T|_U)^*\big) = \begin{pmatrix} \overline{A_{1,1}} & \cdots & \overline{A_{m,1}} \\ \vdots & \ddots & \vdots \\ \overline{A_{1,m}} & \cdots & \overline{A_{m,m}} \end{pmatrix}.$$

Part (b) shows that

$$\mathcal{M}\big((T^*)|_U\big) = \begin{pmatrix} \overline{A_{1,1}} & \cdots & \overline{A_{m,1}} \\ \vdots & \ddots & \vdots \\ \overline{A_{1,m}} & \cdots & \overline{A_{m,m}} \end{pmatrix} = \mathcal{M}\big((T|_U)^*\big).$$

Thus $(T|_U)^* = (T^*)|_U$.

(d) Using part (c), notice that

$$(T|_U)^* T|_U = (T^*)|_U \, T|_U = (T^*T)|_U = (TT^*)|_U = T|_U \, (T^*)|_U = T|_U \, (T|_U)^*.$$

Thus $T|_U$ is normal. Replacing $U$ with $U^\perp$ in this result, which is valid by part (a), shows that $T|_{U^\perp}$ is normal.

**Exercise 7.B.21.** Suppose that $T$ is a self-adjoint operator on a finite-dimensional inner product space and that 2 and 3 are the only eigenvalues of $T$. Prove that

$$T^2 - 5T + 6I = 0.$$

**Solution.** Exercise 7.B.14 if $\mathbf{F} = \mathbf{R}$, or Exercise 7.B.15 if $\mathbf{F} = \mathbf{C}$, shows that

$$V = E(2, T) \oplus E(3, T).$$

Thus any $v \in V$ is of the form $v = x + y$, where $x \in E(2, T)$ and $y \in E(3, T)$. It follows that

$$(T^2 - 5T + 6I)v = (T - 3I)(T - 2I)x + (T - 2I)(T - 3I)y = 0.$$



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**Exercise 7.B.22.** Give an example of an operator $T \in \mathcal{L}(\mathbf{C}^3)$ such that 2 and 3 are the only eigenvalues of $T$ and $T^2 - 5T + 6I \neq 0$.

**Solution.** Let $e_1, e_2, e_3$ be the standard basis of $\mathbf{C}^3$ and define $T \in \mathcal{L}(\mathbf{C}^3)$ by

$$Te_1 = 2e_1, \quad Te_2 = e_1 + 2e_2, \quad Te_3 = 3e_3,$$

so that the matrix of $T$ with respect to $e_1, e_2, e_3$ is

$$\begin{pmatrix} 2 & 1 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{pmatrix}.$$

Since this matrix is upper-triangular, we see that 2 and 3 are the only eigenvalues of $T$. However, notice that

$$(T^2 - 5T + 6I)e_2 = -e_1 \neq 0.$$

**Exercise 7.B.23.** Suppose $T \in \mathcal{L}(V)$ is self-adjoint, $\lambda \in \mathbf{F}$, and $\varepsilon > 0$. Suppose there exists $v \in V$ such that $\|v\| = 1$ and

$$\|Tv - \lambda v\| < \varepsilon.$$

Prove that $T$ has an eigenvalue $\lambda'$ such that $|\lambda - \lambda'| < \varepsilon$.

> *This exercise shows that for a self-adjoint operator, a number that is close to satisfying an equation that would make it an eigenvalue is close to an eigenvalue.*

**Solution.** The relevant spectral theorem (7.29 or 7.31) implies that there is an orthonormal basis $e_1, ..., e_n$ of $V$ consisting of eigenvectors of $T$, so that $Te_k = \lambda_k e_k$ for each $k \in \{1, ..., n\}$, where $\lambda_1, ..., \lambda_n \in \mathbf{F}$ are the eigenvalues of $T$. It follows from 6.30 and that

$$\|Tv - \lambda v\|^2 = |\lambda_1 - \lambda|^2 |\langle v, e_1 \rangle|^2 + \cdots + |\lambda_n - \lambda|^2 |\langle v, e_n \rangle|^2.$$

Let $\lambda'$ be the eigenvalue in $\{\lambda_1, ..., \lambda_n\}$ which minimizes $|\lambda_k - \lambda|$ and observe that

$$\begin{aligned} |\lambda' - \lambda|^2 &= |\lambda' - \lambda|^2 \|v\|^2 \\ &= |\lambda' - \lambda|^2 |\langle v, e_1 \rangle|^2 + \cdots + |\lambda' - \lambda|^2 |\langle v, e_n \rangle|^2 \\ &\leq |\lambda_1 - \lambda|^2 |\langle v, e_1 \rangle|^2 + \cdots + |\lambda_n - \lambda|^2 |\langle v, e_n \rangle|^2 \\ &= \|Tv - \lambda v\|^2 \\ &< \varepsilon^2. \end{aligned}$$

Thus $|\lambda' - \lambda| < \varepsilon$.



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**Exercise 7.B.24.** Suppose $U$ is a finite-dimensional vector space and $T \in \mathcal{L}(U)$.

(a) Suppose $\mathbf{F} = \mathbf{R}$. Prove that $T$ is diagonalizable if and only if there is a basis of $U$ such that the matrix of $T$ with respect to this basis equals its transpose.

(b) Suppose $\mathbf{F} = \mathbf{C}$. Prove that $T$ is diagonalizable if and only if there is a basis of $U$ such that the matrix of $T$ with respect to this basis commutes with its conjugate transpose.

*This exercise adds another equivalence to the list of conditions equivalent to diagonalizability in 5.55.*

**Solution.**

(a) If $T$ is diagonalizable then there is a basis $u_1, ..., u_n$ of $U$ such that

$$A \coloneqq \mathcal{M}(T, (u_1, ..., u_n))$$

is diagonal. It follows that $A = A^{\mathrm{t}}$.

Now suppose that there is a basis $u_1, ..., u_n$ of $U$ such that $A = A^{\mathrm{t}}$, where

$$A = \mathcal{M}(T, (u_1, ..., u_n)).$$

For $u = a_1 u_1 + \cdots + a_n u_n$ and $v = b_1 u_1 + \cdots + b_n u_n$ in $U$, define

$$\langle u, v \rangle = a_1 b_1 + \cdots + a_n b_n;$$

it is straightforward to verify that this defines an inner product on $U$. Notice that $u_1, ..., u_n$ is an orthonormal basis of $U$ with respect to this inner product; it follows that $T$ is self-adjoint with respect to this inner product and so we may apply the real spectral theorem (7.29) to obtain a basis of $U$ consisting of eigenvectors of $T$. Thus $T$ is diagonalizable.

(b) If $T$ is diagonalizable then there is a basis $u_1, ..., u_n$ of $U$ such that

$$A \coloneqq \mathcal{M}(T, (u_1, ..., u_n))$$

is diagonal. It follows that $A^*$ is also diagonal and hence that $AA^* = A^*A$, since diagonal matrices always commute.

Now suppose that there is a basis $u_1, ..., u_n$ of $U$ such that $AA^* = A^*A$, where

$$A = \mathcal{M}(T, (u_1, ..., u_n)).$$

For $u = a_1 u_1 + \cdots + a_n u_n$ and $v = b_1 u_1 + \cdots + b_n u_n$ in $U$, define

$$\langle u, v \rangle = a_1 \overline{b_1} + \cdots + a_n \overline{b_n};$$

it is straightforward to verify that this defines an inner product on $U$. Notice that $u_1, ..., u_n$ is an orthonormal basis of $U$ with respect to this inner product; it follows that $T$ is normal with respect to this inner product and so we may apply the complex



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spectral theorem (7.31) to obtain a basis of $U$ consisting of eigenvectors of $T$. Thus $T$ is diagonalizable.

**Exercise 7.B.25.** Suppose that $T \in \mathcal{L}(V)$ and there is an orthonormal basis $e_1, ..., e_n$ of $V$ consisting of eigenvectors of $T$, with corresponding eigenvalues $\lambda_1, ..., \lambda_n$. Show that if $k \in \{1, ..., n\}$, then the pseudoinverse $T^\dagger$ satisfies the equation

$$T^\dagger e_k = \begin{cases} \frac{1}{\lambda_k} e_k & \text{if } \lambda_k \neq 0, \\ 0 & \text{if } \lambda_k = 0. \end{cases}$$

**Solution.** By the relevant spectral theorem (7.29 or 7.31), $T$ is either self-adjoint if $\mathbf{F} = \mathbf{R}$ or normal if $\mathbf{F} = \mathbf{C}$. In either case, 7.21 shows that

$$\operatorname{null} T = \operatorname{null} T^* \quad \text{and} \quad \operatorname{range} T = \operatorname{range} T^*.$$

Let $k \in \{1, ..., n\}$ be given. If $\lambda_k = 0$ then

$$e_k \in \operatorname{null} T = \operatorname{null} T^* = (\operatorname{range} T)^\perp = \operatorname{null} P_{\operatorname{range} T},$$

where we have used 6.57(e) and 7.6(a). It follows that

$$T^\dagger e_k = (T|_{(\operatorname{null} T)^\perp})^{-1} P_{\operatorname{range} T} e_k = (T|_{(\operatorname{null} T)^\perp})^{-1}(0) = 0.$$

If $\lambda_k \neq 0$ then observe that

$$\lambda_k^{-1} e_k = T(\lambda_k^{-2} e_k) \quad \Rightarrow \quad \lambda_k^{-1} e_k \in \operatorname{range} T = \operatorname{range} T^* = (\operatorname{null} T)^\perp,$$

where we have used 7.6(b). Because the restriction of $T$ to $(\operatorname{null} T)^\perp$ is an isomorphism between $(\operatorname{null} T)^\perp$ and $\operatorname{range} T$ (see 6.67), it follows that $(T|_{(\operatorname{null} T)^\perp})^{-1} T(\lambda_k^{-1} e_k) = \lambda_k^{-1} e_k$ and hence that

$$\begin{aligned} T^\dagger e_k = (T|_{(\operatorname{null} T)^\perp})^{-1} P_{\operatorname{range} T} e_k &= (T|_{(\operatorname{null} T)^\perp})^{-1} P_{\operatorname{range} T} T(\lambda_k^{-1} e_k) \\ &= (T|_{(\operatorname{null} T)^\perp})^{-1} T(\lambda_k^{-1} e_k) = \lambda_k^{-1} e_k. \end{aligned}$$



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## 7.C. Positive Operators

**Exercise 7.C.1.** Suppose $T \in \mathcal{L}(V)$. Prove that if both $T$ and $-T$ are positive operators, then $T = 0$.

**Solution.** Let $v \in V$ be given. Observe that $\langle Tv, v\rangle \geq 0$ since $T$ is positive and

$$\langle -Tv, v\rangle \geq 0 \quad \Leftrightarrow \quad -\langle Tv, v\rangle \geq 0 \quad \Leftrightarrow \quad \langle Tv, v\rangle \leq 0$$

since $-T$ is positive. Thus $\langle Tv, v\rangle = 0$ and it follows from 7.43 that $Tv = 0$. Hence $T = 0$.

**Exercise 7.C.2.** Suppose $T \in \mathcal{L}(\mathbf{F}^4)$ is the operator whose matrix (with respect to the standard basis) is

$$\begin{pmatrix} 2 & -1 & 0 & 0 \\ -1 & 2 & -1 & 0 \\ 0 & -1 & 2 & -1 \\ 0 & 0 & -1 & 2 \end{pmatrix}.$$

Show that $T$ is an invertible positive operator.

**Solution.** Note that the matrix in question equals its conjugate transpose; since the standard basis of $\mathbf{F}^4$ is orthonormal, it follows that $T$ is self-adjoint. Some calculations reveal that $T$ has 4 distinct eigenvalues:

$$\frac{3 \pm \sqrt{5}}{2}, \quad \frac{5 \pm \sqrt{5}}{2}.$$

Notice that each eigenvalue is strictly positive. It follows from 5.7 and 7.38(b) that $T$ is an invertible positive operator.

**Exercise 7.C.3.** Suppose $n$ is a positive integer and $T \in \mathcal{L}(\mathbf{F}^n)$ is the operator whose matrix (with respect to the standard basis) consists of all 1's. Show that $T$ is a positive operator.

**Solution.** The matrix of $T$ with respect to the standard basis of $\mathbf{F}^n$ (which is orthonormal) equals its conjugate transpose; it follows that $T$ is self-adjoint. As shown in Exercise 5.B.3 (a), the eigenvalues of $T$ are contained in $\{0, 1\}$. Thus, by 7.38(b), $T$ is positive.

**Exercise 7.C.4.** Suppose $n$ is a positive integer with $n > 1$. Show that there exists an $n$-by-$n$ matrix $A$ such that all of the entries of $A$ are positive numbers and $A = A^*$, but the operator on $\mathbf{F}^n$ whose matrix (with respect to the standard basis) equals $A$ is not a positive operator.



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**Solution.** Let

$$A = \begin{pmatrix} 1 & 2 & \cdots & 2 \\ 2 & 1 & \cdots & 2 \\ \vdots & \vdots & \ddots & \vdots \\ 2 & 2 & \cdots & 1 \end{pmatrix}.$$

Observe that each entry of $A$ is positive and $A = A^*$. Let $T \in \mathcal{L}(\mathbf{F}^n)$ be the operator on $\mathbf{F}^n$ whose matrix with respect to the standard basis equals $A$, i.e.

$$T(x_1, x_2, ..., x_n) = (x_1 + 2x_2 + \cdots + 2x_n, 2x_1 + x_2 + \cdots + 2x_n, ..., 2x_1 + 2x_2 + \cdots + x_n).$$

Notice that

$$T(-1, 1, 0, ..., 0) = (1, -1, 0, ..., 0).$$

It follows that $-1$ is an eigenvalue of $T$. Thus, by 7.38(b), $T$ is not a positive operator.

**Exercise 7.C.5.** Suppose $T \in \mathcal{L}(V)$ is self-adjoint. Prove that $T$ is a positive operator if and only if for every orthonormal basis $e_1, ..., e_n$ of $V$, all entries on the diagonal of $\mathcal{M}(T, (e_1, ..., e_n))$ are nonnegative numbers.

**Solution.** Suppose that $T$ is a positive operator and let $e_1, ..., e_n$ be an orthonormal basis of $V$. For $k \in \{1, ..., n\}$, 6.30(a) shows that the $k^{\text{th}}$ diagonal entry of $\mathcal{M}(T, (e_1, ..., e_n))$ is equal to $\langle Te_k, e_k \rangle$; this must be non-negative by the positivity of $T$.

Suppose that for every orthonormal basis $e_1, ..., e_n$ of $V$, all entries on the diagonal of $\mathcal{M}(T, (e_1, ..., e_n))$ are nonnegative numbers. Because $T$ is self-adjoint, the relevant spectral theorem (7.29 or 7.31) implies the existence of an orthonormal basis $e_1, ..., e_n$ of $V$ with respect to which the matrix of $T$ is diagonal. By assumption each diagonal entry of this matrix is non-negative and it then follows from 7.38(c) that $T$ is a positive operator.

**Exercise 7.C.6.** Prove that the sum of two positive operators on $V$ is a positive operator.

**Solution.** Suppose that $S, T \in \mathcal{L}(V)$ are positive operators and note that $S + T$ is self-adjoint by 7.5(a). Let $v \in V$ be given and observe that

$$\langle (S + T)v, v \rangle = \langle Sv + Tv, v \rangle = \langle Sv, v \rangle + \langle Tv, v \rangle \geq 0.$$

Thus $S + T$ is positive.

**Exercise 7.C.7.** Suppose $S \in \mathcal{L}(V)$ is an invertible positive operator and $T \in \mathcal{L}(V)$ is a positive operator. Prove that $S + T$ is invertible.

**Solution.** Let us prove the following lemma (see 7.61 also).



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**Lemma L.11.** If $T \in \mathcal{L}(V)$ is a positive operator then $T$ is invertible if and only if $\langle Tv, v\rangle > 0$ for every non-zero $v \in V$.

*Proof.* Suppose that $T$ is invertible and let $v \in V$ be non-zero. It follows that $Tv$ is non-zero and thus, by the contrapositive of 7.43, we have $\langle Tv, v\rangle > 0$. If $T$ is not invertible then there exists some non-zero $v \in V$ such that $Tv = 0$, which gives us $\langle Tv, v\rangle = 0$. □

Let $v \in V$ be non-zero. Lemma L.11 shows that $\langle Sv, v\rangle > 0$ and it follows from the positivity of $T$ that

$$\langle (S+T)v, v\rangle = \langle Sv, v\rangle + \langle Tv, v\rangle > 0.$$

Another application of Lemma L.11 allows us to conclude that $S+T$ is invertible.

**Exercise 7.C.8.** Suppose $T \in \mathcal{L}(V)$. Prove that $T$ is a positive operator if and only if the pseudoinverse $T^\dagger$ is a positive operator.

**Solution.** Suppose that $T$ is a positive operator. By 7.38(c) there is an orthonormal basis $e_1, ..., e_n$ of $V$ such that $Te_k = \lambda_k e_k$ with $\lambda_k \geq 0$. For each $k \in \{1, ..., n\}$ let

$$\mu_k = \begin{cases} \lambda_k^{-1} & \text{if } \lambda_k \neq 0, \\ 0 & \text{if } \lambda_k = 0. \end{cases}$$

It follows from Exercise 7.B.25 that $T^\dagger e_k = \mu_k e_k$ for each $k \in \{1, ..., n\}$. Because each $\mu_k$ is non-negative, 7.38(c) implies that $T^\dagger$ is a positive operator.

Replacing $T$ with $T^\dagger$ in the preceding result and using that $\left(T^\dagger\right)^\dagger = T$ (see Exercise 6.C.23) gives us the converse statement.

**Exercise 7.C.9.** Suppose $T \in \mathcal{L}(V)$ is a positive operator and $S \in \mathcal{L}(W, V)$. Prove that $S^*TS$ is a positive operator on $W$.

**Solution.** For any $w \in W$ observe that

$$\langle S^*TSw, w\rangle = \langle T(Sw), Sw\rangle \geq 0,$$

where we have used the positivity of $T$.

**Exercise 7.C.10.** Suppose $T$ is a positive operator on $V$. Suppose $v, w \in V$ are such that

$$Tv = w \quad \text{and} \quad Tw = v.$$

Prove that $v = w$.

**Solution.** Notice that



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$$\|v-w\|^2 = \langle v-w, v-w\rangle = \langle Tw-Tv, v-w\rangle = -\langle T(v-w), v-w\rangle \leq 0,$$

where we have used that $T$ is a positive operator for the final inequality. It follows that $\|v-w\|^2 = 0$, which is the case if and only if $v = w$.

**Exercise 7.C.11.** Suppose $T$ is a positive operator on $V$ and $U$ is a subspace of $V$ invariant under $T$. Prove that $T|_U \in \mathcal{L}(U)$ is a positive operator on $U$.

**Solution.** Exercise 7.B.19 (b) shows that $T|_U$ is self-adjoint, and for any $u \in U$ we have

$$\langle T|_U(u), u\rangle = \langle Tu, u\rangle \geq 0,$$

where we have used that $T$ is a positive operator. Thus $T|_U$ is a positive operator.

**Exercise 7.C.12.** Suppose $T \in \mathcal{L}(V)$ is a positive operator. Prove that $T^k$ is a positive operator for every positive integer $k$.

**Solution.** By 7.38(c) there is an orthonormal basis $e_1, ..., e_n$ of $V$ such that $Te_j = \lambda_j e_j$ with $\lambda_j \geq 0$ for each $j \in \{1, ..., n\}$. Let $k$ be a positive integer and observe that $T^k e_j = \lambda_j^k e_j$ for each $j \in \{1, ..., n\}$. Because each $\lambda_j^k$ is non-negative, 7.38(c) allows us to conclude that $T^k$ is a positive operator.

**Exercise 7.C.13.** Suppose $T \in \mathcal{L}(V)$ is self-adjoint and $\alpha \in \mathbf{R}$.

(a) Prove that $T - \alpha I$ is a positive operator if and only if $\alpha$ is less than or equal to every eigenvalue of $T$.

(b) Prove that $\alpha I - T$ is a positive operator if and only if $\alpha$ is greater than or equal to every eigenvalue of $T$.

**Solution.**

(a) The relevant spectral theorem (7.29 or 7.31) implies the existence of an orthonormal basis $e_1, ..., e_n$ such that $Te_k = \lambda_k e_k$ for some eigenvalues $\lambda_1, ..., \lambda_n$; note that each $\lambda_k$ is real by 7.12. It follows that the matrix of $T - \alpha I$ with respect to $e_1, ..., e_n$ is

$$\begin{pmatrix} \lambda_1 - \alpha & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & \lambda_n - \alpha \end{pmatrix}.$$

If $\alpha$ is less than or equal to every eigenvalue of $T$ then each diagonal entry of this matrix is non-negative and it follows from 7.38(c) that $T - \alpha I$ is a positive operator; if $\alpha$ is greater than some eigenvalue of $T$ then at least one diagonal entry of this matrix is negative and it follows from Exercise 7.C.5 that $T - \alpha I$ is not a positive operator.

(b) It is straightforward to modify the argument in part (a) to prove part (b).



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**Exercise 7.C.14.** Suppose $T$ is a positive operator on $V$ and $v_1, ..., v_m \in V$. Prove that

$$\sum_{j=1}^{m}\sum_{k=1}^{m}\langle Tv_k, v_j\rangle \geq 0.$$

**Solution.** Using the positivity of $T$, observe that

$$\sum_{j=1}^{m}\sum_{k=1}^{m}\langle Tv_k, v_j\rangle = \langle Tv_1 + \cdots + Tv_m, v_1 + \cdots + v_m\rangle = \langle T(v_1 + \cdots + v_m), v_1 + \cdots + v_m\rangle \geq 0.$$

**Exercise 7.C.15.** Suppose $T \in \mathcal{L}(V)$ is self-adjoint. Prove that there exist positive operators $A, B \in \mathcal{L}(V)$ such that

$$T = A - B \quad \text{and} \quad \sqrt{T^*T} = A + B \quad \text{and} \quad AB = BA = 0.$$

**Solution.** The relevant spectral theorem (7.29 or 7.31) implies the existence of an orthonormal basis $e_1, ..., e_n$ such that $Te_k = \lambda_k e_k$ for some eigenvalues $\lambda_1, ..., \lambda_n$; note that each $\lambda_k$ is real by 7.12. Note further that the operator $R \in \mathcal{L}(V)$ given by $Re_k = |\lambda_k|e_k$ is positive by 7.38(c) and satisfies $R^2e_k = \lambda_k^2 e_k = T^2 e_k$; it follows from the uniqueness in 7.39 that $R = \sqrt{T^2} = \sqrt{T^*T}$.

For each $k \in \{1, ..., n\}$ define

$$\alpha_k = \begin{cases} \lambda_k & \text{if } \lambda_k \geq 0, \\ 0 & \text{if } \lambda_k < 0, \end{cases} \quad \text{and} \quad \beta_k = \begin{cases} -\lambda_k & \text{if } \lambda_k \leq 0, \\ 0 & \text{if } \lambda_k > 0. \end{cases}$$

Notice that, for each $k \in \{1, ..., n\}$,

$$\alpha_k, \beta_k \geq 0, \quad \alpha_k - \beta_k = \lambda_k, \quad \alpha_k + \beta_k = |\lambda_k|, \quad \text{and} \quad \alpha_k\beta_k = 0.$$

Define $A, B \in \mathcal{L}(V)$ by $Ae_k = \alpha_k e_k$ and $Be_k = \beta_k e_k$ and note that $A$ and $B$ are positive operators by 7.38(c). Furthermore,

$$(A - B)e_k = (\alpha_k - \beta_k)e_k = \lambda_k e_k = Te_k, \quad (A + B)e_k = (\alpha_k + \beta_k)e_k = |\lambda_k|e_k = \sqrt{T^*T}e_k,$$
$$ABe_k = BAe_k = \alpha_k\beta_k e_k = 0.$$

It follows that

$$T = A - B, \quad \sqrt{T^*T} = A + B \quad \text{and} \quad AB = BA = 0.$$

**Exercise 7.C.16.** Suppose $T$ is a positive operator on $V$. Prove that

$$\operatorname{null}\sqrt{T} = \operatorname{null} T \quad \text{and} \quad \operatorname{range}\sqrt{T} = \operatorname{range} T.$$

**Solution.** Note that $\sqrt{T}$ is positive, hence self-adjoint, hence normal; it follows from Exercise 7.A.27 that



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$$\operatorname{null}\sqrt{T} = \operatorname{null}\left(\sqrt{T}\right)^2 = \operatorname{null} T \quad \text{and} \quad \operatorname{range}\sqrt{T} = \operatorname{range}\left(\sqrt{T}\right)^2 = \operatorname{range} T.$$

**Exercise 7.C.17.** Suppose that $T \in \mathcal{L}(V)$ is a positive operator. Prove that there exists a polynomial $p$ with real coefficients such that $\sqrt{T} = p(T)$.

**Solution.** By 7.38(c) there is an orthonormal basis $e_1, ..., e_n$ of $V$ such that $Te_k = \lambda_k e_k$ with $\lambda_k \geq 0$, where $\lambda_1, ..., \lambda_n$ are the eigenvalues of $T$; note that $\sqrt{T}e_k = \sqrt{\lambda_k}e_k$. Exercise 4.7 shows that there is a polynomial $p \in \mathcal{P}(\mathbf{R})$ satisfying $p(\lambda_k) = \sqrt{\lambda_k}$ for each $k \in \{1, ..., n\}$ and it follows that

$$p(T)e_k = p(\lambda_k)e_k = \sqrt{\lambda_k}e_k = \sqrt{T}e_k.$$

Thus $\sqrt{T} = p(T)$.

**Exercise 7.C.18.** Suppose $S$ and $T$ are positive operators on $V$. Prove that $ST$ is a positive operator if and only if $S$ and $T$ commute.

**Solution.** If $S$ and $T$ do not commute then, by Exercise 7.A.9, $ST$ is not self-adjoint and hence not a positive operator.

If $S$ and $T$ commute then Exercise 7.A.9 shows that $ST$ is self-adjoint and Exercise 7.B.16/ Exercise 7.B.17 shows that there is an orthonormal basis $e_1, ..., e_n$ of $V$ consisting of eigenvectors of both $S$ and $T$, say $Se_k = \mu_k e_k$ and $Te_k = \lambda_k e_k$; each $\mu_k$ and each $\lambda_k$ is a non-negative real number by 7.38(b). It follows that $STe_k = \mu_k \lambda_k e_k$, so that each $e_k$ is an eigenvector of $ST$ with a corresponding non-negative real eigenvalue. Thus, by 7.38(c), $ST$ is a positive operator.

**Exercise 7.C.19.** Show that the identity operator on $\mathbf{F}^2$ has infinitely many self-adjoint square roots.

**Solution.** For any $t \in (0, 1)$, let $R_t$ be the operator on $\mathbf{F}^2$ whose matrix with respect to the standard orthonormal basis of $\mathbf{F}^2$ is

$$\begin{pmatrix} \sqrt{1-t^2} & t \\ t & -\sqrt{1-t^2} \end{pmatrix};$$

note that each $t \in (0, 1)$ gives a distinct operator $R_t$. Because the matrix above equals its conjugate transpose, $R_t$ is self-adjoint. A calculation shows that $R_t^2 = I$.



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**Exercise 7.C.20.** Suppose $T \in \mathcal{L}(V)$ and $e_1, ..., e_n$ is an orthonormal basis of $V$. Prove that $T$ is a positive operator if and only if there exist $v_1, ..., v_n \in V$ such that

$$\langle Te_k, e_j \rangle = \langle v_k, v_j \rangle$$

for all $j, k = 1, ..., n$.

*The numbers $\{\langle Te_k, e_j \rangle\}_{j,k=1,...,n}$ are the entries in the matrix of $T$ with respect to the orthonormal basis $e_1, ..., e_n$.*

**Solution.** Suppose that $T$ is a positive operator and for each $k \in \{1, ..., n\}$ let $v_k = \sqrt{T}e_k$. It follows that

$$\langle Te_k, e_j \rangle = \langle \sqrt{T}e_k, \sqrt{T}e_j \rangle = \langle v_k, v_j \rangle$$

for all $j, k \in \{1, ..., n\}$.

Now suppose that there exist $v_1, ..., v_n \in V$ such that

$$\langle Te_k, e_j \rangle = \langle v_k, v_j \rangle$$

for all $j, k \in \{1, ..., n\}$. Define $R \in \mathcal{L}(V)$ by $Re_k = v_k$, so that

$$\langle Te_k, e_j \rangle = \langle Re_k, Re_j \rangle = \langle R^*Re_k, e_j \rangle$$

for all $j, k \in \{1, ..., n\}$. It follows that

$$\mathcal{M}(T, (e_1, ..., e_n)) = \mathcal{M}(R^*R, (e_1, ..., e_n))$$

and hence that $T = R^*R$. We may use 7.38(f) to conclude that $T$ is a positive operator.

**Exercise 7.C.21.** Suppose $n$ is a positive integer. The $n$-by-$n$ *Hilbert matrix* is the $n$-by-$n$ matrix whose entry in row $j$, column $k$ is $\frac{1}{j+k-1}$. Suppose $T \in \mathcal{L}(V)$ is an operator whose matrix with respect to some orthonormal basis of $V$ is the $n$-by-$n$ Hilbert matrix. Prove that $T$ is a positive invertible operator.

*Example: The 4-by-4 Hilbert matrix is*

$$\begin{pmatrix} 1 & \frac{1}{2} & \frac{1}{3} & \frac{1}{4} \\ \frac{1}{2} & \frac{1}{3} & \frac{1}{4} & \frac{1}{5} \\ \frac{1}{3} & \frac{1}{4} & \frac{1}{5} & \frac{1}{6} \\ \frac{1}{4} & \frac{1}{5} & \frac{1}{6} & \frac{1}{7} \end{pmatrix}.$$

**Solution.** (This solution uses some integration of complex-valued functions of a real variable.) Suppose that $e_1, ..., e_n$ is the orthonormal basis of $V$ with respect to which the matrix of $T$ is the $n$-by-$n$ Hilbert matrix. For any $v = x_1e_1 + \cdots + x_ne_n \in V$, observe that



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$$\begin{aligned}
\langle Tv, v\rangle &= \left\langle \sum_{k=1}^{n} x_k T e_k, \sum_{j=1}^{n} x_j e_j \right\rangle \\
&= \sum_{j,k=1}^{n} \overline{x_j} x_k \langle T e_k, e_j\rangle \\
&= \sum_{j,k=1}^{n} \frac{\overline{x_j} x_k}{j+k-1} \\
&= \sum_{j,k=1}^{n} \overline{x_j} x_k \int_0^1 t^{j+k-2}\, \mathrm{d}t \\
&= \int_0^1 \sum_{j,k=1}^{n} \overline{x_j} x_k t^{j+k-2}\, \mathrm{d}t \\
&= \int_0^1 \left(\sum_{j=1}^{n} \overline{x_j} t^{j-1}\right)\left(\sum_{k=1}^{n} x_k t^{k-1}\right) \mathrm{d}t \\
&= \int_0^1 \left(\overline{\sum_{j=1}^{n} x_j t^{j-1}}\right)\left(\sum_{k=1}^{n} x_k t^{k-1}\right) \mathrm{d}t \\
&= \int_0^1 \left|\sum_{k=1}^{n} x_k t^{k-1}\right|^2 \mathrm{d}t.
\end{aligned}$$

Observe that the integrand above is a non-negative and continuous function $[0,1] \to \mathbf{R}$. It follows that the integral is non-negative and vanishes if and only if

$$\left|\sum_{k=1}^{n} x_k t^{k-1}\right|^2 = 0 \text{ for all } t \in [0,1] \quad \Leftrightarrow \quad \sum_{k=1}^{n} x_k t^{k-1} = 0 \text{ for all } t \in [0,1]$$

$$\Leftrightarrow \quad x_k = 0 \text{ for all } k \in \{1, \ldots, n\} \quad \Leftrightarrow \quad v = 0.$$

Thus $\langle Tv, v\rangle \geq 0$ for every $v \in V$ and $\langle Tv, v\rangle = 0$ if and only if $v = 0$; it follows from Lemma L.11 that $T$ is an invertible positive operator.

**Exercise 7.C.22.** Suppose $T \in \mathcal{L}(V)$ is a positive operator and $u \in V$ is such that $\|u\| = 1$ and $\|Tu\| \geq \|Tv\|$ for all $v \in V$ with $\|v\| = 1$. Show that $u$ is an eigenvector of $T$ corresponding to the largest eigenvalue of $T$.

**Solution.** Suppose $0 \leq \lambda_1 < \cdots < \lambda_n$ are the distinct eigenvalues of $T$; each $\lambda_k$ is a non-negative real number by 7.38(b). Exercise 7.B.14/Exercise 7.B.15 shows that

$$V = E(\lambda_1, T) \oplus \cdots \oplus E(\lambda_n, T)$$



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and that all pairs of eigenvectors of $T$ corresponding to distinct eigenvalues are orthogonal. Suppose that $u = v_1 + \cdots + v_n$ where each $v_k \in E(\lambda_k, T)$. It follows from the Pythagorean theorem that $1 = \|u\|^2 = \|v_1\|^2 + \cdots + \|v_n\|^2$ and hence that

$$\|Tu\|^2 = \lambda_1^2\|v_1\|^2 + \cdots + \lambda_n^2\|v_n\|^2 \leq \lambda_n^2\big(\|v_1\|^2 + \cdots + \|v_n\|^2\big) = \lambda_n^2.$$

Letting $w$ be a unit eigenvector corresponding to the eigenvalue $\lambda_n$, our hypothesis implies that $\lambda_n^2 = \|Tw\|^2 \leq \|Tu\|^2$. Thus $\|Tu\|^2 = \lambda_n^2$, which gives us

$$\|Tu\|^2 = \lambda_n^2 \quad \Leftrightarrow \quad \lambda_1^2\|v_1\|^2 + \cdots + \lambda_n^2\big(\|v_n\|^2 - 1\big) = 0.$$

Note that if $\lambda_n = 0$ then $T$ must be the zero operator and the desired result is clear. If $\lambda_n \neq 0$ then the equation above shows that $\|v_n\|^2 = 1$; combining this with $1 = \|v_1\|^2 + \cdots + \|v_n\|^2$ shows that $v_1 = \cdots = v_{n-1} = 0$ and hence that $u = v_n \in E(\lambda_n, T)$, as desired.

**Exercise 7.C.23.** For $T \in \mathcal{L}(V)$ and $u, v \in V$, define $\langle u, v\rangle_T$ by $\langle u, v\rangle_T = \langle Tu, v\rangle$.

(a) Suppose $T \in \mathcal{L}(V)$. Prove that $\langle \cdot, \cdot\rangle_T$ is an inner product on $V$ if and only if $T$ is an invertible positive operator (with respect to the original inner product $\langle \cdot, \cdot\rangle$).

(b) Prove that every inner product on $V$ is of the form $\langle \cdot, \cdot\rangle_T$ for some positive invertible operator $T \in \mathcal{L}(V)$.

**Solution.**

(a) Suppose that $T$ is an invertible positive operator. We must verify each property of definition 6.2.

**Positivity.** For any $v \in V$ we have $\langle v, v\rangle_T = \langle Tv, v\rangle \geq 0$ by the positivity of $T$.

**Definiteness.** We have $\langle 0, 0\rangle_T = \langle T(0), 0\rangle = 0$, and for $v \neq 0$ we have

$$\langle v, v\rangle_T = \langle Tv, v\rangle > 0$$

by Lemma L.11.

**Additivity in first slot.** For $u, v, w \in V$ we have

$$\langle u + v, w\rangle_T = \langle T(u + v), w\rangle = \langle Tu + Tv, w\rangle = \langle Tu, w\rangle + \langle Tv, w\rangle = \langle u, w\rangle_T + \langle v, w\rangle_T.$$

**Homoegeneity in first slot.** For $u, v \in V$ and $\lambda \in \mathbf{F}$ we have

$$\langle \lambda u, v\rangle_T = \langle T(\lambda u), v\rangle = \langle \lambda Tu, v\rangle = \lambda\langle Tu, v\rangle = \lambda\langle u, v\rangle_T.$$

**Conjugate symmetry.** For $u, v \in V$ we have

$$\langle u, v\rangle_T = \langle Tu, v\rangle = \overline{\langle v, Tu\rangle} = \overline{\langle Tv, u\rangle} = \overline{\langle v, u\rangle_T},$$

where we have used that $T$ is self-adjoint for the third equality.

Thus $\langle \cdot, \cdot\rangle_T$ is an inner product on $V$.

Now suppose that $\langle \cdot, \cdot\rangle_T$ is an inner product on $V$ and let $u, v \in V$ be given. Observe that



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$$\langle Tu, v\rangle = \langle u, v\rangle_T = \overline{\langle v, u\rangle_T} = \overline{\langle Tv, u\rangle} = \langle u, Tv\rangle.$$

Thus $T$ is self-adjoint. Furthermore, for any $v \in V$,

$$\langle Tv, v\rangle = \langle v, v\rangle_T \geq 0.$$

Thus $T$ is a positive operator. Finally, for $v \neq 0$,

$$\langle Tv, v\rangle = \langle v, v\rangle_T > 0.$$

It follows from Lemma L.11 that $T$ is invertible.

(b) Let $\langle \cdot, \cdot \rangle_1$ be the original inner product on $V$ and let $\langle \cdot, \cdot \rangle_2$ be an arbitrary inner product on $V$; we need to show that there exists an invertible operator $T \in \mathcal{L}(V)$ which is positive with respect to $\langle \cdot, \cdot \rangle_1$ such that $\langle u, v\rangle_2 = \langle Tu, v\rangle_1$ for every $u, v \in V$.

Let $e_1, ..., e_n$ be an orthonormal basis of $V$ with respect to $\langle \cdot, \cdot \rangle_1$ and let $f_1, ..., f_n$ be an orthonormal basis of $V$ with respect to $\langle \cdot, \cdot \rangle_2$. Define $R \in \mathcal{L}(V)$ by $Rf_k = e_k$ and note that $R$ is invertible since $R$ maps a basis to a basis. Now define $T \in \mathcal{L}(V)$ by $T = R^*R$, where $R^*$ is the adjoint of $R$ with respect to $\langle \cdot, \cdot \rangle_1$. Observe that $T$ is invertible by 7.5(f) and Exercise 3.D.2, and $T$ is positive with respect to $\langle \cdot, \cdot \rangle_1$ by 7.38(f). Furthermore, for any $u = x_1 f_1 + \cdots + x_n f_n$ and $v = y_1 f_1 + \cdots + y_n f_n$ in $V$, observe that

$$\langle u, v\rangle_2 = x_1\overline{y_1} + \cdots + x_n\overline{y_n} = \langle Ru, Rv\rangle_1 = \langle R^*Ru, v\rangle_1 = \langle Tu, v\rangle_1.$$

**Exercise 7.C.24.** Suppose $S$ and $T$ are positive operators on $V$. Prove that

$$\operatorname{null}(S + T) = \operatorname{null} S \cap \operatorname{null} T.$$

**Solution.** Exercise 7.C.6 shows that $S + T$ is a positive operator. Observe that

$$Sv = 0 \text{ and } Tv = 0 \quad \Leftrightarrow \quad \langle Sv, v\rangle = 0 \text{ and } \langle Tv, v\rangle = 0$$
$$\Leftrightarrow \quad \langle Sv, v\rangle + \langle Tv, v\rangle = 0 \quad \Leftrightarrow \quad \langle (S+T)v, v\rangle = 0 \quad \Leftrightarrow \quad (S+T)v = 0,$$

where we have used 7.43 several times. Thus $\operatorname{null}(S + T) = \operatorname{null} S \cap \operatorname{null} T$.

**Exercise 7.C.25.** Let $T$ be the second derivative operator in Exercise 31(b) in Section 7A. Show that $-T$ is a positive operator.

**Solution.** Define the orthonormal basis $\mathcal{B} = v, e_1, ..., e_n, f_1, ..., f_n$ as in Exercise 7.A.31 and observe that

$$-Tv = 0, \quad -Te_k = k^2 e_k, \quad -Tf_k = k^2 f_k.$$

It follows that the matrix of $-T$ with respect to $\mathcal{B}$ is diagonal with non-negative diagonal entries; 7.38(c) allows us to conclude that $-T$ is a positive operator.



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## 7.D. Isometries, Unitary Operators, and Matrix Factorization

**Exercise 7.D.1.** Suppose $\dim V \geq 2$ and $S \in \mathcal{L}(V, W)$. Prove that $S$ is an isometry if and only if $Se_1, Se_2$ is an orthonormal list in $W$ for every orthonormal list $e_1, e_2$ of length two in $V$.

**Solution.** Suppose $S$ is an isometry and suppose $e_1, e_2$ is an orthonormal list in $V$. Let $U = \text{span}(e_1, e_2)$ and note that $e_1, e_2$ is an orthonormal basis of $U$. Note further that $S|_U$ is an isometry; it follows from 7.49(d) that $Se_1, Se_2$ is an orthonormal list in $W$.

Now suppose that $Se_1, Se_2$ is an orthonormal list in $W$ for every orthonormal list $e_1, e_2$ in $V$. Let $e_1, e_2, ..., e_n$ be an orthonormal basis of $V$ and let $j < k$ in $\{1, ..., n\}$ be given. Since $e_j, e_k$ is an orthonormal list in $V$, our hypothesis guarantees that $Se_j, Se_k$ is an orthonormal list in $V$. Thus $\|Se_j\| = \|Se_k\| = 1$ and $\langle Se_j, Se_k\rangle = 0$. Because $j < k$ in $\{1, ..., n\}$ were arbitrary, it follows that $Se_1, ..., Se_n$ is an orthonormal list in $W$. We may use 7.49(d) to conclude that $S$ is an isometry.

**Exercise 7.D.2.** Suppose $T \in \mathcal{L}(V, W)$. Prove that $T$ is a scalar multiple of an isometry if and only if $T$ preserves orthogonality.

> *The phrase "$T$ preserves orthogonality" means that $\langle Tu, Tv\rangle = 0$ for all $u, v \in V$ such that $\langle u, v\rangle = 0$.*

**Solution.** Suppose that $T = \lambda S$ for some $\lambda \in \mathbf{F}$ and some isometry $S \in \mathcal{L}(V, W)$. For any $u, v \in V$ such that $\langle u, v\rangle = 0$, observe that

$$\langle Tu, Tv\rangle = \langle \lambda Su, \lambda Sv\rangle = |\lambda|^2 \langle Su, Sv\rangle = |\lambda|^2 \langle u, v\rangle = 0,$$

where we have used 7.49(c) for the third equality. Thus $T$ preserves orthogonality.

Now suppose that $T$ preserves orthogonality. Let $e_1, ..., e_n$ be an orthonormal basis of $V$ and let $k \in \{1, ..., n\}$ be given. Using the identity $\langle u + v, u - v\rangle = \|u\|^2 - \|v\|^2$, observe that

$$\langle e_1 + e_k, e_1 - e_k\rangle = \|e_1\| - \|e_k\| = 0 \quad \Rightarrow \quad \langle Te_1 + Te_k, Te_1 - Te_k\rangle = \|Te_1\| - \|Te_k\| = 0.$$

Thus, letting $\lambda = \|Te_1\|$, we have $\lambda = \|Te_k\|$ for each $k \in \{1, ..., n\}$. If $\lambda = 0$ then $T = 0I$, so that $T$ is a scalar multiple of the identity operator, which is certainly an isometry. If $\lambda \neq 0$ then let $S = \lambda^{-1}T$. Observe that, for any distinct $j, k \in \{1, ..., n\}$,

$$\langle e_j, e_k\rangle = 0 \quad \Rightarrow \quad \langle Te_j, Te_k\rangle = 0 \quad \Leftrightarrow \quad \langle \lambda Se_j, \lambda Se_k\rangle = 0 \quad \Leftrightarrow \quad |\lambda|^2 \langle Se_j, Se_k\rangle = 0.$$

Since $\lambda \neq 0$, this last equation implies that $\langle Se_j, Se_k\rangle = 0$. Furthermore, using that $\lambda$ is a non-negative real number,

$$\|Se_k\| = \lambda^{-1}\|Te_k\| = 1.$$



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Thus $Se_1, ..., Se_n$ is an orthonormal list in $W$. It follows from 7.49(d) that $S$ is an isometry and hence that $T = \lambda S$ is a scalar multiple of an isometry.

**Exercise 7.D.3.**

(a) Show that the product of two unitary operators on $V$ is a unitary operator.

(b) Show that the inverse of a unitary operator on $V$ is a unitary operator.

*This exercise shows that the set of unitary operators on $V$ is a group, where the group operation is the usual product of two operators.*

**Solution.**

(a) Suppose $S, T \in \mathcal{L}(V)$ are unitary operators. For any $v \in V$, observe that

$$\|STv\| = \|Sv\| = \|v\|.$$

Thus $ST$ is a unitary operator.

(b) Suppose that $S \in \mathcal{L}(V)$ is a unitary operator. For any $v \in V$ we have $v = Su$ for some $u \in V$. It follows that

$$\|S^{-1}v\| = \|u\| = \|Su\| = \|v\|.$$

Thus $S^{-1}$ is a unitary operator.

**Exercise 7.D.4.** Suppose $\mathbf{F} = \mathbf{C}$ and $A, B \in \mathcal{L}(V)$ are self-adjoint. Show that $A + iB$ is unitary if and only if $AB = BA$ and $A^2 + B^2 = I$.

**Solution.** Suppose $A + iB$ is unitary. It follows from 7.53(f) that $(A + iB)^* = A - iB$ is also unitary. Thus, for any $v \in V$,

$$\|Av\|^2 + \|Bv\|^2 + i(\langle (AB - BA)v, v\rangle) = \|Av + iBv\|^2 = \|v\|^2,$$
$$\|Av\|^2 + \|Bv\|^2 - i(\langle (AB - BA)v, v\rangle) = \|Av - iBv\|^2 = \|v\|^2.$$

Subtracting the latter of these equations from the former, we see that

$$\langle (AB - BA)v, v\rangle = 0$$

for every $v \in V$. It follows from 7.13 that $AB = BA$. We can now use 7.53(c) to see that

$$I = (A + iB)^*(A + iB) = (A - iB)(A + iB) = A^2 + B^2 + i(AB - BA) = A^2 + B^2.$$

If $AB = BA$ and $A^2 + B^2 = I$ then

$$(A + iB)^*(A - iB) = (A - iB)(A + iB) = A^2 + B^2 + i(AB - BA) = I.$$

Thus, by 7.49(b), $A + iB$ is unitary.



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**Exercise 7.D.5.** Suppose $S \in \mathcal{L}(V)$. Prove that the following are equivalent.

(a) $S$ is a self-adjoint unitary operator.

(b) $S = 2P - I$ for some orthogonal projection $P$ on $V$.

(c) There exists a subspace $U$ of $V$ such that $Su = u$ for every $u \in U$ and $Sw = -w$ for every $w \in U^\perp$.

**Solution.** Suppose that (a) holds, i.e. suppose that $S$ is a self-adjoint unitary operator. Let $P = \frac{1}{2}(S + I)$ and notice that $P$ is self-adjoint and that $S = 2P - I$. Notice further that, by 7.53(b), $S^2 = I$; it follows that

$$P^2 = \frac{S^2 + 2S + I}{4} = \frac{S + I}{2} = P.$$

We may now invoke Exercise 7.A.20 to see that $P$ is an orthogonal projection. Thus (b) holds.

Suppose that (b) holds, so that there is some subspace $U$ of $V$ such that $S = 2P_U - I$. For any $u \in U$ and any $w \in U^\perp$ it follows that

$$Su = 2P_U u - u = u \quad \text{and} \quad Sw = 2P_U w - w = -w,$$

where we have used 6.57. Thus (c) holds.

Suppose that (c) holds, i.e. suppose there exists a subspace $U$ of $V$ such that $Su = u$ for every $u \in U$ and $Sw = -w$ for every $w \in U^\perp$. Let $v_1 = u_1 + w_1$ and $v_2 = u_2 + w_2$ in $V$ be given, where $u_1, u_2 \in U$ and $w_1, w_2 \in U^\perp$. Observe that

$$\langle Sv_1, v_2 \rangle = \langle u_1 - w_1, u_2 + w_2 \rangle = \langle u_1, u_2 \rangle - \langle w_1, w_2 \rangle = \langle u_1 + w_1, u_2 - w_2 \rangle = \langle Sv_1, v_2 \rangle.$$

Thus $S$ is self-adjoint. Furthermore,

$$S^2 v_1 = S(S(u_1 + w_1)) = S(u_1 - w_1) = u_1 + w_1 = v_1.$$

Thus $S$ is its own inverse. Combining this with the fact that $S$ is self-adjoint and 7.53(c), we see that $S$ is a self-adjoint unitary operator, i.e. (a) holds.

**Exercise 7.D.6.** Suppose $T_1, T_2$ are both normal operators on $\mathbf{F}^3$ with $2, 5, 7$ as eigenvalues. Prove that there exists a unitary operator $S \in \mathcal{L}(\mathbf{F}^3)$ such that $T_1 = S^* T_2 S$.

**Solution.** Let $\lambda_1 = 2, \lambda_2 = 5, \lambda_3 = 7$, and let $e_1, e_2, e_3, f_1, f_2, f_3 \in V$ be such that

$$T_1 e_k = \lambda_k e_k \quad \text{and} \quad T_2 f_k = \lambda_k f_k.$$

Without loss of generality, we may assume that each $e_k$ and each $f_k$ is a unit vector. Combining this with 7.22, we see that each list $e_1, e_2, e_3$ and $f_1, f_2, f_3$ is an orthonormal basis of $\mathbf{F}^3$. Define $S \in \mathcal{L}(V)$ by $Se_k = f_k$ and note that $S$ is a unitary operator by 7.53(d). Note further that

$$S^* T_2 S e_k = S^* T_2 f_k = \lambda_k S^* f_k = \lambda_k e_k = T_1 e_k,$$



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where we have used that $S^* = S^{-1}$, which holds by 7.53(c). Thus $T_1 = S^* T_2 S$.

**Exercise 7.D.7.** Give an example of two self-adjoint operators $T_1, T_2 \in \mathcal{L}(\mathbf{F}^4)$ such that the eigenvalues of both operators are $2, 5, 7$ but there does not exist a unitary operator $S \in \mathcal{L}(\mathbf{F}^4)$ such that $T_1 = S^* T_2 S$. Be sure to explain why there is no unitary operator with the required property.

**Solution.** Let $T_1$ and $T_2$ be the operators on $\mathbf{F}^4$ whose matrices with respect to the standard basis of $\mathbf{F}^4$ are

$$\begin{pmatrix} 2 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 5 & 0 \\ 0 & 0 & 0 & 7 \end{pmatrix} \quad \text{and} \quad \begin{pmatrix} 2 & 0 & 0 & 0 \\ 0 & 5 & 0 & 0 \\ 0 & 0 & 5 & 0 \\ 0 & 0 & 0 & 7 \end{pmatrix}.$$

Since these matrices are diagonal and the standard basis of $\mathbf{F}^4$ is orthonormal, we see that $T_1$ and $T_2$ are self-adjoint and that their eigenvalues are precisely $2, 5, 7$. If there was an isometry $S \in \mathcal{L}(\mathbf{F}^4)$ such that $T_1 = S^* T_2 S$ then since $S^* S = I$ (by 7.42), we would have

$$T_1 - 2I = S^* T_2 S - 2I = S^*(T_2 - 2I)S.$$

It would then follow from Exercise 3.D.8 that $\dim \operatorname{null}(T_1 - 2I) = \dim \operatorname{null}(T_2 - 2I)$. However, from the matrices of $T_1$ and $T_2$ above we can see that

$$\dim \operatorname{null}(T_1 - 2I) = 2 \neq 1 = \dim \operatorname{null}(T_2 - 2I).$$

**Exercise 7.D.8.** Prove or give a counterexample: If $S \in \mathcal{L}(V)$ and there exists an orthonormal basis $e_1, \ldots, e_n$ of $V$ such that $\|Se_k\| = 1$ for each $e_k$, then $S$ is a unitary operator.

**Solution.** This is false. Let $e_1, e_2$ be the standard orthonormal basis of $\mathbf{R}^2$ and let $S \in \mathcal{L}(\mathbf{R}^2)$ be given by $Se_1 = Se_2 = e_1$. Observe that $\|Se_1\| = \|Se_2\| = \|e_1\| = 1$, but $S$ is not a unitary operator because $S$ is not injective.

**Exercise 7.D.9.** Suppose $\mathbf{F} = \mathbf{C}$ and $T \in \mathcal{L}(V)$. Suppose every eigenvalue of $T$ has absolute value 1 and $\|Tv\| \leq \|v\|$ for every $v \in V$. Prove that $T$ is a unitary operator.

**Solution.** By Schur's theorem (6.38) there is an orthonormal basis $e_1, \ldots, e_n$ of $V$ with respect to which $A := \mathcal{M}(T, (e_1, \ldots, e_n))$ is upper-triangular. Note that $|A_{j,j}| = 1$ for each $j \in \{1, \ldots, n\}$ since each diagonal entry of $A$ is an eigenvalue of $T$. For any $k \in \{2, \ldots, n\}$ it follows that

$$\sum_{j=1}^{k} |A_{j,k}|^2 = 1 + \sum_{j=1}^{k-1} |A_{j,k}|^2 = \|Te_k\|^2 \leq \|e_k\|^2 = 1.$$



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Thus $\sum_{j=1}^{k-1}|A_{j,k}|^2 = 0$, which gives us $A_{1,k} = \cdots = A_{k-1,k} = 0$. Hence $A$ is a diagonal matrix, i.e. $e_1, \ldots, e_n$ is an orthonormal basis of $V$ consisting of eigenvectors of $T$. By assumption each eigenvalue of $T$ has absolute value 1 and thus, by 7.55, $T$ is a unitary operator.

**Exercise 7.D.10.** Suppose $\mathbf{F} = \mathbf{C}$ and $T \in \mathcal{L}(V)$ is a self-adjoint operator such that $\|Tv\| \leq \|v\|$ for all $v \in V$.

(a) Show that $I - T^2$ is a positive operator.

(b) Show that $T + i\sqrt{I - T^2}$ is a unitary operator.

**Solution.**

(a) For any $v \in V$ observe that

$$\begin{aligned}\langle (I - T^2)v, v\rangle = \langle v - T^2 v, v\rangle &= \langle v, v\rangle - \langle T^2 v, v\rangle \\ &= \langle v, v\rangle - \langle Tv, Tv\rangle = \|v\|^2 - \|Tv\|^2 \geq 0.\end{aligned}$$

Thus $I - T^2$ is a positive operator.

(b) Let us prove the following lemma.

**Lemma L.12.** If $S \in \mathcal{L}(V)$ is positive and $T \in \mathcal{L}(V)$ is such that $ST = TS$, then $\sqrt{S}T = T\sqrt{S}$.

*Proof.* By Exercise 7.C.17 there is a polynomial $p \in \mathcal{P}(\mathbf{R})$ such that $\sqrt{S} = p(S)$. Because $S$ and $T$ commute, we can argue as in Exercise 7.B.12 to see that $\sqrt{S}$ and $T$ commute. □

A straightforward calculation (or 5.17) shows that $T$ and $I - T^2$ commute. It follows from Lemma L.12 that $T$ and $\sqrt{I - T^2}$ commute. Observe that

$$T^2 + \left(\sqrt{I - T^2}\right)^2 = I.$$

Thus, by Exercise 7.D.4, $T + i\sqrt{I - T^2}$ is a unitary operator.

**Exercise 7.D.11.** Suppose $S \in \mathcal{L}(V)$. Prove that $S$ is a unitary operator if and only if

$$\{Sv : v \in V \text{ and } \|v\| \leq 1\} = \{v \in V : \|v\| \leq 1\}.$$

**Solution.** Let $X = \{Sv : v \in V \text{ and } \|v\| \leq 1\}$ and $Y = \{v \in V : \|v\| \leq 1\}$, so that our goal is to show that $S$ is a unitary operator if and only if $X = Y$.

Suppose that $S$ is a unitary operator. If $Sv \in X$ for some $v \in V$ such that $\|v\| \leq 1$, then observe that $\|Sv\| = \|v\| \leq 1$; it follows that $Sv \in Y$ and hence that $X \subseteq Y$. Now suppose $v \in Y$, so that $\|v\| \leq 1$, and note that $S$ is invertible and that $S^{-1}$ is an isometry by 7.53. It follows that



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$$v = S(S^{-1}v) \quad \text{and} \quad \|S^{-1}v\| = \|v\| \leq 1.$$

Thus $v \in X$, so that $Y \subseteq X$. We may conclude that $X = Y$.

Now suppose that $S$ is not unitary. If $S$ is not invertible then $S$ is not surjective and thus there exists some necessarily non-zero $w \notin \operatorname{range} S$, which implies $\|w\|^{-1}w \notin \operatorname{range} S$. Thus $\|w\|^{-1}w \in Y$ but $\|w\|^{-1}w \notin X$, so that $X \neq Y$. Suppose that $S$ is invertible. Because $S$ is not unitary, there must exist some $v \in V$ such that $\|Sv\| \neq \|v\|$; note that $v$ must be non-zero. By replacing $v$ with $\|v\|^{-1}v$ if necessary, we may assume that $\|v\| = 1$. Consider the following cases.

**Case 1.** If $\|Sv\| > 1$ then $Sv \in X$ but $Sv \notin Y$. Thus $X \neq Y$.

**Case 2.** If $\|Sv\| < 1$ then note that $\|Sv\| \neq 0$ since $v \neq 0$ and $S$ is injective. Let $u = \|Sv\|^{-1}Sv$, so that $u \in Y$. We claim that $u \notin X$. Indeed, if $u \in X$ then $u = Sw$ for some $w \in V$ such that $\|w\| \leq 1$. It follows from the injectivity of $S$ that

$$\frac{Sv}{\|Sv\|} = Sw \quad \Rightarrow \quad v = \|Sv\|w \quad \Rightarrow \quad \|v\| = \|Sv\|\|w\| < 1,$$

contradicting that $\|v\| = 1$. Thus $u \notin X$, so that $X \neq Y$.

In any case, we have shown that if $S$ is not unitary then $X \neq Y$. We may conclude that $S$ is a unitary operator if and only if $X = Y$.

**Exercise 7.D.12.** Prove or give a counterexample: If $S \in \mathcal{L}(V)$ is invertible and $\|S^{-1}v\| = \|Sv\|$ for every $v \in V$, then $S$ is unitary.

**Solution.** This is false. For a counterexample, consider the operator $S \in \mathcal{L}(\mathbf{C}^2)$ whose matrix with respect to the standard basis of $\mathbf{C}^2$ is

$$\begin{pmatrix} i & \sqrt{2} \\ \sqrt{2} & -i \end{pmatrix}.$$

A calculation shows that $S$ is its own inverse, so that $\|S^{-1}v\| = \|Sv\|$ for every $v \in V$. However, observe that

$$\|S(1,0)\| = \|(i, \sqrt{2})\| = \sqrt{3} \neq 1 = \|(1,0)\|.$$

Thus $S$ is not a unitary operator.

**Exercise 7.D.13.** Explain why the columns of a square matrix of complex numbers form an orthonormal list in $\mathbf{C}^n$ if and only if the rows of the matrix form an orthonormal list in $\mathbf{C}^n$.

**Solution.** Suppose $A \in \mathbf{C}^{n,n}$ and let $S \in \mathcal{L}(\mathbf{C}^n)$ be such that $A$ is the matrix of $S$ with respect to the standard orthonormal basis of $\mathbf{C}^n$. Observe that



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the columns of $A$ form an orthonormal list in $\mathbf{C}^n$

$$\Leftrightarrow \quad S \text{ is unitary} \quad \Leftrightarrow \quad \text{the rows of } A \text{ form an orthonormal list in } \mathbf{C}^n;$$

the first equivalence is 7.49(e) and the second equivalence is 7.53(e).

**Exercise 7.D.14.** Suppose $v \in V$ with $\|v\| = 1$ and $b \in \mathbf{F}$. Also suppose $\dim V \geq 2$. Prove that there exists a unitary operator $S \in \mathcal{L}(V)$ such that $\langle Sv, v\rangle = b$ if and only if $|b| \leq 1$.

**Solution.** If there exists such a unitary operator then, using the Cauchy-Schwarz inequality,

$$|b| = |\langle Sv, v\rangle| \leq \|Sv\|\|v\| = \|v\|^2 \leq 1.$$

Now suppose that $|b| \leq 1$. Let $e_1 = v$ and extend this to an orthonormal basis $e_1, e_2, ..., e_n$ of $V$. Define $S \in \mathcal{L}(V)$ by

$$Se_1 = be_1 + \sqrt{1 - |b|^2}e_2, \quad Se_2 = -\sqrt{1 - |b|^2}e_1 + be_2, \quad Se_k = e_k \text{ for } k > 2.$$

Given that $e_1, e_2, ..., e_n$ is an orthonormal basis of $V$, some straightforward calculations show that $Se_1, Se_2, ..., Se_n$ is also an orthonormal basis of $V$. Thus $S$ is a unitary operator by 7.53(d). Furthermore,

$$\langle Sv, v\rangle = \left\langle be_1 + \sqrt{1 - |b|^2}e_2, e_1 \right\rangle = b\|e\|^2 = b.$$

**Exercise 7.D.15.** Suppose $T$ is a unitary operator on $V$ such that $T - I$ is invertible.

(a) Prove that $(T + I)(T - I)^{-1}$ is a skew operator (meaning that it equals the negative of its adjoint).

(b) Prove that if $\mathbf{F} = \mathbf{C}$, then $i(T + I)(T - I)^{-1}$ is a self-adjoint operator.

*The function $z \mapsto i(z + 1)(z - 1)^{-1}$ maps the unit circle in $\mathbf{C}$ (except for the point 1) to $\mathbf{R}$. Thus (b) illustrates the analogy between the unitary operators and the unit circle in $\mathbf{C}$, along with the analogy between the self-adjoint operators and $\mathbf{R}$.*

**Solution.**

(a) Using that $T^*T = TT^* = I$, which holds by 7.53(b), observe that

$$\begin{aligned} \left[(T + I)(T - I)^{-1}\right]^* &= R^*R(T - T^*), \\ -(T + I)(T - I)^{-1} &= (T - T^*)R^*R, \end{aligned} \tag{1}$$

where $R = (T - I)^{-1}$. Observe further that

$$(R^*R)^{-1}T = (2I - T - T^*)T = T(2I - T - T^*) = T(R^*R)^{-1}.$$

Since $T^{-1} = T^*$ (by 7.53(c)), taking the inverse of both sides of the equation $(R^*R)^{-1}T = T(R^*R)^{-1}$ shows that $(R^*R)T^* = T^*(R^*R)$. Hence $R^*R$ commutes with



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$T^*$, and we can similarly show that $R^*R$ commutes with $T$. Thus $R^*R$ commutes with $T - T^*$. It follows from the expressions in (1) that

$$\big[(T+I)(T-I)^{-1}\big]^* = -(T+I)(T-I)^{-1}.$$

(b) Using part (a) and 7.5(b), we have

$$\big[i(T+I)(T-I)^{-1}\big]^* = (-i)\big[-(T+I)(T-I)^{-1}\big] = i(T+I)(T-I)^{-1}.$$

**Exercise 7.D.16.** Suppose $\mathbf{F} = \mathbf{C}$ and $T \in \mathcal{L}(V)$ is self-adjoint. Prove that

$$(T+iI)(T-iI)^{-1}$$

is a unitary operator and 1 is not an eigenvalue of this operator.

**Solution.** Let $Q = (T+iI)(T-iI)^{-1}$. The complex spectral theorem (7.31) guarantees the existence of an orthonormal basis $e_1, ..., e_n$ of $V$ such that $Te_k = \lambda_k e_k$ for some eigenvalues $\lambda_1, ..., \lambda_n$, which must be real since $T$ is self-adjoint. For each $k \in \{1, ..., n\}$ a routine calculation shows that

$$Qe_k = \frac{\lambda_k + i}{\lambda_k - i} e_k \quad \text{and} \quad \left|\frac{\lambda_k + i}{\lambda_k - i}\right| = 1.$$

Thus, by 7.55, $Q$ is a unitary operator. Now observe that for any $v \in V$,

$$Qv = v \quad \Rightarrow \quad Q^*v = v \quad \Rightarrow \quad (T-iI)v = (T+iI)v \quad \Rightarrow \quad 2iv = 0 \quad \Rightarrow \quad v = 0.$$

Thus 1 is not an eigenvalue of $Q$.

**Exercise 7.D.17.** Explain why the characterization of unitary matrices given by 7.57 holds.

**Solution.** Let $Q \in \mathbf{F}^{n,n}$ be a matrix and let $S \in \mathcal{L}(\mathbf{F}^n)$ be such that $Q$ is the matrix of $S$ with respect to the standard orthonormal basis of $\mathbf{F}^n$. By definition, $Q$ is a unitary matrix if and only if 7.49(e) holds (we are taking both orthonormal bases $e_1, ..., e_n$ and $f_1, ..., f_m$ in the statement of 7.49 to be the standard orthonormal basis of $\mathbf{F}^n$). Observe that

$$7.57\text{(a)} \quad \Leftrightarrow \quad 7.49\text{(e)} \quad \Leftrightarrow \quad 7.49\text{(a)} \quad \Leftrightarrow \quad 7.53\text{(a)} \quad \Leftrightarrow \quad 7.53\text{(e)} \quad \Leftrightarrow \quad 7.57\text{(b)}.$$

Thus 7.57(a) and 7.57(b) are equivalent. After identifying elements of $\mathbf{F}^n$ with column vectors, i.e. $n$-by-1 matrices, note that $\|Qv\| = \|Sv\|$ for any $v \in \mathbf{F}^n$. As noted before, $Q$ is a unitary matrix if and only if $S$ is a unitary operator. The equivalence of 7.57(a) and 7.57(c) is now immediate from the definition of a unitary operator. Finally, by 7.9, the matrix of $S^*$ is $Q^*$. The equivalence of 7.57(a) and 7.57(d) then follows from the equivalence of 7.53(a) and 7.53(b).



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**Exercise 7.D.18.** A square matrix $A$ is called *symmetric* if it equals its transpose. Prove that if $A$ is a symmetric matrix with real entries, then there exists a unitary matrix $Q$ with real entries such that $Q^*AQ$ is a diagonal matrix.

**Solution.** Suppose $A \in \mathbf{R}^{n,n}$ and let $e_1, ..., e_n$ be the standard orthonormal basis of $\mathbf{R}^n$. Let $T \in \mathcal{L}(\mathbf{R}^n)$ be such that $\mathcal{M}(T, (e_1, ..., e_n)) = A$. By assumption $A = A^{\mathrm{t}}$ and thus $T$ is self-adjoint. It follows from the real spectral theorem (7.29) that there exists an orthonormal basis $f_1, ..., f_n$ of $\mathbf{R}^n$ such that $D \coloneqq \mathcal{M}(T, (f_1, ..., f_n))$ is diagonal. Let

$$Q = \mathcal{M}(I, (f_1, ..., f_n), (e_1, ..., e_n)).$$

Certainly the identity operator on $\mathbf{R}^n$ is an isometry. It follows from the equivalence of 7.49(a) and 7.49(e) that $Q$ is a unitary matrix. On one hand, 7.57 shows that $Q^{-1} = Q^*$. On the other hand, 3.82 shows that

$$Q^{-1} = \mathcal{M}(T, (e_1, ..., e_n), (f_1, ..., f_n)).$$

Thus, using the change-of-basis formula (3.84), we have that $Q^*AQ = D$ is diagonal.

**Exercise 7.D.19.** Suppose $n$ is a positive integer. For this exercise, we adopt the notation that a typical element $z$ of $\mathbf{C}^n$ is denoted by $z = (z_0, z_1, ..., z_{n-1})$. Define linear functionals $\omega_0, \omega_1, ..., \omega_{n-1}$ on $\mathbf{C}^n$ by

$$\omega_j(z_0, z_1, ..., z_{n-1}) = \frac{1}{\sqrt{n}} \sum_{m=0}^{n-1} z_m e^{-2\pi i j m/n}.$$

The *discrete Fourier transform* is the operator $\mathcal{F} : \mathbf{C}^n \to \mathbf{C}^n$ defined by

$$\mathcal{F}z = (\omega_0(z), \omega_1(z), ..., \omega_{n-1}(z)).$$

(a) Show that $\mathcal{F}$ is a unitary operator on $\mathbf{C}^n$.

(b) Show that if $(z_0, ..., z_{n-1}) \in \mathbf{C}^n$ and $z_n$ is defined to equal $z_0$, then

$$\mathcal{F}^{-1}(z_0, z_1, ..., z_{n-1}) = \mathcal{F}(z_n, z_{n-1}, ..., z_1).$$

(c) Show that $\mathcal{F}^4 = I$.

*The discrete Fourier transform has many important applications in data analysis. The usual Fourier transform involves expressions of the form $\int_{-\infty}^{\infty} f(x)e^{-2\pi i t x}dx$ for complex-valued integrable functions $f$ defined on $\mathbf{R}$.*

**Solution.**

(a) For this exercise, let us count from 0 to $n-1$ instead of from 1 to $n$ for columns of matrices etc. Let $e_0, ..., e_{n-1}$ be the standard orthonormal basis of $\mathbf{C}^n$ and observe that

$$\omega_j e_k = \frac{\alpha^{jk}}{\sqrt{n}}, \quad \text{where} \quad \alpha = e^{-2\pi i/n}.$$



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It follows that

$$Q := \mathcal{M}(\mathcal{F}, (e_0, ..., e_{n-1})) = \frac{1}{\sqrt{n}} \begin{pmatrix} 1 & 1 & 1 & \cdots & 1 \\ 1 & \alpha & \alpha^2 & \cdots & \alpha^{n-1} \\ 1 & \alpha^2 & \alpha^4 & \cdots & \alpha^{2(n-1)} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & \alpha^{n-1} & \alpha^{2(n-1)} & \cdots & \alpha^{(n-1)(n-1)} \end{pmatrix}.$$

Thus the Euclidean inner product of the $j^{\text{th}}$ and $k^{\text{th}}$ columns of $Q$ is

$$\frac{1}{n} \sum_{m=0}^{n-1} \left(\alpha^j \overline{\alpha}^k\right)^m = \frac{1}{n} \sum_{m=0}^{n-1} \left(e^{2\pi i(k-j)/n}\right)^m.$$

If $j = k$ then $e^{2\pi i(k-j)/n} = 1$, so that

$$\frac{1}{n} \sum_{m=0}^{n-1} \left(e^{2\pi i(k-j)/n}\right)^m = \frac{1}{n} \sum_{m=0}^{n-1} 1 = 1,$$

and if $j \neq k$ then $e^{2\pi i(k-j)/n} \neq 1$ and the geometric series formula gives us

$$\frac{1}{n} \sum_{m=0}^{n-1} \left(e^{2\pi i(k-j)/n}\right)^m = \frac{1 - \left[e^{2\pi i(k-j)/n}\right]^n}{n\left(1 - e^{2\pi i(k-j)/n}\right)} = \frac{1 - e^{2\pi i(k-j)}}{n\left(1 - e^{2\pi i(k-j)/n}\right)} = 0,$$

where we have used that $e^{2\pi i(k-j)} = 1$ since $k - j$ is an integer. We have now shown that the columns of $Q$ form an orthonormal list in $\mathbf{C}^n$ with respect to the Euclidean inner product. It follows from 7.49 that $\mathcal{F}$ is a unitary operator.

(b) Define $\mathcal{E} \in \mathcal{L}(\mathbf{C}^n)$ by $\mathcal{E}(z_0, z_1, ..., z_{n-1}) = \mathcal{F}(z_0, z_{n-1}, ..., z_1)$ and observe that

$$\mathcal{E}e_0 = \mathcal{F}e_0 \quad \text{and} \quad \mathcal{E}e_k = \mathcal{F}e_{n-k} \text{ for } k \in \{1, ..., n-1\}.$$

Thus, letting $X = \mathcal{M}(\mathcal{E}, (e_0, ..., e_{n-1}))$, we have

$$X_{j,k} = \begin{cases} Q_{j,k} & \text{if } k = 0, \\ Q_{j,n-k} & \text{if } k \in \{1, ..., n-1\}. \end{cases}$$

For $k \in \{1, ..., n-1\}$, observe that

$$\overline{\alpha}^{kj} = e^{2\pi ijk/n} = e^{2\pi ijk/n} e^{-2\pi ij} = e^{-2\pi ij(n-k)/n} = \alpha^{j(n-k)}.$$

It follows that

$$Q^*_{j,k} = \overline{Q_{k,j}} = \frac{\overline{\alpha}^{kj}}{\sqrt{n}} = \frac{\alpha^{j(n-k)}}{\sqrt{n}} = Q_{j,n-k}.$$

By inspection of $Q$, it is also clear that $Q^*_{j,0} = Q_{j,0}$. Thus $Q^*_{j,k} = X_{j,k}$ for all $j, k$ and it follows that $\mathcal{E} = \mathcal{F}^* = \mathcal{F}^{-1}$.

(c) The formula in part (b) shows that

$$\mathcal{F}^2(z_0, z_1, ..., z_{n-1}) = (z_0, z_{n-1}, ..., z_1) \quad \Rightarrow \quad \mathcal{F}^4(z_0, z_1, ..., z_{n-1}) = (z_0, z_1, ..., z_{n-1}).$$



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**Exercise 7.D.20.** Suppose $A$ is a square matrix with linearly independent columns. Prove that there exist unique matrices $R$ and $Q$ such that $R$ is lower triangular with only positive numbers on its diagonal, $Q$ is unitary, and $A = RQ$.

**Solution.** If $A$ has linearly independent columns then Exercise 7.A.7 (b) shows that $A^*$ also has linearly independent columns. It follows from 7.58 that there exists a unitary matrix $P$ and an upper triangular matrix $U$ with only positive numbers on its diagonal such that $A^* = PU$. It follows that $A = RQ$, where $R = U^*$ and $Q = P^*$. Observe that $R$ is lower triangular with only positive numbers on its diagonal and that $Q$ is unitary by 7.57.



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## 7.E. Singular Value Decomposition

**Exercise 7.E.1.** Suppose $T \in \mathcal{L}(V, W)$. Show that $T = 0$ if and only if all singular values of $T$ are 0.

**Solution.** Let $N$ be the number of positive singular values of $T$. It will suffice to show that $T = 0$ if and only if $N = 0$. Indeed, 7.68(b) shows that

$$N = 0 \quad \Leftrightarrow \quad \dim \operatorname{range} T = 0 \quad \Leftrightarrow \quad T = 0.$$

**Exercise 7.E.2.** Suppose $T \in \mathcal{L}(V, W)$ and $s > 0$. Prove that $s$ is a singular value of $T$ if and only if there exist nonzero vectors $v \in V$ and $w \in W$ such that

$$Tv = sw \quad \text{and} \quad T^*w = sv.$$

*The vectors $v, w$ satisfying both equations above are called a **Schmidt pair**. Erhard Schmidt introduced the concept of singular values in 1907.*

**Solution.** Suppose that $s$ is a singular value of $T$. Thus, letting $s_1, ..., s_m$ be the positive singular values of $T$, we have $s = s_k$ for some $k \in \{1, ..., m\}$. Let $e_1, ..., e_m$ and $f_1, ..., f_m$ be the orthonormal lists obtained from the SVD (7.70). It follows from 7.70 and 7.75 that

$$Te_k = s_k f_k \quad \text{and} \quad T^* f_k = s_k e_k.$$

Thus we can take $v = e_k$ and $w = f_k$.

Conversely, suppose there exist non-zero vectors $v \in V$ and $w \in W$ such that

$$Tv = sw \quad \text{and} \quad T^*w = sv.$$

It follows that $T^*Tv = s^2 v$, so that $s^2$ is an eigenvalue of $T^*T$. Thus $\sqrt{s^2} = s$ is a singular value of $T$, where we have used that $s$ is non-negative.

**Exercise 7.E.3.** Give an example of $T \in \mathcal{L}(\mathbf{C}^2)$ such that 0 is the only eigenvalue of $T$ and the singular values of $T$ are $5, 0$.

**Solution.** Let $T \in \mathcal{L}(\mathbf{C}^2)$ be the operator whose matrix with respect to the standard basis of $\mathbf{C}^2$ is

$$\begin{pmatrix} 0 & 5 \\ 0 & 0 \end{pmatrix}$$

and note that 0 is the only eigenvalue of $T$. Note further that the matrix of $T^*T$ is

$$\begin{pmatrix} 0 & 0 \\ 5 & 0 \end{pmatrix} \begin{pmatrix} 0 & 5 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 25 \end{pmatrix}.$$

Thus the singular values of $T$ are $5, 0$.



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**Exercise 7.E.4.** Suppose that $T \in \mathcal{L}(V,W)$, $s_1$ is the largest singular value of $T$, and $s_n$ is the smallest singular value of $T$. Prove that

$$\{\|Tv\| : v \in V \text{ and } \|v\| = 1\} = [s_n, s_1].$$

**Solution.** We consider several cases.

**Case 1.** If $s_1 = s_n = 0$ then all singular values of $T$ equal 0 and Exercise 7.E.1 shows that $T = 0$. Thus

$$\{\|Tv\| : v \in V \text{ and } \|v\| = 1\} = \{0\} = [0,0] = [s_n, s_1].$$

**Case 2.** If $s_1 = s_n > 0$ then note that all singular values of $s_1^{-1}T$ equal 1. It follows from 7.69 that $s_1^{-1}T$ is an isometry and hence that

$$\{\|Tv\| : v \in V \text{ and } \|v\| = 1\} = \{s_1\} = [s_1, s_1] = [s_n, s_1].$$

**Case 3.** Suppose that $s_1 > s_n$, which implies $s_1 > 0$. Let $s_1, ..., s_m$ be the positive singular values of $T$ and let $e_1, ..., e_m$ and $f_1, ..., f_m$ be the orthonormal lists obtained from the SVD (7.70). For any $v \in V$, 7.70 and 6.24 show that

$$Tv = s_1\langle v, e_1\rangle f_1 + \cdots + s_m\langle v, e_m\rangle f_m \quad \Rightarrow \quad \|Tv\|^2 = s_1^2|\langle v, e_1\rangle|^2 + \cdots + s_m^2|\langle v, e_m\rangle|^2.$$

For any $v \in V$ such that $\|v\| = 1$, it follows from Bessel's inequality (6.26) that

$$\|Tv\|^2 \le s_1^2\left(|\langle v, e_1\rangle|^2 + \cdots + |\langle v, e_m\rangle|^2\right) \le s_1^2\|v\|^2 = s_1^2 \quad \Rightarrow \quad \|Tv\| \le s_1.$$

Thus $\{\|Tv\| : v \in V \text{ and } \|v\| = 1\} \subseteq [0, s_1]$. To complete the exercise, we now consider two subcases.

**Case 3.1.** If $s_n > 0$ then it must be that $m = n = \dim V$, so that $e_1, ..., e_n$ is an orthonormal basis of $V$. It follows that

$$\|Tv\|^2 \ge s_n^2\left(|\langle v, e_1\rangle|^2 + \cdots + |\langle v, e_n\rangle|^2\right) = s_n^2\|v\|^2 = s_n^2 \quad \Rightarrow \quad \|Tv\| \ge s_n.$$

Thus $\{\|Tv\| : v \in V \text{ and } \|v\| = 1\} \subseteq [s_n, s_1]$. For $s \in [s_n, s_1]$, let

$$v = \sqrt{\frac{s^2 - s_n^2}{s_1^2 - s_n^2}}e_1 + \sqrt{\frac{s_1^2 - s^2}{s_1^2 - s_n^2}}e_n.$$

A calculation shows that $\|v\| = 1$ and $\|Tv\| = s$. Thus $[s_n, s_1] \subseteq \{\|Tv\| : v \in V \text{ and } \|v\| = 1\}$ and we may conclude that

$$\{\|Tv\| : v \in V \text{ and } \|v\| = 1\} = [s_n, s_1].$$

**Case 3.2.** If $s_n = 0$ then it must be that $m < n = \dim V$. Extend the orthonormal list $e_1, ..., e_m$ to an orthonormal basis $e_1, ..., e_n$ of $V$. As noted in the discussion after the proof of 7.70, it follows that $e_n \in \operatorname{null} T$. For $s \in [0, s_1]$, let



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$$v = \frac{s}{s_1} e_1 + \sqrt{1 - \frac{s^2}{s_1^2}} e_n.$$

A calculation shows that $\|v\| = 1$ and $\|Tv\| = s$. Thus $[0, s_1] \subseteq \{\|Tv\| : v \in V \text{ and } \|v\| = 1\}$ and we may conclude that

$$\{\|Tv\| : v \in V \text{ and } \|v\| = 1\} = [0, s_1] = [s_n, s_1].$$

**Exercise 7.E.5.** Suppose $T \in \mathcal{L}(\mathbf{C}^2)$ is defined by $T(x, y) = (-4y, x)$. Find the singular values of $T$.

**Solution.** The matrix of $T$ with respect to the standard basis of $\mathbf{C}^2$ is

$$A = \begin{pmatrix} 0 & -4 \\ 1 & 0 \end{pmatrix} \quad \Rightarrow \quad A^*A = \begin{pmatrix} 0 & 1 \\ -4 & 0 \end{pmatrix} \begin{pmatrix} 0 & -4 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 16 \end{pmatrix}.$$

Thus the singular values of $T$ are $4, 1$.

**Exercise 7.E.6.** Find the singular values of the differentiation operator $D \in \mathcal{L}(\mathcal{P}_2(\mathbf{R}))$ defined by $Dp = p'$, where the inner product on $\mathcal{P}_2(\mathbf{R})$ is as in Example 6.34.

**Solution.** As shown in Example 6.34, the list

$$e_1 = \sqrt{\tfrac{1}{2}}, \quad e_2 = \sqrt{\tfrac{3}{2}} x, \quad e_3 = \sqrt{\tfrac{45}{8}}(x^2 - \tfrac{1}{3})$$

is an orthonormal basis of $\mathcal{P}_2(\mathbf{R})$ with respect to the inner product given in Example 6.34. Observe that

$$De_1 = 0, \quad De_2 = \sqrt{\tfrac{3}{2}} = \sqrt{3} e_1, \quad De_3 = \sqrt{\tfrac{45}{8}}(2x) = \sqrt{15} e_2.$$

Thus the matrix of $D$ with respect to $e_1, e_2, e_3$ is

$$A = \begin{pmatrix} 0 & \sqrt{3} & 0 \\ 0 & 0 & \sqrt{15} \\ 0 & 0 & 0 \end{pmatrix} \quad \Rightarrow \quad A^*A = \begin{pmatrix} 0 & 0 & 0 \\ \sqrt{3} & 0 & 0 \\ 0 & \sqrt{15} & 0 \end{pmatrix} \begin{pmatrix} 0 & \sqrt{3} & 0 \\ 0 & 0 & \sqrt{15} \\ 0 & 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 15 \end{pmatrix}.$$

It follows that the singular values of $D$ are $\sqrt{15}, \sqrt{3}, 0$.

**Exercise 7.E.7.** Suppose that $T \in \mathcal{L}(V)$ is self-adjoint or that $\mathbf{F} = \mathbf{C}$ and $T \in \mathcal{L}(V)$ is normal. Let $\lambda_1, ..., \lambda_n$ be the eigenvalues of $T$, each included in this list as many times as the dimension of the corresponding eigenspace. Show that the singular values of $T$ are $|\lambda_1|, ..., |\lambda_n|$, after these numbers have been sorted into decreasing order.

**Solution.** By Exercise 7.B.14/Exercise 7.B.15, we know that there is a decomposition of $V$ into a direct sum of mutually orthogonal eigenspaces of $T$. Thus $n = \dim V$ and there exists



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an orthonormal basis $e_1, ..., e_n$ of $V$ such that $Te_k = \lambda_k e_k$ for each $k \in \{1, ..., n\}$. It follows from 7.21(e) that

$$T^* e_k = \overline{\lambda_k} e_k \quad \Rightarrow \quad T^* T e_k = |\lambda_k|^2 e_k.$$

Thus each vector in the list $|\lambda_1|, ..., |\lambda_n|$ is a singular value of $T$. As noted in the table on p. 272, the length of the list of singular values of $T$ is exactly $\dim V = n$. Thus, after being sorted into decreasing order, $|\lambda_1|, ..., |\lambda_n|$ are the singular values of $T$.

**Exercise 7.E.8.** Suppose $T \in \mathcal{L}(V, W)$. Suppose $s_1 \geq s_2 \geq \cdots \geq s_m > 0$ and $e_1, ..., e_m$ is an orthonormal list in $V$ and $f_1, ..., f_m$ is an orthonormal list in $W$ such that

$$Tv = s_1 \langle v, e_1 \rangle f_1 + \cdots + s_m \langle v, e_m \rangle f_m$$

for every $v \in V$.

(a) Prove that $f_1, ..., f_m$ is an orthonormal basis of $\operatorname{range} T$.

(b) Prove that $e_1, ..., e_m$ is an orthonormal basis of $(\operatorname{null} T)^\perp$.

(c) Prove that $s_1, ..., s_m$ are the positive singular values of $T$.

(d) Prove that if $k \in \{1, ..., m\}$, then $e_k$ is an eigenvector of $T^*T$ with corresponding eigenvalue $s_k^2$.

(e) Prove that

$$TT^* w = s_1^2 \langle w, f_1 \rangle f_1 + \cdots + s_m^2 \langle w, f_m \rangle f_m$$

for all $w \in W$.

**Solution.**

(a) Because the equation

$$Tv = s_1 \langle v, e_1 \rangle f_1 + \cdots + s_m \langle v, e_m \rangle f_m$$

holds for all $v \in V$, it is clear that $f_1, ..., f_m$ spans $\operatorname{range} T$. By assumption $f_1, ..., f_m$ is orthonormal and hence linearly independent (6.25). Thus $f_1, ..., f_m$ is an orthonormal basis of $\operatorname{range} T$.

(b) If $v \in \operatorname{null} T$ then

$$0 = Tv = s_1 \langle v, e_1 \rangle f_1 + \cdots + s_m \langle v, e_m \rangle f_m \quad \Rightarrow \quad \langle v, e_1 \rangle = \cdots = \langle v, e_m \rangle = 0,$$

where we have used that $f_1, ..., f_m$ is linearly independent and that each $s_k$ is strictly positive. Thus $\{e_1, ..., e_m\} \subseteq (\operatorname{null} T)^\perp$. By assumption $e_1, ..., e_m$ is orthonormal. Furthermore, by 6.67 we have $\dim (\operatorname{null} T)^\perp = \dim \operatorname{range} T = m$. It follows from 6.28 that $e_1, ..., e_m$ is an orthonormal basis of $(\operatorname{null} T)^\perp$.

(c) Extend the orthonormal list $e_1, ..., e_m$ to an orthonormal basis $e_1, ..., e_{\dim V}$ of $V$ and extend the orthonormal list $f_1, ..., f_m$ to an orthonormal basis $f_1, ..., f_{\dim V}$ of $W$. As in the discussion after the proof of 7.70, we have



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$$Te_k = \begin{cases} s_k f_k & \text{if } k \in \{1, ..., m\}, \\ 0 & \text{otherwise.} \end{cases}$$

For any $j \in \{1, ..., \dim W\}$, it follows that

$$T^* f_j = \sum_{k=1}^{\dim V} \langle T^* f_j, e_k \rangle e_k = \sum_{k=1}^{\dim V} \langle f_j, Te_k \rangle e_k = \begin{cases} s_j e_j & \text{if } j \in \{1, ..., m\}, \\ 0 & \text{otherwise.} \end{cases}$$

Thus

$$T^*Te_k = \begin{cases} s_k^2 e_k & \text{if } k \in \{1, ..., m\}, \\ 0 & \text{otherwise.} \end{cases}$$

From this expression and the fact that $s_1 \geq s_2 \geq \cdots \geq s_m > 0$, we see that the list $s_1^2, ..., s_m^2$ consists of the eigenvalues of $T^*T$ listed in decreasing order with each eigenvalue appearing as many times as the dimension of the corresponding eigenspace of $T^*T$. It follows that the singular values of $T$ are precisely $s_1, ..., s_m$.

(d) We proved this in part (c).

(e) Because the equation

$$Tv = s_1 \langle v, e_1 \rangle f_1 + \cdots + s_m \langle v, e_m \rangle f_m$$

holds for all $v \in V$, the calculation in the proof of 7.75 shows that equation 7.77 holds for all $w \in W$. Thus, for any $w \in W$,

$$TT^*w = T\left(\sum_{k=1}^{m} s_k \langle w, f_k \rangle e_k\right) = \sum_{j=1}^{m} s_j \left\langle \sum_{k=1}^{m} s_k \langle w, f_k \rangle e_k, e_j \right\rangle f_j = \sum_{j=1}^{m} s_j^2 \langle w, f_j \rangle f_j.$$

**Exercise 7.E.9.** Suppose $T \in \mathcal{L}(V, W)$. Show that $T$ and $T^*$ have the same positive singular values.

**Solution.** Let $s_1, ..., s_m$ be the positive singular values of $T^*$. By the SVD (7.70), there exists an orthonormal list $f_1, ..., f_m$ in $W$ and an orthonormal list $e_1, ..., e_m$ in $V$ such that

$$T^*w = s_1 \langle w, f_1 \rangle e_1 + \cdots + s_m \langle w, f_m \rangle e_m$$

for every $w \in W$. It follows from 7.75 that

$$Tv = (T^*)^*v = s_1 \langle v, e_1 \rangle f_1 + \cdots + s_m \langle v, e_m \rangle f_m$$

for every $v \in V$. Thus, by Exercise 7.E.8, the positive singular values of $T$ are $s_1, ..., s_m$. We have now shown that if $s_1, ..., s_m$ are the positive singular values of $T^*$ then $s_1, ..., s_m$ are the positive singular values of $T$. Replacing $T$ with $T^*$ in this result and using that $(T^*)^* = T$ gives us the desired equivalence.



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**Exercise 7.E.10.** Suppose $T \in \mathcal{L}(V, W)$ has singular values $s_1, \dots, s_n$. Prove that if $T$ is an invertible linear map, then $T^{-1}$ has singular values

$$\frac{1}{s_n}, \dots, \frac{1}{s_1}.$$

**Solution.** Because $T$ is invertible, 7.68 shows that each singular value of $T$ must be positive. By the SVD (7.70), there exists an orthonormal list $e_1, \dots, e_n$ in $V$ and an orthonormal list $f_1, \dots, f_n$ in $W$ such that

$$Tv = s_1\langle v, e_1\rangle f_1 + \cdots + s_n\langle v, e_n\rangle f_n$$

for every $v \in V$. Note that $T^{-1} = T^\dagger$ by 6.69(a). It follows from 7.75 that

$$T^{-1}w = \frac{\langle w, f_n\rangle}{s_n} e_n + \cdots + \frac{\langle w, f_1\rangle}{s_1} e_1$$

for every $w \in W$. Since

$$s_1 \geq \cdots \geq s_n > 0 \quad \Rightarrow \quad \frac{1}{s_n} \geq \cdots \geq \frac{1}{s_1} > 0,$$

Exercise 7.E.8 shows that $s_n^{-1}, \dots, s_1^{-1}$ are the singular values of $T^{-1}$ (we are taking the orthonormal lists required by Exercise 7.E.8 to be $f_n, \dots, f_1$ and $e_n, \dots, e_1$).

**Exercise 7.E.11.** Suppose that $T \in \mathcal{L}(V, W)$ and $v_1, \dots, v_n$ is an orthonormal basis of $V$. Let $s_1, \dots, s_n$ denote the singular values of $T$.

(a) Prove that $\|Tv_1\|^2 + \cdots + \|Tv_n\|^2 = s_1^2 + \cdots + s_n^2$.

(b) Prove that if $W = V$ and $T$ is a positive operator, then

$$\langle Tv_1, v_1\rangle + \cdots + \langle Tv_n, v_n\rangle = s_1 + \cdots + s_n.$$

*See the comment after Exercise 5 in Section 7A.*

**Solution.**

(a) Let $s_1, \dots, s_m$ be the positive singular values of $T$. As discussed after the proof of 7.70, there exists an orthonormal basis $e_1, \dots, e_n$ of $V$ and an orthonormal basis $f_1, \dots, f_{\dim W}$ such that

$$Te_k = \begin{cases} s_k f_k & \text{if } k \in \{1, \dots, m\}, \\ 0 & \text{otherwise.} \end{cases}$$

It follows from Exercise 7.A.5 that

$$\sum_{k=1}^n \|Tv_k\|^2 = \sum_{k=1}^n \|Te_k\|^2 = \sum_{k=1}^m \|s_k f_k\|^2 = \sum_{k=1}^m s_k^2 = \sum_{k=1}^n s_k^2,$$

where we have used that $s_k = 0$ if $k > m$ for the last equality.



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(b) Because $T$ is a positive operator, $T$ must be self-adjoint and each of its eigenvalues must be non-negative. Recalling that singular values are non-negative, Exercise 7.E.7 shows that $s_1, ..., s_n$ must be the eigenvalues of $T$. It follows from 7.39 that $\sqrt{s_1}, ..., \sqrt{s_n}$ are the eigenvalues of $\sqrt{T}$ and another application of Exercise 7.E.7 shows that the singular values of $\sqrt{T}$ are $\sqrt{s_1}, ..., \sqrt{s_n}$. Thus, by part (a),

$$\sum_{k=1}^{n} \langle Tv_n, v_n \rangle = \sum_{k=1}^{n} \|\sqrt{T}v_n\|^2 = s_1 + \cdots + s_n.$$

**Exercise 7.E.12.**

(a) Give an example of a finite-dimensional vector space and an operator $T$ on it such that the singular values of $T^2$ do not equal the squares of the singular values of $T$.

(b) Suppose $T \in \mathcal{L}(V)$ is normal. Prove that the singular values of $T^2$ equal the squares of the singular values of $T$.

**Solution.**

(a) Consider the operator $T \in \mathcal{L}(\mathbf{F}^2)$ whose matrix with respect to the standard basis of $\mathbf{F}^2$ is

$$\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}.$$

A calculation shows that the singular values of $T$ are $1, 0$. However, since $T^2 = 0$, the singular values of $T^2$ are $0, 0$.

(b) Suppose that $s_1, ..., s_n$ are the singular values of $T$, so that there is an orthonormal basis $e_1, ..., e_n$ of $V$ such that $T^*Te_k = s_k^2 e_k$. Using that $T$ is normal, observe that

$$(T^2)^* T^2 e_k = (T^*T)^2 e_k = s_k^4 e_k.$$

Thus the singular values of $T^2$ are $s_1^2, ..., s_n^2$.

**Exercise 7.E.13.** Suppose $T_1, T_2 \in \mathcal{L}(V)$. Prove that $T_1$ and $T_2$ have the same singular values if and only if there exist unitary operators $S_1, S_2 \in \mathcal{L}(V)$ such that $T_1 = S_1 T_2 S_2$.

**Solution.** Suppose there exist such unitary operators and observe that

$$\begin{aligned} T_1^* = S_2^* T_2^* S_1^* = S_2^{-1} T_2^* S_1^{-1} \quad &\Rightarrow \quad T_1^* T_1 = S_2^{-1} T_2^* T_2 S_2 \\ &\Rightarrow \quad T_1^* T_1 - \lambda I = S_2^{-1}(T_2^* T_2 - \lambda I) S_2 \text{ for all } \lambda \in \mathbf{F} \\ &\Rightarrow \quad \dim E(\lambda, T_1^* T_1) = \dim E(\lambda, T_2^* T_2) \text{ for all } \lambda \in \mathbf{F}. \end{aligned}$$

Thus $T_1$ and $T_2$ have the same singular values.



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Now suppose that $T_1$ and $T_2$ have the same singular values $s_1, ..., s_m, ..., s_n$, where $s_1, ..., s_m$ are positive and $s_k = 0$ if $k > m$ (it may be the case that $m = n$). By the SVD (7.70), there exist orthonormal bases

$$e_1, ..., e_n, \quad f_1, ..., f_n, \quad g_1, ..., g_n, \quad h_1, ..., h_n$$

of $V$ such that

$$T_1 e_k = \begin{cases} s_k f_k & \text{if } k \in \{1, ..., m\}, \\ 0 & \text{otherwise,} \end{cases} \quad \text{and} \quad T_2 g_k = \begin{cases} s_k h_k & \text{if } k \in \{1, ..., m\}, \\ 0 & \text{otherwise.} \end{cases}$$

Define $S_1, S_2 \in \mathcal{L}(V)$ by $S_1 h_k = f_k$ and $S_2 e_k = g_k$ and note that $S_1$ and $S_2$ are unitary operators by 7.53(d). Furthermore,

$$k \in \{1, ..., m\} \quad \Rightarrow \quad S_1 T_2 S_2 e_k = S_1 T_2 g_k = s_k S_1 h_k = s_k f_k = T_1 e_k,$$
$$k > m \quad \Rightarrow \quad S_1 T_2 S_2 e_k = S_1 T_2 g_k = 0 = T_1 e_k.$$

Thus $T_1 = S_1 T_2 S_2$.

**Exercise 7.E.14.** Suppose $T \in \mathcal{L}(V, W)$. Let $s_n$ denote the smallest singular value of $T$. Prove that $s_n \|v\| \leq \|Tv\|$ for every $v \in V$.

**Solution.** If $v = 0$ then the inequality is clear, so suppose that $v \neq 0$. It follows from Exercise 7.E.4 that

$$s_n \leq \left\| T\left(\frac{v}{\|v\|}\right) \right\| \quad \Rightarrow \quad s_n \|v\| \leq \|Tv\|.$$

**Exercise 7.E.15.** Suppose $T \in \mathcal{L}(V)$ and $s_1 \geq \cdots \geq s_n$ are the singular values of $T$. Prove that if $\lambda$ is an eigenvalue of $T$, then $s_1 \geq |\lambda| \geq s_n$.

**Solution.** Let $v \in V$ be such that $Tv = \lambda v$ and $\|v\| = 1$. It follows from Exercise 7.E.4 that

$$|\lambda| = \|\lambda v\| = \|Tv\| \in [s_n, s_1].$$

**Exercise 7.E.16.** Suppose $T \in \mathcal{L}(V, W)$. Prove that $(T^*)^\dagger = (T^\dagger)^*$.

*Compare the result in this exercise to the analogous result for invertible linear maps [see 7.5(f)].*

**Solution.** Let $s_1, ..., s_m$ be the positive singular values of $T$. The SVD (7.70) and 7.75 imply the existence of an orthonormal list $e_1, ..., e_m$ in $V$ and an orthonormal list $f_1, ..., f_m$ in $W$ such that

$$T^* w = s_1 \langle w, f_1 \rangle e_1 + \cdots + s_m \langle w, f_m \rangle e_m$$



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for every $w \in W$. Since $s_1, \ldots, s_m$ are also the positive singular values of $T^*$ (see Exercise 7.E.9), another application of 7.75 shows that

$$(T^*)^\dagger v = \frac{\langle v, e_1 \rangle}{s_1} f_1 + \cdots + \frac{\langle v, e_m \rangle}{s_m} f_m$$

for every $v \in V$.

Also by 7.75, we have

$$T^\dagger w = \frac{\langle w, f_m \rangle}{s_m} e_m + \cdots + \frac{\langle w, f_1 \rangle}{s_1} e_1$$

for every $w \in W$. It follows from Exercise 7.E.8 that $s_m^{-1}, \ldots, s_m^{-1}$ are the positive singular values of $T^\dagger$ and we can apply 7.75 once more to see that

$$\left(T^\dagger\right)^* v = \frac{\langle v, e_m \rangle}{s_m} f_m + \cdots + \frac{\langle v, e_1 \rangle}{s_1} f_1$$

for every $v \in V$. Thus $(T^*)^\dagger = \left(T^\dagger\right)^*$.

**Exercise 7.E.17.** Suppose $T \in \mathcal{L}(V)$. Prove that $T$ is self-adjoint if and only if $T^\dagger$ is self-adjoint.

**Solution.** Observe that

$$T = T^* \quad \Leftrightarrow \quad T^\dagger = (T^*)^\dagger \quad \Leftrightarrow \quad T^\dagger = \left(T^\dagger\right)^*,$$

where we have used Exercise 6.C.23 for the first equivalence and Exercise 7.E.16 for the second equivalence.



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## 7.F. Consequences of Singular Value Decomposition

**Exercise 7.F.1.** Prove that if $S, T \in \mathcal{L}(V, W)$, then $|\|S\| - \|T\|| \leq \|S - T\|$.

*The inequality above is called the **reverse triangle inequality**.*

**Solution.** The proof is essentially the same as the proof given in Exercise 4.2.

**Exercise 7.F.2.** Suppose that $T \in \mathcal{L}(V)$ is self-adjoint or that $\mathbf{F} = \mathbf{C}$ and $T \in \mathcal{L}(V)$ is normal. Prove that

$$\|T\| = \max\{|\lambda| : \lambda \text{ is an eigenvalue of } T\}.$$

**Solution.** Let $\lambda_1, ..., \lambda_n$ be the eigenvalues of $T$, included in this list as many times as the dimension of the corresponding eigenspace. As shown in Exercise 7.E.7, the singular values of $T$ are $|\lambda_1|, ..., |\lambda_n|$ (sorted in decreasing order). It follows from 7.88(a) that

$$\|T\| = \max\{|\lambda| : \lambda \text{ is an eigenvalue of } T\}.$$

**Exercise 7.F.3.** Suppose $T \in \mathcal{L}(V, W)$ and $v \in V$. Prove that

$$\|Tv\| = \|T\|\|v\| \quad \Leftrightarrow \quad T^*Tv = \|T\|^2 v.$$

**Solution.** Suppose that $T^*Tv = \|T\|^2 v$. Using 7.88(c) and 7.91, it follows that

$$\|T\|^2\|v\| = \|T^*Tv\| \leq \|T^*\|\|Tv\| = \|T\|\|Tv\| \quad \Rightarrow \quad \|T\|\|v\| \leq \|Tv\|.$$

Since $\|Tv\| \leq \|T\|\|v\|$ (by 7.88(c)), we may conclude that $\|Tv\| = \|T\|\|v\|$.

Suppose that $\|Tv\| = \|T\|\|v\|$ and observe that

$$\begin{aligned}
\left\|T^*Tv - \|T\|^2 v\right\|^2 &= \langle T^*Tv - \|T\|^2 v, T^*Tv - \|T\|^2 v\rangle \\
&= \|T^*Tv\|^2 + \|T\|^4\|v\|^2 - 2\,\mathrm{Re}\langle T^*Tv, \|T\|^2 v\rangle \\
&\leq \|T^*\|^2\|Tv\|^2 + \|T\|^2\|Tv\|^2 - 2\|T\|^2\|Tv\|^2 && (7.88(c)) \\
&= 0. && (7.91)
\end{aligned}$$

Thus $T^*Tv = \|T\|^2 v$.

**Exercise 7.F.4.** Suppose $T \in \mathcal{L}(V, W), v \in V$, and $\|Tv\| = \|T\|\|v\|$. Prove that if $u \in V$ and $\langle u, v\rangle = 0$, then $\langle Tu, Tv\rangle = 0$.

**Solution.** By Exercise 7.F.4 we must have $T^*Tv = \|T\|^2 v$. It follows that



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$$\langle Tu, Tv\rangle = \langle u, T^*Tv\rangle = \langle u, \|T\|^2 v\rangle = \|T\|^2\langle u, v\rangle = 0.$$

**Exercise 7.F.5.** Suppose $U$ is a finite-dimensional inner product space, $T \in \mathcal{L}(V, U)$, and $S \in \mathcal{L}(U, W)$. Prove that

$$\|ST\| \le \|S\|\|T\|.$$

**Solution.** By 7.88(c) we have, for any $v \in V$,

$$\|STv\| \le \|S\|\|Tv\| \le \|S\|\|T\|\|v\|.$$

It follows from the minimality of $\|ST\|$ that $\|ST\| \le \|S\|\|T\|$.

**Exercise 7.F.6.** Prove or give a counterexample: If $S, T \in \mathcal{L}(V)$, then $\|ST\| = \|TS\|$.

**Solution.** This is false. For a counterexample, consider the operators $S, T \in \mathcal{L}(\mathbf{R}^2)$ whose matrices with respect to the standard basis of $\mathbf{R}^2$ are

$$\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \quad \text{and} \quad \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}.$$

A routine calculation shows that $ST = 0$ whereas $TS \neq 0$. It follows from 7.87(b) that

$$\|ST\| = 0 \neq \|TS\|.$$

**Exercise 7.F.7.** Show that defining $d(S, T) = \|S - T\|$ for $S, T \in \mathcal{L}(V, W)$ makes $d$ a metric on $\mathcal{L}(V, W)$.

*This exercise is intended for readers who are familiar with metric spaces.*

**Solution.** Certainly $d$ is non-negative and $d(T, T) = 0$ for any $T \in \mathcal{L}(V, W)$. Furthermore, $d(S, T) = 0$ implies $S = T$ by 7.87(b), and we have $d(S, T) = d(T, S)$ for any $S, T \in \mathcal{L}(V, W)$ since $\|Sv - Tv\| = \|Tv - Sv\|$ for any $v \in V$. Finally, 7.87(d) shows that

$$d(R, T) \le d(R, S) + d(S, T)$$

for any $R, S, T \in \mathcal{L}(V, W)$.

**Exercise 7.F.8.**

(a) Prove that if $T \in \mathcal{L}(V)$ and $\|I - T\| < 1$, then $T$ is invertible.

(b) Suppose that $S \in \mathcal{L}(V)$ is invertible. Prove that if $T \in \mathcal{L}(V)$ and

$$\|S - T\| < 1/\left\|S^{-1}\right\|,$$

then $T$ is invertible.

*This exercise shows that the set of invertible operators in $\mathcal{L}(V)$ is an open subset of $\mathcal{L}(V)$, using the metric defined in Exercise 7.*



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**Solution.**

(a) We will prove the contrapositive statement. Suppose that $T$ is not invertible, so that there is some non-zero $v \in V$ such that $Tv = 0$. It follows that

$$\|I - T\|\|v\| \geq \|(I - T)v\| = \|v\| \quad \Rightarrow \quad \|I - T\| \geq 1.$$

(b) The proof is a generalization of the proof in part (a). We will again prove the contrapositive statement. Suppose that $T$ is not invertible, so that there is some non-zero $v \in V$ such that $Tv = 0$. Observe that

$$\|v\| = \|S^{-1}Sv\| \leq \|S^{-1}\|\|Sv\| = \|S^{-1}\|\|(S - T)v\| \leq \|S^{-1}\|\|S - T\|\|v\|$$

$$\Rightarrow \quad \|S - T\| \geq \frac{1}{\|S^{-1}\|}.$$

**Exercise 7.F.9.** Suppose $T \in \mathcal{L}(V)$. Prove that for every $\varepsilon > 0$, there exists an invertible operator $S \in \mathcal{L}(V)$ such that $0 < \|T - S\| < \varepsilon$.

**Solution.** Because $T$ can have only finitely many eigenvalues, we can choose some $\delta \in (0, \varepsilon)$ such that $\delta$ is not an eigenvalue of $T$. Letting $S = T - \delta I$, it follows that $S$ is invertible and that $S \neq T$. Thus

$$0 < \|T - S\| = \|\delta I\| = \delta < \varepsilon.$$

**Exercise 7.F.10.** Suppose $\dim V > 1$ and $T \in \mathcal{L}(V)$ is not invertible. Prove that for every $\varepsilon > 0$, there exists $S \in \mathcal{L}(V)$ such that $0 < \|T - S\| < \varepsilon$ and $S$ is not invertible.

**Solution.** Since $T$ is not invertible, there exists some $e_1 \in \operatorname{null} T$ such that $\|e_1\| = 1$. Extend this to an orthonormal basis $e_1, ..., e_n$ of $V$ and note that $n \geq 2$. Define $S \in \mathcal{L}(V)$ by $Se_1 = 0$ and $Se_k = Te_k - \frac{\varepsilon}{2}e_k$ for $k \geq 2$. Notice that $S$ is not invertible and that $S \neq T$. Notice further that, for any $v \in V$,

$$\|(T - S)v\|^2 = \left\|\tfrac{\varepsilon}{2}(\langle v, e_2\rangle e_2 + \cdots + \langle v, e_n\rangle e_n)\right\|^2$$

$$= \frac{\varepsilon^2}{4}\left(|\langle v, e_2\rangle|^2 + \cdots + |\langle v, e_n\rangle|^2\right) \leq \frac{\varepsilon^2}{4}\|v\|^2.$$

Thus $\|(T - S)v\| \leq \frac{\varepsilon}{2}\|v\|$ for any $v \in V$. It follows that $\|T - S\| \leq \frac{\varepsilon}{2} < \varepsilon$ and hence that $0 < \|T - S\| < \varepsilon$, since $S \neq T$.

**Exercise 7.F.11.** Suppose $\mathbf{F} = \mathbf{C}$ and $T \in \mathcal{L}(V)$. Prove that for every $\varepsilon > 0$ there exists a diagonalizable operator $S \in \mathcal{L}(V)$ such that $0 < \|T - S\| < \varepsilon$.

**Solution.** We will prove that $S$ may be chosen to have $\dim V$ distinct eigenvalues, which implies the desired result by 5.58.



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By Schur's theorem (6.38), there is an orthonormal basis $e_1, ..., e_n$ of $V$ with respect to which the matrix of $T$ is upper-triangular, say

$$\mathcal{M}(T, (e_1, ..., e_n)) = \begin{pmatrix} \lambda_1 & * & \cdots & * \\ 0 & \lambda_2 & \cdots & * \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \lambda_n \end{pmatrix}.$$

Let $D \in \mathcal{L}(V)$ be given by $De_k = ke_k$, so that

$$\mathcal{M}(D, (e_1, ..., e_n)) = \begin{pmatrix} 1 & 0 & \cdots & 0 \\ 0 & 2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & n \end{pmatrix}.$$

It is straightforward to verify that $\|D\| = n$. For $\delta \in \mathbf{R}$ and $j, k \in \{1, ..., n\}$ such that $j < k$, notice that

$$\lambda_j + j\delta = \lambda_k + k\delta \quad \Leftrightarrow \quad \delta = \frac{\lambda_j - \lambda_k}{k - j}.$$

Since there are only finitely many such choices of $j$ and $k$, we may choose a $\delta \in \left(0, \frac{\varepsilon}{n}\right)$ such that $\lambda_j + j\delta \neq \lambda_k + k\delta$ for each $j, k \in \{1, ..., n\}$ satisfying $j < k$. It follows that the diagonal entries of the upper-triangular matrix

$$\mathcal{M}(T + \delta D, (e_1, ..., e_n)) = \begin{pmatrix} \lambda_1 + \delta & * & \cdots & * \\ 0 & \lambda_2 + 2\delta & \cdots & * \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \lambda_n + n\delta \end{pmatrix},$$

and hence the $n$ eigenvalues of the operator $S \coloneqq T + \delta D$, are distinct. Furthermore,

$$0 < \delta \quad \Rightarrow \quad S \neq T \quad \Rightarrow \quad 0 < \|T - S\|,$$
$$\delta < \frac{\varepsilon}{n} \quad \Rightarrow \quad \|T - S\| = \delta\|D\| < \varepsilon.$$

**Exercise 7.F.12.** Suppose $T \in \mathcal{L}(V)$ is a positive operator. Show that $\left\|\sqrt{T}\right\| = \sqrt{\|T\|}$.

**Solution.** Let $s$ be the largest singular value of $\sqrt{T}$, i.e. $\left\|\sqrt{T}\right\| = s$. Since $\sqrt{T}$ is positive and hence normal, Exercise 7.E.12 (b) shows that the singular values of $T$ equal the squares of the singular values of $\sqrt{T}$. Given that singular values are non-negative, it follows that the largest singular value of $T$ is $s^2$, i.e. $\|T\| = s^2$. Thus $\left\|\sqrt{T}\right\| = s = \sqrt{\|T\|}$.

**Exercise 7.F.13.** Suppose $S, T \in \mathcal{L}(V)$ are positive operators. Show that

$$\|S - T\| \leq \max\{\|S\|, \|T\|\} \leq \|S + T\|.$$

**Solution.** Let us prove a couple of useful lemmas.



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**Lemma L.13.** If $A \in \mathcal{L}(V)$ is a self-adjoint operator then $\|A\|I - A$ is a positive operator.

*Proof.* First note that $\|A\|I - A$ is a real-linear combination of self-adjoint operators and hence is itself self-adjoint (see 7.5). Now suppose that $\lambda \in \mathbf{R}$ is an eigenvalue of $\|A\|I - A$, say $\|A\|v - Av = \lambda v$ for some $v \in V$. It follows that $\lambda - \|A\|$ is an eigenvalue of $A$ and hence, by Exercise 7.F.2,

$$|\lambda - \|A\|| \leq \|A\| \quad \Rightarrow \quad \lambda \geq 0.$$

Thus $\|A\|I - A$ is a positive operator by 7.38(b). □

**Lemma L.14.** If $A$ and $B - A$ are positive operators then $\|A\| \leq \|B\|$.

*Proof.* Note that $B = (B - A) + A$ is a positive operator by Exercise 7.C.6. It follows from Lemma L.13 that $\|B\|I - B$ is a positive operator and hence that

$$\|B\|I - A = (\|B\|I - B) + (B - A)$$

is a positive operator. Suppose $\lambda \geq 0$ is an eigenvalue of $A$, say $Av = \lambda v$ for some $v \in V$. Observe that

$$(\|B\|I - A)v = (\|B\| - \lambda)v,$$

so that $\|B\| - \lambda$ is an eigenvalue of the positive operator $\|B\|I - A$. It follows that $\|B\| - \lambda$ is non-negative and hence that $\lambda \leq \|B\|$. Since this was true for any eigenvalue of $A$, and each such eigenvalue is non-negative, Exercise 7.F.2 shows that

$$\|A\| = \max\{|\lambda| : \lambda \text{ is an eigenvalue of } A\} \leq \|B\|,$$

as desired. □

Returning to the exercise, suppose $\lambda \in \mathbf{R}$ is an eigenvalue of the self-adjoint operator $S - T$, say $(S - T)v = \lambda v$ for some $v \in V$. It follows that

$$(\|S\|I - (S - T))v = (\|S\| - \lambda)v$$

and hence that $\|S\| - \lambda$ is an eigenvalue of $\|S\|I - (S - T)$. Notice that $\|S\|I - (S - T)$ is a positive operator by Lemma L.13 and Exercise 7.C.6, so that its eigenvalues are non-negative. Thus $\lambda \leq \|S\|$ and a similar argument with the operator $S - T + \|T\|I$ shows that $-\|T\| \leq \lambda$. It follows that $|\lambda| \leq \max\{\|S\|, \|T\|\}$ and hence, by Exercise 7.F.2,

$$\|S - T\| = \max\{|\lambda| : \lambda \text{ is an eigenvalue of } S - T\} \leq \max\{\|S\|, \|T\|\}.$$

Applying Lemma L.14 twice, first with $A = T$ and $B = S + T$ and then with $A = S$ and $B = S + T$, shows that



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$$\|S\| \le \|S+T\| \quad \text{and} \quad \|T\| \le \|S+T\| \quad \Rightarrow \quad \max\{\|S\|, \|T\|\} \le \|S+T\|.$$

**Exercise 7.F.14.** Suppose $U$ and $W$ are subspaces of $V$ such that $\|P_U - P_W\| < 1$. Prove that $\dim U = \dim W$.

**Solution.** Using the identities $P_U = I - P_{U^\perp}$ and $P_W = I - P_{W^\perp}$ (see Exercise 6.C.5) and Exercise 7.F.8 (a), we see that the operators $P_{U^\perp} + P_W$ and $P_U + P_{W^\perp}$ are invertible. It follows that

$$U \cap W^\perp = (\operatorname{null} P_{U^\perp}) \cap (\operatorname{null} P_W) \subseteq \operatorname{null}(P_{U^\perp} + P_W) = \{0\} \quad \Rightarrow \quad U \cap W^\perp = \{0\},$$

$$V = \operatorname{range}(P_U + P_{W^\perp}) \subseteq (\operatorname{range} P_U) + (\operatorname{range} P_{W^\perp}) = U + W^\perp \quad \Rightarrow \quad U + W^\perp = V.$$

Thus, using 6.51,

$$\dim V = \dim(U + W^\perp) = \dim U + \dim W^\perp - \dim(U \cap W^\perp) = \dim V + \dim U - \dim W$$

$$\Rightarrow \quad \dim U = \dim W.$$

**Exercise 7.F.15.** Define $T \in \mathcal{L}(\mathbf{F}^3)$ by

$$T(z_1, z_2, z_3) = (z_3, 2z_1, 3z_2).$$

Find (explicitly) a unitary operator $S \in \mathcal{L}(\mathbf{F}^3)$ such that $T = S\sqrt{T^*T}$.

**Solution.** Let $e_1, e_2, e_3$ be the standard orthonormal basis of $\mathbf{F}^3$. A routine calculation shows that the singular value decomposition of $T$ is

$$Tv = 3\langle v, e_2\rangle e_3 + 2\langle v, e_1\rangle e_2 + \langle v, e_3\rangle e_1.$$

As the proof of 7.93 shows, if we take $S \in \mathcal{L}(V)$ to be the unitary operator defined by

$$Se_2 = e_3, \quad Se_1 = e_2, \quad \text{and} \quad Se_3 = e_1,$$

then $T = S\sqrt{T^*T}$.

**Exercise 7.F.16.** Suppose $S \in \mathcal{L}(V)$ is a positive invertible operator. Prove that there exists $\delta > 0$ such that $T$ is a positive operator for every self-adjoint operator $T \in \mathcal{L}(V)$ with $\|S - T\| < \delta$.

**Solution.** Let $\mu$ be the least singular value of $S$ and note that $\mu > 0$ since $S$ is invertible. Note further that $\sqrt{\mu} > 0$ is the least singular value of $\sqrt{S}$ by Exercise 7.E.12 (b). Thus, by Exercise 7.E.14,

$$\langle Sv, v\rangle = \left\langle S^{1/2}v, S^{1/2}v\right\rangle = \left\|S^{1/2}v\right\|^2 \ge \mu\|v\|^2 \tag{1}$$

for every $v \in V$. Suppose $T \in \mathcal{L}(V)$ is a self-adjoint operator satisfying $\|S - T\| < \mu$. For any $v \in V$, the Cauchy-Schwarz inequality gives us



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$$\langle (S-T)v, v\rangle \le \|(S-T)v\|\|v\| \le \|S-T\|\|v\|^2 < \mu\|v\|^2$$

$$\Rightarrow \quad \mu\|v\|^2 + \langle (T-S)v, v\rangle > 0. \qquad (2)$$

Combining inequalities (1) and (2) we obtain, for any $v \in V$,

$$\langle Tv, v\rangle = \langle Sv, v\rangle + \langle (T-S)v, v\rangle \ge \mu\|v\|^2 + \langle (T-S)v, v\rangle > 0.$$

Thus $T$ is a positive operator.

**Exercise 7.F.17.** Prove that if $u \in V$ and $\varphi_u$ is the linear functional on $V$ defined by the equation $\varphi_u(v) = \langle v, u\rangle$, then $\|\varphi_u\| = \|u\|$.

*Here we are thinking of the scalar field* $\mathbf{F}$ *as an inner product space with* $\langle \alpha, \beta\rangle = \alpha\overline{\beta}$ *for all* $\alpha, \beta \in \mathbf{F}$. *Thus* $\|\varphi_u\|$ *means the norm of* $\varphi_u$ *as a linear map from* $V$ *to* $\mathbf{F}$.

**Solution.** The Cauchy-Schwarz inequality shows that

$$|\varphi_u(v)| = |\langle v, u\rangle| \le \|u\|\|v\|$$

for any $v \in V$. It follows that $\|\varphi_u\| \le \|u\|$. Since $|\varphi_u(u)| = |\langle u, u\rangle| = \|u\|$, we may conclude that $\|\varphi_u\| = \|u\|$.

**Exercise 7.F.18.** Suppose $e_1, ..., e_n$ is an orthonormal basis of $V$ and $T \in \mathcal{L}(V, W)$.

(a) Prove that $\max\{\|Te_1\|, ..., \|Te_n\|\} \le \|T\| \le \left(\|Te_1\|^2 + \cdots + \|Te_n\|^2\right)^{1/2}$.

(b) Prove that $\|T\| = \left(\|Te_1\|^2 + \cdots + \|Te_n\|^2\right)^{1/2}$ if and only if $\dim \operatorname{range} T \le 1$.

*Here* $e_1, ..., e_n$ *is an arbitrary orthonormal basis of* $V$, *not necessarily connected with a singular value decomposition of* $T$. *If* $s_1, ..., s_n$ *is the list of singular values of* $T$, *then the right side of the inequality above equals* $\left(s_1^2 + \cdots + s_n^2\right)^{1/2}$, *as was shown in Exercise 11(a) in Section 7E.*

**Solution.**

(a) For any $k \in \{1, ..., n\}$ we have $\|Te_k\| \le \|T\|\|e_k\| = \|T\|$ and thus

$$\max\{\|Te_1\|, ..., \|Te_n\|\} \le \|T\|.$$

Let $s_1, ..., s_n$ be the singular values of $T$ and suppose that $s_1, ..., s_m$ are the positive singular values of $T$. Suppose further that

$$Tv = s_1\langle v, f_1\rangle g_1 + \cdots + s_m\langle v, f_m\rangle g_m$$

is a singular value decomposition of $T$. For any $v \in V$, observe that

$$\begin{aligned}\|Tv\|^2 &= s_1^2|\langle v, f_1\rangle|^2 + \cdots + s_m^2|\langle v, f_m\rangle|^2 \\ &\le (s_1^2 + \cdots + s_m^2)\|v\|^2 \\ &\le (s_1^2 + \cdots + s_n^2)\|v\|^2.\end{aligned}$$



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Thus $\|Tv\| \le (s_1^2 + \cdots + s_n^2)^{1/2}\|v\|$ for every $v \in V$, which implies that

$$\|T\| \le (s_1^2 + \cdots + s_n^2)^{1/2} = \left(\|Te_1\|^2 + \cdots + \|Te_n\|^2\right)^{1/2},$$

where we have used Exercise 7.E.11 (a) for the last equality.

(b) Let $s_1, \ldots, s_n$ be the singular values of $T$. By Exercise 7.E.11 (a) and 7.68(b), it will suffice to show that $\|T\| = (s_1^2 + \cdots + s_n^2)^{1/2}$ if and only if $T$ has at most one positive singular value.

If $T$ has at most one positive singular value then $s_k = 0$ for $k \ge 2$. It follows that

$$\|T\| = s_1 = (s_1^2)^{1/2} = (s_1^2 + \cdots + s_n^2)^{1/2}.$$

If $T$ has at least two positive singular values then $s_2 > 0$ and it follows that

$$\|T\| = s_1 = (s_1^2)^{1/2} < (s_1^2 + s_2^2 + \cdots + s_n^2)^{1/2}.$$

**Exercise 7.F.19.** Prove that if $T \in \mathcal{L}(V, W)$, then $\|T^*T\| = \|T\|^2$.

*This formula for $\|T^*T\|$ leads to the important subject of $C^*$-algebras.*

**Solution.** For any $v \in V$, observe that

$$\|(T^*T)^{1/2}\, v\|^2 = \langle (T^*T)^{1/2}\, v, (T^*T)^{1/2}\, v\rangle = \langle T^*Tv, v\rangle = \langle Tv, Tv\rangle = \|Tv\|^2 \le \|T\|^2\|v\|^2.$$

It follows that $\|(T^*T)^{1/2}\| \le \|T\|$. By the polar decomposition (7.93), there exists a unitary operator $S \in \mathcal{L}(V)$ such that $T = S(T^*T)^{1/2}$. Note that

$$\|S\| = \max\{\|Sv\| : v \in V \text{ and } \|V\| = 1\} = \max\{1\} = 1.$$

It then follows from Exercise 7.F.5 that

$$\|T\| = \|S(T^*T)^{1/2}\| \le \|S\|\|(T^*T)^{1/2}\| = \|(T^*T)^{1/2}\|.$$

Thus $\|(T^*T)^{1/2}\| = \|T\|$, which is equivalent to $\|T^*T\| = \|T\|^2$ by Exercise 7.F.12.

**Exercise 7.F.20.** Suppose $T \in \mathcal{L}(V)$ is normal. Prove that $\|T^k\| = \|T\|^k$ for every positive integer $k$.

**Solution.** Let $k$ be a positive integer and suppose $s_1, \ldots, s_n$ are the singular values of $T$, so that $\|T\| = s_1$. As in Exercise 7.E.12 (b), let $e_1, \ldots, e_n$ be an orthonormal basis of $V$ such that $T^*Te_j = s_j^2 e_j$ for each $j \in \{1, \ldots, n\}$. Using the normality of $T$, it follows that

$$(T^k)^*T^k e_j = (T^*T)^k e_j = s_j^{2k} e_j$$

for each $j \in \{1, \ldots, n\}$. Thus the singular values of $T^k$ are $s_1^k, \ldots, s_n^k$; these are still in decreasing order since the function $x \mapsto x^k$ is strictly increasing on $[0, \infty)$. It follows that

$$\|T^k\| = s_1^k = \|T\|^k.$$



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**Exercise 7.F.21.** Suppose $\dim V > 1$ and $\dim W > 1$. Prove that the norm on $\mathcal{L}(V,W)$ does not come from an inner product. In other words, prove that there does not exist an inner product on $\mathcal{L}(V,W)$ such that

$$\max\{\|Tv\| : v \in V \text{ and } \|v\| \le 1\} = \sqrt{\langle T, T\rangle}$$

for all $T \in \mathcal{L}(V,W)$.

**Solution.** Let $v_1, ..., v_m$ be an orthonormal basis of $V$ and let $w_1, ..., w_n$ be an orthonormal basis of $W$; note that $m, n \ge 2$. Let $S, T \in \mathcal{L}(V,W)$ be the linear maps given by

$$Sv_1 = w_1, \quad Sv_2 = w_2, \quad \text{and} \quad Sv_k = 0 \text{ if } k > 2,$$
$$Tv_1 = -w_1, \quad Tv_2 = w_2, \quad \text{and} \quad Tv_k = 0 \text{ if } k > 2.$$

It is straightforward to verify that

$$\|S+T\| = \|S-T\| = 2 \quad \text{and} \quad \|S\| = \|T\| = 1$$
$$\Rightarrow \quad \|S+T\|^2 + \|S-T\|^2 = 8 \ne 4 = 2\big(\|S\|^2 + \|T\|^2\big).$$

Thus the norm on $\mathcal{L}(V,W)$ does not satisfy the parallelogram equality (6.21) and hence does not come from an inner product.

**Exercise 7.F.22.** Suppose $T \in \mathcal{L}(V,W)$. Let $n = \dim V$ and let $s_1 \ge \cdots \ge s_n$ denote the singular values of $T$. Prove that if $1 \le k \le n$, then

$$\min\{\|T|_U\| : U \text{ is a subspace of } V \text{ with } \dim U = k\} = s_{n-k+1}.$$

**Solution.** For $k \in \{1, ..., n\}$, let $E_k = \{\|T|_U\| : U \text{ is a subspace of } V \text{ with } \dim U = k\}$. If $T$ has no positive singular values, i.e. $T = 0$, then

$$\min E_k = \min\{0\} = 0 = s_{n-k+1}$$

for any $k \in \{1, ..., n\}$. Furthermore, since the only subspace of $V$ with dimension $n$ is $V$ itself, we have

$$\min E_n = \min\{\|T\|\} = \|T\| = s_1.$$

We may therefore assume that $T$ has at least one singular value and that $1 \le k < n$.

Suppose that $s_1 \ge \cdots \ge s_m$ are the positive singular values of $T$, where $m \ge 1$, and let

$$Tv = s_1\langle v, e_1\rangle f_1 + \cdots + s_m\langle v, e_m\rangle f_m$$

be a singular value decomposition of $T$. Extend the orthonormal list $e_1, ..., e_m$ to an orthonormal basis $e_1, ..., e_n$ of $V$, so that

$$Te_j = \begin{cases} s_j f_j & \text{if } 1 \le j \le m, \\ 0 & \text{if } m < j \le n. \end{cases}$$



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For $1 \le k < n$ let $X = \operatorname{span}(e_{n-k+1}, ..., e_n)$ and note that $\dim X = k$. We consider two cases.

**Case 1.** If $1 \le k \le n - m$, so that $m + 1 \le n - k + 1$, then $s_{n-k+1} = 0$. Furthermore, $T$ vanishes on each of the basis vectors $e_{n-k+1}, ..., e_n$, so that $T|_X = 0$. Since $E_k$ is bounded below by zero, it follows that $\min E_k = 0 = s_{n-k+1}$.

**Case 2.** Suppose that $n - m < k < n$, so that $1 \le n - k < m$, and let $U$ be any subspace of $V$ satisfying $\dim U = k$. If $v \in V$ is such that $\|v\| \le 1$ then note that $P_U v \in U$ and that $\|P_U v\| \le \|v\| \le 1$. It follows that $\|TP_U v\| \le \|T|_U\|$, since

$$\|T|_U\| = \max\{\|Tu\| : u \in U \text{ and } \|u\| \le 1\}.$$

Thus $\|TP_U\| \le \|T|_U\|$. Next, observe that

$$\dim \operatorname{range}(TP_{U^\perp}) \le \dim \operatorname{range} P_{U^\perp} = \dim U^\perp = \dim V - \dim U = n - k.$$

It follows from 7.92 that $\|T - TP_{U^\perp}\| \ge s_{n-k+1}$. Combining these inequalities with the identity $P_U = I - P_{U^\perp}$ from Exercise 6.C.5, we have

$$\|T|_U\| \ge \|TP_U\| = \|T - TP_{U^\perp}\| \ge s_{n-k+1}.$$

Thus $s_{n-k+1}$ is a lower bound of $E_k$. Now observe that, for any $x \in X$,

$$\|(T|_X)x\|^2 = s_{n-k+1}^2 |\langle x, e_{n-k+1}\rangle|^2 + \cdots + s_m^2 |\langle x, e_m\rangle|^2 \le s_{n-k+1}^2 \|x\|^2.$$

It follows that $\|T|_X\| \le s_{n-k+1}$; in fact, this is an equality since $\|e_{n-k+1}\| = 1$ and

$$(T|_X)(e_{n-k+1}) = s_{n-k+1} f_{n-k+1} \quad \Rightarrow \quad \|(T|_X)(e_{n-k+1})\| = s_{n-k+1}.$$

Thus the lower bound $s_{n-k+1}$ belongs to $E_k$, i.e. $\min E_k = s_{n-k+1}$.

**Exercise 7.F.23.** Suppose $T \in \mathcal{L}(V, W)$. Show that $T$ is uniformly continuous with respect to the metrics on $V$ and $W$ that arise from the norms on those spaces (see Exercise 23 in Section 6B).

**Solution.** Let $\varepsilon > 0$ be given and let $\delta = \varepsilon(1 + \|T\|)^{-1}$. If $u, v \in V$ are such that $\|u - v\| < \delta$, then observe that

$$\|Tu - Tv\| = \|T(u - v)\| \le \|T\|\|u - v\| < \|T\|\delta < \varepsilon.$$

Thus $T$ is uniformly continuous.

**Exercise 7.F.24.** Suppose $T \in \mathcal{L}(V)$ is invertible. Prove that

$$\|T^{-1}\| = \|T\|^{-1} \quad \Leftrightarrow \quad \frac{T}{\|T\|} \text{ is a unitary operator.}$$

**Solution.** Let $s_1 \ge \cdots \ge s_n$ be the singular values of $T$, so that $\|T\| = s_1$; note that each singular value is strictly positive since $T$ is invertible. Note further that $s_n^{-1} \ge \cdots \ge s_1^{-1}$ are



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the singular values of $T^{-1}$ by Exercise 7.E.10, which gives us $\|T^{-1}\| = s_n^{-1}$. Thus we wish to prove that

$$s_n^{-1} = s_1^{-1} \quad \Leftrightarrow \quad s_1^{-1}T \text{ is a unitary operator.}$$

Indeed,

$$\begin{aligned} s_n^{-1} = s_1^{-1} &\Leftrightarrow s_1 = s_n \\ &\Leftrightarrow \text{each singular value of } T \text{ equals } s_1 \\ &\Leftrightarrow \text{each singular value of } s_1^{-1}T \text{ equals } 1 \\ &\Leftrightarrow s_1^{-1}T \text{ is a unitary operator.} \end{aligned} \tag{7.69}$$

**Exercise 7.F.25.** Fix $u, x \in V$ with $u \neq 0$. Define $T \in \mathcal{L}(V)$ by $Tv = \langle v, u\rangle x$ for every $v \in V$. Prove that

$$\sqrt{T^*T}v = \frac{\|x\|}{\|u\|}\langle v, u\rangle u$$

for every $v \in V$.

**Solution.** Let $R \in \mathcal{L}(V)$ be given by

$$Rv = \frac{\|x\|}{\|u\|}\langle v, u\rangle u.$$

Our aim is to show that $R = \sqrt{T^*T}$. Using the formula $T^*v = \langle v, x\rangle u$, shown in example 7.3, a routine calculation shows that

$$R^2v = \|x\|^2\langle v, u\rangle u = T^*Tv.$$

Furthermore, for any $v \in V$,

$$\langle Rv, v\rangle = \frac{\|x\|}{\|u\|}|\langle v, u\rangle|^2 \geq 0.$$

Thus $R$ is a positive square root of $T^*T$. It follows from uniqueness (7.39) that $R = \sqrt{T^*T}$.

**Exercise 7.F.26.** Suppose $T \in \mathcal{L}(V)$. Prove that $T$ is invertible if and only if there exists a unique unitary operator $S \in \mathcal{L}(V)$ such that $T = S\sqrt{T^*T}$.

**Solution.** Suppose that $T$ is invertible. The polar decomposition (7.93) provides us with a unitary operator $S \in \mathcal{L}(V)$ such that $T = S\sqrt{T^*T}$. To see that $S$ is unique, note that $\sqrt{T^*T}$ is invertible, since

$$\operatorname{null}\sqrt{T^*T} = \operatorname{null} T^*T = \operatorname{null} T = \{0\},$$



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where the first equality follows from Exercise 7.C.16 and the second equality follows from 7.64(b). Thus $S$ is uniquely determined by the formula $S = T\big(\sqrt{T^*T}\big)^{-1}$.

Now suppose that $T$ is not invertible. Let $s_1, ..., s_m$ be the positive singular values of $T$, let

$$Tv = s_1\langle v, e_1\rangle f_1 + \cdots + s_m\langle v, e_m\rangle f_m$$

be a singular value decomposition of $T$, and extend $e_1, ..., e_m$ and $f_1, ..., f_m$ to orthonormal bases $e_1, ..., e_n$ and $f_1, ..., f_n$ of $V$. Because $T$ is not invertible, it must be the case that $m$ is strictly less than $n$. Define $R, S \in \mathcal{L}(V)$ by

$$Re_k = Se_k = f_k \text{ for } k < n, \quad Re_n = f_n, \quad \text{and} \quad Se_n = -f_n.$$

As in the proof of the polar decomposition (7.93), $R$ and $S$ are both unitary operators and both satisfy $T = R\sqrt{T^*T} = S\sqrt{T^*T}$, however $R \neq S$.

**Exercise 7.F.27.** Suppose $T \in \mathcal{L}(V)$ and $s_1, ..., s_n$ are the singular values of $T$. Let $e_1, ..., e_n$ and $f_1, ..., f_n$ be orthonormal bases of $V$ such that

$$Tv = s_1\langle v, e_1\rangle f_1 + \cdots + s_n\langle v, e_n\rangle f_n$$

for all $v \in V$. Define $S \in \mathcal{L}(V)$ by

$$Sv = \langle v, e_1\rangle f_1 + \cdots + \langle v, e_n\rangle f_n.$$

(a) Show that $S$ is unitary and $\|T - S\| = \max\{|s_1 - 1|, ..., |s_n - 1|\}$.

(b) Show that if $E \in \mathcal{L}(V)$ is unitary, then $\|T - E\| \geq \|T - S\|$.

*This exercise finds a unitary operator $S$ that is as close as possible (among the unitary operators) to a given operator $T$.*

**Solution.**

(a) $S$ was shown to be unitary in the proof of the polar decomposition (7.93). Suppose that $\max\{|s_1 - 1|, ..., |s_n - 1|\} = |s_j - 1|$, where $j \in \{1, ..., n\}$. For any $v \in V$, observe that

$$\|(T - S)v\|^2 = |s_1 - 1|^2|\langle v, e_1\rangle|^2 + \cdots + |s_n - 1|^2|\langle v, e_n\rangle|^2 \leq |s_j - 1|^2\|v\|^2.$$

Thus $\|T - S\| \leq |s_j - 1|$. Since $\|(T - S)e_j\| = |s_j - 1|$, we must have $\|T - S\| = |s_j - 1|$.

(b) Let $k \in \{1, ..., n\}$ be given and note that, since $E$ is unitary, we must have $\|Ee_k\| = \|e_k\| = 1$. It follows from the reverse triangle inequality (see Exercise 6.A.20) that

$$\|T - E\| \geq \|(T - E)e_k\| \geq |\|Te_k\| - \|Ee_k\|| = |\|s_k f_k\| - 1| = |s_k - 1|.$$

Thus $\|T - E\| \geq \max\{|s_1 - 1|, ..., |s_n - 1|\} = \|T - S\|$.

**Exercise 7.F.28.** Suppose $T \in \mathcal{L}(V)$. Prove that there exists a unitary operator $S \in \mathcal{L}(V)$ such that $T = \sqrt{TT^*}S$.



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**Solution.** Let $s_1, ..., s_m$ be the positive singular values of $T$, let

$$Tv = s_1\langle v, e_1\rangle f_1 + \cdots + s_m\langle v, e_m\rangle f_m$$

be a singular value decomposition of $T$, and extend $e_1, ..., e_m$ and $f_1, ..., f_m$ to orthonormal bases $e_1, ..., e_n$ and $f_1, ..., f_n$ of $V$. Define the unitary operator $S \in \mathcal{L}(V)$ as in the proof of the polar decomposition (7.93). Using the formula for $TT^*$ from Exercise 7.E.8 (e), we see that

$$\sqrt{TT^*}v = s_1\langle v, f_1\rangle f_1 + \cdots + s_m\langle v, f_m\rangle f_m$$

for every $v \in V$. For any $k \in \{1, ..., m\}$ and any $v \in V$, notice that $\langle Sv, f_k\rangle = \langle v, e_k\rangle$. It follows that

$$\sqrt{TT^*}Sv = s_1\langle Sv, f_1\rangle f_1 + \cdots + s_m\langle Sv, f_m\rangle f_m = s_1\langle v, e_1\rangle f_1 + \cdots + s_m\langle v, e_m\rangle f_m = Tv.$$

Thus $S$ is a unitary operator satisfying $T = \sqrt{TT^*}S$.

**Exercise 7.F.29.** Suppose $T \in \mathcal{L}(V)$.

(a) Use the polar decomposition to show that there exists a unitary operator $S \in \mathcal{L}(V)$ such that $TT^* = ST^*TS^*$.

(b) Show how (a) implies that $T$ and $T^*$ have the same singular values.

**Solution.**

(a) The polar decomposition (7.93) gives us a unitary operator $S \in \mathcal{L}(V)$ satisfying $T = S\sqrt{T^*T}$; as we showed in Exercise 7.F.28, $S$ also satisfies $T = \sqrt{TT^*}S$. It follows that

$$\sqrt{TT^*}S = S\sqrt{T^*T} \quad \Rightarrow \quad \sqrt{TT^*} = S\sqrt{T^*T}S^*$$
$$\Rightarrow \quad TT^* = S\sqrt{T^*T}S^*S\sqrt{T^*T}S^* = ST^*TS^*.$$

(b) The identity $TT^* = ST^*TS^*$ and Exercise 5.A.13 show that the list of eigenvalues of $T^*T$ is the same as the list of eigenvalues of $TT^*$. It follows that the list of singular values of $T$ is the same as the list of singular values of $T^*$.

**Exercise 7.F.30.** Suppose $T \in \mathcal{L}(V), S \in \mathcal{L}(V)$ is a unitary operator, and $R \in \mathcal{L}(V)$ is a positive operator such that $T = SR$. Prove that $R = \sqrt{T^*T}$.

*This exercise shows that if we write $T$ as the product of a unitary operator and a positive operator (as in the polar decomposition 7.93), then the positive operator equals $\sqrt{T^*T}$.*

**Solution.** Observe that

$$T^*T = (SR)^*SR = R^*S^*SR = R^2,$$



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where we have used that $R$ is self-adjoint and that $S$ is unitary. Thus $R$ is a positive square root of $T^*T$; by uniqueness (7.39) we must have $R = \sqrt{T^*T}$.

**Exercise 7.F.31.** Suppose $\mathbf{F} = \mathbf{C}$ and $T \in \mathcal{L}(V)$ is normal. Prove that there exists a unitary operator $S \in \mathcal{L}(V)$ such that $T = S\sqrt{T^*T}$ and such that $S$ and $\sqrt{T^*T}$ both have diagonal matrices with respect to the same orthonormal basis of $V$.

**Solution.** As the polar decomposition (7.93) and Exercise 7.F.28 show, there exists a unitary operator $S \in \mathcal{L}(V)$ such that

$$T = S\sqrt{T^*T} = \sqrt{TT^*}S = \sqrt{T^*T}S,$$

where we have used that $T$ is normal for the last equality. Thus $S$ and $\sqrt{T^*T}$ commute, which by Exercise 7.B.16 is equivalent to the existence of an orthonormal basis of $V$ with respect to which $S$ and $\sqrt{T^*T}$ both have diagonal matrices.

**Exercise 7.F.32.** Suppose that $T \in \mathcal{L}(V, W)$ and $T \neq 0$. Let $s_1, ..., s_m$ denote the positive singular values of $T$. Show that there exists an orthonormal basis $e_1, ..., e_m$ of $(\operatorname{null} T)^\perp$ such that

$$T\left(E\left(\frac{e_1}{s_1}, ..., \frac{e_m}{s_m}\right)\right)$$

equals the ball in $\operatorname{range} T$ of radius 1 centered at 0.

**Solution.** Let $B$ be the ball in $\operatorname{range} T$ of radius 1 centered at 0 and let

$$Tv = s_1\langle v, e_1\rangle f_1 + \cdots + s_m\langle v, e_m\rangle f_m$$

be a singular value decomposition of $T$; as we showed in Exercise 7.E.8 (b), $e_1, ..., e_m$ is an orthonormal basis of $(\operatorname{null} T)^\perp$. Note that

$$Tv \in B \quad \Leftrightarrow \quad \|Tv\|^2 = s_1^2|\langle v, e_1\rangle|^2 + \cdots + s_m^2|\langle v, e_m\rangle|^2 \leq 1.$$

Note further that

$$Tv \in T\left(E\left(\frac{e_1}{s_1}, ..., \frac{e_m}{s_m}\right)\right) \quad \Leftrightarrow \quad v \in E\left(\frac{e_1}{s_1}, ..., \frac{e_m}{s_m}\right)$$

$$\Leftrightarrow \quad s_1^2|\langle v, e_1\rangle|^2 + \cdots + s_m^2|\langle v, e_m\rangle|^2 \leq 1.$$

Thus $T(E(s_1^{-1}e_1, ..., s_m^{-1}e_m)) = B$.



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# Chapter 8. Operators on Complex Vector Spaces

## 8.A. Generalized Eigenvectors and Nilpotent Operators

**Exercise 8.A.1.** Suppose $T \in \mathcal{L}(V)$. Prove that if $\dim \operatorname{null} T^4 = 8$ and $\dim \operatorname{null} T^6 = 9$, then $\dim \operatorname{null} T^m = 9$ for all integers $m \geq 5$.

**Solution.** By 8.1 we have $\dim \operatorname{null} T^5 \in \{8, 9\}$. Notice that

$$\dim \operatorname{null} T^5 = 8 \quad \Rightarrow \quad \operatorname{null} T^4 = \operatorname{null} T^5 \quad \Rightarrow \quad \operatorname{null} T^5 = \operatorname{null} T^6 \quad \Rightarrow \quad \dim \operatorname{null} T^6 = 8,$$

where we have used 8.2 for the second implication. Since $\dim \operatorname{null} T^6 = 9$ it must then be the case that $\dim \operatorname{null} T^5 = 9$, whence $\operatorname{null} T^5 = \operatorname{null} T^6$. It follows from 8.2 that, for all integers $m \geq 5$,

$$\operatorname{null} T^m = \operatorname{null} T^5 \quad \Rightarrow \quad \dim \operatorname{null} T^m = \dim \operatorname{null} T^5 = 9.$$

**Exercise 8.A.2.** Suppose $T \in \mathcal{L}(V), m$ is a positive integer, $v \in V$, and $T^{m-1}v \neq 0$ but $T^m v = 0$. Prove that $v, Tv, T^2v, ..., T^{m-1}v$ is linearly independent.

*The result in this exercise is used in the proof of 8.45.*

**Solution.** Suppose $a_0, ..., a_{m-1}$ are scalars such that

$$a_0 v + a_1 Tv + \cdots + a_{m-1} T^{m-1} v = 0.$$

Apply $T^{m-1}$ to both sides of this equation to obtain $a_0 T^{m-1} v = 0$. Since $T^{m-1}v \neq 0$, it must be the case that $a_0 = 0$. Thus we have the equation

$$a_1 Tv + \cdots + a_{m-1} T^{m-1} v = 0.$$

Now apply $T^{m-2}$ to both sides of this equation to obtain $a_1 T^{m-1} v = 0$, which implies $a_1 = 0$. By continuing in this manner, we see that each of the scalars $a_0, ..., a_{m-1}$ is zero. Thus $v, Tv, ..., T^{m-1}v$ is linearly independent.

**Exercise 8.A.3.** Suppose $T \in \mathcal{L}(V)$. Prove that

$$V = \operatorname{null} T \oplus \operatorname{range} T \quad \Leftrightarrow \quad \operatorname{null} T^2 = \operatorname{null} T.$$

**Solution.** By Exercise 5.D.4 it will suffice to show that

$$\operatorname{null} T \cap \operatorname{range} T = \{0\} \quad \Leftrightarrow \quad \operatorname{null} T^2 = \operatorname{null} T.$$

First suppose that $\operatorname{null} T \cap \operatorname{range} T = \{0\}$. To show that $\operatorname{null} T^2 = \operatorname{null} T$, it will suffice to show that $\operatorname{null} T^2 \subseteq \operatorname{null} T$ (by 8.1). Suppose therefore that $v \in \operatorname{null} T^2$, so that $T^2 v = 0$, and



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notice that $Tv \in \operatorname{null} T \cap \operatorname{range} T = \{0\}$. Hence $Tv = 0$, i.e. $v \in \operatorname{null} T$. Thus $\operatorname{null} T^2 \subseteq \operatorname{null} T$, as desired.

Now suppose that $\operatorname{null} T^2 = \operatorname{null} T$ and suppose that $v \in \operatorname{null} T \cap \operatorname{range} T$, so that $Tv = 0$ and $v = Tu$ for some $u \in V$. It follows that

$$T^2u = Tv = 0 \quad \Rightarrow \quad u \in \operatorname{null} T^2 \quad \Rightarrow \quad u \in \operatorname{null} T \quad \Rightarrow \quad Tu = v = 0.$$

Thus $\operatorname{null} T \cap \operatorname{range} T = \{0\}$, as desired.

**Exercise 8.A.4.** Suppose $T \in \mathcal{L}(V), \lambda \in \mathbf{F}$, and $m$ is a positive integer such that the minimal polynomial of $T$ is a polynomial multiple of $(z - \lambda)^m$. Prove that

$$\dim \operatorname{null} (T - \lambda I)^m \geq m.$$

**Solution.** Let $p$ be the minimal polynomial of $T$; by assumption we have $p(z) = (z - \lambda)^m q(z)$ for some polynomial $q$. Consider the chain of inclusions provided by 8.1:

$$\{0\} \subseteq \operatorname{null} (T - \lambda I)^1 \subseteq \cdots \subseteq \operatorname{null} (T - \lambda I)^m.$$

Notice that it will suffice to prove that each of these $m$ inclusions is strict, since this will imply that the dimension increases by at least 1 at each inclusion. Suppose therefore, by way of contradiction, that there is some inclusion in this chain which is not strict, i.e. there exists some $k \in \{0, \ldots, m-1\}$ such that

$$\operatorname{null} (T - \lambda I)^k = \operatorname{null} (T - \lambda I)^{k+1} \quad \Rightarrow \quad \operatorname{null} (T - \lambda I)^k = \operatorname{null} (T - \lambda I)^m \quad \text{(by 8.2)}.$$

Define $r(z) = (z - \lambda)^k q(z)$ and note that $\deg r < \deg p$. For any $v \in V$ observe that, since $0 = p(T)v = (T - \lambda I)^m q(T)v$,

$$q(T)v \in \operatorname{null} (T - \lambda I)^m \quad \Rightarrow \quad q(T)v \in \operatorname{null} (T - \lambda I)^k \quad \Rightarrow \quad r(T)v = 0.$$

Thus $r$ is a polynomial of lesser degree than $p$ which annihilates $T$, contradicting that $p$ is the minimal polynomial of $T$.

**Exercise 8.A.5.** Suppose $T \in \mathcal{L}(V)$ and $m$ is a positive integer. Prove that

$$\dim \operatorname{null} T^m \leq m \dim \operatorname{null} T.$$

*Hint: Exercise 21 in Section 3B may be useful.*

**Solution.** Exercise 3.B.22 shows that

$$\dim \operatorname{null} T^m \leq \dim \operatorname{null} T^{m-1} + \dim \operatorname{null} T \leq \dim \operatorname{null} T^{m-2} + 2 \dim \operatorname{null} T$$
$$\leq \cdots \leq m \dim \operatorname{null} T.$$



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**Exercise 8.A.6.** Suppose $T \in \mathcal{L}(V)$. Show that

$$V = \operatorname{range} T^0 \supseteq \operatorname{range} T^1 \supseteq \cdots \supseteq \operatorname{range} T^k \supseteq \operatorname{range} T^{k+1} \supseteq \cdots.$$

**Solution.** Suppose $k$ is a non-negative integer and $w \in \operatorname{range} T^{k+1}$, so that there is some $v \in V$ such that

$$w = T^{k+1}v = T^k(Tv) \in \operatorname{range} T^k.$$

Thus $\operatorname{range} T^{k+1} \subseteq \operatorname{range} T^k$.

**Exercise 8.A.7.** Suppose $T \in \mathcal{L}(V)$ and $m$ is a nonnegative integer such that

$$\operatorname{range} T^m = \operatorname{range} T^{m+1}.$$

Prove that $\operatorname{range} T^k = \operatorname{range} T^m$ for all $k > m$.

**Solution.** It will suffice to prove that $\operatorname{range} T^{m+\ell} = \operatorname{range} T^{m+\ell+1}$ for all positive integers $\ell$. By Exercise 8.A.6, this is equivalent to showing that $\operatorname{range} T^{m+\ell} \subseteq \operatorname{range} T^{m+\ell+1}$ for all positive integers $\ell$. Suppose therefore that $\ell$ is a positive integer and that $w \in \operatorname{range} T^{m+\ell}$, so that $w = T^{m+\ell}v$ for some $v \in V$. It follows that

$$T^m v \in \operatorname{range} T^m \quad \Rightarrow \quad T^m v \in \operatorname{range} T^{m+1} \quad \Rightarrow \quad T^m v = T^{m+1}u \text{ for some } u \in V.$$

Thus $w = T^{m+\ell+1}u \in \operatorname{range} T^{m+\ell+1}$. Hence $\operatorname{range} T^{m+\ell} \subseteq \operatorname{range} T^{m+\ell+1}$, as desired.

**Exercise 8.A.8.** Suppose $T \in \mathcal{L}(V)$. Prove that

$$\operatorname{range} T^{\dim V} = \operatorname{range} T^{\dim V+1} = \operatorname{range} T^{\dim V+2} = \cdots.$$

**Solution.** The proof is similar to 8.3. By Exercise 8.A.7, we need only prove that

$$\operatorname{range} T^{\dim V} = \operatorname{range} T^{\dim V+1}.$$

Seeking a contradiction, suppose that this is not true. It then follows from Exercise 8.A.6 and Exercise 8.A.7 that

$$V = \operatorname{range} T^0 \supsetneq \operatorname{range} T^1 \supsetneq \cdots \supsetneq \operatorname{range} T^{\dim V} \supsetneq \operatorname{range} T^{\dim V+1}.$$

At each of the strict inclusions in this chain, the dimension decreases by at least 1. Thus

$$\dim \operatorname{range} T^{\dim V+1} \leq -1,$$

which is a contradiction since the dimension of a vector space must be a non-negative integer.

**Exercise 8.A.9.** Suppose $T \in \mathcal{L}(V)$ and $m$ is a nonnegative integer. Prove that

$$\operatorname{null} T^m = \operatorname{null} T^{m+1} \quad \Leftrightarrow \quad \operatorname{range} T^m = \operatorname{range} T^{m+1}.$$



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**Solution.** Since $\operatorname{null} T^m$ is a subspace of $\operatorname{null} T^{m+1}$ and $\operatorname{range} T^{m+1}$ is a subspace of $\operatorname{range} T^m$ (see 8.1 and Exercise 8.A.6), it will suffice to show that

$$\dim \operatorname{null} T^m = \dim \operatorname{null} T^{m+1} \quad \Leftrightarrow \quad \dim \operatorname{range} T^m = \dim \operatorname{range} T^{m+1}.$$

Indeed, by the fundamental theorem of linear maps (3.21), we have

$$\dim \operatorname{null} T^{m+1} - \dim \operatorname{null} T^m = \dim \operatorname{range} T^m - \dim \operatorname{range} T^{m+1}.$$

**Exercise 8.A.10.** Define $T \in \mathcal{L}(\mathbf{C}^2)$ by $T(w, z) = (z, 0)$. Find all generalized eigenvectors of $T$.

**Solution.** Observe that the matrix of $T$ with respect to the standard basis of $\mathbf{C}^2$ is

$$\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}.$$

It follows that the only eigenvalue of $T$ is 0 and that $T^2 = 0$, so that every non-zero $v \in \mathbf{C}^2$ is a generalized eigenvector of $T$ corresponding to the eigenvalue 0.

**Exercise 8.A.11.** Suppose that $T \in \mathcal{L}(V)$. Prove that there is a basis of $V$ consisting of generalized eigenvectors of $T$ if and only if the minimal polynomial of $T$ equals $(z - \lambda_1) \cdots (z - \lambda_m)$ for some $\lambda_1, \ldots, \lambda_m \in \mathbf{F}$.

*Assume $\mathbf{F} = \mathbf{R}$ because the case $\mathbf{F} = \mathbf{C}$ follows from 5.27(b) and 8.9.*

*This exercise states that the condition for there to be a basis of $V$ consisting of generalized eigenvectors of $T$ is the same as the condition for there to be a basis with respect to which $T$ has an upper-triangular matrix (see 5.44).*

***Caution:** If $T$ has an upper-triangular matrix with respect to a basis $v_1, \ldots, v_n$ of $V$, then $v_1$ is an eigenvector of $T$ but it is not necessarily true that $v_2, \ldots, v_n$ are generalized eigenvectors of $T$.*

**Solution.** Suppose that there is a basis $v_1, \ldots, v_m$ of $V$ consisting of generalized eigenvectors of $T$, i.e. there exist scalars $\lambda_1, \ldots, \lambda_m \in \mathbf{F}$ and positive integers $k_1, \ldots, k_m$ such that

$$(T - \lambda_n I)^{k_n} v_n = 0$$

for each $n \in \{1, \ldots, m\}$. Let $p$ be the polynomial given by

$$p(z) = (z - \lambda_1)^{k_1} \cdots (z - \lambda_m)^{k_m}$$

and notice that $p$ is a product of linear factors. Notice further that $p(T)$ annihilates each of the basis vectors $v_1, \ldots, v_m$, so that $p(T) = 0$. It follows from 5.29 that $p$ is a polynomial multiple of the minimal polynomial of $T$, from which it follows that the minimal polynomial of $T$ is a product of linear factors.



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Conversely, if the minimal polynomial of $T$ splits into linear factors, then notice that $T$ has an eigenvalue. The proof of 8.9 then shows that there is a basis of $V$ consisting of generalized eigenvectors of $T$.

**Exercise 8.A.12.** Suppose $T \in \mathcal{L}(V)$ is such that every vector in $V$ is a generalized eigenvector of $T$. Prove that there exists $\lambda \in \mathbf{F}$ such that $T - \lambda I$ is nilpotent.

**Solution.** First let us prove the following lemma.

**Lemma L.15.** If $T \in \mathcal{L}(V)$ is such that $u, v$, and $u + v$ are generalized eigenvectors of $T$ then $u$ and $v$ correspond to the same eigenvalue.

*Proof.* Suppose that $u, v$, and $u + v$ correspond to the eigenvalues $\alpha, \beta$, and $\gamma$ respectively. Since the list $u, v, u + v$ is linearly dependent, 8.12 implies that the eigenvalues $\alpha, \beta$, and $\gamma$ cannot be distinct, i.e. at least two of these eigenvalues are equal. If $\alpha = \beta$ then we are done, so suppose that $\beta = \gamma$; the case where $\alpha = \gamma$ is handled similarly. Observe that

$$\beta = \gamma \quad \Rightarrow \quad u + v, v \in \operatorname{null}(T - \beta I)^{\dim V} \quad \Rightarrow \quad u \in \operatorname{null}(T - \beta I)^{\dim V}.$$

Thus $u$ corresponds to both $\alpha$ and $\beta$; it follows from 8.11 that $\alpha = \beta$. □

Returning to the exercise, note that the desired result is clear if $V = \{0\}$. Suppose therefore that $n := \dim V \geq 1$ and fix a non-zero $u \in V$. By assumption $u$ is a generalized eigenvector of $T$ and thus corresponds to some eigenvalue $\lambda \in \mathbf{F}$, i.e. $(T - \lambda I)^n u = 0$. Let $v \in V$ be non-zero. If $u + v = 0$ then

$$(T - \lambda I)^n v = -(T - \lambda I)^n u = 0,$$

and if $u + v \neq 0$ then, by assumption, $v$ and $u + v$ are generalized eigenvectors of $T$. It follows from Lemma L.15 that $u$ and $v$ correspond to the same eigenvalue, so that $(T - \lambda I)^n v = 0$. Thus $(T - \lambda I)^n = 0$, i.e. $T - \lambda I$ is nilpotent.

**Exercise 8.A.13.** Suppose $S, T \in \mathcal{L}(V)$ and $ST$ is nilpotent. Prove that $TS$ is nilpotent.

**Solution.** There is an integer $k$ such that $(ST)^k = 0$, which implies

$$(TS)^{k+1} = T(ST)^k S = 0.$$

Thus $TS$ is nilpotent.

**Exercise 8.A.14.** Suppose $T \in \mathcal{L}(V)$ is nilpotent and $T \neq 0$. Prove $T$ is not diagonalizable.



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**Solution.** 8.18 shows that the minimal polynomial of $T$ is $z^m$ for some positive integer $m$; since $T \neq 0$ we must have $m \geq 2$. It follows from 5.62 that $T$ is not diagonalizable.

**Exercise 8.A.15.** Suppose $\mathbf{F} = \mathbf{C}$ and $T \in \mathcal{L}(V)$. Prove that $T$ is diagonalizable if and only if every generalized eigenvector of $T$ is an eigenvector of $T$.

*For $\mathbf{F} = \mathbf{C}$, this exercise adds another equivalence to the list of conditions for diagonalizability in 5.55.*

**Solution.** If every generalized eigenvector of $T$ is an eigenvector of $T$, then the basis of $V$ consisting of generalized eigenvectors of $T$ provided by 8.9 is in fact a basis of $V$ consisting of eigenvectors of $T$. That is, $T$ is diagonalizable.

Now suppose that there is some $v \in V$ such that $v$ is a generalized eigenvector of $T$ but $v$ is not an eigenvector of $T$, i.e. there exists $\lambda \in \mathbf{F}$ (we need not require $\mathbf{F} = \mathbf{C}$ for this implication) such that

$$(T - \lambda I)^m v = 0 \quad \text{and} \quad (T - \lambda I)v \neq 0$$

for some integer $m \geq 2$. Let $p$ be the minimal polynomial of $T$ and let $p_v$ be the unique monic polynomial of smallest degree satisfying $p_v(T)v = 0$ (see Exercise 5.C.7.) A small modification of 5.29 shows that $p_v$ must divide $(z - \lambda)^m$, so that $p_v = (z - \lambda)^k$ for some non-negative integer $k$. Since $v = (T - \lambda I)^0 v$ and $(T - \lambda I)^1 v$ are both non-zero, it must be the case that $k \geq 2$. It then follows from Exercise 5.C.7 (b) that the minimal polynomial of $T$ has $(z - \lambda)^k$ as a factor, where $k \geq 2$. We may use 5.62 to conclude that $T$ is not diagonalizable.

**Exercise 8.A.16.**

(a) Give an example of nilpotent operators $S, T$ on the same vector space such that neither $S + T$ nor $ST$ is nilpotent.

(b) Suppose $S, T \in \mathcal{L}(V)$ are nilpotent and $ST = TS$. Prove that $S + T$ and $ST$ are nilpotent.

**Solution.**

(a) Let $S, T \in \mathcal{L}(\mathbf{F}^2)$ be the operators whose matrices with respect to the standard basis of $\mathbf{F}^2$ are

$$\mathcal{M}(S) = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} \quad \text{and} \quad \mathcal{M}(T) = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}.$$

Some straightforward calculations show that $S^2 = T^2 = 0$, so that $S$ and $T$ are nilpotent, and that $(S + T)^2 \neq 0$ and $(ST)^2 \neq 0$, so that neither $S + T$ nor $ST$ is nilpotent (by the contrapositive of 8.16).



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(b) By 8.16 we have $S^n = T^n = 0$, where $n = \dim V$. Using that $S$ and $T$ commute, we may write $(ST)^n = S^n T^n = 0$. Thus $ST$ is nilpotent. Furthermore, since $S$ and $T$ commute, we may apply the binomial theorem:

$$(S+T)^{2n} = \sum_{k=0}^{2n} \binom{2n}{k} S^{2n-k} T^k.$$

Notice that $S^{2n-k} = 0$ for $0 \le k \le n$ and $T^k = 0$ for $n \le k \le 2n$. It follows that each term in the sum above is zero, so that $(S+T)^{2n} = 0$. Thus $S+T$ is nilpotent.

**Exercise 8.A.17.** Suppose $T \in \mathcal{L}(V)$ is nilpotent and $m$ is a positive integer such that $T^m = 0$.

(a) Prove that $I - T$ is invertible and that $(I-T)^{-1} = I + T + \cdots + T^{m-1}$.

(b) Explain how you would guess the formula above.

**Solution.**

(a) A calculation shows that

$$(I-T)(I+T+\cdots+T^{m-1}) = I - T^m = I.$$

(b) We might guess this formula by analogy with the formula

$$(1-z)(1+z+\cdots+z^{m-1}) = 1 - z^m$$

for $z \in \mathbf{F}$.

**Exercise 8.A.18.** Suppose $T \in \mathcal{L}(V)$ is nilpotent. Prove that $T^{1+\dim \operatorname{range} T} = 0$.

*If* $\dim \operatorname{range} T < \dim V - 1$*, then this exercise improves 8.16.*

**Solution.** Let $\dim \operatorname{range} T = k$ and, seeking a contradiction, suppose that $T^{k+1} \neq 0$. By 8.16 we have $T^{\dim V} = 0$ and thus we can let $m$ be the smallest integer such that $T^m = 0$ and $T^{m-1} \neq 0$; note that $k+2 \le m \le n$. Let $v \in V$ be such that $T^{m-1}v \neq 0$ and $T^m v = 0$. It then follows from Exercise 8.A.2 that the list

$$v, Tv, ..., T^{m-1}v$$

is linearly independent, from which it follows that $k \ge m - 1 \ge k + 1$—a contradiction.

**Exercise 8.A.19.** Suppose $T \in \mathcal{L}(V)$ is not nilpotent. Show that

$$V = \operatorname{null} T^{\dim V - 1} \oplus \operatorname{range} T^{\dim V - 1}.$$

*For operators that are not nilpotent, this exercise improves 8.4.*

**Solution.** Observe that

$$T \text{ is not nilpotent} \quad \Rightarrow \quad \operatorname{null} T^{\dim V - 1} = \operatorname{null} T^{\dim V}$$



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$$\Rightarrow \quad \operatorname{null} T^{2(\dim V-1)} = \operatorname{null} T^{\dim V-1} \quad \Rightarrow \quad V = \operatorname{null} T^{\dim V-1} \oplus \operatorname{range} T^{\dim V-1},$$

where we have used Exercise 8.A.21 for the first implication, 8.3 for the second implication, and Exercise 8.A.3 for the third implication.

**Exercise 8.A.20.** Suppose $V$ is an inner product space and $T \in \mathcal{L}(V)$ is normal and nilpotent. Prove that $T = 0$.

**Solution.** Observe that

$$T^{\dim V} = 0 \quad \Leftrightarrow \quad \operatorname{null} T^{\dim V} = V \quad \Leftrightarrow \quad \operatorname{null} T = V \quad \Leftrightarrow \quad T = 0,$$

where we have used Exercise 7.A.27 for the second equivalence.

**Exercise 8.A.21.** Suppose $T \in \mathcal{L}(V)$ is such that $\operatorname{null} T^{\dim V-1} \neq \operatorname{null} T^{\dim V}$. Prove that $T$ is nilpotent and that $\dim \operatorname{null} T^k = k$ for every integer $k$ with $0 \leq k \leq \dim V$.

**Solution.** Consider the chain of inclusions provided by 8.1:

$$\{0\} = \operatorname{null} T^0 \subseteq \operatorname{null} T^1 \subseteq \cdots \subseteq \operatorname{null} T^{\dim V-1} \subseteq \operatorname{null} T^{\dim V}.$$

Since $\operatorname{null} T^{\dim V-1} \neq \operatorname{null} T^{\dim V}$, 8.2 shows that each of these inclusions must be strict. Note that if the dimension increased by more than 1 at some inclusion in this chain then we would have $\operatorname{null} T^{\dim V} > \dim V$, which cannot happen. Hence it must be the case that the dimension increases by exactly 1 at each inclusion in this chain, whence $\dim \operatorname{null} T^k = k$ for every integer $k \in \{0, \ldots, \dim V\}$. It follows from this that $T^{\dim V} = 0$, so that $T$ is nilpotent.

**Exercise 8.A.22.** Suppose $T \in \mathcal{L}(\mathbf{C}^5)$ is such that $\operatorname{range} T^4 \neq \operatorname{range} T^5$. Prove that $T$ is nilpotent.

**Solution.** Exercise 8.A.9 shows that $\operatorname{null} T^4 \neq \operatorname{null} T^5$. It then follows from Exercise 8.A.21 that $T$ is nilpotent.

**Exercise 8.A.23.** Give an example of an operator $T$ on a finite-dimensional real vector space such that 0 is the only eigenvalue of $T$ but $T$ is not nilpotent.

*This exercise shows that the implication (b) $\Rightarrow$ (a) in 8.17 does not hold without the hypothesis that* $\mathbf{F} = \mathbf{C}$.

**Solution.** Let $T \in \mathcal{L}(\mathbf{R}^3)$ be the operator whose matrix with respect to the standard basis of $\mathbf{R}^3$ is

$$\mathcal{M}(T) = \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$



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A straightforward calculation reveals that 0 is the only eigenvalue of $T$ and that $T^3 \neq 0$; it follows from 8.16 that $T$ is not nilpotent.

**Exercise 8.A.24.** For each item in Example 8.15, find a basis of the domain vector space such that the matrix of the nilpotent operator with respect to that basis has the upper-triangular form promised by 8.18(c).

**Solution.**

(a) If $e_1, e_2, e_3, e_4$ is the standard basis of $\mathbf{F}^4$ then observe that the matrix of $T$ with respect to the basis $e_3, e_4, e_1, e_2$ is

$$\begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}.$$

(b) Let $v_1 = \left(1, \frac{1}{3}, \frac{2}{3}\right), v_2 = \left(\frac{1}{6}, \frac{1}{6}, 0\right)$, and $v_3 = \left(0, \frac{1}{54}, 0\right)$. Routine calculations show that $v_1, v_2, v_3$ is a basis of $\mathbf{F}^3$ and that the matrix of the nilpotent operator in part (b) with respect to this basis is

$$\begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}.$$

(c) Observe that the matrix of the differentiation operator on $\mathcal{P}_m(\mathbf{R})$ with respect to the basis $1, x, ..., x^m$ is

$$\begin{pmatrix} 0 & 1 & 0 & \cdots & 0 & 0 \\ 0 & 0 & 2 & \cdots & 0 & 0 \\ 0 & 0 & 0 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & 0 & m \\ 0 & 0 & 0 & \cdots & 0 & 0 \end{pmatrix}.$$

**Exercise 8.A.25.** Suppose that $V$ is an inner product space and $T \in \mathcal{L}(V)$ is nilpotent. Show that there is an orthonormal basis of $V$ with respect to which the matrix of $T$ has the upper-triangular form promised by 8.18(c).

**Solution.** Combining 8.18(b) with 6.37, we see that $T$ has an upper-triangular matrix with respect to some orthonormal basis of $V$, and combining 5.41 with 8.17(a) shows that the diagonal entries of this matrix are zero.



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## 8.B. Generalized Eigenspace Decomposition

**Exercise 8.B.1.** Define $T \in \mathcal{L}(\mathbf{C}^2)$ by $T(w,z) = (-z,w)$. Find the generalized eigenspaces corresponding to the distinct eigenvalues of $T$.

**Solution.** As example 5.9 shows, $T$ has the two distinct eigenvalues $\pm i$ with corresponding eigenspaces

$$E(-i,T) = \operatorname{span}((1,i)) \quad \text{and} \quad E(i,T) = \operatorname{span}((1,-i)).$$

Since $\dim \mathbf{C}^2 = 2$, 8.12 shows that there can be no other generalized eigenvectors linearly independent from the two above. Thus

$$G(-i,T) = E(-i,T) = \operatorname{span}((1,i)) \quad \text{and} \quad G(i,T) = E(i,T) = \operatorname{span}((1,-i)).$$

**Exercise 8.B.2.** Suppose $T \in \mathcal{L}(V)$ is invertible. Prove that $G(\lambda,T) = G\left(\frac{1}{\lambda}, T^{-1}\right)$ for every $\lambda \in \mathbf{F}$ with $\lambda \neq 0$.

**Solution.** By 8.20 it will suffice to show that

$$\operatorname{null}(T-\lambda I)^n = \operatorname{null}(T^{-1}-\lambda^{-1}I)^n,$$

where $n = \dim V$. Suppose therefore that $v \in \operatorname{null}(T-\lambda I)^n$, i.e.

$$0 = (T-\lambda I)^n v = \sum_{k=0}^{n} (-1)^k \lambda^k \binom{n}{k} T^{n-k} v.$$

Applying the operator $(-1)^{-n}\lambda^{-n}T^{-n}$ to both sides of this equation and using that $(-1)^{k-n} = (-1)^{n-k}$ for any $k \in \{0,\dots,n\}$, we find that

$$0 = \sum_{k=0}^{n} (-1)^{n-k} (\lambda^{-1})^{n-k} \binom{n}{k} T^{-k} v = (T^{-1}-\lambda^{-1}I)^n v.$$

Thus $v \in \operatorname{null}(T^{-1}-\lambda^{-1}I)^n$ and it follows that $\operatorname{null}(T-\lambda I)^n \subseteq \operatorname{null}(T^{-1}-\lambda^{-1}I)^n$. Replacing $T$ with $T^{-1}$ and $\lambda$ with $\lambda^{-1}$ in this inclusion gives us the desired result.

**Exercise 8.B.3.** Suppose $T \in \mathcal{L}(V)$. Suppose $S \in \mathcal{L}(V)$ is invertible. Prove that $T$ and $S^{-1}TS$ have the same eigenvalues with the same multiplicities.

**Solution.** We showed in Exercise 5.A.13 (a) that $T$ and $S^{-1}TS$ have the same eigenvalues. Suppose that $\lambda \in \mathbf{F}$ is an eigenvalue of $T$ and $S^{-1}TS$. Using the identity $p(T) = p(S^{-1}TS)$ for a polynomial $p$, observe that

$$S^{-1}(T-\lambda I)^n S = (S^{-1}TS - \lambda I)^n,$$

where $\dim V = n$. It now follows from Exercise 3.D.8 that



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$$\dim \operatorname{null} (T - \lambda I)^n = \dim \operatorname{null} (S^{-1}TS - \lambda I)^n,$$

i.e. the multiplicity of $\lambda$ as an eigenvalue of $T$ equals the multiplicity of $\lambda$ as an eigenvalue of $S^{-1}TS$.

**Exercise 8.B.4.** Suppose $\dim V \geq 2$ and $T \in \mathcal{L}(V)$ is such that

$$\operatorname{null} T^{\dim V - 2} \neq \operatorname{null} T^{\dim V - 1}.$$

Prove that $T$ has at most two distinct eigenvalues.

**Solution.** Let $\dim V = n \geq 2$ and note that our hypothesis implies $\dim \operatorname{null} T^{n-1} \geq n - 1$. There are then two cases.

**Case 1.** If $\dim \operatorname{null} T^{n-1} = n$ then $T^{n-1} = 0$ and it follows from 8.17(a) that 0 is the only eigenvalue of $T$.

**Case 2.** Suppose that $\dim \operatorname{null} T^{n-1} = n - 1$, so that we can find $n - 1$ linearly independent generalized eigenvectors of $T$ corresponding to the eigenvalue 0. It follows from 8.12 that $T$ can have at most one non-zero eigenvalue (otherwise we would have at least $n + 1$ linearly independent vectors in a vector space of dimension $n$).

In either case, $T$ has at most two distinct eigenvalues.

**Exercise 8.B.5.** Suppose $T \in \mathcal{L}(V)$ and 3 and 8 are eigenvalues of $T$. Let $n = \dim V$. Prove that $V = (\operatorname{null} T^{n-2}) \oplus (\operatorname{range} T^{n-2})$.

**Solution.** If 0 is not an eigenvalue of $T$, i.e. $T$ is injective, then $\operatorname{null} T^{n-2} = \{0\}$ and the desired result follows from Exercise 5.D.4.

If 0 is an eigenvalue of $T$ then $T$ has at least three distinct eigenvalues and it follows from Exercise 8.B.4 and 8.2 that

$$\operatorname{null} T^{n-2} = \operatorname{null} T^{n-1} = \operatorname{null} T^{2(n-2)}.$$

Thus, by Exercise 8.A.3, $V = (\operatorname{null} T^{n-2}) \oplus (\operatorname{range} T^{n-2})$.

**Exercise 8.B.6.** Suppose $T \in \mathcal{L}(V)$ and $\lambda$ is an eigenvalue of $T$. Explain why the exponent of $z - \lambda$ in the factorization of the minimal polynomial of $T$ is the smallest positive integer $m$ such that $(T - \lambda I)^m|_{G(\lambda,T)} = 0$.

**Solution.** Let $p$ be the minimal polynomial of $T$ and let $m$ be the exponent of $z - \lambda$ in the factorization of $p$, i.e. $p(z) = (z - \lambda)^m q(z)$ for some polynomial $q$ with $q(\lambda) \neq 0$.

We claim that $\operatorname{null} (T - \lambda I)^m = \operatorname{null} (T - \lambda I)^{m+1}$. By 8.1 it will suffice to prove the inclusion $\operatorname{null} (T - \lambda I)^{m+1} \subseteq \operatorname{null} (T - \lambda I)^m$, so suppose that $v \in \operatorname{null} (T - \lambda I)^{m+1}$, define $w = (T - \lambda I)^m v$, and notice that $Tw = \lambda w$. Now observe that



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$$0 = p(T)v = q(T)(T-\lambda I)^m v = q(T)w = q(\lambda)w,$$

where we have used that $Tw = \lambda w$ for the last equality. Since $q(\lambda) \neq 0$ it must be the case that $w = 0$, i.e. $v \in \operatorname{null}(T-\lambda I)^m$. Thus $\operatorname{null}(T-\lambda I)^{m+1} \subseteq \operatorname{null}(T-\lambda I)^m$, as desired.

It now follows from 8.2 that $\operatorname{null}(T-\lambda I)^m = \operatorname{null}(T-\lambda I)^{\dim V} = G(\lambda, T)$, from which it is clear that $(T-\lambda I)^m|_{G(\lambda,T)} = 0$. To see that $m$ is minimal, let $k$ be any positive integer such that $(T-\lambda I)^k|_{G(\lambda,T)} = 0$ and let $s(z) = (z-\lambda)^k q(z)$. The equation $(T-\lambda I)^k|_{G(\lambda,T)} = 0$ implies that

$$\operatorname{null}(T-\lambda I)^k = G(\lambda, T) = \operatorname{null}(T-\lambda I)^m.$$

Let $v \in V$ be given. Since $0 = p(T)v = (T-\lambda I)^m q(T)v$, we either have $q(T)v = 0$, in which case $s(T)v = 0$, or

$$q(T)v \in \operatorname{null}(T-\lambda I)^m = \operatorname{null}(T-\lambda I)^k,$$

in which case $s(T)v = 0$ also. Thus $s(T) = 0$ and the minimality of the degree of $p$ implies that $\deg p \leq \deg s$, from which it follows that $m \leq k$.

**Exercise 8.B.7.** Suppose $T \in \mathcal{L}(V)$ and $\lambda$ is an eigenvalue of $T$ with multiplicity $d$. Prove that $G(\lambda, T) = \operatorname{null}(T-\lambda I)^d$.

*If $d < \dim V$, then this exercise improves 8.20.*

**Solution.** Let $p$ be the minimal polynomial of $T$ and let $m$ be the positive integer such that $p(z) = (z-\lambda)^m q(z)$ with $q(\lambda) \neq 0$. As we showed in Exercise 8.B.6,

$$G(\lambda, T) = \operatorname{null}(T-\lambda I)^m = \operatorname{null}(T-\lambda I)^{m+1}.$$

It then follows from Exercise 8.A.4 and 8.2 that

$$d = \dim G(\lambda, T) = \dim \operatorname{null}(T-\lambda I)^m \geq m$$
$$\Rightarrow \quad \operatorname{null}(T-\lambda I)^d = \operatorname{null}(T-\lambda I)^m = G(\lambda, T).$$

**Exercise 8.B.8.** Suppose $T \in \mathcal{L}(V)$ and $\lambda_1, \ldots, \lambda_m$ are the distinct eigenvalues of $T$. Prove that

$$V = G(\lambda_1, T) \oplus \cdots \oplus G(\lambda_m, T)$$

if and only if the minimal polynomial of $T$ equals $(z-\lambda_1)^{k_1} \cdots (z-\lambda_m)^{k_m}$ for some positive integers $k_1, \ldots, k_m$.

*The case $\mathbf{F} = \mathbf{C}$ follows immediately from 5.27(b) and the generalized eigenspace decomposition (8.22); thus this exercise is interesting only when $\mathbf{F} = \mathbf{R}$.*

**Solution.** Suppose that

$$V = G(\lambda_1, T) \oplus \cdots \oplus G(\lambda_m, T).$$



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For each $i \in \{1, ..., m\}$ let $d_i = \dim G(\lambda_i, T)$ and let $q(z) = (z - \lambda_1)^{d_1} \cdots (z - \lambda_m)^{d_m}$. For any $v \in V$ we have $v = v_1 + \cdots + v_m$, where each $v_i \in G(\lambda_i, T)$. Fix $i \in \{1, ..., m\}$ and note that $G(\lambda_i, T) = \operatorname{null}(T - \lambda_i I)^{d_i}$ by Exercise 8.B.7; it follows that

$$q(T)v_i = \left(\prod_{j \neq i} (T - \lambda_j I)^{d_j}\right)(T - \lambda_i I)^{d_i} v_i = 0.$$

Thus $q(T)v = 0$, whence $q(T) = 0$. It now follows from 5.29 that the minimal polynomial of $T$ is a factor of $q$ and hence must be of the form $(z - \lambda_1)^{k_1} \cdots (z - \lambda_m)^{k_m}$ for some non-negative integers $k_1, ..., k_m$. In fact each $k_i$ must be positive since each $\lambda_i$ is an eigenvalue of $T$ and the eigenvalues of $T$ are precisely the zeros of the minimal polynomial (by 5.27(a)).

Now suppose that the minimal polynomial of $T$ equals $(z - \lambda_1)^{k_1} \cdots (z - \lambda_m)^{k_m}$ for some positive integers $k_1, ..., k_m$. The proof of 8.22(c) shows that the sum

$$G(\lambda_1, T) \oplus \cdots \oplus G(\lambda_m, T)$$

is direct and Exercise 8.A.11 shows that any vector in $V$ can be expressed as a linear combination of generalized eigenvectors of $T$. Thus

$$V = G(\lambda_1, T) \oplus \cdots \oplus G(\lambda_m, T).$$

**Exercise 8.B.9.** Suppose $\mathbf{F} = \mathbf{C}$ and $T \in \mathcal{L}(V)$. Prove that there exist $D, N \in \mathcal{L}(V)$ such that $T = D + N$, the operator $D$ is diagonalizable, $N$ is nilpotent, and $DN = ND$.

**Solution.** Let $\lambda_1, ..., \lambda_m$ be the distinct eigenvalues of $T$ and let

$$V = G(\lambda_1, T) \oplus \cdots \oplus G(\lambda_m, T)$$

be the generalized eigenspace decomposition of $V$. For any $v \in V$ we have $v = v_1 + \cdots + v_m$, where each $v_k \in G(\lambda_k, T)$. Define $D \in \mathcal{L}(V)$ by $Dv_k = \lambda_k v_k$, so that $D|_{G(\lambda_k, T)} = \lambda_k I$, and let $N = T - D$. Certainly $D$ is diagonalizable. Furthermore, for any $k \in \{1, ..., m\}$,

$$N|_{G(\lambda_k, T)} = (T - D)|_{G(\lambda_k, T)} = (T - \lambda_k I)|_{G(\lambda_k, T)}$$

is nilpotent by 8.22(b). It follows that

$$N^{\dim V} v = N^{\dim V} v_1 + \cdots + N^{\dim V} v_m = 0$$

for any $v = v_1 + \cdots + v_m \in V = G(\lambda_1, T) \oplus \cdots \oplus G(\lambda_m, T)$. Thus $N$ is nilpotent. Now, since $DN = DT - D^2$ and $ND = TD - D^2$, to show that $D$ and $N$ commute it will suffice to show that $D$ and $T$ commute. Indeed, for any $k \in \{1, ..., m\}$ and $v_k \in G(\lambda_k, T)$,

$$TDv_k = \lambda_k T v_k = DTv_k,$$

where we have used that $D|_{G(\lambda_k, T)} = \lambda_k I$ and that $G(\lambda_k, T)$ is invariant under $T$ (by 8.22(a)). It follows that $TDv = DTv$ for any $v = v_1 + \cdots + v_m \in V = G(\lambda_1, T) \oplus \cdots \oplus G(\lambda_m, T)$. Thus $T$ and $D$ commute.



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**Exercise 8.B.10.** Suppose $V$ is a complex inner product space, $e_1, ..., e_n$ is an orthonormal basis of $T$, and $T \in \mathcal{L}(V)$. Let $\lambda_1, ..., \lambda_n$ be the eigenvalues of $T$, each included as many times as its multiplicity. Prove that

$$|\lambda_1|^2 + \cdots + |\lambda_n|^2 \le \|Te_1\|^2 + \cdots + \|Te_n\|^2.$$

*See the comment after Exercise 5 in Section 7A.*

**Solution.** Let $\lambda \in \mathbf{C}$ be an eigenvalue of $T$, let $d = \dim G(\lambda, T)$, and consider the restriction operator $R \coloneqq T|_{G(\lambda,T)}$. Certainly $\lambda$ is an eigenvalue of $R$ and $R$ has no other eigenvalues. By Schur's theorem (6.38) there is an orthonormal basis $f_1, ..., f_d$ of $G(\lambda, T)$ with respect to which the matrix $A \coloneqq \mathcal{M}(R)$ is upper-triangular. For any $k \in \{1, ..., d\}$ it then follows that $|\lambda|^2 = \|\lambda f_k\|^2 \le \|Tf_k\|^2$, whence

$$d|\lambda|^2 \le \|Tf_1\|^2 + \cdots + \|Tf_d\|^2.$$

Summing this inequality over the finitely many distinct eigenvalues of $T$ shows that

$$|\lambda_1|^2 + \cdots + |\lambda_n|^2 \le \|Tg_1\|^2 + \cdots + \|Tg_n\|^2,$$

where $g_1, ..., g_n$ is the orthonormal basis of $V$ obtained by combining the orthonormal bases of the generalized eigenspaces of $T$ found in the previous discussion (the generalized eigenspace decomposition (8.22) shows that this provides an orthonormal basis for all of $V$). As we showed in Exercise 7.A.5, the quantity $\|Te_1\|^2 + \cdots + \|Te_n\|^2$ does not depend on which orthonormal basis of $V$ is used and thus

$$|\lambda_1|^2 + \cdots + |\lambda_n|^2 \le \|Te_1\|^2 + \cdots + \|Te_n\|^2.$$

**Exercise 8.B.11.** Give an example of an operator on $\mathbf{C}^4$ whose characteristic polynomial equals $(z-7)^2(z-8)^2$.

**Solution.** Let $T \in \mathcal{L}(\mathbf{C}^4)$ be the operator whose matrix with respect to the standard basis $e_1, e_2, e_3, e_4$ of $\mathbf{C}^4$ is

$$\begin{pmatrix} 7 & 0 & 0 & 0 \\ 0 & 7 & 0 & 0 \\ 0 & 0 & 8 & 0 \\ 0 & 0 & 0 & 8 \end{pmatrix}.$$

A straightforward calculation shows that 7 and 8 are the only eigenvalues of $T$ and that

$$E(7, T) = G(7, T) = \operatorname{span}(e_1, e_2) \quad \text{and} \quad E(8, T) = G(8, T) = \operatorname{span}(e_3, e_4).$$

Thus the multiplicities of the eigenvalues 7 and 8 both equal 2, from which it follows that the characteristic polynomial of $T$ is $(z-7)^2(z-8)^2$.



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**Exercise 8.B.12.** Give an example of an operator on $\mathbf{C}^4$ whose characteristic polynomial equals $(z-1)(z-5)^3$ and whose minimal polynomial equals $(z-1)(z-5)^2$.

**Solution.** Let $T \in \mathcal{L}(\mathbf{C}^4)$ be the operator whose matrix with respect to the standard basis $e_1, e_2, e_3, e_4$ of $\mathbf{C}^4$ is

$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 5 & 1 & 0 \\ 0 & 0 & 5 & 0 \\ 0 & 0 & 0 & 5 \end{pmatrix}.$$

A straightforward calculation shows that 1 and 5 are the only eigenvalues of $T$ and that

$$E(1,T) = G(1,T) = \operatorname{span}(e_1) \quad \text{and} \quad G(5,T) = \operatorname{span}(e_2, e_3, e_4).$$

Thus the multiplicity of the eigenvalue 1 equals 1 and the multiplicity of the eigenvalue 5 equals 3, from which it follows that the characteristic polynomial of $T$ is $(z-1)(z-5)^3$. Another calculation shows that $(T-I)(T-5I) \neq 0$ and that $(T-I)(T-5I)^2 = 0$. Thus the minimal polynomial of $T$ is $(z-1)(z-5)^2$.

**Exercise 8.B.13.** Give an example of an operator on $\mathbf{C}^4$ whose characteristic and minimal polynomials both equal $z(z-1)^2(z-3)$.

**Solution.** Let $T \in \mathcal{L}(\mathbf{C}^4)$ be the operator whose matrix with respect to the standard basis $e_1, e_2, e_3, e_4$ of $\mathbf{C}^4$ is

$$\begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 3 \end{pmatrix}.$$

A straightforward calculation shows that $0, 1$, and $3$ are the only eigenvalues of $T$ and that

$$E(0,T) = G(0,T) = \operatorname{span}(e_1), \quad E(3,T) = G(3,T) = \operatorname{span}(e_4),$$
$$\text{and} \quad G(1,T) = \operatorname{span}(e_2, e_3).$$

Thus the multiplicities of the eigenvalues 0 and 3 both equal 1 and the multiplicity of the eigenvalue 1 equals 2, from which it follows that the characteristic polynomial of $T$ is $z(z-1)^2(z-3)$. Another calculation shows that

$$T(T-I)(T-3I) \neq 0 \quad \text{and} \quad T(T-I)^2(T-3I) = 0.$$

Thus the minimal polynomial of $T$ is $z(z-1)^2(z-3)$.

**Exercise 8.B.14.** Give an example of an operator on $\mathbf{C}^4$ whose characteristic polynomial equals $z(z-1)^2(z-3)$ and whose minimal polynomial equals $z(z-1)(z-3)$.



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**Solution.** Let $T \in \mathcal{L}(\mathbf{C}^4)$ be the operator whose matrix with respect to the standard basis $e_1, e_2, e_3, e_4$ of $\mathbf{C}^4$ is

$$\begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 3 \end{pmatrix}.$$

A straightforward calculation shows that $0, 1$, and $3$ are the only eigenvalues of $T$ and that

$$E(0,T) = G(0,T) = \operatorname{span}(e_1), \quad E(3,T) = G(3,T) = \operatorname{span}(e_4),$$
$$\text{and} \quad E(1,T) = G(1,T) = \operatorname{span}(e_2, e_3).$$

Thus the multiplicities of the eigenvalues 0 and 3 both equal 1 and the multiplicity of the eigenvalue 1 equals 2, from which it follows that the characteristic polynomial of $T$ is $z(z-1)^2(z-3)$. Another calculation shows that $T(T-I)(T-3I) = 0$. Thus the minimal polynomial of $T$ is $z(z-1)(z-3)$.

**Exercise 8.B.15.** Let $T$ be the operator on $\mathbf{C}^4$ defined by

$$T(z_1, z_2, z_3, z_4) = (0, z_1, z_2, z_3).$$

Find the characteristic polynomial and the minimal polynomial of $T$.

**Solution.** Let $e_1, e_2, e_3, e_4$ be the standard basis of $\mathbf{C}^4$. The matrix of $T$ with respect to the basis $e_4, e_3, e_2, e_1$ is

$$\begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix}.$$

Thus 0 is the only eigenvalue of $T$ and it then follows from 8.28 that the characteristic polynomial of $T$ is $z^4$. Since $T^3 \neq 0$ it must be the case that the minimal polynomial of $T$ is also $z^4$.

**Exercise 8.B.16.** Let $T$ be the operator on $\mathbf{C}^6$ defined by

$$T(z_1, z_2, z_3, z_4, z_5, z_6) = (0, z_1, z_2, 0, z_4, 0).$$

Find the characteristic polynomial and the minimal polynomial of $T$.

**Solution.** Let $e_1, e_2, e_3, e_4, e_5, e_6$ be the standard basis of $\mathbf{C}^4$. The matrix of $T$ with respect to the basis $e_3, e_2, e_1, e_5, e_4, e_6$ is



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$$\begin{pmatrix} 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}.$$

Thus 0 is the only eigenvalue of $T$ and it then follows from 8.28 that the characteristic polynomial of $T$ is $z^6$. A straightforward calculation shows that $T^2 \neq 0$ and $T^3 = 0$. Thus the minimal polynomial of $T$ is $z^3$.

**Exercise 8.B.17.** Suppose $\mathbf{F} = \mathbf{C}$ and $P \in \mathcal{L}(V)$ is such that $P^2 = P$. Prove that the characteristic polynomial of $P$ is $z^m(z-1)^n$, where $m = \dim \operatorname{null} P$ and $n = \dim \operatorname{range} P$.

**Solution.** Exercise 5.A.8 shows that the only possible eigenvalues of $P$ are 0 and 1. It follows from 8.28 that the characteristic polynomial of $P$ is of the form $z^\ell(z-1)^k$, where

$$\ell = \dim G(0, P) \quad \text{and} \quad k = \dim V - \ell.$$

Since $\operatorname{null} P^2 = \operatorname{null} P$, 8.2 shows that

$$G(0, P) = \operatorname{null} P^{\dim V} = \operatorname{null} P \quad \Rightarrow \quad \ell = \dim G(0, P) = \dim \operatorname{null} P = m.$$

It now follows from the fundamental theorem of linear maps that

$$k = \dim V - \ell = \dim V - \dim \operatorname{null} P = \dim \operatorname{range} T = n.$$

**Exercise 8.B.18.** Suppose $T \in \mathcal{L}(V)$ and $\lambda$ is an eigenvalue of $T$. Explain why the following four numbers equal each other.

(a) The exponent of $z - \lambda$ in the factorization of the minimal polynomial of $T$.

(b) The smallest positive integer $m$ such that $(T - \lambda I)^m|_{G(\lambda, T)} = 0$.

(c) The smallest positive integer $m$ such that

$$\operatorname{null} (T - \lambda I)^m = \operatorname{null} (T - \lambda I)^{m+1}.$$

(d) The smallest positive integer $m$ such that

$$\operatorname{range} (T - \lambda I)^m = \operatorname{range} (T - \lambda I)^{m+1}.$$

**Solution.** We showed that (a) and (b) are equal in Exercise 8.B.6, and a very small modification of that argument shows that (a) and (c) are also equal. Finally, the fact that (c) and (d) are equal follows from Exercise 8.A.9.

**Exercise 8.B.19.** Suppose $\mathbf{F} = \mathbf{C}$ and $S \in \mathcal{L}(V)$ is a unitary operator. Prove that the constant term in the characteristic polynomial of $S$ has absolute value 1.



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**Solution.** Suppose $(z-\lambda_1)^{d_1}\cdots(z-\lambda_m)^{d_m}$ is the characteristic polynomial of $S$ and note that the constant term of this polynomial is $\pm\lambda_1^{d_1}\cdots\lambda_m^{d_m}$. Note further that $|\lambda_k|=1$ for each $k\in\{1,...,m\}$ by 7.54. It follows that

$$|\pm\lambda_1^{d_1}\cdots\lambda_m^{d_m}| = |\lambda_1|^{d_1}\cdots|\lambda_m|^{d_m} = 1.$$

**Exercise 8.B.20.** Suppose that $\mathbf{F}=\mathbf{C}$ and $V_1,...,V_m$ are nonzero subspaces of $V$ such that

$$V = V_1\oplus\cdots\oplus V_m.$$

Suppose $T\in\mathcal{L}(V)$ and each $V_k$ is invariant under $T$. For each $k$, let $p_k$ denote the characteristic polynomial of $T|_{V_k}$. Prove that the characteristic polynomial of $T$ equals $p_1\cdots p_m$.

**Solution.** It will suffice to prove the case where $m=2$; a straightforward induction argument will then prove the general statement. Suppose therefore that $V=U\oplus W$, where $U$ and $W$ are non-zero subspaces of $V$ invariant under $T$.

Let $E(T)$ be the collection of eigenvalues of $T$ and define $E(T|_U)$ and $E(T|_W)$ similarly. We claim that $E(T)=E(T|_U)\cup E(T|_W)$. Certainly any eigenvalue of $T|_U$ or of $T|_W$ must also be an eigenvalue of $T$, so that $E(T|_U)\cup E(T|_W)\subseteq E(T)$. For the reverse inclusion, suppose that $\lambda\in\mathbf{C}$ is an eigenvalue of $T$, so that $Tv=\lambda v$ for some non-zero $v\in V$. Let $u\in U$ and $w\in W$ be such that $v=u+w$ and observe that

$$Tu-\lambda u = \lambda w - Tw.$$

Since $U$ and $W$ are invariant under $T$ we have $Tu-\lambda u\in U$ and $\lambda w-Tw\in W$, whence

$$Tu-\lambda u=\lambda w-Tw\in U\cap W=\{0\}.$$

Thus $Tu=\lambda u$ and $Tw=\lambda w$. Note that at least one of $u$ and $w$ must be non-zero, since $v$ is non-zero. It follows that $\lambda\in E(T|_U)\cup E(T|_W)$ and we may conclude that

$$E(T)=E(T|_U)\cup E(T|_W),$$

as claimed.

Next we claim that $G(\lambda,T)=G(\lambda,T|_U)\oplus G(\lambda,T|_W)$ for any $\lambda\in\mathbf{C}$. First, note that

$$G(\lambda,T|_U)\subseteq U \quad\text{and}\quad G(\lambda,T|_W)\subseteq W.$$

Since $U\cap W=\{0\}$, it follows that $G(\lambda,T|_U)\cap G(\lambda,T|_W)=\{0\}$. Thus the sum

$$G(\lambda,T|_U)\oplus G(\lambda,T|_W)$$

is indeed direct.



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Now suppose that

$$u + w \in G(\lambda, T|_U) \oplus G(\lambda, T|_W),$$

$$\text{i.e.} \quad u \in U \text{ and } (T|_U - \lambda I)^{\dim U} u = 0, \quad w \in W \text{ and } (T|_W - \lambda I)^{\dim W} w = 0.$$

It follows that $(T - \lambda I)^{\dim V} u = (T - \lambda I)^{\dim V} w = 0$ and thus $u + w \in G(\lambda, T)$. Hence

$$G(\lambda, T|_U) \oplus G(\lambda, T|_W) \subseteq G(\lambda, T).$$

Now suppose that $v \in G(\lambda, T)$, i.e. $(T - \lambda I)^{\dim V} v = 0$, and let $u \in U$ and $w \in W$ be such that $v = u + w$. Observe that

$$(T - \lambda I)^{\dim V}(u + w) = 0 \quad \Leftrightarrow \quad (T - \lambda I)^{\dim V} u = -(T - \lambda I)^{\dim V} w.$$

Since $U$ and $W$ are invariant under $T$ we have $(T - \lambda I)^{\dim V} u \in U$ and $(T - \lambda I)^{\dim V} w \in W$, whence

$$(T - \lambda I)^{\dim V} u = -(T - \lambda I)^{\dim V} w \in U \cap W = \{0\}.$$

Thus

$$(T - \lambda I)^{\dim V} u = 0 \quad \Rightarrow \quad (T|_U - \lambda I)^{\dim V} u = 0 \quad \Rightarrow \quad (T|_U - \lambda I)^{\dim U} u = 0,$$

where we have used 8.3 for the last implication. It follows that $u \in G(\lambda, T|_U)$ and we can similarly show that $w \in G(\lambda, T|_W)$. Hence $v = u + w \in G(\lambda, T|_U) \oplus G(\lambda, T|_W)$ and we may conclude that

$$G(\lambda, T) = G(\lambda, T|_U) \oplus G(\lambda, T|_W),$$

as claimed.

We have now proved the following:

(i) $E(T) = E(T|_U) \cup E(T|_W)$;

(ii) $G(\lambda, T) = G(\lambda, T|_U) \oplus G(\lambda, T|_W)$ for any $\lambda \in \mathbf{C}$.

Let $\lambda_1, ..., \lambda_n$ denote the distinct eigenvalues of $T$, let

$$m_i = \dim G(\lambda_i, T), \quad k_i = \dim G(\lambda_i, T|_U), \quad \text{and} \quad \ell_i = \dim G(\lambda_i, T|_W),$$

and let $r, p$, and $q$ be the characteristic polynomials of $T, T|_U$, and $T|_W$ respectively; by definition we have $r(z) = (z - \lambda_1)^{m_1} \cdots (z - \lambda_n)^{m_n}$. It follows from (i) that

$$p(z) = (z - \lambda_1)^{k_1} \cdots (z - \lambda_n)^{k_n} \quad \text{and} \quad q(z) = (z - \lambda_1)^{\ell_1} \cdots (z - \lambda_n)^{\ell_n},$$

and it follows from (ii) that $k_i + \ell_i = m_i$ for each $i \in \{1, ..., n\}$. Thus

$$p(z)q(z) = (z - \lambda_1)^{k_1 + \ell_1} \cdots (z - \lambda_n)^{k_n + \ell_n} = (z - \lambda_1)^{m_1} \cdots (z - \lambda_n)^{m_n} = r(z),$$

as desired.



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**Exercise 8.B.21.** Suppose $p, q \in \mathcal{P}(\mathbf{C})$ are monic polynomials with the same zeros and $q$ is a polynomial multiple of $p$. Prove that there exists $T \in \mathcal{L}\big(\mathbf{C}^{\deg q}\big)$ such that the characteristic polynomial of $T$ is $q$ and the minimal polynomial of $T$ is $p$.

*This exercise implies that every monic polynomial is the characteristic polynomial of some operator.*

**Solution.** Let $\lambda_1, \dots, \lambda_m$ be the distinct zeros of $p$ and $q$ and suppose that

$$p(z) = (z - \lambda_1)^{k_1} \cdots (z - \lambda_m)^{k_m} \quad \text{and} \quad q(z) = (z - \lambda_1)^{\ell_1} \cdots (z - \lambda_m)^{\ell_m}$$

for some positive integers $k_1, \dots, k_m, \ell_1, \dots, \ell_m$; note that $k_i \le \ell_i$ for each $i \in \{1, \dots, m\}$ since $q$ is a polynomial multiple of $p$. Fix $i \in \{1, \dots, m\}$. If $k_i = 1$ then let $A_i$ be the $\ell_i \times \ell_i$ diagonal matrix with diagonal entries equal to $\lambda_i$. If $k_i > 1$ and $k_i = \ell_i$ then let $A_i$ be the $\ell_i \times \ell_i$ matrix with diagonal entries equal to $\lambda_i$ and entries directly above the diagonal equal to 1, i.e.

$$A_i = \begin{pmatrix} \lambda_i & 1 & 0 & \cdots & 0 & 0 \\ 0 & \lambda_i & 1 & \cdots & 0 & 0 \\ 0 & 0 & \lambda_i & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & \lambda_i & 1 \\ 0 & 0 & 0 & \cdots & 0 & \lambda_i \end{pmatrix}.$$

If $1 < k_i < \ell_i$ then let $B_i$ be the $(\ell_i - k_i) \times (\ell_i - k_i)$ diagonal matrix with diagonal entries equal to $\lambda_i$, let $C_i$ be the $k_i \times k_i$ matrix with diagonal entries equal to $\lambda_i$ and entries directly above the diagonal equal to 1, and let $A_i$ be the block diagonal matrix

$$\begin{pmatrix} B_i & 0 \\ 0 & C_i \end{pmatrix}.$$

Now let $T \in \mathcal{L}(\mathbf{C}^n)$, where $n = \deg q$, be the operator whose matrix with respect to the standard basis is the block diagonal matrix

$$A := \begin{pmatrix} A_1 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & A_m \end{pmatrix}.$$

It follows that the distinct eigenvalues of $T$ are $\lambda_1, \dots, \lambda_m$, since these are precisely the distinct diagonal elements of $A$. Fix $i \in \{1, \dots, m\}$. For any positive integer $d$, a calculation (see Exercise 8.B.22) shows that



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$$(A-\lambda_i I_n)^d = \begin{pmatrix} (A_1-\lambda_i I_{\ell_1})^d & \cdots & 0 & \cdots & 0 \\ \vdots & \ddots & \vdots & \ddots & \vdots \\ 0 & \cdots & (A_i-\lambda_i I_{\ell_i})^d & \cdots & 0 \\ \vdots & \ddots & \vdots & \ddots & \vdots \\ 0 & \cdots & 0 & \cdots & (A_m-\lambda_i I_{\ell_m})^n \end{pmatrix},$$

where $I_n$ is the $n \times n$ identity matrix and $I_{\ell_j}$ is the $\ell_j \times \ell_j$ identity matrix. For $j \neq i$, note that the upper-triangular matrix $A_j - \lambda_i I_{\ell_j}$ has non-zero diagonal entries since $\lambda_j \neq \lambda_i$; it follows that the upper-triangular matrix $(A_j - \lambda_i I_{\ell_j})^d$ also has non-zero diagonal entries and hence is injective. Another calculation shows that $(A_i - \lambda_i I_{\ell_i})^{\ell_i}$ is the $\ell_i \times \ell_i$ zero matrix. It follows from this discussion that

$$\dim G(\lambda_i, T) = \dim \operatorname{null} (A-\lambda_i I_n)^n = \dim \operatorname{null} (A_i - \lambda_i I_{\ell_i})^n = \ell_i.$$

Hence $q$ is the characteristic polynomial of $T$. It also follows from this discussion that the matrix of $(T-\lambda_i I)|_{G(\lambda_i,T)}$ with respect to the standard basis is $A_i - \lambda_i I_{\ell_i}$. A final calculation shows that $k_i$ is the least integer $k$ such that $(A_i - \lambda_i I_{\ell_i})^k = 0$ and thus, by Exercise 8.B.18, $k_i$ must be the exponent of $z - \lambda_i$ in the factorization of the minimal polynomial of $T$. Hence $p$ is the minimal polynomial of $T$.

**Exercise 8.B.22.** Suppose $A$ and $B$ are block diagonal matrices of the form

$$A = \begin{pmatrix} A_1 & & 0 \\ & \ddots & \\ 0 & & A_m \end{pmatrix}, \quad B = \begin{pmatrix} B_1 & & 0 \\ & \ddots & \\ 0 & & B_m \end{pmatrix},$$

where $A_k$ and $B_k$ are square matrices of the same size for each $k = 1, ..., m$. Show that $AB$ is a block diagonal matrix of the form

$$AB = \begin{pmatrix} A_1B_1 & & 0 \\ & \ddots & \\ 0 & & A_mB_m \end{pmatrix}.$$

**Solution.** For each $k \in \{1, ..., m\}$ suppose that $A_k$ and $B_k$ are $\ell_k \times \ell_k$ matrices and let $n = \ell_1 + \cdots + \ell_m$, so that $A$ and $B$ are $n \times n$ matrices. Let $S, T \in \mathcal{L}(\mathbf{F}^n)$ be the operators whose matrices with respect to the standard basis $e_1, ..., e_n$ are $\mathcal{M}(S) = A$ and $\mathcal{M}(T) = B$, so that $\mathcal{M}(ST) = AB$. Let $E_1$ be the list $e_1, ..., e_{\ell_1}$ and, if $m \geq 2$, for each $k \in \{2, ..., m\}$ let $E_k$ be the list

$$e_{\ell_1+\cdots+\ell_{k-1}+1}, ..., e_{\ell_1+\cdots+\ell_k}.$$

Now let $U_k = \operatorname{span} E_k$ for each $k \in \{1, ..., m\}$ and note that $E_k$ is a basis of $U_k$. Note further that each $U_k$ is invariant under both $S$ and $T$ and that the matrices of $S|_{U_k}$ and $T|_{U_k}$ with respect to $E_k$ are $A_k$ and $B_k$ respectively. It follows that each $U_k$ is invariant under $ST$ and that the matrix of $(ST)|_{U_k}$ with respect to the basis $E_k$ is $A_kB_k$. Thus the $k^{\text{th}}$ block on the



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diagonal of $AB$ equals $A_kB_k$, and all entries of $AB$ off the "block diagonal" must be zero (otherwise $U_k$ would not be invariant under $ST$).

**Exercise 8.B.23.** Suppose $\mathbf{F} = \mathbf{R}, T \in \mathcal{L}(V)$, and $\lambda \in \mathbf{C}$.

(a) Show that $u + iv \in G(\lambda, T_{\mathbf{C}})$ if and only if $u - iv \in G(\overline{\lambda}, T_{\mathbf{C}})$.

(b) Show that the multiplicity of $\lambda$ as an eigenvalue of $T_{\mathbf{C}}$ equals the multiplicity of $\overline{\lambda}$ as an eigenvalue of $T_{\mathbf{C}}$.

(c) Use (b) and the result about the sum of multiplicities (8.25) to show that if $\dim V$ is an odd number, then $T_{\mathbf{C}}$ has a real eigenvalue.

(d) Use (c) and the result about real eigenvalues of $T_{\mathbf{C}}$ (Exercise 17 in Section 5A) to show that if $\dim V$ is an odd number, then $T$ has an eigenvalue (thus giving an alternative proof of 5.34).

*See Exercise 33 in Section 3B for the definition of the complexification $T_{\mathbf{C}}$.*

**Solution.**

(a) Analogously to complex conjugation, for $u, v \in V$ let us define $\overline{u + iv} = u - iv$. This operation has the properties we would expect of it, such as

$$\overline{(u + iv) + (x + iy)} = \overline{u + iv} + \overline{x + iy} \quad \text{and} \quad \overline{\alpha(u + iv)} = \overline{\alpha}\,\overline{u + iv} \text{ for } \alpha \in \mathbf{C}.$$

Furthermore, observe that

$$\overline{T_{\mathbf{C}}(u + iv)} = \overline{Tu + iTv} = Tu - iTv = T_{\mathbf{C}}(u - iv) = T_{\mathbf{C}}\big(\overline{u + iv}\big).$$

It follows from these algebraic properties that, for a non-negative integer $m$,

$$\overline{(T_{\mathbf{C}} - \lambda I)^m (u + iv)} = \big(T_{\mathbf{C}} - \overline{\lambda} I\big)^m (u - iv).$$

Combining this identity with the obvious equation $\overline{0} = 0$ (where 0 is the zero vector in $V_{\mathbf{C}}$) shows that $u + iv \in G(\lambda, T_{\mathbf{C}})$ if and only if $u - iv \in G(\overline{\lambda}, T_{\mathbf{C}})$.

(b) Let $u_1 + iv_1, \ldots, u_m + iv_m$ be a basis of $G(\lambda, T_{\mathbf{C}})$; we claim that $u_1 - iv_1, \ldots, u_1 - iv_m$ is a basis of $G(\overline{\lambda}, T_{\mathbf{C}})$. Part (a) shows that each $u_k - iv_k$ indeed belongs to $G(\overline{\lambda}, T_{\mathbf{C}})$. Suppose that $\alpha_1, \ldots, \alpha_m \in \mathbf{C}$ are such that

$$\alpha_1(u_1 - iv_1) + \cdots + \alpha_m(u_m - iv_m) = 0.$$

Taking the complex conjugate of both sides (see part (a)) shows that

$$\overline{\alpha_1}(u_1 + iv_1) + \cdots + \overline{\alpha_m}(u_m + iv_m) = 0.$$

The linear independence of the list $u_1 + iv_1, \ldots, u_m + iv_m$ now shows that

$$\overline{\alpha_1} = \cdots = \overline{\alpha_m} = 0 \quad \Rightarrow \quad \alpha_1 = \cdots = \alpha_m = 0.$$



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Thus the list $u_1 - iv_1, ..., u_1 - iv_m$ is linearly independent. Now let $u + iv \in G(\overline{\lambda}, T_{\mathbf{C}})$ be given. By part (a) we have $u - iv \in G(\lambda, T)$ and thus there exist scalars $\alpha_1, ..., \alpha_m \in \mathbf{C}$ such that

$$u - iv = \alpha_1(u_1 + iv_1) + \cdots + \alpha_m(u_m + iv_m).$$

Taking the complex conjugate of both sides shows that

$$u + iv = \overline{\alpha_1}(u - iv_1) + \cdots + \overline{\alpha_m}(u_m - iv_m).$$

Thus $u_1 - iv_1, ..., u_m - iv_m$ spans $G(\overline{\lambda}, T_{\mathbf{C}})$. Hence $u_1 - iv_1, ..., u_m - iv_m$ is a basis of $G(\overline{\lambda}, T_{\mathbf{C}})$, as claimed. It follows that $\dim G(\lambda, T_{\mathbf{C}}) = \dim G(\overline{\lambda}, T_{\mathbf{C}})$, i.e. the multiplicity of $\lambda$ as an eigenvalue of $T_{\mathbf{C}}$ equals the multiplicity of $\overline{\lambda}$ as an eigenvalue of $T_{\mathbf{C}}$.

(c) We will prove the contrapositive. Suppose that $T_{\mathbf{C}}$ has no real eigenvalues. Since non-real eigenvalues of $T_{\mathbf{C}}$ come in pairs (by part (a)) and both eigenvalues of this pair have the same multiplicity (by part (b)), the sum of the multiplicities of all the eigenvalues of $T_{\mathbf{C}}$ must be an even number. Thus, by 8.25, $\dim V$ is an even number.

(d) This is immediate from part (c) and Exercise 5.A.17.



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## 8.C. Consequences of Generalized Eigenspace Decomposition

**Exercise 8.C.1.** Suppose $T \in \mathcal{L}(\mathbf{C}^3)$ is the operator defined by

$$T(z_1, z_2, z_3) = (z_2, z_3, 0).$$

Prove that $T$ does not have a square root.

**Solution.** Notice that $T$ is nilpotent. Thus, if $S \in \mathcal{L}(\mathbf{C}^3)$ satisfies $S^2 = T$, then $S$ must also be nilpotent. It follows from 8.16 that $S^3 = 0$. However, note that

$$0 \neq T^2(0, 0, 1) = S^4(0, 0, 1).$$

Thus there can be no $S \in \mathcal{L}(\mathbf{C}^3)$ satisfying $S^2 = T$.

**Exercise 8.C.2.** Define $T \in \mathcal{L}(\mathbf{F}^5)$ by $T(x_1, x_2, x_3, x_4, x_5) = (2x_2, 3x_3, -x_4, 4x_5, 0)$.

(a) Show that $T$ is nilpotent.

(b) Find a square root of $I + T$.

**Solution.**

(a) Notice that the matrix of $T$ with respect to the standard basis of $\mathbf{F}^5$ is

$$\begin{pmatrix} 0 & 2 & 0 & 0 & 0 \\ 0 & 0 & 3 & 0 & 0 \\ 0 & 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 & 4 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}.$$

It follows from 8.18 that $T$ is nilpotent.

(b) Note that $T^4 \neq 0$ and $T^5 = 0$. Following the strategy outlined in the proof of 8.39, we should attempt to solve the equation

$$\left(I + a_1 T + a_2 T^2 + a_3 T^3 + a_4 T^4\right)^2 = I + T$$

for the coefficients $a_1, a_2, a_3, a_4$. After calculating, we find that

$$a_1 = \frac{1}{2}, \quad a_2 = -\frac{1}{8}, \quad a_3 = \frac{1}{16}, \quad a_4 = -\frac{5}{128}.$$

Thus the operator $\frac{1}{128}(128I + 64T - 16T^2 + 8T^3 - 5T^4)$ is a square root of $I + T$ (a matrix calculation, by hand or otherwise, verifies this).

**Exercise 8.C.3.** Suppose $V$ is a complex vector space. Prove that every invertible operator on $V$ has a cube root.



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**Solution.** The proof is almost identical to the proof of 8.41, replacing "square root" with "cube root" where applicable. The crux of the argument is the existence of cube roots for operators of the form $I+T$, where $T$ is nilpotent. As in the proof of 8.39, proving this existence amounts to showing that we can always solve the equation

$$\left(I + a_1 T + a_2 T^2 + \cdots + a_{m-1} T^{m-1}\right)^3 = I + T$$

for $a_1, a_2, ..., a_{m-1}$, where $m$ is some positive integer. By multiplying out the left-hand side, we notice that the coefficient of $T^k$ is always a degree 1 polynomial in $a_k$ with constant term involving sums and products of $a_1, ..., a_{k-1}$. Thus, having found $a_1, ..., a_{k-1}$, we can always solve for $a_k$.

**Exercise 8.C.4.** Suppose $V$ is a real vector space. Prove that the operator $-I$ on $V$ has a square root if and only if $\dim V$ is an even number.

**Solution.** Suppose that $\dim V$ is an even number and let $v_1, ..., v_{2n}$ be a basis of $V$. Define $R \in \mathcal{L}(V)$ by

$$Re_{2k-1} = -e_{2k} \quad \text{and} \quad Re_{2k} = e_{2k-1}$$

for each $k \in \{1, ..., n\}$. It follows that

$$R^2 e_{2k-1} = -e_{2k-1} \quad \text{and} \quad R^2 e_{2k} = -e_{2k}$$

for each $k \in \{1, ..., n\}$, so that $R^2 = -I$, i.e. $R$ is a square root of $-I$.

Now suppose that $\dim V$ is an odd number and let $T \in \mathcal{L}(V)$ be given. By 5.34, there exists an eigenvalue $\lambda \in \mathbf{R}$ of $T$, say $Tv = \lambda v$ for some non-zero $v \in V$. It follows that $T^2 v = \lambda^2 v$ and hence that $T^2 v \neq -v$, since $\lambda^2 \geq 0$. Thus no operator $T \in \mathcal{L}(V)$ satisfies $T^2 = -I$.

**Exercise 8.C.5.** Suppose $T \in \mathcal{L}(\mathbf{C}^2)$ is the operator defined by

$$T(w, z) = (-w - z, 9w + 5z).$$

Find a Jordan basis for $T$.

**Solution.** The matrix of $T$ with respect to the basis $(1, -3), (1, -2)$ is

$$\begin{pmatrix} 2 & 1 \\ 0 & 2 \end{pmatrix}.$$

Thus $(1, -3), (1, -2)$ is a Jordan basis for $T$.

**Exercise 8.C.6.** Find a basis of $\mathcal{P}_4(\mathbf{R})$ that is a Jordan basis for the differentiation operator $D$ on $\mathcal{P}_4(\mathbf{R})$ defined by $Dp = p'$.

**Solution.** The matrix of $D$ with respect to the basis $1, x, \frac{1}{2}x^2, \frac{1}{3}x^3, \frac{1}{4}x^4$ is



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$$\begin{pmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}.$$

Thus $1, x, \frac{1}{2}x^2, \frac{1}{3}x^3, \frac{1}{4}x^4$ is a Jordan basis for $D$.

**Exercise 8.C.7.** Suppose $T \in \mathcal{L}(V)$ is nilpotent and $v_1, ..., v_n$ is a Jordan basis for $T$. Prove that the minimal polynomial of $T$ is $z^{m+1}$, where $m$ is the length of the longest consecutive string of 1's that appears on the line directly above the diagonal in the matrix of $T$ with respect to $v_1, ..., v_n$.

**Solution.** The matrix of $T$ with respect to $v_1, ..., v_n$ is a block diagonal matrix of the form

$$A = \begin{pmatrix} A_1 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & A_p \end{pmatrix},$$

where each $A_k$ is a $d_k \times d_k$ matrix of the form

$$A_k = \begin{pmatrix} 0 & 1 & & 0 \\ & \ddots & \ddots & \\ & & \ddots & 1 \\ 0 & & & 0 \end{pmatrix};$$

the diagonal entries are zero as $T$, being nilpotent, has only zero as an eigenvalue. A calculation shows that $A_k^{d_k - 1} \neq 0$ and $A_k^{d_k} = 0$. Note that the length of the string of 1's appearing on the line directly above the diagonal of $A_k$ is exactly $d_k - 1$. It follows that $d_k \leq m + 1$ for each $k \in \{1, ..., n\}$, so that $A_k^{m+1} = 0$, and $d_\ell = m + 1$ for some $\ell \in \{1, ..., n\}$, so that $A_\ell^m \neq 0$. Thus, by Exercise 8.A.22,

$$A^{m+1} = \begin{pmatrix} A_1^{m+1} & \cdots & 0 & \cdots & 0 \\ \vdots & \ddots & \vdots & \ddots & \vdots \\ 0 & \cdots & A_\ell^{m+1} & \cdots & 0 \\ \vdots & \ddots & \vdots & \ddots & \vdots \\ 0 & \cdots & 0 & \cdots & A_p^{m+1} \end{pmatrix} = \begin{pmatrix} 0 & \cdots & 0 & \cdots & 0 \\ \vdots & \ddots & \vdots & \ddots & \vdots \\ 0 & \cdots & 0 & \cdots & 0 \\ \vdots & \ddots & \vdots & \ddots & \vdots \\ 0 & \cdots & 0 & \cdots & 0 \end{pmatrix} = 0,$$

$$A^m = \begin{pmatrix} A_1^m & \cdots & 0 & \cdots & 0 \\ \vdots & \ddots & \vdots & \ddots & \vdots \\ 0 & \cdots & A_\ell^m & \cdots & 0 \\ \vdots & \ddots & \vdots & \ddots & \vdots \\ 0 & \cdots & 0 & \cdots & A_p^m \end{pmatrix} \neq 0.$$

Hence $T^{m+1} = 0$ and $T^m \neq 0$. Furthermore, zero is the only eigenvalue of $T$ and hence the only root of the minimal polynomial of $T$. We may conclude that the minimal polynomial of $T$ is $z^{m+1}$.



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**Exercise 8.C.8.** Suppose $T \in \mathcal{L}(V)$ and $v_1, ..., v_n$ is a basis of $V$ that is a Jordan basis for $T$. Describe the matrix of $T^2$ with respect to this basis.

**Solution.** The matrix of $T$ with respect to $v_1, ..., v_n$ is a block diagonal matrix of the form

$$A = \begin{pmatrix} A_1 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & A_p \end{pmatrix}, \quad \text{where } A_k = \begin{pmatrix} \lambda_k & 1 & & 0 \\ & \ddots & \ddots & \\ & & \ddots & 1 \\ 0 & & & \lambda_k \end{pmatrix}.$$

A calculation shows that

$$A_k^2 = \begin{pmatrix} \lambda_k^2 & 2\lambda_k & 1 & & 0 \\ & \ddots & \ddots & \ddots & \\ & & \ddots & \ddots & 1 \\ & & & \ddots & 2\lambda_k \\ 0 & & & & \lambda_k^2 \end{pmatrix}.$$

Thus, by Exercise 8.A.22, the matrix of $T^2$ with respect to $v_1, ..., v_n$ is

$$A^2 = \begin{pmatrix} A_1^2 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & A_p^2 \end{pmatrix}.$$

**Exercise 8.C.9.** Suppose $T \in \mathcal{L}(V)$ is nilpotent. Explain why there exist $v_1, ..., v_n \in V$ and non-negative integers $m_1, ..., m_n$ such that (a) and (b) below both hold.

(a) $T^{m_1}v_1, ..., Tv_1, v_1, ..., T^{m_n}v_n, ..., Tv_n, v_n$ is a basis of $V$.

(b) $T^{m_1+1}v_1 = \cdots = T^{m_n+1}v_n = 0$.

**Solution.** By 8.45 there exists a Jordan basis for $T$, i.e. a basis with respect to which the matrix of $T$ is of the form

$$A = \begin{pmatrix} A_1 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & A_n \end{pmatrix}, \quad \text{where } A_k = \begin{pmatrix} 0 & 1 & & 0 \\ & \ddots & \ddots & \\ & & \ddots & 1 \\ 0 & & & 0 \end{pmatrix};$$

the diagonal entries of each $A_k$ are zero as $T$, being nilpotent, has only zero as an eigenvalue. Fix $k \in \{1, ..., n\}$ and suppose that $A_k$ is an $(m_k + 1) \times (m_k + 1)$ matrix, where $m_k$ is some non-negative integer. Suppose that the sub-list of the Jordan basis corresponding to the $A_k$ block on the block diagonal of $A$ is $u_{m_k+1}, u_{m_k}, ..., u_3, u_2, u_1$. The form of $A_k$ shows that

$$Tu_1 = u_2, \quad T^2u_1 = Tu_2 = u_3, \quad ..., \quad T^{m_k}u_1 = u_{m_k+1}, \quad T^{m_k+1}u_1 = 0.$$

Thus we can take $v_k = u_k$ for each $k \in \{1, ..., n\}$ and (a) and (b) will both hold.



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**Exercise 8.C.10.** Suppose $T \in \mathcal{L}(V)$ and $v_1, ..., v_n$ is a basis of $V$ that is a Jordan basis for $T$. Describe the matrix of $T$ with respect to the basis $v_n, ..., v_1$ obtained by reversing the order of the $v$'s.

**Solution.** The matrix of $T$ with respect to $v_1, ..., v_n$ is a block diagonal matrix of the form

$$A = \begin{pmatrix} A_1 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & A_p \end{pmatrix}, \quad \text{where } A_k = \begin{pmatrix} \lambda_k & 1 & & 0 \\ & \ddots & \ddots & \\ & & \ddots & 1 \\ 0 & & & \lambda_k \end{pmatrix}.$$

Reversing the order of the sub-list of $v_1, ..., v_n$ corresponding to $A_k$ has the effect of transposing $A_k$, and reversing the order of the entire basis $v_1, ..., v_n$ has the effect of reversing the order of the blocks on the block diagonal of $A$. Thus the matrix of $T$ with respect to the basis $v_n, ..., v_1$ is

$$\begin{pmatrix} A_p^{\mathrm{t}} & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & A_1^{\mathrm{t}} \end{pmatrix}.$$

**Exercise 8.C.11.** Suppose $T \in \mathcal{L}(V)$. Explain why every vector in each Jordan basis for $T$ is a generalized eigenvector of $T$.

**Solution.** Suppose that $v_1, ..., v_n$ is a Jordan basis for $T$, i.e. the matrix of $T$ with respect to $v_1, ..., v_n$ is of the form

$$A = \begin{pmatrix} A_1 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & A_p \end{pmatrix}, \quad \text{where } A_k = \begin{pmatrix} \lambda_k & 1 & & 0 \\ & \ddots & \ddots & \\ & & \ddots & 1 \\ 0 & & & \lambda_k \end{pmatrix}.$$

Let $v_m$ be a vector in the Jordan basis $v_1, ..., v_n$. The form of $A$ above shows that there is a $k \in \{1, ..., p\}$ such that either $(T - \lambda_k I)v_m = 0$ or

$$Tv_m = \lambda_k v_m + v_{m-1} \quad \Rightarrow \quad (T - \lambda_k I)v_m = v_{m-1}.$$

Similarly, either $(T - \lambda_k I)v_{m-1} = 0$, which gives us $(T - \lambda_k I)^2 v_m = 0$, or

$$Tv_{m-1} = \lambda_k v_{m-1} + v_{m-2} \quad \Rightarrow \quad (T - \lambda_k I)v_{m-1} = (T - \lambda_k I)^2 v_m = v_{m-2}.$$

Continuing in this fashion, we find a positive integer $\ell$, no greater than the size of $A_k$, such that $(T - \lambda_k I)^\ell v_m = 0$. Thus $v_m$ is a generalized eigenvector of $T$.

**Exercise 8.C.12.** Suppose $T \in \mathcal{L}(V)$ is diagonalizable. Show that $\mathcal{M}(T)$ is a diagonal matrix with respect to every Jordan basis for $T$.



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**Solution.** Let $v_1, ..., v_n$ be a Jordan basis for $T$. It follows from Exercise 8.C.11 that $v_1, ..., v_n$ is a basis of $V$ consisting of generalized eigenvectors of $T$. Because $T$ is diagonalizable, every generalized eigenvector of $T$ is an eigenvector of $T$ (as we showed in Exercise 8.A.15; note that the proof of this implication does not use the hypothesis of Exercise 8.A.15 that $\mathbf{F} = \mathbf{C}$). Thus $v_1, ..., v_n$ is a basis of $V$ consisting of eigenvectors of $T$ and hence the matrix of $T$ with respect to this basis is diagonal.

**Exercise 8.C.13.** Suppose $T \in \mathcal{L}(V)$ is nilpotent. Prove that if $v_1, ..., v_n$ are vectors in $V$ and $m_1, ..., m_n$ are nonnegative integers such that

$$T^{m_1}v_1, ..., Tv_1, v_1, ..., T^{m_n}v_n, ..., Tv_n, v_n \text{ is a basis of } V$$

and

$$T^{m_1+1}v_1 = \cdots = T^{m_n+1}v_n = 0,$$

then $T^{m_1}v_1, ..., T^{m_n}v_n$ is a basis of $\operatorname{null} T$.

*This exercise shows that $n = \dim \operatorname{null} T$. Thus the positive integer $n$ that appears above depends only on $T$ and not on the specific Jordan basis chosen for $T$.*

**Solution.** The linear independence of the basis

$$T^{m_1}v_1, ..., Tv_1, v_1, ..., T^{m_n}v_n, ..., Tv_n, v_n$$

gives us the linear independence of the list $T^{m_1}v_1, ..., T^{m_n}v_n$. The condition

$$T^{m_1+1}v_1 = \cdots = T^{m_n+1}v_n = 0$$

shows that each of the vectors $T^{m_1}v_1, ..., T^{m_n}v_n$ belongs to $\operatorname{null} T$. Suppose that $v \in \operatorname{null} T$. There are scalars $a_{j,k}$ such that

$$v = \sum_{j=1}^{n}\sum_{k=0}^{m_j} a_{j,k}T^k v_j \quad \Rightarrow \quad 0 = Tv = \sum_{j=1}^{n}\sum_{k=0}^{m_j} a_{j,k}T^{k+1}v_j = \sum_{\substack{j=1 \\ m_j \neq 0}}^{n}\sum_{k=0}^{m_j-1} a_{j,k}T^{k+1}v_j,$$

where we have used that $T^{m_1+1}v_1 = \cdots = T^{m_n+1}v_n = 0$ for the last equality. It follows from the linear independence of the basis

$$T^{m_1}v_1, ..., Tv_1, v_1, ..., T^{m_n}v_n, ..., Tv_n, v_n$$

that $a_{j,k} = 0$ for all $j \in \{1, ..., n\}$ such that $m_j \geq 1$ and all $k \in \{0, ..., m_j - 1\}$. Thus

$$v = a_{1,m_1}T^{m_1}v_1 + \cdots + a_{n,m_n}T^{m_n}v_n.$$

Hence $T^{m_1}v_1, ..., T^{m_n}v_n$ spans $\operatorname{null} T$ and we may conclude that this list is a basis of $\operatorname{null} T$.

**Exercise 8.C.14.** Suppose $\mathbf{F} = \mathbf{C}$ and $T \in \mathcal{L}(V)$. Prove that there does not exist a direct sum decomposition of $V$ into two nonzero subspaces invariant under $T$ if and only if the minimal polynomial of $T$ is of the form $(z - \lambda)^{\dim V}$ for some $\lambda \in \mathbf{C}$.



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**Solution.** First suppose that there exist non-zero subspaces $U$ and $W$ which are invariant under $T$ and satisfy $V = U \oplus W$. If the union of the set of eigenvalues of $T|_U$ and the set of eigenvalues of $T|_W$ contains at least two complex numbers then $T$ has at least two eigenvalues. It follows that the minimal polynomial of $T$ has at least two roots and hence is not of the form $(z-\lambda)^{\dim V}$ for any $\lambda \in \mathbf{C}$. Suppose therefore that $T|_U$ and $T|_W$ both have a single eigenvalue $\mu \in \mathbf{C}$. As we argued in Exercise 8.B.20, it follows that $\mu$ is the only eigenvalue of $T$ and hence that the minimal polynomial of $T$ is of the form $(z-\mu)^\ell$ for some positive integer $\ell \leq \dim V$. Let $m = \max\{\dim U, \dim W\}$ and notice that $m < \dim V$ since $U$ and $W$ are proper subspaces of $V$. Notice further that $V = G(\mu, T), U = G(\mu, T|_U)$, and $W = G(\mu, T|_W)$ by 8.22(c). It follows from 8.22(b) and 8.16 that

$$(T|_U - \mu I)^m = (T|_W - \mu I)^m = 0.$$

Let $v = u + w \in V = U \oplus W$ be given and observe that

$$(T-\mu I)^m|_{G(\mu,T)}\, v = (T-\mu I)^m(u+w) = (T|_U - \mu I)^m u + (T|_W - \mu I)^m w = 0.$$

Thus $(T-\mu I)^m|_{G(\mu,T)} = 0$. Hence, by Exercise 8.B.6, we must have $\ell \leq m < \dim V$. It follows that the minimal polynomial of $T$ is not of the form $(z-\lambda)^{\dim V}$ for any $\lambda \in \mathbf{C}$.

Now suppose that the minimal polynomial of $T$ is not of the form $(z-\lambda)^{\dim V}$ for any $\lambda \in \mathbf{C}$ and let $\lambda_1, ..., \lambda_m$ be the distinct eigenvalues of $T$. If $m \geq 2$ then, by 8.22(c),

$$V = G(\lambda_1, T) \oplus [G(\lambda_2, T) \oplus \cdots \oplus G(\lambda_m, T)]$$

is a direct sum decomposition of $V$ into two non-zero subspaces of $V$ invariant under $T$. If $m = 1$ then the minimal polynomial of $T$ must be of the form $(z-\lambda_1)^\ell$ where $\ell$ is a positive integer satisfying $\ell < \dim V$. By 8.46 there exists a Jordan basis $v_1, ..., v_{\dim V}$ for $T$, so that the matrix of $T$ with respect to this basis is of the form

$$A = \begin{pmatrix} A_1 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & A_p \end{pmatrix}, \quad \text{where } A_k = \begin{pmatrix} \lambda_1 & 1 & & 0 \\ & \ddots & \ddots & \\ & & \ddots & 1 \\ 0 & & & \lambda_1 \end{pmatrix}.$$

Note that if $p = 1$, so that $A = A_1$ is a $(\dim V) \times (\dim V)$ matrix, then $(A - \lambda_1 I)^\ell \neq 0$, contradicting that $(z-\lambda_1)^\ell$ is the minimal polynomial of $T$. Thus $p \geq 2$. Suppose that $A_1$ is a $d \times d$ matrix; since $p \geq 2$ we must have $d < \dim V$. Let

$$U = \operatorname{span}(v_1, ..., v_d) \quad \text{and} \quad W = \operatorname{span}(v_{d+1}, ..., v_{\dim V}).$$

Then $U$ and $W$ are non-zero, invariant under $T$ (since $A$ is block diagonal), and decompose $V$ as the direct sum $V = U \oplus W$.



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## 8.D. Trace: A Connection Between Matrices and Operators

**Exercise 8.D.1.** Suppose $V$ is an inner product space and $v, w \in V$. Define an operator $T \in \mathcal{L}(V)$ by $Tu = \langle u, v\rangle w$. Find a formula for $\operatorname{tr} T$.

**Solution.** Let $e_1, ..., e_n$ be an orthonormal basis of $V$ and suppose that

$$v = a_1 e_1 + \cdots + a_n e_n \quad \text{and} \quad w = b_1 e_1 + \cdots + b_n e_n.$$

Observe that $\langle Te_k, e_k\rangle = \langle \langle e_k, v\rangle w, e_k\rangle = \overline{a_k} b_k$ for each $k \in \{1, ..., n\}$. It follows from 8.55 that

$$\operatorname{tr} T = \sum_{k=1}^{n} \overline{a_k} b_k = \langle w, v\rangle.$$

**Exercise 8.D.2.** Suppose $P \in \mathcal{L}(V)$ satisfies $P^2 = P$. Prove that

$$\operatorname{tr} P = \dim \operatorname{range} P.$$

**Solution.** Since $P(P-1) = 0$, the minimal polynomial of $P$ is either $z, z-1$, or $z(z-1)$. In any case, the minimal polynomial of $P$ splits into distinct linear factors and hence $P$ is diagonalizable by 5.62. Thus there is a basis of $V$ with respect to which the matrix $A := \mathcal{M}(T)$ is diagonal. Since the only possible eigenvalues of $P$ are 0 or 1, each diagonal entry of $A$ is either 0 or 1. It follows that $\operatorname{tr} P$ is the number of diagonal entries of $A$ equal to 1; denoting this number by $m$, it is clear from the form of $A$ that $\dim \operatorname{range} P = \operatorname{rank} A = m$.

**Exercise 8.D.3.** Suppose $T \in \mathcal{L}(V)$ and $T^5 = T$. Prove that the real and imaginary parts of $\operatorname{tr} T$ are both integers.

**Solution.** First suppose that $\mathbf{F} = \mathbf{C}$. The equation $T^5 = T$ is equivalent to

$$T(T - I)(T + I)(T - iI)(T + iI) = 0.$$

Thus the eigenvalues of $T$ are contained in the set $\{0, \pm 1, \pm i\}$. It then follows from 8.52 that the real and imaginary parts of $\operatorname{tr} T$ are both integers.

Before we proceed, let us prove the following lemma.

**Lemma L.16.** Let $V$ be a real vector space and suppose $T \in \mathcal{L}(V)$. Then $\operatorname{tr} T_{\mathbf{C}} = \operatorname{tr} T$.

*Proof.* Let $v_1, ..., v_n$ be a basis of $V$. As we showed in Exercise 2.B.11, $v_1, ..., v_n$ is also a basis of $V_{\mathbf{C}}$. Observe that $T_{\mathbf{C}} v_k = T v_k$ for each $k \in \{1, ..., n\}$. Thus

$$\mathcal{M}(T_{\mathbf{C}}, (v_1, ..., v_n)) = \mathcal{M}(T, (v_1, ..., v_n)).$$

It is now immediate that $\operatorname{tr} T_{\mathbf{C}} = \operatorname{tr} T$. □



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Now suppose that $\mathbf{F} = \mathbf{R}$. Since $T^5 = T$ we also have $T_{\mathbf{C}}^5 = T_{\mathbf{C}}$ and it follows from our previous discussion that the real and imaginary parts of $\operatorname{tr} T_{\mathbf{C}}$ are integers. Thus, by Lemma L.16, $\operatorname{tr} T$ is an integer.

**Exercise 8.D.4.** Suppose $V$ is an inner product space and $T \in \mathcal{L}(V)$. Prove that

$$\operatorname{tr} T^* = \overline{\operatorname{tr} T}.$$

**Solution.** Let $e_1, ..., e_n$ be an orthonormal basis of $V$ and let $A$ be the matrix of $T$ with respect to $e_1, ..., e_n$. It follows from 7.9 that the matrix of $T^*$ with respect to $e_1, ..., e_n$ is $A^*$. Since the diagonal entries of $A^*$ are the complex conjugates of the diagonal entries of $A$, we obtain the equation $\operatorname{tr} T^* = \overline{\operatorname{tr} T}$.

**Exercise 8.D.5.** Suppose $V$ is an inner product space. Suppose $T \in \mathcal{L}(V)$ is a positive operator and $\operatorname{tr} T = 0$. Prove that $T = 0$.

**Solution.** By 7.38(c) there is an orthonormal basis $e_1, ..., e_n$ of $V$ with respect to which the matrix of $T$ is diagonal with only non-negative numbers on the diagonal. Since $\operatorname{tr} T = 0$ the sum of these non-negative diagonal entries is zero, which can be the case only if each diagonal entry is zero. Thus the matrix of $T$ with respect to $e_1, ..., e_n$ is the zero matrix. Hence $T = 0$.

**Exercise 8.D.6.** Suppose $V$ is an inner product space and $P, Q \in \mathcal{L}(V)$ are orthogonal projections. Prove that $\operatorname{tr}(PQ) \geq 0$.

**Solution.** Suppose that $P$ and $Q$ are orthogonal projections onto subspaces $U$ and $W$ respectively. Let $e_1, ..., e_m$ be an orthonormal basis of $W$ and let $f_1, ..., f_n$ be an orthonormal basis of $W^\perp$, so that $e_1, ..., e_m, f_1, ..., f_n$ is an orthonormal basis of $V$. It follows from 8.55 that

$$\begin{aligned}\operatorname{tr}(PQ) &= \langle PQe_1, e_1\rangle + \cdots + \langle PQe_m, e_m\rangle + \langle PQf_1, f_1\rangle + \cdots + \langle PQf_n, f_n\rangle \\ &= \langle Pe_1, e_1\rangle + \cdots + \langle Pe_m, e_m\rangle,\end{aligned}$$

where we have used that $Q|_W = I$ and $Q|_{W^\perp} = 0$. As noted in 7.35(b), orthogonal projections are positive operators. It follows that

$$\operatorname{tr}(PQ) = \langle Pe_1, e_1\rangle + \cdots + \langle Pe_m, e_m\rangle \geq 0.$$



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**Exercise 8.D.7.** Suppose $T \in \mathcal{L}(\mathbf{C}^3)$ is the operator whose matrix is

$$\begin{pmatrix} 51 & -12 & -21 \\ 60 & -40 & -28 \\ 57 & -68 & 1 \end{pmatrix}.$$

Someone tells you (accurately) that $-48$ and $24$ are eigenvalues of $T$. Without using a computer or writing anything down, find the third eigenvalue of $T$.

**Solution.** On one hand, the trace of the matrix above is $51 - 40 + 1 = 12$. On the other hand, by 8.52, the trace of this matrix is the sum of the eigenvalues of $T$. Letting $x$ be the third eigenvalue of $T$, it follows that

$$12 = -48 + 24 + x \quad \Rightarrow \quad x = 36.$$

**Exercise 8.D.8.** Prove or give a counterexample: If $S, T \in \mathcal{L}(V)$, then $\operatorname{tr}(ST) = (\operatorname{tr} S)(\operatorname{tr} T)$.

**Solution.** This is false. For a counterexample, let $S$ and $T$ be the operators on $\mathbf{F}^2$ whose matrices with respect to the standard basis are

$$\mathcal{M}(S) = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \quad \text{and} \quad \mathcal{M}(T) = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \quad \Rightarrow \quad \mathcal{M}(ST) = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}.$$

Then $\operatorname{tr} S = \operatorname{tr} T = 1$ but $\operatorname{tr}(ST) = 0$. Thus $\operatorname{tr}(ST) \neq (\operatorname{tr} S)(\operatorname{tr} T)$.

**Exercise 8.D.9.** Suppose $T \in \mathcal{L}(V)$ is such that $\operatorname{tr}(ST) = 0$ for all $S \in \mathcal{L}(V)$. Prove that $T = 0$.

**Solution.** We will prove the contrapositive. Suppose that $T \neq 0$, so that there exists some $v_1 \in V$ such that $Tv_1 \neq 0$. Extend $v_1$ to a basis $v_1, ..., v_n$ of $V$. Suppose that

$$Tv_1 = A_{1,1}v_1 + \cdots + A_{n,1}v_n.$$

Since $Tv_1 \neq 0$ there must exist some $i \in \{1, ..., n\}$ such that $A_{i,1} \neq 0$. Define $S \in \mathcal{L}(V)$ by $Sv_i = v_1$ and $Sv_k = 0$ for $k \neq i$. Now observe that the matrix of $ST$ with respect to $v_1, ..., v_n$ is

$$\begin{pmatrix} A_{i,1} & A_{i,2} & \cdots & A_{i,n} \\ 0 & 0 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 0 \end{pmatrix}.$$

Thus $\operatorname{tr}(ST) = A_{i,1} \neq 0$.



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**Exercise 8.D.10.** Prove that the trace is the only linear functional $\tau : \mathcal{L}(V) \to \mathbf{F}$ such that

$$\tau(ST) = \tau(TS)$$

for all $S, T \in \mathcal{L}(V)$ and $\tau(I) = \dim V$.

*Hint: Suppose that $v_1, ..., v_n$ is a basis of $V$. For $j, k \in \{1, ..., n\}$, define $P_{j,k} \in \mathcal{L}(V)$ by $P_{j,k}(a_1 v_1 + \cdots + a_n v_n) = a_k v_j$. Prove that*

$$\tau\big(P_{j,k}\big) = \begin{cases} 1 & \text{if } j = k, \\ 0 & \text{if } j \neq k. \end{cases}$$

*Then for $T \in \mathcal{L}(V)$, use the equation $T = \sum_{k=1}^{n} \sum_{j=1}^{n} \mathcal{M}(T)_{j,k} P_{j,k}$ to show that $\tau(T) = \operatorname{tr} T$.*

**Solution.** Let $v_1, ..., v_n$ be a basis of $V$ and for $j, k \in \{1, ..., n\}$ define $P_{j,k} \in \mathcal{L}(V)$ as in the hint. For any $j, k \in \{1, ..., n\}$ observe that

$$P_{j,k} P_{k,j}(a_1 v_1 + \cdots + a_n v_n) = P_{j,k}\big(a_j v_k\big) = a_j v_j = P_{j,j}(a_1 v_1 + \cdots + a_n v_n).$$

Thus $P_{j,k} P_{k,j} = P_{j,j}$. It follows that

$$\tau\big(P_{j,j}\big) = \tau\big(P_{j,k} P_{k,j}\big) = \tau\big(P_{k,j} P_{j,k}\big) = \tau\big(P_{k,k}\big),$$

which implies

$$n = \tau(I) = \tau\big(P_{1,1} + \cdots + P_{n,n}\big) = \tau\big(P_{1,1}\big) + \cdots + \tau\big(P_{n,n}\big) = n\tau\big(P_{1,1}\big).$$

Thus $\tau\big(P_{1,1}\big) = \cdots = \tau\big(P_{n,n}\big) = 1$. Now observe that, for $j \neq k$,

$$P_{k,k} P_{j,k}(a_1 v_1 + \cdots + a_n v_n) = P_{k,k}\big(a_k v_j\big) = 0,$$

$$\text{and} \quad P_{j,k} P_{k,k}(a_1 v_1 + \cdots + a_n v_n) = P_{j,k}(a_k v_k) = a_k v_j = P_{j,k}(a_1 v_1 + \cdots + a_n v_n).$$

Thus $P_{k,k} P_{j,k} = 0$ and $P_{j,k} P_{k,k} = P_{j,k}$. It follows that

$$\tau\big(P_{j,k}\big) = \tau\big(P_{j,k} P_{k,k}\big) = \tau\big(P_{k,k} P_{j,k}\big) = \tau(0) = 0.$$

We have now shown that

$$\tau\big(P_{j,k}\big) = \begin{cases} 1 & \text{if } j = k, \\ 0 & \text{if } j \neq k. \end{cases}$$

For any $T \in \mathcal{L}(V)$, it follows that

$$\tau(T) = \tau\left(\sum_{k=1}^{n} \sum_{j=1}^{n} \mathcal{M}(T)_{j,k} P_{j,k}\right) = \sum_{k=1}^{n} \sum_{j=1}^{n} \mathcal{M}(T)_{j,k} \tau\big(P_{j,k}\big) = \sum_{k=1}^{n} \mathcal{M}(T)_{k,k} = \operatorname{tr} T.$$



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**Exercise 8.D.11.** Suppose $V$ and $W$ are inner product spaces and $T \in \mathcal{L}(V, W)$. Prove that if $e_1, ..., e_n$ is an orthonormal basis of $V$ and $f_1, ..., f_m$ is an orthonormal basis of $W$, then

$$\operatorname{tr}(T^*T) = \sum_{k=1}^{n} \sum_{j=1}^{m} \left| \langle Te_k, f_j \rangle \right|^2.$$

*The numbers $\langle Te_k, f_j \rangle$ are the entries of the matrix of $T$ with respect to the orthonormal bases $e_1, ..., e_n$ and $f_1, ..., f_m$. These numbers depend on the bases, but $\operatorname{tr}(T^*T)$ does not depend on a choice of bases. Thus this exercise shows that the sum of the squares of the absolute values of the matrix entries does not depend on which orthonormal bases are used.*

**Solution.** The matrix of $T$ with respect to $e_1, ..., e_n$ and $f_1, ..., f_m$ is

$$\begin{pmatrix} \langle Te_1, f_1 \rangle & \langle Te_2, f_1 \rangle & \cdots & \langle Te_n, f_1 \rangle \\ \langle Te_1, f_2 \rangle & \langle Te_2, f_2 \rangle & \cdots & \langle Te_n, f_2 \rangle \\ \vdots & \vdots & \ddots & \vdots \\ \langle Te_1, f_m \rangle & \langle Te_2, f_m \rangle & \cdots & \langle Te_n, f_m \rangle \end{pmatrix}$$

and thus by 7.9 the matrix of $T^*$ with respect to $f_1, ..., f_m$ and $e_1, ..., e_n$ is

$$\begin{pmatrix} \overline{\langle Te_1, f_1 \rangle} & \overline{\langle Te_1, f_2 \rangle} & \cdots & \overline{\langle Te_1, f_m \rangle} \\ \overline{\langle Te_2, f_1 \rangle} & \overline{\langle Te_2, f_2 \rangle} & \cdots & \overline{\langle Te_2, f_m \rangle} \\ \vdots & \vdots & \ddots & \vdots \\ \overline{\langle Te_n, f_1 \rangle} & \overline{\langle Te_n, f_2 \rangle} & \cdots & \overline{\langle Te_n, f_m \rangle} \end{pmatrix}.$$

It follows that the $k^{\text{th}}$ diagonal entry of the matrix of $T^*T$ with respect to $e_1, ..., e_n$ is

$$\sum_{j=1}^{m} \left| \langle Te_k, f_j \rangle \right|^2,$$

from which we obtain the desired formula.



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**Exercise 8.D.12.** Suppose $V$ and $W$ are finite-dimensional inner product spaces.

(a) Prove that $\langle S, T\rangle = \operatorname{tr}(T^* S)$ defines an inner product on $\mathcal{L}(V, W)$.

(b) Suppose $e_1, ..., e_n$ is an orthonormal basis of $V$ and $f_1, ..., f_m$ is an orthonormal basis of $W$. Show that the inner product on $\mathcal{L}(V, W)$ from (a) is the same as the standard inner product on $\mathbf{F}^{mn}$, where we identify each element of $\mathcal{L}(V, W)$ with its matrix (with respect to the bases just mentioned) and then with an element of $\mathbf{F}^{mn}$.

*Caution: The norm of a linear map $T \in \mathcal{L}(V, W)$ as defined by 7.86 is not the same as the norm that comes from the inner product in (a) above. Unless explicitly stated otherwise, always assume that $\|T\|$ refers to the norm as defined by 7.86. The norm that comes from the inner product in (a) is called the* ***Frobenius norm*** *or the* ***Hilbert-Schmidt norm****.*

**Solution.**

(a) We shall verify each property in definition 6.2.

**Positivity.** For any $T \in \mathcal{L}(V)$, Exercise 8.D.11 shows that $\langle T, T\rangle = \operatorname{tr}(T^* T)$ is non-negative.

**Definiteness.** Certainly $\langle 0, 0\rangle = 0$. Suppose that $T \in \mathcal{L}(V)$ is such that $\langle T, T\rangle = 0$, i.e. $\operatorname{tr}(T^* T) = 0$. Since $T^* T$ is a positive operator, Exercise 8.D.5 shows that $T^* T = 0$ and Exercise 7.A.2 then gives us $T = 0$.

**Additivity/homogeneity in first slot.** These properties follow from the linearity of the trace (see 8.56).

**Conjugate symmetry.** Let $S, T \in \mathcal{L}(V)$ be given and observe that

$$\langle S, T\rangle = \operatorname{tr}(T^* S) = \operatorname{tr}\big((S^* T)^*\big) = \overline{\operatorname{tr}(S^* T)} = \overline{\langle T, S\rangle},$$

where we have used Exercise 8.D.4 for the third equality.

(b) As we showed in Exercise 8.D.11, the matrix of $T^*$ with respect to $f_1, ..., f_m$ and $e_1, ..., e_n$ is

$$\begin{pmatrix} \overline{\langle Te_1, f_1\rangle} & \overline{\langle Te_1, f_2\rangle} & \cdots & \overline{\langle Te_1, f_m\rangle} \\ \overline{\langle Te_2, f_1\rangle} & \overline{\langle Te_2, f_2\rangle} & \cdots & \overline{\langle Te_2, f_m\rangle} \\ \vdots & \vdots & \ddots & \vdots \\ \overline{\langle Te_n, f_1\rangle} & \overline{\langle Te_n, f_2\rangle} & \cdots & \overline{\langle Te_n, f_m\rangle} \end{pmatrix}.$$

The matrix of $S$ with respect to $e_1, ..., e_n$ and $f_1, ..., f_m$ is



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$$\begin{pmatrix} \langle Se_1, f_1\rangle & \langle Se_2, f_1\rangle & \cdots & \langle Se_n, f_1\rangle \\ \langle Se_1, f_2\rangle & \langle Se_2, f_2\rangle & \cdots & \langle Se_n, f_2\rangle \\ \vdots & \vdots & \ddots & \vdots \\ \langle Se_1, f_m\rangle & \langle Se_2, f_m\rangle & \cdots & \langle Se_n, f_m\rangle \end{pmatrix}.$$

It follows that the $k^{\text{th}}$ diagonal entry of the matrix of $T^*S$ with respect to $e_1, ..., e_n$ is $\sum_{j=1}^{m} \langle Se_k, f_j\rangle \overline{\langle Te_k, f_j\rangle}$, from which we obtain the desired formula

$$\langle S, T\rangle = \operatorname{tr}(T^*S) = \sum_{k=1}^{n}\sum_{j=1}^{m} \langle Se_k, f_j\rangle \overline{\langle Te_k, f_j\rangle}.$$

**Exercise 8.D.13.** Find $S, T \in \mathcal{L}(\mathcal{P}(\mathbf{F}))$ such that $ST - TS = I$.

*Hint: Make an appropriate modification of the operators in Example 3.9.*

*This exercise shows that additional hypotheses are needed on $S$ and $T$ to extend 8.57 to the setting of infinite-dimensional vector spaces.*

**Solution.** Let $S \in \mathcal{L}(\mathcal{P}(\mathbf{F}))$ be the differentiation operator, i.e. $Sp = p'$, and let $T \in \mathcal{L}(\mathcal{P}(\mathbf{F}))$ be the multiplication by $x$ operator, i.e. $(Tp)(x) = xp(x)$. Observe that

$$((ST - TS)p)(x) = [p(x) + xp'(x)] - xp'(x) = p(x).$$

Thus $ST - TS = I$.



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# Chapter 9. Multilinear Algebra and Determinants

## 9.A. Bilinear Forms and Quadratic Forms

**Exercise 9.A.1.** Prove that if $\beta$ is a bilinear form on $\mathbf{F}$, then there exists $c \in \mathbf{F}$ such that

$$\beta(x, y) = cxy$$

for all $x, y \in \mathbf{F}$.

**Solution.** Using the linearity of $\beta$ in each slot, we have

$$\beta(x, y) = \beta(x \cdot 1, y \cdot 1) = xy\beta(1, 1)$$

for all $x, y \in \mathbf{F}$. Thus we can take $c = \beta(1, 1)$.

**Exercise 9.A.2.** Let $n = \dim V$. Suppose $\beta$ is a bilinear form on $V$. Prove that there exist $\varphi_1, ..., \varphi_n, \tau_1, ..., \tau_n \in V'$ such that

$$\beta(u, v) = \varphi_1(u) \cdot \tau_1(v) + \cdots + \varphi_n(u) \cdot \tau_n(v)$$

for all $u, v \in V$.

*This exercise shows that if $n = \dim V$, then every bilinear form on $V$ is of the form given by the last bullet point of Example 9.2.*

**Solution.** Let $e_1, ..., e_n$ be a basis of $V$ and for $j \in \{1, ..., n\}$ define $\varphi_j, \tau_j \in V'$ by

$$\varphi_j\left(\sum_{k=1}^{n} a_k e_k\right) = a_j \quad \text{and} \quad \tau_j\left(\sum_{k=1}^{n} b_k e_k\right) = \sum_{k=1}^{n} b_k \beta(e_j, e_k).$$

For any $u = \sum_{k=1}^{n} a_k e_k, v = \sum_{k=1}^{n} b_k e_k \in V$, observe that

$$\sum_{j=1}^{n} \varphi_j(u)\tau_j(v) = \sum_{j=1}^{n}\sum_{k=1}^{n} a_j b_k \beta(e_j, e_k) = \beta\left(\sum_{j=1}^{n} a_j e_j, \sum_{k=1}^{n} b_k e_k\right) = \beta(u, v).$$

Thus $\varphi_1, ..., \varphi_n, \tau_1, ..., \tau_n$ are the desired linear functionals.

**Exercise 9.A.3.** Suppose $\beta : V \times V \to \mathbf{F}$ is a bilinear form on $V$ and also is a linear functional on $V \times V$. Prove that $\beta = 0$.



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**Solution.** Let $u, v \in V$ be given. On one hand, since $\beta$ is a linear functional on $V \times V$ we must have $2\beta(u,v) = \beta(2u, 2v)$. On the other hand, since $\beta$ is a bilinear form on $V$ we must have $2\beta(u,v) = \beta(u, 2v)$. Thus

$$0 = \beta(2u, 2v) - \beta(u, 2v) = \beta(u, 0),$$

where we have used that $\beta \in (V \times V)'$ for the last equality. Similarly, we can show that $\beta(0, v) = 0$. The linearity of $\beta$ as a map $V \times V \to \mathbf{F}$ then implies that

$$0 = \beta(u, 0) + \beta(0, v) = \beta(u, v).$$

Thus $\beta = 0$.

**Exercise 9.A.4.** Suppose $V$ is a real inner product space and $\beta$ is a bilinear form on $V$. Show that there exists a unique operator $T \in \mathcal{L}(V)$ such that

$$\beta(u, v) = \langle u, Tv \rangle$$

for all $u, v \in V$.

*This exercise states that if $V$ is a real inner product space, then every bilinear form on $V$ is of the form given by the third bullet point in 9.2.*

**Solution.** Let $e_1, ..., e_n$ be an orthonormal basis of $V$ and define $T \in \mathcal{L}(V)$ by

$$Te_k = \beta(e_1, e_k)e_1 + \cdots + \beta(e_n, e_k)e_n$$

for $k \in \{1, ..., n\}$. Observe that $\langle e_j, Te_k \rangle = \beta(e_j, e_k)$ for any $j, k \in \{1, ..., n\}$. It follows that, for any $u = \sum_{k=1}^n a_k e_k, v = \sum_{k=1}^n b_k e_k \in V$,

$$\langle u, Tv \rangle = \sum_{j=1}^n \sum_{k=1}^n a_j b_k \langle e_j, Te_k \rangle = \sum_{j=1}^n \sum_{k=1}^n a_j b_k \beta(e_j, e_k) = \beta(u, v).$$

Thus $T$ is the desired operator.

**Exercise 9.A.5.** Suppose $\beta$ is a bilinear form on a real inner product space $V$ and $T$ is the unique operator on $V$ such that $\beta(u, v) = \langle u, Tv \rangle$ for all $u, v \in V$ (see Exercise 4). Show that $\beta$ is an inner product on $V$ if and only if $T$ is an invertible positive operator on $V$.

**Solution.** First suppose that $T$ is an invertible positive operator. To show that $\beta$ is an inner product on $V$, we must verify each property in definition 6.2.

**Positivity/definiteness.** 7.61 shows that $\beta(v, v) = \langle v, Tv \rangle = \langle Tv, v \rangle > 0$ for all non-zero $v \in V$ and certainly $\beta(0, 0) = 0$.

**Additivity/homogeneity in first slot/symmetry.** These properties are immediate from the corresponding properties of $\langle \cdot, \cdot \rangle$.

Now suppose that $\beta$ is an inner product on $V$. The symmetry of $\beta$ and $\langle \cdot, \cdot \rangle$ gives us



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$$\langle Tu, v\rangle = \langle v, Tu\rangle = \beta(v,u) = \beta(u,v) = \langle u, Tv\rangle.$$

Thus $T$ is self-adjoint. Now let $v \in V$ be non-zero and observe that, by the positive-definiteness of $\beta$,

$$\langle Tv, v\rangle = \langle v, Tv\rangle = \beta(v,v) > 0.$$

7.61 allows us to conclude that $T$ is a positive invertible operator.

**Exercise 9.A.6.** Prove or give a counterexample: If $\rho$ is a symmetric bilinear form on $V$, then

$$\{v \in V : \rho(v,v) = 0\}$$

is a subspace of $V$.

**Solution.** This is false. Let $\rho$ be the symmetric bilinear form on $\mathbf{F}^2$ given by

$$\rho((x_1,x_2),(y_1,y_2)) = x_1y_1 - x_2y_2.$$

Observe that

$$\rho((1,1),(1,1)) = \rho((1,-1),(1,-1)) = 0 \neq \rho((2,0),(2,0)).$$

Thus $\{v \in V : \rho(v,v) = 0\}$ is not a subspace of $\mathbf{F}^2$.

**Exercise 9.A.7.** Explain why the proof of 9.13 (diagonalization of symmetric bilinear form by an orthonormal basis on a real inner product space) fails if the hypothesis that $\mathbf{F} = \mathbf{R}$ is dropped.

**Solution.** Define $B$ and $T$ as in the proof of 9.13. Note that a symmetric matrix with complex entries need not equal its own conjugate transpose nor commute with its own conjugate transpose, e.g.

$$A = \begin{pmatrix} 1 & i \\ i & -1 \end{pmatrix}$$

satisfies $A^{\mathrm{t}} = A, A \neq A^*$, and $AA^* \neq A^*A$. Thus we cannot necessarily claim that $T$ is self-adjoint or normal and hence cannot necessarily apply the complex spectral theorem.

**Exercise 9.A.8.** Find formulas for $\dim V^{(2)}_{\mathrm{sym}}$ and $\dim V^{(2)}_{\mathrm{alt}}$ in terms of $\dim V$.

**Solution.** Let $e_1, ..., e_n$ be a basis of $V$. As 9.5 shows, the map $\beta \mapsto \mathcal{M}(\beta)$ is an isomorphism of $V^{(2)}$ onto $\mathbf{F}^{n,n}$. Furthermore, by 9.12, under this isomorphism the symmetric bilinear forms in $V^{(2)}$ correspond exactly to the symmetric matrices in $\mathbf{F}^{n,n}$. Thus to calculate $\dim V^{(2)}_{\mathrm{sym}}$ it will suffice to find the dimension of the subspace of $\mathbf{F}^{n,n}$ consisting of the symmetric matrices; as we showed in Exercise 7.A.16 (b), this subspace has dimension $n(n+1)/2$. It now follows from 9.17 that



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$$\dim V_{\text{alt}}^{(2)} = \dim V^{(2)} - \dim V_{\text{sym}}^{(2)} = n^2 - \frac{n(n+1)}{2} = \frac{n(n-1)}{2}.$$

**Exercise 9.A.9.** Suppose that $n$ is a positive integer and

$$V = \{p \in \mathcal{P}_n(\mathbf{R}) : p(0) = p(1)\}.$$

Define $\alpha : V \times V \to \mathbf{R}$ by

$$\alpha(p, q) = \int_0^1 pq'.$$

Show that $\alpha$ is an alternating bilinear form on $V$.

**Solution.** The bilinearity of $\alpha$ follows from the linearity of differentiation and of integration. For any $p \in V$, observe that

$$\int_0^1 pp' = \frac{1}{2}\int_0^1 (p^2)' = \frac{1}{2}\left[(p(1))^2 - (p(0))^2\right] = 0.$$

Thus $\alpha$ is alternating.

**Exercise 9.A.10.** Suppose that $n$ is a positive integer and

$$V = \{p \in \mathcal{P}_n(\mathbf{R}) : p(0) = p(1) \text{ and } p'(0) = p'(1)\}.$$

Define $\rho : V \times V \to \mathbf{R}$ by

$$\rho(p, q) = \int_0^1 pq''.$$

Show that $\rho$ is a symmetric bilinear form on $V$.

**Solution.** The bilinearity of $\rho$ follows from the linearity of differentiation and of integration. For any $p, q \in V$, observe that

$$\int_0^1 pq'' - qp'' = \int_0^1 [pq']' - [qp']' = [p(1)q'(1) - p(0)q'(0)] - [q(1)p'(1) - q(0)p'(0)] = 0.$$

Thus $\rho$ is symmetric.



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## 9.B. Alternating Multilinear Forms

**Exercise 9.B.1.** Suppose $m$ is a positive integer. Show that $\dim V^{(m)} = (\dim V)^m$.

**Solution.** Let $e_1, ..., e_n$ be a basis of $V$ and let $\varphi_1, ..., \varphi_n$ be the corresponding dual basis of $V'$, so that for any $v \in V$ we have, by 3.114,

$$v = \varphi_1(v)e_1 + \cdots + \varphi_n(v)e_n.$$

For each $(i_1, ..., i_m) \in \{1, ..., n\}^m$ define an $m$-linear form $\alpha_{i_1,...,i_m}$ by

$$\alpha_{i_1,...,i_m}(v_1, ..., v_m) = \varphi_{i_1}(v_1) \cdots \varphi_{i_m}(v_m).$$

Let $\mathcal{B}$ be the list of all such $\alpha_{i_1,...,i_m}$; we claim that $\mathcal{B}$ is a basis of $V^{(m)}$. For any $\alpha \in V^{(m)}$, observe that

$$\begin{aligned}
\alpha(v_1, ..., v_m) &= \alpha\left(\sum_{i_1=1}^{n} \varphi_{i_1}(v_1)e_{i_1}, ..., \sum_{i_m=1}^{n} \varphi_{i_m}(v_m)e_{i_m}\right) \\
&= \sum_{i_1=1}^{n} \cdots \sum_{i_m=1}^{n} \alpha\big(e_{i_1}, ..., e_{i_m}\big)\, \varphi_{i_1}(v_1) \cdots \varphi_{i_m}(v_m) \\
&= \sum_{i_1=1}^{n} \cdots \sum_{i_m=1}^{n} \alpha\big(e_{i_1}, ..., e_{i_m}\big)\, \alpha_{i_1,...,i_m}(v_1, ..., v_m).
\end{aligned}$$

Thus $\alpha \in \operatorname{span} \mathcal{B}$ and it follows that $V^{(m)} = \operatorname{span} \mathcal{B}$. Now suppose that a linear combination of $\mathcal{B}$ is zero:

$$\sum_{i_1=1}^{n} \cdots \sum_{i_m=1}^{n} c_{i_1,...,i_m}\, \alpha_{i_1,...,i_m}(v_1, ..., v_m) = 0 \quad \text{for all } v_1, ..., v_m \in V.$$

For any $(k_1, ..., k_m) \in \{1, ..., n\}^m$, note that

$$\alpha_{i_1,...,i_m}\big(e_{k_1}, ..., e_{k_m}\big) = \begin{cases} 1 & \text{if } i_1 = k_1, ..., i_m = k_m, \\ 0 & \text{otherwise.} \end{cases}$$

Thus

$$0 = \sum_{i_1=1}^{n} \cdots \sum_{i_m=1}^{n} c_{i_1,...,i_m}\, \alpha_{i_1,...,i_m}\big(e_{k_1}, ..., e_{k_m}\big) = c_{k_1,...,k_m}.$$

Hence $\mathcal{B}$ is linearly independent and it follows that $\mathcal{B}$ is a basis of $V^{(m)}$, as claimed.

Now observe that $\mathcal{B}$ is a list of length $n^m$, since the set $\{1, ..., n\}^m$ contains $n^m$ elements. Thus $\dim V^{(m)} = n^m = (\dim V)^m$.



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**Exercise 9.B.2.** Suppose $n \geq 3$ and $\alpha : \mathbf{F}^n \times \mathbf{F}^n \times \mathbf{F}^n \to \mathbf{F}$ is defined by

$$\begin{aligned}&\alpha((x_1, ..., x_n), (y_1, ..., y_n), (z_1, ..., z_n))\\ &\quad = x_1y_2z_3 - x_2y_1z_3 - x_3y_2z_1 - x_1y_3z_2 + x_3y_1z_2 + x_2y_3z_1.\end{aligned}$$

Show that $\alpha$ is an alternating 3-linear form on $\mathbf{F}^n$.

**Solution.** A straightforward calculation shows that $\alpha$ is 3-linear. To see that $\alpha$ is alternating, let $x = (x_1, ..., x_n), y = (y_1, ..., y_n)$, and $z = (z_1, ..., z_n)$. Observe that

$$\alpha(x, x, z) = x_1x_2z_3 - x_2x_1z_3 - x_3x_2z_1 - x_1x_3z_2 + x_3x_1z_2 + x_2x_3z_1 = 0.$$

We can similarly show that $\alpha(x, y, x) = \alpha(x, y, y) = 0$. Thus $\alpha$ is alternating.

**Exercise 9.B.3.** Suppose $m$ is a positive integer and $\alpha$ is an $m$-linear form on $V$ such that $\alpha(v_1, ..., v_m) = 0$ whenever $v_1, ..., v_m$ is a list of vectors in $V$ with $v_j = v_{j+1}$ for some $j \in \{1, ..., m-1\}$. Prove that $\alpha$ is an alternating $m$-linear form on $V$.

**Solution.** For any $k \in \{1, ..., m-1\}$, observe that

$$\begin{aligned}0 &= \alpha(v_1, ..., v_k + v_{k+1}, v_k + v_{k+1}, ..., v_m)\\ &= \alpha(v_1, ..., v_k, v_k, ..., v_m) + \alpha(v_1, ..., v_{k+1}, v_{k+1}, ..., v_m)\\ &\quad + \alpha(v_1, ..., v_k, v_{k+1}, ..., v_m) + \alpha(v_1, ..., v_{k+1}, v_k, ..., v_m)\\ &= \alpha(v_1, ..., v_k, v_{k+1}, ..., v_m) + \alpha(v_1, ..., v_{k+1}, v_k, ..., v_m).\end{aligned}$$

Thus swapping the vectors in any two consecutive slots of $\alpha(v_1, ..., v_m)$ changes the value of $\alpha$ by a factor of $-1$. Now suppose that $v_1, ..., v_m$ is a list of vectors in $V$ with $v_j = v_k$ for some $1 \leq j < k \leq m$. By performing consecutive swaps in the slots of $\alpha(v_1, ..., v_m)$ if necessary, which only changes the sign of $\alpha$, we can ensure that $v_k$ appears directly after $v_j$:

$$\alpha(v_1, ..., v_j, ..., v_k, ..., v_m) = \pm\alpha(v_1, ..., v_j, v_k, ..., v_m) = 0.$$

Thus $\alpha$ is alternating.

**Exercise 9.B.4.** Prove or give a counterexample: If $\alpha \in V_{\text{alt}}^{(4)}$, then

$$\{(v_1, v_2, v_3, v_4) \in V^4 : \alpha(v_1, v_2, v_3, v_4) = 0\}$$

is a subspace of $V^4$.

**Solution.** This is false. For a counterexample, consider $V = \mathbf{R}^4$. By 9.37 there exists a non-zero alternating 4-linear form $\alpha$ on $\mathbf{R}^4$. Let $e_1, e_2, e_3, e_4$ be the standard basis of $\mathbf{R}^4$ and observe that $\alpha(e_1, e_2, 0, 0) = \alpha(0, 0, e_3, e_4) = 0$. However, $\alpha(e_1, e_2, e_3, e_4) \neq 0$ by 9.39.



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**Exercise 9.B.5.** Suppose $m$ is a positive integer and $\beta$ is an $m$-linear form on $V$. Define an $m$-linear form $\alpha$ on $V$ by

$$\alpha(v_1, ..., v_m) = \sum_{(j_1,...,j_m) \in \operatorname{perm} m} (\operatorname{sign}(j_1, ..., j_m)) \beta\big(v_{j_1}, ..., v_{j_m}\big)$$

for $v_1, ..., v_m \in V$. Explain why $\alpha \in V_{\text{alt}}^{(m)}$.

**Solution.** Suppose that $v_k = v_\ell$ for some $1 \le k < \ell \le m$. We will show that

$$\alpha(v_1, ..., v_k, ..., v_\ell, ..., v_m) = 0.$$

Let $G \subseteq \operatorname{perm} m$ consist of those permutations with $k$ and $\ell$ in the correct order and let $H = (\operatorname{perm} m) \setminus G$ consist of those permutations with $k$ and $\ell$ in reverse order. Notice that the map $\Phi : G \to H$ which swaps the position of $k$ and $\ell$ is a bijection: it is its own inverse. Thus each permutation in $H$ corresponds via $\Phi$ to exactly one permutation in $G$. Notice further that $\operatorname{sign}(\Phi(j_1, ..., j_m)) = -\operatorname{sign}(j_1, ..., j_m)$ by 9.34 and, because $v_k = v_\ell$,

$$\beta\big(v_{j_1}, ..., v_{j_m}\big) = \beta\big(v_{i_1}, ..., v_{i_m}\big) \quad \text{where } (i_1, ..., i_m) = \Phi(j_1, ..., j_m).$$

To summarize, for each $(i_1, ..., i_m) \in H$ we have $(i_1, ..., i_m) = \Phi(j_1, ..., j_m)$ for a unique $(j_1, ..., j_m) \in G$ such that

$$\operatorname{sign}(i_1, ..., i_m) = -\operatorname{sign}(j_1, ..., j_m) \quad \text{and} \quad \beta\big(v_{i_1}, ..., v_{i_m}\big) = \beta\big(v_{j_1}, ..., v_{j_m}\big).$$

It follows that

$$\sum_{(i_1,...,i_m) \in H} (\operatorname{sign}(i_1, ..., i_m)) \beta\big(v_{i_1}, ..., v_{i_m}\big) = -\sum_{(j_1,...,j_m) \in G} \operatorname{sign}(j_1, ..., j_m) \beta\big(v_{j_1}, ..., v_{j_m}\big).$$

Thus

$$\begin{aligned}
\alpha(v_1, ..., v_m) &= \sum_{(j_1,...,j_m) \in \operatorname{perm} m} (\operatorname{sign}(j_1, ..., j_m)) \beta\big(v_{j_1}, ..., v_{j_m}\big) \\
&= \sum_{(j_1,...,j_m) \in G} \operatorname{sign}(j_1, ..., j_m) \beta\big(v_{j_1}, ..., v_{j_m}\big) \\
&\qquad + \sum_{(i_1,...,i_m) \in H} (\operatorname{sign}(i_1, ..., i_m)) \beta\big(v_{i_1}, ..., v_{i_m}\big) \\
&= \sum_{(j_1,...,j_m) \in G} \operatorname{sign}(j_1, ..., j_m) \beta\big(v_{j_1}, ..., v_{j_m}\big) \\
&\qquad - \sum_{(j_1,...,j_m) \in G} \operatorname{sign}(j_1, ..., j_m) \beta\big(v_{j_1}, ..., v_{j_m}\big) \\
&= 0.
\end{aligned}$$



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**Exercise 9.B.6.** Suppose $m$ is a positive integer and $\beta$ is an $m$-linear form on $V$. Define an $m$-linear form $\alpha$ on $V$ by

$$\alpha(v_1, ..., v_m) = \sum_{(j_1,...,j_m) \in \operatorname{perm} m} \beta\big(v_{j_1}, ..., v_{j_m}\big)$$

for $v_1, ..., v_m \in V$. Explain why

$$\alpha\big(v_{k_1}, ..., v_{k_m}\big) = \alpha(v_1, ..., v_m)$$

for all $v_1, ..., v_m \in V$ and all $(k_1, ..., k_m) \in \operatorname{perm} m$.

**Solution.** The set of all permutations of $(1, ..., m)$ is exactly the same as the set of all permutations of $(k_1, ..., k_m)$.

**Exercise 9.B.7.** Give an example of a nonzero alternating 2-linear form $\alpha$ on $\mathbf{R}^3$ and a linearly independent list $v_1, v_2 \in \mathbf{R}^3$ such that $\alpha(v_1, v_2) = 0$.

*This exercise shows that 9.39 can fail if the hypothesis that $n = \dim V$ is deleted.*

**Solution.** We can take

$$\alpha((x_1, x_2, x_3), (y_1, y_2, y_3)) = x_1 y_2 - x_2 y_1, \quad v_1 = (1, 0, 0), \quad \text{and} \quad v_2 = (0, 0, 1).$$



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## 9.C. Determinants

**Exercise 9.C.1.** Prove or give a counterexample:

$$S, T \in \mathcal{L}(V) \quad \Rightarrow \quad \det(S+T) = \det S + \det T.$$

**Solution.** This is false. Let $S, T \in \mathcal{L}(\mathbf{R}^2)$ be the operators whose matrices with respect to the standard basis are

$$\mathcal{M}(S) = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \quad \text{and} \quad \mathcal{M}(T) = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \quad \Rightarrow \quad \mathcal{M}(S+T) = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}.$$

It follows from 9.48 and 9.53 that

$$\det S + \det T = 0 \neq 1 = \det(S+T).$$

**Exercise 9.C.2.** Suppose the first column of a square matrix $A$ consists of all zeros except possibly the first entry $A_{1,1}$. Let $B$ be the matrix obtained from $A$ by deleting the first row and the first column of $A$. Show that $\det A = A_{1,1} \det B$.

**Solution 1.** Suppose that $A$ is an $n \times n$ matrix; we may as well suppose that $n \geq 2$. Let $T$ be the operator on $\mathbf{F}^n$ whose matrix with respect to the standard basis $e_1, ..., e_n$ is $A^{\mathrm{t}}$. Let $U = \operatorname{span}(e_2, ..., e_n)$ and notice that $U$ is invariant under $T$ since all entries of the first row of $A^{\mathrm{t}}$ are zero, except possibly the first entry. Notice further that the matrix of $T|_U$ with respect to $e_2, ..., e_n$ is $B^{\mathrm{t}}$. Let $\alpha$ be an alternating $n$-linear form on $\mathbf{F}^n$ such that $\alpha(e_1, ..., e_n) = 1$; see 9.37 for the existence of such an $\alpha$. If we define $\beta$ by $\beta(v_2, ..., v_n) = \alpha(Te_1, v_2, ..., v_n)$, then $\beta$ is an alternating $(n-1)$-linear form on $U$. Using 9.56(a), observe that

$$\begin{aligned}
\det A = \det A^{\mathrm{t}} &= \det T \\
&= (\det T)\, \alpha(e_1, e_2, ..., e_n) \\
&= \alpha(Te_1, Te_2, ..., Te_n) \\
&= \beta(Te_2, ..., Te_n) \\
&= (\det T|_U)\, \beta(e_2, ..., e_n) \\
&= (\det B^{\mathrm{t}})\, \alpha(Te_1, e_2, ..., e_n) \\
&= (\det B)\, \alpha\big(A_{1,1}e_1 + \cdots + A_{1,n}e_n, e_2, ..., e_n\big) \\
&= A_{1,1}(\det B)\, \alpha(e_1, ..., e_n) \\
&= A_{1,1} \det B.
\end{aligned}$$

**Solution 2.** Suppose that $A$ is an $n \times n$ matrix; we may as well suppose that $n \geq 2$. Consider the formula for $\det A$ given by 9.46:



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$$\det A = \sum_{(j_1,\dots,j_n)\in \operatorname{perm} n} (\operatorname{sign}(j_1,\dots,j_n)) A_{j_1,1}\cdots A_{j_n,n}.$$

By assumption we have $A_{j_1,1} = 0$ if $j_1 \neq 1$ and thus

$$\det A = A_{1,1} \sum_{(1,j_2,\dots,j_n)\in \operatorname{perm} n} (\operatorname{sign}(1,j_2,\dots,j_n)) A_{j_2,2}\cdots A_{j_n,n}.$$

For $(1,j_2,\dots,j_n) \in \operatorname{perm} n$ notice that $j_k \geq 2$ for each $k \in \{2,\dots,n\}$. Thus if we define $i_k = j_{k+1} - 1$ for $k \in \{1,\dots,n-1\}$, then $(i_1,\dots,i_{n-1}) \in \operatorname{perm}(n-1)$. Observe that:

- the map sending $(1,j_2,\dots,j_n) \mapsto (i_1,\dots,i_{n-1})$ is a bijection between the set of all permutations $(j_1,j_2,\dots,j_n) \in \operatorname{perm} n$ satisfying $j_1 = 1$ and $\operatorname{perm}(n-1)$;
- $B_{i_k,k} = A_{j_{k+1}-1,k+1}$ for each $k \in \{1,\dots,n-1\}$;
- $\operatorname{sign}(i_1,\dots,i_{n-1}) = \operatorname{sign}(1,j_2,\dots,j_n)$ since 1 is in its natural position in the permutation $(1,j_2,\dots,j_n)$ and $i_k < i_\ell$ if and only if $j_{k+1} < j_{\ell+1}$.

It follows that

$$\begin{aligned}\det A &= A_{1,1} \sum_{(1,j_2,\dots,j_n)\in \operatorname{perm} n} (\operatorname{sign}(1,j_2,\dots,j_n)) A_{j_2,2}\cdots A_{j_n,n} \\ &= A_{1,1} \sum_{(i_1,\dots,i_{n-1})\in \operatorname{perm}(n-1)} (\operatorname{sign}(i_1,\dots,i_{n-1})) B_{i_1,1}\cdots B_{i_{n-1},n-1} = A_{1,1} \det B.\end{aligned}$$

**Exercise 9.C.3.** Suppose $T \in \mathcal{L}(V)$ is nilpotent. Prove that $\det(I+T) = 1$.

**Solution.** By 8.18 there is a basis $e_1,\dots,e_n$ of $V$ such that the matrix of $T$ with respect to $e_1,\dots,e_n$ is upper-triangular with each diagonal entry equal to zero. It follows that the matrix of $I+T$ with respect to $e_1,\dots,e_n$ is upper-triangular with each diagonal entry equal to one. 9.48 and 9.53 allow us to conclude that $\det(I+T) = 1$.

**Exercise 9.C.4.** Suppose $S \in \mathcal{L}(V)$. Prove that $S$ is unitary if and only if

$$|\det S| = \|S\| = 1.$$

**Solution.** If $S$ is unitary then $|\det S| = 1$ by 9.58 and $\|S\| = 1$ since $\|Sv\| = \|v\|$ for every $v \in V$.

Conversely, suppose that $|\det S| = \|S\| = 1$. Let $s_1 \geq \cdots \geq s_n \geq 0$ be the singular values of $S$, so that $s_1 = \|S\| = 1$. Observe that, by 9.60,

$$1 = |\det S| = s_1 \cdots s_n.$$

It follows that $s_1 = \cdots = s_n = 1$, otherwise the right-hand side of the equation above would be strictly less than 1. Thus $S$ is unitary by 7.69.



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**Exercise 9.C.5.** Suppose $A$ is a block upper-triangular matrix

$$A = \begin{pmatrix} A_1 & & * \\ & \ddots & \\ 0 & & A_m \end{pmatrix},$$

where each $A_k$ along the diagonal is a square matrix. Prove that

$$\det A = (\det A_1) \cdots (\det A_m).$$

**Solution.** It will suffice to prove the case where $m = 2$. A straightforward induction argument will then prove the general case. Suppose therefore that $A$ is a block upper-triangular matrix of the form

$$A = \begin{pmatrix} B & D \\ 0 & C \end{pmatrix},$$

where $B$ is a $k \times k$ matrix and $C$ is an $\ell \times \ell$ matrix, so that $A$ is a $(k+\ell) \times (k+\ell)$ matrix and $D$ is a $k \times \ell$ matrix. Let $e_1, ..., e_{k+\ell}$ be the standard basis of $\mathbf{F}^{k+\ell}$, let

$$U = \operatorname{span}(e_1, ..., e_k), \quad W = \operatorname{span}(e_{k+1}, ..., e_\ell),$$

and let $T \in \mathcal{L}(V)$ and $S \in \mathcal{L}(W)$ be such that

$$\mathcal{M}(T, (e_1, ..., e_{k+\ell})) = A \quad \text{and} \quad \mathcal{M}(S, (e_{k+1}, ..., e_{k+\ell})) = C.$$

Notice that $U$ is invariant under $T$ and $\mathcal{M}(T|_U, (e_1, ..., e_k)) = B$. Let $\alpha$ be a $(k+\ell)$-linear form on $\mathbf{F}^{k+\ell}$ satisfying $\alpha(e_1, ..., e_{k+\ell}) = 1$; see 9.37 for the existence of such an $\alpha$. If we define $\beta : U \to \mathbf{F}$ and $\gamma : W \to \mathbf{F}$ by

$$\beta(v_1, ..., v_k) = \alpha(v_1, ..., v_k, Te_{k+1}, ..., Te_{k+\ell}),$$
$$\gamma(v_{k+1}, ..., v_{k+\ell}) = \alpha(e_1, ..., e_k, v_{k+1}, ..., v_{k+\ell}),$$

then $\beta$ is an alternating $k$-form on $U$ and $\gamma$ is an alternating $\ell$-form on $W$. Now observe that

$$\begin{aligned}
\alpha(e_1, ..., e_k, Te_{k+1}, ..., Te_{k+\ell}) &= \alpha\Bigg(e_1, ..., e_k, \sum_{i_1=1}^{k} D_{i_1,1} e_{i_1} + \sum_{i_1=1}^{\ell} C_{i_1,1} e_{k+i_1}, \\
&\qquad\qquad ..., \sum_{i_\ell=1}^{k} D_{i_\ell,\ell} e_{i_\ell} + \sum_{i_\ell=1}^{\ell} C_{i_\ell,\ell} e_{k+i_\ell}\Bigg) \\
&= \sum_{i_1=1}^{k} \cdots \sum_{i_\ell=1}^{k} D_{i_1,1} \cdots D_{i_\ell,\ell}\, \alpha\big(e_1, ..., e_k, e_{i_1}, ..., e_{i_\ell}\big) \\
&\quad + \sum_{i_1=1}^{k} \cdots \sum_{i_\ell=1}^{k} C_{i_1,1} \cdots C_{i_\ell,\ell}\, \alpha\big(e_1, ..., e_k, e_{k+i_1}, ..., e_{k+i_\ell}\big) \\
&= \sum_{i_1=1}^{k} \cdots \sum_{i_\ell=1}^{k} C_{i_1,1} \cdots C_{i_\ell,\ell}\, \alpha\big(e_1, ..., e_k, e_{k+i_1}, ..., e_{k+i_\ell}\big);
\end{aligned}$$



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this last line holds since $\alpha\big(e_1, ..., e_k, e_{i_1}, ..., e_{i_\ell}\big) = 0$ for any $(i_1, ..., i_\ell) \in \{1, ..., k\}^\ell$ (because $\alpha$ is alternating). It follows that

$$\begin{aligned}\alpha(e_1, ..., e_k, Te_{k+1}, ..., Te_{k+\ell}) &= \sum_{i_1=1}^{k} \cdots \sum_{i_\ell=1}^{k} C_{i_1,1} \cdots C_{i_\ell,\ell}\, \alpha\big(e_1, ..., e_k, e_{k+i_1}, ..., e_{k+i_\ell}\big) \\ &= \alpha\left(e_1, ..., e_k, \sum_{i_1=1}^{\ell} C_{i_1,1} e_{k+i_1}, ..., \sum_{i_\ell=1}^{\ell} C_{i_\ell,\ell} e_{k+i_\ell}\right) \\ &= \alpha(e_1, ..., e_k, Se_{k+1}, ..., Se_{k+\ell}).\end{aligned}$$

Now observe that

$$\begin{aligned}\det A &= \det T \\ &= (\det T)\, \alpha(e_1, ..., e_k, e_{k+1}, ..., e_{k+\ell}) \\ &= \alpha(Te_1, ..., Te_k, Te_{k+1}, ..., Te_{k+\ell}) \\ &= \beta((T|_U)e_1, ..., (T|_U)e_k) \\ &= (\det(T|_U))\, \beta(e_1, ..., e_n) \\ &= (\det B)\, \alpha(e_1, ..., e_n, Te_{k+1}, ..., Te_{k+\ell}) \\ &= (\det B)\, \alpha(e_1, ..., e_k, Se_{k+1}, ..., Se_{k+\ell}) \\ &= (\det B)\, \gamma(Se_{k+1}, ..., Se_{k+\ell}) \\ &= (\det B)(\det S)\gamma(e_{k+1}, ..., e_{k+\ell}) \\ &= (\det B)(\det C)\, \alpha(e_1, ..., e_k, e_{k+1}, ..., e_{k+\ell}) \\ &= (\det B)(\det C).\end{aligned}$$

**Exercise 9.C.6.** Suppose $A = \begin{pmatrix} v_1 & \cdots & v_n \end{pmatrix}$ is an $n$-by-$n$ matrix, with $v_k$ denoting the $k^{\text{th}}$ column of $A$. Show that if $(m_1, ..., m_n) \in \operatorname{perm} n$, then

$$\det\begin{pmatrix} v_{m_1} & \cdots & v_{m_n} \end{pmatrix} = (\operatorname{sign}(m_1, ..., m_n)) \det A.$$

**Solution.** Swapping pairs of columns of $\begin{pmatrix} v_{m_1} & \cdots & v_{m_n} \end{pmatrix}$ multiplies the determinant by $-1$ (see 9.57(b)). If $N$ is the number of swaps required until the columns are in the correct order, i.e. $\begin{pmatrix} v_1 & \cdots & v_n \end{pmatrix}$, then $\operatorname{sign}(m_1, ..., m_n) = (-1)^N$.



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**Exercise 9.C.7.** Suppose $T \in \mathcal{L}(V)$ is invertible. Let $p$ denote the characteristic polynomial of $T$ and let $q$ denote the characteristic polynomial of $T^{-1}$. Prove that

$$q(z) = \frac{1}{p(0)} z^{\dim V} p\left(\frac{1}{z}\right)$$

for all nonzero $z \in \mathbf{F}$.

**Solution.** By definition we have $p(z) = \det(zI - T)$ and $q(z) = \det(zI - T^{-1})$. Now observe that

$$\begin{aligned}
\frac{1}{p(0)} z^{\dim V} p\left(\frac{1}{z}\right) &= \frac{1}{\det(-T)} z^{\dim V} \det\left(\frac{1}{z} I - T\right) \\
&= \det(-T^{-1}) \det(I - zT) \\
&= \det(zI - T^{-1}) \\
&= q(z),
\end{aligned}$$

where the second line holds by 9.50 and the third bullet point of 9.42 and the second line holds by 9.49(a).

**Exercise 9.C.8.** Suppose $T \in \mathcal{L}(V)$ is an operator with no eigenvalues (which implies that $\mathbf{F} = \mathbf{R}$). Prove that $\det T > 0$.

**Solution.** Since $T$ has no eigenvalues, it must be the case that $\mathbf{F} = \mathbf{R}$ and $n := \dim V$ is even (by 5.34). Furthermore, the characteristic polynomial $p$ of $T$ must have no real roots. Thus the factorization of $p$ over $\mathbf{C}$ is of the form

$$p(z) = (z - \lambda_1)(z - \overline{\lambda_1}) \cdots (z - \lambda_m)(z - \overline{\lambda_m})$$

for some non-zero complex numbers $\lambda_1, ..., \lambda_m$ (see Chapter 4). It follows that the constant term of $p$ is

$$\lambda_1 \overline{\lambda_1} \cdots \lambda_m \overline{\lambda_m} = |\lambda_1|^2 \cdots |\lambda_m|^2 > 0.$$

On the other hand, by 9.65, the constant term of $p$ equals $(-1)^n (\det T) = \det T$. Thus $\det T > 0$.

**Exercise 9.C.9.** Suppose that $V$ is a real vector space of even dimension, $T \in \mathcal{L}(V)$, and $\det T < 0$. Prove that $T$ has at least two distinct eigenvalues.

**Solution.** We will prove the contrapositive, i.e. assuming that $V$ is a real vector space of even dimension and $T \in \mathcal{L}(V)$, we will prove that if $T$ has at most one distinct eigenvalue then $\det T \geq 0$. If $T$ has no eigenvalues then Exercise 9.C.8 shows that $\det T > 0$, so suppose



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that $T$ has exactly one eigenvalue $\lambda$ and let $\dim V = 2n$ for some positive integer $n$. It follows that the characteristic polynomial $p$ of $T$ is given by $p(z) = (z-\lambda)^{2n}$. Now observe that

$$\det T = (-1)^{2n} \det T = \det(-T) = p(0) = (-\lambda)^{2n} \geq 0.$$

**Exercise 9.C.10.** Suppose $V$ is a real vector space of odd dimension and $T \in \mathcal{L}(V)$. Without using the minimal polynomial, prove that $T$ has an eigenvalue.

*This result was previously proved without using determinants or the characteristic polynomial—see 5.34.*

**Solution.** The characteristic polynomial $p$ of $T$ is a polynomial of odd degree with real coefficients. It follows from Exercise 4.9 that $p$ has a real zero, i.e. $T$ has an eigenvalue.

**Exercise 9.C.11.** Prove or give a counterexample: If $\mathbf{F} = \mathbf{R}, T \in \mathcal{L}(V)$, and $\det T > 0$, then $T$ has a square root.

*If $\mathbf{F} = \mathbf{C}, T \in \mathcal{L}(V)$ and $\det T \neq 0$, then $T$ has a square root (see 8.41).*

**Solution.** This is false. For a counterexample, let $T$ be the operator on $\mathbf{R}^2$ whose matrix with respect to the standard basis is

$$A = \begin{pmatrix} -1 & 1 \\ 0 & -1 \end{pmatrix}.$$

Thus $\det T = \det A = 1$ by 9.48. We claim that $T$ has no square root. It will suffice to show that for any matrix

$$M = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathbf{R}^{2,2}$$

we have $M^2 \neq A$. Indeed, observe that

$$M^2 = \begin{pmatrix} a^2 + bc & b(a+d) \\ c(a+d) & bc + d^2. \end{pmatrix}$$

Equating this to $A$, we must have $c(a+d) = 0$. If $c = 0$ then $a^2 = -1$, which cannot hold for $a \in \mathbf{R}$, so it must be the case that $a = -d$. But now $b(a+d) = 0 \neq 1$. Thus there does not exist $M \in \mathbf{R}^{2,2}$ satisfying $M^2 = A$.

**Exercise 9.C.12.** Suppose $S, T \in \mathcal{L}(V)$ and $S$ is invertible. Define $p : \mathbf{F} \to \mathbf{F}$ by

$$p(z) = \det(zS - T).$$

Prove that $p$ is a polynomial of degree $\dim V$ and that the coefficient of $z^{\dim V}$ in this polynomial is $\det S$.

**Solution.** Let $n = \dim V$ and observe that



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$$p(z) = \det(zS - T) = \det\big(S(zI - S^{-1}T)\big) = (\det S)\big(\det(zI - S^{-1}T)\big) = (\det S)q(z),$$

where $q$ is the characteristic polynomial of $S^{-1}T$. Since $\det S \neq 0$ (because $S$ is invertible) we see that $p$ is a polynomial of degree $n$. Furthermore, by 9.65, the coefficient of $z^n$ in the polynomial $p$ is $\det S$.

**Exercise 9.C.13.** Suppose $\mathbf{F} = \mathbf{C}, T \in \mathcal{L}(V)$, and $n = \dim V > 2$. Let $\lambda_1, ..., \lambda_n$ denote the eigenvalues of $T$, with each eigenvalue included as many times as its multiplicity.

(a) Find a formula for the coefficient of $z^{n-2}$ in the characteristic polynomial of $T$ in terms of $\lambda_1, ..., \lambda_n$.

(b) Find a formula for the coefficient of $z$ in the characteristic polynomial of $T$ in terms of $\lambda_1, ..., \lambda_n$.

**Solution.**

(a) By 9.62, the characteristic polynomial $p$ of $T$ is given by

$$p(z) = (z - \lambda_1) \cdots (z - \lambda_n).$$

Multiplying $p$ out involves making a binary choice for each factor (either choose $z$ or $-\lambda_k$), summing over all $2^n$ choices, and collecting like powers. To find the coefficient of $z^{n-2}$, we should choose $z$ from $n-2$ of the factors; equivalently, we should choose $-\lambda_k$ from 2 of the factors. That is, for each choice of $1 \leq i < j \leq n$ we obtain a contribution of $(-\lambda_i)\big(-\lambda_j\big) = \lambda_i\lambda_j$ to the coefficient of $z^{n-2}$. Summing over all such choices, we see that the coefficient of $z^{n-2}$ in $p$ is

$$\sum_{1 \leq i < j \leq n} \lambda_i \lambda_j.$$

(b) As in part (a), to find the coefficient of $z$ we should choose $z$ from exactly one of the factors, which is equivalent to choosing $-\lambda_k$ from $n-1$ of the factors. In other words, we should omit exactly one of the $\lambda_k$'s, i.e. for each choice of $1 \leq k \leq n$ we obtain a contribution of

$$(-\lambda_1) \cdots (-\lambda_{k-1})(-\lambda_{k+1}) \cdots (-\lambda_n) = (-1)^{n-1}(\lambda_1 \cdots \lambda_{k-1}\lambda_{k+1} \cdots \lambda_n)$$

to the coefficient of $z$. If we write $\lambda_1 \cdots \widehat{\lambda_k} \cdots \lambda_n$ to mean the product of all the eigenvalues $\lambda_1, ..., \lambda_n$ except for $\lambda_k$, then the coefficient of $z$ in $p$ is

$$(-1)^{n-1} \sum_{k=1}^{n} \lambda_1 \cdots \widehat{\lambda_k} \cdots \lambda_n.$$

**Exercise 9.C.14.** Suppose $V$ is an inner product space and $T$ is a positive operator on $V$. Prove that

$$\det \sqrt{T} = \sqrt{\det T}.$$



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**Solution.** By 7.38(c) there is an orthonormal basis $e_1, ..., e_n$ of $V$ with respect to which the matrix of $T$ is of the form

$$\begin{pmatrix} \lambda_1 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & \lambda_n \end{pmatrix}$$

for some non-negative real numbers $\lambda_1, ..., \lambda_n$. It is then clear that the matrix of $\sqrt{T}$ with respect to $e_1, ..., e_n$ is

$$\begin{pmatrix} \sqrt{\lambda_1} & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & \sqrt{\lambda_n} \end{pmatrix}.$$

It follows from 9.48 that

$$\det \sqrt{T} = \sqrt{\lambda_1} \cdots \sqrt{\lambda_n} = \sqrt{\lambda_1 \cdots \lambda_n} = \sqrt{\det T}.$$

**Exercise 9.C.15.** Suppose $V$ is an inner product space and $T \in \mathcal{L}(V)$. Use the polar decomposition to give a proof that

$$|\det T| = \sqrt{\det(T^*T)}$$

that is different from the proof given earlier (see 9.60).

**Solution.** By the polar decomposition (7.93) there exists a unitary operator $S \in \mathcal{L}(V)$ such that $T = S\sqrt{T^*T}$. Now observe that

$$|\det T| = \left|\det\left(S\sqrt{T^*T}\right)\right| = \left|(\det S)\left(\det\left(\sqrt{T^*T}\right)\right)\right| = |\det S|\left|\sqrt{\det(T^*T)}\right| = \sqrt{\det(T^*T)},$$

where the second equality follows from 9.49, the third equality follows from Exercise 9.C.14, and the fourth equality follows from 9.58 and 9.59.

**Exercise 9.C.16.** Suppose $T \in \mathcal{L}(V)$. Define $g : \mathbf{F} \to \mathbf{F}$ by $g(x) = \det(I + xT)$. Show that $g'(0) = \operatorname{tr} T$.

*Look for a clean solution to this exercise, without using the explicit but complicated formula for the determinant of a matrix.*

**Solution.** Let $n = \dim V$ and let $p$ be the characteristic polynomial of $-T$. For $x \neq 0$, observe that

$$g(x) = \det(I + xT) = x^n \det\left(\frac{1}{x}I + T\right) = x^n p\left(\frac{1}{x}\right) = 1 + (\operatorname{tr} T)x + \cdots + (\det T)x^n,$$

where we have used 9.65 for the last equality. Since $g(0) = \det I = 1$, we see that this formula holds for all $x \in \mathbf{F}$. Thus $g'(0) = \operatorname{tr} T$.



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**Exercise 9.C.17.** Suppose $a, b, c$ are positive numbers. Find the volume of the ellipsoid

$$\left\{(x, y, z) \in \mathbf{R}^3 : \frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} < 1\right\}$$

by finding a set $\Omega \subseteq \mathbf{R}^3$ whose volume you know and an operator $T$ on $\mathbf{R}^3$ such that $T(\Omega)$ equals the ellipsoid above.

**Solution.** Let $e_1, e_2, e_3$ be the standard orthonormal basis of $\mathbf{R}^3$, let $\Omega = \{v \in \mathbf{R}^3 : \|v\| < 1\}$, and define $T \in \mathcal{L}(\mathbf{R}^3)$ by $Te_1 = ae_1, Te_2 = be_2$, and $Te_3 = ce_3$. Then as the proof of 7.99 shows,

$$T(\Omega) = E(ae_1, be_2, ce_3) = \left\{(x, y, z) \in \mathbf{R}^3 : \frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} < 1\right\}.$$

Thus the volume of the ellipsoid in question is given by 9.61:

$$\text{volume}\, T(\Omega) = |\det T|(\text{volume}\, \Omega) = \frac{4abc\pi}{3}.$$

**Exercise 9.C.18.** Suppose that $A$ is an invertible square matrix. Prove that Hadamard's inequality (9.66) is an equality if and only if each column of $A$ is orthogonal to the other columns.

**Solution.** Let $v_1, ..., v_n$ be the columns of $A$, let $e_1, ..., e_n$ be the result of applying the Gram-Schmidt procedure to $v_1, ..., v_n$, and let $A = QR$ be the QR factorization (see 7.58) of $A$, i.e. $Q$ is the unitary matrix whose columns are $e_1, ..., e_n$ and $R$ is the upper-triangular matrix with positive diagonal entries whose $(j, k)^{\text{th}}$ entry is $R_{j,k} = \langle v_k, e_j \rangle$. By studying the Gram-Schmidt procedure, we see that the columns of $A$ are orthogonal to each other if and only if the list $e_1, ..., e_n$ is given by

$$e_1 = \frac{v_1}{\|v_1\|}, \quad ..., \quad e_n = \frac{v_n}{\|v_n\|}. \tag{1}$$

Furthermore, the proof of Hadamard's inequality (9.66) shows that the inequality is an equality if and only if

$$R_{1,1} \cdots R_{n,n} = \|R_{\cdot,1}\| \cdots \|R_{\cdot,n}\|. \tag{2}$$

Suppose that the columns of $A$ are orthogonal to each other. It follows from (1) and the definition of $R$ that $R$ is a diagonal matrix whose $k^{\text{th}}$ diagonal entry is $\|v_k\|$, from which it follows that both sides of (2) are equal to $\|v_1\| \cdots \|v_n\|$. Thus Hadamard's inequality is an equality.

Now suppose that Hadamard's inequality is an equality, so that (2) holds. It follows that each inequality $R_{k,k} \leq \|R_{\cdot,k}\|$ must be an equality, otherwise the left-hand side of (2) would



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be strictly less than the right-hand side. Thus each off-diagonal entry of $R$ must be zero, i.e. $\langle v_k, e_j \rangle = 0$ for $j \neq k$. It follows that at the $k^{\text{th}}$ stage of the Gram-Schmidt procedure we set $e_k = \|v_k\|^{-1} v_k$, so that (1) holds. Thus the columns of $A$ are orthogonal to each other.

**Exercise 9.C.19.** Suppose $V$ is an inner product space, $e_1, ..., e_n$ is an orthonormal basis of $V$, and $T \in \mathcal{L}(V)$ is a positive operator.

(a) Prove that $\det T \leq \prod_{k=1}^{n} \langle Te_k, e_k \rangle$.

(b) Prove that if $T$ is invertible, then the inequality in (a) is an equality if and only if $e_k$ is an eigenvector of $T$ for each $k = 1, ..., n$.

**Solution.**

(a) Let $B$ be the matrix of $\sqrt{T}$ with respect to $e_1, ..., e_n$ and suppose the columns of $B$ are $v_1, ..., v_n$, so that $\|v_k\| = \|\sqrt{T} e_k\|$ for each $k \in \{1, ..., n\}$ (the norm on the left-hand side is the usual norm on $\mathbf{F}^n$ and the norm on the right-hand side is the norm on $V$). Note that $\det T$ and $\det \sqrt{T}$ are non-negative since $T$ and $\sqrt{T}$ are positive operators (see 9.59). It follows from Hadamard's inequality (9.66) that

$$\det \sqrt{T} = \det B \leq \prod_{k=1}^{n} \|v_k\| = \prod_{k=1}^{n} \|\sqrt{T} e_k\| = \prod_{k=1}^{n} \sqrt{\langle Te_k, e_k \rangle}.$$

Thus

$$\det T = \left(\sqrt{\det T}\right)^2 = \left(\det \sqrt{T}\right)^2 \leq \prod_{k=1}^{n} \langle Te_k, e_k \rangle,$$

where we have used Exercise 9.C.14 for the second equality.

(b) Let $\lambda_1, ..., \lambda_n$ be the eigenvalues of $T$; since $T$ is a positive operator, these are also the singular values of $T$ (once sorted into decreasing order; see Exercise 7.E.7). If each $e_k$ is an eigenvector of $T$ then, by 9.60, both sides of the inequality in (a) equal $\lambda_1 \cdots \lambda_n$.

Now suppose that the inequality in (a) is an equality. Since we only used Hadamard's inequality to derive the inequality in (a) and $T$ being invertible implies $\sqrt{T}$ is invertible, Exercise 9.C.19 shows that the columns of $B$ (as defined in (a)) must be orthogonal to each other. It follows that $B^* B$ is a diagonal matrix, since its $(j,k)^{\text{th}}$ entry is $\langle v_j, v_k \rangle$. Now observe that, by the self-adjointness of $\sqrt{T}$,

$$T = \left(\sqrt{T}\right)^2 = \left(\sqrt{T}\right)^* \left(\sqrt{T}\right) \quad \Rightarrow \quad \mathcal{M}(T, (e_1, ..., e_n)) = B^* B.$$

Thus the matrix of $T$ with respect to $e_1, ..., e_n$ is diagonal, i.e. each $e_k$ is an eigenvector of $T$.



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**Exercise 9.C.20.** Suppose $A$ is an $n$-by-$n$ matrix, and suppose $c$ is such that $|A_{j,k}| \le c$ for all $j, k \in \{1, ..., n\}$. Prove that

$$|\det A| \le c^n n^{n/2}.$$

*The formula for the determinant of a matrix (9.46) shows that $|\det A| \le c^n n!$. However, the estimate given by this exercise is much better. For example, if $c = 1$ and $n = 100$, then $c^n n! \approx 10^{158}$, but the estimate given by this exercise is the much smaller number $10^{100}$. If $n$ is an integer power of 2, then the inequality above is sharp and cannot be improved.*

**Solution.** Suppose the columns of $A$ are $v_1, ..., v_n$. For any $k \in \{1, ..., n\}$, observe that

$$\|v_k\|^2 = \sum_{j=1}^{n} |A_{j,k}|^2 \le c^2 n \quad \Rightarrow \quad \|v_k\| \le c n^{1/2}.$$

It follows from this inequality and Hadamard's inequality (9.66) that

$$|\det A| \le \prod_{k=1}^{n} \|v_k\| \le c^n n^{n/2}.$$

**Exercise 9.C.21.** Suppose $n$ is a positive integer and $\delta : \mathbf{C}^{n,n} \to \mathbf{C}$ is a function such that

$$\delta(AB) = \delta(A) \cdot \delta(B)$$

for all $A, B \in \mathbf{C}^{n,n}$ and $\delta(A)$ equals the product of the diagonal entries of $A$ for each diagonal matrix $A \in \mathbf{C}^{n,n}$. Prove that

$$\delta(A) = \det A$$

for all $A \in \mathbf{C}^{n,n}$.

*Recall that $\mathbf{C}^{n,n}$ denotes the set of $n$-by-$n$ matrices with entries in $\mathbf{C}$. This exercise shows that the determinant is the unique function defined on square matrices that is multiplicative and has the desired behavior on diagonal matrices. This result is analogous to Exercise 10 in Section 8D, which shows that the trace is uniquely determined by its algebraic properties.*

**Solution.** First suppose that $A$ is not invertible. It follows from Exercise 3.C.5 that there are matrices $B, C$, and $D$ such that $A = BDC$, where $D$ is diagonal with at least one diagonal entry equal to zero. Thus $\delta(D) = 0$ and hence

$$\delta(A) = \delta(B)\delta(D)\delta(C) = 0 = \det A,$$

where we have used 9.50 for the last equality.



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To prove that $\delta(A) = \det A$ for invertible $A$, let us first prove the following. For $i, j \in \{1, ..., n\}$ such that $i \neq j$ and $\alpha \in \mathbf{C}$, let $C_{i,j}(\alpha)$ be the identity matrix except for an $\alpha$ in the $i^{\text{th}}$ row and $j^{\text{th}}$ column. For any $A \in \mathbf{C}^{n,n}$, a calculation shows that $AC_{i,j}(\alpha)$ is the matrix obtained from $A$ by adding $\alpha$ times column $i$ to column $j$. We claim that $\delta\big(C_{i,j}(\alpha)\big) = 1$. For ease of notation, let $C = C_{i,j}(\alpha)$. Note that $C$ is invertible (its inverse is $C_{i,j}(-\alpha)$). Let $B$ be the identity matrix except for $B_{j,j} = \frac{1}{2}$ and let $D$ be the identity matrix except for $D_{j,j} = 2$. A calculation shows that $C = BCDC^{-1}$ and it follows that

$$\delta(C) = \delta(B)\delta(C)\delta(D)\delta\big(C^{-1}\big) = \tfrac{1}{2} \cdot \delta(C) \cdot 2 \cdot \delta\big(C^{-1}\big) = \delta\big(CC^{-1}\big) = \delta(I) = 1,$$

as claimed.

Now suppose that $A$ is invertible. By 5.47 and 3.84, there exists an invertible matrix $B$ and an invertible upper-triangular matrix $U$ such that $A = B^{-1}UB$, from which it follows that

$$\delta(A) = \delta\big(B^{-1}\big)\delta(U)\delta(B) = \delta(U)\delta\big(B^{-1}B\big) = \delta(U)\delta(I) = \delta(U).$$

Since $\det A = \det U$ by 9.52, it will suffice to show that $\delta(U) = \det U$. Suppose that the diagonal entries of $U$ are $\lambda_1, ..., \lambda_n$; note that each $\lambda_k$ is non-zero since $U$ is invertible and that $\det U = \lambda_1 \cdots \lambda_n$ by 9.48. For any matrix of the form $C_{i,j}(\alpha)$, observe that

$$\delta\big(UC_{i,j}(\alpha)\big) = \delta(U)\delta\big(C_{i,j}(\alpha)\big) = \delta(U).$$

Given that $U$ is upper-triangular with non-zero diagonal entries, we can multiply $U$ on the right by successive matrices of the form $C_{i,j}(\alpha)$ until we obtain a diagonal matrix $D$ with diagonal entries $\lambda_1, ..., \lambda_n$; as we just showed, this has no effect on the value of $\delta$, i.e. $\delta(D) = \delta(U)$. Thus

$$\delta(U) = \delta(D) = \lambda_1 \cdots \lambda_n = \det U,$$

as desired.



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## 9.D. Tensor Products

**Exercise 9.D.1.** Suppose $v \in V$ and $w \in W$. Prove that $v \otimes w = 0$ if and only if $v = 0$ or $w = 0$.

**Solution.** If $v = 0$ then $\varphi(v) = 0$ for any $\varphi \in V'$ and if $w = 0$ then $\tau(w) = 0$ for any $\tau \in W'$. Thus if $v = 0$ or $w = 0$ then

$$(v \otimes w)(\varphi, \tau) = \varphi(v)\tau(w) = 0$$

for any $\varphi \in V'$ and any $\tau \in W'$. Hence $v \otimes w = 0$.

If $v \neq 0$ and $w \neq 0$ then, by Exercise 3.F.3, there exist $\varphi \in V'$ and $\tau \in W'$ such that $\varphi(v) = \tau(w) = 1$. It follows that

$$(v \otimes w)(\varphi, \tau) = \varphi(v)\tau(w) = 1.$$

Thus $v \otimes w \neq 0$.

**Exercise 9.D.2.** Give an example of six distinct vectors $v_1, v_2, v_3, w_1, w_2, w_3 \in \mathbf{R}^3$ such that

$$v_1 \otimes w_1 + v_2 \otimes w_2 + v_3 \otimes w_3 = 0$$

but none of $v_1 \otimes w_1, v_2 \otimes w_2, v_3 \otimes w_3$ is a scalar multiple of another element of this list.

**Solution.** Let

$$v_1 = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}, \quad v_2 = \begin{pmatrix} -1 \\ -1 \\ -1 \end{pmatrix}, \quad v_3 = \begin{pmatrix} 2 \\ 2 \\ 2 \end{pmatrix}, \quad w_1 = \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix}, \quad w_2 = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}, \quad w_3 = \begin{pmatrix} 0 \\ \frac{1}{2} \\ \frac{1}{2} \end{pmatrix}.$$

By identifying $v_k \otimes w_k$ with the matrix $v_k w_k^{\mathrm{t}}$ (see 9.76), we find that

$$v_1 \otimes w_1 = \begin{pmatrix} 1 & -1 & 0 \\ 1 & -1 & 0 \\ 1 & -1 & 0 \end{pmatrix}, \quad v_2 \otimes w_2 = \begin{pmatrix} -1 & 0 & -1 \\ -1 & 0 & -1 \\ -1 & 0 & -1 \end{pmatrix}, \quad v_3 \otimes w_3 = \begin{pmatrix} 0 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 1 & 1 \end{pmatrix}.$$

Since the sum of these three matrices is zero and none of them is a scalar multiple of another, $v_1 \otimes w_1, v_2 \otimes w_2, v_3 \otimes w_3$ is the desired list.

**Exercise 9.D.3.** Suppose that $v_1, \ldots, v_m$ is a linearly independent list in $V$. Suppose also that $w_1, \ldots, w_m$ is a list in $W$ such that

$$v_1 \otimes w_1 + \cdots + v_m \otimes w_m = 0.$$

Prove that $w_1 = \cdots = w_m = 0$.



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**Solution.** We will prove the contrapositive. Suppose that $w_k \neq 0$ for some $k \in \{1, ..., m\}$. By Exercise 3.F.3 there exists some $\tau \in W'$ such that $\tau(w_k) \neq 0$. As in the proof of 9.74, let $\varphi_1, ..., \varphi_m \in V'$ be such that $\varphi_i(v_j) = 1$ if $i = j$ and $\varphi_i(v_j) = 0$ otherwise. Now observe that

$$[v_1 \otimes w_1 + \cdots + v_m \otimes w_m](\varphi_k, \tau) = \varphi_k(v_1)\tau(w_1) + \cdots + \varphi_k(v_k)\tau(w_k) + \cdots + \varphi_k(v_m)\tau(w_m) = \tau(w_k) \neq 0.$$

Thus $v_1 \otimes w_1 + \cdots + v_m \otimes w_m \neq 0$.

**Exercise 9.D.4.** Suppose $\dim V > 1$ and $\dim W > 1$. Prove that

$$\{v \otimes w : (v, w) \in V \times W\}$$

is not a subspace of $V \otimes W$.

*This exercise implies that if* $\dim V > 1$ *and* $\dim W > 1$, *then*

$$\{v \otimes w : (v, w) \in V \times W\} \neq V \otimes W.$$

**Solution.** Let $e_1, e_2, ..., e_m$ be a basis of $V$ with dual basis $\varphi_1, \varphi_2, ..., \varphi_m$, and let $f_1, f_2, ..., f_n$ be a basis of $W$ with dual basis $\tau_1, \tau_2, ..., \tau_n$. Let

$$U = \{v \otimes w : (v, w) \in V \times W\}.$$

Observe that

$$[e_1 \otimes f_1 + e_2 \otimes f_2](\varphi_i, \tau_j) = \begin{cases} 1 & \text{if } i = j \in \{1, 2\}, \\ 0 & \text{otherwise.} \end{cases}$$

For any $v = a_1e_1 + a_2e_2 + \cdots + a_me_m \in V$ and any $w = b_1f_1 + b_2f_2 + \cdots + b_nf_n \in W$, observe that $(v \otimes w)(\varphi_i, \tau_j) = a_ib_j$. Equating $e_1 \otimes f_1 + e_2 \otimes f_2 = v \otimes w$ then gives us the system of equations

$$a_1b_1 = a_2b_2 = 1 \quad \text{and} \quad a_1b_2 = a_2b_1 = 0,$$

which has no solution. Thus $e_1 \otimes f_1 + e_2 \otimes f_2 \neq v \otimes w$ for any $(v, w) \in V \times W$, which shows that $U$ is not closed under addition and hence is not a subspace of $V \otimes W$.

**Exercise 9.D.5.** Suppose $m$ and $n$ are positive integers. For $v \in \mathbf{F}^m$ and $w \in \mathbf{F}^n$, identify $v \otimes w$ with an $m$-by-$n$ matrix as in Example 9.76. With that identification, show that the set

$$\{v \otimes w : v \in \mathbf{F}^m \text{ and } w \in \mathbf{F}^n\}$$

is the set of $m$-by-$n$ matrices (with entries in $\mathbf{F}$) that have rank at most one.

**Solution.** Thinking of $v$ and $w$ as column vectors, Example 9.76 shows that we can identify $v \otimes w$ with the $m$-by-$n$ matrix $vw^t$. The desired result is now immediate from Exercise 3.C.16.



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**Exercise 9.D.6.** Suppose $m$ and $n$ are positive integers. Give a description, analogous to Exercise 5, of the set of $m$-by-$n$ matrices (with entries in $\mathbf{F}$) that have rank at most two.

**Solution.** Let $M$ be the collection of $m$-by-$n$ matrices with entries in $\mathbf{F}$ that have rank at most two. We claim that $M$ can be identified with the set

$$\{u \otimes w + v \otimes x : u, v \in \mathbf{F}^m \text{ and } w, x \in \mathbf{F}^n\}$$

by identifying $u \otimes w$ with $uw^t$ (thinking of $u$ and $w$ as column vectors) as in example 9.76. This amounts to showing that

$$M = \{uw^t + vx^t : u, v \in \mathbf{F}^{m,1} \text{ and } w, x \in \mathbf{F}^{n,1}\}.$$

For $u, v \in \mathbf{F}^{m,1}$ and $w, x \in \mathbf{F}^{n,1}$, observe that $uw^t + vx^t$ is the matrix whose $k^{\text{th}}$ column is $w_k u + x_k v$, from which it is clear that the span of the columns of $uw^t + vx^t$ is equal to the span of $u$ and $v$. Thus $uw^t + vx^t$ has rank at most two.

**Exercise 9.D.7.** Suppose $\dim V > 2$ and $\dim W > 2$. Prove that

$$\{v_1 \otimes w_1 + v_2 \otimes w_2 : v_1, v_2 \in V \text{ and } w_1, w_2 \in W\} \neq V \otimes W.$$

**Solution.** Suppose $e_1, ..., e_m$ is a basis of $V$ and $f_1, ..., f_n$ is a basis of $W$, where $m, n \geq 3$. By identifying an $m$-by-$n$ matrix $A$ with the bilinear functional

$$\sum_{k=1}^{n} \sum_{j=1}^{m} A_{j,k}(e_j \otimes f_k)$$

as in the proof of 9.74, we can identify $V \otimes W$ with $\mathbf{F}^{m,n}$. Under this identification the set

$$E := \{v_1 \otimes w_1 + v_2 \otimes w_2 : v_1, v_2 \in V \text{ and } w_1, w_2 \in W\}$$

corresponds to the set of $m$-by-$n$ matrices that have rank at most two, as we showed in Exercise 9.D.6. Given that $m, n \geq 3$, there exist matrices in $\mathbf{F}^{m,n}$ with rank strictly greater than two. Thus $E \neq V \otimes W$.

**Exercise 9.D.8.** Suppose $v_1, ..., v_m \in V$ and $w_1, ..., w_m \in W$ are such that

$$v_1 \otimes w_1 + \cdots + v_m \otimes w_m = 0.$$

Suppose that $U$ is a vector space and $\Gamma : V \times W \to U$ is a bilinear map. Show that

$$\Gamma(v_1, w_1) + \cdots + \Gamma(v_m, w_m) = 0.$$

**Solution.** 9.79(a) shows that there exists a linear map $\hat{\Gamma} : V \otimes W \to U$ such that

$$\hat{\Gamma}(v \otimes w) = \Gamma(v, w)$$

for all $v \in V$ and all $w \in W$. It follows that



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$$\Gamma(v_1, w_1) + \cdots + \Gamma(v_m, w_m) = \hat{\Gamma}(v_1 \otimes w_1) + \cdots + \hat{\Gamma}(v_m \otimes w_m)$$
$$= \hat{\Gamma}(v_1 \otimes w_1 + \cdots + v_m \otimes w_m) = \hat{\Gamma}(0) = 0.$$

**Exercise 9.D.9.** Suppose $S \in \mathcal{L}(V)$ and $T \in \mathcal{L}(W)$. Prove that there exists a unique operator on $V \otimes W$ that takes $v \otimes w$ to $Sv \otimes Tw$ for all $v \in V$ and $w \in W$.

*In an abuse of notation, the operator on $V \otimes W$ given by this exercise is often called $S \otimes T$.*

**Solution.** Define a bilinear map $\Gamma : V \times W \to V \otimes W$ by $\Gamma(v, w) = Sv \otimes Tw$. By 9.79, there is a unique operator $\hat{\Gamma} \in \mathcal{L}(V \otimes W)$ such that

$$\hat{\Gamma}(v \otimes w) = \Gamma(v, w) = Sv \otimes Tw.$$

**Exercise 9.D.10.** Suppose $S \in \mathcal{L}(V)$ and $T \in \mathcal{L}(W)$. Prove that $S \otimes T$ is an invertible operator on $V \otimes W$ if and only if both $S$ and $T$ are invertible operators. Also, prove that if both $S$ and $T$ are invertible operators, then $(S \otimes W)^{-1} = S^{-1} \otimes T^{-1}$, where we are using the notation from the comment after Exercise 9.

**Solution.** If $S$ and $T$ are both invertible then

$$(S^{-1} \otimes T^{-1})(S \otimes W)(v \otimes w) = (S^{-1} \otimes T^{-1})(Sv \otimes Tw) = S^{-1}Sv \otimes T^{-1}Tw = v \otimes w$$

for any $(v, w) \in V \times W$. Thus $S \otimes W$ is invertible and $(S \otimes W)^{-1} = S^{-1} \otimes T^{-1}$.

Now suppose that $S$ is not invertible (the case where $T$ is not invertible is handled similarly). There exists a non-zero $v \in V$ such that $Sv = 0$. Let $w \in W$ be non-zero (we may as well assume $W \neq 0$) and note that $v \otimes w \neq 0$ by Exercise 9.D.1. Now observe that

$$(S \otimes T)(v \otimes w) = Sv \otimes Tw = 0 \otimes Tw = 0.$$

Thus $S \otimes T$ is not invertible.

**Exercise 9.D.11.** Suppose $V$ and $W$ are inner product spaces. Prove that if $S \in \mathcal{L}(V)$ and $T \in \mathcal{L}(W)$, then $(S \otimes T)^* = S^* \otimes T^*$, where we are using the notation from the comment after Exercise 9.

**Solution.** For any $u, v \in V$ and any $w, x \in W$, observe that

$$\begin{aligned}\langle (S \otimes T)(v \otimes w), u \otimes x \rangle &= \langle Sv \otimes Tw, u \otimes x \rangle \\ &= \langle Sv, u \rangle \langle Tw, x \rangle \\ &= \langle u, S^*u \rangle \langle w, T^*x \rangle \\ &= \langle v \otimes w, S^*u \otimes T^*x \rangle \\ &= \langle v \otimes w, (S^* \otimes T^*)(u \otimes x) \rangle.\end{aligned}$$



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Thus $(S \otimes T)^* = S^* \otimes T^*$.

**Exercise 9.D.12.** Suppose that $V_1, ..., V_m$ are finite-dimensional inner product spaces. Prove that there is a unique inner product on $V_1 \otimes \cdots \otimes V_m$ such that

$$\langle v_1 \otimes \cdots \otimes v_m, u_1 \otimes \cdots \otimes u_m \rangle = \langle v_1, u_1 \rangle \cdots \langle v_m, u_m \rangle$$

for all $(v_1, ..., v_m)$ and $(u_1, ..., u_m)$ in $V_1 \times \cdots \times V_m$.

*Note that the equation above implies that*

$$\|v_1 \otimes \cdots \otimes v_m\| = \|v_1\| \times \cdots \times \|v_m\|$$

*for all* $(v_1, ..., v_m) \in V_1 \times \cdots \times V_m$.

**Solution.** For each $k \in \{1, ..., m\}$ let $e_1^k, ..., e_{n_k}^k$ be an orthonormal basis of $V_k$. A very tedious calculation shows that

$$\left\langle \sum_{j_1=1}^{n_1} \cdots \sum_{j_m=1}^{n_m} b_{j_1,...,j_m} \left(e_{j_1}^1 \otimes \cdots \otimes e_{j_m}^m\right), \sum_{j_1=1}^{n_1} \cdots \sum_{j_m=1}^{n_m} c_{j_1,...,j_m} \left(e_{j_1}^1 \otimes \cdots \otimes e_{j_m}^m\right) \right\rangle$$

$$\sum_{j_1=1}^{n_1} \cdots \sum_{j_m=1}^{n_m} b_{j_1,...,j_m} \overline{c_{j_1,...,j_m}}$$

is the desired inner product.

**Exercise 9.D.13.** Suppose that $V_1, ..., V_m$ are finite-dimensional inner product spaces and $V_1 \otimes \cdots \otimes V_m$ is made into an inner product space using the inner product from Exercise 12. Suppose $e_1^k, ..., e_{n_k}^k$ is an orthonormal basis of $V_k$ for each $k = 1, ..., m$. Show that the list

$$\left\{e_{j_1}^1 \otimes \cdots \otimes e_{j_m}^m\right\}_{j_1=1,...,n_1;\cdots;j_m=1,...,n_m}$$

is an orthonormal basis of $V_1 \otimes \cdots \otimes V_m$.

**Solution.** The list in question is a basis of $V_1 \otimes \cdots \otimes V_m$ by 9.90. To verify orthonormality, suppose that $i_k, j_k \in \{1, ..., n_k\}$ for each $k \in \{1, ..., m\}$ and observe that

$$\left\langle e_{i_1}^1 \otimes \cdots \otimes e_{i_m}^m, e_{j_1}^1 \otimes \cdots \otimes e_{j_m}^m \right\rangle = \left\langle e_{i_1}^1, e_{j_1}^1 \right\rangle \cdots \left\langle e_{i_m}^m, e_{j_m}^m \right\rangle = \begin{cases} 1 & \text{if } i_1 = j_1, ..., i_m = j_m, \\ 0 & \text{otherwise.} \end{cases}$$



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